

Medical Dictionary by Specialty

A Comprehensive Reference of Medical Terminology

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Preface

Medical dictionaries have traditionally been organised alphabetically, an arrangement well suited for looking up an unfamiliar term when you already know its name. But medicine is not learned alphabetically, nor is it practised that way. A cardiologist thinking through a differential diagnosis reaches for cardiac terms, not a mixture of cardiology wedged between caesarean section and cataract.

This dictionary is organised by medical specialty for precisely that reason. Each of the 43 chapters corresponds to a recognised clinical or preclinical discipline, with terms listed alphabetically within that chapter. The result is a reference that mirrors the structure of medical knowledge itself: renal terms in Nephrology, antiarrhythmics in Cardiology, autoimmune diseases in Rheumatology. Cross-references redirect the reader from variant names and alternative spellings to the main entry, so “Pertussis” still leads to “Whooping Cough” and “Hepatolenticular Degeneration” points to “Wilson Disease.”

The entries are drawn primarily from Black’s Medical Dictionary (42nd Edition) and supplemented with definitions reflecting contemporary clinical practice. Each entry is written in a concise single-line format, with the term displayed in bold red and automatically indexed. The dictionary runs to over 5,500 entries across 960 pages, covering anatomy, physiology, pathology, pharmacology, clinical medicine, surgery, and the allied health sciences.

This work is a living document. The LaTeX source is designed to be easily extended, corrected, and reorganised. Contributions, corrections, and suggestions are welcome.

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Chapter 1

Pocket Dictionary: A Terms

5-Alpha Reductase Inhibitor

- A class of medications that inhibit the enzyme 5-alpha-reductase, which converts testosterone to dihydrotestosterone (DHT), the more potent androgen responsible for prostate growth and male pattern baldness. Two isoforms: Type I (predominant in skin and scalp) and Type II (predominant in the prostate).

A-a Gradient

- PAO₂ minus PaO₂, reflecting gas transfer efficiency. Normal approximately 15 mmHg (increases with age: $< 15 + \text{age}/4$). Elevated in V/Q mismatch, shunt, and diffusion impairment. Normal gradient in hypoventilation.

A1c

- A glycated form of hemoglobin that reflects average blood glucose over approximately the preceding two to three months. It is used to diagnose and monitor diabetes, although anemia, hemoglobin variants, altered red-cell survival, and assay limitations can affect interpretation.

AAA Repair

- See abdominal aortic aneurysm.

Abasia

- Inability to walk despite preserved strength or the ability to move the legs in other circumstances, usually reflecting a disorder of gait coordination or another neurologic condition rather than paralysis alone.

ABCDE Assessment

- A structured emergency assessment of Airway, Breathing, Circulation, Disability, and Exposure. Life-threatening abnormalities are identified and treated immediately before proceeding to the next component, with repeated reassessment after inter-

ventions.

ABCDE Rule for Melanoma

- A clinical screening mnemonic for suspicious pigmented lesions: Asymmetry, Border irregularity, Color variation, Diameter greater than approximately 6 mm, and Evolution or change. It is a screening aid rather than a diagnostic test; suspicious lesions require dermoscopic assessment and biopsy.

Abdomen

- The body region between the thorax and pelvis, containing the digestive organs, liver, gallbladder, pancreas, spleen, kidneys, and major blood vessels. Divided into nine regions (right/left hypochondriac, epigastric, umbilical, right/left lumbar, right/left inguinal) or four quadrants (RUQ, LUQ, RLQ, LLQ) for clinical description.

Abdominal Abscess

- A localized collection of pus within the abdomen, usually caused by bacterial infection after appendicitis, bowel perforation, surgery, or another intra-abdominal infection. Symptoms may include fever, abdominal pain, tenderness, and altered bowel function. Treatment usually requires antibiotics and image-guided or surgical drainage.

Abdominal Aorta

- From diaphragmatic hiatus at T12 to common iliac bifurcation at L4. Major branches: celiac trunk (T12), SMA (L1), IMA (L3), renal arteries (L1-L2), gonadal arteries, lumbar arteries. Anterior to the vertebral column. Abdominal aortic aneurysm is common in elderly males (13th leading cause of death).

Abdominal Aortic Aneurysm

- Abnormal focal dilation of blood vessel. See

aneurysm.

Abdominal Aortic Aneurysm Screening

- Ultrasound screening for AAA recommended for men aged 65-75 with smoking history (USPSTF). A one-time abdominal ultrasound detects infrarenal aortic diameter 3 cm. Surveillance: 3.0-3.9 cm q3 years, 4.0-4.4 cm q12 months, 4.5-5.4 cm q6 months. Surgical referral at 5.5 cm or rapid enlargement (>1 cm/yr).

Abdominal Aortic Ultrasound

- Screening for abdominal aortic aneurysm. One-time screening in men 65-75 who smoked.

Abdominal Bloating

- A sensation of fullness or visible enlargement of the abdomen, commonly caused by intestinal gas, constipation, food intolerance, or functional gastrointestinal disorders. Persistent or severe bloating may require evaluation for organic disease.

Abdominal Bruit

- An abnormal blowing or whooshing sound heard over the abdomen with a stethoscope, usually caused by turbulent blood flow through a narrowed or enlarged artery. It may occur with renal-artery stenosis or an abdominal aortic aneurysm. Further assessment depends on symptoms and vascular risk factors.

Abdominal Colic

- Cramping abdominal pain caused by contractions or obstruction of a hollow organ, such as the intestine, ureter, or bile duct. Pain often comes in waves and may be accompanied by nausea, vomiting, sweating, or altered bowel or urinary function. Severe or persistent pain requires prompt medical assessment.

Abdominal Cavity

- The cavity between the diaphragm and pelvis containing the stomach, liver, gallbladder, spleen, pancreas, kidneys, and intestines.

Abdominal Circumference (AC)

- The circumference of the fetal abdomen measured at the level of the stomach, umbilical vein (portal sinus), and spine (ribs not visible). The best single parameter for estimating fetal weight. AC is also the most sensitive measurement for detecting IUGR (the liver is the first organ affected by placental

insufficiency).

Abdominal Compartment Syndrome

- Condition defined by sustained intra-abdominal pressure greater than twenty millimeters of mercury associated with new organ dysfunction. Causes include massive ascites, bowel distension from obstruction or ileus, hemorrhage, and post-surgical conditions particularly after large abdominal wall hernia repairs.

Abdominal CT

- First-line imaging for acute abdominal pain. Evaluates appendicitis, diverticulitis, bowel obstruction, trauma, renal stones, liver masses. Oral and IV contrast often used.

Abdominal Distention

- Visible or subjective enlargement, tightness, or fullness of the abdomen caused by gas, fluid, stool, pregnancy, organ enlargement, mass, or bowel obstruction. Acute distention with pain, vomiting, obstipation, fever, haemodynamic instability, or peritoneal signs requires urgent evaluation.

Abdominal Wall

- The layered muscular, fascial, and skin-covered boundary surrounding the abdominal cavity. It supports and protects abdominal organs and helps with posture, breathing, coughing, and straining. Weakness or defects may produce abdominal or incisional hernias; trauma, infection, and tumors may also affect the wall.

Abdominal Girth

- The circumference of the abdomen measured at a standardized level. It is used to monitor abdominal distention, fetal growth, obesity, ascites, and changes in intra-abdominal volume.

Abdominal Mass

- A palpable or imaging-detected abnormal collection, enlargement, or lesion within the abdomen. Causes include distended bowel or bladder, organ enlargement, hernia, inflammatory collection, benign tumour, and malignancy; assessment depends on location, mobility, tenderness, systemic features, and appropriate imaging.

Abdominal Pain

- Discomfort within the abdomen classified as visceral (dull, midline, poorly localized), parietal

(sharp, well localized, worsened by movement), or referred from a distant source.

Abdominal Pregnancy

- A very rare ectopic pregnancy (1%) implanted in the peritoneal cavity outside the fallopian tubes, ovaries, and uterus. Risk factors: prior ectopic, tubal damage, ART. Diagnosis: ultrasound (empty uterus, gestational sac outside the uterus, fetal parts adjacent to maternal organs).

Abdominal Radiography

- Plain projectional X-ray imaging of the abdomen, sometimes obtained in supine, erect, or decubitus positions.

Abdominal Reflex

- A superficial cutaneous reflex elicited by stroking the abdominal skin toward the umbilicus, producing contraction of the underlying abdominal muscles. Asymmetry may indicate corticospinal tract dysfunction.

Abdominal Rigidity

- Involuntary tensing of the abdominal wall, usually caused by irritation or inflammation of the parietal peritoneum. It is a characteristic sign of peritonitis and may accompany perforation, severe infection, or intra-abdominal hemorrhage.

Abdominal X-Ray

- Plain radiograph of the abdomen (KUB: kidneys, ureters, bladder). Indications: suspected bowel obstruction (air-fluid levels, dilated loops), perforation (free air under diaphragm), constipation, foreign body, and renal calculi (90% of stones are radiopaque). Sensitivity low for most intra-abdominal pathology.

Abducens Nerve

- The sixth cranial nerve, a motor nerve that supplies the lateral rectus muscle and abducts the eye. Lesion of the nerve causes failure of lateral gaze and horizontal diplopia.

Abduction

- Movement of a limb or other body part away from the median plane of the body.

Abetalipoproteinemia

- A rare inherited disorder of lipoprotein assembly that causes very low circulating apoB-containing lipoproteins, fat malabsorption, deficiency of fat-

soluble vitamins, acanthocytosis, and progressive neurologic or retinal disease.

Abfraction

- Loss of tooth structure at cervical area from biomechanical loading. Controversial etiology. May contribute to sensitivity. Treatment: restoration, occlusal adjustment.

ABI

- Ankle-brachial index. Ratio of ankle to arm systolic pressure. Normal: 1.0-1.4. <0.9 indicates PAD.

Ablation Therapy

- Image-guided destruction of tissue. Radiofrequency ablation, cryoablation, microwave ablation. Used for tumors, arrhythmias, pain.

Abnormal Uterine Bleeding

- Uterine bleeding that is abnormal in frequency, regularity, duration, or volume, occurring outside of normal menstruation or in women who are not pregnant.

ABO Blood Group

- The classification of blood according to the presence or absence of A and B antigens on red blood cells and the corresponding antibodies in plasma. The major groups are A, B, AB, and O.

ABO Blood Typing

- Determination of a patient's ABO and Rh blood group prior to transfusion or organ transplantation. ABO system: A (A antigen, anti-B antibody), B (B antigen, anti-A antibody), AB (both antigens, no antibodies), O (no antigens, both antibodies). Rh: D antigen positive (85%) or negative (15%). Universal donor: O negative.

ABO Incompatibility

- An immune reaction caused by transfusion or pregnancy involving incompatible A, B, or O blood-group antigens. It can cause acute hemolysis, transfusion reactions, or hemolytic disease of the newborn. See also Hemolytic Disease of the Newborn.

Abortion

- The termination of a pregnancy by medical or surgical means before the fetus reaches viability (typically before 24 weeks gestation in the UK). Medical abortion uses mifepristone (antiprogesterone) followed by misoprostol (prostaglandin analog) to induce uterine contractions and expel pregnancy

tissue.

Abrasion

- Mechanical wear from foreign objects. Causes: aggressive brushing, toothpick use, nail-biting. Management: modify behavior, restore defects.

Abruptio Placentae

- Premature separation of the placenta from the uterine wall before delivery, occurring in approximately 0.5-1% of pregnancies. Classified as revealed (vaginal bleeding present, blood tracks downward) or concealed (bleeding remains behind the placenta, no visible bleeding, higher risk of coagulopathy).

Abscess

- A localized collection of pus within a tissue, organ, or cavity, typically caused by bacterial infection. Abscesses form when bacteria (most commonly *Staphylococcus aureus* including MRSA) invade tissue, triggering an intense inflammatory response with neutrophil accumulation and tissue necrosis.

Absence Seizure

- A generalized non-motor seizure characterized by sudden, brief impairment of awareness with behavioral arrest, usually lasting a few seconds and followed by immediate recovery. Typical absence seizures show generalized three-hertz spike-and-wave discharges on electroencephalography and are most common in childhood absence epilepsy.

Absent Breath Sounds

- Complete absence of air movement sounds in a lung field, indicating complete airway obstruction, pneumothorax, or large pleural effusion. It is a medical finding requiring urgent evaluation and intervention.

Absolute Neutrophil Count (ANC)

- The calculated number of circulating neutrophils, including segmented and band forms, per unit volume of blood. It is calculated from the white-cell count and neutrophil differential. An ANC below 1,500/L is neutropenia; below 500/L is severe neutropenia with a high risk of serious bacterial and fungal infection.

Absorption

- The process by which a drug enters the bloodstream from its site of administration, such as the digestive tract, skin, or muscle.

Abusive Head Trauma

- Injury to an infant or young child caused by inflicted blunt impact, violent shaking, or both. Findings may include subdural hemorrhage, retinal hemorrhage, encephalopathy, fractures, or other injuries; diagnosis requires careful multidisciplinary assessment because no single finding is diagnostic.

Acalculia

- An acquired inability to perform previously learned numerical operations or understand numerical relationships, commonly associated with dominant parietal or other cortical dysfunction.

Acalculous Cholecystitis

- Acute inflammation of the gallbladder without gallstones, usually occurring in critically ill, postoperative, traumatised, or fasting patients. Presents with fever, right-upper-quadrant pain, leukocytosis, or unexplained sepsis; diagnosis is supported by ultrasound or cholescintigraphy.

Acanthocytes (Spur Cells)

- Abnormal red blood cells with irregular, spiky, thorn-like membrane projections of varying length and width. Caused by alterations in membrane lipid composition (increased cholesterol-to-phospholipid ratio).

Acanthosis Nigricans

- Hyperpigmented, velvety, thickened plaques typically in skin flexures (neck, axillae, groin) and often associated with insulin resistance, obesity, type 2 diabetes mellitus, and metabolic syndrome.

Acapnia

- An abnormally low partial pressure of carbon dioxide in arterial blood, usually caused by excessive alveolar ventilation. It may accompany respiratory alkalosis, light-headedness, or paresthesias.

Acardia

- Absence or severe malformation of the fetal heart, usually occurring in the setting of a rare twin perfusion sequence. The affected fetus cannot sustain independent circulation.

Acariasis

- Infestation or disease caused by mites or ticks. Clinical manifestations vary with the organism and may include dermatitis, pruritus, tissue invasion, or transmission of an associated pathogen.

Acathisia

- A syndrome of inner restlessness accompanied by a compelling need to move, often with pacing or shifting of the legs. It is a clinically recognized movement disorder characterized by observable motor activity and subjective distress.

Access Needle (Interventional)

- See Access Needle.

Accessory Nerve

- The eleventh cranial nerve, formed by cranial and spinal roots. Its spinal component supplies the sternocleidomastoid and trapezius muscles; injury can cause shoulder droop and weak head rotation.

ACE Inhibitor

- A drug that inhibits angiotensin-converting enzyme, reducing formation of angiotensin II and degradation of bradykinin.

ACE Inhibitors

- Angiotensin-converting enzyme inhibitors are a class of medications that block the conversion of angiotensin I to angiotensin II by inhibiting the angiotensin-converting enzyme (ACE, also known as kininase II).

Acetabular Labrum

- A fibrocartilaginous rim attached to the margin of the acetabulum that deepens the hip socket, helps maintain a fluid seal, and contributes to hip-joint stability.

Acetabular Notch

- The indentation in the inferior margin of the acetabulum, bridged by the transverse ligament of the acetabulum and forming a passage for structures entering the hip joint.

Acetabuloplasty

- A surgical procedure that reshapes or augments the acetabular rim or socket to improve coverage of the femoral head in selected structural hip disorders.

Acetabulum

- Cup-shaped lateral hip socket from fused ilium, ischium, and pubis. Articulates with the femoral head at the ball-and-socket hip joint. The acetabular labrum deepens the socket. Oriented with approximately 15 degrees anteversion and 45 degrees inclination. Fractures are serious high-energy

trauma injuries.

Acetaminophen (Paracetamol)

- An analgesic and antipyretic drug that inhibits central COX enzymes and may have serotonergic and endocannabinoid mechanisms. It is first-line for mild pain and fever with fewer side effects than NSAIDs. Maximum daily dose is 4 grams (reduced to 2-3 grams in liver disease or alcohol use).

Acetaminophen (Paracetamol) Toxicity

- Most common cause of acute liver failure in Western countries. Mechanism: toxic metabolite NAPQI depletes glutathione, causing centrilobular hepatic necrosis. Treatment: N-acetylcysteine (NAC) within 8-10 hours of ingestion. Nomogram (Rumack-Matthew) guides treatment based on serum acetaminophen level at 4+ hours.

Acetaminophen Mechanism

- Acetaminophen, also known as paracetamol, is an analgesic and antipyretic drug with weak anti-inflammatory activity, differing from NSAIDs in its mechanism and side effect profile.

Acetazolamide

- A carbonic anhydrase inhibitor diuretic. Blocks carbonic anhydrase in the proximal tubule, reducing bicarbonate reabsorption, producing a mild diuresis and metabolic acidosis.

Acetyl-CoA Carboxylase

- First enzyme of fatty acid synthesis. Produces malonyl-CoA. Inhibited by palmitoyl-CoA.

Acetyl-Coenzyme A

- A central metabolic intermediate consisting of an acetyl group linked to coenzyme A by a high-energy thioester bond. It enters the citric acid cycle and provides carbon for fatty-acid, cholesterol, and ketone-body synthesis.

Acetylcholine

- A major neurotransmitter in both the central and peripheral nervous systems. In the peripheral nervous system, it is released at the neuromuscular junction (nicotinic receptors) to excite skeletal muscle, at all autonomic ganglia (nicotinic receptors), and at parasympathetic postganglionic terminals (muscarinic receptors).

Achalasia

- See Esophageal Achalasia.

Achenbach's syndrome

- A benign condition of recurrent, painful, spontaneous hematoma formation in the pulp of a finger or thumb, typically affecting middle-aged women. It presents with sudden onset of throbbing pain and blue-black discoloration of the fingertip, resolving spontaneously over one to two weeks without treatment.

Achilles Tendinopathy

- Degeneration/inflammation of Achilles tendon. Symptoms: posterior heel pain, thickening, stiffness. Risk factors: running, improper footwear, tight calf muscles. Treatment: eccentric exercises, heel raises, orthotics, shockwave therapy, surgery if refractory.

Achilles Tendon (Calcaneal Tendon)

- The thickest and strongest tendon in the human body, connecting the gastrocnemius and soleus (triceps surae) muscles to the calcaneus (heel bone). It plays a crucial role in plantarflexion of the foot, enabling walking, running, and jumping.

Achilles Tendon Rupture

- Complete tear of the Achilles tendon, most common in middle-aged men during recreational sports (weekend warrior). Mechanism: sudden push-off with knee extended. Presents with hearing/feeling a pop, acute posterior ankle pain, inability to stand on tiptoes, and palpable gap in the tendon.

Achilles Tendonitis

- Inflammation and degeneration of the Achilles tendon, most common in runners and middle-aged athletes. Insertional: at the calcaneal insertion, associated with bone spurs and Haglund deformity. Non-insertional (mid-portion): 2-6 cm above insertion, due to repetitive microtrauma and tendinosis.

Achlorhydria

- Absence of hydrochloric acid in gastric secretions, defined as a gastric pH persistently 7.

Achondroplasia

- Autosomal dominant skeletal dysplasia (FGFR3 gene). Most common form of dwarfism. Features: short limbs, macrocephaly, frontal bossing, trident hand, spinal stenosis. Homozygous form lethal. Normal intelligence.

Acid

- A substance that donates protons (H^+) or accepts electron pairs (Lewis acid). In aqueous solution, acids increase H^+ concentration ($pH < 7$). Strong acids (HCl , H_2SO_4 , HNO_3) dissociate completely; weak acids (acetic acid, carbonic acid H_2CO_3 , phosphate, amino acid side chains) partially dissociate (K_a , pK_a).

Acid Base Balance

- The physiological regulation of hydrogen ion concentration in body fluids, maintaining arterial pH between 7.35 and 7.45.

Acid-Base Analysis in Poisoning

- Integration of pH, bicarbonate, carbon dioxide, anion gap, delta gap, lactate, ketones, and osmolal gap to identify toxic metabolic disturbances. Mixed disorders are common and a normal pH does not exclude severe poisoning.

Acid-Base Disorder

- Disturbance of the body's acid-base balance, classified by primary disturbance and compensation. Respiratory acidosis: hypoventilation, high $PaCO_2$, low pH. Respiratory alkalosis: hyperventilation, low $PaCO_2$, high pH.

Acid-Fast Bacillus (AFB) Smear

- Microscopic examination of sputum for acid-fast bacilli (*Mycobacterium tuberculosis*), using Ziehl-Neelsen or auramine-rhodamine stain. Positive smear indicates active pulmonary TB and correlates with infectiousness. Sensitivity: 50-80% in culture-positive TB. Three specimens collected on consecutive mornings.

Acidosis

- An acid-base disturbance in which the concentration of hydrogen ions increases, tending to lower arterial blood pH below 7.35.

Acinetobacter baumannii resistance

- *Acinetobacter baumannii* is a gram-negative coccobacillus that has emerged as a significant cause of healthcare-associated infections, particularly in intensive care units. The organism has intrinsic and acquired resistance mechanisms, including beta-lactamases, efflux pumps, and porin mutations.

ACL Injuries

- Anterior cruciate ligament (ACL) tears: non-

contact (pivot, deceleration) > contact. Clinical: Lachman (most sensitive), pivot shift (most specific), anterior drawer. MRI confirms (bone bruise lateral femoral condyle + lateral tibial plateau, ACL discontinuity).

ACL Reconstruction

- Surgical reconstruction of the anterior cruciate ligament (ACL) following rupture, one of the most common knee ligament injuries. The ACL is a primary restraint to anterior tibial translation and rotatory instability, and its rupture frequently occurs during non-contact pivoting sports activities (skiing, soccer, basketball).

Acne

- A chronic inflammatory disorder of the pilosebaceous unit, extremely common during adolescence. Pathogenesis involves sebum production, follicular hyperkeratinization, *Cutibacterium acnes* colonization, and inflammation. Grading: comedonal, papulopustular, nodulocystic.

Acne Rosacea Subtypes

- Rosacea subtypes include erythematotelangiectatic (persistent facial erythema and flushing), papulopustular (red bumps and pus-filled pimples), phymatous (skin thickening, rhinophyma), and ocular (eye irritation, blepharitis). Demodex mite overgrowth contributes to pathogenesis.

Acne Vulgaris

- A chronic inflammatory disease of the pilosebaceous unit (hair follicle and associated sebaceous gland), affecting approximately 85 percent of adolescents and young adults between ages 12 and 24, with a prevalence of 12 to 22 percent in young adult women and 3 percent in men, making it the most common skin condition worldwide.

Acoustic Nerve

- The acoustic nerve, more properly called the auditory or vestibulocochlear nerve (cranial nerve VIII), carries hearing and balance information from the inner ear to the brainstem.

Acoustic Neuroma

- A benign, slow-growing schwannoma arising from the Schwann cells of the vestibular portion of the vestibulocochlear nerve (CN VIII), most commonly at the junction of the internal auditory canal and the

cerebellopontine angle. Also known as vestibular schwannoma. Accounts for approximately 8 percent of intracranial tumors.

Acoustic Neuroma (Vestibular Schwannoma)

- A usually benign, slow-growing tumor of Schwann cells involving the vestibular portion of cranial nerve VIII. It may cause unilateral sensorineural hearing loss, tinnitus, imbalance, or facial sensory symptoms; MRI of the internal auditory canals is the preferred diagnostic study.

Acoustic Trauma

- Acute damage to the auditory system from a brief, intense sound such as an explosion or firearm. It may cause tinnitus, temporary or permanent sensorineural hearing loss, and tympanic or ossicular injury; urgent audiologic assessment is indicated when hearing loss is sudden.

Acquisition Time

- Typical PET acquisition: 2-3 minutes per bed position. Total: 20-30 minutes.

Acral lentiginous melanoma

- A subtype of melanoma arising on acral skin (palms, soles, or under nails), appearing as a dark, irregularly pigmented patch or plaque. It is more common in people with darker skin tones and often presents at a later stage due to less frequent surveillance of these sites. Prognosis depends on thickness and stage at diagnosis.

Acrodermatitis Enteropathica

- A zinc-deficiency disorder causing periorificial and acral dermatitis, alopecia, diarrhoea, poor wound healing, and increased susceptibility to infection. It may be inherited from impaired intestinal zinc absorption or acquired through malnutrition, malabsorption, or prolonged parenteral nutrition.

Acromegaly

- Chronic growth hormone (GH) excess in adulthood, typically from a pituitary somatotroph adenoma.

Acromegaly Complications

- Systemic complications of chronic growth hormone excess include diabetes mellitus from insulin resistance, cardiovascular disease with cardiomyopathy and arrhythmias, sleep apnea from macroglossia and upper airway soft tissue hypertrophy, hypertension,

colon polyps and increased colorectal cancer risk, osteoarthritis, carpal tunnel syndrome

Acromioclavicular Joint

- A plane synovial joint between the acromion of the scapula and the lateral end of the clavicle. It permits gliding movements and is stabilized primarily by the acromioclavicular and coracoclavicular ligaments.

Acromioclavicular Joint Injuries

- AC joint injuries: Rockwood classification: I (sprain AC ligaments), II (AC ligaments torn, CC intact), III (AC+CC torn, 25-100% displacement), IV (posterior displacement into trapezius), V (superior displacement >100%), VI (inferior displacement subcoracoid/subacromial).

Acromion

- Bony process forming the shoulder's highest point, extending laterally from the scapular spine. Articulates with the clavicle at the acromioclavicular joint and arches over the glenohumeral joint. Attachment for deltoid and trapezius. The subacromial space contains the supraspinatus tendon and subacromial bursa.

ACTH Stimulation Test

- A dynamic test of adrenal cortisol reserve using synthetic adrenocorticotrophic hormone, usually cosyntropin. An inadequate cortisol rise supports adrenal insufficiency, although interpretation depends on timing, assay, recent glucocorticoid exposure, and whether disease is acute or chronic.

Actinic granuloma

- A benign granulomatous skin condition arising on sun-damaged skin, presenting as annular or ring-shaped plaques with central atrophy and an elevated, erythematous border. It is thought to represent a foreign-body reaction to degenerated elastic tissue from chronic ultraviolet exposure.

Actinic Keratosis

- A premalignant lesion caused by chronic UV exposure, appearing as rough, scaly, erythematous papules on sun-exposed skin. AK is the precursor to squamous cell carcinoma, with approximately 10 percent progressing to invasive cancer.

Actinic Keratosis Management

- Actinic keratosis management requires field-directed and lesion-directed therapy. Lesion-

directed: cryotherapy (liquid nitrogen, 5 to 10 second freeze-thaw cycle), curettage, electrodesiccation.

Actinomycetoma

- A chronic, progressive granulomatous infection of the skin and subcutaneous tissue caused by aerobic actinomycetes, characterized by swelling, draining sinuses, and dark granules in the discharge. It primarily affects the feet and is endemic in tropical regions. Diagnosis is by culture and histopathology.

Actinomycosis

- Chronic, progressive, granulomatous infection caused by branching Gram-positive anaerobic bacteria of the genus *Actinomyces* (most commonly *A. israelii*), a commensal of the oral cavity, GI tract, and female genital tract. Secretes a "sulfur granule" (yellow colonial mass with characteristic club-shaped terminals) within draining sinuses.

Action Potential

- A rapid, all-or-nothing depolarization of the cell membrane that propagates along excitable membranes (neurons, muscle, some endocrine cells).

Activated Charcoal

- Fine black powder produced by controlled pyrolysis of organic materials, processed to yield a massive micro-porous surface area (500-1500 m²/g). Mechanism: non-specific adsorption of toxins onto carbon surface via Van der Waals forces and hydrophobic interactions.

Activated Partial Thromboplastin Time

- Measures intrinsic coagulation pathway. Prolonged in hemophilia, heparin therapy.

Active Caries

- Caries progressing. Soft, chalky, light-colored. Requires treatment.

Active Management of Third Stage

- A bundle of interventions to reduce the risk of postpartum hemorrhage: (1) administration of oxytocin (10 IU IM/IV) immediately after delivery of the baby, (2) controlled cord traction (Brandt-Andrews maneuver) to deliver the placenta, and (3) uterine massage after placental delivery.

Active Phase of Labor

- The part of the first stage of labor from 6 cm to full dilatation (10 cm). Characterized by more rapid

cervical change (minimum: 1 cm/hour in nulliparas, 1.2 cm/hour in multiparas). The cervix dilates more rapidly as labor progresses.

Active Site

- The specific three-dimensional region of an enzyme where substrate binding and catalysis occur. Formed by amino acid residues that may be distant in primary sequence but brought together by protein folding.

Active Surveillance

- Close, structured monitoring of a cancer or precancerous condition without immediate treatment. It includes scheduled examinations, laboratory tests, imaging, or biopsies, with treatment started if evidence of progression develops.

Active Transport

- Movement of substances across cell membranes against their concentration gradient, requiring energy input, typically from ATP hydrolysis.

Activities of Daily Living

- Ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs). See functional status.

Activities of Daily Living (ADLs)

- Basic self-care activities such as bathing, dressing, toileting, eating, grooming, and transferring. ADL performance is a core measure of independence and rehabilitation need.

Acute

- Sudden onset, short duration. Examples: appendicitis, pneumonia.

Acute Agitation

- A state of heightened motor activity, emotional tension, irritability, and impaired behavioral control that may threaten the patient or others. Management begins with de-escalation and assessment for delirium, intoxication, withdrawal, medical illness, and psychosis.

Acute Alcohol Intoxication

- Neurobehavioral and physiologic impairment caused by recent ethanol exposure. Severity varies with concentration, tolerance, co-ingestants, and comorbidity; dangerous effects include respiratory depression, aspiration, hypoglycemia, hypothermia, trauma, and coma.

Acute Angle-Closure Glaucoma

- Ophthalmologic emergency. Elevated intraocular pressure causing optic nerve damage. Symptoms: severe eye pain, headache, nausea, vomiting, halos around lights, blurred vision. Signs: fixed mid-dilated pupil, corneal edema, rock-hard eye. Treatment: medical (acetazolamide, pilocarpine, mannitol), laser peripheral iridotomy.

Acute Aortic Syndrome

- A spectrum of life-threatening conditions involving the thoracic aorta, including aortic dissection, intramural hematoma (IMH), and penetrating aortic ulcer (PAU). See acute aortic syndromes.

Acute Aortic Syndromes

- A spectrum of life-threatening conditions involving the thoracic aorta, including aortic dissection, intramural hematoma (IMH), and penetrating aortic ulcer (PAU). Aortic dissection involves an intimal tear creating a false lumen.

Acute Bacterial Rhinosinusitis

- Bacterial infection of the paranasal sinuses suspected when symptoms persist without improvement for at least 10 days, become substantially worse after initial improvement, or are severe with high fever and purulent discharge. Treatment decisions depend on severity, complications, resistance risk, and local guidance.

Acute blood loss

- Rapid loss of circulating blood volume that can cause hypovolemic shock, tissue hypoxia, tachycardia, hypotension, and acute anemia.

Acute Bronchitis

- Acute inflammation of the tracheobronchial tree, usually caused by a viral respiratory infection and characterized by cough, with or without sputum production, lasting up to several weeks.

Acute Caries

- Rapidly progressing caries. Soft, light-colored cavity. Common in children, xerostomia.

Acute Cholecystitis

- Acute inflammation of the gallbladder, most often caused by persistent cystic-duct obstruction by a gallstone. It commonly causes sustained right-upper-quadrant pain, fever, nausea, and a positive Murphy sign; ultrasound and timely antimicrobial

and surgical assessment are standard.

Acute Coronary Syndrome

- A clinical spectrum caused by acute myocardial ischemia from plaque disruption, thrombosis, or less commonly coronary spasm. It includes unstable angina, NSTEMI, and STEMI and requires rapid assessment with symptoms, serial ECGs, and cardiac troponin testing.

Acute Coronary Syndrome (ACS)

- See Acute Coronary Syndromes. The term includes unstable angina, NSTEMI, and STEMI and describes acute myocardial ischemia requiring urgent assessment.

Acute Coronary Syndromes

- A spectrum of conditions comprising unstable angina (UA), non-ST-elevation myocardial infarction (NSTEMI), and ST-elevation myocardial infarction (STEMI). Pathophysiology: atherosclerotic plaque rupture or erosion leading to thrombus formation and acute coronary artery occlusion.

Acute Cough

- Cough lasting < 3 weeks. Most common causes: URI, acute bronchitis, pneumonia. Usually self-limited.

Acute dacryocystitis

- Acute infection and inflammation of the lacrimal sac, most commonly caused by obstruction of the nasolacrimal duct leading to stasis and secondary bacterial infection. It presents with painful swelling, redness, and tenderness at the medial canthus with purulent discharge.

Acute Disseminated Encephalomyelitis

- An acute, immune-mediated demyelinating disorder, usually occurring days to weeks after an infection and less often after vaccination. It commonly causes encephalopathy with multifocal neurologic deficits, ataxia, or seizures, particularly in children.

Acute esophageal necrosis

- Ischemic necrosis of the esophageal mucosa, most commonly caused by low-flow states such as shock, sepsis, or thromboembolism, presenting with hematemesis and chest pain. Endoscopy shows circumferential black necrotic mucosa, typically in the distal esophagus.

Acute Exacerbation of Asthma

- A progressive increase in dyspnea, cough, wheezing, or chest tightness with reduced expiratory airflow compared with the patient's baseline. Severe attacks may cause exhaustion, silent chest, hypercapnia, respiratory failure, and need for urgent treatment.

Acute Fatty Liver of Pregnancy (AFLP)

- A rare (1:7000-1:15,000) but life-threatening pregnancy complication characterized by microvesicular fatty infiltration of the liver leading to hepatic failure. Occurs in the third trimester (most commonly 35-37 weeks).

Acute Flaccid Myelitis

- A rare but serious inflammatory disorder of the spinal cord, usually affecting the gray matter and causing sudden flaccid limb weakness, reduced reflexes, and sometimes cranial-nerve or respiratory involvement. MRI and cerebrospinal-fluid studies support the diagnosis; urgent neurologic and respiratory monitoring is required.

Acute Generalized Exanthematous Pustulosis

- An acute drug reaction characterised by dozens to thousands of small, sterile, nonfollicular pustules arising on an erythematous base, often with fever and neutrophilia. It usually begins rapidly after exposure to an antibiotic or other medication and resolves after withdrawal of the trigger.

Acute glomerulonephritis

- Acute inflammation of the renal glomeruli causing hematuria, proteinuria, reduced glomerular filtration, edema, and often hypertension.

Acute Hepatitis

- Acute inflammation and injury of hepatocytes, usually developing over days to months and characterized by elevated aminotransferases, with or without jaundice, nausea, right-upper-quadrant discomfort, or constitutional symptoms.

Acute HIV Syndrome

- The acute retroviral syndrome occurring 2-6 weeks after primary HIV infection, corresponding to high-level viremia and immune activation. Presents with fever (most common), lymphadenopathy, pharyngitis, rash (maculopapular, trunk/face), myalgia, arthralgia, headache, nausea, diarrhea, and oral

ulcers.

Acute Hypercapnic Respiratory Failure

- Inability to eliminate carbon dioxide, producing elevated arterial PaCO_2 and often respiratory acidemia. Causes include COPD exacerbation, severe asthma, neuromuscular weakness, central respiratory depression, and chest-wall restriction.

Acute Hypoxemic Respiratory Failure

- Inability of the respiratory system to maintain adequate arterial oxygenation, usually manifested by low arterial oxygen or oxygen saturation despite appropriate inspired oxygen. Causes include pneumonia, pulmonary edema, ARDS, atelectasis, and pulmonary embolism.

Acute Inflammation

- A rapid, predominantly non-specific immune response to tissue injury or infection that typically lasts hours to days and involves vascular and cellular reactions. The cardinal signs are redness, heat, swelling, pain, and loss of function, resulting from increased blood flow, vascular permeability, and cellular infiltration.

Acute Interstitial Nephritis

- Inflammatory infiltrate in the renal interstitium and tubules most commonly caused by drug reactions including NSAIDs, antibiotics particularly penicillins and cephalosporins, proton pump inhibitors, and diuretics. Also caused by infections and autoimmune diseases.

Acute Ischemic Stroke

- Sudden neurologic dysfunction caused by focal cerebral, spinal, or retinal ischemia. Emergency evaluation distinguishes hemorrhage from ischemia and determines eligibility for reperfusion therapy while supporting airway, glucose, temperature, and blood pressure management.

Acute Kidney Injury

- Rapid decline in renal function over hours to days. KDIGO criteria define AKI as creatinine increase by 0.3 mg/dL within 48 hours, 1.5 times baseline within 7 days, or urine output less than 0.5 mL/kg/hr for 6 hours. Classified into stages 1 through 3. Causes include prerenal, intrinsic renal, and postrenal etiologies.

Acute Kidney Injury Staging

- A standardized classification based on changes in serum creatinine and urine output. KDIGO stages 1 through 3 quantify the severity of abrupt kidney-function decline and support prognosis, monitoring, and treatment decisions.

Acute leukemia

- leukemia characterized by a rapid onset, clonal proliferation of immature blast cells ($>20\%$ of bone marrow cells), and rapid progression without treatment. Subtypes: acute lymphoblastic leukemia (ALL, B-cell or T-cell, most common childhood cancer, peak age 2-5) and acute myeloid leukemia (AML, more common in adults, median age 65-70).

Acute Limb Ischemia

- Sudden reduction in arterial blood flow to a limb that threatens tissue viability, usually caused by embolism, thrombosis, or graft occlusion. The six classic findings are pain, pallor, pulselessness, paresthesia, paralysis, and poikilothermia; urgent vascular assessment and revascularization are required.

Acute Limb Ischemia (ALI)

- Sudden, rapid decrease in lower extremity arterial perfusion threatening limb viability, typically caused by acute arterial thrombosis or embolism. Presentation: the 6 Ps—pain, pallor, pulselessness, paresthesia, paralysis, poikilothermia. Diagnosis: clinical, Doppler ultrasound, CT angiography.

Acute Limb Ischemia Six Ps

- The classic findings of acute arterial perfusion loss: pain, pallor, pulselessness, paresthesia, paralysis, and poikilothermia. Motor or sensory deficit indicates threatened limb and requires urgent vascular assessment and revascularization planning.

Acute Liver Failure

- Rapid deterioration of liver function in a previously healthy individual without prior liver disease. Causes include acetaminophen toxicity, viral hepatitis, drug reactions, autoimmune hepatitis, and Wilson disease. Features include coagulopathy, encephalopathy, and multiorgan failure. Presents with jaundice, confusion, and bleeding.

Acute lymphangitis

- Acute inflammation of lymphatic vessels, typically resulting from streptococcal or staphylococcal in-

fection spreading from a distal wound. It presents with red, tender streaks extending proximally from the site of infection, along with fever, malaise, and regional lymphadenopathy. Treatment requires systemic antibiotics.

Acute Lymphoblastic Leukemia

- A clonal malignancy of lymphoid progenitor cells, most common in children (peak age 2-5 years) but also occurring in adults with a worse prognosis with increasing age.

Acute lymphocytic leukemia

- A malignant proliferation of immature lymphoid precursors in the bone marrow and blood, commonly presenting with cytopenias, fever, bone pain, or lymphadenopathy.

Acute Megakaryoblastic Leukemia

- A rare subtype of acute myeloid leukemia characterized by the proliferation of megakaryoblasts in the bone marrow and peripheral blood.

Acute mesenteric ischemia

- Occlusion or reduced flow in the mesenteric arteries causing bowel ischemia, most commonly from embolism or atherosclerosis. Presents with severe abdominal pain disproportionate to physical findings. Delayed treatment leads to bowel necrosis, perforation, and high mortality.

Acute Mountain Sickness

- A mild altitude illness caused by rapid ascent, usually producing headache with nausea, fatigue, dizziness, or sleep disturbance. Prevention and treatment include gradual acclimatization, descent when symptoms worsen, oxygen, and appropriate medical therapy.

Acute Myeloid Leukemia (AML)

- A clonal hematologic malignancy characterized by proliferation of immature myeloid blasts that suppress normal hematopoiesis. Diagnosis requires the appropriate blast burden or defining genetic abnormality, and presentation commonly includes anemia, infection, and bleeding from marrow failure.

Acute obstructive uropathy

- Sudden impairment of urinary flow caused by obstruction of the urinary tract, potentially producing hydronephrosis, pain, infection, and acute kidney injury.

Acute Otitis Media

- Infection of the middle ear space with acute onset of symptoms and signs of middle ear inflammation. Diagnosis requires middle ear effusion plus acute symptoms (ear pain, fever, irritability) or signs (bulging tympanic membrane, erythema). Most common pathogen: *Streptococcus pneumoniae*.

Acute Pancreatitis

- Acute inflammatory process of the pancreas most commonly caused by gallstones and alcohol accounting for approximately eighty percent of cases. The pathogenesis involves premature trypsin activation within pancreatic acinar cells triggering autodigestion.

Acute Promyelocytic Leukemia

- A distinct subtype of acute myeloid leukemia (AML) characterized by the t(15;17)(q24;q21) translocation, which fuses the promyelocytic leukemia (PML) gene on chromosome 15 with the retinoic acid receptor alpha (RARA) gene on chromosome 17.

Acute Rehabilitation

- Intensive inpatient rehabilitation (3+ hours therapy/day) for acute conditions: stroke, spinal cord injury, traumatic brain injury, major trauma. Length of stay: 1-4 weeks.

Acute Respiratory Distress Syndrome

- ARDS: a syndrome of acute, diffuse inflammatory lung injury causing severe hypoxemic respiratory failure with bilateral infiltrates not due to cardiogenic pulmonary edema (PaO₂/FiO₂ ratio ≤300). Causes: sepsis (most common), pneumonia, trauma, aspiration, pancreatitis, massive transfusion.

Acute Respiratory Distress Syndrome (ARDS)

- Progressive respiratory failure due to diffuse alveolar damage. Diagnostic criteria (Berlin definition): acute onset within 1 week, bilateral opacities on chest imaging, respiratory failure not fully explained by cardiac failure, PaO₂/FiO₂ ratio ≤ 300 mmHg.

Acute Sinusitis

- Acute inflammation of the mucosa of one or more paranasal sinuses, usually following a viral upper-respiratory infection. It causes nasal obstruction, purulent or discoloured discharge, facial pressure

or pain, and reduced smell; persistent, severe, or worsening symptoms may indicate acute bacterial rhinosinusitis or a complication.

Acute Tubular Necrosis

- Most common cause of intrinsic AKI resulting from ischemic or nephrotoxic injury to tubular epithelial cells. Ischemic ATN develops from prolonged prerenal states. Nephrotoxic ATN results from direct injury by aminoglycosides, amphotericin B, cisplatin, contrast, and myoglobin.

Acute viral illness

- See Viral Infections.

Acute-on-Chronic Liver Failure (ACLF)

- Acute deterioration of pre-existing chronic liver disease associated with organ failure and high short-term mortality. It is commonly precipitated by infection, alcohol-related hepatitis, bleeding, or other acute insults and requires urgent multidisciplinary management.

Acyclovir

- An antiviral medicine used to treat herpes simplex infections and varicella-zoster infections, including genital herpes and shingles. It reduces viral replication but does not eradicate latent virus. It is available orally, topically, or intravenously; dose adjustment may be needed in kidney disease.

Adam's Apple

- The common name for the laryngeal prominence, the visible projection at the front of the neck formed by the angle where the two laminae of the thyroid cartilage meet.

Adaptive Equipment

- Devices that help individuals with disabilities perform daily activities independently. Examples: reachers, dressing sticks, button hooks, ergonomic utensils, grab bars, shower chairs, wheelchairs, walkers, canes. Selection based on individual needs and goals. Occupational therapy assessment recommended.

Adaptive Immunity

- Antigen-specific immunity mediated principally by B and T lymphocytes. It develops after antigen exposure, generates immunologic memory, and includes humoral responses from B-cell antibodies and cellular responses from T cells.

Addiction

- A chronic, relapsing brain disorder characterized by compulsive substance use or behavior despite harmful consequences. It involves changes in brain reward, motivation, and memory circuits, leading to impaired control over the behavior. Treatment often combines medication, behavioral therapy, and social support.

Addison Disease

- Primary adrenal insufficiency resulting from destruction or dysfunction of the adrenal cortex, leading to deficiency of cortisol, aldosterone, and adrenal androgens.

Addisonian Crisis

- Life-threatening emergency resulting from acute cortisol deficiency in patients with adrenal insufficiency. Precipitated by infection, trauma, surgery, or abrupt discontinuation of chronic glucocorticoid therapy.

Adduction

- Movement of a limb or other body part toward the median plane of the body.

Adductor Muscles

- A group of muscles on the medial thigh that adduct (pull toward the midline) the hip: adductor longus, adductor brevis, adductor magnus, gracilis, and pectineus. Innervated by the obturator nerve (L2-L4); adductor magnus also receives tibial nerve (sciatic). Adductor longus is a common site of athletic injury ('groin strain').

Adenocarcinoma

- A malignant tumor arising from glandular epithelial tissue, forming gland-like structures. The most common type of carcinoma, occurring in organs such as the colon, breast, lung, pancreas, and prostate.

Adenocarcinoma of the colon

- A malignant gland-forming epithelial tumor of the colon, commonly presenting with altered bowel habits, occult bleeding, anemia, or an obstructing mass.

Adenoid Cystic Carcinoma

- A rare malignant tumor of glandular tissue, most commonly of the salivary glands, characterized by slow growth, perineural spread, and late recurrence.

Adenoidectomy

- The surgical removal of the pharyngeal tonsils (adenoids), which are lymphoid tissue structures located in the nasopharynx, and it is commonly performed in children for obstructive sleep apnea, chronic nasal obstruction, recurrent otitis media, and chronic rhinosinusitis that is refractory to medical management.

Adenoids

- Lymphoid tissue in the posterior wall and roof of the nasopharynx, part of Waldeyer ring of pharyngeal lymphoid tissue. Enlarged adenoids (adenoidal hypertrophy) in children cause nasal obstruction, mouth breathing, snoring, obstructive sleep apnea, serous otitis media (eustachian tube obstruction), and facial changes ('adenoid facies').

Adenoma

- A benign tumour arising from glandular or gland-like epithelial tissue. Some adenomas, particularly certain colorectal adenomas, can develop dysplasia and progress to carcinoma.

Adenoma-Carcinoma Sequence

- Established model of stepwise colorectal carcinogenesis progressing from normal epithelium through adenomas to carcinoma over ten to fifteen years.

Adenomatous Polyp

- A benign gland-forming polyp, most commonly in the colon, that may contain dysplasia and can progress through the adenoma–carcinoma sequence. Tubular, villous, and tubulovillous types are recognized.

Adenomyosis

- A condition in which endometrial glands and stroma are present within the myometrium (uterine muscle). Most common in multiparous women in their 40s and 50s. Presents with heavy menstrual bleeding (menorrhagia), severe dysmenorrhea, chronic pelvic pain, and uterine enlargement.

Adenovirus

- A non-enveloped DNA virus that causes respiratory infections, conjunctivitis, gastroenteritis, and cystitis. Adenoviruses are highly stable in the environment and can cause outbreaks in closed populations. The virus has over 50 serotypes with different tissue tropisms. Diagnosis is made by PCR, antigen

detection, or viral culture.

Adenovirus Infection

- Infection with an adenovirus, commonly causing sore throat, cough, fever, conjunctivitis, diarrhea, or vomiting. Most infections are self-limited, but severe respiratory or disseminated disease can occur in infants and people with weakened immunity. Treatment is usually supportive, with medical assessment for breathing difficulty, dehydration, or severe illness.

Adenovirus Vaccine

- A vaccine used primarily to prevent adenovirus types 4 and 7 among selected military personnel. It is an oral, live vaccine and is not part of routine vaccination for the general public. Vaccine use and eligibility depend on current public-health recommendations.

ADHD

- Attention deficit hyperactivity disorder: a neurodevelopmental disorder with inattention, hyperactivity, and impulsivity inappropriate for age, affecting function in school/home; onset before 12 years, persistence ≥ 6 months in ≥ 2 settings. Subtypes: inattentive, hyperactive-impulsive, combined.

Adherence

- Patient following treatment plan. Barriers: cost, side effects, complexity, beliefs. Strategies: simplification, education, motivational interviewing.

Adherence (Geriatric)

- The degree to which an elderly patient correctly follows medical advice, takes medications as prescribed, and maintains lifestyle recommendations.

Adhesions

- Bands of fibrous scar tissue that form between organs or tissues, most commonly following abdominal or pelvic surgery. In gynecology, adhesions can develop after cesarean section, myomectomy, ovarian cystectomy, or treatment of endometriosis.

Adhesive Capsulitis

- A condition of progressive shoulder stiffness and pain caused by inflammation and fibrosis of the glenohumeral joint capsule. Common name: frozen shoulder. Stages: freezing (painful, 2-9 months), frozen (stiff, 4-12 months), thawing (gradual improvement, 5-24 months). Most common ages 40-

60.

Adhesive Tape

- Secures dressings. Types: paper, cloth, plastic. Skin preparation essential for adhesion.

Adie Tonic Pupil

- Uncommon cause of anisocoria, a benign parasympathetic denervation of the iris sphincter and ciliary muscle, typically unilateral in young women, idiopathic associated with diabetes, herpes zoster, or autoimmune neuropathy.

Adipocere (Grave Wax)

- Waxy, grayish-white substance formed postmortem in warm, moist, anaerobic environments (water, damp soil) by the hydrolysis and hydrogenation of body fat into free fatty acids; retards putrefaction and preserves anatomical details for months to years.

Adiponectin

- Adipose-derived hormone. Increases insulin sensitivity. Anti-inflammatory.

Adipose Tissue

- A specialized connective tissue composed mainly of adipocytes that store energy as triglycerides. It provides thermal insulation, cushioning for organs, and endocrine regulation through hormones and signaling molecules such as leptin and adiponectin.

Adjustment Disorder

- Emotional or behavioral symptoms in response to an identifiable stressor occurring within three months of the onset of the stressor. The symptoms are disproportionate to the severity of the stressor and cause significant distress or impairment. It resolves within six months of the termination of the stressor.

Adjustment disorder with anxiety

- An emotional or behavioral response to an identifiable stressor that causes distress or impairment but does not meet criteria for another mental disorder.

Adjuvant Therapy

- Additional treatment after primary treatment to reduce recurrence risk. Examples: chemotherapy after surgery.

ADL Training

- Training for activities of daily living: dressing,

bathing, grooming, feeding, toileting. Adaptive techniques, equipment.

Administered Activity

- The measured amount of radioactive material given to a patient for imaging or therapy, usually expressed in becquerels. It should be prescribed and recorded with the radionuclide, radiopharmaceutical, route, date, and time of administration.

Adolescent Health

- The health and wellbeing of young people (10-19 years), addressing the physical, psychological, and social transitions of puberty and adolescence.

Adoptive Cell Therapy

- A personalized cellular immunotherapy in which a patient's own lymphocytes are expanded or engineered ex vivo and reinfused to attack the tumor; examples include tumor-infiltrating lymphocyte (TIL) therapy and chimeric antigen receptor (CAR) T-cell therapy.

ADP (Adenosine Diphosphate)

- A nucleotide composed of adenine, ribose, and two phosphate groups. Formed by hydrolysis of ATP ($\text{ATP} \rightarrow \text{ADP} + \text{P}_i$, $\Delta G^\circ = -30.5 \text{ kJ/mol}$). Serves as the immediate precursor for ATP synthesis via oxidative phosphorylation (mitochondrial ATP synthase), substrate-level phosphorylation (glycolysis, TCA cycle), and photophosphorylation.

Adrenal Cortex Physiology

- The adrenal cortex produces three classes of steroid hormones from the three zones. The zona glomerulosa produces mineralocorticoids primarily aldosterone regulating sodium and potassium balance under the influence of angiotensin II and potassium.

Adrenal Cortical Carcinoma

- A rare, aggressive malignancy arising from the adrenal cortex, with an incidence of approximately one to two per million population per year. The tumor may be functional, producing excessive hormones, or non-functional.

Adrenal Crisis

- An acute, life-threatening manifestation of severe adrenal insufficiency. Causes: primary (Addison disease), secondary (pituitary failure), or tertiary (exogenous corticosteroid withdrawal). Precipitants: systemic infection, trauma, surgery, or

abrupt steroid cessation.

Adrenalectomy

- Surgical removal of one or both adrenal glands, performed for selected adrenal tumors, severe hormone excess, or other localized disease. Removal of one gland may leave adequate hormone production; bilateral removal causes permanent adrenal insufficiency and requires lifelong hormone replacement and emergency-steroid planning.

Adrenal Gland

- Paired endocrine glands atop each kidney (suprarenal), composed of cortex and medulla. Cortex (three zones): zona glomerulosa (aldosterone, mineralocorticoids), zona fasciculata (cortisol, glucocorticoids), zona reticularis (androgens). Medulla: chromaffin cells secrete epinephrine and norepinephrine (catecholamines).

Adrenal Hyperplasia

- Enlargement or overgrowth of adrenal tissue that may increase production of adrenal hormones. Causes include congenital adrenal hyperplasia, chronic stimulation by ACTH, and some adrenal tumors. Symptoms depend on the hormone affected and may include abnormal genital development, early puberty, excess body hair, high blood pressure, or electrolyte disturbances.

Adrenal Hemorrhage

- Bleeding into the adrenal gland most commonly occurring in critically ill patients with sepsis, coagulopathy, or anticoagulation therapy. Bilateral adrenal hemorrhage may cause acute adrenal insufficiency with circulatory collapse. Waterhouse-Friderichsen syndrome: adrenal hemorrhage associated with meningococcal sepsis.

Adrenal Insufficiency

- Deficient production of cortisol by the adrenal cortex classified as primary adrenal insufficiency or Addison disease from adrenal gland destruction, secondary from pituitary ACTH deficiency, and tertiary from chronic exogenous glucocorticoid suppression of the hypothalamic-pituitary-adrenal axis.

Adrenal Medulla Physiology

- The adrenal medulla produces catecholamines primarily epinephrine and norepinephrine in response to sympathetic nervous system activation.

Epinephrine accounts for approximately eighty percent and norepinephrine approximately twenty percent of catecholamine output.

Adrenal Myelolipoma

- Benign adrenal tumor composed of mature adipose tissue and hematopoietic elements. Usually asymptomatic and discovered incidentally. No malignant potential. May cause adrenal hemorrhage if large. Diagnosis is by imaging showing fat attenuation on CT. No treatment required unless symptomatic from mass effect or hemorrhage.

Adrenal Tumors

- Adrenal incidentaloma: >1cm on imaging. Workup: hormonal (cortisol after 1mg DST, metanephrines, aldosterone/renin if hypertensive, DHEAS/androgens if virilizing), imaging (CT washout: adenoma >60% absolute washout or >40% relative; MRI chemical shift). Size >4cm or suspicious imaging -> surgery.

Adrenal Venous Sampling

- Interventional radiology procedure for lateralizing aldosterone secretion in primary hyperaldosteronism. Catheterization of both adrenal veins allows measurement of aldosterone and cortisol levels to calculate selectivity index confirming successful cannulation and lateralization ratio.

Adrenocortical Carcinoma (ACC)

- A rare malignant tumor of the adrenal cortex that may secrete cortisol, androgens, or aldosterone or be non-functional. Often large at diagnosis; treatment is en bloc resection, with mitotane for advanced disease.

Adrenocorticotrophic Hormone

- ACTH (corticotropin): a peptide hormone from the anterior pituitary (cleaved from POMC) that stimulates the adrenal cortex to produce cortisol (zona fasciculata) and adrenal androgens; regulated by hypothalamic CRH and cortisol negative feedback (circadian).

Adult Spinal Deformity

- Adult spinal deformity (ASD): scoliosis (coronal Cobb >10 deg), kyphosis (sagittal), sagittal imbalance. SRS-Schwab classification: PI-LL mismatch, SVA (sagittal vertical axis), PT (pelvic tilt). Clinical: back pain, radiculopathy, neurogenic claudica-

tion, disability (ODI, SRS-22).

Adult Still's Disease

- Adult Still's disease, also known as systemic-onset juvenile idiopathic arthritis when occurring in children, is a systemic inflammatory disorder characterized by high-spiking daily fevers, evanescent salmon-colored rash, arthritis, and multiorgan involvement.

Adult-Onset Still Disease

- AOSD: a rare systemic autoinflammatory disorder characterized by daily high-spiking fevers, evanescent salmon-pink rash, arthritis/arthralgia, sore throat, lymphadenopathy, hepatosplenomegaly, and markedly elevated ferritin (with low glycosylated ferritin), ESR/CRP. Complications: macrophage activation syndrome, pericarditis, DIC.

Advance Care Planning

- Discussion of future healthcare preferences. Includes advance directives, surrogate designation.

Advance Directive

- Legal document expressing healthcare wishes. Components: living will (treatment preferences), durable power of attorney for healthcare (surrogate decision-maker). Should be reviewed periodically, discussed with surrogate and healthcare team.

Advanced Cancer

- Cancer that has grown beyond its original site. It may be locally advanced when it has invaded nearby tissues or metastatic when it has spread to distant organs.

Advanced Cardiovascular Life Support (ACLS)

- A set of clinical interventions for the urgent treatment of cardiac arrest, stroke, myocardial infarction, and life-threatening arrhythmias.

Advanced Maternal Age

- Maternal age 35 years at the time of delivery. Associated with increased risks: aneuploidy (trisomy 21: 1:350 at age 35, 1:100 at age 40, 1:25 at age 45), miscarriage, gestational diabetes, preeclampsia, preterm birth, IUGR, placenta previa, and stillbirth.

Adverse Drug Event

- Unintended harm from medical care. See adverse event.

Adverse Drug Reaction

- An adverse drug reaction is any noxious or unintended response to a drug that occurs at doses used for prophylaxis, diagnosis, or therapy.

Adverse Drug Reactions

- Any harmful or unintended response to a drug at normal doses. ADRs are classified as Type A (augmented, dose-dependent, predictable: side effects, toxicity) or Type B (bizarre, dose-independent, unpredictable: allergy, idiosyncrasy). ADRs are a major cause of morbidity and mortality, accounting for 5-7% of hospital admissions.

Adverse Event

- Unintended harm from medical care. Includes medical errors and complications.

Adverse Event (AE)

- Any untoward medical occurrence during or after a treatment or intervention, whether or not it is considered causally related to that treatment. Severity, seriousness, expectedness, and relatedness are assessed separately.

Aerobe

- A microorganism that requires molecular oxygen for growth or uses oxygen efficiently for energy production. Obligate aerobes are unable to grow without oxygen, whereas facultative organisms can grow with or without it.

Aerobic

- Requiring or involving molecular oxygen. In microbiology, aerobic organisms use oxygen for growth or energy production, either obligatorily or as one option among several; in physiology, aerobic metabolism uses oxidative phosphorylation in mitochondria to generate ATP from nutrients.

Aerophagy

- Excessive swallowing of air, also called aerophagia, which can lead to frequent belching, abdominal bloating, distension, and discomfort.

Aesthetic Dentistry

- Focus on appearance of teeth and smile. Veneers, bonding, whitening, crowns.

Affective Flattening

- A significant reduction in the range and intensity of emotional expression. It is a negative symptom

of schizophrenia characterized by reduced facial expression, monotonous voice, and decreased body language. It is distinct from blunted affect (reduced range) and constricted affect (narrowed range).

Aflatoxin

- A group of toxic metabolites produced mainly by *Aspergillus* fungi that contaminate foods such as maize, peanuts, and tree nuts. Chronic exposure can cause liver injury and substantially increases the risk of hepatocellular carcinoma, particularly in people with chronic hepatitis B infection.

AFP (Alpha-Fetoprotein)

- A tumor marker and prenatal screening analyte. Tumor marker: hepatocellular carcinoma (elevated in 60-80%, >400 ng/mL highly specific), yolk sac tumors (germ cell tumors). Non-specific: cirrhosis, hepatitis, pregnancy. Prenatal: elevated in neural tube defects (maternal serum AFP).

Afterload

- The resistance against which the ventricles must eject blood during systole. Left ventricular afterload is primarily determined by systemic vascular resistance (SVR); right ventricular afterload is determined by pulmonary vascular resistance (PVR). Afterload is estimated by blood pressure or calculated as wall stress.

Agar

- A gelatinous polysaccharide from red algae used to solidify culture media. Melting near 85°C and solidifying at 35–40°C, it remains solid at incubation temperatures and is largely undegraded by bacteria, enabling isolation of pure colonies on media such as blood, MacConkey, and Sabouraud agars.

Age-Related Macular Degeneration

- The leading cause of irreversible central vision loss in individuals over 50 in developed countries, affecting approximately 200 million people worldwide with prevalence increasing with age (10 percent over age 65, 30 percent over age 75).

Age-Related Macular Degeneration (AMD)

- Progressive degenerative disease of the central retina (macula) causing bilateral, irreversible loss of central vision. Types: dry (atrophic, 90%) with drusen accumulation; wet (neovascular, 10%) with

choroidal neovascularization. Risk factors: age >50, smoking, family history, Caucasians.

Age-Standardized Rate

- A rate adjusted to a specified standard population age distribution so that populations with different age structures can be compared more fairly. It is commonly used for mortality, cancer, and disease-incidence comparisons.

Ageism

- Stereotyping, prejudice, or discrimination directed at people because of their age, which can impair access to appropriate medical care.

Agensis of the Corpus Callosum

- Complete or partial absence of the corpus callosum, the major commissural fiber bundle connecting the cerebral hemispheres. May be isolated or associated with other CNS anomalies (Dandy-Walker, Chiari, heterotopias) and genetic syndromes (Aicardi, Andermann, acrocallosal syndromes).

Ageusia

- Complete loss of the sense of taste. Causes include viral infections, oral or nasal disease, head injury, damage to the facial or glossopharyngeal nerves, medications, chemotherapy, zinc deficiency, and neurologic disorders.

Agglutination

- The visible clumping of cells or particles caused by antibodies or other binding molecules linking antigens on their surfaces. It is used in blood-grouping and crossmatching, infectious-disease serology, and some laboratory identification tests.

Aggregation, Platelet

- The process by which activated platelets adhere to one another to form a primary platelet plug at a site of vascular injury. Platelet agonists such as ADP, thromboxane A₂, and thrombin activate glycoprotein IIb/IIIa, which binds fibrinogen and links adjacent platelets.

Aggression

- Verbal or physical behavior intended to threaten, harm, or dominate another person or object. It may occur with mania, psychosis, personality disorders, dementia, delirium, substance intoxication or withdrawal, traumatic brain injury, or severe frustration.

Aggressive Cancer

- A cancer that grows, invades, or spreads rapidly and may require prompt diagnosis and treatment. Aggressiveness depends on tumour biology, grade, stage, and response to therapy.

Agility

- Ability to change direction quickly. Essential for sports, functional mobility.

Aging

- Biological process of growing older, involving cumulative changes at molecular, cellular, tissue, organ levels.

Agitation

- A state of excessive motor activity, emotional excitement, and restlessness. In psychiatric settings, it may be a symptom of mania, psychosis, delirium, substance intoxication, or withdrawal.

Agnathia

- A rare congenital absence or severe underdevelopment of the mandible, maxilla, or both. It is usually part of a craniofacial developmental disorder and may cause profound facial deformity, airway obstruction, feeding difficulty, and impaired dentition.

Agnosia

- A higher-order neurological deficit characterized by the inability to recognize or process sensory stimuli (visual, auditory, tactile) despite intact primary sensory organs and pathways. Caused by focal lesions in specific cortical association areas (parietal, temporal, or occipital lobes).

Agonist

- A substance that binds to a receptor and activates it, producing a biological response. A full agonist can produce the receptor's maximum response, whereas a partial agonist produces a lower maximum response even when it occupies all available receptors.

Agonist-Antagonist

- A drug that has mixed pharmacological activity, acting as an agonist at one receptor type and an antagonist at another, or as a partial agonist at a single receptor.

Agoraphobia

- Anxiety disorder characterized by fear and avoid-

ance of situations where escape might be difficult or help unavailable during a panic-like episode. Common feared situations include public transportation, open spaces, enclosed spaces, standing in line, being in crowds, and being outside the home alone.

Agranulocytosis

- A severe reduction or near-absence of circulating neutrophils, usually defined as an absolute neutrophil count below $0.5 \times 10^9/L$. It may be caused by medications, chemotherapy, autoimmune disease, infections, or bone-marrow disorders and presents with fever, sore throat, oral ulcers, or recurrent serious infections.

Agrophesthesia

- Inability to recognize numbers or letters traced on the skin, indicating a lesion of the contralateral parietal lobe (somatosensory association cortex, postcentral gyrus). Tested with a blunt object on the palm or fingertip while the patient's eyes are closed.

Agraphia

- An acquired inability to write or express language in written form despite adequate education and motor ability. It may result from dominant-parietal or language-network lesions, especially stroke, traumatic brain injury, dementia, or brain tumors, and may occur with aphasia, alexia, or acalculia.

Aicardi-Goutieres Syndrome

- Rare autosomal recessive inflammatory disorder mimicking congenital viral infection. Characterized by cerebral calcifications, white matter disease, cerebral atrophy, and cerebrospinal fluid lymphocytosis. Presents with severe neurodevelopmental impairment, seizures, and spasticity.

AIDS

- Acquired immunodeficiency syndrome: the advanced stage of HIV infection, defined by CD4 count <200 cells/ μL or the presence of AIDS-defining illnesses.

Air Abrasion

- Minimally invasive caries removal using abrasive particles. Alternative to drilling for small lesions.

Air bronchogram

- A radiographic sign showing air-filled bronchi visible against opacified lung parenchyma, indicating

that the airways remain patent while the surrounding alveoli are filled with fluid, pus, or cells.

Air Contaminants

- See occupational medicine.

Air Embolism

- Presence of gas in the venous or arterial circulation, often fatal if massive. Causes: iatrogenic (central line insertion, surgery—especially neurosurgery in sitting position, dialysis, pressurized infusion), trauma (penetrating neck/chest wounds, diving), and procedures (abortion with irrigation, insufflation).

Air Polishing

- Removal of stains using air-abrasive device. Glycine or erythritol powder.

Air Quality

- Measurement of air pollutants. Examples: particulate matter, ozone, nitrogen dioxide, sulfur dioxide.

Airborne Disease

- Disease transmitted by airborne particles. Examples: tuberculosis, measles, COVID-19.

Airborne precautions

- Infection control measures used for pathogens transmitted through airborne droplet nuclei. Precautions include negative pressure rooms, N95 respirators, and restricted access. Airborne precautions are used for tuberculosis, measles, varicella, and disseminated herpes zoster.

Airway Clearance Techniques

- Physical therapies to mobilize and remove secretions from the airways. Techniques include postural drainage, percussion, vibration, active cycle of breathing, autogenic drainage, positive expiratory pressure, oscillating PEP, and high-frequency chest wall oscillation. They are essential in CF, bronchiectasis, and neuromuscular disease.

Airway Management

- Airway management encompasses techniques and devices used to maintain ventilation and oxygenation in patients with compromised airways.

Airway Obstruction

- Partial or complete blockage of airflow through the upper or lower respiratory tract. It may result from a foreign body, secretions, swelling, trauma, tumor,

or loss of airway muscle tone and can cause stridor, wheeze, hypoxaemia, respiratory distress, or arrest.

Airway Resistance

- The opposition to airflow through the conducting airways, determined largely by airway radius and flow pattern. Bronchoconstriction, mucus, edema, and reduced elastic traction increase resistance; because resistance varies strongly with radius, small changes can markedly affect work of breathing.

Akathisia

- Subjective feeling of inner restlessness accompanied by a compulsive need to move, most commonly an extrapyramidal side effect of antipsychotic drugs (typical and atypical neuroleptics), also seen with dopamine antagonists (metoclopramide), selective serotonin reuptake inhibitor withdrawal, and in Parkinson disease.

Akinesia

- Impairment or absence of voluntary movement, classically a hypokinetic feature of basal ganglia disorders such as Parkinson disease and related parkinsonian syndromes.

Alagille Syndrome

- An autosomal dominant disorder caused by mutations in JAG1 (95%) or NOTCH2 (<5%), affecting the Notch signaling pathway.

ALARA Principle

- Radiation-protection principle requiring exposure to be kept as low as reasonably achievable while obtaining the necessary diagnostic or therapeutic information. Justification, optimization, collimation, shielding, time reduction, and increased distance support ALARA.

Albuterol

- A short-acting beta-2 agonist inhaler that relaxes airway smooth muscle and provides rapid, temporary relief of bronchospasm in asthma and some forms of chronic obstructive pulmonary disease. Common effects include tremor, nervousness, headache, and a fast heartbeat. Frequent need for rescue doses suggests inadequate control and warrants clinical review.

Albinism (Oculocutaneous Albinism)

- A hereditary, congenital pigmentary disorder characterized by a complete or partial absence of melanin

in the skin, hair, and eyes, most commonly caused by mutations in the tyrosinase gene (e.g., OCA1) or other genes involved in melanin synthesis.

Albright's hereditary osteodystrophy

- A genetic disorder of GNAS gene mutations causing resistance to parathyroid hormone, presenting with short fourth metacarpals, subcutaneous calcifications, and basal ganglia calcification.

Albumin

- The most abundant plasma protein (35–50 g/L, 60% of total protein), synthesized in the liver. A single polypeptide (66.5 kDa, 585 amino acids) with high negative charge at physiological pH.

Albumin-to-Creatinine Ratio (ACR)

- The urinary albumin concentration standardized to urinary creatinine concentration, usually measured in a spot urine sample. Persistent albuminuria is a marker of kidney damage and is categorized as A1 (<30 mg/g), A2 (30–300 mg/g), or A3 (>300 mg/g) in commonly used CKD classification.

Albuminuria

- Abnormally increased excretion of albumin in the urine, indicating disruption of the glomerular filtration barrier or another form of kidney disease. It is commonly detected with a urine albumin-to-creatinine ratio and classified as moderately or severely increased according to the quantity and persistence of excretion.

Alcohol

- See ethanol.

Alcohol dependency

- A chronic relapsing disorder characterized by compulsive ethanol use, loss of control over intake, and negative emotional state during withdrawal, leading to tolerance, dependence, and organ damage over time.

Alcohol Intoxication

- Acute effects of ethanol: CNS depression, disinhibition, ataxia, slurred speech, vomiting, respiratory depression (severe). BAC >0.08% impairs driving. Treatment: supportive, airway protection, thiamine (prevent Wernicke), glucose if hypoglycemic.

Alcohol Swab

- Antiseptic preparation for skin. 70% isopropyl alcohol.

Alcohol Use

- See substance use.

Alcohol Use Disorder

- A problematic pattern of alcohol use leading to significant impairment or distress. Severity is classified as mild (2–3 criteria), moderate (4–5 criteria), or severe (6 or more criteria).

Alcohol Withdrawal

- Symptoms occurring after cessation or reduction of prolonged heavy drinking. Timeline includes: tremor and anxiety at 6–12 hours, perceptual disturbances at 12–24 hours, withdrawal seizures at 24–48 hours, and delirium tremens at 48–96 hours.

Alcoholic Cardiomyopathy

- Dilated cardiomyopathy caused by chronic excessive alcohol consumption, one of the most common reversible causes of dilated cardiomyopathy. Direct toxic effect of ethanol and its metabolite acetaldehyde on cardiomyocytes. Presentation is identical to other dilated cardiomyopathies.

Alcoholic Ketoacidosis

- A high-anion-gap metabolic acidosis caused by prolonged heavy alcohol use, poor nutritional intake, and vomiting. It typically presents with abdominal pain, nausea, vomiting, dehydration, and tachypnea, often with a normal or low blood glucose concentration.

Alcoholic Neuropathy

- A length-dependent peripheral neuropathy associated with chronic excessive alcohol use and nutritional deficiency, especially thiamine deficiency. It causes burning pain, numbness, paraesthesia, and weakness, most prominently in the feet and lower legs.

Aldosterone

- A mineralocorticoid hormone produced by the zona glomerulosa of the adrenal cortex, the final effector of the renin-angiotensin-aldosterone system (RAAS).

Algor Mortis

- Post-mortem cooling of the body following death until it reaches ambient environmental temperature. Serves as one of the physical estimators of the post-mortem interval (time since death).

Aliasing Artifact

- Ultrasound artifact from high velocities. Reduced by increasing scale, using lower frequency transducer.

Alkaline phosphatase

- A group of enzymes that catalyze the hydrolysis of phosphate esters at alkaline pH, found in high concentrations in the liver, bone, placenta, and intestinal mucosa. Elevated serum levels occur in obstructive liver disease, bone disorders including Paget disease and fractures, pregnancy, and certain malignancies.

Alkalosis

- A metabolic (elevated bicarbonate) or respiratory (low PaCO₂) disturbance that raises arterial pH above 7.45. Metabolic alkalosis: vomiting/NG suction (HCl loss), diuretics, mineralocorticoid excess, hypokalemia, and alkali ingestion—compensated by hypoventilation; chloride-responsive vs. chloride-resistant distinguishes causes.

Allele

- Alternative form of a gene at a particular locus. Individuals inherit two alleles for each gene (one from each parent). Alleles can be dominant, recessive, or codominant.

Allergen

- A substance capable of inducing an allergic (IgE-mediated) immune response in susceptible individuals. Allergens are typically proteins or glycoproteins with molecular weights between 10 and 70 kilodaltons, possessing structural features that promote IgE binding including protease resistance and the ability to crosslink IgE on

Allergen Immunotherapy

- Gradual desensitization to specific allergens through repeated subcutaneous or sublingual exposure. It reduces allergic asthma symptoms and medication needs. Treatment requires 3–5 years for full benefit. It is indicated for allergic asthma with identified specific allergens not controlled by standard therapy.

Allergic Bronchopulmonary Aspergillosis

- A hypersensitivity reaction to *Aspergillus fumigatus* in asthma or CF patients. Causes central bronchiectasis, elevated total IgE, and eosinophilia.

Treatment includes corticosteroids and itraconazole.

Allergic Conjunctivitis

- An inflammatory condition of the conjunctiva mediated by IgE-dependent mast cell degranulation in response to environmental allergens, characterized by bilateral ocular itching, tearing, conjunctival injection, and eyelid edema.

Allergic Dermatitis

- An inflammatory skin reaction caused by an immune response to an external allergen, most commonly allergic contact dermatitis. It is a delayed, T-cell-mediated hypersensitivity reaction that typically develops after prior sensitization.

Allergic Reactions

- Hypersensitivity reactions mediated by IgE antibodies in response to otherwise harmless environmental substances (allergens). Type I immediate hypersensitivity reactions involve initial sensitization where dendritic cells present allergen to T helper 2 cells, stimulating B cell class switching to IgE production.

Allergic Rhinitis

- An IgE-mediated inflammatory response of the nasal mucosa to inhaled allergens, causing sneezing, nasal congestion, clear rhinorrhea, nasal itching, and often conjunctival symptoms. Management may include allergen avoidance, intranasal corticosteroids, antihistamines, and allergen immunotherapy when indicated.

Allergic Rhinitis (Hay Fever)

- Allergies occur when the immune system overreacts to harmless environmental substances such as pollen, dust mites, mold, or pet dander. Hay fever (allergic rhinitis) specifically involves nasal symptoms like sneezing, congestion, and itchy eyes triggered by airborne allergens.

Allergies

- Immune reactions to normally harmless substances such as foods, pollens, medicines, insect venom, or animal dander. Reactions range from localized rhinitis or dermatitis to life-threatening anaphylaxis.

Allergy Testing

- Identification of IgE-mediated allergic sensitization. Skin prick test (SPT): allergen extract applied to skin via lancet, read after 15 min (wheal >3mm =

positive). Intradermal test: more sensitive, used when SPT negative but suspicion remains.

Allodynia

- A form of neuropathic pain where a normally non-painful stimulus evokes pain.

Allopurinol

- A xanthine-oxidase inhibitor used to lower uric acid and prevent gout attacks and some uric-acid kidney stones. It does not treat an acute attack immediately. Rash, fever, liver injury, and severe hypersensitivity are uncommon but important adverse effects; dosing should account for kidney function and interacting medicines.

Allograft

- Tissue transplanted between genetically different individuals of the same species. Allografts may provoke immune rejection and require appropriate donor screening, processing, and, in some settings, immunosuppression.

Allosteric Regulation

- The regulation of enzyme activity by the binding of an effector molecule at a site other than the active site (allosteric site), causing a conformational change that alters the enzyme's activity. Allosteric enzymes typically have multiple subunits with cooperative binding kinetics (sigmoidal rather than hyperbolic velocity curves).

Alogia

- Poverty of speech or poverty of content of speech, a negative symptom of schizophrenia. It manifests as brief, empty replies to questions, reduced spontaneous speech, and lack of elaboration. It reflects dysfunction in the mesocortical dopamine pathway and is less responsive to antipsychotic medications than positive symptoms.

Alopecia

- Hair loss from any cause. Androgenetic alopecia (male/female pattern hair loss): most common, related to DHT sensitivity. Alopecia areata: autoimmune, patchy hair loss with potential progression to totalis (scalp) or universalis (entire body).

Alopecia Areata

- A common, autoimmune, non-scarring hair loss disorder affecting approximately 1 to 2 percent of the population worldwide, with a lifetime risk of 2 per-

cent, occurring equally across sexes and ethnicities, with peak onset between 15 and 29 years.

Alpha-1 Antitrypsin Deficiency

- An autosomal codominant genetic disorder caused by mutations in the SERPINA1 gene. It predisposes to early-onset panlobular emphysema (lower lobes), liver disease, and panniculitis. PiZZ genotype is most severe. Treatment includes AAT augmentation therapy and smoking avoidance.

Alpha-Agonists

- Alpha-adrenergic agonists are a class of topical medications used in the treatment of glaucoma that lower intraocular pressure through a dual mechanism: reduction of aqueous humor production by the ciliary body through alpha-2 adrenergic receptor stimulation and enhancement of uveoscleral outflow through remodeling of the extracellular ma

Alpha-Glucosidase Inhibitors

- Oral glucose-lowering medications including acarbose, miglitol, and voglibose that delay carbohydrate digestion and glucose absorption in the small intestine by inhibiting brush border alpha-glucoside enzymes. Reduce postprandial glucose excursions without affecting fasting glucose significantly.

Alpha-Ketoglutarate

- A five-carbon intermediate of the citric acid cycle formed from isocitrate and converted to succinyl-CoA. It also serves as a carbon skeleton for amino-acid metabolism and participates in oxidative deamination and cellular signaling.

Alport Syndrome

- Hereditary nephritis caused by mutations in type four collagen genes leading to progressive sensorineural hearing loss, ocular anomalies, and renal failure. X-linked form from COL4A5 mutations is the most common. Autosomal recessive and dominant forms exist.

ALS

- Amyotrophic lateral sclerosis (motor neuron disease, Lou Gehrig disease): a progressive neurodegenerative disease of upper and lower motor neurons, causing muscle weakness, atrophy, fasciculations, spasticity, and eventually respiratory failure.

Altitude Illness

- Acute mountain sickness (AMS): headache + 1

symptom (nausea, fatigue, dizziness, insomnia) at >2500m. HACE: AMS + ataxia/altered mental status (cerebral edema). HAPE: dyspnea at rest, cough, frothy sputum, crackles, hypoxemia (noncardiogenic pulmonary edema).

Aluminum Phosphide

- Fumigant poison released as phosphine gas upon contact with moisture. Mechanism: inhibits cytochrome c oxidase, cellular hypoxia. Treatment: supportive, cardiac monitoring, magnesium, calcium. No specific antidote.

Alvarado Score

- Clinical scoring system for acute appendicitis. Components (MANTRELS): Migration of pain (1), Anorexia (1), Nausea/vomiting (1), Tenderness in RLQ (2), Rebound pain (1), Elevated temperature (1), Leukocytosis (2), Left shift (1).

Alveolar Bone

- Bone surrounding and supporting tooth roots. Resorbs with tooth loss. Maintained by function. Bone grafting may be needed for implants.

Alveolar Gas Equation

- $PAO_2 = (F_iO_2 \times (P_{atm} - P_{H_2O})) - (P_aCO_2 / R)$. $P_{atm} = 760$ mmHg, $P_{H_2O} = 47$ mmHg, $R = 0.8$. Allows calculation of A-a gradient.

Alveolar Ventilation

- The volume of fresh air reaching the alveoli per minute, representing the portion of minute ventilation that participates in gas exchange. Calculated as (tidal volume - dead space volume) x respiratory rate. Normal alveolar ventilation is approximately 4-5 L/min.

Alveolar Ventilation Equation

- The relationship between alveolar ventilation and arterial carbon-dioxide tension, approximately $P_aCO_2 = 0.863$ times carbon-dioxide production divided by alveolar ventilation. It explains why reduced effective ventilation causes hypercapnia.

Alveolus

- The terminal air sac (200-300 microns) where gas exchange occurs. Approximately 300-500 million per lung, providing 70 m² surface area. Wall: type I pneumocytes (95% surface, thin for diffusion), type II pneumocytes (5%, secrete pulmonary surfactant reducing surface tension), alveolar macrophages

(dust cells, phagocytosis).

Alzheimer Disease

- A progressive neurodegenerative disorder characterized pathologically by amyloid-beta plaques, tau neurofibrillary pathology, and neuronal loss. It most commonly presents with insidious impairment of episodic memory followed by broader cognitive and functional decline.

Amalgam

- Restorative material composed of mercury alloyed with silver, tin, copper, and other metals. Properties: strong, durable, cost-effective. Concerns: mercury toxicity debates, aesthetic limitations. Placement: requires cavity preparation, condensation, carving. Removal: sectioning, cooling to prevent mercury vapor.

Amaurosis Fugax

- Transient monocular vision loss lasting seconds to minutes, classically described as a curtain descending over the field, caused by retinal or optic nerve ischemia.

Ambiguous Genitalia

- External genital anatomy that does not have typical male or female features, usually because of a difference in sex development. Evaluation includes examination, chromosome and hormone testing, pelvic or abdominal imaging, and assessment for congenital adrenal hyperplasia.

Amblyopia

- Amblyopia, commonly known as lazy eye, is a neurodevelopmental disorder of visual acuity that results from abnormal visual experience during the critical period of visual development (birth to approximately eight years of age), leading to reduced best-corrected visual acuity that cannot be attributed to structural ocular pathology.

Ambu Bag

- Manual ventilation bag. Self-inflating, 1-2 L capacity.

Amebiasis

- Infection with the protozoan parasite *Entamoeba histolytica*, acquired mainly by ingestion of faecally contaminated food or water. It may be asymptomatic or cause amoebic colitis with abdominal pain, diarrhoea, or dysentery, and can spread

through the portal circulation to produce a liver abscess.

Amebic Liver Abscess

- A focal collection of necrotic material in the liver caused by *Entamoeba histolytica*, usually following intestinal infection. It commonly causes fever, right-upper-quadrant pain, and tender hepatomegaly; diagnosis is supported by imaging and serology.

Amelanotic melanoma

- A variant of melanoma that lacks significant melanin production, presenting as a pink, red, or flesh-coloured papule or nodule rather than a typical dark lesion. Its paucity of pigment can delay diagnosis, making it a diagnostic challenge. Histopathology with immunohistochemical markers confirms the diagnosis.

Amelia

- A rare congenital anomaly characterized by complete absence of one or more limbs. Unilateral upper limb amelia is the most common form. Causes include vascular disruption, amnion rupture sequence, and teratogens. May be isolated or syndromic (Roberts syndrome, ADAM sequence, amniotic band syndrome). Diagnosis by prenatal

Ameloblastoma

- Locally aggressive odontogenic tumor. Multicystic radiolucency in posterior mandible. High recurrence rate. Treatment: surgical resection with margins.

Amelogenesis

- See dental.

Amelogenesis Imperfecta

- Hereditary enamel defect. Affects all teeth. Types: hypoplastic, hypomaturational, hypocalcified.

Amenorrhea

- Absence of menstruation. Primary amenorrhea: no menarche by age 15 with normal secondary sexual development, or by age 13 without. Secondary amenorrhea: absence of menses for 3 cycles or 6 months in women with prior menstruation.

Amino Acid

- The building blocks of proteins, 20 standard L-amino acids sharing a central carbon (α -carbon) bonded to an amino group ($-\text{NH}_2$), carboxyl group ($-\text{COOH}$), hydrogen, and a unique side chain (R group).

Amino Acid Catabolism

- The breakdown of amino acids after protein digestion, with the amino group removed and the carbon skeleton metabolized for energy or gluconeogenesis/ketogenesis. Transamination: amino groups are transferred to α -ketoglutarate, forming glutamate, by aminotransferases requiring pyridoxal phosphate (PLP, vitamin B6) as a cofactor.

Aminoglycoside nephrotoxicity

- Kidney damage caused by aminoglycoside antibiotics, characterized by acute tubular necrosis. Nephrotoxicity is caused by accumulation of aminoglycosides in the proximal tubular cells. The risk is increased with prolonged therapy, high doses, concurrent nephrotoxic drugs, and renal impairment.

Aminoglycoside ototoxicity

- Damage to the inner ear caused by aminoglycoside antibiotics, resulting in hearing loss and vestibular dysfunction. Ototoxicity is caused by accumulation of aminoglycosides in the inner ear, damaging hair cells. The risk is increased with prolonged therapy, high doses, concurrent ototoxic drugs, and renal impairment.

Aminoglycosides

- A class of bactericidal antibiotics that bind irreversibly to the 30S ribosomal subunit, causing misreading of messenger RNA and inhibiting protein synthesis. They include gentamicin, tobramycin, amikacin, and streptomycin.

Amiodarone

- A class III antiarrhythmic with potassium-channel-blocking effects and additional sodium-channel, calcium-channel, and antiadrenergic actions. It treats selected atrial and ventricular arrhythmias but can cause pulmonary, thyroid, hepatic, ocular, neurologic, and skin toxicity and has numerous drug interactions.

Amiodarone-induced skin discoloration

- A slate-grey to bluish hyperpigmentation of sun-exposed skin caused by long-term amiodarone therapy, due to deposition of the drug and its metabolites in dermal macrophages. It is dose-related, more pronounced in photodistributed areas, and may be partially reversible upon drug cessation.

Amiodarone-Induced Thyroid

Dysfunction

- Thyroid abnormalities occurring in approximately fifteen to twenty percent of patients on amiodarone due to the high iodine content of the molecule. Type one results from iodine-induced increased thyroid hormone synthesis in patients with underlying thyroid disease. Type two results from destructive thyroiditis.

Amiodarone-induced vortex keratopathy

- Whorl-like corneal epithelial deposits caused by amiodarone accumulation, appearing as golden-brown verticillata on slit-lamp examination, usually asymptomatic but reversible upon drug discontinuation.

Ammonia

- Measurement of serum ammonia (NH₃). Normal: 15-45 mcg/dL (9-33 micromol/L). Elevated in hepatic encephalopathy (monitoring and diagnosis), urea cycle disorders (very high, >200 in neonates), Reye syndrome, and valproate therapy.

Amnesia

- Loss or impairment of memory that is greater than ordinary forgetting. It may affect the ability to form new memories (anterograde amnesia), recall past events (retrograde amnesia), or both.

Amniocentesis

- Sampling of amniotic fluid for prenatal diagnosis. Typically performed at 15-20 weeks gestation. Indications: advanced maternal age (35), abnormal quad screen, family history of genetic disorders, and fetal lung maturity assessment.

Amnion

- The innermost fetal membrane, lining the amniotic cavity and containing the amniotic fluid. A thin, avascular membrane composed of a single layer of cuboidal epithelium overlying a basement membrane and connective tissue. The amnion produces and resorbs amniotic fluid. It fuses with the chorion by 12-14 weeks.

Amniotic Fluid

- The clear to pale-yellow fluid surrounding the fetus within the amniotic sac. Early in pregnancy it is derived largely from maternal plasma; later, fetal urine and lung fluid are major sources, and the fetus continuously swallows and recycles it.

Amniotic Fluid Embolism

- A rare, catastrophic obstetric emergency where amniotic fluid enters the maternal circulation, causing acute cardiovascular collapse. Characterized by sudden respiratory distress, hypotension, coagulopathy, and seizure. Mortality rate 20-60% among survivors; 50% of affected mothers die.

Amniotic Fluid Index

- A semi-quantitative assessment of amniotic fluid volume using ultrasound. The uterus divided into four quadrants, and the single deepest vertical pocket measured in each quadrant, then summed. Normal range is 5-24 cm at 16-40 weeks. Less than 5 cm indicates oligohydramnios; greater than 24 cm indicates polyhydramnios.

Amoebiasis

- Infection caused by the protozoan parasite *Entamoeba histolytica*, transmitted via the fecal-oral route (contaminated food or water).

Amoxicillin rash in infectious mononucleosis

- A maculopapular, non-allergic rash that develops in up to 90 percent of patients with Epstein-Barr virus infectious mononucleosis who receive amoxicillin or ampicillin. The rash is often widespread and pruritic, and its presence does not indicate true penicillin allergy.

Amoxicillin

- A penicillin-type antibiotic used for selected bacterial infections, including some respiratory, ear, dental, urinary, and skin infections. It does not treat viral illnesses such as colds or influenza. Common adverse effects include nausea, diarrhea, and rash; immediate hives, facial swelling, or breathing difficulty may indicate a serious allergic reaction.

Ampicillin

- A penicillin antibiotic used for selected serious bacterial infections, often in hospital care and sometimes with other medicines. It is active against several gram-positive and gram-negative bacteria, but resistance is common and treatment should be guided by clinical assessment and susceptibility testing. Allergic reactions and antibiotic-associated diarrhea can occur.

AMP (Adenosine Monophosphate)

- A nucleotide composed of adenine, ribose, and a single phosphate group. Formed by hydrolysis of ADP ($\text{ADP} \rightarrow \text{AMP} + \text{Pi}$) or directly from ATP via adenylate kinase ($2 \text{ADP} \rightleftharpoons \text{ATP} + \text{AMP}$).

AMPK

- AMP-activated protein kinase. Energy sensor. Activated by AMP. Inhibits anabolism, stimulates catabolism.

Amputation

- Surgical removal of a limb or part of a limb. Levels: toe, partial foot (transmetatarsal), below-knee (BKA, preserves knee function, 40-60% walk with prosthesis), through-knee, above-knee (AKA), hip disarticulation, and hemipelvectomy. Upper extremity: digit, ray, transradial, transhumeral, and shoulder disarticulation.

Amputation Stump

- The remaining part of a limb after amputation. It must be monitored for wound healing, infection, poor circulation, neuroma pain, phantom-limb pain, and skin problems caused by prosthetic use. Care includes hygiene, compression or shaping when advised, progressive strengthening, and professional prosthetic and rehabilitation assessment.

AMR (Antimicrobial Resistance)

- Bacteria resisting antibiotic effects. See antibiotic resistance.

Amygdala

- Almond-shaped nucleus cluster in the anterior medial temporal lobe, part of the limbic system, primarily processing emotions (especially fear/anxiety). Receives thalamic/cortical sensory input; projects to hypothalamus, brainstem, and cortex for autonomic/behavioral fear responses.

Amylase

- Digestive enzyme. Pancreatic and salivary sources. Elevated in pancreatitis.

Amylase and Lipase

- Pancreatic enzymes, elevated in acute pancreatitis. Lipase is more specific (pancreas-only) and remains elevated longer than amylase. Normal: amylase 30-110 U/L, lipase 10-140 U/L. Diagnosis: elevation $>3\times$ upper limit of normal, with compatible clinical presentation (epigastric pain, nausea/vomiting).

Amylin Analogues

- Pramlintide is a synthetic analogue of amylin a co-secreted hormone from pancreatic beta cells that slows gastric emptying suppresses postprandial glucagon secretion and promotes satiety. Used as adjunctive therapy with insulin in type one and type two diabetes. Reduces postprandial glucose excursions and A1c.

Amyloid PET

- Tracers: Florbetapir, Flutemetamol, Florbetaben. Detect amyloid plaques in Alzheimer disease.

Amyloid purpura

- Purpuric discoloration of the eyelids and periorbital skin (raccoon eyes) caused by cutaneous amyloid deposition, which weakens vessel walls and leads to extravasation of erythrocytes. It may be a sign of systemic amyloidosis, particularly AL amyloidosis, and warrants further workup including serum free light chains and tissue biopsy.

Amyloidosis

- A group of disorders characterized by the extracellular deposition of misfolded proteins that polymerize into insoluble fibrils with a distinctive beta-pleated sheet configuration. These fibrils bind Congo red dye and show characteristic apple-green birefringence under polarized light microscopy.

Amyotrophic Lateral Sclerosis

- Progressive neurodegenerative disorder affecting both upper and lower motor neurons, causing weakness, atrophy, fasciculations, and eventually respiratory failure. Most cases are sporadic (90 percent); 10 percent are familial. Average age of onset 55-60 years.

Anabolism

- The set of metabolic pathways that construct complex molecules from simpler precursors, requiring energy input (ATP, reducing equivalents NADPH). Examples: protein synthesis from amino acids, glycogen synthesis (glycogenesis), fatty acid synthesis (lipogenesis), cholesterol synthesis, nucleotide synthesis, gluconeogenesis.

Anaemia

- A condition in which the number of red blood cells or the hemoglobin concentration is below the normal range, reducing the oxygen-carrying capacity of the

blood.

Anaerobe

- A microorganism that grows without molecular oxygen. Obligate anaerobes may be injured or killed by oxygen, while aerotolerant anaerobes tolerate oxygen but do not use it for energy production.

Anal Abscess

- Collection of pus in the perianal or ischiorectal space usually from infected anal glands. Presents with throbbing anal pain, swelling, fever, and difficulty sitting. Diagnosed clinically with tenderness and fluctuance. Treatment is incision and drainage under local anesthesia. Antibiotics are not routinely required for simple abscesses.

Anal Canal

- The terminal 3-4 cm of the gastrointestinal tract, extending from the anorectal junction (puborectalis/pectinate line) to the anal verge. Above the pectinate line: lined by columnar epithelium, innervated by autonomic nerves, draining to portal system, supplied by superior rectal artery.

Anal Cancer

- Malignant tumor of the anal canal or margin, most commonly squamous cell carcinoma associated with high-risk human papillomavirus infection. Presents with bleeding, pain, or a mass; primary treatment is chemoradiation, with surgery reserved for residual or recurrent disease.

Anal Fissure

- Linear tear in the distal anal mucosa usually posterior in the midline. Presents with painful defecation and bright red blood on wiping or in the toilet. Associated with constipation, hard stools, and chronic diarrhea. The internal anal sphincter spasm perpetuates the fissure by reducing blood flow to the posterior midline.

Anal Fistula

- Abnormal tract connecting the anal canal to the perianal skin usually following a drained abscess. Presents with recurrent drainage, pain, and pruritus. Parks classification categorizes fistulae by relationship to the sphincter muscles. Diagnosed by examination under anesthesia and MRI.

Anal Intercourse

- The insertion of the penis into the anus and rec-

tum for sexual pleasure. The anal canal is rich in nerve endings, making it a sensitive erogenous zone. The rectum lacks natural lubrication, so artificial lubricants are essential to prevent mucosal trauma.

Analgesic

- A drug that relieves pain without producing loss of consciousness.

Analgesic Nephropathy

- Chronic tubulointerstitial renal injury associated with prolonged or excessive use of analgesic medicines. It may cause impaired urinary concentration, sterile pyuria, haematuria, papillary necrosis, hypertension, and progressive chronic kidney disease.

Analysis of Urine

- Analysis of urine. See Urinalysis.

Analytic Sensitivity

- The smallest amount or concentration of an analyte that a laboratory method can reliably detect. It concerns measurement performance and differs from clinical sensitivity, which describes detection of disease in affected patients.

Anaphylactic Shock

- A severe, potentially life-threatening systemic hypersensitivity reaction characterized by rapid onset of multisystem involvement, primarily affecting the skin, respiratory tract, cardiovascular system, and gastrointestinal tract. See Anaphylaxis.

Anaphylaxis

- A severe, potentially life-threatening systemic hypersensitivity reaction characterized by rapid onset of multisystem involvement, primarily affecting the skin, respiratory tract, cardiovascular system, and gastrointestinal tract.

Anaphylaxis During Anesthesia

- A life-threatening systemic hypersensitivity reaction that may cause sudden hypotension, bronchospasm, cardiovascular collapse, and sometimes urticaria or flushing. Common perioperative triggers include neuromuscular-blocking agents, antibiotics, chlorhexidine, and latex.

Anaplasia

- A lack of differentiation, describing cells that have reverted to a more primitive, embryonic phenotype with loss of the specialized structural and functional

characteristics of their tissue of origin.

Anaplastic Thyroid Cancer

- A rare, rapidly growing, and highly aggressive thyroid cancer that often presents as a large fixed neck mass with dysphagia or airway compromise. Prognosis is poor; management is usually palliative.

Anastomosis (Vascular)

- See Anastomosis Device.

Anastomotic Leak

- Breakdown of a surgical anastomosis in the gastrointestinal tract causing spillage of luminal contents into the peritoneal cavity. Presents with fever, tachycardia, abdominal pain, leukocytosis, and peritonitis. Diagnosed by CT with oral or rectal contrast showing extraluminal contrast or fluid collection.

Anatomical Axis

- An imaginary straight line around which a body part moves or rotates.

Anatomical Planes

- Hypothetical flat surfaces that pass through the body in anatomical position, used to slice the body for anatomical descriptions or medical imaging.

Anatomical Position

- The standard reference position of the body used to describe anatomy, defined as standing erect, facing forward, with arms at the sides and palms facing forward.

Anatomical Snuffbox

- Triangular depression on the dorsal wrist, bounded by extensor pollicis longus medially and extensor pollicis brevis/abductor pollicis longus laterally. Floor contains the scaphoid, trapezium, and radial artery. Tenderness after a fall on an outstretched hand suggests scaphoid fracture. The radial artery pulse is palpable here.

Anatomy

- Study of body structure. Regions: head, neck, thorax, abdomen, pelvis, extremities. Systems: skeletal, muscular, nervous, circulatory, respiratory, digestive, urinary, reproductive, endocrine, immune.

Anatomy Survey (Level II Ultrasound)

- A comprehensive fetal ultrasound performed at 18-22 weeks to evaluate fetal anatomy and screen for structural anomalies.

ANCA-Associated Vasculitis

- A group of small-vessel vasculitides associated with antineutrophil cytoplasmic antibodies. Includes granulomatosis with polyangiitis (c-ANCA/PR3), microscopic polyangiitis (p-ANCA/MPO), and eosinophilic granulomatosis with polyangiitis (p-ANCA/MPO with asthma/eosinophilia).

Ancillary Test (Pathology)

- A test performed in addition to routine morphology to refine a pathologic diagnosis. Examples include IHC, flow cytometry, cytogenetics, molecular testing, special stains, and electron microscopy.

Ancylostoma (Hookworm)

- Nematode infestation by *Ancylostoma duodenale* (Old World) or *Necator americanus* (New World), common in tropical and subtropical regions where sanitation is poor.

Androgen Deficiency

- Hypogonadism in males: reduced testosterone production causing symptoms and signs. Causes: primary (testicular—Klinefelter, mumps orchitis, chemotherapy, trauma) and secondary (pituitary/hypothalamic—prolactinoma, iron overload, pituitary lesions, opioids, obesity, aging).

Androgenetic Alopecia

- The most common cause of hair loss, caused by progressive miniaturization of genetically susceptible hair follicles under androgen influence. Male pattern commonly affects the frontal hairline and vertex; female pattern commonly causes diffuse crown thinning with relative preservation of the frontal hairline.

Andropause

- Age-related testosterone decline in men. Symptoms: fatigue, decreased libido, depression. Treatment controversial.

Anejaculation

- Absence of ejaculation. Causes: neurological, surgical, pharmacological, psychological.

Anemia

- A reduction in the oxygen-carrying capacity of the blood, defined by a hemoglobin concentration below the normal range for age and sex: less than 13.5 g/dL in adult males, less than 12.0 g/dL in adult females.

Anemia (Cancer)

- Low red blood cells. Due to cancer, chemotherapy, chronic disease. Treatment: transfusion, erythropoiesis-stimulating agents.

Anemia of Chronic Disease

- Anemia of inflammation. Normocytic or microcytic. Low iron, low TIBC, normal/high ferritin.

Anencephaly

- A fatal neural tube defect where the cerebral hemispheres and cranial vault fail to develop due to failed closure of the rostral neuropore during the fourth week of gestation. The brainstem and spinal cord are present, and reflex movements may occur, but there is no cognitive function.

Anergy

- State of T-lymphocyte non-responsiveness to nominal antigen despite prior sensitization, reflecting functional inactivation rather than cell death.

Anesthesia

- The use of medications to prevent sensation, particularly pain, during medical procedures or surgery.

Anesthesia for Coagulopathy

- Coagulopathy increases bleeding risk during surgery. Assessment includes history, physical examination, and laboratory tests (INR, PT, aPTT, platelet count, fibrinogen). INR greater than 1.5 or aPTT greater than 1.5 times control increases bleeding risk.

Anesthesia for Epilepsy

- Patients with epilepsy generally require continuation of antiseizure medication perioperatively because missed doses increase seizure risk. Propofol and benzodiazepines are anticonvulsant; high concentrations of some volatile agents may produce epileptiform activity.

Anesthesia for Hemophilia

- Hemophilia A results from factor VIII deficiency and hemophilia B from factor IX deficiency. Perioperative care is coordinated with hematology and uses factor replacement or, in selected mild hemophilia A cases, desmopressin.

Anesthesia for Neurosurgery

- Neuroanesthesia aims to maintain cerebral perfusion pressure (CPP = mean arterial pressure

minus intracranial pressure), optimize brain relaxation, and minimize secondary brain injury. Induction agents (propofol, etomidate) reduce cerebral metabolic rate and intracranial pressure.

Anesthesia for Pheochromocytoma

- Pheochromocytoma is a catecholamine-secreting tumor causing episodic hypertension, tachycardia, headache, sweating, and pallor. Preoperative alpha-blockade (phenoxybenzamine 10 to 14 days) is essential before beta-blockade (propranolol) to prevent unopposed alpha stimulation.

Anesthesia for Porphyria

- Acute intermittent porphyria is a metabolic disorder of heme synthesis causing abdominal pain, neuropathy, psychiatric symptoms, and dark urine. Certain anesthetics are porphyrinogenic (triggering attacks): barbiturates, sulfonamides, carbamazepine, and possibly volatile anesthetics (though sevoflurane is considered safe).

Anesthesia Gas Analyzers

- Anesthesia gas analyzers measure delivered and exhaled concentrations of oxygen, carbon dioxide, and volatile anesthetic agents. The oxygen analyzer measures inspired oxygen concentration; pulse oximetry measures peripheral hemoglobin oxygen saturation.

Anesthesia Informatics

- Anesthesia informatics integrates electronic health records, decision support systems, and data analytics into clinical practice. Automated documentation reduces charting time. Clinical decision support alerts for drug interactions and dosing errors. Big data analytics identify patterns in anesthesia outcomes.

Anesthesia Machine

- The anesthesia machine delivers precise mixtures of oxygen, air, and volatile anesthetics to the patient.

Anesthesia Monitoring

- Anesthesia monitoring encompasses continuous assessment of patient physiology during surgery.

Anesthesia Pharmacogenomics

- Anesthesia pharmacogenomics studies genetic variations affecting drug response. Pseudocholinesterase deficiency causes prolonged succinylcholine paralysis. CYP2D6 polymorphisms affect codeine metabolism

to morphine. Malignant hyperthermia susceptibility is linked to RYR1 gene mutations.

Aneurysm

- Abnormal focal dilation of blood vessel. Types: arterial (aortic, peripheral), venous (rare). Risk factors: atherosclerosis, hypertension, smoking, genetic disorders (Marfan, Ehlers-Danlos), infection. Abdominal aortic aneurysm (AAA): screening by ultrasound in men 65-75 who smoked. Repair indicated if >5.5 cm, symptomatic, or expanding.

Aneurysm Clip

- Surgical clip occluding aneurysm neck. Materials: titanium, stainless steel. Indications: intracranial aneurysm. Placed via craniotomy. Complications: clip migration, vessel occlusion, rebleeding.

Aneurysm Coils

- Platinum coils for aneurysm embolization. Detachable coils, balanced coil formation.

Aneurysm Repair

- Surgical or endovascular treatment of aneurysm. Indications: size, symptoms, expansion rate.

Angelman Syndrome

- A genetic disorder caused by loss of function of the UBE3A gene on chromosome 15 (maternal deletion or paternal uniparental disomy). Features: severe intellectual disability, absent speech, seizures, ataxia, and frequent laughter. Characteristic happy demeanor with hand-flapping movements.

Angina Pectoris

- Chest discomfort caused by myocardial ischemia without infarction. Stable angina occurs predictably with exertion and resolves with rest. Unstable angina occurs at rest or with minimal exertion and is an acute coronary syndrome. Prinzmetal (variant) angina occurs at rest due to coronary vasospasm.

Angiodysplasia

- Vascular ectasia consisting of dilated fragile blood vessels in the gastrointestinal mucosa and submucosa, most commonly found in the cecum and ascending colon. They are the most common cause of acute lower gastrointestinal bleeding in elderly patients over sixty years of age.

Angioedema

- Rapid swelling of the deep dermis/subcutaneous and submucosal tissues (lips, eyelids, tongue, throat,

hands, GI tract) due to mast cell mediator release (histamine—allergic/idiopathic, urticaria associated) or bradykinin-mediated (hereditary angioedema—C1 esterase inhibitor deficiency; ACE-inhibitor-induced).

Angiogenesis

- Formation of new blood vessels. Tumors induce angiogenesis for growth. Targeted by anti-angiogenic drugs (bevacizumab).

Angiography

- Invasive imaging of blood vessels by injecting contrast material under fluoroscopy. Coronary angiography (cardiac catheterization): assesses coronary artery stenosis, guides PCI. Cerebral angiography: evaluates aneurysms, AVMs, and vasculitis. Peripheral angiography: evaluates PAD and guides intervention.

Angiomyolipoma

- A benign renal tumor composed of blood vessels, smooth muscle, and fat; it is associated with tuberous sclerosis and may cause retroperitoneal hemorrhage.

Angioplasty

- Percutaneous coronary intervention (PCI) to open stenotic coronary arteries. Balloon angioplasty: inflation of balloon at the lesion to compress plaque. Bare metal stent (BMS): metal scaffold, restenosis 20-30%. Drug-eluting stent (DES): elutes antiproliferative (sirolimus, paclitaxel, everolimus), restenosis 5-10%.

Angioplasty Balloon (Vascular)

- See Angioplasty Balloon.

Angiosarcoma

- A rare, aggressive malignant tumor of blood vessel or lymphatic endothelial cells. Common sites include skin, breast, liver, and bone. Treatment typically involves wide surgical excision with chemotherapy and radiation.

Angiotensin converting-enzyme inhibitor

- A class of antihypertensive medications that block the conversion of angiotensin I to angiotensin II by inhibiting the angiotensin-converting enzyme, thereby reducing vasoconstriction and aldosterone secretion. Used to treat hypertension, heart failure, and diabetic nephropathy.

Angiotensin Receptor Blockers

- A class of cardiovascular drugs that selectively block the angiotensin II type 1 receptor, preventing the binding of angiotensin II and its effects including vasoconstriction, aldosterone secretion, and sympathetic activation.

Anhedonia

- The inability to experience pleasure from activities that were previously enjoyable. It is a core symptom of major depressive disorder and a negative symptom of schizophrenia. It is mediated by dysfunction in the brain's reward circuitry, particularly the mesolimbic dopamine system.

Animal-Assisted Therapy

- Use of animals for therapeutic purposes. Reduces stress, improves mood, motivates exercise.

Anion Gap

- A calculated estimate of unmeasured serum anions, commonly sodium minus chloride and bicarbonate. An increased anion gap supports the presence of unmeasured acids in metabolic acidosis, but interpretation must account for albumin concentration and laboratory method.

Anisocoria

- Unequal pupillary diameter exceeding 0.4 mm, physiological in about 20 percent of the population. Pathological causes are localized by which pupil is abnormal and whether asymmetry worsens in bright light (iris sphincter damage, Adie pupil) or dim light (Horner syndrome).

Anisocytosis

- Increased variation in red blood cell size on a peripheral blood film (e.g., macrocytes with microcytes together), not a diagnosis in itself. Derived from the Greek "anisokytosis" (unequal cells). Quantified by an elevated red cell distribution width (RDW).

Ankle Arthroplasty

- Total ankle arthroplasty (TAA): indications: end-stage ankle OA (post-traumatic, primary, inflammatory), preserved hindfoot alignment (<10 - 15 deg varus/valgus), no severe osteoporosis, no active infection. Contraindications: severe deformity, Charcot, prior infection, young high-demand.

Ankle Fractures

- Ankle fractures: Weber classification (fibular frac-

ture relative to syndesmosis): A (below syndesmosis, stable), B (at syndesmosis, variable stability), C (above syndesmosis, unstable). Lauge-Hansen: supination-external rotation (SER, most common), supination-adduction, pronation-external rotation, pronation-abduction.

Ankle Joint

- Uniaxial hinge synovial joint between distal tibia, lateral malleolus, and superior talus, allowing dorsiflexion/plantarflexion. Medial (deltoid) ligament resists eversion; lateral complex resists inversion. The ankle mortise provides bony stability.

Ankle Pain

- Pain in or around the ankle, commonly caused by sprain, fracture, tendinitis, arthritis, bursitis, or infection. Inability to bear weight, deformity, marked swelling, or fever requires prompt assessment.

Ankle Sprain

- An injury to the ligaments surrounding the ankle joint, most commonly involving the lateral ligament complex (anterior talofibular ligament [ATFL, most commonly injured], calcaneofibular ligament [CFL], and posterior talofibular ligament [PTFL]) from an inversion mechanism.

Ankle-Brachial Index

- Ratio of systolic BP at the ankle (dorsalis pedis or posterior tibial artery) to the brachial artery, measured with Doppler. Normal: 1.0-1.4. PAD: ABI 0.90 (mild 0.7-0.9, moderate 0.4-0.7, severe <0.4). Non-compressible vessels (ABI >1.4 , shows calcified vessels in diabetes).

Ankyloglossia (Tongue-Tie)

- Short or thick lingual frenulum restricting tongue movement. May cause breastfeeding difficulties, speech problems, oral hygiene issues. Treatment: frenotomy (simple release) or frenuloplasty.

Ankylosing Spondylitis

- A chronic, progressive inflammatory disease primarily affecting the axial skeleton (sacroiliac joints and spine), characterized by inflammation at entheses (where tendons, ligaments, and joint capsules insert into bone), leading to new bone formation, syndesmophyte development, and eventual vertebral fusion (bamboo spine).

Ankylosis

- Fusion of tooth root to bone. No periodontal ligament. Tooth appears submerged. Treated: extraction or orthodontic intrusion.

Annihilation

- Positron-electron annihilation producing two 511 keV photons in opposite directions.

Annular Pancreas

- A congenital anomaly where a ring of pancreatic tissue completely or partially encircles the second portion of the duodenum, causing duodenal obstruction. It results from failure of the ventral pancreatic bud to completely rotate around the duodenum during embryogenesis.

Anodontia

- Complete absence of teeth. Rare, associated with ectodermal dysplasia.

Anophthalmia

- Congenital absence of the eye due to failure of optic vesicle development, often associated with syndromes (e.g., SOX2 mutations). Diagnosis by absence of globe on imaging or clinical examination; management requires early socket expansion and prosthetic fitting for orbital and facial development.

Anorectal Abscess

- A localized collection of pus in the tissues around the anus or rectum, usually caused by infection of an anal gland. It commonly causes severe anal pain, swelling, fever, and difficulty sitting. Prompt incision and drainage is the standard treatment.

Anorectal Malformation

- A spectrum of congenital anomalies involving the anus and rectum, classified by the location of the rectal pouch relative to the puborectalis sling: Low (passes through the levator muscle, good prognosis)
- perineal fistula, vestibular fistula, or covered anus.

Anorectal Manometry

- Measurement of pressures in the anal canal and rectum to assess sphincter function and rectal sensation. A pressure-sensing catheter is inserted into the rectum and slowly withdrawn through the anal canal. Measures resting and squeeze pressures reflecting internal and external sphincter function respectively.

Anorexia

- Loss of appetite, distinct from anorexia nervosa (the eating disorder).

Anorexia Nervosa

- An eating disorder characterized by restriction of energy intake, intense fear of weight gain, and distorted body image. Highest mortality of any psychiatric disorder. Features: emaciation, lanugo, bradycardia, hypotension, amenorrhea, and electrolyte abnormalities. Diagnosis by DSM-5 criteria.

Anorgasmia

- Persistent or recurrent delay in, or absence of, orgasm following a normal sexual excitement phase.

Anoscopy

- Examination of the anal canal with a short lighted instrument called an anoscope. It is used to evaluate rectal bleeding, hemorrhoids, fissures, fistulas, and other anorectal disorders.

Anosmia

- Complete loss of the sense of smell. Causes include post-viral olfactory dysfunction, chronic rhinosinusitis with polyps, head trauma, neurodegenerative disease, congenital disorders such as Kallmann syndrome, medications, and toxic exposure.

Anosognosia

- Denial or unawareness of one's own neurologic deficit, classically hemiplegia following a nondominant (right) parietal lobe lesion. Associated with right middle cerebral artery strokes and sometimes with neglect of the left side of space.

Anoxia

- Complete absence of oxygen supply to tissues, distinguished from hypoxia (reduced oxygen). Anoxia causes rapid cell death, particularly in highly aerobic tissues (brain, heart, kidneys).

Antacids

- Medications that neutralise gastric acid, providing rapid symptomatic relief in dyspepsia, gastroesophageal reflux, and peptic ulcer disease. Commonly contain aluminium hydroxide, magnesium hydroxide, calcium carbonate, or sodium bicarbonate.

Antecubital Fossa

- Triangular depression on the anterior elbow,

bounded laterally by brachioradialis, medially by pronator teres, and superiorly by a line connecting the epicondyles. Floor formed by brachialis and supinator. Contains biceps tendon, brachial artery, and median nerve. Primary site for venipuncture and blood pressure measurement.

Antemortem Injury

- Injury inflicted upon a body prior to death, characterized microscopically and macroscopically by physiological tissue responses including hemorrhage, blood clotting, inflammatory cell infiltration, tissue edema, and active healing reactions (differentiating it from postmortem artifact).

Antenatal

- The period before birth, also called prenatal. The medical supervision and screening provided to pregnant women to monitor the health of the mother and fetus, detect complications early, and provide health education.

Antepartum Fetal Surveillance

- The monitoring of fetal well-being in the third trimester for high-risk pregnancies. Methods: fetal kick counts (daily maternal counting of fetal movements), non-stress test (NST), biophysical profile (BPP, AFI, fetal tone, movement, breathing), and Doppler (umbilical artery, MCA).

Antepartum Hemorrhage

- Bleeding from the genital tract after the threshold of fetal viability and before birth. Important causes include placenta previa, placental abruption, vasa previa, and local cervical or vaginal lesions; maternal stabilization and fetal assessment are priorities.

Anterior

- Relating to the front of the body or positioned closer to the front.

Anterior Chamber Angle

- The anterior chamber angle is the anatomical region where the cornea and iris meet, and it contains the trabecular meshwork, the primary drainage structure for aqueous humor from the eye.

Anterior Cruciate Ligament Injury

- Tear of the ACL, a common sports injury (non-contact deceleration, pivoting, or hyperextension). Presents with a pop, immediate swelling (hemarthrosis), and giving way of the knee. Positive Lachman

test (most sensitive), anterior drawer test, and pivot shift test.

Anterior Cruciate Ligament Tear

- ACL tear: common knee injury, often non-contact pivoting sports. Symptoms: popping sound, swelling, instability, inability to continue play. Diagnosis: Lachman test, anterior drawer, MRI. Treatment: surgical reconstruction (autograft/allograft), rehabilitation.

Anterior Pituitary

- The glandular portion of the pituitary that secretes ACTH, TSH, growth hormone, prolactin, FSH, and LH in response to hypothalamic releasing hormones. Its hormones regulate adrenal, thyroid, growth, lactation, and reproductive function.

Anterior Tibial Artery

- Arises from the popliteal artery, passes through the anterior intermuscular septum to the anterior compartment, continuing to the ankle as the dorsalis pedis artery. Supplies anterior leg and dorsal foot muscles. Palpable anterior to the lateral malleolus as the dorsalis pedis pulse.

Anterior Uveitis

- Anterior uveitis, also known as iritis or iridocyclitis when the ciliary body is involved, is inflammation of the anterior segment of the eye involving the iris and/or ciliary body, and it is the most common form of uveitis.

Anterior/Posterior Drawer Test

- Knee ligament testing. Anterior (ACL), posterior (PCL).

Anthrax

- A serious infectious disease caused by the gram-positive bacterium *Bacillus anthracis*, which forms highly resistant spores that can persist in the environment for decades, and it is primarily a zoonotic disease of herbivores (cattle, sheep, goats) that can be transmitted to humans through contact with infected animals or their products.

Anthropometry

- Body measurements: weight, height, BMI, waist, hip, skinfold thickness. Assessment of nutritional status and body composition.

Anti-CCP

- Anti-cyclic citrullinated peptide antibody: an au-

to antibody against citrullinated proteins, highly specific (90-95 percent) for rheumatoid arthritis and predictive of erosive disease. Positive anti-CCP is a key component of ACR/EULAR RA classification; may be present years before clinical onset.

Anti-CCP Antibody

- Anti-cyclic citrullinated peptide antibody, a highly specific serologic marker for rheumatoid arthritis (specificity >95%, sensitivity 70%). Positive anti-CCP is associated with more aggressive erosive RA. Appears early in disease and may precede clinical symptoms.

Anti-citrullinated peptide antibody

- An autoantibody highly specific for rheumatoid arthritis that may precede clinical disease and is associated with erosive joint damage.

Anti-dsDNA Antibody

- Anti-double-stranded DNA antibody, highly specific for systemic lupus erythematosus (specificity 95%, sensitivity 60-70%). Elevated titers correlate with active disease, particularly lupus nephritis and systemic vasculitis. Serial monitoring used to assess disease activity and treatment response.

Anti-GBM Disease

- Autoimmune disease with antibodies against alpha-3 chain of type IV collagen in glomerular and alveolar basement membranes. Presents with crescentic GN and pulmonary hemorrhage (Goodpasture syndrome). Linear IgG deposits on immunofluorescence. Treatment requires emergent plasmapheresis, cyclophosphamide, and corticosteroids.

Anti-IgE Therapy

- Omalizumab is a monoclonal antibody that binds free IgE, preventing it from binding to mast cells and basophils. It reduces allergic inflammation and is approved for moderate-to-severe persistent allergic asthma. It reduces exacerbations and the need for oral corticosteroids. It is administered by subcutaneous injection every 2–4 weeks.

Anti-IL4 Receptor Therapy

- Dupilumab is a monoclonal antibody blocking the IL-4 receptor alpha subunit, inhibiting both IL-4 and IL-13 signaling. It is approved for moderate-to-severe asthma with eosinophilic phenotype or oral corticosteroid dependence. It reduces exacerbations

and improves lung function.

Anti-IL5 Therapy

- Monoclonal antibodies targeting interleukin-5, a key cytokine in eosinophil production and survival. Mepolizumab, reslizumab, and benralizumab (anti-IL-5 receptor) are approved for severe eosinophilic asthma. They reduce exacerbations and oral corticosteroid requirements in patients with blood eosinophils ≥ 300 .

Anti-inflammatory

- An agent that reduces inflammation.

Anti-MDA5 dermatomyositis

- A clinically distinct subtype of dermatomyositis associated with anti-melanoma differentiation-associated gene 5 antibodies, characterised by skin-predominant disease, mechanic's hands, and a high risk of rapidly progressive interstitial lung disease.

Anti-Mullerian Hormone

- AMH is a glycoprotein produced by granulosa cells of ovarian follicles, reflecting the ovarian reserve (quantity of remaining oocytes). Uses: assessment of ovarian reserve in infertility evaluation, predicting response to ovarian stimulation in IVF, diagnosis of PCOS (elevated), and monitoring for granulosa cell tumors.

Anti-Reflux Surgery

- Surgical procedures designed to restore the anti-reflux barrier at the gastroesophageal junction, primarily indicated for gastroesophageal reflux disease (GERD) refractory to medical therapy, patient preference to discontinue lifelong medication, or large hiatal hernias.

Antianginal

- A class of medications used to prevent or relieve angina pectoris (chest pain from myocardial ischemia).

Antiarrhythmic

- A drug used to treat or prevent cardiac arrhythmias, classified by the Vaughan Williams system: Class I (sodium channel blockers: quinidine, lidocaine, flecainide), Class II (beta-blockers), Class III (potassium channel blockers prolonging repolarization: amiodarone, sotalol), Class IV (calcium channel blockers: verapamil, diltiazem), and

Antibiogram

- A periodic summary of antimicrobial susceptibility patterns of bacterial isolates from a hospital or laboratory. Guides empiric antibiotic therapy by showing local resistance patterns (e.g., % of *E. coli* susceptible to TMP-SMX). Updates annually.

Antibiotic

- A substance that kills or slows the growth of bacteria.

Antibiotic cycling

- The scheduled rotation of antimicrobial agents within a healthcare facility to reduce selective pressure for resistance. The strategy involves periodically changing the preferred antibiotics for specific infections. For example, cycling between carbapenems and fluoroquinolones for gram-negative infections.

Antibiotic de-escalation

- The process of narrowing antimicrobial spectrum based on microbiological results and clinical response. De-escalation aims to reduce selective pressure for resistance while maintaining therapeutic efficacy.

Antibiotic mixing

- A stewardship strategy where different patients with the same indication receive different antibiotics concurrently, with the goal of reducing resistance emergence. For example, using different agents for ventilator-associated pneumonia in different patients within the same unit.

Antibiotic Resistance

- Bacteria resisting antibiotic effects. Mechanisms: enzymatic degradation, target modification, efflux pumps, membrane permeability. Driven by antibiotic overuse.

Antibiotic resistance genes

- Genes that encode mechanisms conferring resistance to antibiotics. These genes can be located on chromosomes, plasmids, or transposons. Major resistance genes include *mecA* (methicillin resistance in *S. aureus*), *vanA/vanB* (vancomycin resistance in *Enterococcus*), and *blaKPC* (carbapenem resistance).

Antibiotic Stewardship

- Program to optimize antibiotic use. Goals: improve

outcomes, reduce resistance, decrease costs.

Antibiotics

- Medications that kill or inhibit the growth of bacteria, used to treat bacterial infections such as pneumonia, strep throat, and urinary tract infections. They are ineffective against viral infections like colds or influenza.

Antibodies

- Immunoglobulin proteins produced by B cells that bind to specific antigens. There are five classes: IgG (most abundant), IgM (first produced), IgA (mucosal immunity), IgE (allergic responses), and IgD. Antibodies neutralise pathogens, opsonise for phagocytosis, activate complement, and mediate antibody-dependent cellular cytotoxicity.

Anticholinergic Burden

- The cumulative effect of taking multiple medications with anticholinergic properties. It can cause dry mouth, constipation, urinary retention, blurred vision, confusion, and delirium, especially in elderly patients. Many psychiatric medications (TCAs, first-generation antipsychotics, some antihistamines) have anticholinergic effects.

Anticholinergic Cognitive Burden Scale

- A medication-review tool that estimates the total anticholinergic load of a patient's medicines. It supports risk assessment and deprescribing but does not replace individualized clinical judgment.

Anticoagulant Monitoring

- Unfractionated heparin is monitored with aPTT (target 1.5-2.5 x control). Low molecular weight heparins generally do not require monitoring. Warfarin is monitored with INR (target 2-3 for most indications, 2.5-3.5 for mechanical mitral valve).

Anticoagulation

- Blood thinning medication. Used for DVT, PE, arterial thrombosis, after stenting.

Anticodon

- A three-nucleotide sequence at one end of transfer RNA that base-pairs with a complementary messenger RNA codon during translation, ensuring the correct amino acid is incorporated into the growing polypeptide.

Anticonvulsant Mood Stabilizers

- Antiepileptic drugs used as mood stabilizers in

bipolar disorder. Valproate (divalproex) is effective for acute mania and rapid cycling. Carbamazepine is effective for acute mania but has numerous drug interactions via CYP3A4 induction. Lamotrigine is effective for bipolar depression prevention but not acute mania.

Anticonvulsants

- Medications used to prevent or control seizures in epilepsy and other seizure disorders.

Antidiabetic

- A class of medications used to manage diabetes mellitus by lowering blood glucose levels through various mechanisms.

Antidiarrheal

- A drug that reduces the frequency or volume of diarrhea. Loperamide, a peripheral mu-opioid receptor agonist, slows intestinal motility and reduces fluid loss without CNS effects; diphenoxylate/atropine combinations and bismuth subsalicylate are alternatives. Antidiarrheals are avoided in invasive/inflammatory diarrhea (dysentery, C.

Antidiuretic Hormone (ADH)

- A hypothalamic hormone released from the posterior pituitary in response to increased plasma osmolality or reduced effective circulating volume. It increases collecting-duct water permeability through V2 receptors and promotes vasoconstriction through V1 receptors.

Antidiuretic Hormone Physiology

- Antidiuretic hormone also known as vasopressin is synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and stored in the posterior pituitary. Released in response to increased serum osmolality detected by hypothalamic osmoreceptors and decreased blood volume detected by baroreceptors.

Antidiuretic Hormone Resistance

- Impaired renal response to vasopressin, causing excessive production of dilute urine and polydipsia. Causes include lithium, hypercalcemia, hypokalemia, chronic kidney disease, and inherited defects of the vasopressin receptor or aquaporin pathway.

Antidote

- Substance that counteracts poison toxicity.

Specific: naloxone (opioids), flumazenil (benzodiazepines), N-acetylcysteine (acetaminophen), digoxin immune Fab (digoxin), chelators (EDTA, deferoxamine, dimercaprol), atropine + pralidoxime (organophosphates), methylene blue (methemoglobinemia), cyanide antidote kit (nitrite + thiosulfate).

Antiemetic

- A medication that prevents or treats nausea and vomiting.

Antiemetics

- Medications that prevent or treat nausea and vomiting.

Antiepileptic Drugs

- Medications used to prevent and control seizures by modulating neuronal excitability through various mechanisms. They can be classified by their primary mechanism of action.

Antifungal

- A drug used to treat fungal infections.

Antifungal drug classes

- Major classes of antifungal agents include polyenes (amphotericin B), azoles (fluconazole, voriconazole), echinocandins (caspofungin, micafungin), and allylamines (terbinafine). Polyenes bind to ergosterol in fungal cell membranes. Azoles inhibit ergosterol synthesis. Echinocandins inhibit cell wall synthesis.

Antifungal prophylaxis

- The use of antifungal agents to prevent invasive fungal infections in high-risk patients. Common indications include prolonged neutropenia, hematopoietic stem cell transplantation, and advanced HIV infection. Fluconazole, posaconazole, and micafungin are used for prophylaxis.

Antifungal resistance

- The ability of fungi to withstand the effects of antifungal agents. Resistance can be intrinsic or acquired. Azole resistance occurs through efflux pumps and target mutations. Echinocandin resistance occurs through FKS gene mutations. Amphotericin B resistance is rare.

Antigen

- A molecule or molecular structure specifically recognized by an antibody, B-cell receptor, or T-cell receptor. An immunogen is an antigen capable

of inducing an immune response; antigenicity and immunogenicity are related but not synonymous properties.

Antigen Detection Test

- A diagnostic assay that identifies a pathogen-derived antigen in a clinical specimen using a specific antibody or related binding reagent. It is generally rapid but may be less sensitive than nucleic-acid amplification, especially when organism burden is low.

Antigen-Presenting Cell (APC)

- Specialized immune cells that process protein antigens and display peptide fragments bound to Major Histocompatibility Complex (MHC) molecules on their surface for recognition by T-cell receptors. Professional APCs: dendritic cells (most potent for naive T cells), macrophages, and B cells.

Antigenic Drift

- The gradual accumulation of point mutations in viral surface antigens (hemagglutinin, neuraminidase) during replication, allowing influenza viruses to evade pre-existing antibodies and cause recurrent seasonal epidemics. Drift is continuous and selected by host immunity; it is why the seasonal vaccine is updated annually.

Antigenic Shift

- A major, abrupt change in the antigenic structure of a virus (especially influenza A) resulting from genetic reassortment between different strains (e.g., human and avian viruses) infecting the same cell.

Antigenic variation

- The process by which pathogens change their surface proteins to evade immune recognition. Antigenic drift: gradual changes in surface proteins. Antigenic shift: major changes through reassortment of genome segments. Influenza virus demonstrates both mechanisms. Trypanosomes and Plasmodium also use antigenic variation.

Antigout

- A class of medications used to treat gout by reducing uric acid levels or controlling acute inflammation.

Antihistamine

- A drug that blocks histamine receptors. H1 antihistamines treat allergic rhinitis, urticaria, pruri-

tus, and (older sedating agents) insomnia and motion sickness. First-generation (diphenhydramine, chlorpheniramine): CNS-penetrant—sedating, anticholinergic (dry mouth, urinary retention, confusion in elderly).

Antihistamines

- Medications that block histamine receptors, primarily H1 and H2 receptors, used to treat allergic conditions and other histamine-mediated disorders. H1 antihistamines are classified as first-generation (sedating, with significant anticholinergic effects) and second-generation (non-sedating, with minimal anticholinergic activity).

Antihypertensive

- A drug that lowers blood pressure, used for hypertension management. Major classes: ACE inhibitors and ARBs (preferred first-line, renoprotective), calcium channel blockers, thiazide diuretics, beta-blockers (with CAD/heart failure), alpha-blockers, centrally acting agents (clonidine), and direct vasodilators (hydralazine, minoxidil).

Antihypertensives in Pregnancy

- Medications used to treat hypertension in pregnancy.

Antimicrobial

- A general term for antibiotics and other drugs that fight microscopic organisms in the body, such as bacteria, viruses, fungi, and parasites.

Antimicrobial adverse effects

- Side effects associated with antimicrobial agents. Beta-lactams can cause allergic reactions. Fluoroquinolones can cause tendon rupture and QT prolongation. Aminoglycosides can cause nephrotoxicity and ototoxicity. Azole antifungals can cause hepatotoxicity.

Antimicrobial desensitization

- A procedure that involves administering gradually increasing doses of an antimicrobial to induce tolerance in patients with allergy. Desensitization is used when the causative agent is the only effective treatment and the patient has a history of hypersensitivity.

Antimicrobial dosing optimization

- Strategies to ensure antimicrobial agents achieve therapeutic concentrations at the site of infection.

Optimization includes appropriate drug selection, dosing interval, route of administration, and duration of therapy. Extended or continuous infusion of beta-lactams may improve outcomes. Weight-based dosing is important for many agents.

Antimicrobial drug interactions

- Interactions between antimicrobial agents and other medications. Fluoroquinolones chelate divalent cations. Macrolides inhibit CYP3A4. Rifampin is a potent enzyme inducer. Azole antifungals inhibit CYP3A4. Understanding drug interactions is essential for safe prescribing and preventing adverse effects.

Antimicrobial hypersensitivity reactions

- Hypersensitivity reactions to antimicrobial agents range from mild rash to life-threatening anaphylaxis. Beta-lactam antibiotics are the most common cause of drug allergy. Cross-reactivity between penicillins and cephalosporins is approximately 2%. Sulfonamide antibiotics can cause severe cutaneous reactions.

Antimicrobial pharmacodynamics

- The relationship between antimicrobial drug concentrations and their effect on microorganisms. Time-dependent antibiotics (beta-lactams) are most effective when concentrations remain above the MIC. Concentration-dependent antibiotics (aminoglycosides) are most effective with high peak concentrations.

Antimicrobial pharmacokinetics

- The absorption, distribution, metabolism, and elimination of antimicrobial agents. Pharmacokinetics varies by drug class, patient factors, and disease state. Renal impairment affects elimination of many antibiotics. Hepatic impairment affects metabolism of many antifungals.

Antimicrobial Resistance

- Microorganisms becoming resistant to antibiotics. Major global health threat. Prevention: appropriate antibiotic use, infection control.

Antimicrobial Resistance Mechanisms

- Beta-lactamases: Ambler class A (TEM, SHV, CTX-M ESBLs, KPC carbapenemases), class B (metallo-beta-lactamases: NDM, VIM, IMP zinc-dependent, not inhibited by avibactam), class

C (AmpC, chromosomal/inducible, hydrolyzes cephalosporins), class D (OXA carbapenemases including OXA-48-like and OXA-23/24/58 in *Acinetobacter*, variable

Antimicrobial resistance surveillance

- The systematic monitoring of antimicrobial resistance patterns in clinical isolates. Surveillance programmes track resistance rates over time and guide empiric therapy. National and international networks share resistance data.

Antimicrobial Stewardship

- Core elements (CDC): leadership commitment, accountability, drug expertise, action (prospective audit/feedback, preauthorization), tracking (DOT—days of therapy per 1,000 patient-days, defined daily doses per 1,000 patient-days), reporting (antibiograms, resistance trends, prescribing patterns), education (prescribers,

Antimicrobial susceptibility testing

- Laboratory methods used to determine the susceptibility of microorganisms to antimicrobial agents. Methods include disk diffusion, broth microdilution, and automated systems. MIC (minimum inhibitory concentration) is the gold standard measure. Results guide antimicrobial therapy and are interpreted using standardised breakpoints.

Antineoplastic

- A drug used to treat cancer (chemotherapy).

Antinuclear Antibody

- ANA: autoantibodies against nuclear components (DNA, histones, ribonucleoproteins, etc.), detected by immunofluorescence (HEp-2 cells), reported with pattern (homogeneous, speckled, nucleolar, centromere) and titer.

Antinuclear Antibody (ANA)

- A screening test for autoimmune connective tissue diseases. Positive in SLE (sensitivity 95%), Sjogren syndrome, systemic sclerosis, mixed connective tissue disease, and many other autoimmune conditions. Titer: 1:40 (low positive, common in healthy), 1:160 (clinically significant), 1:320 (strongly positive).

Antioxidant

- Molecule preventing oxidation damage. Vitamins C and E, beta-carotene, selenium, polyphenols. Diets rich in antioxidants associated with reduced chronic

disease risk.

Antioxidants

- Substances that prevent or slow oxidative damage to cells caused by free radicals (reactive oxygen and nitrogen species). Endogenous antioxidants: glutathione, superoxide dismutase, catalase.

Antiparasitic

- A drug used to treat infections caused by parasites (protozoa and helminths). Antiprotozoals: metronidazole (Giardia, amebiasis, trichomoniasis), tinidazole, chloroquine and artemisinin combinations (malaria), pyrimethamine (toxoplasmosis), pentamidine, and nifurtimox (Chagas).

Antiparkinson

- A class of medications used to manage Parkinson disease symptoms by restoring dopaminergic activity or modulating other neurotransmitter systems.

Antiparkinsonian

- A drug used to treat Parkinson disease and drug-induced parkinsonism.

Antiphospholipid Antibodies

- Include lupus anticoagulant (prolongs phospholipid-dependent coagulation tests), anticardiolipin antibodies (IgG/IgM), and anti-beta-2-glycoprotein I antibodies. Detected on ≥ 2 occasions 12 weeks apart. Clinical criteria: vascular thrombosis or pregnancy morbidity. Can be transient (infection, drugs) or persistent.

Antiphospholipid Antibody

- Autoantibody causing thrombosis and pregnancy morbidity. Associated with SLE.

Antiphospholipid Syndrome

- A systemic autoimmune thrombophilia characterized by arterial and venous thrombosis, pregnancy morbidity, and persistent antiphospholipid antibody positivity.

Antiplatelet Therapy

- Medication preventing platelet aggregation. Aspirin, clopidogrel. Used after stenting, bypass.

Antipruritic

- A medication that relieves itching (pruritus).

Antipsychotic

- A drug used to treat psychotic disorders (schizophrenia, schizoaffective disorder, psychosis

in bipolar/manic states and delirium). Typical (first-generation): haloperidol, chlorpromazine—D2 blockade, high EPS/tardive dyskinesia risk, hyperprolactinemia.

Antipsychotic-Induced Parkinsonism

- Drug-induced bradykinesia, rigidity, tremor, or postural instability caused by dopamine-receptor blockade, usually from antipsychotic treatment. Management involves reviewing the medication, dose, alternatives, and symptomatic treatment when appropriate.

Antipyretic

- A drug that reduces fever by lowering the hypothalamic set point, typically by inhibiting prostaglandin synthesis in the preoptic hypothalamus. Acetaminophen and NSAIDs (ibuprofen, aspirin) are the principal antipyretics.

Antipyretics

- Medications that reduce fever by inhibiting cyclooxygenase (COX) enzymes, reducing hypothalamic prostaglandin E₂ synthesis and resetting the thermoregulatory set point.

Antiretroviral therapy

- Treatment of HIV infection with combinations of antiretroviral agents. ART targets multiple steps in the viral life cycle including entry, reverse transcription, integration, and maturation. Combination ART suppresses viral replication and restores immune function. ART is lifelong therapy.

Antiseptic

- Substances used on wounds to prevent or treat infection; they kill or slow the growth of disease-causing organisms, such as bacteria, on the surface of the body.

Antiseptics

- Chemical agents applied to living tissue to reduce microbial load. Antiseptics are used for hand hygiene, skin preparation before procedures, and wound care. Common antiseptics include alcohols, chlorhexidine, povidone-iodine, and hydrogen peroxide. Antiseptics have lower efficacy than disinfectants due to toxicity concerns.

Antisocial Personality Disorder

- A pattern of disregard for and violation of the rights of others occurring since age 15 years. At

least three of seven criteria: failure to conform to social norms, deceitfulness, impulsivity, irritability and aggressiveness, reckless disregard for safety, consistent irresponsibility, and lack of remorse.

Antispasmodic

- A medication that reduces smooth muscle spasm, primarily in the gastrointestinal tract. Anticholinergic agents: hyoscyamine, dicyclomine, atropine — block muscarinic M3 receptors on GI smooth muscle, reducing contractions and peristalsis, used for irritable bowel syndrome and functional dyspepsia.

Antisynthetase Syndrome

- Defined by the presence of anti-aminoacyl-tRNA synthetase antibodies (anti-Jo-1 most common, 75%; anti-PL-7, anti-PL-12, anti-EJ, anti-OJ, anti-KS, anti-Zo, anti-Ha, anti-YRS, anti-FRS, anti-TRS, anti-ARS) with clinical triad: interstitial lung disease (most common, often progressive), inflammatory myopathy (DM/PM), and inflammatory arthr

Antithrombin

- Natural anticoagulant. Inhibits thrombin, factor Xa. Deficiency causes thrombosis.

Antithrombin Deficiency

- A rare (1 in 2,000-5,000), strongly thrombophilic inherited disorder (autosomal dominant). Antithrombin (previously antithrombin III) is a serine protease inhibitor that inactivates thrombin and factors Xa, IXa, XIa, and XIIa. Heterozygous: 10-50 fold increased VTE risk, more severe than factor V Leiden.

Antithrombin III deficiency

- An inherited or acquired thrombophilia caused by reduced antithrombin activity, leading to increased risk of venous thromboembolism.

Antithyroid

- A class of medications that reduce thyroid hormone synthesis, used to treat hyperthyroidism (Graves disease, toxic nodular goiter). Thioamide derivatives: methimazole (propylthiouracil as second-line) — inhibit thyroid peroxidase, blocking iodination of tyrosine residues and coupling of iodotyrosines, reducing production of T3

Antitubercular Agents

- Antimicrobial drugs used to treat *Mycobacterium tuberculosis* infections.

Antitussive

- A medication that suppresses the cough reflex, used for dry, non-productive cough.

Antiviral

- A class of drugs that inhibit viral replication, used to treat viral infections.

Antiviral drug classes

- Major classes of antiviral agents include nucleoside analogues (acyclovir, tenofovir), protease inhibitors (ritonavir, atazanavir), reverse transcriptase inhibitors (efavirenz, sofosbuvir), and neuraminidase inhibitors (oseltamivir). Each class targets a specific step in the viral replication cycle.

Antivirals

- Drugs that inhibit the replication of viruses, reducing the severity and duration of viral infections. They are used for conditions such as influenza, herpes, HIV, hepatitis B and C, and COVID-19. Unlike antibiotics, antivirals target specific viral enzymes or life cycle stages and often work best when started early in the infection.

Antrochoanal polyp

- A benign, single nasal polyp originating from the maxillary sinus mucosa and extending through the accessory ostium into the choana and nasopharynx. It is most common in children and young adults, presenting with unilateral nasal obstruction and rhinorrhea. Treatment is endoscopic surgical removal.

Anuria

- Absence of urine output (<100 mL/day).

Anus

- The terminal opening of the gastrointestinal tract through which feces are expelled. It is surrounded by the internal smooth-muscle sphincter and external skeletal-muscle sphincter.

Anxiety

- A normal emotional response to perceived danger, but anxiety disorders involve excessive, persistent worry that interferes with daily life. Symptoms can include restlessness, rapid heart rate, muscle tension, and avoidance behaviors.

Anxiety Disorder

- A group of psychiatric conditions characterized by excessive fear, anxiety, and related behavioral

disturbances. Includes generalized anxiety disorder (GAD), panic disorder, social anxiety disorder, specific phobias, and agoraphobia.

Anxiety Disorders

- Group of psychiatric disorders characterized by excessive fear, worry, and anxiety that cause significant distress and impairment.

Anxiolytic

- A medication that reduces anxiety without producing excessive sedation. First-line for generalized anxiety disorder: SSRIs (escitalopram, paroxetine, sertraline) and SNRIs (venlafaxine, duloxetine) — increase serotonin/norepinephrine activity, require 2-4 weeks for therapeutic effect, first-line for chronic anxiety disorders.

Aorta

- Largest artery, from the left ventricle distributing oxygenated blood systemically. Approximately 30 cm long, 2.5 cm diameter. Four segments: ascending aorta, aortic arch (brachiocephalic trunk, left common carotid, left subclavian), descending thoracic aorta, descending abdominal aorta (bifurcating into common iliacs at L4).

Aortic Aneurysm

- Permanent dilation of the aorta exceeding 50% of normal diameter. Abdominal aortic aneurysm (AAA) is more common than thoracic aortic aneurysm and is usually caused by atherosclerosis. AAA >5.5 cm or rapid expansion (>0.5 cm/year) warrants surgical repair (open or endovascular).

Aortic Aneurysm Rupture

- Full-thickness failure of an aortic aneurysm with bleeding into a body cavity or surrounding tissue. Ruptured aneurysm commonly causes sudden pain, shock, and collapse and requires immediate resuscitation and operative or endovascular repair.

Aortic Aneurysm Screening

- Abnormal focal dilation of blood vessel. See aneurysm.

Aortic Arch

- Curves from second right sternocostal joint superiorly and left, then posteriorly and inferiorly to T4. From right to left: brachiocephalic trunk, left common carotid, left subclavian. Related to left recurrent laryngeal nerve (hooking under it), trachea,

esophagus, left phrenic nerve. Arch aneurysms may cause hoarseness and SVC syndrome.

Aortic Coarctation

- Congenital narrowing of the aorta, usually near the insertion of the ductus arteriosus. It causes higher blood pressure and stronger pulses in the arms than in the legs, with a radio-femoral delay; severe neonatal disease may present with shock when the ductus closes.

Aortic Dissection

- A tear in the intima of the aorta allowing blood to dissect through the media, creating a false lumen. It is classified as Stanford Type A (ascending aorta involvement, requiring emergency surgery) or Type B (descending aorta only, managed medically unless complicated).

Aortic embolism

- Obstruction of a systemic artery by material originating in the aorta, such as thrombus or atheromatous debris, causing distal ischemia.

Aortic Graft Infection

- Infection involving a prosthetic or endovascular aortic graft, potentially causing sepsis, pseudoaneurysm, hemorrhage, or aortoenteric fistula. Management usually requires prolonged antimicrobial treatment and individualized surgical or endovascular source control.

Aortic Regurgitation

- Incomplete closure of the aortic valve allowing blood to flow backward from the aorta to the left ventricle during diastole. Causes include bicuspid aortic valve, endocarditis, aortic root dilation, hypertension, and connective tissue disorders. Murmur: early diastolic decrescendo murmur at the left sternal border.

Aortic Regurgitation Murmur

- Early diastolic decrescendo murmur heard best at the left sternal border with the patient sitting up and leaning forward in end-expiration. Wide pulse pressure, water-hammer pulses (Corrigan pulse), de Musset sign (head bobbing with each pulse), and Duroziez sign (femoral bruit with compression).

Aortic Stenosis

- Narrowing of the aortic valve opening, obstructing left ventricular outflow. It is the most com-

mon valvular disease in developed countries, usually caused by calcific degeneration. Symptoms include angina, syncope, and heart failure (SAD triad).

Aortic Stenosis Grading

- Severity grading: mild (valve area >1.5 cm², mean gradient <20 mmHg), moderate (valve area 1.0-1.5 cm², mean gradient 20-40 mmHg), severe (valve area <1.0 cm², mean gradient >40 mmHg). Very severe (valve area <0.6 cm², mean gradient >60 mmHg) indicates critical stenosis. Grading guides timing of intervention.

Aortic Valve

- Semilunar valve between left ventricle and aorta with right, left, and posterior cusps. Each cusp forms a sinus of Valsalva from which coronary arteries originate. Opens during systole; closes during diastole.

Aortography

- Contrast imaging of aorta. Evaluates aneurysm, dissection, occlusion. Largely replaced by CTA/MRA.

APACHE II Score

- Severity scoring system for critically ill patients incorporating physiological variables, age, and chronic health status to generate scores from zero to seventy-one. Higher scores indicate greater severity and higher mortality. In acute pancreatitis, scores of eight or greater indicate severe disease.

APC Tumor Suppressor

- The adenomatous polyposis coli protein is a large multidomain tumor suppressor that functions as a critical negative regulator of the canonical Wnt signaling pathway.

Apert Syndrome

- An autosomal dominant craniosynostosis syndrome caused by FGFR2 mutations (most commonly Ser252Trp or Pro253Arg).

APGAR Score

- A scoring system developed by Virginia Apgar in 1952 to assess the newborn's immediate condition at 1 and 5 minutes after birth. Five components (each scored 0, 1, or 2): Appearance (skin color), Pulse (heart rate), Grimace (reflex irritability), Activity (muscle tone), and Respiration (breathing effort).

Aphasia

- Impairment of language comprehension or production resulting from brain damage, most commonly from stroke affecting the left hemisphere (dominant) language areas.

Apheresis

- A procedure that removes a selected blood component, such as plasma, platelets, or abnormal cells, and returns the remaining components to the patient or donor.

Aphonia

- Complete loss of the ability to phonate, distinct from dysphonia in which voice is present but abnormal. Causes are organic (bilateral vocal cord paralysis, severe laryngitis, laryngeal trauma, structural lesions) or psychogenic (conversion disorder, muscle tension dysphonia). Whispered speech and cough are usually preserved.

Aphthous Stomatitis

- See Aphthous Ulcer.

Aphthous Ulcer

- Recurrent, painful, round or oval ulcers of the oral mucosa, commonly called canker sores. Three types: minor (most common, <1 cm, heal in 7-14 days without scarring), major (>1 cm, deeper, heal over weeks with scarring), and herpetiform (multiple pinhead-sized ulcers that fuse).

Apicoectomy

- Surgical removal of root tip and periapical lesion. Indication: failed root canal therapy.

Aplastic Anemia

- A bone marrow failure syndrome characterized by pancytopenia (anemia, leukopenia, thrombocytopenia) and a hypocellular bone marrow with replacement by fat cells, resulting from immune-mediated destruction of hematopoietic stem cells.

Apnea

- See Apnoea.

Apnea of Prematurity

- Cessation of breathing for more than 20 seconds or shorter episodes with bradycardia or desaturation in infants less than 34-37 weeks gestation. Three types: central (no respiratory effort), obstructive (airway collapse), and mixed. Treatment: caffeine citrate

(first-line), continuous positive airway pressure, and methylxanthine therapy.

Apnea-Hypopnea Index

- Number of apneas plus hypopneas per hour of sleep. Apnea = airflow cessation > 10 seconds. Hypopnea = partial obstruction with desaturation or arousal. AHI < 5 normal, 5–14 mild, 15–29 moderate, >= 30 severe.

Apneustic Breathing

- A breathing pattern characterized by a prolonged inspiratory pause, seen in brainstem lesions (particularly the pons). It indicates damage to the apneustic center and is a poor prognostic sign in neurological injury. It may require mechanical ventilation.

Apnoea

- Temporary cessation of breathing, defined as complete cessation of airflow for 10 seconds. Obstructive apnea: airway collapse despite respiratory effort (common in obstructive sleep apnea).

Apoenzyme

- The protein component of an enzyme without its required cofactor or coenzyme. Inactive or minimally active until the cofactor binds. The complete, catalytically active complex of apoenzyme + cofactor is called the holoenzyme.

Apoptosis

- Programmed cell death. Dysregulated in cancer (avoiding death). Pro-apoptotic genes: BAX, P53. Anti-apoptotic: BCL-2.

Apoptosis Extrinsic Pathway

- The extrinsic pathway of apoptosis is triggered by extracellular ligands binding to death receptors belonging to the tumor necrosis factor receptor superfamily, including Fas (CD95), TNFR1, and TRAIL receptors DR4 and DR5.

Apoptosis Intrinsic Pathway

- The intrinsic, or mitochondrial, pathway of apoptosis is initiated by intracellular stress signals including DNA damage, growth factor withdrawal, endoplasmic reticulum stress, and hypoxia.

Apparent Mineralocorticoid Excess

- Rare autosomal recessive disorder caused by deficiency of 11-beta-hydroxysteroid dehydrogenase type 2 normally converting cortisol to cortisone in the collecting duct. Cortisol activates mineralocor-

ticoid receptors causing severe hypertension, hypokalemia, metabolic alkalosis, and low renin with low aldosterone.

Appendectomy

- Surgical removal of the appendix, the most common emergency surgical procedure performed worldwide. Appendectomy is indicated for acute appendicitis, the most common surgical emergency with a lifetime risk of 7-8 percent and peak incidence between 10 and 30 years.

Appendiceal Tumors

- Appendiceal neoplasms: carcinoid (most common, 50%, tip of appendix, <1cm: appendectomy alone; >2cm or positive margins/lymphovascular invasion: right hemicolectomy), mucinous neoplasm (low-grade appendiceal mucinous neoplasm LAMN, high-grade appendiceal mucinous neoplasm HAMN), appendiceal adenocarcinoma (treated like colon cancer: right

Appendicitis

- Inflammation of the vermiform appendix, most commonly due to luminal obstruction by fecalith, lymphoid hyperplasia, or foreign body. Presents with periumbilical pain migrating to the right lower quadrant, nausea, anorexia, and low-grade fever. Diagnosis: clinical, elevated WBC, CT scan. Treatment: laparoscopic appendectomy.

Appendix

- A blind-ended tubular diverticulum of the cecum, typically 6-10 cm long, rich in lymphoid tissue. Its position is variable (retrocecal, pelvic, or retroileal); the base usually projects near McBurney's point, one-third of the distance from the anterior superior iliac spine to the umbilicus.

Appetite Regulation

- Complex control by hormones (leptin, ghrelin, insulin, CCK), neural signals, psychological factors. Leptin signals satiety, ghrelin signals hunger.

Applied Behavior Analysis

- A behavioral intervention based on learning theory principles that uses systematic reinforcement to shape desired behaviors and reduce problematic behaviors. It is the primary evidence-based intervention for autism spectrum disorder, particularly in young children.

Apraxia

- Inability to perform learned, skilled motor movements despite intact motor function, sensation, coordination, and comprehension.

Aquagenic keratoderma

- Transient, painful whitish thickening and wrinkling of the palms and soles occurring within minutes of water immersion, accompanied by a prickling or burning sensation. It is often associated with cystic fibrosis or other sweat gland disorders but can also be idiopathic or drug-induced. Symptoms resolve shortly after drying the skin.

Aquatic Therapy

- Exercise in water using buoyancy, resistance, warmth. See hydrotherapy.

Aqueous Humor

- A clear, watery fluid that fills the anterior and posterior chambers of the eye, maintaining intraocular pressure and providing nutrients and oxygen to the avascular cornea and lens while removing metabolic waste products.

Arachnoid cyst

- A congenital, benign, cerebrospinal fluid-filled cyst within the arachnoid membrane, most commonly found in the middle cranial fossa. Most are asymptomatic and discovered incidentally on imaging, but large cysts may cause headache, seizures, or focal neurological deficits.

Arachnoid Mater

- The middle of the three meninges, a delicate avascular membrane lying between the dura mater and pia mater. The subarachnoid space between arachnoid and pia contains cerebrospinal fluid (CSF) and major vessels. Arachnoid granulations (villi) project into the dural venous sinuses and mediate CSF resorption.

ARB-Nepriylsin Inhibitor (ARNI)

- Sacubitril/valsartan combines an ARB with a neprilysin inhibitor, increasing natriuretic peptide levels while blocking the renin-angiotensin system. The PARADIGM-HF trial demonstrated superior reduction in cardiovascular death and heart failure hospitalization compared to enalapril.

Arbovirus

- A virus transmitted by mosquitoes or other member

of the arthropod phylum.

ARBs

- Block angiotensin II AT1 receptors providing renoprotection similar to ACE inhibitors. Alternative for ACE inhibitor cough. Do not combine ACE inhibitor and ARB due to increased adverse effects without benefit. Similar renoprotective effects to ACE inhibitors.

Arcus Senilis (Corneal Arcus)

- A white, gray, or bluish opaque ring in the peripheral corneal stroma separated from the limbus by a clear zone (lucid interval of Vogt). Caused by lipid (cholesterol) deposition in the peripheral cornea. Common and benign in elderly individuals (>60 years).

ARDS

- Acute respiratory distress syndrome; see Acute Respiratory Distress Syndrome. Acute diffuse inflammatory lung injury causing severe hypoxemia ($\text{PaO}_2/\text{FiO}_2 \leq 300$), bilateral infiltrates, and non-cardiogenic pulmonary edema. Common causes: sepsis, pneumonia, trauma, aspiration.

Area Under the Concentration-Time Curve (AUC)

- The integral of a drug's plasma concentration over time. AUC estimates total systemic exposure and is used to compare bioavailability, guide dosing for selected medicines, and characterize pharmacokinetic interactions.

Argininosuccinic Aciduria

- Autosomal recessive urea cycle disorder caused by deficiency of argininosuccinate lyase (ASL gene, 7cen-q11.2). Biochemical hallmark: accumulation of argininosuccinic acid in blood, urine, and CSF with hyperammonemia, low citrulline, and elevated argininosuccinic acid.

Argyll Robertson Pupil

- Small, irregular, unequal pupils that accommodate and converge normally but do not react to light ("prostitute's pupil: accommodates but doesn't react"), caused by neurosyphilis affecting the pretectal midbrain Edinger-Westphal nucleus pathways, sparing the more ventral pathways for accommodation.

Armpit Lump

- A lump or swelling in the axilla, commonly caused by an enlarged lymph node, skin infection, cyst, or benign growth. Persistent, hard, fixed, enlarging, or unexplained lumps require medical assessment.

Arnold-Chiari Malformation

- Herniation of cerebellar tonsils through the foramen magnum into the upper cervical spinal canal. Type II (myelomeningocele-associated) is the most common form in pediatrics. Associated with hydrocephalus, syringomyelia, and spinal cord dysfunction. Symptoms include headache, neck pain, and lower cranial nerve palsies.

Aromatase Excess Syndrome

- Rare autosomal dominant condition caused by mutations in the CYP19A1 gene leading to overexpression of aromatase enzyme and excessive estrogen production. Presents in males with gynecomastia, advanced bone age, and accelerated linear growth with early epiphyseal closure. In females with precocious puberty, macroclitoris, and virilization.

Aromatase Inhibitor

- A class of medications that inhibit aromatase, the enzyme that converts androgens (androstenedione, testosterone) to estrogens (estrone, estradiol) in peripheral tissues (adipose, breast, muscle, bone). Used primarily in hormone receptor-positive breast cancer in postmenopausal women.

Arrector Pili Muscle

- A small band of involuntary smooth muscle attached to a hair follicle and the superficial dermis. Sympathetic stimulation causes it to contract, pulling the hair upright and producing temporary piloerection, commonly called goosebumps. Contraction also compresses nearby sebaceous glands and may help release sebum onto the skin

Arrested Caries

- A carious lesion that is no longer progressing, often because the surface has become hard, smooth, and cleansable. Arrested lesions may be monitored and remineralized; treatment depends on activity, cavitation, cleansability, and patient risk.

Arrhythmia

- Any abnormality of cardiac rate, rhythm, or site of impulse origin, arising through enhanced automatic-

ity, reentry, or triggered activity. Classified by rate as tachyarrhythmia or bradyarrhythmia and by site as supraventricular or ventricular. Diagnosis rests on the 12-lead ECG and ambulatory monitoring when intermittent.

Arrhythmias

- Abnormal heart rhythms. Tachyarrhythmias: atrial fibrillation, atrial flutter, ventricular tachycardia, supraventricular tachycardia. Bradyarrhythmias: sinus bradycardia, heart blocks. Management: ACLS algorithms, rate/rhythm control, anticoagulation when indicated.

Arrhythmogenic right ventricular cardiomyopathy

- A genetic cardiomyopathy characterized by fibrofatty replacement of the right ventricular myocardium, predisposing to ventricular arrhythmias and sudden cardiac death, particularly during exercise. ECG shows T-wave inversions in V1-V3 and epsilon waves; diagnosis uses echocardiography, cardiac MRI, and genetic testing.

Arsenic toxicity

- Systemic poisoning from arsenic exposure causing gastrointestinal distress, cardiac arrhythmias, peripheral neuropathy, and characteristic Mees lines on nails, requiring chelation therapy with dimercaprol or succimer.

Art Therapy

- Use of art-making for therapeutic purposes. Emotional expression, coping, self-awareness.

Arterial Blood Gas (ABG)

- Direct measurement of arterial blood for pH, PaCO₂, PaO₂, HCO₃⁻, base excess, and oxygen saturation. Indications: respiratory failure assessment, acid-base disorder evaluation, ventilator management, and critical care. Normal values: pH 7.35-7.45, PaCO₂ 35-45 mmHg, PaO₂ 80-100 mmHg, HCO₃⁻ 22-26 mEq/L, SaO₂ 95-100%.

Arterial Blood Gas (ABG) Analyzer

- An analyzer that measures arterial pH, partial pressures of oxygen and carbon dioxide, and usually bicarbonate or calculated acid-base variables. It is used to assess ventilation, oxygenation, and metabolic or respiratory acid-base disorders.

Arterial Catheter

- A catheter placed in an artery, commonly the radial artery, for continuous blood-pressure monitoring and repeated arterial blood sampling. Complications include hemorrhage, thrombosis, infection, and distal ischemia.

Arterial Catheterization

- Placement of a catheter in an artery, usually the radial artery, for continuous blood-pressure measurement and repeated arterial blood sampling. It is used when rapid hemodynamic or gas-exchange assessment is clinically necessary and carries risks including bleeding, infection, thrombosis, and ischemia.

Arterial Duplex

- Ultrasound technique combining B-mode imaging and Doppler flow assessment. Used for evaluating arterial stenosis, aneurysm, graft patency. Non-invasive, widely available.

Arterial insufficiency

- Reduced arterial blood flow to a tissue or limb, causing ischemic symptoms such as claudication, rest pain, cool skin, delayed capillary refill, and diminished pulses. Most commonly caused by atherosclerosis; assessed by ankle-brachial index and duplex ultrasound.

Arterial Line (A-Line)

- A catheter inserted into a peripheral artery (most commonly radial, also femoral, brachial, dorsalis pedis) for continuous blood pressure monitoring and arterial blood gas sampling. Allen test: assesses ulnar collateral flow before radial artery cannulation.

Arterial Ulcer

- A chronic wound caused by arterial insufficiency (peripheral artery disease). Located on distal extremities: toes, heels, lateral malleolus, and sites of pressure. Presents with punched-out, deep, well-demarcated wound with pale or necrotic base, minimal exudate, and cool, hairless, shiny skin.

Arteriole

- A small artery that delivers blood to a capillary bed. Arteriolar smooth muscle regulates vascular resistance and controls tissue perfusion and systemic blood pressure.

Arteriosclerosis

- Chronic inflammatory disease of medium and large arteries characterized by accumulation of lipids, inflammatory cells, and fibrous tissue within the arterial wall, forming atherosclerotic plaques. See Atherosclerosis.

Arteriovenous Fistula

- Surgical anastomosis between artery and vein for hemodialysis. Preferred access due to lower infection and thrombosis rates. Radiocephalic at wrist or brachiocephalic at elbow. Takes 6 to 8 weeks to mature. Failure to mature occurs in 30%.

Arteriovenous Graft

- Synthetic conduit connecting artery to vein for dialysis. Used when veins inadequate for fistula. Thrombosis and infection more common than AVF. Can be used earlier than fistula. Dec clotting with thrombectomy or angioplasty.

Arteriovenous Malformation

- Abnormal connection between artery and vein without capillary bed. Can bleed or cause symptoms.

Artery

- A blood vessel that carries oxygenated blood away from the heart to the tissues, except for the pulmonary arteries which carry deoxygenated blood to the lungs. Arteries have thick, elastic, muscular walls that withstand high pressure and distribute blood throughout the body. The aorta is the largest artery.

Arthritis

- Inflammation of one or more joints, characterized by joint pain, swelling, warmth, and stiffness.

Arthrocentesis

- Aspiration of joint fluid for diagnostic or therapeutic purposes. Indications: septic arthritis (urgent, crystals), inflammatory arthritis (gout, pseudogout), hemorrhagic effusion (trauma, coagulopathy), and therapeutic drainage for symptomatic relief.

Arthrodesis

- Surgical fusion of a joint by removing or preparing articular surfaces and stabilizing the bones until osseous union occurs. It relieves pain and provides stability at the cost of joint motion.

Arthrography

- Contrast injected into joint for imaging. Used for shoulder, knee, hip evaluation. MR arthrography more common now.

Arthrogryposis

- A group of congenital disorders characterized by multiple joint contractures present at birth, caused by decreased fetal movement (fetal akinesia) during development.

Arthroplasty

- Surgical reconstruction or replacement of a joint to relieve pain, restore alignment, or improve function. It may be partial or total and is commonly performed for advanced degenerative joint disease, fracture, inflammatory arthritis, or other destructive joint conditions.

Arthroscopy

- Minimally invasive joint visualization and intervention via a rigid arthroscope inserted through small portals, used for meniscal and ligamentous repair, chondroplasty, loose body removal, and synovectomy.

Articular Cartilage Injuries

- Chondral/osteochondral lesions: Outerbridge grading (I: softening, II: fibrillation <50%, III: fibrillation >50%, IV: subchondral bone exposed). OCD (osteochondritis dissecans): subchondral bone necrosis, fragment separation (knee: lateral femoral condyle, elbow: capitellum, ankle: talar dome).

Artifacts

- Image distortions not representing true anatomy. Types: motion, beam hardening, metal, aliasing, partial volume. Recognition important for accurate interpretation.

Artificial Pacemaker

- See Cardiac Pacemaker.

Artificial Rupture of Membranes (AROM)

- The intentional rupture of the amniotic sac with a small hook-like instrument (Amnihook) during a cervical exam to induce or augment labor. Indications: induction or augmentation of labor, placement of internal monitors (fetal scalp electrode, intrauterine pressure catheter), and assessment of amniotic fluid (meconium staining).

Artificial Ventricle (Total Artificial Heart)

- A pulsatile, pneumatically or electrically driven biventricular replacement that removes the native ventricles. Indications: end-stage heart failure bridging to transplant or destination therapy. Requirements: surgical implantation, external power source, anticoagulation.

Arytenoid Cartilage

- The arytenoid cartilages are a pair of small, pyramid-shaped cartilages located on the posterior aspect of the larynx, articulating with the superior border of the cricoid cartilage lamina, and they serve as the attachment points for the vocal cords and the intrinsic laryngeal muscles that control vocal cord movement.

ASA Physical Status Classification System

- A system developed by the American Society of Anesthesiologists to assess and communicate a patient's pre-operative physical status and comorbidities. It ranges from ASA I (a normal healthy patient) to ASA VI (a declared brain-dead patient whose organs are being removed for donor purposes).

Asbestosis

- See public health.

Ascariasis

- Infection with *Ascaris lumbricoides*, the most common helminth infection worldwide (soil-transmitted).

Ascending Aorta

- From left ventricle at the aortic valve to the sternal angle (approximately 5 cm), enclosed in pericardium. Gives rise to right/left coronary arteries at the aortic sinuses. Most common site of aneurysm and dissection, especially in Marfan syndrome, bicuspid aortic valve, and hypertension.

Ascending aortic aneurysm

- See Aortic Aneurysm.

Ascending Colon

- Extends from cecum to hepatic flexure below the liver, approximately 15-20 cm, retroperitoneal (fixed to posterior abdominal wall). Anteriorly: abdominal wall. Posteriorly: right kidney. Absorbs water/electrolytes and compacts fecal matter. Tumors

may present as a palpable right-sided mass.

Ascites

- Pathological accumulation of fluid within the peritoneal cavity representing the most common decompensating event in cirrhosis. The pathogenesis involves splanchnic arterial vasodilation reducing effective arterial blood volume and activating neurohormonal mechanisms leading to sodium and water retention.

Aseptic Loosening

- Loss of fixation of an implant without active infection, commonly caused by mechanical stress and a biologic response to wear particles. It may cause pain, radiographic lucency, instability, and need for revision surgery.

Aseptic Technique

- A set of laboratory practices preventing contamination of sterile sites, cultures, and specimens by unwanted microorganisms. Key elements include hand hygiene, sterile fields and instruments, surface disinfection, and minimal exposure of sterile items to air and touch, forming the basis of reliable clinical and culture work.

Asherman Syndrome

- Intrauterine adhesions or scar tissue that partially or completely obliterate the uterine cavity, most often after uterine instrumentation or pregnancy-related curettage. It may cause hypomenorrhea or amenorrhea, infertility, recurrent miscarriage, or abnormal placentation.

Aspergillosis

- A spectrum of diseases caused by the fungus *Aspergillus* (most commonly *A. fumigatus*). Allergic forms: allergic bronchopulmonary aspergillosis (ABPA, in asthma or cystic fibrosis – wheezing, eosinophilia, pulmonary infiltrates), allergic *Aspergillus* sinusitis.

Asphyxia

- A life-threatening lack of oxygen due to drowning, choking, or an obstruction of the airways. Management: remove obstruction, rescue breathing, CPR if needed. Foreign body airway obstruction: Heimlich maneuver (abdominal thrusts) for conscious adults; back blows and chest thrusts for infants.

Aspiration

- The inhalation of foreign material (oropharyngeal secretions, gastric contents, food, liquid, or blood) into the larynx and lower respiratory tract. Aspiration pneumonitis (Mendelson syndrome) is a chemical lung injury caused by acidic gastric fluid.

Aspiration Pneumonia

- Infectious pneumonia from aspiration of oropharyngeal secretions. Risk factors: dysphagia, impaired consciousness, enteral feeding. Common pathogens: anaerobes, gram-negative bacilli. Treatment: antibiotics covering oral flora.

Aspiration Pneumonitis

- Chemical lung injury caused by inhalation of gastric contents, characterized by bronchospasm, hypoxemia, pulmonary edema, and atelectasis. Risk factors include full stomach, emergency surgery, obesity, pregnancy, gastroesophageal reflux, bowel obstruction, and decreased consciousness.

Aspiration Risk

- The likelihood that food, liquid, saliva, or gastric contents will enter the larynx or lower respiratory tract. Risk rises with dysphagia, delirium, stroke, neuromuscular disease, reflux, poor dentition, and reduced consciousness; assessment and prevention are multidisciplinary.

Aspirin

- An antiplatelet drug that irreversibly inhibits platelet cyclooxygenase-1, reducing thromboxane A₂ production for the life of the platelet. It is used in selected acute coronary and atherosclerotic disease settings; bleeding, hypersensitivity, and gastrointestinal toxicity must be considered.

Aspirin (Salicylate) Toxicity

- Mixed respiratory alkalosis and metabolic acidosis. Early: tinnitus, hyperventilation, respiratory alkalosis. Late: metabolic acidosis, fever, seizures, coma. Treatment: activated charcoal, urine alkalization (sodium bicarbonate), hemodialysis if severe (renal failure, pulmonary edema, severe acidosis, very high levels).

Aspirin-Exacerbated Respiratory Disease

- A triad of asthma, nasal polyps, and aspirin sensitivity due to COX-1 inhibition. It involves overproduction of cysteinyl leukotrienes. Leukotriene

receptor antagonists are specifically effective.

Assist-Control Ventilation

- Ventilator delivers a set tidal volume at a set rate. Every breath receives the full set volume. Guarantees minimum rate and minute ventilation. Risk of respiratory alkalosis and auto-PEEP.

Assisted Breech Delivery

- Vaginal breech delivery with operator assistance after spontaneous delivery to the umbilicus.

Assisted Living

- Residential setting with assistance for ADLs. Less intensive than nursing home.

Assisted Reproductive Technology

- ART: fertility treatments involving manipulation of oocytes/sperm/embryos: in vitro fertilization (IVF), intracytoplasmic sperm injection (ICSI), intrauterine insemination (IUI, some definitions), embryo cryopreservation, gamete donation, and surrogacy.

Assistive Device

- See rehabilitation.

Assistive Device Training

- Training for walkers, canes, crutches: fitting, gait patterns, stair negotiation, safety.

Assistive Devices

- Tools enhancing mobility and independence. Categories: mobility (walkers, crutches, canes, wheelchairs), ADL (reachers, dressing aids, adaptive utensils), communication (sign language, communication boards), environmental (retractable showerheads, raised toilet seats). Prescription by PT/OT.

Association Study

- Comparison of allele frequencies between cases and controls. Genome-wide association studies (GWAS) identify common variants with small effects on complex diseases.

Astereognosis

- Inability to identify an object by touch alone despite intact primary somatosensory modalities (touch, vibration, proprioception), indicating dysfunction of the contralateral parietal association cortex (typically the postcentral gyrus region or the dominant inferior parietal lobule).

Asterixis

- Brief, irregular, non-rhythmic lapse of sustained

posture causing a flapping movement, best elicited by asking the patient to hold the arms extended with wrists dorsiflexed ("hold arms up") for 30 seconds.

Asthma

- A chronic inflammatory disorder of the airways characterized by recurrent episodes of wheezing, breathlessness, chest tightness, and cough. Airflow limitation is often reversible. It involves eosinophilic inflammation, mucus hypersecretion, and smooth muscle hypertrophy.

Asthma exacerbation

- An acute or subacute worsening of asthma symptoms and airflow limitation, typically causing increased wheeze, cough, chest tightness, or breathlessness.

Asthma-COPD Overlap

- Features of both asthma and COPD in the same patient. Characterized by persistent airflow limitation with features of both diseases: airway inflammation (eosinophilic and neutrophilic), bronchial hyperresponsiveness, and partially reversible obstruction. Treatment: inhaled corticosteroids plus long-acting bronchodilators.

Astigmatism

- A refractive error caused by an irregular curvature of the cornea or lens, resulting in light rays being focused at multiple points rather than a single point on the retina, leading to blurred or distorted vision at all distances.

Astrocytoma

- A glioma arising from star-shaped cells called astrocytes. Graded I-IV by the WHO; grade IV (glioblastoma) is the most aggressive and common primary malignant brain tumor in adults.

Asymptomatic Bacteriuria in Pregnancy

- The presence of >100,000 CFU/mL of bacteria on a clean-catch urine culture without urinary symptoms. Occurs in 2-10% of pregnancies. Risk factors: prior UTI, diabetes, sickle cell trait, and multiparity. If untreated, 20-40% develop pyelonephritis. Screening: urine culture at the first prenatal visit.

Asystole

- Complete cessation of cardiac electrical activity, presenting as a flat line on the ECG with no QRS complexes, P waves, or other organized electrical

activity.

Ataxia

- Lack of voluntary coordination of muscle movements, resulting from dysfunction of the cerebellum, posterior columns, or peripheral nerves. Types: cerebellar (gait and limb ataxia, dysarthria, nystagmus), sensory (impaired proprioception causing sensory ataxia, positive Romberg sign), and vestibular (vertigo, nystagmus, nausea).

Ataxia-Telangiectasia

- Autosomal recessive (ATM gene mutation, DNA damage response kinase).

Ataxic Breathing

- An irregular, unpredictable breathing pattern with random variations in rate and depth, seen in medullary lesions. It is the most dangerous respiratory pattern and often precedes respiratory arrest. It requires mechanical ventilation.

Atelectasis

- Complete or partial collapse of lung tissue, resulting in loss of aeration. It may be caused by airway obstruction (resorption atelectasis), pleural disease (compressive atelectasis), surfactant dysfunction (dissolution atelectasis), or scarring (cicatrization atelectasis). It impairs gas exchange and may predispose to infection.

Atherectomy

- Mechanical removal of atherosclerotic plaque. Types: directional, laser, rotational, orbital. Used for peripheral arterial disease.

Atherectomy (Vascular)

- See Atherectomy Device.

Atheroembolic Renal Disease

- Cholesterol crystal embolization to kidneys after vascular procedures. AKI with livedo reticularis and peripheral eosinophilia. Renal biopsy shows cholesterol clefts. Poor prognosis with progressive renal failure.

Atherosclerosis

- Chronic inflammatory disease of medium and large arteries characterized by accumulation of lipids, inflammatory cells, and fibrous tissue within the arterial wall, forming atherosclerotic plaques. Initiating event is endothelial dysfunction from risk factors (hypertension, smoking, diabetes, dyslipidemia).

Atherosclerotic Plaque

- An atherosclerotic plaque is a lesion within the arterial intima composed of a necrotic lipid core covered by a fibrous cap, representing the pathological substrate of atherosclerotic cardiovascular disease.

Athetosis

- Slow, involuntary, writhing, sinuous movements predominantly of the distal extremities, sometimes also affecting the face, neck, and trunk, arising from dysfunction of the basal ganglia (putamen and globus pallidus).

Athlete Foot

- Tinea pedis: dermatophyte infection (*Trichophyton rubrum*, *T. mentagrophytes*) of the feet, common in those who wear occlusive footwear or use communal showers. Forms: interdigital (scaly, fissured, macerated between toes—most common), moccasin (diffuse plantar scaling), vesicular.

Athlete's Foot

- A superficial fungal infection of the feet, usually affecting the spaces between the toes and causing itching, scaling, maceration, or fissures. It is also called tinea pedis.

Atlanto-occipital dislocation

- Traumatic separation of the atlas from the occipital condyles, a highly unstable injury with high mortality often associated with spinal cord injury and requiring emergent immobilization.

Atlas (C1)

- First cervical vertebra articulating with occipital condyles at the atlanto-occipital joint for nodding movement. Lacks a body and spinous process; has anterior/posterior arches and lateral masses. Transverse foramina transmit vertebral arteries. The anterior arch has a facet for the axis dens.

Atopic Dermatitis

- A chronic, relapsing inflammatory skin disease characterized by pruritus, xerosis, and eczematous lesions. It reflects epidermal-barrier dysfunction and immune dysregulation, with flares triggered by irritants, allergens, infection, stress, or climate.

Atopic eruption of pregnancy

- Pruritic eczematous skin eruptions occurring in pregnancy, presenting as eczematous patches or pruritic urticarial papules and plaques, typically

resolving after delivery.

Atopy

- The genetic predisposition to develop immediate (IgE-mediated, type I) hypersensitivity reactions to common environmental allergens (pollen, dust mites, foods, pet dander).

Atorvastatin

- A statin medicine that lowers LDL cholesterol by reducing cholesterol production in the liver. It helps reduce the risk of heart attack and stroke when combined with appropriate lifestyle measures. Muscle pain or weakness, dark urine, or yellowing of the skin requires medical attention; it is generally avoided during pregnancy.

Atovaquone–proguanil

- An antimalarial combination used for prevention and treatment of *Plasmodium falciparum* malaria and selected travel-related malaria regimens.

ATP (Adenosine Triphosphate)

- The primary energy currency of the cell, composed of adenine, ribose, and three phosphate groups. Hydrolysis of the terminal phosphoanhydride bond releases $\Delta G^\circ = -30.5 \text{ kJ/mol}$ (-7.3 kcal/mol), driving endergonic cellular processes such as muscle contraction, active transport, and biosynthesis.

ATP Synthase

- Complex V of electron transport chain. Uses proton gradient to synthesize ATP. Rotational catalysis mechanism.

Atransferrinemia

- Very rare autosomal recessive disorder of iron transport caused by mutations in the TF gene (3q22.1) leading to absence or severe deficiency of transferrin.

Atrial Fibrillation

- The most common sustained cardiac arrhythmia, characterized by rapid, irregular atrial activity (350-600 bpm) with irregular ventricular response. ECG shows absent P waves, irregularly irregular RR intervals, and narrow QRS complexes (unless aberrant conduction).

Atrial Fibrillation Ablation

- Catheter-based procedure to eliminate atrial fibrillation by creating pulmonary vein isolation (PVI), electrically disconnecting the pulmonary veins from the left atrium. Indicated for symptomatic AFib

refractory to or intolerant of antiarrhythmic drugs. Success rates: 70-80% for paroxysmal AFib, 50-60% for persistent AFib.

Atrial Flutter

- A rapid, regular atrial rhythm (typically 250-350 bpm) caused by a macro-reentrant circuit, most commonly in the right atrium. ECG shows sawtooth flutter waves best seen in leads II, III, and aVF, with regular or irregular ventricular response depending on AV conduction ratio (commonly 2:1 or 4:1).

Atrial Natriuretic Peptide

- A peptide hormone secreted by atrial myocytes in response to atrial stretch from volume expansion. ANP promotes natriuresis and diuresis by increasing glomerular filtration rate, inhibiting sodium reabsorption in the collecting ducts, and reducing renin and aldosterone secretion.

Atrial Septal Defect

- Congenital opening in the interatrial secundum allowing left-to-right shunting of blood. Small defects may close spontaneously. Larger defects cause right heart volume overload, right atrial and ventricular dilation, pulmonary hypertension, and eventually right heart failure. Paradoxical embolism may occur with right-to-left shunting.

Atrial Tachycardia

- Supraventricular tachycardia arising from an atrial focus outside the sinus node, often focal with rates 100-250 beats per minute with discrete P waves of different morphology from sinus P waves.

Atrioventricular Node

- Located in the posterior-inferior region of the interatrial septum near the coronary sinus. It receives impulses from the SA node via internodal pathways and delays them by approximately 120-200 milliseconds before transmitting them to the ventricles. This delay allows atrial contraction to complete before ventricular contraction.

Atrium

- One of the two upper chambers of the heart that receive blood returning to the heart. The right atrium receives deoxygenated blood from the superior and inferior venae cavae and the coronary sinus; the left atrium receives oxygenated blood from the four pulmonary veins.

Atropine in Organophosphate Poisoning

- A muscarinic antagonist used to reverse bronchorrhea, bronchospasm, bradycardia, and other muscarinic effects of organophosphate toxicity. Treatment is titrated to drying of respiratory secretions and improvement in ventilation, not to pupil size alone.

Attachment (Dental)

- Retentive device connecting denture to implant or tooth. Precision attachments preferred over bar attachments.

Attack Rate

- The proportion of an at-risk population that develops a specified illness during a defined outbreak or exposure period. It is a cumulative incidence measure commonly used for foodborne and other localized outbreaks.

Attention-Deficit/Hyperactivity Disorder

- A neurodevelopmental disorder characterized by persistent patterns of inattention, hyperactivity, and impulsivity that interfere with functioning or development. Three presentations: predominantly inattentive, predominantly hyperactive-impulsive, and combined.

Attention-Deficit/Hyperactivity Disorder (ADHD)

- A neurodevelopmental condition marked by persistent inattention, hyperactivity, and impulsivity that interferes with daily functioning. In children, symptoms often include difficulty staying seated, frequent interrupting, and trouble sustaining attention.

Attenuation Correction

- Compensation for photon absorption in tissue. CT-based (PET/CT), transmission source (SPECT), or MR-based (PET/MRI).

Attributable Risk

- Difference in risk between exposed and unexposed. $AR = \text{incidence exposed} - \text{incidence unexposed}$.

Atypia

- Cytologic or architectural abnormality that deviates from the expected appearance of a cell or tissue. Atypia may be reactive, degenerative, premalignant, or malignant and must be interpreted with architecture, clinical context, and ancillary studies.

Atypical Antipsychotics

- A class of modern psychiatric medications used to treat schizophrenia and bipolar disorder. They work by balancing dopamine and serotonin levels in the brain, which helps reduce symptoms like hallucinations and delusions with fewer movement-related side effects than older antipsychotics.

Atypical Facial Pain

- Chronic facial pain without identifiable cause. Psychological factors may contribute.

Atypical Features

- A specifier for depressive episodes characterized by mood reactivity, significant weight gain or increased appetite, hypersomnia, leaden paralysis, and a long-standing pattern of interpersonal rejection sensitivity. Atypical depression often presents in younger patients and may respond preferentially to MAOIs or SSRIs rather than TCAs.

Atypical Pneumonia

- Pneumonia with a gradual onset, relatively mild chest findings, and prominent systemic or extrapulmonary symptoms. Common causes include *Mycoplasma pneumoniae*, *Chlamydia pneumoniae*, and respiratory viruses.

Audiogram

- A chart that shows a person's ability to hear at different pitches or frequencies.

Audiologist

- A health professional who assesses hearing and fits hearing aids.

Audiometry

- A complete hearing test that involves listening to sounds of different frequencies and volume. See ent.

Auer Rods

- Pathognomonic intracellular crystalline inclusions formed by the fusion of azurophilic granules, seen in the cytoplasm of leukemic myeloblasts. Strongly diagnostic of Acute Myeloid Leukemia (AML), particularly Acute Promyelocytic Leukemia (APL, FAB M3 classification, t(15;17) PML-RARA translocation).

Auger Electrons

- Low-energy electrons emitted during electron capture decay. Short range (nanometers), high linear

energy transfer. Potential for targeted radiotherapy if localized near DNA.

Augmentation of Labor

- The stimulation of spontaneous contractions that are judged to be inadequate to achieve progressive cervical dilatation. Most commonly uses oxytocin (Pitocin) infusion. Also includes artificial rupture of membranes (AROM, amniotomy).

Auricle

- The visible external portion of the ear that projects from the side of the head, designed to collect and channel sound waves into the ear canal.

Auscultation

- The act of listening to sounds produced within the body, typically using a stethoscope. Essential for assessing heart sounds, lung sounds, bowel sounds, and vascular bruits. Normal findings: S1 and S2 heart sounds, vesicular breath sounds, normal bowel sounds. Abnormal: murmurs, crackles, wheezes, rubs, bruits.

Auscultatory Gap

- A transient silence between the first and second sounds of blood pressure measurement, sometimes heard in hypertensive patients. Can lead to underestimation of systolic pressure if not recognized.

Austin Flint Murmur

- A mid-diastolic, low-pitched rumbling murmur heard at the apex, caused by the regurgitant jet from severe aortic regurgitation striking the anterior mitral leaflet.

Autism Spectrum Disorder

- Neurodevelopmental disorder characterized by persistent deficits in social communication and social interaction across multiple contexts, restricted and repetitive patterns of behavior, interests, or activities. Symptoms present in early developmental period and cause clinically significant impairment.

Autoantibody

- An immunoglobulin that recognizes and binds an endogenous antigen. Autoantibodies may be pathogenic, as in autoimmune hemolytic anemia, or may serve as laboratory markers of autoimmune disease.

Autoclave

- Sterilization device using high-pressure steam. Pa-

rameters: 121°C, 15 psi, 15-30 minutes. Essential for infection control.

Autogenic Drainage

- A breathing technique using controlled breathing at different lung volumes to mobilize secretions from peripheral to central airways. It involves breathing at low lung volumes (collect), mid lung volumes (loosen), and high lung volumes (evacuate). It requires patient training and cooperation.

Autograft

- Tissue transplanted from one site to another within the same individual. Autografts are genetically identical to the recipient and generally avoid alloimmune rejection, although donor-site morbidity may occur.

Autoimmune Disease

- Autoimmune diseases result from loss of immunological self-tolerance, leading to immune-mediated attack on the body's own tissues.

Autoimmune Diseases

- Autoimmune diseases occur when the immune system mistakenly attacks the body's own healthy tissues, producing chronic inflammation and organ damage. There are over 80 types, including rheumatoid arthritis, lupus, type 1 diabetes, multiple sclerosis, and Hashimoto's thyroiditis.

Autoimmune Disorder

- A disease caused by an abnormal immune response against the body's own cells, tissues, or organs. Autoimmune disorders may be organ-specific or systemic and can produce inflammation, tissue injury, functional impairment, or circulating autoantibodies.

Autoimmune Enteropathy

- Rare immune-mediated disease causing persistent diarrhoea and small-intestinal mucosal injury, usually with villous atrophy and crypt hyperplasia. It may occur in children or adults and can be associated with other autoimmune disorders or immune dysregulation.

Autoimmune Gastritis

- Chronic atrophic gastritis caused by antibodies against parietal cells leading to achlorhydria and vitamin B12 deficiency. Associated with pernicious anemia and increased risk of gastric carcinoid tumors and type 1 gastric neuroendocrine tumors.

Anti-parietal cell antibodies are present in approximately ninety percent of patients.

Autoimmune Hemolytic Anemia

- AIHA: hemolytic anemia due to antibodies directed against red blood cells. Warm AIHA (IgG, 37C—idiopathic, SLE, CLL, drugs: methyldopa, penicillin): splenic sequestration, spherocytes, positive direct antiglobulin (DAT).

Autoimmune Hepatitis

- Chronic hepatitis caused by immune-mediated destruction of hepatocytes characterized by interface hepatitis with lymphoplasmacytic infiltrate, elevated transaminases, hypergammaglobulinemia, and autoantibodies.

Autoimmune keratitis

- Corneal inflammation caused by the body's own immune system attacking corneal tissue, often associated with systemic autoimmune conditions such as rheumatoid arthritis, lupus, or granulomatosis with polyangiitis. It presents with corneal opacity, pain, and vision loss and requires immunosuppressive therapy.

Autoimmune Pancreatitis

- Chronic inflammatory condition characterized by dense lymphoplasmacytic infiltrate with IgG4-positive plasma cells representing the pancreatic manifestation of IgG4-related disease. Type 1 is more common affecting older men with elevated serum IgG4 and extrapancreatic involvement.

Autoimmune Polyendocrine Syndromes

- APS-1 (APECED, AIRE mutation): autosomal recessive, chronic mucocutaneous candidiasis, hypoparathyroidism, adrenal insufficiency (Addison's), plus type 1 diabetes, autoimmune thyroiditis, vitiligo, alopecia, pernicious anemia, hepatitis, asplenia.

Autoimmunity: Mechanisms

- Loss of tolerance: central (thymic negative selection defects, AIRE mutation -> APS-1) and peripheral (Treg dysfunction, anergy failure, ignorance loss). Molecular mimicry: cross-reactivity between pathogen and self (rheumatic fever: Strep M protein + cardiac myosin; Guillain-Barre: Campylobacter + GM1 ganglioside; reactive arthritis).

Autologous Blood Donation

- Autologous blood donation involves preoperative collection and storage of the patient's own blood for intraoperative or postoperative transfusion. It eliminates risks of alloimmune reactions, transfusion-transmitted infections, and immune modulation.

Autonomic Dysreflexia

- Acute, potentially life-threatening sympathetic discharge in a person with spinal-cord injury at or above approximately T6, triggered by a noxious stimulus below the lesion. It causes sudden hypertension, headache, flushing or sweating above the lesion, and reflex bradycardia or other heart-rate changes.

Autonomic Nervous System

- The part of the peripheral nervous system that regulates involuntary functions, including heart rate, vascular tone, digestion, sweating, pupil size, and bladder activity.

Autophagy

- Cellular self-degradation process delivering cytoplasmic components to lysosomes for recycling. Macroautophagy: phagophore forms, expands to autophagosome (LC3-II on membrane), fuses with lysosome (autolysosome) for degradation. Selective autophagy: mitophagy (damaged mitochondria), xenophagy (pathogens), aggrephagy (protein

Autopsy

- A postmortem examination of a body to determine cause and manner of death, performed by a forensic pathologist (medical examiner/coroner) or, for hospital autopsies, a pathologist.

Autopsy (Medicolegal)

- Systematic external examination and surgical dissection of a deceased body performed under legal authority by a forensic pathologist to determine the Cause, Manner, and Mechanism of Death, document injuries, collect evidence, and assist judicial investigations.

Autopsy (Post-Mortem Examination)

- Specialized surgical procedure performed on a deceased body to determine the cause of death, mechanism of death, manner of death (natural, accidental, suicide, homicide, or undetermined), state of health prior to death, and to document injuries or disease

processes.

Autosomal Dominant PKD

- Most common hereditary kidney disease affecting 1 in 400 to 1000. PKD1 chromosome 16 more severe with earlier ESRD than PKD2 chromosome 4. Progressive cyst growth causes hypertension, pain, hematuria, infections, and eventually ESRD. Screen family members with ultrasound. Tolvaptan may slow cyst growth.

Autosomal dominant polycystic kidney disease

- An inherited disorder causing progressive growth of numerous renal cysts that enlarge the kidneys and destroy parenchyma, leading to hypertension, flank pain, and eventual kidney failure. Associated with hepatic cysts and intracranial aneurysms.

Autosomal Inheritance

- Transmission of a genetic trait through one of the 22 non-sex chromosomes. Autosomal dominant traits may appear in successive generations, whereas autosomal recessive disease generally requires pathogenic variants on both homologous chromosomes.

Autosomal Recessive PKD

- Rare childhood disorder from PKHD1 mutation. Bilateral enlarged kidneys with fusiform dilation of collecting ducts. Associated with congenital hepatic fibrosis. Severe forms cause pulmonary hypoplasia in utero. Variable severity from perinatal lethal to presentation in childhood or adolescence.

Avascular Necrosis

- See vascular surgery. Osteonecrosis of bone due to interrupted blood supply. Common sites: femoral head, humeral head, talus. Risk factors: steroids, alcohol, trauma, sickle cell, SLE. Treatment: core decompression, joint replacement.

Avascular Necrosis (Osteonecrosis)

- Death of bone tissue caused by impaired blood supply. It commonly affects the femoral head and may lead to subchondral collapse, secondary arthritis, and need for joint-preserving surgery or arthroplasty.

Avian Influenza

- Influenza caused by avian-adapted influenza A viruses, acquired mainly through contact with infected birds or contaminated environments. Human

infection can cause severe pneumonia, respiratory failure, and occasionally systemic disease.

Avoidant Personality Disorder

- A pattern of social inhibition, feelings of inadequacy, and hypersensitivity to negative evaluation.

Avolition

- A severe lack of motivation to initiate and perform self-directed, purposeful activities. It is a negative symptom of schizophrenia and is seen in severe depression. It manifests as neglect of personal hygiene, inability to complete tasks, and social withdrawal.

Avulsed Tooth

- A tooth completely displaced from its socket by trauma. It is a dental emergency; management depends on patient age, extraoral dry time, storage medium, root development, and periodontal-ligament viability, and may require urgent replantation by a dental professional.

Avulsion

- An injury in which tissue is forcibly torn from the body by shearing or pulling. Common forms include dental, fingertip, nail, and degloving avulsions. Management focuses on hemorrhage control, debridement, tissue preservation, and urgent consultation.

Avulsion Fracture

- A fracture in which a fragment of bone is pulled away by the tendon or ligament attached to it, usually during sudden forceful contraction or injury. Common sites include the pelvis, ankle, elbow, and fingers; evaluation considers displacement, joint involvement, skeletal maturity, and associated soft-tissue injury.

Awake Fiberoptic Intubation

- The gold standard for anticipated difficult airways, performed with the patient conscious and cooperatively maintaining spontaneous ventilation. Topical anesthesia is achieved through lidocaine nebulization, nerve blocks (glossopharyngeal, superior laryngeal, transtracheal), and spray-as-you-go technique.

Awareness Under Anesthesia

- Unintended recall of events during general anesthesia, most commonly occurring during procedures where neuromuscular blockade prevents movement. Incidence: 0.1-0.2% of all general anesthetics, higher

in cardiac, trauma, and cesarean section. Presents with postoperative recall of conversations, surgical sounds, or pain (worst outcomes).

Axilla

- Pyramidal space between the upper lateral thorax and medial arm. Bounded by anterior fold (pectoralis major/minor), posterior fold (latissimus dorsi/teres major), medial wall (ribs/serratus anterior), and lateral wall (humerus). Contains axillary artery/vein, brachial plexus cords, and lymph nodes.

Axillary Artery

- The continuation of the subclavian artery, running from the outer border of the first rib to the lower border of teres major, where it becomes the brachial artery. Divided into three parts by the pectoralis minor.

Axillary Nerve

- From posterior cord (C5-C6), passing through the quadrangular space with the posterior circumflex humeral artery. Innervates deltoid and teres minor; provides sensation to the regimental badge area (lateral deltoid skin). Vulnerable in anterior shoulder dislocations and surgical neck humeral fractures.

Axis (C2)

- Second cervical vertebra with the dens (odontoid process) projecting superiorly, serving as a pivot for atlas rotation, allowing approximately 50 percent of cervical rotation. Has a large spinous process and flat superior articular facets.

Axon

- The single long process of a neuron that conducts action potentials away from the cell body toward the axon terminal for synaptic transmission.

Azole Antifungals

- A class of antifungal agents that inhibit the enzyme lanosterol 14-alpha-demethylase, a cytochrome P450 enzyme essential for ergosterol synthesis in fungal cell membranes. The depletion of ergosterol and accumulation of toxic methylated sterols disrupt membrane integrity and function.

Azoospermia

- Absence of spermatozoa in the ejaculate after appropriate centrifugation and microscopic examination. It may be obstructive, with sperm production preserved but outflow blocked, or nonobstructive,

with impaired spermatogenesis.

Azygos Vein

- A vein that drains the posterior intercostal veins and empties into the superior vena cava. It forms an important collateral pathway when the SVC is obstructed. On chest imaging, enlargement of the azygos vein may indicate SVC obstruction, heart failure, or portal hypertension.

Azygous vein

- A large vein that drains the posterior thoracic wall and mediastinum, ascending along the right side of the vertebral column to empty into the superior vena cava. It provides collateral venous return when the superior or inferior vena cava is obstructed.

Anhidrosis

- Absent or greatly reduced sweating. It may result from nerve damage, skin disease, dehydration, medications, or inherited conditions. Inability to sweat can cause dangerous overheating, especially during exercise or hot weather.

Anovulation

- Failure to release an egg from the ovary during the menstrual cycle. It may cause irregular or absent periods, infertility, or abnormal bleeding and can result from hormonal disorders, excessive exercise, low body weight, stress, or polycystic ovary syndrome.

Asthenopia

- Eye strain or visual fatigue, often experienced as aching, burning, blurred vision, headache, or difficulty focusing after prolonged visual tasks. Refractive errors, dry eye, poor lighting, and extended screen use may contribute.

Abdominal Wall Hernia

- Protrusion of abdominal tissue through a weak area or opening in the abdominal wall. Common types include inguinal, umbilical, femoral, and incisional hernias. A painful, firm, or non-reducible bulge with vomiting or abdominal distension may indicate incarceration or strangulation and requires urgent assessment.

Abdominal Swelling

- Visible or felt enlargement of the abdomen caused by gas, constipation, pregnancy, fluid accumulation, organ enlargement, mass, or abdominal-wall changes. Persistent, progressive, painful, or sudden

swelling—especially with vomiting, fever, bleeding, or inability to pass stool or gas—requires medical evaluation.

Acute Diarrhea

- Passage of loose or watery stools, usually three or more times daily, lasting less than two weeks. Infection, food-related illness, medicines, and other intestinal disorders can cause it. Oral rehydration is important; blood in stool, severe dehydration, persistent vomiting, or severe pain requires prompt assessment.

Acute Stress Disorder

- A trauma-related mental-health condition beginning after exposure to actual or threatened death, serious injury, or sexual violence and lasting from three days to one month. Symptoms may include intrusive memories, avoidance, negative mood, dissociation, and increased arousal; professional assessment and support are recommended.

Ankle Swelling

- Enlargement around the ankle caused by injury, inflammation, infection, venous disease, lymphatic problems, or fluid retention. One-sided sudden swelling with pain, warmth, or shortness of breath may indicate a blood clot or infection and requires urgent assessment; swelling after trauma may require evaluation for fracture.

Aortic Valve Disease

- Abnormal narrowing, leakage, or structural change of the valve between the left ventricle and aorta. It may cause chest pain, breathlessness, fatigue, fainting, or no symptoms initially. Evaluation commonly includes examination and echocardiography; severe disease may require specialist treatment.

Autonomic Neuropathy

- Damage to nerves that control involuntary functions such as heart rate, blood pressure, digestion, sweating, bladder function, and sexual function. Diabetes is a common cause, but other diseases, toxins, and medicines may contribute. Symptoms depend on the affected organs and require evaluation of the underlying cause.

Chapter 2

Pocket Dictionary: B Terms

B Cell (B Lymphocyte)

- A key cellular component of adaptive humoral immunity that matures in the bone marrow. Upon antigen recognition via B-cell receptor (BCR/surface IgM) and T-cell help (CD40/CD40L interaction), B cells proliferate and differentiate into antibody-secreting plasma cells and long-lived memory B cells.

B Chromosome

- A supernumerary chromosome that is not part of the standard chromosome complement and is not required for normal development. It may occur in certain plants, animals, or tumors and varies in number between cells.

B-Cell Receptor (BCR)

- The membrane-bound immunoglobulin complex on a B lymphocyte that recognizes native antigen. BCR signaling, with appropriate co-stimulation, promotes B-cell activation, proliferation, differentiation, and antibody production.

B-Lynch Suture

- A surgical technique for controlling postpartum hemorrhage due to uterine atony unresponsive to medical management. A compressive suture placed around the uterus to mechanically compress the uterine walls and reduce blood flow. The suture creates a brace-like effect. Can be combined with other compression sutures (Cho, Hayman).

Babesiosis

- A tick-borne malaria-like illness caused by *Babesia* species (*B. microti* in the US, *B. divergens* in Europe), transmitted by Ixodes ticks (same vector as Lyme disease). Parasites infect red blood cells, causing hemolytic anemia.

Babinski Reflex

- An extensor plantar response present from birth to approximately 2 years. When the lateral sole is stroked, the great toe dorsiflexes and other toes fan outward. Disappears as the corticospinal tract myelinates. Persistence beyond age 2 may indicate neurologic abnormality.

Babinski Sign

- The Babinski sign is a pathological reflex elicited by stroking the lateral aspect of the sole of the foot from the heel to the metatarsal heads with a blunt object. A normal response (flexor plantar response) is plantar flexion (curling downward) of all toes.

Bacillary Angiomatosis

- A vascular proliferative disease caused by *Bartonella henselae* or *B. quintana*, occurring in immunocompromised patients (HIV/AIDS, transplant). Presents with erythematous papules, nodules, or pedunculated lesions resembling Kaposi sarcoma or pyogenic granuloma. Can involve skin, liver (peliosis hepatis), spleen, bone, and lymph nodes.

Baclofen

- A muscle relaxant that reduces muscle spasticity by activating GABA-B receptors in the spinal cord. It is used for conditions such as multiple sclerosis and spinal-cord injury. Drowsiness, weakness, dizziness, and confusion can occur; it should not be stopped suddenly after regular use because withdrawal can be dangerous.

Bacilluria

- The presence of bacilli in the urine, usually referring to rod-shaped bacteria detected by microscopy or culture. Its clinical significance depends on symptoms, quantitative culture, and the organism identi-

fied.

Back Pain

- One of the most common medical complaints, affecting up to 80 percent of the population at some point in life, and the leading cause of disability worldwide. Back pain is classified by duration: acute (less than 4 weeks), subacute (4-12 weeks), and chronic (greater than 12 weeks).

Backache

- Pain or discomfort located anywhere in the posterior trunk from neck to buttocks, a symptom rather than a disease. Causes include musculoskeletal strain (most common), disc herniation, spinal stenosis, facet arthropathy, sacroiliac joint dysfunction, and referred pain from renal, pancreatic, or vascular sources.

Bacteremia

- The presence of viable bacteria in the bloodstream, a serious condition that can be transient (after dental procedures, toothbrushing), intermittent (undrained abscess), or continuous (endovascular infection like endocarditis). Bacteremia triggers systemic inflammatory responses ranging from fever and leukocytosis to septic shock.

Bacteria

- Single-celled microscopic organisms lacking a nucleus, which can exist independently or in colonies; some are beneficial to health (microbiome) while others cause infectious disease.

Bacterial adherence

- The process by which bacteria attach to host cell surfaces, a critical first step in colonisation and infection. Adherence is mediated by adhesins, which are surface structures that bind to specific receptors on host cells. Adhesins include pili, fimbriae, outer membrane proteins, and capsules.

Bacterial biofilms

- Structured communities of bacteria attached to surfaces and embedded in a self-produced matrix of extracellular polymeric substances. Biofilms are a major cause of chronic infections, particularly on medical devices including catheters, prosthetic joints, and heart valves.

Bacterial capsule

- A polysaccharide layer surrounding the bacterial

cell wall that protects against phagocytosis and complement-mediated killing. Capsules are major virulence factors for many pathogens including *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Klebsiella pneumoniae*.

Bacterial cell wall

- The rigid outer layer of bacteria that provides structural integrity and determines cell shape. The cell wall is composed primarily of peptidoglycan, a polymer of sugars cross-linked by short peptides.

Bacterial classification

- The organisation of bacteria into taxonomic groups based on genetic, phenotypic, and evolutionary relationships. Bacteria are classified using a hierarchical system: domain, phylum, class, order, family, genus, and species.

Bacterial conjugation

- The transfer of genetic material between bacteria through direct cell-to-cell contact via a pilus. Conjugation is mediated by conjugative plasmids that encode the genes for transfer. The donor cell (F+) forms a conjugative pilus that attaches to the recipient cell (F-).

Bacterial Conjunctivitis

- An infection of the conjunctival mucous membrane caused by various bacterial pathogens, presenting with a purulent or mucopurulent discharge that causes the eyelids to become matted together, particularly upon waking, conjunctival injection, and a gritty or foreign body sensation.

Bacterial dermatitis

- Inflammation of the skin caused by bacterial infection, typically presenting with erythema, edema, warmth, pain, and purulent discharge. Common causative organisms include *Staphylococcus aureus* and *Streptococcus pyogenes*, and the condition may arise de novo or as a superinfection of pre-existing dermatitis.

Bacterial dormancy

- A state of reduced metabolic activity that allows bacteria to survive unfavourable conditions. Dormant forms include spores, persisters, and viable but non-culturable cells. Spores are highly resistant structures formed by *Bacillus* and *Clostridium*. Persisters are phenotypic variants that are tolerant to

antibiotics.

Bacterial Endocarditis

- See Infective Endocarditis.

Bacterial Endocarditis Vegetations

- Vegetations in bacterial endocarditis are complex masses composed of bacteria, fibrin, and platelets that form on heart valve surfaces or endocardium. The pathogenesis begins with bacteremia seeding a site of endothelial damage or a previously abnormal valve surface.

Bacterial endospores

- Highly resistant dormant structures formed by certain gram-positive bacteria including *Bacillus* and *Clostridium* species. Endospores form during nutrient deprivation and can survive for decades. They are resistant to heat, radiation, chemicals, and desiccation.

Bacterial endotoxin

- Lipopolysaccharide from the outer membrane of gram-negative bacteria that activates innate immunity and can contribute to fever, hypotension, and septic shock.

Bacterial exotoxin

- A secreted bacterial protein that produces specific toxic effects in host cells or tissues.

Bacterial flagella

- Thread-like appendages composed of flagellin protein that provide motility to bacteria. Flagella are composed of three parts: the filament, hook, and basal body. The basal body is embedded in the cell wall and functions as a rotary motor.

Bacterial Gastroenteritis

- Acute inflammation of the stomach and intestines caused by pathogenic bacteria such as *Salmonella*, *Campylobacter*, *Shigella*, or pathogenic *Escherichia coli*. It commonly causes diarrhea, abdominal cramps, fever, and sometimes bloody stools or vomiting.

Bacterial genetics

- The study of genes and genetic processes in bacteria. Bacteria have a single circular chromosome and may contain plasmids, which are extrachromosomal DNA elements. Bacterial genes are organized into operons, which are clusters of genes regulated as a unit.

Bacterial growth phases

- The stages of bacterial growth in culture. The lag phase is when bacteria adapt to their environment. The log (exponential) phase is when bacteria multiply at their maximum rate. The stationary phase is when nutrient depletion and waste accumulation limit growth. The death phase is when bacteria die.

Bacterial Infections

- Illnesses caused by pathogenic bacteria proliferating in the body, ranging from mild sore throats and ear infections to life-threatening sepsis and meningitis. They are diagnosed through cultures, PCR testing, or other microbiological methods.

Bacterial Meningitis

- Infection of the meninges by bacteria, a medical emergency. In neonates: Group B *Streptococcus*, *E. coli*, *Listeria*. In older children: *Neisseria meningitidis*, *Streptococcus pneumoniae*. Presents with fever, headache, neck stiffness, altered consciousness, and Kernig/Brudzinski signs. Diagnosis by lumbar puncture.

Bacterial metabolism

- The chemical processes that occur within bacteria to maintain life. Bacteria have diverse metabolic capabilities including aerobic and anaerobic respiration, fermentation, and photosynthesis. Metabolic pathways are targets for antimicrobial drugs.

Bacterial morphology

- The shape and arrangement of bacteria as observed under microscopy. Basic shapes include cocci (spheres), bacilli (rods), and spirilla (spirals). Cocci can be arranged in pairs (diplococci), chains (streptococci), clusters (staphylococci), or tetrads. Bacilli can be single, paired, or arranged in chains.

Bacterial Overgrowth

- Excessive proliferation of bacteria in the small intestine, commonly called small-intestinal bacterial overgrowth. It may result from impaired intestinal motility, anatomical stasis, blind loops, strictures, diverticula, or loss of the normal ileocaecal barrier.

Bacterial pathogenesis

- The mechanisms by which bacteria cause disease. Pathogenesis involves adhesion, invasion, toxin production, and immune evasion. Adhesins mediate attachment to host cells. Invasins allow penetra-

tion of tissues. Toxins damage host cells. Immune evasion mechanisms include capsules, biofilms, and intracellular survival.

Bacterial pili

- Hair-like appendages on the bacterial surface composed of pilin protein. Pili are shorter and thicker than flagella. Type I pili mediate adherence to host cells and are important virulence factors. Type IV pili are involved in twitching motility and DNA uptake during transformation.

Bacterial Pneumonia

- See Pneumonia.

Bacterial Spore

- A metabolically dormant, highly resistant structure produced by certain bacteria, notably *Bacillus* and *Clostridioides* species, in response to environmental stress. Spores resist desiccation, heat, and many disinfectants and require validated sterilization for reliable destruction.

Bacterial Toxin

- A poisonous molecule produced by bacteria that damages host cells or disrupts normal physiology. Exotoxins are secreted or released from viable bacteria, whereas endotoxin refers principally to lipopolysaccharide from the outer membrane of gram-negative bacteria.

Bacterial toxins

- Substances produced by bacteria that cause damage to host cells. There are two major categories: exotoxins and endotoxins. Exotoxins are proteins secreted by bacteria that have specific toxic effects. Endotoxins are lipopolysaccharide components of gram-negative bacterial outer membranes released during cell lysis.

Bacterial Tracheitis

- See Tracheitis.

Bacterial transduction

- The transfer of bacterial DNA from one bacterium to another through bacteriophages (bacterial viruses).

Bacterial transformation

- The uptake of free DNA from the environment by competent bacteria. Transformation occurs naturally in some bacterial species including *Streptococcus pneumoniae*, *Haemophilus influenzae*, and

Neisseria. Competence is the ability to take up DNA and is regulated by specific genes.

Bacterial Vaginosis

- BV: replacement of normal vaginal lactobacilli by an overgrowth of mixed anaerobes (*Gardnerella vaginalis*, *Mobiluncus*, etc.), the most common cause of vaginal discharge in reproductive-age women. Risk: new/multiple partners, douching, smoking.

Bacterial virulence factors

- Molecules produced by bacteria that contribute to their ability to cause disease. Virulence factors include adhesins, invasins, toxins, capsules, and enzymes. Adhesins mediate attachment to host cells. Toxins damage host cells directly. Capsules protect against phagocytosis. Enzymes degrade host tissues.

Bactericide

- An agent that kills bacteria rather than merely inhibiting their growth (bacteriostatic). Bactericidal antibiotics include beta-lactams (penicillins, cephalosporins, carbapenems), aminoglycosides, fluoroquinolones, metronidazole, and vancomycin.

Bacteriology

- The branch of microbiology concerned with the study of bacteria, including their morphology, genetics, physiology, ecology, and clinical significance.

Bacteriophage

- A virus that infects and replicates within bacteria.

Bacteriophage therapy

- The use of bacterial viruses (bacteriophages) to treat bacterial infections. Bacteriophages are viruses that specifically infect and lyse bacteria. They have narrow specificity for target bacteria and do not affect normal flora.

Bacteriophages

- Viruses that specifically infect and replicate within bacteria, also known as phages. Each phage type typically infects a narrow range of bacterial strains. Phages attach to bacterial surface receptors and inject their genome, hijacking bacterial machinery to produce progeny phages that lyse the host cell.

Bacterium

- A unicellular prokaryote lacking a nucleus and membrane-bound organelles, possessing a single circular chromosome and typically a peptidoglycan cell

wall.

Bacteriuria

- The presence of bacteria in urine, detected by urinalysis or culture. It may occur with urinary symptoms, as in a urinary-tract infection, or without symptoms, as in asymptomatic bacteriuria; interpretation depends on specimen quality, bacterial quantity, patient context, and the presence of inflammation or symptoms.

Bag-Mask Ventilation

- Manual ventilation of a non-breathing or inadequately breathing patient using a self-inflating bag (AMBU) attached to a mask sealed over the face. Effective technique: correct mask size, jaw thrust/head extension, two-person technique (two-hand seal), adequate tidal volumes with ETCO₂ monitoring.

Bagassosis

- An occupational hypersensitivity pneumonitis (extrinsic allergic alveolitis) caused by inhalation of dust from moldy bagasse—the fibrous residue of sugarcane after extraction. Caused by thermophilic actinomycetes (*Saccharopolyspora rectivirgula*) that colonize stored bagasse.

Bainbridge Reflex

- An increase in heart rate produced by stretch of the atrial walls, especially during increased venous return. The reflex helps limit venous congestion by adjusting cardiac output.

Baker Cyst

- A fluid-filled cyst in the popliteal fossa, resulting from herniation of the synovial membrane through the posterior knee joint capsule or from a communication between the knee joint and the gastrocnemius-semimembranosus bursa.

Baker's cyst

- A fluid-filled synovial cyst arising from the gastrocnemius-semimembranosus bursa posterior to the knee, typically caused by chronic knee joint effusion from conditions such as osteoarthritis, rheumatoid arthritis, or meniscal tears. It presents as a palpable, tender mass in the popliteal fossa that may rupture causing calf swelling.

Balance

- The ability to maintain the body's center of mass

over its base of support, relying on input from the vestibular system, vision, and proprioception. It is essential for walking, standing, and performing daily activities without falling. Age-related decline, medications, and neurological disorders can impair balance, increasing fall risk.

Balance Assessment

- Tests: Berg Balance Scale, Tinetti, timed up and go, Romberg. Identifies fall risk.

Balance Disorders and Vertigo

- Balance disorders encompass conditions that cause dizziness, unsteadiness, or a sensation of motion, often arising from dysfunction in the inner ear, brain, or sensory pathways.

Balance Impairment

- The ability to maintain the body's center of mass over its base of support, relying on input from the vestibular system, vision, and proprioception. See balance.

Balance Training

- Exercises improving postural stability and preventing falls. Components: static balance, dynamic balance, reactive balance. Techniques: weight shifting, tandem stance, single-leg stance, perturbations, balance boards. Essential for elderly, neurological conditions.

Balanitis

- Inflammation of the glans penis, often involving the prepuce (balanoposthitis). Causes include poor hygiene, irritants, infection, and dermatologic conditions such as lichen planus, psoriasis, and lichen sclerosus. It may cause erythema, swelling, discharge, pruritus, and discomfort; treatment depends on the cause.

Balanoposthitis

- Inflammation of both the glans penis and the prepuce. It may cause pain, erythema, swelling, discharge, or difficulty retracting the foreskin and can result from infection, irritation, or inflammatory skin disease.

Baldness

- The common term for hair loss or alopecia, most frequently referring to androgenetic alopecia (male or female pattern hair loss). Affects up to 50% of men by age 50 and 25% of women by age 60. Driven

by dihydrotestosterone (DHT)-mediated miniaturization of hair follicles in susceptible individuals.

Ballard Score

- Clinical assessment of gestational age in newborns using 6 physical and 6 neuromuscular maturity criteria. Physical parameters: skin texture, ear form and recoil, eye appearance, breast tissue, and genitalia. Neuromuscular: posture, arm and leg recoil, popliteal angle, scarf sign, and heel-to-ear.

Ballism

- A hyperkinetic movement disorder characterized by sudden, involuntary, large-amplitude, flinging movements of a limb or limbs. It is often related to dysfunction of the subthalamic region or connected basal-ganglia circuits.

Ballistocardiography

- The graphic recording of repetitive body movements produced by the ejection of blood with each heartbeat. It has been used to study cardiac mechanical activity and circulation.

Balloon (Vascular)

- Inflatable balloon for angioplasty, stent deployment. See Balloon Catheter.

Balloon Aortic Valvuloplasty

- Catheter-based procedure using balloon inflation to dilate a stenotic aortic valve. Provides temporary hemodynamic improvement in severe AS but restenosis is common within 6-12 months. Used as a bridge to TAVR or surgical AVR in hemodynamically unstable patients, or as palliation in patients not eligible for definitive valve replacement.

Balloon Catheter

- Inflatable balloon for angioplasty, stent deployment. Non-compliant balloons for high-pressure dilation.

Band Keratopathy

- A horizontal band of calcium deposition in the superficial cornea, typically across the interpalpebral area. It may occur with chronic ocular inflammation, hypercalcemia, renal disease, or longstanding eye disease and can impair vision.

Bandage

- Material for wound covering or support. Types: gauze, adhesive, elastic, tubular.

Bandage Scissors

- Blunt-tipped scissors for cutting bandages.

Bankart Lesion

- Injury to inferior glenohumeral ligament/labrum from shoulder dislocation. Associated with recurrent instability. Treatment: surgical repair (arthroscopic/open).

Bannayan-Riley-Ruvalcaba Syndrome

- Autosomal dominant PTEN hamartoma tumor syndrome (PHTS) caused by loss-of-function mutations in PTEN (10q23), allelic to Cowden syndrome.

Banti Disease

- A historical term for congestive splenomegaly associated with portal hypertension and hypersplenism. Modern evaluation usually identifies the underlying hepatic, portal-venous, or hematologic cause.

Banti Syndrome

- Historical eponym for non-cirrhotic portal hypertension with splenomegaly and hypersplenism, originally described by Guido Banti as "splenic anemia with portal hypertension but without hepatic cirrhosis." Causes include extrahepatic portal vein thrombosis, congenital hepatic fibrosis, schistosomiasis, nodular regenerative

Barbiturate

- A class of CNS depressants derived from barbituric acid that enhance GABA-A receptor activity by increasing the duration of chloride channel opening (as opposed to benzodiazepines, which increase the frequency). At low doses: anxiolysis and sedation. At moderate doses: hypnosis and anesthesia.

Bare Stent (Vascular)

- See Bare Stent.

Bariatric Surgery

- Weight loss procedures and surgery, collectively known as bariatric surgery, are invasive interventions for severe obesity that produce substantial and sustained weight loss by altering gastrointestinal anatomy and physiology.

Barium

- A metallic element (atomic number 56, symbol Ba) whose sulfate salt (barium sulfate) is used as a radiocontrast agent due to its high atomic number, which effectively absorbs X-rays.

Barium Enema

- Fluoroscopic imaging of the large intestine after retrograde filling with barium contrast. Double-contrast (barium + air) provides better mucosal detail. Indications: colorectal cancer screening (historical, now CT colonography preferred), evaluation of colonic obstruction, diverticulosis, and postsurgical anatomy.

Barium Study

- X-ray imaging using barium sulfate contrast. Barium swallow, barium enema, upper GI series. Evaluates GI tract morphology and function.

Barium Swallow

- Fluoroscopic examination of the esophagus after ingestion of barium contrast. Used to evaluate esophageal anatomy, motility, strictures, and structural abnormalities. The study demonstrates mucosal detail, filling defects, strictures, and motility disorders. A barium follow-through extends the examination to the small bowel.

Barium-133

- Calibration source. Half-life 10.5 years. Used for energy calibration.

Barognosis

- The ability to recognize and distinguish the weight of objects by handling them, with intact primary sensation. Impaired barognosis indicates a cortical sensory deficit.

Baroreceptor Reflex

- The rapid neural mechanism for short-term blood pressure regulation, involving baroreceptors (mechanoreceptors) in the carotid sinus (innervated by CN IX) and aortic arch (innervated by CN X). When blood pressure rises, baroreceptors increase their firing rate, signaling the cardiovascular center in the medulla.

Barotrauma

- Lung injury from excessive pressure during ventilation, causing alveolar rupture (pneumothorax, pneumomediastinum). Risk factors: high pressures, high PEEP. Prevented by lung-protective ventilation.

Barrett Esophagus

- Metaplastic transformation of the normal stratified squamous epithelium of the distal esophagus

to specialized intestinal-type columnar epithelium containing goblet cells, occurring as a consequence of chronic gastroesophageal reflux disease.

Barrett Esophagus Surveillance

- Systematic endoscopic monitoring of patients with Barrett esophagus to detect dysplasia and early esophageal adenocarcinoma. Patients without dysplasia undergo surveillance every three to five years with four-quadrant biopsies every two centimeters throughout the Barrett segment.

Barrier Creams

- Topical preparations applied to the skin to create a physical protective layer against irritants, moisture, and pathogens. Common formulations include zinc oxide, petrolatum, dimethicone, and titanium dioxide.

Barthel Index

- Standardized measure of functional independence in activities of daily living (ADL). Scores 0-100 based on 10 items: feeding, bathing, grooming, dressing, bowel control, bladder control, toilet use, transfers, mobility, stair climbing. <40: severe dependency; 41-60: moderate dependency; >60: mild dependency.

Bartholin Cyst

- A cyst of the Bartholin gland (paired mucus-secreting glands at the posterolateral introitus) due to obstruction of the duct; if infected, a Bartholin abscess.

Bartholin Gland

- One of two small glands near the vaginal opening that produce lubricating mucus. Blockage of its duct can cause a painless cyst; infection can produce a painful abscess with swelling, redness, and difficulty walking or sitting. Recurrent or atypical masses require gynecologic assessment.

Bartonellosis

- A group of infections caused by Bartonella species. Cat scratch disease (*B. henselae*): regional lymphadenopathy after cat scratch/bite, fever, inoculation papule. Bacillary angiomatosis (*B. henselae*, *B. quintana*): vascular proliferative lesions in immunocompromised (HIV), red papules/nodules on skin, bone, and viscera. Oroya fever (*B.*

Bartter Syndrome

- Inherited autosomal recessive disorder of the thick ascending limb of the loop of Henle causing salt wasting, hypokalemic metabolic alkalosis, and hypercalciuria. Results from mutations in genes encoding the NKCC2 transporter or its regulatory subunits. Presents in childhood with failure to thrive, polyuria, and polydipsia.

Basal Cell Carcinoma

- The most common malignant skin tumor, arising from basal keratinocyte-lineage cells and associated with cumulative ultraviolet exposure and Hedgehog-pathway abnormalities. It commonly presents as a pearly papule or nodule with telangiectasia, ulceration, or a rolled border.

Basal Ganglia

- Deep gray matter nuclei in the cerebral hemispheres for motor control, cognition, and emotion. Key structures: caudate nucleus, putamen, globus pallidus (internal/external segments), subthalamic nucleus, substantia nigra. Direct pathway facilitates movement; indirect inhibits it.

Basal Ganglia Dysfunction

- Abnormal function of the deep cerebral nuclei that regulate movement, cognition, and procedural learning. It may cause tremor, rigidity, bradykinesia, chorea, dystonia, or abnormal involuntary movements, as in Parkinson disease and Huntington disease.

Basal Metabolic Rate (BMR)

- Energy expenditure at rest for basic physiological functions. Accounts for 60-75% of total energy expenditure. Affected by age, sex, body composition, thyroid status, genetics. Measured by indirect calorimetry.

Base

- A substance that accepts protons (H^+) or donates electron pairs (Lewis base). In aqueous solution, bases increase OH^- concentration ($pH > 7$). Strong bases ($NaOH$, KOH) dissociate completely; weak bases (NH_3 , HCO_3^- , protein side chains) partially.

Base Excision Repair

- The repair pathway for small, non-helix-distorting base lesions including oxidized bases (8-oxoguanine), deaminated bases (uracil from cytosine deamina-

tion), alkylated bases, and abasic (AP) sites.

Basement Membrane

- A thin, delicate layer of extracellular matrix that anchors epithelial tissue to the underlying connective tissue.

Basic Life Support (BLS)

- Level of medical care used for victims of life-threatening illnesses or injuries until full medical care can be provided.

Basic Metabolic Panel

- A panel of blood tests measuring serum glucose, sodium, potassium, chloride, bicarbonate, BUN, and creatinine. Also includes calcium in some institutions. Uses: assess hydration status, renal function, acid-base balance, and glucose control.

Basic Reproduction Number (R_0)

- The expected number of secondary infections generated by one typical infectious person in a completely susceptible population, under specified conditions. It is a population and context parameter, not a fixed biological constant.

Basilar artery aneurysm

- Dilation of the basilar artery, most commonly fusiform, often associated with atherosclerosis or hypertension. May compress cranial nerves or brainstem structures causing facial pain, double vision, or hemiparesis. Rupture causes subarachnoid hemorrhage with high morbidity.

Basophil

- A rare (<1 percent of leukocytes) granulocyte with large, basophilic granules containing histamine, heparin, and leukotrienes. Basophils bear high-affinity IgE receptors (Fc-epsilon-RI) and, like mast cells, release histamine and mediators during immediate hypersensitivity (allergic) reactions.

Battered Child Syndrome

- A pattern of repeated, non-accidental physical injury inflicted on a child by a caregiver, formally described by Kempe in 1962. Features include fractures of differing ages, subdural and retinal hemorrhages, inconsistent histories, and delayed presentation; mandatory reporting is required in most jurisdictions.

Bauhin Valve

- The ileocecal valve, a mucosal sphincter at the

junction of the terminal ileum and cecum that helps regulate passage of intestinal contents and limits reflux from the colon.

BCG Vaccine

- Bacille Calmette-Guerin, a live attenuated vaccine derived from *Mycobacterium bovis*, used for TB prevention. Efficacy: 60-80% against disseminated TB and TB meningitis in children; variable efficacy against pulmonary TB. Indications: part of routine childhood immunization in high-TB-burden countries.

Beals Syndrome

- A congenital connective-tissue disorder characterized by congenital contractures, arachnodactyly, kyphoscoliosis, and crumpled ears. It is usually associated with pathogenic variants affecting fibrillin-2.

Beam Hardening Artifact

- CT artifact from dense structures. Reduced by beam hardening correction algorithms.

Beau Lines

- Transverse depressions or grooves across the nail plate caused by temporary interruption of nail-matrix growth. They may follow severe illness, fever, trauma, malnutrition, chemotherapy, or systemic inflammatory disease.

Beau's Lines

- Transverse grooves or depressions across a nail caused by temporary interruption of nail-matrix growth. They may follow severe illness, trauma, nutritional stress, or systemic disease and grow distally with the nail.

Bechterew Disease

- Historical eponym for ankylosing spondylitis (also known as Marie-Strümpell disease), describing the chronic inflammatory arthritis of the axial skeleton first reported by Vladimir Bekhterev (Russian, 1892) and Adolf Strümpell (German, 1884).

Beck Triad

- Classic diagnostic clinical triad of Cardiac Tamponade: (1) Hypotension, (2) Distended neck veins (elevated JVP), and (3) Muffled / distant heart sounds. See Cardiac Tamponade.

Becker Muscular Dystrophy

- An X-linked recessive neuromuscular disorder caused by mutations in the dystrophin gene, simi-

lar to Duchenne muscular dystrophy (DMD), but with in-frame mutations that allow production of some functional dystrophin protein, albeit reduced in quantity or abnormal in size. The result is a milder clinical phenotype compared to DMD.

Becker's nevus

- A benign hamartomatous lesion appearing as a unilateral, hyperpigmented, often hairy patch, typically developing in adolescence on the shoulder or upper trunk of males. It results from a localized androgen-dependent increase in dermal smooth muscle and pigmentation. It is cosmetically concerning but carries no malignant potential.

Beckwith-Wiedemann Syndrome

- An overgrowth disorder associated with 11p15.5 imprinting defects. Features: macrosomia, macroglossia, omphalocele, visceromegaly, and neonatal hypoglycemia. Increased risk of embryonal tumors (Wilms tumor, hepatoblastoma).

Becquerel (Bq)

- The SI unit of radioactive activity, equal to one nuclear transformation per second. Administered activity is commonly reported in megabecquerels or gigabecquerels in nuclear medicine.

Bedbug Bites

- Cutaneous reactions to bites of *Cimex lectularius* (bedbugs), nocturnal blood-feeding insects. Bites are painless but cause erythematous, pruritic papules, often in a linear arrangement (breakfast, lunch, dinner sign) or clusters on exposed skin (face, neck, arms, hands). Delayed hypersensitivity develops over days.

Bedpan

- Bedside toilet for patients.

Bedsore

- Pressure ulcer (decubitus): localized skin/soft tissue injury from prolonged pressure over bony prominences (sacrum, heels, occiput, trochanters), usually in immobile patients.

Bedwetting

- Involuntary urination during sleep after the age at which nighttime bladder control is usually established. It may be primary or secondary to urinary infection, constipation, diabetes, sleep disorders, stress, or other medical conditions and should be

assessed when persistent, new, or accompanied by daytime symptoms.

Beers Criteria

- List of potentially inappropriate medications in elderly. Updated periodically. Guides deprescribing.

Behavioral Rehabilitation

- Behavioral interventions for chronic conditions: pain management, addiction, adjustment.

Behcet Disease

- A chronic, relapsing systemic vasculitis characterized by recurrent oral ulcers, genital ulcers, and ocular inflammation, with a distribution along the ancient Silk Road from the Mediterranean to East Asia.

Belching

- The release of swallowed air or gas from the stomach through the mouth. Frequent belching may occur with aerophagia, gastroesophageal reflux, functional dyspepsia, or carbonated-drink consumption.

Bell Palsy

- Acute, idiopathic, unilateral facial nerve (CN VII) paralysis of lower motor neuron type, with an incidence of approximately 20-30 per 100,000 person-years.

Bell's Palsy

- Idiopathic facial nerve paralysis. Acute onset unilateral facial weakness. May follow viral infection. Treatment: corticosteroids within 72 hours, eye protection. Most recover spontaneously within 3-6 months.

Bence Jones Protein

- Monoclonal immunoglobulin light chains (free kappa or lambda) detectable in urine of patients with monoclonal gammopathies, most characteristically multiple myeloma but also Waldenstrom macroglobulinemia, MGUS, and primary amyloidosis (AL).

Bence-Jones Protein

- A monoclonal immunoglobulin light chain found in urine or blood, classically associated with plasma-cell disorders such as multiple myeloma. Detection contributes to evaluation of monoclonal gammopathy and kidney injury.

Bends

- A manifestation of decompression sickness caused by gas bubbles forming in tissues or blood after a rapid reduction in ambient pressure. It commonly produces deep joint or muscle pain but may also involve the nervous system or lungs.

Beneficence and Non-Maleficence

- Core biomedical ethical principles guiding clinical practice and medical research. Beneficence requires healthcare providers to act in the best interest of the patient, taking positive steps to promote health, prevent harm, and relieve suffering.

Benign Intracranial Hypertension

- An older term for idiopathic intracranial hypertension, a disorder of raised intracranial pressure without an intracranial mass or hydrocephalus. Headache and papilledema are typical, and untreated disease may threaten vision.

Benign Mixed Tumor

- See Pleomorphic Adenoma.

Benign Neoplasm

- Benign neoplasms are proliferations of cells that remain confined to their site of origin, grow by expansion rather than invasion, and do not metastasize to distant sites. They typically grow slowly, pushing aside adjacent normal tissue and often becoming enclosed by a fibrous pseudocapsule formed from compressed connective tissue.

Benign Paroxysmal Positional Vertigo (BPPV)

- A common peripheral vertigo from dislodged otoconia (canalithiasis) within the semicircular canals, most often the posterior canal. Brief episodes of rotational vertigo lasting under one minute provoked by changes in head position.

Benign Positional Vertigo

- Brief episodes of spinning dizziness triggered by changes in head position, usually caused by displaced calcium carbonate crystals in a semicircular canal. It is also called benign paroxysmal positional vertigo or BPPV.

Benign Prostatic Hyperplasia

- Non-malignant enlargement of the prostate gland due to hyperplasia of stromal and glandular elements, affecting >50% of men by age 60 and >90%

by age 85. Presents with lower urinary tract symptoms (LUTS): hesitancy, weak stream, intermittency, straining, incomplete emptying, urgency, frequency, and nocturia.

Benign Prostatic Hyperplasia (BPH)

- Non-cancerous enlargement of prostate gland causing lower urinary tract symptoms (LUTS) in aging men. Prevalence: 50% at age 60, 90% at age 85. Symptoms: hesitancy, weak stream, urgency, frequency, nocturia, incomplete emptying. Diagnosis: DRE, PSA, uroflowmetry, post-void residual.

Benign Prostatic Obstruction (BPO)

- Bladder-outlet obstruction caused by benign prostatic enlargement or increased prostatic smooth-muscle tone. It can produce weak stream, hesitancy, intermittency, straining, incomplete emptying, retention, infection, and upper-tract deterioration.

Benign Tumor

- Non-cancerous growth, slow-growing, well-differentiated, localized, does not invade or metastasize. Examples: lipoma, fibroma, adenoma. May cause problems by compression or hormone secretion.

Bent Spine Syndrome

- A marked, usually reducible forward flexion of the trunk caused by weakness or abnormal contraction of axial extensor muscles. It is also called campocormia and may accompany neuromuscular or movement disorders.

Benzene

- Solvent, industrial chemical. Chronic exposure: aplastic anemia, leukemia (especially AML). Acute: CNS depression, arrhythmias. Prevention: substitution, ventilation, monitoring.

Benzo(a)pyrene

- Polycyclic aromatic hydrocarbon from combustion. Carcinogenic (lung, skin cancer). Found in tobacco smoke, charred meat, air pollution.

Benzodiazepine Use Disorder

- A problematic pattern of benzodiazepine use leading to significant impairment. Benzodiazepines include alprazolam, clonazepam, diazepam, and lorazepam. Dependence can develop rapidly with regular use. Withdrawal can be life-threatening, presenting with anxiety, insomnia, tremor, seizures,

and psychosis.

Benzodiazepine Withdrawal

- Symptoms from cessation or reduction of prolonged benzodiazepine use. Early symptoms include anxiety, insomnia, tremor, and diaphoresis. Severe withdrawal can include seizures, psychosis, and delirium. Withdrawal severity depends on the half-life of the benzodiazepine, duration of use, and dose.

Benzodiazepines

- A class of drugs that enhance the effect of the inhibitory neurotransmitter gamma-aminobutyric acid at the GABA-A receptor, increasing the frequency of chloride channel opening and producing anxiolytic, sedative, anticonvulsant, and muscle relaxant effects.

Bereavement

- Natural grief response to the death of a loved one, marked by sadness, yearning, and preoccupation with the deceased, experienced in waves that diminish over weeks to months with preserved self-esteem. Distinct from major depression by its wave-like quality and preserved self-worth.

Bereavement reaction

- A normal emotional response to the death of a loved one, involving grief that generally follows a variable course without necessarily indicating a mental disorder.

Berger Disease

- Historical name for IgA nephropathy named after Jean Berger who first characterized mesangial IgA deposition in 1968. Characterized by recurrent gross hematuria triggered by mucosal infections with intervals of microscopic hematuria and proteinuria. Generally benign course in most patients but subset progresses to CKD.

Beriberi

- A nutritional deficiency disease caused by inadequate thiamine (vitamin B1), occurring in two forms. Wet beriberi presents with high-output heart failure, peripheral vasodilation, edema, and tachycardia due to impaired cardiac energy metabolism.

Bernard–Soulier syndrome

- An inherited platelet-adhesion disorder caused by deficiency or dysfunction of the glycoprotein Ib-IX-V complex, producing mucocutaneous bleeding and

giant platelets.

Berry Aneurysm

- A small, rounded saccular aneurysm arising at an arterial branch point in the circle of Willis. Rupture is a major cause of nontraumatic subarachnoid hemorrhage.

Berylliosis

- A granulomatous lung disease from beryllium exposure. Chronic berylliosis resembles sarcoidosis. Diagnosis requires beryllium lymphocyte proliferation test.

Beta Blockers

- A class of medications that block the effects of adrenaline on the body's beta receptors. They slow down the heart rate and reduce blood pressure, making them widely used for treating heart conditions, high blood pressure, and anxiety.

Beta-Adrenergic Blocker

- A drug that antagonizes beta-adrenergic receptors, reducing heart rate, myocardial contractility, and renin release. It is used in selected patients with hypertension, angina, arrhythmias, post-myocardial infarction disease, and heart failure; abrupt withdrawal may worsen ischemia.

Beta-Blocker Overdose

- Bradycardia, hypotension, hypoglycemia, bronchospasm (non-selective). Treatment: atropine, glucagon (1-10 mg IV), high-dose insulin euglycemia therapy, calcium, vasopressors, pacing if refractory.

Beta-Blockers

- Beta-adrenergic blocking agents, commonly known as beta-blockers, are topical medications used in the treatment of glaucoma that lower intraocular pressure by reducing the production of aqueous humor by the ciliary body, primarily through blockade of beta-2 adrenergic receptors.

Beta-Lactam Antibiotics

- A large class of antimicrobial agents characterized by a beta-lactam ring in their molecular structure, which is essential for their bactericidal activity.

Beta-lactamase inhibitor combinations

- Combinations of beta-lactam antibiotics with beta-lactamase inhibitors that extend activity against beta-lactamase producing organisms. The major combinations include

amoxicillin-clavulanate, ampicillin-sulbactam, piperacillin-tazobactam, cefoperazone-sulbactam, and ceftolozane-tazobactam.

Beta-Oxidation

- The catabolic process by which fatty acid molecules are broken down in the mitochondria (and peroxisomes) to generate acetyl-CoA, NADH, and FADH₂.

Beta-thalassemia

- An inherited hemoglobin disorder with deficient or absent beta-globin chain synthesis, ranging from mild trait to transfusion-dependent major disease with iron overload, extramedullary hematopoiesis, and skeletal changes.

Bezoar

- See Gastric Bezoar.

BI-RADS

- Breast Imaging-Reporting and Data System. Standardizes mammography reporting. Categories: 0 (incomplete), 1 (negative), 2 (benign), 3 (probably benign), 4 (suspicious), 5 (highly suggestive of malignancy), 6 (known biopsy-proven malignancy).

Bicarbonate

- The principal extracellular buffer anion, formed from carbon dioxide and water and regulated chiefly by the lungs and kidneys. The bicarbonate-carbonic acid system helps maintain arterial pH; serum bicarbonate is commonly used to assess metabolic components of acid-base disorders.

Biceps Brachii Muscle

- Two-headed anterior arm muscle. Long head from supraglenoid tubercle, short head from coracoid process, both inserting on the radial tuberosity. Innervated by musculocutaneous nerve (C5-C6). Most powerful elbow flexor in supination; supinates forearm; weakly flexes shoulder. Distal rupture causes Popeye deformity.

Biceps Reflex

- A tendon reflex produced by tapping the biceps tendon, causing elbow flexion through the musculocutaneous nerve and predominantly C5-C6 spinal segments. It helps assess the corresponding peripheral and central pathways.

Bicipital Groove

- The intertubercular groove of the humerus that transmits the tendon of the long head of the bi-

ceps brachii, with the transverse humeral ligament and adjacent rotator-cuff structures contributing to tendon stability.

Bicipital Tuberosity

- The radial tuberosity, a roughened prominence on the proximal radius where the biceps brachii tendon inserts and produces flexion and supination of the forearm.

Bicuspid Aortic Valve

- The most common congenital cardiac anomaly, present in 1-2% of the population. The aortic valve has two leaflets instead of three, predisposing to aortic stenosis (usually by age 50-60), aortic regurgitation, and infective endocarditis. Associated with aortopathy (aortic root dilation and dissection risk) and coarctation of the aorta.

Bicuspid Valve

- A heart valve with two leaflets rather than the usual three. The term most often refers to congenital bicuspid aortic valve, which can cause aortic stenosis, regurgitation, infective endocarditis, or ascending-aortic disease.

Bietti's Crystalline Dystrophy

- A rare autosomal recessive chorioretinal dystrophy characterized by tiny crystalline deposits in the retina and cornea with progressive chorioretinal atrophy, presenting with night blindness and progressive visual field loss beginning in early adulthood.

Biguanide

- A class of oral antidiabetic drugs derived from guanidine, with metformin being the only biguanide in current clinical use (phenformin and buformin were withdrawn due to lactic acidosis risk). Metformin is the first-line pharmacotherapy for type 2 diabetes mellitus.

Bilateral peritonsillar abscesses

- Suppurative infection in the peritonsillar space on both sides, a rare complication of acute tonsillitis presenting with severe sore throat, dysphagia, trismus, and muffled hot potato voice. Bilateral involvement risks acute upper airway obstruction and requires urgent needle aspiration or incision and drainage.

Bile Acids

- Substances produced in the liver that aid diges-

tion of fats and absorption of fat-soluble vitamins. In pregnancy, elevated bile acids (10 mol/L) indicate intrahepatic cholestasis of pregnancy (obstetric cholestasis), a liver disorder characterized by pruritus (itching, typically palms and soles) and elevated serum bile

Bile Duct

- A channel that carries bile from the liver and gallbladder to the duodenum. The right and left hepatic ducts unite as the common hepatic duct; union with the cystic duct forms the common bile duct, which usually joins the pancreatic duct before entering the duodenum at the major duodenal papilla.

Bile Reflux

- Reflux of duodenal contents (bile acids, lysolecithin, pancreatic enzymes) into the stomach and esophagus, caused by incompetence of the pyloric sphincter, prior gastric surgery (Billroth II, Roux-en-Y, vagotomy with pyloroplasty), cholecystectomy, or functional gastroparesis.

Bile Salt Emulsification

- Bile salts (derived from cholesterol in the liver) are amphipathic molecules with hydrophobic and hydrophilic regions that act as biological detergents, emulsifying dietary lipids into smaller droplets (micelles). This increases the surface area of lipid substrate available for pancreatic lipase action.

Bile Salt Malabsorption

- Condition characterized by impaired reabsorption of bile acids in the terminal ileum leading to excessive bile acids entering the colon where they stimulate water and electrolyte secretion causing chronic watery diarrhea.

Biliary Atresia

- A progressive, obliterative cholangiopathy of the extrahepatic bile ducts in neonates, leading to biliary obstruction, cholestasis, and ultimately biliary cirrhosis if untreated.

Biliary Cirrhosis

- See Primary Biliary Cholangitis.

Biliary Colic

- Episodic right-upper-quadrant or epigastric pain caused by transient obstruction of the cystic duct, usually by a gallstone. Pain is typically steady, may follow meals, and resolves when the obstruction

passes; fever and persistent inflammatory findings suggest a complication.

Biliary Dyskinesia

- Functional gallbladder disorder with impaired gallbladder ejection fraction in the absence of gallstones. Presents with biliary colic-like right upper quadrant pain. Diagnosed by CCK-HIDA scan showing ejection fraction less than thirty-five percent. The diagnosis requires exclusion of other causes of biliary-type pain.

Biliary Fever

- Fever associated with infection or inflammation of the biliary tract, classically acute ascending cholangitis. It may occur with right-upper-quadrant pain and jaundice and can progress rapidly to sepsis.

Biliary Stent

- Tube placed in bile duct to relieve obstruction. Types: plastic (short-term, cheaper, higher occlusion rate), self-expanding metal (longer-term, patent longer). Indications: malignant obstruction (pancreatic cancer, cholangiocarcinoma), benign stricture. Complications: migration, occlusion, cholangitis, pancreatitis.

Biliary Tract Injuries

- Bile duct injury (BDI) occurs in 0.3-0.5% of laparoscopic cholecystectomies. Strasberg classification: A (cystic duct leak/minor hepatic duct leak), B (occlusion of aberrant right hepatic duct), C (transection of aberrant right hepatic duct without ligation), D (lateral injury to major duct), E (major duct injury: E1-E5 based on level).

Bilious Vomiting

- Vomiting of green/yellow, bilious material in the newborn or infant, defined as emerald green or yellow vomitus that contains bile.

Bilirubin

- A yellow pigment formed mainly from heme breakdown. Unconjugated bilirubin is transported bound to albumin to the liver, where it is conjugated with glucuronic acid and excreted in bile; accumulation causes jaundice.

Bilirubin Encephalopathy

- Neurologic injury caused by deposition of unconjugated bilirubin in the brain, especially the basal ganglia. Severe neonatal hyperbilirubinemia can

produce acute lethargy, abnormal tone, feeding difficulty, seizures, and permanent kernicterus.

Bilirubin Metabolism

- Bilirubin is the breakdown product of heme from senescent red blood cells, metabolized primarily by the liver through a series of enzymatic steps. Heme is converted to biliverdin by heme oxygenase, then to unconjugated bilirubin by biliverdin reductase. Unconjugated bilirubin is lipophilic and transported bound to albumin to the liver.

Bilirubinuria

- The presence of conjugated bilirubin in urine. It indicates that water-soluble bilirubin has entered the circulation, often because of hepatocellular disease or biliary obstruction.

Binet-Simon Scale

- An early developmental assessment used to estimate intellectual functioning in children by comparing performance with age-based tasks. Modern evaluation uses standardized, validated developmental and cognitive instruments.

Binge Eating Disorder

- Recurrent episodes of eating large quantities of food with a sense of lack of control, without regular compensatory behavior. Associated with obesity, depression, and anxiety. Diagnosis by DSM-5 criteria. Treatment: cognitive behavioral therapy, interpersonal therapy, and SSRIs. Weight management with dietary counseling and exercise.

Binocular visual loss

- Reduction in vision affecting both eyes simultaneously, typically resulting from bilateral optic neuropathy, chiasmal lesions, or diffuse bilateral retinal disease rather than monocular ocular pathology. Evaluation requires neuroimaging and visual field testing to localise the lesion along the visual pathway.

Bioabsorbable Stent (Vascular)

- See Bioabsorbable Stent.

Bioavailability

- The fraction of an administered drug that reaches the systemic circulation in its unchanged active form. It is expressed as a percentage and is determined by the route of administration and the extent of first-pass metabolism. For intravenous administration,

bioavailability is 100 percent by definition.

Bioburden

- The number and type of viable microorganisms contaminating a surface, medical device, tissue, or pharmaceutical product; a measure of contamination. Bioburden testing assesses the level of microbial contamination (total viable count) before sterilization or in quality control.

Biochemical Assessment

- Blood tests: albumin, prealbumin, transferrin, electrolytes, vitamins, minerals, glucose, lipids. Reflect nutritional status and metabolic function.

Biocompatibility

- Ability of material to function without adverse response. ISO 10993 standards for biocompatibility testing.

Biodegradable Implant

- An implant designed to perform a temporary function and then degrade into products that can be metabolized or eliminated by the body. Degradation rate, local tissue response, and mechanical strength must be matched to the intended clinical use.

Biodistribution

- The time-dependent distribution of a radiopharmaceutical among organs, tissues, blood, and excretory pathways. It determines image quality, target-to-background contrast, therapeutic delivery, and radiation exposure to normal organs.

Bioequivalence

- The property of two drug products (brand-name and generic) that have the same rate and extent of absorption when administered at the same molar dose under similar conditions, resulting in comparable bioavailability and therapeutic effect.

Bioethics

- Interdisciplinary study of the ethical, legal, philosophical, and social implications of advancements in medicine, biology, healthcare technology, and biotechnology.

Bioethics (Geriatric)

- Ethical considerations in elderly care: autonomy vs. paternalism, informed consent capacity, end-of-life decisions, resource allocation, elder abuse. Principles: respect for autonomy, beneficence, non-maleficence, justice. Advance directives: living will,

durable power of attorney for healthcare.

Biofilm

- An organized community of microorganisms attached to a surface and embedded in a self-produced extracellular polymeric matrix (polysaccharides, proteins, DNA).

Biofilm-associated infections

- Infections caused by bacteria growing in biofilms on medical devices or tissues. Biofilms are structured communities of bacteria embedded in extracellular polymeric substances. Bacteria in biofilms are more resistant to antibiotics and host immune responses.

Biologic Therapy

- Treatment with biologic agents targeting specific immune pathways, used in autoimmune and inflammatory disease.

Biologic Therapy in Dermatology

- Treatment with targeted biological agents, usually monoclonal antibodies or receptor proteins, that inhibit selected immune pathways. Biologics are used for diseases such as psoriasis, atopic dermatitis, hidradenitis suppurativa, and urticaria; infection risk, vaccination status, screening, and monitoring must be considered.

Biologic Variation

- Natural fluctuation in a measurable analyte within an individual over time and variation between individuals. It should be considered when interpreting serial results and deciding whether a change is clinically meaningful.

Biological Half-Life

- Time required for the human body to eliminate half the administered quantity of a drug or radiopharmaceutical through metabolic and excretory processes. Different from physical half-life (radioactive decay) and effective half-life (combined biological and physical elimination). Formula: $1/T\%_{\text{effective}} = 1/T\%_{\text{physical}} + 1/T\%_{\text{biological}}$.

Biomarker

- Measurable indicator of disease. Cancer biomarkers: diagnostic, prognostic, predictive.

Biomarker Testing

- Analysis of DNA/RNA for mutations, fusions, expression. See molecular testing.

Biomechanics

- Study of movement mechanics. Gait analysis, joint forces, muscle activation. Applied to rehabilitation device design.

Biomolecule

- Any organic molecule produced by living organisms, essential for structure, function, and regulation of biological processes.

Biomonitoring

- Measurement of substances in body. Examples: blood glucose, drug levels, toxicants.

Biophysical Profile

- A non-stress test combined with ultrasound assessment of five fetal parameters, each scored 0 or 2 for a total of 0-10. Parameters include non-stress test (fetal heart rate reactivity), fetal breathing movements, fetal tone, fetal movement, and amniotic fluid volume.

Biophysical Profile (BPP)

- Antepartum fetal assessment combining ultrasound and NST. Components (2 points each): NST reactivity (reactive), fetal breathing movements (1 episode of 30 seconds in 30 minutes), fetal movement (3 discrete movements in 30 minutes), fetal tone (1 episode of extension/flexion), and amniotic fluid volume (AFI 5 cm).

Biopsy

- Removal of tissue for histological examination to diagnose disease. Types: needle biopsy (fine needle aspiration, core needle), excisional, incisional, punch, endoscopic. Sites: breast, lung, liver, lymph node, bone marrow. Complications: bleeding, infection, seeding (rare).

Biopsy Needle

- A needle designed to obtain cells or tissue for cytologic or histopathologic examination. Fine-needle aspiration retrieves cells, whereas core biopsy obtains a tissue cylinder that preserves architecture.

Biosensor

- A device that combines a biological recognition element with a physical or chemical transducer to detect and measure an analyte. Medical examples include sensors for glucose, pathogens, hormones, and cardiac biomarkers.

Biosimilar

- A biologic medicine highly similar to an approved reference biologic, with no clinically meaningful differences in safety, purity, or potency for its approved uses. Because biologics are complex, biosimilarity is established through a comparative evidence program rather than simple chemical identity.

Biot Breathing

- An irregular breathing pattern with random periods of normal breathing interspersed with apneic episodes, seen in brainstem lesions. It is less organized than Cheyne-Stokes breathing and indicates medullary damage.

Biotelemetry

- The remote measurement and transmission of physiologic data from a person or animal. Clinical applications include ambulatory cardiac rhythm monitoring, temperature monitoring, and intensive-care observation.

Bioterrorism Agents

- Category A: anthrax (*Bacillus anthracis*), smallpox (variola), botulism (*Clostridium botulinum*), plague (*Yersinia pestis*), tularemia (*Francisella tularensis*), viral hemorrhagic fevers (Ebola, Marburg, Lassa, Machupo). Anthrax: inhalational (mediastinal widening, hemorrhagic mediastinitis), cutaneous (eschar), GI.

Biotinidase Deficiency

- An autosomal recessive inborn error of biotin recycling caused by BTD gene mutations, resulting in inability to cleave biotin from biocytin (proteolytic degradation product of biotin-dependent carboxylases).

Biotransformation

- The process by which drugs and other xenobiotics are chemically modified by metabolic enzymes, primarily in the liver, to facilitate elimination from the body. Biotransformation occurs in two phases.

Biotrauma

- Inflammatory response to ventilator-induced lung injury with systemic cytokine release. Can lead to multi-organ dysfunction. Minimized by lung-protective ventilation strategies.

BiPAP

- Bilevel Positive Airway Pressure, a form of non-

invasive ventilation delivering two levels of pressure: IPAP (inspiratory) for ventilation and EPAP (expiratory) for alveolar recruitment. It is particularly effective for COPD exacerbations with hypercapnic respiratory failure. It reduces work of breathing and improves gas exchange.

BiPAP Machine

- Bi-level positive airway pressure. Two pressure settings.

Biparietal Diameter (BPD)

- The transverse diameter of the fetal skull measured at the level of the thalami and cavum septum pellucidum. Used for dating (14-26 weeks, ± 7 -10 days) and assessing growth. Normal growth: 2-4 mm per week. BPD is less reliable after 26 weeks and in abnormal fetal head shapes (dolichocephaly, brachycephaly).

Bipolar Disorder

- A mood disorder characterized by episodes of mania/hypomania and depression. Bipolar I: at least one manic episode (elevated mood, grandiosity, decreased need for sleep, pressured speech, increased goal-directed activity, risky behavior) lasting 1 week or requiring hospitalization.

Bipolar I Disorder

- A mood disorder defined by at least one manic episode lasting at least seven days or requiring hospitalization. A manic episode is a distinct period of abnormally elevated, expansive, or irritable mood and increased goal-directed activity or energy.

Bipolar II Disorder

- A mood disorder characterized by at least one hypomanic episode and at least one major depressive episode, with no history of a full manic episode. The depressive episodes in bipolar II tend to be more frequent and longer-lasting than in bipolar I, leading to greater cumulative disability.

Bird Fancier's Lung

- HP from avian proteins in bird droppings or feathers. Common among pigeon breeders. Chronic exposure leads to progressive fibrosis.

Bird Flu

- Influenza caused by avian-adapted influenza A viruses. Human infection is uncommon but may cause severe respiratory disease, and public-health

assessment depends on exposure history, testing, and infection-control measures.

Birt-Hogg-Dube Syndrome

- Autosomal dominant disorder caused by mutations in the FLCN gene (17p11.2) encoding folliculin.

Birth Control

- Contraception; see Contraception. Methods to prevent unintended pregnancy: hormonal contraceptives (pill, patch, ring, injection, implant), intrauterine devices (copper, levonorgestrel), barrier methods (condoms, diaphragm), emergency contraception, fertility awareness, and sterilization.

Birthmarks

- Congenital or early-appearing marks on the skin, including vascular lesions such as hemangiomas and port-wine stains and pigmented lesions such as nevi and café-au-lait macules. Most are benign, but changes in appearance warrant assessment.

Bisacodyl

- A stimulant laxative (diphenylmethane derivative) that stimulates colonic motility and enhances water/electrolyte secretion, producing a bowel movement within 6-12 hours (oral) or 15-60 minutes (suppository). Used for constipation, bowel preparation, and evacuation before procedures.

Bishop Score

- A pre-induction cervical scoring system that predicts the success of labor induction. Five components: cervical dilatation (0-3), effacement (0-3), station (-3 to +2, scored 0-3), consistency (firm, medium, soft; scored 0-2), and position (posterior, mid, anterior; scored 0-2). Total score: 0-13.

Bismuth Salicylate

- A compound combining bismuth subsalicylate with salicylic acid that provides antimicrobial and antisecretory effects, used for diarrhea and gastric upset but capable of causing salicylate toxicity in overdose.

Bismuth Subsalicylate

- An antidiarrheal and anti-ulcer agent (e.g., Pepto-Bismol) with antimicrobial, anti-inflammatory, and coating actions. Used for traveler's diarrhea (prophylaxis and treatment), dyspepsia, *Helicobacter pylori* eradication (quadruple therapy), and symptoms of indigestion.

Benzocaine

- A topical local anesthetic used for short-term relief of pain or itching on the skin or in the mouth and throat. It can rarely cause methemoglobinemia, especially in young children or after excessive use; seek urgent care for blue or gray skin, shortness of breath, fatigue, or rapid heartbeat. Do not use oral products in children younger than two years unless specifically directed by a clinician.

Bisoprolol

- A beta-1 selective blocker used to treat high blood pressure, angina, and some forms of heart failure. It slows the heart rate and reduces cardiac workload. Dizziness, fatigue, slow pulse, and worsening breathing symptoms may occur; it should generally be tapered rather than stopped abruptly.

Bisphenol A (BPA)

- Endocrine disruptor in plastics. Estrogenic effects, associated with reproductive disorders, obesity, diabetes. Avoid BPA-containing products, especially for children.

Bitemporal Hemianopia

- Loss of vision in the temporal (lateral) half of both visual fields due to lesions that cross the optic chiasm, damaging the decussating fibers from the nasal retina of each eye (which carry temporal field information).

Bitewing

- A dental radiograph that records the crowns of the upper and lower teeth and the adjacent alveolar bone in one image. It is particularly useful for detecting interproximal caries and assessing crestal bone levels.

Bitewing Radiograph

- An intraoral radiograph showing the crowns of the maxillary and mandibular posterior teeth and the alveolar crest. It is particularly useful for detecting interproximal caries and assessing crestal bone levels.

Bitot's Spots

- Foamy, whitish, triangular patches on the conjunctiva, usually near the temporal limbus, associated classically with vitamin A deficiency. They may accompany night blindness and xerosis of the conjunctiva or cornea.

BK Virus Nephropathy

- Polyomavirus infection in transplant recipients. Viral cytopathic changes on biopsy. Reduce immunosuppression as primary treatment. Can lead to graft loss. Monitor viral loads in blood for early detection.

Black Hairy Tongue

- A benign elongation and dark discoloration of the filiform papillae on the dorsal tongue caused by impaired desquamation and trapping of debris or pigment. Predisposing factors include smoking, poor oral hygiene, dry mouth, and some exposures.

Black Lung

- Coal workers' pneumoconiosis caused by prolonged inhalation of coal-mine dust. It ranges from simple anthracosis to progressive massive fibrosis and may cause cough, breathlessness, and permanent lung impairment.

Black Stain

- Extrinsic staining along gumline. Associated with poor hygiene. Polishing removes.

Black Triangle

- Triangular space between teeth at gumline. Caused by papilla loss, tooth shape, bone level. Treatment: orthodontics, restorative, papilla reconstruction.

Blackheads

- Open comedones formed when a hair follicle becomes blocked with sebum and keratin and the surface material oxidizes. They are a common non-inflammatory lesion of acne, especially on the face, chest, and back.

Bladder Cancer

- Urothelial carcinoma. Risk factors: smoking, occupational exposures.

Bladder Carcinoma

- Malignant neoplasm originating from the urothelium lining the urinary bladder, with transition cell carcinoma (urothelial carcinoma) accounting for over 90 percent of cases. Predominant risk factor: cigarette smoking (responsible for 50% of cases) and occupational exposure to aromatic amines or aniline dyes.

Bladder diverticula

- Outpouchings of the bladder mucosa through the

muscular wall, often acquired from chronic bladder outlet obstruction or congenital in origin. Acquired diverticula are associated with benign prostatic hyperplasia and may cause urinary stasis, recurrent infections, and stone formation.

Bladder Diverticulum

- Outpouching of bladder wall. Congenital or acquired (obstruction). Asymptomatic or causes urinary stasis, stones, infection. Treatment: address obstruction, diverticulectomy if symptomatic.

Bladder Exstrophy

- A rare congenital abdominal wall defect where the bladder is exposed on the outside of the body due to failure of the lower abdominal wall and bladder to close during embryonic development. Part of the exstrophy-epispadias complex. Associated with pubic bone diastasis, external genital anomalies, and potential bladder dysfunction.

Bladder Fistula

- An abnormal connection between the bladder and another organ or the skin, most often the vagina, bowel, or uterus. It may cause continuous urine leakage, recurrent infections, gas or stool in the urine, or unusual vaginal wetness. Treatment depends on the cause and may require catheter drainage and surgical repair.

Bladder Health and Disorders

- Bladder health involves the proper storage and voluntary release of urine through coordinated function of the detrusor muscle and sphincters. Common disorders include urinary tract infections, overactive bladder, stress incontinence, and interstitial cystitis.

Bladder Outlet Obstruction

- Blockage of urine flow at the bladder neck or urethra, commonly caused by benign prostatic enlargement, urethral stricture, stones, tumors, or pelvic-organ prolapse. It may cause weak stream, hesitancy, retention, hydronephrosis, infection, and renal impairment.

Bladder Pain Syndrome

- Chronic bladder-related pain, pressure, or discomfort associated with urinary symptoms in the absence of infection or another identifiable cause. It is a diagnosis of exclusion and is managed with multimodal, patient-centered treatment.

Bladder Perforation

- A tear or hole in the bladder wall caused by trauma, pelvic surgery, instrumentation, severe inflammation, or obstruction. Urine may leak into surrounding tissues, causing abdominal pain, distention, blood in the urine, or infection. It requires urgent medical assessment and may need catheter drainage or surgical repair.

Bladder Problems

- A range of conditions affecting urine storage and voiding, including incontinence, overactive bladder, urinary retention, and recurrent infections. Causes may include neurological damage, pelvic floor weakness, prostate enlargement, or bladder outlet obstruction.

Bladder Spasm

- An involuntary contraction of the bladder muscle that can cause sudden urgency, cramping, urine leakage, or pain. It may occur with infection, catheter use, bladder irritation, neurological conditions, or after surgery. Treatment addresses the cause and may include bladder training or medicines.

Bladder Retraining

- See urology.

Bladder Stone

- Calculus forming in bladder. Causes: obstruction, infection, foreign body, stasis. Symptoms: dysuria, hematuria, interrupted stream. Treatment: removal, address underlying cause.

Bladder Stones

- Calculi that form in or migrate into the urinary bladder. They may cause lower abdominal pain, dysuria, urinary frequency, hematuria, recurrent infection, or intermittent urinary obstruction.

Blalock-Taussig Operation

- A surgical systemic-to-pulmonary shunt that increases pulmonary blood flow in selected cyanotic congenital heart disease. The modern modified procedure commonly uses a prosthetic conduit rather than direct sacrifice of a subclavian artery.

Blast Crisis

- Aggressive, terminal phase of myeloid (chiefly CML) or, less commonly, lymphoid malignancies in which immature blasts dominate the marrow and peripheral blood, behaving like acute leukemia.

CML blast crisis is defined by 20% blasts in peripheral blood or bone marrow or by the presence of extramedullary blastic infiltrates (chloromas).

Blast Injury

- Tissue injury caused by an explosion, classified as primary injury from the pressure wave, secondary injury from projectiles, tertiary injury from displacement or impact, and quaternary injury from burns, inhalation, toxins, or crush. Multiple mechanisms may coexist.

Blastic plasmacytoid dendritic-cell neoplasm

- A rare, aggressive hematologic malignancy derived from plasmacytoid dendritic cells. It often involves skin, bone marrow, lymph nodes, and central nervous system, with a poor prognosis.

Blastocyst

- An early stage in embryonic development, consisting of a hollow sphere of cells that implants into the uterine wall.

Blastoma

- Generic descriptor for embryonal (primitive) malignant tumors arising from precursor ("blast") cells that fail to differentiate normally.

Blastomyces dermatitidis

- A dimorphic fungus that causes blastomycosis, a systemic pyogranulomatous infection acquired by inhalation of conidia from soil in endemic regions such as the Ohio and Mississippi River valleys. Cutaneous manifestations include verrucous or ulcerated plaques, and pulmonary involvement is common.

Blastomycosis

- A fungal infection caused by *Blastomyces dermatitidis*, endemic to the Ohio and Mississippi River valleys and Great Lakes region. Inhalation of conidia causes pulmonary infection. Pulmonary: acute pneumonia (often misdiagnosed as bacterial), chronic pneumonia with nodules and cavitation.

Bleeding

- The escape of blood from the vascular system, which may be external (visible hemorrhage from a wound, orifice, or natural opening) or internal (bleeding into a body cavity, organ, or tissue).

Bleeding Disorders

- A heterogeneous group of conditions characterized

by abnormal hemostasis, resulting from defects in blood vessels, platelets, or coagulation factors.

Bleeding Esophageal Varices

- Life-threatening hemorrhage from dilated submucosal veins in the lower esophagus, occurring as a complication of portal hypertension, most commonly from cirrhosis. Varices form when the portal venous pressure gradient exceeds 10 mmHg (the threshold for variceal development).

Bleeding into a thyroid nodule

- Hemorrhage within a thyroid nodule, presenting with sudden onset of neck pain and rapid enlargement of a pre-existing or new thyroid mass. It may mimic thyroid malignancy clinically and on imaging. Ultrasound shows a complex cystic lesion with internal echoes. Fine needle aspiration confirms the benign nature of the hemorrhagic cyst.

Blennorrhea

- A mucopurulent discharge from a mucous membrane, especially the urethra or conjunctiva. The finding requires clinical assessment for infectious and noninfectious causes rather than treatment based on the symptom alone.

Blepharitis

- Chronic inflammation of the eyelid margins, one of the most common ophthalmic conditions, classified into anterior blepharitis (affecting the eyelash follicle and base) and posterior blepharitis (affecting the meibomian glands).

Blepharoconjunctivitis

- Inflammation involving the eyelid margins and conjunctiva. It may cause crusting, itching, redness, tearing, irritation, and recurrent symptoms associated with infection, allergy, or eyelid-gland dysfunction.

Blepharoplasty

- Surgical reconstruction or reshaping of the eyelids, often involving removal or repositioning of excess skin, muscle, or orbital fat. It may be functional, cosmetic, or reconstructive.

Blepharospasm

- Involuntary, bilateral tonic-clonic spasms of the eyelid muscles causing forced eye closure that can be functionally disabling. Most often essential (idiopathic) but can follow ocular irritation or brainstem

lesions; first-line treatment is local botulinum toxin injection.

Blind Loop Syndrome

- Malabsorption syndrome due to stasis and bacterial overgrowth (SIBO) in a non-propulsive ("blind") segment of small bowel, classically caused by surgical bypass loops (Billroth II, end-to-side anastomosis), strictures (Crohn disease, radiation, adhesions), diverticulosis, or impaired motility (diabetes, scleroderma).

Blinding

- Concealing group assignment. Single-blind: participants. Double-blind: participants and investigators. Triple-blind: participants, investigators, analysts.

Blindness

- Severe or complete loss of vision in one or both eyes, caused by disease or injury affecting the cornea, lens, retina, optic nerve, visual pathways, or visual cortex. It may be congenital or acquired, sudden or progressive, and reversible or permanent; sudden visual loss is an ophthalmic emergency.

Blood Alcohol Concentration

- The concentration of ethanol in blood, commonly expressed as mass per volume or as a percentage. Interpretation depends on sampling, timing, metabolism, tolerance, co-ingestants, and local standards; clinical impairment cannot be inferred from a single value without context.

Blood Alcohol Concentration (BAC)

- Measure of ethanol in blood. Legal limit for driving: 0.08% (varies by jurisdiction). Metabolism: zero-order (15-20 mg/dL/hour), liver breakdown. Units: g/dL or mg/dL.

Blood Alcohol Level

- Measurement of ethanol concentration in blood. Legal limit for driving: 0.08 g/dL (80 mg/dL) in most US states. Impaired: 0.04-0.08 g/dL. Severe intoxication: 0.15-0.30 g/dL. Potentially fatal: >0.30 g/dL. With chronic alcoholism: marked tolerance (may appear functional at 0.30+). Elimination rate: 15-25 mg/dL/hour.

Blood Clot

- A mass of coagulated blood within a blood vessel or body cavity. A clot may be protective when it stops bleeding, but an abnormal thrombus can obstruct

blood flow or embolize to the lungs, brain, heart, or other organs.

Blood Clots

- Gel-like masses formed when blood coagulates inside a blood vessel or injured tissue. Clots may stop bleeding but can cause deep vein thrombosis, pulmonary embolism, stroke, heart attack, or arterial ischemia when they obstruct circulation.

Blood Coagulation

- The cascade of enzymatic reactions converting circulating fibrinogen into an insoluble fibrin clot.

Blood Coagulation Disorder

- A disorder in which clot formation is deficient, excessive, or inappropriately regulated, causing abnormal bleeding or thrombosis. Causes include inherited factor deficiencies, platelet disorders, liver disease, vitamin deficiencies, immune conditions, and acquired systemic disease.

Blood Culture

- Inoculation of blood into culture media to detect bacteremia or fungemia. Two sets (two bottles each: aerobic + anaerobic) from separate venipuncture sites, 20-30 mL total. Indications: fever of unknown origin, suspected sepsis, infective endocarditis, and immunocompromised with fever.

Blood Culture Contamination

- Introduction of skin or environmental microorganisms into a blood-culture bottle during collection or handling rather than true bloodstream infection. It can lead to unnecessary antimicrobial treatment and should be assessed using the organism, number of positive sets, time to positivity, and clinical findings.

Blood Film (Peripheral Blood Smear)

- A stained microscopic preparation of peripheral blood used to assess erythrocyte, leukocyte, and platelet morphology and to detect abnormal cells, parasites, blasts, and red-cell fragments.

Blood Flow

- The volume of blood passing through a vessel, organ, or the systemic circulation per unit time, typically expressed in mL/min. Governed by the hemodynamic equivalent of Ohm's law: flow = pressure gradient / resistance. Total cardiac output (5 L/min at rest) equals systemic blood flow.

Blood Gas (Venous)

- Sampling of venous blood (typically from a peripheral vein or central line) for pH, PvCO₂, HCO₃, and base excess. Less painful than ABG, avoids arterial puncture risks. Limitations: no PaO₂ or SpO₂ data; peripheral venous gases in shock may not reflect arterial pH. VBG pH is typically 0.03-0.05 lower than ABG; PvCO₂ is 3-8 mmHg higher.

Blood Glucose Monitoring

- Blood sugar monitoring involves measuring glucose levels in the blood to manage diabetes, typically using a glucometer with finger-stick testing or continuous glucose monitors that track interstitial glucose in real time.

Blood Group

- The classification of blood based on the presence or absence of specific antigens (agglutinogens) on the surface of red blood cells. The ABO system has four groups: A (A antigen), B (B antigen), AB (both A and B), and O (neither A nor B). The Rh system: RhD-positive (85% of population) or RhD-negative (15%).

Blood Loss

- The volume of blood lost during surgery or trauma, estimated by weighing swabs, suction contents, and hemodynamic monitoring, guiding fluid and transfusion management.

Blood Pressure

- The force exerted by blood against arterial walls, measured in millimeters of mercury (mmHg). Systolic pressure (peak during ventricular ejection, normally approximately 120 mmHg) and diastolic pressure (minimum during ventricular relaxation, normally approximately 80 mmHg).

Blood Products

- Blood products used in anesthesia include packed red blood cells (increase oxygen-carrying capacity, each unit raises hemoglobin by approximately 1 g/dL), fresh frozen plasma (contains all coagulation factors, used for warfarin reversal, massive transfusion, and coagulopathy), platelets (prevent or treat thrombocytopenia, 1

Blood Smear

- Peripheral blood film: a thin blood smear on a slide stained (Wright/Giemsa) and examined micro-

scopically for cell morphology, aiding diagnosis.

Blood Spatter Analysis

- The forensic interpretation of bloodstain patterns to reconstruct events surrounding a violent incident. Types: passive (drops, pools, flows), transfer (contact patterns), and impact/expirated spatter.

Blood Sugar Test

- Measurement of blood glucose to detect or monitor diabetes and other disorders of glucose regulation. Testing may be performed fasting, randomly, or after a glucose load, using a laboratory sample or a point-of-care meter.

Blood Transfusion

- The administration of whole blood or blood components (packed red blood cells, platelets, fresh frozen plasma, cryoprecipitate) to restore or maintain adequate circulating blood volume and oxygen-carrying capacity.

Blood Urea Nitrogen

- BUN: serum urea, the end product of protein catabolism, filtered and partially reabsorbed by the tubules. Used with creatinine to assess renal function and volume status.

Blood Viscosity

- The internal friction of blood that resists flow, determined primarily by hematocrit (red cell concentration), with smaller contributions from plasma proteins and temperature. Normal whole-blood viscosity is approximately 3-4 times that of water.

Blood-Borne Disease

- Disease transmitted via blood. Examples: HIV, hepatitis B, hepatitis C.

Blood-Brain Barrier

- The blood-brain barrier is a highly selective semipermeable membrane that separates the circulating blood from the brain extracellular fluid, consisting of endothelial cells connected by tight junctions, astrocyte foot processes, and a basement membrane.

Bloom Syndrome

- Rare autosomal recessive chromosomal instability syndrome caused by biallelic mutations in BLM (15q26.1) encoding a RecQ helicase.

Blount Disease

- A growth disorder of the proximal tibia causing progressive bowing of the legs, usually involving the medial tibial physis. It may occur in infants or adolescents and can lead to gait abnormality, limb-length discrepancy, and early knee degeneration.

Blue Baby

- Generic term for an infant with cyanosis (bluish skin, lips, nail beds) from arterial oxygen desaturation.

Blue Bloater

- A clinical phenotype of chronic bronchitis in COPD. Patients present with chronic cough, sputum production, cyanosis, right heart failure, and airway-centric inflammation. They tend to be younger, have more hypercapnia, and less emphysema on imaging.

Blunt Abdominal Trauma

- Trauma to the abdomen without penetrating injury. Mechanisms: motor vehicle collisions (most common), falls, assaults. Commonly injured organs: spleen (most common, left upper quadrant), liver (right upper quadrant), kidneys (flank), bowel (seatbelt sign, delayed perforation), and pancreas (epigastric).

Blunt Force Trauma

- Injury produced when a blunt object strikes the body, or the body strikes a blunt surface, without a sharp edge penetrating the skin. Patterns include abrasions, contusions, lacerations with tissue bridging, and fractures; patterned injuries may reproduce the impacting object.

BMI (Body Mass Index)

- Weight in kg divided by height in meters squared. Screening tool for weight categories: underweight (<18.5), normal (18.5-24.9), overweight (25-29.9), obese (≥ 30). Limitations: does not distinguish muscle from fat, may misclassify athletes, elderly, certain ethnic groups.

BNP

- B-type natriuretic peptide. Marker of heart failure. Released from ventricles in response to stretch.

Bochdalek Hernia

- A congenital diaphragmatic hernia (CDH) occurring through the posterolateral foramen of Bochdalek, resulting from failure of the pleuroperi-

toneal membrane to close by the 8th week of gestation. The left side is affected in 85% of cases (the right pleuroperitoneal canal closes earlier).

Body Dysmorphic Disorder

- Preoccupation with one or more perceived defects or flaws in physical appearance that are not observable or appear slight to others. The individual performs repetitive behaviors in response to appearance concerns such as mirror checking, excessive grooming, skin picking, reassurance seeking, and comparing.

Body Identification

- The process of establishing the identity of a deceased person, essential in forensic and mass-disaster contexts.

Body Lice

- Infestation of clothing and skin by the human body louse, causing intense itching, excoriations, and sometimes secondary bacterial infection. Treatment requires hygiene, laundering or heat treatment of clothing and bedding, and appropriate antiparasitic measures.

Body Plethysmography

- Technique for measuring lung volumes (TLC, RV, FRC) using a sealed chamber. Gold standard for TLC measurement. Useful for diagnosing restriction and hyperinflation.

Body Ringworm

- Dermatophyte infection of the glabrous skin, also called tinea corporis. It typically produces an expanding annular scaly lesion with a more active border and central clearing.

Body Temperature

- The balance between heat production (metabolism, muscle activity) and heat loss (radiation, conduction, convection, evaporation). See Temperature (Body Temperature).

Boeck's Sarcoid

- An older eponym for sarcoidosis, a multisystem granulomatous disease characterized by noncaseating granulomas and commonly involving the lungs, intrathoracic lymph nodes, skin, or eyes.

Boerhaave Syndrome

- Transmural spontaneous rupture of the esophagus typically occurring after forceful vomiting in the

setting of a full stomach. Most commonly involves the left posterolateral wall of the distal esophagus. Presents with severe chest pain, subcutaneous emphysema, mediastinal emphysema, and rapidly progressive sepsis.

Bohn Nodules

- Small white cysts on gingiva of newborns. Epithelial remnants. Resolve spontaneously.

Bohr Effect

- The decrease in hemoglobin's oxygen affinity (right shift of the dissociation curve) caused by decreased pH (increased H⁺) and increased PCO₂. In metabolically active tissues, CO₂ production and lactic acid accumulation lower the local pH, causing hemoglobin to release more oxygen precisely where it is needed most.

Boil

- Furuncle: an acute, deep infection of a hair follicle (folliculitis that extends to the dermis) by *Staphylococcus aureus*, forming a painful, red, swollen nodule with central purulent core that may spontaneously drain.

Boil (Furuncle)

- A painful, deep-seated infection of a hair follicle caused most commonly by *Staphylococcus aureus*, extending into the surrounding subcutaneous tissue. Presents as a tender, erythematous, fluctuant nodule that eventually discharges pus. When multiple adjacent follicles coalesce, it forms a carbuncle, often with systemic symptoms.

Boils

- Painful, pus-filled infections of a hair follicle and surrounding skin, usually caused by *Staphylococcus aureus*. Recurrent or large boils may require incision, drainage, culture, and evaluation for risk factors such as diabetes or household transmission.

Bone

- A rigid, mineralised connective tissue that forms the skeleton, providing structural support, protection of vital organs, hematopoiesis (bone marrow), and mineral storage (calcium, phosphate). Bone composition: organic matrix (90% type I collagen) and inorganic mineral (hydroxyapatite, calcium phosphate).

Bone Cement

(Polymethylmethacrylate—PMMA)

- Acrylic polymer used to fix prosthetic joints to bone. Mixes as dough, cures exothermically. Used in hip/knee arthroplasty, vertebroplasty, cranial reconstruction. Complications: cement implantation syndrome (hypotension, hypoxia), infection, loosening.

Bone Densitometry

- Dual-energy X-ray absorptiometry measures bone mineral density at the lumbar spine and hip. Results are expressed as T-score comparing to young adult mean and Z-score comparing to age-matched controls. T-score of minus 1.0 to minus 2.5 indicates osteopenia. T-score of minus 2.5 or less indicates osteoporosis.

Bone Density

- Measurement of bone mineral content (BMC) and density (BMD). Assessed by DEXA scan (gold standard). T-score: compares to young adult reference. Normal: >-1.0; osteopenia: -1.0 to -2.5; osteoporosis: <-2.5. Fracture risk increases with decreasing BMD. Screening: women >65, men >70, or younger with risk factors.

Bone Density Screening

- Systemic skeletal disease characterized by low bone mass and microarchitectural deterioration. See osteoporosis.

Bone Graft

- Bone tissue or a bone-substitute material used to fill a defect, promote fusion, support reconstruction, or enhance fracture healing. Autograft provides osteogenic cells and biologic signals; allograft and synthetic substitutes provide variable osteoconductive and osteoinductive properties.

Bone Graft (Dental)

- Material placed to augment alveolar bone. Types: autograft, allograft, xenograft, alloplast. Used for implants, ridge preservation.

Bone Health

- Systemic skeletal disease characterized by low bone mass and microarchitectural deterioration. See osteoporosis.

Bone Infection

- See Osteomyelitis.

Bone Marrow

- Vascular tissue within the medullary cavities and cancellous bone. Red marrow produces blood cells, whereas yellow marrow primarily stores fat and can revert to red marrow when demand increases.

Bone Marrow Aspiration

- A procedure that withdraws liquid marrow, usually from the posterior iliac crest, for cytologic, flow-cytometric, cytogenetic, molecular, and microbiologic evaluation. It complements the core biopsy, which assesses marrow architecture.

Bone Marrow Aspiration and Biopsy

- Sampling of bone marrow, typically from the posterior iliac crest. Aspiration: liquid marrow for cytology (differential, morphology), flow cytometry, cytogenetics, and MRD. Biopsy: core of bone for cellularity, architecture, fibrosis, and infiltrative disease (lymphoma, metastatic cancer, granulomas).

Bone Marrow Biopsy

- A procedure sampling bone marrow (typically posterior iliac crest) via aspiration (cytology, flow cytometry, cultures) and trephine core biopsy (histology, architecture, fibrosis, cellularity).

Bone Marrow Depression

- See myelosuppression.

Bone marrow hyperplasia

- Increased cellularity of the bone marrow resulting from accelerated hematopoiesis, typically compensating for chronic hemolytic anemia, hemorrhage, or systemic hypoxia.

Bone marrow necrosis

- Death of hematopoietic and stromal cells within the bone marrow, often associated with sickle cell crisis, malignancy, or infections, presenting with severe bone pain and cytopenias.

Bone Metastases

- Bone metastases: most common from breast, prostate, lung, kidney, thyroid. Solitary vs multiple. Clinical: pain (worse at night), pathologic fracture, hypercalcemia, spinal cord compression.

Bone Metastasis

- Common in breast, prostate, lung, thyroid, kidney. Pain, fractures, hypercalcemia. Treatment: radiation, bisphosphonates, surgery.

Bone Scan

- Skeletal imaging using Tc-99m MDP. Detects osteoblastic activity. Sensitivity: high; specificity: low. Used for metastasis detection, fracture, infection, metabolic bone disease. "Hot spots" indicate increased uptake; "cold spots" indicate decreased uptake.

Bone Scan (Nuclear Scintigraphy)

- Nuclear medicine imaging using Tc-99m MDP (methylene diphosphonate) that accumulates in areas of increased bone turnover. Three phases: perfusion (immediate), blood pool (5 min), and delayed (2-3 hours).

Bone Turnover Markers

- Biochemical markers reflecting the rate of bone remodeling. Formation markers include bone-specific alkaline phosphatase, osteocalcin, and N-terminal and C-terminal propeptides of type I procollagen. Resorption markers include C-terminal and N-terminal telopeptides of type I collagen and deoxypyridinoline.

Bones and Joints

- Bones and joints form the skeletal system, providing structural support, protection of organs, and enabling movement through articulations between bones. Joints are classified as fibrous, cartilaginous, or synovial, with synovial joints offering the greatest range of motion.

Bony Exostosis

- A benign localized bony protuberance, including torus palatinus of the palate and torus mandibularis of the mandible. Treatment is usually unnecessary unless the lesion causes trauma, interferes with speech or prosthesis, or requires diagnostic clarification.

Borderline Personality Disorder

- A pattern of instability in interpersonal relationships, self-image, and affects, with marked impulsivity.

Bordetella pertussis

- A Gram-negative coccobacillus and the causative agent of whooping cough, a highly contagious respiratory infection. It produces pertussis toxin, which inhibits G-protein signaling in respiratory epithelial cells, and tracheal cytotoxin, which damages ciliated

epithelium.

Bornholm Disease

- An acute epidemic pleurodynia caused most often by coxsackieviruses. It produces sudden attacks of severe chest or upper-abdominal muscle pain, fever, and sometimes headache or vomiting.

Borrelia

- A genus of spiral-shaped bacteria (spirochetes) transmitted by arthropod vectors, most notably ticks and lice. The major human pathogen is *Borrelia burgdorferi* sensu lato, the causative agent of Lyme disease, transmitted by Ixodes ticks.

Borrelia burgdorferi

- See Lyme Disease.

Borreliosis

- Infection caused by bacteria of the genus *Borrelia*. Human disease includes Lyme borreliosis and relapsing fever, with manifestations varying according to species, stage, and organ involvement.

Bosniak Classification

- System classifying renal cysts by CT features. Categories I-II benign, IIF follow-up, III-IV likely malignant.

Bot fly

- A parasitic fly of the family Oestridae whose larvae cause myiasis in humans and animals. *Dermatobia hominis*, the human bot fly, deposits eggs on blood-sucking insects; larvae penetrate the skin and develop subcutaneously, causing a painful boil-like lesion with a central breathing pore. Treatment is manual extraction of the larva.

Botryomycosis

- A chronic bacterial infection characterized by granule-forming lesions that can resemble a fungal infection. It most often affects skin and subcutaneous tissue but may involve viscera in susceptible patients.

Botulism

- A rare, potentially fatal neuroparalytic disease caused by botulinum toxin, a potent presynaptic neurotoxin produced by *Clostridium botulinum*, an anaerobic, spore-forming, gram-positive bacillus.

Bouchard's Node

- A hard, palpable enlargement of the proximal inter-

phalangeal joint caused by osteoarthritis. It reflects osteophyte formation and may be associated with pain, stiffness, and reduced finger motion.

Bougienage

- The use of a bougie to dilate or explore a narrowed tubular structure, such as the esophagus or urethra. It is a procedure that requires assessment of the cause and risks of the stricture.

Boutonneuse Fever

- A tick-borne rickettsial infection, usually caused by *Rickettsia conorii*, characterized by fever, headache, rash, and often a dark inoculation eschar called a tache noire.

Bouveret Syndrome

- Rare variant of gallstone ileus where a large gallstone erodes through a cholecystoduodenal fistula and impacts in the duodenum or pylorus causing gastric outlet obstruction. Represents approximately two to three percent of gallstone ileus cases. Presents with nausea, vomiting, epigastric pain, and anorexia.

Bovine Spongiform Encephalopathy

- A fatal transmissible prion disease of cattle causing progressive neurologic degeneration. It is related to variant Creutzfeldt–Jakob disease in humans through exposure to contaminated tissue.

Bowel Incontinence

- The involuntary loss of solid or liquid stool through the anus, also known as fecal incontinence, affecting approximately 2-15 percent of the general population and increasing in prevalence with age.

Bowel infarction

- Ischemic necrosis of bowel tissue from occlusive or non-occlusive mesenteric ischemia, presenting with severe abdominal pain out of proportion to examination, bloody stools, and peritoneal signs. Causes include mesenteric arterial embolism or thrombosis, venous thrombosis, and low-flow states.

Bowel ischemia

- Reduced blood flow to the intestines causing mucosal injury, most commonly from mesenteric atherosclerosis, low-flow states, or mesenteric venous thrombosis. Presents with abdominal pain, bloody stools, and elevated lactate. Severe cases progress to transmural necrosis.

Bowel Obstruction

- Blockage of intestinal lumen. X-ray, CT diagnosis. Differentiate small vs. large bowel obstruction. Mechanical vs. paralytic ileus.

Bowel Program

- See gastroenterology.

Bowenoid papulosis

- A low-grade squamous intraepithelial neoplasia of the anogenital skin caused by human papillomavirus infection, typically types 16 and 18, presenting as multiple pigmented papules on the genital mucosa. Histologically it resembles squamous cell carcinoma in situ but follows a benign clinical course.

Bowlegs

- Outward bowing of the legs causing the knees to remain apart when the feet and ankles are together, also called genu varum. It is common in early childhood but requires assessment when severe, asymmetric, painful, or persistent beyond the expected age.

Bowman Capsule

- Double-walled epithelial cup surrounding the glomerular capillaries forming the beginning of the nephron. The visceral layer is composed of podocytes whose foot processes interdigitate to form filtration slits. The parietal layer is composed of simple squamous epithelium.

Bowman's Gland

- A serous gland in the olfactory mucosa whose secretion helps dissolve odorant molecules and cleanse the olfactory surface. It is located in the lamina propria beneath the olfactory epithelium.

Boxed Warning

- The most prominent safety warning required in prescribing information for a medicine when there is a serious or life-threatening risk. It communicates major hazards and does not necessarily mean that the medication is never appropriate.

BPH

- Abbreviation for Benign Prostatic Hyperplasia.

Brachial Artery

- The continuation of the axillary artery, running from the lower border of teres major to the cubital fossa, where it divides into radial and ulnar arteries.

Lies medial to the biceps; palpated for blood pressure and brachial pulse. Branches: profunda brachii (deep brachial), superior/inferior ulnar collaterals.

Brachial Plexopathy

- Dysfunction of the brachial plexus causing pain, weakness, sensory loss, or paralysis in the shoulder and upper limb. Causes include trauma, compression, tumors, inflammation, and radiation injury.

Brachial Plexus

- Nerve network (C5-T1) formed by roots, trunks, divisions, and cords, giving rise to upper limb nerves. Roots unite into superior/middle/inferior trunks, dividing into anterior/posterior divisions, recombining into lateral/posterior/medial cords. Major branches: musculocutaneous, axillary, radial, median, ulnar.

Brachial Plexus Cords

- Formed by divisions recombining behind the pectoralis minor. Lateral cord (anterior divisions of upper/middle trunks), posterior cord (all three posterior divisions), medial cord (anterior division of lower trunk). Named for relationship to the axillary artery. Lateral cord: lateral pectoral nerve, contributes to median nerve.

Brachial Plexus Injuries

- Brachial plexus injuries: traction (motorcycle accidents, sports), compression (crutches, thoracic outlet). Classification: supraclavicular (roots/trunks), infraclavicular (divisions/cords). Narakas classification: I (proximal C5-C6, Erb's palsy: waiter's tip), II (C5-C7), III (C5-T1), IV (complete C5-T1), V (complete + root)

Brachial Plexus Injury

- Damage to the brachial plexus nerves during birth, most commonly from shoulder dystocia or excessive traction. Erb's palsy (C5-C6) presents with waiter's tip position (arm adducted, elbow extended, forearm pronated). Klumpke's palsy (C8-T1) affects the hand and may cause Horner syndrome. Most cases resolve spontaneously.

Brachial Plexus Roots

- Ventral rami of C5-T1 emerging from intervertebral foramina, passing between anterior and middle scalene muscles (interscalene triangle). C5-C6 form the upper trunk, C7 continues as middle trunk, C8-T1 form the lower trunk. Root injuries cause specific

dermatome/myotome deficits. C5 avulsion causes loss of deltoid/biceps function.

Brachial Plexus Trunks

- Formed in the posterior neck triangle. Upper (C5-C6), middle (C7), and lower (C8-T1) trunks each divide into anterior/posterior divisions behind the clavicle. Upper trunk is most commonly injured in birth injuries (Erb-Duchenne palsy), causing waiter's tip position.

Brachialis Muscle

- Flat strong muscle deep to the biceps, from distal anterior humerus to ulnar tuberosity and coronoid process. Innervated by musculocutaneous nerve (C5-C6). Primary elbow flexor regardless of forearm position ("workhorse" of elbow flexion). Less visible than biceps but significantly contributes to arm strength.

Brachiocephalic Trunk

- The first and largest branch of the aortic arch. It ascends to the right sternoclavicular joint, where it divides into the right common carotid and right subclavian arteries.

Brachytherapy

- Internal radiation therapy. Sources placed inside or near tumor. Used for prostate, cervical, breast cancer.

Brachytherapy Seed

- Radioactive source implanted directly into tumor. I-125 or Pd-103 seeds for prostate cancer. Permanent or temporary implantation.

Bradycardia

- A heart rate <60 beats per minute. May be physiologic (athletes) or pathologic due to sinus node dysfunction, AV block, or medication effects. Symptomatic bradycardia causes fatigue, dizziness, syncope, or heart failure. Evaluation includes ECG, Holter monitoring, and assessment of reversible causes.

Bradycardia of Prematurity

- Heart rate less than 100 beats per minute in a premature infant, often associated with apnea and oxygen desaturation. May be caused by vagal stimulation, reflux, or immaturity of the cardiac conduction system. Treatment: gentle stimulation, caffeine, and addressing underlying causes.

Bradykinesia

- Slowness and reduced amplitude of voluntary movement, often accompanied by difficulty initiating or sustaining repetitive actions. It is a cardinal feature of parkinsonism and may be associated with rigidity, resting tremor, postural instability, reduced arm swing, masked facial expression, and a shuffling gait.

Bradypnea

- Abnormally slow respiratory rate (adult <12 breaths/min). Causes: opioid/benzodiazepine/sedative overdose, central nervous system depression (stroke, trauma, increased ICP), hypothyroidism, hypothermia, and metabolic (e.g., hypokalemia). May lead to hypoventilation, hypercapnia, and respiratory failure.

Brain

- The central organ of the nervous system, enclosed in the cranial cavity and composed of the cerebrum, cerebellum, and brainstem. The brain receives, processes, and integrates sensory information; initiates and coordinates motor responses; and serves as the seat of cognition, emotion, memory, and consciousness.

Brain Abscess

- A focal, purulent infection within the brain parenchyma, most commonly occurring in the frontal or temporal lobes.

Brain Concussion

- A mild traumatic brain injury caused by a blow, jolt, or force that temporarily changes brain function. Symptoms may include headache, dizziness, confusion, memory difficulty, nausea, and sensitivity to light or noise, sometimes without loss of consciousness. Worsening headache, repeated vomiting, seizure, weakness, or increasing drowsiness requires urgent assessment.

Brain Death

- The irreversible cessation of all brain and brainstem function, including consciousness and brainstem reflexes, with absent respiratory drive—a legal definition of death.

Brain Edema

- Swelling caused by excess fluid in brain tissue,

which can raise pressure inside the skull. Causes include head injury, stroke, infection, tumor, bleeding, and severe metabolic disturbances. Symptoms may include headache, vomiting, confusion, seizures, weakness, or reduced consciousness; severe cases are medical emergencies.

Brain Health

- Brain health encompasses the ability to perform cognitive functions such as memory, learning, decision-making, and emotional regulation across the lifespan. Key factors include cardiovascular health, nutrition, sleep, social engagement, cognitive stimulation, and management of chronic conditions like hypertension and diabetes.

Brain Herniation

- Displacement of brain tissue across rigid intracranial compartments or through the foramen magnum due to mass effect or raised ICP. It can compress the brainstem and cranial nerves and is a life-threatening neurosurgical emergency.

Brain Injuries

- Brain injuries range from mild concussions to severe traumatic brain injury caused by falls, motor vehicle accidents, sports impacts, or penetrating trauma. Symptoms depend on injury location and severity, including headache, confusion, memory loss, seizures, and altered consciousness.

Brain Injury

- See Brain Injuries.

Brain Metastases

- The most common intracranial neoplasms, occurring in approximately ten to forty percent of patients with systemic cancer, and are more common than primary brain tumors.

Brain Metastasis

- Common in lung, breast, melanoma. Symptoms: headache, seizures, focal deficits. Treatment: surgery, radiation, stereotactic radiosurgery.

Brain Monitoring

- Brain monitoring during anesthesia assesses the depth of hypnosis and neurological function. Bispectral index (BIS) processes the frontal EEG into a number from 0 (isoelectric) to 100 (fully awake), with 40 to 60 recommended for general anesthesia. BIS monitoring reduces awareness risk and may

reduce anesthetic requirements.

Brain Natriuretic Peptide (BNP)

- A cardiac neurohormone released from the ventricles in response to volume overload and wall stress. BNP <100 pg/mL makes heart failure unlikely; >400 pg/mL suggests heart failure. NT-proBNP: <125 pg/mL (<75 years) or <450 pg/mL (>75 years) excluding HF. Also prognostic: rising levels indicate worsening HF and higher mortality.

Brain SPECT

- Cerebral perfusion imaging using Tc-99m HMPAO or ECD. Used for epilepsy localization, dementia differentiation, cerebrovascular disease.

Brain Tumor

- An abnormal growth of cells within the brain or its surrounding structures, classified as primary (originating from brain tissue or its coverings) or secondary/metastatic (spread from a cancer elsewhere in the body, which are more common, accounting for approximately 50 percent of all brain tumors).

Brain Ventricles

- The four interconnected fluid-filled cavities within the brain that produce and contain cerebrospinal fluid (CSF).

Brainstem

- Connects cerebrum to spinal cord, with midbrain (superior), pons (middle), and medulla oblongata (inferior). Contains cranial nerve nuclei (III-XII), ascending/descending tracts, and vital autonomic centers for cardiac, respiratory, and vasomotor function. Reticular formation regulates consciousness and sleep-wake cycles.

Branchial Arch

- One of the paired embryonic arches that contribute to development of facial, pharyngeal, laryngeal, muscular, skeletal, vascular, and neural structures of the head and neck.

Branchial Cleft

- An embryonic groove between adjacent branchial arches. Persistence of cleft remnants can produce congenital branchial cysts, sinuses, or fistulas, most commonly along the lateral neck.

Branchial Cleft Anomaly

- A congenital remnant of the branchial apparatus presenting as a cyst, sinus, or fistula in the head

and neck. Recurrent infection or drainage from a persistent opening warrants imaging and specialist management.

Branchial Cleft Cyst

- Congenital cystic mass arising from incomplete obliteration of branchial cleft apparatus during embryogenesis.

Braxton Hicks Contractions

- Irregular, painless uterine contractions occurring throughout pregnancy, becoming more frequent in the third trimester. Usually last 30-60 seconds and are felt as a tightening of the abdomen. Distinguished from true labour contractions by their irregularity, lack of cervical change, and resolution with rest or hydration.

BRCA

- BReast CAncer gene: BRCA1 and BRCA2 are tumor suppressor genes involved in DNA double-strand break repair via homologous recombination. Mutations in these genes confer a significantly increased lifetime risk of breast cancer (up to 70-80% for BRCA1, 40-70% for BRCA2) and ovarian cancer (up to 40-50% for BRCA1, 10-20% for BRCA2).

BRCA1 and BRCA2

- Tumor-suppressor genes involved in homologous-recombination DNA repair. Germline pathogenic variants increase susceptibility to breast, ovarian, fallopian-tube, primary peritoneal, prostate, and pancreatic cancers and may guide surveillance and treatment.

Breakbone Fever

- A traditional name for dengue fever, reflecting the severe myalgia and arthralgia that may accompany the acute viral illness.

Breast Abscess

- A localized collection of pus within the breast tissue, most commonly a complication of lactational mastitis. Presents as a fluctuant, painful mass with overlying erythema and systemic symptoms. Diagnosis by ultrasound. Treatment requires incision and drainage plus antibiotics.

Breast Cancer

- Most common cancer in women. Risk factors: age, genetics, hormones, lifestyle. Screening: mammography. Treatment: surgery, radiation, chemother-

apy, hormone therapy, targeted therapy.

Breast Cancer Genetics

- Hereditary breast cancer: BRCA1/BRCA2 mutations (lifetime risk 45-85%). Other genes: TP53 (Li-Fraumeni), PTEN (Cowden), PALB2, CHEK2. Genetic counseling and testing indicated for early onset, bilateral, male breast cancer, Ashkenazi Jewish heritage.

Breast Engorgement

- Overfilling of the breasts with milk, causing bilateral pain, firmness, warmth, and nipple flattening. Occurs 3-5 days postpartum. Pathophysiology: vascular congestion and edema before milk secretion is established.

Breast Health

- Breast health involves routine self-awareness, clinical examinations, and mammographic screening to detect abnormalities such as lumps, pain, or skin changes that may indicate benign conditions or breast cancer. Risk factors include genetics, age, hormone exposure, and lifestyle factors.

Breast Infection

- Infection and inflammation of breast tissue, most commonly mastitis during breastfeeding. It causes localized pain, warmth, redness, swelling, and sometimes fever; an untreated infection may progress to a breast abscess.

Breast Lump

- A localized thickening, mass, or swelling in breast tissue caused by a cyst, fibroadenoma, infection, trauma, or malignancy. New, persistent, hard, fixed, or changing lumps require clinical and often imaging assessment.

Breast MRI

- Magnetic resonance imaging of the breast with IV gadolinium contrast.

Breast Pain

- Pain or tenderness in one or both breasts, caused by hormonal changes, cysts, infection, trauma, medications, or musculoskeletal disease. Persistent focal pain or an associated lump requires assessment.

Breast Surgery

- Breast cancer surgery: breast-conserving therapy (BCT: lumpectomy + radiation) equivalent survival to mastectomy for early stage. Margins: "no ink on

tumor" for invasive cancer; 2mm for DCIS.

Breast Ultrasound

- Ultrasound imaging of the breast. Indications: evaluating palpable masses (distinguishing solid vs. cystic), characterizing mammographic abnormalities, guiding biopsy (core needle, fine needle, or vacuum-assisted), evaluating implant integrity, and imaging young/dense breast tissue.

Breastfeeding

- The feeding of an infant with breast milk, recommended exclusively for the first 6 months and continued with complementary foods to 2 years and beyond. Benefits: infant (nutrition, immune protection—IgA, reduced infection/allergy/obesity, bonding) and maternal (uterine involution, contraception, reduced breast/ovarian cancer).

Breath Sounds

- The sounds produced by airflow through the respiratory tract during breathing, assessed by auscultation with a stethoscope over the chest wall.

Breath Stacking

- A technique using a manual resuscitation bag or glossopharyngeal breathing to deliver successive breaths without exhaling, stacking tidal volumes to achieve higher lung volumes. It improves cough effectiveness and secretion clearance in patients with weak respiratory muscles.

Breath-Holding Spells

- A benign paroxysmal phenomenon in young children (6 months to 6 years), provoked by frustration, pain, or fear. Cyanotic type (more common): child cries, forcefully exhales, becomes cyanotic, and may lose consciousness briefly.

Breech Delivery

- Vaginal delivery of a breech presentation. Types: assisted breech (the baby delivers spontaneously to the umbilicus, then the operator assists delivery of the shoulders and head) and total breech extraction (the operator delivers the entire baby by grasping the feet and pulling).

Breech Presentation

- A fetal presentation where the buttocks or feet are positioned to deliver first. Types include frank breech (hips flexed, knees extended, most common), complete breech (hips and knees flexed), and

footling breech (one or both feet presenting). Associated with increased risk of cord prolapse, head entrapment, and birth trauma.

Bremsstrahlung

- Braking radiation produced when beta particles decelerate in matter. Continuous spectrum. Detected in I-131 imaging (poor quality). Used for Y-90 bremsstrahlung SPECT.

Brenner Tumor

- Rare ovarian epithelial neoplasm (1-2% of ovarian tumors) composed of nests of transitional (urothelial-like) epithelial cells embedded in a dense fibromatous stroma. Most are unilateral, small (< 2 cm), benign (90%) and incidental; borderline (proliferative) and malignant variants are uncommon.

Breslow Thickness

- Microstaging measurement of melanoma invasion depth from the top of the granular layer of the epidermis to the deepest point of tumor invasion, in millimeters. Most important prognostic factor in primary cutaneous melanoma, used in the AJCC TNM staging system (T category).

Bridge (Dental)

- Fixed prosthetic replacing missing teeth. Retained by adjacent teeth (abutments). Requires preparation of abutments.

Brief Psychotic Disorder

- A disturbance characterized by the sudden onset of at least one psychotic symptom: delusions, hallucinations, disorganized speech, or grossly disorganized or catatonic behavior. The disturbance lasts at least one day but less than one month, after which the individual fully returns to premorbid functioning.

Brief Resolved Unexplained Event (BRUE)

- A sudden, brief, and now-resolved episode in an infant involving abnormal breathing, skin color, muscle tone, or responsiveness, with no explanation after the initial evaluation. Risk assessment is based on age, history, examination findings, and recurrence; higher-risk events require targeted investigation.

Bright's Disease

- A historical term encompassing several forms of kidney disease, particularly glomerular disorders associated with proteinuria, edema, and altered

urinary findings. Modern diagnosis specifies the underlying renal lesion.

Brill's Disease

- A recrudescent form of epidemic typhus caused by reactivation of latent *Rickettsia prowazekii* infection years after the original illness. It is usually milder than the initial infection but can transmit to body lice.

Brittle Asthma

- Rare phenotype of severe asthma characterized by wide, chaotic, unpredictable peak expiratory flow variability (> 40% diurnal variation over weeks), classified as type 1 (chronic, persistent chaotic variability maintained over months despite maximal therapy) and type 2 (sudden onset of severe, life-threatening attacks against a background

Brittle Diabetes

- Labile, unstable type 1 diabetes mellitus characterized by unpredictable, large, recurrent swings in blood glucose between severe hyperglycemia and recurrent hypoglycemia (often with ketoacidosis) despite optimal adherence, significantly disrupting quality of life.

Broca Aphasia

- Acquired language disorder of expressive speech (motor aphasia) caused by lesion of the posterior inferior frontal gyrus (Broca area, Brodmann area 44/45) of the dominant (usually left) hemisphere, most commonly from middle cerebral artery (MCA) superior division infarction.

Broca's Aphasia

- A nonfluent language disorder caused by dysfunction of the dominant inferior frontal language region. Speech is effortful and reduced, while comprehension is relatively preserved compared with repetition and fluent language production.

Broca's Area

- A language-related region of the dominant inferior frontal gyrus involved in speech production, language formulation, and aspects of syntax. Lesions may cause expressive aphasia.

Broken Heart Syndrome

- See Takotsubo Cardiomyopathy.

Bromhidrosis

- Excessive or unpleasant body odor produced when

microbial metabolism acts on sweat or skin secretions. It is often prominent in the axillae or feet and may be associated with hyperhidrosis or local skin disease.

Bromocriptine

- An ergot dopamine agonist (D2) used for hyperprolactinemia (prolactinomas, galactorrhea), acromegaly (adjunct), Parkinson disease (less common now), and postpartum breast engorgement (historical). Also used for neuroleptic malignant syndrome.

Bromoderma

- A skin eruption or dermatitis attributed to exposure to bromine or bromide compounds. It may present with acneiform, pustular, or other inflammatory lesions.

Bronchi

- The major airways conducting air from the trachea into the lungs. The trachea bifurcates at the carina (T4-T5) into right and left main bronchi. The right main bronchus is wider, shorter, and more vertical—aspirated foreign bodies preferentially lodge here.

Bronchial Artery

- An artery arising from the systemic circulation that supplies the walls of the bronchi, connective tissue of the lungs, and visceral pleura. It differs from the pulmonary arteries, which carry blood for gas exchange.

Bronchial Biopsy

- Tissue sampling of the bronchial wall during bronchoscopy. It is used to diagnose inflammatory conditions (sarcoidosis, eosinophilic bronchitis), infections, and malignancy. Endobronchial biopsy of visible lesions and transbronchial biopsy of parenchymal lesions provide complementary diagnostic information.

Bronchial Carcinoid

- Pulmonary neuroendocrine tumor arising from Kulchitsky cells of the bronchial mucosa; subclassified as typical carcinoid (TC, low-grade, < 3 mitoses/2 mm², no necrosis, excellent prognosis with > 90% 5-year survival) and atypical carcinoid (AC, intermediate-grade, 2-10 mitoses or necrosis present, higher recurrence/ metastasis).

Bronchial Stenosis

- Narrowing of a bronchus from trauma, infection, inflammation, or prior intervention. It causes localized wheezing, recurrent infections, and impaired secretion clearance. Diagnosis is by bronchoscopy. Treatment includes balloon dilation, stenting, or surgical resection.

Bronchial Tree

- The branching airway system formed by the trachea, main bronchi, lobar and segmental bronchi, bronchioles, and terminal airways leading toward the alveoli.

Bronchial Vein

- A vein draining portions of the bronchial walls and larger airways. It empties partly into the azygos system and partly into pulmonary veins, creating a small physiologic venous admixture.

Bronchiectasis

- Permanent, abnormal dilation of bronchi caused by destruction of the bronchial wall from chronic airway inflammation and infection. Causes include CF, immune deficiency, primary ciliary dyskinesia, and post-infectious damage. Features include chronic cough, copious purulent sputum, and recurrent hemoptysis.

Bronchiole

- Small airways (less than 2 mm diameter) without cartilage, branching from segmental bronchi to terminal bronchioles (conducting zone) to respiratory bronchioles (transitional, alveoli bud from walls) to alveolar ducts to alveolar sacs to alveoli.

Bronchiolectasis

- Permanent abnormal dilation of bronchioles, usually caused by chronic inflammation, infection, or obstruction. It may accompany bronchiectasis and contribute to impaired mucus clearance and recurrent infection.

Bronchiolitis

- Lower respiratory tract infection causing inflammation and obstruction of small airways. Most common cause is respiratory syncytial virus. Affects infants under 12 months. Presents with wheezing, tachypnea, nasal flaring, and retractions. Diagnosis is clinical.

Bronchiolitis Obliterans

- An uncommon, often irreversible obstructive small-airway disease caused by inflammation and fibrotic narrowing or obliteration of the bronchioles. Causes include severe infection, inhalational injury, autoimmune disease, and complications after lung or bone-marrow transplantation.

Bronchitis

- Inflammation of the bronchial mucosa. Acute bronchitis is typically viral, presents with cough (often productive), and resolves spontaneously. Cough with sputum production for at least 3 months in 2 consecutive years, often due to smoking.

Bronchoalveolar Lavage

- Diagnostic procedure during bronchoscopy where saline is instilled and aspirated. Analyzed for cell counts, microbiology, and cytology. Useful for Pneumocystis pneumonia, pulmonary alveolar proteinosis, and sarcoidosis diagnosis.

Bronchodilator Response

- The change in spirometric airflow after administration of a short-acting bronchodilator. A significant response can support variable airflow limitation, but a negative response does not exclude asthma and a positive response is not specific to one disease.

Bronchodilator Reversibility

- Improvement in FEV1 > 12% and > 200 mL after short-acting bronchodilator. Supports asthma diagnosis. Some COPD patients show partial reversibility.

Bronchoesophageal fistula

- Abnormal communication between the bronchial tree and the esophagus, most commonly acquired from malignancy, mediastinal infection, or tuberculous lymphadenitis. Congenital forms are associated with esophageal atresia.

Bronchogenic Cyst

- A congenital cyst derived from abnormal budding of the ventral foregut. It is typically found in the middle mediastinum or near the carina. It is usually asymptomatic but may cause compression of adjacent structures. Treatment is surgical resection if symptomatic.

Bronchophony

- Increased transmission of spoken sounds to the

stethoscope over consolidated lung tissue. It is normally heard over the central airways. Excessive bronchophony over peripheral lung fields indicates consolidation. It is part of the constellation of physical examination findings in pneumonia.

Bronchopleural Fistula

- An abnormal communication between the bronchial tree and pleural space. It is a complication of pneumonectomy, lung surgery, or empyema. It causes persistent air leak, subcutaneous emphysema, and respiratory compromise. Treatment includes chest tube drainage, bronchoscopic intervention, and surgical repair.

Bronchopneumonia

- Patchy consolidation centered on bronchioles across multiple lobes. Caused by *S. aureus*, gram-negative bacteria, and *Pseudomonas*.

Bronchopulmonary Dysplasia

- A chronic lung disease of premature infants, resulting from lung injury from mechanical ventilation, oxygen toxicity, and inflammation. Most commonly occurs in infants born at less than 28 weeks gestation with birth weight less than 1500 grams.

Bronchoscopy

- Endoscopic examination of the tracheobronchial tree. Flexible bronchoscopy: most common, for diagnosis (lung mass, persistent cough, hemoptysis, suspected infection, ILD) and therapy (foreign body removal, bronchial washing, biopsy, BAL). Rigid bronchoscopy: for massive hemoptysis, large foreign bodies, and stent placement.

Bronchospasm

- Involuntary constriction of bronchial smooth muscle causing airway obstruction, wheezing, and increased airway pressures. Triggers include airway manipulation, volatile anesthetics (sevoflurane is bronchodilating; desflurane can cause bronchospasm), histamine-releasing drugs (morphine, atracurium), and reactive airway disease.

Bronchostenosis

- Narrowing of a bronchus that restricts airflow. Causes include scarring, tumors, foreign bodies, inflammation, external compression, or postinfectious and postprocedural changes.

Bronchus

- One of the two main branches of the trachea that conduct air into the lungs. The right main bronchus is wider, shorter, and more vertical than the left, making it more susceptible to aspiration. Each main bronchus divides into lobar bronchi (secondary), segmental bronchi (tertiary), and progressively smaller branches (bronchioles).

Bronze Diabetes

- Historical eponym for hereditary hemochromatosis (HFE-related HH, C282Y homozygous mutation), characterized by the classic triad of skin pigmentation ("bronze"), cirrhosis, and diabetes mellitus from pancreatic iron deposition. Also known as "Troisier syndrome" or pigment cirrhosis.

Brow Presentation

- A fetal presentation where the fetal head is in a military position between full flexion (vertex) and hyperextension (face). The presenting diameter is the largest (occipitofrontal, 13.5 cm). Incidence: 1:1000-1:2000. Vaginal delivery is impossible with a persistent brow in a term fetus unless it converts to vertex or face.

Brown Lung Disease

- An occupational airway disease associated with inhalation of cotton, flax, or hemp dust, classically causing byssinosis. Symptoms may include chest tightness, cough, wheezing, and work-related airflow limitation.

Brown Sequard Syndrome

- Brown-Sequard syndrome (hemisection of the spinal cord) results from a lesion affecting one half of the spinal cord, producing a characteristic pattern of ipsilateral and contralateral neurological deficits.

Brown Stain

- Extrinsic staining from chromogenic bacteria, diet, tobacco. Removed by prophylaxis.

Brucellosis

- A zoonotic infectious disease caused by gram-negative coccobacilli of the genus *Brucella*, transmitted to humans through contact with infected animals (cattle, goats, pigs, sheep) or consumption of unpasteurized dairy products. *Brucella melitensis* (goats and sheep, most severe), *B. abortus* (cattle), *B. suis* (pigs), and *B.*

Brudzinski Sign

- Flexion of the hips and knees that occurs involuntarily when the neck is passively flexed. It is a meningeal irritation sign that may occur in meningitis but lacks sufficient sensitivity or specificity to establish the diagnosis alone.

Brugada Syndrome

- A genetic channelopathy causing abnormal sodium channel function (SCN5A mutation). ECG pattern: coved ST elevation >2mm in V1-V3 (type 1) followed by negative T wave. Patients are at risk for ventricular fibrillation and sudden cardiac death, often at rest or during sleep. More common in young Southeast Asian men.

Bruise

- A contusion: localized hemorrhage into tissues due to blunt force trauma rupturing small blood vessels, with intact skin.

Bruising

- Ecchymosis/contusion: a purplish-blue discoloration from blood extravasation into skin/subcutaneous tissue following trauma; color changes with age (red-blue purple green/yellow—bilirubin breakdown).

Bruits

- Turbulent blood flow sounds heard with stethoscope. Indicates stenosis.

Brunner's Gland

- A mucus-secreting gland in the submucosa of the duodenum. Its alkaline secretion helps protect the mucosa from gastric acid and provides conditions favorable for intestinal enzymes.

Bruxism

- Grinding or clenching teeth, often during sleep. Causes: stress, occlusion, sleep disorders. Consequences: tooth wear, fractures, muscle pain, TMJ disorders. Treatment: occlusal guard, stress management, behavior modification.

Buccal Gland

- A minor salivary gland located in the cheek. It secretes mucus into the oral cavity to lubricate the buccal mucosa and assist formation and handling of the food bolus.

Budd-Chiari Syndrome

- Obstruction of hepatic venous outflow at any level from small hepatic venules to the inferior vena cava junction with the right atrium. The most common predisposing condition is myeloproliferative disorders accounting for forty to fifty percent of cases. Other risk factors include thrombophilia and oral contraceptive use.

Buerger Disease

- See thromboangiitis obliterans.

Buerger's Disease

- Thromboangiitis obliterans, a nonatherosclerotic inflammatory disease of small and medium arteries and veins, strongly associated with tobacco exposure. It causes distal ischemia, pain, ulcers, and sometimes gangrene.

Buffer

- A solution that resists pH change upon addition of acid or base, composed of a weak acid and its conjugate base (or weak base and conjugate acid). Buffer capacity is maximal when $\text{pH} = \text{pK}_a \pm 1$.

Bulbar Conjunctiva

- The bulbar conjunctiva is the portion of the conjunctival membrane that covers the anterior surface of the sclera, extending from the limbus (the junction of the cornea and sclera) to the fornices where it reflects onto the inner surface of the eyelids.

Bulbar Paralysis

- Weakness or paralysis of muscles supplied by the lower cranial nerves, causing dysarthria, dysphagia, impaired gag or cough, and sometimes respiratory compromise. Causes include brainstem disease, motor-neuron disease, and neuromuscular disorders.

Bulimia Nervosa

- An eating disorder characterized by recurrent episodes of binge eating followed by compensatory behaviors (purging, laxatives, excessive exercise). Self-esteem overly influenced by body shape and weight.

Bullae

- Large (>1 cm) fluid-filled blisters; see also vesicle (<1 cm).

Bullet Track

- The path a projectile traverses through the body,

defined by the entry wound, internal course (through organs, vessels, bones), and exit wound (or retained bullet).

Bullous Impetigo

- A superficial staphylococcal skin infection caused by exfoliative toxins that produce fragile, flaccid bullae and honey-coloured or varnish-like crusts, especially in infants and young children.

Bullous Pemphigoid

- An autoimmune subepidermal blistering disease affecting elderly patients. It presents with tense blisters on erythematous or normal-appearing skin, often preceded by pruritic urticarial plaques. Autoantibodies target BP180 and BP230 (hemidesmosome proteins).

Bumetanide

- A potent loop diuretic inhibiting the Na-K-2Cl cotransporter in the thick ascending limb of the loop of Henle. Used for edema in heart failure, cirrhosis, and renal disease, and for hypertension in renal impairment. Approximately 40 times more potent than furosemide; more bioavailable orally, so equivalent IV-to-oral dosing is possible.

Bundle Branch Block

- Delay or block in the electrical conduction through the bundle branches. Left Bundle Branch Block (LBBB): QRS >120 ms, broad notched R waves in lateral leads (I, V5-V6), deep S waves in V1-V2. Right Bundle Branch Block (RBBB): QRS >120 ms, rSR' pattern in V1-V2, wide S waves in I, V5-V6.

Bundle of His

- A narrow band of specialized cardiac tissue that conducts electrical impulses from the AV node through the fibrous cardiac skeleton to the inter-ventricular septum. It divides into the right and left bundle branches. The bundle of His is the only electrical connection between atria and ventricles through the fibrous skeleton.

Bunion

- A bony prominence (exostosis) at the medial aspect of the first metatarsal head, causing lateral deviation of the great toe (hallux valgus). The deformity is characterized by a structural abnormality at the first metatarsophalangeal joint, with the first metatarsal deviating medially and the proximal

phalanx deviating laterally.

Bunionectomy

- Surgical correction of hallux valgus (bunion) deformity, indicated for symptomatic bunions refractory to conservative measures (proper footwear, orthoses, toe spacers, NSAIDs). Procedure chosen based on severity (hallux valgus angle and intermetatarsal angle) and patient factors (age, activity, comorbidities).

Buphthalmos

- Enlargement of the eye caused by elevated intraocular pressure during early childhood, when the ocular coats are still distensible. It is classically associated with congenital or infantile glaucoma.

Bupropion

- An atypical antidepressant that inhibits dopamine and norepinephrine reuptake. Used for major depressive disorder, seasonal affective disorder, and smoking cessation (Zyban). Low sexual side effects and weight neutrality; energizing. Also used off-label for ADHD.

Burkitt Lymphoma

- A highly aggressive B-cell lymphoma characterized by MYC translocation (t(8;14) IGH-MYC, t(8;22) IGL-MYC, t(2;8) IGK-MYC) and a "starry sky" morphology (tingible body macrophages). Three variants: endemic (African, EBV+, jaw/facial bones), sporadic (abdominal, ileocecal), immunodeficiency-associated (HIV).

Burn

- See Burns.

Burn (Oral)

- Thermal or chemical injury to oral mucosa. Food burn common. Treatment: symptomatic, healing usually rapid.

Burn Surgery

- Burn assessment: rule of 9s (adult), Lund-Browder (children). Depth: superficial (epidermis), superficial partial (dermis, blanch, blister, painful), deep partial (deeper dermis, white, sluggish capillary refill, less painful), full thickness (all dermis, white/charred, anesthetic).

Burning Mouth Syndrome

- A chronic oral pain disorder characterized by burning, scalding, or tingling discomfort of the mouth,

often involving the tongue, lips, or palate, without a sufficient visible mucosal abnormality.

Burns

- Tissue damage from thermal, chemical, electrical, or radiation sources. Classification: first-degree (epidermis, erythema), second-degree (partial thickness, blisters), third-degree (full thickness, eschar).

Bursectomy

- Surgical excision of a bursa, usually performed for persistent painful bursitis, recurrent infection, or another structural problem that has not responded to appropriate conservative management.

Bursitis

- Inflammation of a bursa, a fluid-filled sac that reduces friction between tendons, muscles, bones, and skin. Bursae are lined by synovial cells and normally contain a thin layer of lubricating synovial fluid.

Button magnets in the nasal cavity

- Ingestion or insertion of small magnets into the nose, posing a risk of nasal septal perforation or necrosis if multiple magnets are involved, requiring urgent removal.

Bypass

- Surgical creation of alternative route around blocked vessel. Uses vein or prosthetic graft.

Bypass (Vascular)

- Surgical procedure creating detour around blocked vessel using autologous vein (saphenous) or prosthetic graft (PTFE, Dacron). See Bypass Graft.

Bypass Graft

- Surgical procedure creating detour around blocked vessel using autologous vein (saphenous) or prosthetic graft (PTFE, Dacron). Indications: peripheral arterial disease, coronary artery disease, aortic aneurysm. Long-term patency depends on graft type, inflow/outflow, patient factors.

Bypass Surgery

- A surgical conduit that reroutes flow around an obstructed or diseased segment, applied to coronary arteries (CABG), peripheral arteries (autologous vein or prosthetic bypass), and the gastrointestinal tract (gastric bypass). Principal long-term concerns are conduit patency, graft infection, and anastomotic stenosis.

Byssinosis

- An occupational obstructive airway disease caused by inhalation of textile fibers, especially cotton, flax, or hemp dust. It may cause chest tightness and cough that worsen during or after work exposure.

Biliary Tract

- The network of ducts that carries bile from the liver and gallbladder to the small intestine. Blockage or inflammation can cause abdominal pain, jaundice, infection, or pancreatitis.

Barrel Chest

- A rounded, enlarged appearance of the chest caused by increased lung volume or changes in the chest wall. It may occur with chronic obstructive pulmonary disease, severe asthma, or other long-standing lung conditions.

Beau's Lines

- Horizontal grooves or depressions across a fingernail or toenail caused by temporary interruption of nail growth. They may follow severe illness, injury, high fever, nutritional deficiency, or local trauma; the timing of the groove can help estimate when the event occurred.

Back Injury

- Damage to the muscles, ligaments, joints, discs, vertebrae, or nerves of the back caused by trauma, lifting, or repetitive strain. Pain and stiffness are common; weakness, numbness, loss of bladder or bowel control, or severe pain after major trauma requires urgent assessment.

Bleeding Gums

- Bleeding from the gum tissue, often caused by plaque-related gingivitis, periodontitis, trauma, or vigorous brushing. It can also occur with blood disorders or medicines that affect clotting. Persistent bleeding, swelling, loose teeth, or spontaneous bleeding requires dental or medical evaluation.

Body Aches

- Generalized muscle or body pain that may occur with viral infections, fever, exertion, inflammation, medication effects, or other systemic illness. Severe pain with weakness, dark urine, breathing difficulty, chest pain, or high fever requires prompt medical assessment.

Bone Cancer

- Cancer that begins in bone, such as osteosarcoma or chondrosarcoma, or cancer that has spread to bone from another organ. Possible features include persistent localized bone pain, swelling, or a fracture after minor trauma. Diagnosis requires imaging and tissue evaluation; treatment depends on the type and extent of disease.

Bone Fracture

- A partial or complete break in a bone caused by injury, repeated stress, or weakened bone. Pain, swelling, bruising, deformity, and inability to use the limb may occur. Open fractures, loss of circulation or sensation, or major deformity require emergency care.

Breast Cyst

- A fluid-filled sac within breast tissue, usually benign and sometimes tender or larger before menstruation. Examination and imaging distinguish a cyst from a solid lesion; persistent, growing, bloody, or newly discovered breast lumps should be assessed by a clinician.

Breast Discharge

- Fluid released from the nipple without breastfeeding. It may be caused by hormonal changes, infection, benign duct disease, or cancer. Bloody, clear, spontaneous, one-sided, or persistent discharge, especially with a lump or skin change, requires prompt breast evaluation.

Chapter 3

Pocket Dictionary: C Terms

C-Peptide

- A peptide released in equimolar amounts with endogenous insulin when proinsulin is cleaved. Measurement helps estimate endogenous beta-cell insulin secretion and distinguish endogenous from injected insulin, with interpretation affected by renal function and glucose level.

C-Reactive Protein

- An acute-phase protein produced mainly by the liver whose blood concentration rises with inflammation, infection, tissue injury, and some malignancies.

C-Reactive Protein (CRP)

- An acute-phase reactant produced by the liver in response to inflammation (IL-6). CRP rises within 4-6 hours of inflammatory stimulus and peaks at 36-50 hours. Uses: monitoring inflammatory disease activity (RA, vasculitis), detecting infection post-operatively, and cardiovascular risk stratification (hs-CRP: high-sensitivity).

C-Section

- Cesarean section, surgical delivery of a fetus through incisions in the abdominal wall and uterus.

C. difficile

- *Clostridioides difficile*, a gram-positive, spore-forming bacterium that causes antibiotic-associated diarrhea and colitis. *C. difficile* produces toxins A and B that cause colonic inflammation. Risk factors include antibiotic exposure, proton pump inhibitor use, and advanced age. Treatment includes vancomycin or fidaxomicin.

C282Y Mutation

- Most common genetic cause of hereditary hemochromatosis, a C to T transition in the HFE gene resulting in cysteine to tyrosine substitution at position 282. Homozygosity occurs in approxi-

mately one in two hundred individuals of Northern European descent.

CA 19-9

- A tumor marker for pancreatic cancer. Normal: <37 U/mL. Sensitivity: 70-80% for pancreatic cancer (lower for early-stage). Specificity: limited (elevated in cholangitis, pancreatitis, cholecystitis, biliary obstruction, cirrhosis, and other GI malignancies).

CA-125

- Cancer antigen 125, a tumor marker most commonly used for ovarian cancer. Elevated in 80% of epithelial ovarian cancers (less in mucinous). Non-specific: elevated in endometriosis, PID, pregnancy, fibroids, pancreatitis, cirrhosis, and other malignancies (pancreatic, breast, lung, endometrial). Normal: <35 U/mL.

CA125

- Cancer Antigen 125, a protein tumor marker used primarily in the evaluation of ovarian cancer.

CABG (Coronary Artery Bypass Grafting)

- Open heart surgery to bypass stenotic coronary arteries using autologous grafts. On-pump: cardiopulmonary bypass, cardioplegic arrest. Off-pump (OPCAB): beating heart, no bypass machine.

Cabot Ring

- A threadlike or ring-shaped inclusion in a red blood cell, sometimes seen in severe anemia or disorders of erythropoiesis. It represents residual nuclear material or mitotic spindle remnants.

Cachexia

- A multifactorial syndrome of involuntary weight loss, skeletal-muscle wasting, and systemic inflam-

mation associated with chronic diseases such as cancer, heart failure, COPD, and advanced kidney disease.

Cacosmia

- A distorted or unpleasant perception of smell, in which odors may seem foul or disagreeable. It can occur with sinonasal disease, infection, seizures, trauma, or other neurologic disorders.

Cadaver

- A dead human body used for anatomical dissection, surgical training, and medical research. Cadavers are preserved through embalming (arterial injection of formaldehyde-based fixatives) to prevent decomposition and maintain tissue integrity.

Cadaveric Spasm

- Rare form of instantaneous post-mortem stiffening of a specific group of voluntary muscles at the exact moment of death, without an intervening period of muscle relaxation.

Caecitis

- Inflammation of the cecum. Causes include infection, inflammatory bowel disease, ischemia, neutropenic enterocolitis, and extension of adjacent inflammatory disease.

Caecopexy

- Surgical fixation of the cecum to the abdominal wall or retroperitoneum, performed to prevent or treat abnormal cecal mobility or recurrent cecal volvulus.

Caecostomy

- Surgical creation of an opening from the cecum to the abdominal wall for decompression or diversion of intestinal contents. It may be temporary or part of treatment for selected distal obstruction.

Caecotomy

- Surgical incision into the cecum, usually to remove a foreign body, access the bowel, or facilitate another abdominal procedure.

Caesarean Birth

- An operative procedure in which a baby is delivered through an incision in the abdominal wall and uterus. Indications: fetal distress, failure to progress in labor, breech presentation, placenta praevia, multiple pregnancy, previous cesarean, and maternal request.

Cafe-au-lait Macules

- Flat, uniformly hyperpigmented, light-brown to dark-brown (cafe-au-lait, "coffee with milk") macules, > 5 mm in prepubertal children or > 15 mm in postpubertal individuals.

Caffeine

- A naturally occurring xanthine alkaloid and central nervous system stimulant found in coffee, tea, chocolate, and many soft drinks and energy beverages. Mechanism: adenosine receptor antagonist (primarily A1 and A2A subtypes), increasing neuronal activity and promoting catecholamine release.

Caffeine Withdrawal

- A syndrome occurring after abrupt cessation or reduction of caffeine use. Symptoms include headache, fatigue, depressed mood, difficulty concentrating, and flu-like symptoms. Onset is typically 12–24 hours after last caffeine intake, peak at 20–51 hours, and duration of 2–9 days.

Café au Lait Spot

- A flat, sharply demarcated, light-brown skin macule caused by increased melanin. Multiple lesions may be a feature of neurofibromatosis type 1 or other genetic disorders.

Caisson Disease

- Decompression sickness caused by formation of inert-gas bubbles after a rapid reduction in ambient pressure. It can cause joint pain, neurologic injury, respiratory symptoms, or circulatory collapse.

Calcaneus

- Largest tarsal bone forming the heel. Articulates with talus proximally and cuboid anteriorly. Lever for the calf muscles via the Achilles tendon on its posterior tuberosity. Predominantly trabecular bone, the most commonly fractured tarsal bone from falls from height. Bohler's angle assesses fractures on lateral radiographs.

Calciferol

- Vitamin D is a fat-soluble vitamin (actually a prohormone) essential for calcium and phosphorus homeostasis and bone health. See Vitamin D: Calcitriol.

Calcific Uremic Arteriopathy

- Calcification of small blood vessels in kidney disease. Painful skin necrosis. Treatment: sodium

thiosulfate, wound care.

Calcification

- See Dystrophic Calcification and Metastatic Calcification.

Calcinosis

- Cutaneous calcinosis, the deposition of insoluble calcium phosphate salts within the skin and subcutaneous tissues, occurring in connective tissue diseases such as scleroderma and dermatomyositis, or as a localized phenomenon. Lesions present as firm, white, subcutaneous nodules that may ulcerate and extrude chalky material.

Calcinosis

- The pathological deposition of calcium salts in cutaneous and subcutaneous tissues, presenting as firm, white to yellowish nodules or plaques that may ulcerate and extrude chalky material.

Calciophylaxis

- Calcific uremic arteriolopathy causing painful skin necrosis in CKD. Risk factors include warfarin use, elevated calcium-phosphorus product, and obesity. High mortality rate. Treat with sodium thiosulfate, wound care, and discontinuation of warfarin. Avoid calcium-based binders.

Calcitonin

- A 32-amino-acid peptide hormone secreted by the parafollicular C-cells of the thyroid gland in response to elevated serum calcium. It lowers blood calcium by inhibiting osteoclast-mediated bone resorption and increasing renal calcium excretion.

Calcitriol and Analogues

- Active vitamin D forms suppressing PTH. Calcitriol, paricalcitol, doxercalciferol. Monitor calcium and phosphorus to avoid hypercalcemia and hyperphosphatemia. Part of CKD-MBD management.

Calcium

- A mineral and intracellular second messenger required for bone mineralization, muscle contraction, neurotransmitter release, blood coagulation, and enzyme regulation. Plasma calcium is present as ionized, protein-bound, and complexed fractions.

Calcium Blood Test

- Measurement of calcium in the blood, usually reported as total calcium and interpreted with albumin; ionized calcium may be measured directly

when the biologically active fraction is needed. It helps evaluate parathyroid disorders, bone disease, kidney disease, and abnormal calcium levels.

Calcium Channel Blocker Overdose

- Hypotension, bradycardia, hyperglycemia. Treatment: calcium gluconate/chloride, glucagon, high-dose insulin, vasopressors, lipid emulsion therapy, pacing.

Calcium Channel Blockers

- Agents that block L-type calcium channels, reducing calcium influx into cardiac and smooth muscle cells. Dihydropyridines (amlodipine, nifedipine) cause vasodilation and are used for hypertension and angina.

Calcium phosphate

- The principal mineral constituent of bone and teeth, existing as hydroxyapatite crystals. In pathological states, calcium phosphate deposits form in soft tissues through dystrophic calcification in damaged tissue or metastatic calcification in hypercalcemic conditions.

Calcium Pyrophosphate Arthritis

- Acute or chronic inflammatory arthritis caused by deposition of calcium pyrophosphate crystals in cartilage and joints. Acute attacks, often called pseudogout, commonly affect the knee or wrist and are diagnosed by identifying weakly positively birefringent crystals in synovial fluid.

Calcium Pyrophosphate Deposition Disease

- A crystal arthropathy caused by deposition of calcium pyrophosphate dihydrate crystals in articular cartilage (chondrocalcinosis). Presents with acute pseudogout (most common): acute monoarticular inflammation of the knee or wrist, mimicking gout but less painful.

Calcium Sensing Receptor

- G-protein coupled receptor expressed primarily on parathyroid glands and renal tubular cells that senses extracellular ionized calcium levels. On parathyroid glands activation suppresses PTH secretion. On renal tubular cells activation increases calcium excretion and decreases phosphate reabsorption.

Calcium Stones

- Most common kidney stone type composed of calcium oxalate or calcium phosphate. Risk factors include hypercalciuria, hyperoxaluria, hypocitraturia, and low urine volume. Treat with thiazides for hypercalciuria, dietary modification, and hydration. Potassium citrate for hypocitraturia.

Calculus (Tartar)

- Mineralized dental plaque. Cannot be removed by brushing. Requires professional scaling. Retains plaque, promotes inflammation.

Calculus Removal

- Professional cleaning to remove hardened plaque. Scaling and root planing. Ultrasonic and hand instruments.

Caldwell-Luc Operation

- A surgical approach to the maxillary sinus through the canine fossa of the anterior maxilla. It is used selectively for access to maxillary sinus disease, lesions, or foreign bodies.

Caloric Restriction

- Reduced calorie intake without malnutrition. Associated with longevity, reduced chronic disease risk. Requires careful planning.

Calorie

- A unit of energy measurement used in nutrition to quantify the energy content of foods and energy expenditure of the body. One kilocalorie (kcal, commonly called a Calorie with capital C) equals the energy required to raise 1 kilogram of water by 1 degree Celsius.

Cambridge Classification

- ERCP-based grading system for chronic pancreatitis severity evaluating the main pancreatic duct, branches, and common bile duct. Normal pancreas shows smooth duct without abnormalities. Equivocal shows fewer than three abnormal side branches. Mild shows three or more abnormal side branches. Moderate shows abnormal main duct.

Cameron Lesions

- Linear gastric erosions or ulcers found on the mucosal folds at the diaphragmatic hiatus in patients with large hiatal hernias. They result from mechanical trauma as the gastric mucosa prolapses through the diaphragmatic crura during respiration

and movement.

Campimetry

- Measurement and mapping of the visual field, usually by presenting stimuli at different locations while the patient maintains fixation. It helps detect and characterize defects from retinal, optic-nerve, or retrochiasmal disease.

Camptocormia

- A marked forward flexion of the trunk that increases while standing or walking and improves when lying down. It can occur with axial myopathy, Parkinsonian disorders, dystonia, or other neuromuscular disease.

Camptodactyly

- A congenital or acquired flexion deformity of one or more fingers, usually caused by shortening or abnormality of tendons, muscles, fascia, or joints. It often affects the proximal interphalangeal joints.

Campylobacter Infection

- Bacterial gastroenteritis caused most often by *Campylobacter jejuni* or *C. coli*, usually acquired from contaminated poultry, food, water, or unpasteurized milk. It causes fever, abdominal cramps, and diarrhea that may be bloody and can rarely precede Guillain-Barre syndrome.

Campylobacteriosis

- Infection caused by *Campylobacter* species, commonly producing fever, abdominal pain, and inflammatory diarrhea that may be bloody. Complications can include dehydration, bacteremia, reactive arthritis, or Guillain-Barré syndrome.

Canal of Schlemm

- A circular endothelial-lined channel at the corneoscleral junction that drains aqueous humor from the anterior chamber into the episcleral venous circulation. Impaired outflow contributes to glaucoma.

Canalith Repositioning Maneuver

- See Epley Maneuver.

Canavan Disease

- An autosomal-recessive leukodystrophy caused by aspartoacylase deficiency, leading to accumulation of N-acetylaspartate and progressive spongy degeneration of cerebral white matter. It typically presents in infancy with hypotonia followed by developmental impairment and macrocephaly.

Cancer

- A group of diseases in which abnormal cells grow uncontrollably, invade nearby tissues, or spread to distant sites. Cancer may arise in any tissue and is classified by its site and cell type. Early detection and treatment improve outcomes.

Cancer Care Team

- The multidisciplinary group of health professionals involved in cancer care, which may include oncologists, surgeons, radiologists, pathologists, nurses, pharmacists, rehabilitation specialists, psychologists, and palliative-care professionals.

Cancer Cell

- A cell that has acquired genetic and epigenetic abnormalities allowing uncontrolled growth, impaired normal differentiation, resistance to cell death, invasion of surrounding tissues, or spread to distant sites.

Cancer Genetics

- Inherited cancer syndromes. Examples: BRCA1/2 (breast, ovarian), Lynch syndrome (colon), Li-Fraumeni (various).

Cancer Nutrition

- Malnutrition common in cancer. Hypermetabolism, cachexia, treatment side effects. Nutritional support improves outcomes. High protein, high calorie.

Cancer of Unknown Primary (CUP)

- Metastatic cancer in which the primary site cannot be identified despite investigation. Workup includes histology, immunohistochemistry, and molecular profiling to guide treatment.

Cancer Registry

- System for collecting cancer data. Used for epidemiology, quality improvement, research.

Cancer Screening

- Testing asymptomatic individuals for disease. See screening.

Cancer Pain

- Pain caused by a tumor, its spread, or cancer treatment. It may be continuous or episodic and can involve bone, nerves, or internal organs. Management may include treating the cancer, non-opioid medicines, opioids, nerve-directed treatments, and supportive care.

Cancer Staging

- Classification of cancer extent. TNM system: T (tumor size/extent), N (nodal involvement), M (metastasis). Stages: 0 (in situ), I (early), II-III (advanced), IV (metastatic). Guides treatment and prognosis.

Cancer Surveillance

- The systematic, ongoing collection and analysis of information about cancer incidence, stage, treatment, survival, recurrence, and mortality for patient follow-up, research, and public-health planning.

Cancer Survivor

- Person living with cancer from diagnosis through end of life and beyond.

Cancer Susceptibility Genes

- Genes in which inherited pathogenic variants increase a person's risk of developing one or more cancers. Examples include BRCA1, BRCA2, APC, and mismatch-repair genes; a susceptibility variant does not guarantee that cancer will develop.

Cancer Vaccine

- A vaccine intended either to prevent cancer-causing infections or to stimulate an immune response against cancer cells. Preventive examples target oncogenic infections, while therapeutic vaccines are designed for selected established cancers.

Cancer-Associated Thrombosis

- Venous or arterial thrombosis occurring in the setting of active malignancy or its treatment. Tumor biology, immobilization, surgery, central venous devices, and systemic therapy contribute to risk; anticoagulation is individualized according to bleeding risk and cancer factors.

Cancrum Oris

- Noma, a rapidly destructive gangrenous infection of the mouth and face that occurs mainly in severely malnourished or immunocompromised children. It begins with gingival or mucosal necrosis and can extend through facial tissues.

Candida albicans infection

- See Candidiasis.

Candida Auris

- A multidrug-resistant yeast that can colonize skin and cause invasive bloodstream, wound, or ear in-

fections, especially in hospitalized or immunocompromised patients. It spreads readily in healthcare settings and requires laboratory identification and strict infection-control measures.

Candidal Intertrigo

- A Candida infection of skin folds (axillae, inframammary, inguinal, intergluteal, pannus) characterized by erythematous, macerated plaques with satellite pustules at the periphery. Most common in obese, diabetic, and immunocompromised patients, and with prolonged antibiotic use. Presents with itching, burning, and soreness.

Candidemia

- The presence of Candida species in the bloodstream. It is an invasive fungal infection associated with intravascular devices, surgery, intensive care, immunosuppression, and substantial risk of organ dissemination.

Candidiasis

- Fungal infection caused by Candida species, most commonly *C. albicans*. Mucocutaneous: oral thrush (white plaques on oral mucosa), esophagitis, vulvovaginitis (itching, discharge), and intertrigo. Invasive candidiasis: candidemia and deep organ infection, typically in immunocompromised or ICU patients. Diagnosis: culture, histopathology.

Cane

- Walking aid for balance. Types: quad (four-base), single-point. Held on opposite side of injury.

Canicola Fever

- A form of leptospirosis associated particularly with *Leptospira interrogans* serovar canicola. It may cause fever, headache, myalgia, conjunctival suffusion, and renal or hepatic involvement.

Canker Sore

- A common aphthous ulcer of the oral mucosa, appearing as a painful, shallow, round or oval ulcer with an erythematous border. Most heal spontaneously, but recurrent, severe, or atypical ulcers require assessment.

Canagliflozin

- A sodium-glucose cotransporter-2 (SGLT2) inhibitor used for type 2 diabetes and, in selected patients, heart or kidney disease. It lowers blood glucose by increasing urinary glucose excretion; im-

portant risks include genital infections, dehydration, and ketoacidosis.

Cannabis Use Disorder

- A problematic pattern of cannabis use leading to significant impairment. Criteria include tolerance and withdrawal (irritability, anger, decreased appetite, restlessness, anxiety, sleep disturbance). Cannabis intoxication includes euphoria, impaired judgment, and time distortion.

Cannot Intubate Cannot Oxygenate

- The most critical airway emergency — inability to intubate the trachea and inability to oxygenate the patient by face mask or supraglottic airway despite optimal efforts. CICO is a life-threatening situation requiring immediate surgical airway access. Signs: SpO₂ decreasing despite 100% O₂, absent end-tidal CO₂, no chest rise.

Cannulation

- The insertion of a cannula into a blood vessel, body cavity, duct, or other anatomical passage to permit drainage, infusion, monitoring, sampling, or access for a procedure.

Cantlie Line

- Functional liver division extending from the gallbladder fossa to the IVC, corresponding to the middle hepatic vein plane. Unlike the falciform ligament (anatomical division), it represents the true functional blood supply/biliary drainage division. Right hepatic artery and portal vein supply the right half; left supply the left.

Capacity

- Autonomous voluntary authorization given by a competent patient to undergo a specific medical intervention, procedure, or clinical trial. See informed consent.

Capacity and Competence

- Decision-making capacity is a clinical evaluation performed by a physician to determine whether a patient understands the nature, consequences, risks, and alternatives of a specific proposed medical decision at a given time.

Capgras Syndrome

- A delusional misidentification syndrome in which a person believes that someone familiar has been replaced by an identical impostor. It can occur in

neurodegenerative disease, psychosis, brain injury, or other neurologic conditions.

Capillaroscopy

- Examination of the small vessels in the nail-fold or another accessible surface using magnification. It can reveal capillary dilation, hemorrhage, avascular areas, and abnormal architecture in connective-tissue and microvascular disease.

Capillary

- The smallest blood vessel in the body, forming a microscopic network (capillary bed) that connects arterioles to venules.

Capillary Bed

- A network of interconnected capillaries that links arterioles to venules within a tissue. It provides a large surface area for exchange of oxygen, carbon dioxide, nutrients, hormones, and metabolic waste between blood and interstitial fluid.

Capillary Leak Syndrome

- A disorder of increased capillary permeability in which plasma and proteins move from the intravascular space into surrounding tissues. It may cause hypotension, haemoconcentration, hypoalbuminaemia, oedema, and shock; causes include severe infection, immune-mediated disease, toxins, and rare idiopathic episodes.

Capillary Refill

- A bedside assessment of peripheral perfusion performed by briefly pressing a nail bed or skin and measuring the time for normal color to return. Delayed refill may indicate poor circulation, shock, hypothermia, or vascular disease.

Capillary Refill Time

- Time for color to return to blanched nail bed or skin after pressure. Normal: <2 seconds. Prolonged in shock, dehydration, peripheral vasoconstriction.

Capitate

- Largest carpal bone, centrally located in the distal row as the carpal keystone. Rounded head articulates with scaphoid and lunate; distal surface with the third metacarpal. Last carpal bone to ossify. Wrist joint axis, bearing significant axial load. Avascular necrosis may occur with fractures similar to scaphoid injuries.

Capnocytophaga canimorsus

- A Gram-negative, capnophilic bacillus found in the oral flora of dogs and cats that causes severe sepsis, meningitis, and cellulitis in humans, particularly the immunocompromised, asplenic, or elderly. Infection typically follows dog or cat bites. It is a fastidious organism requiring prolonged incubation for culture.

Capnography

- Continuous measurement and graphical display of exhaled carbon dioxide during anesthesia and sedation. The waveform and end-tidal value help assess ventilation, airway patency, circulation, and confirmation of tracheal tube placement.

Captopril

- An angiotensin-converting enzyme (ACE) inhibitor used to treat hypertension, heart failure, and some kidney complications of diabetes. It may cause cough, low blood pressure, increased potassium, kidney-function changes, or rarely angioedema.

Capsular Contracture

- Abnormal tightening and hardening of the fibrous capsule around an implanted device, especially a breast implant. It may cause pain, distortion, firmness, or displacement.

Capsule Endoscopy

- Non-invasive diagnostic technique using a swallowable wireless endoscopic device to capture images of the small intestinal mucosa. The capsule contains a miniature video camera and transmitter that sends images to a receiver worn on the patient's belt.

Capsule of Glisson

- The connective-tissue capsule surrounding the liver, which sends fibrous septa inward along portal tracts and supports the organ's vessels, ducts, and nerves.

Capsule of Tenon

- A fascial sheath surrounding the globe of the eye and separating it from orbital fat. It contributes to eye movement and forms a potential space used in some ophthalmic procedures.

Capsulectomy

- Surgical removal of part or all of a fibrous capsule surrounding an organ, implant, or other structure. The indication depends on contracture, infection, pathology, or the need to revise a reconstruction.

Caput Medusae

- Dilated periumbilical veins radiating outward from the umbilicus, seen in portal hypertension with recanalization of the umbilical vein. Not strictly a cardiac finding but relevant in right heart failure causing hepatic congestion and portal hypertension. It must be distinguished from caval collateral circulation.

Caput Succedaneum

- Edema of the scalp presenting as a soft, boggy swelling that crosses suture lines. Caused by pressure of the vaginal walls on the fetal head during labor. Resolves within hours to days. Differentiated from cephalohematoma (subperiosteal, does not cross suture lines) and subgaleal hemorrhage (life-threatening, crosses suture lines).

CAR T-Cell

- Chimeric antigen receptor T-cells. Engineered immune cells targeting cancer.

Carbapenem-resistant organisms

- Bacteria that are resistant to carbapenem antibiotics, typically through production of carbapenemases. The most important carbapenemases include KPC (*Klebsiella pneumoniae* carbapenemase), NDM (New Delhi metallo-beta-lactamase), and OXA-48-like enzymes. CRO are a major public health threat due to limited treatment options.

Carbidopa

- A peripheral dopa decarboxylase inhibitor that does not cross the blood-brain barrier; combined with levodopa to prevent its peripheral conversion to dopamine, reducing side effects (nausea, hypotension) and increasing CNS levodopa delivery.

Carbohydrate

- A class of organic compounds composed of carbon, hydrogen, and oxygen (general formula $C_m(H_2O)_n$), serving as the primary energy source for the body.

Carbohydrate Counting

- Method for diabetes management. Matching insulin to carbohydrate intake. Education essential.

Carbolic Acid (Phenol)

- An aromatic organic compound (C_6H_5OH) with antiseptic, disinfectant, and caustic properties. Historically significant as the first surgical antiseptic introduced by Joseph Lister in 1867, dramatically re-

ducing post-operative infection and mortality. Mechanism: denatures bacterial proteins and disrupts cell membranes.

Carbon Dioxide Absorber

- A safety device used in anesthesia breathing circuits that filters out carbon dioxide from a patient's exhaled breath, allowing the remaining anesthetic gases to be safely recycled and rebreathed.

Carbon Dioxide Transport

- The movement of CO_2 from tissues to lungs. CO_2 is carried in blood in three forms: dissolved (approximately 10 percent), bound to hemoglobin as carbamino compounds (approximately 20 percent), and as bicarbonate (approximately 70 percent).

Carbon Ion Therapy

- Radiation using carbon ions. Higher biological effectiveness. Limited availability.

Carbon Monoxide (CO) Poisoning

- Inhibits hemoglobin oxygen binding (affinity 200x oxygen). Causes headache, dizziness, nausea, cherry-red skin, syncope, death. Diagnosis: carboxyhemoglobin level (not routine). Treatment: 100% oxygen, hyperbaric oxygen if severe (neurological symptoms, $COHb > 25\%$, pregnancy).

Carbon Monoxide Poisoning

- Toxicity from inhalation of carbon monoxide (CO), a colorless, odorless gas binding hemoglobin 200-300 times more avidly than oxygen, forming carboxyhemoglobin (COHb) and reducing oxygen delivery. Sources: fires, faulty heaters, engine exhaust, suicide by car exhaust.

Carbonic Anhydrase Inhibitors

- A class of medications used in the treatment of glaucoma that lower intraocular pressure by reducing the production of aqueous humor through inhibition of the enzyme carbonic anhydrase, which catalyzes the conversion of carbon dioxide and water to bicarbonate and hydrogen ions, a process essential for the active secretion of bicarbonate i

Carboxyhemoglobin

- Toxicity from inhalation of carbon monoxide (CO), a colorless, odorless gas binding hemoglobin 200-300 times more avidly than oxygen, forming carboxyhemoglobin (COHb) and reducing oxygen delivery. See carbon monoxide poisoning.

Carbuncle

- A deep, coalescent infection of multiple adjacent hair follicles by *Staphylococcus aureus* (usually MRSA), forming a larger, swollen, erythematous, painful mass with multiple draining sinuses, most often on the neck, back, or thighs; more serious than a furuncle, commonly with fever, malaise, and systemic infection.

Carcinogen

- Substance causing cancer. Types: chemical (benzene, asbestos, aflatoxin), physical (radiation), biological (HPV, HBV, HCV). Mechanisms: DNA damage, mutations, epigenetic changes. IARC classifications: Group 1 (carcinogenic), Group 2A (probably carcinogenic), Group 2B (possibly carcinogenic), Group 3 (not classifiable).

Carcinoid Heart Disease

- Cardiac complication of carcinoid syndrome occurring in approximately fifty percent of patients with hepatic metastases. Serotonin and other vasoactive substances bypass hepatic metabolism and cause endocardial fibrosis affecting the right-sided heart valves particularly the tricuspid and pulmonary valves.

Carcinoid Syndrome

- Serotonin-secreting neuroendocrine tumor causing flushing, diarrhea, bronchospasm, and right-sided heart disease. Occurs when vasoactive substances bypass hepatic metabolism through venous drainage or liver metastases. Diagnosis is with twenty-four hour urine 5-hydroxyindoleacetic acid or serum chromogranin A.

Carcinoid Tumor

- Carcinoid tumors are well-differentiated neuroendocrine neoplasms that arise from enterochromaffin cells throughout the gastrointestinal tract and bronchial system. The most common primary sites are the appendix, small intestine, and rectum.

Carcinoma

- Malignant tumor of epithelial origin. Types: squamous cell carcinoma, adenocarcinoma, basal cell carcinoma, transitional cell carcinoma. Most common cancer type (80-90% of malignancies). Examples: lung carcinoma, breast carcinoma, colon carcinoma.

Carcinoma en cuirasse

- An unusual form of breast cancer characterized by diffuse, firm, indurated skin that resembles armor. It results from extensive dermal lymphatic invasion by tumor cells.

Carcinoma in Situ

- Full-thickness dysplasia without invasion through basement membrane. Stage 0 cancer.

Carcinosarcoma

- A malignant tumor containing both carcinomatous and sarcomatous elements, most commonly of the uterus (malignant mixed Müllerian tumor), with an aggressive course.

Cardiac allograft

- A donor heart transplanted into a recipient with end-stage heart failure. The procedure requires life-long immunosuppression to prevent rejection. Complications include rejection, cardiac allograft vasculopathy, infection, and malignancy; monitored by endomyocardial biopsy and echocardiography.

Cardiac Amyloidosis

- Deposition of amyloid fibrils in the myocardium, causing restrictive cardiomyopathy with thickened, stiff ventricular walls. The most common cardiac amyloidosis is due to transthyretin (ATTR) amyloidosis, which can be wild-type (age-related) or hereditary.

Cardiac Amyloidosis Diagnosis

- Diagnosis involves a combination of clinical suspicion, imaging, and tissue biopsy. Non-invasive diagnosis using Tc-99m pyrophosphate (PYP) or DPD scintigraphy for ATTR cardiac amyloidosis (with negative serum/urine free light chains to exclude AL amyloidosis).

Cardiac Amyloidosis Staging

- Mayo Clinic staging system for ATTR cardiac amyloidosis: Stage I (NT-proBNP <3000 pg/mL), Stage II (NT-proBNP 3000-8499), Stage III (NT-proBNP ≥8500). Median survival: Stage I: 69 months, Stage II: 47 months, Stage III: 14 months. Treatment with tafamidis (TTR stabilizer) improves survival and functional capacity.

Cardiac Arrest

- Sudden cessation of effective cardiac function, resulting in loss of pulse and consciousness. Most

common rhythm in adults is ventricular fibrillation or pulseless ventricular tachycardia. Treatment follows ACLS protocol: high-quality CPR, defibrillation for shockable rhythms, and ACLS medications (epinephrine, amiodarone).

Cardiac Arrest Survival

- Overall survival to discharge for out-of-hospital cardiac arrest is approximately 10%. For witnessed VF/pVT with early defibrillation, survival can exceed 50%. Chain of survival: early recognition and activation of EMS, high-quality CPR, early defibrillation, advanced life support, and post-arrest care.

Cardiac Axis

- The mean direction of ventricular depolarization in the frontal plane, estimated from the QRS complex on a 12-lead ECG. Left or right axis deviation may reflect conduction disease, ventricular hypertrophy, infarction, or altered cardiac position.

Cardiac Biomarkers

- Blood measurements used to assess myocardial injury. Cardiac troponins I and T are the preferred specific biomarkers; CK-MB may assist in selected situations, while myoglobin is sensitive but nonspecific. Results must be interpreted with symptoms, electrocardiography, and serial testing.

Cardiac Catheterization

- Invasive procedure involving insertion of catheters into the heart via peripheral arteries or veins. Diagnostic catheterization measures intracardiac pressures, saturations, and performs coronary angiography. Indications include evaluation of CAD, valvular disease, cardiomyopathy, and congenital heart disease.

Cardiac Conduction System

- The specialized tissue responsible for generating and conducting electrical impulses through the heart. Components include the sinoatrial (SA) node, internodal pathways, atrioventricular (AV) node, bundle of His, right and left bundle branches, and Purkinje fibers.

Cardiac CT

- CT imaging of heart and coronary arteries. Coronary calcium scoring, coronary CT angiography, cardiac function assessment.

Cardiac CT Angiography

- Noninvasive imaging using CT to visualize coronary arteries and cardiac structures. It has high negative predictive value for excluding coronary artery disease, making it useful for low-to-intermediate risk patients with chest pain. Calcium scoring quantifies coronary artery calcification as a marker of atherosclerosis.

Cardiac Cycle

- The sequence of events occurring during one complete heartbeat, consisting of systole (contraction) and diastole (relaxation). During diastole, the ventricles fill with blood; during systole, they contract and eject blood.

Cardiac Defibrillation

- Delivery of an unsynchronized electrical shock for ventricular fibrillation or pulseless ventricular tachycardia. It is combined with high-quality CPR and advanced life support according to the cardiac-arrest rhythm and protocol.

Cardiac Device Infections

- CIED (pacemaker, ICD) infection: pocket infection (erythema, warmth, purulence, erosion), lead infection (bacteremia, endocarditis). Diagnosis: blood cultures, TEE (lead vegetations), pocket culture. Treatment: complete system removal (leads + generator) - CLASS I.

Cardiac Diet

- A dietary pattern emphasizing vegetables, fruits, whole grains, legumes, fish, unsaturated fats, and limited sodium, added sugars, and saturated or trans fats. It supports cardiovascular risk reduction but should be individualized for heart failure, kidney disease, diabetes, and nutritional adequacy.

Cardiac Gated SPECT

- SPECT imaging synchronized with ECG. Assess ventricular function, wall motion, viability.

Cardiac Index

- Cardiac output normalized to body surface area (BSA), calculated as CO / BSA . Normal cardiac index is 2.5-4.2 L/min/m². It provides a more accurate assessment of cardiac function than cardiac output alone, accounting for body size. A cardiac index below 2.2 L/min/m² indicates cardiogenic shock.

Cardiac Metastases

- More common than primary cardiac tumors. Most frequent sources: lung cancer, breast cancer, melanoma, lymphoma, and renal cell carcinoma. Most commonly involve the pericardium (effusion, tamponade), but can affect myocardium and endocardium.

Cardiac MRI

- Magnetic-resonance imaging of the heart that characterizes ventricular function, myocardial tissue, perfusion, scar, edema, infiltration, and congenital anatomy without ionizing radiation. Gadolinium contrast is avoided or used cautiously in selected patients with severe renal dysfunction.

Cardiac Muscle

- Involuntary, striated muscle forming the myocardium. Cardiac muscle cells are electrically connected by intercalated discs, allowing coordinated contraction of the heart.

Cardiac Output

- The volume of blood pumped by the heart per minute, calculated as the heart rate multiplied by the stroke volume.

Cardiac Output Measurement

- Methods include: thermodilution via pulmonary artery catheter (bolus of cold saline and temperature change measurement), Fick principle (O_2 consumption / arteriovenous O_2 difference), echocardiographic Doppler (flow velocity time integral $\times$ valve area $\times$ heart rate), and non-invasive cardiac output monitors (bioimpedance, bioreactance).

Cardiac Pacemaker Types

- Single-chamber: pacing one chamber (atrium or ventricle). Dual-chamber (DDD): paces and senses both atrium and ventricle, maintaining AV synchrony. Biventricular (CRT): paces both ventricles for resynchronization in heart failure.

Cardiac PET

- Viability, perfusion, amyloidosis. Quantitative perfusion measurement.

Cardiac PET Imaging

- Positron emission tomography using radiotracers ($Rb-82$, $N-13$ ammonia, $O-15$ water) for quantitative myocardial perfusion assessment. Advantages over SPECT: superior resolution, shorter scan time, at-

tenuation correction, and absolute myocardial blood flow quantification (mL/g/min).

Cardiac Radiation Injury

- Late complication of mediastinal radiation therapy (Hodgkin lymphoma, breast cancer, lung cancer). Manifestations include pericardial disease (constrictive pericarditis, effusion), coronary artery disease (accelerated atherosclerosis), valvular disease (fibrosis), myocardial fibrosis, and conduction abnormalities.

Cardiac Rehab Outcomes

- Cardiac rehabilitation reduces all-cause mortality by 13% and cardiovascular mortality by 26% post-MI. It improves exercise capacity, lipid profiles, blood pressure control, and glycemic control. Psychosocial benefits include reduced depression and anxiety. Despite Class I recommendation, only 20-30% of eligible patients participate.

Cardiac Rehabilitation

- Supervised program for cardiac patients post-MI, PCI, CABG, heart failure. Phases: inpatient (early mobilization), outpatient (supervised exercise, education), maintenance. Components: exercise training, risk factor modification, psychological support, return to work counseling. Reduces mortality, hospitalizations.

Cardiac Resynchronization Therapy

- Biventricular pacing that synchronizes contraction of the left and right ventricles in patients with heart failure, reduced EF, and prolonged QRS duration (especially LBBB). CRT improves symptoms, exercise capacity, and quality of life, and reduces heart failure hospitalizations and mortality.

Cardiac Resynchronization Therapy (CRT)

- Biventricular pacing for heart failure with dyssynchrony. Paces both ventricles simultaneously. Indications: LBBB, QRS >150 ms, EF $<35\%$. Improves symptoms, reduces hospitalizations, mortality.

Cardiac Sarcoidosis

- Infiltrative disease characterized by non-caseating granulomas in the myocardium, affecting 25-30% of patients with systemic sarcoidosis. Presentation includes conduction abnormalities (AV block), ventricular arrhythmias, and heart failure.

Cardiac Skeleton

- A dense connective tissue framework composed of four fibrous rings that surround the heart valves and provide structural support. The cardiac skeleton includes the annuli fibrosi (valve rings), the membranous septum, and the left and right fibrous trigones.

Cardiac SPECT

- Myocardial perfusion imaging using SPECT. Quantifies ischemia, infarction, viability. Stress-rest protocol. gated SPECT assesses ventricular function.

Cardiac Stress Test

- Evaluation of cardiac function and ischemia under increased cardiac workload. Exercise stress test (treadmill, Bruce protocol): ECG monitoring for ST-segment changes, symptoms, BP response, and arrhythmias. Sensitivity 65-75% for CAD.

Cardiac Stress Testing

- Assessment of cardiac function under physiological or pharmacological stress to detect ischemia. Exercise stress testing involves treadmill or bicycle exercise with continuous ECG monitoring. Pharmacological stress (dobutamine, adenosine, regadenoson) is used when patients cannot exercise.

Cardiac tamponade

- See Cardiac Tamponade Physiology.

Cardiac Tamponade Physiology

- Accumulation of pericardial fluid causing compression of cardiac chambers, impairing diastolic filling and reducing cardiac output. Hemodynamic consequences include equalization of diastolic pressures ($RA = RV \text{ diastolic} = PCWP$), pulsus paradoxus, and decreased cardiac output.

Cardiac Tumors

- Primary cardiac tumors are rare; most common benign is myxoma (75%), typically in the left atrium presenting with embolism, obstruction, or constitutional symptoms. Malignant primary cardiac tumors are rare (angiosarcoma most common).

Cardiac Tumors Management

- Myxoma: surgical excision is curative, with low recurrence rate. Benign tumors (rhabdomyoma, fibroma) may be observed if asymptomatic but require surgery if causing obstruction or arrhythmia. Malignant tumors: prognosis is poor; treatment

may include surgery, chemotherapy, and radiation depending on tumor type.

Cardio-Oncology

- The subspecialty focusing on cardiovascular care of cancer patients, addressing cancer therapy-related cardiac dysfunction, prevention, and management. Key concerns include anthracycline cardiotoxicity, trastuzumab-related cardiomyopathy, immune checkpoint inhibitor myocarditis, and radiation-induced heart disease.

Cardiogenic Shock

- A state of inadequate systemic tissue perfusion due to primary cardiac dysfunction, defined hemodynamically as a persistent systolic blood pressure <90 mmHg, cardiac index <2.2 L/min/m², and elevated pulmonary capillary wedge pressure (PCWP >15 mmHg).

Cardiogenic Shock Management

- Low cardiac output state with end-organ hypoperfusion, typically from acute MI. Inotropes: dobutamine (increases cardiac output, decreases SVR), milrinone (PDE3 inhibitor, vasodilator). Vasopressors: norepinephrine (first-line for hypotension), dopamine.

Cardiology

- The branch of medicine devoted to the study, diagnosis, and treatment of disorders of the heart and circulatory system. Cardiologists manage conditions including coronary artery disease, heart failure, valvular heart disease, arrhythmias, cardiomyopathy, pericardial disease, congenital heart disease, and hypertension.

Cardiomegaly

- Abnormal enlargement of the heart, identified clinically, on chest radiography, or with cardiac imaging. It may reflect chamber dilatation, myocardial hypertrophy, pericardial effusion, or a combination of these; causes include hypertension, valvular disease, cardiomyopathy, congenital heart disease, and high-output states.

Cardiomyopathy

- Diseases of the myocardium that result in cardiac dysfunction, excluding coronary artery disease, valvular disease, and congenital heart disease as primary causes. The three main types are dilated,

hypertrophic, and restrictive cardiomyopathy.

Cardiomyopathy Classification

- The MOGE(S) classification system provides phenotypic and genotypic description: M (morpho-functional phenotype), O (organ/system involvement), G (genetic inheritance), E (etiology), S (functional status/NYHA class).

Cardioplegia

- Deliberate temporary arrest and protection of the heart during cardiac surgery, usually achieved with a cold or blood-based solution that alters myocardial electrical and metabolic activity.

Cardiopulmonary Exercise Testing

- Comprehensive assessment of cardiovascular and pulmonary function during graded exercise on a treadmill or cycle ergometer. Measures: VO₂ max, anaerobic threshold, ventilatory efficiency (VE/VCO₂ slope), and hemodynamic responses.

Cardiopulmonary Exercise Testing (CPET)

- Comprehensive exercise test with breath-by-breath analysis of respiratory gases (VO₂, VCO₂, VE). Indications: unexplained dyspnea, pre-operative risk assessment (lung resection, cardiac surgery), evaluation of exercise tolerance in heart failure (peak VO₂ predicts prognosis and guides transplant listing), and differentiating cardiac vs.

Cardiopulmonary Resuscitation

- A sequence of chest compressions, ventilation, defibrillation when indicated, and advanced life-support interventions for cardiac arrest. Its purpose is to maintain circulation and oxygen delivery until spontaneous circulation is restored or definitive treatment is provided.

Cardiopulmonary Resuscitation (CPR)

- Emergency procedure combining chest compressions and rescue breaths to maintain circulation and oxygenation during cardiac arrest. Compression depth: 2-2.4 inches (5-6 cm) in adults. Rate: 100-120/min. Allow full chest recoil. Minimize interruptions.

Cardiorenal Syndrome

- Dysfunction of the heart and kidneys where acute or chronic failure of one organ contributes to dysfunction of the other. Type 1: acute cardiac dysfunction

causing acute kidney injury (e.g., cardiogenic shock). Type 2: chronic cardiac dysfunction causing progressive CKD. Type 3: acute kidney injury causing cardiac dysfunction.

Cardioselective Beta-Blockers

- Cardioselective beta-blockers selectively block beta-1 adrenergic receptors, which are predominant in the heart, while having minimal effects on beta-2 receptors in the bronchial smooth muscle and peripheral vasculature at therapeutic doses. Common cardioselective beta-blockers include metoprolol, atenolol, bisoprolol, and nebivolol.

Cardiospasm

- Spasm or impaired relaxation of the lower esophageal sphincter, historically used as a term for achalasia. It causes dysphagia, regurgitation, and sometimes chest discomfort.

Cardiotocography (CTG)

- A continuous electronic recording of the fetal heart rate and uterine contractions used to assess fetal well-being during pregnancy and labor. The CTG machine uses two transducers placed on the maternal abdomen: an ultrasound transducer records the fetal heart rate, and a tocodynamometer records uterine contractions.

Cardiotoxicity

- Injury or functional impairment of the heart caused by a toxic exposure, systemic disease, or treatment. Manifestations include arrhythmia, myocardial dysfunction, ischemia, or heart failure.

Cardiotoxicity from Chemotherapy

- Chemotherapy agents that damage the myocardium, most commonly anthracyclines (doxorubicin, daunorubicin). Cardiotoxicity presents as dilated cardiomyopathy with reduced ejection fraction. Risk is dose-dependent (doxorubicin cumulative dose >550 mg/m²).

Cardiovascular Changes in Pregnancy

- Plasma volume increases 40-50% (total volume 5 L), red cell mass increases 20-30% (physiologic anemia of pregnancy due to dilution). Cardiac output increases 30-50% by 32 weeks (stroke volume increases initially, then heart rate).

Cardiovascular disease

- Disease affecting the heart or blood vessels, includ-

ing coronary artery disease, heart failure, arrhythmias, valvular disease, and vascular disorders.

Cardiovascular Health

- Cardio is shorthand for cardiovascular health or exercise that elevates heart rate and respiratory effort to strengthen the heart, lungs, and circulatory system. Aerobic activities such as walking, running, cycling, and swimming improve cardiac output, lower blood pressure, and enhance metabolic efficiency.

Cardiovascular Screening

- See cardiology.

Cardiovascular System

- The organ system comprising the heart and blood vessels (arteries, arterioles, capillaries, venules, veins) responsible for circulating blood throughout the body.

Cardioversion

- Restoration of a normal cardiac rhythm from an arrhythmia by synchronized electrical shock or, in some contexts, a planned pharmacologic intervention. Synchronized shock avoids delivery during vulnerable ventricular repolarization.

Cardioversion (Synchronized)

- Delivery of a low-energy electrical shock synchronized to the QRS complex to terminate arrhythmias. Synchronization: shock delivered on the R wave (not T wave, which induces VF). Energy: atrial fibrillation (100-200 J biphasic), atrial flutter (50-100 J), SVT (50-100 J), stable VT (100 J). Success: 70-90% for AF, >95% for atrial flutter.

Carditis

- Inflammation of the heart or one of its layers, including the myocardium, endocardium, or pericardium. It may result from infection, immune-mediated disease, or systemic inflammation.

Care Coordination

- Deliberate organization and communication of healthcare activities among professionals, services, patients, and caregivers to support safe, continuous, and appropriate care across settings.

Caregiver

- Person providing care to elderly. Can be family, friend, professional. Burnout common.

Caregiver Burden

- Emotional, physical, financial strain from caregiving. See caregiver stress.

Caregiver Stress

- Emotional, physical, financial strain from caregiving. Screening and support essential.

Caries

- Progressive destruction of calcified tooth tissue caused by acid-producing microbial biofilm acting on fermentable carbohydrates. Untreated disease can extend from enamel into dentin and pulp.

Caries (Dental)

- A biofilm-mediated, sugar-driven, multifactorial disease causing net demineralization of enamel, dentin, or cementum. Risk is influenced by fermentable-carbohydrate frequency, fluoride exposure, saliva, oral hygiene, tooth anatomy, and previous disease.

Caries Risk Assessment

- Systematic evaluation of an individual's likelihood of developing dental caries. Low risk: no active caries, good oral hygiene, regular fluoride exposure. Moderate risk: one or more active lesions in past 3 years, fair oral hygiene.

Carina

- The ridge at the base of the trachea where it bifurcates into the right and left main bronchi. It is a landmark for endotracheal tube placement and bronchoscopy. The right main bronchus is wider, shorter, and more vertical than the left, making it the preferred path for aspirated foreign bodies.

Cariogenic

- Capable of promoting dental caries, especially by producing acid from fermentable carbohydrates or supporting a plaque environment that demineralizes tooth structure.

Cariogram

- A computer-based program that illustrates the interaction of caries-related factors and provides an overall caries risk assessment. Factors: diet (content, frequency), bacteria (mutans streptococci, lactobacilli), fluoride exposure, saliva (flow rate, buffering capacity), plaque amount, and clinical/social history.

Carney Complex

- Autosomal dominant multiple endocrine neoplasia syndrome (NAME syndrome / LAMB syndrome), caused by PRKAR1A (17q22-24, protein kinase A regulatory subunit type 1A, 70% of cases) or PRKACA locus.

Carnitine

- Transports long-chain fatty acids into mitochondria for beta-oxidation. Deficiency causes hypoketotic hypoglycemia.

Caroli Disease

- Congenital, autosomal recessive fibropolycystic liver disease (PKHD1 gene, encodes fibrocystin/polyductin; same gene as infantile polycystic kidney disease) characterized by non-obstructive, segmental, saccular dilation of the larger intrahepatic bile ducts.

Carotid Artery

- The paired arterial system supplying the head and neck. Each common carotid artery divides into internal and external carotid arteries; the internal carotid supplies the brain and orbit, while the external carotid supplies most external head and neck structures.

Carotid Artery Disease

- Narrowing or blockage of the carotid arteries, most often caused by atherosclerotic plaque. It can reduce cerebral blood flow or release emboli and is an important cause of transient ischemic attack and ischemic stroke.

Carotid Artery Stenting

- Endovascular placement of a stent across a carotid stenosis, usually with embolic-protection strategies, to improve cerebral blood flow and reduce stroke risk in selected patients. Antiplatelet therapy and peri-procedural neurologic monitoring are required.

Carotid Bruit

- An audible systolic (sometimes continuous) murmur over the carotid artery due to turbulent flow, most commonly from atherosclerotic carotid stenosis. A carotid bruit raises concern for significant stenosis (≥ 50 percent), which confers increased stroke risk.

Carotid cavernous fistula

- An abnormal communication between the carotid

artery or its branches and the cavernous sinus, most commonly traumatic but also spontaneous. Clinical features include pulsatile proptosis, chemosis, bruit over the orbit, and cranial nerve palsies. Diagnosis is by CT or MR angiography.

Carotid cavernous sinus fistula

- See Carotid cavernous fistula.

Carotid Duplex

- Ultrasound combining B-mode and Doppler imaging of carotid arteries. Assesses stenosis, plaque characterization.

Carotid Duplex Ultrasound

- Ultrasound evaluation of the carotid arteries for atherosclerotic plaque and stenosis. Measures peak systolic velocity (PSV) and end-diastolic velocity (EDV) in the internal carotid artery (ICA).

Carotid Endarterectomy

- Surgical removal of atherosclerotic plaque from the carotid bifurcation to prevent stroke. Indications: symptomatic carotid stenosis 50% (NASCET criteria: TIA or non-disabling stroke) and asymptomatic stenosis 70% (if perioperative risk $< 3\%$). Pre-operative: CTA or MRA, antiplatelet therapy.

Carotid Stenosis

- Narrowing of carotid artery, usually from atherosclerosis. Risk: ischemic stroke. Asymptomatic: screening by auscultation (bruit), ultrasound if risk factors. Symptomatic: TIA or stroke. Treatment: carotid endarterectomy (CEA) or carotid artery stenting (CAS) if $> 70\%$ stenosis, performed within 2 weeks of symptoms.

Carpal Tunnel

- Narrow passageway on the palmar wrist, bounded by carpal bones dorsally and the flexor retinaculum as the roof. Transmits the median nerve and flexor tendons. Carpal tunnel syndrome causes numbness in the lateral 3.5 digits and thenar weakness.

Carpal Tunnel Syndrome

- The most common entrapment neuropathy, caused by compression of the median nerve as it passes through the carpal tunnel, a narrow fibro-osseous canal bounded by the carpal bones and the flexor retinaculum (transverse carpal ligament) at the wrist.

Carpal Tunnel Syndrome Tests

- Provocative tests for diagnosing carpal tunnel syndrome. Tinel sign: percuss over median nerve at the carpal tunnel tingling in median distribution. Phalen sign: hold wrists in full flexion for 60 seconds reproduces symptoms. Durkan test (carpal compression): direct compression over carpal tunnel for 30 seconds paresthesias.

Carrier

- Individual heterozygous for a recessive genetic disorder. Typically asymptomatic but can pass mutant allele to offspring. Carrier screening recommended for certain populations (e.g., Ashkenazi Jewish for Tay-Sachs).

Carrier State

- Asymptomatic infection with pathogen capable of transmission. Examples: typhoid carrier, hepatitis B carrier, meningococcal carrier. Importance: public health control, screening, treatment.

Cartilage

- A tough, flexible avascular connective tissue found in joints, the ear, nose, rib cage, and intervertebral discs. Composed of chondrocytes embedded in an extracellular matrix of collagen fibers, proteoglycans, and water, providing resilience and compressive strength.

Carvedilol

- A nonselective beta-blocker with additional alpha-1 adrenergic blockade and antioxidant properties, producing vasodilation with balanced reduction of heart rate, contractility, and afterload.

Case Definition

- Standardized clinical, laboratory, and epidemiologic criteria used to classify a person as a suspected, probable, or confirmed case for surveillance or outbreak investigation. It may differ from the criteria used for individual clinical diagnosis.

Case Fatality Rate

- Proportion of cases dying from disease. Measures severity.

Case Management

- See geriatrics.

Case-Control Study

- Observational study comparing cases with disease

to controls without. Looks back at exposures.

Case-Fatality Rate

- Proportion of cases dying from disease. Formula: deaths / cases. Measures severity.

Caseous Necrosis

- A distinctive form of cell death that characteristically occurs in granulomatous infections, most notably tuberculosis. The term derives from the cheese-like, friable, whitish-yellow appearance of the necrotic tissue on gross examination.

Caspofungin

- An echinocandin antifungal agent used in the treatment of invasive candidiasis, esophageal candidiasis, and invasive aspergillosis (as salvage therapy). Caspofungin inhibits beta-(1,3)-D-glucan synthase, disrupting fungal cell wall synthesis.

Castleman Disease

- Heterogeneous group of lymphoproliferative disorders (angiofollicular lymph node hyperplasia, giant lymph node hyperplasia) sharing lymph node histopathology.

Cat Scratch Disease

- A regional lymphadenitis caused by *Bartonella henselae*, a gramnegative bacillus transmitted to humans through cat scratches, bites, or licks on broken skin, particularly from kittens (which carry higher bacterial loads).

Cat-Scratch Disease

- An infection caused by *Bartonella henselae*, usually transmitted by a scratch or bite from an infected cat. It commonly causes a papule followed by regional lymph-node enlargement, fever, and fatigue; atypical disease may affect the eye, liver, spleen, or nervous system.

Cat-Scratch Fever

- See Cat Scratch Disease.

Catabolism

- The set of metabolic pathways that break down complex molecules into simpler units, releasing energy captured as ATP, NADH, FADH₂, NADPH. Examples: glycolysis, beta-oxidation, proteolysis, glycogenolysis, nucleotide catabolism.

Catalepsy

- See psychiatry.

Catalyst

- A substance that increases the rate of a chemical reaction without being consumed, by lowering the activation energy (providing an alternative reaction pathway). Enzymes are biological catalysts (proteins or ribozymes). Catalysts do not alter the reaction equilibrium (G), only the kinetics.

Cataract

- A cataract is an opacity or clouding of the crystalline lens that impairs the transmission of light to the retina, resulting in blurred vision, glare, decreased contrast sensitivity, and progressive vision loss, and it is the leading cause of preventable blindness worldwide.

Cataracts

- A clouding of the eye's natural lens that develops gradually with aging, causing blurred vision, glare sensitivity, faded colors, and difficulty seeing at night. They are the leading cause of vision loss worldwide but are treatable with surgical removal of the cloudy lens and replacement with an artificial intraocular lens.

Catastrophic Aging

- Rapid functional decline after stressor (illness, surgery, fall). Elderly have decreased reserve, less ability to recover. Prevention: comprehensive assessment, optimization, early intervention.

Catatonia

- A neuropsychiatric syndrome characterized by motor abnormalities including immobility, mutism, posturing, waxy flexibility, stereotypy, mannerisms, grimacing, echolalia, and echopraxia. It can occur in mood disorders, psychotic disorders, medical conditions, or as catatonia due to another medical condition.

Catch-Up Immunization

- Vaccination of a person who is behind the recommended schedule using age- and risk-appropriate intervals. Doses generally do not need to be restarted because of delay, but minimum ages and intervals must be respected.

Catecholamine Crisis

- Acute life-threatening elevation of catecholamines causing severe hypertension, tachycardia, and multi-organ dysfunction. Occurs in pheochromocytoma and paraganglioma precipitated by anesthesia,

surgery, trauma, or medications that stimulate catecholamine release.

Catecholamines (Urine/Plasma)

- Measurement of catecholamines (epinephrine, norepinephrine, dopamine) and their metabolites (metanephrines, VMA) for diagnosis of pheochromocytoma and paraganglioma. Plasma free metanephrines: sensitivity 96-100%. 24-hour urinary fractionated metanephrines: sensitivity 90-95%.

Category I Fetal Heart Tracing

- A normal fetal heart rate tracing by NICHD 2008 criteria. All three characteristics present: baseline 110-160 bpm, moderate variability (6-25 bpm), and no late or variable decelerations. May have accelerations. Category I tracings are associated with normal fetal acid-base status.

Category II Fetal Heart Tracing

- An indeterminate fetal heart rate tracing that falls between Category I and III.

Category III Fetal Heart Tracing

- An abnormal fetal heart rate tracing indicating potential fetal hypoxia or acidosis. Two specific patterns: (1) absent baseline variability with recurrent late decelerations, (2) absent baseline variability with recurrent variable decelerations, and (3) sinusoidal pattern.

Catheter

- Flexible tube for drainage or infusion. Types: urinary (Foley), intravenous, central venous, arterial. Materials: silicone, latex, PVC.

Catheter-Associated UTI (CAUTI)

- Urinary tract infection related to indwelling urinary catheter. Most common healthcare-associated infection. Risk: prolonged catheterization, female sex, diabetes, immunosuppression. Prevention: avoid unnecessary catheters, aseptic insertion, closed drainage, daily hygiene, early removal.

Catheter-Related Infections

- Infections caused by microorganisms entering along or contaminating an intravascular or other indwelling catheter. They may cause local inflammation or bloodstream infection; removal or replacement and antimicrobial treatment depend on the

catheter and severity.

Catheterization (Urinary)

- Insertion of a flexible tube (catheter) through the urethra or suprapubically into the urinary bladder to drain urine. Types: Foley catheter (indwelling balloon catheter), intermittent (in-out) catheter, and suprapubic catheter.

Cauda Equina Syndrome

- A neurosurgical emergency caused by compression of the cauda equina nerve roots below the level of the conus medullaris (typically at L1-L2 and below), resulting in bilateral lower extremity weakness, sensory loss, and bladder and bowel dysfunction.

Caudal

- Positioned toward the tail or lower part of the body.

Cause of Death

- The specific disease, injury, or toxic agent responsible for initiating the lethal physiological sequence leading directly to death (e.g., Gunshot wound to the head, Acute myocardial infarction).

Cavernous Sinus Thrombosis

- Life-threatening septic thrombosis of the cavernous sinus, usually a complication of mid-face, nasal, paranasal sinus (sphenoid, ethmoid), dental, or orbital infection spreading through the valveless ophthalmic veins.

CD4 Count

- Absolute count of CD4+ T-helper lymphocytes, essential for monitoring HIV disease progression and treatment response. Normal: 500-1500 cells/uL. HIV staging: >500 (asymptomatic, start ART regardless), 200-500 (consider prophylaxis for some OIs), <200 (AIDS-defining: start PCP prophylaxis, TB screening, high risk of OIs).

CD4-Positive T Cell

- A T lymphocyte that recognizes peptide presented by major histocompatibility complex class II molecules. Its differentiated subsets coordinate immune responses, including macrophage activation, B-cell help, mucosal defense, and immune regulation.

CD8-Positive T Cell

- A T lymphocyte that recognizes peptide presented by major histocompatibility complex class

I molecules. Cytotoxic CD8-positive cells can induce apoptosis in infected, malignant, or otherwise altered target cells.

CEA (Carcinoembryonic Antigen)

- A tumor marker primarily for colorectal cancer. Normal: <3 ng/mL (non-smokers), <5 ng/mL (smokers). Uses: monitoring treatment response in metastatic colorectal cancer, surveillance for recurrence after curative resection (q3-6 months for 2 years, then q6-12 months for 5 years), and prognosis (preoperative level).

Cecal volvulus

- Axial twisting of the cecum and ascending colon around its mesentery, accounting for approximately one-third of colonic volvulus cases. It presents with acute or subacute large bowel obstruction, abdominal pain, and distension. Diagnosis is by CT scan showing the characteristic whirl sign and a massively dilated cecum.

Cecum

- First and widest large intestinal part in the right iliac fossa below the ileocecal valve, a blind-ended pouch approximately 6 cm long. Appendix arises from its posteromedial wall approximately 2 cm below the valve, containing abundant lymphoid tissue. Cecal volvulus causes large bowel obstruction.

Ceftazidime-avibactam

- A combination of the third-generation cephalosporin ceftazidime with the non-beta-lactam beta-lactamase inhibitor avibactam. Avibactam inhibits class A, class C, and some class D beta-lactamases, including KPC carbapenemases. This combination has activity against many carbapenem-resistant Enterobacteriaceae and *Pseudomonas aeruginosa*.

Ceftriaxone

- A third-generation cephalosporin antibiotic given by injection for serious bacterial infections, including pneumonia, meningitis, sepsis, and gonorrhea. It is not effective against viruses; allergic reactions and antibiotic-associated diarrhea are important risks.

Celecoxib

- A selective cyclooxygenase-2 (COX-2) inhibitor used to relieve pain and inflammation in arthritis and other conditions. It can increase cardiovascular

and gastrointestinal risks, especially at high doses or with prolonged use.

Celiac Artery

- The first major ventral branch of the abdominal aorta at T12, arising from the aorta just below the diaphragm and supplying the foregut (stomach, proximal duodenum, liver, gallbladder, spleen, pancreas). It trifurcates into the left gastric, splenic, and common hepatic arteries, with rich collateral circulation between these branches.

Celiac Crisis

- Acute life-threatening presentation of celiac disease characterized by severe dehydration, electrolyte imbalances, metabolic acidosis, hypocalcemic tetany, and hypovolemic shock. Most commonly occurs in infants and young children after a viral illness or introduction of gluten-containing foods.

Celiac Disease

- An immune-mediated enteropathy triggered by gluten ingestion in genetically predisposed individuals (HLA-DQ2/DQ8). Gliadin peptides from wheat, barley, and rye activate an autoimmune response against tissue transglutaminase (tTG), causing villous atrophy, crypt hyperplasia, and intraepithelial lymphocytosis in the small intestine.

Celiac Trunk

- First major abdominal aortic branch at T12-L1, the foregut artery. Divides into left gastric (lesser curvature), splenic (spleen, pancreas, greater curvature), and common hepatic (proper hepatic/gastroduodenal) arteries. Surrounded by the celiac plexus.

Celiac Trunk Stenosis

- Narrowing of the celiac trunk typically caused by extrinsic compression from the median arcuate ligament of the diaphragm, known as median arcuate ligament syndrome or celiac artery compression syndrome. Causes chronic abdominal pain particularly after eating and weight loss due to postprandial mesenteric ischemia.

Cell

- The fundamental structural and functional unit of all living organisms. Human cells are eukaryotic, characterized by a nucleus containing DNA and membrane-bound organelles.

Cell-Free Fetal DNA (cffDNA) Screening

- A prenatal screening test that analyzes cell-free fetal DNA (cffDNA) circulating in maternal blood, derived from placental trophoblast apoptosis. Performed from 10 weeks onward. Highly sensitive and specific for trisomy 21 (>99%, FPR <0.1%), trisomy 18 (>97%), and trisomy 13 (>90%).

Cell-mediated immunity

- A type of immune response mounted against viruses, certain types of parasites, and perhaps cancer cells.

Cellular Adaptation

- Cellular adaptations are reversible changes in cell morphology and function that occur in response to altered physiological demands or mild pathological conditions.

Cellular Swelling

- An early manifestation of reversible injury caused by failure of ATP-dependent ion pumps and intracellular accumulation of sodium and water. It is also called hydropic change and is often most apparent in cells exposed to ischemia or toxins.

Cellulitis

- An acute, spreading bacterial infection of the deep dermis and subcutaneous tissue, most commonly caused by beta-hemolytic streptococci (particularly *Streptococcus pyogenes*) or *Staphylococcus aureus*.

Cemento-Enamel Junction

- Junction between enamel and cementum. Three relationships: butt junction, enamel overlapping cementum, cementum overlapping enamel.

Cementum

- A mineralized connective tissue covering the tooth root. It provides attachment for periodontal-ligament fibers and is thinner near the cemento-enamel junction and thicker toward the root apex.

Centenarian

- Person aged 100 or older. 100,000 worldwide. More women than men (4:1). Often maintain functional independence, cognitive function. Models for healthy aging research.

Central

- Located near or toward the middle of the body or an organ.

Central Auditory Processing Disorder

- Impaired processing of auditory information by the central nervous system despite adequate peripheral hearing. It may cause difficulty understanding speech, especially in noise, and can affect learning and communication.

Central Deafness

- Loss or severe impairment of hearing caused by disease of the auditory pathways or auditory cortex in the brain, rather than by a disorder of the external, middle, or inner ear.

Central Diabetes Insipidus

- Excessive urination and thirst caused by deficient secretion of antidiuretic hormone from the hypothalamus or posterior pituitary. The kidneys cannot appropriately concentrate urine, producing hypernatremia if water intake is insufficient.

Central Line Placement

- Insertion of a catheter into a central vein (internal jugular, subclavian, or femoral), with the tip in the superior (or inferior) vena cava. Ultrasound guidance is standard. Indications: administration of vasopressors, TPN, chemotherapy, long-term antibiotics, central venous pressure monitoring, hemodialysis, and plasmapheresis.

Central Line-Associated Bloodstream Infection (CLABSI)

- Bloodstream infection associated with central venous catheter. Prevention: maximal sterile barriers, chlorhexidine skin preparation, optimal site selection (subclavian preferred), daily review of line necessity. Diagnosis: paired blood cultures (peripheral and line). Treatment: antibiotics, catheter removal if possible.

Central Nervous System (CNS)

- The part of the nervous system comprising the brain and spinal cord, enclosed within the skull and vertebral column.

Central pontine myelinolysis

- A form of osmotic demyelination syndrome caused by overly rapid correction of hyponatremia, leading to demyelination of the central pons. Clinical features include flaccid quadriplegia, pseudobulbar palsy, dysarthria, dysphagia, and in severe cases locked-in syndrome.

Central Retinal Artery Occlusion (CRAO)

- Acute obstruction of the central retinal artery causing sudden, usually painless, severe monocular visual loss. It is an ocular stroke and requires urgent systemic and vascular evaluation because it may signal a high risk of cerebral ischemia.

Central Serous Chorioretinopathy

- Accumulation of serous fluid beneath the neurosensory retina, usually in the macular region, due to altered choroidal vascular permeability and retinal pigment-epithelium dysfunction. It often causes acute or subacute central visual distortion and may be associated with corticosteroid exposure.

Central Sleep Apnea

- Apneas from variability in respiratory effort without airway obstruction. Associated with heart failure (Cheyne-Stokes respiration), high altitude, and opioid use.

Central Venous Catheter

- A catheter whose tip lies in a central vein, commonly the superior vena cava or cavoatrial junction. It permits administration of selected infusions, venous blood sampling, and in some settings hemodynamic assessment; insertion can cause infection, thrombosis, arterial puncture, or pneumothorax.

Central Venous Catheter (CVC)

- A catheter whose tip terminates in a central vein, usually the lower superior vena cava or cavoatrial junction. It provides reliable access for vasoactive drugs, irritating infusions, hemodynamic monitoring, and prolonged therapy, but may cause bloodstream infection, thrombosis, or mechanical complications.

Central Venous Pressure

- Pressure in the superior vena cava or right atrium, measured via central venous catheter. Normal: 2-8 mmHg. Elevated CVP indicates right heart failure, fluid overload, cardiac tamponade, or constrictive pericarditis. Low CVP indicates hypovolemia. CVP monitoring guides fluid resuscitation and diuretic therapy in heart failure management.

Centriacinar Emphysema

- A subtype of emphysema in which destruction begins in the respiratory bronchioles and extends pe-

ripherally. It predominantly affects the upper lobes and is strongly associated with cigarette smoking. It is the most common form of emphysema and often coexists with chronic bronchitis.

Centrifuge

- Spins samples to separate components by density. Used for blood processing, urine analysis, tissue preparation. Speed: RPM or RCF.

Cephalalgia

- Headache or pain in the head, including pain arising from cranial or related structures. It is a symptom with many possible primary and secondary causes.

Cephalhaematoma

- A collection of blood between the periosteum and the skull bone, caused by rupture of blood vessels during childbirth, most commonly associated with ventouse (vacuum) delivery. Unlike caput succedaneum, a cephalhaematoma does not cross suture lines and is limited by the periosteal attachments.

Cephalohematoma

- A collection of blood beneath the periosteum of a newborn's skull, usually caused by birth trauma. It is confined by suture lines and generally resolves gradually.

Cephalopelvic Disproportion

- A mismatch between the size or shape of the maternal pelvis and the fetal head that prevents or impedes vaginal delivery. It may be suspected from clinical assessment but cannot always be diagnosed reliably before labor.

Cephalosporin

- A class of beta-lactam antibiotics derived from the fungus *Cephalosporium acremonium*, with a mechanism identical to penicillins — inhibition of bacterial cell wall synthesis by binding to penicillin-binding proteins and inhibiting transpeptidase. Cephalosporins are classified into five generations based on their antimicrobial spectrum.

Ceramide

- A sphingolipid composed of a sphingoid base linked to a fatty acid. Ceramides are structural components of cell membranes and the epidermal permeability barrier and also participate in cell signaling.

Cerclage

- A surgical placement of a suture around the cervix

to prevent preterm delivery in patients with cervical insufficiency. Types include transvaginal (McDonald or Shirodkar) and transabdominal. Usually placed at 12-14 weeks gestation and removed at 36-37 weeks.

Cerebellar Ataxia

- A neurological sign characterized by impaired coordination of voluntary movements, resulting from dysfunction of the cerebellum or its connections. The cerebellum coordinates the timing, force, and accuracy of movements, and cerebellar dysfunction produces a characteristic constellation of findings.

Cerebellar tumor

- A neoplasm involving the cerebellum that may cause ataxia, dysmetria, nystagmus, vomiting, or hydrocephalus.

Cerebellitis

- Inflammation of the cerebellum, often after infection and sometimes associated with an autoimmune response. It may produce ataxia, dysarthria, nystagmus, headache, and altered consciousness.

Cerebellopontine Angle

- The angle-shaped region at the junction of the cerebellum, pons, and petrous temporal bone. Lesions in this space may affect cranial nerves and commonly include vestibular schwannoma.

Cerebellopontine Angle Tumor

- Lesion at the cerebellopontine angle (CPA), an anatomic space between the cerebellum and pons, medial to the temporal lobe and lateral to the brainstem, containing CN V (trigeminal), VII (facial), VIII (vestibulocochlear), and the anterior inferior cerebellar artery (AICA).

Cerebellum

- Largest brain portion (approximately 10 percent of volume but over 50 percent of neurons) in the posterior cranial fossa behind the brainstem. Two hemispheres and a vermis, divided into vestibulocerebellum (balance, eye movements), spinocerebellum (posture, gait, coordination), and cerebrocerebellum (motor planning, cognition).

Cerebral Amyloid Angiopathy

- Deposition of amyloid beta in the walls of small and medium cerebral and leptomeningeal arteries. It commonly causes recurrent lobar intracerebral

hemorrhage in older adults and may also cause transient focal neurological episodes or cognitive decline.

Cerebral Aneurysm

- An abnormal focal dilation of a cerebral artery, most commonly at bifurcations of the Circle of Willis. Saccular (berry) aneurysms account for 90% of intracranial aneurysms. Risk factors: smoking, hypertension, female sex, polycystic kidney disease, Ehlers-Danlos, family history. Most are asymptomatic (incidental finding).

Cerebral Angiography

- Contrast imaging of cerebral vessels. Evaluates aneurysms, AVMs, stenosis. Invasive procedure.

Cerebral Arteriovenous Malformation

- An abnormal direct connection between cerebral arteries and veins without a normal capillary bed. It may cause intracranial hemorrhage, seizures, headache, or focal neurological deficits and is evaluated with brain imaging and angiography.

Cerebral Autoregulation

- The intrinsic ability of cerebral arterioles to maintain relatively stable cerebral blood flow across a range of perfusion pressures. It may be impaired by severe brain injury, ischemia, hypercapnia, or anesthetic and vasoactive drugs.

Cerebral cavernous malformations

- Low-flow vascular malformations composed of dilated, thin-walled capillaries without intervening brain parenchyma, often multiple and familial (autosomal dominant with mutations in CCM1, CCM2, or CCM3 genes). They may present with seizures, focal neurological deficits, or intracerebral hemorrhage.

Cerebral Cortex

- The outermost layer of the cerebrum, composed of gray matter (neuronal cell bodies) folded into gyri (ridges) and sulci (grooves), greatly increasing surface area.

Cerebral Edema

- The accumulation of excess fluid within the brain parenchyma, leading to increased intracranial pressure and potentially fatal brain herniation. There are three main types with different pathophysiological mechanisms.

Cerebral Hemorrhage

- Bleeding within the brain parenchyma, a type of hemorrhagic stroke causing mass effect, edema, and neurological deficits. Causes: hypertension (most common—basal ganglia, thalamus), cerebral amyloid angiopathy (lobar, elderly), arteriovenous malformation, aneurysm, anticoagulants, trauma, tumors.

Cerebral Hypoxia

- Insufficient oxygen delivery to the brain, caused by cardiac arrest, respiratory failure, severe hypotension, or another interruption of oxygen supply. Prolonged hypoxia can produce permanent neuronal injury and hypoxic-ischemic encephalopathy.

Cerebral Infarction

- Cerebral infarction, or ischemic stroke, results from occlusion of a cerebral artery causing irreversible neuronal damage. The most common cause is thromboembolism from atherosclerotic carotid or intracranial arteries, or cardioembolism from atrial fibrillation or cardiac thrombus.

Cerebral Lobes

- Four lobes divided by sulci. Frontal (anterior to central sulcus): executive function, motor control (precentral gyrus), Broca's area. Parietal (between central and parieto-occipital sulci): somatosensory (postcentral gyrus), spatial awareness. Temporal (below lateral sulcus): auditory processing, memory (hippocampus), Wernicke's area.

Cerebral Palsy

- A group of permanent disorders of movement and posture development causing activity limitation, attributed to non-progressive disturbances occurring in the developing fetal or infant brain. It is the most common motor disability in childhood, affecting approximately 1 in 500 live births.

Cerebral Peduncle

- Paired structures of the ventral midbrain carrying descending motor tracts (corticospinal, corticobulbar, corticopontine) connecting the cerebrum to the pons and below. Lesions produce contralateral hemiparesis; Weber syndrome (peduncle + CN III) causes ipsilateral CN III palsy with contralateral hemiparesis.

Cerebral Perfusion Pressure (CPP)

- The pressure gradient driving blood flow through the brain, approximated as mean arterial pressure minus intracranial pressure. CPP falls when systemic pressure decreases or ICP rises and must be interpreted with the clinical context.

Cerebral Salt Wasting

- Renal sodium wasting causing hyponatremia and volume depletion occurring after intracranial surgery or in setting of intracranial disease. Distinguished from SIADH by the presence of volume depletion with negative sodium balance and compensatory elevation of renin and aldosterone.

Cerebral tumor

- A neoplasm arising within or involving the brain, producing symptoms according to its location, growth rate, and effect on intracranial pressure.

Cerebral Venous Sinus Thrombosis

- Thrombosis of the cerebral venous sinuses (most commonly the superior sagittal sinus; also transverse, sigmoid, straight; or cortical veins), with resultant venous hypertension, intracranial hypertension, hemorrhagic cortical infarcts, and venous congestion.

Cerebral Visual Impairment (CVI)

- Visual impairment caused by damage to the brain's visual pathways or occipital cortex rather than the eye itself; the leading cause of childhood visual impairment in developed countries. Features include poor visual attention, difficulty recognizing faces or objects, and variable vision; management is educational and rehabilitative.

Cerebriform nevus sebaceous

- A variant of nevus sebaceous presenting as a soft, cerebriform (brain-like), pedunculated or sessile nodule, often found on the scalp or face. It is a benign adnexal hamartoma composed of sebaceous glands, but like other nevus sebaceous lesions, it carries a low risk of secondary neoplasms and may warrant monitoring or excision.

Cerebritis

- Inflammatory process in brain parenchyma without abscess formation. May precede abscess. Causes: bacterial, fungal, parasitic infections. Diagnosis: MRI with contrast showing enhancement. Treat-

ment: antibiotics/antifungals, surgical drainage if abscess forms.

Cerebroplacental Ratio (CPR)

- The ratio of the middle cerebral artery (MCA) pulsatility index to the umbilical artery (UA) pulsatility index. CPR <5th percentile (or <1.0-1.1) suggests fetal brain-sparing effect (vasodilation of cerebral vessels in response to hypoxemia) due to placental insufficiency.

Cerebrospinal Fluid

- A clear, colorless fluid that surrounds and cushions the brain and spinal cord, providing physical protection and chemical stability.

Cerebrospinal Fluid (CSF)

- A clear, colorless fluid that fills the ventricles of the brain, the central canal of the spinal cord, and the subarachnoid space around the brain and spinal cord. Produced primarily by the choroid plexus of the lateral, third, and fourth ventricles (500 mL/day, total volume 150 mL in adults).

Cerebrospinal Fluid Analysis

- Laboratory examination of CSF obtained via lumbar puncture. Indications: meningitis, encephalitis, subarachnoid hemorrhage, demyelinating disease (MS: oligoclonal bands, elevated IgG index), and carcinomatous meningitis. Components: opening pressure (normal 10-20 cmH₂O), appearance (clear vs. cloudy vs.

Cerebrovascular Accident

- Stroke: an acute neurological deficit resulting from cerebral ischemia (80-85 percent) or hemorrhage (15-20 percent). Ischemic: thrombotic (large/small vessel), embolic (cardioembolic—AF), and others; hemorrhagic: intracerebral hemorrhage (hypertension) and subarachnoid hemorrhage (aneurysm).

Cerebrovascular Accident (Stroke)

- Interruption of blood flow to the brain, causing neurological deficits. Types: ischemic (87%, thrombotic or embolic), hemorrhagic (13%, intracerebral or subarachnoid). Ischemic stroke treatment: thrombolysis (alteplase within 4.5 hours), thrombectomy (large vessel occlusion within 24 hours).

Cerebrum

- The largest part of the brain, comprising two cerebral hemispheres connected by the corpus callosum.

Each hemisphere has four lobes: frontal (executive function, motor, Broca's area), parietal (somatosensory, spatial awareness), temporal (auditory, memory, Wernicke's area), occipital (visual processing).

Ceruloplasmin

- The principal copper-carrying protein in blood, synthesized mainly by the liver. Its concentration and copper-binding function are clinically relevant in disorders such as Wilson disease.

Cerumen

- A substance that helps keep dirt out of the ear and lubricates the skin in the ear. More commonly known as earwax.

Cerumen Impaction

- Accumulation of earwax in the external auditory canal that causes symptoms or prevents visualization of the tympanic membrane. It may cause aural fullness, conductive hearing loss, tinnitus, itching, or pain.

Cervical Cancer

- A malignant tumour arising from the cervix, most commonly caused by persistent infection with high-risk human papillomavirus. It may be asymptomatic early or cause abnormal vaginal bleeding, discharge, or pelvic pain; prevention and early detection rely on HPV vaccination, screening, and assessment of abnormal results.

Cervical carcinoma

- A malignant tumor of the uterine cervix, most commonly associated with persistent high-risk human papillomavirus infection.

Cervical Cerclage

- Surgical placement of a non-absorbable suture around the cervix to provide mechanical support and prevent preterm delivery in patients with cervical insufficiency.

Cervical Dilatation

- Progressive opening of the cervical os during labor, measured in centimeters from 0 (closed) to 10 cm (fully dilated). The cervix must also efface (thin out) before significant dilatation occurs. Cervical status assessed by digital vaginal examination during labor.

Cervical Dilation

- The opening of the cervical canal during labor, measured in centimeters from closed (0 cm) to fully

dilated (10 cm); required for vaginal delivery. Occurs with effacement (thinning/shortening of the cervix) during the first stage of labor, driven by effective uterine contractions.

Cervical Disc Herniation

- A common cause of neck pain and radiculopathy, occurring when the nucleus pulposus of an intervertebral disc in the cervical spine herniates through the annulus fibrosus, compressing adjacent nerve roots or the spinal cord.

Cervical Dysplasia

- Abnormal growth of cervical epithelial cells, representing a premalignant condition that may progress to cervical cancer if untreated. Also known as cervical intraepithelial neoplasia (CIN). CIN is caused by persistent infection with high-risk human papillomavirus (HPV) types, most commonly HPV 16 and 18.

Cervical Dystonia

- A focal movement disorder causing involuntary contractions of neck muscles and abnormal positioning of the head. It may produce pain, tremor, and sustained or intermittent twisting of the neck.

Cervical Effacement

- Thinning and shortening of the cervix during labor, measured as a percentage from 0% (thick, normal length) to 100% (paper thin). Effacement precedes and accompanies cervical dilatation. Nulliparous women often efface before dilating; multiparous women may efface and dilate simultaneously.

Cervical Exam in Labor

- Serial digital vaginal examinations to assess cervical dilatation (cm), effacement (%), station (relationship to ischial spines, -5 to +5), consistency (firm, medium, soft), and position (posterior, mid, anterior). Performed every 2-4 hours in active labor. In the second stage, assessing fetal head position and asynclitism.

Cervical Incompetence

- The inability of the cervix to maintain pregnancy in the second trimester due to painless cervical dilatation. Results in recurrent second trimester pregnancy loss or preterm delivery. Risk factors include prior cervical trauma, cone biopsy, and congenital anomalies.

Cervical Length Screening

- Transvaginal ultrasound measurement of the cervix between 18-24 weeks to assess risk of spontaneous preterm birth. Normal cervix measures greater than 25 mm. Short cervix (less than 25 mm) increases preterm delivery risk. Cervical funneling and dilatation are additional concerning findings.

Cervical Mucus

- Secretions from the cervical glands that change throughout the menstrual cycle under hormonal influence. Estrogen produces thin, clear, stretchy mucus favorable for sperm transport around ovulation (spinnbarkeit). Progesterone produces thick, sticky, opaque mucus that forms a plug in the cervical canal during pregnancy.

Cervical Polyps

- Usually benign, finger-like growths arising from the endocervical canal or cervix. They may cause irregular bleeding, postcoital bleeding, or vaginal discharge and are often found during a pelvic examination. Symptomatic or atypical polyps are removed and sent for pathology.

Cervical Pregnancy

- A rare ectopic pregnancy (0.1-1%) implanted in the cervical canal. Presents with painless vaginal bleeding in the first trimester. Diagnosis: ultrasound (hourglass-shaped uterus, gestational sac in the cervix with absent sliding sign, empty uterine cavity). High risk of severe hemorrhage.

Cervical Radiculopathy

- A condition caused by compression, irritation, or inflammation of a spinal nerve root in the cervical spine, resulting in pain, numbness, tingling, and weakness in the distribution of the affected nerve root.

Cervical Rib

- An extra rib arising from a cervical vertebra, usually the seventh. It may be asymptomatic or contribute to thoracic outlet syndrome by compressing the brachial plexus or subclavian vessels.

Cervical Ripening Agents

- Medications or mechanical devices used to soften and dilate the cervix before induction of labor when the Bishop score is unfavorable (5). Pharmacologic: prostaglandin E1 (misoprostol/Cytotec, 25 mcg PV

q4h), prostaglandin E2 (dinoprostone/Cervidil vaginal insert 10 mg over 12 hours, or Prepidil gel 0.5 mg intracervical q6h).

Cervical Screening (Pap Smear)

- Screening for cervical cancer by collecting cells from the transformation zone of the cervix. Liquid-based cytology (ThinPrep) or conventional Pap smear. HPV co-testing (high-risk HPV 16, 18, others) improves sensitivity.

Cervical Spine

- The upper segment of the vertebral column consisting of seven cervical vertebrae, C1 through C7, supporting the head and protecting the cervical spinal cord. It permits substantial flexion, extension, rotation, and lateral flexion while transmitting the vertebral arteries through the transverse foramina of C1-C6.

Cervical Spine Trauma

- Cervical spine trauma: Subaxial (C3-C7) injury classification (SLIC - Subaxial Cervical Spine Injury Classification): morphology (1-4), discoligamentous complex (0-2), neurologic (0-3). Total <4 nonop, >4 op, =4 surgeon discretion.

Cervical spondyloarthropathy

- A degenerative condition of the cervical spine involving disc degeneration, osteophyte formation, and facet joint arthropathy that may compress nerve roots or the spinal cord. It can cause neck pain, radiculopathy, or myelopathy, and is a common cause of cervical spinal stenosis in older adults.

Cervical Spondylosis

- Age-related degenerative changes of the cervical spine—disc degeneration, osteophyte formation, facet joint arthritis, and ligament hypertrophy—affecting the intervertebral discs, vertebrae, and joints.

Cervical Vertebrae (C1-C7)

- The seven vertebrae of the neck. Typical cervical vertebrae have small bodies, bifid spinous processes, and transverse foramina transmitting the vertebral vessels; C1, C2, and C7 are specialized.

Cervicitis

- Inflammation of the uterine cervix, most commonly caused by sexually transmitted infections (Chlamydia trachomatis and Neisseria gonorrhoeae) but also

by other organisms (*Mycoplasma genitalium*, *Trichomonas vaginalis*, herpes simplex virus), or non-infectious causes (chemical irritation, foreign body, allergy).

Cervix

- The lower, narrow part of the uterus that opens into the vagina. The cervix is approximately 3-4 cm long in the non-pregnant state and consists of two parts: the ectocervix (visible portion protruding into the vagina, covered by stratified squamous epithelium) and the endocervical canal (lined by columnar epithelium).

Cesarean Delivery

- Surgical delivery of the fetus through abdominal and uterine incisions. Performed when vaginal delivery is contraindicated or fails. Indications include malpresentation, placenta previa, previous cesarean, fetal distress, and failed induction. Most common uterine incision is low transverse (classical for very preterm or transverse lie).

Cesarean Section

- C-section: surgical delivery of the infant through incisions in the abdominal wall and uterus. Indications: fetal distress, failure to progress, malpresentation (breech, transverse), placenta previa, abruption, previous cesarean (some), multiple pregnancy, preeclampsia, cord prolapse, and maternal conditions.

Cesarean Section (C-section)

- Surgical delivery of a fetus through incisions in the abdominal wall (laparotomy) and uterus (hysterotomy).

Cestodiasis

- Infection or infestation caused by cestodes, commonly called tapeworms. Disease may involve the intestine or tissues, depending on the species and life-cycle stage.

Cetirizine

- A second-generation H1 antihistamine (metabolite of hydroxyzine) used for allergic rhinitis and urticaria. Slightly more sedating than loratadine/fexofenadine (some CNS penetration). Once daily; renally cleared (dose-reduce in renal impairment). Adverse effects: somnolence (dose-related), dry mouth, and fatigue. Available OTC.

CFTR Modulator Therapy

- Targeted therapies correcting the defective CFTR protein. Elexacaftor-tezacaftor-ivacaftor (Trikafta) is the most effective combination for patients with at least one F508del mutation. These therapies have dramatically changed CF prognosis.

Chagas Disease

- A parasitic infection caused by *Trypanosoma cruzi*, transmitted by reduviid bugs (kissing bugs), endemic in Latin America. Acute phase: fever, swelling at bite site (chagoma), Romana sign (peri-orbital edema). Chronic phase (years later): Chagas cardiomyopathy (most common and serious), megasophagus, and megacolon.

Chagoma

- A localized inflammatory lesion at the site where *Trypanosoma cruzi* enters the body. It is an early manifestation of Chagas disease and may be accompanied by regional lymphadenopathy.

Chain of Custody

- Meticulous paper trail and chronological documentation tracking the seizure, custody, control, transfer, analysis, and disposition of physical and electronic evidence in medico-legal investigations (e.g., blood samples for toxicology, bullets, sexual assault kits, biological fluids).

Chalazion

- A chronic, non-infectious granulomatous inflammation of a meibomian gland caused by duct obstruction. Presents as a firm, painless, well-circumscribed nodule in the eyelid, usually away from the lid margin. Different from a sty (which is acute and tender). Most resolve spontaneously over weeks to months.

Chancre

- A usually painless ulcer at the site of inoculation of an infection, especially the primary lesion of syphilis. It may be accompanied by regional lymph-node enlargement and heals without treatment, although infection persists.

Chancroid

- A sexually transmitted infection caused by *Haemophilus ducreyi*, characterized by painful genital ulcers and tender inguinal lymphadenopathy (bubo). Ulcers are ragged, non-indurated, with pu-

rulent exudate and surrounding erythema. More common in men and in resource-limited settings. Associated with increased HIV transmission.

Channelopathy

- A disorder caused by abnormal structure or function of an ion channel or its regulatory proteins. Channelopathies can affect excitable tissues and cause neurologic, muscular, or cardiac disease.

Charcoal

- Fine black powder produced by controlled pyrolysis of organic materials, processed to yield a massive micro-porous surface area (500-1500 m²/g). See activated charcoal.

Charcot Arthropathy

- Progressive destruction and deformity of a joint caused by loss of protective pain or proprioceptive sensation. It is most often associated with peripheral neuropathy, particularly diabetic neuropathy.

Charcot Foot

- A progressive destructive arthropathy of the foot and ankle, usually occurring in people with peripheral neuropathy, especially diabetes. Repeated unnoticed trauma causes warmth, swelling, deformity, instability, and an increased risk of ulceration and amputation.

Charcot Joint

- Progressive, neurotrophic arthropathy caused by loss of sensation (proprioception and pain) leading to joint destruction with bone fragmentation, joint dislocation, deformity, and loss of normal pain and proprioceptive sensation in the affected limb.

Charcot Triad

- Clinical constellation of fever, right upper quadrant pain, and jaundice representing the classic presentation of ascending cholangitis, infection of the biliary tree from common bile duct obstruction by a gallstone. The full triad is found in approximately fifty to seventy percent of patients.

Charcot Triad (Cholangitis)

- Classic diagnostic triad of acute ascending cholangitis: (1) Fever, (2) Right Upper Quadrant abdominal pain, and (3) Jaundice. Reynolds Pentad adds Hypotension and Altered Mental Status. See Acute Cholangitis.

Charcot-Marie-Tooth Disease

- Group of inherited peripheral neuropathies (hereditary motor and sensory neuropathy, HMSN), the most common inherited neuromuscular disorder (prevalence 1 in 2,500). More than 100 causative genes (PMP22, MPZ, GJB1, MFN2, NEFL).

CHARGE Syndrome

- A genetic disorder caused by CHD7 gene mutations (autosomal dominant, most cases de novo) affecting multiple organ systems.

Checkpoint Inhibitor

- Drug blocking immune checkpoints. Examples: pembrolizumab (PD-1), ipilimumab (CTLA-4).

Checkpoints

- Immune regulatory molecules. Cancer exploits checkpoints. Inhibitors block checkpoints.

Chediak-Higashi Syndrome

- Autosomal recessive (LYST mutation, lysosomal trafficking regulator). Giant lysosomal granules in neutrophils, melanocytes, neurons. Partial oculocutaneous albinism, silvery hair, photophobia, nystagmus.

Cheilectropion

- An eversion deformity of the lip, usually involving the lower lip. It may result from scarring, congenital malformation, or loss of tissue support.

Cheilitis

- Inflammation of one or both lips, causing dryness, scaling, fissuring, erythema, swelling, pain, or ulceration. Causes include irritant or allergic contact exposure, infection, nutritional deficiency, chronic sun exposure, and inflammatory skin disease; persistent lesions require evaluation for premalignant or malignant change.

Cheiloplasty

- Plastic or reconstructive surgery of the lip, performed to repair congenital clefting, trauma, tumors, scars, or other deformity.

Cheiloschisis

- Cleft lip, a congenital fissure of the upper lip caused by incomplete fusion of facial processes during embryonic development. It may occur alone or with cleft palate.

Cheilosis

- A disorder of the lips characterized by dryness, scaling, fissuring, or inflammation. Causes include irritation, nutritional deficiency, infection, and systemic disease.

Cheiopompholyx

- A form of dyshidrotic eczema affecting the hands, with intensely itchy small vesicles on the palms or sides of the fingers.

Chelation

- The formation of a stable complex between a metal ion and a molecule with multiple binding sites. In medicine, chelation can be used under supervision to bind and promote elimination of selected toxic metals.

Chelation Study

- Nuclear medicine study using chelating agents. Evaluates renal function, obstruction.

Chelation Therapy

- Use of chelating agents to bind metals for excretion. Agents: EDTA (lead), deferoxamine (iron), dimercaprol (arsenic, mercury, gold), DMSA (lead, mercury), penicillamine (copper, lead). Complications: nephrotoxicity, electrolyte disturbances, redistribution of metals.

Chemical Eye Injury

- Ocular damage caused by exposure to an acidic or alkaline substance. Immediate copious irrigation, removal of particulate material, measurement of ocular-surface pH, and urgent ophthalmic assessment are required; alkali injuries may penetrate deeply and continue to worsen after exposure.

Chemical Injury

- Tissue damage from corrosive substances. Acid burns: coagulation necrosis, eschar formation. Alkali burns: liquefaction necrosis, deeper penetration. Management: copious irrigation, remove contaminated clothing, assess for airway involvement.

Chemical Peels

- Chemical peels use acids to exfoliate skin layers. Superficial peels (glycolic acid 20 to 70 percent, salicylic acid 20 to 30 percent) affect the epidermis. Medium peels (trichloroacetic acid 35 to 50 percent) reach the papillary dermis. Deep peels (phenol 88 percent) reach the reticular dermis.

Chemical Pneumonitis

- Inflammation and lung injury caused by inhalation or aspiration of irritating chemicals, gastric contents, hydrocarbons, or other toxic substances. It may cause cough, dyspnea, hypoxemia, wheezing, and diffuse pulmonary infiltrates.

Chemodectoma

- A tumor arising from paraganglionic tissue, especially a carotid-body tumor. It is usually slow-growing and may compress nearby cranial nerves or vessels.

Chemoembolization

- A procedure that combines delivery of an antineoplastic agent with embolization of the blood supply to a tumor, most commonly for selected liver cancers.

Chemokine

- A family of small secreted cytokines that direct leukocyte migration (chemotaxis) to sites of infection, inflammation, and tissue repair. Classified by cysteine motif: CXC (e.g., IL-8/CXCL8, recruit neutrophils), CC (e.g., MCP-1/CCL2, attract monocytes), C, and CX3C.

Chemokinesis

- Nonspecific stimulation of cell movement by a chemical substance, without directional movement toward or away from a concentration gradient.

Chemophobia

- An excessive or irrational fear of chemicals, sometimes associated with anxiety or obsessive concerns about contamination or toxicity.

Chemoprevention

- The use of drugs or natural agents to prevent, delay, or reverse cancer development, e.g., tamoxifen or raloxifene in high-risk breast cancer, HPV and hepatitis B vaccination, and aspirin in colorectal cancer.

Chemoradiation

- The concurrent combination of chemotherapy and radiation therapy, used to treat locoregionally advanced cancers such as anal, cervical, head and neck, esophageal, and limited-stage small cell lung cancer.

Chemoreceptor

- A sensory receptor that responds to changes in

blood or CSF chemistry, involved in respiratory and cardiovascular regulation.

Chemoreceptor Trigger Zone

- A region near the floor of the fourth ventricle that detects emetic stimuli in blood and cerebrospinal fluid and helps initiate vomiting.

Chemoresistance

- Reduced or absent responsiveness of cells, especially tumor cells, to a chemical treatment that would ordinarily inhibit or destroy them.

Chemosensitivity

- The degree to which cells or a tumor respond to a chemical treatment. Testing may be used experimentally or clinically to help characterize treatment response.

Chemosis

- Edema of the conjunctiva, producing a swollen, often translucent or blister-like appearance. It may occur with allergy, infection, inflammation, trauma, or orbital disease.

Chemotaxis

- Directed movement of a cell or organism toward or away from a chemical stimulus. It is important in immune-cell recruitment and inflammatory responses.

Chemotherapy

- Use of cytotoxic drugs to treat cancer. Mechanisms: DNA damage, microtubule inhibition, antimetabolite action. Common agents: alkylating agents, anthracyclines, taxanes, platins, antimetabolites. Side effects: myelosuppression, nausea, alopecia.

Chemotherapy Extravasation

- Leakage of an intravenous anticancer drug into surrounding tissue. Vesicant extravasation can cause severe local injury, so infusion must stop immediately while the catheter is managed and drug-specific antidote and surgical guidance are obtained.

Cherry Angioma

- A common benign proliferation of small dermal blood vessels that appears as a bright-red to purple papule, usually on the trunk or limbs. Lesions may bleed after trauma but generally require no treatment unless diagnosis is uncertain or removal is desired.

Cherry Eye

- Prolapse of the gland of the third eyelid, producing a rounded red mass at the inner corner of the eye. The term is used especially in veterinary ophthalmology.

Cherry-Red Spot

- A striking fundoscopic finding in which the macula appears bright red against a pale, ischemic, or lipid-laden surrounding retina. Mechanism: the fovea centralis lacks inner retinal layers and receives blood supply from the underlying choroid, preserving its natural red color while adjacent edematous retina turns opaque white.

Cherubism

- A rare inherited disorder characterized by bilateral enlargement of the jaws from expansile giant-cell lesions, usually beginning in childhood and often regressing after puberty.

Chest Pain

- A cardinal symptom with cardiac and non-cardiac causes requiring urgent distinction. Cardiac: acute coronary syndrome (angina, MI), pericarditis, aortic dissection, myocarditis. Pulmonary: pulmonary embolism, pneumothorax, pneumonia, pleurisy. GI: GERD, esophageal spasm, peptic ulcer. Musculoskeletal and psychogenic causes also common.

Chest Physiotherapy

- Techniques to mobilize and clear secretions from the lungs, including postural drainage, percussion, vibration, and huffing. It is used in bronchiectasis, CF, and post-operative patients. Positive expiratory pressure devices and oscillating PEP devices are also used.

Chest Tube Insertion

- Placement of a thoracostomy tube in the safe triangle (latissimus dorsi, pectoralis major, nipple level) to drain air, fluid, or blood from the pleural space. Connected to underwater seal drainage.

Chest Tube Thoracostomy

- Insertion of a tube into the pleural space to drain air, fluid, or blood. Indications: pneumothorax (large, tension, or on mechanical ventilation), hemothorax, pleural effusion (empyema, malignant, transudative), and postoperative (cardiac, thoracic surgery).

Chest Wall

- The musculoskeletal boundary of the thoracic cavity, formed by the skin, fascia, muscles, ribs, costal cartilages, sternum, thoracic vertebrae, and associated neurovascular structures. It protects the thoracic organs and participates in breathing through coordinated movement of the ribs and respiratory muscles.

Chest Wall Deformity

- Abnormalities of the chest wall that can affect lung function. Pectus excavatum (sunken sternum) may compress the heart and lungs. Pectus carinatum (protruding sternum) is usually cosmetic. Scoliosis can cause restrictive lung disease. These may require surgical correction if severe.

Chest X-ray

- Initial imaging for respiratory symptoms. PA and lateral views. Evaluates pneumonia, pneumothorax, pleural effusion, heart size, mediastinal masses. Limited sensitivity for early disease.

Chewing

- Mastication of food. Teeth, muscles, joints coordinated. Difficulties: dental pain, missing teeth, TMJ disorder.

Cheyne-Stokes Respiration

- Crescendo-decrescendo breathing pattern with periods of apnea, seen in heart failure and stroke. It is a form of central sleep apnea.

Chiari I malformation

- See Chiari Malformation.

Chiari Malformation

- Chiari malformations are structural abnormalities of the posterior fossa and cerebellum characterized by downward displacement of the cerebellar tonsils through the foramen magnum. Type I is the most common variant, defined as cerebellar tonsillar ectopia of five millimeters or more below the foramen magnum.

Chiari-Frommel Syndrome

- Persistent postpartum secretion of milk with absence of menstruation, associated with hyperprolactinemia after pregnancy. It is a form of galactorrhea-amenorrhea syndrome.

Chickenpox

- Primary infection with Varicella-Zoster Virus, characterized by pruritic, vesicular rash in different stages (macules, papules, vesicles, crusts). Highly contagious via respiratory droplets and contact with lesions. Incubation 10-21 days. Complications include secondary bacterial infection, pneumonia, encephalitis, and Reye syndrome.

Chignon

- A temporary swelling (edema) on the baby's scalp caused by the vacuum cap used during ventouse (vacuum) assisted delivery. The chignon forms as the vacuum creates negative pressure pulling the scalp tissue into the cup. It settles within 24-48 hours after birth without any treatment.

Chikungunya

- An acute mosquito-borne viral infection characterized by fever and often severe joint pain, with rash and other systemic symptoms in some patients.

Chikungunya Fever

- An acute febrile illness caused by chikungunya virus and transmitted primarily by Aedes mosquitoes. It typically causes abrupt fever, severe symmetrical polyarthralgia, myalgia, headache, and rash; joint symptoms may persist for months or longer, especially in older adults or people with pre-existing rheumatic disease.

Chikungunya virus

- A positive-sense RNA virus belonging to the Togaviridae family that causes fever and severe joint pain. The virus is transmitted by Aedes mosquitoes. Symptoms include fever, rash, and polyarthralgia that can be debilitating. Most patients recover fully, but some develop chronic joint pain lasting months to years.

Chilblain

- An inflammatory skin lesion caused by an abnormal response to cold and damp conditions. It produces itchy, red or violaceous swollen areas, usually on fingers, toes, ears, or nose.

Child Abuse

- Physical, sexual, or emotional harm or neglect of a child by a caregiver. Recognized by inconsistent histories, injury patterns inconsistent with mechanism, and behavioral indicators. Types: physical

abuse, sexual abuse, neglect, and emotional abuse. Mandatory reporting requirements for healthcare providers.

Child Neglect

- The failure of a caregiver to provide adequate food, shelter, clothing, supervision, education, or medical care. The most common form of child maltreatment. Types: physical neglect, educational neglect, emotional neglect, and medical neglect. May result in failure to thrive, developmental delay, and chronic illness.

Child Poisoning

- Poisoning in children. See pediatric toxicology.

Child-Pugh Classification

- Clinical scoring system assessing prognosis of chronic liver disease by evaluating five parameters: total bilirubin, serum albumin, international normalized ratio, ascites, and hepatic encephalopathy. Each parameter is scored one to three points.

Childbirth

- The process of giving birth, encompassing three stages: (1) First stage: from the onset of regular uterine contractions (cervical effacement and dilation) to full dilation (10 cm). See Parturition (Childbirth).

Childhood Absence Epilepsy

- An epilepsy syndrome beginning in childhood with frequent brief absence seizures, characterized by sudden impairment of awareness and a typical generalized spike-and-wave EEG pattern.

Chills

- A subjective sensation of cold, often accompanied by shivering or piloerection, caused by hypothalamic thermoregulatory responses. Chills commonly occur during fever or infection but may also accompany exposure, endocrine or metabolic disturbance, medication effects, or acute systemic illness.

Chimerism

- The presence in one individual of cell populations derived from genetically different zygotes or sources. It may occur naturally, after transplantation, or through exchange of cells between a pregnant person and fetus.

Chionablepsia

- Temporary visual impairment caused by exposure

to intense reflected light from snow or ice, commonly called snow blindness. It is usually related to ultraviolet injury of the cornea.

Chlamydia

- The most common reportable bacterial STI, caused by *Chlamydia trachomatis*. Often asymptomatic in women, causing cervicitis, urethritis, and pelvic inflammatory disease. Complications include tubal factor infertility and ectopic pregnancy. Diagnosis by nucleic acid amplification test. Treatment: azithromycin or doxycycline.

Chlamydia trachomatis infection

- See Sexually Transmitted Infections (STIs).

Chlamydial pneumonia

- See *Chlamydia trachomatis* infection.

Chlamydiosis

- Infection caused by bacteria of the genus *Chlamydia*. Depending on the species and site, it may cause sexually transmitted infection, conjunctivitis, pneumonia, or other disease.

Chlamydophila Pneumonia

- Atypical pneumonia with mild presentation: sore throat, hoarseness, cough. Treatment is macrolides or tetracyclines.

Chloasma

- Patchy brown hyperpigmentation, especially of the face, associated with pregnancy, hormones, sun exposure, or other causes. It is commonly called melasma.

Chloracne

- An acne-like eruption caused by exposure to certain halogenated aromatic compounds, such as dioxins and related industrial chemicals.

Chloramphenicol

- A broad-spectrum antibiotic with activity against gram-positive and gram-negative organisms, as well as *Rickettsia*. Chloramphenicol inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. It is indicated for serious infections when other agents are not appropriate, including bacterial meningitis and typhoid fever.

Chloranemia

- Anemia in which the red blood cells are abnormally pale, usually because of reduced hemoglobin content,

as in iron deficiency.

Chloroleukemia

- A rare form of myeloid leukemia associated with greenish tumor masses composed of myeloid cells, also called granulocytic sarcoma or chloroma.

Chloroma

- A greenish extramedullary mass of immature myeloid cells, usually associated with acute myeloid leukemia. It is also called granulocytic sarcoma or myeloid sarcoma.

Chloropsia

- A visual disturbance in which objects appear green or have a greenish tint. It can occur with retinal disease, medication or toxin exposure, or other visual-pathway disorders.

Choanal Atresia

- A congenital obstruction of the posterior nasal passage (choana) due to failure of the buconasal membrane to rupture during embryogenesis. Bilateral choanal atresia presents as a respiratory emergency in the newborn, who is an obligate nasal breather — cyclic cyanosis that improves with crying.

Choking

- Airway obstruction caused by an object lodged in the pharynx or larynx, producing impaired ventilation. Complete obstruction can cause hypoxic brain injury and death within minutes; treatment depends on the patient's age, consciousness, and effectiveness of coughing.

Cholangitis

- Inflammation of the bile ducts. It may be caused by infection, obstruction, autoimmune disease, or other biliary disorders.

Cholangiocarcinoma

- A malignant tumor arising from the biliary epithelium, classified based on anatomical location as intrahepatic, perihilar, or distal extrahepatic.

Cholangiography

- Contrast imaging of bile ducts. Intraoperative, percutaneous, or ERCP approach. Evaluates biliary obstruction, stones, strictures.

Cholangitis

- Bacterial infection of the biliary tree, most commonly caused by choledocholithiasis (common bile

duct stones) obstructing bile flow, but also from biliary strictures, stent occlusion, and malignancy. The classic clinical presentation is Charcot triad: fever, right upper quadrant pain, and jaundice (present in 50-70 percent of cases).

Cholecystectomy

- Surgical removal of the gallbladder. Laparoscopic cholecystectomy (gold standard, >95%): 4 small incisions, short stay. Open cholecystectomy: for severe inflammation, cirrhosis, or failed laparoscopy. Indications: symptomatic cholelithiasis (biliary colic), acute cholecystitis, gallstone pancreatitis, and gallstone ileus.

Cholecystitis

- Inflammation of the gallbladder, most commonly caused by obstruction of the cystic duct by a gallstone, which occurs in approximately 95 percent of cases. The obstruction leads to bile stasis, increased intraluminal pressure, and ischemia of the gallbladder wall, which may progress to bacterial infection.

Cholecystography

- Radiographic examination of the gallbladder, usually after administration of a contrast substance, to assess its filling, concentration, and emptying.

Cholecystokinin

- CCK is a peptide hormone produced by I-cells in the duodenal and jejunal mucosa in response to fatty acids (long-chain fatty acids are the most potent stimuli) and amino acids in the chyme.

Choledochal Cyst

- Congenital dilatation of the biliary tree classified into five types according to the Todani classification. Type I involving dilation of the common bile duct is the most common. Presents with the classic triad of abdominal pain, jaundice, and palpable mass though all three are present in less than twenty percent of cases.

Choledocholithiasis

- The presence of gallstones within the common bile duct, occurring in approximately 10-20 percent of patients with cholelithiasis.

Cholelithiasis

- Presence of gallstones in the gallbladder. Types: cholesterol stones (80%, radiolucent, due to super-

saturated bile), pigment stones (black: hemolysis, cirrhosis; brown: infection, bile stasis). Risk factors: female, fertile, fat, forty (5 Fs), obesity, rapid weight loss, hemolytic disorders, cirrhosis.

Cholera

- An acute diarrheal illness caused by *Vibrio cholerae*, a gram-negative, comma-shaped, flagellated bacterium, serogroups O1 and O139, that produces cholera toxin, an enterotoxin that irreversibly activates adenylate cyclase in intestinal epithelial cells, causing massive secretion of isotonic fluid into the intestinal lumen.

Cholestasis

- Impairment of bile formation or flow leading to retention of bile acids, bilirubin, and cholesterol in the blood and tissues.

Cholesteatoma

- A pathological accumulation of keratinizing squamous epithelium within the middle ear, mastoid, or temporal bone that behaves like a locally invasive, destructive lesion, causing progressive conductive hearing loss, chronic otorrhea, and potential complications including ossicular erosion, labyrinthine fistula, facial nerve paralysis,

Cholesterol

- A 27-carbon sterol essential for cell membrane structure (modulates fluidity), precursor of steroid hormones (cortisol, aldosterone, sex hormones), bile acids, and vitamin D. Synthesized de novo (1 g/day, primarily liver) from acetyl-CoA via the mevalonate pathway (HMG-CoA reductase is rate-limiting, targeted by statins).

Cholesterol and Lipids

- Fatty substances in the blood, including LDL cholesterol, HDL cholesterol, and triglycerides, that play critical roles in cellular function and energy metabolism. Abnormal lipid levels, particularly high LDL and triglycerides with low HDL, promote atherosclerosis and increase the risk of heart attack and stroke.

Cholesterol crystal embolization

- Atheroembolic disease caused by cholesterol crystal fragments dislodging from proximal atherosclerotic plaques into distal small arteries, often after vascular procedures. Presents with livedo reticularis,

blue-toe syndrome, renal failure, and eosinophilia. Diagnosis by skin, kidney, or muscle biopsy.

Cholesterol crystals

- Needle-shaped, cleft-like spaces seen in histological sections representing dissolved cholesterol esters, commonly found in atherosclerotic plaques, atheroembolic disease, and cholesterol-rich effusions. In atheroembolic disease, cholesterol crystals embolize from ruptured plaques to distal organs causing tissue ischemia.

Cholesterol emboli

- See Cholesterol crystal embolization.

Cholesterol embolism

- See Cholesterol crystal embolization.

Cholesterol embolization

- See Cholesterol crystal embolization.

Cholesterol Synthesis

- Occurs in all nucleated cells (primarily liver and intestine, approximately 1 g/day production), using acetyl-CoA as the precursor through approximately 30 enzymatic steps.

Cholesterolemia

- The presence and concentration of cholesterol in the blood. Abnormal elevation is a risk factor for atherosclerotic cardiovascular disease.

Cholinergic Crisis

- Excess acetylcholine from organophosphate or carbamate poisoning. SLUDGE syndrome: salivation, lacrimation, urination, defecation, GI upset, emesis. Miosis, bradycardia, bronchospasm, muscle fasciculations. Treatment: atropine, pralidoxime.

Cholinesterase Inhibitor

- A class of medications that inhibit acetylcholinesterase, the enzyme that breaks down acetylcholine in the synaptic cleft, thereby increasing cholinergic neurotransmission in the central nervous system and peripheral tissues.

Chondritis

- Inflammation of cartilage, such as the cartilage of the ear, nose, ribs, or larynx. It may result from infection, trauma, or immune-mediated disease.

Chondroblastic osteosarcoma

- A variant of osteosarcoma in which the malignant cells produce a cartilaginous matrix. It typically af-

fects the long bones of adolescents and young adults, with a predilection for the metaphyseal regions.

Chondrocalcinosis

- Calcification of cartilage visible on X-ray. Associated with CPPD disease. Found in knees, wrists, pubic symphysis. May be incidental or symptomatic.

Chondrodysplasia

- A disorder of cartilage and bone development that causes abnormal skeletal growth and, in some forms, disproportionate short stature.

Chondroma

- Benign cartilage-forming tumor composed of mature hyaline cartilage, most often intramedullary (enchondroma) or subperiosteal (juxtacortical / periosteal chondroma).

Chondromalacia

- Softening and degeneration of cartilage, especially articular cartilage. The term is often used for changes affecting the patella and causing anterior knee pain.

Chondrosarcoma

- The second most common primary malignant bone tumor (after osteosarcoma) and the most common primary malignant bone tumor in adults (peak age 40-70 years).

Chopart's Joint

- The transverse tarsal joint, consisting of the talonavicular and calcaneocuboid joints, between the hindfoot and the midfoot.

Chordae Tendineae

- Thin tendinous cords connecting the papillary muscles to the free edges of the atrioventricular valve leaflets (mitral and tricuspid). They tether the leaflets and, with papillary muscle contraction during ventricular systole, prevent valve prolapse and regurgitation.

Chordoma

- Slow-growing, locally aggressive malignant bone tumor originating from embryonic remnants of the notochord along the axial skeleton. Three anatomic locations: (1) sacrococcygeal (50% of cases, most common), (2) sphenoccipital/clival (35%, base of skull), (3) vertebral (cervical thoracolumbar, 15%).

Chorea

- Involuntary, irregular, brief, non-repetitive jerky movements flowing randomly from one body part to another ('dance-like'), due to basal ganglia (striatal) dysfunction.

Chorioamnionitis

- Infection of the chorion and amnion, typically due to ascending bacterial infection from the genital tract. Risk factors include prolonged rupture of membranes, multiple vaginal exams, and preterm labor. Presents with maternal fever, uterine tenderness, and fetal tachycardia. Diagnosis is clinical; amniocentesis for confirmation.

Chorioamnionitis (Intraamniotic Infection)

- Infection or inflammation of the amniotic fluid, membranes, placenta, or decidua, usually due to ascending polymicrobial infection. Maternal fever, uterine tenderness, fetal tachycardia, and maternal leukocytosis may occur; treatment requires broad-spectrum antibiotics and delivery planning.

Choriocarcinoma

- Malignant trophoblastic tumor from abnormal proliferation of syncytiotrophoblast and cytotrophoblast, the most aggressive form of gestational trophoblastic disease (GTD).

Chorion

- The outermost fetal membrane, derived from the trophoblast. The chorion frondosum (villous chorion) develops into the fetal portion of the placenta. The chorion laeve (smooth chorion) is the non-villous portion. Chorionic villi contain fetal blood vessels and are bathed in maternal blood.

Chorionic Villus Sampling

- Prenatal diagnostic test at 10-13 weeks gestation, earlier than amniocentesis. Transabdominal or transcervical biopsy of chorionic villi (placental tissue). Indications: advanced maternal age, abnormal first-trimester screen, carrier screening, or family history of genetic disorders. Analyzed for karyotype, FISH, and microarray.

Chorionic Villus Sampling (CVS)

- A first-trimester invasive prenatal diagnostic procedure (10-13 weeks) that samples chorionic villi for genetic analysis. Transcervical (catheter through the

cervix into the placenta under ultrasound guidance) or transabdominal (needle through the abdominal wall).

Chorioretinitis

- Inflammation involving the choroid and retina, often caused by infection, autoimmune disease, or systemic inflammatory illness. It can produce floaters and impaired vision.

Choroid

- The vascular layer of the eye between the sclera and retina, part of the uveal tract (with ciliary body and iris). Contains dense capillary network (choriocapillaris) supplying outer retina (photoreceptors). Pigmented (melanin) to absorb stray light. Bruchs membrane separates choroid from retinal pigment epithelium.

Choroid Plexus Cyst (CPC)

- A sonolucent fluid-filled cyst within the choroid plexus of the lateral cerebral ventricle, seen in 1-3% of second-trimester fetuses. In isolation, it is a normal variant with no clinical significance. CPC is associated with trisomy 18 (30-50% of trisomy 18 fetuses have CPC, but most fetuses with CPC are normal).

Choroid Plexus Tumor

- A papillary neoplasm of the choroid plexus, occurring most often in children, and frequently presenting with hydrocephalus from excessive cerebrospinal fluid production.

Choroidal Neovascularization

- The pathological growth of new, fragile blood vessels from the choroid through Bruch membrane into the subretinal or sub-retinal pigment epithelium space, and it is the hallmark of wet age-related macular degeneration and the primary cause of severe central vision loss in this condition.

Christmas Disease

- Hemophilia B, an inherited bleeding disorder caused by deficiency or dysfunction of coagulation factor IX.

Chromaffinoma

- A tumor of chromaffin cells, usually a pheochromocytoma of the adrenal medulla or a related extra-adrenal paraganglioma. It may secrete catecholamines and cause episodic hypertension.

Chromatin

- The complex of DNA, histone proteins, and associated molecules that packages the genome in the nucleus. Its organization and chemical modification regulate DNA accessibility, replication, repair, and transcription.

Chromatopsia

- A visual disturbance in which objects appear abnormally colored. It may result from retinal disease, toxic exposure, medication effects, or other disorders of the visual system.

Chromhidrosis

- A rare condition in which sweat is abnormally colored, often yellow, green, blue, or black, because of pigment in the sweat glands or on the skin surface.

Chromoblastomycosis

- See Fungal Infections.

Chromosomal Abnormality

- Change in chromosome number or structure. Numerical: aneuploidy (abnormal number, e.g., trisomy 21), polyploidy (extra sets). Structural: deletion, duplication, inversion, translocation.

Chromosome

- Thread-like structure of nucleic acids and protein found in nucleus, carrying genetic information. Humans have 46 chromosomes (23 pairs): 22 pairs of autosomes, 1 pair of sex chromosomes (XX female, XY male). Chromosomes condense during cell division for visualization.

Chromosomes

- See genetics.

Chronic

- Long-lasting, persistent. Examples: diabetes, hypertension.

Chronic Allograft Nephropathy

- Progressive transplant kidney function decline over months to years. Interstitial fibrosis and tubular atrophy on histology. Can be immunologic or non-immunologic. Optimize immunosuppression and manage risk factors including hypertension and recurrent disease.

Chronic autoimmune Hashimoto thyroiditis

- An autoimmune inflammatory disease of the thy-

roid that progressively destroys thyroid tissue and commonly causes hypothyroidism.

Chronic autoimmune thyroiditis

- Hashimoto thyroiditis, an autoimmune destruction of the thyroid that commonly causes hypothyroidism and may be associated with anti-thyroid antibodies.

Chronic Bronchitis

- Defined clinically as a productive cough for at least three months in each of two consecutive years, after excluding other causes. It is characterized by hypertrophy of mucus-secreting glands in the bronchi (Reid index > 0.5), goblet cell hyperplasia, and chronic airway inflammation.

Chronic Caries

- Slowly progressing caries. Hard, dark-colored cavity. Common in adults.

Chronic Cough

- Cough lasting > 8 weeks. Top 3 causes in non-smokers with normal CXR: upper airway cough syndrome, asthma, GERD. ACE inhibitor use is another common cause.

Chronic Eosinophilic Leukaemia

- A clonal myeloid neoplasm characterized by persistent eosinophilia, increased or dysplastic eosinophils, and evidence of an underlying clonal process or organ injury. Symptoms may include fatigue, fever, weight loss, rash, respiratory symptoms, splenomegaly, or cardiac and neurological complications.

Chronic eosinophilic pneumonia

- An idiopathic inflammatory lung disease characterized by eosinophilic infiltration of the pulmonary parenchyma, presenting with fever, cough, dyspnea, and peripheral airspace opacities on imaging.

Chronic Fatigue Syndrome

- A complex, debilitating disorder characterized by profound fatigue lasting >6 months that is not relieved by rest and is worsened by physical or mental exertion (post-exertional malaise). Associated symptoms: unrefreshing sleep, cognitive impairment (brain fog), orthostatic intolerance, joint pain, headaches, and sore throat.

Chronic Granulomatous Disease

- A primary immunodeficiency caused by defects in

the NADPH oxidase complex in phagocytes (neutrophils, monocytes, macrophages), impairing the respiratory burst and killing of catalase-positive organisms.

Chronic Hypertension in Pregnancy

- Hypertension (BP 140/90 mmHg) that predates pregnancy or is diagnosed before 20 weeks gestation. Affects 1-5% of pregnancies. Increased risk of superimposed preeclampsia (20-50%), IUGR, preterm birth, and placental abruption.

Chronic Inflammation

- A prolonged inflammatory response characterized by simultaneous tissue destruction and attempts at repair, lasting weeks to years. Unlike acute inflammation, it is predominantly mononuclear, involving macrophages, lymphocytes, and plasma cells rather than neutrophils.

Chronic Inflammatory Demyelinating Polyneuropathy

- An acquired, immune-mediated peripheral neuropathy with a relapsing-remitting or progressive course. Presents with symmetric proximal and distal weakness, sensory loss, areflexia, and elevated CSF protein. Distinguishing from Guillain-Barre: progressive course >8 weeks (vs. rapid onset in GBS).

Chronic Intestinal Pseudo-Obstruction

- Chronic condition of recurrent episodes of intestinal obstruction without mechanical blockage due to neuromuscular dysfunction of the gastrointestinal tract. May involve the small bowel, colon, or both. Causes include systemic sclerosis, diabetes, Parkinson disease, amyloidosis, and paraneoplastic syndromes.

Chronic Kidney Disease

- Progressive loss of kidney function over months to years, defined by GFR <60 mL/min/1.73 m² or markers of kidney damage for ≥ 3 months. Stages 1-5 based on GFR. Leading causes: diabetes mellitus (diabetic nephropathy) and hypertension.

Chronic Kidney Disease (CKD)

- Chronic kidney disease is the progressive loss of kidney function over months to years, typically from diabetes, hypertension, or glomerular diseases. It is staged by estimated glomerular filtration rate and

albuminuria, with end-stage renal disease requiring dialysis or transplantation.

Chronic Kidney Disease Staging

- Classification of kidney disease by cause, eGFR category, and albuminuria category. G1 through G5 represent eGFR from normal or high to kidney failure; staging is applied when abnormalities persist for at least 3 months and helps estimate progression and complications.

Chronic lead poisoning

- Prolonged low-level lead exposure causing abdominal pain, anemia with basophilic stippling, peripheral neuropathy, and nephropathy, requiring chelation therapy and removal of the exposure source.

Chronic leukemia

- leukemia characterized by an insidious onset, clonal proliferation of relatively mature blood cells, and a more indolent clinical course.

Chronic Limb-Threatening Ischemia (CLTI)

- The most severe clinical form of peripheral arterial disease, characterized by chronic ischemic rest pain, nonhealing ulceration, or gangrene attributable to arterial insufficiency. It carries high risks of limb loss and cardiovascular mortality and requires vascular evaluation for revascularization and wound management.

Chronic Liver Disease

- Persistent hepatic injury and structural change lasting at least 6 months, caused by viral, metabolic, alcohol-related, autoimmune, cholestatic, vascular, or drug-related disease. Progressive fibrosis can lead to cirrhosis, portal hypertension, liver failure, and hepatocellular carcinoma.

Chronic Lymphocytic Leukemia

- The most common leukemia in Western countries, characterized by the clonal proliferation of mature-appearing but functionally incompetent B lymphocytes that accumulate in blood, bone marrow, and lymphoid tissues.

Chronic Mesenteric Ischemia

- Recurrent abdominal pain occurring after eating resulting from insufficient blood flow to meet the metabolic demands of the gastrointestinal tract. Most commonly caused by atherosclerotic steno-

sis of at least two of the three major mesenteric arteries.

Chronic Mucocutaneous Candidiasis

- Persistent/recurrent *Candida* infections of skin, nails, mucosa (oral, esophageal, vaginal) without systemic dissemination. Autosomal dominant (STAT1 gain-of-function) or recessive (IL17F, IL17RA, ACT1, RORC). Associated with autoimmunity (hypothyroidism, AIHA, ITP, alopecia), endocrinopathies, Staph infections.

Chronic myelogenous leukemia

- A myeloproliferative neoplasm associated with the BCR-ABL1 fusion gene, usually producing marked leukocytosis and splenomegaly.

Chronic Myeloid Leukemia

- A myeloproliferative neoplasm characterized by the Philadelphia chromosome [t(9;22)(q34;q11.2)], which creates the BCR-ABL1 fusion gene encoding a constitutively active tyrosine kinase.

Chronic Myelomonocytic Leukemia

- A rare myelodysplastic/myeloproliferative neoplasm characterized by persistent monocytosis (absolute monocyte count greater than 1,000 per microliter and greater than 10 percent of white blood cells) in the peripheral blood and bone marrow, with dysplastic features and variable blast count.

Chronic Neutrophilic Leukaemia

- A rare clonal myeloproliferative neoplasm characterized by sustained mature neutrophilic leukocytosis without a primary infectious or inflammatory cause. It may cause fatigue, fever, weight loss, splenomegaly, and progressive constitutional symptoms.

Chronic Obstructive Pulmonary Disease

- COPD is a preventable lung disease characterized by persistent respiratory symptoms and airflow limitation due to airway and/or alveolar abnormalities, usually caused by significant exposure to noxious particles or gases (primarily tobacco smoke). Includes emphysema and chronic bronchitis.

Chronic Obstructive Pulmonary Disease (COPD)

- COPD is a progressive lung disease characterized by persistent airflow limitation due to chronic bronchitis and emphysema, most commonly caused

by cigarette smoking. Symptoms include dyspnea, chronic cough, sputum production, and frequent respiratory infections.

Chronic Pain

- Pain persisting longer than three months or beyond normal tissue healing time, often lacking an ongoing identifiable cause. It involves complex interactions between biological, psychological, and social factors that can amplify pain signaling in the nervous system.

Chronic Pain Management

- Chronic pain management extends beyond the perioperative period. Opioid therapy for chronic pain requires careful patient selection, informed consent, and regular reassessment.

Chronic Pancreatitis

- Progressive irreversible inflammatory condition characterized by permanent structural damage causing exocrine and endocrine impairment. The most common cause is chronic alcohol consumption. Other causes include hereditary pancreatitis, autoimmune pancreatitis, and metabolic factors.

Chronic Rhinosinusitis

- A heterogeneous group of disorders characterized by persistent inflammation of the paranasal sinuses lasting more than twelve weeks, with symptoms including nasal obstruction, purulent rhinorrhea (anterior or posterior), facial pain or pressure, and olfactory dysfunction, and it is classified into chronic rhinosinusitis without nasal

Chronic sinusitis

- Persistent inflammation of the paranasal sinuses for at least 12 weeks, causing nasal obstruction, discharge, facial pressure, and reduced smell.

Chronic Thromboembolic Pulmonary Hypertension (CTEPH)

- Pulmonary hypertension caused by organized, persistent thromboembolic obstruction and secondary pulmonary-vessel remodeling after pulmonary embolism. It is potentially curable in selected patients by pulmonary endarterectomy or balloon pulmonary angioplasty.

Chronic Toxicity

- Adverse effects from repeated or prolonged exposure to low doses. Cumulative damage may be irre-

versible. Examples: heavy metals (lead, mercury), asbestos, alcohol, solvent exposure. Monitoring: biomarkers, organ function tests.

Chronic Venous Disease

- A spectrum of venous outflow disorders caused by reflux, obstruction, or calf-muscle-pump dysfunction. Findings range from edema and varicose veins to skin changes, lipodermatosclerosis, and venous ulceration.

Chronic Venous Insufficiency

- Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. Leads to venous hypertension, edema, skin changes (lipodermatosclerosis, hyperpigmentation), and venous stasis ulcers (medial malleolus). Risk factors: varicose veins, DVT, obesity, prolonged standing.

Chronic Venous Insufficiency (CVI)

- Inadequate venous return causing venous hypertension. Symptoms: edema, skin changes (hyperpigmentation, lipodermatosclerosis), venous ulcer. Diagnosis: venous duplex ultrasound. Treatment: compression therapy, wound care, surgical intervention if indicated.

Chronic venous stasis

- See Chronic Venous Insufficiency.

Churg-Strauss Syndrome

- Eosinophilic granulomatosis with polyangiitis, a necrotizing inflammation of small and medium blood vessels associated with asthma, eosinophilia, and possible injury to the lungs, nerves, skin, heart, or kidneys.

Chvostek Sign

- A clinical sign of latent tetany caused by hypocalcemia (ionized $\text{Ca}^{2+} < 4.4 \text{ mg/dL}$). Elicited by tapping the facial nerve over the parotid gland just anterior to the ear canal. Positive response: involuntary twitching or contraction of ipsilateral facial muscles (lip twitching or facial spasm).

Chylangioma

- A benign malformation or tumor-like lesion of lymphatic vessels containing chyle or lymph. It may occur in the neck, mediastinum, or other tissues.

Chylomicron

- Transports dietary lipids. Largest, least dense lipoprotein.

Chylothorax

- Pleural effusion with chyle from thoracic duct disruption. Triglycerides > 110 mg/dL. Treatment includes dietary modification, octreotide, and surgical ligation if refractory.

Chyluria

- Passage of lymphatic fluid into the urine, giving it a milky appearance. It usually results from an abnormal communication between lymphatic vessels and the urinary tract.

Ciclopia

- A rare severe congenital malformation in which the eyes are fused or a single median eye develops, usually as part of holoprosencephaly.

CIDP

- Chronic inflammatory demyelinating polyneuropathy (CIDP) is a chronic, immune-mediated, peripheral neuropathy characterized by progressive or relapsing symmetrical weakness and sensory loss in all four limbs, with areflexia or hyporeflexia, lasting for more than eight weeks.

Ciguatera

- Food poisoning caused by eating reef fish containing ciguatoxins. It may cause gastrointestinal illness followed by neurologic symptoms such as paresthesias and temperature-sensation disturbance.

Ciliary Body

- Circumferential structure between iris and choroid, with ciliary muscle (accommodation control via zonular fibers) and ciliary processes (aqueous humor production). Part of the uveal tract (iris, ciliary body, choroid).

Ciliopathy

- A disorder caused by abnormal structure or function of cellular cilia. Ciliopathies may affect development, vision, kidney function, respiration, or multiple organ systems.

Cineangiocardiology

- Radiographic recording of the heart and blood vessels during passage of contrast material, used to visualize cardiac chambers, valves, and coronary or great vessels.

Cinefluorography

- Motion-picture recording of fluoroscopic images,

used to document movement of organs or contrast material during a diagnostic examination.

Cineplasty

- Surgical creation of a muscle-controlled stump or related reconstruction, historically used to improve control of a prosthetic limb.

Cingulotomy

- Surgical interruption of selected fibers of the cingulum, sometimes performed as a psychosurgical treatment for severe, treatment-resistant psychiatric illness or chronic pain.

Circadian Rhythm

- The approximately 24-hour endogenous biological cycle regulating sleep-wake behavior, hormone secretion, body temperature, metabolism, and other physiological processes. Generated by the suprachiasmatic nucleus of the hypothalamus, which is entrained to the environment primarily by light via the retinohypothalamic tract.

Circle of Willis

- A ring-like arterial anastomosis at the base of the brain that provides collateral blood flow between the anterior (internal carotid) and posterior (vertebrobasilar) circulations. It is formed by the anterior cerebral, anterior communicating, internal carotid, posterior cerebral, and posterior communicating arteries.

Circulating Tumor DNA (ctDNA)

- Fragments of tumor-derived DNA detectable in blood, enabling molecular profiling, treatment monitoring, and detection of minimal residual disease without tissue biopsy (liquid biopsy).

Circumcision

- Surgical removal of the foreskin (prepuce) covering the glans penis. Indications: medical reasons such as severe phimosis, recurrent balanoposthitis, paraphimosis, or lichen sclerosus (balanitis xerotica obliterans); elective infant circumcision for religious, cultural, or preventive health reasons.

Circumflex Artery

- A branch of the left coronary artery that travels in the left atrioventricular groove. It supplies the lateral wall of the left ventricle, the left atrium, and in left-dominant systems, the inferior wall and posteromedial papillary muscle.

Cirrhosis

- The end stage of chronic liver disease, characterized by diffuse fibrosis with distortion of the hepatic architecture and formation of regenerative nodules. The most common causes include chronic hepatitis C infection, alcoholic liver disease, non-alcoholic steatohepatitis, chronic hepatitis B infection, and autoimmune hepatitis.

Citalopram

- A selective serotonin reuptake inhibitor (SSRI) used for major depressive disorder and (off-label) anxiety disorders. Well tolerated; lowest cost among SSRIs. Adverse effects: nausea, insomnia, sexual dysfunction, and dose-dependent QT prolongation (maximum 40 mg/day; 20 mg in elderly/CYP2C19 poor metabolizers—torsades risk).

Citrate Synthase

- First enzyme of TCA cycle. Condenses oxaloacetate and acetyl-CoA to citrate.

Citric Acid Cycle (Krebs Cycle)

- A central metabolic pathway occurring in the mitochondrial matrix that oxidizes acetyl-CoA (derived from carbohydrates, lipids, and proteins) into CO₂, generating 3 NADH, 1 FADH₂, and 1 GTP (ATP) per turn. Rate-limiting enzyme: isocitrate dehydrogenase (inhibited by ATP and NADH; activated by ADP).

CKD

- Abbreviation for Chronic Kidney Disease: a progressive loss of kidney function over months to years, defined by a glomerular filtration rate (GFR) less than 60 mL/min/1.73m² or markers of kidney damage (albuminuria, haematuria, structural abnormalities) persisting for at least 3 months.

CKD-Mineral Bone Disorder

- Systemic disorder of mineral and bone metabolism in CKD. Includes calcium, phosphorus, PTH, and vitamin D abnormalities with vascular calcification and bone disease. Renal osteodystrophy encompasses osteitis fibrosa, osteomalacia, adynamic bone disease, and mixed uremic osteodystrophy.

Class IA Antiarrhythmics

- Sodium channel blockers that slow phase 0 depolarization and prolong the action potential duration and refractory period. Examples include quinidine,

procainamide, and disopyramide. Used for atrial and ventricular arrhythmias. Quinidine can cause QT prolongation and torsades de pointes.

Class IB Antiarrhythmics

- Sodium channel blockers that shorten the action potential duration and are preferentially taken up by ischemic tissue. Examples include lidocaine and mexiletine. Lidocaine is used for ventricular tachycardia and ventricular fibrillation, particularly in the setting of acute myocardial infarction.

Class IC Antiarrhythmics

- Sodium channel blockers that significantly slow phase 0 depolarization without affecting action potential duration. Examples include flecainide and propafenone. Used for supraventricular arrhythmias in patients without structural heart disease.

Class III Antiarrhythmics

- Potassium channel blockers that prolong the action potential duration and refractory period. Examples include amiodarone, sotalol, dofetilide, and ibutilide. Amiodarone is the most commonly used but has significant toxicity. Sotalol has additional beta-blocking properties and can cause torsades de pointes.

Claudication

- Atherosclerotic occlusive disease of arteries supplying the lower limbs, causing reduced blood flow. See peripheral arterial disease.

Claudication Distance

- Distance walked before pain onset. Improves with exercise, medication, revascularization.

Claustrophobia

- An anxiety disorder characterized by an intense, irrational fear of enclosed or confined spaces (e.g., lifts, tunnels, MRI scanners, windowless rooms). Individuals experience marked anxiety, panic attacks, and avoidance behaviour. Prevalence: approximately 5-10% of the general population.

Clavicle

- Only horizontally oriented long bone, serving as a shoulder-trunk strut. Medial end articulates with the manubrium (sternoclavicular joint); lateral end with the acromion (acromioclavicular joint). Most common fracture at the middle-lateral third junction from falls on an outstretched hand.

Clavicle Fracture

- The most common birth-related fracture, occurring in approximately 1-2% of vaginal deliveries. Risk factors include shoulder dystocia, macrosomia, and breech presentation. Presents with asymmetric arm movements, palpable crepitus, and reluctance to move the affected arm. Diagnosis by X-ray.

Clavicle Fracture (Intentional)

- Intentional fracture of the fetal clavicle as a last resort to reduce the bisacromial diameter in shoulder dystocia. The operator applies upward pressure on the midportion of the anterior fetal clavicle with a finger, fracturing it. The clavicle heals without long-term sequelae.

Clavicle Fractures

- Clavicle fractures: 80% midshaft (Group I), 15% lateral (Group II: Neer I nondisplaced, II displaced with CC ligament disruption, III intra-articular), 5% medial (Group III). Midshaft: nondisplaced/shortening <2cm: sling, figure-8; displaced/comminuted/shortening >2cm/skin tenting: ORIF (superior plate, anteroinferior

Claw Foot

- A deformity in which the toes are hyperextended at the metatarsophalangeal joints and flexed at the interphalangeal joints. It may result from muscle imbalance or neurologic disease.

Claw Hand

- A hand deformity with hyperextension at the metacarpophalangeal joints and flexion at the interphalangeal joints, classically caused by ulnar-nerve palsy or lower brachial-plexus injury.

Claw Toe

- A toe deformity characterized by extension at the metatarsophalangeal joint and flexion at the proximal and distal interphalangeal joints.

Clear Margins

- A histological finding indicating that no abnormal cells (e.g., cervical intraepithelial neoplasia or cancer cells) are present at the edge of a tissue specimen removed during a procedure such as LLETZ, cone biopsy, or surgical excision.

Clearance

- The volume of plasma from which a drug is completely removed per unit time, typically expressed

in milliliters per minute or liters per hour. Total clearance is the sum of all clearance mechanisms, primarily hepatic and renal clearance.

Cleft Lip

- A congenital malformation: incomplete fusion of the lip (and often palate) during embryologic development, causing a facial cleft; cleft lip and/or cleft palate are the most common craniofacial birth defects.

Cleft Lip and Palate

- A congenital anomaly caused by failure of fusion of the facial prominences during embryonic development (4-10 weeks gestation). Cleft lip: failure of maxillary and medial nasal processes to fuse. Cleft palate: failure of palatal shelves to fuse. Can occur together or separately.

Cleft Palate

- A congenital malformation characterized by an opening (cleft) in the palate (roof of the mouth) due to failure of fusion of the palatal shelves during embryonic development (weeks 6–9 of gestation). Incidence: 1 in 1,500–2,500 live births. Cleft palate may occur in isolation or with cleft lip (cleft lip and palate—CLP).

Cleidocranial Dysplasia

- An inherited skeletal disorder characterized by defective development or ossification of the clavicles and skull, delayed closure of cranial sutures, dental abnormalities, and short stature.

Climacteric

- The period of physiologic change around the end of reproductive capacity, including the menopausal transition. It may involve changes in menstruation, vasomotor symptoms, and other endocrine effects.

Clinical Communication

- Exchange of information between healthcare professionals, patients, and caregivers to support diagnosis, treatment, safety, and informed decisions.

Clinical Counseling

- Guidance provided in healthcare to help patients make informed decisions about health, prevention, treatment, and self-management.

Clinical Depression

- A depressive disorder marked by persistent depressed mood or loss of interest with associated

cognitive, emotional, or physical symptoms that cause clinically significant distress or impairment.

Clinical Deterioration

- Worsening of physiologic function or symptoms that increases risk of organ failure, disability, or death. Changes in mental status, respiratory effort, perfusion, urine output, and vital signs may precede overt collapse.

Clinical Documentation

- Recording of patient history, examination, assessment, treatment, and follow-up in the medical record.

Clinical Frailty Scale

- A clinician-rated scale that summarizes frailty from very fit to severely frail or terminally ill using activity, function, and disease burden. It supports communication and prognostic assessment but should be interpreted in the patient's clinical context.

Clinical Guideline

- Systematically developed statements to assist practice. Examples: CDC guidelines, WHO guidelines, specialty society guidelines.

Clinical Handover

- Transfer of responsibility and relevant patient information from one healthcare professional or team to another. Effective handover communicates diagnosis, current status, risks, pending tasks, contingency plans, and accountability.

Clinical Practice Guideline

- Systematically developed statements to assist practice. See clinical guideline.

Clinical Presentation

- Signs and symptoms of disease. Varies by disease, severity, patient factors.

Clinical Reasoning

- The cognitive process of collecting, weighting, and integrating clinical information to formulate diagnostic and management decisions. It includes problem representation, differential diagnosis, probability assessment, and recognition of cognitive bias.

Clinical Research

- Research study testing new cancer treatments. See clinical trial.

Clinical Research Bias

- Systematic error in clinical study design or analysis, including selection bias, information bias, and confounding.

Clinical Trial

- Research study testing new cancer treatments. Phases: I (safety), II (efficacy), III (comparison), IV (post-marketing).

Clinical Trial Phases

- The stages of drug development: Phase I (safety, dosing, pharmacokinetics in 20-100 healthy volunteers); Phase II (efficacy and side effects in 100-300 patients with the target condition); Phase III (large-scale efficacy and safety comparison with standard treatment in 1,000-3,000 patients); Phase IV (post-marketing surveillance for long-

Clinical Trials

- Research study testing new cancer treatments. See clinical trial.

Clinicopathologic Correlation

- Integration of clinical history, examination, imaging, laboratory data, and pathologic findings to reach a diagnosis. It is essential when tissue findings are limited, nonspecific, or potentially affected by sampling or artifact.

Clinodactyly

- A congenital or developmental lateral curvature of a finger or toe, most often affecting the fifth finger and resulting from abnormal shape of a phalanx.

Clinoid Process

- One of several bony projections of the sphenoid bone that form part of the cranial base and support structures around the optic canal and cavernous sinus.

Clip (Vascular)

- See Vascular Clip.

Clitoridectomy

- Surgical removal of part or all of the clitoris. In medicine it may be performed for selected tumors or severe disease; nonmedical removal is a harmful practice with no health benefit.

Clitoris

- A small, sensitive erectile organ located at the anterior junction of the labia minora, beneath the

clitoral hood (prepuce). The clitoris is the primary source of female sexual pleasure and is homologous to the male penis.

Cloacal Exstrophy

- The most severe congenital anomaly of the ventral abdominal wall, also called the OEIS complex (Omphalocele, Exstrophy of the bladder, Imperforate anus, Spinal defects). Incidence 1 in 200,000-400,000 births.

Cloacitis

- Inflammation of a cloaca or cloacal region, particularly in disorders affecting the common terminal outlet of the urinary, genital, and intestinal tracts.

Clonal Evolution

- The selection and expansion of tumor-cell subclones with biologic or treatment-related advantages over time. It contributes to progression, metastasis, and acquired resistance to anticancer therapy.

Clonic Seizure

- A seizure characterized by rhythmic, repetitive jerking caused by recurrent involuntary muscle contractions. It may be focal or generalized, with or without impaired awareness, and is classified using the clinical pattern, electroencephalography, and the underlying cause.

Clindamycin

- An antibiotic used for selected bacterial infections, including skin, dental, bone, and anaerobic infections. It can cause severe antibiotic-associated diarrhea from *Clostridioides difficile*; use should follow appropriate clinical assessment.

Clonidine

- A medicine that lowers blood pressure by reducing sympathetic nerve signals. It is also used in some patients for withdrawal symptoms, pain, or attention disorders. Sudden discontinuation can cause a dangerous rebound increase in blood pressure.

Clonorchiasis

- Infection with the Chinese liver fluke, *Clonorchis sinensis*. It affects the biliary tract and may cause inflammation, obstruction, and long-term hepatobiliary complications.

Clonus

- A rhythmic, involuntary oscillation of a muscle (usually ankle or patellar) produced by sustained

stretch, indicating upper motor neuron (UMN) disease—hyperreflexia. Elicited by rapid dorsiflexion of the foot (sustained clonus—5 or more beats) or tapping. Associated with spasticity, hyperreflexia, and UMN signs (Babinski).

Closed Comedo

- A small skin lesion formed when a follicular opening is blocked by sebum and keratin but remains covered by the epidermis. It is commonly called a whitehead.

Closed Fracture

- A fracture in which the broken bone does not communicate through a wound with the external environment.

Closed Reduction

- Nonoperative realignment of a fractured or dislocated bone or joint by manipulation without surgically opening the site.

Closed-Angle Glaucoma

- Glaucoma caused by obstruction of the drainage angle of the anterior chamber, preventing aqueous humor outflow and producing potentially rapid elevation of intraocular pressure.

Closed-Loop Insulin Delivery

- Automated insulin delivery systems that combine continuous glucose monitoring with an insulin pump and control algorithm. The algorithm adjusts basal insulin delivery based on sensor glucose values in real time. Hybrid closed-loop systems require user-initiated meal boluses but automatically manage basal rates.

Clostridioides difficile

- An anaerobic, spore-forming, toxin-producing gram-positive bacillus causing antibiotic-associated diarrhea and pseudomembranous colitis. Risk factors: antibiotics (especially clindamycin, fluoroquinolones, cephalosporins), hospitalization, age >65, PPIs, and immunosuppression. Toxins A and B damage colonic epithelium.

Clostridioides difficile infection

- An infectious diarrhea caused by the bacterium *Clostridioides difficile*, typically occurring after disruption of normal gastrointestinal flora by antibiotics. *C. difficile* produces toxins A and B that cause colonic inflammation and diarrhea. Risk factors include antibiotic exposure, proton pump inhibitor

use, and advanced age.

Clostridioides difficile pathogenesis

- The mechanisms by which *C. difficile* causes disease. The organism produces toxins A and B that damage the colonic mucosa. Antibiotic disruption of normal gut flora allows *C. difficile* to colonize and produce toxins. The organism forms spores that are resistant to disinfectants and can persist in the environment. Understanding *C.*

Clostridium

- A genus of anaerobic, usually spore-forming bacteria that includes species capable of causing diseases such as tetanus, botulism, gas gangrene, and antibiotic-associated colitis.

Clostridium botulinum

- A gram-positive, spore-forming, anaerobic rod that produces botulinum toxin, the most potent toxin known. The toxin causes flaccid paralysis by blocking acetylcholine release at neuromuscular junctions. Infant botulism occurs from ingestion of spores in honey or soil. Wound botulism occurs from injection of contaminated heroin.

Clostridium Difficile

- *Clostridioides difficile* infection (CDI): a toxin-producing anaerobic gram-positive rod causing antibiotic-associated diarrhea and pseudomembranous colitis, most commonly after broad-spectrum antibiotics (clindamycin, cephalosporins, fluoroquinolones).

Clostridium difficile Infection (CDI)

- An anaerobic, spore-forming, toxin-producing gram-positive bacillus causing antibiotic-associated diarrhea and pseudomembranous colitis. See *Clostridioides difficile*.

Clostridium perfringens

- A gram-positive, spore-forming, anaerobic rod that causes gas gangrene (clostridial myonecrosis) and food poisoning. The organism produces alpha-toxin, which causes tissue necrosis. Gas gangrene presents with crepitus, tissue necrosis, and systemic toxicity. Food poisoning presents with acute onset of diarrhea and abdominal cramps.

Clostridium tetani

- A gram-positive, spore-forming, anaerobic rod that causes tetanus. The organism produces tetanospas-

min, a neurotoxin that causes spastic paralysis. Spores are found in soil and enter through wound contamination. The toxin blocks inhibitory neurotransmitters causing uncontrolled muscle contractions.

Clotting

- The cascade of enzymatic reactions converting circulating fibrinogen into an insoluble fibrin clot. See Blood Coagulation.

Clubbing

- Bulbous enlargement of the distal fingers or toes with increased nail curvature and loss of the normal nail-fold angle. It reflects proliferation of connective tissue in the distal digits and may be associated with chronic pulmonary disease, cyanotic heart disease, inflammatory bowel disease, or malignancy.

Clubfoot

- A congenital deformity in which the foot is fixed in equinus, hindfoot varus, forefoot adduction, and cavus; the medical term is congenital talipes equinovarus. It is usually idiopathic but may be associated with neuromuscular or syndromic conditions and requires early orthopaedic assessment.

Clubfoot (Talipes Equinovarus)

- A congenital foot deformity with an incidence of approximately 1 in 1,000 live births, characterized by four components: equinus (ankle plantarflexion), varus (hindfoot inversion), adductus (forefoot adduction), and cavus (high arch).

Clubhand

- A congenital deformity of the hand in which the wrist and hand deviate toward the radial or ulnar side, often associated with deficiency or abnormal development of forearm bones.

Cluster Headache

- A primary headache disorder characterized by severe, unilateral, strictly orbital or periorbital pain attacks lasting fifteen to one hundred eighty minutes, occurring with one of three autonomic symptoms on the same side as the pain: conjunctival injection, lacrimation, nasal congestion or rhinorrhea, eyelid edema, forehead or facial swea

Clutton's Joints

- Painless bilateral swelling of the knees or other large joints associated with congenital syphilis, caused by

chronic synovitis.

CMV

- A widespread herpesvirus infection. See Cytomegalovirus Infection.

CMV Colitis

- Colonic inflammation caused by cytomegalovirus reactivation in immunocompromised patients including those with HIV and organ transplant recipients. Presents with bloody diarrhea and abdominal pain. Diagnosed by colonoscopy with biopsy showing characteristic owl-eye intranuclear inclusions in enlarged cells.

CO Poisoning

- Toxicity from inhalation of carbon monoxide (CO), a colorless, odorless gas binding hemoglobin 200-300 times more avidly than oxygen, forming carboxyhemoglobin (COHb) and reducing oxygen delivery. See carbon monoxide poisoning.

Coagulation Cascade

- The coagulation cascade is a series of enzymatic reactions leading to fibrin clot formation, involving serine protease activation cascades. The intrinsic pathway is initiated by contact activation when factor XII binds exposed collagen, sequentially activating factors XI, IX, and the cofactor VIII.

Coagulation Factor Deficiencies

- Rare factor deficiencies: Factor VII (most common rare bleeding disorder, prolonged PT, normal aPTT, mucosal bleeding, hemarthrosis; recombinant FVIIa for treatment), Factor X (prolonged PT/aPTT), Factor XI (Ashkenazi Jews, prolonged aPTT, bleeding with surgery/trauma, FFP for replacement), Factor XIII (normal PT/aPTT, clot solubility test)

Coagulation Profile

- Panel of tests assessing the coagulation cascade. PT (prothrombin time, 11-14 sec): measures extrinsic and common pathways; elevated in vitamin K deficiency, warfarin, liver disease, DIC. INR standardizes PT for warfarin monitoring.

Coagulative Necrosis

- The most common form of necrosis, typically resulting from ischemic injury or infarction in most solid organs except the brain. The fundamental process involves denaturation of both structural proteins and enzymatic proteins, which prevents

the immediate liquefaction of the dead tissue.

Coagulopathy

- A disorder of haemostasis that impairs normal clot formation or regulation and predisposes to abnormal bleeding or thrombosis. It may result from coagulation-factor deficiency, platelet dysfunction, anticoagulant exposure, liver disease, consumptive coagulopathy, or systemic illness.

Coal Worker's Pneumoconiosis

- Pneumoconiosis from coal dust. Simple CWP shows upper lobe macules. Complicated CWP (progressive massive fibrosis) involves coalescent macules. May be associated with Caplan syndrome.

Coarctation

- A localized narrowing of a tubular anatomical structure, especially coarctation of the aorta, which obstructs blood flow and produces characteristic pressure differences.

Coarctation of the Aorta

- A congenital narrowing of the aorta, typically near the ductus arteriosus. Presents with upper extremity hypertension, diminished femoral pulses, and a blood pressure gradient between arms and legs. Associated with bicuspid aortic valve and other congenital heart defects.

Coarse Crackles

- Lower-pitched, longer-duration crackles heard throughout inspiration, caused by fluid or secretions in larger airways. They are heard in COPD, bronchiectasis, pneumonia, and pulmonary edema. They are less specific than fine crackles.

Coats Disease

- An uncommon retinal disorder characterized by abnormal retinal blood vessels, leakage, exudation, and sometimes retinal detachment, usually affecting one eye in childhood.

Cobalamin Deficiency

- Deficiency of vitamin B12 (cobalamin), a cofactor for methionine synthase (methylcobalamin) and methylmalonyl-CoA mutase (adenosylcobalamin).

Cobalt-57

- Calibration source for gamma cameras. Half-life 271 days. Used for quality control.

Cobalt-Chromium

- Base metal alloy for cast restorations. Strong, corrosion-resistant. Used for partial dentures, crowns.

Cocaine

- A potent central nervous system stimulant and sympathomimetic causing vasoconstriction, tachycardia, hypertension, and euphoria, with acute toxicity manifesting as myocardial ischemia, seizures, and hyperthermia.

Cocaine Cardiotoxicity

- Cocaine blocks norepinephrine and dopamine reuptake, causing sympathomimetic effects: tachycardia, hypertension, and coronary vasospasm. Acute myocardial infarction can occur even with normal coronary arteries due to vasospasm, accelerated atherosclerosis, and prothrombotic effects. Treatment: benzodiazepines for sympatholysis.

Cocaine Toxicity

- Sympathomimetic toxidrome: hypertension, tachycardia, hyperthermia, agitation, seizures, myocardial infarction, stroke. Management: benzodiazepines first-line, cooling, activated charcoal if recent ingestion. Avoid beta-blockers (unopposed alpha stimulation).

Coccidioidomycosis

- A fungal infection caused by *Coccidioides immitis* or *C. posadasii*, endemic to the southwestern US (San Joaquin Valley fever). Inhalation of arthroconidia causes primary pulmonary infection. Most cases are asymptomatic or mild (flu-like illness with cough, fever, erythema nodosum).

Coccidiosis

- Infection by coccidian protozoa. In humans it may cause intestinal disease or, in immunocompromised patients, infection of other organs.

Coccydynia

- Pain in the region of the coccyx, often worsened by sitting. Causes include trauma, childbirth, degenerative change, and less commonly infection or tumor.

Coccygectomy

- Surgical removal of the coccyx, reserved for selected cases of severe persistent coccygeal pain or a lesion that has not responded to conservative treatment.

Coccyx

- Terminal vertebral column part with 3-5 fused vertebrae. Articulates with the sacrum, allowing limited flexion important during childbirth. Provides attachment for gluteus maximus, coccygeus, and levator ani (pelvic floor). A common site of pain (coccydynia) following falls, direct trauma, and prolonged sitting.

Cochlea

- Snail-shaped hearing organ in the inner ear, coiled approximately 2.5 turns around the modiolus. Three fluid-filled chambers: scala vestibuli (perilymph), scala media (endolymph, organ of Corti), scala tympani (perilymph).

Cochlear Implant

- A cochlear implant is an electronic medical device that bypasses damaged or absent inner ear hair cells to provide direct electrical stimulation of the auditory nerve, restoring a perception of sound in individuals with severe to profound sensorineural hearing loss who derive limited benefit from hearing aids.

Cockayne Syndrome

- A rare inherited disorder of DNA repair characterized by progressive neurologic dysfunction, growth failure, developmental impairment, photosensitivity, and a prematurely aged appearance.

Code Blue

- A hospital emergency signal indicating that a person may be in cardiopulmonary arrest and requires immediate resuscitation response.

Codman Triangle

- A characteristic periosteal reaction seen on plain radiographs, formed when an aggressive bone lesion elevates the periosteum away from the bone cortex, creating a triangular area of new reactive bone at the angle between the raised periosteum and underlying cortex.

Codon

- Sequence of three nucleotides coding for one amino acid or stop signal. 64 possible codons: 61 code for amino acids, 3 are stop codons (UAA, UAG, UGA). Degeneracy: multiple codons code for same amino acid.

Coenzyme

- A non-protein organic molecule that binds to an enzyme (apoenzyme) and is required for its catalytic activity. Coenzymes often act as transient carriers of specific chemical groups or electrons.

Coenzyme Q

- See Ubiquinone.

Cofactor

- A non-protein chemical compound or metallic ion required for an enzyme's catalytic activity. Cofactors include: (1) Metal ions (Mg^{2+} , Zn^{2+} , Fe^{2+}/Fe^{3+} , Cu^{2+} , Mn^{2+} , Ca^{2+} , Mo, Se) that stabilize structure, participate in redox, or act as Lewis acids; (2) Coenzymes (organic cofactors, often vitamin-derived).

Cogan's syndrome

- An autoimmune vasculitis characterized by interstitial keratitis and vestibular dysfunction including vertigo, hearing loss, and tinnitus. May involve large vessels including the aorta, causing aortic aneurysm or regurgitation. Treatment includes corticosteroids and immunosuppressive agents.

Cognitive Behavioral Therapy

- A structured, time-limited psychotherapy that identifies and modifies maladaptive thought patterns (cognitive distortions) and behaviors. It is based on the cognitive model that thoughts, feelings, and behaviors are interconnected. CBT uses techniques including cognitive restructuring, behavioral activation, exposure, and skills training.

Cognitive Behavioral Therapy for Insomnia (CBT-I)

- Cognitive behavioral therapy for insomnia is a structured, evidence-based program that targets the thoughts and behaviors perpetuating poor sleep. Components include sleep restriction, stimulus control, cognitive restructuring, and sleep hygiene education.

Cognitive Decline

- Decline in one or more cognitive domains: memory, attention, language, visuospatial, executive function. See cognitive impairment.

Cognitive Distortions

- Systematic errors in thinking that perpetuate negative emotional states and maladaptive behaviors.

Common distortions include all-or-nothing thinking, catastrophizing, mind reading, overgeneralization, emotional reasoning, and should statements. CBT focuses on identifying and modifying these patterns.

Cognitive Fitness

- The active maintenance of mental sharpness and brain health through stimulating activities, physical exercise, social engagement, and proper nutrition. Regular challenges such as learning new skills, puzzles, reading, and social interaction support neuroplasticity and cognitive reserve.

Cognitive Health and Memory

- Cognitive health encompasses the ability to think clearly, learn, remember, and make decisions, with memory being a core component that involves encoding, storing, and retrieving information. Age-related changes in processing speed and recall are normal, but persistent decline may signal mild cognitive impairment or dementia.

Cognitive Impairment

- Decline in one or more cognitive domains: memory, attention, language, visuospatial, executive function. Types: subjective cognitive decline, mild cognitive impairment (MCI), dementia. MCI: functional independence preserved, impairment on testing. Dementia: functional impairment.

Cognitive Rehabilitation

- See neurology.

Cognitive Screening

- Decline in one or more cognitive domains: memory, attention, language, visuospatial, executive function. See cognitive impairment.

Cohort Study

- Follows exposed and unexposed groups over time. Prospective or retrospective. Measures incidence, relative risk.

Coil (Vascular)

- See Coil.

Coincidence Detection

- Principle of PET imaging. Two 511 keV photons detected simultaneously (within nanoseconds) by opposing detectors. Line of response defined. Time-of-flight PET improves localization.

Cold Abscess

- An abscess that develops slowly with little heat, redness, or acute inflammation, classically in tuberculosis or other chronic infections.

Cold Agglutinin

- An antibody that causes red blood cells to agglutinate at relatively low temperatures. Pathologic cold agglutinins can cause autoimmune hemolytic anemia.

Cold Agglutinin Disease

- Autoimmune hemolytic anemia caused by IgM autoantibodies directed against the I/i antigens on RBC membranes (maximum reactivity at $< 30^{\circ}\text{C}$, optimal 4°C).

Cold agglutinin syndrome

- See Cold Agglutinin Disease.

Cold agglutinins

- IgM autoantibodies that bind red blood cell antigens at temperatures below 37 degrees Celsius, causing agglutination and potentially complement-mediated hemolysis in cooler peripheral circulation.

Cold Injury

- Tissue or systemic injury caused by exposure to low temperature, including hypothermia, frostbite, chilblains, and non-freezing cold injury. Severity depends on temperature, exposure duration, moisture, wind, and perfusion; affected patients require assessment for core-temperature disturbance and local tissue damage.

Cold intolerance

- Abnormal sensitivity to cold that may occur with hypothyroidism, low body weight, anemia, vascular disease, or other systemic conditions.

Cold Sore

- Herpes labialis: recurrent herpes simplex virus type 1 (HSV-1) infection of the lips/around mouth, presenting as prodromal tingling, then grouped vesicles that crust and heal in 7-14 days. Triggered by stress, fever, sunlight, immunosuppression, illness. Recurrence due to reactivation of latent virus in the trigeminal ganglion.

Cold Sores

- Recurrent clusters of painful vesicles caused by herpes simplex virus, usually HSV-1, at or around

the lips. Lesions progress from tingling to vesicles, erosions, and crusting and are contagious from the prodrome until healing.

Colectomy

- Surgical resection of part or all of the colon. Right hemicolectomy: for cecal/ascending colon cancer. Left hemicolectomy: for descending/sigmoid cancer. Sigmoidectomy: for diverticulitis or sigmoid cancer. Total colectomy: for FAP, UC, or Lynch syndrome.

Colibacillosis

- Disease caused by pathogenic strains of *Escherichia coli*, which may produce intestinal infection, urinary-tract infection, sepsis, or other illness.

Colic

- See Infantile Colic.

Colchicine

- An anti-inflammatory medicine used mainly to treat gout flares and prevent recurrent gout or pericarditis in selected patients. Excess dosing can cause severe gastrointestinal symptoms, bone-marrow suppression, muscle injury, or organ toxicity.

Colitis

- Inflammation of the colon, with a broad differential diagnosis and diverse clinical presentations. Acute colitis presents with diarrhea (often bloody), abdominal pain, tenesmus, and urgency. The most common causes include infectious colitis (bacterial: *Shigella*, *Salmonella*, *Campylobacter*, *E. coli* O157:H7, *C.*

Collagen

- The major structural protein of connective tissue, composed of three polypeptide chains arranged in a triple helix. Different collagen types provide tensile strength to skin, bone, cartilage, tendons, blood vessels, and basement membranes.

Collagen Disease

- A systemic disorder involving connective tissue and often immune-mediated injury to collagen-rich structures. The term has historically included diseases such as systemic lupus erythematosus and systemic sclerosis.

Collagen Implant

- Biocompatible scaffold for tissue engineering. Types: bovine, porcine, synthetic. Used for hernia repair, soft tissue augmentation.

Collagenous Colitis

- Chronic inflammatory bowel disease characterized by thickening of the subepithelial collagen band in the colonic mucosa to greater than ten micrometers normally less than five micrometers with preservation of the epithelial surface. Presents with chronic watery non-bloody diarrhea primarily in middle-aged women.

Collapsed Lung

- A common description of pneumothorax, in which air enters the pleural space and partially or completely collapses the lung. It may cause sudden chest pain and shortness of breath and can be life-threatening when tension develops. See also Pneumothorax.

Collateral Circulation

- Alternative vascular pathways that develop when coronary arteries become chronically narrowed. Collaterals connect different coronary artery territories and can partially compensate for stenosis. They develop gradually in response to ischemia and are more prominent in patients with chronic coronary artery disease.

Collateral Ligament Injury

- MCL (medial) or LCL (lateral) knee ligament injury. MCL more common, often from valgus stress. Treatment: braces, physical therapy, surgical repair if severe.

Collecting Duct

- Final segment of the nephron where fine-tuning of urine concentration and composition occurs. Principal cells respond to antidiuretic hormone by inserting aquaporin-2 water channels into the luminal membrane, enabling water reabsorption and urine concentration.

Colles Fracture

- A fracture of the distal radius with dorsal displacement and angulation of the distal fragment, usually caused by falling on an outstretched hand.

Collimator

- Lead device with holes to define gamma ray direction. Types: parallel-hole, pinhole, converging, diverging. Resolution vs. sensitivity trade-off.

Colloids

- Solutions containing large molecules that remain

in the intravascular space, providing greater volume expansion per milliliter than crystalloids. Albumin (5 percent, 25 percent) is a natural plasma protein used for volume expansion and specific indications (cirrhosis with paracentesis, nephrotic syndrome).

Coloboma

- A congenital defect caused by failure of the optic fissure to close during embryonic development, resulting in a keyhole-shaped gap in any ocular structure including the iris, retina, choroid, or optic nerve. Iris colobomas are the most common and appear as a comma-shaped defect in the inferior iris.

Colocolonic intussusception

- telescoping of a segment of colon into an adjacent segment, less common in adults than the ileocolic type and usually associated with a pathological lead point such as a polyp, tumor, or lipoma. Adults typically present with intermittent abdominal pain, bloating, and change in bowel habits.

Colon

- The largest part of the large intestine, extending from the cecum to the rectum. Anatomically divided into ascending colon (right side), transverse colon (across the abdomen), descending colon (left side), and sigmoid colon (S-shaped, leading to the rectum).

Colon cancer

- Malignant neoplasm arising from the colonic epithelium, the third most common cancer worldwide and a leading cause of cancer-related mortality. Most arise from adenomatous polyps through the adenoma-carcinoma sequence over ten to fifteen years.

Colonic Ischemia

- Decreased blood flow to the colon causing mucosal or transmural injury. Most commonly affects watershed areas particularly the splenic flexure and rectosigmoid junction. Causes include non-occlusive hypoperfusion from cardiogenic shock, vasoconstrictive medications, and vasculitis.

Colonization

- Presence and multiplication of microorganisms on or in a body site without tissue invasion or clinical disease. Colonization may precede infection but does not by itself establish that symptoms are

caused by the organism.

Colonoscopy

- Visualization of the entire colon and terminal ileum using a flexible colonoscope. It is the gold standard for colorectal cancer screening and the preferred method for evaluating colonic pathology. The procedure requires bowel preparation with purgative solution to clear the colon.

Colony-Forming Unit (CFU)

- A unit used to estimate the number of viable microorganisms in a sample, derived from counting colonies grown on agar after plating a diluted sample. Each colony is assumed to arise from a single viable microorganism (or a cluster); results are reported as CFU/mL or CFU/g.

Color Blindness

- Inability or decreased ability to perceive color differences under normal lighting conditions. Most commonly X-linked recessive, affecting 8% of males and 0.5% of females.

Color Doppler

- Ultrasound technique combining B-mode imaging and Doppler flow assessment. See arterial duplex.

Color Vision Deficiency

- Color vision deficiency, commonly known as color blindness, is a condition in which the ability to perceive and discriminate certain colors is reduced or absent, most commonly affecting the perception of red and green hues, and it is caused by abnormalities in the cone photoreceptors responsible for color vision.

Colorado Tick Fever

- An acute viral infection transmitted by ticks and characterized by fever, headache, myalgia, and sometimes leukopenia or rash.

Colorectal Adenocarcinoma

- The third most common cancer worldwide and the second leading cause of cancer death, arising from the colonic epithelium through the adenoma-carcinoma sequence over 10 to 15 years. The pathogenesis involves sequential accumulation of genetic mutations beginning with APC loss, followed by KRAS activation, TP53 loss, and other alterations.

Colorectal Cancer

- Third most common cancer. Screening:

colonoscopy. Prevention: polyp removal. Treatment: surgery, chemotherapy, radiation.

Colorectal Cancer Genetics

- Lynch syndrome (HNPCC): MLH1, MSH2, MSH6, PMS2 mutations. Autosomal dominant, early-onset CRC, endometrial cancer. Screening: colonoscopy every 1-2 years starting age 20-25. Familial adenomatous polyposis (FAP): APC gene, hundreds-thousands of polyps, near 100% CRC risk if prophylactic colectomy not performed.

Colorectal Cancer Screening

- Public health strategy to detect colorectal cancer and precancerous polyps in asymptomatic individuals. Current guidelines recommend screening beginning at age forty-five.

Colorectal Cancer Screening Tests

- Multiple screening modalities. Stool-based: fecal immunochemical test (FIT, annual), high-sensitivity guaiac FOBT (annual), and multitarget stool DNA (FIT-DNA/Cologuard, q3y). Direct visualization: colonoscopy (q10y preferred), CT colonography (q5y), flexible sigmoidoscopy (q5-10y), and capsule colonoscopy (q5y).

Colorectal Surgery

- Colon cancer: right hemicolectomy (cecum to transverse), extended right (to mid-transverse), transverse colectomy, left hemicolectomy (splenic flexure to sigmoid), sigmoid colectomy, anterior resection (low anterior resection - LAR for upper/mid rectal), abdominoperineal resection (APR for low rectal with sphincter involvement),

Colostomy

- A surgically created stoma formed by bringing the colon to the abdominal wall for fecal diversion, indicated for distal obstruction, trauma, perforation, or abdominoperineal resection. Types include end, loop, and double-barrel. Complications include parastomal hernia, prolapse, retraction, and high-output dehydration.

Colostrum

- The thick, yellowish fluid produced by the mammary glands in late pregnancy and the first few days postpartum. Rich in immunoglobulins (especially IgA), proteins, leukocytes, and growth factors.

Colpitis

- Inflammation of the vagina, often due to infection, altered vaginal flora, irritation, or hormonal change.

Colposcopy

- Magnified visualization of the cervix, vagina, and vulva after application of acetic acid. Indication: abnormal Pap smear (ASCUS with high-risk HPV, LSIL, HSIL, AGC). Acetic acid causes acetowhite lesions in dysplastic epithelium. Lugol iodine (Schiller test): normal glycogenated epithelium stains brown; abnormal epithelium does not.

Coma

- See emergency medicine.

Coma Scale

- See Glasgow Coma Scale.

Combination Inhalers

- Inhalers containing both an inhaled corticosteroid and a long-acting beta-2 agonist. Examples include fluticasone/salmeterol (Advair), budesonide/formoterol (Symbicort), and fluticasone furoate/vilanterol (Breo Ellipta). They simplify asthma and COPD maintenance therapy and improve adherence.

Combination Therapy

- Using multiple treatments together. Improves outcomes, reduces resistance.

Combined Spinal-Epidural (CSE) Analgesia

- A neuraxial analgesia technique for labor that combines a spinal injection (short-acting opioid or opioid + low-dose local anesthetic) with an epidural catheter. Advantages: rapid onset of pain relief (3-5 minutes from the spinal component), minimal motor block, and the ability to dose through the epidural catheter for prolonged labor.

Comedocarcinoma

- A form of ductal carcinoma in situ of the breast characterized by central necrosis within malignant ducts, which may produce a comedo-like appearance.

Comfort Measures Only

- CMO: focus on symptom relief, not life prolongation.

Comminuted

- Describing a fracture in which a bone is broken into multiple fragments.

Comminuted Fracture

- A fracture in which the bone is broken into three or more fragments. It may result from high-energy trauma and can be associated with soft-tissue injury, instability, and impaired healing.

Commissurotomy

- Surgical incision or separation of a commissure, especially a fused cardiac valve commissure, to relieve obstruction.

Commode

- Portable toilet for patients.

Common Bile Duct

- The duct formed by union of the common hepatic and cystic ducts. It conveys bile to the duodenum, usually joining the pancreatic duct at the hepatopancreatic ampulla.

Common Cold

- An acute, self-limited viral infection of the upper respiratory tract, one of the most frequent illnesses in humans, with adults experiencing 2-4 episodes annually and children 6-8 episodes.

Common Iliac Arteries

- Terminal abdominal aortic branches at L4, each approximately 4 cm long, dividing into internal and external iliac arteries. They supply the pelvis and lower extremities. The right common iliac artery crosses anterior to the right common iliac vein.

Common Peroneal Nerve

- Sciatic division (L4-S2) winding around the fibular neck, the body's most vulnerable major nerve to external compression. Divides into superficial peroneal (peroneus longus/brevis, dorsal foot sensation) and deep peroneal (tibialis anterior, extensors, first web space sensation).

Common peroneal neuropathy

- A peripheral nerve disorder causing weakness of ankle dorsiflexion and eversion, sensory loss over the anterolateral leg and dorsum of the foot, and possible foot drop.

Common Variable Immunodeficiency

- The most common symptomatic primary immun-

odeficiency in adults, affecting approximately 1 in 25,000 individuals. It is characterized by hypogammaglobulinemia (low IgG with low IgA and/or IgM), impaired antibody response to vaccines, and increased susceptibility to infections.

Common wart

- A benign epidermal proliferation caused by human papillomavirus, presenting as a rough hyperkeratotic papule, commonly on the hands or fingers.

Commotio Cordis

- Sudden cardiac arrest caused by a blunt, nonpenetrating impact to the chest at a vulnerable moment in the cardiac cycle, without structural heart injury.

Communicable disease

- Any disease caused by bacteria, viruses, or other pathogens that is spread from person to person.

Communicating Hydrocephalus

- A type of hydrocephalus in which there is free flow of cerebrospinal fluid (CSF) between the ventricles, but impaired absorption of CSF at the arachnoid granulations or increased CSF production. The ventricles are uniformly dilated, including the fourth ventricle.

Community Acquired Infection

- An infection present or incubating at hospital admission, or developing within 48 hours of admission without recent healthcare exposure. Common examples include pneumonia, urinary tract infection, and meningitis caused by organisms such as *Streptococcus pneumoniae* and *Escherichia coli*, generally susceptible to narrow-spectrum antibiotics.

Community Health Worker

- Lay health worker from community. Bridges community and healthcare. Examples: promotoras, CHWs.

Community Rehabilitation

- Rehabilitation services in community settings.

Community Reintegration

- Return to community roles: work, leisure, social. Vocational rehabilitation, accessibility modifications.

Community-Acquired

- Obtained outside hospital. Examples: community-acquired pneumonia.

Community-Acquired Pneumonia

- CAP: acquired outside hospital. CURB-65 (confusion, uremia >7, respiratory rate ≥ 30 , BP <90/60, age ≥ 65) and PSI/PORT score for severity/site of care. Outpatient: amoxicillin 1g TID, doxycycline 100 mg BID, or macrolide (if resistance <25 percent); respiratory fluoroquinolone (levofloxacin, moxifloxacin) as alternative.

Comorbidity

- Presence of two or more chronic conditions in same individual. Common in elderly: hypertension, diabetes, arthritis, COPD, depression. Increases complexity of care, polypharmacy risk, functional decline. Comprehensive assessment needed for each condition and interactions.

Comparison

- Reference to prior imaging in radiology report.

Compartment Pressure

- The pressure within a closed osteofascial compartment. Markedly elevated pressure can impair capillary perfusion and cause acute compartment syndrome; pressure measurements support but do not replace clinical assessment.

Compartment Syndrome

- A surgical emergency caused by increased pressure within a closed osteofascial compartment, leading to tissue ischemia, necrosis, and permanent functional loss. The lower leg (anterior, lateral, deep posterior, and superficial posterior compartments) and forearm (flexor and extensor compartments) are the most commonly affected sites.

Competence

- Autonomous voluntary authorization given by a competent patient to undergo a specific medical intervention, procedure, or clinical trial. See informed consent.

Competitive Antagonist

- A competitive antagonist is a drug that binds to the same receptor site as an agonist and competes for binding, reducing the effect of the agonist without producing an effect of its own.

Competitive Inhibition

- A type of enzyme inhibition where the inhibitor (I) competes with the substrate (S) for binding to the enzyme's active site. The inhibitor structurally

resembles the substrate and binds reversibly to the free enzyme (E), forming an EI complex that cannot proceed to product.

Complement C1q Deficiency

- Autosomal recessive (C1QA, C1QB, C1QC). C1q essential for classical pathway activation and immune complex clearance. 85-90% develop SLE-like disease (malar rash, photosensitivity, nephritis, anti-nuclear antibodies). Recurrent infections with encapsulated bacteria. Low CH50, low C1q, normal C3/C4.

Complement C2/C4 Deficiency

- C2 deficiency: most common early complement deficiency (1:20,000 Caucasians). SLE-like disease (10-30%), recurrent sinopulmonary infections. C4 deficiency: similar, strong SLE association (C4A/C4B null alleles). Low CH50, low C2 or C4, normal C3. Treatment: SLE management, prophylactic antibiotics, IVIG for recurrent infections.

Complement C3

- The central complement component cleaved into C3a and C3b during activation of the classical, lectin, or alternative pathway. C3b promotes opsonization and formation of downstream convertases; severe C3 deficiency causes recurrent severe infections, especially with encapsulated bacteria.

Complement Cascade

- The ordered activation of 30 serum and membrane proteins (complement) that opsonize pathogens, lyse microbes (membrane attack complex), and promote inflammation.

Complement Deficiencies

- Rare primary immunodeficiencies caused by genetic defects in complement pathway proteins, increasing susceptibility to specific infections and autoimmune diseases.

Complete Blood Count

- A laboratory test measuring blood-cell numbers and indices, commonly including erythrocytes, hemoglobin, hematocrit, leukocytes, and platelets.

Complete Blood Count (CBC)

- A panel of blood tests evaluating cellular components: WBC (leukocytes: normal 4.5-11.0 K/uL), RBC (erythrocytes: male 4.5-5.9, female 4.1-5.1 M/uL), hemoglobin (male 13.5-17.5, female 12.0-

16.0 g/dL), hematocrit (male 41-53%, female 36-46%), platelets (150-450 K/uL), RBC indices (MCV, MCH, MCHC), and differential (neutrophils, lymphocyte

Complete Denture

- Removable prosthesis replacing all teeth in arch. Maxillary and mandibular. Stability challenging in edentulous patients.

Complete Response

- All detectable cancer disappears.

Complex Carbohydrates

- Carbohydrates made up of longer chains of sugars, chiefly starches and dietary fiber, found in foods such as whole grains, legumes, vegetables, and potatoes. Fiber is not fully digested and supports bowel function, satiety, and metabolic health, while starch is digested into glucose at a rate that depends on food structure and processing.

Complex Regional Pain Syndrome

- A chronic pain condition characterized by regional pain and sensory changes disproportionate to the inciting event. Type I (reflex sympathetic dystrophy): no identifiable nerve injury. Type II (causalgia): follows a major nerve injury.

Compliance

- Following medical advice. Preferred term: adherence.

Complicated Grief

- Grief after loss. See bereavement.

Complicated Urinary Tract Infection

- A urinary infection associated with functional or structural abnormality, obstruction, catheterization, renal impairment, male anatomy, pregnancy, or another factor increasing treatment failure or complication risk. Management requires culture-directed therapy and correction of the underlying problem when possible.

Complication

- Any adverse event arising during or after a surgical procedure, graded by the Clavien-Dindo classification from minor deviation (Grade I) through organ failure (Grade IV) to death (Grade V). Common types include surgical site infection, hemorrhage, wound dehiscence, anastomotic leak, ileus, and venous thromboembolism.

Complications (Vascular)

- Post-procedural issues: bleeding, infection, thrombosis, embolism, limb ischemia.

Composite Resin

- Tooth-colored restorative material. Bis-GMA or UDMA resin matrix with silica filler. Adhesive bonding. Esthetic but less durable than ceramic.

Compound Presentation

- A fetal presentation where an extremity (hand, arm, foot) presents alongside the vertex or breech. Most common: hand alongside the head. Incidence: 1:300-1:1000. Fetal prognosis is generally good; the prolapsed extremity often retracts as labor progresses. Management: expectant management in most cases.

Compounding

- Preparation of radiopharmaceuticals for individual patients. USP <797> standards.

Comprehensive Geriatric Assessment

- Comprehensive multidimensional evaluation of elderly person. See geriatric assessment.

Comprehensive Metabolic Panel

- Extended version of BMP including albumin, total protein, alkaline phosphatase, ALT, AST, and bilirubin (total and direct). Evaluates liver function, kidney function, electrolytes, glucose, calcium, and protein status. Uses: preoperative assessment, medication monitoring, and liver/kidney disease evaluation.

Compression Therapy

- External pressure on veins. Used for DVT prevention, venous insufficiency, varicose veins.

Compression Ultrasound

- Ultrasound technique applying pressure. Used for DVT diagnosis, testicular evaluation.

Compressive Atelectasis

- Atelectasis caused by external compression of the lung by pleural fluid, air, or a mass. The lung volume is reduced by the external force. It is reversible when the compressing force is relieved (e.g., drainage of pleural effusion).

Compton Scattering

- Photon interaction with loosely bound electron. Major interaction in diagnostic energy range.

Compulsions

- Repetitive behaviors or mental acts that the individual feels driven to perform in response to an obsession or according to rigid rules. They are aimed at preventing or reducing anxiety or preventing some dreaded event, but are not connected in a realistic way with what they are designed to neutralize.

Computed Tomography (CT)

- Cross-sectional imaging using rotating X-ray tube and detectors. Produces axial images, reconstructable into sagittal, coronal, 3D. CT number measured in Hounsfield units (HU): water = 0, air = -1000, bone = +1000. Contrast: iodine-based, enhances vascular structures and perfused organs. Speed: seconds.

Computed Tomography (CT) Scan

- Cross-sectional imaging using X-rays and computer processing. Emergency uses: head trauma (hemorrhage, fracture), chest (pulmonary embolism, aortic dissection), abdomen (appendicitis, bowel obstruction), CT pulmonary angiography (CTPA), CT angiography (CTA). Fast, widely available, high sensitivity for acute conditions.

Concato's Disease

- A historical term for primary polycythemia, now generally called polycythemia vera, a myeloproliferative neoplasm causing increased red-cell mass and often elevated blood viscosity.

Concealed Pregnancy

- Pregnancy that is hidden or not recognized until delivery, often by denial (psychogenic) or deliberate concealment; a medicolegal concept relevant to infanticide.

Concentration

- Amount of substance per unit volume. Units: mg/dL, mmol/L, mEq/L, ppm. Conversion: molecular weight needed. Critical for toxicity assessment.

Conception (Fertilization)

- The fusion of a sperm and an ovum in the ampulla of the fallopian tube, forming a zygote. The sperm penetrates the corona radiata and zona pellucida (acrosome reaction). After fertilization, the zygote undergoes cleavage (mitotic division) as it travels toward the uterus. Implantation occurs approximately 6-10 days after ovulation.

Conchae

- See Turbinates.

Conclusion

- Summary of key findings and recommendations in radiology report. See impression.

Concomitant

- Occurring simultaneously. Concomitant medication: additional drug taken with primary treatment.

Concordance

- Probability both twins have same trait. Monozygotic (identical) twins > dizygotic (fraternal) for genetic traits. High concordance suggests genetic influence.

Concrescence

- Union of adjacent teeth by cementum only. Usually posterior teeth.

Concretio Cordis

- Abnormal adhesion or fusion of structures within or around the heart, including adhesion of the pericardial surfaces or fusion of cardiac components.

Concussion

- A form of mild traumatic brain injury caused by a biomechanical force (direct blow to the head, face, neck, or elsewhere on the body with impulsive force transmitted to the head), resulting in a transient disturbance of brain function without structural damage visible on conventional neuroimaging.

Condition

- A general term for a state of bodily health or, in medical usage, a specific deviation from normal in structure or function that may or may not constitute a disease, including complaints lacking a discrete pathologic basis and overt diagnoses.

Conduct Disorder

- A childhood/adolescent disorder characterized by a persistent pattern of violating others' basic rights or major age-appropriate norms. Behaviors: aggression to people/animals, destruction of property, deceitfulness/theft, and serious rule violations. Diagnosis requires 3 criteria in the past 12 months with functional

Conduction Deafness

- Hearing loss caused by impaired transmission of sound through the external or middle ear, with the

inner ear and neural auditory pathways relatively preserved.

Conductive Deafness

- Hearing loss resulting from a problem in the external auditory canal, tympanic membrane, or middle ear that reduces conduction of sound to the inner ear.

Conductive Hearing Loss

- A type of hearing loss caused by impairment of the transmission of sound through the outer ear, middle ear, or both, preventing sound vibrations from reaching the cochlea efficiently, and it is typically characterized by a conductive gap (an air-bone gap on audiometry indicating a difference between air conduction and bone conduction threshold).

Condylar Process

- The rounded articular projection of a bone, especially the mandibular process that articulates with the temporal bone to form the temporomandibular joint.

Condyloma Acuminata

- See Genital Warts (Condylomata Acuminata).

Condyloma acuminatum

- A benign, exophytic, papillomatous growth of the anogenital skin caused by human papillomavirus, most commonly types 6 and 11. They present as soft, flesh-colored, verrucous papules or plaques in the genital and perianal region. Condylomata are sexually transmitted and may be associated with cervical dysplasia.

Condyloma Latum

- Mucocutaneous lesion of secondary syphilis caused by *Treponema pallidum*, distinct from condyloma acuminata (genital wart).

Condylomata lata

- Broad, flat, moist, grayish-white papules occurring in secondary syphilis, typically found in the anogenital region and other intertriginous areas. They are highly infectious and contain abundant *Treponema pallidum* organisms. Diagnosis is by darkfield microscopy or serological testing for syphilis.

Cone-Beam Computed Tomography

- Three-dimensional dental imaging using a cone-shaped X-ray beam and a rotating detector. It provides high-resolution views of teeth and max-

illofacial bone with generally less radiation than conventional medical CT, but should be used only when the additional information affects care.

Confabulation

- Production of false or inaccurate statements or memories without conscious intent to deceive, often associated with memory impairment or frontal-lobe dysfunction.

Confidentiality

- The ethical and legal duty of healthcare professionals to protect patient information from unauthorized disclosure. Founded on the principle of respect for patient autonomy and trust, confidentiality is a core requirement of medical ethics (GMC guidelines) and law (GDPR, Data Protection Act, Caldicott principles).

Confirmatory Test

- A test used to verify a preliminary or screening result, often with a different method or greater specificity. Confirmation should be selected according to the condition, specimen, timing, and accepted clinical guidelines.

Confounding

- Distortion of association by third variable. Examples: age confounding relationship between exposure and disease. Controlled by matching, stratification, regression.

Confusion (Delirium)

- Acute onset fluctuating disturbance in attention and awareness. Usually reversible. Precipitants: infection, medications, metabolic derangement, pain, constipation, urinary retention. Differentiate from dementia (chronic, progressive). Treatment: identify and treat underlying cause, environmental modification, avoid anticholinergics.

Confusion Assessment Method

- A standardized bedside algorithm for detecting delirium. It identifies acute or fluctuating change, inattention, disorganized thinking, and altered level of consciousness; delirium requires the first two features plus either of the latter two.

Congenital Adrenal Hyperplasia

- Group of autosomal recessive disorders caused by enzyme deficiencies in the adrenal steroidogenesis pathway. Twenty-one hydroxylase deficiency is the

most common accounting for approximately ninety-five percent of cases.

Congenital Cataract

- Opacification of the crystalline lens present at birth, a leading cause of treatable childhood blindness worldwide.

Congenital CMV

- Cytomegalovirus infection acquired in utero, the most common congenital viral infection. Maternal primary infection poses highest risk of fetal transmission and disease. Can cause microcephaly, sensorineural hearing loss, chorioretinitis, hepatosplenomegaly, and petechiae. Diagnosis by CMV PCR in urine or saliva.

Congenital Contractural Arachnodactyly

- An inherited connective-tissue disorder characterized by long slender fingers and toes, congenital flexion contractures, muscular hypoplasia, and sometimes kyphoscoliosis.

Congenital cytomegalovirus infection

- See Cytomegalovirus Infection.

Congenital Diaphragmatic Hernia

- A defect in the diaphragm allowing abdominal organs to herniate into the thoracic cavity, causing pulmonary hypoplasia and respiratory distress at birth. Most commonly a posterolateral (Bochdalek) defect on the left side. Presents with cyanosis, scaphoid abdomen, and bowel sounds in the chest.

Congenital Disorder

- A medical condition or structural abnormality present in an individual from birth, which may be caused by genetic factors or environmental influences in the womb.

Congenital Glaucoma

- A rare form of glaucoma (1 in 10,000-20,000 births) caused by developmental anomalies of the trabecular meshwork and anterior chamber angle, impairing aqueous humor outflow. Most cases are sporadic; familial forms are autosomal recessive (CYP1B1 gene mutations).

Congenital Heart Disease

- Structural heart defects present at birth, the most common congenital anomaly (1% of live births). Common defects: ventricular septal defect (VSD, 30%), atrial septal defect (ASD), patent ductus ar-

teriosus (PDA), tetralogy of Fallot (TOF), transposition of great arteries (TGA), coarctation of aorta, and hypoplastic left heart syndrome.

Congenital Heart Murmur

- A heart murmur detected in the newborn period, present in approximately 50-60% of healthy newborns. Most are innocent (functional) murmurs that disappear by 6 months of age. Pathologic murmurs may indicate congenital heart defects. Differentiation based on murmur characteristics, timing, and associated findings.

Congenital Hip Dislocation

- See Hip Dysplasia.

Congenital Hyperinsulinism

- Condition of inappropriate insulin secretion in neonates and infants causing severe hypoglycemia. Most commonly caused by mutations in the KATP channel genes ABCC8 and KCNJ11. Histologically classified into focal and diffuse forms. Presents with severe symptomatic hypoglycemia in the first days of life.

Congenital Hyperthyroidism

- Thyrotoxicosis present at birth most commonly caused by maternal Graves disease with transplacental passage of thyroid-stimulating immunoglobulins. Presents with tachycardia, irritability, poor feeding, weight loss, and goiter. Transient neonatal hyperthyroidism resolves as maternal antibodies are cleared over weeks to months.

Congenital Hypothyroidism

- Thyroid hormone deficiency present at birth occurring in approximately one in two thousand to four thousand live births. Most common cause is thyroid dysgenesis including aplasia, ectopia, and hypoplasia. Universal newborn screening with TSH or T4 allows early detection and treatment preventing intellectual disability.

Congenital Megacolon

- Hirschsprung disease, a congenital absence of enteric ganglion cells in a segment of the bowel that causes functional obstruction and enlargement of the proximal colon.

Congenital melanocytic nevus

- A melanocytic nevus present at birth or appearing within the first few weeks of life, ranging from

small (< 1.5 cm) to giant lesions covering large body surface areas. It results from proliferation of melanocytes during embryonic development and carries a small but important risk of melanoma, which is higher in larger lesions.

Congenital nevus

- See congenital melanocytic nevus. A melanocytic nevus present at birth or arising in early infancy, composed of proliferating melanocytes within the epidermis and dermis. Small congenital nevi are common and carry minimal malignant risk, while medium and giant lesions require closer surveillance due to increased melanoma potential.

Congenital Pneumonia

- Lung infection acquired in utero, often from ascending chorioamnionitis. Presents with respiratory distress, fever or hypothermia, and sepsis. Common organisms include Group B Streptococcus, E. coli, and Ureaplasma. Diagnosis by chest X-ray showing infiltrates and blood culture.

Congenital Rubella Syndrome

- Fetal infection with rubella virus, particularly dangerous in the first trimester. Classic triad: congenital heart disease (patent ductus arteriosus, pulmonary stenosis), cataracts, and sensorineural hearing loss. Also causes hepatosplenomegaly, thrombocytopenia, and blueberry muffin rash.

Congenital Scoliosis

- A lateral curvature of the spine caused by congenital vertebral anomalies that arise during embryogenesis (failure of formation or segmentation). Failure of formation: hemivertebra (wedge-shaped vertebra causing progressive curvature). Failure of segmentation: unilateral unsegmented bar (block of bone on one side).

Congenital Syphilis

- Treponema pallidum infection acquired transplacentally. Can cause stillbirth, prematurity, hepatosplenomegaly, rash, snuffles, osteochondritis, and Hutchinson teeth. Diagnosed by darkfield microscopy, VDRL/RPR in neonatal serum, or PCR.

Congenital Zika Syndrome

- Fetal brain infection with Zika virus, causing microcephaly, brain abnormalities, eye defects, hearing loss, and limb contractures. Transmission occurs via

mosquito bite or sexual contact. Diagnosis by Zika virus PCR in maternal serum or amniotic fluid. No specific treatment. Prevention by avoiding endemic areas during pregnancy.

Congestion

- A passive vascular phenomenon of engorgement of a tissue or organ with blood from impaired venous or lymphatic drainage. Chronic venous congestion produces stasis, hypoxia, dilated vessels, edema, hemorrhage, and hemosiderin-laden macrophages, exemplified by the lungs (heart failure cells) and liver (nutmeg liver).

Congestive Heart Failure

- A historical term for heart failure with clinical fluid congestion. Current terminology is heart failure, which may occur with or without congestion and is classified by left-ventricular ejection fraction, symptoms, and disease stage.

Conization

- Surgical removal of a cone-shaped portion of the cervix for diagnosis or treatment of cervical dysplasia or selected early cervical cancers.

Conjoined Twins

- Monozygotic, monochorionic, monoamniotic twins who are physically connected at birth, resulting from incomplete fission of the fertilized ovum after day 13 of fertilization. Incidence: 1 in 50,000-100,000 births.

Conjunctiva

- The conjunctiva is a thin, transparent, mucous membrane that lines the inner surface of the eyelids (palpebral conjunctiva) and covers the anterior surface of the sclera (bulbar conjunctiva), forming a continuous sheet that protects the eye and facilitates smooth eyelid movement.

Conjunctival squamous-cell carcinoma

- A malignant tumor of the conjunctival epithelium, typically arising from the limbal area. It is associated with UV radiation exposure and can invade locally if not treated promptly.

Conjunctivitis

- Inflammation of the conjunctiva, the thin, transparent mucous membrane lining the inner surface of the eyelids and the anterior surface of the sclera. Conjunctivitis is the most common cause of red eye

and is classified by etiology.

Conjunctivoplasty

- Plastic or reconstructive surgery of the conjunctiva, performed to repair defects, scarring, adhesions, or other abnormalities of the ocular surface.

Conn's Syndrome

- Primary hyperaldosteronism caused by autonomous secretion of aldosterone, leading to hypertension and often low serum potassium.

Connective Tissue

- A diverse type of tissue that supports, binds, protects, and insulates organs and structures throughout the body.

Connective Tissue Disease

- Autoimmune diseases affecting connective tissue. Examples: SLE, scleroderma, RA, Sjogren syndrome, mixed connective tissue disease. Overlap syndromes common.

Conscious Sedation

- A drug-induced depression of consciousness in which patients respond purposefully to verbal or tactile stimulation, maintain a patent airway, and retain protective reflexes. Typical agents include midazolam, fentanyl, or low-dose propofol. Indications include endoscopic and minor surgical procedures.

Consolidation

- Lung parenchyma filled with fluid, detected on imaging. Pneumonia, edema, hemorrhage, malignancy.

Consolidation Therapy

- Additional treatment given after remission to eliminate residual disease and reduce the risk of relapse.

Constipation

- A common symptom-based disorder defined by the Rome IV criteria as having two or more of the following for at least 3 months: straining during more than 25 percent of bowel movements, lumpy or hard stools, sensation of incomplete evacuation, sensation of anorectal obstruction, manual maneuvers to facilitate defecation, or fewer than 3 bow

Constitutional delay in puberty

- A normal variant in which pubertal development is delayed but eventually progresses normally; growth and bone age are typically delayed.

Constraint-Induced Movement Therapy

- CIMT: restrain unaffected limb, force use of affected limb. Based on neuroplasticity. Effective for stroke hemiparesis.

Constrictive Pericarditis

- Thickening and fibrosis of the pericardium restricting diastolic ventricular filling. Presents with jugular venous distension, peripheral edema, hepatomegaly, and Kussmaul sign. Causes include idiopathic, post-viral, radiation, and tuberculosis. Pericardiectomy is the definitive treatment.

Consultation-Liaison Psychiatry

- The subspecialty of psychiatry focused on the interface between psychiatry and general medical illness. Psychiatrists provide consultation on medical-surgical wards for patients with psychiatric symptoms arising from or coexisting with medical conditions.

Contact Allergen Testing

- Patch testing identifies allergens causing allergic contact dermatitis. Standard panel includes 35 common allergens (nickel, fragrance mix, balsam of Peru, formaldehyde, rubber accelerators, topical antibiotics, preservatives). Patches are applied to the back for 48 hours and read at 48, 72, and 96 hours.

Contact Dermatitis

- An inflammatory skin reaction caused by direct contact with an external agent, divided into two main types: allergic contact dermatitis (ACD, type IV delayed hypersensitivity reaction, mediated by sensitized T lymphocytes) and irritant contact dermatitis (ICD, non-immunologic, dose-dependent direct chemical or physical damage to t

Contact precautions

- Infection control measures used for patients colonised or infected with multidrug-resistant organisms. Precautions include gown and glove use, patient placement in single rooms, and dedicated equipment. Contact precautions are used for MRSA, VRE, CRE, and other resistant organisms.

Contact Tracing

- Identifying and monitoring persons exposed to infectious disease. Critical for outbreak control. Examples: COVID-19, Ebola, TB.

Contagious Disease

- An infectious disease that can be transmitted directly or indirectly from one person, animal, or source to another.

Contamination

- Accidental presence of microorganisms in a specimen, on a surface, or in a normally sterile material without true infection of the host. Contamination can produce misleading culture results and must be distinguished from colonization and invasive disease.

Continuing Medical Education

- Postgraduate education that maintains and updates a clinician's knowledge, skills, and professional competence.

Continuity of Care

- Consistent and coordinated care over time and across clinicians or settings, integrating the patient's history, current needs, treatment plan, and preferences. It supports safety, adherence, prevention, and chronic-disease outcomes.

Continuous Electronic Fetal Monitoring (EFM)

- The continuous recording of fetal heart rate (FHR) and uterine contractions using external transducers (Doppler ultrasound and tocodynamometer) or internal devices (fetal scalp electrode and intrauterine pressure catheter). Used intrapartum to detect fetal hypoxia.

Continuous Glucose Monitoring

- Technology measuring interstitial glucose levels continuously providing real-time glucose data, trends, and alerts. Sensors are inserted subcutaneously typically in the abdomen or arm and measure glucose every five minutes. Devices include Dexcom, FreeStyle Libre, and Medtronic Guardian systems.

Continuous Kidney Replacement Therapy (CKRT)

- A prolonged extracorporeal blood-purification therapy, usually delivered continuously in an intensive-care setting. It provides gradual fluid and solute removal and is useful for hemodynamically unstable patients with severe AKI.

Continuous Murmurs

- Murmurs that persist throughout systole and diastole without interruption, produced by blood flow-

ing continuously through an abnormal connection. The most common cause is patent ductus arteriosus (PDA), where blood flows from the aorta to pulmonary artery throughout the cardiac cycle.

Continuous Positive Airway Pressure

- Noninvasive respiratory support that delivers a constant positive pressure throughout the breathing cycle to keep the upper airway or alveoli open.

Contraception

- Methods to prevent pregnancy.

Contraception and Birth Control

- Contraception encompasses methods used to prevent pregnancy, including hormonal options like oral contraceptives, patches, rings, injections, and implants, as well as non-hormonal methods like copper IUDs, barrier methods, diaphragms, and permanent sterilization.

Contraception for Adolescents

- Methods to prevent unintended pregnancy. Options: combined oral contraceptives, progestin-only pills, depot medroxyprogesterone acetate, contraceptive implant (Nexplanon), intrauterine devices (IUDs), condoms, and abstinence.

Contracted Pelvis

- A pelvis whose size or shape is reduced or abnormal enough to interfere with passage of the fetus during labor.

Contraction Stress Test

- An antepartum fetal surveillance technique assessing fetal heart rate response to induced uterine contractions. Oxytocin or nipple stimulation produces contractions. A negative test shows no late decelerations. A positive CST demonstrates late decelerations with more than 50% of contractions.

Contraction Stress Test (CST)

- An antepartum test that evaluates the fetal heart rate response to uterine contractions. Contractions are induced with oxytocin or nipple stimulation (until 3 contractions in 10 minutes). Negative (normal): no late or significant variable decelerations. Positive (abnormal): late decelerations with 50% of contractions.

Contractions

- Rhythmic uterine muscular contractions of labor that produce cervical dilation and effacement, driv-

ing descent of the fetus. Characterized by frequency, duration, and intensity; progress is monitored by partogram and tocography (CTG). Braxton Hicks contractions: irregular, non-progressive, painless prelabor contractions.

Contracture

- Permanent or semi-permanent limitation of passive joint range of motion caused by shortening or structural change in muscle, tendon, capsule, fascia, skin, or other periarticular tissue. Prevention and treatment depend on cause and stage.

Contracture Prevention

- Positioning, stretching, splinting, orthotics, serial casting, surgery.

Contraindication

- A condition or factor that renders a particular procedure, treatment, or medication inadvisable. Absolute contraindication: condition where intervention should never be performed. Relative contraindication: risks outweigh benefits but may be acceptable in specific circumstances.

Contralateral

- Located on the opposite side of the body.

Contrast Agent

- Substance enhancing image visibility. X-ray/CT: iodine-based (positive contrast). MRI: gadolinium-based (shortens T1, bright on T1). Ultrasound: microbubble agents. Barium sulfate for GI studies.

Contrast Echocardiography

- Use of microbubble contrast agents to enhance endocardial border definition and assess myocardial perfusion. Indicated for suboptimal non-contrast echo images and assessment of myocardial viability. Microbubbles remain entirely within the intravascular space, making them ideal for perfusion assessment.

Contrast Reaction

- Adverse reaction to contrast agent. Mild: nausea, urticaria. Severe: anaphylaxis, bronchospasm. Pre-medication for prior reactions.

Contrast-Associated Acute Kidney Injury (CA-AKI)

- Acute worsening of kidney function occurring after administration of contrast, whether or not contrast is causal. Risk assessment includes baseline kidney

function, hemodynamic status, diabetes, nephrotoxins, urgency, and the need to avoid unnecessary contrast exposure.

Contrast-Enhanced Ultrasound

- Ultrasound using microbubble contrast agents. Improves lesion characterization, perfusion assessment.

Contrast-Induced Nephropathy

- Acute kidney injury after iodinated contrast administration. Risk factors: pre-existing renal impairment, diabetes, dehydration. Prevention: hydration, lowest contrast volume, N-acetylcysteine controversial.

Contrecoup

- An injury occurring on the side of an organ opposite the site of impact, especially a brain injury produced when the brain moves within the skull after trauma.

Control Group

- Group not receiving experimental intervention. Used for comparison.

Controlled Cord Traction (CCT)

- A technique for active management of third-stage labor: gentle, steady traction on the umbilical cord with one hand while the other hand applies suprapubic counter-pressure (to prevent uterine inversion). Performed after signs of placental separation. The Brandt-Andrews maneuver and the Finocchietti technique are CCT methods.

Controlled Substance

- A drug or chemical whose manufacture, possession, and use are regulated by government authorities due to its potential for abuse, dependence, or harm.

Contusion

- A closed soft-tissue injury caused by blunt force, producing bleeding into tissues without an open wound. It commonly causes pain, swelling, and ecchymosis; severe contusions may involve muscle necrosis, compartment syndrome, internal bleeding, or associated bone and organ injury.

Conus Medullaris Syndrome

- A clinical condition resulting from damage to the conus medullaris, the tapered, terminal end of the spinal cord located at approximately the L1-L2 vertebral level.

Convergence Insufficiency

- Inability to maintain binocular convergence on near objects, causing eyestrain, headaches, blurred near vision, and transient diplopia during reading. Common in children and young adults; treated with convergence exercises and base-in prism, with surgery rarely needed.

Coombs Test

- A laboratory test that detects antibodies or complement attached to red blood cells, or antibodies in serum that can react with red cells. It is used in evaluating immune hemolysis and blood compatibility.

Cooper's Ligament

- Fibrous connective-tissue septa extending from the deep surface of the breast toward the skin and pectoral fascia. Tumor invasion or fibrosis may cause skin retraction.

Coordination

- Smooth, efficient movement. Tests: finger-to-nose, heel-to-shin, rapid alternating movements.

COPD

- Chronic obstructive pulmonary disease is a progressive, largely irreversible airflow limitation caused by chronic bronchitis, emphysema, or both, most commonly due to cigarette smoking. It presents with chronic cough, sputum production, and exertional dyspnea. Spirometry shows a reduced FEV1/FVC ratio.

COPD Exacerbation

- Acute worsening of respiratory symptoms beyond normal day-to-day variation in a patient with COPD that leads to additional therapy. Common triggers include viral or bacterial infection, air pollution, and cardiopulmonary comorbidity.

COPD Rehabilitation

- Comprehensive intervention for COPD patients including exercise training, education, behavior change. Components: aerobic exercise, strength training, breathing techniques, nutrition, psychosocial support. Improves exercise capacity, quality of life, reduces hospitalizations.

Copper

- An essential trace element required for iron transport, connective-tissue cross-linking, energy

metabolism, neurotransmitter synthesis, and antioxidant enzymes. Deficiency can cause anemia, neutropenia, impaired neurologic function, and bone abnormalities; excess is toxic and accumulates in Wilson disease.

copy number variation (CNV)

- Deletions or duplications of DNA segments >1 kb. Contribute to genetic diversity and disease. Examples: 22q11.2 deletion, Williams syndrome (7q11.23 deletion).

Cor Pulmonale

- Right ventricular hypertrophy and dilation from pulmonary hypertension due to lung disease. Features include peripheral edema, JVD, and hepatomegaly.

Cor Triatriatum

- A rare congenital heart defect where the left atrium is divided into two chambers by a fibromuscular membrane, separating the pulmonary venous chamber (posterior) from the true left atrium (anterior) containing the left atrial appendage and mitral valve. The membrane obstructs pulmonary venous return, mimicking mitral stenosis.

Coracoid Process

- Hook-like scapular projection extending anteriorly and laterally. Attachment for 3 muscles (pectoralis minor, coracobrachialis, short head of biceps) and 3 ligaments. Important surgical landmark. Fractures uncommon but may occur with direct trauma or shoulder dislocations.

Cord Presentation

- The presence of the umbilical cord between the presenting fetal part and the cervix, with intact membranes. Risk factors: malpresentation, unstable lie, low-lying placenta, and long cord. If the membranes rupture, the cord may prolapse (umbilical cord prolapse).

Cordocentesis

- A prenatal diagnostic procedure in which fetal blood is sampled from the umbilical cord, usually under ultrasound guidance.

Cordotomy

- Surgical division of selected nerve fibers, especially a spinal cord tract, to interrupt transmission of severe chronic pain.

Core Biopsy

- Removal of a cylindrical sample of tissue with a hollow cutting needle for microscopic examination, commonly used for breast, liver, prostate, and other lesions.

Core Needle Biopsy

- Biopsy using core needle. CT, ultrasound, or MRI guidance.

Cori Cycle

- Lactate produced by muscles transported to liver, converted to glucose via gluconeogenesis, returned to muscles.

Cornea

- Transparent, avascular, dome-shaped anterior eye surface providing approximately two-thirds of total refractive power. Five layers: epithelium (stratified squamous, rapidly regenerating), Bowman's membrane, stroma (thickest, regular collagen), Descemet's membrane, endothelium (single cell layer maintaining transparency).

Corneal Abrasion

- Loss of corneal epithelium caused by mechanical trauma — contact with fingernail, makeup brush, foreign body, contact lens overwear, dry eye, sports injury, paper edge, or surgical instrumentation. Most common eye injury presenting to emergency departments.

Corneal Cross-Linking

- Corneal collagen cross-linking is a minimally invasive therapeutic procedure that strengthens the corneal stroma by creating additional chemical bonds between collagen fibrils, thereby stiffening the cornea and halting or slowing the progression of corneal ectatic disorders, particularly keratoconus and post-refractive surgery ectasia.

Corneal Edema

- Excess fluid within the corneal stroma or epithelium causing corneal thickening, reduced transparency, glare, and blurred vision. Causes include endothelial dysfunction, acute glaucoma, inflammation, trauma, and postoperative injury.

Corneal Implant

- An implanted device or tissue substitute placed in or on the cornea to restore vision, replace diseased tissue, or alter corneal shape.

Corneal Ulcer

- A corneal ulcer is a defect in the corneal epithelium with associated stromal inflammation and necrosis, typically caused by infectious or non-infectious mechanisms, and it is a sight-threatening condition that requires prompt diagnosis and treatment.

Corneitis

- Inflammation of the cornea, often producing pain, photophobia, tearing, and impaired vision. It may be infectious, immune-mediated, traumatic, or related to exposure.

Cornelia de Lange Syndrome

- A genetic disorder caused by mutations in genes of the cohesin complex (NIPBL 60%, SMC1A, SMC3, RAD21, HDAC8).

Corns and Calluses

- Corns (heloma) are focal areas of hyperkeratosis caused by pressure and friction, typically on toes. Hard corns (heloma durum) occur on dorsal toes; soft corns (heloma molle) occur between toes. Calluses (tyloma) are diffuse hyperkeratosis on weight-bearing surfaces.

Coronal Caries

- Caries on clinical crown. Smooth surface or pits/fissures.

Coronal Plane

- A vertical plane that divides the body or an organ into front (anterior) and back (posterior) portions.

Coronary Angiography

- The invasive imaging of coronary arteries via contrast injection and X-ray fluoroscopy, performed by catheterizing the femoral, radial, or brachial artery and advancing a catheter to the coronary ostia. Gold standard for assessing coronary artery disease, stenosis severity, and anatomy before revascularization (PCI or CABG).

Coronary Arteries

- The right and left arteries arising from the aortic sinuses that supply the myocardium. Their branches include the right coronary, anterior interventricular, and circumflex arteries.

Coronary Artery

- One of the blood vessels branching from the aorta that supplies oxygen-rich blood directly to the heart

muscle.

Coronary Artery Aneurysm

- A localized dilatation of a coronary artery exceeding the normal vessel diameter by approximately 1.5 times. It may be caused by atherosclerosis, Kawasaki disease, connective-tissue disorders, infection, trauma, or an iatrogenic procedure, and can lead to thrombosis, distal embolization, myocardial ischemia, or rupture.

Coronary Artery Bypass

- A surgical procedure that creates a new route for blood flow around a narrowed or blocked coronary artery using a vascular graft.

Coronary Artery Bypass Grafting (CABG)

- A surgical revascularization procedure for severe coronary artery disease (CAD). Grafts, most commonly the left internal mammary artery (LIMA) and saphenous vein grafts (SVG), are used to bypass stenotic coronary arterial segments. Indicated for multi-vessel CAD, left main coronary artery stenosis, or severe CAD in diabetic patients.

Coronary Artery Disease

- Narrowing or occlusion of coronary arteries, usually due to atherosclerosis, reducing blood flow to the myocardium. Risk factors include hypertension, diabetes, hyperlipidemia, smoking, family history, and obesity. CAD can present as stable angina, acute coronary syndromes (unstable angina, NSTEMI, STEMI), or sudden cardiac death.

Coronary Artery Spasm

- Transient intense constriction of an epicardial coronary artery that reduces myocardial blood flow. It may cause rest angina and transient ST-segment elevation and can be triggered by smoking, stimulants, or endothelial dysfunction; calcium-channel blockers and nitrates are commonly used for prevention and relief.

Coronary Calcium Score

- A CT-based measurement of coronary artery calcification, quantified as the Agatston score. A score of 0 indicates very low risk of cardiovascular events; >400 indicates high burden and significant atherosclerosis. It is used for cardiovascular risk stratification in asymptomatic individuals and may

guide statin therapy decisions.

Coronary Calcium Scoring

- Non-contrast CT measuring calcified plaque in coronary arteries. Agatston score predicts cardiovascular risk.

Coronary Circulation

- The vascular system supplying blood to the myocardium. The left coronary artery (LCA) divides into the left anterior descending (LAD) and circumflex (LCx) arteries. The right coronary artery (RCA) supplies the right ventricle, SA node (in 60% of people), and inferior left ventricular wall.

Coronary Dominance

- Determined by which artery gives rise to the posterior descending artery (PDA). Right coronary dominance (85% of people): RCA gives rise to PDA. Left coronary dominance (8%): LCx gives rise to PDA. Codominance (7%): both RCA and LCx contribute.

Coronary Heart Disease

- Disease caused by reduced blood flow through the coronary arteries, usually due to atherosclerotic plaque. It may cause angina, myocardial infarction, heart failure, or sudden cardiac death. Also called coronary artery disease.

Coronary Insufficiency

- Inadequate blood flow through the coronary circulation to meet myocardial oxygen demand, usually because of coronary artery disease.

Coronary Microvascular Dysfunction

- Impairment of the small coronary arteries and arterioles causing impaired coronary flow reserve despite normal epicardial coronary arteries. Common in women, patients with diabetes, and hypertension. Causes chest pain (angina equivalent) and may contribute to heart failure with preserved ejection fraction.

Coronary Occlusion

- Obstruction of a coronary artery, commonly by atherosclerotic plaque and thrombus, that can cause myocardial ischemia or infarction.

Coronary Sinus

- A large vein on the back of the heart that collects deoxygenated blood from the heart muscle and drains it into the right atrium.

Coronary Stent

- Mesh tube keeping coronary artery open after angioplasty. Types: bare-metal (higher restenosis), drug-eluting (releases antiproliferative drug, lower restenosis but prolonged DAPT). Complications: stent thrombosis, restenosis, bleeding (from DAPT).

Coronary Thrombosis

- See Myocardial Infarction (Heart Attack).

Coronavirus Disease 2019

- A respiratory and systemic infectious disease caused by SARS-CoV-2. Illness ranges from asymptomatic infection to pneumonia, respiratory failure, and multisystem complications.

Corpse

- The dead body of a human being; the subject of medicolegal examination. Forensic assessment of a corpse includes identification, determination of death and time of death (rigor mortis, livor mortis, algor mortis, vitreous potassium, gastric emptying), external and internal examination, and collection of evidence.

Corpus Callosum

- Brain's largest white matter tract with approximately 200 million axons connecting left and right hemispheres. Four parts: rostrum, genu, body, splenium. Facilitates interhemispheric communication for sensory, motor, and cognitive integration.

Corrected Age

- The age of a preterm infant calculated by subtracting the number of weeks born before 40 weeks of gestation from chronological age. Corrected age helps interpret growth and developmental milestones during early childhood, commonly until about 2 years of age.

Corrigan Pulse

- A forceful, rapidly rising and collapsing arterial pulse, classically associated with aortic regurgitation.

Corrosive

- Substance causing tissue destruction by chemical action. Acids: coagulation necrosis, eschar. Alkalis: liquefaction necrosis, deep penetration. Ingestion: avoid emesis, endoscopy, surgical consultation.

Corti's Ganglion

- The spiral ganglion of the cochlea, containing the cell bodies of sensory neurons that transmit auditory information from the organ of Corti.

Cortical Cataract

- A type of age-related lens opacity that begins in the peripheral lens cortex as wedge-shaped or spoke-like opacities that extend centripetally toward the visual axis, caused by disruption of the normally orderly arrangement of lens fibers and the accumulation of fluid between them.

Corticobasal Degeneration

- A rare, progressive neurodegenerative disorder characterized by asymmetric cortical and basal ganglia dysfunction. Presents with asymmetric parkinsonism (rigidity, bradykinesia), apraxia (loss of learned motor skills), alien limb phenomenon (involuntary, purposeless movements), cortical sensory loss, myoclonus, and dystonia.

Corticosteroids

- Synthetic glucocorticoids with potent anti-inflammatory and immunosuppressive effects. Examples include prednisone, prednisolone, methylprednisolone, dexamethasone, and hydrocortisone. They suppress inflammatory gene expression and inhibit immune cell function.

Corticosteroids Mechanism

- Corticosteroids exert their anti-inflammatory and immunosuppressive effects primarily by binding to intracellular glucocorticoid receptors, which then translocate to the nucleus and modulate gene transcription.

Cortisol

- The primary glucocorticoid produced by the adrenal cortex (zona fasciculata) under ACTH control; essential for stress response, metabolism, and homeostasis. Functions: gluconeogenesis, glycogenolysis, protein catabolism, lipolysis, anti-inflammatory/immunosuppressive effects, and maintenance of vascular tone.

Coryza

- Acute inflammation of the nasal mucous membranes with nasal discharge and obstruction, commonly occurring in the common cold or other upper-respiratory infection.

Cosmetic Dermatology

- Cosmetic dermatology encompasses aesthetic procedures. Botulinum toxin injections (Botox, Dysport, Xeomin) relax facial muscles reducing wrinkles. Dermal fillers (hyaluronic acid, calcium hydroxylapatite, poly-L-lactic acid, polymethylmethacrylate) restore volume.

Cost-Effectiveness

- Comparison of costs and outcomes of interventions. Used for resource allocation.

Costochondritis

- Inflammation of the costochondral or costosternal junctions, the most common cause of chest wall pain. Presents with sharp, reproducible tenderness on palpation of the anterior chest, usually unilateral involving the 2nd to 5th ribs. May follow respiratory infection, trauma, or vigorous coughing, but often idiopathic.

Cot Death

- Sudden infant death syndrome, the sudden unexplained death of an apparently healthy infant, usually during sleep, that remains unexplained after investigation.

Cough

- A complex reflex arc protecting the airway by clearing secretions, irritants, and foreign material from the tracheobronchial tree.

Cough Augmentation

- Techniques to improve cough effectiveness in patients with neuromuscular disease or chest wall weakness. Methods include manually assisted cough (abdominal thrust timed with cough), mechanical insufflation-exsufflation (CoughAssist device), and breath stacking with a resuscitation bag.

Count Rate

- Number of detected events per second. Dead time losses at high count rates.

Countercurrent Multiplication

- The renal medullary process in which opposing tubular fluid flow and selective permeability along the loop of Henle generate a corticomedullary osmotic gradient. This gradient permits collecting-duct water reabsorption when antidiuretic hormone is present.

Countertransference

- The therapist’s emotional reactions to the patient, which may be influenced by the therapist’s own unresolved conflicts or may be a response to the patient’s transference. Awareness of countertransference is essential for effective therapy and can provide valuable clinical information about the patient’s interpersonal style.

Coup-Contrecoup Injury

- Brain injury from acceleration-deceleration trauma. Coup: injury at site of impact. Contrecoup: injury on opposite side. Common in falls, motor vehicle accidents.

Courvoisier Sign

- Clinical finding of a palpable non-tender gallbladder in obstructive jaundice suggesting malignant biliary obstruction. In gallstone disease, the gallbladder is typically fibrotic and not palpable, whereas in malignant obstruction it is distended because the wall is initially normal and compliant.

Covered Stent (Vascular)

- See Covered Stent.

COVID Toe

- A chilblain-like skin lesion reported in association with COVID-19, usually affecting the toes or fingers and appearing red, purple, or swollen.

COVID-19

- Coronavirus disease 2019 caused by SARS-CoV-2, first identified in December 2019. Ranges from asymptomatic to severe respiratory failure and death. Common symptoms: fever, cough, dyspnea, loss of taste/smell (anosmia), myalgia, and fatigue. Severe disease: pneumonia, ARDS, hyperinflammatory state, and thromboembolic complications.

COVID-19 in Pregnancy

- Pregnant women with COVID-19 are at increased risk of severe illness (ICU admission, mechanical ventilation, death) compared to non-pregnant women. Risks to the fetus: preterm birth, cesarean delivery, and possibly stillbirth. Vertical transmission is rare but possible.

COVID-19 Test (SARS-CoV-2)

- Detection of SARS-CoV-2. NAAT (RT-PCR): gold standard, nasopharyngeal/nasal swab, highly sensitive/specific, results in 1-48 hours. Rapid antigen

test: lateral flow, results in 15 minutes, sensitive in high viral load (early symptomatic), less sensitive in asymptomatic.

Cowden Syndrome

- Autosomal dominant PTEN hamartoma tumor syndrome (PHTS), caused by loss-of-function mutations in PTEN (10q23) — a negative regulator of the PI3K/AKT/mTOR pathway. Multiple hamartomas and increased risk of benign and malignant tumors.

Cowper’s Gland

- One of a pair of small male accessory sex glands below the prostate. Its secretion contributes mucus and fluid to the urethra before ejaculation.

Cowpox

- A zoonotic viral infection caused by cowpox virus or related orthopoxviruses, producing localized pustular or ulcerative skin lesions and sometimes systemic illness.

COX Inhibitor

- A class of drugs that inhibit cyclooxygenase enzymes (COX-1 and COX-2), which catalyze the conversion of arachidonic acid to prostaglandins and thromboxanes. COX-1 is constitutively expressed in most tissues, producing prostaglandins that protect the gastric mucosa, maintain renal blood flow, and support platelet aggregation.

Coxa Vara

- A deformity in which the angle between the femoral neck and shaft is abnormally reduced, potentially causing a limp, shortened limb, or altered hip mechanics.

Coxalgia

- Pain in the hip or hip joint.

Coxiella burnetii

- An obligate intracellular bacterium that causes Q fever, a zoonotic infection transmitted from animals to humans. The organism is transmitted through inhalation of contaminated aerosols or ingestion of contaminated dairy products. Acute Q fever presents with fever, headache, and pneumonia. Chronic Q fever can cause endocarditis.

Coxsackie virus type A16

- A member of the Enterovirus genus and the most common cause of hand, foot, and mouth disease, a

mild, self-limited viral illness affecting children. It presents with fever, oral vesicles and ulcers, and a maculopapular or vesicular rash on the hands and feet. Complications are rare but may include aseptic meningitis.

Coxsackievirus

- A group of enteroviruses that cause a range of human diseases. Group A viruses commonly cause hand-foot-and-mouth disease and herpangina, whereas group B viruses are associated with myocarditis, pericarditis, pleurodynia, and aseptic meningitis.

CPAP

- Continuous Positive Airway Pressure, first-line for OSA. Delivers constant pressure through a mask to prevent airway collapse. Eliminates apneas and improves outcomes. Adherence is the main challenge.

CPAP Machine

- Continuous positive airway pressure for sleep apnea.

Cracked Tooth Syndrome

- Incomplete fracture of tooth causing pain on biting. Diagnosis difficult. Treatment: crown, root canal if pulpal involvement.

Crackles (Rales)

- Discontinuous, non-musical adventitious lung sounds produced by the sudden explosive opening of collapsed small airways and alveoli during inspiration or the bubbling of air through fluid in the bronchioles.

Cradle Cap

- A common form of seborrheic dermatitis in infants, causing greasy or flaky scales on the scalp with mild redness. It is usually harmless and improves with gentle scalp care; persistent inflammation or signs of infection require assessment.

Cranial Arteritis

- Giant-cell arteritis, an inflammatory disease of medium and large arteries, especially branches of the external carotid artery, that can cause headache, visual loss, and systemic symptoms.

Cranial Cavity

- The hollow space inside the skull that contains and protects the brain.

Cranial Implant

- See neurosurgical implants.

Cranial Nerve

- Twelve paired nerves (CN I-XII) emerging directly from the brain/brainstem, primarily serving head and neck.

Cranial Nerve Palsy

- Weakness or paralysis of one or more cranial nerves (CN III-XII) causing specific deficits: CN III (ptosis, mydriasis, 'down and out' eye—aneurysm, diabetes, herniation), CN IV (vertical diplopia—trauma), CN VI (lateral rectus weakness, horizontal diplopia—increased ICP, diabetes, microvascular), CN VII (facial palsy—Bell palsy,

Craniectomy

- Surgical removal of part of the skull without immediate replacement, performed to relieve intracranial pressure or provide access to the brain.

Craniocerebral Injury

- Trauma to scalp, skull, or brain. See head injury.

Craniofacial Dysostosis

- A congenital disorder involving premature fusion of cranial sutures and characteristic abnormalities of the face and skull, such as those seen in Crouzon syndrome.

Craniopharyngioma

- Histologically benign but locally aggressive epithelial tumor arising embryologically from Rathke pouch remnants; the most common supratentorial non-glial pediatric brain tumor (peak 5-14 years, second peak 50-74 years); bimodal age distribution.

Cranioplasty

- Surgical repair or reconstruction of a skull defect using bone, an implant, or another suitable material.

Craniosynostosis

- Premature fusion of one or more cranial sutures, restricting skull growth perpendicular to the affected suture. Presents with abnormal head shape (scaphocephaly from sagittal fusion, plagiocephaly from coronal fusion). May increase intracranial pressure. Diagnosis by physical examination and CT with 3D reconstruction.

Craniotomy

- Surgical opening of the skull (bone flap) for ac-

cess to intracranial contents. Indications: brain tumor resection (glioblastoma, meningioma, mets), hematoma evacuation (SDH, EDH, ICH), aneurysm clipping, AVM resection, and epilepsy surgery (lobectomy).

Crazy Paving Pattern

- A HRCT finding characterized by ground-glass opacification with superimposed interlobular septal thickening and intralobular lines. It is seen in pulmonary alveolar proteinosis, lipoid pneumonia, ARDS, pulmonary hemorrhage, and Pneumocystis pneumonia. Despite its name, it is not specific to any single diagnosis.

CRE

- Carbapenem-resistant Enterobacteriaceae, resistant to carbapenem antibiotics through production of carbapenemases including KPC, NDM, and OXA-48-like enzymes. CRE are a major public health threat due to limited treatment options. Infections are associated with high mortality rates.

Cream

- A semisolid emulsion (oil in water or water in oil) for topical application, containing a drug in an emulsion base. Creams are lighter, less greasy, and easier to spread than ointments, and are preferred for moist or oozing skin lesions. They penetrate less deeply than ointments and are cosmetically acceptable.

Creatine Kinase

- An enzyme found largely in skeletal muscle, heart muscle, and brain. Blood levels may rise after muscle injury, myocardial injury, or strenuous exertion.

Creatine Kinase (CK)

- An enzyme found in muscle, brain, and heart. CK-MB: cardiac isoenzyme (used historically for MI diagnosis, largely replaced by troponin). CK-MM: skeletal muscle (elevated in rhabdomyolysis, trauma, statins, myopathy). Total CK: elevated in muscle injury, myocardial infarction, seizures, hypothyroidism, and statin myopathy.

Creatinine

- A breakdown product of muscle creatine phosphate produced at a constant rate, filtered by the glomerulus and minimally reabsorbed; serum creatinine is the most used marker of kidney function.

CREST Syndrome

- Limited cutaneous form of systemic sclerosis (scleroderma), characterized by the eponymic features: Calcinosis cutis (multiple subcutaneous calcium deposits, often in extremities — can cause pain, inflammation, ulceration), Raynaud phenomenon (usually long-standing; often precedes other features by years), Esophageal dysmotility (lower tw

Cretinism

- An outdated term for severe congenital or early-childhood hypothyroidism causing impaired physical and intellectual development; modern clinical language uses congenital hypothyroidism.

Creutzfeldt-Jakob disease

- See Prion Disease.

Creutzfeldt Jakob Disease

- Creutzfeldt-Jakob disease (CJD) is a rare, fatal, transmissible spongiform encephalopathy caused by the accumulation of misfolded prion protein (PrP^{Sc}) in the brain, leading to rapid, progressive dementia and myoclonus.

Creutzfeldt-Jakob Disease

- A rapidly progressive fatal neurodegenerative disease caused by abnormal prion protein, characterized by dementia, myoclonus, ataxia, and other neurologic signs.

Cri du Chat Syndrome

- A congenital genetic disorder caused by deletion of part of the short arm of chromosome 5, characterized in infancy by a high-pitched cry, developmental delay, and characteristic facial features.

Cri-du-Chat Syndrome

- A genetic disorder caused by a deletion of the short arm of chromosome 5 (5p-), with the size of the deletion correlating with severity. Incidence 1 in 15,000-50,000.

Cricoid Cartilage

- The cricoid cartilage is a complete, ring-shaped cartilage located at the base of the larynx, directly above the trachea and below the thyroid cartilage, and it is the only complete cartilaginous ring in the airway, providing structural support and maintaining airway patency.

Cricoid Pressure

- Cricoid pressure (Sellick maneuver) applies posterior displacement of the cricoid cartilage against the cervical vertebrae, occluding the esophagus and reducing aspiration risk during rapid sequence intubation.

Cricothyrotomy

- An emergency airway procedure that creates an opening through the cricothyroid membrane to provide oxygenation and ventilation when a patient cannot be intubated and cannot be oxygenated by less invasive methods.

Criminal Abortion

- Abortion performed illegally, outside permitted legal indications and procedures (e.g., unsafe methods, unqualified practitioners, or non-medical settings), carrying high maternal risk.

CRISPR-Cas9

- Genome editing tool derived from bacterial immune system. Cas9 enzyme guided by RNA to specific DNA sequence, creating double-strand break. Repaired by non-homologous end joining (NHEJ) or homology-directed repair (HDR). Applications: research, therapy.

Critical Care

- Specialty managing critically ill patients. See intensive care medicine.

Critical Findings

- Findings requiring immediate communication. Examples: pneumothorax, brain herniation, aortic dissection.

Critical Limb Ischemia

- See acute limb ischemia.

Critical Results

- Unexpected significant findings require immediate notification. See critical findings.

Critical Value

- A laboratory result indicating a potentially life-threatening condition that requires prompt communication to a responsible clinician or care team. The threshold is test- and laboratory-specific and should be defined by institutional policy.

Crohn Disease

- Chronic transmural inflammatory condition that

can affect any segment of the gastrointestinal tract from the oral cavity to the perianal region, though the terminal ileum and ileocolonic region are most commonly involved.

Crohn Disease Cobblestoning

- Endoscopic appearance of the colonic or small bowel mucosa in Crohn disease characterized by a bumpy, irregular surface resembling cobblestones. This distinctive appearance results from intersecting lines of deep linear and serpiginous ulceration that surround areas of relatively preserved or edematous but intact mucosa.

Cross-Sectional Study

- Measures exposure and outcome at same time. Prevalence study. Cannot establish temporality.

Cross-Tolerance

- Tolerance to one substance that extends to another substance with a similar mechanism of action. For example, tolerance to one benzodiazepine extends to all benzodiazepines, and tolerance to alcohol extends partially to benzodiazepines due to shared GABA-A receptor mechanisms.

Croup

- Acute laryngotracheobronchitis, most commonly caused by parainfluenza virus. Affects children 6 months to 3 years. Presents with barking cough, stridor, hoarseness, and inspiratory distress, worse at night. Diagnosis is clinical; X-ray shows subglottic narrowing (steeple sign).

Crouzon Syndrome

- An autosomal dominant craniosynostosis syndrome caused by mutations in the FGFR2 or FGFR3 genes, leading to premature fusion of multiple cranial sutures. Features: brachycephaly (short, broad head from bilateral coronal synostosis), proptosis (shallow orbits), hypertelorism, midface hypoplasia, beaked nose, and Class III malocclusion.

Crown (Dental)

- Cap covering damaged tooth. Materials: metal, porcelain-fused-to-metal, all-ceramic, zirconia. Indications: large caries, fractured tooth, root canal treated tooth.

Crown Lengthening

- Surgical procedure exposing more tooth structure. Involves gingivectomy and bone removal. For

restorative or aesthetic purposes.

Crown-Rump Length (CRL)

- The measurement from the top of the fetal head (crown) to the bottom of the buttocks (rump), the most accurate method for establishing gestational age in the first trimester. Measured at 7-13 weeks. Accuracy: ± 3 -5 days. The EDD is set using CRL if the difference from LMP dating exceeds 7 days. CRL growth: approximately 1 mm per day.

Crowning

- The stage of the second stage of labor when the fetal head (widest diameter) distends the perineum and becomes visible at the introitus and is no longer receding between contractions.

CRP

- C-reactive protein. Acute phase reactant. Marker of inflammation.

Crush Syndrome

- Systemic manifestations of prolonged muscle crush injury (rhabdomyolysis) following trauma — typically from collapsed buildings, landslides, entrapment under debris, or excessive compression of limbs. Bywaters syndrome.

Crushing Injury

- Damage caused when a body part is compressed between heavy objects. It can produce fractures, nerve and blood-vessel injury, tissue death, and rhabdomyolysis. Severe limb compression or sudden release after prolonged entrapment requires emergency treatment.

Crusted scabies

- A severe, highly contagious form of scabies (Norwegian scabies) characterised by thick, hyperkeratotic crusts teeming with enormous numbers of *Sarcoptes scabiei* mites, typically occurring in immunocompromised, elderly, or neurologically impaired individuals.

Crutches

- Walking aids. Types: axillary, forearm (Loftstrand). Axillary: under arm. Forearm: through loop.

Cryoablation

- Image-guided destruction of tissue. See ablation therapy.

Cryobiopsy

- A bronchoscopic technique using rapid freezing to obtain larger lung tissue samples than conventional transbronchial biopsy. It provides better histopathological specimens for diagnosing interstitial lung diseases. It requires general anesthesia and bronchoscopic expertise. It is less invasive than surgical lung biopsy.

Cryoglobulinemic vasculitis

- Systemic vasculitis caused by cryoglobulin-containing immune complexes that deposit in small and medium vessels, classically presenting with purpura, arthralgia, and glomerulonephritis. Most cases are associated with hepatitis C virus infection.

Cryoglobulins

- Immunoglobulins that precipitate at cold temperatures ($< 37^{\circ}\text{C}$) and redissolve on warming. Types: Type I (monoclonal IgM, in MGUS, Waldenstrom, myeloma), Type II (mixed: monoclonal + polyclonal, in HCV infection), Type III (mixed polyclonal, in autoimmune disease).

Cryopexy

- A procedure that uses controlled freezing to destroy or seal abnormal tissue, especially retinal breaks or small retinal detachments.

Cryoprecipitate

- A plasma fraction rich in fibrinogen, factor VIII, von Willebrand factor, and factor XIII, used selectively to treat certain bleeding disorders.

Cryopreservation

- Preservation of cells, tissues, or embryos by cooling them to very low temperatures, usually with protective substances that limit ice-related injury.

Cryopyrin-Associated Periodic Syndromes

- Autosomal dominant autoinflammatory disorders (NLRP3 mutations, gain-of-function) causing constitutive inflammasome activation and IL-1 β overproduction.

Cryosurgery

- Surgery that destroys abnormal tissue by controlled freezing, used in selected skin lesions, tumors, and other conditions.

Cryotherapy

- The therapeutic application of extreme cold (typically using liquid nitrogen at -196°C) to destroy abnormal or diseased tissue.

Cryptitis

- Inflammation of a crypt or small anatomical recess, especially inflammation of the anal glands and crypts.

Cryptococcal meningitis

- See Cryptococcosis.

Cryptococcosis

- A fungal infection caused by *Cryptococcus neoformans* or *C. gattii*, primarily affecting immunocompromised patients (HIV/AIDS, organ transplant, hematologic malignancy). Pulmonary cryptococcosis: cough, dyspnea, pulmonary nodules. Meningoencephalitis: fever, headache, altered mental status, and elevated intracranial pressure.

Cryptococcus neoformans

- See Cryptococcosis.

Cryptogenic Organizing Pneumonia

- An inflammatory lung disease with granulation tissue within alveolar ducts. May be idiopathic or secondary to infection, drugs, or connective tissue disease. Responds well to corticosteroids.

Cryptorchidism

- Failure of one or both testes to descend into the scrotum, the most common congenital genitourinary anomaly in boys (3% at birth, 1% at 1 year). 70% of undescended testes are palpable (inguinal), 30% non-palpable (intra-abdominal, absent, or atrophic). Spontaneous descent by 6 months occurs in about one-third.

Cryptosporidiosis

- A diarrhoeal disease caused by the protozoan parasite *Cryptosporidium parvum* (zoonotic) or *C. hominis* (anthroponotic), transmitted via contaminated water (chlorine-resistant oocysts), food, or direct contact. Presents with watery diarrhea, abdominal cramps, nausea, and low-grade fever, lasting 1-2 weeks.

Crystal Arthropathy

- Joint disease caused by crystal deposition. Gout (monosodium urate) and pseudogout (calcium py-

rophosphate). Diagnosis by joint aspiration and polarized microscopy.

Crystalloids

- Aqueous solutions of mineral salts that distribute between intravascular and interstitial compartments. Normal saline (0.9 percent NaCl) has 154 mEq/L each of sodium and chloride, pH 5.4, and is associated with hyperchloremic metabolic acidosis at large volumes.

Crystalluria

- The presence of crystals in urine, which may reflect precipitation of salts, medication effects, metabolic disease, or a tendency to form urinary stones.

CSF Analysis

- See neurology.

CSF Rhinorrhea

- Cerebrospinal fluid rhinorrhea is the leakage of cerebrospinal fluid from the subarachnoid space through a defect in the skull base into the nasal cavity or paranasal sinuses, resulting in a clear, watery nasal discharge that may be mistaken for allergic rhinitis or sinusitis.

CT Angiography (CTA)

- CT imaging with IV contrast timed to visualize arteries. Coronary CTA: assesses coronary artery stenosis and plaque, calcium scoring (Agatston). CTA chest: pulmonary embolism (PE protocol), aortic dissection. CTA head/neck: stroke evaluation, aneurysm, dissection, and vasculitis.

CT Chest

- Detailed lung and mediastinal imaging. High-resolution CT (HRCT) for interstitial lung disease. CT pulmonary angiography (CTPA) for pulmonary embolism. Lung cancer screening with low-dose CT.

CT Colonography

- CT imaging of the colon after bowel preparation and insufflation with air or carbon dioxide. Used for colorectal cancer screening in patients unable to undergo optical colonoscopy and for evaluating complete colonoscopy when the cecum was not reached. Detects polyps greater than six millimeters with high sensitivity.

CT Colonography (Virtual Colonoscopy)

- CT imaging of the colon after bowel preparation and air insufflation. Indications: colorectal cancer

screening (alternative to colonoscopy), incomplete colonoscopy due to obstructing mass, and patients unable to undergo colonoscopy. Sensitivity >90% for polyps >10mm. Requires same bowel prep as colonoscopy.

CT Contrast

- Substance enhancing image visibility. See contrast agent.

CT Coronary Angiography

- CT imaging of coronary arteries after contrast. Detects stenosis, plaque. Alternative to invasive angiography in low-intermediate risk.

CT Dose-Length Product (DLP)

- A CT dose metric calculated from the volume CT dose index and scan length. It estimates the total scanner output for an examination and is used for protocol optimization and dose monitoring, not as a direct patient-specific risk estimate.

CT Enterography

- CT imaging of the small bowel after administration of large volumes of neutral oral contrast distending the small bowel with IV contrast. Excellent for evaluating Crohn disease, small bowel tumors, and obscure gastrointestinal bleeding.

CT Perfusion

- CT technique measuring brain or organ perfusion. Used for acute stroke, tumor evaluation.

CT Pulmonary Angiography

- Gold standard for PE diagnosis, showing thrombi as filling defects. Sensitivity > 95%, specificity > 97% for central emboli.

CT Scan

- See radiology.

CT Scan (Computed Tomography)

- Cross-sectional imaging using X-rays and computer reconstruction. Non-contrast: evaluate hemorrhage, stones, and bone. IV contrast (iodinated): evaluate inflammation, infection, tumor, and vascular structures. Oral contrast: opacify bowel.

Cubital Tunnel Syndrome

- Cubital tunnel syndrome: ulnar nerve compression at elbow (cubital tunnel: Osborne's ligament, FCU aponeurosis, medial epicondyle, olecranon). Clinical: paresthesias ulnar 1.5 digits, nocturnal, Tinel

at elbow, Froment sign (IP flexion of thumb with key pinch), Wartenberg sign (abduction of small finger), thenar/hypothenar wasting late.

Cubitus Valgus

- An abnormal outward angulation of the forearm relative to the upper arm when the elbow is extended.

Cubitus Varus

- An abnormal inward angulation of the forearm relative to the upper arm when the elbow is extended, often developing after a childhood elbow fracture.

Cuboid

- Cube-shaped lateral foot tarsal bone articulating with calcaneus proximally and 4th/5th metatarsals distally. Attachment for the long plantar ligament and peroneus longus tendon. Keystone of the lateral longitudinal arch. Cuboid syndrome (subluxation) is a common lateral foot pain cause in athletes from excessive pronation.

Culdocentesis

- Surgical puncture of the posterior vaginal fornix to sample or drain fluid from the rectouterine pouch.

Cullen's Sign

- Bluish or violaceous periumbilical ecchymosis caused by blood tracking through the fascial planes to the umbilicus. It is a sign of intra-abdominal or retroperitoneal haemorrhage, classically severe haemorrhagic pancreatitis, and is associated with serious underlying disease.

Cultural Psychiatry

- The study and clinical practice of how cultural factors influence mental health, illness expression, diagnosis, and treatment. Cultural concepts of distress, idioms of distress, and cultural syndromes must be considered. The DSM-5 includes a Cultural Formulation Interview.

Culture and Sensitivity

- Microbiological culture of a clinical specimen (blood, urine, sputum, wound swab, CSF) to identify pathogens and determine antimicrobial susceptibility. Culture media: blood agar (most bacteria), MacConkey (Gram-negative), chocolate agar (fastidious: *N. gonorrhoeae*, *H. influenzae*), and selective media.

Culture Media

- Nutrient preparations used to grow microorganisms in the laboratory, classified by composition and purpose.

Culture Swab

- Swab for microbial culture. Types: bacterial, viral, fungal.

Cumulative Dose

- Total amount of substance received over time. Important for carcinogens, teratogens, cumulative toxins.

Cumulative Trauma Disorder

- A disorder caused by repeated forceful or awkward movements that progressively injure muscles, tendons, nerves, or other soft tissues.

Cup-to-Disc Ratio

- The cup-to-disc ratio is a clinical measurement used to assess the severity of glaucomatous optic nerve damage, comparing the diameter of the optic cup (the central depression within the optic nerve head where neural tissue is absent) to the diameter of the optic disc (the entire optic nerve head).

CURB-65 Score

- One point each for: Confusion, Urea > 7 mmol/L, RR > 30, BP < 90/60, age ≥ 65. 0–1: outpatient. 2: consider hospitalization. 3–5: severe.

Curettage

- Removal of tissue from a body cavity or surface with a curette, used for diagnosis, treatment, or removal of abnormal tissue.

Curettage Specimen

- Tissue obtained by scraping or scooping a surface or cavity, such as the endometrium or a bone lesion, for histologic examination. The limited and fragmented nature of the specimen may restrict assessment of invasion or architecture.

Curette

- Spoon-shaped instrument for scraping tissue.

Curling Ulcer

- Stress ulceration of the stomach or duodenum associated with severe burns. Part of the Curling triad with hypovolemia and sepsis. The pathogenesis involves reduced mucosal blood flow during the systemic inflammatory response to burn injury.

Curling's Ulcer

- An acute peptic ulcer associated with severe burns, often occurring in the duodenum or stomach during physiologic stress.

Curvularia alcornii

- A dematiaceous fungal species causing phaeohyphomycosis, an opportunistic infection of the skin, sinuses, and occasionally the central nervous system in immunocompromised patients. It produces dark, septate hyphae in tissue. Diagnosis is by culture and histopathology showing pigmented fungal elements.

Cushing Disease

- Hypercortisolism caused by a pituitary ACTH-secreting adenoma, the most common cause of endogenous Cushing syndrome (70%). Presents with central obesity, moon face, buffalo hump, violaceous striae, proximal myopathy, easy bruising, osteoporosis, hypertension, hyperglycemia, depression, and menstrual irregularities.

Cushing Syndrome

- Clinical state resulting from chronic exposure to excess glucocorticoids regardless of cause. Most common endogenous cause is Cushing disease from pituitary ACTH-secreting adenoma accounting for approximately seventy percent.

Cushing Syndrome Complications

- Untreated Cushing syndrome carries significant morbidity and mortality from cardiovascular disease, infections, thromboembolic events, and metabolic derangements. Cardiovascular complications include accelerated atherosclerosis, hypertension, and dyslipidemia. Infection risk is increased from immunosuppression.

Cushing Ulcer

- Stress ulceration associated with central nervous system disease, trauma, or surgery. Caused by vagally mediated gastric acid hypersecretion, distinguishing it from Curling ulcer which results from mucosal ischemia. The ulcers are typically solitary, located in the stomach or duodenum, and present with gastrointestinal bleeding.

Cushing's Disease

- Hypercortisolism caused specifically by an adrenocorticotrophic-hormone-secreting pituitary adenoma.

Cushing's Syndrome

- The clinical state caused by prolonged excessive exposure to cortisol, from endogenous overproduction or exogenous corticosteroid use.

Cusp Fracture

- Breakage of tooth cusp. Often due to large restoration weakening tooth. Treatment: crown or onlay.

Cutaneous diphtheria

- See Diphtheria.

Cutaneous Drug Eruptions

- a spectrum of reactions. Exanthematous (maculopapular) is the most common, appearing 4 to 14 days after drug initiation. Urticarial appears within hours. Photoallergic occurs at sun-exposed sites. Lichenoid resembles lichen planus. Pigmented purpuric dermatosis presents as cayenne pepper spots.

Cutaneous larva migrans

- A creeping eruption caused by hookworm larvae (*Ancylostoma braziliense*, *A. caninum*) from dog/cat feces penetrating the skin, typically in tropical/subtropical regions (walking barefoot on contaminated sand). Larvae migrate erratically under the skin causing intensely pruritic, serpiginous, erythematous tracks.

Cutaneous leishmaniasis

- See Leishmaniasis.

Cutaneous lupus

- A form of lupus erythematosus limited to the skin, manifesting as discoid lesions, subacute cutaneous plaques, or acute malar erythema. It may occur in isolation or as a feature of systemic lupus erythematosus.

Cutaneous Lupus Erythematosus

- Autoimmune skin disease associated with lupus-spectrum illness and characterized by inflammatory lesions that may scar, alter pigmentation, or cause photosensitivity. Subacute and chronic forms have different morphologies and systemic associations; evaluation may include biopsy, direct immunofluorescence, and autoimmune testing.

Cutaneous Lymphoma

- Cutaneous lymphomas are heterogeneous group of T-cell and B-cell malignancies primarily involving the skin. Primary cutaneous B-cell lymphomas

include marginal zone lymphoma, follicle center lymphoma, and diffuse large B-cell lymphoma leg type. Primary cutaneous T-cell lymphomas include mycosis fungoides and Sézary syndrome.

Cutaneous metastases

- Secondary deposits of cancer in the skin, originating from a primary tumor elsewhere in the body. They may present as nodules, plaques, or ulcers, and often indicate advanced or recurrent disease.

Cutaneous Small-Vessel Vasculitis

- Inflammation and destruction of small dermal blood vessels, commonly presenting as palpable purpura on dependent areas. Causes include infections, medications, autoimmune disease, and systemic vasculitis; skin biopsy with appropriate specimen handling is central to diagnosis.

Cutaneous T-Cell Lymphoma

- A malignancy of skin-homing T lymphocytes. Mycosis fungoides is the most common subtype, presenting as patches, plaques, and tumors in a clothing distribution. Sézary syndrome is the leukemic variant with erythroderma, lymphadenopathy, and circulating malignant cells.

Cutting Balloon (Vascular)

- See Cutting Balloon.

CVA

- Interruption of blood flow to the brain, causing neurological deficits. See Cerebrovascular Accident (Stroke).

Cyanide Poisoning

- Toxicity from cyanide compounds (hydrogen cyanide, potassium/sodium cyanide) that inhibit cytochrome oxidase (complex IV), blocking oxidative phosphorylation and cellular respiration despite adequate oxygen—histotoxic hypoxia. Sources: smoke inhalation (fires—plastic/acrylic), ingestion (industrial, suicidal), occupational exposure.

Cyanosis

- A bluish discoloration of the skin and mucous membranes resulting from excessive amounts of deoxygenated hemoglobin in the capillary blood. It becomes clinically apparent when the concentration of deoxygenated hemoglobin exceeds 5 grams per deciliter, which corresponds to an oxygen saturation of approximately 85 percent.

Cycle Time

- Time for complete image acquisition. PET: minutes. SPECT: 20-30 minutes.

Cyclosporine

- A calcineurin inhibitor immunosuppressant that blocks T-cell activation by preventing NFAT-mediated cytokine (IL-2) transcription. Used to prevent organ transplant rejection, graft-versus-host disease, and in autoimmune conditions (psoriasis, rheumatoid arthritis, nephrotic syndrome, ulcerative colitis).

Cyclothymic Disorder

- A chronic mood disorder characterized by numerous periods of hypomanic symptoms and periods of depressive symptoms over at least two years (one year in children and adolescents). The hypomanic and depressive symptoms do not meet criteria for a hypomanic or major depressive episode.

CYP2C19 Polymorphism

- CYP2C19 is a cytochrome P450 enzyme involved in the metabolism of approximately 10 percent of drugs, including the antiplatelet agent clopidogrel, proton pump inhibitors, some antidepressants, and the antiepileptic drug phenytoin.

CYP2D6 Polymorphism

- CYP2D6 is a cytochrome P450 enzyme responsible for the metabolism of approximately 25 percent of commonly prescribed drugs, including codeine, tamoxifen, many antidepressants, and antipsychotics.

CYP450 System

- A superfamily of heme-containing enzymes primarily in the liver that metabolize drugs and xenobiotics. Major isoforms include CYP3A4 (metabolizes 50% of drugs), CYP2D6, CYP2C9, CYP2C19, CYP1A2, and CYP2E1.

Cyst

- An abnormal closed cavity or sac lined by epithelium, containing liquid, semisolid, or gaseous material, and distinguished from an abscess (which contains pus) and a pseudocyst (which lacks an epithelial lining).

Cystadenoma

- A usually benign glandular epithelial tumor that forms cystic spaces or cyst-like structures.

Cystic echinococcosis

- Hydatid disease: infection with larval cysts of *Echinococcus granulosus* (dog tapeworm), transmitted by ingestion of eggs from dog feces. Cysts grow slowly in liver (most common, 60-70 percent), lungs (20-30 percent), and other organs. Hepatic: RUQ pain, hepatomegaly, biliary obstruction. Pulmonary: cough, chest pain, hemoptysis.

Cystic Fibrosis

- Autosomal recessive disorder caused by CFTR gene mutations (most common: F508). Defective chloride channel causes thick mucus in lungs, pancreas, liver, reproductive tract. Symptoms: chronic sinusitis, bronchiectasis, pancreatic insufficiency, male infertility (absent vas deferens), meconium ileus.

Cystic Hygroma

- Congenital lymphatic malformation composed of thin-walled lymphatic cysts, manifesting as a soft, fluid-filled, painless, translucent (transilluminates brilliantly) mass, typically in the posterior triangle of the neck (75%, left-sided), axilla (20%), mediastinum, retroperitoneum, or groin.

Cysticercosis

- Tissue infection caused by larval cysts (cysticerci) of the pork tapeworm *Taenia solium*, acquired by ingestion of tapeworm eggs (not by eating pork itself) from contaminated food, water, or fecal-oral autoinfection.

Cystine and triple phosphate (struvite) crystals

- Crystalline substances found in urine sediments with diagnostic significance. Cystine crystals are hexagonal and colorless, diagnostic of cystinuria, an inherited disorder of renal tubular transport.

Cystine Stones

- Rare stones from inherited cystinuria with defective cystine transport. Hexagonal crystals on urine microscopy. Difficult to dissolve. Treat with aggressive hydration, urine alkalization to pH greater than 7.5, and thiol agents including tiopronin and D-penicillamine that form soluble complexes with cystine.

Cystinuria

- Autosomal recessive inherited disorder (incidence 1 in 7,000 to 1 in 15,000) of proximal renal tubu-

lar reabsorption of the dibasic amino acids cystine, ornithine, arginine, and lysine (the "COLA" amino acids), caused by mutations in SLC3A1 (type A, autosomal recessive) or SLC7A9 (type B, sometimes autosomal dominant with incomplete

Cystitis

- Inflammation of the urinary bladder, typically from bacterial infection, one of the most common bacterial infections in women.

Cystocele

- Anterior vaginal wall prolapse: herniation of the bladder into the anterior vaginal wall due to weakening of the pubocervical fascia, most commonly after childbirth and menopause.

Cystography

- Contrast imaging of bladder. Evaluates trauma, fistulas, vesicoureteral reflux.

Cystometrogram

- Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases. See urodynamic testing.

Cystometry

- See cystometrogram.

Cystoscopy

- Endoscopic examination of the urethra, bladder, and ureteral orifices. Flexible cystoscopy: office-based, local anesthesia. Rigid cystoscopy: OR, allows larger instruments for interventions.

Cystostomy

- Surgical creation of an opening from the urinary bladder to the skin or another structure to permit drainage of urine.

Cystourethrography

- Radiographic imaging of the urinary bladder and urethra, usually during filling and voiding after contrast has been introduced.

Cytochrome c

- Mobile electron carrier between Complex III and IV. Also involved in apoptosis.

Cytochrome P450

- The cytochrome P450 system is a superfamily of heme-containing monooxygenase enzymes primarily

located in the endoplasmic reticulum of hepatocytes, responsible for Phase I drug metabolism through oxidation, reduction, and hydrolysis reactions.

Cytogenetics

- Study of chromosomes. Detects abnormalities in cancer.

Cytogenetics (Karyotype)

- Visualization of chromosomes under light microscopy from dividing cells (bone marrow, blood, amniotic fluid, tissue). Staining: G-banding (Giemsa, light and dark bands). Resolution: 400-550 bands per haploid genome.

Cytokine

- Small signaling proteins (5-25 kDa) mediating cell-to-cell communication in immunity, inflammation, and hematopoiesis.

Cytokine Release Syndrome

- CRS: systemic inflammatory response from massive cytokine release (IL-6, IFN-gamma, TNF, IL-10) after T cell engaging therapies (CAR-T, bispecific antibodies, TGN1412).

Cytokine Storm

- A severe, dysregulated systemic inflammatory response involving excessive cytokine release that can cause tissue injury, organ dysfunction, and shock.

Cytology

- The microscopic study of individual cells, distinct from histology in that tissue architecture is not examined, used for screening and diagnosis of neoplastic and infectious disease. Specimens are exfoliative (cervical smears, sputum) or aspirated (fine needle aspiration, body fluids), interpreted with Papanicolaou or special stains.

Cytomegalovirus

- A herpesvirus that causes significant disease in immunocompromised patients including transplant recipients and HIV-infected individuals. CMV infects a wide range of cell types and establishes latent infections in myeloid progenitor cells. Primary infection is usually asymptomatic in immunocompetent individuals.

Cytomegalovirus Infection

- A widespread herpesvirus infection. Primary infection in immunocompetent individuals is usually asymptomatic or causes mononucleosis-like syn-

drome. Congenital CMV: most common congenital infection, causes sensorineural hearing loss, microcephaly, and developmental delay.

Cytopathology

- The branch of pathology using cytologic examination of cells to diagnose disease, particularly neoplastic and infectious processes, built on standardized reporting systems that guide management.

Cytopenia

- A reduction in the number of one or more circulating blood-cell lineages: red cells, white cells, or platelets.

Cytoreductive Surgery

- Surgical removal of as much visible tumor as possible to reduce disease burden, a cornerstone of ovarian cancer management and combined with hyperthermic intraperitoneal chemotherapy for peritoneal carcinomatosis.

Cytoskeleton

- Dynamic protein filament network providing structural support, cell shape, intracellular transport, and cell division. Three main components: (1) Microfilaments (actin, 7 nm): cell cortex, stress fibers, contractile ring (cytokinesis), lamellipodia/filopodia (motility), myosin interaction (muscle contraction).

Cytotoxic T Cell

- The effector cells of cell-mediated immunity that kill virus-infected cells and tumor cells. Recognize antigenic peptides presented on MHC class I molecules via the T-cell receptor; co-stimulation by CD4+ helper cells and cytokines (IL-2) is needed for activation.

Cytotoxicity

- The capacity of an immune effector cell or immune mediator to injure or kill a target cell. T lymphocytes and natural killer cells commonly use perforin-granzyme pathways or death-receptor signaling to induce target-cell apoptosis.

Chronic Migraine

- A migraine disorder with headache on at least 15 days per month for more than three months, with migraine features on at least eight days per month. Symptoms may include throbbing pain, nausea, vomiting, and light or sound sensitivity; frequent or disabling headaches require clinical eval-

uation.

Chronic Tension-Type Headache

- A recurring headache occurring on at least 15 days per month for more than three months, usually causing bilateral pressing or tightening pain rather than throbbing. It may be associated with scalp or neck tenderness but typically lacks prominent nausea; a new or changing pattern requires clinical assessment.

Chapter 4

Pocket Dictionary: D Terms

D and C

- Dilation and curettage, a procedure in which the cervix is dilated and tissue is removed from the uterine cavity for diagnosis, treatment, or management of pregnancy-related conditions.

D and E

- Dilation and evacuation, a surgical procedure in which the cervix is dilated and uterine contents are removed by suction and instruments.

D&C Instruments

- Dilators and curettes for dilation and curettage.

D-Dimer

- A fibrin degradation product, elevated in thrombosis (PE, DVT, DIC). High sensitivity but low specificity. D-dimer <500 ng/mL (ELISA) has high negative predictive value for VTE; used to rule out PE/DVT in patients with low or moderate pretest probability (Wells criteria).

Dabigatran

- A direct oral anticoagulant that reversibly inhibits thrombin (factor IIa). Used for stroke prevention in nonvalvular atrial fibrillation and treatment/prophylaxis of venous thromboembolism. Fixed dosing, rapid onset, predictable pharmacokinetics, no routine monitoring. Prodrug requiring acid environment; capsules must not be opened.

Dacrocytes (Teardrop Cells)

- Teardrop-shaped red blood cells formed as erythrocytes are squeezed and deformed while exiting a fibrotic or infiltrated bone marrow space. Characteristic feature of Primary Myelofibrosis (JAK2 V617F mutation), bone marrow infiltration (myelophthitic anemia from metastatic cancer, granulomas, or leukemia), and severe thalassemia.

Dacryoadenitis

- Inflammation of the lacrimal gland, presenting as tender swelling in the superolateral aspect of the upper eyelid with characteristic S-shaped ptosis. It can be acute or chronic in presentation.

Dacryocystitis

- Inflammation of the lacrimal sac, most commonly caused by obstruction of the nasolacrimal duct leading to stagnation of tears and secondary bacterial infection (*Staphylococcus aureus*, *Streptococcus pneumoniae*, *Haemophilus influenzae*).

Dacryocystography

- Contrast imaging of lacrimal drainage system. Evaluates obstruction.

Dacryocystorhinostomy

- A surgical procedure creating a passage between the lacrimal sac and the nasal cavity to bypass an obstructed nasolacrimal duct.

Dacryostenosis

- Narrowing or obstruction of the lacrimal drainage passages, which can cause excessive tearing and recurrent infection of the lacrimal sac.

Dactylitis

- Inflammation of an entire digit (finger or toe), producing a diffuse, fusiform swelling often described as sausage-shaped.

Dactylosymphysis

- Abnormal fusion or webbing of adjacent digits, usually present from birth.

Daily Value

- Nutrition labeling reference values. Based on 2000 calorie diet. Useful for comparing products.

Daltonism

- Color-vision deficiency, especially difficulty dis-

tinguishing certain colors because of abnormal or absent retinal cone function.

DALY

- Disability-adjusted life year. Measure of disease burden.

Damage-Control Resuscitation

- A trauma strategy prioritizing rapid hemorrhage control, permissive approaches to blood pressure when appropriate, prevention of hypothermia, correction of coagulopathy, and balanced blood-product resuscitation rather than large-volume crystalloid administration.

Damage-Control Surgery

- A staged operative strategy for critically ill or severely injured patients in whom definitive repair is unsafe because of hypothermia, acidosis, coagulopathy, or physiologic instability. Initial surgery controls hemorrhage and contamination, followed by resuscitation and later definitive reconstruction.

Dandruff

- Scaling and shedding of the scalp skin, commonly associated with seborrheic dermatitis or dry skin.

Dandruff and Scalp Conditions

- Dandruff is a common scalp condition presenting as flaking, itching, and mild inflammation caused by overgrowth of the yeast *Malassezia*, dry skin, or seborrheic dermatitis. Other scalp conditions include psoriasis, contact dermatitis, ringworm tinea capitis, and head lice.

Dandy Fever

- Dengue fever, an acute mosquito-borne viral illness causing fever, headache, severe muscle or joint pain, and sometimes hemorrhage or shock.

Dandy-Walker Malformation

- A congenital hindbrain malformation characterized by hypoplasia or aplasia of the cerebellar vermis, cystic dilation of the fourth ventricle, and posterior fossa enlargement with upward displacement of the tentorium, transverse sinuses, and torcular herophili.

Dandy-Walker Syndrome

- See Dandy-Walker Malformation.

Dane Particle

- The complete infectious particle of hepatitis B

virus, consisting of the viral core and envelope.

Daptomycin resistance

- Resistance to daptomycin can develop through mutations in genes encoding cell membrane phospholipid synthesis. The mechanism involves changes in the bacterial cell membrane that reduce daptomycin binding. Resistance is more common in VRE than MRSA. Resistance can emerge during therapy, particularly in endocarditis and bacteremia.

Darier's Disease

- An inherited skin disorder characterized by abnormal keratinization, greasy crusted papules, and nail changes, often worsened by heat and sweating.

Dark Adaptation

- The process by which the eyes become increasingly sensitive to dim light after moving from bright illumination into darkness.

Dartos

- A layer of smooth muscle and connective tissue beneath the skin of the scrotum that wrinkles the scrotal skin in response to temperature and other stimuli.

Darwin's Tubercle

- A variable small thickening or projection on the rim of the external ear, regarded as a remnant of an ancestral mammalian ear form.

DASH Diet

- Dietary Approaches to Stop Hypertension. Emphasizes: fruits, vegetables, whole grains, low-fat dairy, reduced saturated fat, sodium <2300 mg/day. Lowers blood pressure effectively.

Daubenton's Plane

- An anatomical reference plane extending from the lower margin of the orbit to the upper margin of the external auditory opening, used in describing skull orientation.

Dawn Phenomenon

- Physiological increase in blood glucose occurring in the early morning hours between approximately three and eight AM driven by counter-regulatory hormone surges including growth hormone, cortisol, and catecholamines. More pronounced in patients with diabetes who lack the compensatory insulin response.

De Novo Variant

- A genetic variant present in an affected individual that was not detected in either biological parent, usually arising in a parental gamete or early embryogenesis. It may cause a dominant disorder and can have implications for recurrence risk in future pregnancies.

De Quervain Tendinitis

- Painful stenosing inflammation of the tendons of the abductor pollicis longus and extensor pollicis brevis in the first dorsal wrist compartment. It causes radial wrist pain worsened by thumb movement or grasping and is assessed clinically.

De Quervain Tenosynovitis

- De Quervain tenosynovitis: first dorsal compartment (APL, EPB) stenosis. Clinical: radial wrist pain, Finkelstein test (thumb in palm, ulnar deviation), swelling over radial styloid.

De-Escalation

- A set of verbal and nonverbal techniques used to reduce agitation and prevent violence while preserving dignity and safety. It includes calm communication, appropriate personal space, clear limits, environmental modification, and offering choices.

Dead Space

- The volume of ventilated air that does not participate in gas exchange. Anatomic dead space: the conducting airways (nose, pharynx, trachea, bronchi, terminal bronchioles) that warm, humidify, and filter inspired air but contain no alveoli for gas exchange. Approximately 150 mL in a 70 kg adult.

Dead Space Ventilation

- Ventilation of alveoli without perfusion ($V/Q = \text{infinity}$). Anatomic dead space is the conducting airways (150 mL). Physiologic dead space includes alveoli without perfusion. Increased in PE, hypotension, and emphysema. Measured by the Bohr equation.

Dead Time

- Period after each event during which system cannot process another event. Limits count rate. Corrected electronically.

Deafferentation

- Loss or interruption of sensory nerve input from a body region to the central nervous system, which

may produce abnormal sensation or deafferentation pain.

Deafness

- Complete or partial loss of the ability to hear, classified by type: conductive deafness (impaired sound transmission through the external or middle ear, causes include cerumen impaction, otitis media, otosclerosis, tympanic membrane perforation), sensorineural deafness (damage to the cochlea or auditory nerve, causes include presbycusis,

Death

- End of life. Cultural variations in attitudes.

Deauville Criteria

- 5-point scale for PET response in lymphoma.

Debridement

- The surgical removal of dead, damaged, or infected tissue from a wound to promote healing and prevent infection.

Debulking

- Surgical removal of as much tumor as possible when complete resection is not feasible, aiming to reduce tumor burden (cytoreduction) and improve response to subsequent therapy; central to ovarian cancer management.

Decapitation

- Separation of the head from the body, seen in severe trauma (vehicle accidents, train injuries, explosions), industrial accidents, hanging with judicial execution, or homicide. Complete decapitation is almost always rapidly fatal due to exsanguination and brain ischemia.

Decay Correction

- Mathematical adjustment for radioactive decay during imaging. Essential for quantitative comparison between studies.

Decerebrate Posture

- Abnormal extension of the arms and legs caused by severe dysfunction of the brainstem below the red nucleus. It is a sign of serious neurologic injury and may occur with traumatic brain injury, hemorrhage, or raised intracranial pressure.

Decerebrate Posturing

- Abnormal extension posturing indicating severe brain injury. Arms extended, legs extended, internal

rotation. Poor prognosis.

Decerebration

- Severe loss of cerebral function, often with extension and rigid posturing of the limbs from brainstem dysfunction; it may result from major injury or increased intracranial pressure.

Decibel

- A logarithmic unit (dB) used to measure the intensity of sound. The decibel scale compares a given sound pressure level to a reference level (0 dB = 20 micropascals, the threshold of normal hearing at 1 kHz). The scale is logarithmic: an increase of 10 dB represents a tenfold increase in sound intensity. Normal conversation: 60 dB.

Decidua

- The transformed endometrium of pregnancy, shed after delivery. Three regions: decidua basalis (under the placenta, site of trophoblast invasion), decidua capsularis (over the gestational sac, covering the embryo), and decidua parietalis (the remainder of the uterine lining).

Deciduitis

- Inflammation of the decidua, the specialized uterine lining of pregnancy.

Deciduosis

- The presence of extensive or ectopic decidual tissue, usually associated with pregnancy and sometimes involving the cervix, peritoneum, or other sites.

Decision-Making Capacity

- The clinical ability to understand relevant information, appreciate its consequences, reason about options, and communicate a stable choice regarding a specific decision. Capacity is decision-specific and may fluctuate; it is distinct from legal competence.

Decompensated Cirrhosis

- Cirrhosis complicated by clinically significant liver dysfunction or portal hypertension, commonly manifested by ascites, variceal hemorrhage, hepatic encephalopathy, or jaundice. It carries substantially higher mortality than compensated cirrhosis and warrants assessment for transplant and complication prevention.

Decomposition

- Postmortem breakdown of the body through autolysis and putrefaction. It begins within 24 to 72

hours at ambient temperature, producing green ceal discoloration, venous marbling, gas distension, skin slippage, and purge fluid.

Decompression sickness

- A systemic illness caused by nitrogen bubble formation in tissues during rapid decompression, presenting with joint pain, skin mottling, and neurological deficits, treated with hyperbaric oxygen therapy.

Deconditioning

- Physiological decline from reduced activity or immobilization. Effects: muscle atrophy, weakness, decreased cardiorespiratory fitness, balance impairment, increased fall risk. Preventable with early mobilization, exercise. Reversible with rehabilitation.

Decongestants

- Medications that reduce nasal congestion by constricting blood vessels in the nasal mucosa, decreasing swelling and mucus production. Systemic decongestants: pseudoephedrine and phenylephrine (oral), which stimulate alpha-adrenergic receptors causing vasoconstriction.

Decontamination

- Measures that reduce continued absorption of a toxic substance from the skin, eyes, gastrointestinal tract, or clothing. Interventions are exposure-specific and may include removal from the source, irrigation, washing, activated charcoal, or specialist procedures.

Decorticate Posture

- Abnormal flexion of the arms with extension of the legs, indicating severe damage affecting the cerebral hemispheres, internal capsule, or thalamus. It is a sign of major brain injury and impaired consciousness.

Decorticate Posturing

- Abnormal flexion posturing indicating brain injury. Arms flexed, legs extended. Better prognosis than decerebrate.

Decreased Breath Sounds

- Diminished air movement heard on auscultation, indicating decreased ventilation. Causes include pneumothorax, pleural effusion, atelectasis, and mucus plugging. Unilateral decreased breath sounds may indicate tension pneumothorax or large effu-

sion.

Decubitus Ulcer

- Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences. See pressure injury.

Deep

- Located further inward from the surface of the body.

Deep Tendon Reflex

- The monosynaptic stretch reflex elicited by tapping a tendon with a reflex hammer, testing the integrity of the reflex arc (sensory afferent, spinal cord, motor efferent). Common reflexes: biceps (C5-6), triceps (C7), brachioradialis (C5-6), knee/patellar (L3-4), ankle/Achilles (S1). Graded 0 (absent) to 4+ (hyperactive, clonus).

Deep Vein Thrombosis

- Formation of a blood clot in a deep vein, most commonly in the lower extremities. Risk factors include immobility, surgery, cancer, pregnancy, and thrombophilia. Presents with unilateral leg swelling, pain, and erythema. Diagnosis by venous ultrasound and D-dimer testing. Treatment: anticoagulation (heparin, DOACs, warfarin).

Deep Vein Thrombosis (DVT)

- Blood clot in deep veins, usually lower extremity. Risk factors: Virchow triad (stasis, hypercoagulability, endothelial injury). Symptoms: leg swelling, pain, warmth, redness. Diagnosis: D-dimer, compression ultrasound. Treatment: anticoagulation (heparin, DOACs), compression stockings, IVC filter if contraindication to anticoagulation.

Deep Vein Thrombosis Ultrasound

- Venous duplex ultrasound for DVT diagnosis. Complete compression ultrasound (CUS): venous segments are compressed; non-compressible = DVT (96% sensitivity for proximal DVT). Whole-leg ultrasound (proximal + distal): for symptomatic patients.

Deep Venous Reflux

- Retrograde flow through incompetent venous valves in the deep venous system. It contributes to chronic venous hypertension, edema, skin changes, and venous ulcer formation.

Deep venous thrombosis

- See Deep Vein Thrombosis.

Defecation

- The process of eliminating feces from the body. Filling of the rectum triggers stretch receptors that initiate the defecation reflex via the sacral spinal cord (S2-S4), causing contraction of the rectal wall and relaxation of the internal anal sphincter.

Defecography

- Dynamic imaging of the rectum and pelvic floor during defecation using fluoroscopy or MRI. The rectum is filled with contrast paste and the patient performs defecation while images are captured. Evaluates rectal prolapse, intussusception, rectocele, enterocele, and pelvic floor dyssynergia.

Defense Mechanisms

- Unconscious psychological strategies used to manage anxiety and protect the ego from distressing thoughts or feelings. Mature defenses include sublimation, humor, and altruism. Immature defenses include projection, denial, splitting, and acting out. Understanding defense mechanisms helps clinicians conceptualize personality

Defense Mechanisms (Forensic)

- Physical evidence patterns on a body demonstrating active resistance against an assailant prior to incapacitation.

Defense Wounds

- Injuries sustained by a victim attempting to ward off an attack, typically found on the flexor surfaces of hands, palms, wrists, and forearms (incised/stab wounds from knives or contusions/abrasions from blunt force).

Defense Wounds (Inscribed)

- See Defense Wounds. The distribution and morphology of defensive injuries may help reconstruct an assault but are not independently proof of a particular event.

Defibrillation

- Delivery of an unsynchronized electrical shock to terminate ventricular fibrillation or pulseless ventricular tachycardia by depolarizing a critical mass of myocardium. It is distinct from synchronized cardioversion, which times the shock to the QRS complex.

Defibrillator

- Device delivering electric shock to heart. Types: external (AED, manual), internal (implantable). Used for ventricular fibrillation, pulseless ventricular tachycardia. Energy: biphasic 120-200 J.

Defibrillator Implant

- See pacemaker/ICD.

Degenerative Spine Disease

- Lumbar spinal stenosis: central (neurogenic claudication: bilateral leg pain with walking, relieved by sitting/flexion), lateral recess, foraminal. Imaging: MRI (central canal AP diameter <10mm, cross-sectional area <70mm²).

Dehydration

- Deficit of total body water. Hypertonic (water loss > sodium), isotonic (loss proportional), hypotonic (sodium loss > water). Causes: vomiting, diarrhea, fever, inadequate intake, diabetes insipidus. Treatment: oral rehydration or IV fluids.

Dehydration in Children

- A deficit of total body water, usually caused by gastrointestinal losses, inadequate intake, fever, or renal losses. Clinical assessment includes mental status, heart rate, mucous membranes, tears, capillary refill, urine output, weight change, and perfusion rather than a single sign.

Delamanid

- A nitroimidazole derivative used in the treatment of multidrug-resistant tuberculosis. Delamanid inhibits mycolic acid synthesis in *Mycobacterium tuberculosis*, disrupting the cell wall. It is indicated for pulmonary multidrug-resistant tuberculosis in combination with other agents. The drug is administered orally twice daily.

Delavirdine

- A non-nucleoside reverse transcriptase inhibitor (NNRTI) that was previously used in combination antiretroviral therapy for HIV-1 infection but is now rarely used due to the availability of more effective and better-tolerated agents.

Delayed Cord Clamping

- The practice of delaying umbilical cord clamping for 30-60 seconds after delivery, allowing placental transfusion of blood to the newborn. Benefits: increases neonatal blood volume by 30%, improves iron stores

(ferritin, hemoglobin) for the first 6 months, reduces anemia, and improves transitional circulation.

Delayed Ejaculation

- A sexual dysfunction characterized by a persistent or recurrent delay in, infrequency of, or absence of ejaculation during partnered sexual activity, lasting 6 months and causing marked distress. Prevalence: 1-5% of men. Lifelong vs. acquired.

Delayed Emergence

- Prolonged time to regain consciousness after discontinuation of anesthetic agents. Failure to regain consciousness within 30-60 minutes of completing surgery.

Delayed Puberty

- Failure to develop secondary sexual characteristics by age thirteen in girls and age fourteen in boys. Causes include constitutional delay which is the most common, chronic illness, malnutrition, hypogonadotropic hypogonadism from Kallmann syndrome or pituitary disease, and hypergonadotropic hypogonadism from gonadal failure.

Deletion

- Loss of chromosomal segment. Terminal deletion: loss from end. Interstitial deletion: loss from middle. Microdeletion syndromes: 22q11.2 deletion (DiGeorge), 5p deletion (Cri-du-chat).

Delirium

- Acute onset fluctuating disturbance in attention and awareness. Usually reversible. Precipitants: infection, medications, metabolic derangement, pain, constipation, urinary retention. Differentiate from dementia (chronic, progressive). Treatment: identify and treat underlying cause, environmental modification, avoid anticholinergics.

Delirium in Anesthesia

- Delirium is an acute confusional state characterized by fluctuating consciousness, inattention, disorganized thinking, and altered level of awareness. Postoperative delirium occurs in 15 to 50 percent of elderly surgical patients, with higher rates in hip fracture and cardiac surgery.

Delirium Tremens

- The most severe form of alcohol withdrawal syndrome, a medical emergency with mortality of 5-15% if untreated. Typically occurs 48-96 hours after

the last drink in individuals with chronic heavy alcohol use.

Delivery

- The process of childbirth: vaginal delivery (spontaneous vertex most common; assisted—vacuum, forceps) or cesarean delivery (abdominal). Stages of labor culminate in delivery of the infant and placenta.

Delivery of Posterior Arm

- A shoulder dystocia maneuver: the operator's hand enters the posterior vagina along the fetal chest, finds the fetal hand, grasps the forearm, sweeps it across the chest and face, and delivers the posterior arm and hand over the perineum. This reduces the bisacromial diameter by 20% and is often highly effective.

Deltoid Muscle

- Large triangular muscle forming the shoulder contour, from lateral clavicle, acromion, and scapular spine to the deltoid tuberosity. Innervated by the axillary nerve (C5-C6). Anterior fibers flex/medially rotate, middle fibers abduct (primary abductor beyond 15 degrees), posterior fibers extend/laterally rotate.

Delusion

- A fixed belief that is not consistent with available evidence or the person's sociocultural context and is not readily changed by reasoned discussion. Delusions may be persecutory, grandiose, referential, somatic, nihilistic, or erotomanic.

Delusional Disorder

- A disorder characterized by the presence of one or more delusions lasting one month or more in the context of relatively preserved functioning. The delusions are non-bizarre (involving real-life situations that could conceivably occur, such as being followed, infected, loved from afar, or conspired against).

Dementia (Major Neurocognitive Disorder)

- Dementia is the common clinical term for an acquired syndrome of decline in one or more cognitive domains that interferes with independence in daily activities. Major neurocognitive disorder is the preferred diagnostic classification term.

Demyelinating Disease

- A disease in which the myelin sheath surrounding nerve fibres is damaged, slowing or blocking neural conduction.

Dendrite

- The branched, tapering neuronal processes that receive synaptic input from other neurons and conduct graded potentials toward the cell body. Dendrites contain receptors, ion channels, and postsynaptic densities.

Dendritic Cell

- The most potent professional antigen-presenting cell, capturing antigens in tissues and migrating to lymph nodes to present peptides on MHC molecules to T cells, initiating adaptive immunity.

Dendritic Ulcer

- A characteristic branching (dendritic) epithelial ulcer of the cornea caused by herpes simplex virus (HSV) infection. It is the most common infectious cause of corneal blindness in developed countries. Presents with unilateral red eye, photophobia, tearing, foreign body sensation, and decreased vision.

Dengue

- See Dengue Fever.

Dengue fever

- An acute mosquito-borne viral illness caused by dengue virus (DENV-1, -2, -3, -4), a flavivirus transmitted primarily by *Aedes aegypti* mosquitoes.

Dengue virus

- A positive-sense RNA virus belonging to the Flaviviridae family that causes dengue fever. The virus has four serotypes, and infection with one serotype provides lifelong immunity to that serotype but only temporary cross-protection against others.

Denial

- A defense mechanism in which a person refuses to accept an unpleasant reality. It is a common initial response to traumatic events and terminal illness diagnoses. While adaptive in the short term, chronic denial can prevent appropriate medical treatment and psychological processing.

Denosumab for Osteoporosis

- Fully human monoclonal antibody targeting RANK ligand preventing osteoclast formation activation

and survival. Reduces fracture risk at the spine hip and non-vertebral sites. Administered as subcutaneous injection every six months. Advantages include no renal dose adjustment and effectiveness even with oral bisphosphonate intolerance.

Dens

- See Odontoid Process.

Dens Evaginatus

- Extra cusp or tubercle on tooth. Premolar most common. May contain pulp, fracture risk.

Dens Invaginatus

- Enamel-dentin invagination into pulp. Retains debris, high caries risk. Often mandibular lateral incisor.

Dental Abscess

- A localized collection of pus arising from infection of a tooth or the surrounding periodontal tissues. It may cause toothache, percussion tenderness, gingival swelling, facial swelling, fever, trismus, or regional lymphadenopathy; spread into deep facial spaces can threaten the airway and requires urgent assessment.

Dental Age

- Dental development stage used for forensic age estimation in children. Teeth develop at predictable times.

Dental Anesthesia

- Techniques for pain control in dentistry. Local anesthesia (lidocaine, articaine), nitrous oxide, conscious sedation, general anesthesia.

Dental Cavities

- Permanent areas of tooth decay caused by acid-producing bacteria acting on dietary carbohydrates. Cavities progress from enamel demineralization to dentin and pulp involvement, causing sensitivity, pain, or infection. Also called dental caries.

Dental Ceramic

- Firing ceramic material for crowns and veneers. Feldspathic, leucite-reinforced, lithium disilicate, zirconia.

Dental Extraction

- Removal of a tooth from its alveolar socket. Indications include unrestorable disease, advanced periodontal destruction, impaction, orthodontic treat-

ment, or selected prosthodontic needs; risks include bleeding, infection, nerve injury, fracture, and alveolar osteitis.

Dental Implant

- Titanium fixture placed in jawbone for tooth replacement. Osseointegration: bone fusion around implant. Success rate: >95% at 10 years.

Dental Plaque

- Biofilm of bacteria on tooth surfaces. Initiated by acquired pellicle. Maturation leads to disease if not removed. Disrupted by mechanical cleaning.

Dental Pulp Necrosis

- Death of the dental pulp, usually following untreated inflammation, trauma, or loss of blood supply. The tooth generally fails sensibility testing and may develop apical periodontitis or a periapical abscess.

Dental Pulpitis

- Inflammation of the dental pulp caused by caries, trauma, restorative procedures, cracks, or other irritation. Reversible pulpitis produces brief provoked pain, whereas irreversible pulpitis causes spontaneous or lingering pain and may require pulpal treatment or extraction.

Dental Radiographs

- Imaging for dental diagnosis. Intraoral: periapical (tooth root and surrounding bone), bitewing (interproximal caries detection, crestal bone level), and occlusal (larger area, localization).

Dental Radiography

- Imaging techniques for dental diagnosis: bitewing (caries, bone levels), periapical (root, periapical tissues), panoramic (full mouth overview), CBCT (3D imaging). Radiation dose: dental radiography low exposure. ALARA principle applies.

Dental Restorative Materials

- Materials used to restore teeth: amalgam, composite resin, glass ionomer, ceramic, gold alloy. Selection based on location, aesthetics, strength, cost, patient factors. Composite: tooth-colored, adhesive, conservative preparation. Ceramic: excellent aesthetics, brittle, expensive. Gold: durable, biocompatible, aesthetic concern.

Dental Screening

- See dentistry.

Dental Sealant

- Thin protective coating applied to chewing surfaces of molars. Prevents caries by sealing pits and fissures.

Dental Traumatology

- Management of dental injuries. Types: enamel fractures, enamel-dentin fractures, pulp exposure, root fractures, avulsion. Immediate care critical for avulsed teeth.

Dentin

- Calcified tissue beneath enamel forming bulk of tooth. Contains dentinal tubules with fluid and processes. Sensitive to stimuli when exposed. Reparative dentin formed in response to irritation.

Dentinogenesis Imperfecta

- Hereditary dentin defect. Teeth appear opalescent, blue-gray or amber. Weak teeth, wear rapidly. Associated with osteogenesis imperfecta.

Dentistry (Geriatric)

- Oral health challenges in elderly: xerostomia, root caries, periodontal disease, denture problems, oral cancer, medication side effects. Assessment: complete oral exam, cancer screening, denture fit, caries risk. Prevention: fluoride, oral hygiene, regular dental visits. Adaptations: handholds, magnification, positioning.

Dentition

- See dentistry.

Denture

- A removable dental prosthesis that replaces some or all missing teeth and associated oral tissues. Partial dentures replace selected teeth while complete dentures replace an entire dental arch; fit, oral hygiene, tissue health, and regular review are important for function and prevention of mucosal injury.

Dependence

- A chronic, relapsing brain disorder characterized by compulsive substance use or behavior despite harmful consequences. See addiction.

Dependent Personality Disorder

- A pattern of submissive and clinging behavior related to an excessive need to be taken care of.

Depersonalization-Derealization Disorder

- Persistent or recurrent experiences of depersonal-

ization, derealization, or both. Depersonalization includes experiences of unreality, detachment, or being an outside observer with respect to one's thoughts, feelings, sensations, body, or actions. Derealization includes experiences of unreality or detachment with respect to

Depolarization

- The change in membrane potential toward a less negative (more positive) value, typically caused by inward sodium or calcium current. In neurons, depolarization to threshold triggers an action potential.

Depot Implant

- Long-acting drug delivery implant. Hormonal contraceptives, antipsychotics. Subdermal placement.

Deprescribing

- A planned, supervised process of reducing the dose or stopping a medication for which harms, burden, or lack of benefit outweigh expected benefit. It requires review of indication, withdrawal risk, disease recurrence, and patient goals.

Depression

- Major depressive disorder is a common, disabling mood disorder characterized by persistent sadness and loss of interest or pleasure in activities, affecting approximately 7 percent of adults worldwide annually.

Depression (Geriatric)

- Mood disorder common in elderly, often underdiagnosed. Symptoms: sad mood, anhedonia, sleep/appetite changes, fatigue, guilt, concentration difficulty, suicidal ideation. May present as cognitive impairment ("pseudodementia"). Risk factors: chronic illness, social isolation, bereavement, medications.

Depression Screening

- Major depressive disorder is a common, disabling mood disorder characterized by persistent sadness and loss of interest or pleasure in activities, affecting approximately 7 percent of adults worldwide annually. See depression.

Dermal Structure

- The dermis has two layers: papillary dermis (superficial, thin, containing capillaries, lymphatics, and Meissner corpuscles) and reticular dermis (deep, thick, containing dense irregular connective tissue,

collagen, elastin, hair follicles, sweat glands, and sebaceous glands).

Dermatitis

- Inflammation of the skin, often used interchangeably with eczema. Atopic dermatitis: chronic pruritic inflammation in flexural areas, associated with asthma and allergies, driven by skin barrier dysfunction and type 2 inflammation. Contact dermatitis: allergic (type IV hypersensitivity) or irritant (direct toxicity).

Dermatitis Herpetiformis

- A chronic, intensely pruritic, autoimmune blistering skin condition characterized by grouped, symmetric erythematous papules, urticarial plaques, and vesicles occurring primarily on extensor surfaces (elbows, knees, buttocks, scalp). It is a cutaneous manifestation of gluten-sensitive enteropathy (celiac disease).

Dermatofibroma

- A common benign dermal nodule, typically on the lower extremities of young adults. It presents as a firm, hyperpigmented papule that dimples when pinched (Fitzpatrick sign). It is thought to result from reactive fibrosis following minor trauma or insect bites. Dermatofibromas are asymptomatic and require no treatment.

Dermatofibrosarcoma

- Dermatofibrosarcoma protuberans is a locally aggressive fibrohistiocytic tumor of the skin. It typically presents as a slowly enlarging, indurated plaque on the trunk, which may develop nodules over time. It has a high local recurrence rate (up to 50 percent with simple excision) but low metastatic potential.

Dermatofibrosarcoma Protuberans

- A rare locally aggressive cutaneous sarcoma arising from dermal fibroblasts. It presents as a slow-growing, firm, reddish-purple plaque or nodule, typically on the trunk. It has a high local recurrence rate but low metastatic potential. Treatment is wide local excision with Mohs micrographic surgery or imatinib for unresectable disease.

Dermatographism

- Urticaria factitia: the development of linear wheals (hives) within minutes of stroking or scratching the

skin, the most common form of physical urticaria. Mast cell hyper-reactivity releases histamine after minor mechanical pressure. Associated with chronic spontaneous urticaria and atopy.

Dermatomyositis

- An inflammatory myopathy with characteristic skin manifestations: Gottron papules (over knuckles), heliotrope rash (purple periorbital), shawl/V sign erythema, and proximal muscle weakness; associated with interstitial lung disease, and in adults an increased risk of malignancy (ovary, lung, GI).

Dermatopathology

- The microscopic examination of skin specimens. Histopathologic features include acanthosis (epidermal thickening), spongiosis (intercellular edema), parakeratosis (retained nuclei in stratum corneum), hyperkeratosis (thickened stratum corneum), dyskeratosis (individual cell keratinization), and Civatte bodies (apoptotic

Dermatophytosis

- A superficial infection of keratinized skin, hair, or nails caused by dermatophyte fungi, producing conditions such as tinea corporis, tinea pedis, and onychomycosis.

Dermis

- Thick connective tissue layer beneath the epidermis providing structural support, elasticity, and nourishment. Two layers: papillary (superficial, loose connective tissue with dermal papillae interdigitating with epidermal ridges) and reticular (deep, dense irregular connective tissue).

Dermoid

- A benign cystic teratoma containing elements derived from ectoderm, including skin, hair follicles, sebaceous glands, and sometimes teeth. Ovarian dermoid cysts are the most common germ cell tumor of the ovary, typically occurring in young women. They are usually asymptomatic but may cause torsion, rupture, or malignant transformation.

Dermoid Cyst

- See Dermoid.

Dermoscopy

- Noninvasive examination of skin lesions with magnification and polarized or nonpolarized illumination that reveals structures not visible to the unaided

eye. It improves assessment of pigmented lesions and selected inflammatory or infectious conditions but does not replace biopsy when malignancy is suspected.

Descending Colon

- From the splenic flexure to the left iliac fossa, approximately 25-30 cm, and mostly retroperitoneal and fixed to the posterior abdominal wall. The phrenicocolic ligament supports the splenic flexure. Anteriorly it is related to small intestine; posteriorly to the left kidney. It absorbs water and electrolytes and compacts fecal matter.

Descending Thoracic Aorta

- From aortic arch at T4 to diaphragm at T12. Gives off bronchial, esophageal, mediastinal, and posterior intercostal arteries. In the posterior mediastinum, initially left of the vertebral column. Related to esophagus, thoracic duct, and azygos vein. Aneurysms are mostly atherosclerotic, treatable with thoracic endovascular aortic

Desmoid Tumor

- A locally aggressive but non-metastasizing fibrous tumor (fibromatosis), associated with familial adenomatous polyposis. Treatment includes surgery, radiation, and systemic therapy; recurrence after resection is common.

Desquamative Interstitial Pneumonia

- A rare ILD characterized by uniform filling of alveoli with macrophages containing brown pigment (hemosiderin-laden). It is strongly associated with smoking. HRCT shows bilateral ground-glass opacities with lower lobe predominance. Smoking cessation and corticosteroids are the main treatments.

Detached Retina

- Separation of the neurosensory retina from the underlying retinal pigment epithelium (RPE), a medical emergency requiring prompt surgical repair to prevent permanent vision loss.

Deterministic Effects

- Radiation effects with threshold dose. Cataracts, skin erythema, bone marrow suppression. Severity increases with dose.

Detoxification

- Removal of toxic substance from body. Methods: activated charcoal, gastric lavage (rarely used),

whole bowel irrigation, hemodialysis, hemoperfusion, chelation, supportive care.

Developmental Dysplasia of the Hip

- A spectrum of hip abnormalities in infants and children, ranging from mild acetabular dysplasia to complete hip dislocation. The femoral head is partially or completely displaced from the acetabulum due to capsular laxity and acetabular underdevelopment.

Developmental Milestones

- Skills and abilities that children typically acquire at predictable ages. Examples: social smile at 2 months, rolling over at 4-6 months, sitting independently at 6-8 months, walking at 12 months, and speaking two-word phrases at 2 years. Delays in reaching milestones may indicate developmental delay, autism, or neurologic impairment.

Developmental Regression

- Loss of previously acquired developmental skills. Regression is a neurologic red flag requiring prompt evaluation for genetic, metabolic, neurodegenerative, epileptic, infectious, or structural causes.

Developmental Surveillance

- The ongoing process of eliciting caregiver concerns, observing the child, documenting developmental history, and monitoring milestones during routine care. It differs from formal developmental screening, which uses standardized instruments at specified ages.

Deviated Septum

- Displacement of the nasal septum from the midline, congenital or acquired from trauma, and present to some degree in most adults. Significant deviation narrows one nasal cavity, impairing airflow and sinus drainage and producing obstruction, congestion, recurrent sinusitis, epistaxis, and sleep disturbance.

Devic Disease

- See Neuromyelitis Optica.

DEXA (Dual-Energy X-ray Absorptiometry)

- Measures bone mineral density (BMD) at spine and hip. T-score compares to young adult reference. Osteoporosis: T-score $\leq$ -2.5. Osteopenia: -1.0 to -2.5. Low radiation dose.

Dexamethasone Suppression Test

- A dynamic test used to assess cortisol regulation. In normal physiology, dexamethasone suppresses ACTH and cortisol; failure of suppression supports endogenous hypercortisolism but requires interpretation with medication, absorption, and comorbidity considerations.

Dexmedetomidine

- A relatively selective alpha-2 adrenergic agonist used for sedation and as an adjunct to analgesia. It can produce cooperative sedation with limited respiratory depression but may cause bradycardia and hypotension.

Dextroamphetamine

- A central nervous system stimulant (d-amphetamine isomer) increasing dopamine and norepinephrine. Used for ADHD and narcolepsy. Available as immediate and extended-release, and in mixed amphetamine salts (Adderall) and lisdexamfetamine (prodrug).

Dextrocardia

- A congenital cardiac positional anomaly where the heart is located in the right hemithorax with the apex pointing to the right.

Diabetes

- A group of disorders characterized by persistently elevated blood glucose caused by inadequate insulin secretion, impaired insulin action, or both. Chronic hyperglycemia can damage the eyes, kidneys, nerves, heart, and blood vessels. Major forms include type 1, type 2, and gestational diabetes.

Diabetes and Metabolic Health

- Closely linked through insulin resistance, obesity, dyslipidemia, and hypertension, collectively known as the metabolic syndrome. Poor metabolic health increases the risk of developing type 2 diabetes and worsens glycemic control in existing diabetes.

Diabetes Complications

- Diabetes complications arise from chronic hyperglycemia damaging blood vessels and nerves over time. Microvascular complications include diabetic retinopathy leading to vision loss, nephropathy progressing to kidney failure, and neuropathy causing pain and loss of sensation.

Diabetes insipidus

- A disorder characterized by polyuria (excessive urine output exceeding 40 mL/kg/24 hours) and polydipsia due to deficient secretion of antidiuretic hormone (ADH, vasopressin) from the posterior pituitary (central diabetes insipidus) or renal resistance to ADH action (nephrogenic diabetes insipidus).

Diabetes Management

- Tight glucose control reduces complications. Foot examination essential.

Diabetes Medications

- Diabetes medications lower blood glucose through various mechanisms: metformin reduces hepatic glucose production; sulfonylureas stimulate insulin secretion; SGLT2 inhibitors increase urinary glucose excretion; GLP-1 receptor agonists enhance incretin effects; and insulin replaces deficient hormone.

Diabetes Mellitus

- A metabolic disorder characterized by chronic hyperglycemia resulting from insufficient insulin production, insulin resistance, or both. Type 1 diabetes results from autoimmune destruction of pancreatic beta cells, while type 2 is driven by genetic and lifestyle factors leading to insulin resistance.

Diabetes Mellitus Type 1

- An autoimmune disease characterized by destruction of pancreatic beta cells leading to absolute insulin deficiency. Usually presents in childhood or young adulthood with acute onset of polyuria, polydipsia, weight loss, and diabetic ketoacidosis (DKA) at diagnosis in many.

Diabetes Mellitus Type 2

- A progressive metabolic disorder characterized by insulin resistance and relative insulin deficiency, strongly associated with obesity and physical inactivity. Typically presents in adults but increasingly seen in younger populations. Diagnosis: fasting glucose 126 mg/dL, HbA1c 6.5%, or OGTT 200 mg/dL.

Diabetes Prevention

- Preventing diabetes focuses on reducing the risk of developing type 2 diabetes through lifestyle interventions and risk factor management. Maintaining a healthy body weight, engaging in regular physical

activity, and following a balanced diet low in refined sugars and saturated fats are cornerstone strategies.

Diabetic Autonomic Neuropathy

- Dysfunction of autonomic nerves caused by diabetes, affecting cardiovascular, gastrointestinal, genitourinary, sudomotor, and pupillary systems. Manifestations include orthostatic hypotension, resting tachycardia, gastroparesis, bladder dysfunction, erectile dysfunction, and impaired hypoglycemia awareness.

Diabetic Complications

- The chronic and acute consequences of diabetes mellitus. Microvascular: retinopathy (leading cause of blindness), nephropathy (leading cause of ESRD—albuminuria, declining GFR), neuropathy (peripheral—stocking-glove, autonomic—gastroparesis, erectile dysfunction, postural hypotension, cardiac autonomic).

Diabetic Diet

- Carbohydrate counting, glycemic index consideration, balanced meals. Medical nutrition therapy improves glycemic control.

Diabetic Foot

- See endocrinology.

Diabetic Foot Disease

- Complex multifactorial condition resulting from the combination of peripheral neuropathy, peripheral arterial disease, and impaired immune function in diabetes. Neuropathy leads to loss of protective sensation and foot deformities. Peripheral arterial disease impairs healing. Minor trauma progresses to ulceration and infection.

Diabetic Foot Ulcer

- A chronic wound of the foot in a patient with diabetes, resulting from peripheral neuropathy, peripheral artery disease, and impaired wound healing. Neuropathic ulcers: painless, at pressure points (plantar metatarsal heads, heel), punched-out appearance, surrounded by callus.

Diabetic Hyperosmolar Hyperglycemic State (HHS)

- Severe hyperglycemia (>600 mg/dL), hyperosmolality (>320 mOsm/kg), dehydration without significant ketoacidosis. Type 2 diabetes. Mortality: 10-20%. Treatment: aggressive fluid resuscitation,

insulin, electrolyte replacement.

Diabetic Ketoacidosis

- An acute, life-threatening complication of diabetes mellitus, more common in type 1. Characterized by hyperglycemia (>250 mg/dL), metabolic acidosis ($\text{pH} < 7.3$, bicarbonate < 15 mEq/L), and ketonemia. Precipitating factors: infection, missed insulin doses, MI, pancreatitis, and stress.

Diabetic Ketoacidosis (DKA)

- Life-threatening complication of diabetes mellitus (usually type 1) characterized by hyperglycemia (>250 mg/dL), ketonemia, and metabolic acidosis ($\text{pH} < 7.3$, bicarbonate < 18 mEq/L). Precipitated by infection, missed insulin, new-onset diabetes.

Diabetic Ketoacidosis Monitoring

- Monitoring during DKA management includes hourly blood glucose measurement, serum electrolytes every two to four hours, venous blood gas assessment, beta-hydroxybutyrate levels, and fluid balance.

Diabetic Macular Edema

- The accumulation of fluid within the macula due to breakdown of the blood-retinal barrier in diabetic retinopathy, resulting in thickening of the macula and central vision loss, and it can occur at any stage of diabetic retinopathy, from mild non-proliferative to proliferative disease.

Diabetic Nephropathy

- Kidney disease caused by diabetes mellitus, the leading cause of ESRD worldwide. Pathophysiology: hyperglycemia-induced glomerular hyperfiltration, basement membrane thickening, mesangial expansion, and glomerulosclerosis. Presents with progressive albuminuria (microalbuminuria macroalbuminuria) and declining GFR.

Diabetic Neuropathy

- Most common complication of diabetes affecting approximately fifty percent of patients. Distal symmetric polyneuropathy presents with stocking-glove sensory loss, burning pain, and loss of proprioception. Autonomic neuropathy causes gastroparesis, orthostatic hypotension, erectile dysfunction, and sudomotor dysfunction.

Diabetic Peripheral Neuropathy

- The most common complication of diabetes melli-

tus, affecting approximately fifty percent of patients with long-standing disease. It is a length-dependent, sensorimotor polyneuropathy that affects the distal lower extremities first, in a glove-and-stocking distribution, and progresses proximally over time.

Diabetic Retinopathy

- A microvascular complication of diabetes mellitus and a leading cause of blindness in working-age adults. Non-proliferative DR (NPDR): microaneurysms, dot-blot hemorrhages, exudates (hard and cotton-wool spots), and venous beading. Proliferative DR (PDR): retinal neovascularization, vitreous hemorrhage, and tractional retinal detachment.

Diagnosis

- Identification of disease. Based on history, examination, tests. Differential diagnosis considers multiple possibilities.

Diagnostic microbiology

- The laboratory identification of microorganisms causing disease. Methods include microscopy, culture, biochemical testing, serology, and molecular methods. Rapid diagnostic tests provide same-day results. MALDI-TOF mass spectrometry revolutionized bacterial identification.

Diagnostic Radiology

- Medical term; definition not provided in source.

Diagnostic Test

- A test used to evaluate a person with symptoms, signs, or an abnormal screening result to establish, exclude, or characterize a disease. Its usefulness depends on analytic validity, clinical performance, and the consequences of the result.

Diagnostic Uncertainty

- The state in which available information does not establish a single diagnosis with sufficient confidence. It should be communicated transparently and managed with safety-netting, planned reassessment, and targeted additional evaluation.

Dialectical Behavior Therapy

- A comprehensive cognitive-behavioral treatment originally developed for borderline personality disorder. It combines cognitive-behavioral techniques with mindfulness and acceptance strategies. Core modules include mindfulness, distress tolerance, emotion regulation, and interpersonal effectiveness.

Dialysis

- See nephrology.

Dialysis Access

- Vascular access for hemodialysis. Types: arteriovenous fistula (AVF, gold standard), arteriovenous graft (AVG), central venous catheter. AVF: autogenous, durable, lower complication rate. AVG: prosthetic, used when veins unsuitable. Complications: stenosis, thrombosis, infection, steal syndrome.

Dialysis Adequacy

- The degree to which a dialysis prescription removes uremic solutes and controls fluid, electrolyte, and acid-base balance while meeting patient-centered treatment goals. Measures such as Kt/V and urea reduction ratio are used for hemodialysis, but clinical status remains essential.

Dialysis Disequilibrium Syndrome

- Neurological complications during hemodialysis in severely uremic patients. Headache, nausea, confusion, seizures, and coma. Results from rapid solute removal causing cerebral edema. Prevent with gradual dialysis initiation and shorter initial sessions.

Diaper Rash

- Irritant inflammation of the skin covered by a diaper, commonly caused by prolonged contact with urine or stool, friction, moisture, or *Candida* infection. It causes redness and soreness; persistent, severe, or spreading rash requires evaluation for infection or another skin condition.

Diaphragm

- Dome-shaped musculotendinous partition separating thoracic/abdominal cavities, primary inspiration muscle. Central tendon with muscular periphery from xiphoid, lower six costal cartilages, and lumbar vertebrae via crura and arcuate ligaments. Innervated by phrenic nerves (C3-C5). Contraction flattens the dome increasing thoracic volume.

Diaphragmatic Hernia

- Protrusion of abdominal contents through a defect or opening in the diaphragm into the thoracic cavity. It may be congenital or acquired after trauma and can cause respiratory compromise, bowel obstruction, or strangulation; diagnosis is based on imaging and management depends on symptoms, contents, and the risk of organ compromise.

Diaphragmatic rupture

- A tear in the diaphragm most commonly caused by blunt or penetrating thoracoabdominal trauma, with the left side affected in approximately 75 per cent of cases due to the protective effect of the liver on the right. Herniation of abdominal viscera into the thorax may cause respiratory distress, bowel obstruction, or strangulation.

Diaphysis

- The shaft of a long bone, comprising thick cortical (compact) bone surrounding the medullary cavity, which contains bone marrow in children and is largely yellow (fatty) marrow in adults. Enclosed by periosteum; the nutrient artery enters here. Fractures of the diaphysis heal by callus formation (enchondral and intramembranous).

Diarrhea

- Frequent passage of loose, watery stools, typically defined as more than 3 bowel movements per day with stool weight exceeding 200 grams. Acute diarrhea (less than 14 days) is most commonly infectious in etiology: viral (norovirus, rotavirus, adenovirus), bacterial (*Campylobacter*, *Salmonella*, *Shigella*, *E. coli*, *C.*

Diarrhea, Constipation, Gas, and Bloating

- These common gastrointestinal symptoms arise from disruptions in normal digestive function caused by diet, infections, medications, or underlying conditions like irritable bowel syndrome. Diarrhea involves increased stool frequency and water content, while constipation involves infrequent or difficult passage of stool.

Diastasis Recti

- Separation and thinning of the linea alba between the two rectus abdominis muscles, commonly occurring during or after pregnancy. It produces a midline abdominal bulge that becomes prominent with raised intra-abdominal pressure.

Diastole

- The phase of the cardiac cycle when the heart muscle relaxes, allowing the chambers to fill with blood.

Diastolic Dysfunction

- Impaired relaxation, compliance, or filling of the

left or right ventricle during diastole. It can raise filling pressures and cause exertional dyspnoea, pulmonary congestion, or heart failure with preserved ejection fraction; echocardiography assesses relaxation, chamber structure, and estimated filling pressure.

Diastolic Murmurs

- Murmurs occurring between S2 and the next S1. Early diastolic decrescendo murmurs indicate semilunar valve regurgitation (aortic regurgitation, pulmonary regurgitation). Mid-diastolic low-pitched rumbles indicate AV valve stenosis (mitral stenosis, tricuspid stenosis). Late diastolic murmurs may indicate mitral stenosis with sinus rhythm.

Diathermy

- The use of high-frequency electrical current to generate heat for surgical purposes, including cutting tissue, coagulating blood vessels to stop bleeding (hemostasis), and destroying abnormal tissue (electrocautery, electrocoagulation).

Diatom Test

- Forensic technique used to support the diagnosis of drowning; microscopic silica-shelled diatoms present in drowning water enter lungs, penetrate alveolar capillaries, and circulate to distant closed organs (femur bone marrow, brain) only if active circulation was present at time of submersion.

Dichorionic Twins

- Twins each having their own chorion and amnion, occurring in approximately 30% of monozygotic twins and all dizygotic twins. Each twin has its own placenta, though placentas may be adjacent. Complications include growth restriction, preterm delivery, and preeclampsia, but not twin-to-twin transfusion syndrome.

DICOM

- Digital Imaging and Communications in Medicine. Standard for medical image exchange.

Didanosine

- A nucleoside reverse transcriptase inhibitor (NRTI) used in the treatment of HIV-1 infection, typically as part of combination antiretroviral therapy. Didanosine is an adenosine analogue that inhibits viral reverse transcriptase, leading to chain termination of viral DNA synthesis.

Dietary Assessment

- Methods: 24-hour recall, food frequency questionnaire, food diary. Evaluation of intake adequacy, patterns, preferences.

Dietary Fat

- A macronutrient composed mainly of triglycerides and fatty acids that provides approximately 9 kcal per gram. Fat supports cell membranes, energy storage, absorption of vitamins A, D, E, and K, and synthesis of signaling molecules; the type of fat affects cardiometabolic risk.

Dietary Reference Intakes

- Reference values for nutrient intake: RDA (Recommended Dietary Allowance), AI (Adequate Intake), UL (Tolerable Upper Intake Level), EAR (Estimated Average Requirement).

Dieulafoy Lesion

- Large caliber persistent submucosal artery in the stomach wall that can cause massive gastrointestinal bleeding. The artery is abnormally large and superficial, lacking the normal tapering into smaller arterioles. Presents with hematemesis or melena without a visible ulcer.

Differential Diagnosis

- The process of distinguishing a particular disease or condition from others that present with similar symptoms. Systematic approach: consider life-threatening conditions first, use pretest probability, apply diagnostic criteria, and order targeted tests.

Differentiation

- Degree of cellular differentiation. See grade.

Difficult Airway

- A clinical situation in which a trained anesthesiologist or emergency physician encounters difficulty with bag-mask ventilation, supraglottic airway placement, laryngoscopy, or endotracheal intubation. Difficult mask ventilation: inability to maintain SpO₂ >92% using 100% oxygen and positive pressure with a mask.

Difficult Intubation

- Inability to perform direct laryngoscopy or endotracheal intubation despite appropriate expertise and equipment. Predictors include Mallampati Class III or IV, limited mouth opening (less than 3 cm), limited neck mobility, short thyromental distance

(less than 6.5 cm), obesity, beard, facial trauma, and prior difficult intubation.

Diffuse Axonal Injury

- Shearing of axons from acceleration-deceleration trauma. Causes immediate coma. Poor prognosis. Diagnosis: MRI shows punctate hemorrhages at gray-white junction, brainstem.

Diffuse Esophageal Spasm

- A motility disorder characterised by premature, uncoordinated esophageal contractions, causing intermittent chest pain and dysphagia for liquids and solids. Endoscopy excludes structural disease; high-resolution manometry confirms premature contractions.

Diffusing Capacity (DLCO)

- Ability of lungs to transfer gas from alveoli to blood. Reduced in emphysema, ILD, and anemia. Increased in pulmonary hemorrhage and left-to-right shunts.

Diffusing Capacity of the Lung

- The capacity of the respiratory membrane to transfer gas from alveolar air into pulmonary capillary blood. It depends on surface area, membrane thickness, gas properties, hemoglobin, and pulmonary capillary blood volume and is commonly measured with carbon monoxide as DLCO.

Diffusion

- The net movement of molecules from an area of higher concentration to lower concentration due to random thermal motion (Brownian motion).

Diffusion Restriction

- Reduced movement of water molecules detected by diffusion-sensitive MRI sequences, usually appearing bright on DWI with corresponding low ADC. It may indicate acute ischemia but can also occur with abscess, hypercellular tumor, seizure, and other processes.

Diffusion-Weighted Imaging (DWI)

- MRI technique measuring water molecule diffusion. Restricted diffusion bright. Used for stroke, abscess, tumor characterization.

DiGeorge Syndrome

- 22q11.2 deletion syndrome (most common microdeletion syndrome, 1:4000).

Digestion

- The process by which the body breaks down ingested food into smaller molecules that can be absorbed and utilized. Digestion begins in the mouth (mechanical–mastication; enzymatic–salivary amylase for starch, lingual lipase).

Digestive Health

- The efficient functioning of the gastrointestinal tract in breaking down food, absorbing nutrients, and eliminating waste. Digestive health depends on factors such as diet, hydration, gut microbiome balance, and the integrity of the digestive mucosa.

Digital Clubbing

- Bulbous enlargement of distal phalanges with loss of nail-bed angle (> 180 degrees). Associated with lung cancer, ILD, bronchiectasis, and cyanotic heart disease.

Digital ischemia

- Reduced arterial blood flow to the fingers causing pain, pallor, coolness, and potential tissue necrosis. Causes include Raynaud phenomenon, thromboangiitis obliterans, cholesterol emboli, and arterial thrombosis. Severe cases may require amputation of affected digits.

Digital Mucous Cyst

- A benign ganglion cyst arising from the distal interphalangeal joint, presenting as a translucent, dome-shaped nodule on the dorsal aspect of the finger near the nail fold. It is associated with osteoarthritis of the DIP joint.

Digital Subtraction Angiography (DSA)

- Technique subtracting pre-contrast images from post-contrast images to visualize blood vessels. Gold standard for vascular imaging. Used for diagnosing and treating vascular diseases.

Digitalis (Digoxin) Toxicity

- Nausea, vomiting, visual changes (yellow halos), arrhythmias. Risk: narrow therapeutic index, renal impairment, hypokalemia. Treatment: digoxin immune Fab, cardiac monitoring, correct electrolytes.

Digitalis toxicity

- A potentially life-threatening adverse effect of cardiac glycoside therapy (digoxin, digitoxin) characterized by a narrow therapeutic index (0.5-2.0 ng/mL for digoxin).

Digoxin Immune Fab

- Antibody fragments that bind free digoxin and related cardiac glycosides, reducing their tissue activity. Indications include life-threatening dysrhythmia, severe hyperkalemia from acute toxicity, or a potentially massive ingestion.

Dilated Cardiomyopathy

- A cardiomyopathy characterized by ventricular dilation and impaired systolic function, resulting in reduced ejection fraction. It is the most common cause of heart failure and the leading indication for heart transplantation.

Dilated Cardiomyopathy Prognosis

- 5-year survival approximately 50%. Risk factors for poor prognosis: low EF, advanced NYHA class, elevated BNP, ventricular arrhythmias, family history of sudden death, and failure to respond to medical therapy. Sudden cardiac death accounts for 30-50% of mortality.

Dilation and Curettage (D&C)

- Surgical dilation of the cervix and curettage (scraping) of the uterine lining. Indications: incomplete or missed abortion (miscarriage management), therapeutic abortion, endometrial sampling (abnormal uterine bleeding, postmenopausal bleeding, endometrial hyperplasia/cancer), and removal of retained products of conception (RPOC).

Dilator

- Instrument for widening opening.

Dilute Russell Viper Venom Time

- A phospholipid-dependent clotting assay used mainly to detect lupus anticoagulant. It is commonly performed as part of laboratory evaluation for antiphospholipid syndrome, with confirmatory testing required because anticoagulants and other factor abnormalities can affect the result.

Dimercaprol (BAL)

- Chelating agent for arsenic, mercury, gold. Given IM. Side effects: hypertension, tachycardia, nausea, pain at injection site.

Dinoprostone (Cervidil, Prepidil)

- A synthetic prostaglandin E₂ (PGE₂) used for cervical ripening and induction. Forms: Cervidil (10 mg vaginal insert, slow-release over 12 hours, removed if tachysystole occurs), Prepidil (0.5 mg

intracervical gel, may be repeated q6h), and Prostin E2 (vaginal suppository). Indications: unfavorable cervix (Bishop 5).

Dipeptidyl Peptidase-4 Inhibitors

- Oral glucose-lowering medications including sitagliptin, saxagliptin, linagliptin, and alogliptin that inhibit the enzyme degrading incretin hormones GLP-1 and GIP. Resulting increase in active incretin levels enhances glucose-dependent insulin secretion and suppresses glucagon. Neutral effect on weight and low hypoglycemia risk.

Diphenhydramine

- A first-generation H1 antihistamine with potent anticholinergic and sedative effects. Used for allergic reactions, urticaria/pruritus, motion sickness, insomnia (OTC sleep aid), and as an antitussive; IV form for anaphylaxis (adjunct) and acute dystonic reactions.

Diphtheria

- An acute, toxin-mediated infectious disease caused by *Corynebacterium diphtheriae*, a gram-positive aerobic bacillus that produces a potent exotoxin that inhibits protein synthesis via ADP-ribosylation of elongation factor 2, causing local tissue necrosis and systemic toxicity.

Diphtheria Vaccine

- Toxoid vaccine containing formalin-inactivated diphtheria toxin, providing immunity against the exotoxin produced by *Corynebacterium diphtheriae* rather than the bacterium itself.

Dipyllobothrium latum

- The fish tapeworm, a large cestode infecting humans through consumption of raw or undercooked freshwater fish. It can grow up to 10 meters in length and absorbs vitamin B12 in the intestine, potentially causing megaloblastic anemia. Diagnosis is by identification of eggs or proglottids in stool. Treatment is with praziquantel.

Diplopia

- See Diplopia (Double Vision).

Diplopia (Double Vision)

- The perception of two images of a single object, classified by onset, duration, and whether it resolves with covering one eye. Monocular diplopia (persists with one eye covered): caused by ocular surface ab-

normalities (dry eye, irregular astigmatism, cataract, corneal scar, lens dislocation)—usually benign.

Direct Antiglobulin Test (DAT)

- An immunohematologic test that detects immunoglobulin and/or complement bound to the surface of a patient's erythrocytes. A positive result supports in-vivo red-cell coating, as in autoimmune hemolytic anemia, hemolytic transfusion reactions, and hemolytic disease of the fetus and newborn.

Direct Coombs Test (DAT)

- Direct antiglobulin test: detects antibodies or complement (C3) bound to the surface of RBCs in vivo. Positive in: autoimmune hemolytic anemia (warm: IgG, cold: C3), drug-induced hemolytic anemia, hemolytic transfusion reaction, and hemolytic disease of the newborn (HDN). Grading: 1+ to 4+.

Direct Oral Anticoagulants

- DOACs directly inhibit specific coagulation factors. Direct thrombin inhibitors: dabigatran. Factor Xa inhibitors: rivaroxaban, apixaban, edoxaban, betrixaban. They have predictable pharmacokinetics, fixed dosing, and routine monitoring is not required. They have fewer drug interactions than warfarin.

Direct Veneer

- Composite restoration placed chairside for aesthetic improvement. Conservative alternative to laboratory veneer.

Directly Observed Therapy

- Supervision of medication ingestion to ensure adherence. Standard of care for TB treatment. Ensures completion of the full treatment course and reduces development of drug resistance.

Dirofilaria repens

- A filarial nematode transmitted by mosquitoes that causes subcutaneous or ocular dirofilariasis in humans, an accidental dead-end host. It presents as a solitary, migratory subcutaneous nodule, often on the trunk or extremities, or as a conjunctival or periorbital mass.

Disability

- Ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs). See functional status.

Disability Evaluation

- Assessment of functional limitations for legal, in-

surance, or benefits purposes. Objective measures, standardized tests.

Disability-Adjusted Life Year (DALY)

- A population-health measure combining years of life lost because of premature mortality with years lived with disability. One DALY represents one lost year of healthy life under the chosen methods and disability weights.

Disaster Preparedness

- Planning for disasters. Examples: hurricane, earthquake, pandemic, terrorism. Components: response, recovery, mitigation.

Discectomy

- Surgical removal of a herniated intervertebral disc fragment compressing a nerve root. Microscopic discectomy (most common): small incision, microscope, minimal muscle dissection.

Discharge Planning

- A multidisciplinary process that prepares a patient for safe transition from hospital or facility care to the next setting. It includes medication review, follow-up, education, equipment, self-management support, and communication of pending results.

Discitis

- Inflammation or infection of an intervertebral disc, often involving adjacent vertebral endplates. It may cause severe localized back or neck pain, stiffness, fever, and reduced mobility; causes include haematogenous infection, postoperative infection, and inflammatory disease, with MRI being the preferred imaging study.

Discography

- Contrast injection into intervertebral disc. Evaluates disc pathology. Controversial, limited use.

Disease

- Abnormal condition affecting body. Can be infectious, degenerative, neoplastic, etc.

Disease Reporting

- Mandatory notification of specified diseases or health events to public-health authorities.

Disease-Free Survival

- The length of time after treatment during which a patient remains alive without detectable evidence of the disease. The precise definition depends on

the cancer and the study protocol.

Disease-Modifying Antirheumatic Drug (DMARD)

- A medication that reduces inflammatory disease activity and helps prevent structural damage or disability rather than providing only symptomatic relief. Conventional synthetic, biologic, and targeted synthetic DMARDs are used according to disease and patient factors.

Disinfection

- The elimination of most pathogenic microorganisms on inanimate surfaces. Disinfection does not eliminate all forms of microbial life, particularly spores. High-level disinfection eliminates all microorganisms except spores. Intermediate-level disinfection eliminates mycobacteria, most viruses, and bacteria.

Disk Diffusion Susceptibility Test

- A standardized method in which antimicrobial-impregnated disks are placed on an inoculated agar surface and zones of growth inhibition are measured. Zone diameters are interpreted using validated organism- and drug-specific breakpoints.

Diskectomy

- The surgical removal of herniated intervertebral disk material that compresses nerve roots or the spinal cord, causing radiculopathy or myelopathy. Indicated in severe disk herniation causing unremitting pain, progressive motor weakness, or cauda equina syndrome.

Diskitis

- Infection or inflammation of an intervertebral disc and adjacent vertebral endplates, usually caused by bacterial infection. It causes localized back pain, stiffness, fever, and reluctance to move, particularly in children.

Dislocation

- Complete displacement of the articular surfaces of a joint, resulting in loss of normal bony alignment and often associated with ligamentous, capsular, and neurovascular injury.

Dislocation of the lens

- Partial or complete displacement of the crystalline lens from its normal position within the pupillary axis due to disruption or weakness of the zonular fibers. Causes include Marfan syndrome, homo-

cystinuria, trauma, and prior intraocular surgery; it may cause acute angle-closure glaucoma or blurred vision.

Disofenin Scan

- Hepatobiliary imaging using Tc-99m disofenin. Similar to mebrofenin scan.

Disorder

- Functional abnormality. Less severe than disease. Examples: sleep disorder, eating disorder.

Dissection

- A tear in the intima of the aorta allowing blood to dissect through the media, creating a false lumen. See aortic dissection.

Disseminated coccidioidomycosis

- See Coccidioidomycosis.

Disseminated cutaneous blastomycosis

- See Blastomycosis.

Disseminated Intravascular Coagulation

- A pathological state of simultaneous systemic activation of coagulation and fibrinolysis, leading to widespread microvascular thrombosis that causes organ ischemia and dysfunction, paradoxically accompanied by consumptive coagulopathy and increased bleeding risk.

Disseminated Mycobacterium avium–intracellulare infection

- See Non-Tuberculous Mycobacteria.

Disseminated varicella infection

- See Varicella.

Dissociation

- A disruption in the normal integration of consciousness, memory, identity, emotion, perception, behavior, and sense of self. It exists on a spectrum from normal (daydreaming) to pathological (DID, depersonalization-derealization).

Dissociative Amnesia

- Inability to recall important autobiographical information, usually of a traumatic or stressful nature, that is inconsistent with ordinary forgetting. The amnesia may be localized, selective, generalized, or continuous.

Dissociative Disorder

- DSM-5 category comprising disruption of the nor-

mally integrated functions of consciousness, memory, identity, perception, and behaviour, including dissociative amnesia, dissociative identity disorder, and depersonalization-derealization disorder. Strongly associated with severe early-life trauma.

Dissociative Identity Disorder

- The presence of two or more distinct personality states or an experience of possession involving marked discontinuity in sense of self, agency, affect, behavior, memory, perception, cognition, and/or sensory-motor functioning.

Distal

- Positioned further away from the trunk of the body or point of origin of a limb.

Distal Radius Fractures

- The most common upper extremity fracture. Classification: AO/OTA: A (extra-articular), B (partial articular), C (complete articular); Frykman. Colles fracture: dorsal angulation/displacement (fall on outstretched hand). Smith fracture: volar angulation (fall on flexed wrist).

Distal Tubule

- Segment of the nephron extending from the thick ascending limb of the loop of Henle to the collecting duct. The early distal tubule reabsorbs sodium and chloride through the NCC cotransporter, which is the target of thiazide diuretics.

Distraction Osteogenesis

- Surgical bone lengthening by gradual distraction. Used for jaw deformities, trauma.

Distributive Shock

- Shock caused by pathologic vasodilation and maldistribution of blood flow, producing inadequate effective circulating volume despite variable cardiac output. Causes include sepsis, anaphylaxis, neurogenic shock, and some toxic syndromes.

Disulfiram

- An aversive agent used to treat alcohol use disorder. Inhibits acetaldehyde dehydrogenase, causing accumulation of acetaldehyde when alcohol is consumed—producing flushing, tachycardia, nausea, vomiting, and hypotension ('acetaldehyde syndrome'). Provides a strong deterrent to drinking but requires motivation and abstinence.

Diuresis

- Increased urine production. Can be physiological (fluid overload) or pharmacological (diuretics). Osmotic diuresis (hyperglycemia), water diuresis (DI), saline diuresis (hypertonic saline).

Diuretic

- A drug that increases urine production, used for edema (heart failure, cirrhosis, renal disease), hypertension, and electrolyte disorders.

Diverticular Disease

- Condition characterized by the presence of diverticula, which are outpouchings of the colonic mucosa and submucosa through defects in the muscularis propria at sites where vasa recta penetrate the bowel wall.

Diverticulitis

- Inflammation and infection of colonic diverticula, a common complication of diverticulosis affecting 10-25 percent of individuals with diverticula. The pathophysiology involves microperforation of a diverticulum with fecal contents entering the pericolic tissues, inciting inflammation and potential abscess formation.

Diverticulosis

- The presence of diverticula, small mucosal outpouchings through the colonic wall at sites of vascular penetration, most commonly in the sigmoid colon (90 percent of cases). Prevalence increases with age, affecting over 50 percent of individuals older than 60 years and nearly 70 percent by age 85.

Diverticulosis and Diverticulitis

- Diverticulosis is a condition in which small, bulging pouches (diverticula) form in the lining of the digestive tract, most commonly in the colon. Diverticulitis occurs when these pouches become inflamed or infected, causing abdominal pain, fever, and changes in bowel habits.

Dix-Hallpike Maneuver

- The Dix-Hallpike maneuver is a diagnostic test used to identify benign paroxysmal positional vertigo by detecting the characteristic nystagmus and vertigo that occur when displaced otoconia within the posterior semicircular canal move in response to gravity during specific head positioning.

Dizziness

- A common symptom encompassing multiple distinct sensations requiring careful differentiation: vertigo (false sense of motion—vestibular: BPPV, vestibular neuritis, Ménière, CNS—brainstem/cerebellar), presyncope (feeling about to faint—cardiac/orthostatic), disequilibrium (unsteadiness with walking—neuro, vestibular,

DMFT Index

- An epidemiologic index recording the number of permanent teeth that are decayed, missing because of caries, or filled. It estimates cumulative caries experience but does not measure current lesion activity or distinguish all causes of tooth loss.

DMSA Scan

- Nuclear medicine scan evaluating renal cortical function and scarring. Gold standard for detecting pyelonephritis and renal scars in children. Shows photopenic areas corresponding to cortical damage. Important for long-term follow-up of childhood UTI.

DNA

- Deoxyribonucleic acid, the hereditary material in humans and most organisms. Double helix of two antiparallel strands (sugar-phosphate backbone, nitrogenous bases: A-T, G-C). Human genome: 3 billion base pairs, 20,000 protein-coding genes.

DNA (Deoxyribonucleic Acid)

- Double-stranded helical molecule carrying genetic information. Composed of nucleotides (sugar-phosphate backbone with bases adenine, thymine, guanine, cytosine). Replication is semi-conservative. Genes are specific sequences encoding proteins or functional RNAs. Chromosomes are highly condensed DNA-protein complexes.

DNA Double Helix

- The molecular structure of DNA described by Watson and Crick in 1953, consisting of two antiparallel polynucleotide chains wound around each other in a right-handed helix. The sugar-phosphate backbone is on the exterior; nitrogenous bases project inward.

DNA Profiling

- Analysis of polymorphic DNA markers, commonly short tandem repeats, to compare biological samples for human identification or kinship assessment.

Results are statistical associations and require validated laboratory procedures, appropriate controls, and documented sample integrity.

DNA Repair

- See base excision repair. Also: nucleotide excision repair, mismatch repair, double-strand break repair.

DNA Repair Mechanisms

- Cellular processes that correct DNA damage caused by replication errors, chemicals, radiation, and reactive oxygen species. Mismatch repair (MMR): corrects base-base mismatches and insertion-deletion loops that escape proofreading during replication.

DNA Replication

- The semi-conservative process by which DNA duplicates before cell division, producing two identical daughter molecules each containing one original and one new strand.

DNA Sequencing

- Determination of nucleotide order in DNA. Sanger sequencing: chain termination method, gold standard for small targets. Next-generation sequencing (NGS): massively parallel sequencing, whole genome, whole exome, targeted panels.

Do Not Resuscitate

- DNR: order not to perform CPR. Can be combined with DNI (do not intubate).

DOACs

- Direct oral anticoagulants are a class of anticoagulant drugs that directly inhibit specific coagulation factors, providing an alternative to warfarin with more predictable pharmacokinetics and no routine monitoring.

Dobutamine Stress Echocardiography

- Pharmacological stress test using dobutamine (beta-1 agonist) to increase heart rate, contractility, and myocardial oxygen demand. New or worsening wall motion abnormalities indicate ischemia. Preferred when patients cannot exercise. Beta-blockers should be held before testing.

Docusate

- A stool softener (surfactant laxative) that increases water and fat incorporation into stool by lowering surface tension, facilitating defecation in patients who should avoid straining (post-MI, post-surgery, hemorrhoids) and for opioid-related constipation.

Onset 12-72 hours.

Domestic Violence

- A pattern of coercive control and physical, sexual, psychological, or economic abuse by one intimate partner or household member against another. Forensic evaluation documents patterned injuries, defensive wounds, strangulation signs, burns, and injuries of differing ages.

Dominant

- Allele whose phenotype is expressed in heterozygotes. Autosomal dominant disorders: Huntington disease, marfan syndrome, achondroplasia. Requires only one copy for expression.

Donor Insemination

- A fertility treatment in which sperm from a donor (anonymous or known) is placed into a woman's vagina, cervix, or uterus to achieve pregnancy. Indications: male factor infertility (azoospermia, severe oligospermia), genetic disorders in the male partner, single women, and same-sex female couples.

Donovanosis

- A chronic, progressive, ulcerative genital infection caused by the intracellular gram-negative bacterium *Klebsiella granulomatis* (formerly *Calymmatobacterium granulomatis*). Also known as granuloma inguinale.

Dopamine Agonist

- A class of medications that stimulate dopamine receptors (D1-D5), mimicking the effects of endogenous dopamine.

Doppler Ultrasound

- Ultrasound technique using the Doppler effect to evaluate blood flow. Color Doppler: direction and velocity of flow (red = toward transducer, blue = away). Spectral Doppler: waveform analysis (arterial: pulsatile; venous: continuous). Pulsed-wave: measures flow at a specific depth.

Doppler Velocimetry

- Ultrasound measuring blood flow velocity in umbilical and fetal vessels. Assesses fetal well-being.

Doravirine

- A non-nucleoside reverse transcriptase inhibitor (NNRTI) used in combination antiretroviral therapy for HIV-1 infection. Doravirine binds to the HIV reverse transcriptase enzyme and inhibits its activity

through a unique mechanism that maintains activity against strains resistant to first-generation NNRTIs.

Dornase Alfa

- A recombinant human DNase that cleaves extracellular DNA in airway secretions, reducing sputum viscosity. It is used in CF to improve lung function and reduce exacerbations. It is administered as a nebulized solution once or twice daily. It has become a standard part of CF airway clearance regimens.

Dorsal

- Relating to the back side of the body; synonym for posterior in human anatomy.

Dorsal Root Ganglion

- A swelling on the dorsal root of a spinal nerve containing the cell bodies of primary sensory neurons. Its pseudounipolar neurons convey peripheral sensory information to the spinal cord.

Dosage

- The amount and frequency of a drug administered to achieve a therapeutic effect while minimizing toxicity. Components: dose (amount, e.g., mg), dosing interval (frequency), route, and duration.

Dose

- Amount of substance administered. Factors: weight, age, renal/hepatic function, drug interactions. Toxic dose varies by substance and individual.

Dose-Limiting Toxicity (DLT)

- An adverse effect of an anticancer treatment severe enough to prevent further dose escalation or require dose reduction according to a predefined protocol. DLTs are used in early-phase clinical trials to estimate a tolerable dose.

Dose-Response Curve

- A graphical representation of the relationship between the dose of a drug and the magnitude of its biological effect. The x-axis represents dose (typically on a logarithmic scale), and the y-axis represents the response (as percentage of maximum effect).

Dose-Response Relationship

- The dose-response relationship describes the quantitative relationship between the dose of a drug and the magnitude of the pharmacological response it produces. This relationship is typically represented as a sigmoidal curve when response is plotted against the log of the dose.

Dosimetry

- Measurement and calculation of radiation dose. Patient dosimetry for therapy.

DOTATATE

- Somatostatin analog labeled with Ga-68 for PET imaging of neuroendocrine tumors.

Double Outlet Right Ventricle

- A congenital heart defect where more than 50% of both great arteries arise from the right ventricle, with a VSD as the only outlet from the left ventricle.

Double-Balloon Enteroscopy

- Advanced endoscopic technique using an enteroscope equipped with balloons at the tip of both the endoscope and an overtube allowing systematic advancement and visualization of the entire small bowel. The balloons are alternately inflated and deflated to anchor the bowel wall and advance the endoscope.

Douche

- A stream of water or solution directed into a body cavity, most commonly the vagina, for cleansing, therapeutic, or contraceptive purposes.

Down Syndrome

- Trisomy 21, most common viable trisomy. Features: intellectual disability, characteristic facies (flat face, upslanting palpebral fissures), hypotonia, heart defects, increased risk of leukemia, Alzheimer disease. Maternal age is major risk factor.

Doxazosin

- A selective alpha-1 adrenergic blocker causing vascular smooth muscle relaxation and reduced peripheral resistance. Used for hypertension and benign prostatic hyperplasia (relaxes prostatic smooth muscle). Extended-release formulation reduces first-dose hypotension.

DPP-4 Inhibitors

- Dipeptidyl peptidase-4 inhibitors that prevent degradation of endogenous GLP-1 and GIP, enhancing incretin effects on insulin secretion. Examples include sitagliptin, saxagliptin, linagliptin, and alogliptin. They are well-tolerated with low hypoglycemia risk and weight neutrality. They have modest glucose-lowering effects.

Drain (Surgical)

- Tube for fluid evacuation. Types: closed suction (Jackson-Pratt, Hemovac), open (Penrose), pigtail. Indications: abscess, hematoma, seroma, biliary, urinary. Complications: infection, blockage, migration.

Drainage Procedures

- Image-guided drainage of fluid collections. Abscess, hematoma, biloma, urinoma. CT or ultrasound guidance.

Dreams (Geriatric)

- Sleep disturbances common. Nightmares, insomnia. Evaluate for delirium, depression, medications.

DRESS Syndrome

- Drug Reaction with Eosinophilia and Systemic Symptoms, a severe, potentially life-threatening adverse drug reaction. Typically occurs 2-8 weeks after drug initiation (latency longer than SJS/TEN). Causative drugs: anticonvulsants (phenytoin, carbamazepine, lamotrigine), allopurinol, sulfonamides, dapsone, minocycline.

Dressler Syndrome

- A delayed pericarditis occurring 2-10 weeks after myocardial infarction or cardiac surgery. It is an autoimmune-mediated inflammation caused by antibodies against myocardial antigens released during tissue injury. Symptoms include fever, pleuritic chest pain, and pericardial friction rub. ECG may show diffuse ST elevation.

Drift (Sit-to-Stand)

- Functional test measuring lower extremity strength and balance. Timed Up and Go (TUG): stand, walk 3 meters, turn, return, sit. Normal: <10 seconds; fall risk: >12 seconds. Three-step squat test, chair rise test alternative.

Driving Evaluation

- Assess driving fitness after neurological, musculoskeletal conditions. On-road testing, vehicle modifications.

Driving Rehabilitation

- Training for adaptive driving: hand controls, left-foot accelerator, other modifications.

Drooling

- Excessive flow of saliva from the mouth, caused

by increased production, impaired swallowing, or inability to keep saliva in the mouth. It may occur with neurologic disease, oral disorders, medication effects, or developmental conditions.

Droplet precautions

- Infection control measures used for pathogens transmitted through respiratory droplets. Precautions include surgical masks, patient placement in single rooms, and spatial separation. Droplet precautions are used for influenza, pertussis, and meningococcal infections. The distance for spatial separation is typically 3-6 feet.

Drowning

- Respiratory impairment from submersion/immersion in liquid. Drowning can be fatal or non-fatal. Pathophysiology: laryngospasm followed by aspiration of fluid, surfactant washout, pulmonary edema, and hypoxemia. Fresh water: washes out surfactant, causes alveolar collapse. Salt water: draws fluid into alveoli (pulmonary edema).

Drowning (Dry/Wet Signs)

- Cause of death resulting from submersion or immersion in liquid causing primary respiratory impairment.

Drowsiness

- An increased tendency to fall asleep or reduced alertness, ranging from mild sleepiness to difficulty staying awake. Causes include inadequate sleep, medical illness, sleep disorders, substance use, and medication effects; sudden or severe drowsiness may indicate an urgent neurologic or metabolic problem.

Drug Allergy

- An adverse immune-mediated reaction to a medication, representing a hypersensitivity response that is distinct from side effects, toxicities, and idiosyncratic reactions.

Drug Antagonism

- An interaction where one drug blocks or decreases the effectiveness of another, such as when an antidote blocks a poison.

Drug Bioavailability

- The fraction of an administered drug that reaches systemic circulation unchanged. Oral bioavailability is affected by first-pass metabolism, absorption, and drug formulation. IV administration has 100%

bioavailability. Bioavailability influences drug dosing and route selection.

Drug Classification Systems

- Drugs are classified using multiple complementary systems.

Drug Clearance

- The volume of plasma completely cleared of drug per unit time. Hepatic clearance depends on hepatic blood flow and enzyme activity; renal clearance depends on glomerular filtration, tubular secretion, and reabsorption. Total clearance determines maintenance dose.

Drug Dependence

- A state of physiological or psychological adaptation to a drug, characterized by a withdrawal syndrome upon discontinuation or dose reduction. Physical dependence: the body has adapted to the presence of the drug, and abrupt cessation produces a predictable withdrawal syndrome specific to the drug class.

Drug Hypersensitivity Reactions

- Immune-mediated adverse drug reactions classified by Gell and Coombs.

Drug Hypersensitivity Syndrome

- A severe delayed drug reaction occurring 2 to 8 weeks after drug initiation. It presents with fever, rash (morbilliform, facial edema, exfoliative dermatitis), lymphadenopathy, eosinophilia, and internal organ involvement (hepatitis most common, also nephritis, pneumonitis, myocarditis).

Drug Interaction

- See pharmacology.

Drug Interactions

- Drug interactions occur when one drug affects the activity of another drug when both are administered together, potentially leading to therapeutic failure or toxicity. Pharmacokinetic interactions alter the absorption, distribution, metabolism, or excretion of a drug.

Drug Metabolism Pharmacogenomics

- Pharmacogenomics of drug metabolism examines how genetic variations affect individual responses to drugs, enabling personalized medicine approaches to prescribing.

Drug Overdose

- Excessive ingestion of medication. Common: opioids, benzodiazepines, acetaminophen, salicylates, tricyclic antidepressants. Management: ABCs, activated charcoal, specific antidotes (naloxone, flumazenil, N-acetylcysteine), supportive care.

Drug Overdose Management

- Treatment of drug overdose involves supportive care (airway, breathing, circulation), decontamination (gastric lavage, activated charcoal within 1 hour), enhanced elimination (urinary alkalization, hemodialysis), and specific antidotes when available.

Drug reaction with eosinophilia and systemic symptoms

- A severe hypersensitivity syndrome characterized by fever, rash, lymphadenopathy, and multiorgan involvement triggered by medications, requiring prompt drug cessation and systemic corticosteroids.

Drug Resistance

- The ability of microorganisms, cancer cells, or parasites to survive and proliferate despite exposure to drugs that were previously effective.

Drug Screening

- Urine, blood, hair testing for substances of abuse. Panels: opiates, cocaine, amphetamines, cannabinoids, PCP, benzodiazepines, barbiturates. False positives possible. Confirmation by GC/MS or LC-MS/MS.

Drug Testing

- Laboratory analysis of biological specimens to detect drugs, metabolites, or poisons, used in clinical, workplace, sports, and forensic contexts. Specimens include blood, urine, oral fluid, hair, and post-mortem tissues.

Drug Therapeutics

- See pharmacology.

Drug Tolerance

- A decreased pharmacological response to a drug after repeated administration, requiring dose escalation to achieve the same effect.

Drug Toxicity

- Adverse effects from drugs at normal or elevated doses. Idiosyncratic reactions, cumulative toxicity, organ damage.

Drug Withdrawal

- A clinically significant syndrome or return of symptoms caused by abrupt reduction or cessation of a drug after physiologic adaptation or dependence has developed. Tapering is required for some medications to reduce withdrawal and rebound effects.

Drug-Drug Interactions

- When one drug alters the pharmacokinetics or pharmacodynamics of another. Pharmacokinetic interactions affect absorption, distribution, metabolism, or excretion (e.g., CYP inhibitors increasing warfarin levels).

Drug-Eluting Balloon (Vascular)

- See Drug-Eluting Balloon.

Drug-Eluting Stent

- Mesh tube keeping coronary artery open after angioplasty. See coronary stent.

Drug-Eluting Stent (Vascular)

- See Drug-Eluting Stent.

Drug-Facilitated Sexual Assault

- Sexual contact occurring when a person's capacity to consent or resist is impaired by alcohol, medication, or another drug, whether administered covertly or knowingly consumed. Medical care includes urgent safety assessment, treatment of injuries, toxicology sampling when indicated, prophylaxis, and trauma-informed support.

Drug-Induced Autoimmunity

- Drug-induced lupus (DIL): procainamide, hydralazine, isoniazid, minocycline, TNF inhibitors, anti-TNF; anti-histone antibodies, positive ANA, negative anti-dsDNA/anti-Sm; symptoms resolve after drug cessation.

Drug-Induced Liver Injury

- Liver damage caused by medications, herbal products, or dietary supplements, representing the most common cause of acute liver failure in the United States.

Drusen

- Yellowish-white deposits of extracellular material that accumulate beneath the retinal pigment epithelium between the retinal pigment epithelium basement membrane and Bruch membrane, and they are the hallmark clinical finding in dry age-related

macular degeneration.

Dry Eye Disease

- Dry eye disease, also known as keratoconjunctivitis sicca, is a multifactorial disorder of the ocular surface characterized by loss of homeostasis of the tear film, resulting in symptoms of ocular discomfort, visual disturbance, and tear film instability, with potential damage to the ocular surface.

Dry Eye Syndrome

- A multifactorial disease of the ocular surface characterized by tear film instability, hyperosmolarity, and inflammation. Aqueous-deficient dry eye: Sjogren syndrome (autoimmune), non-Sjogren (lacrimal gland dysfunction, age). Evaporative dry eye: meibomian gland dysfunction (most common), incomplete blink, environmental factors.

Dry Powder Inhaler

- A breath-actuated inhaler delivering powdered medication. Does not require hand-breath coordination. Requires adequate inspiratory flow. Examples include Advair Diskus and Spiriva HandiHaler.

Dry Skin

- Abnormally dry, rough, or scaly skin caused by reduced moisture and disruption of the skin barrier. It is common with aging, low humidity, frequent washing, and eczema and may cause itching or fissuring.

Dry Socket (Alveolar Osteitis)

- A painful postoperative complication of tooth extraction, most commonly occurring after the removal of mandibular third molars (wisdom teeth). It is caused by the premature dissolution or loss of the blood clot in the extraction socket, exposing the underlying alveolar bone and nerve endings to the oral cavity.

DTaP Vaccine

- Diphtheria, Tetanus, and acellular Pertussis vaccine given at 2, 4, 6, and 15-18 months, and 4-6 years. Protects against three serious bacterial diseases. Diphtheria causes respiratory and cardiac complications. Tetanus causes lockjaw and muscle rigidity. Pertussis causes whooping cough, most severe in young infants.

Dual Antiplatelet Therapy

- The combination of aspirin and a P2Y₁₂ inhibitor

(clopidogrel, prasugrel, or ticagrelor) for prevention of thrombotic events after acute coronary syndrome and percutaneous coronary intervention. Duration varies: 1 month after bare-metal stent, 6-12 months after drug-eluting stent, and 12 months after ACS.

Dual-Energy CT

- CT using two different energy spectra. Material differentiation, virtual non-contrast images, gout detection.

Dubin-Johnson syndrome

- A benign, autosomal recessive disorder of conjugated hyperbilirubinemia caused by mutations in the *ABCC2* gene encoding the multidrug resistance-associated protein 2 (MRP2), which transports conjugated bilirubin from hepatocytes into bile.

Duchenne Muscular Dystrophy

- An X-linked recessive neuromuscular disorder caused by mutations in the dystrophin gene (*DMD* gene) on chromosome Xp21.2, resulting in absent or severely reduced dystrophin protein, which is essential for maintaining the structural integrity of the sarcolemma during muscle contraction.

Ductal-Dependent Lesions

- Congenital heart defects that require the ductus arteriosus to remain patent for adequate systemic or pulmonary blood flow. Includes pulmonary atresia, critical aortic stenosis, coarctation, hypoplastic left heart syndrome, and transposition of the great arteries.

Ductus Arteriosus

- A fetal blood vessel connecting the pulmonary artery to the descending aorta, allowing blood to bypass the non-functioning fetal lungs. In utero, the ductus arteriosus is kept patent by prostaglandin E2 and low oxygen tension.

Duloxetine

- A serotonin-norepinephrine reuptake inhibitor (SNRI) used for major depressive disorder, generalized anxiety disorder, diabetic peripheral neuropathy, fibromyalgia, chronic musculoskeletal pain (including osteoarthritis), and stress urinary incontinence.

Dumping Syndrome

- Rapid gastric emptying of hypertonic contents into the small bowel after gastrectomy or gastric bypass

surgery. Early dumping occurs within thirty minutes with vasomotor symptoms including flushing, palpitations, diarrhea, and dizziness from fluid shifts into the intestinal lumen.

Duodenal Atresia

- A congenital obstruction of the duodenum caused by failure of recanalization during the 8th-10th week of gestation. Associated with Down syndrome (30%), polyhydramnios (due to impaired fetal swallowing of amniotic fluid), and other anomalies (VACTERL, congenital heart disease, annular pancreas, malrotation).

Duodenal perforation

- Full-thickness breach of the duodenal wall, most commonly caused by penetrating peptic ulcer disease, trauma, or iatrogenic injury during endoscopy. Presents with sudden severe epigastric pain, board-like abdominal rigidity, and peritonitis. Free air under the diaphragm on upright chest radiograph is diagnostic.

Duodenal Ulcer

- Peptic ulcer in the duodenal mucosa strongly associated with *H. pylori* infection. Presents with epigastric pain relieved by food typically occurring several hours after meals and at night. Complications include hemorrhage, perforation, and gastric outlet obstruction. Diagnosis is by upper endoscopy.

Duodenitis

- Inflammation of the duodenal mucosa, caused by *Helicobacter pylori* infection, acid-related disease, nonsteroidal anti-inflammatory drugs, infection, coeliac disease, or inflammatory bowel disease. Symptoms may include epigastric pain, nausea, vomiting, and gastrointestinal bleeding.

Duodenum

- Shortest first small intestinal segment (25 cm), C-shaped around the pancreatic head. Four parts. Descending part has the major duodenal papilla for bile/pancreatic duct entry via the hepatopancreatic ampulla. Brunner's glands produce alkaline mucus. Most common peptic ulcer site. Primary iron and calcium absorption site.

Duplex Surveillance

- Scheduled clinical and ultrasound monitoring of an artery, vein, graft, stent, aneurysm, or dialysis

access to detect restenosis, thrombosis, endoleak, or other failure before symptoms or limb threat develop.

Duplex Ultrasound

- Ultrasound technique combining B-mode imaging and Doppler flow assessment. See arterial duplex.

Duplication

- Presence of extra copy of chromosomal segment. Can cause gene dosage effects. Examples: Charcot-Marie-Tooth disease type 1A (PMP22 duplication).

Dupuyten's contracture

- See Dupuytren contracture.

Dupuytren contracture

- A benign, progressive fibroproliferative disorder of the palmar fascia causing nodules, cords, and contractures of the fingers, most commonly affecting the ring and small fingers.

Dura Mater

- The outermost, thickest meningeal layer, a tough fibrous membrane. In the cranium it is two layers: periosteal (outer, adherent to skull) and meningeal (inner), which separate to form dural venous sinuses (e.g., superior sagittal, cavernous). Dural folds: falx cerebri, tentorium cerebelli, falx cerebelli.

Durable Power of Attorney

- Legal document expressing healthcare wishes. See advance directive.

Dural Venous Sinuses

- Venous channels between dural layers, draining brain blood and CSF from the subarachnoid space into the internal jugular veins. Major sinuses: superior sagittal, transverse, sigmoid, cavernous, straight. Valveless with bidirectional flow possible.

DVT

- Formation of a blood clot in a deep vein, most commonly in the lower extremities. See Deep Vein Thrombosis.

DVT Prevention

- Formation of a blood clot in a deep vein, most commonly in the lower extremities. See deep vein thrombosis.

Dwarfism

- Short stature resulting from a medical condition, most commonly achondroplasia (the most common

form of skeletal dysplasia, caused by gain-of-function mutation in FGFR3, presenting with rhizomelic limb shortening, macrocephaly, frontal bossing, and trident hands).

DWI FLAIR Mismatch

- Diffusion-weighted imaging (DWI) and fluid-attenuated inversion recovery (FLAIR) MRI sequences are used together to estimate the age of an ischemic stroke lesion and guide treatment decisions.

Dx

- Identification of disease. See diagnosis.

DXA Scan (Dual-Energy X-ray Absorptiometry)

- The gold standard for measuring bone mineral density (BMD) and diagnosing osteoporosis. Measures at the lumbar spine, total hip, and femoral neck. T-score: normal (-1.0), osteopenia (-1.0 to -2.5), osteoporosis (-2.5). Z-score (compared to age-matched) used in premenopausal women and men <50.

Dying

- Process approaching death. Physical, psychological, social changes.

Dysarthria

- A motor speech disorder resulting from neurological impairment of the muscular components of speech production, including the lips, tongue, palate, vocal cords, and respiratory muscles.

Dysautonomia

- Dysfunction of the autonomic nervous system, causing abnormal regulation of blood pressure, heart rate, sweating, temperature, gastrointestinal motility, bladder function, or sexual function. It may be primary or secondary to diabetes, neurodegenerative disease, autoimmune disease, infection, medication effects, or other

Dysbiosis

- An alteration in microbiome composition, diversity, or function relative to a healthy state, permitting overgrowth of pathobionts or loss of beneficial microbes. Causes include antibiotics, dietary change, stress, and infection. It is associated with inflammatory bowel disease, obesity, allergy, and *Clostridioides difficile* colitis.

Dyscalculia

- A persistent difficulty with understanding or manipulating numerical quantities and performing arithmetic that is not explained by inadequate instruction alone. It may occur as a developmental learning disorder or result from acquired brain injury, particularly involving dominant parietal networks.

Dyscrasia

- An historic term broadly referring to any abnormal condition of the blood or bone marrow, now used primarily in the context of plasma cell dyscrasias (also called monoclonal gammopathies).

Dysentery

- Severe diarrheal illness with blood and mucus in stool, typically caused by *Shigella*, *Salmonella*, *Campylobacter*, or *Entamoeba histolytica*. Presents with frequent bloody stools, abdominal cramps, tenesmus, and fever. Diagnosis: stool culture, microscopy, PCR.

Dysesthesia

- An unpleasant, abnormal sensation produced by a normally non-painful stimulus, or an unpleasant, exaggerated response to a normal stimulus. It is distinguished from paresthesia (which is simply an abnormal sensation that may or may not be unpleasant) by its consistently unpleasant quality.

Dysgeusia

- Distortion, reduction, or loss of the normal sense of taste. It may result from oral or nasal disease, viral infection, neurological injury, nutritional deficiency, systemic illness, or medication exposure and may occur with altered smell, dry mouth, or mucosal inflammation.

Dysgraphia

- A developmental or acquired impairment of written expression involving handwriting, spelling, written composition, or the motor and language processes required for writing. It may occur alone or with other learning, language, attention, or neurological disorders.

Dyslexia

- A specific learning disorder with neurobiological origin, characterized by difficulties with accurate and fluent word recognition, spelling, and decoding ability, despite adequate intelligence, motivation,

and educational opportunity. Affects approximately 5–10% of the population.

Dyslipidemia

- An abnormal concentration or composition of circulating lipoproteins, including elevated low-density lipoprotein cholesterol, elevated triglycerides, reduced high-density lipoprotein cholesterol, or mixed abnormalities.

Dysmenorrhea

- Painful menstruation with cramping pelvic pain. Primary dysmenorrhea: no underlying pathology, associated with prostaglandin release, starts within 3 years of menarche. Secondary dysmenorrhea: due to underlying pathology (endometriosis, fibroids, adenomyosis, PID).

Dysmenorrhoea

- Painful menstruation, classified as primary (no identifiable pelvic pathology) or secondary (underlying structural or inflammatory condition).

Dyspareunia

- Painful sexual intercourse.

Dyspepsia

- Chronic or recurrent pain or discomfort centered in the upper abdomen. Organic causes: peptic ulcer disease, GERD, gastritis, gastric cancer. Functional dyspepsia is diagnosed after exclusion of organic pathology with normal upper endoscopy. Rome IV criteria define subtypes: epigastric pain syndrome and postprandial distress syndrome.

Dysphagia

- Difficulty swallowing, classified as oropharyngeal (difficulty initiating a swallow, choking, coughing, nasal regurgitation) or esophageal (sensation of food sticking retrosternally).

Dysphagia (Geriatric)

- Difficulty swallowing, common in elderly. Causes: neurological (stroke, Parkinson's), structural (cancer, stricture), muscular (presbyesophagus), medications. Symptoms: coughing with eating, choking, pocketing food, weight loss, aspiration. Assessment: clinical exam, videofluoroscopy, endoscopy.

Dysphagia Diet

- A texture- and consistency-modified diet prescribed after assessment of swallowing safety. Liquids and foods may be thickened or altered to reduce aspira-

tion risk while maintaining adequate energy, protein, hydration, and patient acceptability.

Dysphagia Rehabilitation

- Medical term; definition not provided in source.

Dysphasia

- A language disorder caused by damage to the language-dominant hemisphere of the brain (typically left), involving difficulty with expression, comprehension, reading, or writing. Dysphasia (often used interchangeably with aphasia): partial loss of language function.

Dysphonia

- An abnormal voice quality encompassing hoarseness, breathiness, roughness, vocal fatigue, pitch breaks, and strained or whispered speech. It is organic when due to structural laryngeal pathology (nodules, polyps, laryngitis, laryngeal cancer) and functional when due to voice misuse, muscle tension dysphonia, or psychogenic causes.

Dysplasia

- Disordered cellular growth and maturation characterized by structural abnormalities within a tissue, representing a pre-neoplastic condition that may progress to invasive carcinoma.

Dysplastic Nevus

- Atypical nevus: a melanocytic lesion with clinical and histologic atypia, larger (>5-6 mm), with irregular borders, variable color (pink-tan-brown), and a macular/papular component; often multiple in the dysplastic nevus syndrome (familial atypical mole-melanoma, FAMMM).

Dyspnea

- See pulmonology.

Dyspnea on Exertion

- Shortness of breath precipitated by physical activity, a cardinal symptom of cardiac and pulmonary disease. Cardiac causes: heart failure, coronary artery disease (angina equivalent), valvular disease; pulmonary: COPD, asthma, ILD, anemia, deconditioning. Quantified by NYHA functional class (heart failure) and mMRC scale (dyspnea).

Dyspnoea

- The subjective sensation of difficult or labored breathing, also known as shortness of breath (SOB). It is a symptom common to many cardiopulmonary

and systemic conditions.

Dysthymia (Persistent Depressive Disorder)

- A chronic form of depression with depressed mood for most of the day, more days than not, for 2 years (1 year in children/adolescents). Symptoms: poor appetite or overeating, insomnia or hypersomnia, low energy, low self-esteem, poor concentration, and hopelessness.

Dystonia

- A movement disorder characterized by sustained or intermittent muscle contractions causing abnormal, often repetitive, movements and postures.

Dystrophic Calcification

- The deposition of calcium phosphate salts in damaged, necrotic, or degenerating tissues despite normal serum calcium levels. It occurs in areas of tissue injury, inflammation, or chronic degeneration where the local tissue environment favors calcium precipitation.

Dysuria

- Painful, burning, or uncomfortable sensation experienced during urination. Most commonly indicates inflammation or infection of the lower urinary tract. Causes in females: acute cystitis, urethritis (Chlamydia trachomatis, Neisseria gonorrhoeae), vaginitis (Candida, Trichomonas), and vulvar atrophy.

Dapsone

- A medicine used for leprosy, dermatitis herpetiformis, and selected bacterial or inflammatory conditions. It can cause hemolytic anemia, methemoglobinemia, or severe skin reactions; testing for glucose-6-phosphate dehydrogenase deficiency may be needed before treatment.

Dantrolene

- A skeletal-muscle relaxant used urgently to treat malignant hyperthermia and sometimes used for severe muscle spasticity. It reduces calcium release inside muscle cells; liver injury and muscle weakness are important adverse effects.

Delayed Hypersensitivity

- A delayed immune reaction, usually mediated by T cells rather than antibodies, that develops hours to days after exposure. Examples include contact

dermatitis, the tuberculin reaction, and some drug eruptions.

Dental Disease

- Disorders affecting the teeth, gums, or supporting tissues, including tooth decay, gum disease, oral infections, and tooth loss. Prevention includes fluoride, regular brushing and flossing, limiting frequent sugar exposure, and routine dental examinations.

Dexamethasone

- A potent corticosteroid used to reduce inflammation and suppress excessive immune activity. It is used for selected allergic, respiratory, neurologic, cancer-related, and endocrine conditions. Longer or higher-dose courses can increase infection risk, blood glucose, and bone loss.

Diabetic Cardiomyopathy

- Damage to the heart muscle associated with diabetes that occurs independently of coronary artery disease or hypertension. It may cause impaired relaxation, reduced pumping function, heart failure, and rhythm problems; glucose and cardiovascular risk control can reduce risk.

Diclofenac

- A nonsteroidal anti-inflammatory drug (NSAID) used for short-term treatment of pain and inflammation. It may cause stomach bleeding, kidney injury, fluid retention, or cardiovascular events; topical forms may have fewer systemic effects.

Dicyclomine

- An anticholinergic medicine used to reduce intestinal smooth-muscle spasms in irritable bowel syndrome. It may cause dry mouth, blurred vision, constipation, urinary retention, or confusion and is not suitable for everyone with bowel symptoms.

Digoxin

- A cardiac glycoside used in selected patients with heart failure or atrial fibrillation to improve symptoms or slow the heart rate. It has a narrow safety margin; nausea, visual changes, confusion, or irregular heartbeat may indicate toxicity.

Diltiazem

- A calcium-channel blocker used to treat hypertension, angina, and some rapid heart rhythms. It slows conduction through the atrioventricular node and may cause low blood pressure, slow heart rate,

constipation, or worsening heart failure.

Doxycycline

- A tetracycline antibiotic used for many bacterial infections, acne, and some tick-borne diseases. It can cause stomach upset, sun sensitivity, and esophageal irritation; use in pregnancy and young children requires specific clinical consideration.

Disorientation

- Difficulty knowing one's time, place, situation, or identity. It may occur with delirium, infection, metabolic disturbances, intoxication or withdrawal, brain injury, or dementia. Sudden disorientation is an urgent medical symptom.

Dementia

- A group of symptoms involving progressive decline in memory, thinking, communication, or daily functioning. Causes include Alzheimer disease, vascular disease, Lewy body disease, frontotemporal disorders, and other conditions. Evaluation is important because some causes of cognitive decline may be treatable.

Chapter 5

Pocket Dictionary: E Terms

E Coli Enteritis

- Intestinal infection caused by pathogenic strains of *Escherichia coli*. Depending on the strain, it may cause watery or bloody diarrhea, abdominal cramps, vomiting, or systemic complications such as hemolytic uremic syndrome.

E Coli Infection

- Infections caused by *Escherichia coli*, a gram-negative enteric bacillus. Causes: UTI (most common—cystitis, pyelonephritis), diarrhea (enterotoxigenic ETEC—traveler’s diarrhea; enteropathogenic; enterohemorrhagic EHEC O157:H7—bloody diarrhea, HUS; enteroinvasive), neonatal meningitis, sepsis, and intra-abdominal infections.

Ear

- Organ of hearing and balance, divided into external, middle, and inner ear. External: auricle (pinna), external auditory canal, tympanic membrane. Middle: air-filled cavity in temporal bone, auditory ossicles (malleus, incus, stapes), Eustachian tube (to nasopharynx).

Ear Barotrauma

- Injury to the ear caused by a pressure difference between the middle ear and the surrounding environment, commonly during air travel or diving. It may cause ear pain, fullness, hearing loss, tinnitus, vertigo, or tympanic-membrane rupture.

Ear canal

- A tube leading from the eardrum to the outer ear.

Ear Diseases

- Any disorder affecting the structures of the ear—external ear (pinna, ear canal), middle ear (tympanic membrane, ossicles, Eustachian tube), or inner ear (cochlea, vestibular apparatus).

Ear Infection

- Infection or inflammation of the ear, commonly involving the external canal or middle ear. It may cause ear pain, fever, reduced hearing, discharge, or irritability in children. Also called otitis.

Ear Infections

- Infections of the ear, most commonly affecting the middle ear (otitis media) or the outer ear canal (otitis externa). They are often caused by bacteria or viruses and present with ear pain, drainage, hearing loss, and sometimes fever.

Ear, Nose, and Throat (ENT)

- A medical specialty (otolaryngology) focused on the diagnosis and treatment of disorders affecting the ears, nose, throat, head, and neck. Conditions managed include hearing loss, sinusitis, tonsillitis, voice disorders, and head and neck cancers.

Earache

- Pain in or around the ear, caused by otitis, blockage, trauma, dental disease, temporomandibular-joint disorders, or referred throat pathology. Persistent, severe, or discharge-associated pain requires evaluation.

Earache (Otalgia)

- Pain originating from the ear, a common symptom with diverse causes. Otalgia is classified as primary (originating within the ear itself) or secondary/referred (pain perceived in the ear but arising from another site).

Eardrum

- A thin membrane separating the ear canal and middle ear.

Early Decelerations

- A fetal heart rate pattern with a gradual decrease

(onset to nadir >30 seconds) that mirrors the contraction and returns to baseline as the contraction ends. Caused by fetal head compression during contractions (vagal response). A benign, physiologic finding associated with normal fetal acid-base status. No intervention required.

Early Detection

– Testing asymptomatic individuals for disease. See screening.

Early Intervention

– Rehabilitation for infants, toddlers with developmental delays or disabilities.

Earwax

– Cerumen, a waxy secretion produced by the ceruminous glands in the outer third of the external auditory canal. Earwax lubricates and protects the skin of the ear canal, trapping dust, debris, and microorganisms to prevent them from reaching the tympanic membrane.

Eating Disorder

– A group of conditions characterized by abnormal eating habits and distorted body image, including anorexia nervosa, bulimia nervosa, binge eating disorder, and avoidant/restrictive food intake disorder (ARFID). Anorexia: severe calorie restriction, intense fear of weight gain, and body image disturbance with dangerously low weight.

Ebola

– Ebola virus disease (EVD): a severe, often fatal viral hemorrhagic fever caused by Ebola virus (Filoviridae, RNA), transmitted by direct contact with blood/body fluids of infected humans/animals. Incubation 2-21 days.

Ebola virus

– A filovirus that causes Ebola virus disease, a severe hemorrhagic fever. The virus is transmitted through contact with infected body fluids. The incubation period is 2-21 days. Symptoms include fever, headache, myalgia, and hemorrhage. Diagnosis is made by PCR and antigen detection.

Ebola Virus Disease

– See Ebola.

Ebstein Anomaly

– A rare congenital heart defect accounting for less than 1 percent of all congenital heart disease, char-

acterized by apical displacement of the septal and posterior leaflets of the tricuspid valve into the right ventricle, resulting in atrialization of the proximal right ventricle and a small functional right ventricular chamber.

Ecchymosis

– An ecchymosis is a flat, purple or blue discoloration of the skin larger than 10 millimeters caused by extravasation of blood from damaged blood vessels into the subcutaneous tissue. Unlike petechiae, which result from platelet disorders, ecchymoses may result from coagulopathies, trauma, or vascular fragility.

Eccrine Gland

– A coiled sweat gland distributed throughout the skin that secretes a watery fluid directly onto the skin surface through a duct. Eccrine sweating supports thermoregulation and is stimulated by sympathetic cholinergic fibres; the glands are especially numerous on the palms, soles, and forehead.

ECG/EKG Machine

– Records electrical activity of heart. 12-lead ECG standard. Components: electrodes, leads, amplifier, recorder. Essential for cardiac diagnosis.

Echinocandins

– A class of antifungal agents that inhibit beta-1,3-glucan synthase, an enzyme essential for synthesis of beta-1,3-glucan, a major component of the fungal cell wall. This unique mechanism targets a structure absent in mammalian cells, resulting in favorable toxicity profiles.

Echinococcal cysts

– See Echinococcosis.

Echinococcosis

– See Cystic echinococcosis.

Echinococcus granulosus

– A tapeworm that causes hydatid disease (cystic echinococcosis). The adult worm infects dogs, and humans are accidental intermediate hosts. Infection occurs through ingestion of eggs from dog feces. Larval cysts form in the liver and lungs, causing space-occupying lesions.

Echinocytes (Burr Cells)

– Red blood cells with uniform, small, evenly spaced serrated or spiky membrane projections along the

entire erythrocyte circumference. Most commonly caused by End-Stage Renal Disease (Uremia), pyruvate kinase deficiency, severe hypokalemia, or in vitro artifact (crenation due to glass slides or prolonged storage).

Echo

- Ultrasound imaging of the heart, the primary modality for assessing cardiac structure and function. See echocardiography.

Echocardiogram

- An ultrasound examination of the heart that shows its chambers, valves, wall motion, and blood flow. It is used to evaluate heart failure, valve disease, congenital defects, and other cardiac disorders.

Echocardiography

- Ultrasound imaging of the heart, the primary modality for assessing cardiac structure and function. Transthoracic echocardiography (TTE) is the most common approach; transesophageal echocardiography (TEE) provides better views of posterior structures (left atrium, mitral valve, aorta).

Echogenic Bowel

- Increased echogenicity of the fetal bowel relative to adjacent bone (iliac wing). Identified in 0.1-0.5% of second-trimester ultrasounds.

Echogenic Intracardiac Focus (EIF)

- A small bright spot (calcification) in the fetal heart, typically in the left ventricle (papillary muscle). Seen in 3-5% of normal pregnancies. In isolation, it is a normal variant with minimal aneuploidy risk. However, EIF is more common in trisomy 21 fetuses (10-30%), particularly when combined with other soft markers.

Echolalia

- The involuntary, automatic repetition of words or phrases spoken by another person, a speech phenomenon observed in various neurologic and psychiatric conditions. Immediate echolalia involves repeating sounds immediately after hearing them; delayed echolalia involves repeating phrases heard hours, days, or weeks earlier.

Echovirus

- A group of enteroviruses that commonly cause asymptomatic or mild infection but may also cause aseptic meningitis, encephalitis, rash, respiratory ill-

ness, myocarditis, or neonatal sepsis. Transmission is mainly faecal-oral or respiratory, and diagnosis is usually by nucleic-acid testing or viral culture in selected cases.

Eclampsia

- The occurrence of generalized tonic-clonic seizures in a patient with preeclampsia, representing a medical emergency. Complications include aspiration, placental abruption, disseminated intravascular coagulation, and maternal/fetal death.

ECOG Performance Status

- A five-point scale describing functional impairment, from 0 for fully active to 5 for death. It is widely used in clinical trials, prognostic assessment, and decisions about systemic anticancer treatment.

Ecthyma gangrenosum

- A severe cutaneous infection caused by *Pseudomonas aeruginosa*, characterized by painless, hemorrhagic vesicles that rapidly progress to necrotic ulcers with black eschars. It occurs almost exclusively in immunocompromised or neutropenic patients and results from bacterial invasion of blood vessel walls.

Ectoderm

- The outermost of the three primary germ layers in an embryo, which develops into the skin, nervous system, and sensory organs.

Ectodermal Dysplasia

- A group of inherited disorders affecting tissues derived from the embryonic ectoderm, including hair, teeth, nails, skin, and sweat glands. Findings vary but may include sparse hair, abnormal teeth, reduced sweating, heat intolerance, and recurrent infections.

Ectopia lentis

- Displacement or malposition of the crystalline lens from its normal position, most commonly due to trauma but also associated with systemic conditions including Marfan syndrome (superior displacement), homocystinuria (inferior displacement), and Weill-Marchesani syndrome.

Ectopic ACTH Syndrome

- Adrenocorticotrophic hormone secretion by non-pituitary tumors causing Cushing syndrome. Most commonly caused by small cell lung cancer and bronchial carcinoid tumors. Other sources include

thymic carcinoid, pancreatic neuroendocrine tumors, and medullary thyroid carcinoma.

Ectopic Cushing Syndrome

- Hypercortisolism caused by an extrapituitary tumor producing adrenocorticotrophic hormone or, rarely, corticotropin-releasing hormone. It may cause hypertension, diabetes, hypokalemia, muscle weakness, infection risk, and rapid features of cortisol excess.

Ectopic Heartbeat

- A heartbeat originating outside the normal sinoatrial-node rhythm. Premature atrial or ventricular beats may feel like a skipped beat, flutter, or sudden thump and can occur in healthy people or with structural heart disease.

Ectopic Pregnancy

- Implantation of a fertilized ovum outside the uterine cavity, most commonly in the fallopian tube (95%). Risk factors: prior ectopic, PID, tubal surgery, IUD, smoking, and assisted reproductive technology. Presents with lower abdominal pain, amenorrhea, and vaginal bleeding. Rupture causes hemodynamic instability.

Ectopic thyroid

- Thyroid tissue located outside its normal pretracheal position, most commonly along the path of thyroid descent from the foramen cecum to the normal cervical location. The most common site is the base of the tongue (lingual thyroid).

Ectopic Thyroid Tissue

- Thyroid tissue located outside the normal pretracheal position resulting from failure of normal thyroid descent during embryological development. Most commonly found at the base of the tongue termed lingual thyroid. May be the only functioning thyroid tissue. Symptoms include dysphagia, dysphonia, and upper airway obstruction.

Ectrodactyly

- A congenital limb malformation characterized by a deep median cleft of the hand or foot with missing central digits, resulting in a claw-like or lobster-claw appearance. Also known as split hand/foot malformation (SHFM). Most cases are autosomal dominant with incomplete penetrance, linked to mutations in TP63, DLX5, or WNT10B genes.

Ectropion

- Outward turning of the eyelid margin, most often the lower lid, exposing the conjunctiva and causing tearing, irritation, dryness, and incomplete eyelid closure. It may be involutional, paralytic, cicatricial, or mechanical; persistent exposure can damage the cornea.

Eczema

- A group of inflammatory skin conditions characterized by pruritic, erythematous, scaly patches. Atopic dermatitis is the most common form. Nummular eczema presents as coin-shaped lesions. Dyshidrotic eczema affects palms and soles with vesicles. Eczema herpeticum is a disseminated HSV infection in eczema patients, a medical emergency.

Eczema (Atopic Dermatitis)

- A chronic inflammatory skin condition characterized by dry, itchy, and inflamed patches of skin. It is driven by a combination of genetic predisposition, immune system dysfunction, and environmental triggers.

Eczema herpeticum

- A widespread viral skin infection caused by herpes simplex virus occurring in patients with pre-existing skin barrier defects, most commonly atopic dermatitis. It presents with monomorphic, umbilicated vesicles, pustules, and hemorrhagic crusted erosions, often with fever and lymphadenopathy.

Edema

- Swelling caused by excess fluid trapped in body tissues. Types: pitting (press leaves indentation), non-pitting (myxedema, lymphedema). Causes: heart failure, kidney disease, liver cirrhosis, venous insufficiency, lymphatic obstruction, medications (calcium channel blockers, NSAIDs).

Edema (historically Dropsy)

- An historic term for edema, referring to the abnormal accumulation of serous fluid in the tissues or body cavities.

Edentulism

- Complete or partial absence of natural teeth. It can impair mastication, speech, nutrition, facial support, and quality of life and may be managed with removable, fixed, or implant-supported prostheses.

Edward Syndrome

- Trisomy 18. Features: severe intellectual disability, clenched hands, rocker-bottom feet, cardiac defects, feeding difficulties. Most die within first year.

Edwards Syndrome

- Trisomy 18, the second most common autosomal trisomy in live births (1 in 5,000). Caused by complete trisomy 18 (94%), mosaicism, or partial trisomy (translocation).

EFAST Examination

- Extended focused assessment with sonography in trauma, a bedside ultrasound examination for intraperitoneal or pericardial free fluid and pneumothorax or hemothorax. It rapidly informs trauma resuscitation but does not exclude all internal injuries when negative.

Effacement

- Thinning (shortening) of the cervix during labor, expressed as a percentage (0-100 percent), as the cervix is taken up into the lower uterine segment, enabling dilation. Occurs with cervical dilation in the first stage of labor, driven by contractions and the presenting part.

Effective Dose

- A radiation-protection quantity that estimates the population-averaged stochastic health risk from nonuniform exposure by weighting absorbed doses to different organs and tissues. It is not a patient-specific prediction of cancer risk.

Effective Half-Life

- The time required for the activity of a radionuclide in a tissue or body to fall by one half through both physical radioactive decay and biologic elimination. It determines the duration of radiation exposure in that compartment.

Effective Reproduction Number (R_t)

- The average number of secondary infections caused by an infectious person at a particular time, accounting for existing immunity, behavior, interventions, and population structure. Sustained transmission generally requires R_t above 1.

Efficacy

- The maximum therapeutic effect a drug or intervention can produce under ideal conditions (e.g., in a clinical trial). Efficacy is determined by the drug's

mechanism of action and the receptor-effector system; it is measured by the E_{max} (maximal effect) on the dose-response curve.

EGD

- Esophagogastroduodenoscopy: upper endoscopy visualizing the esophagus, stomach, and duodenum using a flexible endoscope. Indications: dyspepsia with alarm features, dysphagia, GI bleeding (diagnosis and therapy—*injection, clips, banding, cautery*), biopsy (*celiac, H.*

EGFR inhibition

- A targeted cancer therapy using monoclonal antibodies or small-molecule tyrosine kinase inhibitors that block the epidermal growth factor receptor signaling pathway, inhibiting tumor cell proliferation, survival, and metastasis. Used in colorectal cancer, non-small cell lung cancer, and head and neck squamous cell carcinoma.

Egophony

- An abnormal vocal sound heard during auscultation in which spoken “E” sounds are heard as “A” (the “E-to-A” change). It is a sign of lung consolidation, as in lobar pneumonia. The consolidated lung tissue conducts high-frequency sounds better than air-filled lung.

Ehlers-Danlos Syndrome

- A group of hereditary connective tissue disorders caused by defects in collagen synthesis or processing.

EHR

- Electronic Health Record. Broader than EMR, includes population health.

Ehrlichia and Anaplasma

- Obligate intracellular bacteria transmitted by ticks that cause ehrlichiosis and anaplasmosis. *Ehrlichia chaffeensis* causes human monocytic ehrlichiosis. *Anaplasma phagocytophilum* causes human granulocytic anaplasmosis. The organisms infect white blood cells (*Ehrlichia*) or granulocytes (*Anaplasma*).

Ehrlichiosis

- A tick-borne bacterial infection caused by *Ehrlichia chaffeensis* (human monocytic ehrlichiosis) and *E. ewingii* (granulocytic). Endemic in the southeastern and south-central US. Presents with fever, headache, myalgia, malaise, and thrombocytopenia/leukopenia. Rash less common than RMSF.

Eisenmenger Syndrome

- Reversal of a left-to-right cardiac shunt due to irreversible pulmonary hypertension, resulting in right-to-left shunting and cyanosis. It occurs in untreated congenital heart defects with large left-to-right shunts (VSD, ASD, PDA) that cause progressive pulmonary vascular disease.

Eisenmenger's syndrome

- See Eisenmenger Syndrome.

Ejaculation

- The expulsion of seminal fluid from the male urethra during sexual climax, a reflex coordinated by the sympathetic nervous system (T10-L2) and involving contraction of the epididymis, vas deferens, seminal vesicles, prostate, and bulbospongiosus muscle.

Ejection Click

- A high-pitched sound occurring at the beginning of systole, caused by opening of a stenotic semilunar valve. Aortic ejection click is heard in bicuspid aortic valve and aortic stenosis with pliable leaflets. Pulmonary ejection click is heard in pulmonary stenosis and disappears as the valve becomes more calcified and immobile.

Ejection Fraction

- The percentage of blood ejected from the left ventricle during systole, calculated as (stroke volume / end-diastolic volume) x 100. Normal EF is 55-70%. Heart failure with reduced ejection fraction (HFrEF): EF 40% or less; heart failure with preserved ejection fraction (HFpEF): EF 50% or greater.

Elastic Cartilage

- Flexible cartilage containing abundant elastic fibers in its matrix. It supports structures that require shape retention with flexibility, including the external ear, epiglottis, and auditory tube.

Elastography

- Imaging tissue stiffness. Used for liver fibrosis assessment, breast lesion characterization, thyroid nodule evaluation.

Elbow Joint

- A synovial hinge complex comprising the humeroulnar and humeroradial joints. It primarily permits flexion and extension, while the proximal radioulnar

component permits forearm rotation.

Elder Abuse

- Intentional or neglectful act causing harm to older person. Types: physical, emotional, sexual, financial, neglect. Risk factors: social isolation, caregiver stress, cognitive impairment, dependency. Mandatory reporting required. Assessment: direct question, observation, collateral history.

Elderly

- Person aged 65 or older. Divisions: young-old (65-74), middle-old (75-84), old-old (85+). Heterogeneous population with varying health status, needs, preferences.

Electrical Cardioversion

- Delivery of a synchronized electrical shock timed to the R wave to terminate selected tachyarrhythmias, such as atrial fibrillation, atrial flutter, or certain supraventricular tachycardias. Sedation, thromboembolic risk, and the patient's hemodynamic status must be assessed.

Electrical Injury

- Tissue damage caused by passage of electrical current through the body. Severity depends on voltage, current (alternating vs. direct), pathway, duration, and resistance. High-voltage (>1000V): extensive internal tissue damage (muscle necrosis, compartment syndrome, rhabdomyolysis) with relatively small skin burns (entry/exit wounds).

Electrical Stimulation

- TENS, NMES for pain, muscle re-education, spasticity.

Electrocardiogram

- A recording of the heart's electrical activity through electrodes placed on the skin. It helps detect arrhythmias, conduction abnormalities, myocardial ischemia or infarction, and some structural heart conditions. Also called ECG or EKG.

Electrocardiogram (ECG / EKG)

- A recording of the heart's electrical activity using electrodes placed on the body surface. A standard 12-lead ECG provides views of the heart from different angles: limb leads (I, II, III, aVR, aVL, aVF) and precordial leads (V1-V6).

Electrocoagulation

- A surgical technique using high-frequency electri-

cal current to destroy tissue or stop bleeding by coagulating proteins, also known as diathermy or cauterization.

Electroconvulsive Therapy

- A neuromodulation treatment involving the induction of a generalized seizure under general anesthesia using electrical current applied to the scalp. It is the most effective treatment for severe depression, especially with psychotic features, catatonia, or suicidal ideation.

Electrocution

- Injury from electrical current. Types: low voltage (<1000V), high voltage (>1000V). Effects: cardiac arrhythmias, muscle contractions, burns, neurological damage. Management: ABCs, cardiac monitoring, tetanus prophylaxis, wound care, check for compartment syndrome.

Electroencephalogram

- EEG: a recording of electrical activity of the brain via scalp electrodes, reflecting cortical neuronal activity. Uses: diagnosis and classification of seizures/epilepsy (interictal epileptiform discharges, spikes/waves), status epilepticus monitoring, assessment of encephalopathy, sleep disorders, and determination of brain death.

Electroencephalogram (EEG)

- Recording of the brain's electrical activity from scalp electrodes. Background rhythms: alpha (8-13 Hz, relaxed, eyes closed, occipital), beta (13-30 Hz, alert/active), theta (4-8 Hz, drowsy/light sleep, pathological in awake adults), delta (<4 Hz, deep sleep, pathological in awake).

Electroencephalography

- A non-invasive neurophysiological test that records the electrical activity of the brain using electrodes placed on the scalp according to the international 10-20 system.

Electroencephalography (EEG)

- Recording of electrical activity from scalp electrodes. Used for seizure disorders, encephalopathy, brain death.

Electrolyte Balance

- Homeostasis of sodium, potassium, calcium, magnesium, chloride, phosphate. Regulated by kidneys, hormones (aldosterone, ADH, PTH). Imbalance

causes neuromuscular, cardiac, metabolic dysfunction.

Electrolyte Imbalance

- Abnormal serum concentration of electrolytes (sodium, potassium, calcium, magnesium, phosphate, chloride). Hyponatremia (<135 mEq/L): most common electrolyte disorder; causes include SIADH, heart failure, hypovolemia, and polydipsia. Hypernatremia (>145 mEq/L): free water deficit from poor intake, diabetes insipidus, or diuretics.

Electrolytes

- Ions in body fluids that carry an electrical charge and are essential for normal physiologic function.

Electromagnetic Navigation Bronchoscopy

- A guidance system for bronchoscopy that uses electromagnetic fields to navigate to peripheral lung lesions. It allows biopsy of lesions not visible on standard bronchoscopy. It is more accurate than conventional transbronchial biopsy for peripheral lesions. It may reduce the need for CT-guided biopsy.

Electromyography

- Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases. See urodynamic testing.

Electromyography (EMG)

- Recording of electrical activity from muscles. Used for neuropathy, myopathy, radiculopathy.

Electron Transport Chain

- Series of protein complexes (I-IV) and mobile carriers (ubiquinone, cytochrome c) in the inner mitochondrial membrane that transfer electrons from NADH and FADH₂ to oxygen, pumping protons to create electrochemical gradient used by ATP synthase. Complex I (NADH dehydrogenase): NADH → ubiquinone.

Electrophysiology Study (EPS)

- Invasive cardiac catheterization with multielectrode catheters placed in the heart to map the cardiac conduction system. Measures: sinus node function (SNRT, SACT), AV node function (AH interval, HV interval), and inducibility of arrhythmias (programmed electrical stimulation).

Electrosurgical Unit (ESU)

- A generator that delivers high-frequency electrical energy through an active electrode to cut tissue or achieve hemostasis. Monopolar and bipolar configurations differ in current pathway and safety considerations; burns, unintended tissue injury, and interference with implanted devices are possible complications.

Elephantiasis

- Massive enlargement of a body part, typically a limb or the scrotum, caused by chronic lymphedema from obstruction of lymphatic vessels. The most common cause worldwide is lymphatic filariasis, a mosquito-borne parasitic infection by *Wuchereria bancrofti*, *Brugia malayi*, or *Brugia timori*.

Elevation

- Superior movement of a body part, such as raising the mandible or shoulder girdle.

Elimination Half-Life

- Time for plasma concentration to decrease by 50%. Determines dosing interval, accumulation, withdrawal from exposure. Affected by renal/hepatic function, age, drug interactions.

Embolectomy

- Surgical removal of embolus from blood vessel. Indications: acute arterial occlusion, pulmonary embolism. Techniques: Fogarty catheter, open surgical extraction. Often combined with thrombolysis.

Embolectomy (Vascular)

- See Embolectomy Catheter.

Embolism

- An embolism is the occlusion of a blood vessel by a detached mass that has traveled through the bloodstream from its site of origin. The most common form is thromboembolism, where a fragment of a thrombus dislodges and travels to a distant site through the venous or arterial circulation.

Embolization

- Deliberate occlusion of a blood vessel or vascular malformation using materials such as coils, particles, glue, or plugs. It is used to control hemorrhage, treat aneurysms or malformations, reduce tumor vascularity, or induce organ infarction in selected therapies.

Embolization (Vascular)

- See Embolization Device.

Embolization Therapy

- Image-guided occlusion of vessels. Used for hemorrhage control, tumor devascularization, aneurysm treatment, varices.

Embryo

- The developing human organism from fertilization (conception) through the eighth week of gestation (10th week by last menstrual period).

Emergence Agitation

- A state of confusion, restlessness, and inconsolable crying during emergence from anesthesia, occurring in 10 to 80 percent of pediatric patients and 5 to 10 percent of adults. Risk factors include sevoflurane induction, short procedures, parental separation, and preexisting behavioral disorders.

Emergency Contraception

- Postcoital contraception to prevent pregnancy after unprotected intercourse or contraceptive failure. Levonorgestrel (Plan B): effective within 72 hours, available over the counter. Ulipristal acetate (Ella): effective within 120 hours, prescription only. Copper IUD: most effective emergency contraception, placed within 5 days.

Emergency Department

- Hospital department providing immediate medical care for acute illnesses and injuries. Staffed by emergency physicians and supported by specialists. Key principles: triage (prioritize by severity), stabilization, diagnosis, treatment, disposition.

Emergency Procedures

- Spill response, contamination control, radiation exposure.

Emerging Infectious Disease

- New or reemerging infectious diseases. Examples: COVID-19, Zika, Ebola, MERS.

Emesis (Vomiting)

- The forceful expulsion of gastric contents through the mouth, a complex reflex coordinated by the vomiting center in the medulla oblongata.

Emetics

- Reflex expulsion of gastric contents. See vomiting.

Emission

- The discharge or release of a substance, usually a fluid. In male reproduction, emission refers to the movement of semen into the posterior urethra in preparation for ejaculation.

Emotional Eating

- Eating in response to emotions rather than hunger. Stress, boredom, sadness triggers. Cognitive behavioral therapy may help.

Emphysema

- A form of COPD characterized by permanent, abnormal enlargement of airspaces distal to the terminal bronchioles, accompanied by destruction of their walls. Centriacinar emphysema affects the respiratory bronchioles and is associated with smoking.

Emphysematous aortitis due to Clostridium septicum

- A rare, life-threatening infection of the aortic wall by *Clostridium septicum*, characterized by gas formation within the aortic wall visible on imaging. It typically occurs in immunocompromised patients and may present with sepsis, back pain, or aortoenteric fistula.

Emphysematous cholecystitis

- A severe, gas-forming infection of the gallbladder wall caused by anaerobic organisms, most commonly *Clostridium* species and *Escherichia coli*, occurring predominantly in diabetic men. Gas within the gallbladder wall or lumen is visible on imaging. It is a surgical emergency requiring urgent cholecystectomy and broad-spectrum antibiotics.

Emphysematous hepatitis

- A rare, life-threatening gas-forming infection of the liver parenchyma, most commonly caused by *Clostridium* species or mixed enteric organisms in immunocompromised or diabetic patients. Imaging reveals gas within the liver parenchyma.

Emphysematous prostatitis

- A rare, severe gas-forming infection of the prostate gland, most commonly caused by *Escherichia coli* in diabetic or immunocompromised patients. It presents with fever, perineal pain, urinary retention, and sepsis. Imaging shows gas within the prostatic parenchyma.

Emphysematous Pyelonephritis

- Severe necrotizing infection of renal parenchyma with gas formation. Almost exclusively in diabetic patients. Presents with high fever, flank pain, and sepsis. CT shows gas within renal parenchyma. A urological emergency requiring aggressive antibiotics and usually nephrectomy. Mortality is high even with treatment.

Empty Calories

- Foods providing energy but few nutrients. Sugary drinks, candies, alcohol. Contribute to obesity, nutrient deficiency.

Empty Sella Syndrome

- Herniation of subarachnoid space into the sella turcica, compressing the pituitary gland and flattening it against the sellar floor. Primary: due to congenital weakness of the diaphragma sella. Secondary: post-surgery, post-radiation, or pituitary apoplexy (Sheehan syndrome). Most are asymptomatic.

Empyema

- Pus in the pleural space. CT shows peripheral enhancement (split pleura sign). Treatment requires chest tube drainage, intrapleural fibrinolytics, or VATS decortication.

Empyema Necessitans

- Extension of pleural empyema through the chest wall. Presents as a painful, fluctuant mass. Requires drainage and antibiotics.

EMR

- Electronic Medical Record. Patient medical records.

Enamel

- Hard, highly mineralized outer layer of tooth crown. Avascular, aneural, cannot regenerate. Enamel hypoplasia causes defects. Dental caries begins in enamel. Fluoride strengthens enamel.

Enamel Hypoplasia

- Defective enamel quantity due to systemic disturbance during development. Pits, grooves, or thin enamel.

Enamel Pearl

- Small mound of enamel on root surface. Usually bifurcation area of molars. May interfere with periodontal attachment.

Encapsulating peritoneal sclerosis

- Rare condition characterized by a thick fibro-collagenous membrane encasing the small intestine, leading to recurrent bowel obstruction. Most commonly associated with long-term peritoneal dialysis but also seen after abdominal surgery, radiation, and beta-blocker use.

Encephalitis

- Inflammation of the brain parenchyma, most commonly caused by viral infections (herpes simplex virus, arboviruses, enteroviruses). Presents with fever, headache, altered mental status, seizures, and focal neurological deficits. CSF shows lymphocytic pleocytosis. Diagnosis: CSF PCR, MRI, EEG. Treatment: acyclovir for HSV, supportive

Encephalocele

- A neural tube defect where brain tissue and meninges herniate through a skull defect, most commonly in the occipital region. The herniated sac contains cerebrospinal fluid, meninges, and often brain tissue. Associated with other CNS anomalies including hydrocephalus and Chiari malformation. Presents as a visible, pulsatile midline mass.

Encephalopathy

- Any diffuse disease of the brain that alters its structure or function, presenting with a global change in mental status (confusion, lethargy, personality change, coma) rather than focal neurologic deficits.

Encopresis

- Recurrent passage of stool in inappropriate places by a child who is developmentally old enough for bowel control. It is commonly associated with chronic constipation and fecal retention, with overflow soiling around impacted stool.

End-of-Life Care

- See hospice, palliative care.

End-Stage Kidney Disease

- The advanced stage of chronic kidney disease in which the kidneys can no longer adequately maintain fluid, electrolyte, and waste balance. Treatment requires dialysis or kidney transplantation, together with management of complications.

End-Stage Renal Disease

- The final stage of chronic kidney disease (CKD stage 5), defined by GFR <15 mL/min/1.73 m²

or the need for renal replacement therapy. Uremia develops with accumulation of toxins causing nausea, anorexia, pruritus, pericarditis, encephalopathy, and bleeding diathesis.

Endemic

- Disease constantly present in population. Examples: malaria in tropics, influenza in winter.

Endemic fungi

- Fungi that are found in specific geographic regions and cause systemic infections in immunocompetent individuals.

Endobronchial Foreign Body

- A foreign object lodged in a bronchus, causing partial or complete airway obstruction. It may cause localized wheezing, atelectasis, and post-obstructive pneumonia. It is most common in children and may be initially misdiagnosed as asthma. Bronchoscopy is both diagnostic and therapeutic.

Endobronchial Tumor

- A tumor growing within the bronchial lumen, causing obstruction, atelectasis, and post-obstructive pneumonia. It may be malignant (lung cancer, carcinoid tumor) or benign (hamartoma, papilloma). Symptoms include cough, hemoptysis, and wheezing. Treatment is bronchoscopic removal or debulking plus definitive therapy.

Endobronchial Ultrasound

- Bronchoscopy with ultrasound guidance for trans-bronchial needle aspiration of mediastinal and hilar lymph nodes. Essential for lung cancer staging. Less invasive than mediastinoscopy.

Endobronchial Valve

- A one-way valve placed bronchoscopically in a target airway to achieve lobar atelectasis in emphysema. It allows air and secretions to exit but prevents air from entering. It is used for bronchoscopic lung volume reduction in selected patients with heterogeneous emphysema and intact fissures.

Endocardial Cushion Defect

- A congenital heart defect caused by abnormal development of the atrioventricular cushions, producing an atrioventricular septal defect with abnormal atrioventricular valves. It is strongly associated with Down syndrome and may cause heart failure and pulmonary hypertension.

Endocarditis

- Infection of the endocardial surface of the heart, most commonly involving the valves. Acute bacterial endocarditis: *Staph aureus* (most common in IV drug users and prosthetic valves), rapid destruction. Subacute: viridans group streptococci (pre-existing valve disease).

Endocarditis Prophylaxis

- Infective endocarditis prophylaxis with antibiotics is recommended for patients at highest risk of adverse outcomes from endocarditis who are undergoing dental procedures that involve manipulation of gingival tissue, the periapical region of teeth, or perforation of the oral mucosa.

Endocarditis Surgical Indications

- Surgery indicated for: heart failure from valve dysfunction, uncontrolled infection (abscess, fistula, vegetation >10 mm with persistent emboli), fungal endocarditis, prosthetic valve dehiscence, and large vegetations (>15 mm) with embolic events.

Endocardium

- The innermost layer of the heart wall, a thin membrane of endothelial cells and underlying connective tissue that lines the cardiac chambers and covers the heart valves. The endocardium is continuous with the endothelium of blood vessels and provides a smooth, non-thrombogenic surface.

Endocrine Disrupting Chemicals

- Environmental chemicals that interfere with endocrine function including bisphenol A, phthalates, pesticides, and polychlorinated biphenyls. Potential associations with diabetes, obesity, thyroid dysfunction, reproductive disorders, and hormone-sensitive cancers. Evidence is strongest for associations with type two diabetes and obesity.

Endocrine Disruptor

- Chemical interfering with hormone systems. Examples: BPA, phthalates, DDT, PCBs. Effects: reproductive disorders, developmental problems, cancer.

Endocrine Glands

- Ductless glands that secrete hormones directly into the bloodstream to regulate distant target organs.

Endocrine Therapy

- Systemic treatment that manipulates hormones or their receptors in hormone-sensitive cancers, in-

cluding tamoxifen, aromatase inhibitors, luteinizing hormone-releasing hormone agonists, and antiandrogens.

Endocrinology

- The branch of medicine concerned with the study, diagnosis, and treatment of disorders of the endocrine system—the network of glands that secrete hormones regulating metabolism, growth and development, tissue function, sexual function, reproduction, sleep, and mood.

Endocytosis

- Process by which cells internalize extracellular material by engulfing it with the cell membrane, forming intracellular vesicles. Phagocytosis ("cell eating") involves engulfing large particles (>0.5 micrometers) such as bacteria or debris, forming a phagosome that fuses with lysosomes for digestion.

Endoderm

- The innermost germ layer in an embryo, which develops into the linings of the digestive and respiratory tracts, as well as the liver and pancreas.

Endodontic Therapy (Root Canal)

- Treatment of dental pulp disease or necrosis. Indications: irreversible pulpitis, pulp necrosis, periapical abscess. Procedure: access cavity, locate canals, clean and shape, obturate with gutta-percha, seal. Success rate: >90%. Alternative: extraction.

Endoleak

- Persistent blood flow into an aneurysm sac after endovascular aneurysm repair. Type I and III endoleaks reflect inadequate sealing or device failure and generally require urgent correction; other types are managed according to sac expansion and risk.

Endometrial Cancer

- The most common gynecologic malignancy in developed countries, arising from the lining of the uterine cavity, typically in postmenopausal women.

Endometrioma

- A chocolate cyst of the ovary formed by ectopic endometrial tissue that hemorrhages cyclically, appearing as a homogenous low-level echo mass on ultrasound and associated with chronic pelvic pain and infertility.

Endometriosis

- A chronic, estrogen-dependent gynecologic disorder

characterized by the presence of endometrial-like tissue (glands and stroma) outside the uterine cavity, most commonly implanted on the pelvic peritoneum, ovaries, uterosacral ligaments, and rectovaginal septum.

Endometritis

- Infection of the endometrium, the most common cause of puerperal fever after cesarean delivery. Risk factors include prolonged labor, multiple vaginal examinations, premature rupture of membranes, and cesarean section. Presents with fever, uterine tenderness, foul-smelling lochia, and leukocytosis.

Endometrium

- The lining of the uterus.

Endophthalmitis

- A severe, sight-threatening intraocular inflammation involving the vitreous cavity, aqueous humor, and adjacent structures, representing an ophthalmologic emergency.

Endoplasmic Reticulum

- Interconnected network of membranous tubules and sacs continuous with the nuclear envelope. Rough ER: studded with ribosomes, site of synthesis of secretory, membrane, and lysosomal proteins; co-translational insertion into ER lumen.

Endoprosthesis

- Internal prosthetic device. Vascular stent-grafts, airway stents.

Endorphin

- Endogenous opioid peptides (beta-endorphin, enkephalins, dynorphins) produced by the pituitary, hypothalamus, and immune cells that bind opioid receptors (mu, delta, kappa) to produce analgesia, euphoria, and modulation of pain perception and stress responses.

Endoscope

- Flexible or rigid tube with camera for visualizing internal structures. Types: bronchoscope, colonoscope, gastroscope, arthroscope, cystoscope. Essential for diagnosis and intervention.

Endoscopic Mucosal Resection

- Advanced endoscopic technique for removal of superficial gastrointestinal neoplasms by lifting the lesion through submucosal injection and resecting with an electrosurgical snare.

Endoscopic Retrograde Cholangiopancreatography

- Combined endoscopic and fluoroscopic procedure used to visualize and treat disorders of the biliary and pancreatic ducts. A duodenoscope is advanced to the second portion of the duodenum and the ampulla of Vater is cannulated. Contrast dye is injected under fluoroscopic guidance creating cholangiograms and pancreatograms.

Endoscopic Submucosal Dissection

- Advanced endoscopic technique for en bloc resection of large gastrointestinal neoplasms using specialized electrosurgical knives to dissect through the submucosa. The technique allows complete histological assessment and is particularly valuable for early gastric and colorectal cancers.

Endoscopic Ultrasound

- Diagnostic and therapeutic endoscopic procedure combining endoscopy with high-frequency ultrasound to provide detailed imaging of the gastrointestinal wall layers and adjacent organs.

Endoscopic Ultrasound (EUS)

- Combined endoscopy and ultrasound for high-resolution imaging of the GI wall and adjacent structures. Linear echoendoscope: allows FNA of mediastinal, pancreatic, and peripancreatic lesions. Radial echoendoscope: 360-degree view for staging GI cancers (TNM staging).

Endoscopy

- Visualization of the gastrointestinal tract using flexible fiber-optic or video endoscopes. Upper endoscopy examines the esophagus, stomach, and duodenum. Colonoscopy examines the colon and terminal ileum. Both allow direct visualization, biopsy, and therapeutic interventions including polypectomy, hemostasis, and dilation.

Endosteum

- The thin connective tissue membrane lining the inner surface of bone, including the medullary cavity, Haversian canals, and trabeculae. Contains osteoprogenitor cells and osteoblasts, contributing to bone formation, remodeling, and repair (fracture healing).

Endothelin Receptor Antagonists

- Medications that block endothelin-1, a potent vaso-

constrictor implicated in pulmonary arterial hypertension. Bosentan (dual ERA), ambrisentan (selective ETA), and macitentan are used for PAH. Side effects include hepatotoxicity (bosentan), anemia, and teratogenicity.

Endotoxin

- Lipopolysaccharide (LPS) located in the outer membrane of gram-negative bacteria, released on cell lysis or death; the classic pyrogen. Composed of lipid A (active/toxic component), core polysaccharide, and O antigen.

Endotoxins

- Lipopolysaccharide components of gram-negative bacterial outer membranes released during cell lysis or bacterial growth. Endotoxin is composed of lipid A (the toxic component), core oligosaccharide, and O-antigen. It activates macrophages through toll-like receptor 4, leading to cytokine release.

Endotracheal Intubation

- Placement of a cuffed tube through the mouth or nose into the trachea to maintain airway patency, deliver positive-pressure ventilation, administer inhaled anesthetics, and protect the airway in selected patients. Correct position must be confirmed clinically and with exhaled carbon dioxide when circulation is present.

Endotracheal Tube

- Tube for airway management. Sizes: 6.0-8.0 mm internal diameter. Cuffed for ventilation.

Endovascular

- Minimally invasive vascular procedure via catheter. Examples: angioplasty, stenting, aneurysm repair.

Endovascular Aneurysm Repair (EVAR)

- Minimally invasive technique for abdominal aortic aneurysm repair using stent graft. Accessed via femoral arteries. Advantages: lower perioperative mortality, faster recovery. Disadvantages: need for surveillance, endoleak risk, may require reintervention.

Endovascular Graft

- Stent-graft for aortic aneurysm repair (EVAR/TEVAR). Fabric tube on metal stent frame. Delivered via femoral arteries. Indications: AAA >5.5 cm, symptomatic aneurysm, descending thoracic aneurysm. Surveillance: CT angiography

annually.

Endovascular Repair

- See EVAR.

Endurance

- Aerobic capacity, muscular endurance. Cardiorespiratory fitness testing.

Enema

- The introduction of fluid into the rectum and colon through the anus for diagnostic or therapeutic purposes.

Energy Conservation

- Strategies that reduce unnecessary energy expenditure and fatigue during daily activities, such as pacing, prioritizing, planning, positioning, and using adaptive equipment.

Energy Resolution

- Ability to distinguish different gamma energies. FWHM of photopeak. NaI(Tl): 10% at 140 keV.

Energy Window

- Range of photon energies accepted by gamma camera. Typically 20% centered on photopeak. Reduces scatter, improves image quality.

Enfuvirtide

- A fusion inhibitor used in combination antiretroviral therapy for HIV-1 infection in treatment-experienced patients with multidrug-resistant virus.

Engagement

- The descent of the fetal presenting part into the pelvis, typically occurring 2-3 weeks before labor in nulliparas or at the onset of labor in multiparas.

Enhanced Recovery After Surgery (ERAS)

- A multimodal, evidence-based perioperative pathway that reduces surgical stress and supports early recovery through preoperative counseling, appropriate fasting, opioid-sparing analgesia, early mobilization, early nutrition, and prevention of complications.

Enhancement Artifact

- Ultrasound artifact from fluid-filled structures. Enhanced by harmonic imaging.

Enhancer

- DNA sequence increasing transcription of tar-

get gene, can be distant from gene, orientation-independent. Binds activator proteins. Examples: LCR (locus control region) for beta-globin.

Enoxaparin

- A low-molecular-weight heparin inhibiting factor Xa (and to a lesser extent thrombin) via antithrombin. Used for prophylaxis and treatment of deep vein thrombosis and pulmonary embolism, acute coronary syndromes, and as a bridge during anticoagulation interruption.

Enrichment

- Restoration of nutrients lost during processing. Examples: enriched flour, restored B vitamins.

Entacapone

- A peripheral COMT (catechol-O-methyltransferase) inhibitor used as an adjunct to levodopa/carbidopa in Parkinson disease, extending levodopa's duration of action and reducing 'wearing-off'. Does not cross the blood-brain barrier.

Entamoeba histolytica

- A protozoan parasite that causes amoebiasis, a gastrointestinal and extra-intestinal infection transmitted via the fecal-oral route. The organism exists in two forms: the infectious cyst (transmitted in contaminated food or water) and the invasive trophozoite.

Enteral Formula

- A nutritionally complete or modular liquid preparation delivered into the gastrointestinal tract. Formulas may be polymeric, semi-elemental, or elemental and are selected according to digestion, absorption, fluid, renal, hepatic, and metabolic requirements.

Enteral Nutrition

- Delivery of nutrients via gastrointestinal tract. Routes: nasogastric, nasoduodenal, gastrostomy (PEG), jejunostomy. Indications: dysphagia, neurological disease, head/neck cancer, prolonged inability to eat. Complications: aspiration, diarrhea, tube displacement, sinusitis.

Enteric duplication cyst

- A congenital, smooth-muscle-lined cyst arising from the gastrointestinal tract, most commonly in the ileum, containing ectopic gastric or intestinal mucosa. It may present in childhood with abdominal pain, obstruction, or gastrointestinal bleeding from

ectopic gastric acid secretion.

Enteritis

- Inflammation of the small intestine, most commonly caused by infectious agents (viral, bacterial, or parasitic). Viral enteritis (norovirus, rotavirus, adenovirus, astrovirus) presents with acute onset of watery diarrhea, nausea, vomiting, and abdominal cramps, typically self-limited over 2–7 days.

Enterobacteriaceae carbapenem resistance

- Resistance to carbapenem antibiotics in Enterobacteriaceae, primarily through production of carbapenemases. The most important carbapenemases include KPC, NDM, and OXA-48-like enzymes. Carbapenem-resistant Enterobacteriaceae (CRE) are a major public health threat due to limited treatment options.

Enterobiasis

- See Pinworm (Threadworm).

Enterobius vermicularis

- See Pinworm (Threadworm).

Enterocolitis

- Inflammation of both the small intestine and colon, caused by infection, immune-mediated disease, reduced blood flow, toxins, or other injury. It may produce abdominal pain, diarrhoea, fever, vomiting, bleeding, dehydration, or systemic illness; diagnosis and management depend on the cause and severity.

Enterohepatic Circulation

- Recycling of toxins between liver and intestine via bile. Prolonged exposure, increased toxicity. Interrupted by activated charcoal, cholestyramine. Examples: morphine, digoxin, estrogen, thyroxine.

Enteropathic Arthritis

- Spondyloarthritis associated with inflammatory bowel disease (Crohn disease, ulcerative colitis), affecting 10-20% of IBD patients.

Enterotoxin

- A toxin acting on the intestinal tract to cause diarrhea and vomiting by altering epithelial fluid secretion. Examples include cholera toxin, enterotoxigenic *Escherichia coli* toxins, and *Staphylococcus aureus* enterotoxins causing food poisoning; some, such as enterotoxin B, also function as superantigens.

Enterovirus D68

- A respiratory enterovirus that usually causes a mild cold-like illness but can produce severe respiratory disease and, rarely, acute flaccid myelitis. Children with asthma or other respiratory conditions are at increased risk of severe illness.

Enthesis

- Attachment site of tendon/ligament to bone. Implant fixation often targets entheses.

Enthesitis

- Inflammation at the site where a tendon, ligament, or joint capsule inserts into bone (entheses), a hallmark of spondyloarthritis (ankylosing spondylitis, psoriatic arthritis, reactive arthritis, IBD-associated). Affects Achilles tendon, plantar fascia, patellar, and costochondral junctions.

Entrance Wound

- A skin defect created where a projectile enters the body, distinguished from an exit wound by smaller size, rounded shape, and abraded inward margin. Contact wounds show a muzzle imprint and soot; close-range wounds show soot and powder tattooing; intermediate-range wounds show stippling; distant wounds show a clean abrasion ring.

Entropion

- An eyelid malposition in which the eyelid margin rolls inward, causing the eyelashes and skin to rub against the ocular surface, resulting in corneal irritation, conjunctival injection, tearing, photophobia, and a foreign body sensation. The condition most commonly affects the lower eyelid and is classified into four types.

Enuresis

- Repeated involuntary urination, usually during sleep in a child who is developmentally expected to remain dry. It may be primary or secondary and requires assessment for constipation, urinary infection, diabetes, sleep or developmental disorders, psychosocial stressors, and lower-urinary-tract dysfunction when clinically indicated.

Enuresis (Bedwetting)

- Involuntary urination during sleep in a child aged 5 years or older, affecting approximately 5–10% of 7-year-olds. Primary enuresis: the child has never achieved sustained nighttime dryness. Secondary

enuresis: wetting recurs after at least 6 months of dryness, often triggered by psychosocial stress, urinary tract infection, or

Environmental Health

- Study of environmental factors affecting health. Examples: air pollution, water quality, occupational hazards, climate change.

Enzyme

- Biological catalyst (protein or RNA) that accelerates chemical reactions by lowering activation energy. Specificity: lock-and-key or induced fit models. Kinetics: Michaelis-Menten equation (V_{max} , K_m). Regulation: allosteric control, covalent modification (phosphorylation), proteolytic activation, feedback inhibition.

Enzyme Induction

- A process by which a drug or xenobiotic increases the synthesis or activity of drug-metabolizing enzymes, primarily cytochrome P450 enzymes in the liver, accelerating the metabolism of itself (autoinduction) or other drugs. Enzyme induction typically occurs over days to weeks as it requires new protein synthesis.

Enzyme Inhibition

- A process by which a drug inhibits the activity of drug-metabolizing enzymes, leading to increased plasma concentrations of co-administered drugs that are metabolized by the same enzyme. Enzyme inhibition occurs rapidly (hours to days) as it does not require new protein synthesis.

Enzyme Kinetics

- Study of reaction rates and factors affecting them. Michaelis-Menten kinetics: $V = V_{max}[S]/(K_m + [S])$. K_m (Michaelis constant): substrate concentration at half V_{max} , inversely related to enzyme affinity. V_{max} : maximum reaction rate at saturating substrate. Lineweaver-Burk plot: double-reciprocal plot ($1/V$ vs.

Enzyme-Linked Immunosorbent Assay (ELISA)

- An immunoassay that uses an enzyme-labeled antibody or antigen and a substrate-generated signal to detect or quantify a target analyte. It is used for measurement of antibodies, antigens, hormones, cytokines, and infectious disease markers.

Eosinophil

- A granulocyte (1-4 percent of leukocytes) with bilobed nucleus and eosinophilic granules containing major basic protein, eosinophil cationic protein, and peroxidase. Roles: defense against parasitic (helminth) infections and modulation of allergic inflammation; contain histaminase and aryl sulfatase.

Eosinophilia

- An absolute eosinophil count exceeding 500 cells per microliter in peripheral blood, most commonly caused by parasitic infections, allergic disorders, drug reactions, and autoimmune diseases. Hypereosinophilia, with counts above 1500 per microliter, may cause end-organ damage to the heart, lungs, skin, and nervous system.

Eosinophilic bronchitis

- Inflammation of the bronchial airways with eosinophilic infiltration causing chronic cough, with normal spirometry and no airway hyperresponsiveness, distinguishing it from asthma.

Eosinophilic Esophagitis

- Chronic immune-mediated condition characterized by dense eosinophilic infiltration of the esophageal mucosa defined as fifteen or more eosinophils per high-power field, in the absence of other causes of esophageal eosinophilia.

Eosinophilic fasciitis

- A rare connective tissue disorder characterized by symmetrical inflammation and thickening of the deep fascia and adjacent muscle, presenting with painful swelling and induration of the extremities and a peau d'orange skin appearance. Peripheral eosinophilia and elevated inflammatory markers are typical.

Eosinophilic Gastroenteritis

- Rare chronic inflammatory condition characterized by eosinophilic infiltration of the gastrointestinal tract beyond the esophagus involving any segment from stomach to rectum and extending beyond the mucosa to involve the muscularis propria or serosa.

Eosinophilic Granulomatosis with Polyangiitis

- EGPA (formerly Churg-Strauss syndrome): an ANCA-associated vasculitis characterized by asthma, peripheral and tissue eosinophilia, and

necrotizing vasculitis.

Ependymoma

- A glioma arising from the ependymal cells lining the ventricles and spinal canal; more common in children and frequently associated with hydrocephalus.

Epicardium

- The outermost layer of the heart wall, which is the visceral layer of the serous pericardium. It consists of a single layer of mesothelial cells and underlying connective tissue containing blood vessels, nerves, and adipose tissue. The epicardium produces pericardial fluid that lubricates the heart surface.

Epidemic

- Sudden increase in disease cases above expected level. Examples: cholera outbreak, measles outbreak.

Epidemiology

- Study of distribution and determinants of health-related states in populations. Branches: descriptive (who, when, where), analytic (why, how), experimental (intervention trials). Measures: incidence, prevalence, mortality rates, case-fatality rate.

Epidermal Inclusion Cyst

- A common benign cyst lined by stratified squamous epithelium with keratin debris. It presents as a firm, mobile, subcutaneous nodule, typically on the face, neck, and back. A central punctum is characteristic. It may become inflamed or infected. Treatment includes complete excision (including the cyst wall) to prevent recurrence.

Epidermal Layers

- The epidermis has five layers from deep to superficial: stratum basale (germinative layer with keratinocyte stem cells and melanocytes), stratum spinosum (prickle cell layer with desmosomes), stratum granulosum (keratohyalin granules), stratum lucidum (transparent layer, only in thick skin), and stratum corneum (dead, flattened keratinocytes).

Epidermis

- Outermost skin layer, stratified squamous keratinized epithelium approximately 0.1 mm thick, with five layers (superficial to deep): stratum corneum (dead keratinized cells), lucidum (clear layer, thick skin only), granulosum (keratinization begins), spinosum (spiny layer, Langerhans cells),

basale (stem cells, melanocytes, Merkel cells).

Epidermoid Cyst

- A benign, slow-growing cyst lined by squamous epithelium and filled with keratin, usually presenting as a firm subcutaneous nodule. Rupture can cause inflammation, tenderness, and a foul-smelling discharge.

Epidermolysis Bullosa

- A group of genetic disorders causing skin fragility and blistering with minimal trauma. Simplex (intraepidermal blistering), junctional (blistering at lamina lucida), dystrophic (sublamina densa blistering), and Kindler syndrome (mixed). Dystrophic EB can cause scarring, contractures, and squamous cell carcinoma.

Epididymal Cyst

- Simple fluid-filled cyst in epididymis. Transilluminates. Asymptomatic, requires no treatment unless large or symptomatic.

Epididymis

- Coiled duct approximately 6 meters long on the posterolateral aspect of each testis, divided into head (caput, receiving efferent tubules from the testis), body (corpus), and tail (cauda, continuing as the vas deferens). Functions include sperm maturation (gaining motility over approximately 12 days), storage (in the tail), and transport.

Epididymitis

- Inflammation of the epididymis, a common cause of acute scrotal pain in adults, typically resulting from infection. In sexually active men under 35 years, the most common pathogens are sexually transmitted: *Chlamydia trachomatis* (most common) and *Neisseria gonorrhoeae*, with epididymitis often accompanied by urethritis.

Epidural

- Regional analgesia (or anesthesia) delivering local anesthetic/opioid via a catheter in the epidural space (lumbar), used for pain relief during labor and for cesarean (higher dose) anesthesia. Provides excellent analgesia while allowing maternal mobility and pushing; reduces catecholamine stress.

Epidural Abscess

- A collection of infected material between the dura mater and the skull or vertebral canal, capable of

compressing the brain, spinal cord, or nerve roots. Spinal epidural abscess may cause severe focal back pain, fever, neurological deficit, or sphincter dysfunction and requires urgent imaging and specialist assessment.

Epidural Analgesia

- Epidural analgesia provides segmental pain relief by infusing local anesthetic (bupivacaine 0.1 to 0.2 percent) with or without opioids (fentanyl 2 to 4 mcg/mL, hydromorphone 10 to 20 mcg/mL) through an epidural catheter. Indications include major abdominal, thoracic, and orthopedic surgery, and labor analgesia.

Epidural Analgesia for Labor

- A neuraxial technique for labor analgesia: a catheter is placed in the epidural space (L3-L4 or L2-L3 interspace) and local anesthetic (bupivacaine 0.0625-0.125%) with opioid (fentanyl 2 µg/mL) is infused continuously or in patient-controlled epidural analgesia (PCEA) boluses.

Epidural Anesthesia

- A regional anesthetic technique involving injection of local anesthetic and/or opioid into the epidural space (the potential space between the dura mater and the ligamentum flavum within the spinal canal), producing segmental analgesia and sympathetic blockade.

Epidural Blood Patch

- A minimally invasive procedure used to treat post-dural puncture headache (PDPH) resulting from accidental dural puncture during epidural placement or spinal anesthesia. It involves injecting a small volume of the patient's autologous blood (typically 15-20 mL) into the epidural space at or near the site of the previous puncture.

Epidural Hematoma

- Accumulation of blood between the skull and the dura mater, typically from arterial bleeding (middle meningeal artery) following temporal bone fracture. Presents with loss of consciousness, then a lucid interval (hours), followed by rapid neurological deterioration due to mass effect.

Epi gastric Region

- Central superior region between the hypochondriac regions and above the umbilical region. Overlies the

stomach, duodenum, pancreas, and portions of the liver. The aorta and inferior vena cava pass through it. Epigastric pain associates with gastric ulcers, pancreatitis, GERD, and aortic aneurysms.

Epigenetics

- Heritable changes in gene expression without DNA sequence alteration. Mechanisms: DNA methylation, histone modification, non-coding RNA. Influenced by environment, diet, stress. Reversible.

Epiglottis

- A leaf-shaped elastic cartilage at the base of the tongue that covers the laryngeal inlet during swallowing, directing food and liquid into the esophagus and preventing aspiration into the airway. Elevates during swallowing (with the larynx) under glossopharyngeal/vagal innervation.

Epiglottitis

- Inflammation of the epiglottis, a life-threatening emergency. Most commonly caused by *Haemophilus influenzae* type b (now rare due to vaccination). Presents with high fever, drooling, dysphagia, tripod position, and rapid progression to airway obstruction. Do not examine the throat.

Epilepsy

- A neurological disorder characterized by recurrent, unprovoked seizures due to abnormal electrical activity in the brain. Classification includes focal, generalized, and unknown onset seizures. Etiology: genetic, structural, metabolic, infectious, immune. Diagnosis: clinical history, EEG, neuroimaging.

Epilepsy Surgery

- Indicated for patients with drug-resistant epilepsy who have a concordant, resectable seizure focus identified on comprehensive presurgical evaluation. Failure of two or more appropriately chosen and adequately dosed antiepileptic drugs to achieve sustained seizure freedom.

Epinephrin (Adrenaline)

- Alternate spelling of epinephrine (adrenaline). See Epinephrine (Adrenaline).

Epinephrine

- Fight-or-flight hormone. Increases heart rate, blood glucose, lipolysis.

Epinephrine (Adrenaline)

- A catecholamine hormone and neurotransmitter

produced by the adrenal medulla and sympathetic nerve terminals, acting on alpha-1, alpha-2, beta-1, beta-2, and beta-3 adrenergic receptors.

Epiphora

- Excessive tearing caused by impaired tear drainage or reflex hypersecretion. Causes: nasolacrimal duct obstruction (most common in infants and elderly), punctal stenosis, eyelid malposition (ectropion, entropion), and reflex tearing from dry eye or ocular surface irritation.

Epiphyseal Plate

- A layer of hyaline cartilage, also called the growth plate or physis, located between the epiphysis and metaphysis of a growing long bone. It permits longitudinal bone growth through endochondral ossification, progressing from resting cartilage through proliferation and hypertrophy to calcification and replacement by bone.

Epiphysis

- The expanded end of a long bone, usually covered by articular cartilage where it forms a synovial joint. It contains spongy (trabecular) bone with a thin outer shell of compact bone and contributes to the shape and function of the joint.

Episcleritis

- A benign, self-limited inflammatory condition of the episclera, the thin layer of connective tissue between the sclera and conjunctiva.

Episiotomy

- A surgical incision of the perineum to enlarge the vaginal outlet during delivery. Two types: mediolateral (most common worldwide) and median/midline. Previously performed routinely; current evidence supports selective use. Indications include operative vaginal delivery, fetal distress requiring expedited delivery, and shoulder dystocia.

Episiotomy Repair

- Surgical repair of an episiotomy incision after delivery. Technique: continuous suturing of the vaginal mucosa (2-0 chromic), interrupted sutures for the perineal muscles, and subcuticular closure of the perineal skin. The rectum should be examined after repair (suture inadvertently through the rectal mucosa).

Epispadias

- A congenital anomaly where the urethral meatus opens on the dorsal (top) surface of the penis in boys or in a split, open position on the clitoral hood in girls. Caused by abnormal development of the genital tubercle and cloacal membrane.

Epistasis

- Interaction between genes where one gene masks or modifies expression of another. Examples: coat color in mice, ABO blood group modifications.

Epistaxis

- Epistaxis, commonly known as a nosebleed, is hemorrhage from the nasal mucosa that can range from minor, self-limited bleeding to severe, lifethreatening hemorrhage requiring emergency intervention, and it is a common presenting complaint in both primary care and emergency departments.

Epithelial Tissue

- The tissue that covers external body surfaces, lines internal cavities and pathways, and forms glands.

Epithelium

- One of the four primary tissue types, composed of cells that line the surfaces of the body (skin, mucous membranes, serous membranes) and form glands.

Eplerenone

- A selective aldosterone (mineralocorticoid) receptor antagonist, a potassium-sparing diuretic. Used to reduce mortality in heart failure with reduced ejection fraction (post-MI and chronic HFrEF, NYHA II-IV) and for resistant hypertension. More selective than spironolactone with less gynecomastia/breast tenderness.

Epley Maneuver

- The Epley maneuver, also known as the canalith repositioning maneuver, is a therapeutic procedure used to treat posterior canal benign paroxysmal positional vertigo by repositioning displaced calcium carbonate crystals (otoconia) from the posterior semicircular canal back into the utricle through a series of sequential head and body

Epstein Pearls

- Small white cysts on palate of newborn. Keratin cysts. Resolve spontaneously.

Epstein-Barr Virus

- EBV (human herpesvirus 4): ubiquitous virus causing infectious mononucleosis, transmitted via saliva ('kissing disease').

Eravacycline

- A fluorocycline antibiotic with activity against gram-positive and gram-negative organisms, including multidrug-resistant strains. Eravacycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It has activity against MRSA, VRE, ESBL-producing Enterobacteriaceae, and anaerobes.

Erb's Palsy

- Upper brachial plexus injury (C5-C6) causing weakness or paralysis of the shoulder and arm muscles. Presents with the characteristic waiter's tip position: arm adducted, elbow extended, forearm pronated, wrist flexed. Most commonly caused by shoulder dystocia during delivery. See also Brachial Plexus Injury.

Erdheim-Chester disease

- A rare, non-Langerhans cell histiocytosis characterized by xanthogranulomatous infiltration of multiple organ systems, most commonly the long bones, retroperitoneum, and cardiovascular system. It causes symmetric osteosclerosis of the diaphyses and metaphyses of long bones, exophthalmos, and xanthelasma.

Erectile Disorder

- A sexual dysfunction characterized by persistent or recurrent difficulty achieving or maintaining an erection sufficient for sexual activity, or a marked decrease in erectile rigidity, lasting 6 months and causing marked distress. Prevalence: 10-20% of men under 60, rising to 40-70% over age 70.

Erectile Dysfunction

- The persistent inability to achieve or maintain an erection sufficient for satisfactory sexual performance. Affects up to 50% of men aged 40-70. Etiology: organic (vascular disease, diabetes, hypogonadism, neurological disorders, medications, Peyronie disease) and psychogenic (anxiety, depression, relationship issues).

Erectile Dysfunction (ED)

- Inability to attain or maintain a penile erection

sufficient for satisfactory sexual performance.

Ergonomics

- Design for human use. Workstation setup, tool design, injury prevention.

Ergotism

- A toxidrome caused by prolonged exposure to ergot alkaloids from the fungus *Claviceps purpurea* contaminating rye and other grains, or from ergot-derived medications.

Erosion

- Partial-thickness loss of epidermis that does not extend through the entire epidermis and therefore usually heals without scarring. It may result from trauma, blister rupture, maceration, or inflammatory disease.

Erosion (Tooth)

- Chemical wear without bacterial involvement. Causes: acidic foods/drinks, gastric acid (GERD, bulimia), environmental acids. Management: dietary modification, fluoride, restoration.

Erosive Gastritis

- Inflammation of the stomach lining with superficial erosions but no full-thickness ulceration. Commonly caused by NSAIDs, alcohol, or critical illness. May present with epigastric pain or gastrointestinal bleeding. Diagnosis is by upper endoscopy showing erosions and erythema.

Eruption Cyst

- Fluid-filled swelling over erupting tooth. Blue or purple color. Resolves spontaneously.

Eruptive xanthomas

- Sudden eruption of numerous small, yellow-orange papules with an erythematous halo, typically on the buttocks, elbows, knees, and trunk, caused by severe hypertriglyceridemia. They represent lipid-laden macrophages in the dermis and are a marker of significantly elevated serum triglycerides, often exceeding 1000 mg/dL.

Erysipelas

- A superficial form of cellulitis involving the upper dermis and superficial lymphatics, most commonly caused by *Streptococcus pyogenes*. It presents as a bright red, well-demarcated, raised plaque with sharp borders, typically on the face or lower extremities. Fever and chills precede the rash.

Erythema

- Redness of the skin or mucous membranes caused by increased blood flow to dilated superficial capillaries, often a sign of inflammation, infection, or other underlying conditions. Erythema blanch on pressure (distinguishes it from purpura, which does not blanch).

Erythema ab igne

- A reticulated, hyperpigmented erythematous patch caused by chronic, repeated exposure to moderate heat or infrared radiation, such as from heating pads, laptops, or fireplaces. Histologically it shows increased melanin and dermal amyloid deposition.

Erythema gyratum repens

- A distinctive, rapidly moving, concentric erythematous eruption with a wood-grain or serpiginous pattern, which is almost always a cutaneous paraneoplastic marker. The lesions migrate daily and are most commonly associated with underlying malignancy, particularly lung cancer.

Erythema Infectiosum

- A common childhood exanthem caused by parvovirus B19, also called fifth disease. A mild prodrome is followed by a slapped-cheek facial eruption and a lacy rash on the trunk and limbs.

Erythema marginatum

- A migratory, non-pruritic, erythematous rash with a serpiginous or annular pattern that blanches centrally, occurring as a major Jones criterion in acute rheumatic fever. It typically appears on the trunk and proximal extremities and may come and go over days or weeks.

Erythema Migrans

- The expanding annular skin lesion of early Lyme disease, appearing days to weeks after infection with *Borrelia* transmitted by Ixodes ticks. It may be uniformly erythematous rather than a classic bull's-eye lesion and is usually diagnosed clinically.

Erythema Multiforme

- An acute, self-limited immune-mediated reaction, most commonly triggered by herpes simplex virus (90 percent of cases) and *Mycoplasma pneumoniae*. It presents as target lesions (iris lesions with three zones: central duskiness, middle pale edematous ring, outer erythematous halo) on extremities.

Erythema Nodosum

- The most common form of panniculitis, characterized by painful, erythematous, tender nodules and plaques on the bilateral shins, representing an inflammatory reaction of the subcutaneous fat.

Erythrasma

- A superficial infection caused by *Corynebacterium minutissimum* in moist intertriginous areas, producing well-demarcated reddish-brown patches with fine scale and sometimes mild pruritus. Wood-lamp examination may show coral-red fluorescence.

Erythrocyte

- A mature red blood cell specialized to transport oxygen and carbon dioxide through the circulation. Its biconcave, anucleate form maximizes surface area and flexibility; haemoglobin provides the principal oxygen-binding capacity.

Erythrocyte Sedimentation Rate

- ESR: a nonspecific inflammatory marker measuring the rate of red blood cell sedimentation in a vertical tube over 1 hour (mm/h), elevated by inflammation, infection, malignancy, and anemia. Extremely elevated (>100 mm/h): myeloma, giant cell arteritis/polymyalgia rheumatica, infection, SLE, malignancy.

Erythrocyte Sedimentation Rate (ESR)

- A non-specific inflammatory marker measuring the rate at which erythrocytes settle in 1 hour. Normals: males $\text{age}/2$, females $(\text{age}+10)/2$. Markedly elevated >100 mm/hr: suggests infection (endocarditis), autoimmune (giant cell arteritis, polymyalgia rheumatica, SLE, RA), malignancy (multiple myeloma), and chronic kidney disease.

Erythrocytosis

- An increased red-cell mass, usually reflected by elevated haemoglobin and haematocrit. It may be relative because of reduced plasma volume or absolute because of increased erythropoiesis, which can be primary from a marrow disorder or secondary to hypoxia, erythropoietin production, or other causes.

Erythrodermic psoriasis

- A severe, life-threatening variant of psoriasis characterised by widespread erythema and scaling involving more than 90% of the body surface area, often with systemic symptoms including fever, chills,

and hemodynamic instability.

Erythromelalgia

- Episodic burning pain, redness, and warmth of the extremities triggered by warmth or exercise and relieved by cooling, caused by arteriolar inflammation and platelet-mediated microvascular occlusion.

Erythromycin Ophthalmic Prophylaxis

- Erythromycin 0.5% ophthalmic ointment administered to all newborns within 1 hour of birth to prevent ophthalmia neonatorum (neonatal conjunctivitis) from *Neisseria gonorrhoeae* and *Chlamydia trachomatis*. Instilled into both conjunctival sacs. Silver nitrate (Credé's method) was used historically but is less effective for chlamydia.

Erythroplakia

- Red patch with high malignancy risk. More concerning than leukoplakia. Biopsy essential.

Erythropoiesis

- The process by which haematopoietic stem and progenitor cells develop into mature red blood cells. It occurs primarily in adult bone marrow and is regulated chiefly by erythropoietin in response to tissue oxygenation; maturation requires iron, vitamin B12, folate, and appropriate marrow function.

Erythropoiesis-Stimulating Agents

- Epoetin alfa and darbepoetin alfa stimulate red blood cell production in CKD anemia. Target hemoglobin 10 to 11.5 g/dL. Avoid exceeding 13 g/dL due to increased cardiovascular risk. Ensure adequate iron stores before initiating therapy.

Erythropoietin

- A glycoprotein hormone (EPO) that regulates erythropoiesis—the production of red blood cells in the bone marrow. Produced primarily by interstitial peritubular cells in the kidney (90%) and hepatocytes (10%) in response to tissue hypoxia.

ESBL

- Extended-spectrum beta-lactamase, enzymes that hydrolyse penicillins, cephalosporins, and monobactams. ESBLs are common in Enterobacteriaceae, particularly *Klebsiella pneumoniae* and *E. coli*. ESBL production confers resistance to most beta-lactam antibiotics except carbapenems.

ESBL-producing organisms

- Bacteria that produce extended-spectrum beta-

lactamases (ESBLs), enzymes that hydrolyze penicillins, cephalosporins, and monobactams. ESBLs are common in Enterobacteriaceae, particularly *Klebsiella pneumoniae* and *E. coli*. ESBL production confers resistance to most beta-lactam antibiotics except carbapenems.

Escherichia coli

- A gram-negative rod that is part of the normal intestinal flora but can cause various infections. Uropathogenic *E. coli* causes urinary tract infections. Enterotoxigenic *E. coli* causes traveller's diarrhea. Enterohemorrhagic *E. coli* (O157:H7) causes bloody diarrhea and hemolytic uraemic syndrome. ESBL-producing *E.*

Escitalopram

- A selective serotonin reuptake inhibitor (SSRI), the S-enantiomer of citalopram, used for major depressive disorder and generalized anxiety disorder. Increases synaptic serotonin; well tolerated, low drug-interaction profile (less QT prolongation than citalopram).

ESKD

- End-stage kidney disease defined as GFR less than 15 or need for dialysis. Medicare eligibility at GFR less than 15 or on dialysis. Palliative care option for those not wanting renal replacement therapy. Comprehensive advance care planning essential.

Esomeprazole

- The S-enantiomer of omeprazole, a proton pump inhibitor irreversibly inhibiting gastric H⁺/K⁺ ATPase. Used for GERD, erosive esophagitis, peptic ulcer disease, *H. pylori* eradication, and hypersecretory states. Greater acid suppression than omeprazole at equivalent doses; once daily, with an IV formulation for acute GI bleeding.

Esophageal Achalasia

- Primary esophageal motility disorder characterized by failure of the lower esophageal sphincter to relax during swallowing and absent peristalsis in the distal esophagus, resulting from selective loss of inhibitory neurons in the myenteric plexus.

Esophageal Atresia

- A congenital anomaly where the esophagus ends in a blind pouch, often with a tracheoesophageal fistula connecting the distal esophagus to the tra-

chea. Presents with drooling, inability to pass a nasogastric tube, and choking with feedings. The 3 Es association: Esophageal atresia, Esophageal anomalies, and Extremity anomalies.

Esophageal Cancer

- Malignant tumor of the esophagus. Squamous cell carcinoma predominates in the upper and middle third (smoking, alcohol); adenocarcinoma of the lower third arises from Barrett esophagus. Presents with progressive dysphagia and weight loss; treatment depends on stage and includes surgery, chemoradiation, and systemic therapy.

Esophageal Candidiasis

- Fungal infection of the esophagus caused by *Candida* species common in immunocompromised patients including those with HIV, organ transplant recipients, and patients on inhaled corticosteroids. Presents with odynophagia and dysphagia. Diagnosed by endoscopy showing white adherent plaques and biopsy confirming fungal hyphae.

Esophageal Dilation

- Endoscopic passage of bougies or balloons to widen esophageal strictures. Used for symptomatic strictures from peptic disease, radiation, caustic injury, or eosinophilic esophagitis. Savary bougies are passed over a guidewire with progressive dilation. Through-the-scope balloons allow controlled radial dilation under direct visualization.

Esophageal Diverticulum

- Outpouching of the esophageal wall classified by location. Zenker diverticulum occurs in the hypopharynx and is the most common esophageal diverticulum. Mid-esophageal diverticula are traction diverticula caused by extrinsic inflammatory disease including tuberculosis and histoplasmosis.

Esophageal Foreign Body

- Ingestion of a foreign object lodged in the esophagus. Most common in children (coins, batteries, toys) and adults with dentures (food bolus), psychiatric illness, or prisoners. Common sites: cricopharyngeus (C6, 60%), aortic arch (T4), and gastroesophageal junction. Symptoms: dysphagia, odynophagia, drooling, and foreign body

Esophageal Intramural

Pseudodiverticulosis

- Condition characterized by multiple small outpouchings of the esophageal submucosal glands seen on barium swallow as flask-shaped contrast collections oriented longitudinally along the esophagus. The pseudodiverticula are not true diverticula but rather dilated excretory ducts of submucosal glands.

Esophageal Leukoplakia

- White plaques on the esophageal mucosa representing squamous cell hyperplasia with hyperkeratosis. Similar to oral leukoplakia it is considered a potentially premalignant condition. Associated with chronic irritation from smoking, alcohol, and gastroesophageal reflux.

Esophageal Manometry

- Diagnostic test measuring esophageal motor function by recording intraluminal pressures during swallowing. A catheter with pressure sensors is passed through the nose into the stomach and withdrawn to record pressures at multiple levels. High-resolution manometry using closely spaced sensors provides a spatiotemporal color plot.

Esophageal Perforation

- Full-thickness rupture of the esophageal wall, most commonly Boerhaave syndrome from forceful vomiting occurring on the left posterolateral distal esophagus. Iatrogenic perforation during endoscopy is another common cause. Presents with chest pain, subcutaneous emphysema, and mediastinitis.

Esophageal pH Monitoring

- 24-hour ambulatory monitoring of esophageal acid exposure. Catheter-based (transnasal pH probe placed 5 cm above LES) or wireless (Bravo capsule endoscopically attached to esophageal mucosa). Measures: % time pH <4 (total, upright, supine), DeMeester score (>14.7 abnormal), and symptom-reflux correlation.

Esophageal Spasm

- Esophageal motility disorder characterized by uncoordinated or excessive contractions impairing bolus transport. Diffuse esophageal spasm shows simultaneous contractions on manometry. Jackhammer esophagus features extremely high-amplitude peristaltic contractions.

Esophageal Stent

- Self-expanding metal stent for esophageal obstruction. Palliative for malignancy. Migration, tumor ingrowth common complications.

Esophageal Stricture

- Narrowing of the esophageal lumen from chronic gastroesophageal reflux disease, caustic ingestion, or eosinophilic esophagitis. Presents with progressive dysphagia initially to solids then liquids. Diagnosed by upper endoscopy and barium swallow.

Esophageal Surgery

- GERD surgery: Nissen fundoplication (360° wrap, gold standard), partial wraps (Toupet 270°, Dor anterior 180-200°) for dysmotility. LINX device (magnetic sphincter augmentation) alternative.

Esophageal Varices

- Dilated, tortuous submucosal veins in the lower esophagus, most commonly caused by portal hypertension (cirrhosis—hepatic fibrosis with increased portal venous pressure >10 mmHg). They develop as portosystemic collaterals connecting the portal venous system (left gastric vein) to the systemic venous system (azygos vein).

Esophageal Web

- Thin mucosal fold protruding into the esophageal lumen most commonly in the postcricoid region or upper esophagus. Webs may be asymptomatic or cause dysphagia particularly to solids. Multiple webs can occur in association with iron deficiency anemia as part of Plummer-Vinson syndrome or Paterson-Kelly syndrome.

Esophagitis

- Inflammation of the esophageal mucosa.

Esophagogastric bypass

- Surgical creation of a conduit between the esophagus and stomach to restore gastrointestinal continuity, most commonly performed as part of esophagectomy or to palliate unresectable esophageal malignancy. Techniques include gastric pull-up with esophagogastrostomy and jejunal interposition.

Esophagogastroduodenoscopy (EGD)

- Upper endoscopy: flexible endoscopic examination of the esophagus, stomach, and duodenum.

Esophagus

- A muscular tube that transports food and liquids from the pharynx (throat) to the stomach.

Esophagus (esophagus)

- A muscular tube approximately 25 cm long connecting the pharynx to the stomach, extending from the cricopharyngeus muscle (at the level of C6) to the gastroesophageal junction (T11).

Esotropia

- A type of strabismus in which one eye deviates inward (toward the nose). Congenital/infantile esotropia: large-angle, constant, onset before 6 months; requires early surgical correction. Accommodative esotropia: associated with hyperopia, onset age 2-3 years; treatment with corrective glasses improves alignment.

ESR

- Erythrocyte sedimentation rate. Non-specific inflammation marker.

Essential Hypertension

- Persistent elevation of systemic arterial blood pressure without an identifiable single secondary cause. It results from interacting genetic, vascular, renal, metabolic, and environmental factors and increases the risk of stroke, coronary disease, heart failure, kidney disease, and retinopathy.

Essential Medicines

- Drugs that satisfy the priority healthcare needs of a population, as defined by the WHO Model List of Essential Medicines. Selection criteria include efficacy, safety, comparative cost-effectiveness, and suitability for use in resource-limited settings.

Essential Nutrient

- Nutrient that cannot be synthesized by body, must be obtained from diet. Essential amino acids, fatty acids, vitamins, minerals.

Essential Thrombocythemia

- A chronic myeloproliferative neoplasm characterized by sustained platelet count elevation (greater than 450,000 per microliter) due to clonal megakaryocytic proliferation, with a risk of both thrombotic and hemorrhagic complications.

Essential Tremor

- The most common movement disorder, charac-

terized by a bilateral, symmetrical, postural and kinetic tremor predominantly affecting the upper extremities, with a frequency of four to twelve hertz. It typically begins insidiously in the second or third decade of life and slowly progresses over years.

Estimated Due Date (EDD)

- The expected date of delivery, calculated as 40 weeks (280 days) from the first day of the last menstrual period. Confirmed or adjusted by first-trimester ultrasound crown-rump length measurement (most accurate ± 5 -7 days). Second-trimester ultrasound is less accurate (± 10 -14 days). Third-trimester is least accurate (± 21 -30 days).

Estimated Fetal Weight (EFW)

- An ultrasound-derived estimate of fetal weight, calculated using formulas that combine BPD, HC, AC, and FL. The Hadlock formula (using HC, AC, FL) is the most commonly used. Accuracy: ± 10 -15%. Indications: assess fetal growth, determine size for gestational age, and plan delivery. EFW <10th percentile: SGA/IUGR.

Estimated Glomerular Filtration Rate (eGFR)

- An estimate of glomerular filtration derived from serum creatinine and, depending on the validated equation, age and sex. It is used to stage chronic kidney disease and guide medication dosing, but is less reliable when creatinine is changing rapidly or in unusual body composition.

Estrogen

- See Oestrogen.

Ethmoid Bone

- Delicate bone between the orbits forming parts of the nasal septum and orbital/nasal roofs. Has the cribriform plate (olfactory nerve filaments), perpendicular plate (superior nasal septum), and labyrinths (nasal conchae, ethmoid air cells). The crista galli provides falx cerebri attachment.

Ethmoidal Air Cells

- See Ethmoidal Sinus.

Ethmoidal Sinus

- The ethmoidal sinuses, also known as ethmoidal air cells, are a collection of multiple small, thin-walled, air-containing cavities within the ethmoid bone between the orbits and above the nasal cavity,

and they are divided into anterior and posterior groups based on their drainage sites.

Ethylene Glycol

- Antifreeze poisoning. See ethylene glycol poisoning.

Ethylene Glycol Poisoning

- Antifreeze poisoning. Metabolized to toxic acids (glycolic, oxalic). Causes high anion gap metabolic acidosis, calcium oxalate crystals in urine, renal failure. Treatment: fomepizole (alcohol dehydrogenase inhibitor), hemodialysis, calcium gluconate if hypocalcemia.

Etiology

- Cause or origin of disease. Examples: genetic, infectious, environmental, idiopathic.

Etomidate

- An intravenous hypnotic used primarily for induction of anesthesia. It generally causes relatively little cardiovascular depression but may produce myoclonus, nausea, pain on injection, and transient inhibition of adrenal steroid synthesis.

Etravirine

- A second-generation non-nucleoside reverse transcriptase inhibitor (NNRTI) used in combination antiretroviral therapy for HIV-1 infection, particularly in treatment-experienced patients with NNRTI resistance.

Euphoria

- An intense feeling of well-being, elation, or happiness, which can be a normal emotional response or a pathologic symptom.

Eustachian Tube

- Auditory/pharyngotympanic tube connecting the middle ear to the nasopharynx, approximately 36 mm long in adults. Cartilaginous medial two-thirds and bony lateral third. Functions: equalizing middle ear/atmospheric pressure, draining middle ear secretions, and protecting from nasopharyngeal pathogens.

Eustachian-Tube Dysfunction

- Failure of the auditory tube to equalize middle-ear pressure or clear secretions. It can cause aural fullness, popping, conductive hearing loss, and retracted tympanic membrane and is common with upper-respiratory infection, allergic rhinitis, or craniofacial disorders.

Euthanasia

- The deliberate act of ending a person's life to relieve suffering, typically performed by a physician. Active euthanasia involves directly administering a lethal substance (e.g., barbiturate overdose followed by neuromuscular blockade) at the patient's explicit request.

Euthanasia and Assisted Dying

- Ethical and legal concepts regarding intentional termination of life to relieve intractable pain and suffering. Active Euthanasia: deliberate administration of a lethal agent (e.g., lethal dose of barbiturates) by a physician to terminate a patient's life upon their explicit request.

Euthyroid Sick Syndrome

- Abnormal thyroid function tests in patients with non-thyroidal illness (critical illness, starvation, major surgery) without intrinsic thyroid disease. Pattern: low T3 (most common), low T4, reverse T3 elevated, normal or low TSH. The severity correlates with illness severity.

Event Monitor

- Extended ECG monitoring for infrequent symptoms. Worn for weeks to months.

Event-Free Survival

- The time from randomization or treatment start to a specified event such as disease progression, relapse, or death from any cause.

Eversion

- Turning the sole of the foot laterally, so that it faces away from the median plane.

Evidence-Based Medicine

- Integration of best research evidence with clinical expertise and patient values. Foundation for clinical guidelines.

Evidence-Based Practice

- Integration of best research evidence, clinical expertise, patient values. Rehabilitation guidelines based on systematic reviews.

Evisceration

- Protrusion of viscera through a dehiscence abdominal wound, typically occurring 5-8 days postoperatively and constituting a surgical emergency. Risk factors include obesity, wound infection, and inadequate fascial closure.

cial closure. Management requires sterile coverage, broad-spectrum antibiotics, and emergent operative reclosure.

Evoked Potentials

- Measurement of electrical responses to stimuli. Visual, auditory, somatosensory. Used for demyelinating diseases.

Evoked Potentials (VEP, BAER, SSEP)

- Electrophysiologic tests measuring the CNS electrical response to specific sensory stimuli. Visual evoked potentials (VEP): checkboard pattern reversal, measures P100 latency; delayed in optic neuritis/MS. Brainstem auditory evoked responses (BAER): click stimuli, waves I-V; delayed in acoustic neuroma, MS, brainstem stroke.

Ewart Sign

- Dullness to percussion, decreased breath sounds, and bronchial breathing at the left infrascapular region caused by compressive atelectasis from a large pericardial effusion. It is a physical examination finding of pericardial effusion. The effusion compresses the adjacent left lower lobe, creating a zone of consolidation.

Ewing Sarcoma

- A highly malignant small round blue cell tumor of bone and soft tissue, representing the second most common primary bone malignancy in children and adolescents (peak age 10-20 years), with a strong predilection for males and Caucasian populations.

Exacerbation of COPD

- An acute worsening of respiratory symptoms beyond normal day-to-day variations, requiring change in medication. It is commonly triggered by respiratory infections (viral, bacterial), air pollution, or non-compliance. Treatment includes increased bronchodilators, systemic corticosteroids, antibiotics if indicated, and oxygen therapy.

Examination

- Systematic assessment of a patient's health status. Components: inspection, palpation, percussion, auscultation (IPPA). General survey: appearance, posture, gait, vital signs. Focused examination based on chief complaint.

Exanthema

- A widespread skin eruption, usually macular or

papular, associated with an acute systemic illness. Viral infections are common causes, but drug reactions, bacterial disease, immune disorders, and other systemic conditions may also produce exanthema; fever, mucosal involvement, blistering, or skin pain may indicate a severe process.

Exanthematous pustulosis

- See acute generalised exanthematous pustulosis. A severe drug-induced eruption with acute onset of hundreds of small, non-follicular, sterile pustules on an erythematous base, often accompanied by fever and neutrophilia. It typically resolves within two weeks after withdrawal of the offending medication.

Exchange Transfusion

- Emergent procedure to rapidly lower severe hyperbilirubinemia by removing small volumes of the infant's blood and replacing with compatible donor blood. Used when phototherapy fails or bilirubin approaches toxic levels (greater than 25 mg/dL in term infants or above exchange threshold on Bhutani curve).

Excision

- The surgical removal of a piece of tissue, an organ, or a lesion. Excisional biopsy removes the entire lesion with a margin of normal tissue, used for definitive diagnosis and treatment of small tumors (suspected melanoma, breast lump, lymph node).

Excisional Biopsy

- Surgical removal of entire lesion. Diagnosis and treatment.

Excitation-Contraction Coupling

- The sequence of events linking an action potential to muscle contraction. In skeletal muscle: the action potential travels down the T-tubule, triggering calcium release from the sarcoplasmic reticulum via ryanodine receptors (calcium-induced calcium release mediated by voltage-gated dihydropyridine receptors).

Excoriation Disorder

- Recurrent skin picking resulting in skin lesions. The individual has made repeated attempts to decrease or stop the skin picking. The skin picking causes clinically significant distress or impairment. It is not attributable to a medical condition or substance.

Common sites include face, arms, and hands.

Excretion

- Removal of substance from body. Routes: renal (urine), hepatic (bile), pulmonary (gas), sweat, breast milk. Important for toxicity and drug clearance.

Exercise Intolerance

- Reduced ability to perform physical activity at the expected level because of exertional fatigue, breathlessness, chest discomfort, leg pain, or other symptoms. Causes include cardiac, pulmonary, vascular, metabolic, neuromuscular, haematological, and deconditioning states.

Exercise Stress Test

- A noninvasive cardiac diagnostic test that evaluates the cardiovascular response to graded physical exertion, primarily used to diagnose coronary artery disease, assess functional capacity, and guide prognosis.

Exercise-Induced Arrhythmias

- Catecholamine-sensitive ventricular tachycardia and right ventricular outflow tract VT can be triggered by exercise. HCM-related sudden death risk increases during vigorous exercise. Exercise-induced myocardial ischemia may trigger ventricular arrhythmias in CAD.

Exercise-Induced Asthma

- Bronchoconstriction occurring during or shortly after physical exertion, causing cough, wheezing, chest tightness, or shortness of breath. It is associated with airway hyperresponsiveness and may occur in people with or without persistent asthma.

Exercise-Induced Bronchoconstriction

- Airway narrowing during or after exercise, presenting with cough, wheeze, and dyspnea. It occurs in up to 90% of asthmatics and is triggered by heat and water loss from the airways. Prevention includes pre-exercise SABA.

Exhaled Nitric Oxide

- A non-invasive measure of airway eosinophilic inflammation. FeNO > 25 ppb in adults suggests eosinophilic airway inflammation and predicts response to inhaled corticosteroids. It is useful for diagnosing eosinophilic asthma, monitoring treatment response, and differentiating asthma from

other causes of cough.

Exhibitionistic Disorder

- A paraphilic disorder involving recurrent, intense sexual arousal from exposing one's genitals to an unsuspecting person. The individual has acted on these urges or they cause marked distress or interpersonal difficulty. Duration 6 months. Onset typically before age 18. Treatment: CBT and SSRIs.

Exhumation

- The lawful disinterment and removal of a buried body or remains for re-examination, re-testing, or identification. Indications: new evidence, suspicion of poisoning or foul play, DNA testing, paternity/maternity testing, exhumation of a wrongly buried body, or repatriation.

Existential Distress

- See psychological.

Exit Wound

- A skin defect produced where a projectile leaves the body, typically larger, irregular, and stellate, with everted edges, no abrasion ring or soot. Size depends on deformation, yaw, and fragmentation; small non-deformed bullets may produce shored exits, and an exit may be absent if the projectile is retained.

Exocytosis

- Process by which intracellular vesicles fuse with the plasma membrane, releasing their contents into the extracellular space and incorporating membrane proteins/lipids into the cell membrane. The SNARE complex (v-SNARE on vesicles and t-SNARE on target membrane) mediates vesicle docking and fusion.

Exon

- A segment of a DNA or RNA molecule containing information coding for a protein or peptide sequence.

Exophthalmos

- Abnormal protrusion of the eyeball (globe) from the orbit, most commonly caused by thyroid eye disease (Graves ophthalmopathy—an autoimmune condition associated with hyperthyroidism). The proptosis results from inflammation, edema, and fibrosis of the extraocular muscles and proliferation of orbital fat within the fixed bony orbit.

Exotoxin

- A heat-labile, protein toxin secreted or released by bacteria (mostly gram-positive; also some gram-negative), often encoded on plasmids or phages, that acts at distant sites.

Exotoxins

- Proteins secreted by bacteria that have specific toxic effects on host cells. Exotoxins are generally more potent than endotoxins and have specific mechanisms of action. A-B toxins consist of an active (A) component and a binding (B) component. Examples include diphtheria toxin, cholera toxin, and botulinum toxin.

Exotropia

- A type of strabismus in which one eye deviates outward (toward the temple). Intermittent exotropia: most common type, often noticed when tired or daydreaming, can be controlled with fusion. Constant exotropia: requires treatment to prevent amblyopia. Congenital exotropia: rare, requires early evaluation.

Expectorants

- Medications that promote the clearance of mucus from the respiratory tract by increasing sputum volume and reducing its viscosity, facilitating productive cough. The most common expectorant is guaifenesin (glyceryl guaiacolate), which increases respiratory tract fluid secretion and reduces sputum adhesiveness.

Expectoration

- The act of coughing up and spitting out mucus, sputum, or phlegm from the respiratory tract; sputum production.

Expiratory Grunting

- A forced exhalation against a partially closed glottis, creating an audible sound. It is an end-expiratory maneuver that generates intrinsic PEEP, helping to maintain alveolar recruitment and improve oxygenation. It is commonly seen in children with respiratory distress.

Explantation

- Surgical removal of an implanted device or prosthesis. It may be required for infection, malfunction, intolerance, migration, fracture, tissue injury, or completion of a temporary therapy.

Exploratory Laparotomy

- Open surgical entry into the abdominal cavity to diagnose or treat suspected intra-abdominal pathology, such as hemorrhage, perforation, ischemia, or obstruction, when less invasive evaluation is inadequate or inappropriate.

Exposure Route

- Path by which substance enters body. Routes: inhalation, ingestion, dermal, injection, ocular. Determines absorption rate and toxicity.

Expressivity

- Degree to which a genotype is expressed in phenotype. Variable expressivity: same genotype produces different phenotypic severity in different individuals. Examples: neurofibromatosis type 1, Marfan syndrome.

Extension

- Movement that increases the angle between two bones or straightens a joint.

Extensively Drug-Resistant TB

- TB resistant to isoniazid, rifampin, fluoroquinolones, and at least one injectable agent. Very limited treatment options and poor outcomes.

Extensor Digitorum

- Posterior forearm muscle from lateral epicondyle via the common extensor tendon, dividing into 4 tendons inserting on extensor expansions of digits 2-5. Innervated by posterior interosseous nerve (C7-C8). Extends MCP joints, assists PIP/DIP extension. Lateral epicondylitis (tennis elbow) is an overuse injury of its common origin.

External Auditory Canal

- Tube from the concha to the tympanic membrane, approximately 2.5 cm long in adults. Lateral third is cartilaginous; medial two-thirds are bony (temporal bone). Skin-lined with ceruminous glands (producing cerumen/earwax) and hair follicles. Cerumen has antimicrobial and insect-repellent properties.

External Beam Radiation Therapy

- Radiation from outside body. 3D conformal, IMRT, stereotactic body radiation therapy (SBRT).

External Cephalic Version (ECV)

- A procedure to manually rotate a breech or transverse fetus to a cephalic (head-down) presentation

from outside the abdomen. Performed after 36-37 weeks (term, with adequate amniotic fluid). Success rate: 30-80% (higher with prior vaginal delivery, transverse lie, adequate fluid, and non-anterior placenta).

External Fixation

- Stabilization of a fracture or limb using pins or wires attached to an external frame. It may provide temporary damage-control stabilization or definitive treatment, particularly when soft-tissue injury or contamination limits internal fixation.

External Iliac Artery

- Larger common iliac terminal branch, continuing as the femoral artery below the inguinal ligament. Gives off inferior epigastric and deep circumflex iliac arteries. Related to external iliac vein medially and ureter anteriorly. Commonly used for cardiac catheterization and endovascular access.

External Jugular Vein

- Formed by posterior auricular and retromandibular vein divisions, crossing the sternocleidomastoid superficially to drain into the subclavian vein. Visible through the skin; used for peripheral IV access when other sites are unavailable. Distended in right heart failure and SVC obstruction. A useful physical examination landmark.

External Oblique Muscle

- Most superficial flat abdominal muscle from outer lower eight ribs to linea alba, pubic tubercle, and iliac crest. Innervated by T7-T12. Contralateral trunk rotation and flexion; compresses abdomen. Forms the inguinal ligament from its free inferior border. Superficial inguinal ring is an opening in its aponeurosis.

External Resorption

- Loss of tooth structure from outside. Inflammatory, replacement, surface, cervical resorption. Treatment depends on type and extent.

Extracellular Fluid (ECF)

- The fluid outside cells, including plasma and interstitial fluid. Sodium, chloride, and bicarbonate are its principal effective solutes; ECF volume and composition are regulated by the kidneys, cardiovascular system, and endocrine mechanisms.

Extracellular Matrix

- A complex network of proteins and carbohydrates secreted by cells that provides structural and chemical support to surrounding cells.

Extracellular pathogens

- Microorganisms that survive and replicate outside host cells. Extracellular bacteria include Streptococcus, Staphylococcus, and Escherichia coli. These pathogens are typically killed by antibodies and complement. They produce toxins and enzymes to damage host tissues.

Extracorporeal Membrane Oxygenation

- A temporary mechanical circulatory support system that provides cardiac and/or respiratory support. VA-ECMO (veno-arterial) provides both cardiac and respiratory support; VV-ECMO (veno-venous) provides respiratory support only.

Extracorporeal Shock Wave Lithotripsy (ESWL)

- Non-invasive treatment method for urinary calculi that uses focused acoustic shock waves generated outside the body to fragment kidney or ureteral stones into smaller gravel-like pieces (dust/sand) that pass spontaneously in urine.

Extranodal natural killer T-cell lymphoma

- An aggressive lymphoma arising from natural killer cells, most commonly involving the nasal cavity and upper aerodigestive tract. It is strongly associated with Epstein-Barr virus infection.

Extraocular Muscle Function Testing

- Assessment of the six extraocular muscles innervated by CN III (medial rectus: adduction; superior rectus: elevation; inferior rectus: depression; inferior oblique: extorsion), CN IV (superior oblique: intorsion and depression), and CN VI (lateral rectus: abduction).

Extraocular Muscles

- Six muscles controlling eye movement: four rectus (superior, inferior, medial, lateral) and two oblique (superior, inferior).

Extrapulmonary manifestation of tuberculosis

- See Tuberculosis (TB).

Extrapyramidal Symptoms

- Drug-induced movement disorders caused by D2 receptor blockade in the nigrostriatal pathway.

Extrauterine Pregnancy

- A pregnancy in which the fertilized ovum implants outside the uterine cavity, an alternative term for ectopic pregnancy. By far the most common site is the fallopian tube (ampulla 70%, isthmus 12%, fimbria 11%, cornual/interstitial 2–4%).

Extravasation

- Leakage of urine outside urinary tract. Causes: trauma, surgery, obstruction. Pseudovesical extravasation (rectus sheath), extraperitoneal, intraperitoneal. Treatment: catheter drainage, surgical repair.

Extubation Criteria

- Extubation requires the patient to meet specific criteria: fully awake and following commands, adequate tidal volume and respiratory rate, strong cough and gag reflex, adequate oxygenation (SpO2 greater than 95 percent on room air or baseline), hemodynamic stability, and return of neuromuscular function (TOF ratio greater than 0.9, sustain

Exudate

- An exudate is an inflammatory fluid with high protein content and abundant cells, resulting from increased vascular permeability during inflammation. Exudates are typically caused by bacterial infections, autoimmune diseases, viral infections, or malignancy.

Exudative Pleural Effusion

- From local pleural disease. Causes include pneumonia, malignancy, TB, and PE. Elevated protein, LDH, and cell count.

Eye

- The organ of vision, a spherical structure (approximately 24 mm diameter) in the bony orbit. Three layers: fibrous (sclera, cornea), vascular/uveal (choroid, ciliary body, iris), neural (retina). Anterior segment: cornea, anterior chamber (aqueous humor), iris/pupil, lens.

Eye Disorders

- The sensory ability to perceive light and interpret visual information processed by the eyes and brain, enabling recognition of shapes, colors, depth, and

motion. See Vision.

Eye Floaters

- Spots, threads, or cobweb-like shapes that move across the visual field, usually caused by opacities in the vitreous. Sudden new floaters, flashes, or a curtain-like shadow require urgent evaluation for retinal tear or detachment. See also Floaters.

Eye Health

- The maintenance of optimal vision and prevention of ocular disease through regular eye examinations, protective measures, and timely treatment. Common conditions affecting eye health include refractive errors, cataracts, glaucoma, diabetic retinopathy, and age-related macular degeneration.

Eye Irritation

- Chemical injury to eye. Immediate irrigation with water or saline for 15–20 minutes. Alkali burns more dangerous than acid. Ophthalmology consultation.

Eye Pain

- Pain in or around the eye caused by surface irritation, infection, trauma, inflammation, glaucoma, or orbital disease. Severe pain with vision loss, photophobia, trauma, or nausea requires urgent assessment.

Eye Redness

- Red discoloration of the eye caused by dilation or inflammation of conjunctival or deeper ocular blood vessels. Causes range from irritation and conjunctivitis to keratitis, uveitis, and acute glaucoma.

Eye Trauma

- Ocular injuries ranging from corneal abrasions to globe rupture. Chemical burns: irrigate immediately with saline or water for 30 minutes. Penetrating injuries: shield eye, no pressure, emergent ophthalmology consultation. Foreign bodies: remove superficial items; embedded objects require surgical removal.

Ezetimibe

- A cholesterol absorption inhibitor that blocks intestinal absorption of dietary and biliary cholesterol at the Niemann-Pick C1-like 1 (NPC1L1) transporter in the small intestine.

Eosinophilic Bronchitis

- A cause of chronic cough in which eosinophilic inflammation affects the airways without the vari-

able airflow obstruction typical of asthma. Cough-variant asthma and reflux may mimic it; diagnosis relies on clinical assessment and airway inflammation testing.

Eosinophilic Fasciitis

- A rare inflammatory disorder causing painful swelling, induration, and progressive tightening of the fascia and skin, usually in the limbs. Blood eosinophilia may occur. Diagnosis often requires imaging or deep biopsy to distinguish it from scleroderma and other causes.

Eggshell Nail

- A nail that becomes unusually thin, soft, flexible, and sometimes whitish or concave. It may be associated with nutritional deficiency, chronic illness, aging, or repeated trauma; persistent nail changes should be evaluated with the skin and other symptoms in mind.

Endometrial Hyperplasia

- Abnormal overgrowth of the lining of the uterus, often related to prolonged estrogen exposure without enough progesterone. It may cause heavy, irregular, or postmenopausal bleeding; atypical forms can precede endometrial cancer and require gynecologic evaluation, often including endometrial sampling.

Chapter 6

Pocket Dictionary: F Terms

Fabry Disease

- An X-linked lysosomal storage disorder caused by deficient alpha-galactosidase A activity and accumulation of globotriaosylceramide in multiple organs.

Face Presentation

- A fetal presentation where the neck is hyperextended and the face is the presenting part. Incidence: 1:500-1:1000. Types: mentum anterior (chin toward pubis, delivers vaginally), mentum posterior (chin toward the sacrum, cannot deliver vaginally unless rotates to anterior).

Facet Joints

- Zygapophysial joints connecting adjacent vertebral articular processes. Orientation varies: cervical (45 degrees, allowing rotation), thoracic (coronal, allowing rotation but limiting flexion/extension), lumbar (sagittal, allowing flexion/extension but limiting rotation). Innervated by medial branches of dorsal rami.

Facial Pain

- Pain in face. Dental causes must be ruled out. Trigeminal neuralgia, TMJ disorder, sinusitis, neuropathic pain.

Facial paralysis

- Loss of voluntary motor function of the facial muscles, most commonly from peripheral facial nerve (CN VII) dysfunction. The most frequent cause is Bell palsy (idiopathic peripheral facial paralysis), accounting for 60-70 percent of cases, with an incidence of 20-30 per 100,000 annually.

Facilitated Diffusion

- Passive transport of molecules across cell membranes via specific transmembrane protein carriers or channels, moving substances down their concentration gradient without energy expenditure.

Channel-mediated facilitated diffusion uses ion channels (voltage-gated, ligand-gated, or mechanically-gated) for rapid ion movement.

Factitious Disorder

- Presentation of physical or psychological symptoms or imposing injury or disease on another, with the deceptive intention of assuming the sick role. In factitious disorder imposed on self, the individual falsifies signs or symptoms, induces injury or disease, and presents to medical providers.

Factor H Deficiency

- Autosomal recessive (CFH mutation). Factor H regulates alternative pathway (cofactor for factor I, accelerates decay). Uncontrolled alternative pathway activation -> C3 consumption -> low C3, normal C4. Atypical HUS (aHUS): microangiopathic hemolytic anemia, thrombocytopenia, acute kidney injury.

Factor I Deficiency

- Autosomal recessive (CFI mutation). Factor I cleaves C3b/C4b (cofactor: factor H, MCP, C4BP). Uncontrolled complement -> low C3, low C4, low factor H. Recurrent pyogenic infections, aHUS, C3 glomerulopathy, SLE-like disease. Treatment: eculizumab for aHUS/C3G; plasma infusion (contains factor I); IVIG.

Factor V Deficiency

- An inherited or acquired bleeding disorder caused by reduced factor V activity, impairing thrombin generation and fibrin formation. It may cause easy bruising, nosebleeds, heavy menstrual bleeding, postoperative bleeding, or severe bleeding in rare cases.

Factor V Leiden

- The most common inherited thrombophilia, caused

by a point mutation in the factor V gene (G1691A, R506Q) resulting in resistance to activated protein C (APC resistance). Heterozygous: 5-fold increased VTE risk; homozygous: 50-80 fold increased risk. Most common in Caucasians (3-7% prevalence).

Factor V Leiden mutation

- A factor V variant resistant to activated protein C, causing inherited thrombophilia and increased venous thrombosis risk.

Facultative Anaerobe

- A microorganism capable of growth under both aerobic and anaerobic conditions, using respiration when a suitable terminal electron acceptor is present and fermentation or anaerobic respiration when it is not.

FAD (Flavin Adenine Dinucleotide)

- A flavin-derived redox coenzyme, usually bound tightly to enzymes as a prosthetic group. FAD accepts two hydrogen atoms to form FADH₂, which donates electrons to the respiratory chain through selected dehydrogenases.

Failed Induction of Labor

- The inability to achieve active labor and progressive cervical change after an adequate trial of induction. Definition: no active labor (cervical change) after 12-18 hours of oxytocin with ruptured membranes and adequate contractions. Management: the latent phase may be prolonged (up to 24 hours or more).

Failed Intubation

- Inability to successfully place an endotracheal tube after multiple attempts. In the operating room, a failed intubation algorithm is followed: call for help, maintain oxygenation (face mask, supraglottic airway), and decide to wake the patient (if non-emergent) or proceed with a supraglottic airway (if surgery is urgent).

Failure to Thrive

- Inadequate growth in children, defined as weight below the 5th percentile or crossing two major percentile lines on growth charts. Causes include caloric insufficiency (neglect, poor feeding), organic disease (congenital heart disease, renal disease), and psychosocial factors.

Fainting

- A transient loss of consciousness (TLOC) due

to global cerebral hypoperfusion, characterized by rapid onset, short duration, and spontaneous complete recovery. See Syncope (Fainting).

Falciform ligament

- A sickle-shaped peritoneal fold that attaches the liver to the anterior abdominal wall and diaphragm, separating the left and right lobes of the liver. It contains the ligamentum teres, the obliterated remnant of the umbilical vein. It may be implicated in rare complications such as internal herniation or necrosis following trauma.

Fall

- Unintended change in position to ground. Major cause of morbidity/mortality in elderly. Risk factors: gait/balance impairment, muscle weakness, visual impairment, polypharmacy, environmental hazards, orthostatic hypotension. Prevention: exercise, medication review, vision correction, home modification, hip protectors.

Fall Prevention

- Unintended change in position to ground. See fall.

Fall Risk

- Unintended change in position to ground. See fall.

Fall Screening

- Unintended change in position to ground. See fall.

Fallopian Tube

- The paired muscular tubes (10-12 cm) connecting the ovaries to the uterus, conducting oocytes and sperm and the site of fertilization (usually ampulla). Segments: infundibulum (fimbriae—sweep the oocyte), ampulla, isthmus, intramural.

Fallopian Tube Cancer

- A rare tumor histologically similar to high-grade serous ovarian carcinoma. Presents with a pelvic mass, pain, or watery discharge; staging and treatment follow the principles of ovarian cancer.

Fallopian tube torsion

- Twisting of the fallopian tube around its pedicle, usually around an underlying mass such as a hydrosalpinx or paraovarian cyst, causing acute pelvic pain and requiring urgent surgical intervention.

Falls

- Unintended change in position to ground. See fall.

False Ribs

- Ribs 8-10 not articulating directly with the sternum; their cartilages join rib 7's cartilage forming the costal margin. Greater mobility and flexibility of the lower thoracic wall. More susceptible to fractures at their angles due to greater length and less rigid attachment.

False-Negative Result

- A negative test result in a person who does have the target condition. Causes include inadequate specimen, early or late testing, low organism or biomarker concentration, prior treatment, technical error, and limited test sensitivity.

False-Positive Result

- A positive test result in a person who does not have the target condition. Causes include limited specificity, cross-reactivity, technical error, contamination, and testing in a low-prevalence population.

Familial Adenomatous Polyposis

- Hereditary condition with hundreds to thousands of adenomas throughout the colon caused by germline APC gene mutations with autosomal dominant inheritance. Polyps develop during the second decade and colorectal cancer occurs by age forty if untreated. Diagnosis is by colonoscopy showing numerous polyps and genetic testing.

Familial Hypercholesterolemia

- An autosomal dominant genetic disorder causing severely elevated LDL cholesterol from birth, most commonly due to LDL receptor mutations. Heterozygous FH affects 1 in 250 people, causing premature coronary artery disease (MI in 40s-50s). Diagnosis: LDL >190 mg/dL in adults, or >160 mg/dL in children.

Familial Hypocalciuric Hypercalcemia

- Autosomal dominant condition caused by inactivating mutations in the calcium-sensing receptor gene leading to a higher set point for calcium-mediated PTH suppression. Presents with mild hypercalcemia, inappropriately normal or mildly elevated PTH, and low urinary calcium excretion with calcium-to-creatinine clearance ratio less than 0.01.

Familial Mediterranean Fever

- An autosomal recessive autoinflammatory disorder

(MEFV gene mutations, pyrin/inflammasome dysregulation) causing recurrent self-limited febrile attacks with serositis (peritonitis, pleuritis, synovitis), erysipelas-like erythema, lasting 12-72 hours.

Familial multiple lipomatosis

- An autosomal dominant condition characterized by the development of numerous, symmetrically distributed subcutaneous lipomas, typically appearing in early adulthood. The lipomas are encapsulated, painless, and slow-growing, affecting the trunk and extremities. Unlike Dercum disease, they are not painful.

Family Involvement

- Family support crucial for rehabilitation success. Training family as therapists, addressing caregiver burden.

Fanconi Anemia

- An autosomal recessive (or X-linked) bone marrow failure syndrome caused by mutations in at least 22 FANC genes involved in DNA repair (the FA/BRCA pathway).

Fanconi Syndrome

- Generalized proximal tubular dysfunction with impaired reabsorption of glucose, amino acids, phosphate, uric acid, and bicarbonate. Causes glycosuria, aminoaciduria, phosphaturia, and metabolic acidosis. Associated with proximal RTA. Causes include myeloma, heavy metal toxicity, Wilson disease, and genetic disorders.

Farmer's Lung

- The most common HP, caused by thermophilic actinomycetes in moldy hay. Presents acutely 4–8 hours after exposure with fever, cough, and dyspnea.

Farsightedness

- A refractive error, also called hyperopia, in which light focuses behind the retina when the eye is relaxed. It can cause blurred near vision, eyestrain, and headaches and is corrected with converging lenses or refractive procedures.

Fascial Loop

- Autologous or synthetic material for surgical reinforcement. Midurethral sling uses synthetic tape.

Fasciitis

- Inflammation of fascia. Examples: plantar fasciitis (heel pain), eosinophilic fasciitis (Shulman dis-

ease—skin thickening, eosinophilia). Treatment: NSAIDs, corticosteroids, physical therapy.

Fascioliasis

- A parasitic infection caused by liver flukes of the genus *Fasciola*, acquired by ingesting metacercariae on contaminated aquatic plants or in water. Acute hepatic migration may cause fever, right-upper-quadrant pain, hepatomegaly, and eosinophilia; chronic biliary infection may cause colic, cholangitis, or biliary obstruction.

Fasciotomy

- Surgical incision through the fascia to relieve compartment syndrome. Indications: acute compartment syndrome (trauma, reperfusion, burns, crush injury). Classic signs: pain out of proportion, pain with passive stretch, paresthesias, pallor, pulselessness (late).

Fast Heart Rate

- A heart rate faster than expected for age or activity, usually called tachycardia. It may be a normal response to exercise, fever, pain, or stress, or a sign of arrhythmia or another medical disorder.

Fasting Guidelines

- Preoperative fasting guidelines minimize aspiration risk while reducing patient discomfort and catabolism. Clear liquids (water, clear juice without pulp, black coffee, sports drinks) are allowed up to 2 hours before surgery. Breast milk up to 4 hours. Infant formula up to 6 hours. Light meal (toast and clear liquid) up to 6 hours.

Fat Embolism

- Fat embolism occurs when fat globules enter the bloodstream, typically following long bone fractures, orthopedic surgery, or severe trauma involving adipose tissue.

Fat Necrosis

- A specialized form of necrosis that occurs in adipose tissue, most commonly following acute pancreatitis when activated pancreatic lipases escape into the surrounding tissues and digest triglycerides in mesenteric or subcutaneous fat cells.

Fat-Soluble Toxins

- Lipophilic toxins stored in fat, prolonged elimination. Examples: organochlorine pesticides, PCBs, dioxins, tetrachlorodiphenylmethane.

Fatigue

- A persistent feeling of tiredness, exhaustion, or lack of energy that is not fully relieved by rest. It may result from sleep disturbance, anemia, infection, endocrine disease, chronic organ disease, medication exposure, or psychological and neurological conditions.

Fatigue (Cancer)

- Most common cancer symptom. Multi-factorial: tumor, treatment, anemia, psychological. Management: exercise, energy conservation, treat underlying causes.

Fatty Acid

- A carboxylic acid with a hydrocarbon chain that may be saturated or contain one or more double bonds. Fatty acids provide energy, form membrane lipids, and serve as precursors for signaling molecules.

Fatty Acid Synthase

- Multi-enzyme complex for fatty acid synthesis. Produces palmitate.

Fatty Liver Disease (MASLD)

- A condition characterized by the accumulation of excess fat in liver cells, most commonly associated with obesity, insulin resistance, and metabolic syndrome.

Febrile

- Feverish; having a high body temperature.

Febrile Neutropenia

- Fever in a patient with a sufficiently low absolute neutrophil count to create a high risk of serious infection, generally defined as a single temperature of at least 38.3°C or sustained temperature of at least 38.0°C with ANC below 500/L or expected to fall below that level.

Febrile Seizure

- A seizure occurring in a child usually between 6 months and 5 years of age with fever but without evidence of central nervous system infection or a previous afebrile seizure. Simple febrile seizures are generalized, brief, and nonrecurrent within the same illness; prolonged or focal events require further evaluation.

Febrile Seizures

- Seizures occurring in children 6 months to 5 years triggered by fever (temperature greater than 38 degrees Celsius), without intracranial infection or defined cause. Occurs in 2-5% of children. Simple: generalized, single seizure lasting less than 15 minutes, no recurrence within 24 hours.

Fecal Calprotectin

- Neutrophil-derived protein measured in stool as a marker of intestinal inflammation. It is stable in stool at room temperature for up to seven days. Useful for differentiating inflammatory from functional bowel disease and monitoring inflammatory bowel disease activity.

Fecal Elastase

- Pancreatic enzyme measured in stool as a marker of exocrine pancreatic function. Low levels below two hundred micrograms per gram indicate pancreatic insufficiency, and levels below one hundred are consistent with severe exocrine insufficiency.

Fecal Immunochemical Test (FIT)

- A stool-based screening test for colorectal cancer that detects human hemoglobin in a single stool sample using antibodies. Annual screening. Sensitivity: 74-80% for cancer, 25-30% for advanced adenomas. Specificity: 94-96%. No dietary restrictions required. Positive test requires follow-up colonoscopy. More specific than guaiac FOBT.

Fecal Impaction

- A large, firm mass of stool retained in the rectum or colon that cannot be evacuated normally. It may cause abdominal or rectal pain, distention, overflow diarrhoea, urinary symptoms, nausea, or bowel obstruction and is associated with constipation, immobility, dehydration, neurological disease, and medications.

Fecal Incontinence

- Inability to control bowel movements ranging from minor soiling to complete loss of continence. Causes include sphincter injury from obstetric trauma or surgery, neuropathy from diabetes or spinal cord injury, diarrhea, rectal prolapse, and cognitive impairment.

Fecal microbiota transplantation

- The transfer of fecal material from a healthy donor

to a patient with recurrent *Clostridioides difficile* infection. The procedure aims to restore normal gastrointestinal flora and eliminate *C. difficile* colonization. FMT is highly effective, with cure rates exceeding 90% for recurrent *C. difficile* infection.

Fecundity

- The biological ability to conceive and produce a live birth. Fecundity is influenced by age (declines significantly after 35), ovarian reserve, tubal patency, uterine factors, and male fertility (sperm count, motility, morphology). Fecundability: the probability of achieving pregnancy in a single menstrual cycle.

Federal Register

- See regulatory affairs.

Feedback Inhibition

- Regulatory mechanism where the end product of a metabolic pathway inhibits an earlier enzyme (usually the first committed step). Examples: isoleucine inhibiting threonine deaminase, CTP inhibiting aspartate transcarbamoylase, ATP inhibiting phosphofructokinase. Allows efficient resource utilization and prevents overproduction.

Feeding Jejunostomy

- Surgical or endoscopic placement of a feeding tube into the jejunum for enteral nutrition when upper gastrointestinal access is not feasible. Used in patients with head and neck cancers, esophageal obstruction, or severe dysphagia. Provides long-term nutritional support and avoids the complications of parenteral nutrition.

Feeding Tube

- Delivery of nutrients via gastrointestinal tract. See enteral nutrition.

Felty Syndrome

- Triad of rheumatoid arthritis, splenomegaly, and neutropenia (ANC <2000). Complication of longstanding, severe, seropositive RA. Increased infection risk. Treatment: DMARDs (methotrexate, TNF inhibitors), G-CSF for neutropenia, splenectomy in refractory cases.

Female Genital Mutilation

- FGM: the partial or total removal of female external genitalia for non-medical reasons, a human rights violation practiced in parts of Africa, the Middle

East, and Asia (types I-IV). Immediate complications: severe pain, hemorrhage, infection, urinary retention.

Female Infertility

- See obstetrics and gynecology.

Female Orgasmic Disorder (Anorgasmia)

- A sexual dysfunction characterized by a persistent or recurrent delay in, infrequency of, or absence of orgasm, or markedly reduced intensity of orgasmic sensation, in 75% of sexual encounters, lasting 6 months and causing marked distress. Lifelong (primary, never achieved orgasm by any means) vs. acquired (previously able)

Female Sexual Interest/Arousal Disorder

- A sexual dysfunction in women characterized by absent or reduced sexual interest, thoughts, or receptivity, and/or absent or reduced sexual arousal (genital or nongenital), persisting for 6 months and causing marked distress. Combines the former DSM-IV diagnoses of hypoactive sexual desire disorder and female sexual arousal

Femoral Artery

- Continues from external iliac through the femoral triangle and adductor canal to become the popliteal artery. Branches: superficial/deep circumflex femoral, deep femoral (profunda femoris), muscular. Palpable in the femoral triangle; most common site for arterial catheterization. Related to femoral vein medially and nerve laterally.

Femoral Hernia

- A hernia of abdominal contents through the femoral ring, into the femoral canal, medial to the femoral vessels and below the inguinal ligament. More common in women (3:1) and more prone to strangulation (40%) than inguinal hernias due to narrow, rigid femoral ring.

Femoral Nerve

- Largest lumbar plexus nerve (L2-L4), passing through psoas major and entering the thigh under the inguinal ligament lateral to the femoral artery. Innervates quadriceps, sartorius, pectineus; sensation to anterior thigh and medial leg (saphenous nerve). Used for knee surgery nerve block anesthesia.

Femoral Shaft Fractures

- Femoral shaft fractures: high energy (MVC, fall from height) or low energy (elderly, pathologic). Winquist-Hansen classification of comminution: I (none), II (butterfly <50% width), III (butterfly >50%), IV (segmental).

Femoral Triangle

- Upper anterior thigh space bounded superiorly by the inguinal ligament, laterally by sartorius, and medially by adductor longus. Contents from lateral to medial: femoral nerve, artery, vein, and deep inguinal lymph nodes. The femoral artery is used for arterial catheterization. The femoral vein is accessed for central venous lines.

Femoral Vein

- The major deep vein of the thigh, continuation of the popliteal vein (through the adductor hiatus), running with the femoral artery in the femoral sheath (medial to artery) through the femoral triangle, becoming the external iliac vein at the inguinal ligament. Receives the great saphenous vein.

Femur

- Longest, heaviest, strongest body bone. Proximal: head, neck, trochanters. Shaft: linea aspera. Distal: condyles, epicondyles. The neck connects head to shaft at approximately 125 degrees. Neck fractures are common in elderly osteoporotic patients with avascular necrosis risk. Bears entire upper body weight during standing.

Femur Length (FL)

- The length of the fetal femur (diaphysis), measured from the greater trochanter to the distal femoral metaphysis. Used for dating (14-26 weeks) and growth assessment. Normal growth: 2 mm per week. Short femur: can be constitutional (especially with short parents), associated with aneuploidy (trisomy 21, skeletal dysplasias), or IUGR.

Fencer's Pose

- See Tonic Neck Reflex.

FeNO (Fractional Exhaled Nitric Oxide)

- A non-invasive biomarker of type 2 airway inflammation (eosinophilic). Normal: <25 ppb (adults), <35 ppb (children). Elevated (>50 ppb): eosinophilic asthma, good response to inhaled corticosteroids. Low (<25 ppb): non-eosinophilic

asthma, neutrophilic inflammation.

Fenofibrate

- A fibric acid derivative (fibrate) activating PPAR-alpha, increasing HDL and reducing triglycerides, with modest LDL effects. Used for severe hypertriglyceridemia (pancreatitis risk), mixed dyslipidemia, and as an adjunct in residual hypertriglyceridemia.

Fentanyl Derivatives

- Opioid derivatives used in anesthesia include fentanyl (most common, 50 to 100 mcg IV), sufentanil (5 to 10 times more potent than fentanyl, 5 to 10 mcg IV, used in cardiac anesthesia), remifentanyl (ultra-short-acting, metabolized by esterases, 0.1 to 0.5 mcg/kg/min infusion), alfentanil (shorter duration than fentanyl, used for brief pr

Ferritin

- Intracellular iron storage protein reflecting total body iron stores. Low ferritin: iron deficiency anemia (<30 ng/mL, absolute; <100 can be consistent with iron deficiency in inflammation).

Fertilisation

- The union of a sperm and an egg (oocyte) to form a zygote, the first cell of a new individual.

Fertility

- The biological capacity to conceive and produce offspring. Fertility depends on the proper functioning of the reproductive system in both males and females, including ovulation, sperm production, and hormonal regulation.

Fertility Preservation

- Measures taken before gonadotoxic cancer therapy to preserve reproductive potential, including embryo or oocyte cryopreservation, sperm banking, and ovarian or testicular tissue cryopreservation.

FESS

- Functional endoscopic sinus surgery is a minimally invasive surgical technique that uses nasal endoscopes and specialized instruments to restore normal sinus drainage and ventilation by removing obstructive tissue and opening blocked sinus ostia, while preserving as much normal sinus mucosa and anatomy as possible.

Fetal Acid-Base Status

- The assessment of fetal oxygenation and acid-

base balance during labor, reflecting fetal metabolic condition. Normal: umbilical artery pH 7.25-7.35, pCO₂ 40-50 mmHg, base excess 0 to -5. Fetal acidosis: pH <7.20 (mild), pH <7.00 (severe, associated with neonatal encephalopathy).

Fetal Alcohol Spectrum Disorder (FASD)

- A broad spectrum of physical, behavioral, and cognitive impairments caused by prenatal alcohol exposure.

Fetal Alcohol Syndrome

- A collection of birth defects resulting from exposure of the fetus to alcohol during pregnancy, representing the most severe form of fetal alcohol spectrum disorder (FASD).

Fetal Attitude

- The relationship of fetal body parts to one another, describing the degree of flexion or extension of the fetal head and extremities. Normal attitude is full flexion (vertex presentation with chin tucked to chest). attitudes include deflexion (less flexion), military (neutral position), brow, face, or complete extension.

Fetal Bradycardia

- A baseline fetal heart rate <110 bpm for 10 minutes. Moderate (100-110 bpm): often benign, especially in post-term fetuses. Severe (<100 bpm): concerning for fetal hypoxia, acidosis, congenital heart block (SLE, anti-Ro/anti-La antibodies), cord accident, placental abruption, or maternal seizures.

Fetal Distress

- Abnormal fetal status during labor, indicated by abnormal fetal heart rate patterns (late or prolonged decelerations, minimal variability, absent accelerations). May indicate fetal hypoxia, cord compression, or placental insufficiency.

Fetal Echocardiography

- Ultrasound imaging of fetal heart. Evaluates congenital heart disease.

Fetal Growth Restriction (FGR)

- Failure of a fetus to reach its biologic growth potential, often associated with placental insufficiency. It is assessed using serial growth measurements, amniotic fluid, Doppler studies, and fetal surveillance rather than a single estimated-weight threshold alone.

Fetal Heart Rate

- Normal fetal heart rate ranges from 110-160 beats per minute. Monitored via auscultation with a fetoscope, Doppler ultrasound, or continuous electronic fetal monitoring. Baseline FHR is determined over a 10-minute window and is modulated by the autonomic nervous system.

Fetal Kick Counts (Fetal Movement Counting)

- A method of antepartum fetal surveillance where the mother counts fetal movements (kicks, rolls, stretches) daily. Cardiff count (10 kicks in 2 hours): lie on the left side, record the time until 10 movements are felt. Normal: 10 movements in 2 hours. Decreased fetal movement is associated with increased risk of stillbirth and IUGR.

Fetal Lie

- The relationship of the long axis of the fetus to the long axis of the mother. Longitudinal lie (parallel, 99%) or transverse lie (perpendicular, 1%). Most common is left occiput transverse or left occiput anterior. Transverse lie at term is abnormal and requires cesarean delivery if not corrected.

Fetal Macrosomia

- Excessive fetal size, often defined as estimated or birth weight of at least 4,000 or 4,500 g depending on the clinical context. Diabetes, excessive maternal weight gain, and prior macrosomia are risk factors; complications include shoulder dystocia, birth trauma, and cesarean delivery.

Fetal Presentation

- The relationship of the fetal presenting part to the maternal pelvis. Includes cephalic (head first, most common at 97%), breech (buttocks or feet first), shoulder/oblique, and transverse lie. Cephalic presentations include vertex, face, brow, and compound. Breech presentations include complete, frank, and footling.

Fetal Scalp Electrode (FSE)

- A spiral wire electrode placed on the fetal scalp through the cervix to directly record the fetal heart rate. Provides a more accurate FHR signal than external Doppler, particularly in obese patients or when external monitoring is inadequate. Indications: poor-quality external tracing, need for internal mon-

itoring.

Fetal Scalp Stimulation

- A test of fetal well-being during labor by digitally rubbing or pinching the fetal scalp during a vaginal exam. A reactive acceleration (increase of 15 bpm for 15 seconds) in the FHR in response to stimulation is a reassuring sign of normal fetal acid-base status (negative predictive value 95% for fetal acidosis).

Fetal Tachycardia

- A baseline fetal heart rate >160 bpm for 10 minutes. Mild (161-180 bpm) or severe (>180 bpm).

Fetal Ultrasound

- Ultrasound imaging during pregnancy. First trimester (6-12 weeks): confirm pregnancy, gestational age (crown-rump length), number of fetuses, viability (heart activity), nuchal translucency (screening for aneuploidy at 11-13 weeks).

Fetal Viability

- The gestational age at which a fetus has the potential to survive outside the uterus with medical support, generally considered to be 23-24 weeks of gestation. Survival rates increase significantly with each additional week: approximately 30-50% at 23 weeks, 50-70% at 24 weeks, and over 90% by 27-28 weeks.

Fetishistic Disorder

- A paraphilic disorder characterized by recurrent, intense sexual arousal from either the use of nonliving objects (fetish) or a highly specific focus on nongenital body parts (partialism), lasting 6 months and causing distress or impairment or involving acting on these urges.

Fetomaternal Hemorrhage

- Transplacental passage of fetal red blood cells into the maternal circulation. Occurs in small amounts during normal pregnancy but can be massive during trauma, placental abruption, or cesarean delivery.

Fetus

- An unborn offspring from the 9th week of gestation (after the embryonic period) until birth. The fetal period is characterized by rapid growth, maturation of organ systems, and increasing complexity of function.

FEV1/FVC Ratio

- Fraction of air exhaled in first second relative to total. < 0.70 post-bronchodilator indicates obstruction. Normal or elevated in restrictive disease.

Fever

- Elevated body temperature (>38 degrees C or 100.4 degrees F). Workup depends on duration, height, and patient factors. Common causes: infections (bacterial, viral), autoimmune disorders, malignancies, drug reactions. Management: antipyretics (acetaminophen, ibuprofen), treat underlying cause.

Fever of Unknown Origin

- Persistent or recurrent fever for which an appropriate initial evaluation has not established a diagnosis.

Fexofenadine

- A second-generation (non-sedating) H1 antihistamine used for allergic rhinitis and chronic urticaria. Peripherally selective with minimal sedation and anticholinergic effects; no significant CNS penetration. Adverse effects: headache, dizziness, and dry mouth.

Fiber

- Indigestible carbohydrate from plant cell walls. Soluble (oats, beans, apples) lowers cholesterol, blood glucose. Insoluble (whole grains, vegetables) adds bulk, prevents constipation. Recommended: 25-30 g/day.

Fiber Optic

- See radiology.

Fibrin Sealant

- Biological adhesive for surgical closure. Thrombin + fibrinogen. Hemostasis, sealant.

Fibrinogen

- Coagulation factor. Acute phase reactant. Low in liver disease, DIC.

Fibrinoid Necrosis

- A specialized pattern of vascular necrosis characterized by the deposition of immune complexes, fibrin, and plasma proteins within the walls of blood vessels, creating an amorphous, bright pink, smudgy eosinophilic appearance on hematoxylin and eosin staining.

Fibrinolysis

- The enzymatic breakdown of fibrin within a blood

clot, principally mediated by plasmin generated from plasminogen. It limits clot extension and supports restoration of vascular patency; excessive fibrinolysis can cause bleeding, whereas impaired fibrinolysis can promote thrombosis.

Fibrinous Pericarditis

- Inflammation of the pericardium characterized by deposition of fibrin on the pericardial surfaces, giving them a rough, shaggy appearance described as bread-and-butter pericarditis.

Fibroadenoma

- The most common benign tumor of the female breast, a proliferation of stromal and epithelial elements presenting as a firm, rubbery, smooth, mobile, painless mass, most often in women aged fifteen to thirty. The mass slips easily under the fingers (breast mouse), and ultrasonography shows a well-defined oval hypoechoic lesion.

Fibrocartilage

- Strong cartilage containing dense bundles of type I collagen, adapted to resist compression and tension. It forms intervertebral discs, the pubic symphysis, and menisci.

Fibrocystic Breast Changes

- A benign condition characterized by lumpy, painful breast tissue, most common in women aged 20-50, driven by hormonal fluctuations during the menstrual cycle (estrogen and progesterone). Microscopic features include stromal fibrosis, cyst formation (macrocyts and microcyts), and adenosis (increased glandular tissue).

Fibrocystic Breasts

- Benign breast changes involving fibrosis and cyst formation, often producing lumpy or tender breasts that fluctuate with the menstrual cycle. They are common and are not themselves a premalignant condition.

Fibroids, Cysts, and Polyps

- Benign growths that can develop in the reproductive organs. Fibroids are smooth muscle tumors of the uterus; cysts are fluid-filled sacs that commonly form on the ovaries; and polyps are tissue overgrowths in the endometrium or cervix.

Fibroma

- A benign cardiac tumour composed of fibroblasts

and collagen, usually arising within the ventricular myocardium or interventricular septum in children. It may cause ventricular obstruction or clinically important arrhythmias; echocardiography and cardiac MRI help define its location and extent.

Fibromuscular Dysplasia

- Non-atherosclerotic arterial disease affecting renal arteries. Causes renovascular hypertension in young women. String-of-beads on angiography from medial fibroplasia. Treated with angioplasty alone which is highly effective. Renal function normalizes after successful procedure.

Fibromyalgia

- A chronic centralized pain syndrome characterized by widespread musculoskeletal pain (>3 months), fatigue, sleep disturbance, cognitive dysfunction (fibro fog), and somatic symptoms, affecting 2-4% of population (female:male 9:1).

Fibrosarcoma

- A malignant tumor of fibroblasts, a soft tissue sarcoma of adulthood or infancy, arising in deep soft tissues, bones, or paranasal sinuses.

FibroScan

- Non-invasive transient elastography technique measuring liver stiffness to assess hepatic fibrosis. Values correlate with fibrosis stage with higher values indicating more advanced fibrosis. The technique avoids the need for liver biopsy in many patients. Results may be unreliable in obesity, ascites, and narrow intercostal spaces.

Fibrosing alopecia

- A group of primary cicatricial (scarring) alopecias in which chronic inflammation leads to permanent destruction of hair follicles and replacement by fibrous tract remnants. It includes conditions such as lichen planopilaris, frontal fibrosing alopecia, and central centrifugal cicatricial alopecia.

Fibrosis

- The excessive deposition of extracellular matrix components, primarily collagen types I and III, in response to chronic tissue injury. It represents a pathological wound healing response that leads to organ scarring and dysfunction.

Fibrous Dysplasia

- A benign skeletal disorder in which normal bone

is replaced by fibro-osseous tissue, producing weakened or misshapen bone. It may affect one bone or multiple bones and can cause pain, deformity, pathological fracture, or nerve compression; imaging often shows a characteristic ground-glass appearance.

Fibula

- Slender lateral leg bone, not weight-bearing, serving for muscle attachment and forming the lateral malleolus. Head provides biceps femoris and lateral collateral ligament attachment; the common peroneal nerve wraps around its neck, vulnerable to injury. Lateral malleolus extends more distally and posteriorly than the medial malleolus.

Field of View

- Area imaged simultaneously. PET: larger FOV than SPECT.

Fifth Disease

- Erythema infectiosum caused by Parvovirus B19. Characterized by bright red cheeks (slapped cheek appearance) followed by a lacy, reticulated rash on the trunk and extremities. Mild febrile illness in children. Concern in pregnancy (hydrops fetalis risk) and immunocompromised patients (pure red cell aplasia). Diagnosis by serology.

Filariasis

- A group of nematode infections transmitted by mosquitoes and other vectors. The major species causing lymphatic filariasis are *Wuchereria bancrofti*, *Brugia malayi*, and *Brugia timori*. Adult worms reside in lymphatic vessels causing lymphoedema and elephantiasis.

Filoviruses

- A family of negative-sense RNA viruses that includes Ebola and Marburg viruses. These viruses cause severe hemorrhagic fever with high mortality rates. The virus is transmitted through contact with infected body fluids. The incubation period is 2-21 days. Symptoms include fever, headache, myalgia, and hemorrhage.

Filter (Vascular)

- Device placed in inferior vena cava to prevent pulmonary embolism. See IVC Filter.

Filtered Back Projection

- Traditional image reconstruction method. Fast but noisy.

Filtration Fraction

- The fraction of renal plasma flow that is filtered at the glomerulus, calculated as GFR divided by renal plasma flow (RPF). Normal GFR is approximately 125 mL/min and RPF is approximately 625 mL/min, yielding a filtration fraction of approximately 0.20 (20 percent).

Fimbriae

- The finger-like projections at the distal end of each fallopian tube that sweep over the ovary during ovulation to capture the released egg and guide it into the fallopian tube. The fimbriae are lined with ciliated epithelium that creates a current to draw the egg into the tube.

Financial Abuse

- Intentional or neglectful act causing harm to older person. See elder abuse.

Findings

- Detailed description of imaging observations in radiology report.

Fine Crackles

- High-pitched, brief, discontinuous sounds heard at the end of inspiration, caused by opening of small airways. They are characteristic of ILD, pulmonary fibrosis, and heart failure. They are also called Velcro crackles due to their resemblance to the sound of Velcro being pulled apart.

Fine Motor Skills

- The ability to use small muscles (hands/fingers) for precise movements, developing with age: reaching (3-4 months), grasping (5-6 months), pincer grasp (9-12 months), self-feeding (12-18 months), building towers/scribbling (18-24 months), drawing shapes and using utensils (3-5 years), and handwriting.

Fine Needle Aspiration

- A diagnostic procedure that uses a thin, hollow needle to extract cells or fluid from a lump or mass for microscopic examination.

Fine Needle Aspiration (FNA)

- A percutaneous sampling technique in which a 22 to 25-gauge needle, often ultrasound- or palpation-guided, aspirates cellular material from a palpable or imaging-detected lesion for cytologic examination, with minimal trauma and lower morbidity than core or open biopsy.

Finkelstein Test

- Wrist deviation test for De Quervain tenosynovitis. Pain over radial styloid.

First Stage of Labor

- From the onset of regular contractions to full cervical dilatation. Two phases: latent (0-6 cm, slow progress, 6-12 hours in nulliparas) and active (6-10 cm, faster progress, 1-2 cm/hour in nulliparas). Active phase begins at 6 cm (the Friedman curve was revised: the old 4 cm threshold was updated to 6 cm by ACOG/NICHD in 2014).

First Trimester

- The period from conception to 12 weeks gestational age. Characterized by rapid embryonic organogenesis, rising hCG levels, and establishment of the placenta. Common symptoms include nausea, vomiting, fatigue, and breast tenderness. Dating ultrasound performed to confirm gestational age. Risk of miscarriage highest during this trimester.

First-Degree AV Block

- A conduction delay in which every atrial impulse reaches the ventricles but with a prolonged PR interval (>0.20 seconds). It is caused by delayed conduction through the AV node or His-Purkinje system. First-degree block is usually benign and asymptomatic.

First-Degree Burn

- A superficial burn affecting only the epidermis: erythema, pain, and dryness without blistering (e.g., mild sunburn); heals in 3-7 days without scarring. Assessment: size (percent BSA—rule of nines), depth, and cause (thermal, chemical, electrical).

First-Episode Psychosis

- The first clinically recognized episode of psychosis, regardless of eventual diagnosis. It requires urgent assessment for substance exposure, medication effects, mood symptoms, neurologic disease, systemic illness, safety risk, and functional decline.

First-Pass Effect

- The metabolism of a drug by the liver and intestinal wall before it reaches the systemic circulation after oral administration.

First-Pass Metabolism

- The metabolism of a drug by the liver before it reaches systemic circulation after oral administration.

tion. Drugs with high first-pass metabolism (e.g., morphine, nitroglycerin, lidocaine) have low oral bioavailability and may require alternative routes (sublingual, transdermal, IV).

FISH (Fluorescence In Situ Hybridization)

- A molecular cytogenetic technique using fluorescent DNA probes that hybridize to specific chromosomal regions. Faster than karyotype (24-48 hours, vs. 7-14 days).

Fistula

- Abnormal connection involving the urinary tract. See Fistula (Pathology).

Fistula (Pathology)

- An abnormal epithelial-lined connection between two body surfaces or organs that normally do not communicate. Fistulas may be congenital, such as tracheoesophageal fistula, or acquired, resulting from inflammation, infection, surgery, trauma, or malignancy.

Fistula-in-Ano

- Abnormal epithelialized tract connecting the anal canal to the perianal skin, representing a common and distressing complication of Crohn disease and also occurring following perianal abscess drainage or from cryptoglandular infection.

Fitz-Hugh-Curtis Syndrome

- Perihepatitis associated with pelvic inflammatory disease caused by ascension of *Neisseria gonorrhoeae* or *Chlamydia trachomatis* from the pelvis to the liver capsule along the right paracolic gutter. Presents with right upper quadrant pain, fever, and tender hepatomegaly often in young sexually active women.

Five-Year Survival Rate

- The proportion of patients alive five years after diagnosis or treatment; a standard benchmark for comparing cancer outcomes.

Fixed Drug Eruption

- A recurrent drug reaction appearing at the same site with each exposure to the offending drug. It presents as well-demarcated, round to oval, erythematous to violaceous plaques that may blister. Common culprits include sulfonamides, tetracyclines, NSAIDs, and barbiturates.

Flail Chest

- A clinical condition caused by three or more consecutive ribs fractured in two or more places, creating a free-floating segment of chest wall. The segment moves paradoxically inward during inspiration and outward during expiration. It causes severe pain, impaired ventilation, and may require mechanical ventilation.

Flank Pain

- Pain arising from the lateral back between the ribs and iliac crest, often reflecting disease of the kidney, ureter, musculoskeletal structures, or adjacent abdominal organs. Associated fever, haematuria, vomiting, urinary symptoms, trauma, or hypotension helps distinguish infection, obstruction, stones, injury, and other causes.

Flashes

- Perception of flickering light or lightning-like streaks from mechanoelectrical stimulation of the retina rather than an external source. The leading cause is posterior vitreous detachment, often with floaters; retinal tears and detachment, migraine aura, and optic neuritis also produce flashes.

Flat Feet

- Reduction or absence of the medial longitudinal arch, causing the sole to approach or contact the ground during standing. Flexible flat feet are common and often asymptomatic; rigid or painful flat feet may result from structural deformity, tendon dysfunction, or arthritis.

Flatfoot

- A reduction or collapse of the medial longitudinal arch of the foot, with increased contact of the sole with the ground. It may be flexible and asymptomatic or rigid and painful; causes include normal development, ligamentous laxity, tendon dysfunction, trauma, inflammatory disease, and structural deformity.

Flatulence

- Passage of intestinal gas through the anus. It reflects swallowed air and bacterial fermentation of undigested carbohydrates and may be increased by dietary factors, constipation, malabsorption, altered gut motility, or functional gastrointestinal disorders; persistent symptoms require assessment for associ-

ated disease.

Flecainide

- A class IC sodium channel blocker that slows conduction velocity with little effect on refractoriness. Used for suppression of supraventricular arrhythmias (atrial fibrillation in patients without structural heart disease—'pill-in-the-pocket', paroxysmal SVT) and ventricular arrhythmias.

Flexibility

- Tissue extensibility. Stretching programs improve flexibility.

Flexible Nasolaryngoscopy

- Office examination of the nasal cavity, nasopharynx, pharynx, and larynx using a flexible fiberoptic or digital endoscope passed through the nose. It evaluates voice change, swallowing symptoms, nasal obstruction, lesions, vocal-fold movement, and airway anatomy.

Flexible Sigmoidoscopy

- Endoscopic examination of the rectum and sigmoid colon (up to 60 cm). Indications: colorectal cancer screening (q5-10 years, combined with FIT), evaluation of rectal bleeding, chronic diarrhea, and surveillance of known distal colonic disease (ulcerative colitis).

Flexion

- Movement that decreases the angle between two bones or bends a joint, generally in the sagittal plane.

Flexor Digitorum Profundus

- Deep anterior forearm muscle with ulnar and humeroulnar heads, inserting on distal phalanges bases of digits 2-5. Medial half (digits 4-5) innervated by ulnar nerve; lateral half (digits 2-3) by anterior interosseous nerve (C8-T1). Flexes DIP, PIP, MCP joints, and wrist.

Flexor Digitorum Superficialis

- Large anterior forearm muscle with humeroulnar, radial, and ulnar heads, inserting on middle phalanges of digits 2-5. Innervated by median nerve (C8-T1). Flexes PIP joints, MCP joints, and wrist. Important for median nerve function assessment. Tendons pass through carpal tunnel, susceptible to stenosing tenosynovitis (trigger finger).

Floaters

- Mobile spots, threads, cobwebs, or rings in the visual field caused by vitreous opacities casting shadows on the retina. Most arise from posterior vitreous detachment, an age-related separation of the vitreous gel from the retina; hemorrhage, vitritis, and asteroid hyalosis are other causes.

Floating Ribs

- Ribs 11-12 with short costal cartilages terminating in abdominal wall musculature, not articulating with the sternum or other ribs. Most mobile and least protected, susceptible to fracture from direct trauma. Fractures may injure underlying kidneys, requiring hematuria and flank tenderness assessment in trauma.

Flossing

- Interdental cleaning with dental floss. Removes plaque between teeth. Essential for periodontal health.

Flow Cytometry

- Technique that measures physical and chemical characteristics of cells or particles in a fluid stream, using laser-based detection. Commonly used in hematology: immunophenotyping of leukemia and lymphoma (CD markers: CD3, CD19, CD20, CD34, CD33, CD13), detection of minimal residual disease, and evaluation of immunodeficiency

Flow-Volume Loop

- Graphical representation of airflow during forced breathing. Obstructive: scooped expiratory limb. Restrictive: narrowed loop with preserved shape. Variable extrathoracic obstruction: flattening of inspiratory limb.

Flu

- Influenza: a highly contagious acute respiratory illness caused by influenza A and B viruses (orthomyxovirus, RNA, frequent antigenic drift/shift). Features: sudden fever, myalgia, headache, cough, sore throat, malaise; complications: pneumonia (viral or bacterial—*S. aureus*, *S.*

Fludeoxyglucose F-18

- See nuclear medicine.

Fludeoxyglucose F-18 (FDG)

- Radiotracer used in PET imaging, analog of glucose labeled with fluorine-18. Taken up by cells

via glucose transporters and phosphorylated by hexokinase, trapped intracellularly. Uptake reflects metabolic activity. Used for oncology (tumor detection, staging, response assessment), neurology (epilepsy, dementia), cardiology (viability).

Fluid Imbalance

- An abnormal excess or deficiency of body water or electrolytes such as sodium, potassium, calcium, or bicarbonate. It may result from dehydration, kidney or heart disease, endocrine disorders, gastrointestinal losses, or inappropriate fluid administration.

Fluid Overload

- Excess total body fluid due to impaired renal sodium/water excretion or excessive intake, leading to edema, weight gain, hypertension, and pulmonary congestion. Causes: heart failure, renal failure (acute/chronic), nephrotic syndrome, cirrhosis, iatrogenic (IV fluids), and hypoalbuminemia.

Fluorescein Angiography

- An ophthalmological diagnostic procedure that involves intravenous injection of sodium fluorescein dye and sequential photography of the retinal vasculature using a specialized fundus camera fitted with excitation and barrier filters, providing detailed information about retinal blood flow, vascular permeability, and retinal pigment epith

Fluorescence In-Situ Hybridization (FISH)

- A targeted cytogenetic technique using fluorescent DNA probes to detect or localize specific chromosome sequences. It can identify selected aneuploidies, microdeletions, gene rearrangements, or amplifications but does not survey the entire genome.

Fluoride

- Mineral that strengthens tooth enamel and inhibits bacterial acid production. Sources: water fluoridation (0.7 ppm optimal), toothpaste, professional applications. Mechanism: incorporates into hydroxyapatite forming fluorapatite (more acid-resistant), promotes remineralization, inhibits bacterial enzymes.

Fluoride Varnish

- Concentrated fluoride application professional use. Reduces caries by 30-40%. Applied 2-4 times yearly for high-risk patients.

Fluorine-18

- PET radionuclide, half-life 110 minutes. Produced in cyclotron via O-18(p,n)F-18 reaction. Used in FDG and other tracers. Requires on-site or nearby cyclotron due to short half-life.

Fluoroquinolones

- Broad-spectrum bactericidal antibiotics that inhibit bacterial DNA gyrase and topoisomerase IV, enzymes essential for DNA replication, transcription, and repair. They are synthetic derivatives of nalidixic acid and include ciprofloxacin, levofloxacin, moxifloxacin, and others.

Fluoroscopy

- Real-time X-ray imaging used to visualize dynamic processes.

Fluorosis

- Enamel hypomineralization from excess fluoride during development. Mild: white spots. Severe: brown stains, pitting. Prevention: monitor fluoride intake in children.

Fluorouracil side effect

- Toxicity from the antimetabolite 5-fluorouracil causing mucositis, diarrhea, hand-foot syndrome, myelosuppression, and potential cardiotoxicity, requiring dose adjustment or discontinuation.

Fluoxetine

- A selective serotonin reuptake inhibitor (SSRI) used for major depressive disorder, obsessive-compulsive disorder, bulimia, panic disorder, and premenstrual dysphoric disorder.

Flushing

- Transient reddening and warmth of the skin caused by increased cutaneous blood flow. It may result from heat, emotion, exertion, fever, alcohol, endocrine or neuroendocrine disease, mast-cell activation, medication exposure, or autonomic dysfunction; associated diarrhoea, wheeze, hypotension, or palpitations may indicate systemic disease.

Flutter Valve

- A handheld device combining PEP therapy with oscillations during exhalation. The oscillations create shear forces that loosen mucus from airway walls, while the PEP keeps airways open. It is used for airway clearance in bronchiectasis, CF, and COPD.

FNA

- A percutaneous sampling technique in which a 22 to 25-gauge needle, often ultrasound- or palpation-guided, aspirates cellular material from a palpable or imaging-detected lesion for cytologic examination, with minimal trauma and lower morbidity than core or open biopsy. See Fine Needle Aspiration (FNA).

Focal Nodular Hyperplasia

- Benign liver lesion with a characteristic central stellate scar visible on imaging. The lesion represents a hyperplastic response to a pre-existing vascular malformation rather than a true neoplasm. Associated with oral contraceptive use though it also occurs in men and children. No risk of hemorrhage or malignant transformation.

Focal Segmental Glomerulosclerosis

- Pattern of glomerular injury with sclerosis in some focal glomeruli and segments. Can be primary idiopathic or secondary to HIV, obesity, heroin use, or adaptive responses to nephron loss. Histology shows segmental sclerosis with hyalinosis. Less responsive to steroids than minimal change with approximately 50% achieving remission.

Focal Segmental Glomerulosclerosis (FSGS)

- The most common cause of nephrotic syndrome in African American adults and Hispanic populations. Associated with HIV infection (HIV-associated nephropathy), heroin use, sickle cell disease, massive obesity, interferon therapy, and renal hyperfiltration (solitary kidney).

Focal Seizure

- A seizure arising from networks limited to one cerebral hemisphere at onset. Awareness may be preserved or impaired, and motor, sensory, autonomic, cognitive, or emotional features depend on the involved region; a focal seizure may spread to become bilateral tonic-clonic.

Focal-Onset Seizure

- A seizure that begins within networks limited to one cerebral hemisphere. Awareness may be preserved or impaired, and motor, sensory, autonomic, cognitive, or emotional features reflect the involved cortical region; it may evolve to bilateral tonic-clonic activity.

FODMAP Diet

- Restriction of Fermentable Oligo-, Di-, Mono-saccharides And Polyols. Used for IBS management. Low FODMAP diet followed by gradual reintroduction.

Folate Deficiency

- Folate (vitamin B9) deficiency causes megaloblastic anemia identical to B12 deficiency but without neurologic manifestations (except possible cognitive changes). Dietary sources: leafy greens, legumes, liver; daily requirement 400 mcg. Absorption in jejunum.

Folate-deficiency anemia

- A type of megaloblastic anemia caused by insufficient folic acid, a B vitamin essential for DNA synthesis, cell division, and red blood cell production.

Foley Catheter

- Indwelling urinary catheter with balloon. For urine drainage. Various sizes: 12-24 Fr.

Folic Acid Supplementation

- Daily folic acid (400-800 mcg) taken before conception and during the first trimester to prevent neural tube defects (NTDs: spina bifida, anencephaly, encephalocele). Folic acid is a B vitamin (folate) essential for DNA synthesis and cell division.

Follicle Stimulating Hormone

- FSH: a gonadotropin secreted by the anterior pituitary, regulating gonadal function. In females: stimulates ovarian follicle growth and estradiol production; LH surge triggers ovulation. In males: stimulates Sertoli cells and spermatogenesis. Controlled by GnRH and feedback (estrogen, inhibin).

Follicular Lymphoma

- The most common indolent non-Hodgkin lymphoma, accounting for 20-25 percent of NHL cases, characterized by t(14;18) translocation leading to BCL2 overexpression and follicular growth pattern. Clinical presentation: painless lymphadenopathy, often widespread, with bone marrow involvement in 60-70 percent.

Follicular Thyroid Carcinoma

- Second most common thyroid carcinoma accounting for approximately ten percent of cases. Distinguished from follicular adenoma by capsular or vas-

cular invasion on histological examination which cannot be determined by fine-needle aspiration. Metastasizes hematogenously to bone, lung, and liver.

Folliculitis

- Inflammation of hair follicles, presenting as pustules centered on follicles. Bacterial folliculitis (*Staphylococcus aureus*) is most common. Hot tub folliculitis (*Pseudomonas*) occurs after hot tub exposure. Gram-negative folliculitis complicates long-term antibiotic use. Pseudofolliculitis barbae (razor bumps) results from ingrown hairs.

Follow-Up

- Post-rehabilitation follow-up. Monitor progress, adjust treatment, prevent complications.

Follow-Up Recommendations

- Recommendations for further imaging or evaluation. Based on guidelines (Fleischner Society for lung nodules).

Fomepizole

- An alcohol dehydrogenase inhibitor used as an antidote for toxic alcohol exposure, particularly methanol and ethylene glycol. It prevents formation of toxic metabolites and is used with supportive care and hemodialysis when indicated.

Fomite

- An inanimate object or surface capable of transmitting infectious agents (viruses, bacteria, fungi) between people (e.g., doorknobs, towels, medical equipment, toys, needles). Transmission occurs when contaminated fomites contact mucous membranes or skin, or are involved in procedures.

Fondaparinux

- A synthetic pentasaccharide that selectively inhibits factor Xa via antithrombin, with no direct thrombin inhibition. Used for prophylaxis and treatment of venous thromboembolism and acute coronary syndromes (as an alternative to LMWH).

Fontanelle

- Soft membranous spaces between skull bones that allow brain growth and skull molding during delivery. Anterior fontanelle diamond-shaped, closes by 12-18 months. Posterior fontanelle triangular, closes by 2-3 months. Bulging fontanelle suggests increased intracranial pressure; sunken fontanelle indicates dehydration.

Food Allergy

- Immune-mediated reaction to food protein. IgE-mediated: anaphylaxis. Non-IgE: EoE, food protein-induced enterocolitis. Common allergens: milk, egg, peanut, tree nut, fish, shellfish, soy, wheat.

Food Intolerance

- Adverse reaction not immune-mediated. Lactose intolerance, gluten sensitivity (non-celiac), histamine intolerance, FODMAP sensitivity. Diagnosis: elimination diet, challenge.

Food Poisoning

- Illness caused by consuming food contaminated with bacteria, viruses, parasites, or their toxins. Common pathogens include *Salmonella*, *Escherichia coli*, *Listeria*, and norovirus, with symptoms such as nausea, vomiting, diarrhea, abdominal cramps, and fever.

Food-Borne Disease

- Disease from contaminated food. Examples: salmonella, *E. coli*, listeria.

Food-Drug Interaction

- An alteration in drug absorption, metabolism, distribution, or effect caused by a food, beverage, nutrient, or supplement. Examples include vitamin K affecting warfarin response, grapefruit affecting selected CYP3A4 substrates, and calcium impairing absorption of some medicines.

Foodborne Illness

- Any illness resulting from the ingestion of contaminated food, encompassing infections and intoxications caused by pathogens, toxins, or chemical contaminants. It affects millions annually and poses particular risks to pregnant women, young children, older adults, and immunocompromised individuals.

Foot Drop

- Weakness or paralysis of ankle dorsiflexion causing the forefoot to drag or the patient to lift the knee abnormally while walking. Common causes include common peroneal nerve injury, L5 radiculopathy, motor-neuron disease, and peripheral neuropathy.

Force Plates

- Measure ground reaction forces during gait. Assess balance, symmetry, loading.

Forced Expiratory Volume

- The volume of air forcibly exhaled in a specified time during a maximal expiratory effort. FEV1 is the volume exhaled in the first second; normally at least 70-80 percent of the forced vital capacity (FEV1/FVC ratio).

Forced Oscillation Technique

- Similar to impulse oscillometry, applying small pressure oscillations at various frequencies during tidal breathing to measure respiratory resistance and reactance. It provides information about airway mechanics without requiring forced expiratory maneuvers.

Forced Vital Capacity

- Total air exhaled after maximal inspiration. Reduced in both obstructive and restrictive disease. In restrictive disease, FVC reduced with preserved ratio.

Forceps

- Grasping instrument. Tissue forceps: with teeth. Dressing forceps: without teeth.

Forceps (Obstetric)

- See Labor Instruments.

Forceps (Surgical)

- Instruments for surgery. See surgical instruments.

Forceps (Vascular)

- See Forceps.

Forceps Delivery

- Assisted vaginal delivery using forceps (curved blades applied to the fetal head) to expedite delivery for maternal or fetal indications (prolonged second stage, fetal distress, maternal exhaustion, medical conditions limiting pushing).

Forceps-Assisted Delivery

- Operative vaginal delivery using forceps (two curved blades applied to the fetal head). Types: outlet (forceps applied at the introitus, fetal scalp visible), low (station +2, not at outlet), midpelvic (station +2 but head above ischial spines), and rotational (forceps used to rotate from OP to OA).

Fordyce's angiokeratomas

- Benign vascular ectasias of the superficial dermal capillaries covered by an epidermal reaction, appearing as small, dark red to purple, asymptomatic

papules on the scrotum, vulva, or oral mucosa. They are common, increase with age, and are clinically insignificant. They may bleed if traumatized.

Foreign Body

- Any object lodged in body. Types: organic (wood, thorn), inorganic (metal, glass, plastic). Radiopaque vs. radiolucent. Removal: surgical, endoscopic, spontaneous passage. Complications: infection, granuloma, migration.

Foreign Body Aspiration

- Inhalation of object into airway. Common in children (peanuts, coins, small toys). Signs: choking, coughing, stridor, wheezing, decreased breath sounds. Diagnosis: chest X-ray (may be negative), bronchoscopy.

Foreign-Body Aspiration

- Entry of an object into the larynx, trachea, or bronchi, most often in toddlers. Sudden cough, choking, unilateral wheeze, or focal air trapping may occur; complete airway obstruction is an immediate emergency and requires age-appropriate resuscitation.

Forensic Anthropology

- The application of anthropological methods to identify human remains and analyze evidence, mainly skeletal material. Tasks: determination of biological profile (age at death, sex, ancestry, stature), distinguishing human from non-human bone, estimation of time since death (bone weathering, DNA), identifying trauma (perimortem vs.

Forensic Entomology

- The study of insects and other arthropods in legal investigations, principally to estimate the post-mortem interval (PMI) and to link suspects to scenes.

Forensic Evidence

- Physical, biological, documentary, or testimonial material gathered, preserved, and analyzed to establish facts in a legal investigation. Categories include biological evidence (blood, semen, hair), trace evidence (fibers, glass, gunshot residue), pattern evidence (fingerprints, toolmarks, bloodstain patterns), and digital evidence.

Forensic Examination

- Structured medicolegal evaluation of a living or

deceased person, scene, or evidence to document findings relevant to a legal matter. Components include sexual assault examination with evidence collection, child abuse assessment, injury documentation, driving-impairment evaluation, and autopsy.

Forensic Medicine (Legal Medicine)

- Branch of medicine that applies medical, biological, and scientific knowledge to legal questions and the administration of justice.

Forensic Odontology

- The application of dental science in legal contexts. Main uses: human identification through comparison of dental records (restorations, radiographs, unique anatomy—antemortem vs.

Forensic Pathology

- A branch of pathology and medicine applying medical and pathological knowledge to legal questions, primarily determining cause and manner of death and documenting injuries. The forensic pathologist performs autopsies in sudden, unnatural, suspicious, or violent deaths, interprets injuries (blunt vs. sharp vs. gunshot vs.

Forensic Psychiatry

- The medical specialty at the interface of psychiatry and the law. Core activities: competency to stand trial, criminal responsibility/insanity defense (mental state at time of offense), dangerousness/risk assessment, civil commitment, testamentary capacity, and personal injury claims.

Forensic Radiography

- The use of imaging (X-ray, CT, MRI, ultrasound) in forensic and medico-legal investigation.

Forensic Toxicology

- The analysis of drugs, poisons, and their metabolites in biological specimens (blood, urine, vitreous humor, hair, tissues) to determine their role in intoxication, poisoning, or death.

Forensic Toxicology Screen

- Laboratory analysis of biological specimens for drugs, alcohols, poisons, and their metabolites. A screening result is presumptive until confirmed and quantified by a validated method such as mass spectrometry, with interpretation based on specimen type, timing, therapeutic use, and postmortem changes.

Formaldehyde

- Preservative, industrial chemical. Irritant, carcinogen (nasopharyngeal cancer). Sensitizer. Use in fume hood, PPE required.

Formalin-Fixed Paraffin-Embedded (FFPE) Tissue

- Tissue preserved in formalin, processed, and embedded in paraffin wax for sectioning and microscopic examination. FFPE blocks support histology, immunohistochemistry, and many molecular tests but may contain fixation-related nucleic-acid or antigen alterations.

Formula Feeding

- Alternative to breastfeeding using commercially prepared infant formula. Types include cow's milk-based, soy-based, hydrolyzed protein, and amino acid-based formulas. Iron-fortified formulas standard to prevent iron deficiency anemia. Preparation requires clean water and proper sterilization of bottles.

Fornix

- The conjunctival fornix is the redundant, dome-shaped fold of conjunctiva where the palpebral conjunctiva transitions to the bulbar conjunctiva, forming a deep recess superiorly (superior fornix), inferiorly (inferior fornix), and medially and laterally (medial and lateral fornices).

Fortification

- Addition of nutrients to food. Examples: iodized salt, vitamin D milk, folic acid grains. Public health strategy for deficiency prevention.

Four-Dimensional Ultrasound

- Real-time 3D ultrasound. Used in obstetrics for fetal imaging.

Fournier Gangrene

- A rapidly progressive necrotizing infection of the perineum, external genitalia, or perianal region. It causes severe pain, swelling, fever, systemic toxicity, and tissue necrosis and requires immediate surgical debridement and broad-spectrum antimicrobial treatment.

Fourth-Degree Tear

- The most severe type of perineal tear during childbirth, extending through the anal sphincter (both internal and external) and into the rectal mucosa.

Fourth-degree tears occur in approximately 0.5-1% of vaginal deliveries and are associated with forceps delivery, shoulder dystocia, large baby, and episiotomy.

Fovea

- A tiny depression in the center of the retina that provides the sharpest and most detailed visual acuity.

Fovea Centralis

- Central macular pit (approximately 1.5 mm) with the highest cone density (approximately 200,000 per square millimeter) and no rods, providing the sharpest visual acuity and color vision. Avascular with overlying retinal layers displaced laterally for direct light access to cones.

Fractional Excretion of Sodium (FENa)

- The percentage of filtered sodium excreted in urine, calculated from serum and urine sodium and creatinine. It may help distinguish prerenal physiology from tubular injury, although diuretics, CKD, sepsis, and other conditions limit interpretation.

Fractional Excretion of Urea (FEUrea)

- The percentage of filtered urea excreted in urine. It may be used as an adjunct when diuretics make FENa less interpretable, but it is also affected by catabolism, sepsis, gastrointestinal bleeding, and chronic kidney disease.

Fractional Flow Reserve

- An invasive physiological measurement during coronary catheterization using a pressure wire to assess the hemodynamic significance of a coronary stenosis. $FFR = Pa/Pd$ (aortic pressure / distal coronary pressure) during maximal hyperemia (adenosine).

Fractional IPV (fIPV)

- A reduced-dose schedule of inactivated polio vaccine used in the WHO and India Universal Immunization Schedule, where one-fifth of the standard intramuscular IPV dose (0.1 mL instead of 0.5 mL) is given by intradermal injection at each of two deltoid sites (total 0.2 mL per fractional dose), rather than the full 0.5 mL intramuscular dose.

Fracture

- A break in the continuity of bone, classified by location, pattern (transverse, oblique, spiral, comminuted), displacement, and whether the fracture

communicates with the external environment (open vs. closed). Fracture healing occurs through inflammatory, reparative, and remodeling phases.

Fracture Displacement

- Loss of normal anatomic alignment between fracture fragments, described by translation, angulation, shortening, or rotation. The acceptable degree depends on the bone, joint involvement, patient age, and functional demands.

Fracture Liaison Service

- A coordinated program that identifies patients with fragility fractures, evaluates bone health and secondary causes, initiates fracture prevention, and ensures follow-up to reduce recurrent fractures.

Fracture of the temporal bone

- A break in the thin bones forming the skull base and ear structures, potentially injuring the facial nerve, carotid artery, or middle ear and presenting with hearing loss, vertigo, or cerebrospinal fluid leak.

Fracture Prevention

- Systemic skeletal disease characterized by low bone mass and microarchitectural deterioration. See osteoporosis.

Fracture-Fixation Device

- An implant such as a plate, screw, intramedullary nail, or external-fixation component used to stabilize a broken bone and maintain alignment during healing. Selection depends on fracture pattern, bone quality, and soft-tissue condition.

Fragile X Syndrome

- X-linked dominant disorder caused by CGG trinucleotide repeat expansion in the FMR1 gene. Most common inherited cause of intellectual disability. Features: long face, prominent ears, macroorchidism in post-pubertal males, and behavioral problems including autism spectrum features. Diagnosis by DNA testing for CGG repeat expansion.

FRAIL Scale

- A brief frailty screen assessing Fatigue, Resistance, Ambulation, Illnesses, and Loss of weight. It identifies risk of frailty but is not a substitute for a comprehensive geriatric assessment.

Frailty

- Clinical syndrome of decreased reserve and re-

silience to stressors. Components: unintentional weight loss, exhaustion, low physical activity, slow gait speed, weak grip strength. Identification: Fried phenotype, frailty index. Consequences: adverse outcomes (falls, hospitalization, mortality), polypharmacy, functional decline.

Frameshift Mutation

- Insertion or deletion of nucleotides not in multiples of three, shifting reading frame. Usually produces nonfunctional protein. Examples: some cystic fibrosis mutations, Tay-Sachs disease.

Francisella tularensis

- A gram-negative coccobacillus that causes tularemia, a zoonotic infection transmitted by ticks, deer flies, or direct contact with infected animals. The organism can be acquired through the skin, respiratory tract, or gastrointestinal tract. Ulceroglandular tularemia presents with skin ulcer and regional lymphadenopathy.

Frank-Starling Mechanism

- The physiological principle stating that stroke volume increases as end-diastolic volume (preload) increases, up to an optimal point. Within physiological limits, greater ventricular stretch leads to stronger contraction due to increased calcium sensitivity of myofilaments and optimal overlap of actin and myosin.

Free Radical

- Molecule with unpaired electron. Highly reactive. Causes cellular damage.

Free Radical Injury

- Free radicals are highly reactive molecules with unpaired electrons that can damage cellular components through oxidative stress. Reactive oxygen species include the superoxide anion, hydrogen peroxide, and the hydroxyl radical generated through the Fenton reaction.

Free Water Clearance

- The rate at which the kidneys excrete water in excess of or less than the amount needed to produce urine isotonic to plasma. Positive clearance indicates excretion of dilute urine; negative clearance indicates retention of water through concentration of urine.

Frenectomy

- Surgical removal or repositioning of frenulum. Indications: diastema, gingival recession, tongue-tie, interference with denture.

Frequency

- Increased daytime micturition, typically more than seven episodes per day, distinguished from polyuria (high urine volume) and nocturia (nighttime voiding). Common causes include urinary tract infection, benign prostatic hyperplasia, overactive bladder, and diabetes mellitus.

Frey's syndrome

- Aberrant regeneration of parasympathetic nerve fibers to sweat glands after injury to the auriculotemporal nerve, most commonly following parotid surgery or trauma. It results in gustatory sweating and facial flushing while eating. Diagnosis is confirmed by the starch-iodine test.

Friedman Curve

- The classic labor curve described by Emanuel Friedman in the 1950s, plotting cervical dilatation against time. Defined the latent and active phases, with active phase beginning at 4 cm. The Friedman curve was the standard for labor management for decades.

Friedreich Ataxia

- See genetics.

Frontal Bone

- Forms the forehead and superior orbits, with the squama frontalis, orbital plates, and nasal part. Contains frontal sinuses draining into the nasal cavity via the frontonasal duct. Articulates with parietal bones at the coronal suture. The metopic suture fuses by age six.

Frontal Lobe

- The largest lobe of the cerebral cortex, lying anterior to the central sulcus, comprising the primary motor cortex (precentral gyrus, Brodmann 4), premotor and supplementary motor areas, prefrontal cortex, and Broca's area (dominant hemisphere, expressive language).

Frontal Sinus

- The frontal sinuses are paired, air-filled cavities located within the frontal bone above the orbits and the root of the nose, separated by a bony septum that may be deviated to one side, and they are the

last of the paranasal sinuses to develop, typically appearing around age seven and reaching full size by late adolescence.

Frontotemporal Dementia

- Dementia affecting frontal and temporal lobes. Behavioral variant (personality changes) or language variant (aphasia). Younger onset (50-60).

Frostbite

- Tissue injury caused by freezing of skin and underlying tissues. Severity grading: first degree (superficial, numbness, erythema), second degree (blisters, clear fluid, pain), third degree (hemorrhagic blisters, necrosis, full-thickness skin loss), fourth degree (deep tissue necrosis, mummification, autoamputation).

Frotteuristic Disorder

- A paraphilic disorder involving recurrent, intense sexual arousal from touching or rubbing against a nonconsenting person, lasting 6 months and causing distress or impairment or involving acting on these urges. Typically occurs in crowded public places (subways, buses, concerts).

Frozen Section

- An intraoperative pathology consultation in which a fresh tissue specimen is rapidly frozen, sectioned on a cryostat, and stained with hematoxylin and eosin within 10 to 20 minutes to guide immediate surgical decision-making. It is used to establish benign versus malignant, assess margins, and identify lymph node metastases.

Frozen Shoulder

- Adhesive capsulitis: progressive shoulder pain and stiffness. Stages: freezing (pain), frozen (stiffness), thawing (recovery). Associated with diabetes, thyroid disease, immobilization. Treatment: physical therapy, corticosteroid injection, manipulation under anesthesia, arthroscopic capsular release.

Fructosamine

- A measurement of glycated serum proteins, primarily albumin, that reflects average glycemia over approximately two to three weeks. It is useful when HbA1c is unreliable because of altered red-cell survival, but it is affected by changes in protein concentration and turnover.

Fructose Malabsorption

- Poor absorption of fructose leading to GI symptoms. Fermentation by bacteria produces gas, osmotic diarrhea. Low-fructose diet beneficial.

Fructose Metabolism

- The enzymatic conversion of fructose primarily in the liver through fructokinase, aldolase B, and downstream glycolytic intermediates. Essential fructosuria is usually benign, whereas aldolase B deficiency causes hereditary fructose intolerance and hypoglycemia after fructose ingestion.

Fuchs Dystrophy

- A progressive corneal endothelial disorder in which endothelial cell loss impairs fluid regulation and causes corneal oedema. Early disease may cause morning blur and glare; advanced disease produces persistent oedema, bullae, pain, and reduced vision. Diagnosis is clinical and supported by slit-lamp examination and corneal imaging.

Full Agonist

- A full agonist is a drug that binds to a receptor and produces the maximum possible biological response that the receptor can generate, achieving 100 percent of the maximal effect. Full agonists have both high affinity for the receptor and high intrinsic activity, meaning they are very effective at activating the receptor once bound.

Functional Asplenia

- Loss of splenic function due to sickling and infarction in sickle cell disease, typically by age 5 years. Increases risk of overwhelming bacterial infection, particularly from encapsulated organisms (*Streptococcus pneumoniae*, *Haemophilus influenzae*).

Functional Assessment

- Evaluation of ability to perform daily activities. Tools: Barthel Index, Lawton IADL Scale, Katz Index. Identifies needs for assistance, predicts outcomes, guides care planning.

Functional Capacity

- The maximum level of physical, cognitive, communication, or self-care performance a person can safely achieve under specified conditions. It is assessed in relation to activities, participation, environment, and personal goals.

Functional Dyspepsia

- Chronic epigastric pain or discomfort without identifiable structural cause. The Rome IV criteria include postprandial distress syndrome with early satiety and postprandial fullness, and epigastric pain syndrome with epigastric pain or burning unrelated to meals.

Functional Electrical Stimulation (FES)

- Application of electrical impulses to peripheral nerves or muscles to produce purposeful contraction during a functional task, such as foot clearance during gait or cycling after neurologic injury.

Functional Independence Measure

- FIM: standardized assessment of disability. Scores 18 (total assistance) to 7 (independent). 18 items: self-care, sphincter control, transfers, locomotion, communication, social cognition.

Functional MRI (fMRI)

- MRI measuring brain activity by detecting blood flow changes. Used for pre-surgical planning, research.

Functional Neurological Disorder

- Neurologic symptoms such as weakness, abnormal movements, sensory loss, or seizures that are incompatible with recognized patterns of structural neurologic disease and arise from altered nervous-system function. Diagnosis is based on positive clinical signs, not merely on normal investigations.

Functional Neurological Symptom Disorder

- A condition in which neurological symptoms such as weakness, abnormal movement, sensory loss, or episodes resembling seizures are incompatible with recognized neurological disease patterns and cause significant distress or impairment.

Functional Neurological Symptom Disorder (Conversion Disorder)

- A condition in which patients present with neurological symptoms (weakness, paralysis, non-epileptic seizures, sensory loss, gait abnormalities, dystonia, speech difficulties) that cannot be explained by structural neurological disease. The symptoms are not intentionally produced (unlike malingering or factitious disorder).

Functional Residual Capacity

- The volume of air remaining in the lungs at the end of a normal expiration, equal to the sum of expiratory reserve volume plus residual volume (approximately 2.2-3 L in adults). FRC represents the lung volume at which the outward recoil of the chest wall balances the inward recoil of the lungs, the resting equilibrium position.

Functional Screening

- Evaluation of ability to perform daily activities. See functional assessment.

Functional Status

- Ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs). ADLs: bathing, dressing, toileting, transferring, continence, feeding. IADLs: shopping, cooking, managing medications, finances, transportation, housekeeping. Decline indicates need for assistance.

Functional Urinary Incontinence

- Urinary incontinence due to physical or cognitive impairment preventing toilet access.

Fundal Height

- The distance from the symphysis pubis to the top of the uterus measured in centimeters. Used to estimate gestational age and fetal growth. Generally correlates with weeks of gestation between 20-36 weeks (e.g., 24 cm at 24 weeks). Discrepancies may suggest fetal growth restriction, macrosomia, polyhydramnios, or multiple gestation.

Fundus Photography

- The documentation of the appearance of the retina, optic nerve, and retinal vasculature using a specialized camera that illuminates and photographs the posterior segment of the eye, providing a permanent, objective record of retinal findings that can be used for comparison over time, patient education, and telemedicine consultations.

Fungal endophthalmitis

- See Fungal Infections.

Fungal Infection

- See Antifungal. Infections caused by fungi ranging from superficial (dermatophytes: tinea, candidiasis) to invasive/systemic disease (aspergillosis, candidemia, cryptococcosis, histoplasmosis) in immunocompromised hosts. Risk factors: neutropenia,

transplantation, HIV/AIDS, corticosteroids, and diabetes.

Fungal Infection (Tinea/Dermatophytosis)

- Superficial fungal infections of skin, hair, or nails caused by dermatophytes (Trichophyton, Microsporum, Epidermophyton). Tinea corporis (ringworm): annular scaling plaque with central clearing. Tinea pedis (athlete's foot): interdigital scaling, maceration. Tinea cruris (jock itch): groin rash sparing scrotum.

Fungal Infections

- Candidiasis: mucosal (thrush, esophagitis, vulvovaginal) - fluconazole; invasive (candidemia, intraabdominal, UTI) - echinocandin (caspofungin, micafungin, anidulafungin) first-line, step-down to fluconazole if susceptible.

Fungal Nail Infection

- Infection of a fingernail or toenail by dermatophytes or other fungi, causing thickening, discoloration, brittleness, separation from the nail bed, or subungual debris.

Fungal pathogenesis

- The mechanisms by which fungi cause disease. Fungal pathogenesis involves adhesion, invasion, and evasion of host defences. Adhesins mediate attachment to host cells. Invasive hyphae penetrate tissues. Capsules and melanin protect against immune responses. Biofilm formation provides protection against antifungal agents.

Fungal resistance mechanisms

- Mechanisms by which fungi develop resistance to antifungal agents. Azole resistance occurs through upregulation of efflux pumps (CDR1, CDR2, MDR1), mutations in the target enzyme ERG11, and overexpression of ERG11. Echinocandin resistance is primarily due to mutations in the FKS1 and FKS2 genes encoding beta-(1,3)-D-glucan synthase.

Fungal Skin Infections

- Skin infections caused by dermatophytes or yeasts, including ringworm (tinea), athlete's foot, and candidiasis. These infections thrive in warm, moist environments and present with red, itchy, scaly, or ring-shaped lesions. Treatment typically involves

topical or oral antifungal medications and measures to keep the skin clean and dry.

Fungemia

- The presence of viable fungi in the bloodstream. Candida species are the most common cause in hospitalized patients; fungemia requires prompt evaluation for a source, metastatic infection, and appropriate antifungal treatment.

Fungus

- A eukaryotic microorganism with a chitin cell wall and ergosterol-containing membrane, classified as yeasts, moulds, or dimorphic fungi. Reproducing by spores, fungi cause superficial, mucosal, subcutaneous, and systemic infections, especially in immunocompromised hosts.

Furosemide

- A potent high-ceiling loop diuretic that reversibly inhibits the $\text{Na}^+/\text{K}^+/2\text{Cl}^-$ cotransporter (NKCC2) in the thick ascending limb of the loop of Henle. Induces rapid diuresis and natriuresis by disrupting the hypertonic medullary osmotic gradient.

Furuncle

- See Boil. A deep follicular infection by Staphylococcus aureus forming a painful, erythematous nodule with a central necrotic core, often requiring incision and drainage. Recurrent furunculosis warrants screening for nasal carriage and immunosuppression (diabetes). Complications: cellulitis, bacteremia, scarring.

Furuncle and Carbuncle

- Furuncle (boil) is a deep infection of the hair follicle caused by Staphylococcus aureus, presenting as a painful, erythematous nodule that may drain pus. Carbuncle is a cluster of interconnected furuncles causing a larger, more deeply invasive infection.

Furuncular myiasis

- A cutaneous infestation caused by larvae of certain fly species (such as Dermatobia hominis or Cuterebra) that develop within boil-like furuncular lesions in the skin, presenting with a painful, erythematous nodule with a central punctum through which the larva breathes.

Furunculosis

- Recurrent or multiple furuncles, which are deep infections of hair follicles and surrounding tissue,

usually caused by *Staphylococcus aureus*. Lesions are painful, erythematous, and purulent and may occur with colonization, close-contact transmission, skin disease, diabetes, immune dysfunction, or other risk factors.

Fusariosis

- An opportunistic invasive fungal infection caused by *Fusarium* species, affecting predominantly immunocompromised patients, particularly those with prolonged neutropenia or hematological malignancies. It causes disseminated infection with skin lesions, bloodstream infection, and involvement of multiple organs.

Fusion

- Union of two tooth buds. Results in large tooth with reduced tooth count.

Facial Edema

- Swelling caused by excess fluid in the tissues of the face. Causes include allergy, infection, trauma, kidney or heart disease, and venous obstruction. Sudden swelling of the lips, tongue, or throat with breathing difficulty is an emergency.

Factor VII

- A blood-clotting protein made mainly in the liver that begins the tissue-factor pathway of coagulation. Inherited deficiency can cause abnormal bleeding, ranging from easy bruising to serious bleeding after injury or procedures.

Factor VIII

- A clotting protein that works with factor IX to help form a stable blood clot. Inherited deficiency causes hemophilia A, characterized by prolonged or spontaneous bleeding into joints, muscles, or other tissues.

Fetal Hypoxia

- Inadequate oxygen available to the fetus before or during birth. It may result from placental problems, reduced maternal oxygenation, cord compression, or prolonged labor. Abnormal fetal heart-rate patterns can indicate risk and require prompt obstetric assessment.

Fissured Tongue

- A usually harmless condition in which grooves or cracks develop on the surface of the tongue. Food and debris may collect in the fissures, causing irri-

tation or odor. Gentle brushing of the tongue and routine oral care help prevent symptoms.

Foot Pain

- Pain in the foot caused by injury, overuse, arthritis, nerve disease, infection, or circulation problems. Persistent pain, inability to bear weight, numbness, discoloration, fever, or an open wound requires medical assessment.

Foreign Body Reaction

- Inflammation and immune response to material that remains in tissue, such as a splinter, suture, implant, or other medical device. It may cause a lump, redness, drainage, pain, or scarring; treatment depends on the material and whether infection is present.

Formication

- A sensation that insects are crawling on or under the skin when none are present. It may occur with nerve disorders, menopause, substance withdrawal, medication effects, or psychiatric illness. Persistent or new symptoms should be medically evaluated.

Frontal Sinusitis

- Inflammation or infection of the frontal sinuses, causing forehead pain or pressure, nasal congestion, and drainage. Severe headache, swelling around the eye, high fever, confusion, or neurologic symptoms may indicate a complication requiring urgent care.

Fetal Hydrops

- Abnormal accumulation of fluid in at least two fetal compartments, such as the skin, chest, abdomen, or around the heart. Causes include fetal anemia, heart disease, infection, chromosome conditions, and immune incompatibility; it requires specialist obstetric evaluation.

Chapter 7

Pocket Dictionary: G Terms

G-Protein Coupled Receptors

- The largest family of cell surface receptors with seven transmembrane alpha-helical domains. Upon ligand binding, the receptor undergoes a conformational change that activates an associated heterotrimeric G-protein (G α , G β , G γ). The G α subunit exchanges GDP for GTP and dissociates from the G β - γ complex.

G6PD Deficiency

- Glucose-6-phosphate dehydrogenase (G6PD) deficiency is an X-linked enzymopathy affecting the pentose phosphate pathway, leading to impaired NADPH production and reduced glutathione regeneration, rendering erythrocytes susceptible to oxidative hemolysis. Common variants: African A- (mild), Mediterranean (severe), Mahidol (SE Asia).

GABA (Gamma-Aminobutyric Acid)

- The primary inhibitory neurotransmitter in the mammalian central nervous system, synthesised from glutamate via glutamic acid decarboxylase (GAD) with pyridoxal phosphate (vitamin B6) as cofactor.

Gadolinium-Based Contrast Agent

- A paramagnetic contrast agent used in MRI to alter tissue signal and improve lesion conspicuity or vascular assessment. Agent selection and dosing consider kidney function, pregnancy, prior reactions, and the rare risk of nephrogenic systemic fibrosis.

Gadolinium-Induced Nephrogenic Systemic Fibrosis

- Rare but serious condition in patients with severe renal impairment receiving gadolinium-based MRI contrast. Causes skin thickening, organ fibrosis. Risk reduced with macrocyclic agents.

Gagging

- A protective retching response triggered by stimulation of the pharynx, posterior tongue, or nearby structures. It may occur during swallowing, examination, aspiration, reflux, infection, or emotional distress; persistent gagging with dysphagia, vomiting, respiratory symptoms, or weight loss requires assessment.

Gait

- The pattern or style of walking. Gait is a complex motor function involving coordination of the central and peripheral nervous systems, muscles, and joints. Gait assessment evaluates stance, stride length, cadence, arm swing, truncal stability, and turning.

Gait Abnormality

- A deviation from normal walking caused by musculoskeletal, neurological, vestibular, visual, cardiovascular, or systemic disease. Assessment considers speed, base, symmetry, balance, stride, turning, assistive-device use, and associated weakness, sensory loss, pain, or cognitive impairment.

Gait Assessment

- A crucial component of the neurological examination that provides significant diagnostic information about the underlying neurological condition. Normal gait requires intact motor, sensory, cerebellar, and vestibular systems, as well as intact cognition.

Gait Disturbance

- An abnormality of walking—a common presenting symptom with diverse causes.

Gait Speed

- Walking velocity measured over a standardized distance. It is a simple marker of mobility, frailty, disability, and mortality risk and can track func-

tional change over time.

Gait Training

- Teaching or retraining walking pattern. Techniques: parallel bars, walker, cane, stair training. Addresses abnormalities: limping, antalgic gait, hemiplegic gait, parkinsonian gait. Goals: safety, efficiency, independence.

Galactocele

- A milk-filled retention cyst of the breast, occurring during lactation or after weaning. Results from obstruction of a milk duct. Presents as a painless, fluctuant breast mass. Diagnosis by ultrasound showing a well-defined, hypoechoic or anechoic mass. Treatment includes aspiration. May become infected, requiring antibiotics.

Galactagogue

- A substance that promotes breast milk production.

Galactorrhea

- Milky nipple discharge not associated with normal lactation, caused by hyperprolactinemia or nipple/breast stimulation. Causes: prolactinoma (pituitary adenoma), medications (antipsychotics, metoclopramide, estrogens, opiates, verapamil, SSRIs), hypothyroidism (TRH stimulates prolactin), chest wall trauma, renal failure, and

Galactorrhoea

- The spontaneous flow of milk from the breast not associated with childbirth or breastfeeding. Galactorrhoea can be unilateral or bilateral and is caused by hyperprolactinaemia (elevated prolactin).

Galactosaemia

- A rare autosomal recessive disorder of galactose metabolism caused by deficiency of one of three enzymes: galactose-1-phosphate uridylyltransferase (GALT, classic galactosaemia, type I), galactokinase (GALK, type II), or UDP-galactose epimerase (GALE, type III).

Galactose Metabolism

- Conversion of galactose to glucose-1-phosphate through the Leloir pathway involving galactokinase, galactose-1-phosphate uridylyltransferase, and UDP-galactose 4-epimerase. Defects cause galactosemia, which can produce liver disease, cataracts, and developmental toxicity.

Galactosemia

- An inherited disorder of galactose metabolism, most often caused by galactose-1-phosphate uridylyltransferase deficiency. Untreated classic disease can cause jaundice, liver dysfunction, vomiting, cataracts, sepsis, and neurodevelopmental impairment after milk feeding.

Galant Reflex

- A primitive reflex present from birth to 4 months, elicited by stroking the skin along the side of the infant's back. The infant curves toward the stimulated side. Assessed by running a finger from the lateral back toward the hip while the infant is in ventral suspension. Absence may indicate neurologic abnormality.

Gall-Bladder

- A pear-shaped sac located beneath the liver on the visceral surface, in the gallbladder fossa between the right and quadrate lobes of the liver. Capacity: approximately 30-50 mL.

Gall-Bladder, Diseases of

- Pathological conditions affecting the gallbladder. Types: (1) Cholelithiasis (gallstones)—most common, affects 10-15% of adults in Western countries. (2) Acute cholecystitis—inflammation of the gallbladder, usually due to stone obstruction of the cystic duct (90%).

Gallavardin Phenomenon

- The dissociation of the murmur of aortic stenosis into two components: a musical quality heard at the apex and a harsh, rough quality heard at the right upper sternal border. This occurs because high-frequency components are transmitted to the apex while low-frequency components are heard at the base.

Gallbladder

- A small, pear-shaped sac under the liver that stores and concentrates bile before releasing it into the small intestine to aid fat digestion.

Gallbladder Cancer

- Aggressive, often diagnosed incidentally after cholecystectomy for cholelithiasis (1-2% of cholecystectomies). Risk factors: gallstones (chronic inflammation), porcelain gallbladder (calcified wall, 10-25% cancer risk), anomalous pancreaticobiliary junction,

gallbladder polyps >1cm, typhoid carrier state.

Gallbladder volvulus

- Axial torsion of the gallbladder around its mesentery and cystic duct, most commonly affecting elderly women, leading to acute gallbladder ischemia. Presents with sudden onset severe right upper quadrant pain, nausea, vomiting, and signs of peritoneal irritation.

Gallium Scan

- Nuclear medicine imaging using Ga-67. Detects inflammation, infection, tumors. Largely replaced by FDG-PET.

Gallium-68

- PET radionuclide, half-life 68 minutes. Produced from Ge-68/Ga-68 generator. Used in DOTATATE, PSMA, and other theranostic applications.

Gallstone Ileus

- Large gallstone eroding through the gallbladder wall into the duodenum causing mechanical small bowel obstruction. Diagnosed by CT showing Rigler triad of small bowel obstruction, pneumobilia, and ectopic gallstone. The most common site of obstruction is the ileum where the bowel caliber narrows.

Gallstones

- Crystalline deposits that form in the gallbladder, composed predominantly of cholesterol (80 percent of stones in Western populations, yellow-green, radiolucent on plain x-ray, develop when bile is supersaturated with cholesterol) or bilirubin pigment (black pigment stones: sterile bile with high unconjugated bilirubin from hemolysis, cir

Gallstones (Cholelithiasis)

- Crystalline deposits that form in the gallbladder, composed predominantly of cholesterol (80 percent of stones in Western populations, yellow-green, radiolucent on plain x-ray, develop when bile is supersaturated with cholesterol) or bilirubin pigment (black pigment stones: sterile bile with high unconjugated bilirubin from hemolysis, cir

Galvanism

- The application of a direct electric current for therapeutic or diagnostic purposes. Named after Luigi Galvani (1737-1798). Historical uses: galvanic stimulation of muscles, iontophoresis (introduction of ionised drugs through skin), electroconvulsive

therapy (modified).

Gamma Camera

- Imaging device using NaI(Tl) scintillation crystal, photomultiplier tubes, collimator. Detects gamma rays from radiopharmaceuticals. Anger camera principle.

Gamma-Globulin

- A fraction of plasma proteins containing antibodies (immunoglobulins). Historically the term referred to a group of serum proteins with the slowest electrophoretic mobility (gamma region on serum protein electrophoresis). Currently synonymous with immunoglobulin preparations used for passive immunisation and immunotherapy.

Ganglion

- A collection of neuronal cell bodies located outside the central nervous system (CNS).

Ganglion Cysts

- Ganglion cysts: mucin-filled cysts from joint capsule/tendon sheath. Dorsal wrist (60-70%, scapholunate interval), volar wrist (radial artery), flexor tendon sheath (volar retinaculum, "seed ganglion"), DIP (mucous cyst, Heberden node association).

Ganglioneuroblastoma

- A pediatric tumor of the sympathetic nervous system containing both neuroblastic and ganglion-cell elements. It most often arises in the adrenal or paraspinal region and may present with an abdominal mass, pain, hypertension, or systemic symptoms.

Ganglionic Blocker

- A class of drugs that block nicotinic acetylcholine receptors at autonomic ganglia (both sympathetic and parasympathetic), inhibiting neurotransmission in both divisions of the autonomic nervous system.

Gangrene

- Death of tissue due to loss of blood supply, typically affecting the extremities and bowel, and represents a form of coagulative necrosis. Dry gangrene occurs in tissues with gradual arterial occlusion, where the dead tissue becomes dry, shrunken, and dark brown or black.

Gargalesthesia

- The sensation of tickling. A complex sensory experience mediated by light tactile stimulation of sensitive skin areas (axillae, soles, flanks, ribcage).

Involves both spinal reflexes (goosebumps, withdrawal) and emotional processing in the anterior cingulate cortex and somatosensory cortex.

Gargle

- The act of rinsing the mouth and throat with a liquid, holding it at the back of the throat while exhaling to create a bubbling action. Used for relief of sore throat or as a delivery method for antiseptics, anaesthetics, and anti-inflammatory agents.

Gartner duct cyst

- A benign cyst arising from remnants of the mesonephric (Wolffian) duct along the lateral wall of the vagina or uterus. They are typically asymptomatic and discovered incidentally on pelvic examination or imaging. Most require no treatment unless symptomatic, in which case surgical excision is performed.

Gas Gangrene

- A life-threatening bacterial infection characterized by the presence of gas in tissues (subcutaneous emphysema) and rapid myonecrosis. Caused by *Clostridium perfringens* (type A, 80% of cases), *C. novyi*, *C. septicum*, and *C.*

Gaskin Maneuver (All-Fours Position)

- A maneuver for shoulder dystocia resolution: the mother is rolled onto her hands and knees (all-fours position). Gravity and the change in pelvic dimensions (increased anteroposterior diameter of the pelvic outlet) may dislodge the impacted shoulder. The posterior arm may also be delivered more easily.

Gastralgia

- Pain in the stomach or epigastric region.

Gastrectomy

- Surgical removal of all (total gastrectomy) or part (partial/subtotal gastrectomy) of the stomach. Indications: gastric cancer (most common), severe peptic ulcer disease (rarely needed now due to PPIs and *H. pylori* treatment), gastric lymphoma, gastrointestinal stromal tumor (GIST), perforation, bleeding.

Gastric Acid

- Hydrochloric acid secreted by gastric parietal cells, producing an intragastric pH of 1-3. Secretion is stimulated by histamine (H₂ receptors), gastrin (CCK-B/gastrin receptors), and acetylcholine (M₃

muscarinic receptors), and inhibited by somatostatin, prostaglandins, and luminal acid (negative feedback).

Gastric Bezoar

- Accumulation of indigestible material in the stomach. Phytobezoars consist of vegetable fiber, trichobezoars of hair, and pharmacobezoars of medications. Associated with gastroparesis, prior surgery, and impaired gastric motility. Diagnosed by endoscopy.

Gastric Bypass

- Roux-en-Y gastric bypass creating a small proximal pouch drained by a Roux limb, combining restrictive and mildly malabsorptive mechanisms. Indicated for BMI over 40, or over 35 with obesity-related comorbidity. Mean excess weight loss approaches 60-70 percent.

Gastric cancer

- See Stomach Cancer.

Gastric Carcinoid Tumor

- Neuroendocrine tumor arising from enterochromaffin-like cells in the gastric fundus. Type I gastric carcinoids are associated with autoimmune atrophic gastritis and hypergastrinemia, are typically small and multifocal, and carry a good prognosis.

Gastric Emptying

- See physiology.

Gastric Emptying Study

- Nuclear medicine diagnostic test measuring the rate of gastric emptying. The protocol involves ingestion of a radiolabeled meal followed by serial gamma camera images obtained at one, two, and four hours. Gastric emptying is considered delayed if more than ten percent of the meal remains at four hours.

Gastric Lymphoma

- Malignant proliferation of lymphoid tissue in the stomach most commonly mucosa-associated lymphoid tissue lymphoma. Strongly associated with *H. pylori* infection. Early disease may regress completely with *H. pylori* eradication therapy alone.

Gastric Malignancy

- See stomach cancer.

Gastric Outlet Obstruction

- Blockage of gastric outflow at the pylorus or duodenum. Causes include peptic ulcer scarring, gastric cancer, and pyloric stenosis in infants. Presents with projectile vomiting, dehydration, and metabolic alkalosis. Diagnosis is by upper endoscopy and CT scan.

Gastric pneumatosis

- Presence of gas within the gastric wall, also called gastric emphysema, which may be benign (gastric emphysema) or life-threatening (emphysematous gastritis). Benign forms result from barotrauma, chronic obstructive pulmonary disease, or mechanical causes.

Gastric Polyps

- Protruding lesions of the gastric mucosa classified by histological type. Fundic gland polyps are the most common type associated with long-term proton pump inhibitor use and familial adenomatous polyposis. Hyperplastic polyps are associated with chronic gastritis and *H. pylori* infection and carry a small risk of dysplasia.

Gastric Surgery

- Gastric cancer: D2 lymphadenectomy standard (stations 1-12a, 12b). Gastrectomy types: total, distal subtotal (Billroth I gastroduodenostomy, Billroth II gastrojejunostomy, Roux-en-Y), proximal, pylorus-preserving. Reconstruction: Roux-en-Y preferred (prevents bile reflux).

Gastric Ulcer

- Peptic ulcer in the gastric mucosa associated with *H. pylori*, NSAIDs, and smoking. Pain is typically worsened by food. There is a higher risk of malignancy than duodenal ulcers. Diagnosis is by upper endoscopy with biopsy to exclude malignancy. Treatment involves proton pump inhibitors and *H. pylori* eradication.

Gastric Ulcer (see Peptic Ulcer)

- See Peptic Ulcer.

Gastric Volvulus

- Rotation of the stomach along its axis causing obstruction. It may present with Borchardt triad of epigastric pain, retching, and inability to pass a nasogastric tube. It is a surgical emergency requiring decompression and gastropexy. Organoaxial volvu-

lus is the most common type where the stomach rotates along its longitudinal axis.

Gastrin

- A peptide hormone produced by G cells in the gastric antrum and duodenum. Secreted in response to: (1) gastric distension, (2) amino acids and peptides (especially from protein-rich meals), (3) vagal stimulation (via gastrin-releasing peptide, GRP), (4) hypercalcaemia.

Gastrin Secretion

- Gastrin is a peptide hormone produced by G-cells in the gastric antrum and duodenum, stimulating hydrochloric acid secretion by parietal cells, pepsinogen secretion by chief cells, and growth of the gastric mucosa.

Gastrinoma

- Gastrin-secreting neuroendocrine tumor causing Zollinger-Ellison syndrome characterized by gastric acid hypersecretion leading to severe peptic ulcer disease, diarrhea, and esophagitis. Approximately sixty percent are malignant with liver metastases. Multiple endocrine neoplasia type 1 association in approximately twenty percent.

Gastritis

- Inflammation of the gastric mucosa, classified by duration (acute or chronic), etiology, and histologic pattern. Acute gastritis is most commonly caused by *Helicobacter pylori* infection (acute H.

Gastrocnemius

- The large superficial muscle of the posterior calf, forming the bulk of the calf contour. Origin: medial and lateral femoral condyles. Insertion: calcaneus via the Achilles tendon. Nerve: tibial nerve (S1-S2). Action: plantar flexion of the ankle (standing, walking, running, jumping); also flexes the knee.

Gastrocnemius Muscle

- Most superficial posterior leg muscle forming the calf prominence, with medial/lateral femoral condyle heads merging with soleus to form the Achilles tendon on the calcaneus. Innervated by tibial nerve (S1-S2). Plantarflexes ankle and flexes knee (two-joint muscle). Active in walking/running/jumping.

Gastroduodenal Crohn Disease

- Involvement of the stomach and duodenum by

Crohn disease occurring in approximately five percent of patients with Crohn. Gastric Crohn typically involves the antrum and pylorus and can cause gastric outlet obstruction from edema, fibrosis, or stricture. Duodenal Crohn presents with epigastric pain, nausea, vomiting, and weight loss.

Gastroenteritis

- Inflammation of the stomach and intestinal mucosa, most commonly caused by viral infections (norovirus, rotavirus, adenovirus), bacterial pathogens (*E. coli*, *Salmonella*, *Campylobacter*), or parasites. Presents with diarrhea, vomiting, abdominal cramps, and dehydration. Self-limited in most cases.

Gastroenterology

- The medical specialty focused on the diagnosis and management of diseases of the gastrointestinal tract (esophagus, stomach, small intestine, colon, rectum, pancreas, gallbladder, bile ducts, liver). Subspecialties: hepatology (liver), pancreatology, inflammatory bowel disease, gastrointestinal oncology, neurogastroenterology.

Gastroesophageal Reflux

- Backward flow of gastric contents into the esophagus, caused by transient or persistent relaxation of the lower esophageal sphincter or other mechanical factors.

Gastroesophageal Reflux Disease (GERD)

- A chronic condition in which gastric contents reflux backward into the esophagus, causing troublesome symptoms and/or complications. Epidemiology: affects 10–20% of adults in Western countries.

Gastrogastic intussusception

- Rare form of intussusception where one segment of the stomach telescopes into an adjacent gastric segment, typically associated with a lead point such as a gastric polyp, leiomyoma, or carcinoma. Presents with abdominal pain, nausea, vomiting, and occasionally upper gastrointestinal bleeding.

Gastrografin Study

- X-ray imaging using water-soluble contrast. Used when barium contraindicated (perforation risk). Less detailed than barium.

Gastrointestinal amyloidosis

- Deposition of misfolded amyloid fibrils in the gastrointestinal tract, either as part of systemic amyloidosis (AL, AA, or ATTR types) or rarely as primary localized disease.

Gastrointestinal Bleeding

- Hemorrhage from any part of the gastrointestinal tract classified as upper or lower based on location relative to the ligament of Treitz. Upper gastrointestinal bleeding originating proximal to this landmark typically presents with hematemesis or melena.

Gastrointestinal Bleeding Scan

- Detection of GI bleeding using Tc-99m labeled RBCs. Sensitive for active bleeding (>0.1 mL/min). Localizes bleeding site. Alternative: tagged RBC scan, CT angiography.

Gastrointestinal Decontamination

- Methods: activated charcoal, gastric lavage (rare), whole bowel irrigation (sorbitol, PEG). Timing critical for charcoal (1-2 hours post-ingestion).

Gastrointestinal Obstruction

- Partial or complete blockage of the passage of intestinal contents through the gastrointestinal tract. It may be mechanical or functional and can cause colicky pain, vomiting, distention, constipation, obstipation, dehydration, bowel ischaemia, perforation, or sepsis.

Gastrointestinal Perforation

- A full-thickness defect in the gastrointestinal wall allowing luminal contents to enter the peritoneal or adjacent space. It can cause peritonitis, sepsis, and shock and usually requires urgent imaging, antimicrobial therapy, resuscitation, and source control.

Gastrointestinal Stromal Tumor

- Mesenchymal tumor arising from interstitial cells of Cajal in the gastrointestinal wall most commonly in the stomach. Often driven by gain-of-function mutations in the KIT proto-oncogene or PDGFRA gene. The majority are but approximately twenty to thirty percent are malignant.

Gastrointestinal Stromal Tumor (GIST)

- Mesenchymal tumor of GI tract. CT diagnosis. KIT mutation common.

Gastrointestinal Villi

- Finger-like projections of the small-intestinal mucosa that greatly increase absorptive surface area. Each villus contains blood capillaries and a central lacteal for transport of absorbed nutrients and lipids.

Gastroparesis

- Delayed gastric emptying without mechanical obstruction most commonly caused by diabetes mellitus, post-surgical changes, or idiopathic causes. Presents with nausea, vomiting of undigested food, early satiety, bloating, and abdominal pain. Diagnosed by gastric emptying scintigraphy showing more than ten percent retention at four hours.

Gastroschisis

- A full-thickness abdominal wall defect, usually to the right of the umbilicus, through which bowel herniates without a covering membrane. Not associated with chromosomal abnormalities. Presents at birth with exposed bowel loops that may be edematous, thickened, and discolored from amniotic fluid exposure.

Gastrospenic fistulization

- Abnormal communication between the stomach and spleen, most commonly arising from gastric ulcer erosion into the splenic hilum or as a complication of splenic abscess. Presents with abdominal pain, gastrointestinal hemorrhage, and sepsis.

Gastrostomy

- Surgical creation of an opening (stoma) through the abdominal wall into the stomach for enteral feeding or decompression.

Gastrostomy Tube (PEG)

- Tube through abdominal wall into stomach for feeding. Percutaneous endoscopic gastrostomy. Indications: prolonged enteral feeding, dysphagia, neurological disorders. Complications: infection, leakage, granulation tissue, burial bumper.

Gastrulation

- A crucial early phase in embryonic development where the single-layered blastula reorganizes into a multi-layered structure with three germ layers.

Gaucher Disease

- See genetics.

Gaucher's disease

- See Storage Diseases.

Gauze

- Sterile cloth for wound care. Types: 4x4 pads, roller gauze, lap pads.

Gavage

- Feeding through a tube passed into the stomach (nasogastric, orogastric, or gastrostomy tube). Used when oral feeding is impossible or inadequate. In neonates: gavage feeding is used for premature infants unable to coordinate suck-swallow-breathe.

Gel

- A semisolid pharmaceutical formulation in which a liquid (usually water) is immobilized within a three-dimensional network of a gelling agent (e.g., carbomer, cellulose, alginate).

Gelfoam

- An absorbable hemostatic agent made from purified porcine gelatin. Used in surgical settings to control bleeding where conventional hemostasis is difficult (liver, bone, neurosurgery). Mechanism: provides a physical matrix that promotes platelet aggregation and clot formation. Absorbed within 4-6 weeks.

Gemfibrozil

- A fibrate (PPAR-alpha agonist) used primarily for severe hypertriglyceridemia (reducing pancreatitis risk) and mixed hyperlipidemia. Raises HDL and lowers triglycerides markedly; modest LDL lowering.

Gemination

- Attempted division of single tooth bud. Results in large tooth with bifid crown, normal tooth count.

Gender Dysphoria

- Psychological distress resulting from incongruence between one's experienced gender and assigned gender at birth. Persists for 6 months and causes significant impairment. Onset in childhood (may resolve) or adolescence/adulthood (persistent).

Gene

- Basic unit of heredity, segment of DNA encoding a functional product (protein or RNA). Structure: promoter, coding sequence (exons), introns, terminator. Genes are located at specific positions (loci) on chromosomes.

Gene Expression

- Process by which information from a gene is used to synthesize functional product (protein or RNA). Steps: transcription, RNA processing, translation, post-translational modification. Regulated at multiple levels.

Gene Panel

- A laboratory test that sequences a selected group of genes associated with a clinical phenotype. Panels can provide efficient, focused analysis but may miss variants outside the selected genes or changes poorly detected by the platform.

Gene Therapy

- Treatment by introducing, removing, or changing genetic material. Approaches: gene addition (functional copy), gene silencing (siRNA, antisense), gene editing (CRISPR-Cas9). Examples: Luxturna (RPE65 mutation), Zolgensma (SMN1 in spinal muscular atrophy).

General Anesthesia

- A controlled, reversible state of unconsciousness, analgesia, amnesia, and muscle relaxation induced by pharmacologic agents, enabling surgical procedures without patient awareness or pain.

General Anesthetic

- An agent producing reversible loss of consciousness, amnesia, analgesia, and muscle relaxation sufficient for surgery.

Generalized Anxiety Disorder

- Characterized by excessive, difficult-to-control anxiety and worry about multiple events or activities occurring more days than not for at least six months. Associated symptoms include restlessness, being easily fatigued, difficulty concentrating, irritability, muscle tension, and sleep disturbance.

Generalized granuloma annulare

- A widespread variant of granuloma annulare presenting with numerous small, flesh-coloured to erythematous papules distributed over the trunk and extremities, lacking the classic annular arrangement of localized disease. It is often associated with diabetes mellitus and may be more persistent than localized forms.

Generalized Tonic-Clonic Seizure

- A generalized tonic-clonic seizure (formerly grand

mal seizure) is a seizure involving bilateral tonic stiffening followed by rhythmic clonic jerking of the extremities, with loss of consciousness.

Generalized-Onset Seizure

- A seizure that begins in bilaterally distributed networks and rapidly engages both cerebral hemispheres. Generalized absence, myoclonic, tonic, clonic, atonic, and tonic-clonic seizures are recognized clinical types.

Generator

- System for producing short-lived radionuclide from longer-lived parent. Tc-99m from Mo-99. Elution with saline. Quality control: Mo-99 breakthrough, Al-3+ breakthrough, sterility.

Generic Medicine

- A medicine containing the same active ingredient, strength, dosage form, route, and intended clinical effect as an approved reference product. Regulatory requirements generally include demonstration of pharmaceutical equivalence and bioequivalence.

Genetic Counseling

- Process of helping individuals understand and adapt to genetic aspects of disease. Includes risk assessment, education, psychosocial support, informed decision-making.

Genetic Counselor

- A health professional who evaluates inherited cancer risk, arranges and interprets genetic testing, and provides counseling to patients and families, including cascade testing of relatives.

Genetic Heterogeneity

- The occurrence of similar or related phenotypes caused by different genetic mechanisms. Locus heterogeneity refers to variants in different genes causing the same phenotype; allelic heterogeneity refers to different variants in the same gene.

Genetic Screening

- Testing of asymptomatic individuals for genetic disorders. Newborn screening: metabolic disorders, hemoglobinopathies. Carrier screening: before pregnancy or during pregnancy. Prenatal screening: non-invasive prenatal testing (NIPT), maternal serum screening.

Genetic Sonogram

- A targeted ultrasound performed at 18-22 weeks

that combines standard anatomic survey with specific soft markers for aneuploidy.

Genetic Testing

- Analysis of DNA for disease risk or diagnosis. Examples: BRCA testing, carrier screening.

Genetic Testing (Sequencing)

- Analysis of DNA to identify pathogenic variants associated with genetic disorders. Sanger sequencing: gold standard for single-gene testing (CFTR for cystic fibrosis, BRCA1/2 for hereditary breast/ovarian cancer).

Genetic Variant of Uncertain Significance

- A DNA variant for which available evidence is insufficient to classify it as benign or pathogenic. It should not generally be used alone to make predictive or treatment decisions and may be reclassified as evidence evolves.

Genital Herpes

- Herpes simplex virus infection of the genital region (HSV-2 most commonly; HSV-1 increasingly), transmitted sexually, causing painful grouped vesicles/ulcers, urethritis, dysuria, and lymphadenopathy; primary infection may cause fever and systemic symptoms. Recurrent episodes are shorter and milder.

Genital Injury

- Trauma to the genital or anogenital tissues caused by sexual contact, assault, childbirth, accident, or disease. Examination should document location, morphology, healing stage, pain, bleeding, and associated injuries without assuming mechanism from appearance alone.

Genital Warts

- See Genital Warts (Condylomata Acuminata).

Genital Warts (Condylomata Acuminata)

- Anogenital warts caused by human papillomavirus (HPV), most commonly types 6 and 11 (low-risk, >90% of genital warts, not associated with malignancy). Transmission: direct skin-to-skin contact during vaginal, anal, or oral sex; highly infectious (65% transmission rate per sexual encounter).

Genitalia

- The external and internal reproductive organs. The term includes the vulva and vagina, or the penis, scrotum, and associated structures, together

with the gonads and reproductive ducts when used broadly; anatomy varies with sex, age, and developmental stage.

Genitals

- The external and internal reproductive organs. Female genitals: external (vulva — mons pubis, labia majora, labia minora, clitoris, vaginal opening, Bartholin glands) and internal (vagina, cervix, uterus, fallopian tubes, ovaries).

Genito-Pelvic Pain/Penetration Disorder

- A sexual dysfunction characterized by persistent or recurrent difficulties with vaginal penetration, marked vulvovaginal or pelvic pain during intercourse, fear or anxiety about penetration, and/or tensing of the pelvic floor muscles, lasting 6 months and causing marked distress.

Genodermatoses

- Genetic skin disorders. Inherited ichthyoses (lamellar, X-linked, congenital ichthyosiform erythroderma). Epidermolysis bullosa (simplex, junctional, dystrophic). Neurocutaneous syndromes (neurofibromatosis, tuberous sclerosis, Sturge-Weber). Ectodermal dysplasias (hypohidrotic, hypotrichosis, nail dystrophy).

Genomic Imprinting

- Parent-of-origin-specific gene expression. Only one allele (maternal or paternal) is expressed. Examples: Prader-Willi (paternal deletion 15q11-q13), Angelman (maternal deletion 15q11-q13).

Genotoxicity

- Ability of substance to damage DNA. Tests: Ames test, chromosomal aberration, micronucleus. Predicts carcinogenicity.

Genotype

- Genetic makeup of an individual, the specific alleles present at one or more loci. Determines phenotype along with environmental factors.

Geographic Atrophy

- The advanced form of dry age-related macular degeneration, characterized by well-demarcated, progressively enlarging areas of complete loss of the retinal pigment epithelium, choriocapillaris, and overlying photoreceptors in the macula, resulting in irreversible central vision loss.

Geographic tongue

- A benign inflammatory condition of the tongue featuring migratory red patches with white or yellow borders, alternating with areas of depapillation, often asymptomatic and of unknown cause.

Geriatric Assessment

- Comprehensive multidimensional evaluation of elderly person. Domains: medical, functional, cognitive, psychological, social, environmental, pharmacological. Performed by interdisciplinary team. Guides care planning, identifies problems, predicts outcomes.

Geriatric Dentistry

- Dental care for elderly. Management of age-related changes, polypharmacy effects, xerostomia, root caries, periodontal disease, oral cancer screening.

Geriatric Depression Scale

- A screening questionnaire designed to detect depressive symptoms in older adults while minimizing emphasis on somatic symptoms that may overlap with medical illness. A positive screen requires clinical diagnostic assessment.

Geriatric Medical Ethics

- Ethical principles applied to care of older adults, including autonomy, capacity, beneficence, non-maleficence, and justice.

Geriatric Nutrition

- Nutrition care addressing the effects of aging, chronic disease, medications, frailty, oral and swallowing problems, sensory changes, and social factors on intake and nutritional status. Goals include preserving function, muscle mass, hydration, and quality of life.

Geriatric Rehabilitation

- See geriatrics.

Geriatric Syndrome

- Multifactorial condition common in elderly, not attributable to single disease. Examples: falls, incontinence, frailty, delirium, immobility, pressure ulcers. Require comprehensive assessment and multidisciplinary management.

Geriatrician

- Physician specializing in geriatrics.

Geriatrics

- Medical care of elderly.

Geriatric

- Outdated term for elderly female patient. Prefer "older woman" or "elderly patient."

Germ

- A colloquial term for a microorganism that causes disease.

Germ Cell Tumor

- A tumor arising from germ cells (precursors of sperm and eggs), including seminoma, dysgerminoma, teratoma, yolk sac tumor, embryonal carcinoma, and choriocarcinoma. Occurs in the testis, ovary, and extragonadal sites (mediastinum, retroperitoneum, pineal); many are highly curable with platinum-based chemotherapy.

Germ Theory of Disease

- The scientific principle that specific microorganisms cause specific infectious diseases; established in the late 19th century by Koch (postulates—organism present in all cases, isolated in pure culture, reproduces disease when inoculated, re-isolated) and promoted by Pasteur and Lister.

German Measles (Rubella)

- German measles caused by Rubivirus. See Rubella.

Germinoma

- A germ-cell tumor most commonly arising in the pineal or suprasellar region, highly sensitive to radiation and chemotherapy. Ovarian (dysgerminoma) and testicular (seminoma) counterparts exist.

Germline Mutation

- A genetic variant present in the egg or sperm and therefore present in essentially every cell of the individual. Pathogenic germline variants may confer inherited cancer susceptibility and can have implications for relatives and treatment.

Germline Testing

- Genetic testing performed to identify inherited or potentially heritable variants, usually from blood or saliva. Results may affect the patient, biological relatives, cancer surveillance, reproductive planning, and genetic counseling.

Germline Variant

- A DNA variant present in the cells that give rise

to eggs or sperm and therefore potentially transmissible to offspring. It may also be present in most cells of the individual, unlike a variant restricted to a somatic tissue.

Gerontologist

- Specialist in gerontology.

Gerontology

- Study of aging. Biological, psychological, social aspects.

Gestation

- The period of development of a fetus from fertilization to birth. Human gestation averages 280 days (40 weeks) from the first day of the last menstrual period (LMP) or 266 days (38 weeks) from conception.

Gestational Age

- The age of pregnancy measured in weeks from the first day of the last menstrual period (LMP). Also called menstrual age. Typically 2 weeks longer than fertilization age. Calculated using LMP or firsttrimester ultrasound. Used to estimate due date (40 weeks from LMP). Preterm is less than 37 weeks; post-term is beyond 42 weeks.

Gestational Diabetes

- Glucose intolerance first diagnosed during pregnancy, affecting 6-9% of pregnancies. Risk factors include obesity, advanced maternal age, family history, and prior gestational diabetes. Complications include macrosomia, birth trauma, neonatal hypoglycemia, and preeclampsia.

Gestational Diabetes Mellitus

- Glucose intolerance first recognized during pregnancy affecting approximately six to nine percent of pregnancies. Results from insulin resistance induced by placental hormones including human placental lactogen, cortisol, and progesterone combined with insufficient beta cell compensation.

Gestational Hypertension

- New-onset hypertension (BP 140/90 mmHg) after 20 weeks of gestation in a woman with previously normal blood pressure, without proteinuria or other features of preeclampsia. Affects 6-8% of pregnancies. Management: monitoring blood pressure, screening for preeclampsia progression (proteinuria, labs, symptoms).

Gestational Sac

- The first sonographic sign of an intrauterine pregnancy, visible on transvaginal ultrasound at approximately 4.5-5 weeks gestational age. A fluid-filled structure within the decidua, surrounded by the chorionic ring. Mean sac diameter (MSD) correlates with gestational age.

Gestational Trophoblastic Disease

- A spectrum of conditions arising from abnormal trophoblast proliferation, including complete and partial hydatidiform moles and gestational choriocarcinoma. Complete mole has 46,XX karyotype (all paternal origin); partial mole is triploid (69 chromosomes). Presents with elevated hCG, uterus large for dates, and vaginal bleeding.

Gestational Trophoblastic Neoplasia

- Persistent or malignant trophoblastic disease after a pregnancy event, identified by abnormal or rising beta-hCG, metastases, or histologic criteria. It includes invasive mole and choriocarcinoma and is highly responsive to risk-adapted chemotherapy.

Gettler Test

- Historical quantitative chemical test for drowning, comparing chloride concentration in blood harvested from the left and right heart ventricles.

Ghrelin

- Hunger hormone from stomach. Stimulates appetite.

Giant Cell Arteritis

- A large-vessel vasculitis of adults >50 (peak 70-80) affecting cranial branches of carotid arteries, with risk of permanent vision loss from anterior ischemic optic neuropathy. Clinical: new-onset headache (temporal), scalp tenderness, jaw claudication, visual disturbances (amaurosis fugax, diplopia), polymyalgia rheumatica symptoms.

Giant Cell Arteritis Emergency

- Suspected giant cell arteritis with visual symptoms, jaw or tongue claudication, or neurologic ischemia requires immediate glucocorticoid treatment to reduce risk of irreversible visual loss while diagnostic evaluation proceeds.

Giant Coronary Aneurysm

- See Coronary Artery Aneurysm.

Giant right atrium

- Marked dilatation of the right atrium, most commonly caused by chronic severe tricuspid regurgitation, Ebstein anomaly, or long-standing atrial fibrillation with right heart failure. The enlarged atrium may cause compression of adjacent structures, arrhythmias, and thrombus formation. Diagnosis is by echocardiography or cardiac imaging.

Giardia Infection

- Intestinal infection caused by the protozoan *Giardia duodenalis*, usually acquired from contaminated water, food, or person-to-person spread. It commonly causes foul-smelling diarrhea, bloating, abdominal cramps, and malabsorption.

Giardia lamblia

- A protozoan parasite that causes giardiasis, a diarrhoeal illness transmitted through the fecal-oral route. The organism has a trophozoite and cyst stage. Infection occurs through ingestion of cysts in contaminated water. Symptoms include foul-smelling diarrhea, abdominal cramps, and bloating.

Giardiasis

- Infection with *Giardia lamblia* (*G. intestinalis*), a protozoan parasite causing diarrheal illness. Transmission via contaminated water (fecal-oral). Presents with foul-smelling, greasy diarrhea, bloating, abdominal cramps, weight loss, and steatorrhea. Diagnosis: stool antigen testing or microscopy. Treatment: metronidazole or tinidazole.

Giemsa Stain

- A Romanowsky-type stain of methylene blue and eosin dyes used to visualise blood cells and microorganisms in smears and tissue sections. In microbiology it stains *Plasmodium*, *Babesia*, *Trypanosoma*, *Leishmania*, *Borrelia*, and intracellular cytomegalovirus inclusions, supporting rapid diagnosis of blood-borne parasitic and other infections.

Gigantism

- A rare condition of excessive growth and tall stature caused by growth hormone (GH) hypersecretion that begins before epiphyseal plate closure in childhood, before the onset of puberty.

Gilbert Syndrome

- A common, benign autosomal recessive hereditary hyperbilirubinemia caused by a promoter mutation

in the UGT1A1 gene leading to reduced activity (30% of normal) of UDP-glucuronosyltransferase.

Gingival Hyperplasia

- An abnormal overgrowth of gingival tissue around the teeth, which can impair plaque control, increase the risk of periodontal disease, and affect aesthetics. It can be caused by chronic inflammatory conditions (associated with poor oral hygiene) or systemic factors.

Gingival melanoma

- A rare and aggressive mucosal melanoma arising from melanocytes in the gingival mucosa, presenting as a pigmented or amelanotic nodule or ulcer on the gums. Mucosal melanomas of the head and neck carry a poor prognosis compared to cutaneous melanoma, often presenting at an advanced stage.

Gingival Recession

- Apical displacement of the gingival margin from the cemento-enamel junction, exposing the root surface. Causes include periodontal inflammation, traumatic toothbrushing, thin periodontal tissue, tooth position, and iatrogenic factors; it may produce sensitivity and root-carries risk.

Gingivectomy

- Surgical removal of gingival tissue. Indications: gingival hyperplasia, cosmetic recontouring, access to caries.

Gingivitis

- Inflammation of gums without bone loss. Caused by dental plaque accumulation. Signs: redness, swelling, bleeding on probing. Reversible with improved oral hygiene and professional cleaning. If untreated, progresses to periodontitis.

Gingivoplasty

- Surgical reshaping of gingiva. Often combined with gingivectomy.

Gingivostomatitis

- Inflammation of the gums and oral mucosa, commonly caused by primary herpes simplex infection but also associated with other infections, trauma, or systemic disease. It may cause painful ulcers, swollen bleeding gums, fever, and difficulty eating or drinking.

Gitelman Syndrome

- Inherited autosomal recessive disorder of the dis-

tal convoluted tubule caused by mutations in the NCC sodium-chloride cotransporter gene. Milder than Bartter syndrome typically presenting in adolescence or adulthood.

Gland

- An organized group of epithelial cells specialized for secretion.

Glandular Fever (Infectious Mononucleosis)

- An acute viral syndrome caused by Epstein-Barr virus (EBV), primarily affecting adolescents and young adults. See Mononucleosis.

Glanzmann thrombasthenia

- An inherited platelet aggregation disorder caused by deficient or dysfunctional glycoprotein IIb/IIIa, producing mucocutaneous bleeding despite a normal platelet count.

Glasgow Coma Scale (GCS)

- A standardized assessment of consciousness based on eye opening, verbal response, and motor response, with a total score from 3 to 15. It supports serial neurologic assessment but should be recorded with the individual component scores and confounders.

Glass Ionomer

- Restorative material releasing fluoride. Bonds to tooth structure. Used for pediatric dentistry, cervical lesions, bases.

Glaucoma

- A group of progressive optic neuropathies characterized by the degeneration of retinal ganglion cells and their axons, resulting in characteristic optic nerve head changes and visual field loss, with elevated intraocular pressure being the most important modifiable risk factor.

Glenohumeral Joint

- Ball-and-socket synovial joint between humeral head and glenoid cavity. Most mobile body joint allowing flexion, extension, abduction, adduction, medial/lateral rotation, circumduction. Mobility trades for stability, making it the most commonly dislocated joint. Stabilized by glenoid labrum, glenohemoral ligaments, and rotator cuff.

Glenoid Cavity

- Shallow pear-shaped depression on the lateral scapular angle articulating with the humeral head.

The glenoid labrum deepens the socket by approximately 50 percent. Due to its shallowness, the glenohumeral joint is the most commonly dislocated joint.

GLIM Criteria

- A consensus framework for diagnosing malnutrition using at least one phenotypic criterion, such as weight loss, low body-mass index, or reduced muscle mass, and at least one etiologic criterion, such as reduced intake or assimilation or disease-related inflammation. Severity is graded using phenotypic thresholds.

Glioblastoma

- Grade IV astrocytoma (WHO), the most common and aggressive primary malignant brain tumor in adults. Arises de novo (primary GBM, IDH-wildtype) or from low-grade astrocytoma (secondary, IDH-mutant—better prognosis).

Glioma

- A tumor arising from glial cells (astrocytes, oligodendrocytes, ependymal cells) in the brain or spinal cord. Includes astrocytomas, oligodendrogliomas, and ependymomas; behavior ranges from low-grade to highly aggressive (glioblastoma).

Glisson's Capsule

- Thin fibrous connective tissue envelope surrounding the liver, sending septa inward dividing it into lobules. Richly innervated by phrenic and intercostal nerve pain fibers, sensitive to distension and inflammation. Covered by visceral peritoneum except at the bare area.

Global Health

- Health issues transcending national boundaries. Examples: pandemics, health disparities, climate change.

Globin

- A family of heme-binding proteins involved in oxygen transport and storage. Structure: globin fold (8 alpha-helices arranged in a globular structure), binds one heme molecule (iron-protoporphyrin IX) per subunit.

Globulin

- A group of serum proteins (globular proteins) that are less soluble in water but soluble in dilute salt solutions. On serum protein electrophoresis, globulins

migrate in the alpha-1, alpha-2, beta, and gamma regions. Alpha-1 globulins: alpha-1-antitrypsin, alpha-fetoprotein.

Globus Sensation

- The persistent or intermittent sensation of a lump or foreign body in the throat without an anatomic lesion or dysphagia, also termed globus pharyngeus, typically noticed between meals. There is no impedance to swallowing and no weight loss; the feeling worsens with stress, anxiety, or laryngopharyngeal reflux.

Glomerular Filtration Rate

- Volume of plasma filtered by the glomeruli per unit time normally ninety to one hundred twenty milliliters per minute per one point seven three square meters body surface area. Estimated by the CKD-EPI equation using serum creatinine, age, sex, and race.

Glomerulonephritis

- Inflammation of the glomeruli, classified by clinical presentation (nephritic vs. nephrotic syndrome) and histopathology (e.g., IgA nephropathy, post-streptococcal, membranoproliferative, crescentic). Presents with hematuria, proteinuria, hypertension, and reduced GFR.

Glomerulopathy

- Any disease affecting the renal glomeruli, the capillary structures responsible for blood filtration. It may produce albuminuria or proteinuria, haematuria, reduced filtration, oedema, hypertension, or kidney failure and may be primary or secondary to systemic, metabolic, infectious, or immune disease.

Glomerulus

- The tuft of capillaries in the renal corpuscle (Bowman's capsule) where blood is filtered to form urine.

Glomus Tumor

- A rare benign neoplasm arising from glomus bodies (thermoregulatory shunts), presenting as a small, blue-red, exquisitely tender subungual or digital nodule. The classic triad is severe pain, point tenderness, and cold sensitivity. Diagnosis is clinical with positive Love pin test and Hildreth sign (tumor disappears with ischemia).

Glomus tympanicum

- A benign vascular tumor arising from the glomus

bodies on the promontory of the middle ear, presenting with pulsatile tinnitus, hearing loss, and a reddish mass behind the tympanic membrane.

Glossitis

- Inflammation of the tongue. Causes: (1) Infection: bacterial (*Streptococcus*, *Actinomyces*), viral (HSV, Coxsackie–herpangina, HIV), fungal (*Candida*–oral thrush).

Glossopharyngeal Neuralgia

- Recurrent, brief attacks of severe stabbing or electric-shock-like pain in the throat, tonsillar region, base of the tongue, or ear, caused by irritation of the glossopharyngeal nerve. Swallowing, coughing, or talking may trigger attacks.

Glottis

- The opening between the vocal folds (true vocal cords) in the larynx, through which air passes during respiration and phonation. Boundaries: anterior (anterior commissure, attachment of vocal folds to thyroid cartilage), posterior (arytenoid cartilages and posterior commissure).

Gloves

- Protective hand covering. Types: examination (nitrile, latex, vinyl), surgical. Sterile vs. non-sterile.

GLP-1 Receptor Agonist

- A glucose-lowering drug that activates the glucagon-like peptide-1 receptor, enhancing glucose-dependent insulin secretion, suppressing inappropriate glucagon secretion, slowing gastric emptying, and increasing satiety.

GLP-1 Receptor Agonists

- Glucagon-like peptide-1 receptor agonists are a class of injectable antidiabetic agents that mimic the effects of the incretin hormone GLP-1, which is secreted by intestinal L-cells in response to food intake.

Glucagon

- A peptide hormone produced by the alpha cells of the pancreatic islets (Langerhans).

Glucagon Signaling

- Glucagon binds GPCR, activates adenylate cyclase, increases cAMP, activates PKA. Promotes glycogenolysis, gluconeogenesis.

Glucagonoma

- A rare pancreatic neuroendocrine tumor secreting glucagon. Classic presentation: 4 Ds — diabetes mellitus (mild, new onset), dermatitis (necrolytic migratory erythema, characteristic painful, erythematous, blistering rash on groin, perineum, extremities), diarrhea, and deep vein thrombosis.

Glucocorticoid Replacement Therapy

- Replacement of cortisol deficiency in adrenal insufficiency using hydrocortisone in divided doses. Typical regimen is ten to fifteen milligrams in the morning and five to ten milligrams in the early afternoon mimicking the physiological cortisol diurnal rhythm.

Glucocorticoid-Sparing Therapy

- Use of a non-glucocorticoid treatment strategy to control inflammation while reducing cumulative corticosteroid exposure. It aims to limit osteoporosis, infection, diabetes, hypertension, adrenal suppression, and other glucocorticoid complications.

Glucocorticoids

- A class of steroid hormones produced by the zona fasciculata of the adrenal cortex, regulated by the hypothalamic-pituitary-adrenal (HPA) axis.

Glucometer

- Device measuring blood glucose. Capillary blood from fingerstick.

Gluconeogenesis

- Metabolic pathway generating glucose from non-carbohydrate precursors (lactate, glycerol, gluconeogenic amino acids). Occurs primarily in liver (90%) and kidney (10%).

Glucose

- A monosaccharide ($C_6H_{12}O_6$) that is the primary energy source for cells. Types: D-glucose (dextrose, naturally occurring enantiomer) and L-glucose (synthetic). Routes of entry: dietary carbohydrates (digested to glucose), gluconeogenesis (synthesis from non-carbohydrate precursors), glycogenolysis (breakdown of stored glycogen).

Glucose Intolerance

- Impaired handling of glucose resulting in blood-glucose concentrations above normal but below the diagnostic threshold for diabetes in a defined testing context. It often reflects insulin resistance or

inadequate insulin secretion and is associated with increased risk of type 2 diabetes and cardiovascular disease.

Glucose Management Indicator

- Derived metric from continuous glucose monitoring that estimates the hemoglobin A1c equivalent based on mean sensor glucose values. Formula converts mean glucose to an A1c-equivalent value allowing comparison between CGM data and traditional A1c.

Glucose Sensor

- Continuous glucose monitor (CGM) for diabetes. Subcutaneous sensor measures interstitial glucose. Transmitter sends data to receiver/phone. Examples: Dexcom, Freestyle Libre, Medtronic. Reduces hypoglycemia, improves A1c.

Glucose Tolerance Test

- Testing for diabetes mellitus and gestational diabetes. 75g OGTT (oral glucose tolerance test): measure plasma glucose at 0, 60, and 120 minutes. Diagnostic: fasting 126 mg/dL, 2-hour 200 mg/dL. Gestational diabetes screening: 50g glucose challenge test (GCT) at 24-28 weeks; if >140 mg/dL at 1 hour, proceed to 100g 3-hour OGTT.

Glucose-6-Phosphatase

- Enzyme converting glucose-6-phosphate to glucose. Located in liver and kidney. Deficiency causes Von Gierke disease.

Glucose-6-phosphate dehydrogenase deficiency

- An X-linked enzymopathy affecting approximately 400 million people worldwide, causing episodes of hemolytic anemia triggered by oxidative stress from infections, certain drugs including primaquine and sulfonamides, and fava beans. G6PD is essential for generating NADPH to protect red blood cells from oxidative damage.

Glue (Vascular)

- A vascular embolization material used to occlude selected vessels under image guidance.

Glutamate

- The principal excitatory neurotransmitter in the central nervous system. Synthesized from glutamine or alpha-ketoglutarate and acting on ionotropic receptors (AMPA, kainate, NMDA) for fast synaptic transmission and metabotropic receptors (mGluR1-

8) for modulation.

Glutathione

- Antioxidant tripeptide (Glu-Cys-Gly). Detoxifies reactive metabolites (e.g., NAPQI from acetaminophen). Depletion causes toxicity. N-acetylcysteine replenishes glutathione.

Gluten

- A complex of storage proteins (prolamins and glutelins) found in wheat (gliadin), barley (hordein), and rye (secalin). Gluten provides elasticity to dough. In susceptible individuals, gluten triggers an autoimmune response (celiac disease) or non-autoimmune sensitivity (non-celiac gluten sensitivity, NCGS).

Gluten-Free Diet

- Eliminates gluten (wheat, barley, rye). Essential for celiac disease, non-celiac gluten sensitivity. Risk of nutrient deficiency if not supplemented. May be unnecessarily restrictive for others.

Gluten-Sensitive Enteropathy (Celiac Disease)

- An autoimmune enteropathy triggered by ingestion of gluten (wheat, barley, rye) in genetically predisposed individuals. Incidence: 1% worldwide, higher in HLA-DQ2 (90%) and HLA-DQ8 carriers.

Gluteus Maximus Muscle

- Largest, most superficial gluteal muscle forming the buttock bulk, from posterior iliac gluteal line, sacrum, coccyx, and sacrotuberous ligament to the iliotibial tract and gluteal tuberosity. Innervated by inferior gluteal nerve (L5-S2). Primary hip extensor and lateral rotator.

Gluteus Medius Muscle

- Thick fan-shaped muscle between maximus and minimus, from external ilium to the greater trochanter. Innervated by superior gluteal nerve (L4-S1). Primary hip abductor essential for pelvic stability during single-leg stance. Superior gluteal nerve injury causes Trendelenburg sign with contralateral pelvic drop.

Gluteus Minimus Muscle

- Smallest, deepest gluteal muscle from external ilium to the greater trochanter. Innervated by superior gluteal nerve (L4-S1). Hip abductor and medial rotator stabilizing the hip during walking.

Works synergistically with medius for pelvic stability. Tendinopathy commonly causes lateral hip pain mimicking trochanteric

Glycaemia

- The concentration of glucose in the blood, normally maintained within a narrow range (fasting: 3.9-5.5 mmol/L or 70-100 mg/dL) through the coordinated action of insulin (lowers glucose) and counter-regulatory hormones (glucagon, cortisol, catecholamines, growth hormone).

Glycemic Control

- Maintenance of blood glucose within target range. In elderly, goals less stringent to avoid hypoglycemia. HbA1c target: <7.5-8.0% depending on health status. Monitoring: self-monitoring, continuous glucose monitoring. Medications: metformin first-line, avoid sulfonylureas if hypoglycemia risk.

Glycemic Index

- Ranking of carbohydrates by effect on blood glucose. High GI foods spike glucose. Low GI foods provide sustained energy. Useful for diabetes management.

Glycemic Load

- Combines glycemic index with carbohydrate amount. More practical measure of blood glucose impact.

Glycerin

- A colorless, odorless, sweet-tasting trihydroxy alcohol (glycerol) with humectant, osmotic, and solvent properties.

Glycerol (Glycerine)

- A triol compound ($\text{CH}_2\text{OH}-\text{CHOH}-\text{CH}_2\text{OH}$), a colourless, odourless, viscous liquid with a sweet taste.

Glycogen

- A branched polysaccharide composed of glucose units linked by alpha-1,4 glycosidic bonds (linear) and alpha-1,6 bonds (branching points, every 8-10 residues). Glycogen serves as the main storage form of glucose in animals.

Glycogen Storage Disease

- A group of inherited metabolic disorders caused by defects in glycogen synthesis, breakdown, or regulation, resulting in abnormal glycogen accumulation in tissues. Major types: Type I (von Gierke,

G6PC deficiency) — fasting hypoglycemia, lactic acidosis, hepatomegaly, hyperuricemia, and hypertriglyceridemia.

Glycogen Storage Diseases

- A group of inherited disorders caused by deficiencies in enzymes involved in glycogen metabolism, affecting the liver, muscle, or both.

Glycogenesis

- The synthesis of glycogen from glucose, primarily in liver and skeletal muscle. Glucose-1-phosphate is converted to UDP-glucose, and glycogen synthase and the branching enzyme extend and branch the polymer; insulin promotes the pathway.

Glycogenolysis

- Breakdown of glycogen to glucose-1-phosphate (converted to glucose-6-phosphate). Liver glycogenolysis maintains blood glucose during fasting; muscle glycogenolysis provides energy for contraction. Regulated by hormonal signals (glucagon, epinephrine) and allosteric effectors (AMP activates phosphorylase).

Glycolysis

- Metabolic pathway breaking down glucose (6C) to pyruvate (3C), generating 2 ATP (net) and 2 NADH. Occurs in cytoplasm. Ten enzymatic steps: investment phase (uses 2 ATP) and payoff phase (produces 4 ATP, 2 NADH). Key regulatory enzymes: hexokinase, phosphofructokinase-1 (PFK-1, rate-limiting), pyruvate kinase.

Glycosuria

- The presence of glucose in the urine. Normally, the renal tubules reabsorb filtered glucose via sodium-glucose co-transporters (SGLT2 in proximal tubule). Glycosuria occurs when blood glucose exceeds the renal threshold (approximately 10 mmol/L or 180 mg/dL), overwhelming reabsorptive capacity.

GnRH Agonist

- Gonadotropin-releasing hormone agonists are a class of medications that initially stimulate (flare effect, 1-3 weeks) and then suppress pituitary gonadotropin (LH and FSH) secretion through receptor downregulation, leading to profound suppression of sex steroid production (testosterone in men, estradiol in women).

GnRH Antagonist

- A class of medications that competitively block GnRH receptors on pituitary gonadotrophs, producing immediate, reversible suppression of LH and FSH secretion without the initial flare effect seen with GnRH agonists.

Goal Attainment Scaling

- An individualized outcome method in which patient-specific goals are defined with graded levels of expected attainment and scored after intervention. It captures meaningful changes that standardized measures may miss.

Goal Setting

- SMART goals: Specific, Measurable, Achievable, Relevant, Time-bound. Patient-centered goals improve adherence.

Goals-of-Care Discussion

- A clinical conversation that aligns treatment recommendations with a patient's values, priorities, prognosis, acceptable burdens, and preferred outcomes. It may include decisions about hospitalization, resuscitation, artificial nutrition, and comfort-focused care.

Goiter

- Enlargement of the thyroid gland, which can be diffuse or nodular. Causes: iodine deficiency (most common worldwide), Hashimoto thyroiditis, Graves disease, multinodular goiter, and thyroid cancer. May be associated with normal, hyperthyroid, or hypothyroid function.

Gold Alloy

- Noble metal alloy for dental restorations. Excellent biocompatibility, durability, marginal adaptation. Expensive.

Goldenhar Syndrome

- A variant of hemifacial microsomia characterized by the triad: epibulbar dermoids (benign conjunctival tumors), preauricular skin tags or sinuses, and vertebral anomalies (hemivertebrae, fused vertebrae, scoliosis). Also involves facial asymmetry, microtia, conductive hearing loss, and macrostomia.

Goldmann Tonometry

- Goldmann applanation tonometry is the gold standard method for measuring intraocular pressure, the most important modifiable risk factor for glaucoma,

and it involves applanating (flattening) a fixed area of the cornea using a biprism attached to a slit lamp, measuring the force required to achieve this applanation.

Golfer Elbow

- Medial epicondylitis, a tendinopathy of the common flexor/pronator tendon at the medial epicondyle. Less common than tennis elbow. Caused by repetitive wrist flexion and forearm pronation. Presents with medial elbow pain worsened by wrist flexion and gripping. Tenderness over the medial epicondyle. Positive Golfer elbow test.

Golgi Apparatus

- Membrane-bound organelle (Golgi complex) consisting of flattened cisternae (cis, medial, trans) that modifies, sorts, and packages proteins and lipids from the ER for secretion or delivery to other organelles.

Golgi Tendon Organ

- A mechanoreceptor located at the junction between a skeletal muscle and its tendon, innervated by Ib afferent fibers. The Golgi tendon organ detects tension generated by muscle contraction or passive stretch.

Gonadal Dysgenesis

- Incomplete or abnormal development of the ovaries or testes, often resulting from chromosomal abnormalities, gene variants, or disordered sex development. It may cause atypical genital development, delayed or absent puberty, infertility, and abnormal gonadal hormone production.

Gonadotrophins

- Hormones that stimulate the gonads. The anterior pituitary secretes follicle-stimulating hormone (FSH) and luteinising hormone (LH), both glycoprotein hormones sharing a common alpha subunit (identical to TSH and hCG alpha) with a unique beta subunit.

Gonads

- The primary reproductive organs: testes in males and ovaries in females. Functions: (1) gamete production (spermatogenesis in testes, oogenesis in ovaries); (2) sex steroid hormone secretion (testosterone from testicular Leydig cells; estrogen and progesterone from ovarian follicular cells and corpus

luteum).

Gonioscopy

- An ophthalmological examination technique that allows direct visualization of the anterior chamber angle, the anatomical region where the cornea and iris meet and where the trabecular meshwork is located, and it is essential for distinguishing between open-angle and angle-closure forms of glaucoma.

Gonococcal Arthritis

- A disseminated infection of joints caused by *Neisseria gonorrhoeae*, often with fever, migratory joint or tendon pain, skin pustules, and tenosynovitis. It requires prompt antimicrobial treatment and evaluation for concomitant sexually transmitted infection.

Gonococcal pharyngitis

- See Gonorrhea.

Gonorrhea

- A sexually transmitted infection caused by the Gram-negative diplococcus *Neisseria gonorrhoeae*. Epidemiology: second most common bacterial STI (after Chlamydia), highest incidence in 15-24 year olds.

Goodpasture Syndrome

- Anti-GBM disease affecting both kidneys and lungs. Pulmonary hemorrhage with hemoptysis and crescentic GN. Linear IgG deposits on renal biopsy. Emergent plasmapheresis, cyclophosphamide, and steroids. Early treatment essential to prevent irreversible renal and pulmonary damage.

Gotttron Papules

- Pathognomonic dermatological sign of dermatomyositis, consisting of erythematous to violaceous lichenoid papules located over the dorsal aspects of the interphalangeal (PIP/DIP) and metacarpophalangeal (MCP) joints. May undergo necrosis or ulceration.

Gout

- A crystal deposition disease caused by monosodium urate (MSU) crystal formation in joints and tissues due to chronic hyperuricemia (serum urate >6.8 mg/dL). Pathophysiology: urate overproduction (10%) or underexcretion (90%); crystals activate NLRP3 inflammasome -> IL-1beta release -> neutrophilic inflammation.

Gout Flare

- An acute inflammatory response to monosodium urate crystal deposition in or around a joint. It commonly causes abrupt severe pain, swelling, warmth, and erythema, often involving the first metatarsophalangeal joint; aspiration can confirm crystals and exclude infection.

Gout nodulosis

- The formation of tophi—large, firm, subcutaneous nodules composed of monosodium urate crystals—resulting from chronic, poorly controlled hyperuricemia. Tophi most commonly occur at the ears, fingers, toes, olecranon bursae, and Achilles tendon, and may ulcerate and discharge chalky white material.

Gown

- Protective clothing. Types: examination, surgical, isolation.

Grade

- Degree of cellular differentiation. Grade 1: well-differentiated. Grade 2: moderately differentiated. Grade 3: poorly differentiated. Grade 4: undifferentiated. Higher grade = more aggressive.

Graded Potentials

- Local changes in membrane potential that vary in magnitude proportional to stimulus strength, unlike all-or-none action potentials. They include receptor potentials (in sensory receptors), postsynaptic potentials (EPSPs and IPSPs at synapses), and pacemaker potentials (in cardiac/specialized cells).

Graft

- Surgical procedure creating detour around blocked vessel using autologous vein (saphenous) or prosthetic graft (PTFE, Dacron). See bypass graft.

Graft (Periodontal)

- Tissue used to cover exposed root or augment gingiva. Connective tissue graft (gold standard), free gingival graft, allograft, xenograft.

Graft-Versus-Host Disease

- Immunological reaction where donor T cells attack recipient tissues after allogeneic bone marrow transplant. Affects the skin, liver, and gastrointestinal tract. GI manifestations include diarrhea, abdominal pain, nausea, anorexia, and gastrointestinal bleeding.

Graft-Versus-Host Disease (GVHD)

- An immune-mediated complication of allogeneic hematopoietic transplantation in which donor-derived immunocompetent T cells attack recipient tissues. Acute or chronic GVHD commonly affects skin, liver, and gastrointestinal tract and requires prevention and immunosuppressive treatment.

Grafting

- The surgical transfer of tissue from one site to another, or from one individual to another. Types: (1) Autograft: same individual (gold standard, e.g., skin graft for burns, bone graft for non-union, saphenous vein graft for CABG). (2) Allograft: same species, different individual (cadaveric bone, cornea, heart valve, kidney).

Gram Negative Bacilli

- Rod-shaped gram-negative bacteria appearing pink on Gram stain owing to a thin peptidoglycan layer and lipopolysaccharide outer membrane.

Gram Positive Cocci

- Spherical gram-positive bacteria staining purple owing to a thick peptidoglycan layer and absent outer membrane. Clinically important genera include *Staphylococcus*, *Streptococcus*, and *Enterococcus*, identifiable by arrangement (clusters, chains, pairs) and hemolysis, and causing skin, pharyngeal, respiratory, and bloodstream infections.

Gram stain

- A differential staining technique that classifies bacteria into two major groups based on cell wall structure. The procedure involves crystal violet staining, iodine mordant, alcohol decolorisation, and safranin counterstain.

Gram-Negative Bacteria

- Bacteria that do not retain crystal violet in the Gram stain (appear pink/red), characterized by a thin peptidoglycan layer and an outer membrane containing lipopolysaccharide (endotoxin). Examples: *Enterobacteriaceae* (E).

Gram-Negative Meningitis

- Meningitis caused by gram-negative bacteria, including *Escherichia coli*, *Klebsiella*, and *Neisseria* species. It can progress rapidly to sepsis, neurologic injury, and death and requires urgent cerebrospinal-fluid evaluation and antimicrobial treatment.

Gram-Positive Bacteria

- Bacteria that retain crystal violet in the Gram stain (appear purple/blue), characterized by a thick peptidoglycan cell wall and no outer membrane.

Grand Mal

- A type of generalised seizure characterized by bilateral synchronous epileptic activity involving the entire brain. Formerly called grand mal, now classified as bilateral tonic-clonic seizure according to the ILAE 2017 classification.

Granular Casts

- Degenerated cellular debris within tubular lumen appearing as coarse or fine-grained structures. Coarse casts represent more acute process, fine casts indicate chronic degenerative process. Common in acute tubular necrosis reflecting tubular epithelial cell damage. Also seen in chronic kidney disease, nephrotic syndrome, and hypertension.

Granulocyte

- A type of white blood cell (leukocyte) containing cytoplasmic granules that are visible on light microscopy. Granulocytes are produced in the bone marrow (myelopoiesis, 3-4 days maturation) and are part of the innate immune system.

Granuloma

- A distinctive pattern of chronic inflammation characterized by the aggregation of modified macrophages (epithelioid histiocytes), often with multinucleated giant cells (Langhans or foreign-body type), surrounded by lymphocytes. Granulomas form when macrophages cannot eliminate a persistent antigen or pathogen.

Granuloma Annulare

- A benign inflammatory condition characterized by annular, erythematous to skin-colored plaques with raised borders and central clearing. It most commonly affects the hands, feet, and elbows. Localized form is most common; generalized form is associated with diabetes mellitus.

Granulomatosis with Polyangiitis

- GPA (formerly Wegener granulomatosis): an ANCA-associated small-vessel necrotizing vasculitis affecting the upper and lower respiratory tract and kidneys.

Granulomatous amebic encephalitis

- A rare, usually fatal central nervous system infection caused by free-living amoebae such as *Acanthamoeba* species, typically affecting immunocompromised individuals. It presents with a subacute to chronic meningoencephalitis and forms granulomatous lesions in the brain.

Granulomatous Inflammation

- A distinctive pattern of chronic inflammation characterized by organized collections of activated macrophages that have undergone morphological transformation into epithelioid cells, often surrounded by a rim of lymphocytes and occasionally fibroblasts.

Granulosa Cell Tumor

- An estrogen-secreting ovarian sex cord-stromal tumor that can cause precocious puberty, endometrial hyperplasia, or carcinoma. Generally low-grade with a tendency to late recurrence.

Graves Disease

- Autoimmune disorder and most common cause of hyperthyroidism characterized by thyroid-stimulating immunoglobulins that bind and activate the thyroid-stimulating hormone receptor causing unregulated thyroid hormone synthesis and secretion. The diffuse goiter reflects TSH receptor stimulation.

Gray (Gy)

- The SI unit of absorbed radiation dose, equal to one joule of ionizing radiation energy deposited per kilogram of matter. It describes physical energy absorption and differs from the sievert, which incorporates biologic weighting.

Gray Matter

- Regions of the brain and spinal cord composed primarily of nerve cell bodies, dendrites, and synapses, responsible for processing information.

Greater Curvature

- Convex lateral stomach border from gastroesophageal junction to pylorus, approximately four times longer than the lesser curvature. Attachment site for the greater omentum. Blood supply from short gastric and left gastroepiploic vessels. Common site for gastric ulcers and cancer.

Greater Trochanter

- Large quadrilateral lateral proximal femoral prominence. Insertion for gluteus medius/minimus and piriformis. Palpable landmark and site of trochanteric bursa. Intertrochanteric fractures between trochanters are extracapsular with lower avascular necrosis risk because periosteal blood supply is preserved.

Grey Turner's Sign

- Bluish or violaceous ecchymosis of the flanks caused by retroperitoneal bleeding tracking through the abdominal wall. It is classically associated with severe haemorrhagic pancreatitis but may occur with other retroperitoneal haemorrhage.

Grid Implant

- See mesh implant.

Grief

- A natural emotional response to loss, particularly the death of a loved one, but also following other significant losses such as divorce, job loss, or declining health. It encompasses a range of feelings including sadness, anger, guilt, and numbness, often occurring in waves rather than linear stages.

Gripping (see Colic)

- Severe abdominal pain, typically spasmodic in nature. See Colic.

Grippe (see Influenza)

- An older term for influenza. See Influenza.

Griscelli Syndrome

- Three types: Type 1 (MYO5A mutation): primary neurologic impairment, no immunodeficiency; Type 2 (RAB27A mutation): immunodeficiency + partial albinism, HLH risk; Type 3 (MLPH mutation): only partial albinism. Type 2: silvery hair, large melanosomes in melanocytes, defective cytotoxic granule release -> HLH.

Gross Examination

- The macroscopic, naked-eye inspection of a specimen in the pathology laboratory, performed before microscopic sectioning and the essential first step in pathologic diagnosis of resected tissues and organs.

Gross Motor Skills

- The ability to use large muscle groups for movement and posture, developing in a predictable sequence in

infancy/childhood: head control (3 months), sitting (6 months), crawling (9 months), standing (10-12 months), walking (12-18 months), running/jumping (2-3 years), hopping and balance (3-5 years).

Ground-Glass Opacity

- Hazy lung opacity on CT without obscuring vessels. Non-specific, may represent inflammation, edema, fibrosis, malignancy.

Group A Strep

- Group A Streptococcus (*Streptococcus pyogenes*), a gram-positive coccus in chains, beta-hemolytic. Infections: pharyngitis/tonsillitis, impetigo, erysipelas, cellulitis, scarlet fever, necrotizing fasciitis, streptococcal toxic shock syndrome, pneumonia, and puerperal sepsis.

Group A streptococcus

- See Group A Strep.

Group B Strep

- Group B Streptococcus (GBS, *Streptococcus agalactiae*): a gram-positive coccus colonizing the GI and genitourinary tract in 10-30 percent of women; major cause of neonatal sepsis and meningitis.

Group B Streptococcus

- A gram-positive bacterium (*Streptococcus agalactiae*) colonizing the genitourinary and gastrointestinal tracts. Asymptomatic vaginal colonization occurs in 20-30% of pregnant women. Screening is performed at 36-37 weeks gestation via rectovaginal swab culture.

Growing Pains

- Recurrent, bilateral leg pain in children ages 3-12, typically occurring in the late afternoon or evening and resolving by morning. Pain is deep, aching in the calves, shins, and thighs (not joints). No limping, fever, swelling, or functional impairment.

Growth Chart

- A graphical representation of growth (weight, height/length, head circumference, BMI) plotted against age, standardized by population (WHO growth standards 0-5 years; CDC growth charts 2-20 years; percentiles/z-scores).

Growth Factor

- Proteins that stimulate cell growth, proliferation, differentiation, and survival. Major families: EGF

(epidermal growth factor), FGF (fibroblast growth factor), PDGF (platelet-derived), VEGF (vascular endothelial), TGF-beta (transforming), NGF (nerve), IGF (insulin-like), HGF (hepatocyte).

Growth Hormone

- Pituitary hormone. Stimulates IGF-1 production. Lipolytic, anti-insulin effects.

Growth Hormone Deficiency

- Insufficient secretion of growth hormone from the anterior pituitary, causing short stature in children and metabolic effects in adults. In children: delayed growth, short stature, delayed bone age, and increased fat mass. In adults: decreased muscle mass, increased fat mass, reduced bone density, fatigue, and dyslipidemia.

Growth Hormone Physiology

- Growth hormone is secreted in a pulsatile pattern by pituitary somatotroph cells regulated by hypothalamic GHRH which stimulates release and somatostatin which inhibits release. GH acts directly on tissues and indirectly through IGF-1 produced primarily by the liver.

Growth Medium

- A nutrient preparation supporting microbial multiplication in vitro, supplying carbon, nitrogen, minerals, and growth factors in liquid or solid form.

Growth Retardation

- Abnormally slow physical growth or failure to reach expected height or weight for age and developmental stage. Causes include inadequate nutrition, chronic systemic disease, endocrine or genetic disorders, malabsorption, psychosocial deprivation, and constitutional variation; assessment uses growth charts and clinical context.

Growth Scan

- An ultrasound performed in the third trimester (usually 28-32 weeks and/or 36-38 weeks) to assess fetal growth. Measurements: BPD, HC, AC, FL, and estimated fetal weight (EFW). Growth percentiles plotted on a growth curve. Abnormal: <10th percentile (SGA/IUGR) or >90th percentile (LGA/macrosomia).

Guaifenesin

- An expectorant that thins and loosens bronchial mucus by increasing respiratory tract fluid and

reducing mucus viscosity, facilitating productive cough. Used for symptomatic relief of chest congestion and productive cough in upper respiratory infections and COPD. Typical adult dose 200-400 mg every 4 hours.

Guided Tissue Regeneration

- Periodontal procedure using barrier membrane to promote regeneration of bone, cementum, periodontal ligament.

Guillain-Barre Syndrome

- An acute, immune-mediated polyneuropathy typically presenting with ascending paralysis following an antecedent infection (Campylobacter jejuni, CMV, EBV, Zika). Variants: acute inflammatory demyelinating polyneuropathy (AIDP, most common in North America), acute motor axonal neuropathy (AMAN, more common in Asia).

Gummy Smile

- Excessive gingival display when smiling. Caused by hypermobile lip, vertical maxillary excess, altered passive eruption. Treatment: surgery, Botox, orthodontics.

Gunshot Wound

- An injury produced by a projectile discharged from a firearm; forensic classification by range and direction. Entrance wound: rounded/ovoid defect with an abraded ring and often a zone of soot (close range) or stippling/tattooing (intermediate range); contact wounds show muzzle imprint.

Gut Microbiome

- The complex community of trillions of microorganisms, including bacteria, viruses, and fungi, residing in the gastrointestinal tract. The gut microbiome plays a crucial role in digestion, immune function, vitamin synthesis, and protection against pathogens.

Guttate Psoriasis

- An acute eruption of numerous small, scaly, drop-like psoriatic papules and plaques, often following streptococcal pharyngitis in children or young adults. Diagnosis is clinical; the eruption may resolve spontaneously or evolve into chronic plaque psoriasis.

Gynaecologist

- A medical doctor who specializes in the health of the female reproductive system (vagina, cervix,

uterus, fallopian tubes, ovaries) and breasts.

Gynaecomastia

- The benign enlargement of breast tissue in males. Pathophysiology: an imbalance between estrogen (stimulates breast tissue growth) and androgen (suppresses) activity, resulting in increased estrogen-to-testosterone ratio.

gynecology

- The medical specialty concerned with the health of the female reproductive system (uterus, ovaries, fallopian tubes, cervix, vagina, and vulva).

Gynecomastia

- Enlargement of breast tissue in males, often bilateral, occurring physiologically in neonates, during puberty (40-60% of boys), and in aging men. Pathologic causes include Klinefelter syndrome, testicular tumors, medications (spironolactone, cannabis), and liver disease. Persistent or asymmetric gynecomastia warrants evaluation.

Gyrus

- A ridge or convolution on the surface of the cerebral cortex, separated by sulci (grooves). The folded structure triples the surface area of the brain (approximately 2500 cm²) while fitting within the cranial vault.

Gastrointestinal Pain

- Pain arising from the digestive tract or related organs. Causes include indigestion, infection, ulcer disease, inflammation, obstruction, gallstones, and disorders of the liver or pancreas. Severe, persistent, or worsening pain requires medical assessment.

Chapter 8

Pocket Dictionary: H Terms

21-Hydroxylase deficiency

- An inherited defect in CYP21A2 that reduces cortisol and aldosterone synthesis, increases 17-hydroxyprogesterone, and causes congenital adrenal hyperplasia with androgen excess.

H Influenzae Meningitis

- Bacterial meningitis caused by *Haemophilus influenzae*, especially type b in unvaccinated children. It may cause fever, headache, neck stiffness, altered consciousness, seizures, and serious neurologic complications.

H1N1 Influenza

- Influenza A infection caused by an H1N1 virus strain. It usually produces fever, cough, sore throat, myalgia, and fatigue but can cause viral pneumonia and severe respiratory failure in high-risk patients.

H2 Blocker (Histamine-2 Receptor Antagonist)

- A drug that blocks histamine H2 receptors on gastric parietal cells, reducing gastric acid secretion. Agents: cimetidine, ranitidine, famotidine, nizatidine. Used for GERD, peptic ulcer disease, and dyspepsia—less potent than PPIs. Adverse effects: headache, diarrhea, confusion (elderly, cimetidine), and rare thrombocytopenia.

Habenula

- A paired epithalamic structure forming part of the dorsal diencephalon. It integrates limbic and basal-forebrain signals with midbrain monoaminergic systems and contributes to reward, aversion, stress, and motivational behavior.

Haem- / Haema- / Haemato- / Haemo- (Prefixes relating to blood)

- Prefixes derived from the Greek word *haima* (blood), used in medical terminology to denote

blood-related conditions and procedures.

Haemangioma

- A benign vascular tumor composed of proliferating endothelial cells, the most common soft tissue tumor of infancy.

Haemarthrosis

- Bleeding into a joint cavity. Causes: trauma (most common), hemophilia (spontaneous haemarthrosis is characteristic, especially in severe deficiency of factor VIII or IX, common in knees, elbows, ankles), anticoagulant therapy, pigmented villonodular synovitis (PVNS, recurrent haemarthrosis), joint surgery, and vascular malformations.

Haematocrit

- The proportion of blood volume occupied by red blood cells (RBCs), expressed as a percentage or decimal fraction. Normal reference range: men 42-50%, women 37-47% (varies by altitude, age, and laboratory).

Haematologist

- A medical doctor who specializes in the diagnosis, treatment, and prevention of diseases of the blood and blood-forming organs.

Haematoma

- A localised collection of blood outside blood vessels, usually clotted, within an organ or tissue. Haematomas form when trauma or disease causes a blood vessel to rupture.

Haematuria

- The presence of blood in the urine. Classification: gross (visible to the naked eye, indicates bleeding beyond the nephron) vs microscopic (detected only on urinalysis or microscopy, >3 RBCs per high-power field).

Haemodialysis

- A medical procedure that filters waste products, toxins, and excess fluid from the blood when the kidneys are unable to do so (end-stage renal disease, ESRD). Mechanism: blood is pumped through an extracorporeal circuit and passes through a dialyser (artificial kidney) containing thousands of hollow fibers with a semipermeable membrane.

Haemolytic Disease of the Newborn

- See blood-related entries in hematology.

Haemophilia

- An X-linked recessive bleeding disorder caused by deficiency of coagulation factor VIII (hemophilia A, 80-85%) or factor IX (hemophilia B, Christmas disease, 15-20%). Incidence: 1 in 5,000 male births (hemophilia A), 1 in 30,000 (hemophilia B).

Haemophilus influenzae

- A gram-negative coccobacillus that causes meningitis, epiglottitis, pneumonia, and other infections. The encapsulated type b (Hib) strain was the most virulent, causing invasive disease in children. The Hib conjugate vaccine has dramatically reduced the incidence of invasive Hib disease.

Haemoptysis

- Coughing up blood or blood-stained sputum from the lower respiratory tract.

Haemorrhoids

- Enlarged symptomatic vascular cushions of the anal canal that may cause painless bright-red rectal bleeding, prolapse, itching, or discomfort.

Hair Cells

- The sensory receptor cells of the inner ear that convert mechanical sound vibrations into electrical neural signals, and they are named for the bundle of stereocilia projecting from their apical surface.

Hair Follicle

- Complex dermal/hypodermal mini-organ producing hair. Has infundibulum (surface opening), isthmus (from sebaceous duct to arrector pili insertion), and lower segment (bulb with matrix and dermal papilla). Hair cycle: anagen (active growth, 2-7 years), catagen (regression, 2-3 weeks), telogen (resting, 2-3 months).

Hair Loss

- Alopecia: reduction of scalp or body hair. Types: nonscarring (androgenetic—pattern hair loss, telogen effluvium—stress/illness/drugs, anagen effluvium—chemotherapy, alopecia areata—autoimmune patches, tinea capitis, trichotillomania) and scarring (lupus, lichen planopilaris, folliculitis decalvans). Female pattern hair loss is common.

Hair Loss (Alopecia)

- The partial or complete loss of hair from areas where it normally grows, most commonly the scalp. Causes include genetic predisposition (androgenetic alopecia), autoimmune conditions (alopecia areata), hormonal changes, medications, stress, and nutritional deficiencies.

Hair Loss Classification

- Hair loss (alopecia) is classified as scarring (cicatricial) or non-scarring. Non-scarring alopecias are potentially reversible: androgenetic alopecia (male and female pattern), alopecia areata, telogen effluvium, anagen effluvium, and tinea capitis.

Hairy Cell Leukemia

- A rare, indolent B-cell chronic lymphoproliferative disorder characterized by the accumulation of mature B lymphocytes with characteristic cytoplasmic projections ("hairy cells") in the bone marrow, peripheral blood, and spleen.

Haldane Effect

- The decrease in hemoglobin's CO₂ carrying capacity when oxygen binds to it, facilitating CO₂ release at the lungs. Deoxygenated hemoglobin (deoxyhemoglobin) is a better buffer for H⁺ ions and forms more carbaminohemoglobin than oxyhemoglobin.

Half-Life

- Elimination half-life is the time required for the plasma concentration of a drug to decrease by 50 percent after reaching equilibrium. It is calculated from the elimination rate constant using the equation $t_{1/2} \text{ equals } 0.693 \text{ divided by the elimination rate constant}$.

Half-life (Physical)

- Time for radioactive isotope to decay to half its original activity. F-18: 110 min; Tc-99m: 6 hours; I-131: 8 days; Ga-68: 68 min. Affects imaging timing,

radiation dose.

Half-Life ($t_{1/2}$)

- The time required for plasma drug concentration to decrease by 50%. It determines dosing frequency and time to steady state (4-5 half-lives). Drugs with long half-lives (e.g., amiodarone: 40-55 days, warfarin: 40 hours) can be given once daily but take longer to reach steady state and to be eliminated.

Halitosis

- An unpleasant odour of the breath that may be socially distressing. The most common cause (85-90%) is poor oral hygiene—accumulation of bacteria on the tongue, dental plaque, periodontitis, and gingivitis.

Hallucination

- A perception in the absence of an external stimulus, experienced as real and occurring in objective space.

Hallucinations

- Perceptions occurring without an external sensory stimulus, such as hearing voices, seeing images, or feeling sensations that are not present. They may result from psychiatric illness, neurological disease, delirium, substance exposure, sensory loss, or severe sleep disruption.

Hallux Valgus

- Lateral deviation of the great toe (hallux) with medial prominence of the first metatarsal head (bunion). More common in women, associated with narrow-toed shoes, high heels, and genetic predisposition. Presents with pain over the medial eminence, difficulty with shoe wear, and progressive deformity.

Hallux Valgus (Bunion)

- A progressive deformity of the first metatarsophalangeal (MTP) joint, characterized by lateral deviation of the great toe (hallux valgus) and medial deviation of the first metatarsal head (metatarsus primus varus), resulting in a bony prominence on the medial side of the forefoot (the bunion).

Halo nevus

- A benign melanocytic nevus surrounded by a rim of depigmentation (Sutton nevus), caused by an immune-mediated T-cell response directed against melanocytes in the nevus. It is most common on the trunk of children and young adults and typically resolves by central depigmentation, leaving a white

macule.

Hamartoma

- A benign growth composed of a disorganized mixture of tissue elements normally present at that site, e.g., pulmonary hamartoma, Peutz-Jeghers polyps, and PTEN-related (Cowden) lesions.

Hamate

- Wedge-shaped distal row carpal bone on the ulnar side with the hook of the hamate. The hook forms Guyon's canal medial border and serves as flexor retinaculum attachment. Articulates with 4th/5th metacarpals. Hook fractures associated with racquet sports and ulnar nerve compression.

Hamman's Sign

- A crunching or crackling sound heard on auscultation synchronized with the heartbeat, caused by air in the mediastinum (pneumomediastinum). It is a rare but specific finding. It is also described as a mediastinal crunch heard in spontaneous pneumomediastinum.

Hammer Toe

- A deformity of the second, third, or fourth toe characterized by flexion at the PIP joint and extension at the MTP and DIP joints. Caused by muscle imbalance (intrinsic foot muscle weakness) and narrow-toed shoes. Presents with pain over the dorsal PIP joint (corn/callus) and the tip of the toe. Flexible hammer toe: correctable passively.

Hampton Hump

- Wedge-shaped opacification with its base abutting the pleura on chest radiography, representing pulmonary infarction secondary to Pulmonary Embolism. See Pulmonary Embolism (PE).

Hampton's Hump

- Wedge-shaped, pleural-based opacity on CXR representing pulmonary infarction distal to PE.

Hamstring

- One of three muscles forming the posterior compartment of the thigh: semitendinosus, semimembranosus, and biceps femoris (long and short heads). Common origin: ischial tuberosity (except short head of biceps femoris which originates from the linea aspera of the femur).

Hamstring Muscles

- Three posterior thigh muscles: biceps femoris

(long/short heads), semitendinosus, semimembranosus, mostly from the ischial tuberosity (short head of biceps from linea aspera). Long head/semimembranosus/semitendinosus innervated by tibial sciatic nerve; short head by common peroneal division. Flex knee, extend hip.

Hand Foot and Mouth Disease

- A common viral illness (mostly children <10 years) caused by coxsackievirus A16 (and enterovirus 71), transmitted by fecal-oral/respiratory.

Hand hygiene

- The most effective measure for preventing healthcare-associated infections. Hand hygiene includes handwashing with soap and water and hand rubbing with alcohol-based rubs. Alcohol-based rubs are preferred for routine hand hygiene as they are faster and more effective than soap and water.

Hand Prolapse

- Prolapse of a fetal hand or arm alongside or below the presenting part. May occur spontaneously or after manual rotation. The prolapsed arm may be mistaken for a cord or foot. Management: if the hand is above the presenting part (compound presentation), it may be gently pushed back during contractions.

Hand, Foot, and Mouth Disease

- See pediatrics.

Hand-Foot-Mouth Disease

- A common enteroviral infection, usually in young children, causing fever, painful oral ulcers, and a vesicular rash on the hands, feet, and sometimes buttocks. It is highly contagious and usually self-limited.

Hanging (Complete vs Incomplete)

- Form of violent asphyxial death produced by suspension of the body by a ligature around the neck. Complete Hanging: feet off ground, total body weight exerts pressure. Incomplete Hanging: part of body touches ground. Asphyxia results from occlusion of jugular veins (2 kg force) or carotid arteries (5 kg force).

Hanging and Strangulation Signs

- Medico-legal findings resulting from mechanical constriction of the neck structures. Hanging: neck constriction by a ligature tightened by the weight

of the victim's body. Ligature mark: pale, hard, parchment-like groove running obliquely upward toward the knot, usually situated above the thyroid cartilage.

Hansen's Disease (Leprosy)

- A chronic infectious disease caused by *Mycobacterium leprae*, an acid-fast, obligate intracellular bacillus. Transmission: respiratory droplets (prolonged close contact). Incubation: 2-10 years (average 5). Epidemiology: 200,000 new cases per year globally (mostly India, Brazil, Indonesia). Pathophysiology: M.

Hantavirus

- A genus of enveloped negative-sense RNA viruses of the Bunyaviridae family, transmitted to humans primarily through inhalation of aerosolized rodent excreta (urine, feces, saliva).

Haplotype

- Set of alleles inherited together on same chromosome. Used in linkage analysis, population genetics, disease association studies.

Haptoglobin

- An alpha-2 globulin that binds free hemoglobin, preventing oxidative damage and renal excretion. Low/undetectable: acute hemolysis (intravascular and extravascular), liver disease (decreased production), and hereditary deficiency. Elevated: acute phase reaction (inflammation, infection, malignancy).

Harlequin color change

- A transient, benign skin color change in newborns where one half of the body becomes flushed while the other half remains pale, due to asymmetric vasomotor tone. It occurs in up to 10 percent of healthy neonates, typically when the infant is placed on their side.

Harm Reduction

- Evidence-based strategies that decrease health, social, and legal harms associated with substance use without requiring immediate abstinence. Examples include naloxone access, sterile injecting equipment, drug-checking where available, safer-use education, and linkage to treatment.

Hartmann Procedure

- Surgical resection of the rectosigmoid colon with

creation of an end colostomy and rectal stump. Used for emergency resection of complicated diverticulitis with peritonitis and colorectal cancer with obstruction or perforation. The procedure avoids primary anastomosis in a contaminated field.

Hartnup Disease

- An autosomal-recessive disorder of neutral-amino acid transport, especially in the intestine and renal tubules. Defective tryptophan handling can cause pellagra-like photosensitive dermatitis, cerebellar ataxia, and neuropsychiatric symptoms, often intermittently.

Hashimoto Thyroiditis

- Autoimmune thyroiditis characterized by lymphocytic infiltration and progressive destruction of thyroid follicles leading to hypothyroidism. Most common cause of hypothyroidism in iodine-sufficient areas. Anti-thyroid peroxidase and anti-thyroglobulin antibodies are present. The thyroid may be diffusely enlarged and

Hay Fever (Allergic Rhinitis)

- See Allergic Rhinitis. Hay fever is seasonal allergic rhinitis caused by airborne allergens such as pollens; perennial allergic rhinitis occurs throughout the year with triggers such as dust mites, animal dander, or mold.

HbA1c (Glycated Hemoglobin)

- Hemoglobin containing glucose-derived adducts formed by nonenzymatic glycation. It reflects average glycemic exposure over approximately two to three months, with interpretation affected by altered red-cell survival and some hemoglobin variants.

hCG (Human Chorionic Gonadotropin)

- A glycoprotein hormone produced by the placenta, detected in urine and serum for pregnancy diagnosis. Qualitative: positive 25 mIU/mL (urine). Quantitative: serum hCG levels — <5 mIU/mL (negative), 5-25 (indeterminate), >25 (positive). In early pregnancy: hCG doubles every 48-72 hours until 8-10 weeks.

HDL

- High-density lipoprotein; the 'good' cholesterol that mediates reverse cholesterol transport, removing cholesterol from peripheral tissues and arterial walls and returning it to the liver for excretion. High

HDL is associated with reduced cardiovascular risk; low HDL (<40 mg/dL men, <50 mg/dL women) is a risk factor.

Head and Neck Cancer

- Cancers arising in the oral cavity, pharynx, larynx, sinonasal tract, and salivary glands, most commonly squamous cell carcinoma. Risk factors include tobacco, alcohol, and HPV (oropharyngeal). Treatment involves surgery, radiation, and chemotherapy, often combined to preserve function.

Head Circumference

- Measured at the largest circumference above the ears and eyebrows, reflecting brain growth. Plotted on growth charts to detect microcephaly (below 3rd percentile) or macrocephaly (above 98th percentile). Rapid increase may indicate hydrocephalus or intracranial hemorrhage; failure to grow may indicate microcephaly or brain atrophy.

Head Circumference (HC)

- The circumference of the fetal skull, measured at the same level as BPD (thalami, cavum septum pellucidum). Used for dating (second trimester) and for assessing head growth. HC is more reliable than BPD for dating when head shape is abnormal. Microcephaly: HC <3rd percentile (workup for infection, aneuploidy, genetic syndromes).

Head Injury

- Trauma to scalp, skull, or brain. Scalp lacerations: bleed profusely, require suturing. Skull fracture: basilar (Battle sign, raccoon eyes) or depressed (surgical emergency). Concussion: transient neurological dysfunction, manage with rest, gradual return to activity.

Head lice

- An infestation of the scalp by the obligate human ectoparasite *Pediculus humanus capitis*, a wingless, blood-sucking insect that feeds on human blood, most common in children aged 3-12. It is extremely common, especially in school-aged children aged 3-11 years, with an estimated 6-12 million infestations annually in the United States.

Headache

- One of the most common neurologic symptoms, classified as primary (migraine, tension-type, cluster, and trigeminal autonomic cephalalgias) or secondary

(attributed to an underlying structural, infectious, vascular, or metabolic cause).

Headache and Migraine

- Headaches are pain in the head or upper neck region, classified as primary (tension-type, migraine, cluster) or secondary to underlying conditions. Migraines are a neurological disorder characterized by recurrent episodes of moderate to severe throbbing pain, often accompanied by nausea, photophobia, and aura.

Health Behavior

- Actions affecting health. Examples: diet, exercise, smoking, alcohol, sexual behavior, medication adherence.

Health Behavior Change

- Modification of behaviors that affect health, such as tobacco use, diet, physical activity, and adherence to treatment.

Health Behavior Motivation

- Factors that influence initiation and maintenance of health-related behaviors, including adherence, physical activity, and smoking cessation.

Health Disparities

- See social determinants.

Health Disparity

- Difference in health outcomes between populations. Examples: racial disparities, socioeconomic disparities, geographic disparities.

Health Equity

- Attainment of highest possible health for all people. Requires addressing disparities.

Health Inequity

- A systematic, avoidable, and unjust difference in health or access to health resources between social groups. It differs from health variation that is not preventable or not linked to unfair social conditions.

Health Information Technology

- Use of technology for health information. Examples: electronic health records, telemedicine, mobile health, decision support.

Health Literacy

- Ability to obtain, process, understand health information. Examples: reading labels, following instructions, communicating with providers.

Health Promotion

- Activities maintaining/improving health in elderly. Components: vaccination, screening, exercise, nutrition, cognitive stimulation, social engagement, fall prevention. Tailored to individual capacity and preferences.

Health Surveillance

- Ongoing collection, analysis, interpretation of health data. See surveillance.

Health-Related QoL

- Perception of well-being including physical, psychological, social, spiritual domains. See quality of life.

Healthcare Associated Infection

- An infection developing as a consequence of healthcare delivery, typically beyond 48 hours after admission or within 30 days of a procedure.

Healthcare Provider

- A healthcare professional or organization that provides medical care, such as a physician, nurse, therapist, or clinic.

Healthcare Proxy

- Legal document expressing healthcare wishes. See advance directive.

Healthcare-Associated Pneumonia

- HCAP concept retired (2016 IDSA/ATS); risk factors for MDR now assessed individually. Risk factors for MDR pathogens (prior antibiotics, structural lung disease, frequent hospitalization) should be assessed individually rather than using the HCAP category.

Healthy Aging

- Process of developing and maintaining functional ability. WHO definition.

Healthy Eating for Diabetes

- A dietary approach designed to maintain stable blood glucose levels in individuals with diabetes mellitus. It emphasizes carbohydrate consistency, high fiber intake, lean proteins, and healthy fats while minimizing refined sugars and simple carbohydrates.

Healthy Years

- Years of life lost due to premature death or disability. See healthy years lost.

Healthy Years Lost

- Years of life lost due to premature death or disability. DALYs combine YLL and YLD.

Hearing Aid

- Amplification device for hearing loss. Types: behind-the-ear (BTE), in-the-ear (ITE), receiver-in-canal (RIC). Components: microphone, amplifier, speaker, battery. Fitting: audiogram-based. Adjustments: volume, programs.

Hearing Aids and Devices

- Electronic devices worn in or behind the ear that amplify sound for individuals with hearing loss. Modern hearing aids include digital signal processing, directional microphones, and connectivity to smartphones and other audio sources.

Hearing Impairment

- Partial or complete reduction in the ability to detect or understand sound. It may be conductive, sensorineural, or mixed and can result from congenital disorders, infection, noise, ototoxic injury, trauma, ageing, or disease of the ear or auditory pathways.

Hearing Loss

- Reduced ability to hear, classified as conductive (outer/middle ear pathology: cerumen impaction, otitis media, otosclerosis, tympanic membrane perforation), sensorineural (inner ear or auditory nerve: presbycusis, noise exposure, ototoxic drugs, Meniere disease, acoustic neuroma), or mixed.

Hearing Screening

- Newborn hearing screening (universal): otoacoustic emissions (OAE) and/or automated auditory brainstem response (AABR) before discharge; referral for comprehensive audiology if failed.

Heart

- A four-chambered muscular pump (two atria, two ventricles) located in the mediastinum, responsible for circulating blood throughout the body.

Heart Attack

- Acute myocardial injury caused most often by blockage of a coronary artery, usually from rupture of an atherosclerotic plaque and thrombosis. It may cause chest pressure, shortness of breath, sweating, nausea, or atypical symptoms and requires emergency treatment.

Heart Attack (Myocardial Infarction)

- The death of heart muscle tissue due to sudden blockage of a coronary artery, typically by a blood clot. It presents with chest pain or pressure, shortness of breath, nausea, and diaphoresis, though symptoms may be atypical in women and diabetics.

Heart Block

- A delay or interruption in the conduction of cardiac impulses from the atria to the ventricles through the atrioventricular (AV) node and His-Purkinje system. Classification: first-degree AV block: PR interval >200 ms, each atrial impulse conducts to the ventricles, typically asymptomatic.

Heart Disease

- A broad term encompassing various conditions affecting the heart and blood vessels, most commonly coronary artery disease, heart failure, arrhythmias, and valvular heart disease. Atherosclerosis, the buildup of plaque in the arteries, is the underlying cause of most cardiovascular events.

Heart Failure

- A clinical syndrome resulting from the heart's inability to pump sufficient blood to meet metabolic demands. It is classified by ejection fraction (HFrEF: EF 40% or less; HFpEF: EF 50% or greater) and by symptoms (NYHA class I-IV).

Heart Failure Stages

- ACC/AHA staging: Stage A (at risk for HF but no structural disease or symptoms); Stage B (structural heart disease but no symptoms); Stage C (structural disease with prior or current symptoms); Stage D (advanced HF requiring specialized interventions). This staging system emphasizes prevention and progression.

Heart Failure with Preserved Ejection Fraction (HFpEF)

- A clinical syndrome of heart failure characterized by left ventricular diastolic dysfunction (impaired relaxation and increased stiffness) with a preserved ejection fraction (50%). Most common in elderly, female, hypertensive, obese, and diabetic patients.

Heart Failure with Reduced Ejection Fraction (HFrEF)

- A clinical syndrome of heart failure characterized by left ventricular systolic dysfunction with a reduced

ejection fraction (40%). Causes: ischemic heart disease (prior MI, 60-70%), dilated cardiomyopathy, hypertension, valvular heart disease.

Heart Health

- The state of cardiovascular well-being, characterized by efficient pumping function, clear arteries, normal blood pressure, and regular heart rhythm. Maintaining heart health involves a heart-healthy diet, regular aerobic and resistance exercise, weight management, stress reduction, and avoidance of tobacco.

Heart Murmur

- An audible extra sound produced by turbulent blood flow within the heart or great vessels, heard on auscultation.

Heart Palpitations

- An awareness of the heartbeat as pounding, racing, fluttering, or irregular. Palpitations may occur with normal rhythm or arrhythmia and can be triggered by stress, exertion, fever, stimulants, endocrine disease, or structural heart disease.

Heart Rate

- The number of cardiac contractions (heartbeats) per minute. Normal resting heart rate (sinus rhythm): 60-100 bpm in adults; higher in children (70-120 at age 1-2 years, 80-130 in neonates).

Heart Sounds

- Audible sounds produced by the heart during the cardiac cycle, normally heard as two sounds (lub-dub). S1 (lub) is caused by closure of the mitral and tricuspid valves at the beginning of systole. S2 (dub) is caused by closure of the aortic and pulmonary valves at the end of systole. Additional sounds (S3, S4) may indicate pathology.

Heart Surgery

- Surgical procedures performed on the heart or great vessels to treat structural or functional cardiac conditions. Common operations include coronary artery bypass grafting (CABG), valve repair or replacement, and repair of congenital heart defects.

Heart Valve

- Replacement heart valve. See heart valve prosthesis.

Heart Valve Disorders

- Disorders affecting any of the four heart valves,

including stenosis (narrowing) and regurgitation (leaking), which impair blood flow through the heart. Causes include congenital malformations, age-related degeneration, rheumatic fever, and endocarditis.

Heart Valve Prosthesis

- Replacement heart valve. Types: mechanical (titanium, carbon, durable, requires lifelong anticoagulation), bioprosthetic (cow/pig tissue, no anticoagulation, degenerates). Mechanical: more durable, higher thrombotic risk. Bioprosthetic: better quality of life, reoperation risk.

Heart Valves

- The four cardiac valves ensure unidirectional blood flow through the heart.

Heart Ventricles

- One of the two lower, thicker-walled chambers of the heart that pump blood out of the heart. See Ventricle.

Heart-Healthy Eating

- A dietary approach emphasizing foods that support cardiovascular function and reduce the risk of heart disease. Key principles include limiting saturated fats, trans fats, sodium, and added sugars while prioritizing fruits, vegetables, whole grains, lean proteins, and healthy fats like those from olive oil and fatty fish.

Heartburn

- A burning sensation, typically retrosternal (behind the sternum), that may radiate to the throat and is often associated with regurgitation of sour or bitter-tasting fluid into the mouth. Despite its name, heartburn is a symptom of gastro-oesophageal reflux disease (GORD), not a cardiac symptom.

Heat Exhaustion

- See environmental emergencies.

Heat Intolerance

- An abnormal inability to tolerate warm environments or increased body heat, often causing excessive sweating, flushing, fatigue, or dizziness. Common causes include hyperthyroidism, menopause, fever, medications, autonomic disorders, and dehydration.

Heat Pack

- Warm therapy for muscle pain. Increases blood

flow, reduces stiffness.

Heat Stroke

- See environmental emergencies.

Heavy Metal Poisoning

- Lead, mercury, arsenic, cadmium, thallium. Chelation therapy, supportive care, prevention. Sources: paint, water pipes, fish, occupational exposure.

Heberden Node

- A firm bony enlargement of the distal interphalangeal joint, usually caused by hand osteoarthritis. It may be tender during progression and can produce deformity, reduced fine-motor function, or restricted finger movement.

Heel Pain

- Pain beneath or around the heel, commonly caused by plantar fasciitis, Achilles tendinopathy, bursitis, stress fracture, nerve compression, or inflammatory arthritis. Persistent pain, swelling, inability to bear weight, or trauma requires assessment.

Heimlich Manoeuvre

- An emergency procedure for relieving upper airway obstruction caused by a foreign body (choking). The maneuver is performed on a conscious, choking adult who cannot speak, cough effectively, or breathe.

Heinz Bodies

- Inclusions within red blood cells composed of denatured, oxidized hemoglobin. Formed under severe oxidative stress when protective reducing agents (glutathione) are depleted. Classically seen in Glucose-6-Phosphate Dehydrogenase (G6PD) deficiency following exposure to oxidative triggers (fava beans, primaquine, sulfonamides,

Helicobacter pylori

- Gram-negative, microaerophilic, spiral-shaped bacterium specifically adapted to colonize the human gastric mucosa. It is estimated that approximately half the world's population is infected, making it one of the most common chronic bacterial infections globally. H.

Helicobacter Pylori Infection

- Infection of the stomach by *H. pylori*, a spiral-shaped bacterium associated with chronic gastritis, peptic ulcer disease, gastric adenocarcinoma, and gastric MALT lymphoma. It is diagnosed with noninvasive or endoscopic testing and treated with

combination eradication therapy.

HELLP Syndrome

- A severe variant of preeclampsia characterized by Hemolysis, Elevated Liver enzymes, and Low Platelets. Occurs in 10-20% of patients with severe preeclampsia. Diagnostic criteria include haemolysis (LDH >600 U/L, schistocytes on peripheral smear), elevated liver enzymes (AST 70 U/L), and low platelets (<100,000/ μ L).

Helminths

- Parasitic worms that cause disease in humans. The major groups are nematodes (roundworms), cestodes (tapeworms), and trematodes (flukes). Helminths have complex life cycles often involving intermediate hosts. Infection occurs through ingestion of eggs or larvae, penetration of skin, or insect vectors.

Helper T Cell

- Orchestrators of the adaptive immune response.

HELPERR Mnemonic

- A mnemonic for the sequential management of shoulder dystocia: H — Help (call for additional personnel, anesthesia, and pediatrics). E — Evaluate for episiotomy (generous episiotomy to provide more room). L — Legs (McRoberts maneuver: hyperflex maternal legs onto the abdomen).

Hemangioblastoma

- A benign, well-circumscribed vascular tumor of the central nervous system, most commonly the cerebellum, often cystic with a mural nodule. Associated with von Hippel-Lindau disease; treated by surgical resection.

Hemangioma

- Benign vascular tumor. Port-wine stain (capillary malformation), strawberry hemangioma. Treatment: observation, laser, surgery.

Hemangiosarcoma

- A malignant tumor of vascular endothelial cells, an aggressive sarcoma that may arise in the skin, liver, or bone.

Hematemesis

- Vomiting of blood, indicating upper gastrointestinal bleeding proximal to the ligament of Treitz. Blood may be bright red (acute, brisk bleeding) or resemble coffee grounds (altered blood that has

been in contact with gastric acid).

Hematochezia

- The passage of bright red or maroon blood per rectum, usually indicating active lower gastrointestinal (LGI) bleeding distal to the ligament of Treitz (e.g., diverticulosis, arteriovenous malformations/angiodysplasia, colorectal polyps/cancer, inflammatory bowel disease, hemorrhoids).

Hematocrit

- The percentage of total blood volume composed of packed red blood cells, one of the most commonly ordered laboratory tests in clinical medicine.

Hematologic Changes in Pregnancy

- Plasma volume increases 40-50% while red cell mass increases 20-30%, resulting in physiologic anemia (Hb 10-11 g/dL). Leukocytosis (WBC up to 15,000-16,000/μL, predominantly neutrophils) is normal.

hematology

- The branch of medicine concerned with the study, diagnosis, treatment, and prevention of disorders of the blood and blood-forming organs (bone marrow, lymph nodes, spleen).

Hematometocolpos

- Accumulation of blood in the uterine cavity and vagina due to outflow obstruction such as an imperforate hymen or transverse vaginal septum, presenting with cyclic pelvic pain and primary amenorrhea in adolescents.

Hematopoiesis

- The production and maturation of blood cells from multipotent hematopoietic stem cells, principally in the bone marrow after birth. It generates erythrocytes, leukocytes, and platelets through regulated progenitor-cell differentiation.

Hematopoietic Stem Cell Transplantation (HSCT)

- Infusion of hematopoietic stem cells after a conditioning regimen to restore marrow function or replace diseased hematopoiesis. Autologous HSCT uses the patient's own cells; allogeneic HSCT uses cells from a donor and carries risks including graft-versus-host disease and graft failure.

Hematuria

- Presence of red blood cells in the urine, categorized as microscopic (>3 RBCs per high-power field on

urine microscopy) or gross (macroscopic, visible color change from pink to dark red). Microscopic hematuria is often asymptomatic and detected on routine urinalysis.

Hemianopia

- Loss of vision in half the visual field of one or both eyes. Homonymous hemianopia affects the same side in both eyes and localizes to a contralateral lesion posterior to the optic chiasm, most often stroke or tumor.

Hemicolecotomy

- Surgical resection of part or all of the colon. See Colectomy.

Hemifacial Microsomia

- A congenital craniofacial disorder involving asymmetric hypoplasia of the structures derived from the first and second branchial arches, most commonly affecting the mandible, ear, and soft tissues on one side. Also known as craniofacial microsomia or Goldenhar syndrome when epibulbar dermoids and vertebral anomalies are present.

Hemifacial Spasm

- Involuntary, unilateral, intermittent twitching of the facial muscles innervated by the facial nerve (CN VII). Typically begins around the eye (orbicularis oculi) and spreads to the lower face.

Hemiparesis

- Weakness affecting one side of the body, less severe than hemiplegia (paralysis). It results from a lesion of the corticospinal tract anywhere from the motor cortex to the spinal cord, most commonly from stroke, but also from tumors, trauma, or demyelination. The pattern of weakness helps localize the lesion.

Hemiplegia

- Paralysis of one side of the body (the face, arm, and leg on the same side), caused by damage to the corticospinal tract (pyramidal tract) at any level from the cerebral cortex to the spinal cord.

Hemochromatosis

- Iron overload disorder characterized by excessive intestinal iron absorption leading to progressive iron deposition in parenchymal organs. Hereditary hemochromatosis is most commonly caused by homozygous C282Y mutations in the HFE gene affect-

ing approximately one in two hundred individuals of Northern European descent.

Hemochromatosis Cardiomyopathy

- Iron overload cardiomyopathy from hereditary (HFE gene mutations, most common) or secondary causes. Iron deposition in the myocardium causes restrictive then dilated cardiomyopathy. Diagnosis: elevated transferrin saturation (>45%), ferritin, and genetic testing. Cardiac MRI T2* <20 ms indicates myocardial iron overload.

Hemodialysis

- Extracorporeal blood purification using semipermeable membrane. Three sessions per week for 3 to 4 hours. Requires vascular access with AV fistula preferred. Adequacy assessed by Kt/V greater than 1.4. Complications include hypotension, access infection, and dialyzer reactions.

Hemofiltration

- An extracorporeal process that removes plasma water and dissolved solutes by convection across a semipermeable membrane, with replacement fluid used to maintain volume. It is used in continuous and intermittent kidney-replacement modalities.

Hemoglobin

- Tetrameric protein in red blood cells (4 subunits: 2 alpha, 2 beta in adults) containing heme groups that bind oxygen. Each heme contains iron (Fe²⁺) that reversibly binds O₂. Cooperative binding produces sigmoidal oxygen dissociation curve. P50 (pO₂ at 50% saturation) 27 mmHg.

Hemoglobin A1c

- Glycated hemoglobin reflecting average blood glucose concentration over the preceding two to three months corresponding to the red blood cell lifespan. Formed by non-enzymatic glycation of hemoglobin at the N-terminal valine of the beta chain. The percent of glycated hemoglobin is directly proportional to mean glucose levels.

Hemoglobin A1c Variability

- Factors that affect hemoglobin A1c accuracy include hemoglobinopathies such as sickle cell disease and thalassemias which alter red blood cell lifespan and glycation rates, iron deficiency anemia which shortens red blood cell survival lowering A1c, vitamin B12 and folate deficiency, pregnancy, chronic

kidney disease, erythropoietin therapy

Hemoglobin C Disease

- An inherited hemoglobinopathy caused by a beta-globin variant that produces hemoglobin C. It may cause mild hemolytic anemia, jaundice, splenomegaly, or, when combined with hemoglobin S, clinically significant sickling disease.

Hemoglobin Electrophoresis

- Separation of hemoglobin variants by electrophoresis or HPLC. Normal adult: HbA (95-98%), HbA₂ (2-3%), HbF (<1%). Sickle cell disease (HbSS): HbS only, no HbA. Sickle cell trait (HbAS): HbA > HbS. Thalassemia: elevated HbA₂ or HbF. Beta thalassemia trait: HbA₂ 3.5-7%.

Hemoglobinopathies: Thalassemias

- Thalassemias are inherited disorders of globin chain synthesis imbalance: alpha-thalassemia (HBA1/HBA2 deletions) and beta-thalassemia (HBB mutations).

Hemoglobinopathy

- An inherited disorder affecting the structure or production of haemoglobin. Examples include sickle-cell disease and thalassemia; manifestations range from asymptomatic microcytosis to haemolytic anaemia, vaso-occlusion, and organ injury.

hemolysis

- The premature destruction of red blood cells, releasing hemoglobin into the plasma. Extravascular hemolysis (most common): RBCs are destroyed by macrophages in the spleen and liver — causes unconjugated hyperbilirubinemia, increased LDH, and decreased haptoglobin.

Hemolytic Anemia

- Anemia due to accelerated destruction of red blood cells (RBCs), classified as intrinsic (RBC defect) or extrinsic (external factors). Intrinsic: hereditary spherocytosis, G6PD deficiency, sickle cell disease, thalassemia.

Hemolytic Anemias: Autoimmune

- Autoimmune hemolytic anemia (AIHA): warm (IgG, 70-80%, extravascular hemolysis in spleen, DAT positive for IgG +/- C3, associated with CLL, SLE, drugs, idiopathic), cold agglutinin disease (IgM, complement-mediated, intravascular/extravascular, DAT positive for C3 only, associ-

ated with lymphoproliferative, Mycoplasma), mixed (both warm and cold).

Hemolytic Anemias: Hereditary

- Hereditary spherocytosis (HS): cytoskeletal protein defects (ankyrin, band 3, protein 4.2, alpha/beta spectrin) -> spherical RBCs, splenic trapping, hemolysis. AD (75%) or AR. Clinical: anemia, jaundice, splenomegaly, gallstones.

hemolytic Disease of the Newborn

- A condition in which maternal antibodies destroy fetal/neonatal red blood cells (RBCs), causing hemolytic anemia, hyperbilirubinaemia, and in severe cases hydrops fetalis.

Hemolytic Jaundice

- Jaundice resulting from excessive hemolysis (breakdown of red blood cells) causing overproduction of unconjugated bilirubin that exceeds the liver's capacity for conjugation and excretion. See Jaundice for complete overview.

Hemolytic Uremic Syndrome

- A thrombotic microangiopathy characterized by microangiopathic hemolytic anemia, thrombocytopenia, and acute kidney injury. Typical (Shiga toxin-producing E. coli).

Hemolytic-Uremic Syndrome

- A potentially life-threatening disorder characterized by microangiopathic hemolytic anemia, thrombocytopenia, and acute kidney injury. It often follows Shiga-toxin-producing bacterial gastroenteritis but may also result from complement dysregulation or other triggers.

Hemoperfusion

- Blood passed through adsorbent column (charcoal, resin) to remove toxins. Used for drug overdoses with high protein binding, large volume of distribution.

Hemophagocytic Lymphohistiocytosis

- HLH: uncontrolled immune activation with cytokine storm (IFN-gamma, IL-6, IL-10, TNF). Familial HLH (FHL): PRF1 (FHL2), UNC13D (FHL3), STX11 (FHL4), STXBP2 (FHL5), RAB27A (FHL6/Griscelli type 2). Secondary HLH: infections (EBV, CMV), malignancy (lymphoma), autoimmune (SJIA/MAS, SLE), immunodeficiency.

Hemophagocytic Lymphohistiocytosis

(HLH)

- A life-threatening hyperinflammatory syndrome caused by uncontrolled activation of cytotoxic lymphocytes and macrophages.

Hemophagocytosis of sickle cells

- Activation of macrophages that engulf erythrocytes, leukocytes, and platelets within the bone marrow, occurring as a complication of sickle cell disease during severe hemolytic crises or infection.

Hemophilia

- An inherited bleeding disorder caused by deficiency or dysfunction of clotting factor VIII, IX, or another coagulation component. It causes prolonged bleeding, particularly into joints and muscles, after injury or procedures.

Hemophilia A

- Factor VIII deficiency. X-linked recessive. Hemarthrosis, deep bleeding.

Hemophilia B

- Factor IX deficiency. X-linked recessive. Similar to hemophilia A.

Hemoptysis

- Expectoration of blood from the lower respiratory tract. Must be differentiated from hematemesis and epistaxis. Causes: bronchitis, pneumonia, TB, bronchiectasis, lung cancer, PE.

Hemorrhage

- Extravasation of blood from the cardiovascular system (heart, arteries, veins, capillaries).

Hemorrhagic

- See hematology.

Hemorrhagic Shock

- Loss of >20% blood volume (>1L in adult). Signs: tachycardia, hypotension, cool clammy skin, altered mental status, oliguria. Classification: Class I (<15%), Class II (15-30%), Class III (30-40%), Class IV (>40%). Management: control bleeding, IV fluids, blood transfusion (1:1:1 ratio for massive transfusion), vasopressors if needed.

Hemorrhagic Stroke

- A neurological deficit caused by rupture of a cerebral blood vessel, accounting for 13% of all strokes but 40% of stroke mortality. Intracerebral hemorrhage: hypertensive (basal ganglia, thalamus, pons,

cerebellum), amyloid angiopathy (lobar, elderly), vascular malformations, anticoagulation. Subarachnoid hemorrhage: aneurysm rupture.

Hemorrhoidectomy

- Surgical excision of symptomatic hemorrhoids. Indications: Grade III-IV internal hemorrhoids (prolapsed, irreducible), external hemorrhoids with uncontrolled bleeding or recurrent thrombosis, mixed hemorrhoids.

Hemorrhoids

- Dilated veins in the rectal and anal canal. Internal hemorrhoids are above the dentate line, innervated by visceral nerves, and present with painless bright red bleeding. External hemorrhoids are below the dentate line, innervated by somatic nerves, and may thrombose causing severe pain.

hemorrhoids (Piles)

- Vascular cushions in the anal canal composed of arteriovenous anastomoses (normal structures present from birth), which become symptomatic when they bleed, prolapse, or thrombose.

Hemosiderin

- A golden-brown, granular, iron-containing pigment derived from the degradation of hemoglobin and stored within macrophages as an insoluble complex. It forms when red blood cells are phagocytosed and degraded, typically in conditions of hemorrhage, hemolysis, or excessive red blood cell destruction.

hemostasis

- The physiological process that stops bleeding at the site of vascular injury.

Hemostat

- Clamping vessel to control bleeding. Types: Mosquito (small), Kelly (medium), Satinsky (vascular).

Hemostat (Perineal)

- See Hemostat.

Hemostat (Surgical)

- Instruments for surgery. See surgical instruments.

Hemostat (Vascular)

- See Hemostat.

Hemothorax

- Blood in pleural space. Traumatic or spontaneous. Symptoms: chest pain, dyspnea, decreased breath

sounds, dullness to percussion. Diagnosis: chest X-ray, ultrasound, CT. Treatment: tube thoracostomy, blood transfusion, surgery if ongoing bleeding.

Henoch-Schonlein Purpura

- IgA vasculitis: palpable purpura on buttocks/legs, abdominal pain, arthralgia, renal involvement (IgA nephropathy), most common in children. Self-limited in most cases; corticosteroids for severe GI/renal involvement.

Henoch-Schönlein purpura

- See IgA vasculitis. A small-vessel leukocytoclastic vasculitis characterised by palpable purpura predominantly on the lower extremities and buttocks, often accompanied by arthralgia, abdominal pain, and glomerulonephritis.

Heparin

- An anticoagulant that accelerates antithrombin-mediated inactivation of thrombin and factor Xa. Unfractionated heparin is monitored commonly with aPTT or anti-Xa activity; complications include bleeding, heparin-induced thrombocytopenia, and osteoporosis with prolonged exposure.

Heparin-Induced Thrombocytopenia

- An immune-mediated adverse drug reaction caused by IgG antibodies against platelet factor 4 (PF4)-heparin complexes, leading to platelet activation, thrombocytopenia, and a prothrombotic state. HIT typically occurs 5-14 days after heparin exposure (unfractionated heparin > low molecular weight heparin).

Hepatectomy

- Surgical removal of part or all of the liver for primary or metastatic tumors, requiring an adequate future liver remnant; portal vein embolization can hypertrophy the remnant before resection.

Hepatic Cyst

- Fluid-filled cavity in the liver most commonly congenital and asymptomatic. Simple hepatic cysts are lined by biliary epithelium and contain clear fluid. May cause pain or compressive symptoms if large or complicated by hemorrhage, infection, or rupture. Simple cysts require no treatment.

Hepatic Dose Adjustment

- Modification of medication selection or dosing because impaired liver function may reduce

metabolism, transport, protein synthesis, or biliary excretion. Standardized liver tests do not always predict drug clearance, so drug-specific guidance is required.

Hepatic Drug Metabolism

- Phase I reactions (oxidation, reduction, hydrolysis) primarily by CYP450 enzymes introduce or expose functional groups. Phase II reactions (conjugation) with glucuronide, sulfate, or acetyl groups increase water solubility for renal excretion. Hepatic metabolism can activate prodrugs, inactivate active drugs, or produce toxic

Hepatic Encephalopathy

- Neuropsychiatric syndrome occurring in advanced liver disease resulting from accumulation of neurotoxic substances, particularly ammonia. Ammonia crosses the blood-brain barrier and is metabolized by astrocytes into glutamine causing osmotic swelling and impaired neurotransmission.

Hepatic Hemangioma

- Most common benign liver tumor composed of dilated blood-filled vascular spaces lined by endothelial cells. Usually asymptomatic and found incidentally on imaging. Most are small and require no treatment.

Hepatic Hydrothorax

- Large pleural effusion occurring in patients with cirrhosis and portal hypertension in the absence of primary cardiac or pulmonary disease. Most commonly right-sided resulting from transit of ascitic fluid through small defects in the diaphragm into the pleural space driven by the pressure gradient between the abdomen and thorax.

Hepatic Ischemia

- Liver injury caused by inadequate hepatic blood flow or oxygen delivery, often during shock, severe hypoxemia, cardiac failure, or hepatic vascular occlusion. It can produce a rapid, marked rise in aminotransferases and may progress to acute liver failure.

Hepatic Portal System

- The venous network draining blood from the gastrointestinal tract, spleen, and pancreas to the liver before it reaches the systemic circulation. Formed by the union of the superior mesenteric and splenic

veins behind the pancreatic neck; receives inferior mesenteric, left gastric, and other tributaries.

Hepatic Surgery

- Liver resection: Couinaud segments. Indications: colorectal liver metastases (CRLM, up to 70% 5-year survival with R0), HCC (Child-Pugh A, portal pressure <10 mmHg, future liver remnant >20% normal, >40% cirrhotic), cholangiocarcinoma, benign tumors (adenoma >5cm, symptomatic hemangioma, FNH diagnostic uncertainty).

Hepatic Synthetic Dysfunction

- Impaired hepatic production of proteins and other molecules, including albumin and coagulation factors. It may cause hypoalbuminemia, edema, prolonged prothrombin time, and increased INR, although INR can also be altered by vitamin K deficiency, anticoagulants, and other conditions.

Hepatic Trauma

- Liver injury from trauma. CT staging guides management. Grades I-V based on injury severity.

Hepatic Vein

- Veins draining the liver into the inferior vena cava. The right, middle, and left hepatic veins emerge from the posterior surface of the liver at the bare area and drain into the IVC just below the diaphragm. They carry deoxygenated blood leaving the liver (after the hepatic portal system delivers blood for processing).

Hepatitis

- Inflammation of the liver caused by hepatotropic viruses. See Hepatitis (Viral Hepatitis) for complete definition.

Hepatitis (Viral Hepatitis)

- Inflammation of the liver caused by hepatotropic viruses. Hepatitis A (HAV): fecal-oral, acute self-limited, no chronic carrier state; vaccine-preventable. Hepatitis B (HBV): blood/sexual/perinatal transmission; acute and chronic infection; leading cause of cirrhosis and HCC.

Hepatitis A

- An acute viral infection of the liver caused by hepatitis A virus (HAV), transmitted via fecal-oral route (contaminated food or water). Incubation 15-50 days. Presents with fever, malaise, nausea, jaundice, and elevated transaminases. Self-limited;

does not cause chronic infection. Fulminant hepatitis is rare. Diagnosis: IgM anti-HAV.

Hepatitis A Virus

– Picornavirus transmitted through the fecal-oral route most commonly via contaminated food or water. The incubation period is fifteen to fifty days. Clinical presentation ranges from asymptomatic infection in young children to symptomatic acute hepatitis with jaundice, fatigue, nausea, elevated transaminases, dark urine, and pale stools.

Hepatitis A, B, C, D, E

– The five hepatotropic viruses that cause inflammation of the liver. See individual entries: Hepatitis A, Hepatitis B, Hepatitis C, Hepatitis D, Hepatitis E.

Hepatitis B

– A viral infection caused by hepatitis B virus (HBV), transmitted via blood, sexual contact, and perinatally. Can cause acute (self-limited or fulminant) and chronic infection (defined by HBsAg positivity >6 months). Chronic HBV leads to cirrhosis and hepatocellular carcinoma.

Hepatitis B Vaccine

– Recombinant vaccine given at birth, 1-2 months, and 6-18 months. Protects against hepatitis B virus, which can cause chronic liver disease, cirrhosis, and hepatocellular carcinoma. Also prevents perinatal transmission when given to infants of HBsAg-positive mothers within 12 hours of birth, plus hepatitis B immune globulin.

Hepatitis B Vaccine (Birth Dose)

– The first dose of the hepatitis B vaccine is recommended within 24 hours of birth for all medically stable infants. Administered IM in the vastus lateralis. Prevents perinatal transmission of HBV. For infants born to HBsAg-positive mothers: HBIG (0.5 mL IM) is given within 12 hours of birth along with the vaccine.

Hepatitis B Virus

– Hepadnavirus transmitted through percutaneous exposure, sexual contact, and perinatally causing acute and potentially chronic hepatitis. The virus has a partially double-stranded circular DNA genome.

Hepatitis C

– A viral infection caused by hepatitis C virus (HCV), primarily transmitted through blood exposure (IV drug use, blood transfusion before 1992). Acute infection is often asymptomatic; 50-80% progress to chronic hepatitis. Chronic HCV can lead to cirrhosis and hepatocellular carcinoma over decades. Diagnosis: anti-HCV antibody, HCV RNA.

Hepatitis C Virus

– Flavivirus transmitted primarily through percutaneous exposure with injection drug use being the most common route. Approximately seventy to eighty percent of acute infections progress to chronic hepatitis C, the leading cause of cirrhosis, hepatocellular carcinoma, and liver transplantation in many Western countries.

Hepatitis D Virus

– Defective RNA virus requiring co-infection with hepatitis B to replicate, using hepatitis B surface antigen for its envelope. Hepatitis D causes more severe liver disease than hepatitis B alone with accelerated progression to cirrhosis and higher risk of hepatocellular carcinoma.

Hepatitis E Virus

– Non-enveloped RNA virus transmitted through the fecal-oral route most commonly via contaminated drinking water. It is the most common cause of acute viral hepatitis worldwide. Clinical presentation is similar to hepatitis A with jaundice and elevated transaminases.

Hepatitis Serology

– Panel of serologic markers for hepatitis A, B, C, and E. HAV: anti-HAV IgM (acute), anti-HAV IgG (past infection/vaccination). HBV: HBsAg (active infection), anti-HBs (immunity/vaccination), anti-HBc IgM (acute), anti-HBc IgG (past), HBeAg (high replication), anti-HBe (low replication). HBV DNA: quantifies viral load.

Hepatitis Viruses

– Hepatitis A: fecal-oral, acute self-limited, vaccine (inactivated). Hepatitis B: blood/sexual/perinatal. Acute: HBsAg, IgM anti-HBc. Chronic: HBsAg >6 months, HBV DNA, HBeAg. Treatment: entecavir, tenofovir (TAF/TDF) - suppress DNA, prevent cirrhosis/HCC. HDV: requires HBV, pegylated

interferon. Hepatitis C: blood-borne. Acute: HCV RNA.

Hepatobiliary Scintigraphy

- Liver/gallbladder imaging using Tc-99m mebrofenin or disofenin. Assess biliary patency, gallbladder function, bile leak, acute cholecystitis (non-visualization of gallbladder).

Hepatoblastoma

- A malignant embryonal liver tumor occurring almost exclusively in infants and young children, associated with elevated AFP and prematurity.

Hepatocellular Carcinoma (HCC)

- Primary liver cancer arising from hepatocytes. Risk factors: cirrhosis, chronic hepatitis B/C, alcohol, and metabolic dysfunction-associated steatotic liver disease. Surveillance with ultrasound and AFP; treatment includes resection, transplantation, ablation, TACE, and systemic therapy.

Hepatocellular Jaundice

- Jaundice caused by dysfunction of hepatocytes, impairing the liver's ability to take up, conjugate, or excrete bilirubin. See Jaundice for complete overview. Causes: viral hepatitis (A, B, C, D, E), alcoholic hepatitis, drug-induced liver injury (paracetamol, isoniazid), cirrhosis, autoimmune hepatitis, Wilson disease, haemochromatosis.

Hepatojugular Reflux

- Sustained pressure applied over the liver for 15-30 seconds causes a sustained rise in JVP of more than 3 cm above baseline. It indicates right heart failure or fluid overload, reflecting the inability of the right ventricle to accommodate increased venous return.

Hepatoma

- See Hepatocellular Carcinoma.

Hepatomegaly

- Enlarged liver detected on imaging. Causes: fatty liver, hepatitis, congestion, infiltration, malignancy.

Hepatopulmonary Syndrome

- Triad of liver disease, intrapulmonary vascular dilatation, and impaired oxygenation causing hypoxemia. Presents with platypnea which is dyspnea worsening in the upright position and orthodeoxia which is oxygen desaturation in the upright position.

Hepatorenal Syndrome

- Functional renal failure in advanced liver disease characterized by intense renal vasoconstriction in the setting of splanchnic vasodilation. Type 1 hepatorenal syndrome is rapidly progressive with median survival of approximately two weeks without treatment. Type 2 is slowly progressive associated with refractory ascites.

Hepatosplenomegaly

- Enlargement of both the liver and spleen. It may result from infection, portal hypertension, storage disease, malignancy, haemolysis, or systemic inflammatory and infiltrative disorders; evaluation depends on associated symptoms and laboratory and imaging findings.

Hepatotoxicity

- Liver damage from toxins, drugs, chemicals. Acetaminophen most common cause of acute liver failure. Monitoring: LFTs, INR, clinical assessment.

HER2-Positive

- Cancer overexpressing HER2 protein. Treated with trastuzumab.

Herd Immunity

- Resistance of population to invasion by infectious agent when sufficient proportion immune. Threshold varies by disease (measles 95%, polio 80%).

Hereditary Angioedema

- Recurrent episodes of nonpitting swelling of the skin, gastrointestinal tract, or upper airway caused most often by deficiency or dysfunction of C1 inhibitor. Attacks can cause abdominal pain, vomiting, airway obstruction, or life-threatening laryngeal edema without typical urticaria.

Hereditary Cancer

- Cancer caused by inherited genetic mutations. Genetic counseling recommended.

Hereditary Hemorrhagic Telangiectasia

- An autosomal dominant vascular disorder (also called Osler-Weber-Rendu syndrome) characterized by telangiectasias on skin and mucous membranes, pulmonary AVMs, hepatic AVMs, and cerebral AVMs. Pulmonary AVMs can cause hypoxemia and paradoxical emboli. Treatment is embolization of pulmonary AVMs.

Hereditary Hemorrhagic Telangiectasia

(Osler-Weber-Rendu Disease)

- An autosomal dominant vascular dysplasia characterized by malformed blood vessels lacking smooth muscle layers. Mutations in ENG (endoglin) or ACVRL1 (ALK1) genes.

Hereditary Spherocytosis

- The most common inherited hemolytic anemia in Northern Europeans (autosomal dominant 75%, recessive 25%). Caused by defects in RBC membrane proteins (spectrin, ankyrin, band 3, protein 4.2), leading to loss of membrane surface area, reduced deformability, and premature destruction in the spleen.

Hermansky-Pudlak Syndrome

- Autosomal recessive (10+ genes: HPS1-10, BLOC/AP-3 complexes). Oculocutaneous albinism, bleeding disorder (absent dense bodies in platelets -> prolonged bleeding time), pulmonary fibrosis (HPS-1, HPS-4), granulomatous colitis, neutropenia. Diagnosis: electron microscopy of platelets (absent dense bodies), genetic testing.

Hernia

- The protrusion of an organ or tissue through an abnormal opening in the wall of the cavity that normally contains it. Most hernias occur in the abdominal wall.

Hernia Mesh

- Synthetic or biologic material for hernia repair. Polypropylene most common. Complications: infection, recurrence, chronic pain, mesh migration.

Hernia Repair (Herniorrhaphy)

- Surgical repair of a hernia (inguinal, femoral, umbilical, incisional, hiatal). Inguinal hernia repair: open (Lichtenstein tension-free mesh, Shouldice, Bassini) or laparoscopic (TEP: totally extraperitoneal, TAPP: transabdominal preperitoneal). Mesh reinforcement reduces recurrence (1-3% for mesh vs. 5-15% for primary tissue repair).

Herniated Disc

- Protrusion of the nucleus pulposus through a tear in the annulus fibrosus of the intervertebral disc, causing compression or inflammation of adjacent nerve roots. Most common at L4-L5 and L5-S1 (lumbar) and C5-C6, C6-C7 (cervical). Lumbar: sciatica (radiating leg pain), paresthesias, sensory

loss, and motor weakness (foot drop).

Herniated disk

- Displacement of nucleus pulposus material through a tear in the annulus fibrosus of an intervertebral disc, most commonly occurring in the lumbar spine (L4-L5 and L5-S1 account for 95 percent of lumbar herniations) and cervical spine (C5-C6 and C6-C7), potentially compressing spinal nerves.

Herpangina

- An acute viral infection, usually caused by enteroviruses, that produces fever, sore throat, and small vesicles or ulcers on the soft palate and posterior pharynx. It is most common in children and spreads through respiratory and fecal-oral secretions.

Herpes

- See Herpes Simplex (Cold Sores / Genital Herpes).

Herpes gestationis

- An autoimmune blistering skin disease of pregnancy mediated by anti-basement membrane zone antibodies, presenting with pruritic urticarial plaques and vesicles, typically in the second or third trimester.

Herpes Simplex

- See Herpes Simplex (Cold Sores / Genital Herpes).

Herpes Simplex (Cold Sores / Genital Herpes)

- A viral infection caused by herpes simplex virus type 1 (HSV-1, typically orofacial, though increasingly anogenital) and type 2 (HSV-2, predominantly anogenital). HSV is a double-stranded DNA virus of the Herpesviridae family with latency in sensory ganglia and periodic reactivation.

Herpes Simplex Encephalitis

- The most common cause of sporadic fatal encephalitis, caused by herpes simplex virus type 1 (HSV-1) in over ninety percent of cases. The virus typically infects the temporal lobes, orbitofrontal cortex, and insular cortex, causing hemorrhagic necrosis.

Herpes Simplex Skin Infection

- Infection of skin or mucosa by herpes simplex virus, producing grouped vesicles or erosions on an erythematous base. Recurrent disease commonly affects the oral or genital region; widespread infection in patients with eczema is eczema herpeticum and requires urgent medical assessment.

Herpes simplex virus

- A double-stranded DNA virus belonging to the Herpesviridae family that causes oral and genital herpes. HSV has two serotypes: HSV-1, which typically causes oral herpes, and HSV-2, which typically causes genital herpes. The virus establishes latent infections in sensory ganglia and can reactivate periodically.

Herpes simplex virus infection

- See Herpes Simplex (Cold Sores / Genital Herpes).

Herpes vegetans

- A rare, chronic, verrucous or vegetative form of herpes simplex virus infection occurring in immunocompromised patients, particularly those with HIV/AIDS. It presents as exophytic, proliferative, ulcerative plaques on the genital or perianal skin, often refractory to standard antiviral therapy.

Herpes Zoster

- A painful, vesicular eruption resulting from reactivation of latent varicella-zoster virus (VZV, human herpesvirus 3) in the dorsal root ganglia or cranial nerve ganglia, where the virus has remained dormant since the primary varicella (chickenpox) infection, with an incidence of approximately 1 to 3 per 1,000 person-years overall and up

Herpes Zoster (Shingles)

- A painful, vesicular rash caused by reactivation of latent varicella-zoster virus (VZV, human herpesvirus 3) in the dorsal root ganglia.

Herpes zoster ophthalmicus

- Reactivation of varicella-zoster virus latency in the trigeminal ganglion, with viral spread along the ophthalmic division (V1) causing painful vesicular skin eruption on the forehead, eyelid, and nose, with potential ocular involvement including keratitis, uveitis, and secondary glaucoma.

Herpetic glossitis

- Inflammation of the tongue caused by herpes simplex virus infection, presenting with painful vesicles and ulceration on the tongue surface, often recurring in immunocompromised patients.

Herpetic Whitlow

- A painful viral infection of the finger caused by the herpes simplex virus (typically HSV-1), characterized by swelling, redness, and painful vesicular

lesions on the finger.

Hesitation Wounds

- Multiple superficial, tentative parallel incised cuts surrounding a deeper primary fatal cut wound, classically observed in suicides involving sharp instruments (wrists, neck).

heteroplasmy

- Inheritance through mitochondrial DNA (mtDNA), passed only from mother. See mitochondrial inheritance.

Heterotopic Pregnancy

- Simultaneous intrauterine and ectopic pregnancy. Rare in spontaneous conception (1:30,000) but more common with ART (1:100-1:500). Risk factors: ovulation induction, IVF with multiple embryo transfer. The presence of an intrauterine pregnancy does not exclude an ectopic pregnancy.

Heterozygous

- Having two different alleles of a particular gene (Aa). The dominant allele is typically expressed in the phenotype.

Hexokinase

- Phosphorylates glucose to glucose-6-phosphate. Low K_m , inhibited by product. Glucokinase (liver) has high K_m , not inhibited.

Hiatal Hernia

- Protrusion of part of the stomach through the esophageal hiatus into the thoracic cavity. Type I (sliding, 95%): gastroesophageal junction above the diaphragm. Types II-IV (paraesophageal): fundus or other portions herniate alongside the esophagus. Sliding hernias are associated with GERD.

Hiatus Hernia

- The protrusion of part of the stomach through the esophageal hiatus of the diaphragm into the chest cavity. Types: (1) Sliding hernia (type I, 95%): the gastro-esophageal junction (GOJ) and a portion of the gastric cardia slide above the diaphragm through the hiatus.

Hib

- *Haemophilus influenzae* type b: an encapsulated gram-negative bacterium causing serious invasive disease in unvaccinated children—meningitis, epiglottitis, pneumonia, septic arthritis, cellulitis, and sepsis. Highly prevented by the Hib conjugate

vaccine (given in infancy, part of routine schedule).

Hiccough (Hiccup)

- An involuntary, repetitive, spasmodic contraction of the diaphragm followed by sudden closure of the glottis, producing the characteristic 'hic' sound. Hiccups are caused by irritation or stimulation of the phrenic nerve (C3-C5, innervates the diaphragm) or the vagus nerve (CN X).

Hiccups

- Recurrent involuntary spasms of the diaphragm followed by sudden closure of the glottis, producing a characteristic sound. Persistent or intractable hiccups may reflect irritation of the diaphragm or phrenic and vagal pathways, metabolic disease, medication exposure, or central nervous system disease.

Hiccups (Singultus)

- Involuntary, spasmodic contractions of the diaphragm and intercostal muscles followed by sudden glottic closure, producing the characteristic "hic" sound. The reflex arc involves the phrenic nerve (C3-C5), vagus nerve, and brainstem centers.

Hidradenitis

- Hidradenitis suppurativa is a chronic inflammatory follicular occlusion disease affecting apocrine gland-bearing areas (axillae, groin, inframammary). It presents with recurrent painful nodules, abscesses, sinus tracts, and scarring.

Hidradenitis Suppurativa

- A chronic, inflammatory, follicular occlusion disease characterized by recurrent, painful nodules, abscesses, sinus tracts, and scarring in apocrine gland-bearing areas (axillae [most common, 70 to 80 percent], groin and inguinal folds [30 to 40 percent], perineum and perianal region [20 to 35 percent], buttocks [20 to 30 percent], and in

High Arch

- An abnormally elevated medial longitudinal arch of the foot, also called pes cavus. It may be idiopathic or associated with neuromuscular disease and can cause overload of the heel and forefoot, calluses, pain, instability, or difficulty fitting footwear.

High-Dependency Unit

- A ward or area within a hospital that provides a level of care between a general ward and an intensive care unit, for patients who require intensive obser-

vation or treatment but not mechanical ventilation.

High-Flow Nasal Cannula

- A device delivering heated and humidified oxygen at flow rates up to 60 L/min through a nasal cannula. It provides a modest amount of PEEP, reduces dead space ventilation, and improves patient comfort compared to non-invasive ventilation. It is used for acute hypoxemic respiratory failure and post-extubation support.

High-Intensity Focused Ultrasound (HIFU)

- Ultrasound ablation using focused energy. Used for uterine fibroids, prostate cancer, bone metastases.

High-Level Disinfection

- Disinfection that inactivates all microorganisms except high numbers of bacterial spores. It is commonly used for heat-sensitive semicritical equipment such as some flexible endoscopes after meticulous cleaning.

Hill-Sachs Lesion

- Impaction fracture of posterolateral humeral head from anterior shoulder dislocation. Associated with Bankart lesion.

Hip Dislocation

- Loss of normal congruence between the femoral head and acetabulum. It is usually traumatic in adults and may damage the sciatic nerve or femoral-head blood supply; congenital or developmental dislocation occurs in developmental dysplasia of the hip.

Hip Disorders

- The hip joints are ball-and-socket joints connecting the femur to the pelvis, providing stability and a wide range of motion for walking, running, and weight-bearing activities. Common hip conditions include osteoarthritis, fractures, bursitis, and labral tears.

Hip Dysplasia

- Developmental dysplasia of the hip (DDH): abnormal development of the acetabulum/femoral head causing instability, subluxation, or dislocation. Risk factors: female, breech presentation, family history, firstborn, and tight swaddling.

Hip Fracture

- Fracture of proximal femur, usually femoral neck

or intertrochanteric region. Mechanism: low-energy fall in osteoporotic bone. Symptoms: pain, inability to bear weight, shortened/externally rotated leg. Diagnosis: X-ray, MRI if X-ray negative. Treatment: surgical fixation (internal fixation, arthroplasty).

Hip Fractures

- Common in elderly (osteoporosis, falls), classified by location: femoral neck (intracapsular), intertrochanteric (extracapsular), subtrochanteric. Garden classification for femoral neck: I (incomplete/impacted), II (complete nondisplaced), III (complete partially displaced), IV (complete displaced).

Hip Joint

- Multiaxial ball-and-socket synovial joint between femoral head and acetabulum. Designed for stability and mobility, bearing entire upper body weight. Acetabular labrum deepens the socket; strong ligaments (iliofemoral, pubofemoral, ischiofemoral) reinforce it. Ligamentum teres contains the femoral head artery.

Hip Prosthesis

- Total hip replacement components: femoral stem (metal), head (metal/ceramic), acetabular cup (polyethylene/metal/ceramic). Fixation: cemented (older patients), cementless (younger, porous coating for bone ingrowth). Materials: cobalt-chrome, titanium, ceramics.

Hip Replacement

- A surgical procedure in which the diseased hip joint is replaced with a prosthetic implant (arthroplasty). Total hip replacement (THR) replaces both the femoral head and the acetabulum.

Hip-Fracture Rehabilitation

- Multidisciplinary postoperative management aimed at restoring mobility, preventing complications, and returning the older person to the highest feasible level of independence. It includes early mobilization, pain control, osteoporosis treatment, delirium prevention, nutrition, and discharge planning.

Hippocampus

- Curved medial temporal lobe structure, part of the limbic system, critical for new declarative memory formation (episodic and spatial) and short-to-long-term memory conversion. Has dentate gyrus,

hippocampus proper (CA1-CA4), subiculum, and entorhinal cortex. One of few adult neurogenesis sites (dentate gyrus).

Hippotherapy

- Therapeutic horseback riding for neurological, musculoskeletal conditions. Improves balance, posture, coordination.

Hirschsprung Disease

- Congenital condition characterized by the absence of enteric ganglion cells in a segment of the distal colon, resulting from failure of neural crest cell migration during embryonic development.

Hirschsprung's Disease

- A congenital disorder characterized by the absence of parasympathetic ganglion cells (aganglionosis) in the distal colon, resulting in failure of relaxation and functional obstruction. Incidence: 1 in 5,000 live births, male predominance (4:1).

Hirsutism

- Excess terminal hair growth in women in a male-pattern distribution (face—upper lip, chin, sideburns; chest, back, lower abdomen, inner thighs). Hirsutism affects 5-10% of women of reproductive age. Pathophysiology: increased androgen production or increased end-organ sensitivity.

Histamine

- A biogenic amine (imidazoleethylamine) synthesised from L-histidine by histidine decarboxylase, stored in basophils, mast cells, and platelets, and released during allergic and inflammatory responses.

Histiocytosis

- A group of rare disorders of abnormal histiocyte (macrophage/dendritic cell) proliferation, including Langerhans cell histiocytosis, Erdheim-Chester disease, and hemophagocytic lymphohistiocytosis.

Histology

- The microscopic study of the structure of tissues and cells. Histology involves preparing thin tissue sections (3-10 micrometres) by fixation (formalin, paraffin embedding), sectioning (microtome), staining (haematoxylin and eosin, H&E), and examination under a light or electron microscope.

Histopathology

- The laboratory-based diagnosis of disease through microscopic examination of tissue sections from

biopsy, excision, or autopsy, built on fixation, paraffin embedding, sectioning, and hematoxylin and eosin staining.

Histoplasma capsulatum

- See Histoplasmosis.

Histoplasmosis

- A systemic fungal infection caused by *Histoplasma capsulatum*, a dimorphic fungus found in nitrogen-rich soil enriched with bat or bird guano, most commonly in the Ohio and Mississippi River valleys of the United States, Central and South America, and parts of Africa and Asia.

Histrionic Personality Disorder

- A pattern of excessive emotionality and attention-seeking.

HIV

- Human immunodeficiency virus: a retrovirus (HIV-1, HIV-2) causing progressive CD4+ T-cell depletion and AIDS if untreated. Transmitted via sexual contact, blood/blood products, mother-to-child.

HIV (Human Immunodeficiency Virus)

- A retrovirus (family Retroviridae, genus Lentivirus) that causes acquired immunodeficiency syndrome (AIDS) by infecting and destroying CD4+ T lymphocytes (helper T cells). Two types: HIV-1 (pandemic, most common worldwide) and HIV-2 (less virulent, West Africa).

HIV infection

- See HIV.

HIV Test

- Screening and confirmatory testing for HIV. Fourth-generation HIV antigen/antibody combo test (p24 antigen + HIV-1/2 antibodies): window period 2-4 weeks. Positive screening requires confirmatory HIV-1/2 antibody differentiation assay. Rapid point-of-care tests (fingerstick, oral swab): results in 20 minutes.

HIV/AIDS

- HIV-1/2: retrovirus, CD4+ T cell depletion. Transmission: sexual, blood, perinatal, breast milk. Acute retroviral syndrome: fever, lymphadenopathy, pharyngitis, rash, high viral load. Clinical latency: CD4 decline over years. AIDS: CD4 <200 or AIDS-defining illness (PJP, toxo, CMV, MAC, crypto, KS, lymphoma, TB).

HIV/AIDS Immunopathogenesis

- HIV-1 infects CD4+ T cells, macrophages, dendritic cells via gp120 binding to CD4 + CCR5 (R5-tropic, early) or CXCR4 (X4-tropic, late). Reverse transcription -> integration -> transcription/translation -> budding. Acute retroviral syndrome (2-4 weeks): fever, lymphadenopathy, pharyngitis, rash, high viral load.

HIV/AIDS Nutrition

- Malnutrition due to increased need/requirement, malabsorption, side effects. Supplementation, opportunistic infection management.

Hives

- Transient, itchy, raised wheals caused by localized dermal edema. Urticaria may be acute or chronic and can result from infection, foods, medications, physical triggers, or autoimmune mechanisms; swelling of the airway is an emergency.

Hives (Urticaria)

- A common skin condition characterized by transient, well-circumscribed, intensely pruritic wheals that blanch with pressure and resolve within 24 hours without scarring, caused by mast-cell mediator release in the dermis. Acute urticaria lasts less than 6 weeks; chronic urticaria recurs for 6 weeks or longer and is often spontaneous.

HLA-B27

- Human leukocyte antigen B27, a class I MHC molecule strongly associated with spondyloarthropathies. Present in 90% of ankylosing spondylitis patients, 60-80% of reactive arthritis, and associated with acute anterior uveitis, psoriatic arthritis, and IBD-associated arthritis. Not diagnostic alone—present in 8% of healthy populations.

HMG-CoA Reductase

- Rate-limiting enzyme of cholesterol synthesis. Target of statin drugs.

Hoarding Disorder

- Persistent difficulty discarding or parting with possessions, regardless of their actual value. The individual perceives a need to save items and experiences distress at the thought of discarding them. This results in accumulation of possessions that congest and clutter active living areas.

Hoarseness

- A subjective change in voice quality characterized by roughness, breathiness, strain, or altered pitch; a subset of dysphonia rather than a diagnosis. Common causes include acute viral laryngitis, voice overuse, smoking, reflux, and vocal cord lesions (nodules, polyps, papillomas, laryngeal carcinoma).

Hodgkin Lymphoma

- A B-cell lymphoma characterized by the presence of Reed-Sternberg cells (large, binucleate or multinucleate cells with owl-eye nucleoli) in a background of reactive inflammatory cells.

Hodgkin's Lymphoma

- A B-cell lymphoma characterized by the presence of Reed-Sternberg cells (large, binucleate or multinucleate cells with owl-eye nucleoli) in a background of reactive inflammatory cells. See Hodgkin Lymphoma.

Holoprosencephaly

- A structural brain malformation resulting from incomplete or failed division of the forebrain (prosencephalon) into the cerebral hemispheres during the fifth week of gestation.

Holosystolic Murmur Differential

- Pansystolic (holosystolic) murmurs are plateau-shaped and extend from S1 to S2.

Holt-Oram Syndrome

- An autosomal dominant disorder caused by TBX5 gene mutations, characterized by congenital heart disease and upper limb malformations. Cardiac anomalies: atrial septal defect (most common), VSD, and conduction abnormalities (heart block, atrial fibrillation).

Holter Monitor

- Continuous ECG recording for 24-48 hours to detect arrhythmias, quantify burden (atrial fibrillation, PVCs, PACs), and correlate symptoms with rhythm. Patient diary logs symptoms. Indications: palpitations, syncope, presyncope, and assessment of arrhythmia control. Reveals paroxysmal arrhythmias not captured on standard 12-lead ECG.

Holter Monitoring

- Continuous ECG recording for 24-48 hours. Detects arrhythmias, ischemia.

Home Care

- Medical care provided at home. Includes nursing, therapy, personal care.

Home Health

- Medical care provided at home. See home care.

Home Health Rehabilitation

- Therapy provided at home. For patients unable to travel. Home exercise program emphasized.

Home Rehabilitation

- Therapy provided at home. See home health rehabilitation.

Homeostasis

- The maintenance of stable internal conditions despite external changes. Key regulated variables: temperature, pH, glucose, electrolytes, blood pressure, fluid balance. Mechanisms: negative feedback loops (most common), positive feedback (rare, e.g., labor, blood clotting).

Homocystinuria

- An autosomal recessive inborn error of methionine metabolism caused most commonly by deficiency of cystathionine beta-synthase (CBS), leading to accumulation of homocysteine and methionine.

Homozygous

- Having two identical alleles of a particular gene (AA or aa). Can be homozygous dominant or homozygous recessive.

Hookworm

- Intestinal nematode infection by *Ancylostoma duodenale* or *Necator americanus*, acquired by skin penetration of filariform larvae (walking barefoot on contaminated soil).

Hookworm Infection

- Intestinal infection caused by hookworms acquired when larvae in contaminated soil penetrate the skin or are ingested. Chronic infection can cause abdominal symptoms, iron-deficiency anemia, protein loss, and impaired growth in children.

Hookworm infestation

- See Hookworm.

Hookworms

- Nematodes including *Ancylostoma duodenale* and *Necator americanus* that cause hookworm infection. Infection occurs through skin penetration by filariform larvae.

iform larvae, typically when walking barefoot on contaminated soil. Adult worms attach to the intestinal mucosa and feed on blood, causing iron deficiency anemia.

Hordeolum (Stye)

- A common infection of the eyelid. See Stye.

Horizontal gene transfer

- The transfer of genetic material between bacteria through mechanisms other than parent-to-offspring transmission. The three mechanisms are transformation (uptake of free DNA), transduction (transfer via bacteriophages), and conjugation (direct cell-to-cell transfer).

Hormone

- Chemical messenger secreted by endocrine glands into blood, acting on distant target organs with specific receptors. Classes: peptides/proteins (insulin, GH), steroids (cortisol, estrogen), amines (thyroid hormone, catecholamines). Mechanisms: cell surface receptors (GPCRs, RTKs) or intracellular receptors (nuclear receptors).

Hormone Replacement Therapy

- HRT (menopause hormone therapy): estrogen $\pm$ progestin to relieve menopausal symptoms (vasomotor, vaginal atrophy, sleep, mood) and prevent osteoporosis. Regimens: estrogen-only (post-hysterectomy) or combined (with progestin for endometrial protection in uterus-intact women); routes: oral, transdermal, vaginal.

Hormone Replacement Therapy (HRT)

- The administration of exogenous estrogen (with or without progestogen) to relieve symptoms of estrogen deficiency and prevent long-term health consequences in postmenopausal women.

Hormone Therapy

- Treatment blocking hormone effects in hormone-sensitive cancers. Examples: tamoxifen (breast cancer), leuprolide (prostate cancer), aromatase inhibitors.

Hormone-Receptor Positive

- Cancer expressing hormone receptors (ER+, PR+, AR+). Treated with hormone therapy.

Hormone-Sensitive Lipase

- Breaks down triglycerides in adipose tissue. Activated by glucagon, epinephrine via cAMP.

Horner Syndrome

- A clinical triad caused by disruption of the sympathetic pathway to the eye and face, resulting in ipsilateral miosis (constricted pupil), ptosis (drooping of the upper eyelid due to paralysis of the superior tarsal muscle), and anhidrosis (loss of sweating on the affected side of the face).

Horseshoe Kidney

- The most common renal fusion anomaly (1 in 400-500), where the lower poles of both kidneys are fused across the midline by an isthmus of renal parenchyma or fibrous tissue.

Hospice

- Palliative care for terminal illness with prognosis <6 months. Focus on comfort, quality of life, support for patient and family. Provided at home, hospice facility, or hospital. Interdisciplinary team: physician, nurse, social worker, chaplain, volunteer.

Hospice Rehabilitation

- Rehabilitation in hospice setting. Symptom management, positioning, caregiver education.

Hospital-Acquired Infection

- Infection acquired during hospitalization. Also: healthcare-associated infection (HAI). Examples: CLABSI, CAUTI, surgical site infection.

Hospital-acquired infections

- Infections that develop during or after hospitalization. Major types include central line-associated bloodstream infections, catheter-associated urinary tract infections, ventilator-associated pneumonia, and surgical site infections. Risk factors include invasive devices, prolonged hospitalisation, and antimicrobial use.

Hospital-Acquired Pneumonia

- Pneumonia developing > 48 hours after hospital admission. Common pathogens include *Pseudomonas* and MRSA. Empiric treatment covers resistant organisms.

Hospitalization (Geriatric)

- Inpatient stay for elderly patients. High risk for adverse outcomes: delirium, functional decline, institutionalization. Precautions: fall prevention, early mobilization, medication reconciliation, delirium prevention, nutrition support. Geriatrician consultation beneficial.

Host

- A person or other living organism that can be infected by a virus or other pathogen under natural conditions. The host may be a human, animal, or plant. In medical contexts, the host is the patient in whom an infection or disease develops.

Host Response

- Immunological reaction to implant. Foreign body response, encapsulation, rejection.

Hot Flashes

- Vasomotor symptoms: sudden sensations of intense heat, sweating, flushing, and palpitations, most commonly from estrogen withdrawal around menopause; also in hypogonadism, chemotherapy-induced ovarian failure, and some medications (SERMs, GnRH agonists). Mechanism: hypothalamic thermoregulatory dysfunction.

Hounsfield Units (HU)

- CT number scale referencing water (0 HU) and air (-1000 HU). Fat: -50 to -100 HU. Water: 0 HU. Soft tissue: +20 to +50 HU. Bone: +400 to +3000 HU. Blood: +40 to +60 HU (acute), +20 to +40 HU (chronic).

Howell-Jolly Bodies

- Basophilic nuclear remnants (clusters of DNA) normally removed from circulating red blood cells by splenic macrophages (pitting). Presence on peripheral blood smear indicates functional asplenia or anatomical splenectomy (e.g., trauma, sickle cell auto-splenectomy). Also seen in severe megaloblastic anemia.

HPLC

- High-performance liquid chromatography for radiochemical purity analysis.

HPV

- Human papillomavirus: a DNA virus with >100 types causing epithelial infections. Low-risk types (6, 11) cause genital warts (condyloma acuminata) and laryngeal papillomas; high-risk types (16, 18—70 percent of cervical cancer; 31, 33, 45, etc.) cause cervical, anal, penile, vulvar, vaginal, oropharyngeal cancers and precancerous

HPV Vaccine

- Human Papillomavirus vaccine given at 11-12 years (can start at 9), with series completion by 15 years.

Two-dose series if started before 15; three-dose series if started at 15 or older. Protects against HPV types causing cervical cancer, oropharyngeal cancer, and genital warts. Gardasil 9 covers nine HPV types.

HRT

- Hormone Replacement Therapy, the use of estrogen (with or without progesterone) to treat menopausal symptoms caused by declining ovarian hormone production. HRT is the most effective treatment for vasomotor symptoms (hot flashes, night sweats), prevents bone loss (osteoporosis), and improves urogenital atrophy.

Human Chorionic Gonadotropin

- A glycoprotein hormone produced by trophoblast cells after implantation. Detectable in serum 8-11 days post-conception and in urine 12-14 days. Levels double approximately every 48-72 hours in early pregnancy, peaking at 8-11 weeks then plateauing.

Human immunodeficiency virus

- A retrovirus belonging to the Lentivirus genus that causes acquired immunodeficiency syndrome (AIDS). HIV has two species: HIV-1 and HIV-2. The virus targets CD4+ T cells, macrophages, and dendritic cells. HIV has a complex genome encoding structural proteins, enzymes (reverse transcriptase, integrase, protease), and regulatory proteins.

Human microbiome

- The collection of microorganisms that live on and in the human body. The microbiome includes bacteria, viruses, fungi, and archaea. The gut microbiome is the most studied and plays a role in digestion, immune function, and disease.

Human Papillomavirus

- The most common STI, with over 100 types. High-risk types (16, 18) cause cervical, oropharyngeal, anal, and genital cancers. Low-risk types (6, 11) cause genital warts. Prevention: HPV vaccine (Gardasil 9) protects against 9 HPV types. Screening: Pap smear and HPV testing for cervical cancer prevention.

Human Placental Lactogen

- A peptide hormone produced by the placenta that contributes to mammary-gland development and shifts maternal metabolism toward increased lipolysis and reduced insulin sensitivity during pregnancy.

It supports nutrient availability for the fetus and contributes to physiologic gestational insulin resistance.

Humeroradial Joint

- Between capitulum and radial head, allowing flexion-extension and contributing to pronation-supination via radial head rotation. Radial collateral ligament provides lateral stability. Capitulum has articular cartilage only anteriorly.

Humeroulnar Joint

- Uniaxial hinge between trochlea and trochlear notch of the ulna, allowing only flexion/extension (0 to approximately 145 degrees). Ulnar collateral ligament is the primary valgus restraint. Trochlear notch provides inherent bony stability, making it the most congruent elbow joint.

Humerus

- Single upper arm bone, the longest of the upper limb. Proximal end: head, anatomical/surgical necks, tubercles. Shaft: deltoid tuberosity, radial groove, epicondyles. Distal end: trochlea, capitulum, olecranon fossa. The surgical neck is the common fracture site potentially injuring the axillary nerve.

Humoral Immunity

- The component of adaptive immunity mediated by antibodies produced by B lymphocytes (plasma cells) that neutralize extracellular pathogens and toxins. Humoral responses defend against bacteria, viruses outside cells, and parasites.

Hunger

- Physiological drive to eat. Regulated by ghrelin, blood glucose, stomach distension. Psychological factors also influence.

Hungry Bone Syndrome

- Severe hypocalcemia occurring after parathyroidectomy particularly in patients with severe preoperative hyperparathyroidism and high bone turnover. Rapid bone remineralization after removal of excess PTH sequesters calcium and phosphorus from the serum causing symptomatic hypocalcemia.

Hunter Syndrome

- Mucopolysaccharidosis type II (MPS II), an X-linked recessive lysosomal storage disorder caused by iduronate-2-sulfatase deficiency (IDS gene mu-

tations). Accumulation of dermatan sulfate and heparan sulfate (same as Hurler).

Huntington Disease

- Autosomal dominant neurodegenerative disorder caused by CAG trinucleotide repeat expansion in HTT gene. Onset: 30-50 years. Symptoms: chorea, dementia, psychiatric disease. Anticipation: earlier onset in successive generations (paternal transmission). No cure; symptomatic treatment.

Huntington's Disease

- An autosomal dominant neurodegenerative disorder caused by a CAG trinucleotide repeat expansion in the huntingtin gene on chromosome 4p16.3. The disease is characterized by the triad of progressive chorea, cognitive decline, and psychiatric manifestations.

Huntington's Disease (Chorea)

- An autosomal dominant neurodegenerative disorder caused by a CAG trinucleotide repeat expansion (>36 repeats, normally <26) in the HTT gene on chromosome 4 (4p16.3), which encodes the huntingtin protein.

Hurler Syndrome

- Mucopolysaccharidosis type I (MPS I), the severe form, an autosomal recessive lysosomal storage disorder caused by alpha-L-iduronidase deficiency (IDUA gene mutations). Accumulation of dermatan sulfate and heparan sulfate leads to progressive multisystem involvement.

Hyaline Cartilage

- The most common type of cartilage, containing fine type II collagen fibers in a firm, glassy matrix. It supports airways, covers articular surfaces, and forms much of the fetal skeleton and growth plates.

Hyaline Casts

- Transparent colorless cylindrical structures composed primarily of Tamm-Horsfall protein secreted by the thick ascending limb. Most common type of urine cast. Normal in concentrated urine and after exercise. Increased in dehydration, prerenal states, and volume depletion.

Hyaline Membrane

- See pediatrics.

Hydatid cyst

- A cyst caused by the larval stage of *Echinococcus*

granulosus, a tapeworm transmitted from dogs to humans through ingestion of contaminated food. The liver is the most commonly affected organ, followed by the lungs. Cysts grow slowly and may become symptomatic from mass effect or rupture, causing anaphylaxis.

Hydatidiform Mole

- A type of gestational trophoblastic disease characterized by abnormal proliferation of trophoblastic tissue. Complete mole: 46,XX (all paternal), no fetal tissue, diffuse vesicular swelling; risk of malignant transformation (15-20%). Partial mole: triploid (69,XXX), some fetal tissue, focal swelling; lower risk.

Hydralazine

- A direct arteriolar vasodilator used for hypertension, particularly in preeclampsia/eclampsia (IV), and in combination with isosorbide dinitrate for heart failure with reduced ejection fraction in African Americans (improves survival). Mechanism: smooth muscle relaxation (nitric oxide pathway), reducing afterload.

Hydranencephaly

- A rare congenital condition where the cerebral hemispheres are nearly completely absent, replaced by thin-walled, fluid-filled sacs, while the brainstem, basal ganglia, and cerebellum are relatively preserved. Caused by bilateral internal carotid artery occlusion in utero (vascular insult, infection, or twin-twin transfusion).

Hydrocele

- Accumulation of serous fluid within the tunica vaginalis surrounding the testis. Congenital hydrocele results from patent processus vaginalis; acquired hydrocele may follow infection, trauma, or tumor. Transilluminates on examination. Ultrasound confirms fluid collection.

Hydrocephalus

- Abnormal accumulation of cerebrospinal fluid (CSF) within the ventricles of the brain, leading to increased intracranial pressure. Communicating hydrocephalus: impaired CSF absorption (subarachnoid hemorrhage, meningitis, leptomeningeal carcinomatosis).

Hydrochlorothiazide

- A thiazide diuretic inhibiting the sodium-chloride cotransporter in the distal convoluted tubule, promoting natriuresis and reducing plasma volume, with long-term vasodilation. Used for hypertension and mild edema. Lowers blood pressure effectively; useful in calcium stones and osteoporosis (reduces calciuria).

Hydrocolpos

- Distension of the vagina with fluid due to obstruction of the vaginal outlet, often from an imperforate hymen or transverse septum, presenting as a pelvic mass in neonates or adolescents.

Hydrofluoric Acid Exposure

- See chemical injuries.

Hydrogen Bond

- Weak electrostatic interaction between a hydrogen atom covalently bonded to an electronegative atom (N, O, F) and another electronegative atom. Critical for protein structure (alpha helix, beta sheet), DNA double helix (base pairing), enzyme-substrate binding, and water properties (high boiling point, solvent properties).

Hydrogen Breath Test

- Non-invasive test for carbohydrate malabsorption (lactose, fructose) and small intestinal bacterial overgrowth (SIBO). Patient ingests a substrate (lactose, glucose, or lactulose). Breath hydrogen is measured at baseline and q15-30 minutes for 2-3 hours. Rise in H₂ >20 ppm from baseline: positive (bacterial fermentation).

Hydronephrosis

- Dilation of renal collecting system proximal to obstruction. Causes include stones, strictures, malignancy, and BPH. Graded on ultrasound from mild to severe. Prolonged obstruction leads to cortical thinning and renal failure. Bilateral more clinically significant. Urgent decompression required for obstructive uropathy with infection.

Hydrophobia

- A fatal viral encephalitis caused by rabies virus, transmitted through the bite of infected animals. See Rabies.

Hydropneumothorax

- The simultaneous presence of both fluid and air in

the pleural space, causing respiratory distress and dullness to percussion with absent breath sounds, typically requiring chest tube drainage.

Hydrops Fetalis

- Abnormal accumulation of fluid in two or more fetal compartments, including pleural effusion, pericardial effusion, ascites, and subcutaneous edema. Causes include Rh isoimmunization, chromosomal abnormalities, cardiac defects, infections (parvovirus B19), and twin-to-twin transfusion syndrome.

Hydrotherapy

- Exercise in water using buoyancy, resistance, warmth. Benefits: reduced weight-bearing, improved circulation, pain relief, increased range of motion. Used for arthritis, spinal cord injury, amputation, neurological conditions.

Hydroxocobalamin

- A cobalt-containing form of vitamin B12 used as an antidote for cyanide poisoning. It binds cyanide to form cyanocobalamin, which is excreted in urine; it can cause red discoloration of skin and body fluids and interfere with some laboratory tests.

Hydroxychloroquine

- An antimalarial DMARD used in SLE (first-line for skin/joint/mild disease, reduces flares and improves survival), rheumatoid arthritis, and antiphospholipid syndrome. Mechanisms: interference with antigen processing, Toll-like receptor inhibition, immunomodulation.

Hymen

- A thin fold of mucous membrane that partially covers the external opening of the vagina. The hymen is a remnant of fetal development, with no known physiological function in postnatal life.

Hyoid Bone

- U-shaped bone in the anterior neck at C3, unique for not directly articulating with any other bone. Suspended by muscles and the stylohyoid ligament. Provides attachment for suprahyoid (digastric, stylohyoid, mylohyoid, geniohyoid) and infrahyoid (sternohyoid, sternothyroid, thyrohyoid, omohyoid) muscles.

Hyper-IgM Syndromes

- A group of disorders characterized by normal or

elevated IgM with low IgG, IgA, and IgE, due to defective class switch recombination (CSR) and somatic hypermutation (SHM).

Hyperaldosteronism

- Hyperaldosteronism, also known as Conn syndrome when caused by an adrenal adenoma, is characterized by excessive aldosterone production causing hypertension, hypokalemia, and metabolic alkalosis.

Hyperaldosteronism Complications

- Beyond hypertension and hypokalemia, primary hyperaldosteronism causes target organ damage including left ventricular hypertrophy, fibrosis, stroke, coronary artery disease, atrial fibrillation, and chronic kidney disease. These complications occur at higher rates than in essential hypertension of similar severity.

Hyperammonemia

- An abnormally elevated concentration of ammonia in the blood. It may result from an inherited urea-cycle or organic-acid disorder, severe liver dysfunction, or other acquired causes and can produce encephalopathy, cerebral oedema, seizures, and coma.

Hyperbaric Oxygen Therapy

- Administration of 100% oxygen at increased atmospheric pressure. Indications: carbon monoxide poisoning, decompression sickness, gas gangrene, necrotizing soft tissue infections, delayed radiation injury, crush injury, compromised grafts. Mechanism: increases dissolved oxygen, vasoconstriction, antimicrobial effect.

Hypercalcemia

- Serum calcium greater than 10.5 milligrams per deciliter. Causes summarized by mnemonic MALIGNANT: Myeloma, Metastases, Metabolic bone disease, Hyperparathyroidism, Adrenal insufficiency, Paget disease, Immobilization, Thiazide diuretics, Addison disease, Neonatal, Tumor-derived PTHrP.

Hypercalcemia of Malignancy

- Elevated serum calcium caused by cancer, a common paraneoplastic and oncologic emergency. Mechanisms: PTHrP secretion (most common—squamous cell lung, breast, renal, ovarian), osteolytic metastases (breast, myeloma), 1,25-

dihydroxyvitamin D (lymphoma), and ectopic PTH (rare).

Hypercapnia

- An abnormally elevated arterial partial pressure of carbon dioxide, usually caused by alveolar hypoventilation. It may produce headache, somnolence, confusion, asterixis, respiratory acidosis, and, when severe or rapidly developing, coma.

Hypercholesterolemia

- Elevated serum total cholesterol and LDL cholesterol levels, a major modifiable risk factor for atherosclerotic cardiovascular disease. It may be primary, due to familial hypercholesterolemia from LDL receptor mutations, or secondary to diet, obesity, hypothyroidism, or nephrotic syndrome.

Hypercoagulable State

- Condition increasing thrombosis risk. Inherited: Factor V Leiden, prothrombin mutation, protein C/S deficiency, antithrombin deficiency. Acquired: cancer, pregnancy, oral contraceptives, antiphospholipid syndrome, myeloproliferative disorders.

Hypercyanotic Tetralogy

- A complex congenital heart defect with four features: ventricular septal defect, overriding aorta, right ventricular outflow tract obstruction (pulmonary stenosis), and right ventricular hypertrophy. It causes right-to-left shunting and cyanosis. Treatment is surgical repair, typically in infancy.

Hyperemesis

- See gastroenterology.

Hyperemesis Gravidarum

- Severe, persistent nausea and vomiting during pregnancy resulting in weight loss greater than 5%, dehydration, electrolyte imbalances, and ketosis. Occurs in 0.3-2% of pregnancies, typically presenting in the first trimester. Differentiated from normal morning sickness by severity and metabolic derangement.

Hyperesthesia

- Increased sensitivity to sensory stimulation—touch (tactile), pain (hyperalgesia), or other stimuli—so that normally non-painful or mildly painful stimuli produce exaggerated discomfort.

Hyperglycemia

- Elevated blood glucose levels above normal range (>140 mg/dL postprandial or >126 mg/dL fasting).

Acute hyperglycemia can cause osmotic diuresis, dehydration, and electrolyte disturbances. Chronic hyperglycemia leads to microvascular (retinopathy, nephropathy, neuropathy) and macrovascular complications.

Hypergonadotropic Hypogonadism

- Primary testicular failure with elevated LH/FSH. Causes: Klinefelter syndrome, cryptorchidism, orchitis, chemotherapy, radiation.

Hyperhidrosis

- Excessive sweating beyond thermoregulatory needs. Primary focal hyperhidrosis affects palms, soles, axillae, and face without underlying cause. Secondary generalized hyperhidrosis is caused by medications, endocrine disorders, infections, or malignancies.

Hyperkalemia

- Serum potassium >5.0 mEq/L. ECG changes: peaked T waves, widened QRS, sine wave. Causes: renal failure, medications (ACE inhibitors, potassium-sparing diuretics), tissue breakdown, acidosis.

Hyperlipidemia

- Elevated levels of lipids in the blood—total cholesterol, LDL, triglycerides, and/or reduced HDL—a major modifiable risk factor for atherosclerosis and cardiovascular disease. Primary (genetic: familial hypercholesterolemia) or secondary (hypothyroidism, diabetes, nephrotic syndrome, obesity, alcohol).

Hypernatremia

- Serum sodium greater than 145 mEq/L. Always indicates free water deficit relative to sodium. Causes include inadequate water intake particularly in elderly and cognitively impaired patients, diabetes insipidus, osmotic diuresis from hyperglycemia, and GI or renal water losses.

Hyperopia

- Hyperopia, commonly known as farsightedness, is a refractive error in which parallel light rays from distant objects are focused posterior to the retina rather than directly on it, resulting in blurred near vision that may be accompanied by blurred distance vision in moderate to high degrees.

Hyperosmolar Hyperglycemic State

- Hyperglycemic emergency characterized by se-

vere hyperglycemia greater than 600 milligrams per deciliter, hyperosmolality greater than 320 milliosmoles per kilogram, and profound dehydration without significant ketosis.

Hyperparathyroidism

- A disorder of excessive parathyroid hormone (PTH) secretion from one or more of the four parathyroid glands, classified as primary, secondary, or tertiary.

Hyperphosphatemia

- Serum phosphate above normal range (>4.5 mg/dL), usually due to impaired renal excretion (CKD—most common), hypoparathyroidism, tumor lysis syndrome, rhabdomyolysis, or excessive intake/absorbable laxatives. Consequences: secondary hyperparathyroidism, vascular/metastatic calcification, and when combined with hypocalcemia, tetany.

Hyperpigmentation

- Darkening of the skin caused by increased melanin production, deposition of pigment in the dermis, or accumulation of other pigments.

Hyperplasia

- Increased cell number. Can be physiological or pathological.

Hyperpolarization

- A change in membrane potential to a more negative value, moving it further from threshold and making the cell less excitable. Occurs when potassium leaves the cell (opening of voltage-gated potassium channels, as in the afterhyperpolarization after an action potential) or when chloride enters (GABA-A receptor activation).

Hyperprolactinaemia

- Elevated levels of the hormone prolactin in the blood, above the normal range (typically >500 mIU/L in women).

Hyperprolactinemia

- Elevated serum prolactin levels most commonly caused by prolactin-secreting pituitary adenomas termed prolactinomas. Other causes include medications particularly antipsychotics and metoclopramide, hypothyroidism, stalk effect from non-functioning pituitary adenomas, renal failure, and chest wall lesions.

Hypersensitivity

- See pharmacology.

Hypersensitivity Pneumonitis

- An immune-mediated lung disease from environmental antigen inhalation (molds, bird proteins). Acute, subacute, and chronic forms exist. HRCT shows ground-glass opacities, air trapping, and centrilobular nodules. Chronic HP progresses to fibrosis.

Hypersensitivity Reactions

- Inappropriate or excessive immune responses causing tissue damage or disease.

Hypersensitivity Type I

- Type I hypersensitivity, also known as immediate hypersensitivity or allergic reactions, is mediated by IgE antibodies and mast cells, producing symptoms within minutes of allergen re-exposure.

Hypersensitivity Type II

- Type II hypersensitivity, also known as cytotoxic hypersensitivity, is mediated by IgG or IgM antibodies directed against antigens on cell surfaces or the extracellular matrix, causing tissue damage through complement activation, antibody-dependent cellular cytotoxicity, and phagocytosis.

Hypersensitivity Type III

- Type III hypersensitivity is caused by deposition of antigen-antibody immune complexes in tissues and blood vessel walls, activating complement and recruiting neutrophils that release destructive enzymes and reactive oxygen species.

Hypersensitivity Type IV

- Type IV hypersensitivity, also known as delayed-type hypersensitivity, is mediated by T lymphocytes rather than antibodies and takes 48 to 72 hours to develop after antigen exposure.

Hypertelorism

- Increased distance between the orbits (the bony eye sockets), measured as an increased interorbital distance on imaging.

Hypertension

- Persistently elevated blood pressure, defined as systolic BP >130 mmHg or diastolic BP >80 mmHg (ACC/AHA guidelines). It is the most common modifiable risk factor for cardiovascular disease,

stroke, heart failure, and chronic kidney disease.

Hypertension (High Blood Pressure)

- Persistently elevated blood pressure, defined as systolic BP >130 mmHg or diastolic BP >80 mmHg (ACC/AHA guidelines). See Hypertension.

Hypertension in Anesthesia

- Hypertension during anesthesia causes increased bleeding, myocardial ischemia, and stroke risk. Common causes include light anesthesia, pain, hypercarbia, hypoxia, bladder distension, shivering, and drug withdrawal (beta-blockers, clonidine).

Hypertensive

- See cardiology.

Hypertensive Crisis

- Severe elevation in blood pressure (SBP >180 or DBP >120) requiring urgent medical attention. Two types: hypertensive emergency (with end-organ damage) and hypertensive urgency (no acute end-organ damage). Hypertensive urgency: can be managed with oral antihypertensives (captopril, clonidine, labetalol) and careful outpatient follow-up.

Hypertensive Emergency

- Severe hypertension (SBP >180 or DBP >120) with acute target organ damage. Organs involved: brain (hypertensive encephalopathy, stroke), heart (acute coronary syndrome, acute pulmonary edema, aortic dissection), kidneys (acute kidney injury), and eyes (papilledema).

Hypertensive Heart Disease

- Left ventricular hypertrophy and diastolic/systolic dysfunction resulting from chronic hypertension. LVH is an independent cardiovascular risk factor and precursor to heart failure. Diagnosis by ECG (voltage criteria) or echocardiography (wall thickness >1.1 cm). Treatment of hypertension can regress LVH.

Hypertensive Nephropathy

- Kidney damage caused by chronic hypertension, leading to nephrosclerosis (arteriolar hyalinosis and glomerulosclerosis). Presents with slowly progressive proteinuria (usually subnephrotic), hypertension, and gradual decline in kidney function. More common in African American patients. Diagnosis of exclusion.

Hypertensive Nephrosclerosis

- Kidney damage from chronic hypertension. Benign form with hyaline arteriolosclerosis. Malignant form with hyperplastic arteriolosclerosis and onion-skinning. Second leading cause of ESRD after diabetes. Tight BP control with ACE inhibitors or ARBs essential to slow progression.

Hypertensive Retinopathy

- Retinal changes caused by chronic or acute hypertension. Grading (Keith-Wagener-Barker): Grade I: generalized arteriolar narrowing; Grade II: arteriovenous nicking; Grade III: cotton-wool spots, flame hemorrhages, hard exudates; Grade IV: papilledema (malignant hypertension).

Hyperthermia in Anesthesia

- Perioperative hyperthermia (temperature greater than 38 degrees Celsius) has multiple causes. Malignant hyperthermia (discussed separately) is the most urgent. Sepsis, thyroid storm, serotonin syndrome, neuroleptic malignant syndrome, and central fever are other causes.

Hyperthermic Intraperitoneal Chemotherapy (HIPEC)

- Heated chemotherapy circulated through the peritoneal cavity during surgery, following cytoreductive surgery, for peritoneal carcinomatosis from appendiceal, colorectal, and ovarian cancers and peritoneal mesothelioma.

Hyperthyroidism

- A condition of excess thyroid hormone production and secretion from the thyroid gland, resulting in thyrotoxicosis (elevated serum T4 and T3 concentrations).

Hyperthyroidism (Thyrotoxicosis)

- A condition of excess thyroid hormone production and secretion from the thyroid gland, resulting in thyrotoxicosis (elevated serum T4 and T3 concentrations). See Hyperthyroidism.

Hypertonic Saline

- Intravenous saline with a sodium concentration greater than plasma, commonly 3% sodium chloride. It is used selectively for severe symptomatic hyponatremia or other urgent indications, with careful monitoring to avoid overly rapid correction and osmotic demyelination.

Hypertonic Saline Inhalation

- Nebulized 3–7% sodium chloride solution that draws water into the airway surface liquid by osmosis, improving mucociliary clearance. It is used in CF and bronchiectasis. It may cause bronchospasm and should be preceded by a bronchodilator. It reduces exacerbations in CF.

Hypertrichosis

- Excessive hair growth in androgen-independent areas, affecting both sexes, either congenital or acquired. Generalized: genetic syndromes, medications (minoxidil, cyclosporine, phenytoin, glucocorticoids), malnutrition, porphyria, hypothyroidism, and malignancy (paraneoplastic).

Hypertriglyceridemia

- Elevated serum triglyceride levels, classified as mild at 150 to 499 mg per dL, moderate at 500 to 999, and severe above 1000. Severe hypertriglyceridemia increases the risk of acute pancreatitis. Causes include obesity, diabetes, alcohol use, and genetic disorders of lipoprotein metabolism.

Hypertrophic Cardiomyopathy

- An autosomal dominant genetic disorder characterized by asymmetric left ventricular hypertrophy (typically septal), myocyte disarray, and fibrosis. It is the most common cause of sudden cardiac death in young athletes. Many patients are asymptomatic; symptoms include dyspnea, chest pain, syncope, and palpitations.

Hypertrophic Cardiomyopathy Genetics

- HCM is caused by mutations in genes encoding sarcomeric proteins, most commonly beta-myosin heavy chain (MYH7) and myosin-binding protein C (MYBPC3). Inheritance is autosomal dominant with variable penetrance. Genetic testing identifies causative mutations in 40–60% of cases.

Hypertrophic Cardiomyopathy Treatment

- Symptomatic management: beta-blockers (first-line for exertional symptoms), non-dihydropyridine calcium channel blockers (alternative or additive), disopyramide (reduces LVOT gradient).

Hypertrophic Pyloric Stenosis

- Hypertrophy and hyperplasia of the pyloric muscle causing gastric outlet obstruction, typically in young infants. Progressive nonbilious projectile

vomiting, hunger after vomiting, dehydration, and hypochloremic hypokalemic metabolic alkalosis may occur.

Hypertrophic Scar

- Raised scar tissue confined to the original wound margins, caused by excessive collagen production during healing. Unlike keloids, hypertrophic scars regress over time and do not extend beyond wound boundaries. They are common after burns, surgery, and trauma.

Hypertrophy

- Increased cell size. Physiological (muscle) or pathological.

Hyperventilation

- Breathing in excess of metabolic requirements, resulting in decreased partial pressure of arterial carbon dioxide (PaCO₂) and respiratory alkalosis.

Hyperventilation Syndrome

- Recurrent or persistent overbreathing that exceeds metabolic requirements, usually producing reduced arterial carbon dioxide and respiratory alkalosis in the absence of primary pulmonary disease. Symptoms may include breathlessness, chest tightness, dizziness, palpitations, paraesthesia, and carpopedal spasm.

Hyphema

- Bleeding into the anterior chamber of the eye, between the cornea and iris, typically resulting from blunt ocular trauma that tears the blood vessels of the iris or ciliary body.

Hypnosis

- A state of focused attention, reduced peripheral awareness, and heightened suggestibility, resembling a trance-like state but distinct from sleep. Hypnosis is induced by a procedure (hypnotic induction) involving focused concentration, relaxation, and guided imagery.

Hypnotic

- A medication that induces, maintains, or improves sleep, used for insomnia.

Hypoactive Sexual Desire Disorder (HSDD)

- A sexual dysfunction characterized by persistently or recurrently deficient or absent sexual thoughts, fantasies, and desire for sexual activity, causing

marked distress or interpersonal difficulty. In DSM-5, HSDD in women is merged with female sexual arousal disorder into female sexual interest/arousal disorder.

Hypocalcemia

- Serum calcium less than 8.5 milligrams per deciliter or ionized calcium less than 4.5 milligrams per deciliter. Causes include hypoparathyroidism, vitamin D deficiency, hypomagnesemia, hyperphosphatemia, and critical illness.

Hypocalcemia, hypophosphatemia, and hyperkalemia

- The classic electrolyte triad of acute hypoparathyroidism, most commonly occurring after thyroid or parathyroid surgery from inadvertent removal or devascularization of the parathyroid glands. Hypocalcemia causes tetany, paresthesias, and Chvostek and Trousseau signs.

Hypodermis

- Subcutaneous tissue/superficial fascia, the deepest skin layer with loose connective tissue and adipose tissue. Attaches skin to underlying muscles/bones via fibrous septa. Provides thermal insulation, energy storage (fat), mechanical cushioning, and contains blood vessels, lymphatics, and nerves.

Hypodontia

- Congenital absence of one or more teeth. Common: third molars, maxillary lateral incisors, second premolars.

Hypogammaglobulinemia

- A reduction in circulating immunoglobulins that predisposes to recurrent infections, especially with encapsulated bacteria; it may be inherited or acquired.

Hypogastric Region

- Central inferior region below the umbilical region, containing the urinary bladder, sigmoid colon, rectum, and in females the uterus and ovaries. Assessed for bladder distension, uterine enlargement, and pelvic masses. Suprapubic pain may indicate cystitis, calculi, or pelvic pathology.

Hypoglossal nerve injury

- Damage to the twelfth cranial nerve (CN XII), which innervates the ipsilateral intrinsic and extrinsic tongue muscles. Causes include skull base

fractures, surgical trauma, tumors, or infection.

Hypoglossal nerve palsy

- Weakness or paralysis of the hypoglossal nerve (CN XII), leading to ipsilateral tongue weakness, atrophy, and fasciculations. Causes include tumors at the skull base or in the hypoglossal canal, surgical injury, or central lesions. Unilateral palsy causes the tongue to deviate toward the affected side on protrusion.

Hypoglycemia

- Blood glucose <70 mg/dL. Symptoms: tremor, palpitations, sweating, hunger, confusion, seizures, coma. Causes: insulin or sulfonylurea overdose, missed meals, alcohol, critical illness. Treatment: conscious patient—oral glucose; unconscious—IV dextrose (D50W) or IM glucagon. Recurrent hypoglycemia requires medication adjustment.

Hypoglycemia in Diabetes

- Blood glucose less than 70 milligrams per deciliter in patients with diabetes. Classification by level: level one less than 70, level two less than 54 which is clinically significant hypoglycemia, and level three severe hypoglycemia requiring assistance.

Hypoglycemia Unawareness

- Loss of the warning symptoms of hypoglycemia including tremor, sweating, and palpitations occurring in patients with recurrent hypoglycemia particularly type one diabetes. Results from attenuated counter-regulatory hormone responses particularly impaired glucagon and epinephrine responses.

Hypogonadism

- Decreased functional activity of the gonads (testes or ovaries) with reduced sex hormone production. Male hypogonadism: low testosterone causing decreased libido, erectile dysfunction, infertility, reduced muscle mass, and osteoporosis. Female: estrogen deficiency causing amenorrhea, hot flashes, and osteoporosis.

Hypogonadism in Men

- Testosterone deficiency resulting from inadequate testicular function classified as primary testicular failure or secondary hypothalamic-pituitary dysfunction. Primary causes include Klinefelter syndrome, cryptorchidism, orchitis, trauma, and chemotherapy.

Hypogonadism in Women

- Testosterone deficiency in women causing decreased libido, fatigue, depressed mood, and reduced sense of well-being. Causes include oophorectomy, premature ovarian insufficiency, adrenal insufficiency, hypopituitarism, and medications.

Hypogonadotropic Hypogonadism

- Secondary testicular failure with low/normal LH/FSH. Causes: Kallmann syndrome, pituitary tumors, obesity, medications, systemic illness.

Hypohidrosis

- Reduced or absent sweating, leading to heat intolerance and risk of hyperthermia. Causes include autonomic neuropathy (diabetes, Parkinson disease), skin diseases (scleroderma, burns), medications (anticholinergics), and genetic disorders (hypohidrotic ectodermal dysplasia).

Hypokalemia

- Serum potassium less than 3.5 mEq/L. Causes include gastrointestinal losses from vomiting and diarrhea, renal losses from diuretics, RTA, and hyperaldosteronism, intracellular shift from alkalosis and insulin, and inadequate intake.

Hypomagnesemia

- Serum magnesium less than 1.7 milligrams per deciliter. Causes include gastrointestinal losses from diarrhea, renal losses from diuretics and alcohol, malnutrition, and medications including proton pump inhibitors.

Hypomania

- Distinct period of abnormally elevated, expansive, or irritable mood with increased energy lasting at least four consecutive days, observable by others but without marked functional impairment, hospitalisation, or psychotic features.

Hypomanic Episode

- A distinct period of abnormally elevated, expansive, or irritable mood with increased energy lasting at least four consecutive days. The episode is not severe enough to cause marked impairment in functioning or necessitate hospitalization, and there are no psychotic features. The change in functioning is observable by others.

Hyponatremia

- Serum sodium less than 135 mEq/L. Most com-

mon electrolyte disorder in hospitalized patients. Causes are classified by volume status: hypovolemic from GI losses and diuretics, euvolemic from SIADH and hypothyroidism, and hypervolemic from heart failure, cirrhosis, and nephrotic syndrome.

Hypoparathyroidism

- Deficiency or absence of parathyroid hormone causing hypocalcemia and hyperphosphatemia. Most commonly occurs after thyroid or parathyroid surgery from inadvertent removal of parathyroid glands. Autoimmune hypoparathyroidism may occur alone or as part of autoimmune polyendocrine syndrome type one.

Hypoparathyroidism Management

- Treatment of chronic hypocalcemia from hypoparathyroidism includes oral calcium supplementation in divided doses and active vitamin D calcitriol to enhance calcium absorption. Goal is to maintain serum calcium in the low-normal range to prevent hypercalciuria and renal complications. Thiazide diuretics reduce urinary calcium

Hypophosphatemia

- An abnormally low serum phosphate concentration caused by reduced intake or absorption, intracellular shifting, or renal phosphate wasting. Severe deficiency can cause muscle weakness, respiratory failure, hemolysis, impaired myocardial function, and neurologic dysfunction.

Hypopituitarism

- Deficiency of one or more pituitary hormones most commonly from pituitary adenomas, craniopharyngiomas, pituitary surgery or radiation, traumatic brain injury, or infiltrative diseases. Growth hormone deficiency is typically the first hormone lost followed by gonadotropins then TSH and ACTH.

Hypoplasia

- Incomplete or underdevelopment of a tissue, organ, or body structure, resulting in reduced size or functional capacity. It may be congenital, developmental, genetic, or acquired and ranges from a mild anatomical variant to severe organ dysfunction.

Hypoplastic Left Heart Syndrome

- A severe congenital heart defect where the left heart structures (mitral valve, left ventricle, aortic valve, ascending aorta) are underdeveloped and unable to

support systemic circulation. The right ventricle pumps blood to both the pulmonary and systemic circulations via a patent ductus arteriosus.

Hypopyon

- A layering of inflammatory cells (purulent exudate/leukocytes) in the anterior chamber of the eye, forming a visible white-to-yellow fluid level in the lower portion of the cornea. Caused by severe anterior uveitis (HLA-B27 spondyloarthropathies, Behçet disease), infectious keratitis (bacterial/fungal corneal ulcers), or endophthalmitis.

Hyposmia

- Decreased sensitivity to odors in which olfaction is reduced but not absent, distinct from anosmia and parosmia, and the most common olfactory complaint.

Hypospadias

- Congenital anomaly of male urogenital development characterized by an abnormally proximal urethral meatus located on the ventral surface of the penis, scrotum, or perineum, rather than at the tip of the glans. Classified by meatal location: anterior (glandular, subcoronal—60-70%), middle (penile), or posterior (equinoscrotal),

Hypotension

- Abnormally low blood pressure, generally systolic <90 mmHg or mean arterial pressure <65 mmHg. Causes include hypovolemia, cardiac failure, sepsis, anaphylaxis, and medications. Orthostatic hypotension is a drop upon standing. Severe hypotension (shock) requires urgent fluid resuscitation and vasopressors.

Hypotension in Anesthesia

- Mean arterial pressure less than 65 mmHg or a decrease greater than 20 percent from baseline during anesthesia.

Hypothalamic-Pituitary-Adrenal Axis (HPA Axis)

- An endocrine regulatory pathway in which hypothalamic corticotropin-releasing hormone stimulates pituitary ACTH secretion, which stimulates adrenal cortisol production. Cortisol exerts negative feedback and coordinates metabolic, immune, and stress responses.

Hypothalamic-Pituitary-Adrenal Axis

Physiology

- The hypothalamus releases CRH in a diurnal rhythm peaking in the early morning stimulating pituitary corticotrophs to secrete ACTH which stimulates the adrenal cortex to produce cortisol. Cortisol exerts negative feedback on both the hypothalamus and pituitary suppressing CRH and ACTH release.

Hypothalamic-Pituitary-Gonadal Axis Physiology

- The hypothalamus releases GnRH in a pulsatile fashion stimulating pituitary gonadotrophs to secrete LH and FSH. In males LH stimulates Leydig cells to produce testosterone while FSH stimulates Sertoli cells to support spermatogenesis.

Hypothalamic-Pituitary-Thyroid Axis Physiology

- The hypothalamus secretes TRH stimulating pituitary thyrotrophs to release TSH which stimulates the thyroid gland to produce T4 and T3. T3 the more biologically active form is produced by peripheral conversion of T4 by deiodinase enzymes.

Hypothalamus

- Small region below the thalamus forming the third ventricle's floor and lower lateral walls, weighing approximately 4 grams, the master homeostasis regulator. Controls autonomic and endocrine systems (via the pituitary through the hypophyseal portal system), temperature, hunger, thirst, sleep-wake cycles, and emotion.

Hypothermia

- Core temperature <35 degrees C (95 degrees F). Stages: mild (32-35), moderate (28-32), severe (<28). J-wave (Osborn wave) on ECG. Management: passive rewarming (mild), active external rewarming (moderate), active internal rewarming (severe). Avoid rough handling (risk of VF).

Hypothermia in Anesthesia

- Hypothermia (core temperature less than 36 degrees Celsius) occurs in 50 to 80 percent of surgical patients due to anesthetic-induced thermoregulatory dysfunction and heat redistribution. Initial phase (first 30 to 60 minutes) involves core-to-peripheral heat redistribution causing 0.5 to 1.5 degree Celsius drop.

Hypothyroidism

- A common condition of inadequate thyroid hormone production, with a prevalence of 4-10 percent in the general population and up to 20 percent in women over 60 years.

Hypotonia

- Abnormally reduced muscle tone, causing decreased resistance to passive movement and a floppy or poorly supported posture. It may result from muscle, peripheral nerve, neuromuscular-junction, spinal-cord, brain, genetic, or metabolic disorders and requires evaluation when persistent or associated with weakness or developmental delay.

Hypovolemic Shock

- See emergency medicine.

Hypoxemia

- $\text{PaO}_2 < 80$ mmHg on room air. Causes: hypoventilation, V/Q mismatch, shunt, diffusion impairment, low FiO_2 . A-a gradient helps distinguish causes.

Hypoxia

- Reduced oxygen availability at the tissue level, defined as arterial partial pressure of oxygen (PaO_2) < 60 mmHg or arterial oxygen saturation (SpO_2) $< 90\%$ on room air.

Hypoxic Cell Injury

- Cellular injury caused by inadequate oxygen availability, reducing oxidative phosphorylation and ATP production. Causes include ischemia, hypoxemia, anemia, and impaired oxygen utilization; severity and duration determine reversibility.

Hypoxic Pulmonary Vasoconstriction

- The unique response of pulmonary arterioles to low alveolar oxygen tension (PO_2 less than approximately 60 mmHg), causing vasoconstriction.

Hypoxic-Ischemic Brain Injury

- Brain injury caused by inadequate oxygen delivery or cerebral blood flow. It may follow cardiac arrest, asphyxia, shock, respiratory failure, or severe hypoglycemia and can produce coma, seizures, cognitive impairment, or death.

Hysterectomy

- The surgical removal of the uterus, the second most common major surgical procedure in women after cesarean section in the United States.

Hysterectomy and Surgical Menopause

- Hysterectomy is the surgical removal of the uterus, and when the ovaries are also removed (oophorectomy), it induces surgical menopause regardless of age.

Hysterosalpingo-Contrast-Sonography

- An ultrasound-based technique for evaluating the uterine cavity and fallopian tube patency. Sterile saline or contrast medium is injected through the cervix while transvaginal ultrasound is performed.

Hysterosalpingo-Contrast-Sonography (HyCoSy)

- Ultrasound evaluation of tubal patency using contrast. Alternative to hysterosalpingography.

Hysterosalpingogram (HSG)

- An X-ray procedure used to evaluate the uterine cavity and fallopian tube patency. A radiopaque contrast medium is injected through the cervix, and serial X-ray images are taken as the contrast fills the uterine cavity and spills from the fallopian tubes.

Hysterosalpingography

- Contrast imaging of uterus and fallopian tubes. Evaluates infertility, uterine anomalies, tubal patency.

Hysterosalpingography (HSG)

- Fluoroscopic imaging after injection of iodinated contrast into the uterine cavity through a cervical cannula. Used for infertility evaluation: assesses tubal patency (normal: contrast spills freely into peritoneal cavity), uterine cavity shape (fibroids, polyps, synechiae, Mullerian anomalies), and proximal/distal tubal occlusion.

Hairy Tongue

- A usually harmless overgrowth and discoloration of the tongue's papillae, often appearing black, brown, or white. It may be associated with smoking, poor oral hygiene, dry mouth, or antibiotics; gentle tongue cleaning and removal of contributing factors usually help.

Heel Spur

- A bony growth arising from the heel bone, often associated with plantar-fascia strain. The spur itself may cause no symptoms; heel pain is more often related to surrounding soft-tissue inflammation.

Hypoesthesia

- Reduced sensitivity to touch, temperature, pain, or other sensations. It may result from nerve compression, neuropathy, spinal disease, stroke, injury, or other neurologic conditions; sudden or spreading numbness requires prompt assessment.

Hypermobile Ehlers-Danlos Syndrome

- A heritable connective-tissue disorder characterized mainly by generalized joint hypermobility, instability, chronic musculoskeletal pain, and other variable features involving skin, autonomic function, or digestion. Diagnosis is clinical and requires assessment for other connective-tissue disorders and related conditions.

Chapter 9

Pocket Dictionary: I Terms

IADL Training

- Training for instrumental ADLs: cooking, cleaning, shopping, managing medications, finances.

Iatrogenic

- An illness or condition caused by medical examination or treatment. Examples: hospital-acquired infections, drug side effects, surgical complications, radiation-induced damage. Prevention: proper technique, informed consent, monitoring, reporting systems.

Iatrogenic Injury

- Urological injury from medical procedures. Ureteral injury most common. Recognition and repair essential.

ICD Shock Therapy

- Anti-tachycardia pacing (ATP) for slow VT delivers rapid pacing pulses to terminate the arrhythmia painlessly. Cardioversion shock synchronously terminates VT. Defibrillation shock treats VFib asynchronously. The MADIT-RIT trial showed that higher-rate programming (detection >180-200 bpm) reduced inappropriate shocks and mortality.

Ice Pack

- Cold therapy for injury. Reduces swelling, pain. Apply 20 minutes on, 20 off.

ICF Framework

- International Classification of Functioning, Disability and Health. Biopsychosocial model: body functions/structures, activities, participation, environmental factors, personal factors.

Ichthyosis

- A group of disorders characterized by dry, scaly skin resembling fish scales. Ichthyosis vulgaris is the most common, autosomal dominant, associated with

filaggrin mutation and atopy. X-linked ichthyosis is caused by steroid sulfatase deficiency. Lamellar ichthyosis is a severe congenital form.

Ichthyosis Vulgaris

- A common inherited disorder of skin keratinization characterized by dry, scaling skin, especially on the extensor limbs. Symptoms often improve with age and with regular emollient and keratolytic skin care.

Ideas of Reference

- The false belief that unrelated events, objects, or people have a particular and unusual significance specifically for the patient. For example, believing that TV newscasters are sending personal messages or that strangers are conspiring against them.

Idiopathic

- Of unknown cause. Examples: idiopathic pulmonary fibrosis, idiopathic thrombocytopenia.

Idiopathic Hypersomnia

- A sleep disorder characterized by excessive daytime sleepiness, prolonged or unrefreshing sleep, difficulty awakening, and impaired alertness despite adequate opportunity for sleep. It is diagnosed after excluding other medical, psychiatric, circadian, and substance-related causes.

Idiopathic Intracranial Hypertension

- Elevated intracranial pressure without mass lesion. Symptoms: headache, pulsatile tinnitus, visual changes, papilledema. Treatment: acetazolamide, weight loss, surgical options.

Idiopathic Pulmonary Fibrosis

- A chronic, progressive fibrosing interstitial pneumonia of unknown cause. Histopathology shows usual interstitial pneumonia (UIP). Features include exertional dyspnea, dry cough, bibasilar crackles, and

digital clubbing. Median survival is 3–5 years.

Idiosyncratic Reaction

- An uncommon, unexpected, and often genetically determined adverse drug reaction that is unrelated to the drug's known pharmacology and usually dose-independent (a Type B reaction).

Idioventricular Rhythm

- A ventricular rhythm arising from an ectopic focus in the His-Purkinje system or ventricular myocardium, typically at a slow rate with wide QRS complexes. It may occur during escape from higher pacemakers, reperfusion, conduction block, or severe cardiac disease.

IgA

- Immunoglobulin A, the predominant antibody in mucosal secretions (saliva, tears, breast milk, gastrointestinal and respiratory tract). Exists mainly as a dimer linked by a J chain and secretory component, which protects it from proteolysis.

IgA Nephropathy

- The most common primary glomerulonephritis worldwide, characterized by mesangial IgA deposition. Presents with hematuria (macroscopic with mucosal infections, especially upper respiratory), proteinuria, and hypertension. Most common in young adults, with male predominance.

IgA Vasculitis

- An immune-complex small-vessel vasculitis characterized by palpable purpura, joint pain, abdominal symptoms, and kidney involvement. It is common in children and may follow an infection; renal disease determines long-term risk.

IgA Vasculitis (Henoch-Schonlein Purpura)

- The most common systemic vasculitis in children (peak age 2-8 years), characterized by IgA1-containing immune complex deposition in small vessels. Often preceded by an upper respiratory tract infection (e.g., group A Streptococcus).

IgD

- Immunoglobulin D, a membrane-bound antibody expressed on the surface of mature naive B cells (with IgM) serving as a B-cell receptor. Serum IgD is present in trace amounts and its secreted function is unclear. Its role in B-cell activation and signaling

is an area of research.

IgE

- Immunoglobulin E, the antibody class mediating immediate (type I) hypersensitivity and defense against parasites. Binds high-affinity Fc-epsilon-RI receptors on mast cells and basophils; allergen cross-linking triggers degranulation and release of histamine and leukotrienes, causing allergic rhinitis, asthma, urticaria, and anaphylaxis.

IgG

- Immunoglobulin G, the most abundant antibody class in serum (75-80 percent), produced during secondary immune responses. Functions: neutralization, opsonization, complement activation (classical pathway), and antibody-dependent cellular cytotoxicity.

IgG and complement deposition at the dermis-epidermis junction

- A pathognomonic immunofluorescence finding in systemic lupus erythematosus, where immune complexes of IgG and complement components deposit at the basement membrane zone in a continuous, linear pattern.

IgG4-Related Disease

- A fibro-inflammatory condition characterized by tumefactive lesions in multiple organs, dense lymphoplasmacytic infiltrate with >10 IgG4+ plasma cells/HPF and IgG4+/IgG+ ratio >40%, elevated serum IgG4 (>135 mg/dL), obliterative phlebitis, storiform fibrosis.

IgG4-related disease arteritis

- A form of IgG4-related disease affecting large and medium-sized arteries, characterized by transmural inflammation with abundant IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis. It may involve the aorta and its branches, causing stenosis or aneurysm formation.

IgG4-related ophthalmic disease

- A fibroinflammatory condition affecting orbital and adnexal structures, characterised by lymphoplasmacytic infiltration enriched in IgG4-positive plasma cells, storiform fibrosis, and obliterative phlebitis. It commonly presents as lacrimal gland enlargement, orbital mass, or extraocular muscle involvement and responds to

IgM

- Immunoglobulin M, the largest antibody (pentamer), produced during the primary immune response (first antibody to appear). Functions: potent complement activation (classical pathway) and agglutination; expressed as a membrane-bound B-cell receptor on naive B cells.

IGRA (Interferon-Gamma Release Assay)

- Blood test for *Mycobacterium tuberculosis* infection (latent or active). T-SPOT.TB (ELISPOT) or QuantiFERON-TB Gold Plus (ELISA). Measures IFN-gamma released by T cells after exposure to MTB-specific antigens (ESAT-6, CFP-10, TB7.7).

Ileal Atresia

- Congenital complete interruption or absence of a segment of the ileal lumen. It causes neonatal intestinal obstruction with abdominal distension, bilious vomiting, and failure to pass meconium and generally requires surgical management.

Ileal Pouch-Anal Anastomosis

- Surgical creation of a J-pouch from the terminal ileum after proctocolectomy to preserve intestinal continuity for ulcerative colitis. Avoids permanent ileostomy by allowing defecation through the anal route. The procedure involves removal of the entire colon and rectum with construction of a reservoir from distal ileum.

Ileitis

- Inflammation of the ileum, the distal segment of the small intestine. Causes include Crohn disease, infection, ischemia, medication-related injury, and other inflammatory disorders; symptoms may include right-lower-quadrant pain, diarrhea, fever, or bleeding.

Ileocecal Valve

- A sphincter-like fold at the junction of the terminal ileum and caecum. It regulates passage of intestinal contents into the colon and helps limit reflux of colonic contents into the small intestine.

Ileostomy

- A surgically created stoma from the terminal ileum to the abdominal wall, performed after proctocolectomy for ulcerative colitis or familial adenomatous polyposis, or to protect a distal anastomosis. Output is liquid and enzyme-rich, requiring careful fluid

and skin management.

Ileum

- Longest distal segment (3.5 m) with thinner walls, shorter villi, and Peyer's patches on the antimesenteric border. Absorbs vitamin B12, bile salts, remaining nutrients. Ileocecal valve prevents colonic backflow. Crohn's disease most commonly affects the terminal ileum; resection may cause B12 deficiency and bile salt malabsorption.

Ileus

- Failure of coordinated intestinal propulsion without a mechanical blockage, usually from postoperative bowel inhibition, inflammation, metabolic disturbance, or medication effect. It causes distension, nausea, vomiting, and reduced passage of flatus or stool.

Iliotibial Band Syndrome

- Lateral knee pain from IT band friction over lateral femoral epicondyle. Common in runners. Treatment: rest, stretching, strengthening, biomechanical correction.

Ilium

- Largest hip bone forming the superior portion. Body contributes to acetabulum; wing has a concave iliac fossa. Iliac crest runs ASIS to PSIS, a palpable landmark for lumbar puncture level. Sacroiliac joint articulates with sacrum. Gluteal muscles originate from its external surface; iliacus fills its fossa.

Illness Anxiety Disorder

- Preoccupation with having or acquiring a serious illness, with minimal or no somatic symptoms present. The individual has a high level of anxiety about health, performs excessive health-related behaviors, and experiences maladaptive avoidance. Symptoms persist for at least six months. The individual is not delusional and insight may vary.

Image Display

- Transaxial, coronal, sagittal, MIP, fused images. Windowing for optimal visualization.

Image Processing

- Post-acquisition manipulation: filtering, reconstruction, correction.

Image Quality

- Assessment of diagnostic adequacy. Factors: resolution, contrast, noise, artifacts.

Image Reconstruction

- Mathematical process converting projection data to cross-sectional images. Iterative methods (OSEM) superior to filtered back projection. Corrects for attenuation, scatter, resolution.

Image-Guided Ablation

- Image-guided destruction of tissue. See ablation therapy.

Image-Guided Biopsy

- Biopsy performed under imaging guidance. CT, ultrasound, MRI, or fluoroscopy. Improves accuracy, safety.

Image-Guided Drainage

- Placement of a catheter or needle under imaging guidance to evacuate fluid, pus, blood, or bile from a pathologic collection. Indications include abscess, biloma, urinoma, pleural collection, and symptomatic cysts.

Image-Guided Therapy

- Image-guided minimally invasive procedures. See interventional radiology.

Imaging

- See radiology.

Imaging-Guided Biopsy

- Tissue acquisition directed by ultrasound, CT, MRI, or fluoroscopy to target a lesion while avoiding critical structures. The procedure requires appropriate specimen handling, informed consent, bleeding-risk assessment, and post-procedure monitoring.

Iminoacetic Acid Scan

- Renal imaging using Tc-99m iminodiacetic acid. Evaluates renal function, secretion.

Immediate Hypersensitivity

- A rapid, typically immunoglobulin-E-mediated hypersensitivity reaction in which allergen cross-linking of mast-cell or basophil receptors releases inflammatory mediators. It can cause urticaria, bronchospasm, gastrointestinal symptoms, or anaphylaxis.

Immune and Infectious Diseases

- A medical field encompassing disorders of the immune system and illnesses caused by pathogenic microorganisms. Immune system disorders include autoimmune diseases (e.g., lupus, rheumatoid arthritis), immunodeficiencies, and hypersensitivity reactions.

Immune Checkpoint

- An inhibitory or stimulatory receptor-ligand pathway that regulates lymphocyte activation and limits tissue injury. Therapeutic blockade of inhibitory checkpoints such as CTLA-4 or PD-1 can enhance antitumor immunity but may cause immune-related adverse effects.

Immune Complex

- A soluble or tissue-associated aggregate formed by antigen bound to antibody. Persistent or inadequately cleared immune complexes may activate complement and deposit in tissues, producing type III hypersensitivity and vasculitic or renal injury.

Immune Evasion

- Cancer mechanisms to avoid immune detection. Examples: PD-L1 expression, antigen loss.

Immune Hemolytic Anemia

- Anemia caused by immune-mediated destruction of red blood cells, either by autoantibodies or antibodies directed against foreign red-cell antigens. It may cause fatigue, jaundice, dark urine, splenomegaly, and laboratory evidence of hemolysis.

Immune Response

- The coordinated reaction of the immune system to foreign substances (antigens), pathogens, or abnormal cells. The innate immune response acts rapidly and non-specifically (physical barriers, phagocytes, complement, NK cells, inflammation).

Immune Thrombocytopenia

- An autoimmune disorder characterized by isolated thrombocytopenia (platelet count $<100 \times 10^9/L$) due to autoantibody-mediated platelet destruction and impaired platelet production. Primary ITP (no identifiable cause) vs secondary (SLE, CLL, HIV, HCV, drugs).

Immune Thrombocytopenic Purpura

- Autoimmune destruction of platelets by antiplatelet antibodies, causing isolated thrombocytopenia. Most common in children 2-6 years, often following viral infection. Presents with petechiae, purpura, and mucosal bleeding. Diagnosis of exclusion.

Immune Thrombocytopenic Purpura

(ITP)

- An autoimmune bleeding disorder characterized by isolated thrombocytopenia (platelet count $<100,000/L$) with normal white blood cell count and hemoglobin. Caused by autoantibodies (primarily IgG) directed against platelet surface glycoproteins (IIb/IIIa or Ib/IX), leading to splenic macrophage phagocytosis of platelets.

Immune Tolerance

- The state of unresponsiveness of the immune system to self-antigens and to specific foreign antigens, preventing autoimmunity while preserving responses to pathogens. Central tolerance: deletion of self-reactive lymphocytes in the thymus (T cells) and bone marrow (B cells).

Immune-Mediated Necrotizing Myopathy

- A subacute severe proximal myopathy with very high CK (often $>10,000$), minimal inflammation on biopsy, necrotic fibers. Two major autoantibodies: anti-HMGCR (3-hydroxy-3-methylglutaryl-CoA reductase, statin-associated or spontaneous) and anti-SRP (signal recognition particle).

Immune-Related Adverse Event (irAE)

- An inflammatory or autoimmune toxicity caused by immune checkpoint blockade or another immune-activating cancer treatment. It may affect skin, bowel, liver, endocrine organs, lungs, heart, kidneys, or nervous system and often requires treatment-specific immunosuppression.

Immunization

- Routine childhood (CDC schedule): HepB, RV, DTaP, Hib, PCV, IPV, Influenza, MMR, VAR, HepA, HPV, MenACWY, MenB, COVID19. Adult: Tdap/Td q10y, influenza annually, zoster (RZV 2-dose), pneumococcal (PCV20 or PCV15+PPSV23), HPV (through 26, shared decision 27-45), HepA/B, MenACWY/MenB (riskbased), COVID-19.

Immunocompromised Host

- A person whose immune defenses are impaired by congenital disease, infection, malignancy, immunosuppressive treatment, organ failure, or another condition. Such patients have increased susceptibility to opportunistic infection, severe disease, and atypical presentations.

Immunodeficiency

- Impaired development or function of one or more components of the immune system, resulting in increased susceptibility to infection, malignancy, or autoimmunity. It may be primary and genetic or secondary to disease, medication, malnutrition, or another acquired cause.

Immunodeficiency and infections

- Immunocompromised individuals are at increased risk for opportunistic infections. Primary immunodeficiencies are genetic disorders affecting immune function. Secondary immunodeficiencies include HIV/AIDS, organ transplantation, chemotherapy, and corticosteroid use.

Immunodeficiency Diseases

- Disorders in which one or more components of the immune system are absent, deficient, or dysfunctional. They may be inherited or acquired and can cause recurrent, severe, unusual, or persistent infections, autoimmune disease, and increased malignancy risk.

Immunofluorescence

- A technique using fluorescently labeled antibodies to detect specific antigens in tissues, cells, or microorganisms; visualized by fluorescence microscopy.

Immunoglobulin (Antibody)

- Y-shaped glycoprotein produced by plasma cells that specifically binds antigens. Composed of two identical heavy chains and two identical light chains linked by disulfide bonds. Variable regions (Fab) bind antigen; constant region (Fc) mediates effector functions (complement activation, Fc receptor binding).

Immunoglobulin Replacement Therapy

- Indications: primary antibody deficiencies (XLA, CVID, hyper-IgM, specific antibody deficiency), secondary hypogammaglobulinemia (CLL, myeloma, post-rituximab, post-transplant) with recurrent infections.

Immunohistochemistry

- Technique using antibodies to detect specific antigens in tissue sections.

Immunohistochemistry (IHC)

- A technique that uses labeled antibodies to detect specific antigens in tissue sections. It supports

tumor classification, identification of lineage, assessment of predictive biomarkers, and detection of infectious or immune-mediated processes.

Immunological Memory

- The ability of the adaptive immune system to mount a faster, larger, and more effective response upon re-exposure to an antigen, conferred by long-lived memory B cells and memory T cells generated during the primary response.

Immunomodulators in Dermatology

- Topical immunomodulators treat inflammatory skin diseases without corticosteroid side effects. Pimecrolimus 1 percent cream and tacrolimus 0.03 and 0.1 percent ointments are calcineurin inhibitors used for atopic dermatitis, facial eczema, and intertriginous areas. They are steroid-sparing agents.

Immunophenotyping

- Identification and characterization of cells according to their surface, cytoplasmic, or nuclear antigens, usually by flow cytometry or immunohistochemistry. It is used to classify leukemias, lymphomas, immune-cell subsets, and some immunodeficiency disorders.

Immunosenescence

- Age-related changes in innate and adaptive immunity that alter infection susceptibility, vaccine responses, inflammation, and cancer surveillance. It is influenced by chronic disease, nutrition, medications, and functional status.

Immunosuppressant

- A drug that reduces immune system activity, used to prevent organ transplant rejection and to treat autoimmune and inflammatory diseases.

Immunosuppression in Transplant

- Medications to prevent rejection. Induction with thymoglobulin or basiliximab. Maintenance with tacrolimus, mycophenolate, and prednisone. Monitor drug levels and renal function. Calcineurin inhibitor nephrotoxicity is major concern requiring dose optimization.

Immunosuppression Screening

- Pre-treatment evaluation for latent or active infection, vaccination status, malignancy risk, and organ dysfunction before immunosuppressive therapy. Testing commonly includes tuberculosis, hepatitis B and C, HIV, blood counts, liver and kidney function,

and pregnancy when relevant.

Immunosuppressive Drug

- A drug that suppresses immune responses, used for organ transplantation, graft-versus-host disease, and autoimmune/inflammatory diseases.

Immunotherapy

- Treatment using immune system to fight cancer. Types: checkpoint inhibitors (PD-1, PD-L1, CTLA-4), CAR T-cell therapy, cytokines, cancer vaccines. Examples: pembrolizumab, nivolumab, ipilimumab, axicabtagene ciloleucel.

Immunotherapy and Biologic Treatments

- Advanced therapies that harness or modify the immune system to treat diseases, particularly cancer, autoimmune disorders, and inflammatory conditions. Biologics are large-molecule drugs derived from living organisms that target specific immune pathways, such as TNF inhibitors, monoclonal antibodies, and checkpoint inhibitors.

Immunotherapy in Dermatology

- Cancer immunotherapy has revolutionized skin cancer treatment. Checkpoint inhibitors (pembrolizumab, nivolumab, ipilimumab) block PD-1 or CTLA-4, enhancing T-cell mediated tumor killing. They are first-line for advanced melanoma and metastatic SCC. Cemiplimab is approved for advanced BCC and SCC.

Immunotoxicity

- Immune system damage from chemicals. Effects: immunosuppression, hypersensitivity, autoimmunity. Examples: benzene, chemotherapy, biologics.

Impacted Tooth

- Tooth unable to erupt into functional position. Most common: mandibular third molars, maxillary canines. May cause pathology: cyst, root resorption, caries. Treatment: monitoring or extraction.

Impedance hearing testing

- A test that sends sound waves to the eardrum to determine if a problem in the middle ear is causing hearing loss.

Impella Device

- Percutaneous microaxial left ventricular assist device that actively pumps blood from the LV into the aorta, reducing LV workload and increasing cardiac output. Types: Impella 2.5 (up to 2.5 L/min),

Impella CP (up to 3.7 L/min), Impella 5.0 (up to 5 L/min, surgically placed).

Imperforate Anus

- A congenital anomaly where the anus is completely absent or fails to open at the perineum. Classified as low (distal to the levator ani complex) or high (proximal to the levator ani). Often associated with other anomalies including vertebral, anal, cardiac, tracheoesophageal, renal, and limb defects.

Imperforate Hymen

- A congenital obstruction of the vaginal opening by an intact hymen without a normal central opening. After puberty it may cause absent menstruation, cyclic pelvic pain, urinary retention, or a pelvic mass from retained menstrual blood.

Impetigo

- A highly contagious, superficial bacterial infection of the epidermis, primarily affecting children.

Implant (Dental)

- Titanium fixture placed in jawbone to support prosthetic tooth. Osseointegration occurs over 3-6 months. Indications: single tooth replacement, partial edentulism, complete edentulism (implant-supported denture). Success rate: >95% at 10 years. Complications: peri-implantitis, mechanical failure, aesthetic issues.

Implant Failure

- Loss of an implant's intended structural, biologic, or functional performance. Causes include infection, fracture, loosening, wear, malposition, thrombosis, material degradation, and patient or tissue factors.

Implant Sterilization

- Validated processing that removes or inactivates all viable microorganisms, including bacterial spores, from an implant. Methods include moist heat, ethylene oxide, radiation, and low-temperature plasma according to material and device requirements.

Implant-Associated Infection

- Infection involving an implanted medical device or the tissue surrounding it. Microbial adherence and biofilm formation can impair antimicrobial penetration and often require device-directed therapy, source control, or removal.

Implant-Supported Prosthesis

- Prosthetic tooth supported by dental implants.

Fixed or removable. Superior to conventional dentures.

Implantable Cardioverter Defibrillator (ICD)

- An implanted electronic device that delivers electrical impulses to the heart to regulate heart rate and rhythm. See pacemaker.

Implantable Cardioverter-Defibrillator

- An implanted device that monitors heart rhythm and delivers electrical therapy (antitachycardia pacing or defibrillation) for life-threatening ventricular arrhythmias.

Implantable Loop Recorder

- Subcutaneous device providing continuous ECG monitoring for up to 3 years. Used for evaluation of unexplained syncope, recurrent palpitations, cryptogenic stroke, and occult atrial fibrillation detection. Small size (coin-sized), inserted under local anesthesia in the chest.

Implantable Port

- A subcutaneously placed vascular-access device consisting of a reservoir connected to a catheter, used for repeated administration of medication, fluids, blood products, or parenteral nutrition. It requires sterile access and may be complicated by infection, thrombosis, occlusion, or mechanical failure.

Implantation

- The embedding of the blastocyst into the endometrium, occurring approximately 6-10 days after ovulation. The blastocyst hatches from the zona pellucida, and the trophoblast invades the decidua. Implantation triggers hCG production, which maintains the corpus luteum. Failure of implantation is a common cause of early pregnancy loss.

Impression

- Summary of key findings and recommendations in radiology report.

imprinting

- Parent-of-origin-specific gene expression. See genomic imprinting.

Impulse Oscillometry

- A pulmonary function testing technique that measures respiratory impedance by applying small pressure oscillations during tidal breathing. It assesses large airway resistance (R5), small airway resistance

(R5-R20), and reactance.

In Situ Carcinoma

- Full-thickness dysplasia without invasion through basement membrane. See carcinoma in situ.

In Vitro Fertilisation (IVF)

- An assisted reproductive technology in which eggs are retrieved from the ovaries, fertilized with sperm in a laboratory, and resulting embryos are transferred to the uterus.

Inactivated Polio Vaccine (IPV)

- Salk vaccine, a formalin-inactivated preparation containing all three poliovirus serotypes (1, 2, and 3), administered by intramuscular injection. The only polio vaccine used in the United States since 2000, and the recommended vaccine for immunocompromised individuals and pregnant women worldwide.

Inborn Error of Metabolism

- See biochemistry.

Incentive Spirometry

- A device that encourages deep breathing to prevent atelectasis after surgery. The patient inhales slowly and deeply through the device, maintaining maximal inspiration for several seconds. It promotes alveolar recruitment and prevents mucus plugging. It is standard post-operative care.

Incidence

- Number of new cases of a disease in a population during a specified time period. Measures risk of developing disease. Incidence rate = new cases / population at risk x time unit.

Incidental Finding

- Incidental imaging finding of uncertain significance. See incidentaloma.

Incidental Findings

- See incidental finding.

Incidentaloma

- Incidental imaging finding of uncertain significance. May require further evaluation.

Incised Wound

- Clean-cut wound produced by a sharp-edged instrument (scalpel, knife, glass) that is longer on the skin surface than it is deep, featuring clean edges without tissue bridging or marginal abrasion.

Incisional Biopsy

- Surgical removal of part of lesion. Diagnosis before definitive treatment.

Incisional Hernia

- A hernia through a previous surgical incision scar, the most common long-term complication of abdominal surgery (10-20% of laparotomies). Risk factors: wound infection, obesity, smoking, malnutrition, COPD, and multiple previous surgeries. Present as a bulge at or near the incision site, worse with standing or straining.

Inclusion Body Myositis

- The most common acquired myopathy after age 50, characterized by asymmetric distal and proximal weakness (finger flexors, quadriceps), rimmed vacuoles, 15-18 nm tubulofilaments, TDP-43 pathology, endomysial inflammation with CD8+ T cells invading non-necrotic fibers. Refractory to immunotherapy.

Incontinence

- The involuntary loss of control over urination (urinary incontinence) or defecation (fecal incontinence).

Incontinence (Fecal)

- Involuntary fecal leakage. Causes: diarrhea, constipation with overflow, sphincter damage, neurological disease, medications. Assessment: history, examination, anorectal manometry. Treatment: bowel regimen, sphincter exercises, biofeedback, surgery if indicated.

Incontinence (Urinary)

- Involuntary urine leakage. Types: stress (cough/sneeze), urge (overactive bladder), overflow (retention), functional (immobility/cognition), mixed. Assessment: history, bladder diary, urinalysis, post-void residual.

Incontinentia Pigmenti

- An X-linked dominant disorder affecting skin, teeth, eyes, and the nervous system, caused by pathogenic variants in IKBKG. Skin lesions evolve through vesicular, verrucous, hyperpigmented, and atrophic stages and may be associated with retinal disease or seizures.

Increased Intracranial Pressure

- Abnormally elevated pressure within the skull

caused by brain swelling, hemorrhage, mass lesion, impaired cerebrospinal-fluid circulation, or venous obstruction. It can cause headache, vomiting, visual disturbance, reduced consciousness, and brain herniation.

Increased Vascular Permeability

- A fundamental feature of acute inflammation that allows plasma proteins, fluid, and leukocytes to exit the vascular space and enter the extravascular tissue.

Incubation period

- The time between when a person is exposed to an infection and when symptoms appear.

Incubator

- A controlled-environment chamber culturing microorganisms, cells, or tissues at defined temperature, humidity, and atmospheric composition.

Incus

- The incus, commonly known as the anvil, is the middle ossicle of the three bones in the middle ear, articulating with the malleus at the incudomalleolar joint and with the stapes at the incudostapedial joint.

Independence

- Ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs). See functional status.

India Immunization Schedule

- The Universal Immunization Schedule (UIP) of the Government of India, providing free vaccines to all children, adolescents, and pregnant women through the public health system.

Indication

- Clinical reason for imaging in radiology report.

Indigestion

- Dyspepsia characterized by recurrent upper-abdominal discomfort, early satiety, postprandial fullness, bloating, or nausea. It may result from functional dyspepsia, gastroesophageal reflux, peptic ulcer disease, medications, or other gastrointestinal disorders.

Indirect Antiglobulin Test (IAT)

- A test that detects free, clinically significant antibodies in a patient's serum or plasma by incubating

it with reagent erythrocytes. It is used for antibody screening, compatibility testing, and prenatal red-cell antibody evaluation.

Indirect Calorimetry

- Measurement of oxygen consumption and carbon-dioxide production to estimate resting energy expenditure. It is the preferred method for determining energy needs when prediction equations are unreliable, particularly in critically ill or metabolically complex patients.

Indirect Veneer

- Ceramic restoration fabricated in lab. Porcelain or composite. Aesthetic enhancement.

Indium-111

- SPECT radionuclide, half-life 2.8 days. Used in WBC labeling, octreotide scanning. Emits gamma rays at 171 and 245 keV.

Indium-111 Scan

- Nuclear medicine imaging using In-111. Used for WBC labeling, octreotide scanning.

Induced Sputum Analysis

- A non-invasive method of assessing airway inflammation by examining sputum cell counts and differentials obtained after hypertonic saline inhalation. Sputum eosinophils > 3% indicate eosinophilic airway inflammation (asthma, EGPA). Neutrophilic inflammation suggests infection, COPD, or bronchiectasis.

Induction of Labor

- The artificial initiation of uterine contractions before the onset of spontaneous labor.

Induction Therapy

- Intensive initial treatment aimed at achieving remission, especially in acute leukemias.

Infant Botulism

- A neuromuscular illness caused by intestinal colonization with *Clostridium botulinum* and production of botulinum neurotoxin in infants. It causes constipation, poor feeding, weak cry, hypotonia, facial weakness, and descending paralysis.

Infant Feeding

- Provision of breast milk, appropriately prepared formula, or both to meet an infant's energy, protein, fluid, and micronutrient requirements. Feeding

recommendations depend on age, growth, medical conditions, developmental readiness, and safe preparation practices.

Infant harlequin syndrome

- A transient, benign cutaneous vascular phenomenon in neonates characterized by asymmetric flushing of one half of the body with sharp midline demarcation, caused by transient asymmetric sympathetic tone. The affected side is red and warm while the contralateral side is pale.

Infant Sleep Position

- The supine sleep position (back to sleep) is recommended by the AAP to reduce the risk of sudden infant death syndrome (SIDS). The Back to Sleep campaign (1994) reduced SIDS rates by >50%. Always place the baby on the back for sleep (naps and nighttime). Use a firm sleep surface with no soft bedding, pillows, bumpers, or stuffed animals.

Infanticide

- The killing of a child, in law typically defined as the killing of a newborn or infant (usually within the first year, or specifically within 24 hours of birth in some jurisdictions) by its mother while mentally disturbed by the effects of childbirth or lactation.

Infantile Colic

- A behavioral syndrome in healthy infants characterized by episodes of excessive, inconsolable crying for >3 hours per day, >3 days per week, for >3 weeks (Wessel criteria). Onset typically at 2-3 weeks, peaks at 6-8 weeks, and resolves by 3-4 months of age.

Infantile hepatic hemangiomas

- Benign vascular tumors of the liver occurring in infants, typically presenting within the first six months of life with hepatomegaly, high-output cardiac failure, consumptive hypothyroidism, or cutaneous hemangiomas. Large tumors may cause abdominal compartment syndrome and thrombocytopenia from Kasabach-Merritt phenomenon.

Infarction

- Tissue death from prolonged ischemia. Examples: myocardial infarction, cerebral infarction.

Infection

- Measures to prevent healthcare-associated infections: hand hygiene, personal protective equipment (PPE), isolation precautions (contact, droplet, air-

borne), sterilization, disinfection. See infection control.

Infection (Geriatric)

- Infectious diseases in elderly often present atypically: fever may be absent, confusion may be sole symptom. Common infections: UTI, pneumonia, skin/soft tissue, bloodstream. Prevention: vaccination, hand hygiene, aspiration precautions. Treatment: earlier presentation, broader differential, drug interactions, renal dosing.

Infection Control

- Measures to prevent healthcare-associated infections: hand hygiene, personal protective equipment (PPE), isolation precautions (contact, droplet, airborne), sterilization, disinfection. Multidrug-resistant organisms: MRSA, VRE, *C. difficile*, carbapenem-resistant Enterobacteriaceae.

Infection Imaging

- WBC scanning, gallium scanning, FDG PET. Detection of occult infection.

Infection Prevention

- Measures to prevent healthcare-associated infections: hand hygiene, personal protective equipment (PPE), isolation precautions (contact, droplet, airborne), sterilization, disinfection. See infection control.

Infections in Pregnancy

- Listeriosis: *Listeria monocytogenes* (deli meats, soft cheese), meningoenitis, fetal loss, neonatal sepsis. Ampicillin + gentamicin. GBS: vaginal colonization 20-30 percent of pregnant women; universal screening at 35-37 weeks; intrapartum prophylaxis (penicillin G) for GBS-positive women to prevent early-onset neonatal disease.

Infectious Disease

- Disease caused by pathogen. Examples: bacterial, viral, fungal, parasitic.

Infectious Dose

- The number of viable organisms required to establish infection in a host under specified conditions. The dose depends on the pathogen, route of exposure, host defenses, and organism viability.

Infectious Esophagitis

- Inflammation of the esophagus caused by an infectious organism, most often *Candida*, herpes simplex

virus, or cytomegalovirus in immunocompromised patients. It commonly causes painful swallowing, difficulty swallowing, and retrosternal pain.

Infectious Myringitis

- Inflammation and infection of the tympanic membrane, often associated with viral or bacterial respiratory infection. It can cause severe ear pain, fever, hearing reduction, and small fluid-filled blisters on the eardrum.

Infective Endocarditis

- Infection of the endocardial surface of the heart, most commonly affecting the heart valves. It is diagnosed using the Duke criteria, which include major criteria (positive blood cultures, echocardiographic evidence of vegetations) and minor criteria (predisposing heart condition, fever, vascular phenomena, immunologic phenomena).

Inferior

- Positioned lower or closer to the feet.

Inferior Alveolar Nerve Block

- A regional anesthetic injection near the mandibular foramen to anesthetize the inferior alveolar nerve before it enters the mandibular canal. It usually anesthetizes mandibular teeth on that side; the lingual nerve is often anesthetized incidentally.

Inferior hemifield loss

- Loss of vision in the lower half of the visual field in one or both eyes, which localises to lesions of the superior retina or superior optic radiations passing through the parietal lobe. Causes include superior retinal detachment, branch retinal artery occlusion, and parietal lobe stroke.

Inferior Mesenteric Artery

- Third major abdominal aortic branch at L3, the hindgut artery. Supplies distal transverse colon through rectum. Branches: left colic, sigmoid, superior rectal. The marginal artery of Drummond provides SMA-IMA collateral circulation. Occlusion may cause ischemic colitis at the splenic flexure watershed (Griffiths' point).

Inferior Vena Cava

- Body's largest vein, formed by common iliac vein union at L5, returning lower body deoxygenated blood to the right atrium. Ascends right of the aorta, pierces the diaphragm at T8, enters the pericardium.

Receives hepatic, renal, gonadal, and lumbar veins. IVC thrombosis/compression causes bilateral lower extremity edema and DVT.

Infertility

- Inability to conceive after 12 months of regular unprotected intercourse (6 months if woman >35). Affects 15% of couples. Female causes: ovulation disorders (PCOS, hypothalamic amenorrhea), tubal factor (blockage, PID, endometriosis), uterine (fibroids, polyps, adhesions), and diminished ovarian reserve.

Infiltrate

- Patchy lung opacity detected on imaging. Non-specific, may represent infection, edema, hemorrhage, inflammation.

Inflammaging

- Persistent, low-grade, age-associated inflammatory activity linked to cellular senescence, immune dysregulation, chronic infection, tissue injury, and comorbidity. It contributes to frailty, atherosclerosis, sarcopenia, and neurodegenerative disease.

Inflammasomes

- Multiprotein complexes activating caspase-1 → cleavage of pro-IL-1β/pro-IL-18 → mature IL-1β/IL-18 + pyroptosis (gasdermin D pores). Canonical: NLRP3 (crystals, ATP, K⁺ efflux, ROS, lysosomal rupture), NLRC4 (flagellin, T3SS), AIM2 (cytosolic DNA), Pyrin (RhoA inactivation).

Inflammation

- The body's protective response to injury or infection. Cardinal signs: redness (rubor), heat (calor), swelling (tumor), pain (dolor), loss of function (functio laesa). Mediators: histamine, prostaglandins, cytokines, complement. Acute: neutrophil-predominant. Chronic: lymphocyte/macrophage-predominant.

Inflammation (Dietary)

- Chronic low-grade inflammation linked to diet. Western diet pro-inflammatory (saturated fat, sugar, refined grains). Mediterranean diet anti-inflammatory.

Inflammation Imaging

- FDG PET sensitive for inflammatory processes. Differentiate infection from malignancy challenging.

Inflammatory Back Pain

- A pattern of chronic back pain characterized by insidious onset, improvement with exercise, lack of improvement with rest, nocturnal pain, and prolonged morning stiffness. It suggests axial spondyloarthritis but is not diagnostic without supportive clinical, laboratory, and imaging findings.

Inflammatory Bowel Disease

- Chronic relapsing inflammatory conditions of the gastrointestinal tract encompassing Crohn disease and ulcerative colitis, resulting from a dysregulated immune response in genetically susceptible individuals triggered by environmental factors and alterations in the gut microbiome.

Inflammatory Bowel Disease (IBD)

- A group of chronic inflammatory conditions affecting the gastrointestinal tract, primarily Crohn's disease and ulcerative colitis. Crohn's disease can involve any part of the digestive tract with transmural inflammation, while ulcerative colitis is confined to the colonic mucosa.

Inflammatory Bowel Disease Diet

- Crohn disease, ulcerative colitis. Nutritional support during flares. Exclusive enteral nutrition for Crohn induction. Individual triggers vary.

Inflammatory Myopathies

- A group of autoimmune disorders characterized by chronic muscle inflammation and progressive proximal muscle weakness. The three main types are dermatomyositis, polymyositis, and inclusion body myositis.

Influenza

- A contagious respiratory illness caused by influenza A or B viruses, characterized by sudden onset of fever, myalgia, headache, malaise, nonproductive cough, sore throat, and nasal congestion. Seasonal epidemics occur annually. Complications: pneumonia (primary viral or secondary bacterial), exacerbation of underlying conditions.

Influenza Test

- Detection of influenza A and B viruses. Rapid antigen test: nasal/throat swab, results in 15 minutes, specificity >95%, sensitivity 50-70% (high false negative rate). RT-PCR: gold standard, highly sensitive and specific, results in 1-4 hours, distinguishes

influenza A subtypes (H1N1, H3N2).

Influenza Vaccination in Pregnancy

- Annual inactivated influenza vaccine recommended for all pregnant women. Pregnancy increases the risk of severe influenza (pneumonia, ICU admission, death) due to changes in immune function and cardiopulmonary physiology.

Influenza Vaccine

- Inactivated influenza vaccine given annually starting at 6 months of age. Two doses needed in the first season for children under 9. Live attenuated nasal spray vaccine (FluMist) available for children over 2. Updated annually to match circulating strains. Reduces influenza complications, hospitalizations, and deaths.

Influenza virus

- A segmented, negative-sense RNA virus belonging to the Orthomyxoviridae family. Influenza causes seasonal epidemics and occasional pandemics of respiratory illness. The virus has two surface glycoproteins: hemagglutinin (HA), which mediates attachment, and neuraminidase (NA), which facilitates release.

Informed Consent

- Autonomous voluntary authorization given by a competent patient to undergo a specific medical intervention, procedure, or clinical trial.

Informed Consent (Geriatric)

- Process of obtaining voluntary authorization from elderly patient with decision-making capacity. Capacity assessment: understanding, appreciation, reasoning, communication of choice. Impaired capacity may require surrogate decision-maker.

Infusion Pump

- Device controlling IV fluid delivery rate. Types: syringe pump, volumetric pump, elastomer. Precision: mL/hour. Alarms: occlusion, air embolism, completion.

Ingestion

- Harmful effect from exposure to toxic substance. See poisoning.

Ingrown Toenail

- Ingrown toenail (onychocryptosis) occurs when the nail edge grows into the surrounding skin, causing pain, redness, and swelling. It is common in

adolescents and adults, exacerbated by improper nail trimming, tight shoes, and trauma. Treatment includes partial nail avulsion with matrixectomy (phenol application) for recurrent cases.

Inguinal Canal

- An oblique passage (4 cm) through the anterior abdominal wall, from the deep (internal) inguinal ring to the superficial (external) inguinal ring, transmitting the spermatic cord in males and round ligament in females, plus ilioinguinal and genital branch of genitofemoral nerves.

Inguinal Hernia

- Protrusion of abdominal contents through the inguinal canal. Indirect (lateral to inferior epigastric vessels, through deep inguinal ring): congenital, patent processus vaginalis; most common overall (60-70%), male predominance.

Inguinal Region

- Area at the thigh-trunk junction traversed by the inguinal ligament (ASIS to pubic tubercle). Clinically significant for inguinal hernias: indirect (through deep inguinal ring, most common) and direct (through Hesselbach's triangle).

Inhalation Injury

- Thermal or chemical injury to the airway or lungs caused by smoke, heated gases, or toxic combustion products. It can produce upper-airway edema, bronchospasm, impaired mucociliary clearance, and carbon-monoxide or cyanide toxicity and may worsen after the initial burn.

Inhalational Anesthesia

- Anesthetic maintenance or induction using volatile liquids or gases delivered through the respiratory tract. Depth is influenced by inspired concentration, alveolar ventilation, uptake, distribution, and the agent's blood-gas solubility.

Inhaled Corticosteroids

- Anti-inflammatory medications delivered directly to the airways, reducing airway inflammation, mucus production, and bronchial hyperresponsiveness. Examples include fluticasone, budesonide, beclomethasone, and mometasone. They are the cornerstone of asthma maintenance therapy. Side effects include oral candidiasis and dysphonia.

Inhaled Insulin

- Rapid-acting insulin delivered via inhalation as a powder using the Afrezza device. Approved for mealtime insulin coverage in adults with diabetes. Onset of action within twelve minutes and duration of approximately three hours providing a more physiological postprandial insulin profile than subcutaneous rapid-acting analogs.

Inhaled Nitric Oxide

- A selective pulmonary vasodilator used in acute settings (acute respiratory failure, right heart failure, post-cardiac surgery). It reduces pulmonary vascular resistance and improves oxygenation by dilating well-ventilated pulmonary vessels (improving V/Q matching). It has a short half-life and requires continuous delivery.

Inhaler

- Device for administering aerosolized medication. Types: MDI (metered-dose), DPI (dry powder).

Iniencephaly

- A rare, usually lethal neural-tube defect characterized by occipital bone defect, severe retroflexion of the head, and marked shortening or distortion of the cervical and thoracic spine. It is often identified prenatally and may occur with other congenital anomalies.

Injury Classification (Abrasion, Laceration, Contusion, Incised Wounds)

- Forensic categorization of mechanical wounds. Abrasion: superficial friction injury affecting only the epidermis (e.g., scrape, graze, pattern abrasion).

Injury Dating

- Estimation of the age of an injury using clinical appearance, histopathology, imaging, and the history of symptoms. Exact dating is often uncertain, particularly for bruises and soft-tissue injuries, and findings should be reported with appropriate limitations.

Innate immune evasion

- Mechanisms by which pathogens evade the innate immune system. Strategies include capsules that prevent phagocytosis, biofilms that protect against immune cells, and intracellular survival within macrophages. Some bacteria produce enzymes that degrade complement components. Oth-

ers modify their surface molecules to avoid recognition.

Innate Immunity

- First line of defense against pathogens, present from birth. Rapid (minutes to hours), germline-encoded, lacks immunological memory.

Innate Lymphoid Cells

- ILCs: lymphoid lineage, no rearranged antigen receptors, mirror Th subsets. ILC1 (T-bet, IFN-gamma): intracellular pathogens, IBD. ILC2 (GATA3, IL-5/13): helminths, allergy, asthma, tissue repair. ILC3 (ROR-gammat, IL-17/22): extracellular bacteria/fungi, lymphoid organogenesis.

Inner ear

- The deepest part of the ear, consisting of the cochlea and the labyrinth.

Inoculation

- The introduction of microorganisms or clinical material into a culture medium, host, or sterile site to initiate growth, infection, or immune response.

Inotrope

- A drug that increases myocardial contractility (positive inotropy), used for cardiogenic shock, decompensated heart failure, and perioperative low-output states. Agents: dobutamine (beta-1 agonist), milrinone (PDE3 inhibitor), dopamine, epinephrine, norepinephrine (also vasopressors), and digoxin (chronic).

Inotropes in Anesthesia

- Inotropes increase myocardial contractility. Dobutamine is a beta-1 and beta-2 agonist increasing cardiac output with minimal effect on heart rate. Dose: 2 to 20 mcg/kg/min infusion. Milrinone is a phosphodiesterase-3 inhibitor increasing contractility and causing vasodilation.

Inpatient Rehabilitation

- Intensive inpatient rehabilitation (3+ hours therapy/day) for acute conditions: stroke, spinal cord injury, traumatic brain injury, major trauma. See acute rehabilitation.

INR

- International normalized ratio: a standardized measure of the prothrombin time (PT), calculated as $(\text{patient PT}/\text{mean normal PT})^{\text{ISI}}$, to ensure consistent monitoring of vitamin K antagonist (warfarin)

therapy across laboratories. Target INR: 2-3 for most indications (AF, VTE); 2.5-3.5 for mechanical heart valves.

Insect bite

- A cutaneous reaction resulting from the injection of salivary contents from an insect during feeding, causing a localized or systemic hypersensitivity response. Common presentations include a pruritic papule or wheal with surrounding erythema. Severe reactions may include anaphylaxis, particularly with Hymenoptera stings.

Insomnia

- A sleep disorder characterized by difficulty falling asleep, staying asleep, or waking too early despite adequate opportunity for sleep, with resulting daytime impairment. It may be acute or chronic and is often triggered by stress, anxiety, depression, or poor sleep habits.

Insomnia Disorder

- Difficulty initiating or maintaining sleep, or early-morning awakening with inability to return to sleep, at least three nights per week for at least three months. The individual experiences clinically significant distress or impairment in functioning.

Inspection

- The first step of the physical examination, involving systematic visual observation of the patient's overall appearance, posture, body habitus, gait, skin color, nutritional status, respiratory effort, and specific anatomical regions. Requires adequate lighting and proper patient exposure.

Inspiratory Crackles

- Fine, discontinuous sounds heard during inspiration, caused by the sudden opening of collapsed airways. They are a hallmark of interstitial lung disease (bibasilar, Velcro-like crackles) and pulmonary fibrosis. They are also heard in pneumonia, atelectasis, and heart failure.

Instrument Sterilization

- A validated process that destroys all forms of microbial life, including bacterial spores, on or within a medical instrument. Common methods include steam under pressure, ethylene oxide, hydrogen-peroxide plasma, and dry heat, selected according to device material and design.

Instrumental Activities of Daily Living (IADLs)

- More complex activities required for independent community living, including medication management, shopping, cooking, finances, communication, transportation, and household tasks.

Instrumental ADL

- Ability to perform activities of daily living (ADLs) and instrumental ADLs (IADLs). See functional status.

Insufficient Cervix

- Painless cervical shortening and dilation during the second trimester, also called cervical insufficiency. It can cause recurrent mid-trimester pregnancy loss or spontaneous preterm birth without regular uterine contractions.

Insulin

- Peptide hormone (51 amino acids) produced by beta cells of pancreatic islets. Secreted in response to hyperglycemia. Actions: promotes glucose uptake (GLUT4 translocation in muscle/adipose), glycogen synthesis (activates glycogen synthase), lipogenesis, protein synthesis; inhibits gluconeogenesis, glycogenolysis, lipolysis, proteolysis.

Insulin Aspart

- A rapid-acting insulin analog (modified amino acid sequence) with onset 10-20 minutes, peak 1-3 hours, duration 3-5 hours. Used to cover prandial glucose (injected at or just after meals) in type 1 and type 2 diabetes; also used in continuous subcutaneous insulin infusion (pump) and IV in acute settings.

Insulin Autoantibodies

- Autoantibodies directed against insulin or other pancreatic beta-cell antigens, including GAD, IA-2, and ZnT8 antibodies. Their presence supports autoimmune type 1 diabetes but a negative panel does not exclude it.

Insulin Glargine

- A long-acting basal insulin analog with a flat, peakless profile providing near-24-hour coverage, administered once daily (or twice in some patients). Formed by modification of human insulin producing slow, steady absorption from the injection depot. Used for type 1 and type 2 diabetes (basal component).

Insulin Injection Technique

- Proper technique for subcutaneous insulin injection includes selecting appropriate injection sites rotating between abdomen thighs buttocks and upper arms using appropriate needle length four to eight millimeters for most adults pinching skin for thin patients to avoid intramuscular injection rotating injection sites within an area

Insulin lipohypertrophy

- See Insulin-induced lipohypertrophy.

Insulin Pen Devices

- Pre-filled or reusable insulin pen devices providing convenient and accurate insulin delivery. Benefits include portability precision dosing discrete injection and reduced needle phobia.

Insulin Preparations

- Hormone synthesized by pancreatic beta cells that regulates glucose homeostasis by promoting glucose uptake in skeletal muscle and adipose tissue, suppressing hepatic glucose production, and inhibiting lipolysis.

Insulin Pump Therapy

- Continuous subcutaneous insulin infusion delivering rapid-acting insulin analog through a catheter placed subcutaneously typically in the abdomen. Provides basal insulin at programmed rates and bolus doses at mealtimes.

Insulin Resistance

- Diminished biological response to physiological insulin concentrations resulting in impaired glucose disposal in peripheral tissues and failure to suppress hepatic glucose production. Pathophysiological hallmark of type two diabetes and metabolic syndrome.

Insulin Resistance Syndromes

- Rare inherited severe insulin resistance syndromes include Donohue syndrome also known as leprechaunism, Rabson-Mendenhall syndrome, and type A insulin resistance syndrome caused by mutations in the insulin receptor gene. Donohue syndrome is the most severe with death in infancy.

Insulin Secretion

- Release of insulin from pancreatic beta cells in response primarily to increased plasma glucose, with amplification by incretin hormones and nutrients. Insulin promotes glucose uptake, glycogen synthesis,

lipogenesis, and protein synthesis while suppressing hepatic glucose production.

Insulin Signaling

- Insulin binds receptor, activates IRS proteins, PI3K, Akt pathway. Promotes glucose uptake, glycogen synthesis, lipogenesis.

Insulin-induced lipohypertrophy

- A common complication of subcutaneous insulin therapy characterized by localized accumulation of adipose tissue at injection sites, caused by the lipogenic effect of insulin on subcutaneous fat. It presents as firm, rubbery, non-tender subcutaneous nodules that can cause unpredictable insulin absorption and glycemic variability.

Insulin-like Growth Factor

- IGF-1 promotes growth. Produced in liver in response to GH.

Insulinoma

- A usually small pancreatic neuroendocrine tumour that secretes insulin inappropriately. It causes episodes of fasting or exertional hypoglycaemia with autonomic and neuroglycopenic symptoms that improve when blood glucose is raised.

Integrative Rehabilitation

- Combination of conventional and complementary therapies: acupuncture, massage, yoga, meditation.

Integrans

- Genetic elements that capture and express gene cassettes through site-specific recombination. Integrans contain an integrase gene and a recombination site for gene cassette insertion. They are important vehicles for the spread of antibiotic resistance genes, particularly in gram-negative bacteria.

Integumentary System

- The body system comprising the skin and its appendages: hair, nails, sweat glands, and sebaceous glands. It forms a protective barrier, limits fluid loss, contributes to thermoregulation and sensation, supports immune defence, and participates in vitamin D synthesis.

Intellectual Disability

- Neurodevelopmental disorder with onset in the developmental period, characterised by deficits in intellectual and adaptive functioning across conceptual, social, and practical domains. DSM-5 grades

severity by adaptive functioning: mild, moderate, severe, profound.

Intensity-Modulated Radiation Therapy (IMRT)

- A computer-optimized form of three-dimensional radiation therapy that shapes and modulates the radiation dose to conform tightly to the target while sparing adjacent normal tissue.

Intensive Care Medicine

- Specialty managing critically ill patients. Components: monitoring, ventilatory support, hemodynamic support, infection control, nutrition, sedation, pain management. Common conditions: sepsis, ARDS, traumatic brain injury, post-cardiac arrest, multiorgan failure.

Intensive Care Rehabilitation

- Early rehabilitation in ICU setting. Early mobilization, positioning, passive range of motion. Improves outcomes, reduces complications.

Intensive Care Unit

- A specialist hospital unit providing continuous monitoring and life support for critically ill patients. In obstetrics, women may require ICU admission for: severe pre-eclampsia or eclampsia, massive postpartum hemorrhage, amniotic fluid embolism, sepsis, cardiac disease in pregnancy, or complications of anesthesia.

Interatrial Septum

- The wall separating the right and left atria. It contains the fossa ovalis, a remnant of the foramen ovale from fetal circulation. The interatrial septum has a thinner area where transseptal puncture can be performed during cardiac catheterization.

Intercostal Retraction

- Inward movement of the skin and soft tissues between the ribs during inspiration, indicating increased work of breathing. It is common in infants and children with respiratory distress and may accompany airway obstruction, bronchiolitis, pneumonia, asthma, or other pulmonary disease.

Intercostal Retractions

- Inward movement of the skin between the ribs during inspiration, indicating increased work of breathing. It is commonly seen in infants and children with airway obstruction, bronchiolitis, asthma, pneumo-

nia, or respiratory failure.

Interdental Brush

- Small brush for cleaning between teeth. Used where spaces exist.

Interference

- See pharmacology.

Interferon Signaling

- Type I IFN (IFN-alpha/beta): all nucleated cells produce; receptor IFNAR (ubiquitous); JAK1/TYK2 -> STAT1/2/IRF9 (ISGF3) -> ISGs (hundreds: antiviral, antiproliferative, immunomodulatory). Type II IFN (IFN-gamma): T/NK cells; receptor IFNGR; JAK1/JAK2 -> STAT1 -> GAS elements. Type III IFN (IFN-lambda): mucosal epithelium; receptor IFNLR.

Interferon-Gamma Release Assay

- Blood tests measuring IFN-gamma release in response to M. tuberculosis antigens (ESAT-6, CFP-10). More specific than PPD and unaffected by BCG. Cannot distinguish latent from active TB.

Interferonopathies

- Type I interferonopathies: Aicardi-Goutieres syndrome (TREX1, RNASEH2, SAMHD1 mutations, neonatal encephalopathy, calcifications, chilblains, SLE-like); STING-associated vasculopathy with onset in infancy (SAVI, TMEM173 mutations, severe interstitial lung disease, digital ischemia); Proteasome-associated autoinflammatory syndromes (PRAAS,

Interleukin

- A family of cytokines (interleukins) produced mainly by leukocytes that regulate immune cell growth, differentiation, activation, and communication. Examples: IL-1 and IL-6 (inflammation, fever), IL-2 (T-cell growth), IL-4/IL-13 (Th2, IgE), IL-5 (eosinophils), IL-10 (anti-inflammatory), IL-12 (Th1), IL-17 (Th17, autoimmunity), IL-23.

Interleukin-6

- Pro-inflammatory cytokine. Acute phase response.

Intermediate Cuneiform

- Smallest cuneiform, centrally located between medial and lateral cuneiforms. Articulates with navicular, medial/lateral cuneiforms, and second metatarsal. Forms the transverse arch apex. Relatively fixed, contributing to midfoot stability. In-

juries are typically associated with high-energy mid-foot trauma.

Intermediate Uveitis

- Inflammation primarily affecting the vitreous body, peripheral retina, and ciliary body, with the classic presentation being pars planitis, which is characterized by the presence of snowbanking (a white, inflammatory exudate on the inferior pars plana) and snowballs (vitreous inflammatory aggregates).

Intermittent Auscultation (IA)

- The periodic auscultation of the fetal heart rate with a fetoscope or Doppler ultrasound during labor, without continuous electronic monitoring. Protocol: auscultate after a contraction for 60 seconds, during the active phase every 15-30 minutes in the first stage and every 5-15 minutes in the second stage.

Intermittent Claudication

- Reproducible muscle discomfort, usually in the calf, thigh, or buttock, induced by walking and relieved by rest, caused by exertional arterial insufficiency. It is distinguished from rest pain and nonvascular musculoskeletal pain by history and vascular testing.

Intermittent Explosive Disorder

- A disorder characterized by recurrent, impulsive, problematic outbursts of verbal or physical aggression that are grossly disproportionate to the provocation. The outbursts are not premeditated and cause marked distress or functional impairment. Age >6 years.

Intermittent Fasting

- Cycling between fasting and eating periods. Methods: alternate-day fasting, 5:2 diet, time-restricted eating. Potential benefits: weight loss, metabolic health. Evidence emerging.

Internal Hernia

- Herniation of bowel through a mesenteric defect common after Roux-en-Y gastric bypass creating a defect at the jejunojejunostomy. May cause intermittent or acute small bowel obstruction. Presents with episodic abdominal pain, nausea, and vomiting.

Internal Iliac Artery

- Smaller common iliac terminal branch (hypogastric), supplying the pelvis, perineum, gluteal region, and medial thigh. Posterior division: iliolumbar, lateral sacral, superior gluteal. Anterior division: in-

ferior gluteal, internal pudendal, obturator, vesical, uterine, vaginal, middle rectal arteries.

Internal Jugular Vein

- Descends in the carotid sheath with the internal carotid artery and vagus nerve, draining brain, face, and neck. Begins at the jugular foramen as a sigmoid sinus continuation, joining the subclavian vein to form the brachiocephalic vein. Contains backflow-preventing valves. Used for central venous catheterization.

Internal Oblique Muscle

- Middle flat abdominal layer deep to external oblique, from thoracolumbar fascia, iliac crest, and lateral inguinal ligament to lower ribs, linea alba, and pubic crest. Innervated by T7-T12 and L1. Ipsilateral rotation (opposite of external oblique). Contributes to inguinal canal and abdominal wall integrity.

Internal Podalic Version

- A maneuver where the operator inserts a hand into the uterus, grasps one or both fetal feet, and converts a transverse lie or cephalic presentation to a breech, then delivers by total breech extraction. Indication: delivery of the second twin (transverse lie) or severe fetal distress in the second stage. Rare in modern obstetrics.

Internal Resorption

- Loss of dentin from pulp cavity inward. Pink spot may be visible. Treatment: root canal therapy.

Internuclear Ophthalmoplegia

- A conjugate horizontal gaze disorder caused by a lesion of the medial longitudinal fasciculus (MLF), a heavily myelinated fiber tract that connects the abducens nucleus (CN VI) on one side to the oculomotor nucleus (CN III) on the contralateral side, coordinating horizontal gaze.

Interpersonal Psychotherapy

- A time-limited, evidence-based psychotherapy for depression that focuses on the interpersonal context of mood symptoms. It identifies four problem areas: grief, role disputes, role transitions, and interpersonal deficits. IPT helps patients improve interpersonal functioning and social support.

Interpreter

- Exchange of information between provider and

patient. See clinical communication.

Interproximal Caries

- Dental caries occurring on the contact surfaces between adjacent teeth. It may be difficult to detect clinically and is commonly assessed with bitewing radiographs when imaging is indicated.

Interstitial (Cornual) Pregnancy

- A rare ectopic pregnancy (2-4%) implanted in the intramural portion of the fallopian tube within the uterine wall. Higher morbidity than tubal ectopic because the myometrial layer expands, allowing later rupture (12-16 weeks) with massive hemorrhage. Risk factors: prior ipsilateral salpingectomy, IVF.

Interstitial Cystitis

- Chronic bladder pain syndrome. Symptoms: suprapubic pain, frequency, urgency. Diagnosis: exclusion, cystoscopy with hydrodistension. Treatment: lifestyle modification, medications, bladder instillations, surgery.

Interstitial Keratitis

- Inflammation of the corneal stroma without primary involvement of the corneal surface. It may follow infection, autoimmune disease, or trauma and can cause pain, photophobia, redness, and reduced vision.

Interstitial Lung Disease

- A heterogeneous group of disorders with inflammation and fibrosis of lung parenchyma, causing restrictive physiology (reduced FVC and TLC with normal or elevated FEV1/FVC ratio).

Interstitial Nephritis

- Inflammation of the renal interstitium and tubules, which can be acute or chronic. Acute interstitial nephritis is most commonly caused by medications, particularly NSAIDs, antibiotics including penicillins and cephalosporins, and proton pump inhibitors.

Interstitial Pneumonia

- Inflammation centered on the interstitium. Caused by Mycoplasma, Chlamydophila, Legionella, and viruses. Presents with dry cough and diffuse crackles.

Intertrigo

- An inflammatory rash of opposing skin surfaces (skin folds) caused by friction, moisture, heat, and

lack of air circulation. Affects intertriginous areas: axillae, inframammary, inguinal, abdominal folds, and intergluteal cleft. Presents with erythema, maceration, fissuring, and erosions.

Interventional Procedures

- Image-guided treatments: angioplasty, stenting, embolization, ablation, biopsy, drain placement. Advantages: minimally invasive, faster recovery, lower complication rates.

Interventional Radiologist

- Physician performing image-guided procedures.

Interventional Radiology

- Image-guided minimally invasive procedures. Examples: angioplasty, stenting, embolization, biopsy, ablation, drain placement. Reduces need for open surgery.

Interventricular Septum

- The thick muscular wall separating the left and right ventricles. It has two portions: the muscular (inferior, thicker) and membranous (superior, thinner) portions. The septum contains parts of the cardiac conduction system, including the bundle of His.

Intervertebral Disc

- A fibrocartilaginous cushion between adjacent vertebral bodies, composed of a gelatinous nucleus pulposus surrounded by a tough annulus fibrosus. It absorbs shock and permits limited movement of the vertebral column.

Intestinal Atresia

- A congenital obstruction of the small intestine (jejunum or ileum) caused by an intrauterine vascular accident (mesenteric ischemia) leading to aseptic necrosis and resorption of a segment of bowel. In contrast to duodenal atresia, it is not associated with Down syndrome.

Intestinal Ganglioneuromatosis

- Benign proliferation of ganglion cells and nerve fibers in the gastrointestinal tract wall. May be diffuse involving the entire gastrointestinal tract or localized as ganglioneuromas. Associated with multiple endocrine neoplasia type 2B where intestinal ganglioneuromatosis may precede the development of medullary thyroid carcinoma.

Intestinal lipomatosis

- Benign overgrowth of adipose tissue within the intestinal wall, most commonly affecting the ileum and colon. Often asymptomatic and discovered incidentally, but large lipomas can cause intussusception, gastrointestinal bleeding, or bowel obstruction.

Intestinal Lymphoma

- Malignant proliferation of lymphoid cells arising in the small or large intestine. Important types include enteropathy-associated T-cell lymphoma, monomorphic epitheliotropic intestinal T-cell lymphoma, and diffuse large B-cell lymphoma.

Intestinal Malrotation

- Congenital failure of normal midgut rotation and fixation, predisposing to a narrow mesenteric base, volvulus, and duodenal obstruction. Infants may present with bilious vomiting, abdominal distension, or bowel ischemia; sudden symptoms require emergency assessment.

Intestinal Obstruction

- A mechanical or functional blockage of intestinal contents. Small bowel obstruction (SBO): most common cause is adhesions (60%), followed by hernias, tumors, and Crohn disease. Large bowel obstruction (LBO): most common is colorectal cancer, volvulus (sigmoid most common), and diverticulitis.

Intestinal Perforation

- Full-thickness breach in the bowel wall allowing luminal contents to enter the peritoneal cavity. Causes include diverticulitis, appendicitis, peptic ulcer disease, and trauma. Presents with acute abdomen, peritonitis, and free air on imaging.

Intestinal Pseudo-Obstruction

- Acute or chronic dilation of the bowel without mechanical obstruction due to neuromuscular dysfunction. Acute colonic pseudo-obstruction also called Ogilvie syndrome occurs in hospitalized or post-surgical patients.

Intimal Hyperplasia

- Proliferation and migration of vascular smooth-muscle cells with extracellular-matrix deposition within the intima after vascular injury or graft placement. It can produce restenosis at anastomoses, stents, dialysis access, or bypass grafts.

Intimate Partner Violence (IPV) in

Pregnancy

- Physical, sexual, or psychological abuse by a current or former partner during pregnancy. Prevalence: 3-9% of pregnancies. Risk factors: unplanned pregnancy, late prenatal care, substance use, and young age. IPV increases the risk of miscarriage, preterm birth, low birth weight, placental abruption, and maternal death.

Intoxication

- Harmful effect from exposure to toxic substance. See poisoning.

Intra-Abdominal Abscess

- Localized collection of pus within the abdomen commonly in the pelvis, between bowel loops, or beneath the diaphragm. Results from intra-abdominal infection or surgery. Presents with fever, abdominal pain, and leukocytosis. Diagnosed by CT with contrast showing rim-enhancing fluid collection.

Intra-Abdominal Hypertension

- Sustained elevation of intra-abdominal pressure above twelve millimeters of mercury measured through the urinary bladder. Graded as grade one twelve to fifteen, grade two sixteen to twenty, grade three twenty-one to twenty-five, and grade four greater than twenty-five millimeters of mercury.

Intra-Aortic Balloon Pump

- A mechanical circulatory support device that increases myocardial perfusion and reduces afterload. The balloon inflates during diastole (augmenting coronary perfusion) and deflates during systole (reducing afterload). Inserted percutaneously into the femoral artery.

Intra-aortic Balloon Pump (IABP)

- Mechanical circulatory support. Inflatable catheter in descending aorta. Inflates during diastole (increases coronary perfusion), deflates during systole (reduces afterload). Indications: cardiogenic shock, unstable angina, high-risk PCI. Complications: limb ischemia, bleeding, infection.

Intraabdominal cancer

- A broad term for malignancies arising within the abdominal cavity, including cancers of the colon, stomach, pancreas, liver, ovaries, and peritoneum. Presentation varies widely depending on the specific organ involved.

Intraabdominal hernia

- Herniation of abdominal viscera through a congenital or acquired defect or abnormal mesenteric aperture within the peritoneal cavity, distinct from ventral or inguinal hernias. Types include paraduodenal, foramen of Winslow, transmesocolic, and transomental hernias.

Intracellular Fluid (ICF)

- The fluid contained within cells, comprising approximately two thirds of total body water in a typical adult. Its major solutes include potassium, magnesium, phosphate, and intracellular proteins, and its composition is maintained by selective membrane transport.

Intracellular pathogens

- Microorganisms that survive and replicate within host cells. Intracellular bacteria include *Mycobacterium*, *Listeria*, and *Chlamydia*. Viruses are obligate intracellular pathogens. Intracellular survival protects pathogens from antibiotics and immune responses.

Intracerebral Hemorrhage

- Bleeding directly into the brain parenchyma, accounting for approximately ten to fifteen percent of all strokes and carrying a high mortality rate of approximately forty percent at thirty days.

Intracranial Hemorrhage

- Bleeding within the cranial cavity. Epidural hematoma: arterial bleeding (middle meningeal artery), typically from temporal bone fracture; lucid interval then rapid deterioration. Subdural hematoma: venous bleeding from bridging veins; acute (<72h), subacute, or chronic (elderly, alcoholics).

Intracranial Pressure (ICP)

- The pressure within the cranial vault generated by brain tissue, cerebrospinal fluid, and intracranial blood. Sustained elevation can reduce cerebral perfusion and cause headache, vomiting, altered consciousness, pupillary abnormalities, and brain herniation.

Intraductal Papilloma

- A benign papillary tumor arising within a breast duct, often presenting with spontaneous unilateral bloody or clear nipple discharge. It may be solitary

near the nipple or multiple in the peripheral breast.

Intrahepatic Cholestasis of Pregnancy (ICP)

- A pregnancy-specific liver disorder characterized by pruritus (palms and soles, often nocturnal, with no rash) and elevated serum bile acids ($>10\text{--}40\text{ }\mu\text{mol/L}$). Incidence: 0.5-2% of pregnancies. Risk factors: prior ICP, multiple gestation, hepatitis C, and high estrogen states.

Intramedullary Nail

- A metal implant inserted into the medullary canal of a long bone to align and stabilize a fracture. It shares load along the bone axis and is commonly used for femoral and tibial shaft fractures.

Intramural Hematoma

- Collection of blood within the wall of the gastrointestinal tract most commonly the duodenum. Caused by trauma, anticoagulation, or coagulopathy. Presents with abdominal pain, vomiting, and sometimes obstructive symptoms. Diagnosis is by CT showing circumferential wall thickening with high-attenuation hematoma.

Intramuscular (IM)

- A route of drug administration in which the drug is injected into skeletal muscle (deltoid, vastus lateralis, gluteus maximus).

Intraocular Lens

- An intraocular lens is an artificial, biocompatible, transparent lens implanted during cataract surgery to replace the natural crystalline lens that has been removed, restoring the eye's refractive power and enabling clear vision.

Intraocular Lens (IOL)

- Artificial lens replacing natural lens after cataract extraction. Types: monofocal (distance vision), multifocal/bifocal (near and distance), toric (astigmatism), extended depth of focus. Implantation via small incision, folds for insertion. Complications: posterior capsule opacification, dysphotopsia, infection, retinal detachment.

Intraocular lens dislocation

- Postoperative displacement of an implanted artificial intraocular lens from its intended position within the capsular bag or ciliary sulcus into the vitreous cavity or anterior chamber. It is most com-

monly caused by progressive zonular dehiscence and may require surgical repositioning or exchange.

Intraocular Pressure (IOP)

- The pressure generated by aqueous humor within the eye, normally maintained by a balance between aqueous production and outflow. Abnormal IOP is an important risk factor for glaucomatous optic neuropathy, but glaucoma can occur at statistically normal pressures.

Intraocular tumor

- Any neoplasm arising within the eye, with uveal melanoma being the most common primary malignancy in adults and retinoblastoma the most common in children. Diagnosis requires dilated fundus examination, ultrasound, and MRI; treatment depends on tumour type, size, and location.

Intraoperative Blood Transfusion

- Intraoperative blood transfusion replaces lost blood volume and maintains oxygen-carrying capacity during major surgery. Indications include hemoglobin less than 7 g/dL in stable patients, less than 8 g/dL in cardiac disease, or active hemorrhage with hemodynamic instability.

Intraoperative Neuromonitoring

- Intraoperative neuromonitoring (IONM) uses electrophysiological techniques to monitor neural pathway integrity during surgery. Somatosensory evoked potentials (SSEP) monitor sensory pathways. Motor evoked potentials (MEP) monitor motor pathways. Electromyography (EMG) monitors cranial nerves and spinal roots.

Intraoperative Temperature Monitoring

- Core temperature monitoring is essential during prolonged or major surgery. Sites include esophageal (most accurate for cardiac surgery), nasopharyngeal, bladder (with continuous irrigation), tympanic, and pulmonary artery (gold standard, requires PA catheter).

Intraosseous Access

- Placement of a vascular-access needle into the medullary cavity of a suitable bone to permit rapid infusion of fluids, blood products, and medications when intravenous access is delayed or impossible. It is a temporary emergency route and can cause extravasation, infection, or compartment injury.

Intraosseous Infusion

- Administration of fluids, blood products, or medications through an intraosseous access device. Infusion should be monitored for resistance, swelling, pain, and extravasation and replaced with definitive vascular access as soon as practical.

Intrapartum

- The period during labor and childbirth, from the onset of regular uterine contractions leading to cervical dilation until delivery of the baby and placenta.

Intrapulmonary Shunt

- Perfusion of alveoli without ventilation ($V/Q = 0$). Blood passes through the pulmonary circulation without gas exchange, mixing with oxygenated blood and causing hypoxemia. It is seen in ARDS, pneumonia, and atelectasis. It is refractory to supplemental oxygen.

Intrathecal Chemotherapy

- Administration of chemotherapy directly into the cerebrospinal fluid via lumbar puncture or an Ommaya reservoir, used for leptomeningeal metastasis and central nervous system prophylaxis in leukemias and lymphomas.

Intrathoracic localized obstruction

- A focal blockage within the thoracic cavity affecting the airways, esophagus, or great vessels, leading to localized symptoms depending on the structure involved. Causes include endobronchial tumors, foreign bodies, esophageal strictures, or vascular anomalies. Diagnosis is by imaging and endoscopy.

Intrauterine Device

- IUD: a long-acting reversible contraceptive (LARC) device placed in the uterus. Types: copper IUD (ParaGard—hormone-free, up to 10-12 years, also emergency contraception) and levonorgestrel-releasing IUS (Mirena/Skyla—progestin, up to 3-8 years, reduces bleeding).

Intrauterine Fetal Demise (IUFD) / Stillbirth

- Fetal death before complete expulsion or extraction from the mother, at 20 weeks gestation. Affects 1:160 pregnancies in the US (25,000 annually). Causes: placental abruption, umbilical cord accidents, infection, genetic abnormalities, maternal medical conditions (diabetes, hypertension, SLE),

and unexplained (>25%).

Intrauterine Growth Restriction

- IUGR: a fetus with estimated weight below the 10th percentile (or growth velocity decline), usually with placental insufficiency or maternal/fetal factors.

Intrauterine Pressure Catheter (IUPC)

- A fluid-filled catheter inserted into the uterine cavity alongside the fetus to measure the intensity and frequency of contractions in mm Hg (Montevideo units, MVU).

Intrauterine Transfusion

- Ultrasonically guided intravascular or intraperitoneal transfusion of fetal red blood cells to treat severe fetal anemia, most commonly from Rh isoimmunization. Performed through the umbilical vein using a 22-25 gauge needle. Packed red blood cells, CMV-negative and crossmatched, are transfused slowly.

Intravascular Catheter Infections

- CLABSI: central line-associated bloodstream infection. Diagnosis: quantitative blood cultures (catheter vs peripheral >5:1), differential time to positivity (>2h), catheter tip culture (>15 CFU). Catheter-related BSIs: coagulase-negative staphylococci (most common), *S. aureus*, enterococci, gram-negatives, *Candida*.

Intravascular Filter

- Device placed in inferior vena cava to prevent pulmonary embolism. See IVC filter.

Intravascular lymphoma

- A rare aggressive B-cell non-Hodgkin lymphoma characterized by proliferation of neoplastic lymphocytes within the lumina of small blood vessels, predominantly affecting the skin and central nervous system. Presents with livedo reticularis, skin nodules, neurological deficits, and fever of unknown origin.

Intravascular Ultrasound

- Catheter-based imaging using a miniature ultrasound probe inside coronary arteries to visualize vessel wall layers, plaque composition, and stent deployment. Provides more detailed information than angiography alone, including vessel remodeling, plaque burden, and true vessel diameter.

Intravenous Anesthesia

- Anesthetic induction or maintenance using drugs delivered into a vein. It permits rapid titration but requires attention to dose, context-sensitive recovery, cardiovascular and respiratory effects, and the availability of a secure airway.

Intravenous Lipid Emulsion Therapy

- Administration of a lipid emulsion as a rescue treatment for selected severe lipophilic drug toxicities, especially local-anesthetic systemic toxicity. It may act as a lipid sink and through metabolic and cardiovascular effects and requires critical-care monitoring.

Intravenous pyelogram

- A radiographic imaging study of the urinary tract following intravenous administration of iodinated contrast material. The contrast is filtered by the glomeruli and excreted through the collecting system, allowing sequential imaging of the renal parenchyma, calyces, renal pelvis, ureters, and bladder.

Intravenous Pyelogram (IVP)

- Radiographic imaging of the urinary tract after IV contrast administration. Used to evaluate the kidneys, ureters, and bladder for obstruction, stones, tumors, and anatomic abnormalities. Sequential films show nephrogram (renal parenchyma), pyelogram (collecting system), and ureterogram (ureters).

Intraventricular Hemorrhage

- Bleeding into the germinal matrix and ventricular system, primarily in premature infants less than 32 weeks gestation. Graded I-IV by severity on cranial ultrasound. Grade I (germinal matrix only) and Grade II (intraventricular without dilation) have good prognosis.

Intravitreal Injection

- An ophthalmological procedure in which medication is injected directly into the vitreous cavity of the eye through the pars plana, a safe entry zone located approximately 3.5 to 4 millimeters posterior to the limbus, to deliver therapeutic agents directly to the retina and vitreous for the treatment of various retinal diseases.

Intrinsic Renal AKI

- AKI caused by damage to renal parenchyma. Ma-

– jor categories include acute tubular necrosis from ischemic or nephrotoxic injury, acute interstitial nephritis from drug reactions, glomerulonephritis, and vascular causes including thrombotic microangiopathy. FENa typically greater than 2% indicating tubular dysfunction.

Intron

- A non-coding segment of a gene that is transcribed into RNA but spliced out before the messenger RNA is translated into a protein.

Intussusception

- Condition in which a segment of intestine telescopes into the adjacent distal segment causing bowel obstruction and compromise of the mesenteric blood supply. It is the most common cause of intestinal obstruction in infants and children between six months and three years of age.

Invasive ductal carcinoma

- The most common type of breast cancer, characterized by malignant cells that have broken through the duct wall and invaded surrounding breast tissue. It can metastasize via lymphatic or hematogenous spread.

Invasive Lobular Carcinoma

- A breast cancer characterized by discohesive cells infiltrating in single-file lines. More frequently bilateral and multifocal than ductal carcinoma and harder to detect on imaging; usually hormone receptor positive.

Inverse Agonist

- A type of drug that binds to a cell receptor and actively produces the opposite biological effect of an agonist, effectively turning down the receptor's baseline activity.

Inversion

- Chromosomal segment reversed end-to-end. Paracentric: does not include centromere. Pericentric: includes centromere. Usually benign unless breakpoint disrupts gene.

Involuntary Psychiatric Treatment

- Assessment, hospitalization, or treatment without the patient's consent under applicable law when serious mental disorder creates a substantial risk of harm or severe inability to meet basic needs. The least restrictive alternative, due process, and

ongoing reassessment are required.

Involution

- The process by which the uterus returns to its prepregnancy size after delivery. The uterus weighs approximately 1000 grams immediately postpartum and returns to approximately 60 grams by 6 weeks. Fundal height decreases approximately 1 cm per day.

Iodide-associated sialadenitis

- Inflammation and swelling of the salivary glands caused by excessive iodide intake, typically from medications such as amiodarone or iodine-containing contrast, resolving with dose reduction.

Iodinated Contrast Media

- Contrast agents containing iodine used primarily for radiography, CT, angiography, and some fluoroscopic procedures. Important risks include immediate hypersensitivity-like reactions, extravasation, thyroid effects, and kidney injury in susceptible patients.

Iodine

- Required for thyroid hormones. Deficiency: goiter, hypothyroidism, cretinism. Prevention: iodized salt, seafood. Excess can cause thyroid dysfunction.

Iodine Swab

- Antiseptic for skin preparation. Povidone-iodine.

Iodine-123

- Pure gamma emitter, half-life 13 hours. Used for thyroid imaging. No beta emission, lower radiation dose than I-131.

Iodine-131

- See radioactive iodine.

Iododerma

- A papulopustular or acneiform skin eruption caused by iodine excess, presenting on the face and extensor surfaces, resolving with iodine restriction and symptomatic treatment.

Ionizing Radiation

- Radiation with enough energy to remove electrons from atoms and create ions, including X-rays, gamma rays, and certain particulate emissions. It can cause deterministic tissue effects at sufficient doses and stochastic cancer risk at lower doses.

Ipecac

- Emetic no longer recommended for routine poisoning management. Risk of aspiration, delays charcoal administration.

Ipratropium Bromide

- An inhaled anticholinergic (antimuscarinic) bronchodilator that blocks acetylcholine at airway smooth muscle muscarinic receptors. It reduces bronchospasm and mucus secretion. It is used in COPD and acute asthma exacerbations. It is less effective than beta-2 agonists in asthma but more effective in COPD.

Ipsilateral

- Located on the same side of the body.

Iridocyclitis

- Inflammation involving the iris and ciliary body, a form of anterior uveitis. It may cause eye pain, photophobia, redness, blurred vision, and cells or flare in the anterior chamber and may be associated with autoimmune disease, infection, trauma, or other causes.

Iridotomy

- Laser peripheral iridotomy is a minimally invasive laser procedure used to treat or prevent angle-closure glaucoma by creating a small full-thickness hole in the peripheral iris, allowing aqueous humor to flow directly from the posterior chamber to the anterior chamber, bypassing the pupil and equalizing the pressure between the two chambers.

Iris

- Colored circular muscular diaphragm between cornea and lens, controlling pupil size and light entry. Two muscles: sphincter pupillae (parasympathetic CN III, constricts in bright light) and dilator pupillae (sympathetic, dilates in darkness). Iris root attaches to the ciliary body.

Iris atrophy

- Thinning or loss of iris tissue resulting from trauma, chronic inflammation, ischaemia, or neurodegenerative disease, leading to a translucent or patched iris appearance. It may be sectoral or diffuse and can be associated with decreased pupillary response, photophobia, or secondary glaucoma.

Iritis

- Inflammation of the iris, the most common form of

anterior uveitis. Presents with acute onset eye pain, photophobia, blurred vision, and redness (ciliary flush). Signs: cells and flare in the anterior chamber, keratic precipitates (KP), synechiae (adhesions between iris and lens).

Iron

- Essential for hemoglobin, myoglobin, and electron transport. Heme iron from animal foods is generally more bioavailable than non-heme iron from plant foods. Deficiency is the most common nutritional deficiency worldwide and may result from blood loss, increased requirements, poor intake, or malabsorption.

Iron deficiency

- The most common nutritional deficiency worldwide, resulting from inadequate iron intake, impaired absorption, or chronic blood loss, leading to reduced hemoglobin synthesis and microcytic hypochromic anemia. Clinical features include fatigue, pallor, koilonychia, pica, and angular cheilitis.

Iron Deficiency Anemia

- Most common anemia worldwide. Microcytic, hypochromic. Treated with iron supplementation.

Irregular Sleep-Wake Syndrome

- A circadian rhythm sleep-wake disorder with multiple irregular periods of sleep and wakefulness across the 24-hour day rather than one consolidated nighttime sleep period. It is associated with neurological disease, dementia, aging, and reduced exposure to environmental time cues.

Irreversible Cell Injury

- Cell damage beyond the point of recovery, resulting in death by necrosis, apoptosis, or another regulated cell-death pathway. Key features include inability to reverse mitochondrial dysfunction and severe membrane or nuclear damage.

Irritable Bowel Syndrome

- Functional gastrointestinal disorder characterized by recurrent abdominal pain associated with altered bowel habits without identifiable structural abnormalities. The Rome IV criteria define IBS as recurrent abdominal pain occurring at least one day per week associated with changes in defecation or stool form.

Irritable Bowel Syndrome (IBS)

- A chronic functional gastrointestinal disorder characterized by recurrent abdominal pain and altered bowel habits (diarrhea, constipation, or mixed) without identifiable structural pathology. Pathophysiology involves dysregulation of the gut–brain axis, visceral hypersensitivity, altered motility, and imbalances in the gut microbiome.

Irritable Bowel Syndrome Diet

- See gastroenterology. Low FODMAP diet effective for symptom control.

Isaacs Syndrome

- A rare peripheral nerve hyperexcitability disorder causing continuous muscle-fibre activity, stiffness, cramps, fasciculations, myokymia, and delayed muscle relaxation. It may be autoimmune, paraneoplastic, or associated with inherited or acquired peripheral-nerve disorders.

Isavuconazole

- A triazole antifungal agent used in the treatment of invasive aspergillosis and mucormycosis. Isavuconazole inhibits fungal lanosterol 14- α demethylase, disrupting ergosterol synthesis and fungal cell membrane integrity.

Ischemia

- Reduced blood flow causing oxygen deprivation. Can lead to tissue damage or death.

Ischemia-Reperfusion Injury

- The paradoxical tissue damage that occurs when blood flow is restored to previously ischemic tissue, a phenomenon with major clinical significance in myocardial infarction treatment, stroke management, organ transplantation, and shock resuscitation.

Ischemic Cardiomyopathy

- Dilated cardiomyopathy resulting from extensive myocardial infarction with significant scar burden. EF is typically severely reduced ($<35\%$). The infarcted myocardium is replaced by fibrotic scar tissue, impairing contractile function and promoting ventricular arrhythmias.

Ischemic Colitis

- Colonic mucosal ischemia resulting from reduced blood flow through the marginal artery and end-arteries, most commonly occurring at watershed areas between vascular territories, particularly the

splenic flexure.

Ischemic gangrene

- Death of tissue due to prolonged severe arterial insufficiency, most commonly in the extremities. Presents as dry, black, mummified tissue with a clear line of demarcation from viable tissue. Requires revascularization when possible and surgical debridement or amputation of necrotic tissue.

Ischemic Stroke

- A neurological deficit caused by focal cerebral ischemia due to thrombotic or embolic occlusion of a cerebral artery (87% of all strokes). Thrombotic: in situ occlusion of atherosclerotic vessel. Embolic: embolus from cardiac source (atrial fibrillation, valvular disease, patent foramen ovale) or large artery (carotid/vertebral).

Ischium

- Inferior-posterior hip bone. Ischial tuberosity bears seated weight and serves as hamstring attachment. Ischial spine is a pudendal nerve block landmark and determines pelvic outlet level. Lesser sciatic notch lies between spine and tuberosity.

Isocitrate Dehydrogenase

- A regulated enzyme of the citric acid cycle that converts isocitrate to alpha-ketoglutarate while producing NADH and releasing carbon dioxide. Mitochondrial isoforms are stimulated by ADP and calcium and inhibited by ATP and NADH. Mutations in IDH1 or IDH2 can produce the oncometabolite 2-hydroxyglutarate.

Isokinetic Exercise

- Muscle contraction at constant speed throughout range of motion. Uses specialized equipment. Assesses and improves strength.

Isolation

- Separation of sick persons with contagious disease. Examples: tuberculosis isolation, airborne infection isolation rooms.

Isolation (Social)

- Lack of social contacts and interactions. Common in elderly: bereavement, mobility limitation, sensory impairment, retirement. Consequences: depression, cognitive decline, functional impairment, mortality. Interventions: social programs, transportation, technology, volunteer visits.

Isolation Precautions

- Measures for disease transmission. Types: contact, droplet, airborne. PLUS standard precautions.

Isoniazid Toxicity

- Hepatotoxicity, peripheral neuropathy, seizures. Treatment: pyridoxine (vitamin B6) supplementation, activated charcoal, seizure control.

Isosorbide Dinitrate

- An organic nitrate (and prodrug) that releases nitric oxide, relaxing vascular smooth muscle—predominantly venous (preload reduction) with coronary dilation.

Isosporiasis

- Intestinal infection caused by the protozoan *Cystoisospora belli*. It typically causes watery diarrhea, abdominal cramps, nausea, fever, and weight loss and may be prolonged or severe in people with impaired cellular immunity.

Isotope

- See radionuclide.

Isotretinoin

- A systemic retinoid used for severe, recalcitrant nodular acne. It reduces sebum production, normalizes follicular keratinization, decreases *Propionibacterium acnes*, and has anti-inflammatory effects. Dose is 0.5 to 1 mg/kg/day for 15 to 20 weeks, targeting cumulative dose of 120 to 150 mg/kg.

Itching

- Pruritus, the unpleasant cutaneous sensation that provokes the desire to scratch. It may be acute or chronic and may arise from skin disease, systemic illness, medication, or neuropathic mechanisms.

Iterative Reconstruction

- Modern reconstruction method. OSEM (Ordered Subsets Expectation Maximization). Better quality, longer computation.

IV Fluids

- Intravenous fluid therapy for volume resuscitation and maintenance. Crystalloids: normal saline (0.9% NaCl, increases chloride, risk of hyperchloremic metabolic acidosis), lactated Ringer (balanced, contains K⁺, Ca²⁺, lactate buffer, safer in large volumes), Plasma-Lyte (balanced, closest to plasma).

IVC Filter

- Device placed in inferior vena cava to prevent pulmonary embolism. Indications: DVT/PE with contraindication to anticoagulation, recurrent PE despite anticoagulation. Complications: filter thrombosis, migration, perforation, IVC occlusion.

IVF

- In vitro fertilization: the assisted reproductive technology in which oocytes are retrieved, fertilized by sperm outside the body (in vitro), and the resulting embryos transferred to the uterus. Indications: tubal factor, male factor (with ICSI), endometriosis, unexplained, ovulation disorders, genetic testing, and same-sex couples or individuals using donor sperm or eggs.

Ileal Obstruction

- Partial or complete blockage of the ileum, the final portion of the small intestine. Causes include adhesions, hernia, inflammatory bowel disease, tumors, and twisting of the bowel. Symptoms may include cramping pain, vomiting, abdominal distension, and inability to pass stool or gas.

Infectious Arthritis

- Inflammation of a joint caused by infection, usually bacterial and sometimes viral or fungal. Acute pain, swelling, warmth, fever, and restricted movement require urgent evaluation because joint damage can develop quickly.

Infectious Meningitis

- Infection and inflammation of the membranes surrounding the brain and spinal cord. Bacterial meningitis is a medical emergency; headache, fever, neck stiffness, vomiting, confusion, or a new rash requires immediate medical attention.

Invasive Candidiasis

- A serious *Candida* infection that enters the bloodstream or deep organs, most often in people who are critically ill or immunocompromised. It may cause persistent fever or organ dysfunction and requires prompt medical diagnosis and treatment.

Iron-Deficiency Anemia

- Anemia caused by insufficient iron for normal hemoglobin production. It may cause fatigue, weakness, shortness of breath, paleness, headache, or unusual cravings such as ice; the cause of iron loss

or inadequate intake should be identified.

Chapter 10

Pocket Dictionary: J Terms

Jaccoud's arthropathy

- A non-erosive, reducible deformity of the metacarpophalangeal joints caused by chronic periarticular fibrosis and tendon laxity, most commonly associated with systemic lupus erythematosus.

Jacksonian Epilepsy (Focal Seizures)

- A type of focal (partial) seizure originating in the primary motor cortex, characterized by clonic (rhythmic jerking) or tonic (sustained contraction) movements that spread sequentially to adjacent body parts along the motor homunculus (Jacksonian march). Named after John Hughlings Jackson (1835-1911), who first described the phenomenon.

JAK-STAT Pathway

- A rapid signaling pathway for cytokines and growth factors (interferons, interleukins, erythropoietin, growth hormone) that lack intrinsic enzymatic activity. Ligand binding to the cytokine receptor causes receptor dimerization and activation of associated Janus kinases (JAK1, JAK2, JAK3, TYK2) by trans-phosphorylation.

JAK2

- Janus kinase 2 is a tyrosine kinase essential for hematopoietic cell signaling.

Janeway Lesions

- Painless, erythematous or hemorrhagic macules on the palms and soles, caused by septic microemboli in infective endocarditis. They are a minor criterion in the Duke criteria. Unlike Osler nodes, Janeway lesions are painless and represent microabscesses in the dermis. They are transient and may disappear quickly.

Janus Kinase Inhibitors

- JAK inhibitors (tofacitinib, baricitinib, upadacitinib, filgotinib, abrocitinib) block JAK-STAT

signaling downstream of cytokine receptors (IL-6, IFN-gamma, IL-2, IL-7, IL-15, IL-21, GM-CSF, erythropoietin, thrombopoietin). Selectivity: JAK1/JAK3 (tofacitinib), JAK1/JAK2 (baricitinib), JAK1 (upadacitinib, filgotinib, abrocitinib).

Japanese Encephalitis

- A mosquito-borne viral infection of the central nervous system caused by Japanese encephalitis virus. Most infections are asymptomatic, but severe disease can cause fever, altered consciousness, seizures, movement disorders, and lasting neurological disability.

Japanese Encephalitis (JE) Vaccine

- An inactivated (SA 14-14-2 strain, widely used) or live attenuated vaccine protecting against Japanese encephalitis, the most common cause of viral encephalitis in Asia, transmitted by Culex mosquitoes in rural rice-growing and pig-farming areas.

Jaundice

- Yellowish discoloration of the skin, sclerae, and mucous membranes caused by hyperbilirubinemia (serum bilirubin greater than 2-3 mg/dL), resulting from an imbalance between bilirubin production and clearance.

Jaundice (Icterus)

- Yellowish discoloration of the skin, sclerae, and mucous membranes caused by hyperbilirubinemia (serum bilirubin greater than 2-3 mg/dL), resulting from an imbalance between bilirubin production and clearance. See Jaundice.

Jaundice in newborns

- See Neonatal Jaundice (Jaundice of the Newborn).

Jejunal Atresia

- Congenital complete interruption or absence of a

segment of the jejunal lumen. It presents in a newborn with intestinal obstruction, commonly causing bilious vomiting, abdominal distension, and failure to pass meconium, and generally requires surgery.

Jejunostomy

- Surgical or endoscopic placement of a feeding tube into the jejunum for enteral nutrition when upper gastrointestinal access is not feasible. See Feeding Jejunostomy.

Jejunum

- Middle segment (2.5 m) with prominent circular folds and tall villi for a velvety appearance. Richer blood supply and thicker walls than the ileum. Primary nutrient absorption site for carbohydrates, amino acids, fatty acids, vitamins, minerals. Fewer Peyer's patches. More commonly affected by Crohn's disease.

Job Modification

- Adjust job duties, equipment, schedule for disability.

Job Syndrome

- A primary immunodeficiency, usually caused by pathogenic variants affecting STAT3, characterized by eczema, recurrent staphylococcal skin and lung infections, and markedly elevated immunoglobulin E. Skeletal, dental, and characteristic facial findings may also occur.

Joint

- Articulation between two or more bones permitting movement. Structural classification: fibrous (sutures, syndesmoses, gomphoses – immovable/slightly movable), cartilaginous (synchondroses, symphyses – slightly movable), synovial (freely movable – plane, hinge, pivot, condyloid, saddle, ball-and-socket).

Joint Contracture

- Persistent limitation of passive joint movement caused by shortening or fibrosis of muscle, tendon, capsule, ligament, fascia, skin, or other periarticular tissue. It may follow immobilization, neurological disease, trauma, inflammation, burns, or congenital disorders.

Joint Diseases

- A broad category of pathological conditions affecting the structure and function of joints. Classifi-

cation by etiology: (1) Degenerative: osteoarthritis (most common, non-inflammatory, affects load-bearing joints–knee, hip, spine).

Joint Effusion

- Excess fluid within a joint space. Common sites: knee (most accessible), ankle, wrist, shoulder. Causes: trauma, infection, inflammatory arthritis, crystal arthropathy. Detected by ballottement or bulge sign.

Joint Pain

- Discomfort arising from articular structures including cartilage, synovium, ligaments, and periarticular soft tissues due to inflammation, degeneration, injury, or infection.

Joint Prosthesis

- An artificial joint or implant designed to replace a damaged or diseased biological joint, restoring mobility and relieving chronic pain.

Joint Replacement

- See arthroplasty.

Joint Replacement (Arthroplasty)

- A surgical procedure in which a damaged or arthritic joint surface is removed and replaced with a prosthetic implant, most commonly performed for the hip (total hip arthroplasty) and knee (total knee arthroplasty).

Joint Replacement Rehabilitation

- See orthopedics.

Joint Swelling

- Enlargement of a joint caused by excess synovial fluid, blood, inflammation, infection, trauma, or structural disease. It may be accompanied by pain, warmth, stiffness, or restricted movement and can indicate inflammatory or septic arthritis.

Joints

- The sites where two or more bones meet, providing structural support and enabling movement. Classification by structure and function: (1) Fibrous joints (synarthroses, immovable): sutures of the skull, gomphoses (tooth in socket), syndesmoses (tibia-fibula).

Jugular Vein

- One of the major veins draining blood from the head, face, and neck. There are two pairs: internal

jugular vein (IJV) and external jugular vein (EJV).

Juvenile angiofibroma

- A rare, benign but locally aggressive vascular tumor of the nasopharynx, accounting for 0.05-0.5 percent of all head and neck tumors, occurring almost exclusively in adolescent males (mean age 14 years, 80-90 percent male predominance).

Juvenile Arthritis

- The most common chronic rheumatic disease of childhood (onset <16 years), encompassing 7 subtypes: systemic JIA (Still disease of childhood, quotidian fever, salmon rash, arthritis, hepatosplenomegaly, serositis, macrophage activation syndrome risk), oligoarticular JIA (4 or fewer joints in first 6 months, high risk of chronic anterior

Juvenile Astrocytoma

- An astrocytic glioma arising in a child or adolescent. Many are low-grade and may cause seizures, headache, visual symptoms, endocrine disturbance, or focal neurological deficits depending on location; diagnosis requires neuroimaging and tumour evaluation.

Juvenile Dermatomyositis

- Juvenile dermatomyositis (JDM) presents with proximal weakness, heliotrope rash, Gottron papules, calcinosis (20-30%), lipodystrophy, vasculopathy (nailfold capillary changes). Anti-Mi-2 (classic DM, good prognosis), anti-NXP2 (calcinosis, malignancy risk low in children), anti-TIF1-gamma (malignancy risk in adults).

Juvenile Idiopathic Arthritis

- The most common chronic rheumatic disease of childhood (onset <16 years), encompassing 7 subtypes: systemic JIA (Still disease of childhood, quotidian fever, salmon rash, arthritis, hepatosplenomegaly, serositis, macrophage activation syndrome risk), oligoarticular JIA (4 or fewer joints in first 6 months, high risk of chronic anterior

Juvenile Idiopathic Arthritis (JIA)

- A heterogeneous group of autoimmune and inflammatory conditions causing persistent arthritis (joint swelling, pain, limited range of motion) in children aged <16 years, lasting >6 weeks.

Juvenile Osteoporosis

- Reduced bone mass or impaired bone strength

occurring during childhood or adolescence, increasing fracture risk. It may be primary or secondary to chronic disease, nutritional deficiency, endocrine disorder, immobility, or medication exposure.

Juvenile Rheumatoid Arthritis

- See rheumatology.

Juvenile Xanthogranuloma

- A benign non-Langerhans-cell histiocytic disorder that most often presents in infancy or early childhood as a yellow-red papule or nodule. Skin lesions are usually self-limited, but ocular or visceral involvement can occur and requires appropriate assessment.

Juxta-articular

- Near joint. Periprosthetic fracture, periprosthetic infection.

Juxtaglomerular Apparatus

- Specialized structure located at the vascular pole of the glomerulus where the distal tubule comes into contact with the afferent arteriole. Composed of macula densa cells in the tubular wall, juxtaglomerular cells in the afferent arteriolar wall that produce renin, and extraglomerular mesangial cells.

Juxtavesical

- Near bladder. Juxtavesical ureterocele, juxtavesical mass.

Joint Stiffness

- Reduced ease or range of joint movement, often worse after rest. Causes include arthritis, injury, inflammation, prolonged immobility, or muscle and tendon disorders. Sudden swelling, redness, fever, severe pain, or inability to use the joint requires assessment.

Jaw Pain

- Pain in the jaw, jaw joint, or chewing muscles caused by temporomandibular disorders, dental disease, infection, injury, or less commonly heart or nerve disease. Pain with chest discomfort, sweating, breathlessness, or exertion requires urgent assessment; persistent pain or restricted jaw movement requires dental or medical review.

Jock Itch

- A fungal infection of the groin and inner thighs, also called tinea cruris. It causes an itchy, scaly, ring-shaped or spreading rash, often worsened by warmth

and moisture. Pain, fever, extensive involvement, or failure to improve with appropriate care requires clinical evaluation.

Chapter 11

Pocket Dictionary: K Terms

Kallmann Syndrome

- Congenital form of hypogonadotropic hypogonadism caused by failure of GnRH neuron migration from the olfactory placode to the hypothalamus during embryonic development. Characterized by delayed puberty with low LH and FSH and anosmia or hyposmia from olfactory bulb hypoplasia.

Kangaroo Care (Skin-to-Skin Contact)

- Holding a newborn against the mother's bare chest, typically for at least 60 minutes, promoting physiologic stability, thermoregulation, bonding, and breastfeeding initiation. Especially beneficial for preterm and low-birth-weight infants: reduces mortality (by 40%), hypothermia, and hospital-acquired infections.

Kaposi Sarcoma

- A vascular tumor caused by human herpesvirus 8 (HHV-8), presenting as violaceous plaques and nodules on skin and mucous membranes. Four types: classic (elderly Mediterranean men), endemic (African), iatrogenic (immunosuppression post-transplant), and AIDS-associated. Cutaneous lesions are the most common presentation in AIDS.

Kaposi's sarcoma

- See AIDS.

Karnofsky Performance Status

- The Karnofsky Performance Status (KPS) scale is a widely used clinical tool for assessing the functional status of patients, particularly those with cancer, and quantifies the patient's ability to perform activities of daily living and tolerate treatments.

Kartagener Syndrome

- A subset of primary ciliary dyskinesia with situs inversus, chronic sinusitis, and bronchiectasis. Caused by dynein defects impairing both mucociliary clear-

ance and embryonic nodal cilia rotation.

Kartagener Triad

- The classic triad of situs inversus, chronic sinusitis, and bronchiectasis, occurring in approximately 50% of patients with primary ciliary dyskinesia. It is caused by dynein defects impairing both mucociliary clearance and embryonic nodal cilia rotation.

Karyotype

- Display of chromosomes arranged in pairs, ordered by size and banding pattern. Used to detect chromosomal abnormalities. Resolution: 5-10 Mb.

Karyotyping

- A laboratory technique that produces an image (karyotype) of an individual's complete set of chromosomes, arranged in pairs by size, centromere position, and banding pattern.

Katz Index

- Assessment of ADLs: bathing, dressing, toileting, transferring, continence, feeding.

Kawasaki Disease

- An acute vasculitis of childhood affecting medium-sized arteries, particularly coronary arteries. Diagnosis requires fever greater than 5 days plus 4 of 5 criteria: bilateral conjunctivitis, oral mucosal changes, cervical lymphadenopathy, extremity changes, and polymorphous rash. Complications include coronary artery aneurysms.

Kawasaki's disease

- An acute, self-limited systemic vasculitis of childhood, also known as mucocutaneous lymph node syndrome, primarily affecting medium-sized arteries with a predilection for the coronary arteries.

Kayser-Fleischer Rings

- Golden-brown copper deposition rings in Descemet

membrane of the cornea representing a pathognomonic finding of Wilson disease. These rings are visible on slit lamp examination and present in virtually all patients with neurological manifestations of Wilson disease though they may be absent in isolated hepatic disease.

Kearns-Sayre Syndrome

- A mitochondrial cytopathy caused by large-scale mitochondrial DNA deletions or rearrangements. It classically begins before age 20 with progressive external ophthalmoplegia, pigmentary retinopathy, and cardiac-conduction disease, and may involve endocrine or neurological systems.

Kegel

- An exercise that helps prevent and treat incontinence by strengthening pelvic floor muscles.

Kegel Exercise

- Pelvic floor muscle strengthening exercises. Treatment for stress urinary incontinence, post-prostatectomy incontinence.

Keloid

- An overgrowth of scar tissue extending beyond the original wound margins, caused by excessive collagen deposition during wound healing. Keloids are more common in darker skin types (Fitzpatrick IV to VI) and occur at sites of trauma, surgery, piercings, and burns. They are pruritic and painful.

Keloids

- Firm, raised scars caused by excessive collagen deposition extending beyond the original wound. They may develop after surgery, trauma, piercing, burns, or acne and are more common in people with darker skin.

Keratinocyte

- Primary cell type in epidermis. Used in wound healing research, skin grafts.

Keratitis

- Inflammation of the cornea. Infectious keratitis: bacterial (contact lens wear, trauma: *Pseudomonas*, *Staph aureus*), viral (HSV, VZV, adenovirus), fungal (*Candida*, *Fusarium*, *Aspergillus*), and *Acanthamoeba* (contact lens-related). Presents with eye pain, photophobia, tearing, conjunctival injection, and corneal opacity/ulcer.

Keratoacanthoma

- A low-grade or well-differentiated variant of squamous cell carcinoma (SCC) originating from hair follicles. Characterized by rapid growth over 4-6 weeks into a dome-shaped, flesh-colored or erythematous nodular lesion with a central keratin-filled crater. Often resolves spontaneously over months via involution, leaving a depressed scar.

Keratoconus

- A progressive, non-inflammatory ectatic corneal disorder characterized by bilateral (90 percent of cases) thinning and cone-like protrusion of the cornea, leading to irregular astigmatism, myopia, and decreased visual acuity.

Keratoderma

- A group of disorders characterised by excessive thickening of the skin of the palms and soles, which may be hereditary or acquired. Acquired forms can be diffuse, focal, or punctate and may be associated with systemic diseases, malignancy, or medications.

Keratoderma blennorrhagicum

- A keratoderma associated with reactive arthritis (formerly Reiter syndrome), characterised by waxy, yellowish, vesiculopustular papules and plaques on the palms and soles that may coalesce and extend to the extremities. It mimics pustular psoriasis histologically and clinically.

Keratoderma blenorrhagicum

- Alternate spelling of keratoderma blennorrhagicum, a cutaneous manifestation of reactive arthritis with waxy, crusty, vesiculopustular lesions on the palms and soles. It is a reactive phenomenon typically occurring in the context of genitourinary or gastrointestinal infection.

Keratoplasty

- Surgical transplantation or reshaping of corneal tissue. Penetrating keratoplasty replaces the full corneal thickness, whereas lamellar techniques replace selected layers and may reduce rejection and recovery time.

Keratoprosthesis

- Artificial cornea for severe corneal blindness. Boston KPro most common. Indications: chemical burns, limbal stem cell deficiency.

Keratosis Pilaris

- A common, benign genetic follicular condition characterized by keratotic follicular papules. Small, rough, flesh-colored or red bumps on the extensor surfaces of proximal arms, thighs, and sometimes buttocks and face. Caused by abnormal keratinization and plugging of hair follicles with retained keratin.

Kernicterus

- Bilirubin encephalopathy resulting from severe unconjugated hyperbilirubinemia (typically greater than 25 mg/dL in term infants). Bilirubin deposits in the basal ganglia, hippocampus, and brainstem nuclei. Presents with lethargy, poor feeding, high-pitched cry, opisthotonus, seizures, and choreoathetosis.

Kernig Sign

- Sign of meningeal irritation. Patient supine, hip and knee flexed 90 degrees; inability to extend knee indicates positive sign. Seen in meningitis, subarachnoid hemorrhage.

Keshan disease

- Heart disease caused by a lack of selenium, an element that the body needs to function properly.

Ketamine

- A dissociative anesthetic that produces analgesia, amnesia, and dose-dependent unconsciousness primarily through NMDA-receptor antagonism. It often preserves spontaneous ventilation and airway reflexes better than many hypnotics but can cause emergence reactions, hypersalivation, nausea, and sympathetic stimulation.

Ketogenesis

- See biochemistry.

Ketogenic Diet

- High fat, adequate protein, very low carbohydrate diet (<50g/day). Induces ketosis. Medical uses: epilepsy, obesity, metabolic syndrome. Risks: nutrient deficiency, kidney stones, dyslipidemia.

Ketone Bodies

- Water-soluble molecules produced mainly by the liver from fatty-acid-derived acetyl-CoA during fasting, starvation, prolonged exercise, or insulin deficiency. The principal ketone bodies are acetoacetate, beta-hydroxybutyrate, and acetone.

Ketones

- Acidic byproducts produced when the body breaks down fat for energy instead of glucose (ketogenesis). Ketones can be detected in blood and urine.

Ketosis

- Metabolic state with elevated ketone bodies. Physiological: fasting, ketogenic diet. Pathological: diabetic ketoacidosis. Distinguish from ketonemia vs ketonuria.

Kidney

- A paired, bean-shaped retroperitoneal organ responsible for filtering blood, excreting waste products as urine, and regulating fluid and electrolyte balance, acid-base homeostasis, and blood pressure (via the renin-angiotensin-aldosterone system). Each kidney contains approximately 1 million nephrons, the functional filtering units.

Kidney and Urinary Health

- Kidney and urinary health encompasses the function of the kidneys, ureters, bladder, and urethra in filtering waste, regulating fluid/electrolyte balance, and eliminating urine.

Kidney Cancer

- Renal cell carcinoma. Risk factors: smoking, obesity, hypertension.

Kidney Failure

- The inability of the kidneys to maintain fluid, electrolyte, and acid-base homeostasis and to excrete metabolic wastes. Acute kidney injury (AKI): sudden decline in GFR over hours-days—prerenal (hypoperfusion), intrinsic (acute tubular necrosis, glomerulonephritis, interstitial nephritis), postrenal (obstruction).

Kidney Stone

- Stone formation in the urinary tract. See nephrolithiasis.

Kidney stones

- Crystalline mineral deposits that form within the renal pelvis and urinary tract, affecting approximately 10 percent of the population over a lifetime, with a male-to-female ratio of 2:1 and peak incidence in the 4th-6th decades.

Kidney Stones (Nephrolithiasis) /

Urolithiasis)

- Crystalline mineral deposits formed within the renal pelvis and calyces due to supersaturation of urine.

Kidney Transplant Rejection

- Immune-mediated injury to a transplanted kidney caused by recipient antibodies, T cells, or both. Hyperacute, acute, and chronic rejection differ in timing and pathology and require correlation of graft function, imaging, serology, and often biopsy.

Kidney Transplantation

- Best treatment for ESRD with improved survival over dialysis. Living donor superior to deceased donor. Lifelong immunosuppression required. Rejection types include hyperacute, acute, and chronic. Complications include rejection, infection, vascular complications, and recurrent disease.

Kienbock Disease

- Kienbock disease (lunatomalacia): AVN of lunate. Lichtman classification: I (normal XR, MRI edema), II (lunate sclerosis), IIIA (lunate collapse, no scaphoid rotation), IIIB (lunate collapse, fixed scaphoid rotation, DISI), IV (perilunate arthritis). Etiology: negative ulnar variance, trauma, vascular variation.

Kiesselbach Plexus

- Kiesselbach plexus, also known as Little area, is a highly vascularized region located on the anterior nasal septum where five arterial anastomoses converge, making it the most common site of epistaxis (nosebleed), accounting for approximately ninety.

Killian's triangle

- An area of potential weakness in the pharyngeal wall between the thyropharyngeus and cricopharyngeus muscles, predisposing to Zenker diverticulum formation.

Killip Classification

- A clinical classification system for assessing the severity of myocardial infarction based on physical examination findings. Class I: no heart failure signs. Class II: mild heart failure (crackles, S3 gallop, elevated JVD). Class III: pulmonary edema. Class IV: cardiogenic shock.

Kimmelstiel-Wilson Nodules

- Characteristic round eosinophilic nodular mesan-

gial expansions in diabetic nephropathy. Represent advanced mesangial matrix accumulation. Found in peripheral mesangium associated with diffuse mesangial expansion, GBM thickening, and arteriolar hyalinosis.

Kimura Disease

- A chronic inflammatory disorder characterized by painless subcutaneous masses, commonly in the head and neck, regional lymphadenopathy, eosinophilia, and elevated immunoglobulin E. It is distinct from angiolymphoid hyperplasia with eosinophilia and may involve the kidneys.

Kinesio Taping

- See physical therapy.

Kinesiology

- Study of human movement. Applied to rehabilitation, sports, ergonomics.

Kinetic Modeling

- Mathematical description of tracer behavior. Compartmental models, graphical analysis.

Kinetics

- See pharmacology.

Klebsiella Pneumonia

- Severe, necrotizing pneumonia in alcoholic or debilitated patients. Upper lobe involvement with “bulging fissure sign” and “currant jelly” sputum.

Klebsiella rhinoscleromatis

- A gram-negative bacillus causing chronic granulomatous infection of the nasal passages and upper airway, leading to rhinoscleroma with progressive nasal obstruction and tissue destruction.

Klinefelter Syndrome

- Most common sex chromosome aneuploidy in males with a karyotype of 47,XXY occurring in approximately one in six hundred live male births. Characterized by small firm testes, gynecomastia, tall stature with long extremities, and infertility from azoospermia.

Klippel-Feil Syndrome

- A congenital condition characterized by fusion of two or more cervical vertebrae, resulting from failure of normal segmentation of the cervical somites during weeks 3-8 of gestation. Classic triad: short neck, low posterior hairline, and limited cervical

range of motion (present in <50% of cases).

Klippel-Trenaunay Syndrome

- A congenital vascular disorder characterized by the triad: capillary malformations (port-wine stains), venous malformations (varicose veins, persistent embryonic veins), and soft tissue or bone hypertrophy of the affected limb (typically a lower extremity).

Klumpke Paralysis

- Lower-trunk brachial-plexus injury affecting primarily the C8 and T1 nerve roots. It causes weakness or paralysis of the intrinsic hand muscles, sensory loss along the medial forearm and hand, and occasionally Horner syndrome when sympathetic fibres are involved.

Knee Disorders

- The knee is a complex hinge-type synovial joint connecting the femur, tibia, and patella, stabilized by ligaments (ACL, PCL, MCL, LCL), menisci (medial and lateral), and surrounding muscles/tendons.

Knee Joint

- Largest, most complex joint with three compartments: medial/lateral tibiofemoral and patellofemoral. Tibiofemoral is a modified hinge allowing flexion-extension with rotation when flexed. Stabilized by cruciate (ACL/PCL) and collateral (MCL/LCL) ligaments; menisci cushion and distribute load. The suprapatellar bursa is the body's

Knee Pain

- Pain felt in or around the knee, commonly caused by injury, osteoarthritis, bursitis, tendinitis, or inflammatory disease. Persistent swelling, instability, inability to bear weight, or fever requires medical assessment.

Knee Prosthesis

- Total knee replacement. Components: femoral component (metal), tibial component (metal + polyethylene insert), patellar button (optional). Fixation: cemented or cementless. Indications: end-stage osteoarthritis, rheumatoid arthritis, post-traumatic. Outcomes: pain relief, function improvement, longevity 15-20+ years.

Kneecap

- The largest sesamoid bone in the human body, embedded within the quadriceps tendon anterior to the knee joint. See Patella (Kneecap).

Kniest Dysplasia

- A rare skeletal dysplasia caused most often by pathogenic variants in COL2A1. It produces disproportionate short stature, enlarged joints, progressive joint disease, distinctive facial features, and sometimes severe ocular, auditory, or respiratory complications.

Knife-cut ulcers from Crohn's disease

- Deep, linear, serpiginous ulcerations of the intestinal mucosa in Crohn disease, resembling cuts made by a knife. These longitudinal ulcers with heaped-up margins are characteristic of Crohn disease and help distinguish it from ulcerative colitis on endoscopy. They may cause strictures, fistulae, and bleeding.

Knock Knees

- Medial angulation of the knees causing the knees to touch while the ankles remain apart, also called genu valgum. It is common during childhood development but may be pathologic when severe, asymmetric, persistent, or associated with pain or growth disturbance.

Knotless Anchor

- Suture anchor for soft tissue repair without tying knots. Used in orthopedic surgery.

KOH Preparation

- Microscopic examination of skin, hair, or nail scrapings treated with potassium hydroxide to dissolve keratin and reveal fungal hyphae or yeast. A negative preparation does not exclude infection when sampling is inadequate or prior treatment has been given.

Koilonychia

- Spoon nails: nails with a concave (spoon-shaped) contour, most commonly associated with iron deficiency anemia; also seen in hemochromatosis, Plummer-Vinson syndrome, malnutrition, and trauma (nail dystrophy). The nail becomes thin, brittle, and turns up at the edges so a drop of water rests in the concavity.

Koplik Spots

- Pathognomonic enanthem of Measles (Rubeola virus infection). Consists of small, 1-2 mm, bluish-white spots on an erythematous background located on the buccal mucosa opposite the lower molars ("grains of salt on a red background"). Appear 1-2

days prior to the onset of the maculopapular rash and disappear 1-2 days after rash onset.

Krabbe Disease

- An autosomal-recessive lysosomal storage disorder caused by deficient galactocerebrosidase activity and accumulation of psychosine. Infantile disease causes irritability, stiffness, developmental regression, seizures, and progressive loss of neurological function; later-onset forms are variable.

Kt/V

- Dialysis adequacy measure representing the fractional clearance of urea. K is dialyzer clearance, t is treatment time, V is volume of distribution of urea. Target Kt/V greater than 1.4 for thrice-weekly hemodialysis. Urea reduction ratio greater than 70% is alternative adequacy measure.

Kussmaul Breathing

- A deep, rapid breathing pattern seen in severe metabolic acidosis (diabetic ketoacidosis, lactic acidosis, uremia). It represents the body's attempt to compensate by blowing off CO₂. The increased respiratory rate and depth create the characteristic labored breathing.

Kussmaul Respirations

- Deep, labored breathing pattern associated with diabetic ketoacidosis. Compensatory hyperventilation to blow off CO₂ and correct metabolic acidosis. Name: Adolf Kussmaul, German physician.

Kussmaul Sign

- Paradoxical increase in jugular venous pressure with inspiration, normally the JVP decreases with inspiration. It indicates impaired right ventricular filling during inspiration. Seen in constrictive pericarditis, restrictive cardiomyopathy, severe right heart failure, cardiac tamponade, and massive pulmonary embolism.

Kwashiorkor

- A severe form of protein-energy malnutrition (PEM) characterized by a severe protein deficiency relative to calorie intake (often occurring in children when weaned from breast milk to a high-carbohydrate starch diet).

Kyphoplasty

- Similar to vertebroplasty but uses balloon to create cavity before cement injection.

Kyphoscoliosis

- An abnormal curvature of the spine in both the coronal (scoliosis) and sagittal (kyphosis) planes. Severe forms cause restrictive lung disease by reducing chest wall compliance and lung volumes. It can lead to chronic respiratory failure and pulmonary hypertension.

Kyphosis

- An excessive anterior-posterior curvature of the thoracic spine, measured by the Cobb angle on lateral spine x-ray (normal thoracic kyphosis is 20-40 degrees).

Keratoconjunctivitis Sicca

- Chronic dryness and inflammation of the cornea and conjunctiva, commonly called dry-eye syndrome. Symptoms include burning, grittiness, redness, fluctuating vision, and light sensitivity; causes include reduced tear production, excessive evaporation, medicines, and autoimmune disease.

Kidney Function Tests

- Blood and urine tests used to assess how well the kidneys filter waste and regulate fluid and electrolytes. Common measures include creatinine, estimated glomerular filtration rate, urine albumin, and urinalysis; results are interpreted with the clinical context.

Chapter 12

Pocket Dictionary: L Terms

Labor

- The process of uterine contractions leading to progressive cervical dilation and effacement and delivery of the fetus. Stages: first (latent—cervical dilation to 3-4 cm, then active—3-10 cm), second (full dilation to delivery of the infant), third (delivery of the placenta), fourth (first hour postpartum).

Labor Dystocia

- Abnormally slow or difficult labor caused by inadequate uterine forces, abnormal fetal position or size, or an inadequate maternal pelvis. Assessment considers the powers, passenger, and passage and guides appropriate augmentation or operative delivery.

Laboratory Test

- Analysis of body fluids or tissues. Includes hematology, chemistry, microbiology, immunology.

Labour

- The process of childbirth, consisting of regular uterine contractions that cause progressive cervical effacement and dilation, leading to delivery of the baby and placenta.

Labyrinth

- The inner ear. It contains the cochlea, which is responsible for hearing, as well as structures that are needed for balance.

Labyrinthitis

- Inflammation of the labyrinth, the inner ear structure containing the cochlea and vestibular apparatus, resulting in both auditory and vestibular symptoms including sudden, severe vertigo, sensorineural hearing loss, tinnitus, nausea, and vomiting.

Laceration

- A wound caused by tearing of body tissue, typically caused by trauma from a sharp object (knife,

glass, metal) or blunt force that splits the skin. Lacerations involve the skin and may extend into subcutaneous tissue, muscle, tendon, nerve, or blood vessels.

Laceration (Forensic)

- Tear or disruption of tissue produced by blunt force trauma exceeding skin elasticity, featuring irregular bruised edges, tissue bridging (intact nerves, vessels across wound floor), and hair crushing.

Lachman Test

- ACL integrity test. Knee 20-30 degrees flexed, anterior drawer. Positive = ACL tear.

Lacrimal Apparatus

- Produces, distributes, and drains tears for corneal moisture and protection. Components: lacrimal gland (superolateral orbit, parasympathetic CN V1, produces aqueous tear layer), superior/inferior puncta (medial eyelid openings), canaliculi, lacrimal sac (lacrimal groove), and nasolacrimal duct (drains into inferior meatus).

Lacrimal Gland

- The paired almond-shaped exocrine glands that secrete the aqueous layer of the tear film, located in the lacrimal fossa of the superior lateral orbit (orbital part) and the upper eyelid (palpebral part).

Lactate

- A product of anaerobic metabolism, elevated in tissue hypoperfusion and hypoxia. Normal: <2.0 mmol/L. Lactate >2.0: hyperlactatemia. Lactate >4.0: severe sepsis/septic shock (associated with increased mortality).

Lactate Dehydrogenase

- Non-specific enzyme. Elevated in tissue damage. Isoenzymes differentiate source.

Lactation

- The physiological process of milk production and secretion from the mammary glands, occurring after childbirth (puerperium). See Lactation (Breastfeeding).

Lactation (Breastfeeding)

- The production of milk by the mammary glands stimulated by the hormone prolactin. Colostrum is produced for the first 2-5 days (rich in IgA, proteins, and growth factors). Mature milk appears at day 5-7. The milk ejection reflex (let-down) is mediated by oxytocin.

Lactation Nutrition

- Nutrition and hydration during breastfeeding sufficient to support maternal health and milk production. Requirements vary with maternal stores, dietary pattern, and the degree of breastfeeding; selected supplements may be required according to local guidance and clinical assessment.

Lactic Acid

- A product of anaerobic metabolism, elevated in tissue hypoperfusion and hypoxia. See Lactate.

Lactic Acidosis

- Lactate >4 mmol/L with acidemia. Types: Type A (tissue hypoxia: shock, sepsis, hypoxia), Type B (non-hypoxic: metformin, liver failure, malignancy, drugs). Treatment: address underlying cause, supportive care.

Lactose

- See biochemistry.

Lactose Fermentation

- The ability of bacteria to ferment lactose to acid (and gas), a key differential characteristic on MacConkey agar. Lactose-fermenting enteric gram-negative bacteria produce pink/red colonies (*E. coli*, *Klebsiella*, *Enterobacter*, *Citrobacter*); non-lactose fermenters form colorless colonies (*Salmonella*, *Shigella*, *Proteus*, *Pseudomonas*).

Lactose intolerance

- A common digestive disorder caused by lactase deficiency leading to malabsorption of lactose, the primary disaccharide in milk and dairy products. The lactase enzyme is produced by enterocytes of the small intestinal brush border and hydrolyzes lactose into glucose and galactose for absorption.

Lactose Tolerance Test

- Non-invasive test for carbohydrate malabsorption (lactose, fructose) and small intestinal bacterial overgrowth (SIBO). See Hydrogen Breath Test.

Lafora Disease

- A progressive autosomal-recessive myoclonic epilepsy caused by pathogenic variants affecting glycogen metabolism. It usually begins in adolescence with generalized seizures, visual symptoms, myoclonus, cognitive decline, and progressive neurological deterioration.

Lambert-Eaton Myasthenic Syndrome

- Autoantibodies against presynaptic calcium channels at the neuromuscular junction, associated with SCLC. Proximal weakness, reduced reflexes, facilitation with exercise.

Laminectomy

- Surgical removal of the lamina (posterior vertebral arch) to decompress the spinal canal. Indications: spinal stenosis (lumbar and cervical), herniated disc (with central stenosis), and spinal tumor. Decompression of the entire spinal canal. Often combined with discectomy (lumbar) or foraminotomy (nerve root decompression).

Lancet

- Small needle for fingerstick. Used for blood glucose testing.

Langerhans-cell histiocytosis

- A group of disorders characterized by clonal proliferation of Langerhans cells (antigen-presenting dendritic cells), which can infiltrate bone, skin, and central nervous system. CNS involvement may cause diabetes insipidus, hypothalamic-pituitary dysfunction, or cerebellar degeneration.

Language Barrier

- Exchange of information between provider and patient. See clinical communication.

Language Therapy

- See speech-language pathology.

Lanugo

- Fine, soft hair that grows all over the body of a fetus and is typically shed before birth.

Laparoscopy

- A minimally invasive surgical technique using

a laparoscope (a thin, lighted telescope) inserted through a small incision (typically at the umbilicus) to visualize the abdominal and pelvic organs. Carbon dioxide gas is insufflated to create a working space. Additional small incisions (ports) allow insertion of instruments.

Laparotomy

- A surgical incision into the abdominal cavity, typically a midline vertical or transverse (Pfannenstiel) incision, providing wide access to the abdominal and pelvic organs.

Large Airway Obstruction

- Obstruction of airways with diameter > 2 mm (trachea and main bronchi). It causes significant symptoms including stridor, dyspnea, and impaired secretion clearance. It may be fixed (stenosis, tumor) or variable (tracheomalacia, vocal cord dysfunction). Flow-volume loops show characteristic patterns.

Large Bowel Obstruction

- Mechanical blockage of the colon most commonly caused by colorectal cancer, volvulus, or diverticular stricture. Clinical presentation includes progressive abdominal distension, constipation, and obstipation. The cecum is at highest risk of perforation with distal obstruction. Diagnosis is by CT scan identifying the site and cause.

Large for Gestational Age

- Birth weight or fetal size above the 90th percentile for gestational age. It is associated with maternal diabetes, excessive maternal weight gain, and genetic or constitutional factors and increases the risk of shoulder dystocia, birth trauma, and neonatal hypoglycemia.

Large Intestine

- Final GI segment (approximately 1.5 meters) from ileocecal valve to rectum. Six segments: cecum, ascending, transverse, descending, sigmoid colon, rectum. Absorbs water, electrolytes, and vitamins (K, biotin) from colonic bacteria. Characterized by haustra, taeniae coli, and epiploic appendages.

Laron Syndrome

- An inherited disorder of growth-hormone signaling, usually caused by growth-hormone-receptor defects. It produces severe postnatal growth failure, char-

acteristic facial features, and very low insulin-like growth factor 1 despite elevated circulating growth hormone.

Larva currens from Strongyloides

- See Strongyloidiasis.

Laryngeal Cancer

- Malignant tumor arising in the larynx, most commonly squamous cell carcinoma. Risk factors include tobacco, alcohol, and selected HPV-related disease; persistent hoarseness, dysphagia, odynophagia, stridor, or neck mass requires laryngoscopic evaluation and biopsy.

Laryngeal candidiasis

- Fungal infection of the larynx by *Candida* species, causing hoarseness, dysphagia, and throat pain, most commonly in immunocompromised patients or those using inhaled corticosteroids.

Laryngeal Nerve Damage

- Injury to the recurrent or superior laryngeal nerves causing hoarseness, vocal-cord weakness, swallowing difficulty, or airway compromise. Common causes include thyroid or neck surgery, malignancy, trauma, and compression along the nerve pathway.

Laryngeal Prosthesis

- Voice prosthesis after laryngectomy. Tracheoesophageal puncture, one-way valve.

Laryngectomy

- Surgical removal of part or all of the larynx, most often for squamous cell carcinoma of the larynx or hypopharynx and less often for severe trauma or refractory aspiration. Total laryngectomy requires a permanent tracheostomy, so the patient breathes through a neck stoma; partial laryngectomy preserves voice and swallowing.

Laryngectomy Tube

- Tube maintaining airway after total laryngectomy. Types: silicone, metal. Indications: laryngeal cancer, trauma. Complications: obstruction, infection, granulation.

Laryngitis

- Inflammation of the larynx and vocal cords that presents with hoarseness, voice loss, sore throat, dry cough, and a sensation of a lump in the throat (globus sensation), and it may be acute or chronic.

Laryngocele

- Abnormal air-filled dilation of the laryngeal sacculus that may be internal, external, or combined, presenting with hoarseness, neck swelling, or airway obstruction.

Laryngomalacia

- Dynamic collapse of supraglottic laryngeal tissue during inspiration, causing congenital stridor in infants. Most cases improve with growth, but feeding difficulty, apnea, hypoxemia, or failure to thrive requires specialist assessment and possible intervention.

Laryngopharyngeal Reflux (LPR)

- Retrograde movement of gastric contents above the upper esophageal sphincter into the larynx and pharynx, producing chronic hoarseness, throat clearing, globus sensation, cough, excessive throat mucus, and laryngospasm, often without heartburn.

Laryngoscope

- Device for visualizing vocal cords during intubation. Blades: Mac (curved), Miller (straight).

Laryngoscopy

- A medical procedure used to visualise the larynx (voice box) and adjacent structures of the throat. Types: (1) Indirect laryngoscopy: the simplest technique, using a laryngeal mirror angled at 120 degrees to reflect the laryngeal image, with a headlight or external light source.

Laryngospasm

- A protective reflex causing violent adduction of the vocal cords, resulting in complete airway obstruction. It is most common during light anesthesia, emergence, and in pediatric patients. Triggers include secretions, blood, airway manipulation, and suctioning.

Larynx

- The larynx, commonly known as the voice box, is a complex organ located in the anterior neck between the pharynx and the trachea that serves three primary functions: phonation (voice production), protection of the lower airway from aspiration during swallowing, and regulation of airflow during respiration.

Laser (Vascular)

- See Laser Catheter.

Laser Ablation

- Laser energy for vessel closure. Endovenous laser ablation for varicose veins.

Laser Dentistry

- Use of lasers for various procedures: caries detection, soft tissue surgery, whitening, periodontal therapy.

Laser Fiber

- Optical fiber for laser delivery in surgery. Used in lithotripsy, tumor ablation.

Laser Trabeculoplasty

- A minimally invasive laser procedure used in the treatment of open-angle glaucoma that lowers intraocular pressure by improving the outflow of aqueous humor through the trabecular meshwork.

LASIK

- LASIK, or laser-assisted in situ keratomileusis, is a refractive surgical procedure that corrects myopia, hyperopia, and astigmatism by reshaping the corneal stroma using an excimer laser, after creating a thin corneal flap with either a microkeratome (a mechanical blade) or a femtosecond laser.

LASIK (Laser-Assisted In Situ Keratomileusis)

- A refractive surgical procedure that reshapes the cornea using an excimer laser to correct myopia, hyperopia, and astigmatism, thereby reducing dependence on glasses or contact lenses.

Lassa Fever

- An acute viral hemorrhagic illness caused by Lassa virus, an arenavirus, endemic in West Africa (Nigeria, Sierra Leone, Guinea, Liberia). Transmission: exposure to excreta (urine, feces) of the multimammate rat (*Mastomys natalensis*), the natural reservoir. Incubation: 6-21 days.

Last Menstrual Period (LMP)

- The first day of the last normal menstrual period, used to estimate gestational age and the due date. Requires regular menstrual cycles and reliable recall. Limitations: irregular cycles, recent oral contraceptive use, early pregnancy bleeding mistaken for menses, and poor recall.

Latarjet Procedure

- Surgical procedure for shoulder instability using

coracoid process transfer. Provides bony block to prevent recurrent dislocation.

Late Decelerations

- A fetal heart rate pattern with a gradual decrease (onset to nadir >30 seconds) that starts at or after the peak of the contraction and returns to baseline after the contraction ends. Associated with uteroplacental insufficiency (reduced oxygen transfer across the placenta).

Late Effects

- Health problems occurring months to years after cancer treatment. Cardiac, pulmonary, endocrine, secondary cancers.

Late Preterm (34-36 weeks)

- Preterm births between 34 weeks 0 days and 36 weeks 6 days, accounting for 70% of all preterm births. Late preterm infants are at increased risk compared to term infants: respiratory distress (TTN, RDS), temperature instability, hypoglycemia, jaundice, feeding difficulties, and hospital readmission.

Latent Infection

- Persistent infection in which an infectious agent remains in a host in a nonreplicating or intermittently replicating state without active clinical disease. Reactivation can occur when immune control is impaired, as in latent tuberculosis.

Latent Phase of Labor

- The early part of the first stage of labor, from the onset of regular contractions to 6 cm cervical dilatation. Characterized by slow, progressive cervical change (usually <1 cm/hour). Duration is highly variable: up to 20 hours in nulliparas and 14 hours in multiparas.

Latent Tuberculosis Infection

- Persistent immune response to *M. tuberculosis* without active disease. Asymptomatic and non-infectious. 5–10% of immunocompetent individuals develop active TB without treatment. Treatment reduces risk by 60–90%.

Lateral

- Positioned away from the midline of the body. Opposite of medial. Example: the thumb is lateral to the little finger in anatomical position.

Lateral Collateral Ligament Injury

- Tear of the LCL, less common than MCL injury but more significant. Mechanism: varus stress to the knee (direct blow to medial knee). Presents with lateral knee pain, swelling, and varus laxity. Part of the posterolateral corner (PLC) — when injured, causes significant rotational instability.

Lateral Cuneiform

- Located between intermediate cuneiform and cuboid, articulating with navicular, intermediate cuneiform, cuboid, and third metatarsal. Contributes to the transverse arch and lateral foot column. Injuries typically from high-energy midfoot trauma.

Lateral Rotation

- Rotation of a limb away from the median plane of the body.

Latex Allergy

- An immune-mediated hypersensitivity reaction to natural rubber latex. Contact may cause dermatitis, urticaria, rhinitis, asthma, or anaphylaxis, especially in healthcare and occupational settings.

Latissimus Dorsi Muscle

- Broadest back muscle forming the posterior axillary wall, from T7-L5 spinous processes, thoracolumbar fascia, iliac crest, and lower ribs to the intertubercular groove floor. Innervated by thoracodorsal nerve (C6-C8). Adducts, medially rotates, extends the arm powerfully in swimming/rowing/climbing. Common reconstructive surgery flap.

Lawton IADL Scale

- Instrumental ADLs: phone use, shopping, food preparation, housekeeping, laundry, transportation, medication management, finance management.

Laxative

- A drug that promotes bowel evacuation, used for constipation and bowel preparation. Classes: bulk-forming (psyllium, methylcellulose), osmotic (polyethylene glycol, lactulose, magnesium salts), stimulant (senna, bisacodyl), stool softener (docusate), and lubricant (mineral oil). Choice depends on constipation type and patient factors.

Laxatives

- Medications used to treat constipation by stimulating bowel movements or softening stool. In ob-

stetrics, constipation is common in pregnancy (progesterone slows bowel motility, iron supplements, and later mechanical compression) and postpartum (perineal pain, fear of damaging sutures, opioid pain relief).

LD50

- Lethal dose for 50% of test population. Measure of acute toxicity. Lower LD50 = more toxic. Measured in mg/kg body weight. Ethical concerns limit animal testing.

LDL

- Low-density lipoprotein; the 'bad' cholesterol that transports cholesterol to peripheral tissues and is the principal driver of atherosclerosis. Elevated LDL is a major modifiable cardiovascular risk factor; guidelines target LDL levels based on global risk (statin intensity).

Lead Poisoning

- Neurotoxic, hematological, renal effects. Sources: paint, water, gasoline, occupational. Children more vulnerable. Treatment: chelation (succimer, EDTA, dimercaprol), remove exposure. Blood lead level guides treatment.

Lead-Time Bias

- The appearance of improved survival after screening because diagnosis occurs earlier, even when the time of death is unchanged. It can make screening appear beneficial without reducing mortality.

Leber Congenital Amaurosis

- The most severe inherited retinal dystrophy, presenting at birth or within the first few months of life with profound visual impairment, nystagmus, and pupillary light-near dissociation.

Left Anterior Descending Artery

- A major branch of the left coronary artery that travels in the anterior interventricular groove and supplies the anterior left ventricle, much of the interventricular septum, and the apex. Its occlusion can cause extensive anterior myocardial infarction.

Left Atrial Myxoma

- Most common primary cardiac tumor (75%), typically originating from the interatrial septum in the left atrium. Presents with embolic events (25%), intracardiac obstruction (mitral valve obstruction mimicking mitral stenosis), and constitutional symp-

toms (fever, weight loss, elevated ESR mimicking infection or malignancy).

Left Atrium

- Receives oxygenated blood from 4 pulmonary veins. Most posterior chamber forming the cardiac base on chest radiography. Smooth walls from absorbed pulmonary veins; small auricle with pectinate muscles. Mitral valve separates it from the left ventricle.

Left Bundle Branch

- A continuation of the bundle of His that divides into anterior and posterior fascicles. The anterior fascicle supplies the anterior and superior left ventricular wall; the posterior fascicle supplies the inferior and posterior wall.

Left hemiparesis, gaze deviated to the right

- A clinical finding indicating a right hemisphere lesion, likely a right middle cerebral artery stroke. Gaze deviation toward the right suggests the lesion is in the right frontal eye field, which normally drives gaze to the contralateral (left) side; when damaged, the intact left frontal eye field drives gaze ipsilaterally (to the right).

Left Hypochondriac Region

- Upper left abdomen lateral to the epigastric region, containing the spleen, stomach fundus, splenic flexure, and pancreatic tail. Important for evaluating splenomegaly, gastric pathology, and splenic infarction. The spleen is normally not palpable below the costal margin; enlargement into this region is a significant finding.

Left Iliac Region

- Lower left abdomen containing the descending colon, sigmoid colon, and left ureter. Important for evaluating diverticulosis and diverticulitis, most commonly affecting the sigmoid colon here. Palpation may reveal masses from inflammatory bowel disease or colorectal cancer.

Left Main Coronary Artery

- The short vessel arising from the left aortic sinus and dividing principally into the left anterior descending and circumflex arteries. Significant left-main disease jeopardizes a large proportion of left-ventricular myocardium.

Left testicular torsion

- Acute rotation of the left testis on its spermatic cord, obstructing blood flow and causing ischemia. It is a urological emergency presenting with sudden severe left scrotal pain, nausea, vomiting, and a high-riding, horizontally oriented testis with absent cremasteric reflex.

Left Ventricle

- Thickest cardiac chamber (13-15 mm), pumping oxygenated blood into the aorta for systemic distribution. Generates approximately 120 mmHg systolic pressure. Receives blood through the mitral valve; ejects through the aortic valve.

Left Ventricular Noncompaction

- A rare cardiomyopathy characterized by prominent myocardial trabeculations with deep intertrabecular recesses, thought to result from arrested myocardial compaction during embryogenesis. It may present with heart failure, arrhythmias, or thromboembolism.

Left-sided superior vena cava

- A congenital venous anomaly where the left superior vena cava drains into the coronary sinus instead of the right atrium via the right SVC. It is the most common thoracic venous anomaly, present in 0.3–0.5% of the general population. Usually asymptomatic but relevant during central venous catheterization or cardiac surgery.

Legg-Calve-Perthes Disease

- Idiopathic avascular necrosis of the femoral head in children, typically ages 4-8 years. More common in boys (4:1) and Caucasian children. Presents with limping (antalgic gait), hip/knee pain, and limited hip range of motion (especially abduction and internal rotation). Etiology: temporary disruption of blood supply to the femoral head.

Legg-Calvé-Perthes disease

- Idiopathic avascular necrosis of the femoral head in children, typically between ages two and twelve, resulting from disrupted blood supply to the capital femoral epiphysis. It presents with insidious onset of hip pain, limp, and restricted range of motion.

Legionella Pneumonia

- From *Legionella pneumophila* in water systems. Presents with high fever, diarrhea, confusion, and

hyponatremia. Diagnosis by urinary antigen test. Treatment is fluoroquinolones or macrolides.

Legionella pneumophila

- A gram-negative rod that causes Legionnaires' disease, a severe pneumonia, and Pontiac fever, a milder illness. The organism is found in aquatic environments and is transmitted through inhalation of contaminated water droplets. Risk factors include age, smoking, and immunosuppression.

Legionnaires' Disease

- A severe form of pneumonia caused by *Legionella pneumophila*, a Gram-negative aerobic bacillus that is found naturally in freshwater environments (lakes, streams) and artificially in water systems (air conditioning cooling towers, hot tubs, decorative fountains, plumbing systems, showers, humidifiers).

Leiomyoma

- Uterine fibroid: a benign smooth muscle tumor of the uterus (estrogen/progesterone-sensitive), the most common pelvic tumor in women. Locations: subserosal, intramural, submucosal (most symptomatic).

Leiomyoma (Fibroid)

- A benign smooth muscle tumor of the uterus, arising from the myometrium. See Fibroids (Uterine Leiomyomas).

Leiomyosarcoma

- A malignant tumor of smooth muscle, most often arising in the uterus, retroperitoneum, or gastrointestinal tract.

Leishmania donovani

- A protozoan parasite that causes visceral leishmaniasis (kala-azar), transmitted by the bite of infected female phlebotomine sandflies. The organism exists as flagellated promastigotes in the sandfly gut and as non-flagellated amastigotes within human macrophages.

Leishmaniasis

- A parasitic disease caused by protozoa of the genus *Leishmania*, transmitted by the bite of infected female phlebotomine sandflies. Visceral leishmaniasis (kala-azar): caused by *L. donovani*/*L. infantum* – fever, hepatosplenomegaly, pancytopenia, weight loss, hypergammaglobulinaemia; fatal if untreated.

Length-Time Bias

- The tendency of screening to preferentially detect slower-growing or less aggressive disease because it remains detectable for a longer preclinical period. It can exaggerate the apparent benefit of screening.

Lens

- Transparent biconvex avascular structure behind the iris/pupil, suspended by zonular fibers from the ciliary body. Provides approximately one-third of refractive power and allows accommodation (focusing on near objects by changing shape). Concentric lens fiber layers from surface epithelium, surrounded by a capsule.

Lentigo maligna

- A melanoma in situ that occurs on chronically sun-damaged skin, most commonly on the face of elderly individuals. It presents as a large, irregularly pigmented, slow-growing macule with variegated shades of brown and black. The radial growth phase may persist for years before invasion occurs.

Leopold Maneuvers

- A systematic abdominal palpation technique used to determine fetal position and presentation. Four maneuvers: (1) fundal grip identifies fetal lie (head vs breech), (2) umbilical grip locates the fetal back, (3) Pawlik grip identifies the presenting part and its position, (4) pelvic grip confirms presentation and attitude.

Lepromatous leprosy

- The disseminated, multibacillary pole of the leprosy spectrum characterised by poor cell-mediated immunity to *Mycobacterium leprae*, resulting in widespread skin nodules, plaques, and diffuse infiltration with thickening of the skin and facial features (leonine facies).

Leprosy

- See Hansen's Disease (Leprosy).

Leprosy (Hansen's Disease)

- A chronic infectious disease caused by *Mycobacterium leprae*, an acid-fast, obligate intracellular bacillus. See Hansen's Disease (Leprosy).

Leptin

- Satiety hormone from adipose tissue. Signals energy sufficiency to brain.

Leptomeningeal Metastasis

- Spread of cancer to the leptomeninges and cerebrospinal fluid (carcinomatous meningitis), causing multifocal neurologic deficits; managed with intrathecal chemotherapy and involved-field radiation.

Leptospirosis

- A zoonotic bacterial infection caused by spirochaetes of the genus *Leptospira*, transmitted through contact with water or soil contaminated with the urine of infected animals (rodents, cattle, dogs). Incubation period 5-14 days. Mild form: fever, headache, myalgia (especially calves), conjunctival suffusion.

Leriche Syndrome

- Aortoiliac occlusive disease causing reduced blood flow to the pelvis and lower limbs. The classic presentation includes buttock or thigh claudication, diminished femoral pulses, and erectile dysfunction, although symptoms vary with the extent of collateral circulation.

Lesion

- A general term for any abnormal change in tissue structure due to disease or injury.

Lesser Curvature

- Concave medial stomach border from gastroesophageal junction to pylorus. Attachment site for the lesser omentum connecting to the liver, containing right/left gastric arteries. Most common gastric ulcer site. The angular incisure marks the body-antrum junction. Cancer here may cause early obstruction from the narrow lumen.

Lesser Trochanter

- Small conical posteromedial proximal femoral prominence. Insertion for iliopsoas (primary hip flexor). Isolated fractures in adults may indicate pathologic fracture from metastatic disease; in adolescents may result from iliopsoas avulsion. Visible on lateral hip radiographs.

Lethal-Means Counseling

- A preventive intervention that reduces access to firearms, medications, toxic substances, or other methods of self-harm during periods of elevated risk. It is conducted collaboratively and is an important component of suicide prevention.

Leucocyte (White Blood Cell)

- See White Blood Cell / Leucocyte.

Leucopenia

- A decreased total white blood cell count below the normal reference range ($< 4.0 \times 10^9/L$). Leucopenia is often due to a decrease in neutrophils (neutropenia) or lymphocytes (lymphopenia).

Leukemia

- A group of hematological malignancies characterized by the clonal proliferation of abnormal white blood cells (blasts) in the bone marrow and peripheral blood. leukemia is classified by: (1) clinical course: acute (rapid onset, $>20\%$ blasts in bone marrow) vs chronic (insidious onset, more differentiated cells).

Leukemia cutis

- Infiltration of the skin by leukemic cells, presenting as papules, nodules, or plaques. It is more common in acute myeloid leukemia and indicates a poorer prognosis.

Leukemia Genetics

- Acute lymphoblastic leukemia (ALL): Philadelphia chromosome (t9;22), Hyperdiploidy, ETV6-RUNX1 fusion. Acute myeloid leukemia (AML): t8;21, t15;17 (APL), FLT3 mutations. Chronic myeloid leukemia (CML): Philadelphia chromosome (BCR-ABL).

Leukemic infiltration of the gingiva

- Gingival enlargement caused by infiltration of leukemic cells, most commonly in acute myeloid leukemia, particularly the monocytic subtype. The gingiva becomes swollen, erythematous, friable, and may bleed spontaneously. It results from leukemic cell proliferation within the gingival connective tissue.

Leukemoid Reaction

- A marked reactive increase in the white-cell count, usually with prominent neutrophilia and immature granulocytes, caused by severe infection, inflammation, malignancy, or tissue injury. It resembles leukemia but is not a clonal hematologic malignancy.

Leukocyte Adhesion Deficiency

- LAD-I (ITGB2 mutation, CD18 deficiency): absent CD11/CD18 integrins $\rightarrow$ impaired neutrophil adhesion/transmigration $\rightarrow$ delayed umbilical cord

separation (>30 days), recurrent soft tissue infections without pus formation (leukocytosis $>20,000$), poor wound healing, periodontitis. Severe: death by age 2 without HSCT.

Leukocyte Alkaline Phosphatase

- An enzyme activity measured in mature neutrophils using a cytochemical stain, historically reported as a leukocyte alkaline phosphatase (LAP) score. The score is typically low in chronic myeloid leukaemia and increased in many leukemoid reactions, infections, pregnancy, and other reactive states.

Leukocytoclastic Vasculitis

- A histopathologic pattern of small-vessel vasculitis characterized by neutrophilic debris, fibrinoid change, and vessel-wall inflammation. Clinically it often presents with palpable purpura; causes may be cutaneous-limited or associated with systemic disease.

Leukocytosis

- An increased total white blood cell count above the normal reference range ($> 11 \times 10^9/L$ in adults).

Leukonychia

- White discoloration of the nails. True leukonychia: involves the nail matrix and moves distally with nail growth (punctate, striate/Mees lines from arsenic poisoning, or total).

Leukoplakia

- A clinical term for a white patch or plaque on the oral mucosa that cannot be rubbed off and cannot be characterized clinically as any other definable lesion (WHO). Leukoplakia is a potentially malignant disorder of the oral cavity, with a risk of malignant transformation to squamous cell carcinoma of approximately 1-5% per year.

Leukopoiesis

- The production and development of white blood cells (leukocytes) from hematopoietic stem cells in the bone marrow, regulated by colony-stimulating factors and interleukins.

Leukotomy

- A neurosurgical procedure involving the cutting of white matter fibers in the prefrontal cortex, also known as frontal lobotomy or prefrontal leukotomy. Developed by Egas Moniz (Nobel Prize 1949) and Walter Freeman.

Levator Ani

- The primary pelvic floor muscle complex supporting the pelvic organs, forming a hammock from the pubic symphysis to the coccyx. Composed of the pubococcygeus, puborectalis, and iliococcygeus muscles. Innervated by the pudendal nerve (S2-S4).

Levodopa

- The precursor of dopamine that crosses the blood-brain barrier; the most effective symptomatic treatment for Parkinson disease. Converted to dopamine in the brain by aromatic amino acid decarboxylase; always combined with carbidopa (peripheral decarboxylase inhibitor) to prevent peripheral conversion and side effects.

Lewy Body Dementia

- Dementia with visual hallucinations, parkinsonism, fluctuating cognition. Alpha-synuclein pathology. Treatment: cholinesterase inhibitors, avoid antipsychotics.

Lewy Body Disease

- A neurodegenerative disorder associated with abnormal neuronal alpha-synuclein aggregates. Dementia with Lewy bodies commonly features fluctuating cognition, recurrent visual hallucinations, REM sleep behavior disorder, and spontaneous parkinsonism; sensitivity to antipsychotics may occur.

Leydig Cell Tumor

- A testicular sex cord-stromal tumor that secretes testosterone or estrogen, causing precocious puberty or gynecomastia; most are benign and treated by orchidectomy.

Lfgren Syndrome

- Acute sarcoidosis with bilateral hilar lymphadenopathy, erythema nodosum, and ankle polyarthralgia. Excellent prognosis with spontaneous resolution. Associated with HLA-DR3.

Li-Fraumeni Syndrome

- An inherited cancer-predisposition syndrome most often caused by a pathogenic TP53 variant. It confers high lifetime risks of sarcomas, breast cancer, brain tumors, adrenocortical carcinoma, leukemia, and other early-onset malignancies.

LI-RADS

- Liver Imaging-Reporting and Data System. Stan-

dardizes reporting of liver imaging in patients with cirrhosis. Used for hepatocellular carcinoma diagnosis.

Libido

- Sexual desire or drive, influenced by biological, psychological, and social factors.

Libido and Sexual Performance

- Libido (sexual desire) is modulated by a complex interplay of hormonal (testosterone, estrogen, prolactin), neurologic, psychologic, and relational factors.

Lice

- Pediculosis: infestation by blood-sucking lice—head lice (*Pediculus humanus capitis*, common in children, scalp), body lice (*Pediculus humanus corporis*, clothing/seams, associated with poor hygiene and epidemic typhus transmission), and pubic/crab lice (*Phthirus pubis*, sexually transmitted).

Lice Infestations

- Pediculosis is infestation by head, body, or pubic lice. Head lice cause scalp pruritus and nits attached to hair shafts; body lice live mainly in clothing; pubic lice affect coarse hair-bearing regions.

Lichen

- A descriptive dermatological term referring to a flat-topped, firm, papular skin lesion resembling the lichen plant (a symbiotic organism of algae and fungi growing on rocks and trees). Lichenoid describes lesions that resemble lichen planus or lichenoid drug eruptions.

Lichen Planus

- A T-cell mediated inflammatory condition affecting skin, mucous membranes, nails, and hair. Classic presentation: the 6 Ps (pruritic, purple, polygonal, planar, papules, plaques), often with Wickham striae (white lacy network on surface). Oral lichen planus affects buccal mucosa and may be erosive.

Lichen Sclerosus

- A chronic inflammatory skin disease most commonly affecting anogenital skin. It presents as ivory-white, atrophic plaques with wrinkled paper-like appearance. In women, it predominantly affects the vulva and perianal region (figure-eight distribution). In men, it affects the glans penis and foreskin

(balanitis xerotica obliterans).

Lichen Simplex Chronicus

- A thickened, leathery plaque caused by chronic scratching and rubbing (itch-scratch cycle). It commonly affects the neck, ankles, lower legs, and genital area. It is a form of neurodermatitis associated with stress and anxiety.

Liddle Syndrome

- Autosomal dominant condition caused by gain-of-function mutations in the ENaC sodium channel subunits leading to constitutive sodium reabsorption and potassium secretion in the collecting duct. Presents with severe hypertension, hypokalemia, and metabolic alkalosis occurring in childhood or young adulthood.

Liddle Syndrome Treatment

- Liddle syndrome is an autosomal dominant cause of early-onset hypertension with hypokalemia and metabolic alkalosis caused by gain-of-function mutations in ENaC sodium channel subunits. Aldosterone levels are suppressed.

Lidocaine

- An amide local anesthetic that blocks voltage-gated sodium channels and prevents propagation of action potentials. It is used for local infiltration, nerve blocks, airway topicalization, and selected intravenous or antiarrhythmic indications; systemic toxicity can cause neurologic and cardiovascular effects.

Life Expectancy

- Average years remaining for person at given age. Varies by country, sex, socioeconomic status.

Ligament

- A dense, fibrous connective tissue band that connects bone to bone, providing passive joint stability and guiding joint motion. Ligaments are composed predominantly of type I collagen fibers (parallel orientation, crimped pattern) and elastin fibers (10-20% in some ligaments), arranged in bundles.

Ligand

- Any molecule that binds specifically to a biological receptor, macromolecule, or active site. In pharmacology and physiology, ligands include hormones, neurotransmitters, drugs, and toxins that bind receptors to initiate (agonist), block (antagonist), or

modulate signaling.

Ligation

- The surgical process of tying off a blood vessel or duct using a suture or clip to prevent bleeding or block fluid flow.

Ligation (Vascular)

- See Ligation Instrument.

Ligature

- Any cord, wire, rope, fabric, or flexible object used to constrict the neck, bind limbs, suspend the body, or inflict injury. The resulting pale groove-shaped abrasion may be horizontal with petechiae in homicidal strangulation, or oblique and upward in hanging.

Light Therapy

- Phototherapy for skin conditions, seasonal affective disorder, jaundice.

Light's Criteria

- An effusion is exudative if any one is met: pleural protein/serum protein > 0.5 , pleural LDH/serum LDH > 0.6 , or pleural LDH $> 2/3$ upper limit of normal serum LDH.

Light-near dissociation

- A pupillary abnormality in which the pupil does not constrict to light but does constrict during accommodation (near response). Classically seen in Argyll Robertson pupils (neurosyphilis) and also in midbrain lesions, diabetes mellitus, and optic nerve disease. Differentiates from tonic pupil (Adie pupil) by preserved near response.

Likelihood Ratio

- Probability of test result in disease vs. non-disease. $LR+ > 1$ increases probability; $LR- < 1$ decreases.

Likelihood Ratio (LR)

- A ratio that describes how much a test result changes the odds of disease. The positive likelihood ratio is sensitivity divided by one minus specificity; the negative likelihood ratio is one minus sensitivity divided by specificity.

Limbic Encephalitis

- Inflammation of the limbic system, often presenting subacutely with memory impairment, seizures, psychiatric or behavioral change, and sleep disturbance. It may be autoimmune, paraneoplastic, infectious,

or of unknown cause.

Limbic System

- A set of interconnected brain structures located in the medial temporal lobe and around the corpus callosum, involved in emotion, memory, motivation, behaviour, and autonomic regulation. The term was coined by Paul Broca (1878, *le grand lobe limbique*) and expanded by Papez (1937, Papez circuit) and MacLean (1952, 'visceral brain').

Linea Aspera

- Prominent longitudinal posterior femoral shaft ridge from converging muscle attachment lines. Insertion for adductor magnus, vastus medialis/lateralis, and short head of biceps femoris. Thickest femoral cortex, resisting bending forces. Diverges distally into supracondylar lines.

Linea Nigra

- A dark brown or black vertical line appearing on the abdomen during pregnancy, extending from the pubic symphysis to the umbilicus or beyond. Caused by hormonal stimulation of melanocytes, resulting in hyperpigmentation. More prominent in darker-skinned women.

Linear Energy Transfer (LET)

- Energy deposited per unit path length. High LET (alpha, neutrons) more biologically effective.

Linear IgA bullous dermatosis

- An autoimmune blistering disease characterised by linear deposition of IgA antibodies at the basement membrane zone, presenting with tense vesicles and bullae arranged in a "string of pearls" configuration on the trunk and extremities. It may be idiopathic or drug-induced, most commonly by vancomycin. Dapsone is the first-line treatment.

Lingua villosa nigra

- A benign condition characterized by elongation and dark pigmentation of the filiform papillae of the tongue, often associated with tobacco use, poor oral hygiene, or antimicrobial therapy.

Lingual artery

- A branch of the external carotid artery supplying the tongue, floor of mouth, and sublingual structures. It arises anterior to the hypoglossal nerve and courses deep to the hyoglossus muscle. Lingual artery aneurysm or thrombosis is rare but can cause

life-threatening hemorrhage or tongue ischemia.

Linitis Plastica

- A diffuse infiltrative form of gastric carcinoma in which tumour cells spread through the stomach wall, producing marked thickening and rigidity with loss of normal distensibility. It may cause early satiety, weight loss, nausea, vomiting, or vague upper-abdominal symptoms.

Linkage

- Co-inheritance of nearby genetic markers and disease gene. Measured by recombination fraction (theta). LOD score >3 indicates linkage.

Lipase

- Pancreatic enzyme. More specific than amylase for pancreatitis.

Lipedema

- A chronic disorder of abnormal, usually symmetrical deposition of painful subcutaneous adipose tissue, predominantly affecting the legs and sometimes the arms. The feet and hands are typically spared, and easy bruising, tenderness, and progressive functional limitation may occur.

Lipemia retinalis

- A creamy, pinkish-white discolouration of the retinal vessels caused by severely elevated serum triglycerides, typically exceeding 2,000 mg/dL. It is a sign of extreme hypertriglyceridaemia and indicates a risk for acute pancreatitis, requiring urgent lipid-lowering therapy and dietary management.

Lipid

- Diverse group of hydrophobic molecules including triglycerides (energy storage), phospholipids (membranes), cholesterol (membranes, steroid precursor), fat-soluble vitamins (A, D, E, K). Triglycerides: glycerol + 3 fatty acids. Fatty acids: saturated (no double bonds) vs. unsaturated (one or more double bonds).

Lipid Emulsion Therapy

- The antidote for local anesthetic systemic toxicity. The mechanism involves the lipid sink theory (lipid phase sequesters lipophilic local anesthetics from neural and cardiac tissue) and metabolic theory (lipid provides fatty acid substrate to ischemic myocardium).

Lipid Panel

- Measurement of total cholesterol, LDL-C, HDL-C, and triglycerides. Fasting (9-12 hours) or non-fasting acceptable for screening. Desirable levels: total cholesterol <200 mg/dL, LDL <100 mg/dL (<70 for very high risk), HDL >40 (men) or >50 (women) mg/dL, triglycerides <150 mg/dL.

Lipoatrophy

- Loss of subcutaneous adipose tissue at specific sites, most commonly caused by repeated subcutaneous injections of insulin or other medications, or associated with autoimmune conditions and HIV lipodystrophy.

Lipodystrophy

- See Lipoatrophy.

Lipodystrophy Syndromes

- Rare conditions characterized by selective loss of adipose tissue leading to severe insulin resistance, diabetes, and hypertriglyceridemia. Congenital generalized lipodystrophy involves loss of all body fat from birth. Familial partial lipodystrophy involves loss of limb and truncal fat with fat accumulation in face and neck.

Lipofuscin

- A yellow-brown granular pigment composed of oxidized lipids and cross-linked proteins that accumulates progressively within cells with age, particularly in post-mitotic cells such as neurons, cardiac myocytes, and hepatocytes.

Lipogenesis

- The synthesis of fatty acids and triglycerides from acetyl-CoA and other carbon sources. It occurs mainly in liver and adipose tissue when energy and carbohydrate are abundant and is stimulated by insulin.

Lipoid Nephrosis

- A historical term for minimal change disease, a glomerular disorder causing nephrotic syndrome through increased glomerular permeability to proteins. It is especially common in children and typically shows diffuse podocyte foot-process effacement on electron microscopy.

Lipoid Pneumonia

- Lung inflammation caused by accumulation of lipid-laden macrophages in the alveoli. Exogenous lipid

pneumonia results from aspiration of mineral oil, petroleum jelly, or oil-based nose drops. Endogenous lipid pneumonia occurs distal to bronchial obstruction. HRCT shows ground-glass opacities with low-attenuation areas (fat density).

Lipolysis

- Hydrolysis of stored triglycerides in adipocytes to release non-esterified fatty acids and glycerol. It is stimulated by catecholamines and fasting-related hormonal signals and inhibited by insulin.

Lipoma

- The most common benign soft tissue tumor, composed of mature adipocytes. It presents as a soft, mobile, painless subcutaneous nodule, commonly on the trunk, neck, and proximal extremities. Lipomas grow slowly and may be multiple (familial multiple lipomatosis). Diagnosis is clinical; ultrasound or MRI for deep or atypical lesions.

Lipomatosis

- A rare condition characterized by the diffuse, symmetric, non-encapsulated proliferation of mature adipose tissue, most commonly affecting the pelvis and lower extremities. Unlike simple lipomas, lipomatosis involves infiltration of fat into surrounding structures without encapsulation, making surgical excision difficult.

Lipoprotein

- Complex of lipids and proteins transporting hydrophobic lipids in aqueous blood. Classes: chylomicrons (dietary lipids), VLDL (endogenous triglycerides), LDL (cholesterol delivery), HDL (reverse cholesterol transport). ApoB-100 in LDL; apoA-1 in HDL. Elevated LDL cholesterol is a major risk factor for atherosclerosis.

Lipoprotein Lipase

- Hydrolyzes triglycerides in chylomicrons and VLDL. Activated by insulin.

Lipoprotein Metabolism

- See biochemistry.

Liposarcoma

- A malignant tumor of fat cells, usually arising in the deep soft tissues of the thigh or retroperitoneum; well-differentiated, myxoid, and pleomorphic subtypes are recognized.

Liposomal amphotericin B

- A liposomal formulation of amphotericin B used in the treatment of severe systemic fungal infections, with reduced nephrotoxicity compared to conventional amphotericin B. The liposomal formulation encapsulates amphotericin B within lipid vesicles, which alters the drug's biodistribution and reduces renal tubular damage.

Liquefactive Necrosis

- Characterized by the complete enzymatic digestion of dead cells, resulting in a liquid, viscous mass that eventually liquefies and forms a cystic cavity.

Liquid Biopsy

- Detection of circulating tumor DNA or cells in blood. Minimally invasive monitoring.

Lisch Nodules

- Pathognomonic ocular finding of Neurofibromatosis Type 1 (NF1, von Recklinghausen disease). Consists of clear, yellow-brown, dome-shaped melanocytic hamartomas located on the surface of the iris. Asymptomatic and do not affect vision. Present in >90% of adult NF1 patients.

Lissencephaly

- A severe brain malformation characterized by a smooth cerebral surface due to absent (agyria) or decreased (pachygyria) sulcation and gyration, caused by impaired neuronal migration during 12-16 weeks of gestation.

Listeria monocytogenes

- A gram-positive, facultatively anaerobic rod that causes listeriosis, a foodborne infection. The organism is transmitted through contaminated food, particularly deli meats, soft cheeses, and ready-to-eat foods. It can grow at refrigeration temperatures.

Listeriosis

- A bacterial infection caused by *Listeria monocytogenes*, transmitted through contaminated food (deli meats, soft cheeses, unpasteurized dairy, produce). Risk groups: pregnant women (fever, flu-like illness), neonates (sepsis, meningitis), immunocompromised, and elderly.

Listeriosis in Pregnancy

- Infection with *Listeria monocytogenes*, a foodborne Gram-positive rod. Pregnant women are 10-20 times more susceptible than the general population. Inges-

tion of contaminated food (deli meats, soft cheeses, unpasteurized dairy, raw sprouts). Maternal symptoms: fever, myalgias, GI symptoms (nausea, diarrhea).

Lithiasis

- Stone formation. Nephrolithiasis, ureterolithiasis, cystolithiasis, urethrolithiasis.

Lithium

- A mood-stabilising medication used primarily for bipolar disorder (mania, hypomania, and prophylaxis of recurrent episodes). See Lithium Toxicity and Bipolar Disorder.

Lithium Toxicity

- A potentially life-threatening condition caused by lithium levels exceeding the therapeutic range (0.6–1.2 mEq/L). Mild toxicity (1.5–2.0 mEq/L): nausea, vomiting, diarrhea, coarse tremor, muscle weakness. Moderate toxicity (2.0–2.5 mEq/L): confusion, ataxia, blurred vision, dysarthria.

Lithotripsy

- A medical procedure that uses shock waves to fragment kidney stones (renal calculi) into smaller pieces that can pass through the urinary tract. See Shock Wave Lithotripsy.

Lithotripsy Probe

- See vascular surgery.

Lithotripter Probe

- See medical instruments.

Little Area

- See Kiesselbach Plexus.

Livedo Reticularis

- A net-like (reticulate) violaceous discoloration of the skin due to reduced blood flow in the dermal vasculature. Physiologic (cold-induced, benign) vs.

Liver

- The largest internal organ, located in the right upper quadrant of the abdomen beneath the diaphragm.

Liver Abscess

- Focal collection of pus within the liver parenchyma. Pyogenic abscesses are most commonly from biliary tract infection or hematogenous spread from portal or systemic infection. Amoebic abscesses are caused by *Entamoeba histolytica* predominantly in tropical

regions. Presents with fever, right upper quadrant pain, and hepatomegaly.

Liver Biopsy

- Percutaneous, transjugular, or laparoscopic sampling of liver tissue for histological evaluation. Indicated for staging fibrosis, diagnosing specific liver diseases, and assessing treatment response. The procedure is guided by ultrasound for accuracy. Complications include bleeding, pain, and pneumothorax.

Liver Cancer

- Hepatocellular carcinoma, often in cirrhosis. Risk factors: hepatitis, alcohol, aflatoxin.

Liver Cirrhosis

- Advanced hepatic fibrosis with regenerative nodules surrounded by dense fibrous septa representing the end result of chronic liver injury. Common etiologies include chronic viral hepatitis, alcoholic liver disease, and nonalcoholic steatohepatitis.

Liver Disease Diet

- Protein restriction in hepatic encephalopathy. Sodium restriction in cirrhosis with ascites. Vitamin supplementation common.

Liver Diseases

- A broad category of pathological conditions affecting the structure and function of the liver. See individual entries: Hepatitis, Liver Cirrhosis, Fatty Liver, Hepatocellular Carcinoma, etc.

Liver Enzymes

- ALT, AST, ALP, GGT. Markers of liver injury. ALT more specific for liver.

Liver Failure

- A life-threatening condition resulting from severe impairment of hepatic synthetic, metabolic, and detoxification functions.

Liver Function Tests

- A panel of blood tests that assess the health and function of the liver, used to screen for, diagnose, and monitor liver disease and to monitor response to treatment with hepatotoxic drugs.

Liver Function Tests (LFTs)

- A panel of blood tests assessing liver health: ALT (alanine aminotransferase, primarily hepatocyte injury), AST (aspartate aminotransferase, also released by muscle/heart), ALP (alkaline phosphatase,

biliary obstruction, bile duct injury), GGT (gamma-glutamyl transferase, confirms hepatobiliary source of elevated ALP), total and direct bi

Liver Health

- Essential for metabolism, detoxification, protein synthesis, bile production, and immune function.

Liver Lobule

- The functional unit of the liver: a hexagonal structural unit centered on a central (hepatic) venule, with portal triads (portal vein, hepatic artery, bile ductule) at the corners.

Liver Mass

- Abnormality in liver detected on imaging. Benign: hemangioma, focal nodular hyperplasia, adenoma. Malignant: hepatocellular carcinoma, metastasis, cholangiocarcinoma.

Liver Transplant

- See gastroenterology.

Liver Transplantation

- Surgical replacement of a diseased liver with a healthy donor organ indicated for end-stage liver disease, acute liver failure, and select hepatocellular carcinomas within Milan criteria. Requires lifelong immunosuppression with calcineurin inhibitors, antimetabolites, and corticosteroids.

Living Will

- Legal document expressing healthcare wishes. See advance directive.

Living Will (Advance Statements)

- A legally binding written document that specifies a person's preferences for medical treatment if they become unable to make decisions for themselves in the future (loss of capacity). Living wills are a type of advance directive (or advance decision/advance statement in the UK).

Livor Mortis (Lividity)

- Post-mortem purplish-red discoloration of dependent body parts resulting from gravitational settling of blood in uncompressed capillaries and small veins following cessation of circulation.

Livor Mortis (Postmortem Lividity)

- Purplish-red discoloration of dependent body parts following death, caused by gravitational pooling of blood in uncompressed capillaries and venules.

Starts within 30 min–2 hours, becomes fixed after 6–12 hours.

LLETZ

- Large Loop Excision of the Transformation Zone, also called LEEP (Loop Electrosurgical Excision Procedure). A common outpatient procedure used to diagnose and treat cervical intraepithelial neoplasia (CIN).

Loa loa

- The African eye worm, a filarial nematode transmitted by Chrysops deer flies, endemic to the rainforests of West and Central Africa. Adult worms migrate subcutaneously, causing Calabar swellings, and may cross the subconjunctiva of the eye, causing visible worm migration. Microfilariae are detected in daytime peripheral blood.

Loading Dose

- A higher initial dose of a drug administered to rapidly achieve therapeutic plasma concentrations, followed by lower maintenance doses to sustain the steady state. The loading dose is calculated as: Loading dose = (Vd x desired concentration) / bioavailability.

Lobar Pneumonia

- Consolidation involving an entire lobe. Most commonly caused by *S. pneumoniae*. Classic progression: congestion, red hepatization, gray hepatization, resolution.

Lobectomy

- Surgical removal of a lobe of an organ. Most commonly lung lobectomy (removal of one lobe of the lung, performed for lung cancer, bronchiectasis, abscess, or benign tumor) or temporal lobectomy (removal of the anterior temporal lobe for drug-resistant temporal lobe epilepsy).

Local Anesthesia

- The reversible abolition of sensation in a circumscribed area by infiltration or topical application of local anesthetic agents (lidocaine, bupivacaine) without loss of consciousness. Maximum safe dosing must be observed to avoid systemic toxicity. Epinephrine prolongs duration but is avoided in end-arterial territories.

Local Anesthetic

- A medication that causes reversible loss of sensa-

tion in a specific, localized area of the body without affecting consciousness. Local anesthetics block voltage-gated sodium channels in nerve cell membranes, preventing action potential generation and propagation.

Local Anesthetic Systemic Toxicity

- A life-threatening complication caused by elevated plasma concentration of local anesthetic, most commonly from inadvertent intravascular injection or rapid absorption.

Local Anesthetic Toxicity

- CNS toxicity: tinnitus, perioral numbness, seizures. Cardiovascular toxicity: arrhythmias, hypotension, cardiac arrest. Treatment: lipid emulsion therapy (Intralipid), seizure control, ACLS.

Local Invasion

- Tumor growth into surrounding tissues. Distinguishes malignant from benign.

Lochia

- Vaginal discharge following delivery that undergoes characteristic changes over the puerperium. Lochia rubra (days 1-4): red, bloody discharge containing blood, decidual tissue, and fetal membranes. Lochia serosa (days 5-10): pinkish-brown discharge with serous fluid, leukocytes, and organisms.

Locked-in Syndrome

- A neurological condition in which the patient is fully conscious and aware but cannot move or speak due to complete paralysis of all voluntary muscles except for vertical eye movements and blinking (total locked-in syndrome: loss of all eye movements).

Lockjaw (Tetanus)

- An acute neurological disease caused by tetanospasmin toxin from *Clostridium tetani*, introduced through wounds. See Tetanus.

Locomotor Ataxia (Tabes Dorsalis)

- A late manifestation of untreated tertiary syphilis (neurosyphilis) caused by infection with *Treponema pallidum*, involving progressive degeneration of the dorsal columns (fasciculus gracilis and cuneatus) and dorsal roots of the spinal cord.

Locus

- Specific position of gene on chromosome. Designated by chromosome number, arm (p/q), band, subband. Example: 7q31.2 (CFTR gene).

Loneliness

- Subjective feeling of isolation. Associated with depression, cognitive decline, mortality.

Long COVID

- Persistent or recurrent symptoms following acute SARS-CoV-2 infection, lasting at least several weeks and sometimes months or years. Common manifestations include fatigue, post-exertional malaise, cognitive difficulty, breathlessness, dysautonomia, and altered smell or taste.

Long QT Syndrome

- A channelopathy characterized by prolonged QT interval (QTc >450 ms in men, >460 ms in women) with risk of torsades de pointes and sudden cardiac death. Acquired: drugs (antiarrhythmics, antipsychotics, antibiotics), electrolyte abnormalities (hypokalemia, hypomagnesemia), bradycardia.

Long-Acting Beta-2 Agonists

- Bronchodilators that provide sustained airway smooth muscle relaxation for 12–24 hours. Examples include salmeterol, formoterol, indacaterol, and olodaterol. They are used for maintenance therapy in asthma and COPD, always in combination with inhaled corticosteroids in asthma.

Long-Term Care

- Assistance with ADLs/IADLs for extended period. Settings: home, assisted living, nursing home. Funding: Medicaid, long-term care insurance, out-of-pocket. Planning essential for elderly and families.

Long-Term Effect

- Adverse effects from repeated or prolonged exposure to low doses. See chronic toxicity.

Long-Term Rehabilitation

- Chronic condition management. Focus on maintaining function, preventing complications.

Loop Diuretics

- A class of potent diuretics that act on the thick ascending limb of the loop of Henle by inhibiting the sodium-potassium-chloride cotransporter, blocking sodium and chloride reabsorption and producing a powerful natriuresis and diuresis. The prototypical loop diuretic is furosemide, with other agents including bumetanide and torsemide.

Loop of Henle

- U-shaped tubular segment consisting of a thin descending limb, thin ascending limb, and thick ascending limb. The descending limb is permeable to water but not solutes, allowing water reabsorption into the hypertonic medullary interstitium.

Loperamide

- A peripheral mu-opioid receptor agonist used as an antidiarrheal. It slows intestinal motility, reduces stool frequency and volume, and increases intestinal transit time without significant CNS effects (does not cross the blood-brain barrier at therapeutic doses) or analgesic activity.

Loratadine

- A second-generation (non-sedating) H1 antihistamine used for allergic rhinitis and chronic urticaria. Peripherally selective—minimal CNS penetration, minimal anticholinergic effects. Once daily; prodrug (desloratadine is active). Adverse effects: headache, dry mouth, and rare sedation. Available OTC.

Lordosis

- The normal inward curvature of the cervical and lumbar spine. Excessive lumbar lordosis (hyperlordosis): increased pelvic tilt, commonly due to hip flexor tightness, weak abdominals, or pregnancy. Presents with low back pain and forward protrusion of the abdomen and buttocks.

Loss of Appetite

- See gastroenterology.

Loss-of-Function Variant

- A genetic variant that reduces or abolishes the normal activity of the encoded gene product. Disease risk depends on the gene, inheritance pattern, residual function, and whether one or both copies are affected.

Low Birth Weight

- Birth weight <2500 g, often due to prematurity and/or intrauterine growth restriction; further classified: very low (<1500 g), extremely low (<1000 g), and small for gestational age.

Low FODMAP Diet

- Dietary intervention for irritable bowel syndrome that restricts fermentable oligosaccharides, disaccharides, monosaccharides, and polyols, which are short-chain carbohydrates poorly absorbed in the

small intestine and rapidly fermented by colonic bacteria producing gas and drawing water into the intestinal lumen.

Low Molecular Weight Heparin

- See hematology.

Low Testosterone (Hypogonadism)

- Low testosterone (male hypogonadism) is a clinical syndrome of deficient testosterone production by the testes (primary) or inadequate stimulation by the pituitary/hypothalamus (secondary), defined as total testosterone <300 ng/dL on two morning measurements with consistent symptoms.

Low Vision

- Visual impairment that cannot be fully corrected with spectacles, contact lenses, medication, or surgery, impairing daily activities. Managed with low-vision aids (magnifiers, telescopes, adaptive devices), lighting, and occupational therapy.

Low-Dose CT Scan

- CT protocol using reduced radiation for lung cancer screening. Detects small nodules at earlier stages. False positives require follow-up per Fleischner guidelines.

Low-Phosphorus Diet

- A dietary pattern limiting phosphorus intake for selected patients with chronic kidney disease or persistent hyperphosphatemia. It emphasizes avoidance of phosphate additives while maintaining adequate protein and overall nutritional status.

Low-Potassium Diet

- A dietary pattern limiting potassium intake when clinically indicated, usually for hyperkalemia or advanced kidney disease. It must be individualized because unnecessary restriction can reduce diet quality and cause inadequate intake of fruits and vegetables.

Low-Residue Diet

- A short-term diet limiting dietary fiber and foods that leave substantial undigested material in the bowel. It may be used in selected gastrointestinal conditions or around certain procedures but is not a universal treatment for inflammatory bowel disease.

Lower Airway Obstruction

- Narrowing of airways below the carina, causing expiratory wheezing and airflow obstruction on spirometry. Causes include asthma, COPD, bronchiectasis,

and foreign body. It is characterized by scooped-out expiratory limb on flow-volume loops and reduced FEV1/FVC ratio.

Lower Extremity Venous Doppler

- Ultrasound assessing deep vein thrombosis. Compression ultrasound, Doppler flow assessment.

Lower Motor Neuron (LMN) Syndrome

- A pattern caused by injury to an anterior horn cell, motor cranial-nerve nucleus, nerve root, plexus, or peripheral motor nerve. It typically causes weakness, reduced tone and reflexes, muscle atrophy, and fasciculations in affected muscles.

Lower Urinary Tract Symptoms (LUTS)

- Urinary symptoms arising from the bladder, bladder outlet, or urethra, classified as storage symptoms, voiding symptoms, and post-micturition symptoms. Causes include benign prostatic obstruction, overactive bladder, infection, neurologic disease, stricture, and malignancy.

Ludwig Angina

- A rapidly spreading cellulitis of the floor of the mouth and submandibular spaces, usually arising from an infected lower molar. Tongue elevation, dysphagia, drooling, and airway obstruction can develop rapidly and require emergency management.

Ludwig's Angina

- A rapidly progressive, life-threatening cellulitis of the submandibular, sublingual, and submental spaces, typically originating from an odontogenic infection of the lower second or third molars.

Lugano Classification

- Staging and response assessment for lymphoma.

Lumbago

- A non-specific term for low back pain, typically referring to acute or chronic pain in the lumbar region (between the lower ribs and the gluteal folds). Lumbago is a symptom, not a diagnosis.

Lumbar Disc Herniation

- The most common cause of lumbar radiculopathy (sciatica), occurring when the nucleus pulposus of a lumbar intervertebral disc herniates through the annulus fibrosus, compressing adjacent nerve roots. The most commonly affected level is L4-L5 and L5-S1, accounting for approximately ninety-five percent of lumbar disc herniations.

Lumbar puncture

- See Lumbar Puncture.

Lumbar Puncture (Spinal Tap)

- Insertion of a needle into the subarachnoid space at L3-L4 or L4-L5 to obtain CSF. Indications: suspected meningitis/encephalitis, subarachnoid hemorrhage, demyelinating disease, and intrathecal therapy. Contraindications: increased intracranial pressure (papilledema, mass lesion on CT), coagulopathy, and infection at the puncture site.

Lumbar Radiculopathy

- A condition caused by compression, irritation, or inflammation of a spinal nerve root in the lumbar spine, resulting in pain radiating into the lower extremity along the distribution of the affected nerve root (sciatica). The most commonly affected nerve roots are L4, L5, and S1.

Lumbar Vertebrae (L1-L5)

- Largest and strongest mobile vertebrae for weight-bearing. Large kidney-shaped bodies, short hatchet-shaped spinous processes, sagittally-oriented facets allowing primarily flexion/extension. Lordotic curvature. Most common disc herniation site, particularly L4-L5 and L5-S1 causing sciatica.

Lumpectomy

- Breast-conserving surgery excising a malignant lesion with a margin of normal tissue, combined with whole-breast radiation to yield survival equivalent to mastectomy for early-stage cancer. An acceptable margin is no ink on tumor for invasive cancer and 2 mm for ductal carcinoma in situ.

Lunate

- Crescent-shaped proximal row carpal bone between scaphoid and triquetrum. Most commonly dislocated carpal bone from falls on extended wrists. Dislocation may compress the median nerve causing acute carpal tunnel syndrome. Vulnerable to avascular necrosis (Kienbock disease) from its precarious blood supply.

Lung

- The paired, cone-shaped respiratory organs located in the thoracic cavity, responsible for gas exchange (oxygen uptake, carbon dioxide elimination). Each lung is divided into lobes: the right lung has three (superior, middle, inferior), the left lung has two

(superior, inferior) with a cardiac notch accommodating the heart.

Lung Abscess

- Localized necrosis with cavitation containing purulent material. Causes include aspiration and anaerobic bacteria. CXR shows thick-walled cavity with air-fluid level.

Lung Cancer

- Leading cancer death worldwide. Types: small cell, non-small cell. Risk factors: smoking, radon, asbestos.

Lung Compliance

- The change in lung volume produced by a given change in transpulmonary pressure. Reduced compliance increases the work of breathing, whereas excessive compliance may occur when elastic recoil is lost, as in emphysema.

Lung Disease

- Lung disease encompasses disorders affecting the airways, parenchyma, pleura, pulmonary vasculature, and chest wall.

Lung Function Tests

- A group of physiological tests that assess the function and capacity of the lungs. See Pulmonary Function Tests.

Lung herniation

- Protrusion of lung tissue through a defect in the chest wall, most commonly occurring after thoracic trauma, thoracotomy, or in association with rib fractures and chronic obstructive pulmonary disease. Presents with a reducible chest wall mass that increases with coughing or straining and may cause pain and respiratory compromise.

Lung Protective Ventilation

- A ventilation strategy using low tidal volumes (6 mL/kg ideal body weight), plateau pressure < 30 cmH₂O, appropriate PEEP, and permissive hypercapnia to minimize ventilator-induced lung injury. It is the standard of care for ARDS and is used in all mechanically ventilated patients to reduce barotrauma, volutrauma, and biotrauma.

Lung Transplant

- Surgical replacement of one or both lungs from a donor for end-stage lung disease: COPD (most common), idiopathic pulmonary fibrosis, cystic fi-

brosis, pulmonary hypertension, and other advanced conditions. Types: single, bilateral, heart-lung.

Lung Transplantation

- Surgical replacement of a diseased lung with a healthy lung from a donor. Indications include IPF, COPD, CF, PAH, and bronchiectasis. Options include single lung transplant, double lung transplant, or heart-lung transplant. Post-transplant immunosuppression is required lifelong. Complications include rejection and infection.

Lupus

- A group of autoimmune diseases in which the immune system attacks the body's own tissues. Systemic lupus erythematosus may affect the skin, joints, kidneys, blood cells, nervous system, and other organs. Also called SLE.

Lupus Erythematosus

- Systemic lupus erythematosus is a chronic autoimmune disease characterized by production of autoantibodies against nuclear antigens, leading to immune complex-mediated inflammation and tissue damage in multiple organ systems. The disease predominantly affects women of childbearing age, with a striking female predominance of 9 to 1.

Lupus Erythematosus (Systemic Lupus Erythematosus - SLE)

- A chronic multisystem autoimmune disease predominantly affecting women (9:1 female:male) of childbearing age, characterized by autoantibody production (ANA 98%, anti-dsDNA 70%, anti-Sm 30%) and immune complex deposition causing tissue injury.

Lupus Nephritis

- Renal involvement in SLE, classified into 6 classes by ISN/RPS: Class I (minimal mesangial), II (mesangial proliferative), III (focal proliferative, <50% glomeruli), IV (diffuse proliferative, ≥50%—most severe), V (membranous), VI (advanced sclerosing). Treatment: Class III/IV: cyclophosphamide or mycophenolate + corticosteroids.

Lupus pernio

- A chronic, violaceous skin lesion of the face and extremities associated with sarcoidosis, indicating a higher risk of systemic involvement and chronic

disease.

Lupus Vulgaris

- A chronic, progressive form of cutaneous tuberculosis (TB of the skin) caused by *Mycobacterium tuberculosis*. Once common in Europe, now rare in developed countries but still seen in developing regions.

Luteinizing Hormone

- LH: a gonadotropin secreted by the anterior pituitary, regulating gonadal steroidogenesis. In females: stimulates theca cells to produce androgens (aromatized to estradiol); mid-cycle LH surge triggers ovulation (used in ovulation prediction and timing of intercourse/IVF). In males: stimulates Leydig cells to produce testosterone.

Lutetium-177

- Beta and gamma emitter, half-life 6.7 days. Used in peptide receptor radionuclide therapy (PRRT) and PSMA therapy. Both therapeutic and imaging capabilities.

Lyme Disease

- A tick-borne infection caused by *Borrelia burgdorferi* (*Ixodes scapularis*, deer tick), endemic to the northeastern and upper midwestern US. Early localized: erythema migrans (bullseye rash, >5cm) at tick bite site, fatigue, fever, myalgia.

Lymph

- A clear, watery fluid that circulates through the lymphatic system, derived from interstitial fluid that has entered the lymphatic capillaries.

Lymph Node / Gland

- Small, bean-shaped organs of the lymphatic system that filter lymph and serve as sites for immune cell activation. Lymph nodes are distributed throughout the body (approximately 500-600 in total) and are concentrated in regions: cervical, axillary, inguinal, mediastinal, mesenteric, para-aortic, and iliac chains.

Lymph Node Dissection

- Removal of lymph nodes for staging or treatment. Sentinel node biopsy less invasive alternative.

Lymph Node Structure

- Small bean-shaped lymphoid organs (1-25 mm) distributed along lymphatic vessels, filtering lymph and mounting immune responses. Each has a cortex

(outer, containing B-cell follicles with germinal centers), paracortex (deep cortex, containing T-cells and dendritic cells), and medulla (inner, containing medullary cords and sinuses).

Lymphadenectomy

- Surgical removal of lymph nodes, either sentinel or regional, for staging or therapeutic purposes; examples include axillary, inguinal, cervical, and retroperitoneal lymph node dissection.

Lymphadenitis

- Inflammation and enlargement of lymph nodes, usually caused by bacterial or viral infection. Affected nodes may be tender, warm, or fluctuant, and the condition may be localized or generalized.

Lymphadenopathy

- Enlargement of lymph nodes (greater than 1 cm in diameter, or >1.5 cm for inguinal nodes). Lymphadenopathy can be localised (single or contiguous region) or generalised (three or more non-contiguous regions).

Lymphangiectasis

- Abnormal dilatation of lymphatic vessels. It may be congenital or acquired and can cause lymph leakage, oedema, protein-losing enteropathy, chylous effusions, or pulmonary and pleural complications depending on the affected site.

Lymphangiography

- Contrast imaging of lymphatic vessels. Evaluates lymphatic obstruction, fistulas. Largely replaced by MRI/CT.

Lymphangioleiomyomatosis

- A rare, progressive cystic lung disease caused by proliferation of abnormal smooth muscle-like cells (LAM cells) in the lungs, lymphatics, and kidneys. It occurs almost exclusively in women of childbearing age. HRCT shows thin-walled cysts throughout both lungs. It may be associated with tuberous sclerosis complex.

Lymphangioma

- A benign congenital malformation of the lymphatic system, composed of dilated lymphatic channels. Lymphangiomas are typically present at birth (90% diagnosed by age 2) and result from failure of the lymphatic system to develop normal connections with the venous system.

Lymphangitis

- Inflammation of the lymphatic channels, most commonly due to acute bacterial infection spreading from a peripheral wound or cellulitis. The most common causative organism is group A *Streptococcus pyogenes* (90%), followed by *Staphylococcus aureus*. Lymphangitis is a sign that infection is spreading and requires urgent medical attention.

Lymphatic System

- A network of lymphatic vessels, lymph nodes, and lymphoid organs (tonsils, thymus, spleen, Peyer patches) that returns interstitial fluid (lymph) to the bloodstream, transports dietary fat (chyle), and serves as the conduit for immune surveillance.

Lymphedema

- See geriatrics.

Lymphedema (Lymphoedema)

- Chronic swelling of a body part due to impaired lymphatic drainage, leading to accumulation of protein-rich interstitial fluid. Primary lymphedema: caused by congenital or hereditary lymphatic maldevelopment.

Lymphedema Management

- See geriatrics.

Lymphocyte

- A type of white blood cell (leukocyte) that is the primary cellular component of the adaptive immune system. Lymphocytes account for 20-40% of circulating white blood cells ($1.0\text{--}4.8 \times 10^9/\text{L}$). Three main types: (1) T lymphocytes (T cells): develop in the thymus, responsible for cell-mediated immunity.

Lymphocytic Colitis

- Chronic inflammatory bowel disease characterized by increased intraepithelial lymphocytes exceeding twenty per one hundred surface epithelial cells in the colonic mucosa without significant subepithelial collagen band thickening.

Lymphocytic Interstitial Pneumonia

- A rare ILD characterized by diffuse polyclonal lymphocytic infiltration of the lung parenchyma. It is strongly associated with Sjgren syndrome and HIV. HRCT shows ground-glass opacities, thin-walled cysts, and lymphadenopathy. Treatment is corticosteroids for symptomatic disease.

Lymphocytopenia

- A reduced lymphocyte count ($< 1.0 \times 10^9/L$). Causes: acute (stress, steroids, sepsis, trauma, burns), infections (HIV, viral, tuberculosis, typhoid), immunosuppressive drugs, radiation, marrow failure (aplasia, leukemia, lymphoma), autoimmune (SLE), and congenital immunodeficiencies.

Lymphogranuloma venereum

- A sexually transmitted infection caused by invasive *Chlamydia trachomatis* serovars L1–L3, classically producing transient genital lesions and painful inguinal lymphadenopathy.

Lymphokine

- A cytokine produced by lymphocytes (a subset of cytokines; the term lymphokine is largely historical, now 'cytokine'). Examples: IL-2, IL-4, IFN-gamma (from T cells), and B-cell growth factors. Mediate lymphocyte communication, activation, and regulation of immune responses.

Lymphoma

- A malignancy of lymphocytes, divided into Hodgkin lymphoma (HL—Reed-Sternberg cells, cervical/supraclavicular nodes, B symptoms, younger adults) and non-Hodgkin lymphoma (NHL—heterogeneous, more common, aggressive or indolent).

Lymphoma (Hodgkin's and Non-Hodgkin's)

- See Hodgkin Lymphoma and Non-Hodgkin Lymphoma.

Lymphopoiesis

- The production and maturation of lymphocytes from hematopoietic stem and progenitor cells. It occurs in primary lymphoid organs such as bone marrow and thymus and continues in peripheral lymphoid tissues during immune responses.

Lymphoscintigraphy

- Lymphatic system imaging using Tc-99m nanocolloid injected near lesion. Used for sentinel lymph node biopsy (melanoma, breast cancer), lymphedema evaluation.

Lynch Syndrome

- Hereditary nonpolyposis colorectal cancer syndrome caused by mismatch repair gene mutations including MLH1, MSH2, MSH6, and PMS2 with

autosomal dominant inheritance. It increases the risk of colorectal, endometrial, ovarian, gastric, and urinary tract cancers.

Lysergic Acid Diethylamide (LSD)

- A potent hallucinogenic drug (psychedelic) classified as a Schedule 1 controlled substance (high abuse potential, no accepted medical use). LSD acts as a partial agonist at serotonin 5-HT_{2A} receptors in the prefrontal cortex, leading to altered sensory perception, mood, and cognition.

Lysogenic Conversion

- The acquisition of new properties (genes) by a bacterium when a temperate bacteriophage integrates its genome as a prophage and expresses phage-encoded genes.

Lysosome

- Membrane-bound organelles containing hydrolytic enzymes (acid hydrolases: proteases, lipases, nucleases, glycosidases) functioning at low pH (4.5–5.0) maintained by V-ATPase proton pump. Degrade macromolecules from endocytosis, phagocytosis, and autophagy.

Lacrimal Duct

- A drainage channel that carries tears from the lacrimal sac into the nasal cavity. Blockage can cause watering of the eye, recurrent infection, swelling near the inner corner, or discharge.

Leg Ulcer

- An open sore on the leg that fails to heal normally, often caused by poor venous circulation, arterial disease, diabetes, pressure, or inflammation. Pain, spreading redness, fever, or black tissue requires prompt assessment.

Macroscopic Hematuria

- Visible blood in the urine, which may appear pink, red, or brown. Causes include urinary infection, stones, prostate disease, kidney disease, injury, and cancer; persistent or unexplained bleeding requires medical evaluation.

Low Back Pain

- Pain or discomfort in the lower back, commonly caused by muscle or ligament strain, disc disease, arthritis, or spinal nerve irritation. Weakness, numbness, new bladder or bowel problems, fever, major trauma, or unexplained weight loss are warning

signs requiring prompt medical assessment.

Chapter 13

Pocket Dictionary: M Terms

MacConkey Agar

- A selective and differential medium for culturing gram-negative bacilli. Selectivity: bile salts and crystal violet inhibit gram-positive organisms. Differential: lactose (with neutral red pH indicator)—lactose fermenters produce pink/red colonies (E).

Macrobiotics

- Diet emphasizing whole grains, vegetables, beans. May be nutritionally inadequate if restrictive. Long-term followers generally healthy but may have deficiencies.

Macrodonia

- Abnormally large teeth. Microdonia: abnormally small teeth.

Macroglossia

- Abnormal enlargement of the tongue that may cause speech or swallowing difficulty, airway obstruction, drooling, or dental malocclusion. Causes include congenital syndromes, endocrine disease, infiltrative disorders, vascular or lymphatic malformations, and tumors.

Macrolides

- A class of bacteriostatic antibiotics that bind to the 50S ribosomal subunit and inhibit translocation of peptidyl-tRNA during protein synthesis. The major macrolides include erythromycin, clarithromycin, and azithromycin.

Macronutrients

- Nutrients needed in large amounts: carbohydrates, protein, fat. Provide energy and structural components.

Macrophage Activation Syndrome

- A life-threatening complication of rheumatic dis-

eases (especially systemic JIA and SLE) characterized by uncontrolled cytokine release and hemophagocytosis.

Macrosonia

- Birth weight exceeding 4000 grams (8 pounds 13 ounces) regardless of gestational age. Risk factors include maternal diabetes, obesity, post-term pregnancy, and prior macrosomia. Complications include shoulder dystocia, brachial plexus injury, clavicular fracture, and neonatal hypoglycemia.

Macula

- The macula is a specialized region of the central retina approximately five and a half millimeters in diameter that is responsible for high-acuity central vision, color perception, and fine detail discrimination, and it contains the highest density of cone photoreceptors in the eye.

Macula Lutea

- Small oval yellowish central retinal area approximately 5.5 mm in diameter for high-acuity vision. Yellow color from xanthophyll pigments (lutein, zeaxanthin) filtering blue light and protecting photoreceptors. The fovea centralis is the central depression (1.5 mm) with highest cone density and no rods.

Macular

- A macule: a flat, circumscribed change in skin color <1 cm with no change in texture or elevation (non-palpable). Color varies: brown (melanin—freckles, lentigines, cafe-au-lait, nevus), red (erythema—inflammatory, vascular—telangiectasia, petechiae), white (vitiligo), blue/gray. Macules are distinguished from papules (elevated).

Macular degeneration

- A progressive degenerative disease of the central retina (macula) and the leading cause of irreversible vision loss in individuals over age 50 in developed countries, affecting approximately 10 percent of those over 65 and 30 percent over 80.

Macular Hole

- A full-thickness defect of the fovea usually caused by vitreomacular traction, presenting with central scotoma and distorted or reduced central vision. Staged by OCT; surgical vitrectomy with internal limiting membrane peel restores closure in most cases.

Macular Pucker

- An epiretinal membrane of fibrocellular tissue on the inner retinal surface that contracts to distort the macula, causing metamorphopsia and reduced vision. Diagnosed by OCT; vitrectomy with membrane peeling is reserved for visually significant symptoms.

Macule

- A flat, circumscribed area of altered skin color without change in surface texture or thickness, conventionally less than 1 cm in diameter. Larger flat lesions are called patches.

Maffucci Syndrome

- A rare congenital disorder characterized by multiple enchondromas together with soft-tissue hemangiomas or lymphangiomas. Skeletal deformity, limb shortening, pathological fractures, and increased risk of cartilage or other malignancy may occur.

MAG3 Renal Scan

- Nuclear medicine scan evaluating renal function and obstruction. Differential function between kidneys. Non-obstructive dilation does not show delayed excretion. Useful for assessing split renal function and planning surgical interventions.

Magnesium

- A predominantly intracellular cation that stabilizes ATP, supports hundreds of enzymes, and contributes to neuromuscular transmission, nucleic-acid synthesis, and bone metabolism. Deficiency may cause weakness, tremor, arrhythmias, hypokalemia, or hypocalcemia.

Magnesium Deficiency

- An abnormally low body magnesium level, often caused by poor intake, gastrointestinal loss, renal wasting, or medications. It may cause weakness, tremor, muscle cramps, seizures, cardiac arrhythmias, hypokalemia, and hypocalcemia.

Magnesium Sulfate for Neuroprotection

- Intravenous magnesium sulfate administered to women in imminent preterm labor (<32 weeks) to reduce the risk of cerebral palsy and major motor dysfunction in the child. Dose: 4-6 g IV bolus, followed by 1-2 g/hour maintenance for up to 24 hours or until delivery.

Magnetic Resonance Angiography

- See MR angiography.

Magnetic Resonance Angiography (MRA)

- MRI techniques used to depict arteries or veins, with or without contrast. It may evaluate stenosis, aneurysm, dissection, thrombosis, and vascular malformation without ionizing radiation.

Magnetic Resonance**Cholangiopancreatography (MRCP)**

- MRI technique visualizing biliary and pancreatic ducts without contrast. Non-invasive alternative to ERCP. Used for choledocholithiasis, biliary strictures, pancreatic pathology.

Magnetic Resonance Imaging

- See radiology.

Magnetic Resonance Imaging (MRI)

- Uses strong magnetic field and radiofrequency pulses to generate images from hydrogen protons. Excellent soft tissue contrast. No ionizing radiation. Sequences: T1-weighted (fat bright, fluid dark), T2-weighted (fluid bright), FLAIR (suppresses CSF), DWI (diffusion restriction). Contrast: gadolinium-based.

Magnetic Resonance Safety

- The practices used to prevent injury from the static magnetic field, gradient fields, radiofrequency energy, and implanted or ferromagnetic objects in the MRI environment. Screening must address implants, foreign bodies, devices, pregnancy, burns, hearing protection, and projectile risk.

Magnetic Resonance Spectroscopy

(MRS)

- MRI technique measuring metabolic compounds. Used for tumor characterization, metabolic disorders.

Magnetic Seed

- Sentinel lymph node localization using magnetic seed. Alternative to radioactive tracer.

Magnetic Therapy

- PEMF for bone healing, pain. Evidence mixed.

Mahvash Disease

- A rare inherited disorder of glucagon signaling caused by inactivating variants in the glucagon receptor. It may cause pancreatic alpha-cell hyperplasia, enlarged pancreas, abnormal glucose regulation, and elevated circulating glucagon.

Maintenance Dose

- The dose of a drug administered at regular intervals to maintain a desired steady-state concentration, balancing the rate of drug administration with the rate of elimination. The maintenance dose is calculated as: Maintenance dose rate = (Clearance x desired concentration) / Bioavailability.

Maintenance Therapy

- Long-term treatment to prevent recurrence. Examples: hormonal therapy, targeted therapy.

Major Depressive Disorder

- A mood disorder characterized by at least one major depressive episode consisting of five or more symptoms during the same two-week period. At least one symptom must be either depressed mood or loss of interest or pleasure (anhedonia).

Major Histocompatibility Complex (MHC)

- A locus of highly polymorphic genes encoding cell-surface glycoproteins that present antigen peptides to T cells. Human Leukocyte Antigen (HLA) in humans. MHC Class I (HLA-A, B, C): expressed on all nucleated cells, presents endogenous antigens (viral/tumor) to CD8⁺ cytotoxic T cells.

Malabsorption

- Defective absorption of nutrients from the small intestine. Causes: celiac disease, pancreatic insufficiency, bacterial overgrowth, Crohn disease, short bowel syndrome, and infections. Presents with di-

arrhea, steatorrhea, weight loss, and deficiency of specific nutrients (iron, B12, vitamin D, calcium).

Malacoplakia

- A chronic granulomatous inflammatory reaction characterized by macrophages containing basophilic Michaelis-Gutmann bodies. It most often affects the urinary or gastrointestinal tract and may form mass-like lesions that mimic malignancy.

Malakoplakia

- Rare chronic inflammatory condition characterized by accumulation of macrophages containing calcified inclusion bodies called Michaelis-Gutmann bodies, most commonly affecting the urinary bladder but also occurring in the gastrointestinal tract.

Malaria

- A life-threatening mosquito-borne disease caused by Plasmodium parasites (*P. falciparum*, *P. vivax*, *P. ovale*, *P. malariae*, *P. knowlesi*), transmitted by Anopheles mosquitoes. Presents with cyclical fevers, chills, headache, myalgia, and anemia. Severe P.

Malate Dehydrogenase

- Converts malate to oxaloacetate in TCA cycle. Produces NADH.

MALDI-TOF Mass Spectrometry

- Matrix-assisted laser desorption/ionization time-of-flight mass spectrometry, a method that identifies microorganisms by comparing their protein spectral fingerprints with a reference database. It provides rapid identification from an isolated colony and does not by itself establish antimicrobial susceptibility.

Male Hypoactive Sexual Desire Disorder

- A sexual dysfunction in men characterized by persistently deficient or absent sexual/erotic thoughts or desire for sexual activity, causing marked distress. Prevalence: 5-15% of men. Distinguish from erectile disorder (men may have ED but intact desire).

Male Infertility

- Male infertility—the inability to conceive after 12 months of unprotected intercourse—contributes to approximately 50% of couple infertility cases.

Malignancy

- See cancer.

Malignant acanthosis nigricans

- A paraneoplastic syndrome in which velvety, hy-

perpigmented, verrucous plaques develop rapidly in flexural and mucosal areas as a marker of an underlying internal malignancy, most commonly gastric adenocarcinoma. Unlike benign acanthosis nigricans, the malignant form is extensive, abrupt in onset, and may involve the oral mucosa.

Malignant Ascites

- Intraperitoneal fluid accumulation caused by peritoneal tumor spread, most often ovarian or gastrointestinal cancer. Management includes therapeutic paracentesis and, in selected cases, intraperitoneal chemotherapy or shunt placement.

Malignant Hypertension

- Severely elevated BP greater than 180/120 with end-organ damage. Renal findings include hyperplastic arteriolosclerosis with onion-skinning and fibrinoid necrosis. Treat with IV antihypertensives. Poor prognosis if untreated. May cause AKI, encephalopathy, and retinopathy.

Malignant Hyperthermia

- A rare, life-threatening pharmacogenetic disorder of skeletal muscle triggered by volatile inhalation anesthetics and succinylcholine, with incidence of approximately 1 in 15,000 to 50,000 anesthetics.

Malignant melanoma

- A highly aggressive malignancy arising from melanocytes, characterised by uncontrolled proliferation and potential for early metastasis via lymphatic and haematogenous spread.

Malignant Mesothelioma

- A cancer of the mesothelial lining of the pleura (most common), peritoneum, or pericardium. Strongly associated with asbestos exposure, presenting with effusion, chest pain, and dyspnea; prognosis is generally poor.

Malignant Neoplasm

- Malignant neoplasms, commonly called cancers, are uncontrolled proliferations of cells that invade surrounding tissues and have the capacity to metastasize to distant sites through lymphatic, hematogenous, or transcoelomic routes.

Malignant Pleural Effusion

- Pleural fluid accumulation from pleural or metastatic disease, causing dyspnea. Managed with thoracentesis, pleurodesis, or an indwelling pleural

catheter.

Malignant syphilis

- A severe, destructive form of secondary syphilis presenting with necrotic, gummatous, or ulcerative skin lesions, often in immunocompromised patients, particularly those with HIV co-infection. It represents an aberrant immune response to heavy *Treponema pallidum* infection and requires aggressive intravenous penicillin therapy.

Malignant Tumor

- Cancerous growth, invasive, can metastasize, poorly differentiated. Requires aggressive treatment.

Malingering

- The intentional fabrication or exaggeration of physical or psychological symptoms for external incentives (avoiding work, obtaining financial compensation, evading criminal prosecution, obtaining drugs). Not a psychiatric illness (coded as a V-code/Z-code).

Mallampati Classification

- A simple scoring system used to predict the ease of endotracheal intubation based on the anatomical structures visible in the oral cavity when the patient opens their mouth wide and protrudes their tongue without phonating. Class I: soft palate, fauces, uvula, and tonsillar pillars visible.

Malleus

- The malleus, commonly known as the hammer, is the largest of the three ossicles in the middle ear and is the most laterally positioned, with its handle (manubrium) embedded in the fibrous layer of the tympanic membrane and its head articulating with the incus at the incudomalleolar joint.

Mallory-Weiss Tear

- Longitudinal mucosal laceration at the gastroesophageal junction caused by forceful vomiting or retching. It typically presents with hematemesis. Most bleeds spontaneously within twenty-four to forty-eight hours. Diagnosis is made by upper endoscopy showing a linear mucosal tear.

Malnutrition

- Deficiency, excess, or imbalance of nutrients leading to adverse effects on tissue, body form, and function. Undernutrition: protein-energy malnutrition, micronutrient deficiencies. Overnutrition:

overweight, obesity, related comorbidities. Screen with MNA (Mini Nutritional Assessment), MUST (Malnutrition Universal Screening Tool).

Malocclusion

- Misalignment of teeth or incorrect relation between dental arches. Types: Class I (normal molar relationship, crowding), Class II (retrognathic mandible), Class III (prognathic mandible).

Malpractice

- Professional negligence by healthcare provider. Requires duty, breach, causation, damages.

Malpresentation

- Any fetal presentation other than vertex (cephalic) at term. Includes breech (buttocks or feet first), face, brow, and shoulder presentations. Breech presentation occurs in 3-4% of term pregnancies and increases the risk of cord prolapse and difficult delivery. External cephalic version may be attempted at 36-38 weeks.

Malrotation

- See Intestinal Malrotation.

Maltol

- Substance added to food for preservation, flavor, color. See food additive.

Mammary Gland

- Modified sweat (apocrine) gland in the breast, overlying pectoralis major, with 15-20 lactiferous ducts converging on the nipple. Composed of glandular lobules (alveoli), fibrous stroma (Cooper's ligaments), and fat. Development regulated by estrogen, progesterone, prolactin, and oxytocin (milk let-down).

Mammary Implant

- Breast augmentation/reconstruction device. Types: saline (silicone shell, saline fill), silicone gel (cohesive gel). Incisions: inframammary, periareolar, axillary. Placement: subglandular or submuscular. Complications: capsular contracture, rupture, infection, BIA-ALCL (anaplastic large cell lymphoma).

Mammogram

- See Mammography.

Mammography

- X-ray imaging of the breast for cancer screening and diagnosis. Screening mammography: annual

for women 40 (USPSTF: biennial 50-74). Views: CC (craniocaudal) and MLO (mediolateral oblique). Digital mammography with CAD (computer-aided detection) improves sensitivity.

Mandible

- Largest facial bone forming the lower jaw, with horseshoe-shaped body and two rami ending in condylar (TMJ) and coronoid processes. Mental foramen transmits the mental nerve. Mandibular foramen is the entry point for inferior alveolar nerve blocks. Most commonly fractured facial bone, typically at the condyle, angle, or symphysis.

Mandibular fracture

- A break in the lower jaw bone typically caused by trauma, presenting with pain, swelling, malocclusion, and difficulty opening the mouth; may require surgical fixation.

Manganese

- Neurotoxic at high exposure. Manganism: Parkinson-like symptoms. Sources: welding, batteries, contaminated water.

Mania

- A distinct period of abnormally elevated, expansive, or irritable mood with increased goal-directed activity or energy, lasting 1 week (or any duration if hospitalization required).

Manic Episode

- A distinct period lasting at least seven days (or any duration if hospitalization is required) of abnormally elevated, expansive, or irritable mood with increased energy or activity.

Manifestation

- Signs and symptoms of disease. See clinical presentation.

Manner of Death

- Legal classification of the circumstances surrounding a death, categorized into 5 official types: Natural, Accident, Suicide, Homicide, or Undetermined.

Mannitol

- An osmotic diuretic that is freely filtered at the glomerulus but not reabsorbed, increasing tubular osmotic pressure and water excretion. Used to reduce intracranial pressure (cerebral edema) and intraocular pressure (acute glaucoma), and to promote diuresis in rhabdomyolysis/contrast nephropa-

thy prophylaxis.

Mannose-Binding Lectin Deficiency

- MBL2 polymorphisms (common, 10-30% heterozygotes, 1-5% homozygotes). MBL activates lectin pathway. Low MBL associated with increased infections in immunocompromised (chemotherapy, transplant), recurrent infections in children, autoimmune disease susceptibility. Diagnosis: serum MBL level, MBL2 genotyping.

Mantle Cell Lymphoma

- An aggressive B-cell lymphoma characterized by t(11;14) translocation leading to cyclin D1 overexpression and CCND1-IGH fusion. Clinical: older males, widespread lymphadenopathy, splenomegaly, GI involvement (lymphomatous polyposis), bone marrow, peripheral blood (leukemic phase).

Manual Removal of Placenta

- The manual extraction of a retained placenta from the uterine cavity. Indication: retained placenta beyond 30-60 minutes.

MAOIs

- Monoamine oxidase inhibitors are a class of antidepressant drugs that inhibit the enzyme monoamine oxidase, which breaks down serotonin, norepinephrine, and dopamine in the brain. By inhibiting this enzyme, MAOIs increase the levels of these neurotransmitters in the synaptic cleft.

Maple Syrup Urine Disease

- An autosomal recessive inborn error of branched-chain amino acid metabolism caused by deficiency of the branched-chain alpha-ketoacid dehydrogenase complex (BCKDC) — mutations in BCKDHA, BCKDHB, DBT, or DLD.

Marasmus

- A severe form of protein-energy malnutrition (PEM) resulting from chronic total calorie and protein deprivation (starvation). Clinical features include severe muscle wasting, loss of subcutaneous fat ("skin and bones" appearance), stunted growth, a "monkey-like" facial appearance, and alertness.

Maraviroc

- A CCR5 antagonist (entry inhibitor) used in combination antiretroviral therapy for HIV-1 infection in patients with CCR5-tropic virus. Maraviroc binds to the human CCR5 coreceptor on CD4+ T cells,

preventing HIV-1 gp120 binding and viral entry.

Marfan Syndrome

- An autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene (fibrillin-1). Cardiovascular manifestations include aortic root dilation and dissection, mitral valve prolapse, and aortic regurgitation. Skeletal features include tall stature, arachnodactyly, pectus deformity, and joint hypermobility.

Marfanoid Habitus

- A body habitus characterized by tall stature, long limbs, long fingers (arachnodactyly), and joint laxity. Seen in Marfan syndrome, but also in other connective tissue disorders and homocystinuria. Differentiated from Marfan syndrome by absence of aortic root dilation and lens subluxation.

Marfan's syndrome

- An autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene encoding fibrillin-1. Neurological manifestations include dural ectasia (widening of the dural sac, causing back pain and headaches), spontaneous intracranial hypotension from CSF leaks, and increased risk of stroke from arterial dissection.

Marrow hyperplasia

- Increased cellularity of the bone marrow due to accelerated production of blood cells, typically seen as a compensatory response to chronic hemolysis, hemorrhage, or hypoxia.

Mask

- Face covering for infection control. Types: surgical, N95 respirator.

Mass

- Large lung lesion detected on imaging. Differentiate nodule (small) from mass (large). Malignancy more likely with mass.

Mass Casualty Incident

- Event overwhelming available resources, requiring triage and resource allocation. Triage systems: START (Simple Triage and Rapid Treatment), SALT. Categories: immediate (red), delayed (yellow), minor (green), expectant (black). Incident command system (ICS) coordinates response.

Masseter Muscle

- Thick rectangular muscle of mastication on the

mandibular ramus, from the zygomatic arch to mandibular angle/ramus. Innervated by the masseteric nerve (V3). Most powerful masticatory muscle, elevating the mandible with up to 70 kg force. Involved in bruxism, TM disorders, and myofascial pain. Hypertrophy can widen facial appearance.

Massive Hemoptysis

- > 600 mL in 24 hours or causing hemodynamic compromise. Medical emergency. Treatment: airway protection, bronchial artery embolization (first-line), or surgery.

Massive Transfusion Protocol

- A coordinated protocol for life-threatening hemorrhage that provides rapid, balanced delivery of red cells, plasma, and platelets with monitoring and correction of fibrinogen, calcium, temperature, and acid-base abnormalities.

Mast Cell

- A tissue-resident immune cell with large granules containing histamine, heparin, tryptase, and cytokines; bears high-affinity IgE receptors (Fc-epsilon-RI).

Mastectomy

- Surgical removal of the breast. Total (simple) mastectomy: removal of breast tissue, areola, and nipple. Modified radical mastectomy (MRM): total mastectomy + axillary lymph node dissection (levels I-III), for invasive breast cancer.

Mastitis

- Inflammation of the breast tissue, most commonly occurring during lactation (lactational mastitis, affecting 5-20 percent of breastfeeding women), typically within the first 6-12 weeks postpartum.

Mastocytosis

- A clonal disorder in which abnormal mast cells accumulate in the skin or internal organs. Cutaneous disease commonly causes urticaria pigmentosa: tan-brown macules or papules that swell and itch after rubbing (Darier sign).

Mastoidectomy

- A surgical procedure that involves the removal of diseased mastoid air cells and bone from the mastoid portion of the temporal bone, performed for chronic otitis media with cholesteatoma, mastoiditis, or as a component of surgery for acoustic neuroma and

other skull base tumors.

Mastoiditis

- Bacterial infection of the mastoid air cells, typically a complication of acute otitis media. Presents with postauricular pain, swelling, erythema, and tenderness, fever, and otorrhea. Diagnosis: CT temporal bone shows mastoid air cell opacification with bony destruction.

Masturbation

- Self-stimulation of the genitalia for sexual pleasure, a normal and common human behavior across all ages and genders. Prevalence: >90% of males and >60% of females report having masturbated at some point in their lives. It is a healthy component of psychosexual development, stress relief, and sexual self-exploration.

Maternal Physiologic Changes in Pregnancy

- Extensive adaptive changes across all organ systems to support the growing fetus.

Maternal Pushing (Second Stage)

- Maternal expulsive efforts during the second stage of labor to facilitate fetal descent. Directed (closed-glottis, Valsalva) pushing: sustained breath-hold for 10 seconds with three pushes per contraction. Spontaneous (open-glottis) pushing: shorter pushes with the mother pushing when she feels an urge.

Maternal Sepsis

- Organ dysfunction caused by a dysregulated maternal response to infection during pregnancy, childbirth, or the postpartum period. It may arise from uterine, urinary, respiratory, wound, or other infection and requires prompt cultures, antimicrobial therapy, source control, and hemodynamic support.

Maturity Onset Diabetes of the Young

- A monogenic form of diabetes with autosomal dominant inheritance, typically presenting before age 25 with non-insulin-dependent hyperglycemia. Caused by mutations in transcription factors or enzymes of beta-cell function.

Maxilla

- Paired bone forming the upper jaw, central face, and parts of orbit, nasal cavity, and hard palate. Body contains the maxillary sinus (largest paranasal sinus). Four processes: frontal, zygomatic, alveolar

(upper teeth), and palatine (anterior hard palate). Fractures classified using the Le Fort system.

Maxillofacial Prosthesis

– Artificial replacement of maxillofacial structures. Obturator for cleft palate or surgical defect. Ocular, nasal, auricular prostheses.

Maximum Plasma Concentration (Cmax)

– The highest measured plasma concentration after a dose. Cmax and the time to reach it are used to describe absorption rate, peak-related toxicity, and comparative pharmacokinetics.

Maximum Tolerated Dose

– The highest dose of a drug that produces acceptable, manageable toxicity without causing unacceptable adverse effects, determined through dose-escalation studies, most commonly in Phase I clinical trials. The MTD identifies the dose range for subsequent efficacy studies.

May-Thurner Syndrome

– Compression of left common iliac vein by right common iliac artery. Causes left lower extremity DVT. Diagnosis: CT/MR venography, phlebography. Treatment: angioplasty and stenting.

MCA Peak Systolic Velocity (PSV)

– Doppler measurement of the peak systolic velocity in the fetal middle cerebral artery. An indicator of fetal anemia: elevated MCA PSV (>1.5 MoM) correlates with moderate-to-severe anemia. The most important application is surveillance of Rh-sensitized pregnancies (detect anemia before hydrops develops).

McCune-Albright Syndrome

– Sporadic condition caused by activating mutations in the GNAS1 gene encoding the Gs alpha subunit leading to constitutive activation of adenylyl cyclase.

McDonald Cerclage

– A transvaginal cervical cerclage using a purse-string suture (Mersilene tape or Prolene) placed at the cervicovaginal junction, without bladder dissection. The most common cerclage technique. Indications: cervical insufficiency (history-indicated or ultrasound-indicated short cervix <25 mm before 24 weeks). Placed at 12-14 weeks.

McMurray Test

– Meniscal tear test. Knee flexed, rotated, extended.

Click/pain indicates tear.

McRoberts Maneuver

– A positioning maneuver for managing shoulder dystocia where the mother's legs are hyperflexed onto her abdomen, with thighs against the abdomen. This flattens the sacrum, increases the anteroposterior diameter of the pelvis, and rotates the symphysis pubis superiorly. Often combined with suprapubic pressure.

Mean Arterial Pressure

– The average pressure driving systemic blood flow during one cardiac cycle. It is influenced by cardiac output and systemic vascular resistance and is approximated clinically as diastolic pressure plus one-third of pulse pressure at normal heart rates.

Mean Corpuscular Hemoglobin Concentration (MCHC)

– The mean concentration of hemoglobin within packed erythrocytes, calculated from hemoglobin and hematocrit. A reduced MCHC indicates hypochromia; an increased value may occur with spherocytosis or analytical interference.

Mean Corpuscular Volume (MCV)

– The average volume of an erythrocyte, reported in femtoliters as part of the CBC. A low MCV indicates microcytosis, whereas a high MCV indicates macrocytosis; interpretation requires correlation with hemoglobin, reticulocytes, and other indices.

Measles

– A highly contagious viral illness caused by Morbillivirus, transmitted by respiratory droplets. Prodrome of fever, coryza, cough, and conjunctivitis, followed by Koplik spots (white buccal mucosal lesions) and maculopapular rash spreading cephalocaudally. Complications: pneumonia, encephalitis, and subacute sclerosing panencephalitis.

Measles virus

– A negative-sense RNA virus that causes measles, a highly contagious respiratory illness. The virus is transmitted by respiratory droplets and is one of the most infectious pathogens known. Symptoms include fever, cough, coryza, conjunctivitis, and rash.

Measles-Rubella (MR) Vaccine

- A combined live attenuated vaccine containing measles and rubella antigens without the mumps component, widely used in countries where mumps is not a significant public health burden, including India's Universal Immunization Schedule.

Measurable Residual Disease

- The burden of cancer remaining after therapy that can be quantified using a validated assay. The term is increasingly preferred to minimal residual disease because residual disease may be clinically important even when not minimal.

Meatuses

- The nasal meatuses are the air passages located beneath each corresponding turbinate on the lateral wall of the nasal cavity, and they serve as the drainage pathways for the paranasal sinuses and the nasolacrimal duct.

Mebrofenin Scan

- Hepatobiliary imaging using Tc-99m mebrofenin. Evaluates liver function, biliary obstruction.

Mechanical Circulatory Support

- Devices that provide temporary or permanent cardiac support. Options include intra-aortic balloon pump (IABP), Impella (percutaneous ventricular assist device), TandemHeart, extracorporeal membrane oxygenation (ECMO), and durable left ventricular assist devices (LVADs).

Mechanical Insufflation-Exsufflation

- A device that delivers positive pressure (insufflation) followed by rapid negative pressure (exsufflation) to simulate a cough. It is used in neuromuscular disease (Duchenne muscular dystrophy, spinal cord injury, ALS) to augment cough effectiveness and clear secretions.

Mechanical Ventilation

- Mechanical ventilation provides respiratory support by delivering controlled or spontaneous breaths through an endotracheal tube or tracheostomy.

Mechanism of Death

- The physiologic or biochemical derangement that causes death (the final failure) as opposed to the underlying disease or injury (cause of death) and the circumstances (manner). Examples: ventricular fibrillation, cerebral hemorrhage, sepsis, respiratory

failure, exsanguination, cardiac tamponade.

Mechanism of Toxicity

- How substance causes harm. Examples: enzyme inhibition, membrane disruption, receptor binding, oxidative stress, DNA damage.

Meckel Diverticulum

- Congenital remnant of the omphalomesenteric duct located on the antimesenteric border of the distal ileum approximately sixty centimeters from the ileocecal valve.

Meckel's diverticulum

- See Meckel Diverticulum.

Meconium Aspiration

- Meconium aspiration syndrome (MAS): aspiration of meconium-stained amniotic fluid into the fetal/neonatal airway causing airway obstruction, chemical pneumonitis, and surfactant dysfunction—respiratory distress in term/post-term infants. Risk: fetal distress/hypoxia (passage of meconium in utero).

Meconium Aspiration Syndrome

- Respiratory distress in newborns born through meconium-stained amniotic fluid, caused by aspiration of meconium into the airways. Presents with respiratory distress, barrel-shaped chest, and coarse crackles. Meconium is thick, greenish-black fetal stool passed in utero. Risk factors include post-term pregnancy and fetal distress.

Meconium Ileus

- Intestinal obstruction in neonates caused by inspissated meconium most commonly associated with cystic fibrosis. Presents with abdominal distension, bilious vomiting, and failure to pass meconium within forty-eight hours.

Meconium Peritonitis

- Sterile chemical peritonitis resulting from in utero intestinal perforation allowing meconium to spill into the peritoneal cavity. Most commonly associated with cystic fibrosis causing inspissated meconium obstruction, intestinal atresia, and volvulus. May be detected prenatally as fetal ascites with echogenic peritoneal calcifications.

Meconium-Stained Amniotic Fluid

- Greenish or yellowish discoloration of amniotic fluid due to fetal meconium passage in utero. Associated

with fetal distress, post-term pregnancy, and fetal growth restriction. Thick meconium increases risk of meconium aspiration syndrome. Management of non-vigorous newborns: tracheal suctioning before positive pressure ventilation.

Medial

- Positioned closer to the midline or middle of the body.

Medial Collateral Ligament Injury

- Tear of the MCL, the most common knee ligament injury. Mechanism: valgus stress to the knee (direct blow to lateral knee). Presents with medial knee pain, swelling, and valgus laxity. Grading: I (pain, no laxity), II (partial tear, some laxity), III (complete tear, significant laxity).

Medial Cuneiform

- Largest cuneiform, on the medial foot between navicular and first metatarsal. Insertion for tibialis anterior and posterior tendons supporting the medial longitudinal arch. The first tarsometatarsal joint (Lisfranc joint) is an important landmark; Lisfranc injuries involve fracture-dislocation at this level.

Medial Rotation

- Rotation of a limb toward the median plane of the body.

Median Nerve

- From lateral and medial cords (C5-T1), passing through the arm without branching, through the cubital fossa, into the forearm innervating most flexor muscles. In the hand innervates thenar muscles (opponens, abductor brevis, flexor brevis pollicis) and first two lumbricals. Sensation to lateral 3.5 digits.

Median Plane

- A sagittal plane that passes directly through the midline of the body, dividing it into equal right and left halves.

Median rhomboid glossitis

- A smooth, red, diamond-shaped depapillated area on the dorsum of the tongue caused by chronic *Candida* infection, usually asymptomatic and managed with antifungal therapy.

Median Survival

- The time at which 50% of patients have died; a common measure of prognosis in advanced cancer.

Mediastinitis

- Inflammation of mediastinum. Acute: post-sternotomy infection (serious, mortality 10-50%). Chronic: mediastinal masses, granulomatous diseases. Diagnosis: CT chest, cultures. Treatment: surgical debridement, antibiotics.

Mediastinoscopy

- Surgical procedure to sample mediastinal lymph nodes through a small incision above the sternum. Gold standard for mediastinal lymph node staging in lung cancer. Being replaced by EBUS in many centers.

Mediastinum

- The central compartment of the thoracic cavity between the lungs, containing the heart, major blood vessels, esophagus, and trachea.

Mediators of Inflammation

- Inflammatory mediators are soluble molecules that initiate, amplify, or regulate the inflammatory response.

Medical Cannabis and Cannabidiol (CBD)

- Cannabis (marijuana) and cannabidiol (CBD) are cannabinoid compounds derived from *Cannabis sativa* that interact with the endocannabinoid system (CB1 and CB2 receptors) to modulate pain, appetite, mood, and inflammation.

Medical Condition

- A state affecting health or bodily function, whether acute, chronic, symptomatic, or asymptomatic.

Medical Ethics

- Moral principles guiding healthcare practice, including autonomy, beneficence, non-maleficence, and justice.

Medical Examiner

- An appointed physician, usually a board-certified forensic pathologist, charged with investigating sudden, unexpected, violent, or suspicious deaths, ordering autopsies, determining cause and manner of death, and certifying the death certificate. Unlike the elected coroner, a medical examiner must meet professional qualifications.

Medical Illness

- The subjective experience of being unwell, which

may or may not correspond to an identifiable disease.

Medical Jargon

- Specialized terminology used by healthcare professionals. Unexplained medical jargon can hinder communication with patients.

Medical Jurisprudence

- The branch of law (and its study) that deals with the rights and duties of physicians and the legal aspects of medical practice, distinct from but overlapping with forensic medicine.

Medical Negligence (Malpractice)

- Legal tort occurring when a healthcare provider fails to exercise the standard of reasonable care, skill, or diligence expected of a prudent practitioner of similar standing, resulting in injury, harm, or death to a patient.

Medical Nutrition Therapy

- Individualized nutrition intervention by registered dietitian. Conditions: diabetes, renal disease, GI disorders, cancer, obesity.

Medical Physicist

- Expert in radiation physics and safety.

Medical Physics

- Physics of radiation imaging and therapy. See radiation physics.

Medical Privacy

- Protection of identifiable patient information and control of its disclosure in healthcare.

Medical Records

- A documented account of a patient's clinical history, examination, investigations, treatment, and outcome maintained by providers. In medicolegal practice, records are contemporaneous evidence in malpractice, injury, criminal, and death cases, so accuracy, timeliness, and unaltered amendments are essential.

Medical Tattooing

- Medical tattooing (permanent makeup, micropigmentation) covers depigmented skin (vitiligo, vitiligo camouflage), restores areolas after mastectomy, and masks scars. Trichopigmentation simulates hair on bald scalp. Medical tattooing uses specialized pigments and techniques.

Medical Tests

- Diagnostic procedures used to screen for, confirm, or monitor disease through analysis of biologic specimens, imaging, physiologic measurements, or genetic material.

Medication Adherence

- The extent to which a patient's medication-taking behavior corresponds with the agreed treatment plan. Adherence is influenced by understanding, adverse effects, cost, access, regimen complexity, beliefs, and cognitive or social factors.

Medication Adverse-Effect Profile

- The clinically relevant pattern of predictable pharmacologic effects and harmful reactions associated with a medication. It includes common dose-related effects, serious rare events, organ-specific toxicity, interactions, and monitoring requirements.

Medication Cascade

- A prescribing sequence in which an adverse drug effect is misinterpreted as a new medical condition and another drug is prescribed to treat it. Recognizing cascades is an important goal of medication reconciliation and deprescribing.

Medication Deprescribing

- A planned, supervised process of reducing the dose or stopping a medicine when its current or future harms outweigh its benefits or when treatment goals change. Tapering may be required to prevent withdrawal or rebound disease.

Medication Error

- A preventable event that may cause or lead to inappropriate medication use or patient harm while the medication is in the control of a healthcare professional, patient, or consumer. Errors can occur during prescribing, transcribing, dispensing, administration, or monitoring.

Medication Management

- The systematic process of optimizing pharmacotherapy to achieve therapeutic goals while minimizing adverse effects, encompassing prescribing, dispensing, administration, monitoring, and deprescribing.

Medication Reconciliation

- A structured process of obtaining and comparing the most accurate medication list with current pre-

scriptions at transitions of care. It identifies omissions, duplications, incorrect doses, interactions, and unintended changes.

Medication Review

- Systematic examination of all medications for appropriateness. Components: indication, efficacy, safety, adherence, interactions. Tools: Beers Criteria, STOPP/START. Goals: deprescribing, optimizing therapy, reducing adverse effects.

Medication Safety

- Systematic examination of all medications for appropriateness. See medication review.

Medication Side Effects

- Unintended harmful responses to medications occurring at standard doses, ranging from mild (nausea, rash, dry mouth) to severe (anaphylaxis, Stevens-Johnson syndrome, QT prolongation, hepatotoxicity).

Medications and Treatments

- all pharmacologic and therapeutic interventions used to prevent, manage, or cure disease, classified by mechanism of action, therapeutic class, and route of administration.

Medicolegal

- Pertaining to the application of medical knowledge to legal questions and the intersection of medicine and the law. Covers forensic pathology, toxicology, psychiatry, and clinical issues: informed consent, malpractice, medical records, disability, personal injury, occupational disease, and public health law.

Mediterranean Diet

- Traditional eating pattern of Greece, Southern Italy. Emphasizes: fruits, vegetables, whole grains, legumes, nuts, olive oil, moderate fish/poultry, limited red meat. Associated with reduced cardiovascular disease, diabetes, cognitive decline.

Medulla Oblongata

- Most inferior brainstem segment, continuous with the spinal cord at the foramen magnum. Contains vital autonomic centers: cardiovascular center (heart rate, blood pressure), respiratory center (dorsal/ventral respiratory groups), and area postrema (vomiting center).

Medullary Sponge Kidney

- Congenital malformation characterized by cystic

dilatation of the collecting ducts in the renal papillae predisposing to nephrolithiasis and nephrocalcinosis. Presents with recurrent kidney stones, hematuria, and urinary tract infections.

Medullary Thyroid Carcinoma

- Neuroendocrine tumor arising from parafollicular C cells that produce calcitonin. Accounts for approximately five percent of thyroid carcinomas. May be sporadic or hereditary as part of multiple endocrine neoplasia type two A or two B. Serum calcitonin serves as a tumor marker. Diagnosis requires FNA with calcitonin immunohistochemistry.

Medulloblastoma

- A fast-growing embryonal tumor of the cerebellum and the most common malignant brain tumor of childhood. Treated with surgical resection, craniospinal radiation, and chemotherapy.

Meige Syndrome

- A segmental cranial dystonia characterized by involuntary bilateral blepharospasm with dystonic contractions of the lower face, jaw, or tongue. Symptoms fluctuate and may be worsened by stress, speech, chewing, or sensory stimuli.

Meigs Syndrome

- The triad of benign ovarian fibroma, ascites, and pleural effusion. The effusions typically resolve after removal of the ovarian tumour and should be distinguished from malignant ovarian disease.

Meiosis

- See genetics.

Melancholic Features

- A specifier for major depressive episodes characterized by a loss of interest or pleasure in all or almost all activities, a depressed mood that is qualitatively distinct from grief, prominent psychomotor retardation or agitation, significant anorexia or weight loss, excessive guilt, and worsening of mood in the morning.

Melanocytes

- Dendritic pigment-producing neural crest-derived cells in the epidermal basal layer and hair follicles, comprising approximately 5-10 percent of basal keratinocytes. Produce melanin (eumelanin, brown-black; pheomelanin, yellow-red) in melanosomes, transferred to keratinocytes via dendritic processes

for UV protection.

Melanoma

- A malignant neoplasm arising from melanocytes (pigment-producing cells derived from the neural crest, located in the basal layer of the epidermis, hair follicles, and mucous membranes), representing the most lethal form of skin cancer, with an incidence of approximately 20 to 30 per 100,000 persons per year in fair-skinned populations and

Melanoma Surveillance

- regular skin examination, patient education on self-examination, and dermoscopic monitoring of atypical nevi. High-risk patients (personal or family history, numerous atypical nevi, immunosuppression) require 6-month follow-up. Standard patients require annual examination. Total body photography establishes baseline for change detection.

Melanoma Treatment

- Stage-dependent. Stage 0 to IA: wide local excision. Stage IB to IIA: wide excision with sentinel lymph node biopsy. Stage IIB to IIC: wide excision, SLNB, and adjuvant therapy.

Melasma

- A common hyperpigmentation disorder characterized by brown or gray-brown patches on the face, particularly the cheeks, forehead, nose, and upper lip. Also called chloasma or the mask of pregnancy.

Melasma (Chloasma)

- An acquired, chronic pigmentary disorder characterized by symmetric, hyperpigmented, light-to-dark brown macules and patches on sun-exposed areas of the face (malar, forehead, upper lip, chin). It primarily affects women of childbearing age and individuals with darker skin types (Fitzpatrick types III-V).

Melatonin Receptor Agonists

- Medications that act on MT1 and MT2 receptors in the suprachiasmatic nucleus to regulate circadian rhythms. Ramelteon is approved for insomnia with sleep onset difficulty. It has no abuse potential and does not cause dependence. It is a useful alternative to benzodiazepines for sleep initiation problems.

MELD Score

- Model for End-Stage Liver Disease predicting three-month mortality in advanced liver disease used for

liver transplant allocation. The score is calculated using serum bilirubin, creatinine, and international normalized ratio generating a continuous score from approximately six to forty points.

melena

- Passage of black, tarry, foul-smelling stools due to the presence of digested blood, indicating upper gastrointestinal bleeding (usually proximal to the ligament of Treitz). The black color results from oxidation of haemoglobin to haematin by intestinal enzymes and bacteria.

Melioidosis

- An infectious disease caused by *Burkholderia pseudomallei*, usually acquired from contaminated soil or water through inoculation or inhalation. It may cause pneumonia, skin infection, abscesses, sepsis, or chronic disease and is more severe in people with diabetes or other risk factors.

Melorheostosis

- A rare sclerosing bone dysplasia producing irregular, flowing cortical hyperostosis on imaging. It may cause pain, stiffness, limb deformity, restricted movement, contractures, or soft-tissue fibrosis and can affect one or several contiguous bones.

Memantine

- An NMDA receptor antagonist (uncompetitive, low-to-moderate affinity) used for moderate to severe Alzheimer disease, often added to an acetylcholinesterase inhibitor. Blocks excessive glutamate-mediated excitotoxicity while preserving normal synaptic transmission. Once or twice daily oral.

Membrane Potential

- The electrical potential difference across a cell membrane, approximately -70 mV in most neurons (resting membrane potential). Established by the unequal distribution of ions (primarily K⁺, Na⁺, Cl⁻) across the membrane and selective permeability.

Membranoproliferative Glomerulonephritis

- Pattern of glomerular injury with mesangial proliferation and capillary wall thickening creating tram-track appearance on silver stain. Hepatitis C is common cause of immune complex MPGN. C3 glomerulopathy represents complement-mediated disease. Associated with monoclonal gammopathies.

Membranous Nephropathy

- A common cause of nephrotic syndrome in adults, characterized by immune complex deposition in the subepithelial space of the glomerular basement membrane. Primary (idiopathic, 75%): associated with anti-PLA2R antibodies (diagnostic).

Memory

- The cognitive process of encoding, storing, and retrieving information, mediated by distributed neural networks involving the hippocampus, medial temporal lobe, prefrontal cortex, and limbic system.

Memory Cell

- Long-lived lymphocytes (memory B cells and memory T cells) that survive after an immune response and mount rapid, enhanced responses to re-exposure to the same antigen (immunological memory).

Memory Loss

- See cognitive impairment/dementia.

Mendelson Syndrome

- See Aspiration Pneumonitis. Mendelson syndrome is a historical eponym for aspiration pneumonitis, originally described in obstetric patients.

Meniere Disease

- An inner ear disorder characterized by recurrent episodes of vertigo lasting 20 minutes to 12 hours, with fluctuating sensorineural hearing loss, tinnitus, and aural fullness. Due to endolymphatic hydrops. Diagnosis: clinical criteria; audiometry shows low-frequency SNHL during attacks.

Meninges

- The three membranes enveloping the brain and spinal cord: dura mater (outermost, tough), arachnoid mater (middle, avascular), and pia mater (innermost, adherent to the CNS surface). The subarachnoid space (between arachnoid and pia) contains cerebrospinal fluid. The epidural and subdural spaces are sites of hemorrhage.

Meningioma

- The most common primary intracranial tumor, accounting for approximately thirty-seven percent of all primary brain tumors, and arises from arachnoid cap cells of the meninges. Most meningiomas are benign (WHO grade I), slow-growing, and well-circumscribed, with a whorled histopathological pattern and psammoma bodies.

Meningitis

- Inflammation of the meninges most commonly caused by bacterial infection (*Neisseria meningitidis*, *Streptococcus pneumoniae*, *Haemophilus influenzae*, Group B *Streptococcus*). Presents with fever, nuchal rigidity, photophobia, and altered mental status. Diagnosis: lumbar puncture with CSF analysis.

Meningococcal Disease

- A bacterial infection caused by *Neisseria meningitidis* (serogroups A, B, C, W, Y), transmitted via respiratory droplets. Meningococcemia: sudden onset fever, petechial/purpuric rash, hypotension, and DIC. Meningococcal meningitis: fever, nuchal rigidity, headache, photophobia.

Meningococcal infection

- See Meningococcal Disease.

Meningococcal Vaccine

- Conjugate vaccine (MenACWY) and serogroup B vaccine (MenB) given at 11-12 years with booster at 16 years. Protects against *Neisseria meningitidis*, causing meningitis and septicemia. MenACWY booster at 16 years covers the high-risk period for serogroup C and Y disease.

Meniscal Injuries

- Meniscal tears: traumatic (vertical longitudinal, bucket handle, radial, horizontal cleavage, flap) vs degenerative (complex, horizontal). Clinical: joint line tenderness, McMurray, Apley, Thessaly test. MRI: sensitivity 90%+.

Meniscal Tear

- Tear of the C-shaped fibrocartilage (medial or lateral meniscus) that provides shock absorption and stability in the knee. Medial meniscus tears more common (attached to capsule). Mechanism: twisting injury with weight-bearing. Presents with joint line tenderness, pain with deep squatting, catching, locking, and giving way.

Menkes Disease

- An X-linked recessive disorder of copper metabolism caused by ATP7A mutations, resulting in impaired intestinal copper absorption and defective copper transport across the blood-brain barrier. Copper accumulates in intestinal and renal cells but is deficient in the liver and brain.

Menkes Kinky Hair Syndrome

- An X-linked disorder of copper transport caused by ATP7A dysfunction. It produces progressive neurodegeneration, seizures, developmental delay, hypopigmented or brittle kinky hair, connective-tissue abnormalities, and failure to thrive.

Menopausal Hormone Therapy (MHT)

- Menopausal hormone therapy (MHT), formerly called hormone replacement therapy (HRT), is the use of estrogen with or without progestogen to alleviate moderate-to-severe menopausal vasomotor symptoms (hot flashes, night sweats), genitourinary syndrome of menopause (vaginal dryness, dyspareunia, urinary urgency), and prevent osteoporosis.

Menopause

- The point marking the end of menstruation, officially designated as one year after a woman's final period.

Menopause and Perimenopause

- Menopause is the permanent cessation of menstruation for 12 consecutive months, reflecting ovarian follicle depletion and estrogen decline, occurring at a median age of 51–52 years in North America.

Menorrhagia

- Excessive or prolonged menstrual bleeding (>80 mL blood loss or >7 days duration). Causes: uterine fibroids, adenomyosis, endometrial polyps, coagulation disorders, thyroid dysfunction, and anovulation. Presents with heavy bleeding, flooding, passage of clots, and iron deficiency anemia.

Menstrual Cycle

- The monthly hormonal and uterine changes preparing for pregnancy, typically 21–35 days.

Menstruation

- Menses: the cyclic shedding of the endometrial lining (menstrual period) as part of the menstrual cycle, occurring with the fall in estrogen/progesterone when implantation does not occur. Normal: interval 21–35 days, duration 2–7 days, blood loss 30–80 mL. First menses = menarche (12–13 years); cessation = menopause (51 years).

Mental Health

- Mental health encompasses emotional, psychologic, and social well-being, affecting how individuals think, feel, behave, and handle stress, relationships,

and decision-making.

Mental Status Examination

- A structured assessment of the patient's current psychological functioning performed during the clinical interview. It evaluates appearance, behaviour, speech, mood, affect, thought process, thought content, perceptions, cognition, insight, and judgment.

Mercury

- Liquid metal combining with other metals to form amalgam. Concerns about toxicity, proper handling essential.

Mercury Poisoning

- Three forms: elemental (inhalation, neurological), inorganic (renal, GI), organic (methylmercury, neurological). Minamata disease from methylmercury. Treatment: chelation (dimercaprol, DMSA), remove exposure.

Merkel Cell Carcinoma

- An aggressive neuroendocrine skin cancer arising from Merkel cells. It presents as a rapidly growing, painless, red-violet nodule on sun-exposed skin of elderly patients. Approximately 80 percent are associated with Merkel cell polyomavirus. Risk factors include immunosuppression, UV exposure, and fair skin.

Meropenem-vaborbactam

- A combination of the carbapenem meropenem with the cyclic boronate beta-lactamase inhibitor vaborbactam. Vaborbactam inhibits class A and class C beta-lactamases, including KPC carbapenemases, but does not inhibit class D or metallo-beta-lactamases. This combination has activity against many carbapenem-resistant Enterobacteriaceae.

MERRF Syndrome

- Myoclonic epilepsy with ragged-red fibres, a mitochondrial disorder usually caused by a pathogenic mitochondrial DNA variant. It features myoclonus, generalized epilepsy, ataxia, and skeletal-muscle mitochondrial abnormalities, with variable multisystem involvement.

Mesenteric Artery Ischemia

- Inadequate blood flow through the mesenteric arteries, usually caused by embolism, thrombosis, or severe low-flow states. Acute disease presents with severe abdominal pain out of proportion to exami-

nation and can progress to bowel infarction.

Mesenteric Ischemia

- Reduced blood flow to intestines. Acute: embolism, thrombosis. Chronic: atherosclerosis. CT angiography diagnosis.

Mesentery

- The fan-shaped double layer of peritoneum attaching the small intestine (jejunum and ileum) to the posterior abdominal wall, carrying the superior mesenteric artery, vein, lymphatics, and autonomic nerves between its layers. Root extends from duodenojejunal junction to ileocecal junction.

Mesh

- A synthetic or biological implant used to provide structural support in pelvic organ prolapse and stress urinary incontinence surgery. Types: polypropylene (most common, mid-urethral slings for SUI, vaginal mesh for prolapse), biological (porcine, bovine, or cadaveric dermis).

Mesoderm

- The middle germ layer in an embryo, which develops into muscles, bones, cartilage, blood vessels, and the kidneys.

Messenger RNA (mRNA)

- Single-stranded RNA molecule transcribed from DNA template, carrying genetic code from nucleus to ribosomes for protein synthesis. Contains 5' cap (methylguanosine), 5' UTR, coding sequence (codons), 3' UTR, and poly-A tail. Eukaryotic mRNA undergoes processing: splicing (intron removal, exon ligation), capping, polyadenylation.

Meta-Analysis

- Statistical combination of results from multiple studies. Increases power, precision. Forest plot visualization.

Metabolic Acidosis

- A primary acid-base disorder characterized by decreased serum bicarbonate (<22 mEq/L) and pH <7.35 . Causes are classified by anion gap (AG). High AG: diabetic ketoacidosis, lactic acidosis, renal failure, starvation, toxins (methanol, ethylene glycol, salicylates).

Metabolic Alkalosis

- A primary increase in plasma bicarbonate or loss of hydrogen ions. Causes include vomiting, diuretic

use, mineralocorticoid excess, and chloride depletion; respiratory compensation increases carbon dioxide but is limited by hypoxemia and acidosis.

Metabolic Rate

- See basal metabolic rate.

Metabolic Syndrome

- Cluster of conditions: abdominal obesity, hypertension, hyperglycemia, high triglycerides, low HDL. Increases cardiovascular disease and diabetes risk.

Metabolic Tumor Volume

- Total volume of metabolically active tumor on PET. Prognostic in lymphoma.

Metabolism

- Sum of all chemical reactions in the body: catabolism (breakdown, releases energy) and anabolism (synthesis, consumes energy). Central pathways: glycolysis, TCA cycle, oxidative phosphorylation, pentose phosphate pathway. Metabolic regulation ensures energy homeostasis.

Metacarpals

- Five long bones (I-V) forming the palm framework. Each has base, shaft, and head. The first is shortest and most mobile, articulating with the trapezium at a saddle joint for opposition. Second/third are firmly fixed; fourth/fifth more mobile. Bennett's fracture affects the first metacarpal base; boxer's fracture affects the fifth neck.

Metacarpophalangeal Joint

- A condyloid synovial joint between a metacarpal head and the base of a proximal phalanx. It permits flexion, extension, abduction, adduction, and limited circumduction.

Metagenomic Sequencing

- Sequencing of nucleic acid recovered directly from a clinical or environmental specimen to identify microbial genetic material without requiring growth in culture. It can detect unusual or polymicrobial infections but is vulnerable to contamination and requires clinical interpretation.

Metal Artifact

- Image degradation from metal implants. Reduced by metal artifact reduction sequences (MRI), iterative reconstruction (CT).

Metaphyseal Corner Fractures

- Bucket-handle or corner fractures at the metaphysis of long bones, characteristic of child abuse. Caused by twisting or pulling forces on extremities. Most common in the knee, ankle, and elbow. Seen on skeletal survey; may not be visible initially. Highly specific for non-accidental trauma in infants.

Metaphysis

- The region of a long bone between the epiphysis (end) and diaphysis (shaft), containing the epiphyseal growth plate (physis) in children—the zone of active cartilage growth (endochondral ossification).

Metaplasia

- A reversible adaptive change in which one differentiated adult cell type is replaced by another, representing the body's attempt to better withstand a chronic stress or irritation. The process involves reprogramming of tissue stem cells to differentiate along a different lineage that is more resilient to the new environment.

Metastasectomy

- Surgical removal of one or more metastases, which may be curative in selected oligometastatic disease, e.g., colorectal liver and lung metastases and sarcoma pulmonary metastases.

Metastasis

- Spread of cancer cells from primary site to distant locations. Routes: lymphatic, hematogenous, direct extension, transcoelomic. Common sites: lungs, liver, bone, brain. Distant metastasis indicates Stage IV disease.

Metastasis of colorectal cancer

- Spread of colorectal malignancy to distant organs via lymphatic, hematogenous, or peritoneal routes, most commonly involving the liver, lungs, and peritoneum. The liver is the first site of metastasis due to portal venous drainage, and hepatic metastases may be resectable with curative intent.

Metastasis-Free Survival

- The time from treatment until the development of distant metastasis or death; an important endpoint in adjuvant therapy trials.

Metastatic adenocarcinoma

- An adenocarcinoma that has spread from its primary site to distant organs, commonly the liver,

lungs, bones, or brain. The prognosis and treatment depend on the primary tumor origin and the extent of metastatic disease.

Metastatic Calcification

- The deposition of calcium salts in otherwise normal tissues due to systemic hypercalcemia, where elevated serum calcium levels exceed the tissue ability to maintain calcium homeostasis.

Metastatic Cascade

- Sequence of events in metastasis: local invasion, intravasation, circulation, extravasation, colonization. Many cells die during process.

Metastatic colon cancer

- Advanced colonic malignancy that has spread beyond the bowel to distant sites, most commonly the liver, lungs, and peritoneum, via lymphatic and hematogenous dissemination. Staging uses CT of chest, abdomen, and pelvis, and PET-CT may identify occult metastases.

Metastatic inguinal lymphadenopathy

- Enlargement of inguinal lymph nodes due to metastatic cancer cells, often from primary tumors of the lower extremities, pelvis, or anorectal region. It may indicate advanced disease requiring systemic therapy.

Metastatic lung cancer

- Lung cancer that has spread beyond the lungs to other organs, such as the brain, bones, liver, or adrenal glands. It represents advanced stage disease with limited treatment options.

Metatarsals

- Five long bones (I-V) forming the forefoot. Each has base, shaft, and head. First is shortest/thickest, bearing approximately one-third of forefoot load. Stress fractures commonly affect the second and third (march fractures). Morton's neuroma typically affects the nerve between 3rd/4th metatarsal heads.

Metered-Dose Inhaler

- A pressurized inhaler delivering a metered dose of medication. Requires proper technique for effective delivery. Spacer devices improve drug delivery and reduce oropharyngeal deposition. Most common delivery device for asthma and COPD maintenance.

Metformin in Prediabetes

- Metformin reduces progression from prediabetes

to type two diabetes by approximately thirty-one percent in high-risk individuals. Most effective in younger patients under sixty with BMI greater than 35 and women with prior gestational diabetes. The Diabetes Prevention Program demonstrated sustained benefit over ten years of follow-up.

Methacholine Challenge Test

- A bronchoprovocation test for diagnosing asthma when spirometry is normal. Inhaled methacholine (a cholinergic agonist) induces bronchoconstriction. Spirometry is repeated after each escalating dose. Positive: 20% fall in FEV1 (PC20) at 16 mg/mL. Sensitivity >90% for asthma.

Methanol Poisoning

- Toxic ingestion or exposure to methanol, whose metabolites cause high-anion-gap metabolic acidosis and visual or neurologic injury. Treatment requires urgent toxicology consultation, antidotal fomepizole or selected ethanol therapy, correction of acidosis, and hemodialysis in severe cases.

Methemoglobinemia

- A condition in which hemoglobin iron is oxidized to the ferric state, reducing oxygen delivery and causing functional anemia. Cyanosis, chocolate-brown blood, headache, dyspnea, neurologic symptoms, and cardiovascular compromise may occur depending on level and comorbidity.

Methicillin-resistant Staphylococcus aureus

- Staphylococcus aureus resistant to beta-lactam antibiotics through acquisition of the *mecA* gene encoding penicillin-binding protein 2a (PBP2a). MRSA is a major cause of healthcare-associated and community-associated infections. Treatment options include vancomycin, daptomycin, linezolid, and trimethoprim-sulfamethoxazole.

Methotrexate

- An antimetabolite that inhibits dihydrofolate reductase and adenosine signaling; used as a disease-modifying antirheumatic drug (rheumatoid arthritis, psoriasis, psoriatic arthritis, IBD, vasculitis) and as chemotherapy (leukemia, lymphoma, choriocarcinoma).

Methotrexate in Dermatology

- An immunomodulatory antimetabolite used in se-

lected inflammatory and autoimmune skin diseases, including psoriasis, eczema, and some connective-tissue disorders. Treatment requires attention to liver disease, blood counts, infection risk, drug interactions, and strict pregnancy precautions.

Methylation

- Addition of methyl group to DNA (cytosine) or histones. DNA methylation at CpG islands typically silences genes. Hypomethylation associated with cancer.

Methylene Blue

- An oxidizing-reducing agent used primarily to treat clinically significant methemoglobinemia by facilitating conversion of methemoglobin to functional hemoglobin. It may be ineffective or hazardous in glucose-6-phosphate dehydrogenase deficiency and can interact with serotonergic drugs.

Methylmalonic Acid Level

- A blood or urine measurement used to assess functional vitamin B12 status. Elevated methylmalonic acid supports vitamin B12 deficiency, particularly when the serum vitamin B12 concentration is borderline. Renal impairment can also increase the level and should be considered during interpretation.

Metoclopramide

- A dopamine (D2) receptor antagonist and 5-HT3 antagonist used as a prokinetic antiemetic for gastroparesis, chemotherapy-induced and postoperative nausea, and GERD (facilitates gastric emptying).

Metronidazole

- A nitroimidazole antibiotic and antiprotozoal agent that is selectively toxic to anaerobic organisms and certain parasites. The drug is activated by reduction of its nitro group within anaerobic cells, generating toxic free radicals that damage DNA.

Metronidazole-associated encephalopathy

- A rare complication of prolonged or high-dose metronidazole therapy, causing cerebellar dysfunction (ataxia, dysarthria, nystagmus) and sometimes confusion or seizures. MRI may show T2-hyperintense lesions in the cerebellar dentate nuclei, corpus callosum splenium, and brainstem. Symptoms typically resolve with drug discontinuation.

Metrorrhagia

- Uterine bleeding at irregular intervals, particu-

larly between expected menstrual periods. Causes: anovulation, endometrial polyps, submucosal fibroids, endometrial hyperplasia/cancer, cervical pathology, and breakthrough bleeding with hormonal contraceptives. Diagnosis: transvaginal ultrasound, endometrial biopsy, and

MIBG Scan

- Nuclear medicine imaging using I-123 or In-111 MIBG. Detects pheochromocytoma, neuroblastoma.

Micafungin

- An echinocandin antifungal agent used in the treatment and prophylaxis of invasive candidiasis and esophageal candidiasis. Micafungin inhibits beta-(1,3)-D-glucan synthase, disrupting fungal cell wall synthesis and leading to osmotic instability and cell death.

Michaelis-Menten Kinetics

- The mathematical model describing the rate of enzyme-catalyzed reactions as a function of substrate concentration.

Microalbuminuria

- Excretion of 30 to 300 mg of albumin daily undetectable by standard dipstick. Earliest marker of diabetic nephropathy and cardiovascular risk predictor. Detected by urine albumin-to-creatinine ratio on spot specimen. Normal less than 30 mg/g creatinine. Signals early glomerular damage in diabetes warranting RAAS blockade initiation.

Microangiopathic hemolysis

- Red-cell fragmentation and intravascular hemolysis caused by passage through abnormal small vessels, producing schistocytes, elevated LDH, and reduced haptoglobin.

Microarray

- Technology measuring expression of thousands of genes simultaneously or detecting copy number variations. Gene expression microarrays: compare expression patterns. SNP arrays: detect genetic variations. Comparative genomic hybridization (CGH): detect deletions/duplications.

Microbiome

- The collection of microorganisms (bacteria, archaea, fungi, viruses—the microbiota) and their genetic material inhabiting a host's body surfaces, particularly the gastrointestinal tract.

Microcephaly

- A head size substantially smaller than expected for age and sex, usually defined by an occipitofrontal circumference below the third percentile or more than two standard deviations below the mean. Causes include genetic disorders, congenital infections, brain injury, malnutrition, and impaired brain development.

Micrognathia

- A condition where the mandible is abnormally small, resulting in a retruded chin and decreased size of the oral cavity. May be isolated or syndromic (Pierre Robin sequence, Treacher Collins, Nager syndrome, Stickler syndrome). Presents at birth with potential airway obstruction (glossoptosis) and feeding difficulties.

Micrograms

- See units.

Micrometastasis

- A small deposit of cancer cells that has spread from the primary tumor but is not detectable clinically or on standard imaging, e.g., isolated tumor cells in a sentinel lymph node; relevant to staging and prognosis.

Micromolar

- See units.

Micronutrients

- Vitamins and minerals required in small amounts for enzymatic, hormonal, structural, hematologic, neurologic, and immune functions. They include fat-soluble vitamins (A, D, E, and K), water-soluble vitamins, and essential minerals.

Microphthalmia

- A congenitally small eye, ranging from mild (reduced axial length) to severe with structural anomalies. Associated with coloboma, congenital cataract, and genetic syndromes; can cause vision loss and requires optical correction, amblyopia therapy, and sometimes socket rehabilitation.

MicroRNA

- Small non-coding RNA regulating gene expression. Post-transcriptional regulation.

Microsatellite Instability (MSI)

- A molecular pattern caused by deficient DNA

mismatch-repair function, resulting in length changes in short repetitive DNA sequences. MSI-high tumors may have increased mutation burden and can be eligible for particular immunotherapies.

Microscope

- Magnifies small structures. Types: light, electron, fluorescent. Essential for histology, hematology, microbiology.

Microscopic Colitis

- Umbrella term encompassing both collagenous colitis and lymphocytic colitis which share the clinical presentation of chronic watery non-bloody diarrhea and the pathophysiological feature of colonic mucosal inflammation. Endoscopic appearance is typically normal necessitating random biopsies for diagnosis.

Microscopic Examination

- The pathologic evaluation of tissue or cells under the light or electron microscope, complementing the gross examination to form the complete morphologic diagnosis. Routine light microscopy uses hematoxylin and eosin sections to assess architecture, cellular morphology, mitotic activity, and relation to margins.

Microscopic Polyangiitis

- A necrotizing vasculitis affecting small blood vessels (capillaries, venules, arterioles) with few or no immune deposits (pauci-immune). Part of the ANCA-associated vasculitis group. Strongly associated with MPO-ANCA (p-ANCA, 80%).

Microwave Ablation

- Image-guided destruction of tissue. See ablation therapy.

Micturition

- The process of urination, coordinated by the pontine micturition center, parasympathetic (pelvic nerve, S2-S4) contraction of the detrusor muscle, and relaxation of the internal (involuntary) and external (voluntary, pudendal nerve) urethral sphincters.

Mid-Systolic Click

- A sharp, high-pitched sound occurring in mid-to-late systole, associated with mitral valve prolapse. The click represents sudden tensing of the elongated chordae tendineae as the prolapsing leaflet billows

into the left atrium. It may be followed by a late systolic murmur of mitral regurgitation.

Mid-Upper-Arm Circumference

- The circumference of the upper arm measured at the midpoint between the acromion and olecranon processes. It is a practical anthropometric measure of muscle and fat stores, particularly useful when body weight or height cannot be measured reliably.

Midaxillary Line

- An imaginary vertical line on the side of the body passing through the apex of the axilla (armpit).

Midbrain

- Most superior brainstem segment connecting pons to diencephalon. Contains cerebral peduncles (corticospinal/corticobulbar tracts), substantia nigra (dopaminergic neurons degenerated in Parkinson's), red nucleus (rubrospinal tract for coordination), and periaqueductal gray (pain modulation).

Midclavicular Line

- An imaginary vertical line on the front of the body passing through the midpoint of the clavicle (collarbone).

Middle Ear

- The middle ear, also known as the tympanic cavity, is an air-filled, mucosa-lined space within the temporal bone that lies between the tympanic membrane laterally and the medial wall of the middle ear (abrynthine wall) medially, and it contains the three ossicles (malleus, incus, and stapes) that transmit sound vibrations from the tympan

Middle East Respiratory Syndrome

- A severe respiratory infection caused by Middle East respiratory syndrome coronavirus, usually acquired through close contact with infected camels or people. It can cause pneumonia, respiratory failure, and renal or multiorgan complications.

Middle Lobe Syndrome

- Chronic or recurrent atelectasis and infection of the right middle lobe due to bronchial obstruction. The long, narrow right middle lobe bronchus is susceptible to compression by lymph nodes, tumors, or mucous plugging. It may be caused by bronchiectasis, malignancy, or granulomatous disease. Treatment depends on the underlying cause.

Midtrimester

- The middle third of pregnancy, from 14 to 28 weeks gestation. The midtrimester is when many women feel their best: nausea and fatigue of the first trimester resolve, and the physical discomforts of the third trimester have not yet developed.

Migraine

- A primary headache disorder characterized by recurrent episodes of moderate to severe, typically unilateral, pulsating headache lasting 4-72 hours, associated with nausea, vomiting, photophobia, and phonophobia.

Migraine With Aura

- A primary headache disorder characterized by recurrent attacks of fully reversible visual, sensory, or speech symptoms that develop gradually over at least five minutes and last no more than sixty minutes, followed by headache with the characteristics of migraine.

Migraine Without Aura

- A primary headache disorder characterized by recurrent attacks of moderate to severe, pulsating, unilateral headache lasting four to seventy-two hours, associated with nausea, vomiting, photophobia, and phonophobia.

Migrating Motor Complex

- Cyclical patterns of intense gastrointestinal motility occurring during fasting (interdigestive period), sweeping residual food, bacteria, and debris from the stomach through the small intestine to the colon.

Mild Cognitive Impairment

- A transitional cognitive state between normal aging and dementia, characterized by subjective cognitive complaints confirmed by objective cognitive impairment that is not severe enough to significantly interfere with daily activities.

Milia

- Small, white, keratin-filled cysts appearing on the face, particularly around the eyes. They are common in newborns (neonatal milia) and adults (primary milia). Secondary milia occur after blistering injuries, dermabrasion, or laser therapy. Milia are asymptomatic and may resolve spontaneously.

Miliaria crystallina

- A benign, asymptomatic eruption of small, clear,

superficial vesicles caused by obstruction of eccrine sweat ducts in the stratum corneum, occurring in hot, humid environments or during high fevers. The vesicles resemble dewdrops on the skin and rupture easily, leaving fine scales.

Miliary TB

- Hematogenous dissemination of *M. tuberculosis* causing innumerable 1-3 mm granulomas throughout lungs and other organs. CXR shows “millet seed” pattern. More common in immunocompromised patients.

Miliary Tuberculosis

- Disseminated tuberculosis caused by hematogenous spread of *Mycobacterium tuberculosis*, producing numerous tiny granulomatous lesions in the lungs and potentially multiple organs. Symptoms may include prolonged fever, weight loss, night sweats, cough, and organ-specific manifestations.

Milk-Alkali Syndrome

- Hypercalcemia, metabolic alkalosis, and renal dysfunction caused by excessive intake of calcium and absorbable alkali. It may occur with high-dose calcium supplements or antacid use.

Milliequivalent

- See units.

Millimolar

- See units.

Milrinone

- A phosphodiesterase-3 inhibitor with positive inotropic and vasodilatory (inodilator) actions, increasing cAMP in cardiac and vascular smooth muscle.

Milroy Disease

- An inherited primary lymphedema caused most often by pathogenic variants affecting VEGFR3 signaling. It typically presents with chronic swelling of the lower limbs from birth or early infancy and may be associated with papillomatosis, hydrocele, or recurrent skin infection.

Mindful Eating

- Paying attention to hunger cues, eating slowly, savoring food. May reduce overeating, improve digestion.

Mineralocorticoid Receptor

- An intracellular steroid receptor activated primarily by aldosterone in the kidney, colon, sweat glands, and other tissues. Its activation increases sodium reabsorption and potassium and hydrogen secretion and contributes to blood-pressure regulation and tissue remodeling.

Mineralocorticoid Receptor Antagonist

- A drug that blocks aldosterone receptors, promoting natriuresis while limiting potassium loss and adverse cardiac remodeling. Spironolactone and eplerenone improve outcomes in selected heart-failure patients; hyperkalemia and renal dysfunction require monitoring.

Mineralocorticoid Receptor Antagonists

- Spironolactone and eplerenone block aldosterone receptors. Antifibrotic effects in CKD. Finerenone is non-steroidal MRA with cardiovascular and renal benefits in diabetic CKD. Monitor for hyperkalemia especially when combined with RAAS inhibitors.

Mini-Mental State Examination

- A brief structured cognitive screening instrument assessing orientation, registration, attention, calculation, recall, language, and visuospatial ability. Scores are affected by education, language, sensory impairment, culture, and delirium and do not establish a dementia diagnosis alone.

Minimal Change Disease

- Most common cause of nephrotic syndrome in children accounting for approximately 80%. Glomeruli appear normal on light microscopy but podocyte foot processes are diffusely effaced on electron microscopy. Circulating factors from dysregulated T cells may increase glomerular permeability. Presents with sudden onset nephrotic syndrome.

Minimal Residual Disease (MRD)

- The small number of malignant cells that remain after treatment and are below detection by conventional clinical examination or routine microscopy. Sensitive flow-cytometric, molecular, or imaging methods may detect MRD and refine relapse-risk assessment.

Minimum Alveolar Concentration

- See pharmacology.

Minimum Alveolar Concentration (MAC)

- The alveolar concentration of an inhalational anesthetic at 1 atmosphere of pressure that prevents movement in 50% of patients in response to a noxious surgical stimulus. It serves as the standard measure of potency for volatile anesthetics (e.g., sevoflurane has a MAC of 2.0%, while desflurane has a MAC of 6.0% in adults).

Minimum Bactericidal Concentration (MBC)

- The lowest concentration of an antimicrobial that kills a defined proportion of the original bacterial inoculum, commonly demonstrated by subculture without growth. MBC is distinct from MIC and is not routinely required for every infection.

Minimum Effect Concentration

- See pharmacology.

Minimum inhibitory concentration

- The lowest concentration of an antimicrobial agent that inhibits visible growth of a microorganism under standardized in vitro conditions. MIC is the gold standard for measuring antimicrobial susceptibility and is determined by broth microdilution or agar dilution methods.

Minimum Inhibitory Concentration (MIC)

- The lowest concentration of an antimicrobial that prevents visible growth of a microorganism under standardized in-vitro conditions. The MIC is interpreted with organism-specific breakpoints to classify susceptibility.

Minimum Lethal Concentration

- See pharmacology.

Minoxidil

- A direct arteriolar vasodilator (potassium channel opener) used for severe refractory hypertension (usually with a beta-blocker and loop diuretic to counter reflex tachycardia and fluid retention), and topically for androgenetic alopecia (promotes hair regrowth).

Minute Ventilation

- The total volume of air breathed in one minute (tidal volume x respiratory rate). Normal is 6–8 L/min. Elevated in increased metabolic demand, pain, anxiety, and sepsis. Reduced in neuromuscular

disease and sedation.

Miosis

- Constriction of the pupil of the eye (<2 mm diameter). Controlled by parasympathetic nerve fibers via the oculomotor nerve (CN III) acting on the sphincter pupillae muscle.

MIRD Schema

- Medical Internal Radiation Dose. Method for calculating radiation absorbed doses.

Mirizzi Syndrome

- Condition where a gallstone impacted in the cystic duct or Hartmann pouch extrinsic compression of the common hepatic duct causing obstructive jaundice. Classified into four types based on the degree of erosion and fistula formation. Type one is extrinsic compression without fistula.

Mirror Therapy

- Mirror reflection of unaffected limb movement creates visual illusion of affected limb moving. Reduces phantom limb pain, improves stroke recovery.

Miscarriage

- The spontaneous loss of a pregnancy before 20 weeks of gestation, occurring in 15-20 percent of clinically recognized pregnancies. The majority (80 percent) occur in the first trimester, with risk decreasing significantly after visualization of fetal cardiac activity.

Mismatch Repair Deficiency (dMMR)

- Loss of DNA mismatch-repair function that produces microsatellite instability and a high tumor mutational burden. Seen in Lynch syndrome and some sporadic colorectal, endometrial, and other cancers, and associated with responsiveness to immune checkpoint inhibitors.

Misoprostol (Cytotec)

- A synthetic prostaglandin E1 analog used off-label for cervical ripening and labor induction. Dose: 25-50 mcg vaginally or 25-100 mcg orally q4-6h. Advantages: inexpensive, stable at room temperature, and multiple routes of administration. Risks: uterine tachysystole (with and without FHR changes), the highest rate among induction

Missense Variant

- A single-nucleotide substitution that changes one amino acid in a protein. Its clinical effect ranges

from benign to pathogenic and must be assessed using molecular, functional, population, and clinical evidence.

Mitochondria

- Double-membrane organelles known as the "powerhouse of the cell," generating ATP through oxidative phosphorylation. Outer membrane is permeable; inner membrane is highly folded (cristae) and contains the electron transport chain and ATP synthase.

Mitochondrial Diabetes

- Diabetes caused by mutations in mitochondrial DNA particularly the m.3243A>G mutation in the MT-TL1 gene. Maternally inherited. Often associated with sensorineural hearing loss termed maternally inherited diabetes and deafness. Patients are typically lean with impaired insulin secretion. Diagnosis requires genetic testing.

Mitochondrial Inheritance

- Inheritance through mitochondrial DNA (mtDNA), passed only from mother. Examples: Leber hereditary optic neuropathy, MELAS syndrome. Heteroplasmy: mixture of mutant and normal mtDNA.

Mitosis

- See genetics.

Mitotic Figure

- A cell undergoing nuclear division that can be identified microscopically. Mitotic count contributes to assessment of proliferative activity and tumor grade, although counting method and area must be standardized.

MitraClip

- A transcatheter device that creates a double orifice in the mitral valve by clipping the anterior and posterior leaflets together, reducing mitral regurgitation. It is indicated for patients with significant (3+ or 4+) mitral regurgitation who are at high risk for surgical mitral valve repair.

Mitral Regurgitation

- Incomplete closure of the mitral valve allowing blood to flow backward from the left ventricle to the left atrium during systole. Causes include mitral valve prolapse, papillary muscle dysfunction, rheumatic disease, and ischemic heart disease. Murmur: pansystolic (holosystolic) blowing murmur at the apex radiating to the axilla.

Mitral Stenosis

- Narrowing of the mitral valve opening, obstructing left atrial to left ventricular blood flow. Most commonly caused by rheumatic heart disease. Symptoms include dyspnea, fatigue, and atrial fibrillation. Murmur: low-pitched, rumbling mid-diastolic murmur at the apex with opening snap.

Mitral Stenosis Murmur

- Low-pitched, rumbling mid-diastolic murmur at the apex, best heard with the bell of the stethoscope in the left lateral decubitus position. Accompanied by an opening snap (early diastolic high-pitched click) and loud S1 in pliable valves. Severe MS may show a diastolic murmur without a preceding S2 gap.

Mitral Valve

- Left atrioventricular (bicuspid) valve with anterior and posterior cusps supported by chordae tendineae and papillary muscles. Anterior cusp is larger and proximate to the aortic valve. Mitral valve prolapse is the most common valvular abnormality. Mitral stenosis (most commonly rheumatic) and regurgitation are significant.

Mitral Valve Prolapse

- A valvular heart disease where the mitral valve leaflets prolapse into the left atrium during systole. Most patients are asymptomatic; some develop palpitations, atypical chest pain, or dyspnea. Diagnosis by echocardiography. Complications include mitral regurgitation, infective endocarditis, and arrhythmias.

Mitral-valve repair

- A surgical or transcatheter procedure to restore mitral valve competence without replacement, typically for regurgitation from prolapse, chordal rupture, or ischemic dysfunction. Repair preserves native valve tissue, maintains left ventricular function, and avoids lifelong anticoagulation required by mechanical prostheses.

Mixed Connective Tissue Disease

- An overlap syndrome with features of systemic lupus erythematosus, systemic sclerosis, and polymyositis/dermatomyositis. Highly associated with anti-U1 RNP antibodies. Presents with Raynaud phenomenon (nearly universal), puffy hands,

arthritis, myositis, esophageal dysmotility, and serositis. Renal disease is less common than in SLE.

Mixed Hearing Loss

- A combination of both conductive and sensorineural hearing loss occurring in the same ear, resulting in a hearing deficit that involves impaired sound transmission through the outer or middle ear in addition to cochlear or neural dysfunction.

Mixed Urinary Incontinence

- Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

MMR Vaccine

- Measles, Mumps, and Rubella live attenuated vaccine given at 12-15 months and 4-6 years. Two doses provide 97% protection against measles and mumps, and 98% against rubella. Contraindicated in pregnancy and immunocompromised individuals. Post-exposure prophylaxis within 72 hours.

Mobile genetic elements

- Genetic elements that can move within and between genomes. These include plasmids, transposons, and integrons. Mobile genetic elements can carry antibiotic resistance genes and virulence factors. They facilitate horizontal gene transfer and contribute to the rapid evolution of bacteria.

Mobile Health (mHealth)

- Use of mobile devices for health. Examples: health apps, text message reminders, wearable devices.

Modafinil

- A wakefulness-promoting agent (mechanism not fully understood, likely dopaminergic) used for narcolepsy, obstructive sleep apnea residual sleepiness, and shift work sleep disorder; used off-label for fatigue and ADHD. Less abuse potential and fewer sympathomimetic effects than amphetamines.

Modified Biophysical Profile (Modified BPP)

- A simplified antepartum test combining the non-stress test (NST) with the amniotic fluid index (AFI). The NST assesses acute fetal status (heart rate reactivity over 20-40 minutes) and the AFI assesses chronic uteroplacental function (oligohydramnios suggests chronic hypoxia or placental in-

sufficiency).

Mohs Surgery

- A micrographically controlled technique in which skin cancer is excised in thin layers and examined margin-by-margin, achieving high cure rates for basal cell and squamous cell carcinoma while sparing normal tissue.

Molar Pregnancy

- A type of gestational trophoblastic disease characterized by abnormal proliferation of trophoblastic tissue. See Hydatidiform Mole.

Mole

- Nevus: a benign proliferation of melanocytes, the most common skin lesion; also used for other nevi (moles are melanocytic nevi). Types: junctional (flat, brown—childhood), compound (slightly raised), intradermal (dome-shaped, flesh/pigmented).

Molecular Pathology

- The investigation of disease using nucleic-acid, genomic, epigenomic, or molecular protein alterations in cells and tissues. It may establish diagnosis, estimate prognosis, identify inherited risk, or select targeted therapy.

Molecular Testing

- Analysis of DNA/RNA for mutations, fusions, expression. Guides targeted therapy.

Molecular Weight

- Mass of molecule. Important for dosing, toxicity calculations.

Molluscum Contagiosum

- A viral skin infection caused by a poxvirus, presenting as firm, dome-shaped, umbilicated papules with central dimpling, 2 to 5 mm in diameter. It is transmitted by direct skin contact, autoinoculation, and sexual contact in adults. Immunocompetent hosts clear lesions spontaneously within 6 to 12 months.

Molybdenum-99

- Parent isotope of Tc-99m, half-life 66 hours. Produced in nuclear reactors. Generator systems provide Tc-99m for 1 week.

Mondor's disease

- Superficial thrombophlebitis of the anterior chest wall veins, typically the lateral thoracic, thoracoepi-

gastric, or superior epigastric veins. Presents as a tender, cord-like subcutaneous structure with skin retraction. Usually self-limited; associated with breast surgery, trauma, or hypercoagulable states.

Monitored Anesthesia Care

- A distinct service where an anesthesia professional provides premedication, administration of analgesics and/or sedatives, continuous monitoring of the patient's cardiorespiratory status, and intervention for any emergent or unavoidable complications that may arise during a diagnostic or therapeutic procedure.

Monitoring Test

- Test to track disease progression or treatment response.

Monoamine Oxidase Inhibitors

- Antidepressants that inhibit monoamine oxidase, the enzyme responsible for breaking down serotonin, norepinephrine, and dopamine. MAOIs include phenelzine, tranylcypromine, isocarboxazid, and selegiline (transdermal). They are effective for atypical depression and treatment-resistant depression.

Monoclonal Antibody

- An antibody produced by a single cloned B-cell line, specific for one epitope, generated by hybridoma technology or recombinant methods (name suffix -mab).

Monocyte

- A circulating leukocyte (2-8 percent) that differentiates into macrophages and dendritic cells in tissues. Monocytes phagocytose pathogens, present antigens, and secrete cytokines (TNF, IL-1, IL-6); participate in innate immunity, inflammation, and tissue repair. Subsets differ in function (classical, nonclassical, intermediate).

Mononucleosis

- An acute viral syndrome caused by Epstein-Barr virus (EBV), primarily affecting adolescents and young adults. Classic triad: fever, pharyngitis/exudative tonsillitis, and cervical lymphadenopathy. Associated with splenomegaly, fatigue, and atypical lymphocytosis. Diagnosis: positive heterophile antibody (Monospot) and EBV serology.

Monosodium urate

- The crystalline salt that precipitates in tissues

when serum uric acid levels exceed the saturation point, forming needle-shaped, negatively birefringent crystals under polarized light microscopy. It is the pathological hallmark of gout, depositing in joints, soft tissues, and kidneys.

Monosomy

- Presence of only one copy of a chromosome instead of two. Only compatible with life is monosomy X (Turner syndrome): short stature, webbed neck, ovarian dysgenesis, normal intelligence.

Monospot (EBV Heterophile Antibody)

- Rapid test for infectious mononucleosis caused by Epstein-Barr virus. Detects heterophile antibodies (IgM) that agglutinate horse or sheep red blood cells. Sensitivity: 70-95% in symptomatic patients. False negative in early disease (<1 week) and children <5. False positive in autoimmune disease, HIV, lymphoma, and other viral

Monounsaturated Fat

- Liquid at room temperature. Sources: olive oil, avocado, nuts. Cardiovascular benefits.

Montelukast

- A leukotriene receptor antagonist (LTRA) blocking cysteinyl leukotrienes (LTC₄, LTD₄, LTE₄) that cause bronchoconstriction, mucus secretion, and airway inflammation. Used as add-on maintenance therapy for asthma (particularly exercise-induced and aspirin-sensitive), allergic rhinitis, and bronchospasm prophylaxis. Oral, once daily.

Montevideo Units (MVU)

- A quantitative measure of uterine contraction intensity, calculated by summing the peak contraction pressures (in mm Hg) minus the resting tone for each contraction in a 10-minute window. Adequate labor: 200 MVU (spontaneous) or 250 MVU (augmented). Measured with an intrauterine pressure catheter (IUPC).

Montgomery Tubercles

- Small, raised sebaceous glands visible on the areola of the breast during pregnancy and lactation. They appear as small bumps surrounding the nipple and secrete an oily substance that lubricates the nipple and areola, protecting against infection. Their prominence increases during pregnancy under hormonal influence.

Montreal Cognitive Assessment

- A brief cognitive screening instrument with greater emphasis on executive function, visuospatial ability, attention, memory, language, and orientation than many shorter screens. Interpretation requires consideration of education, language, sensory deficits, and acute illness.

Mood Stabilizers

- Medications used to treat mood episodes in bipolar disorder and prevent recurrence. The primary mood stabilizers are lithium, valproate, carbamazepine, and lamotrigine. Each has a different mechanism and side effect profile. Lithium is the only agent with demonstrated anti-suicide properties.

Morbid Obesity

- Severe obesity defined clinically as BMI 40 kg/m² or BMI 35 kg/m² with obesity-related comorbidities (type 2 diabetes, hypertension, obstructive sleep apnea, non-alcoholic fatty liver disease, osteoarthritis, cardiovascular disease).

Morbidity

- Disease or illness in population. Morbidity rate: new cases per population per time.

Morgagni Hernia

- A congenital diaphragmatic hernia occurring through the anterior retrosternal foramen of Morgagni (space of Larrey), resulting from failure of the sternal and costal portions of the diaphragm to fuse. The right side is affected in 90% of cases (the hernia sac is present in 90%).

Morning Sickness

- Nausea and vomiting of pregnancy (NVP): nausea (and sometimes vomiting) commonly in the first trimester (6-12 weeks), typically resolving by 14-16 weeks; multifactorial (hCG, estrogen, progesterone, odor sensitivity).

Moro Reflex

- A primitive reflex present from birth to 3-6 months, elicited by suddenly startling the infant (loud noise, head drop). The infant extends both arms outward with fingers spread, then draws arms inward as if embracing. Absence at birth suggests central nervous system injury or brachial plexus palsy.

Mortality

- Death rate in population. Elderly mortality: higher

rates, common causes: cardiovascular disease, cancer, dementia, respiratory disease, fall-related injuries. Life expectancy varies by sex, race, socioeconomic status.

Mortality Rate

- Deaths from disease in population. Crude: total deaths / total population. Specific: disease-specific deaths / population at risk.

Morton Neuroma

- A perineural fibrosis of the plantar digital nerve, most commonly between the third and fourth metatarsal heads. It presents as burning pain, numbness, and a feeling of walking on a pebble. It is not a true neuroma but a mechanical entrapment neuropathy.

Mosaicism

- Presence of two or more cell lines with different genotypes in one individual. Can be somatic (affecting body cells) or germline (affecting gametes). Origin: post-zygotic mutation.

Motion Analysis

- Assessment of movement patterns. 3D gait analysis, video analysis. Identifies abnormalities, guides treatment.

Motion Artifact

- Image degradation from patient movement. Reduced by faster acquisition, breath-hold, sedation.

Motion Correction

- Correction for patient motion during acquisition. Respiratory gating, navigation.

Motivational Interviewing

- A client-centered, directive counseling method for enhancing intrinsic motivation to change by exploring and resolving ambivalence. Developed by Miller and Rollnick, it uses the spirit of partnership, acceptance, compassion, and evocation. Core skills include open-ended questions, affirmations, reflections, and summaries.

Motor Cortex

- The frontal-lobe regions that plan and initiate voluntary movement. The primary motor cortex lies in the precentral gyrus and controls the opposite side of the body in an orderly somatotopic map.

Motor Learning

- A relatively lasting change in the ability to perform a motor skill resulting from practice, experience, feedback, and adaptation. Rehabilitation uses task-specific repetition and meaningful practice to promote motor learning.

Motor Unit

- A single alpha motor neuron and all the skeletal muscle fibers it innervates. The number of muscle fibers per motor unit determines precision of control: small motor units (few fibers, e.g., extraocular and hand muscles) provide fine movement; large motor units (hundreds to thousands of fibers, e.g., gastrocnemius) provide gross power.

Mouth

- The oral cavity that serves as the entry point of the digestive tract, used for eating, breathing, and speaking.

Mouth Ulcers

- Open sores affecting the oral mucosa, commonly caused by aphthous ulcers, trauma, infection, irritation, or systemic inflammatory disease. Most resolve within two weeks; persistent, recurrent, unusually large, or indurated ulcers require evaluation.

Mouthwash

- Antimicrobial or cosmetic liquid for oral rinsing. Chlorhexidine (prescription), essential oils, cetylpyridinium chloride.

Movement Disorder

- A neurologic condition causing excessive, reduced, or otherwise abnormal movement. Hyperkinetic disorders include tremor, dystonia, chorea, myoclonus, tics, and dyskinesia; hypokinetic disorders include bradykinesia and parkinsonism.

Moyamoya Disease

- A progressive occlusive arteriopathy of the terminal internal carotid arteries and their proximal branches, with development of fragile collateral vessels at the brain base. It can cause transient ischemic attacks, strokes, seizures, headaches, or cognitive impairment.

Mpox

- A zoonotic infection caused by monkeypox virus, characterized by fever, swollen lymph nodes, and a distinctive rash that may include deep-seated vesi-

cles or pustules.

Mpox (monkeypox)

- A zoonotic viral disease caused by the monkeypox virus, an orthopoxvirus in the same genus as smallpox (variola) and vaccinia. The virus has two clades: clade I (formerly Congo Basin, Central African) with higher virulence and mortality (up to 10 percent), and clade II (formerly West African) with lower mortality (less than 1 percent).

MR Angiography (MRA)

- Vascular imaging using MRI without ionizing radiation. Time-of-flight (TOF) or contrast-enhanced (CE-MRA). Used for cerebral aneurysms, renal artery stenosis, peripheral arterial disease.

MR Enterography

- MRI of the small bowel after oral contrast administration. Superior soft tissue contrast for evaluating Crohn disease activity, fistulas, and abscesses without ionizing radiation. Demonstrates mural thickening, edema, enhancement patterns, and extraintestinal complications. Diffusion-weighted imaging helps assess disease activity.

MR Neurography

- MRI technique visualizing peripheral nerves. See neurography.

MRCP (Magnetic Resonance Cholangiopancreatography)

- Non-invasive MRI of the biliary and pancreatic ducts using heavily T2-weighted sequences that highlight static fluid. Used for: choledocholithiasis (CBD stones), biliary strictures, primary sclerosing cholangitis, pancreatic duct anatomy, and congenital biliary anomalies.

MRI

- See radiology.

MRI (Magnetic Resonance Imaging)

- Imaging using strong magnetic fields and radiofrequency pulses to generate detailed cross-sectional images. No ionizing radiation. Superior soft-tissue contrast compared to CT.

MRI Brain

- Excellent for brain pathology: tumors, stroke, demyelination, infection, degeneration. Sequences: T1, T2, FLAIR, DWI, SWI, contrast-enhanced. DWI detects acute ischemia within minutes.

MRI Contrast

- Substance enhancing image visibility. See contrast agent.

MRI Joint

- Evaluation of musculoskeletal structures: cartilage, ligaments, tendons, menisci, labrum. Knee: ACL, meniscus tears. Shoulder: rotator cuff, labrum. Hip: labral tears, FAI.

MRI Sequences

- T1-weighted: anatomy, fat bright. T2-weighted: pathology, fluid bright. FLAIR: suppresses CSF, lesions bright. DWI: restricted diffusion bright (acute stroke, abscess). STIR: fat suppression, fluid bright. Post-contrast T1: enhancement bright.

mRNA Vaccine

- A vaccine delivering messenger RNA encoding a pathogen antigen that host cells translate into protein, triggering an immune response. Approved for COVID-19 (Pfizer-BioNTech, Moderna) and RSV; under study for influenza, cancer, and other infections.

MRSA

- Methicillin-resistant *Staphylococcus aureus*, resistant to beta-lactam antibiotics through acquisition of the *mecA* gene encoding PBP2a. MRSA is a major cause of healthcare-associated and community-associated infections. Treatment options include vancomycin, daptomycin, and linezolid.

MRSA decolonization

- Strategies to eliminate MRSA carriage from the nares and skin of colonized individuals. Decolonization reduces the risk of infection and transmission. The most common regimen is intranasal mupirocin for 5 days. Chlorhexidine bathing is used for body decolonization.

MRV (Magnetic Resonance Venography)

- MRI technique to visualize veins, using time-of-flight (TOF) or contrast-enhanced sequences. Indications: cerebral venous sinus thrombosis, dural sinus stenosis, abdominal/pelvic venous thrombosis, portal vein thrombosis, and renal vein thrombosis. Differentiates acute vs. chronic thrombus.

mTOR

- Mammalian target of rapamycin. Controls cell growth, proliferation. Inhibited by rapamycin.

Mucocele

- Mucus extravasation cyst of minor salivary gland. Lower lip most common. Treatment: excision.

Mucolytic Therapy

- Medications that reduce mucus viscosity and elasticity, facilitating clearance. N-acetylcysteine (NAC) breaks disulfide bonds in mucus glycoproteins. Dornase alfa (Pulmozyme) cleaves extracellular DNA in CF sputum. Hypertonic saline aerosol draws water into the airway surface liquid.

Mucormycosis

- A life-threatening fungal infection caused by Mucorales order fungi including Rhizopus, Mucor, and Rhizomucor species. The infection primarily affects immunocompromised patients, particularly those with uncontrolled diabetes mellitus and hematologic malignancies. The fungi invade blood vessels causing thrombosis and tissue infarction.

Mucosal Irritation

- Chemical injury to mucous membranes. Eyes, nose, throat, GI tract. Treatment: irrigation, supportive care.

Mucositis

- Inflammation and ulceration of the oral or gastrointestinal mucosa caused by chemotherapy or radiation, causing pain, dysphagia, and poor intake and predisposing to local infection.

Mucus Plugging

- Accumulation of thick, tenacious mucus within airways causing obstruction. It is common in asthma, CF, bronchiectasis, and chronic bronchitis. It impairs gas exchange, promotes infection, and causes atelectasis. Treatment includes airway clearance techniques, mucolytics, and bronchodilators.

Muenke Syndrome

- A craniosynostosis syndrome caused by a pathogenic FGFR3 variant. Premature fusion of one or more skull sutures may produce an abnormal head shape and can be accompanied by developmental, hearing, limb, or neurological findings.

MUGA Scan (Multigated Acquisition Scan)

- Radionuclide ventriculography using Tc-99m-labeled RBCs to assess left ventricular function. ECG-gated acquisition synchronized to the QRS

complex. Measures: LVEF (most accurate non-invasive method), wall motion, and ventricular volumes. Gold standard for LVEF assessment prior to cardiotoxic chemotherapy (doxorubicin, trastuzumab).

Multicystic Dysplastic Kidney

- A congenital renal anomaly where one or both kidneys are replaced by multiple non-communicating cysts of varying sizes, with no functioning renal parenchyma. The most common cause of an abdominal mass in a newborn (unilateral). The ureter is typically atretic. The contralateral kidney may have VUR (20%) or UPJ obstruction (10%).

Multidisciplinary Disaster Victim Identification

- The coordinated process of identifying victims of mass-fatality incidents (natural disasters, transport crashes, terrorist attacks) using multiple disciplines: fingerprinting, dental records, DNA profiling, forensic anthropology/radiology, and personal effects, with ante-mortem data collection and comparison.

Multidisciplinary Team

- Rehabilitation team includes: physiatrist, physical therapist, occupational therapist, speech-language pathologist, psychologist, social worker, case manager, nursing, recreation therapist.

Multidrug-Resistant TB

- TB resistant to at least isoniazid and rifampin. Requires second-line drugs for 18–24 months. Worse outcomes and higher mortality than drug-susceptible TB.

Multifocal Atrial Tachycardia

- An atrial tachyarrhythmia with at least three distinct P-wave morphologies, irregular rhythm, and a rate above 100 beats per minute. It is strongly associated with severe pulmonary disease, hypoxia, and metabolic stress.

Multimodal Analgesia

- Use of analgesic techniques with different mechanisms to improve pain control while reducing reliance on any single drug, especially opioids. Components may include acetaminophen, nonsteroidal anti-inflammatory drugs, regional anesthesia, local anesthetics, and selected adjuvants.

Multimorbidity

- The coexistence of two or more chronic health conditions in the same person, without requiring one to be considered the primary disease. It complicates treatment, increases polypharmacy risk, and requires coordinated patient-centered care.

Multiparametric MRI of the Prostate

- MRI protocol combining T2-weighted, diffusion-weighted, and dynamic contrast-enhanced sequences to detect and characterize clinically significant prostate cancer. Findings are commonly reported using the PI-RADS system.

Multiple bacterial abscesses

- Numerous circumscribed collections of pus within an organ or tissue, caused by hematogenous seeding or direct extension of bacterial infection. They appear as multiple, low-attenuation lesions on imaging, often with peripheral enhancement. Common sites include the liver, brain, and kidneys.

Multiple biliary hamartomas

- Benign, congenital malformations of the intrahepatic bile ducts appearing as multiple, small, well-defined cystic lesions scattered throughout the liver parenchyma on imaging. They are typically asymptomatic and discovered incidentally. They must be distinguished from metastatic disease and Caroli disease.

Multiple Endocrine Neoplasia Type 1

- An autosomal dominant tumor syndrome caused by MEN1 gene mutation (menin).

Multiple Endocrine Neoplasia Type 2

- An autosomal dominant tumor syndrome caused by RET proto-oncogene mutations. Type 2A (95%): medullary thyroid carcinoma (MTC, nearly 100% penetrance), pheochromocytoma (50%), and hyperparathyroidism (20-30%). Type 2B: MTC, pheochromocytoma, mucosal neuromas, marfanoid habitus; more aggressive MTC.

Multiple endocrine neoplasia type 2B

- An autosomal dominant syndrome caused by RET proto-oncogene mutations, characterized by medullary thyroid carcinoma, pheochromocytoma, mucosal neuromas of the lips and tongue, and a marfanoid habitus. Medullary thyroid carcinoma in MEN 2B is aggressive and occurs at an early

age. Prophylactic thyroidectomy is recommended in infancy.

Multiple Myeloma

- A plasma cell neoplasm characterized by clonal proliferation of malignant plasma cells in the bone marrow, producing monoclonal immunoglobulin (M-protein) and causing end-organ damage defined by the CRAB criteria: hypercalcemia (calcium greater than 11 mg/dL), renal insufficiency (creatinine greater than 2 mg/dL), anemia (hemoglobin less

Multiple Pregnancy

- Gestation with more than one fetus: twins (most common), triplets, higher order. Types: dizygotic (two eggs, two placentas—fraternal, more common, familial and increased with IVF) vs. monozygotic (one egg split—identical, sharing determined by timing: dichorionic-diamniotic, monochorionic-diamniotic, monoamniotic, conjoined).

Multiple Sclerosis

- A chronic autoimmune demyelinating disease of the central nervous system, most commonly presenting in young adults. Characterized by relapsing-remitting episodes of neurological dysfunction (optic neuritis, transverse myelitis, brainstem syndromes). Diagnosis: McDonald criteria with MRI dissemination in space and time.

Multiple System Atrophy

- A progressive, fatal neurodegenerative disorder characterized by varying combinations of autonomic failure, cerebellar ataxia, parkinsonism, and pyramidal signs.

Multiplex PCR (BioFire, FilmArray)

- A rapid molecular diagnostic system using nested multiplex PCR to detect multiple pathogens from a single sample. Panels: respiratory (adenovirus, coronavirus, hMPV, influenza, parainfluenza, RSV, rhinovirus, Bordetella, Chlamydia, Mycoplasma), GI (Campylobacter, Salmonella, Shigella, E.

Mummification

- Postmortem alteration occurring in hot, dry, arid, ventilated environments where rapid desiccation of tissues occurs before putrefaction can proceed, resulting in dry, leathery, dark brown skin clinging to underlying skeleton.

Mumps

- Paramyxovirus infection causing bilateral parotitis, fever, and headache. Complications include orchitis (rarely causing infertility), meningitis, pancreatitis, and sensorineural hearing loss. Prevention by MMR vaccination. Outbreaks occur in unvaccinated populations. Diagnosis by serology or RT-PCR from saliva.

Mumps virus

- A paramyxovirus that causes mumps, a contagious respiratory illness. The virus is transmitted by respiratory droplets. Symptoms include fever, headache, and parotitis. Complications include orchitis, oophoritis, meningitis, and deafness. Diagnosis involves serology and PCR. Treatment is supportive.

Mural Thrombus

- A mural thrombus is a blood clot that forms on the wall of a heart chamber or large blood vessel, typically in areas of damaged endothelium, stasis, or turbulent flow.

Murmur

- Heart sound caused by turbulent blood flow. Grades I–VI. Systolic murmurs more common; diastolic murmurs always pathological. Evaluation: timing, location, radiation, intensity, quality, position changes.

Murmur Classification

- Heart murmurs are classified by timing, grade, shape, location, radiation, and pitch. Timing: systolic (between S1 and S2), diastolic (between S2 and S1), or continuous. Grade: I (barely audible) to VI (heard without stethoscope). Shape: crescendo, decrescendo, or plateau.

Muscle

- A specialized contractile tissue responsible for movement, posture, and heat production.

Muscle Contraction

- The shortening or tension generation of muscle produced by the interaction of actin and myosin myofilaments. In skeletal and cardiac muscle, contraction follows excitation-contraction coupling: calcium binds troponin, tropomyosin moves aside, and myosin cross-bridges cycle against actin (sliding filament mechanism) using ATP.

Muscle Pain (Myalgia)

- Myalgia—muscle pain—is a common symptom arising from skeletal muscle injury, inflammation, overuse, infection, or systemic disease.

Muscle Tissue

- A highly specialized tissue capable of contracting to produce body movement, posture, and heat.

Muscle Weakness

- Age-related loss of muscle mass and function. See sarcopenia.

Muscular Dystrophy

- A group of inherited disorders characterized by progressive skeletal-muscle weakness and degeneration. The pattern, age of onset, and complications vary by genetic type; some forms also affect the heart, respiratory muscles, or other organs.

Musculocutaneous Nerve

- From lateral cord (C5–C7), piercing coracobrachialis, running between biceps and brachialis to supply these muscles and lateral forearm skin (lateral antebrachial cutaneous nerve). Sole motor supply to the anterior arm compartment. Injury causes weakness of elbow flexion/forearm supination with lateral forearm sensation loss.

Music Therapy

- Use of music for therapeutic purposes. Pain reduction, emotional expression, cognitive stimulation.

Mutagen

- Substance causing genetic mutations. Tests: Ames test, chromosomal aberration. Correlated with carcinogenicity.

Mutation

- A heritable or acquired alteration in the DNA sequence. Variants may be single-nucleotide substitutions, insertions, deletions, copy-number changes, or larger rearrangements and may be benign, pathogenic, or of uncertain significance.

Mutation Rate

- Frequency of new mutations per generation or per base pair per replication. Varies by gene, genomic region, organism. High mutation rate in repetitive sequences.

Myalgic Encephalomyelitis

- See Chronic Fatigue Syndrome.

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome

- ME/CFS: debilitating condition characterized by profound fatigue, post-exertional malaise, unrefreshing sleep, cognitive impairment. Diagnosis: exclusion. Treatment: pacing, symptom management, graded exercise therapy controversial.

Myasthenia Gravis

- An autoimmune disorder caused by antibodies against the nicotinic acetylcholine receptor (AChR, 85%) or MuSK at the neuromuscular junction, leading to fluctuating muscle weakness that worsens with use (fatigability). Presents with ptosis, diplopia, dysphagia, dysarthria, and proximal limb weakness. Ocular MG: limited to eye muscles.

Myasthenic (Lambert-Eaton) syndrome

- An autoimmune presynaptic neuromuscular-junction disorder caused by antibodies against voltage-gated calcium channels, producing proximal weakness and autonomic symptoms.

Mycobacterium abscessus

- A rapidly growing nontuberculous mycobacterium that can cause pulmonary, skin, and soft-tissue infections and is often difficult to treat because of multidrug resistance.

Mycobacterium avium complex

- A group of slowly growing mycobacteria that cause pulmonary disease and disseminated infection in immunocompromised patients. MAC is the most common nontuberculous mycobacterium causing pulmonary disease. Treatment for pulmonary MAC requires a macrolide (azithromycin or clarithromycin) combined with ethambutol and a rifamycin.

Mycobacterium leprae

- An acid-fast bacillus that causes leprosy (Hansen's disease), a chronic infection affecting the skin and peripheral nerves. The organism is transmitted through respiratory droplets, though the exact mechanism is not fully understood. The incubation period is long (2-5 years).

Mycobacterium marinum

- See Non-Tuberculous Mycobacteria.

Mycobacterium tuberculosis

- An acid-fast bacillus that causes tuberculosis, a chronic respiratory infection. The organism is trans-

mitted by respiratory droplets and primarily affects the lungs but can disseminate to other organs. Latent TB infection is asymptomatic and non-transmissible. Active TB presents with cough, haemoptysis, fever, and weight loss.

Mycology

- The branch of microbiology that studies fungi, including yeasts, moulds, and mushrooms. Medical mycology focuses on fungal pathogens that cause disease in humans. Fungal infections (mycoses) are classified as superficial, cutaneous, subcutaneous, or systemic.

Mycoplasma genitalium

- A sexually transmitted bacterium that can cause urethritis, cervicitis, pelvic inflammatory disease, and, less commonly, proctitis. Many infections are asymptomatic. Diagnosis is usually by a nucleic acid amplification test, and macrolide resistance is common, so treatment should follow current susceptibility-informed guidance.

Mycoplasma Pneumonia

- Atypical pneumonia most common in ages 5–35. Dry cough, low-grade fever, headache. Patchy infiltrates on CXR disproportionate to symptoms (“walking pneumonia”).

Mycoplasma pneumoniae-associated mucositis

- Inflammatory lesions of the oral, ocular, or genital mucosa occurring as an extrapulmonary manifestation of Mycoplasma pneumoniae infection, often presenting with target-like skin lesions.

Mycosis Fungoides

- The most common cutaneous T-cell lymphoma, presenting with progressive patches, plaques, and tumors on the skin with an indolent course. Staging and therapy depend on skin involvement and extracutaneous spread.

Mydriasis

- Dilation of the pupil of the eye (>5 mm diameter). Controlled by sympathetic nerve fibers acting on the dilator pupillae muscle.

Myelin

- A lipid-rich protective sheath that wraps around nerve fibers (axons), insulating them and speeding up electrical signal transmission.

Myelin Sheath

- The lipid-rich insulating layer around axons formed by Schwann cells (peripheral nervous system) and oligodendrocytes (central nervous system), which wrap the axon in concentric membrane layers.

Myelodysplastic Syndrome

- MDS: a group of clonal stem cell disorders characterized by ineffective hematopoiesis, cytopenias, and risk of transformation to acute myeloid leukemia (AML). Features: anemia (most common), neutropenia, thrombocytopenia; dysplasia (abnormal morphology) in marrow and blood; cytogenetic abnormalities (del5q, trisomy 8, del7q).

Myelodysplastic Syndromes

- A heterogeneous group of clonal hematopoietic stem cell disorders characterized by ineffective hematopoiesis, dysplastic morphology in one or more myeloid lineages, cytopenias, and risk of transformation to acute myeloid leukemia.

Myelofibrosis

- A myeloproliferative neoplasm characterized by bone marrow fibrosis, extramedullary hematopoiesis, and JAK2, CALR, or MPL mutations. Presents with anemia, massive splenomegaly, constitutional symptoms (fever, night sweats, weight loss), and leukoerythroblastic blood smear.

Myelography

- Contrast injected into subarachnoid space for spinal imaging. Used when MRI contraindicated. CT myelography more common now.

Myeloma

- Multiple myeloma: plasma cell malignancy. Features: bone pain, anemia, renal failure, hypercalcemia.

Myeloma Cast Nephropathy

- Light chain cast formation in tubules causing AKI in multiple myeloma. Friable waxy casts with giant cells. Bence Jones proteinuria. Treat myeloma and consider plasmapheresis to reduce light chain burden.

Myelomeningocele

- The most severe form of spina bifida, where the spinal cord and meninges herniate through a vertebral defect into a sac on the back. Presents with a visible, fluid-filled sac at birth. Associated with

hydrocephalus, Arnold-Chiari malformation, and lower extremity paralysis.

Myelopathy

- A general term for any neurological deficit resulting from spinal cord pathology, encompassing a wide range of causes and clinical presentations.

Myelosuppression

- Reduced bone marrow production of red cells, white cells, and platelets, causing anemia, neutropenia, and thrombocytopenia; a common dose-limiting toxicity of chemotherapy and radiation that increases infection and bleeding risk.

Myiasis

- Infestation of living tissue by larvae (maggots) of various fly species, which feed on host tissue and can cause pain, secondary infection, and tissue destruction. It may be cutaneous, involving the skin and subcutaneous tissue, or affect other body sites including the nose, ears, and wounds.

Myocardial Infarction

- The irreversible necrosis of myocardial tissue resulting from prolonged ischemia, most commonly due to atherosclerotic plaque rupture with superimposed thrombosis in a coronary artery.

Myocardial Infarction (MI)

- Death of myocardium due to prolonged ischemia. Types: STEMI (ST elevation), NSTEMI (troponin elevation without ST elevation). Diagnosis: ECG changes, troponin elevation. Treatment: MONA (morphine, oxygen, nitroglycerin, aspirin), reperfusion (PCI or fibrinolysis).

Myocardial Perfusion Imaging

- Nuclear cardiology technique using radiotracers (Tc-99m sestamibi, Tl-201) to assess blood flow to the myocardium. Performed at rest and during stress (exercise or pharmacological) to detect reversible ischemia (stress defect with rest redistribution) or fixed defects (scar from prior MI).

Myocardial Viability

- PET assessment of hibernating myocardium. FDG uptake indicates viable myocardium.

Myocarditis

- Inflammation of the myocardium, most commonly caused by viral infections (coxsackievirus, adenovirus, SARS-CoV-2). Presents with chest pain,

dyspnea, arrhythmias, and elevated troponin. Diagnosis: ECG, echocardiography, cardiac MRI with Lake Louise criteria, and endomyocardial biopsy.

Myocarditis Diagnosis

- Cardiac MRI (Lake Louise criteria) shows myocardial edema (T2-weighted imaging), late gadolinium enhancement (typically epicardial/mid-wall in non-ischemic distribution), and increased myocardial T1/T2 mapping values.

Myocarditis Treatment

- No specific antiviral therapy for most cases. Treatment is supportive: heart failure therapy (diuretics, ACE inhibitors, beta-blockers) for reduced EF, arrhythmia management, and avoidance of exercise during acute phase (risk of sudden death). Immunosuppression may be considered in autoimmune myocarditis or cardiac sarcoidosis.

Myocardium

- The thick muscular middle layer of the heart wall composed of cardiac muscle cells (cardiomyocytes). The myocardium is responsible for the contractile function of the heart. It contains intercalated discs with gap junctions that allow electrical coupling between cells.

Myopia

- Myopia, commonly known as nearsightedness, is a refractive error in which parallel light rays from distant objects are focused anterior to the retina rather than directly on it, resulting in blurred distance vision while near vision remains relatively clear.

Myositis

- Inflammation of skeletal muscle. Includes polymyositis, dermatomyositis, inclusion body myositis, immune-mediated necrotizing myopathy. Diagnosis: CK elevation, EMG, muscle MRI, muscle biopsy. Treatment: immunosuppression with corticosteroids, methotrexate, azathioprine, mycophenolate, IVIG, rituximab.

Myotonia Congenita

- An inherited skeletal-muscle channelopathy causing delayed relaxation after voluntary contraction. Stiffness often improves with repeated movement and may be accompanied by muscle hypertrophy; onset and severity vary between genetic forms.

Myotonic Dystrophy

- The most common adult-onset muscular dystrophy, an autosomal dominant multisystem disorder caused by CTG trinucleotide repeat expansions in the DMPK gene (type 1, DM1) or CCTG tetranucleotide repeat expansions in the CNBP gene (type 2, DM2) on chromosome 19q13.3. DM1 is more common and more severe than DM2.

Myringotomy with Tympanostomy Tube

- Incision of the tympanic membrane with placement of a ventilation tube to equalize middle-ear pressure and permit drainage. It is used selectively for persistent middle-ear effusion, recurrent acute otitis media, or associated hearing and developmental problems.

Myxedema Coma

- Severe decompensated hypothyroidism characterized by altered mental status, hypothermia, bradycardia, hypoventilation, hyponatremia, and hypoglycemia. Most commonly occurs in elderly patients with long-standing untreated hypothyroidism precipitated by infection, surgery, or sedative medications. Mortality rates exceed fifty percent.

Myxoma

- A usually benign primary cardiac tumour composed of myxoid tissue, most often arising in the left atrium from the interatrial septum. It may cause positional obstruction of the mitral valve, constitutional symptoms, or systemic embolism; echocardiography is the usual initial diagnostic test.

Ménétrier's disease

- A rare acquired gastropathy characterized by massive hypertrophy of the gastric mucosal folds, particularly in the body and fundus, with foveolar hyperplasia, cystic dilation of gastric glands, and glandular atrophy. It presents with epigastric pain, nausea, vomiting, weight loss, and protein-losing enteropathy causing hypoalbuminemia.

Malaise

- A general feeling of discomfort, weakness, or being unwell. It is a nonspecific symptom that may accompany infection, inflammation, anemia, chronic disease, medication effects, or emotional stress.

Malar Rash

- A red or violaceous rash across the cheeks and

bridge of the nose, often sparing the folds beside the nose. It is classically associated with systemic lupus erythematosus but can occur with other conditions; appearance alone does not establish a diagnosis.

Male Hypogonadism

- Reduced production of testosterone, impaired sperm production, or both due to disease of the testes or the hypothalamic-pituitary system. Possible features include low libido, erectile difficulty, infertility, reduced muscle mass, fatigue, and loss of bone density.

Meningoencephalitis

- Inflammation of both the meninges surrounding the brain and the brain tissue itself, usually caused by infection or autoimmune disease. Fever, headache, neck stiffness, confusion, seizures, or reduced consciousness requires urgent medical assessment.

Mitochondrial Disease

- A group of genetic disorders caused by impaired mitochondrial energy production. Symptoms vary and may affect the brain, muscles, nerves, heart, eyes, or multiple organs; onset and severity can differ widely even within one family.

Muscle Atrophy

- Loss or wasting of muscle tissue caused by disuse, nerve injury, inflammation, malnutrition, or systemic disease. It may lead to weakness and reduced function; treatment focuses on the cause, nutrition, and appropriately supervised rehabilitation.

Musculoskeletal Pain

- Pain arising from muscles, bones, joints, ligaments, tendons, or related soft tissues. Common causes include injury, overuse, arthritis, and inflammation; persistent pain, weakness, swelling, or neurologic symptoms should be assessed.

Meningocele

- A congenital defect in which the membranes surrounding the spinal cord or brain protrude through an opening in the skull or spine, forming a fluid-filled sac. It may occur with other neural-tube defects and can cause neurologic problems; evaluation commonly includes imaging and specialist assessment.

Menstrual Cramps

- Cramping pain in the lower abdomen during or around menstruation, medically called dysmenor-

rhea. Pain may radiate to the back or thighs and may occur with nausea, diarrhea, headache, or fatigue. Severe, worsening, or newly developed pain requires evaluation for causes such as endometriosis or fibroids.

Motion Sickness

- Nausea, dizziness, sweating, or vomiting caused by sensory mismatch during travel or other motion. It can occur in cars, boats, aircraft, amusement rides, or virtual-reality environments and usually improves when motion stops. Severe or persistent symptoms require assessment for other causes.

Mast Cell Activation Syndrome

- A condition involving recurrent symptoms from inappropriate release of mast-cell mediators, such as flushing, itching, hives, abdominal symptoms, wheezing, or low blood pressure. Diagnosis requires compatible episodes, objective evidence of mediator release, and response to appropriate treatment; many similar conditions must be excluded.

Chapter 14

Pocket Dictionary: N Terms

N-Acetylcysteine

- An antidote that replenishes glutathione and enhances detoxification of the toxic metabolite of acetaminophen. It is most effective when started early but may benefit patients with delayed presentation or established liver injury under specialist guidance.

Na⁺/K⁺ ATPase

- The sodium-potassium pump is a P-type ATPase found in the plasma membrane of all animal cells, directly consuming approximately 20-30 percent of cellular ATP. It pumps 3 Na⁺ ions out of the cell and 2 K⁺ ions in per ATP hydrolyzed, establishing steep electrochemical gradients: high Na⁺ outside, high K⁺ inside.

Nabothian Cyst

- A benign mucus-retention cyst of the uterine cervix caused by obstruction of an endocervical gland. It is common, usually asymptomatic, and generally requires no treatment.

NAD⁺

- See biochemistry.

NAD⁺ (Nicotinamide Adenine Dinucleotide)

- Coenzyme involved in redox reactions, existing in oxidized (NAD⁺) and reduced (NADH) forms. Accepts electrons and hydrogen (NAD⁺ + 2e⁻ + H⁺ → NADH). Key roles: glycolysis (glyceraldehyde-3-phosphate dehydrogenase), TCA cycle (isocitrate, alpha-ketoglutarate, malate dehydrogenases), beta-oxidation.

Nadir

- The lowest point, particularly referring to the lowest white blood cell count after chemotherapy. Bone marrow suppression typically reaches nadir 7-14 days post-treatment. Important for infection risk

assessment.

Naegle Rule

- A formula for estimating the due date (estimated date of delivery, EDD): add 7 days to the first day of the last menstrual period (LMP), then add 9 months (or subtract 3 months). Example: LMP January 1 yields EDD October 8. Assumes a 28-day cycle with ovulation on day 14. The rule is most accurate with regular cycles and confirmed LMP.

Naegleria Fowleri

- A free-living amoeba that can cause primary amoebic meningoencephalitis after contaminated water enters the nose. The infection is rare but rapidly progressive, with fever, severe headache, meningism, altered consciousness, seizures, and high mortality.

Nail

- A protective plate of hardened keratin covering the dorsal surface of the distal phalanges of the fingers and toes.

Nail Changes

- Abnormalities of the nails indicating local or systemic disease: koilonychia (spooning—iron deficiency), clubbing (chronic hypoxia—lung cancer, bronchiectasis, cyanotic heart disease, IBD), onycholysis (separation—psoriasis, thyroid, trauma), Beau lines (transverse grooves—acute illness, chemotherapy), Terry nails (white with distal

Nail Clubbing

- Abnormal enlargement of the distal digit with increased nail curvature and loss of the normal nail-fold angle. It may be familial or associated with pulmonary, cardiovascular, gastrointestinal, or malignant disease and should be interpreted with the clinical context.

Nail Pits

- Small, sharply circumscribed depressions in the nail plate caused by abnormal focal keratinization of the proximal nail matrix. They are classically associated with psoriasis but may also occur in alopecia areata, eczema, lichen planus, or other nail disorders.

Nail Psoriasis

- Nail psoriasis affects 50 percent of patients with psoriasis and 80 percent with nail involvement having joint disease. Features include pitting (most common), onycholysis (nail separation), oil drop sign (salmon patch), subungual hyperkeratosis, and nail plate crumbling. Nail psoriasis is associated with psoriatic arthritis.

Naloxone

- A competitive opioid-receptor antagonist used to reverse opioid-induced respiratory depression and central nervous-system depression. Repeated dosing or continuous infusion may be required because naloxone can have a shorter duration than the opioid involved.

Nanocolloid

- Tc-99m labeled colloidal particles for lymphatic imaging. Particle size 10-100 nm. Used for sentinel node detection.

Narcissistic Personality Disorder

- A pattern of grandiosity, need for admiration, and lack of empathy.

Narcolepsy

- A chronic sleep disorder characterized by excessive daytime sleepiness and irresistible sleep attacks, with or without cataplexy (sudden bilateral loss of muscle tone triggered by strong emotions). Narcolepsy type 1: with cataplexy, caused by loss of hypocretin/orexin neurons in the hypothalamus.

Narrow angle glaucoma

- Anatomical narrowing of the anterior chamber angle due to relative pupillary block or a shallow anterior chamber, creating risk for angle closure and acute intraocular pressure elevation. Identified on gonioscopy, it may be managed with laser peripheral iridotomy to equalise pressure between the posterior and anterior chambers.

Narrow Therapeutic Index

- A drug characteristic where there is a small difference between the therapeutic dose and the toxic dose, requiring careful dosing and therapeutic drug monitoring to ensure efficacy while avoiding toxicity. Narrow therapeutic index drugs have a therapeutic index $<2-3$ (the ratio of toxic dose to effective dose).

Narrow-Complex Tachycardia

- See cardiology.

Narrowband UVB Phototherapy

- Controlled exposure to ultraviolet B light centered near 311–313 nm for inflammatory or pigmentary skin disease. It is used for psoriasis, vitiligo, eczema, and other selected conditions with eye protection, cumulative-dose monitoring, and skin-cancer risk assessment.

Nasal Cannula

- Dual-pronged oxygen delivery. Low flow: 1-6 L/min.

Nasal Cavity

- The nasal cavity is the paired, air-filled space located between the nasal septum medially and the lateral nasal wall, extending from the nares (nostrils) anteriorly to the choanae (posterior openings into the nasopharynx) posteriorly, and it serves three primary functions: warming, humidifying, and filtering inspired air; olfaction (smell)

Nasal Congestion

- A sensation of nasal stuffiness or obstruction from engorgement of the venous sinusoids within the nasal mucosa, especially the inferior turbinates; a symptom shared by many conditions.

Nasal Endoscopy

- Direct endoscopic inspection of the nasal passages, turbinates, septum, sinus drainage pathways, and nasopharynx. It is useful for chronic rhinosinusitis, unilateral obstruction, epistaxis, polyps, foreign bodies, and suspected neoplasia.

Nasal Flaring

- Widening and outward movement of the nostrils during breathing. It is a visible sign of increased work of breathing, particularly in infants and children, and may accompany airway obstruction, bronchiolitis, pneumonia, asthma, or severe systemic illness.

Nasal Implant

- Support for nasal septum/airway. See nasal stent.

Nasal Polyps

- Benign, pedunculated, edematous outgrowths of the sinus and nasal mucosa that arise most commonly from the ethmoidal sinuses and protrude into the nasal cavity through the middle meatus, causing nasal obstruction, anosmia (loss of smell), and rhinorrhea.

Nasal Septal Hematoma

- A collection of blood between the nasal septal cartilage and its perichondrium, usually after nasal trauma. It causes nasal obstruction and a soft fluctuant septal swelling and requires urgent drainage to prevent cartilage necrosis and saddle-nose deformity.

Nasal Septum

- The nasal septum is the midline structure that divides the nasal cavity into two halves, consisting of an anterior cartilaginous portion (the septal cartilage or quadrilateral cartilage), a posterior bony portion (the perpendicular plate of the ethmoid bone and the vomer), and the membranous portion (the columella) at the anterior nasal o

Nasal Stent

- Support for nasal septum/airway. Materials: silicone, metal. Indications: septal hematoma, post-septoplasty support, nasal valve collapse. Complications: crusting, infection, pressure necrosis.

Nasogastric Tube

- Tube through nose to stomach. For feeding, decompression, medication. Sizes: 12-18 Fr.

Nasopharyngeal Airway

- A soft tube inserted through the nose to maintain a passage between the nares and pharynx. It may be useful in a spontaneously breathing patient with partial upper-airway obstruction but is relatively contraindicated in suspected basal skull fracture or significant nasal trauma.

Nasopharyngeal Carcinoma

- A malignancy arising from the epithelial lining of the nasopharynx, and it has a unique geographical and ethnic distribution, with highest incidence in southern China, Southeast Asia, and North Africa.

Natal Teeth

- Teeth present at birth, most often involving the lower central incisors. They may be loose, interfere with feeding, or injure the infant's tongue and require dental assessment.

Natural Killer (NK) Cell

- A cytotoxic innate lymphoid cell that kills virus-infected and tumor cells without prior sensitization or MHC restriction. Recognition: activating receptors (NKG2D) and inhibitory receptors (KIR, NKG2A) for MHC class I—'missing self' (reduced MHC) triggers killing.

Natural Killer Cell

- A type of innate lymphoid cell (lymphocyte) that provides early defense against virus-infected cells and tumor cells without prior sensitization. NK cells recognize stressed or missing-self cells (absent or altered MHC class I) and kill them by releasing perforin and granzymes, or through antibody-dependent cellular cytotoxicity.

Nausea

- An unpleasant sensation of an impending need to vomit, often accompanied by autonomic symptoms such as salivation, sweating, pallor, or dizziness. Common causes include gastrointestinal illness, pregnancy, medication effects, migraine, vestibular disorders, metabolic disease, and serious abdominal or neurologic conditions.

Navicular

- Boat-shaped tarsal bone between talus and three cuneiforms. Navicular tuberosity is a palpable medial prominence serving as posterior tibial tendon insertion, key medial longitudinal arch stabilizer. Central third has a precarious blood supply, susceptible to stress fractures and avascular necrosis (Kohler disease in children).

Near Miss

- A patient-safety event that could have caused harm but did not, either by chance or because it was intercepted. Reporting and analyzing near misses can identify system vulnerabilities before harm occurs.

Nearsightedness

- A refractive error, also called myopia, in which distant objects appear blurred because light focuses in front of the retina. It is corrected with diverging

lenses, contact lenses, or refractive surgery.

Nebulizer

- A device that converts liquid medications into a mist for inhalation. Used for bronchodilators (albuterol, ipratropium) and corticosteroids in acute exacerbations. More effective than MDI in patients with severe obstruction.

Neck Cancer

- Human papillomavirus, particularly HPV-16, is an increasingly important etiological factor in head and neck squamous cell carcinoma, particularly oropharyngeal cancer, and it is responsible for the rising incidence of oropharyngeal cancer in younger patients without traditional risk factors of tobacco and alcohol use.

Neck Pain

- Pain or stiffness in the cervical region, commonly caused by muscle strain, degenerative disease, poor posture, or injury. Severe pain after trauma or pain with weakness, numbness, fever, or neurologic deficits requires prompt assessment.

Necrobiosis Lipoidica

- A chronic granulomatous skin disorder strongly associated with Diabetes Mellitus (present in 0.3-1.2% of diabetic patients, though 20% of affected patients do not have diabetes).

Necrobiosis lipoidica diabetorum

- A chronic granulomatous skin disorder most commonly associated with diabetes mellitus, presenting as well-demarcated, atrophic, yellow-brown plaques with telangiectasia, typically on the pretibial area. The plaques have a waxy, glazed appearance with a violaceous border. Ulceration occurs in approximately 35 percent of cases.

Necrolytic migratory erythema

- A recurrent, erosive, erythematous eruption with a migrating, annular or serpiginous pattern, characteristically involving the perineum, groin, and extremities, and almost always associated with an underlying glucagon-secreting pancreatic alpha-cell tumour (glucagonoma).

Necropsy

- See Post-Mortem.

Necroptosis

- A regulated, programmed form of necrosis that

shares features of both classical necrosis and apoptosis, serving as a backup cell death mechanism when apoptosis is inhibited. It is particularly important during viral infections, as many viruses encode proteins that block caspase-mediated apoptosis to ensure their own replication.

Necrosis

- A pathological, unregulated form of cell death resulting from severe, irreversible injury that invariably provokes an intense inflammatory response.

Necrotizing anterior scleritis

- A severe, sight-threatening form of scleral inflammation characterised by ischaemic necrosis of the sclera with visible underlying uvea through thinned, grey-white scleral patches.

Necrotizing Enterocolitis

- An acquired necrotic inflammatory disease of the intestine, primarily affecting premature infants. Risk factors include prematurity, enteral feeding, and bacterial colonization. Presents with abdominal distension, bloody stools, bilious vomiting, and feeding intolerance. Pneumatosis intestinalis on X-ray is pathognomonic.

Necrotizing Fasciitis

- A life-threatening soft tissue infection characterized by rapid necrosis of the subcutaneous fascia and subcutaneous tissue, with relative sparing of the overlying skin in early stages. It is a surgical emergency requiring immediate wide surgical debridement and broad-spectrum antibiotics.

Necrotizing Soft Tissue Infection

- A rapidly progressive infection of fascia, subcutaneous tissue, or muscle causing severe pain, swelling, systemic toxicity, and tissue necrosis. It is a surgical emergency requiring immediate assessment, broad antimicrobial therapy, and aggressive debridement.

Needle

- Hollow metal tube for fluid withdrawal or injection. Gauge: larger number = smaller diameter. Common gauges: 18G (blood draw), 22G (IV), 25-27G (injection).

Needle Driver

- Instrument for passing suture needle through tissue.

Needle Driver (Perineal)

- See Needle Driver.

Needle Driver (Vascular)

- See Needle Driver.

Needle Holder

- A thread or strand of material used to approximate tissues (bring wound edges together) and hold them in place until healing occurs. See suture.

Negative Feedback

- A regulatory mechanism in which the output of a system feeds back to decrease or reverse the initiating stimulus, maintaining homeostasis around a set point.

Negative Predictive Value

- Probability that negative test indicates no disease. NPV = true negatives / all negatives.

Negative Predictive Value (NPV)

- The probability that a person with a negative test result truly does not have the target condition. NPV varies with disease prevalence or pretest probability and is not an intrinsic fixed property of a test.

Negative Pressure Ventilation

- Ventilation using negative pressure applied around the chest wall to expand the lungs, as opposed to positive pressure applied to the airway. The “iron lung” is a historical example. Modern negative pressure ventilators include the cuirass and wrap. They are rarely used today but may be useful in neuromuscular disease.

Negative Symptoms

- Reduced or absent normal emotional, motivational, cognitive, or social functions, including affective flattening, alogia, avolition, anhedonia, and asociality. Persistent negative symptoms can substantially impair functional recovery.

Neglect

- A form of child maltreatment: the failure to provide for a child’s basic needs—physical (food, clothing, shelter, medical care), educational, or emotional—causing harm or risk of harm. Features: failure to thrive, poor hygiene, untreated illness/injury, missed healthcare appointments, inadequate supervision, and developmental delay.

Neglect Syndrome

- Hemispatial neglect (unilateral neglect, hemineglect) is a neurological condition characterized by

failure to attend to, respond to, or orient toward stimuli on the side of space contralateral to the brain lesion, in the absence of primary sensory or motor deficits sufficient to explain the neglect.

Neisseria gonorrhoeae

- A gram-negative diplococcus that causes gonorrhea, a sexually transmitted infection. The organism has developed resistance to many antibiotics including fluoroquinolones and macrolides.

Neisseria meningitidis

- A gram-negative diplococcus that causes meningitis and meningococemia. The organism has a polysaccharide capsule that is the target of conjugate vaccines. Serogroups A, B, C, W, X, and Y cause most disease. Meningitis presents with fever, headache, neck stiffness, and altered mental status.

Nelson Syndrome

- Enlargement and hypersecretion of an adrenocorticotrophic-hormone-producing pituitary tumour after bilateral adrenalectomy, usually for Cushing disease. It may cause rapid skin hyperpigmentation, headache, visual-field loss, and other effects of an expanding sellar mass.

Neoadjuvant Therapy

- Treatment before primary treatment to shrink tumor. Examples: chemotherapy before surgery.

Neonatal Abstinence Syndrome

- A withdrawal syndrome in neonates following in utero exposure to opioids (most common) or other substances (benzodiazepines, SSRIs, alcohol). Onset within 24-72 hours of birth (varies by half-life of substance).

Neonatal Cephalohematoma

- A subperiosteal hematoma of the skull, occurring in approximately 1-2% of births, usually from forceps or vacuum delivery. Presents as a boggy, fluctuant swelling over the skull that does not cross suture lines (differentiating from caput succedaneum). Resolves spontaneously over weeks to months.

Neonatal Conjunctivitis

- Inflammation of the conjunctiva during the first month of life, caused by chemical irritation, bacteria, or viruses acquired during delivery. Purulent discharge, eyelid swelling, and corneal involvement require urgent evaluation.

Neonatal Diabetes Mellitus

- Diabetes diagnosed within the first six months of life. Can be transient or permanent. Transient neonatal diabetes mellitus is caused by abnormalities at chromosome 6q24 and usually resolves but may relapse in adolescence.

Neonatal Herpes Simplex

- Herpes simplex virus infection in the first 28 days of life, usually HSV-2 from maternal genital infection. Three forms: skin, eye, and mouth disease; CNS disease; and disseminated disease. Disseminated form mimics sepsis with high mortality. Diagnosis by viral PCR, culture, or Tzanck smear. Treatment: IV acyclovir for 14-21 days.

Neonatal herpes simplex virus

- A severe neonatal infection caused by herpes simplex virus, typically acquired during passage through the birth canal, presenting within the first month of life as skin, eye, and mouth disease, central nervous system disease, or disseminated disease with multiorgan involvement.

Neonatal hyperbilirubinemia

- Elevated unconjugated bilirubin levels in newborns exceeding 5 mg per deciliter, potentially causing jaundice and, if severe, kernicterus with neurological damage. Phototherapy is the standard treatment.

Neonatal Hypoglycemia

- Blood glucose less than 40-45 mg/dL in term infants or less than 25 mg/dL in preterm infants. Risk factors include prematurity, small for gestational age, large for gestational age, and maternal diabetes. Presents with jitteriness, lethargy, poor feeding, and seizures. Treatment: early feeding, IV dextrose infusion, and monitoring.

Neonatal Hypothermia

- Core body temperature below 36.5 degrees Celsius in a newborn. Classified as mild (36-36.4), moderate (32-35.9), or severe (below 32). Occurs due to rapid heat loss through evaporation, convection, conduction, and radiation. Prevention: skin-to-skin contact, drying, warm environment, and delayed bathing.

Neonatal Hypoxic-Ischemic Encephalopathy (HIE)

- Brain dysfunction in a newborn caused by in-

adequate oxygenation and/or cerebral blood flow around the time of birth. Moderate or severe HIE may produce altered consciousness, abnormal tone, seizures, and multiorgan injury; selected term or near-term infants may benefit from early therapeutic hypothermia.

Neonatal Infections

- Early-onset sepsis (EOS, <72h): GBS, E. coli, Listeria. Risk factors: maternal GBS+, chorioamnionitis, PROM >18h. Kaiser sepsis calculator for management. Late-onset sepsis (LOS, >72h): CoNS, S. aureus, gram-negatives, Candida. NEC: prematurity, formula feeding, ischemic gut. Pneumatosis intestinalis, portal venous gas.

Neonatal Jaundice

- See Neonatal Jaundice (Jaundice of the Newborn).

Neonatal Jaundice (Jaundice of the Newborn)

- Yellowish discolouration of the skin and sclerae in newborn infants due to elevated unconjugated bilirubin. Occurs in 60% of term infants and 80% of preterm infants. Physiological jaundice: appears after 24 hours of life, peaks at day 3-5 in term infants (day 5-7 in preterm), resolves by 2 weeks.

Neonatal Resuscitation

- The systematic approach to stabilizing a newborn after delivery, guided by the Neonatal Resuscitation Program (NRP).

Neonatal Sepsis

- Systemic bacterial infection in the first 28 days of life, classified as early-onset (within 72 hours, usually from vertical transmission) or late-onset (after 72 hours, often nosocomial). Common organisms: Group B Streptococcus, E. coli, Listeria.

Neonatal Skin Conditions

- Common and usually benign. Erythema toxicum neonatorum (red papules and pustules on erythematous base, appears day 2 to 3, resolves by 2 weeks). Milia (tiny white cysts on nose and cheeks). Miliaria crystallina (clear vesicles from sweat duct obstruction). Sebaceous hyperplasia (yellow papules on nose and cheeks).

Neonatal thyrotoxicosis

- Transient hyperthyroidism in newborns caused by transplacental transfer of maternal thyroid-

stimulating immunoglobulins, presenting with tachycardia, irritability, and poor weight gain in the first weeks of life.

Neonatal Transition

- The physiologic adaptation from placental gas exchange to pulmonary respiration after birth. It includes lung aeration, falling pulmonary vascular resistance, closure of fetal circulatory shunts, and changes in cardiac output and thermoregulation.

Neonatologist

- A pediatrician who specializes in the medical care of newborn infants, particularly premature and critically ill neonates.

Neoplasm

- Abnormal growth of tissue, may be benign or malignant. Benign neoplasms: localized, slow-growing, do not metastasize. Malignant neoplasms (cancer): invasive, metastasize, can be fatal. Classification: carcinoma (epithelial origin), sarcoma (mesenchymal origin), leukemia (hematopoietic), lymphoma (lymphoid).

Nephrectomy

- Surgical removal of a kidney. Radical nephrectomy: kidney + Gerota fascia + adrenal gland (for renal cell carcinoma). Partial nephrectomy (nephron-sparing): for small renal tumors (<7 cm, renal function preservation). Simple nephrectomy: for benign disease (non-functioning kidney, chronic infection, massive hydronephrosis).

Nephritic Syndrome

- Characterized by hematuria with dysmorphic RBCs and RBC casts, oliguria, hypertension, and mild proteinuria less than 3.5 g/day. Results from glomerular inflammation increasing permeability to cells and reducing GFR. Hypertension from sodium and water retention.

Nephroblastoma

- Wilms tumor, a malignant embryonal renal tumor of childhood that commonly presents as an asymptomatic abdominal mass.

Nephrocalcinosis

- Deposition of calcium salts within the renal parenchyma, often caused by hypercalcemia, hypercalciuria, distal renal tubular acidosis, prematurity, or inherited disorders. It may impair renal con-

centrating ability and progress to chronic kidney disease.

Nephrogenic Diabetes Insipidus

- Inability of the renal collecting ducts to respond adequately to antidiuretic hormone, causing excessive dilute urine production and intense thirst. Causes include inherited defects, chronic kidney disease, electrolyte disorders, and medication exposure.

Nephrogenic fibrosing dermopathy

- A rare fibrosing disorder of the skin causing thickening and hardening, primarily of the extremities and trunk, associated with kidney failure and gadolinium-based contrast exposure.

Nephrogenic Systemic Fibrosis

- See gadolinium-induced nephrogenic systemic fibrosis.

Nephrolithiasis

- Stone formation in the urinary tract. Calcium oxalate most common at 80%, followed by struvite, uric acid, and cystine. Risk factors include dehydration, hypercalciuria, hyperoxaluria, hypocitraturia, and low urine volume. Evaluation includes 24-hour urine analysis.

Nephron

- Functional unit of the kidney comprising the glomerulus, proximal tubule, loop of Henle, distal tubule, and collecting duct. Each kidney contains approximately one million nephrons responsible for filtering blood, reabsorbing solutes and water, and excreting waste to form urine.

Nephron Function

- The nephron is the functional unit of the kidney, with approximately 1 million per kidney. Each nephron consists of a renal corpuscle (glomerulus and Bowman's capsule) and a renal tubule (proximal convoluted tubule, loop of Henle, distal convoluted tubule, and collecting duct).

Nephronophthisis

- An autosomal-recessive tubulointerstitial kidney disease causing progressive loss of concentrating ability, polyuria, polydipsia, anemia, and chronic kidney failure. It may be isolated or part of a ciliopathy with retinal, skeletal, hepatic, or neurological manifestations.

Nephropathy

- See nephrology.

Nephrostomy

- An artificial opening created between the kidney and the skin to drain urine directly when the normal pathway is blocked. See urology.

Nephrostomy Tube

- A percutaneously placed catheter in the renal pelvis to drain urine, bypassing a ureteral obstruction. Indications: ureteral obstruction (calculus, tumor, stricture) causing hydronephrosis, pyonephrosis (infected, obstructed kidney — emergency drainage), and as a temporary measure before definitive surgery.

Nephrotic Range Proteinuria

- Protein excretion greater than 3.5 g/day or urine protein-to-creatinine ratio greater than 3.5 g/g. Reflects severe glomerular damage with significant protein loss. Degree correlates with severity of injury and risk of complications including thromboembolism, infection, and progression to ESRD.

Nephrotic Syndrome

- Clinical constellation of heavy proteinuria greater than 3.5 g/day, hypoalbuminemia less than 3.0 g/dL, generalized edema, hyperlipidemia, and lipiduria. Results from increased glomerular permeability to albumin. Reduced oncotic pressure causes fluid extravasation. Hepatic lipoprotein synthesis increases causing hyperlipidemia.

Nephrotoxicity

- Kidney damage from toxins, drugs, contrast. Aminoglycosides, cisplatin, contrast media, ethylene glycol. Monitoring: creatinine, urine output. Prevention: hydration, dose adjustment.

Nerve

- A bundle of axons (nerve fibers) surrounded by connective tissue sheaths that transmits electrical impulses between the central nervous system and the periphery. Structure: epineurium (outer layer), perineurium (surrounds fascicles), endoneurium (surrounds individual axons).

Nerve Conduction Studies

- Measurement of the speed and amplitude of electrical impulses along peripheral nerves. Sensory NCS: stimulate at one point, record distally. Motor NCS:

stimulate at two or more points along the nerve, record from muscle. Parameters: conduction velocity (m/s), amplitude (microvolts/millivolts), and distal latency.

Nerve Conduction Study

- A diagnostic test that measures the speed and strength of electrical signals traveling through a nerve to detect nerve damage or dysfunction.

Nervous Tissue

- The tissue of the nervous system, composed of neurons and supporting cells, that detects stimuli and transmits electrical signals.

Nettle Rash

- See Urticaria (Hives / Nettle Rash).

Neural Tube Defects

- Congenital malformations resulting from failure of the neural tube to close during the third to fourth week of gestation. Include anencephaly (incomplete cranial closure), encephalocele (herniation through skull defect), and spina bifida (incomplete spinal closure).

Neuralgia

- Pain caused by irritation or damage to a peripheral or cranial nerve, typically described as sharp, burning, stabbing, or electric-shock-like and occurring along the nerve's distribution. It may be triggered by light touch or movement and can result from compression, trauma, infection, metabolic disease, or demyelination.

Neuraminidase Inhibitors

- A class of antiviral drugs that inhibit the neuraminidase enzyme on the surface of influenza viruses, preventing the release of newly formed viral particles from infected cells and thereby limiting viral spread. The two available agents are oseltamivir, administered orally, and zanamivir, administered by inhalation.

Neuroblastoma

- Childhood sympathetic nervous system tumor. MYCN amplification poor prognosis. Presents as abdominal mass, opsoclonus-myoclonus syndrome. Spontaneous regression possible in infants.

Neurocognitive Disorder

- Chronic progressive cognitive decline interfering with daily function. See dementia.

Neurocysticercosis

- The most common parasitic infection of the central nervous system worldwide and is caused by the larval stage (cysticercus) of the pork tapeworm *Taenia solium*. It is endemic in Latin America, Asia, and Africa, and is an important cause of epilepsy in endemic regions.

Neurodegeneration

- The progressive loss of structure and function of neurons, culminating in their death, and represents the underlying mechanism of many devastating neurological disorders.

Neuroendocrine Tumor

- Tumor from neuroendocrine cells. CT, PET diagnosis. Octreotide scan useful.

Neuroendocrine Tumors

- Heterogeneous group of neoplasms arising from neuroendocrine cells throughout the gastrointestinal tract and pancreas. May be functional producing hormones or non-functional. Graded by Ki-67 index and mitotic count into well-differentiated low-grade, intermediate-grade, and poorly differentiated high-grade tumors.

Neuroendocrine Tumors Grading

- Classification of neuroendocrine tumors by proliferative rate using Ki-67 index and mitotic count. Well-differentiated low-grade tumors have Ki-67 less than three percent and fewer than two mitoses per ten high-power fields. Intermediate-grade tumors have Ki-67 of three to twenty percent.

Neurofibroma

- A benign peripheral nerve sheath tumor. Solitary neurofibromas are common and present as soft, flesh-colored papules or nodules. Multiple neurofibromas are characteristic of neurofibromatosis type 1 (NF1).

Neurofibromatosis

- A group of autosomal dominant genetic disorders causing tumor growth along nerves. Type 1 (NF1, chromosome 17) causes café-au-lait spots, neurofibromas, optic gliomas, and learning disabilities. Type 2 (NF2, chromosome 22) causes bilateral vestibular schwannomas, meningiomas, and ependymomas.

Neurofibromatosis Type 1

- Autosomal dominant disorder caused by mutations

in the NF1 gene encoding neurofibromin on chromosome seventeen.

Neurofibromatosis Type 2

- An autosomal dominant disorder caused by NF2 gene mutation on chromosome 22. Characterized by bilateral vestibular schwannomas (acoustic neuromas), meningiomas, and ependymomas. Features: hearing loss, tinnitus, and balance problems. Diagnosis: MRI showing bilateral vestibular schwannomas.

Neurogenic Bladder

- Lower urinary-tract dysfunction caused by a neurologic lesion affecting central or peripheral control of storage and voiding. It may produce urgency, retention, incontinence, high bladder pressures, recurrent infection, and upper-tract damage.

Neurogenic megacolon

- Massive dilation of the colon due to loss of autonomic innervation, most commonly from Chagas disease, Hirschsprung disease, or diabetic autonomic neuropathy. It presents with chronic constipation, abdominal distension, fecal impaction, and risk of toxic megacolon.

Neurogenic Shock

- Distributive shock caused by loss of sympathetic vascular tone, usually after acute spinal-cord injury above T6. It is characterized by hypotension with relative bradycardia and warm skin, distinguishing it from hemorrhagic shock when the clinical findings are complete.

Neurography

- MRI technique visualizing peripheral nerves. Used for nerve tumors, entrapment, trauma.

Neuroleptic

- An older term for antipsychotic medications, derived from their ability to produce a state of mental indifference and motor slowing. The term is now largely replaced by “antipsychotic,” though it is still used in specific contexts such as neuroleptic malignant syndrome. Typical neuroleptics are first-generation antipsychotics.

Neuroleptic Malignant Syndrome

- A rare but life-threatening idiosyncratic reaction to dopamine receptor-blocking agents, typically antipsychotics, characterized by hyperthermia, severe

muscle rigidity, altered mental status, and autonomic dysfunction.

Neurologic Localization

- The process of using the pattern of deficits in strength, sensation, reflexes, coordination, cranial-nerve function, and cognition to identify the anatomic site of nervous-system dysfunction. Localization guides differential diagnosis and selection of investigations.

Neurological Examination

- The systematic clinical assessment of the nervous system: mental status/cognition, cranial nerves (I–XII), motor system (bulk, tone, power, pronator drift), reflexes (deep tendon reflexes, plantar/Babinski), coordination (finger–nose, heel–shin), gait, sensory examination (light touch, pinprick, vibration, proprioception, temperature), and

Neurological Rehabilitation

- Rehabilitation for stroke, spinal cord injury, traumatic brain injury, Parkinson disease, multiple sclerosis. Goals: maximize function, independence, quality of life. Multidisciplinary team: physiatrist, PT, OT, speech therapy, psychology.

Neurology PET

- Dementia evaluation, epilepsy localization, Parkinson plus syndromes.

Neuromuscular Blocking Agents (NMBAs)

- Medications administered intraoperatively to induce skeletal muscle relaxation and facilitate endotracheal intubation and surgical exposure.

Neuromuscular Junction

- The neuromuscular junction is the synapse between motor neuron and skeletal muscle fiber, where acetylcholine released from presynaptic terminals binds to nicotinic acetylcholine receptors on the postsynaptic motor end plate, causing depolarization and muscle contraction.

Neuromyelitis Optica

- Neuromyelitis optica spectrum disorder (NMOSD), formerly known as Devic disease, is an autoimmune, inflammatory demyelinating disorder of the central nervous system that preferentially targets the optic nerves and spinal cord.

Neuron

- The fundamental excitable cell of the nervous system specialized for receiving, integrating, and transmitting electrical and chemical signals. Structural components: cell body (soma), dendrites (receptive), axon (conducting), and synaptic terminals (releasing neurotransmitters).

Neuropathic Pain

- Chronic pain caused by damage or disease affecting the somatosensory nervous system, including the peripheral nerves, dorsal root ganglia, spinal cord, or brain.

Neuropathic Ulcer

- A chronic wound of the foot in a patient with diabetes, resulting from peripheral neuropathy, peripheral artery disease, and impaired wound healing. See diabetic foot ulcer.

Neuropathy

- Disease or dysfunction of peripheral nerves causing pain, numbness, tingling, weakness, or altered reflexes. Common causes include diabetes, alcohol, nutritional deficiency, toxins, compression, autoimmune disease, and inherited disorders.

Neuropathy (Peripheral Neuropathy)

- Damage to peripheral nerves causing sensory, motor, or autonomic dysfunction. Common causes: diabetes mellitus, alcohol use disorder, vitamin B12 deficiency, chemotherapy, autoimmune diseases, and hereditary conditions (Charcot-Marie-Tooth disease). Presents with numbness, tingling, burning pain, weakness, and gait instability.

Neuroplasticity

- The brain's ability to reorganize its structure, function, and connections in response to experience, learning, and injury. It is relevant to psychiatric treatment because antidepressants, psychotherapy, and ECT are thought to promote neuroplastic changes including neurogenesis in the hippocampus and synaptic remodeling.

Neurosarcoidosis

- Neurological involvement by sarcoidosis, caused by non-caseating granulomatous inflammation of the brain, meninges, cranial nerves, spinal cord, peripheral nerves, or muscles. Presentations include cranial neuropathies, aseptic meningitis, seizures,

hypothalamic or pituitary dysfunction, myelopathy, and peripheral neuropathy.

Neurosyphilis

- Infection of the central or peripheral nervous system by *Treponema pallidum*. It may occur at any stage of syphilis and cause meningitis, stroke, cranial-nerve dysfunction, sensory ataxia, cognitive decline, or tabes dorsalis.

Neurotoxin

- Substance toxic to nervous system. Examples: lead, mercury, organophosphates, botulinum toxin, tetrodotoxin, snake venoms.

Neurotransmitter

- Neurotransmitters are endogenous chemical messengers that transmit signals across synapses from one neuron to another or to target cells such as muscle or gland cells.

Neurovascular Examination

- Assessment of distal motor function, sensation, pulses, capillary refill, skin temperature, and color in an injured limb. Serial examination is essential for detecting nerve injury, vascular compromise, compartment syndrome, and evolving ischemia.

Neutropenia

- A reduced neutrophil count ($< 1.5 \times 10^9/L$; severe < 0.5), increasing risk of bacterial and fungal infection, especially febrile neutropenia.

Neutrophil Extracellular Traps

- NETs: web-like structures of chromatin + granular proteins (MPO, NE, histone) released by neutrophils (NETosis: suicidal vs vital).

Neutrophil Recruitment

- A multistep process essential for acute inflammation and host defense, involving sequential adhesive interactions between neutrophils and endothelial cells. The process begins with the rolling phase, where neutrophils loosely attach to endothelial cells through selectin-mediated interactions.

Neutrophilic dermatosis

- A group of inflammatory skin conditions characterised by dense neutrophilic infiltrate in the dermis without evidence of infection, which includes Sweet syndrome (acute febrile neutrophilic dermatosis), pyoderma gangrenosum, and Behçet disease.

Neutrophilic eccrine hidradenitis

- A neutrophilic dermatosis characterized by neutrophilic infiltration of the eccrine glands and surrounding dermis, typically occurring in patients receiving chemotherapy, particularly cytarabine. It presents with painful, erythematous plaques or nodules, most commonly on the trunk and extremities.

Nevus

- A benign hamartoma of the skin; most commonly melanocytic nevus (mole). Other nevi: congenital, dysplastic/atypical, Spitz, blue nevus, halo nevus; non-melanocytic: epidermal, sebaceous, vascular (strawberry hemangioma, port-wine stain).

Nevus comedonicus

- A congenital hamartoma of the pilosebaceous unit presenting as a localised cluster of dilated follicular ostia filled with keratinous plugs, resembling closely grouped open comedones. It typically appears at birth or in early childhood on the face, neck, or trunk, and may be associated with other developmental anomalies.

Newborn Circumcision

- Surgical removal of the foreskin from the penis, typically performed within the first few days of life. Medical benefits include reduced risk of urinary tract infections, sexually transmitted infections, and penile cancer. Complications include bleeding, infection, and meatal stenosis.

Newborn Hearing Screen

- Universal screening for congenital hearing loss, performed before hospital discharge. Two-step protocol: otoacoustic emissions (OAE: measures cochlear outer hair cell response to click stimulus) and automated auditory brainstem response (AABR: measures neural response from the cochlea to the brainstem). Pass/refer.

Newborn Metabolic Screen

- A dried blood spot test (heel stick) collected 24-48 hours after birth to screen for congenital disorders before symptoms develop.

Newborn Screening

- A public health program testing newborns for treatable genetic, metabolic, and endocrine disorders 24-48 hours after birth. Blood spot (heel stick) on filter paper (Guthrie card). Conditions vary by

state (Recommended Uniform Screening Panel: 35 core conditions).

Next-Generation Sequencing

- Massively parallel sequencing technology that analyzes many DNA molecules simultaneously. It supports gene panels, whole-exome sequencing, whole-genome sequencing, and tumor sequencing, with limitations involving coverage, structural variants, mosaicism, and interpretation.

Niacin (Vitamin B3)

- A water-soluble vitamin that forms NAD⁺ and NADP⁺, essential electron carriers in oxidation-reduction reactions and biosynthetic pathways. Severe deficiency causes pellagra, characterized by dermatitis, diarrhea, and dementia.

Nickel-Chromium

- Base metal alloy for crowns. Less expensive than noble metals. Nickel allergy concern.

Nicotine

- See tobacco.

Niemann-Pick Disease

- A group of autosomal recessive lysosomal storage disorders caused by sphingomyelinase deficiency (ASM, types A and B, SMPD1 mutations) or NPC1/NPC2 cholesterol transport defects (type C).

Night Blindness

- Impaired vision in dim light (nyctalopia). Causes: vitamin A deficiency (most common cause worldwide, leading to retinol deficiency and rod dysfunction), retinitis pigmentosa (progressive rod-cone degeneration, genetic), congenital stationary night blindness, advanced glaucoma, and cataract.

Night Eating Syndrome

- Delayed circadian pattern of eating. Evening hyperphagia, morning anorexia, insomnia. Associated with obesity, depression.

NIH Stroke Scale

- The National Institutes of Health Stroke Scale (NIHSS) is a standardized, validated clinical assessment tool used to quantify the severity of acute ischemic stroke.

Nikolsky Sign

- A clinical dermatological sign in which light lateral

pressure applied to normal-appearing skin causes epidermal detachment or sloughing. Positive sign indicates intraepidermal loss of cell-cell cohesion (acantholysis).

Nitrates

- Vasodilator drugs that release nitric oxide, causing relaxation of vascular smooth muscle through activation of guanylyl cyclase and increased cyclic GMP. They primarily cause venodilation at low doses, reducing preload and myocardial oxygen demand, and arterial dilation at higher doses, reducing afterload.

Nitrogen Balance

- The difference between dietary nitrogen intake, estimated mainly from protein intake, and nitrogen loss, primarily in urine. Positive balance supports growth or tissue repair; negative balance occurs with inadequate intake, inflammation, trauma, or catabolic disease.

Nitroglycerin

- An organic nitrate that releases nitric oxide, relaxing vascular smooth muscle—predominantly venous. Used for acute angina (sublingual spray/tablet, IV for ACS/heart failure), angina prophylaxis, and preload reduction in acute pulmonary edema. Rapid onset (1-3 min sublingual).

Nitroprusside

- A direct-acting vasodilator (arterial and venous) releasing nitric oxide, used for hypertensive emergencies and controlled hypotension during surgery. IV infusion with rapid onset and offset; reduces both preload and afterload.

Nitrous Oxide

- An inhaled anesthetic gas with analgesic and anesthetic-sparing properties. It has rapid onset and offset but relatively low potency, expands closed gas spaces, and may contribute to diffusion hypoxia or postoperative nausea.

Nitrous Oxide for Labor

- A 50:50 mixture of nitrous oxide and oxygen (Entonox) inhaled by the mother during labor for analgesia. Self-administered via a mask or mouthpiece; the woman controls her own dosing, taking a breath 30 seconds before a contraction.

Nocardia asteroides infection

- See Nocardiosis.

Nocardiosis

- A bacterial infection caused by *Nocardia* species (*N. asteroides*, *N. brasiliensis*), found in soil. Most common in immunocompromised patients (transplant, HIV, chronic steroid use). Pulmonary nocardiosis (most common): cough, fever, necrotizing pneumonia, cavitary lesions.

Nocebo

- Negative placebo effect. Adverse events from negative expectations.

Nociception

- Neural processing of noxious thermal, mechanical, or chemical stimuli that may result in the perception of pain. It includes transduction, transmission, modulation, and projection; nociception and pain are related but not identical concepts.

Nocturia

- Complaint of waking one or more times during the night to void, with each void preceded and followed by sleep. Highly prevalent in older adults and significantly impairs sleep quality and quality of life.

Nodular lymphoid hyperplasia

- Benign aggregates of lymphoid tissue occurring in the gastrointestinal tract, most commonly in the small intestine, presenting as multiple small nodules on endoscopy or imaging. It may be associated with immunodeficiency states, particularly common variable immunodeficiency, and *Giardia lamblia* infection.

Nodule

- Small, palpable mass <2 cm. Can be solid or cystic. Examples: thyroid nodule, rheumatoid nodule, pulmonary nodule. Evaluation depends on location.

Nomogram

- Graphical calculation tool. Rumack-Matthew nomogram for acetaminophen toxicity. Guides NAC treatment.

Non communicating Hydrocephalus

- Non-communicating (obstructive) hydrocephalus is a type of hydrocephalus in which there is obstruction to the flow of cerebrospinal fluid (CSF) within the ventricular system, preventing CSF from reaching the arachnoid granulations for absorption.

Non-Accidental Trauma

- Injuries inflicted on a child, suspected when history and examination findings are inconsistent with the reported mechanism. Red flags: multiple fractures at different stages of healing, posterior rib fractures, metaphyseal corner fractures, and retinal hemorrhages.

Non-Adherence

- Failure to follow treatment. Major cause of treatment failure.

Non-Alcoholic Fatty Liver Disease

- Hepatic steatosis in the absence of significant alcohol consumption, strongly associated with metabolic syndrome, obesity, and type 2 diabetes. NAFLD includes simple steatosis and non-alcoholic steatohepatitis (NASH) with inflammation and fibrosis. NASH can progress to cirrhosis and hepatocellular carcinoma.

Non-Allergic Rhinitis

- Non-allergic rhinitis, also known as vasomotor rhinitis, is a chronic nasal condition characterized by nasal symptoms including congestion, rhinorrhea, and sneezing that are not caused by an IgE-mediated allergic response, and it is distinguished from allergic rhinitis by the absence of allergen-specific IgE, negative allergy testing, and

Non-Asthmatic Eosinophilic Bronchitis

- Chronic cough with airway eosinophilia (induced sputum eosinophils > 3%) but without bronchial hyperresponsiveness. It is a common cause of chronic cough and does not respond to bronchodilators. It responds well to inhaled corticosteroids. It is distinct from asthma in the absence of variable airflow obstruction.

Non-Celiac Gluten Sensitivity

- Symptom improvement with gluten-free diet without celiac disease or wheat allergy. Diagnosis of exclusion.

Non-Communicable Disease

- Disease not transmitted between persons. Examples: cancer, heart disease, diabetes.

Non-Contact Tonometry

- Non-contact tonometry, commonly known as the air puff test, is a method of measuring intraocular pressure that uses a calibrated air pulse to appanate

the cornea, with the force of the air pulse determined by a sensor that detects the moment of appplanation, providing a rapid, non-invasive measurement without requiring direct corneal contact

Non-Distensible Balloon (Vascular)

- See Non-Distensible Balloon.

Non-Functioning Pituitary Adenoma

- Pituitary adenomas that do not secrete clinically significant hormones. The most common pituitary adenomas to present with mass effects.

Non-Hodgkin Lymphoma

- Hematologic malignancy of B or T/NK cells. B-cell NHL (85%) includes diffuse large B-cell lymphoma (DLBCL, most common aggressive), follicular lymphoma (most common indolent), mantle cell lymphoma, marginal zone lymphoma, and Burkitt lymphoma. T-cell NHL: peripheral T-cell lymphoma, anaplastic large cell lymphoma.

Non-Invasive Prenatal Testing

- Screening of cell-free placental DNA in maternal blood for selected fetal chromosomal abnormalities, especially trisomies 21, 18, and 13. It is highly sensitive screening, not a definitive diagnosis; positive results require confirmatory invasive testing.

Non-Invasive Ventilation

- Positive pressure ventilation delivered via a face mask or nasal mask without endotracheal intubation. BiPAP (bilevel positive airway pressure) is most commonly used for acute hypercapnic respiratory failure (COPD exacerbation) and cardiogenic pulmonary edema. It reduces the need for intubation and improves outcomes in selected patients.

Non-Proliferative Diabetic Retinopathy

- The early stage of diabetic retinopathy, characterized by structural abnormalities of the retinal vasculature resulting from chronic hyperglycemia-induced damage to the endothelial cells and pericytes of retinal capillaries, without neovascularization.

Non-Rebreather Mask

- High-flow oxygen delivery. Reservoir bag, one-way valves.

Non-Small Cell Lung Cancer

- Most common lung cancer type. Includes adenocarcinoma, squamous cell, large cell.

Non-Stress Test

- A fetal heart rate monitoring technique assessing fetal heart rate reactivity over 20-40 minutes. Normal (reactive) tracing shows at least 2 fetal heart rate accelerations of 15 beats per minute above baseline lasting at least 15 seconds within a 20-minute period. Non-reactive tracings may indicate fetal hypoxia or sleep cycle.

Non-Stress Test (NST)

- Antepartum fetal monitoring assessing fetal heart rate reactivity. Reactive NST: 2 accelerations of 15 bpm for 15 seconds (32 weeks) or 10 bpm for 10 seconds (<32 weeks) within 20 minutes. Non-reactive: insufficient accelerations after 40 minutes (may indicate fetal sleep, acidosis, or CNS depression). False positive rate high.

Non-Tuberculous Mycobacteria

- NTM: environmental mycobacteria (*M. avium* complex - MAC, *M. kansasii*, *M. abscessus*). Pulmonary NTM: ATS/IDSA criteria (clinical + radiographic + microbiologic: 2+ sputum or 1 BAL + nodules/cavities/bronchiectasis on CT). MAC: nodular bronchiectatic (middle-aged women, "Lady Windermere syndrome") or fibrocavitary (older men, COPD).

Nonalcoholic Fatty Liver Disease

- NAFLD: hepatic steatosis (>5 percent of hepatocytes) in the absence of significant alcohol use, associated with the metabolic syndrome (obesity, type 2 diabetes, dyslipidemia, hypertension).

Nonalcoholic Steatohepatitis

- Progressive form of NAFLD characterized by hepatic steatosis with lobular inflammation and hepatocyte ballooning with significant risk of progression to fibrosis, cirrhosis, and hepatocellular carcinoma. NASH affects approximately three to five percent of the general population.

Noncompetitive Antagonist

- A noncompetitive antagonist is a drug that reduces the effect of an agonist by a mechanism that cannot be overcome by increasing the agonist concentration.

Noncompetitive Inhibition

- A type of enzyme inhibition where the inhibitor binds at a site distinct from the active site (allosteric site), and can bind to both the free enzyme (E)

and the enzyme-substrate complex (ES) with equal affinity. The inhibitor does not compete with the substrate and cannot be overcome by increasing substrate concentration.

Noninvasive Prenatal Testing (NIPT)

- See Cell-Free Fetal DNA (cfDNA) Screening.

Noninvasive Ventilation

- Positive-pressure ventilatory support delivered without an endotracheal tube, commonly by a mask or helmet. CPAP provides continuous positive pressure; bilevel support provides separate inspiratory and expiratory pressures. It is used selectively when the patient can protect the airway and cooperate.

Nonmelanoma Skin Cancer

- A group of skin cancers that arise from the cells of the epidermis, excluding melanoma. The two most common types are basal cell carcinoma (BCC) and squamous cell carcinoma (SCC), which together account for the vast majority of all skin cancers diagnosed globally.

Nonselective Beta-Blockers

- Nonselective beta-adrenergic blocking agents block both beta-1 and beta-2 adrenergic receptors, reducing heart rate, myocardial contractility, and cardiac output through beta-1 blockade, while also blocking beta-2 receptors in the bronchial smooth muscle and peripheral vasculature.

Nonsense Variant

- A sequence variant that creates a premature stop codon, potentially producing a truncated protein or triggering nonsense-mediated mRNA decay. The effect depends on the gene, location, transcript, and residual protein production.

Nonspecific Interstitial Pneumonia

- A pattern with uniform, temporally homogeneous inflammation and/or fibrosis. Most common ILD associated with connective tissue disease. HRCT shows bilateral ground-glass opacities with subpleural sparing.

Nonsteroidal Anti-Inflammatory Drug

- NSAID: drugs inhibiting cyclooxygenase (COX-1/COX-2), reducing prostaglandin synthesis—anti-inflammatory, analgesic, antipyretic. Types: non-selective (ibuprofen, naproxen, diclofenac, indomethacin, ketorolac) and COX-2 selective (cele-

coxib, etoricoxib). Uses: arthritis, musculoskeletal pain, gout, dysmenorrhea, postoperative

Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

- Analgesic and anti-inflammatory drugs that inhibit cyclooxygenase enzymes and prostaglandin synthesis. Perioperative use requires assessment of renal function, gastrointestinal bleeding risk, platelet effects, cardiovascular risk, and surgical bleeding considerations.

Nonunion and Malunion

- Nonunion is failure of a fractured bone to heal after an adequate period (generally 6-9 months for long bones) without clinical or radiographic evidence of progression toward union for 3 months. Classification: hypertrophic (viable ends, inadequate stability), atrophic (devitalized ends, poor biology), oligotrophic (intermediate).

Noonan Syndrome

- An autosomal dominant multisystem disorder caused by mutations in the RAS-MAPK pathway genes (PTPN11 50%, SOS1 10-15%, RAF1 5-10%, RIT1, KRAS).

Norepinephrine

- Sympathetic neurotransmitter. Vasoconstriction, increased heart rate.

Normal Flora

- The resident microorganisms (commensals) colonizing body surfaces—skin, oropharynx, gastrointestinal tract, genitourinary tract—without causing disease under normal conditions. They provide protection (colonization resistance, immune priming), synthesis of vitamins, and nutrient metabolism.

Normal Pressure Hydrocephalus

- See geriatrics.

Normalization

- Correction for detector efficiency variations. Performed daily with uniform source.

Normocalcemic Hyperparathyroidism

- Condition of elevated parathyroid hormone with persistently normal serum calcium and normal vitamin D levels. Exclusion of secondary causes of elevated PTH including vitamin D deficiency, renal insufficiency, and medications is required. Represents an early or mild form of primary hyperparathy-

roidism. May progress to overt hypercalcemia.

Norovirus

- A single-stranded RNA virus that is the leading cause of gastroenteritis outbreaks worldwide. The virus is highly contagious and can cause large outbreaks in closed settings including hospitals, cruise ships, and nursing homes. Norovirus causes acute onset of vomiting and diarrhea. Diagnosis is made by PCR or antigen detection in stool.

Norrie Disease

- An X-linked retinal disorder, usually caused by pathogenic variants in NDP, characterized by congenital or early-onset blindness from abnormal retinal development. Hearing loss, developmental differences, and peripheral vascular or vitreoretinal complications may occur.

Nose

- The external nasal structure and nasal cavities, the primary organ of olfaction and the entry to the respiratory tract. External nose: bone (nasal bones, frontal process of maxilla) and cartilage (septal, upper/lower lateral cartilages).

Nosebleed

- Bleeding from the nasal mucosa, also called epistaxis. Most cases arise from the anterior nasal septum and result from local trauma, dryness, inflammation, or vascular disease; persistent or severe bleeding may require medical intervention.

Nosocomial

- Hospital-acquired. Nosocomial infection: infection acquired in hospital.

Nosocomial Infection

- A hospital-acquired infection developing during or after hospitalization that was not present or incubating at admission, classically defined by onset beyond 48 hours. Common sites are bloodstream, urinary tract, lungs, and surgical wounds; causative organisms are frequently multidrug-resistant, prevented by hand hygiene and stewardship.

Notifiable Disease

- Disease required by law to be reported to health authorities. Examples: tuberculosis, measles, HIV, COVID-19.

NSAIDs

- Nonsteroidal anti-inflammatory drugs are a class of

drugs that inhibit cyclooxygenase enzymes, COX-1 and COX-2, reducing the synthesis of prostaglandins and thromboxanes from arachidonic acid. They produce analgesic, anti-inflammatory, antipyretic, and antiplatelet effects.

NSAIDs (Nonsteroidal Anti-Inflammatory Drugs)

- A class of analgesic, antipyretic, and anti-inflammatory drugs that inhibit cyclooxygenase (COX-1 and/or COX-2), reducing prostaglandin and thromboxane synthesis.

NSTEMI

- Non-ST-elevation myocardial infarction: acute myocardial necrosis (elevated cardiac troponin) without persistent ST-segment elevation on ECG, caused by subtotal occlusion or distal embolization of a coronary artery. Features: chest pain, ECG changes (ST depression, T-wave inversion), elevated troponin.

Nuchal Translucency (NT) Screening

- An ultrasound measurement of the fluid-filled space at the posterior fetal nuchal region, performed at 11-14 weeks (crown-rump length 45-84 mm). Increased NT (>95th percentile or >3.0-3.5 mm) is associated with trisomies 21, 18, 13, Turner syndrome, and cardiac defects.

Nuclear Medicine

- A medical specialty that uses small amounts of radioactive substances (radiotracers) to diagnose, stage, and treat various diseases, including cancer and heart disease.

Nuclear Medicine Documentation

- Record keeping for radioactive materials, including inventory, disposal, administered activity, and patient release.

Nuclear Medicine Reporting

- Structured interpretation of nuclear medicine studies using standardized terminology and clinically relevant findings.

Nuclear Medicine Resolution

- Ability of a PET or SPECT system to distinguish small details, limited by factors such as positron range, non-collinearity, and detector geometry.

Nuclear Sclerosis

- The most common type of age-related cataract, characterized by progressive yellowing and harden-

ing of the lens nucleus due to the accumulation and aggregation of insoluble crystallin proteins, resulting in a gradual decrease in vision that is often noticed as increased difficulty with distance vision and a phenomenon called second-sight v

Nuclear Stress Test

- Myocardial perfusion imaging during stress. Detects coronary artery disease.

Nucleoside Analogs

- A class of antiviral drugs that structurally resemble natural nucleosides and inhibit viral replication by interfering with DNA or RNA synthesis.

Nucleus

- The membrane-bound organelle containing the genetic material (DNA) in eukaryotic cells. Surrounded by the nuclear envelope (double membrane with nuclear pores for transport). Contains chromatin (DNA + histone proteins), nucleolus (ribosome biogenesis), and nuclear matrix.

Number Needed to Harm (NNH)

- The number of people exposed to an intervention for one additional person to experience a specified adverse outcome compared with the control over a defined time. It is the reciprocal of absolute risk increase.

Number Needed to Treat (NNT)

- The number of people who must receive an intervention for one additional person to experience the specified beneficial outcome compared with the control over a defined time. It is the reciprocal of absolute risk reduction.

Numbness and Tingling

- Abnormal sensations, also called paresthesias, that may result from nerve compression, peripheral neuropathy, spinal disease, vitamin deficiency, vascular problems, or neurologic disorders. Sudden one-sided symptoms require urgent evaluation for stroke.

Nummular eczema

- A chronic inflammatory dermatosis presenting as coin-shaped, well-demarcated, erythematous, vesicular, or crusted plaques, typically on the extremities. The lesions are intensely pruritic and may be triggered by dry skin, skin trauma, or allergic contact dermatitis. It is more common in adults with atopic predisposition.

Nursing

- Health of workers. See occupational health.

Nursing Home

- Assistance with ADLs/IADLs for extended period. See long-term care.

Nutcracker esophagus

- Esophageal motility disorder characterized by high-amplitude peristaltic contractions exceeding 180 mmHg in the distal esophagus on manometry, with preserved peristaltic coordination. Presents with intermittent chest pain and dysphagia to solids and liquids.

Nutcracker Syndrome

- Compression of left renal vein between aorta and SMA causing hematuria and flank pain. Managed conservatively. Stenting or surgery for severe cases.

Nutrient Density

- Amount of nutrients per calorie. Whole foods more nutrient-dense than processed foods. Importance for weight management, health.

Nutrition

- Oral role in nutrition. Chewing efficiency affects food selection. Edentulism associated with poor nutrition.

Nutrition (Geriatric)

- Nutritional assessment and intervention in elderly. Risks: malnutrition (undernutrition, obesity), micronutrient deficiencies (vitamin D, B12, calcium), dehydration. Assessment: weight, BMI, dietary intake, lab values. Interventions: supplementation, meal delivery, appetite stimulants, feeding assistance.

Nutrition (Oncology)

- Cancer and cancer treatment affect nutrition. Malnutrition common. Dietitian referral recommended.

Nutrition Education

- Teaching patients about healthy eating. Cultural sensitivity, health literacy appropriate. Motivational interviewing effective.

Nutrition in Pregnancy

- A balanced diet providing adequate calories, protein, vitamins, and minerals to support maternal and fetal health. Additional caloric needs: 300 kcal/day in the second and third trimesters. Key nu-

trients: folic acid (400 mcg/day), iron (27 mg/day), calcium (1000 mg/day), vitamin D (600 IU/day), iodine (220 mcg/day), DHA (200-300 mg/day).

Nutrition Risk Screening 2002

- A validated screening tool for hospitalized adults that considers nutritional status and disease severity to identify patients likely to benefit from nutrition support. Screening is not itself a complete malnutrition diagnosis.

Nutrition-Focused Physical Examination

- A structured examination for physical findings associated with malnutrition, including loss of subcutaneous fat, muscle wasting, edema, ascites, reduced functional status, and micronutrient-related signs. It complements dietary history, anthropometry, laboratory data, and disease assessment.

Nutritional Assessment

- Comprehensive evaluation: dietary intake, anthropometrics, biochemical tests, clinical examination. Dietitian involvement essential.

Nutritional Dermatoses

- Nutritional deficiencies cause distinctive skin findings. Vitamin C deficiency (scurvy): perifollicular hemorrhage, corkscrew hairs, gingival bleeding. Niacin deficiency (pellagra): photosensitive dermatitis in sun-exposed areas, diarrhea, dementia.

Nutritional Screening

- Rapid assessment for malnutrition risk. Tools: MST, MNA, MUST. Screening all elderly patients recommended. Identifies need for comprehensive assessment.

Nyctalopia

- Difficulty seeing in dim light or darkness, commonly called night blindness. Causes include retinal disorders, vitamin-A deficiency, high refractive error, cataract, glaucoma, or medication and systemic disease affecting visual adaptation.

NYHA Functional Classification

- A four-class system grading heart failure severity based on symptoms and exercise tolerance. Class I: no limitation of physical activity. Class II: slight limitation; comfortable at rest but ordinary activity causes symptoms. Class III: marked limitation; comfortable at rest but less than ordinary activity causes symptoms.

Nystagmus

- A rhythmic, involuntary oscillation of the eyes that can be horizontal, vertical, rotational, or mixed in pattern, and it is classified based on the underlying cause as infantile (developing in the first six months of life), acquired (developing later in life), or periodic alternating.

Nail Infection

- Infection of the nail or surrounding tissue, commonly caused by fungi or bacteria. It may cause thickening, discoloration, separation of the nail, pain, swelling, or drainage; diabetes and poor circulation increase the risk of complications.

Nasal Obstruction

- Blocked airflow through one or both nasal passages. Causes include congestion, allergy, infection, a deviated septum, polyps, trauma, or a mass. Persistent one-sided obstruction or bleeding requires medical evaluation.

Night Sweats

- Episodes of excessive sweating during sleep that soak clothing or bedding. They may occur with infection, hormonal changes, cancer, medication effects, or other systemic illness; persistent unexplained night sweats warrant assessment.

New Daily Persistent Headache

- A headache that begins on a clearly remembered day and becomes continuous within 24 hours, persisting for at least three months. It may resemble migraine or tension-type headache; a new persistent headache requires medical evaluation to exclude secondary causes such as infection, pressure disorders, or vascular disease.

Chapter 15

Pocket Dictionary: O Terms

Obesity

- Abnormal or excessive fat accumulation that presents a health risk. Defined by BMI: $\geq 30 \text{ kg/m}^2$ (class I: 30-34.9, class II: 35-39.9, class III: ≥ 40). Major risk factor for type 2 diabetes, hypertension, cardiovascular disease, sleep apnea, osteoarthritis, certain cancers, and reduced life expectancy.

Obesity Hypoventilation Syndrome

- Chronic daytime hypercapnia in obese individuals without other causes. Reduced chest wall compliance and blunted respiratory drive. Often coexists with OSA. Treatment: weight loss and BiPAP.

Obex Region

- The caudal tip of the fourth ventricle where it narrows into the central canal of the spinal cord. It lies near the dorsal medulla and is an important anatomical landmark in the floor of the ventricular system.

Objective Response Rate (ORR)

- The proportion of patients whose tumor shrinks by a predefined amount (complete or partial response) during treatment; a common efficacy endpoint in oncology trials.

Oblique Plane

- A plane that passes through the body or an organ at an angle other than a right angle (diagonal plane).

Obsessions

- Recurrent, persistent, intrusive, and unwanted thoughts, urges, or images that are experienced as intrusive and cause marked anxiety or distress. Common obsessions include contamination fears, fears of harming oneself or others, need for symmetry, taboo thoughts (sexual, religious, or aggressive), and doubt.

Obsessive-Compulsive Disorder

- Characterized by the presence of obsessions, compulsions, or both. Obsessions are recurrent, persistent, intrusive, and unwanted thoughts, urges, or images that cause marked anxiety or distress. The individual attempts to ignore or suppress them or neutralize them with some other thought or action.

Obsessive-Compulsive Personality Disorder

- A pattern of preoccupation with orderliness, perfectionism, and mental and interpersonal control, at the expense of flexibility and efficiency.

Obstetric Anal Sphincter Injury (OASIS)

- Third- and fourth-degree perineal lacerations involving the anal sphincter complex. Incidence: 0.5-5% of vaginal deliveries. Risk factors: midline episiotomy (increases risk 2-3 fold), operative vaginal delivery, primiparity, macrosomia, and occiput posterior position.

Obstetric Ultrasound

- Ultrasound imaging during pregnancy. First trimester: dating, viability. Second trimester: anatomy survey. Third trimester: growth, position.

Obstetrician

- A medical doctor who specializes in the care of women during pregnancy, labor, and the postpartum period (obstetrics). Obstetricians manage normal and high-risk pregnancies, perform cesarean deliveries and assisted vaginal deliveries, and treat pregnancy complications (pre-eclampsia, gestational diabetes, hemorrhage, preterm labor).

Obstetrics

- The branch of medicine and surgery concerned with the care of women during pregnancy, childbirth, and the postpartum period.

Obstructed Infected Urinary System

- Urinary tract obstruction accompanied by infection, pyonephrosis, or sepsis. It is a urologic emergency requiring immediate resuscitation, antimicrobial therapy, and urgent decompression by nephrostomy or ureteral stent.

Obstruction

- Blockage of urinary flow. Ureteral, urethral, bladder outlet. Causes: stones, stricture, tumor, BPH. Complications: hydronephrosis, infection, renal failure.

Obstructive Jaundice

- Jaundice caused by obstruction of bile flow from the liver to the duodenum, resulting in accumulation of conjugated bilirubin in the blood. See Jaundice for complete overview.

Obstructive parotitis

- Inflammation of the parotid gland resulting from obstruction of the parotid duct, most commonly by sialolithiasis, presenting with painful glandular swelling that worsens with meals. The obstruction leads to stasis, secondary bacterial infection, and recurrent episodes of swelling and tenderness.

Obstructive Shock

- A form of shock caused by physical obstruction of blood flow outside the heart, reducing cardiac output despite normal cardiac function.

Obstructive Sleep Apnea

- Recurrent upper airway collapse during sleep. Symptoms: snoring, witnessed apneas, daytime sleepiness, morning headaches. Risk factors: obesity, male sex, older age, neck circumference >17 inches. Diagnosis: polysomnography. Treatment: CPAP, weight loss, oral appliances, surgery.

Obstructive Uropathy

- Urinary tract obstruction at any level from the renal pelvis to the urethra, leading to impaired urine flow and potential renal damage. Causes include nephrolithiasis, benign prostatic hyperplasia, ureteral strictures, tumors of the bladder, prostate, or cervix, and external compression from lymphadenopathy or retroperitoneal fibrosis.

Obturator

- A structure that occludes an opening. In anatomy, the obturator foramen is a large opening in the

pelvic bone, closed by the obturator membrane. The obturator nerve (L2–L4) innervates the adductor muscles of the thigh and provides sensory innervation to the medial thigh. The obturator artery supplies the adductor compartment.

Obturator hernia

- Herniation of abdominal contents through the obturator canal, most commonly occurring in thin elderly women with a history of multiple pregnancies. It is a rare cause of small bowel obstruction and may present with the Howship-Romberg sign of medial thigh pain from obturator nerve compression.

Obtuse Marginal Artery

- A branch of the circumflex coronary artery that runs along the obtuse margin of the heart and supplies the lateral wall of the left ventricle. Its size and number vary with coronary anatomy.

Occipital Bone

- Forms the posterior-inferior cranium surrounding the foramen magnum (spinal cord passage). Has basilar, lateral, and squamous parts. Occipital condyles articulate with C1 for nodding movement. The external occipital protuberance provides muscle attachment.

Occipital Lobe

- The most posterior of the four cerebral lobes, located behind the parieto-occipital sulcus and above the tentorium cerebelli. The occipital lobe is the primary visual processing center of the brain.

Occipital Neuralgia

- A paroxysmal, severe, stabbing or electric shock-like pain in the distribution of the greater, lesser, or third occipital nerves (posterior scalp, from the nuchal line to the vertex).

Occiput Posterior (OP) Position

- The fetal head presenting with the occiput oriented toward the maternal sacrum (back-to-back position). Occurs in 15-20% of labors, about half rotate spontaneously to OA by delivery.

Occiput Transverse (OT) Position

- The fetal head presenting with the sagittal suture oriented transversely. Usually the normal descent position; most rotate to OA with contractions. When persistent, it can cause protracted descent and prolonged second stage.

Occlusal Guard

- Appliance protecting teeth from attrition. Night guard for bruxism. Sports mouthguard.

Occlusion

- Complete blockage of vessel. May be natural (thrombosis) or iatrogenic (embolization).

Occult Blood

- Blood present in feces that is not visible to the naked eye, detectable only by chemical or immunochemical testing. fecal occult blood testing (FOBT) is used for colorectal cancer screening. Guaiac-based FOBT (gFOBT) detects haem peroxidase activity (requires dietary restriction—avoid red meat, horseradish, vitamin C).

Occult Gastrointestinal Bleeding

- Chronic or intermittent gastrointestinal blood loss that is not visible to the patient and may not produce melaena or haematochezia. It commonly presents with iron-deficiency anaemia or a positive faecal blood test.

Occupational Asthma

- Asthma from workplace exposures to sensitizers (isocyanates, flour, latex) or irritants. Improves away from work and worsens at work. Serial peak flow monitoring supports diagnosis.

Occupational COPD

- COPD from workplace dust, chemical, or fume exposure independent of smoking. Accounts for 15–20% of COPD cases. High-risk occupations: mining, construction, farming.

Occupational Dermatoses

- Work-related skin diseases. Occupational contact dermatitis accounts for 90 percent of occupational skin diseases. Irritant contact dermatitis is more common (hands, wet work) than allergic contact dermatitis. Occupations at risk include healthcare workers, hairdressers, construction workers, food handlers, and metalworkers.

Occupational Exposure

- Radiation exposure to nuclear medicine staff. Monitored with dosimeters. Limits: 50 mSv/year whole body, 150 mSv/year lens, 500 mSv/year extremities.

Occupational Exposure Limit

- A regulatory or advisory concentration or dose limit

intended to reduce worker exposure to a hazardous substance over a defined averaging period. Limits vary by substance, jurisdiction, route, and whether the value is a ceiling, short-term limit, or time-weighted average.

Occupational Health

- Health of workers. Examples: workplace safety, occupational diseases, ergonomics.

Occupational Hearing Loss

- Permanent sensorineural hearing impairment caused by repeated or prolonged exposure to workplace noise or ototoxic hazards. Prevention depends on reducing exposure and using effective hearing protection.

Occupational Medicine

- Health of workers. See occupational health.

Occupational Rehabilitation

- See occupational medicine.

Occupational Therapy

- OT helps patients perform daily activities (ADLs, IADLs) through therapeutic activities, adaptive equipment, environmental modification. Focus on upper extremity function, cognition, sensory integration.

Occupational Therapy Assessment

- OT helps patients perform daily activities (ADLs, IADLs) through therapeutic activities, adaptive equipment, environmental modification. See occupational therapy.

OCD

- Characterized by the presence of obsessions, compulsions, or both. See Obsessive-Compulsive Disorder.

Ochronosis

- A blue-black or brownish pigmentation of connective tissues (cartilage, skin, sclerae, tendons) caused by accumulation of homogentisic acid and its oxidation products.

Octreoscan

- Nuclear medicine imaging using In-111 octreotide. Detects neuroendocrine tumors.

Ocular Histoplasmosis Syndrome (OHS)

- Ocular disease following prior subclinical Histoplasma capsulatum infection, characterized by

punched-out peripheral chorioretinal scars, peripapillary atrophy, and choroidal neovascularization that causes central vision loss. Active choroidal neovascularization is treated with anti-VEGF therapy.

Ocular Hypertension

- Persistently elevated intraocular pressure without demonstrable glaucomatous optic-nerve damage or corresponding visual-field loss. It increases future glaucoma risk and requires individualized assessment of corneal thickness, optic-nerve appearance, pressure, and other risk factors.

Ocular myiasis

- Infestation of the eye or ocular adnexa by fly larvae, which may involve the conjunctiva, cornea, or intraocular structures, presenting with pain, irritation, tearing, and visual disturbance. In severe cases, larvae can penetrate the globe causing endophthalmitis and blindness.

Ocular Surface Disease

- A group of disorders affecting the cornea, conjunctiva, eyelids, and tear film, including dry eye, blepharitis, meibomian-gland dysfunction, and exposure-related disease. Symptoms and signs reflect tear-film instability, inflammation, epithelial injury, or abnormal lid function.

Oculocerebrorenal Syndrome

- An inherited peroxisomal disorder, also called Zellweger spectrum disorder, affecting the brain, eyes, kidneys, liver, and other organs. Severe forms present neonatally with hypotonia, seizures, feeding difficulty, distinctive facial features, and profound developmental impairment.

Odds Ratio

- Ratio of odds of exposure in cases vs. controls. Used in case-control studies. OR approximates RR for rare diseases.

Odontogenic cutaneous fistula

- An abnormal tract connecting the skin surface to a periapical dental infection, typically presenting as a draining sinus tract on the chin, cheek, or submandibular region. It results from chronic dental infection eroding through the bone and soft tissue to the skin surface.

Odontogenic Keratocyst

- Aggressive cyst with high recurrence. Associated

with nevoid basal cell carcinoma syndrome (Gorlin syndrome). Treatment: enucleation, resection, or decompression.

Odontoid Process

- A tooth-like projection superiorly from the body of the axis (C2 vertebra). The odontoid process projects into the atlas (C1) and is held in place by the transverse ligament of the atlas, allowing rotation of the head (the atlantoaxial joint accounts for 50% of cervical rotation).

Odontoma

- Hamartomatous dental tissue mass. Compound (multiple tooth-like structures) or complex (amorphous calcified mass). Treatment: surgical removal.

Odor Threshold

- Lowest concentration detectable by smell. Important for warning properties of toxins.

Odynophagia

- Painful swallowing, characterized by a burning, squeezing, or sticking sensation behind the sternum during deglutition. Strongly suggests mucosal inflammation or ulceration of the esophagus.

Oesophageal Ulcer

- A mucosal ulceration of the oesophagus caused by gastro-oesophageal reflux, infection, medication or corrosive injury, inflammatory disease, or malignancy. It may cause odynophagia, dysphagia, chest pain, haematemesis, or melaena.

Oesophageal Varices

- See bleeding esophageal varices.

Oesophagitis

- Inflammation of the oesophageal mucosa. Common causes include gastro-oesophageal reflux, infection in immunocompromised patients, medication-related injury, eosinophilic inflammation, radiation, and caustic ingestion. Symptoms include heartburn, odynophagia, dysphagia, and chest pain.

Oestrogen

- The primary female sex hormone, produced mainly by the ovaries (granulosa cells of developing follicles) and in smaller amounts by adipose tissue, the adrenal glands, and the placenta during pregnancy.

Oestrogen Replacement Therapy

- The administration of exogenous estrogen (with or

without progestogen) to relieve symptoms of estrogen deficiency and prevent long-term health consequences in postmenopausal women. See Hormone Replacement Therapy (HRT).

Off-Label Use

- The use of a licensed medication for an indication, dose, route, or patient population not included in its marketing authorisation (product licence).

Ogilvie Syndrome

- Acute colonic pseudo-obstruction characterized by massive colonic dilation in the absence of mechanical obstruction typically occurring in hospitalized patients with serious comorbidities including post-surgical states, severe infections, and medication effects.

Ogilvie's syndrome

- Acute colonic pseudo-obstruction characterized by massive dilation of the colon, particularly the cecum and right colon, without mechanical obstruction. It occurs in hospitalized patients with serious medical or surgical conditions, often postoperatively, and results from autonomic nervous system imbalance.

Ointment

- A semi-solid, greasy topical preparation for application to the skin, mucous membranes, or eyes. Ointments are typically composed of a hydrophobic base (petrolatum, lanolin, paraffin, beeswax, vegetable oils) and may contain dissolved or suspended active pharmaceutical ingredients.

Okihiro Syndrome

- A congenital disorder, usually caused by pathogenic variants in *SALL4*, combining radial-ray limb abnormalities with Duane retraction anomaly or other eye-movement defects. Hearing, cardiac, renal, and vertebral anomalies may also occur.

Oleothorax

- Historical surgical obliteration of the pleural space using infused oil to collapse the lung, once used for tuberculosis treatment before effective chemotherapy, now largely obsolete.

Olfactory Nerve

- Cranial nerve I (CN I), responsible for the sense of smell. Olfactory receptor neurons in the olfactory epithelium of the nasal roof project axons through the cribriform plate of the ethmoid bone to the

olfactory bulb, then via the olfactory tract to the primary olfactory cortex (piriform cortex, amygdala, entorhinal cortex).

Oligoclonal Bands

- Immunoglobulin G (IgG) proteins produced by a limited number of B-cell clones in the cerebrospinal fluid (CSF), detected by isoelectric focusing of CSF and serum samples.

Oligodendroglioma

- A slow-growing glioma arising from oligodendrocytes, most often in the cerebral hemispheres. Associated with IDH mutation and 1p/19q codeletion, which confer a better prognosis and chemosensitivity.

Oligohydramnios

- Insufficient amniotic fluid volume, defined as amniotic fluid index less than 5 cm or single deepest pocket less than 2 cm. Causes include fetal renal anomalies, post-term pregnancy, premature rupture of membranes, maternal dehydration, and placental insufficiency.

Oligometastasis

- A limited number of metastatic deposits, often defined as up to five, amenable to aggressive local treatment (surgery, stereotactic radiation, or ablation) with the potential for prolonged control.

Oliguria

- Reduced urine output, defined as <400 mL/day or <0.5 mL/kg/hour in adults. Oliguria is an important sign of acute kidney injury (AKI) and prerenal azotaemia.

Olive Pontocerebellar Atrophy

- A group of neurodegenerative disorders characterized by progressive loss of neurons in the inferior olives, pons, and cerebellum. They cause worsening ataxia, dysarthria, incoordination, and sometimes pyramidal, autonomic, or cognitive features.

Omadacycline

- A newer aminomethylcycline antibiotic with activity against gram-positive, gram-negative, and atypical organisms. Omadacycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It has activity against MRSA, vancomycin-resistant enterococci, community-acquired pneumonia pathogens, and skin

Omega-3

- Polyunsaturated fatty acids (EPA, DHA, ALA). Anti-inflammatory, cardiovascular benefits. Sources: fatty fish, flaxseed, walnuts. Supplementation common.

Omega-6

- Polyunsaturated fatty acids (linoleic acid). Pro-inflammatory in excess. Balance with omega-3 important. Sources: vegetable oils, nuts, seeds.

Omenn Syndrome

- A severe combined immunodeficiency caused by hypomorphic defects affecting lymphocyte-receptor gene rearrangement or related pathways. It presents in infancy with erythroderma, lymphadenopathy, hepatosplenomegaly, diarrhea, failure to thrive, and recurrent or opportunistic infections.

Omentectomy

- Removal of part or all of the greater omentum, commonly performed during staging and cytoreduction for ovarian and other intra-abdominal cancers.

Omentum

- A double layer of peritoneum connecting the stomach to other abdominal organs. The greater omentum (gastrocolic omentum) hangs from the greater curvature of the stomach, covers the transverse colon and small intestine, and is rich in fat and immune cells (milky spots—aggregates of macrophages and lymphocytes).

Omeprazole

- A proton pump inhibitor (PPI) that irreversibly inhibits the gastric H⁺/K⁺ ATPase (proton pump) on parietal cells, profoundly reducing acid secretion. Used for GERD, peptic ulcer disease, *H. pylori* eradication (triple therapy), Zollinger-Ellison syndrome, and stress ulcer prophylaxis.

Omphalitis

- Infection of the umbilical stump in neonates, typically occurring within the first 1–2 weeks of life. Omphalitis is a potentially serious condition requiring prompt treatment due to the risk of spread to the liver (portal vein thrombosis, portal hypertension) and peritoneum via the umbilical vessels.

Omphalocele

- A congenital abdominal wall defect where abdominal organs herniate into the base of the umbilical

cord, covered by a peritoneal-amniotic membrane. Associated with chromosomal abnormalities (trisomies 13, 18) and other anomalies. Differentiated from gastroschisis by membrane coverage and midline location.

Onchocerca volvulus

- A filarial nematode that causes onchocerciasis (river blindness), transmitted by repeated bites of infected blackflies (*Simulium* species) breeding near fast-flowing rivers. Adult worms reside in subcutaneous nodules (onchocercomata), where they can survive for years, producing microfilariae that migrate through the skin and eyes.

Onchocerciasis

- A parasitic disease caused by the filarial nematode *Onchocerca volvulus*, transmitted by repeated bites of infected blackflies (*Simulium* species) breeding near fast-flowing rivers. Adult worms form subcutaneous nodules (onchocercomata).

Oncocytoma

- A usually benign epithelial tumor, commonly arising in the kidney and composed of cells with abundant eosinophilic mitochondria-rich cytoplasm.

Oncogene

- A gene that, when activated, overexpressed, or altered, can promote uncontrolled cell proliferation or survival. Oncogenes commonly arise from normal proto-oncogenes through mutation, amplification, translocation, or abnormal regulation.

Oncogenes

- Mutated, amplified, or overexpressed versions of normal cellular genes called proto-oncogenes that encode proteins involved in cell growth, proliferation, and survival. Proto-oncogenes encode growth factors, growth factor receptors, signal transduction molecules, and transcription factors that normally regulate controlled cell

Oncogenic

- Cancer-causing. See carcinogen.

Oncologic Emergency

- A cancer-related condition that threatens life, organ function, or neurologic integrity and requires urgent treatment. Examples include malignant spinal-cord compression, superior vena cava syndrome, tumor lysis syndrome, hypercalcemia, airway obstruction,

and febrile neutropenia.

Oncologist

- A physician who specializes in the diagnosis and treatment of cancer. Includes medical oncologists (systemic therapy), radiation oncologists (radiotherapy), and surgical oncologists (operative management).

Oncology PET

- FDG PET/CT standard for many cancers. Staging, response assessment, restaging.

Oncology Rehabilitation

- Cancer rehabilitation: fatigue, pain, neuropathy, lymphedema, functional impairment. Pre- and post-treatment support.

Oncolytic Virus

- A genetically modified virus that selectively replicates in and destroys cancer cells while sparing normal tissue, e.g., talimogene laherparepvec (T-VEC) for melanoma.

Oncosis

- A form of cell death characterized by cellular swelling, derived from the Greek word *onkos* meaning swelling, representing a pre-lethal state that precedes necrosis. It is typically caused by ischemic injury leading to ATP depletion, which impairs the sodium-potassium pump and causes osmotic water influx.

Oncotic Pressure

- The osmotic pressure exerted by plasma proteins (predominantly albumin) that do not freely cross the capillary wall. Normal plasma oncotic pressure is approximately 25-28 mmHg. It opposes the hydrostatic pressure that drives fluid out of capillaries, favoring fluid reabsorption at the venular end.

One Health

- Approach recognizing health connections between humans, animals, environment. Examples: zoonotic diseases, antibiotic resistance, food safety.

One-Hour Glucose Challenge Test (GCT)

- A screening test for gestational diabetes mellitus (GDM) performed at 24-28 weeks. The patient drinks 50 g of glucose (regardless of fasting state), and plasma glucose is measured at 1 hour. Threshold: 130-140 mg/dL (7.2-7.8 mmol/L) is considered positive, warranting diagnostic 3-hour oral glucose

tolerance test (OGTT).

Onycholysis

- Separation of the nail plate from the underlying nail bed, beginning distally or laterally. Causes include trauma, psoriasis, fungal infection, thyroid disease, medications, and chemical exposure; the underlying cause should be identified.

Onychomycosis

- Fungal nail infection, most commonly caused by dermatophytes (*Trichophyton rubrum*), affecting toenails more than fingernails. It presents as nail thickening, discoloration (yellow-brown), subungual debris, and nail separation. Diagnosis is confirmed by KOH preparation or fungal culture.

Oocyte

- The female gamete (egg cell) produced in the ovary through oogenesis. Oocytes develop from primordial germ cells, which undergo mitosis to form oogonia, then enter meiosis I before birth and arrest in prophase I until puberty.

Oophorectomy

- Surgical removal of one (unilateral) or both (bilateral) ovaries.

Opacity

- Lack of transparency of a normally transparent ocular structure (cornea, lens, vitreous, or aqueous humour). Corneal opacity results from scarring (trauma, infection—herpes simplex keratitis, trachoma), dystrophies (Fuchs endothelial dystrophy, granular/lattice/macular dystrophies), or edema.

Open Fractures

- Fractures with communication between the fracture site and the external environment, classified by Gustilo-Anderson: Grade I (<1 cm wound, clean), Grade II (1-10 cm wound, moderate soft tissue injury), Grade IIIA (>10 cm wound, adequate soft tissue coverage), Grade IIIB (>10 cm wound, extensive soft tissue loss requiring flap coverage), G

Open Reduction and Internal Fixation (ORIF)

- Operative exposure and realignment of a fracture followed by fixation with implants such as plates, screws, wires, or nails. It is used when closed treatment cannot provide acceptable alignment, stability, or joint restoration.

Open Surgery

- Traditional surgical approach with incision. Used for bypass, aneurysm repair, thrombectomy.

Open-book fracture

- A pelvic ring disruption with diastasis of the pubic symphysis or sacroiliac joints, often from high-energy trauma, carrying risk of massive hemorrhage and organ injury.

Open-Heart Surgery

- Any surgical procedure involving opening the chest wall (sternotomy or thoracotomy) and operating on the heart, usually with cardiopulmonary bypass (the heart–lung machine).

Operative Vaginal Delivery

- Vaginal delivery assisted by vacuum or forceps. Indications: prolonged second stage, suspected fetal compromise, need to shorten the second stage for maternal benefit (exhaustion, cardiac disease, hypertensive crisis).

Ophthalmic Emergency

- An eye or visual condition requiring urgent evaluation to prevent irreversible vision loss or severe ocular injury. Examples include acute angle closure, retinal detachment, central retinal artery occlusion, endophthalmitis, chemical injury, orbital cellulitis, and open-globe trauma.

Ophthalmomyiasis

- Infestation of the eye by fly larvae (most commonly Oestridae), which may be superficial (conjunctival) or deep (intraocular), with the latter carrying a risk of retinal detachment, endophthalmitis, and permanent vision loss.

Ophthalmoscope

- Instrument for examining retina and optic nerve. Direct: larger, brighter image. Indirect: wider field. Essential for fundoscopic examination.

Ophthalmoscopy

- Examination of the interior of the eye, especially the retina, optic disc, and retinal blood vessels, using an ophthalmoscope. It helps detect retinal disease, optic-nerve abnormalities, glaucoma, and systemic vascular disease.

Opiate

- A natural alkaloid derived from the opium poppy

(*Papaver somniferum*) or a semi-synthetic derivative that binds to opioid receptors.

Opioid Agonist Treatment

- Treatment of opioid use disorder with a prescribed opioid agonist or partial agonist, most commonly methadone or buprenorphine, together with psychosocial and medical support. It reduces withdrawal, craving, illicit opioid use, overdose, and mortality.

Opioid Analgesics

- A class of drugs that bind to opioid receptors in the central and peripheral nervous systems, producing analgesia and other effects.

Opioid Antagonist

- See naloxone.

Opioid Overdose

- A medical emergency characterized by the classic triad of respiratory depression, miosis (pinpoint pupils), and decreased level of consciousness. It can progress to respiratory arrest, cardiovascular collapse, and death.

Opioid Reversal

- Achieved with naloxone, a competitive mu-opioid receptor antagonist. Dose: 0.04 to 0.1 mg IV, titrated to effect, repeat every 2 to 3 minutes. Maximum typical dose is 2 to 4 mg. Naloxone has shorter half-life (30 to 80 minutes) than most opioids, risking recurrent respiratory depression.

Opioid Toxicity

- Toxic effects of opioid agonists, classically respiratory depression, central nervous system depression, and miosis. Management prioritizes ventilation and airway support; naloxone is titrated to restore adequate breathing while recognizing the risk of recurrent toxicity and withdrawal.

Opioid Use Disorder

- A problematic pattern of opioid use leading to significant impairment or distress. Opioids include heroin, fentanyl, morphine, oxycodone, and hydrocodone. Criteria are the same eleven as for substance use disorders generally. Overdose presents with respiratory depression, miosis, and decreased level of consciousness.

Opioids

- A class of drugs that bind to opioid receptors in the

central and peripheral nervous systems, producing analgesia and other effects. See Opioid Analgesics.

Opisthotonos

- Severe backward arching of the head, neck, and spine caused by sustained extensor muscle spasm. It is classically associated with tetanus but may also occur in meningitis, seizures, or severe brain injury.

Opportunistic Infection

- An infection caused by organisms that ordinarily do not cause disease in immunocompetent hosts but cause significant illness when host defences are impaired.

Opposition

- Movement of the thumb across the palm to contact the fingertips. It combines abduction, flexion, and medial rotation at the first carpometacarpal joint.

Oppositional Defiant Disorder

- A childhood/adolescent disorder characterized by a persistent pattern (6 months) of angry/irritable mood, argumentative/defiant behavior, and vindictiveness.

Opsoclonus

- Rapid, involuntary, multidirectional, conjugate eye movements without an intersaccadic interval. It may occur with myoclonus and ataxia as part of an autoimmune or paraneoplastic syndrome, particularly in children with neuroblastoma.

Oposonization

- The process by which pathogens or particles are coated with opsonins (antibodies—IgG, complement—C3b, or acute-phase proteins) that enhance recognition and phagocytosis by phagocytes bearing corresponding receptors (Fc receptors, complement receptors).

Optic Atrophy

- End-stage degeneration of the optic nerve fibers resulting from any optic neuropathy. Causes: glaucoma, ischemic optic neuropathy, multiple sclerosis, compressive lesions (pituitary tumor, aneurysm), toxins (methanol, ethambutol), and hereditary (Leber hereditary optic neuropathy).

Optic Disc

- The point on the retina where the optic nerve exits the eye, lacking photoreceptors and creating the natural blind spot.

Optic Disc Edema

- Swelling of the optic-nerve head caused by axoplasmic-flow disruption. Bilateral optic-disc edema from raised intracranial pressure is papilledema; unilateral or non-pressure causes require separate evaluation for optic-nerve, retinal, inflammatory, ischemic, or infiltrative disease.

Optic Glioma

- A usually slow-growing glioma of the optic nerve or optic pathway, occurring most often in children. It may cause progressive visual loss, optic atrophy, proptosis, or visual-field defects and is associated with neurofibromatosis type 1.

Optic Nerve

- CN II exiting through the optic disc (blind spot), passing through the optic canal, and partially decussating at the optic chiasm. Technically a CNS tract with oligodendrocyte myelination and meningeal coverings, making it susceptible to demyelinating disease (optic neuritis in MS).

Optic Nerve Atrophy

- Pallor and degeneration of the optic nerve caused by loss of retinal ganglion-cell axons. It produces permanent visual impairment and may result from ischemia, inflammation, compression, glaucoma, trauma, or raised intracranial pressure.

Optic Neuritis

- An inflammatory demyelinating condition of the optic nerve that typically presents with acute or subacute unilateral vision loss, pain with eye movement, and a relative afferent pupillary defect, and it is most commonly associated with multiple sclerosis, occurring in approximately fifty percent of patients at some point during their disease.

Optical Coherence Tomography

- A non-invasive imaging technique that uses low-coherence interferometry to produce high-resolution, cross-sectional images of the retina and other ocular structures, providing quantitative measurements of retinal layer thickness and detecting structural abnormalities with micrometer-level resolution.

Optometry

- A healthcare profession concerned with examining the eyes and visual system for defects and abnormalities, prescribing corrective lenses (spectacles,

contact lenses), diagnosing eye disease, and providing pre- and post-operative care for surgical eye treatments.

Oral Cancer

- Malignancy of oral cavity, including lips, tongue, floor of mouth, buccal mucosa, palate, tonsils, oropharynx. Major risk factors: tobacco use, alcohol consumption, human papillomavirus (HPV, especially p16-positive oropharyngeal cancer). Presentation: persistent ulcer, leukoplakia, erythroplakia, mass, pain, difficulty swallowing.

Oral Candidiasis

- Infection or overgrowth of *Candida* species on the oral mucosa, often presenting as removable white plaques, erythematous patches, or angular cheilitis. Risk factors include antibiotics, inhaled corticosteroids, immunosuppression, diabetes, xerostomia, and dentures.

Oral Cavity

- The space extending from the lips to the oropharynx, bounded by the cheeks, palate, and tongue. It receives food, begins mechanical and chemical digestion, and contributes to speech.

Oral Cavity Cancer

- Squamous cell carcinoma of the lip, tongue, buccal mucosa, floor of mouth, or palate, associated with tobacco, alcohol, and betel quid use. Treatment is primarily surgical resection, with adjuvant radiation or chemoradiation for adverse features.

Oral Contraceptive

- Birth control pill: combined oral contraceptive (COC—ethinylestradiol + progestin) or progestin-only pill (POP/minipill). Mechanisms: ovulation inhibition, cervical mucus thickening, endometrial change.

Oral Contraceptives

- Hormonal medications taken orally to prevent pregnancy. Combined oral contraceptive pills (COCP) contain oestrogen (ethinylestradiol 20–35 g) and a progestogen (levonorgestrel, norethisterone, drospirenone, desogestrel).

Oral Glucose Tolerance Test

- A diagnostic test measuring plasma glucose response after a standardized oral glucose load. It is used in selected evaluations of diabetes, impaired

glucose tolerance, and gestational diabetes when fasting glucose or HbA1c is insufficient.

Oral Hygiene

- Practices maintaining oral health: brushing twice daily with fluoride toothpaste, flossing daily, mouthwash (optional), regular dental visits. Proper brushing technique: soft brush, 45-degree angle to gumline, gentle circular motions. Interdental cleaning essential for plaque removal.

Oral Hypoglycemic

- A class of drugs used to treat type 2 diabetes (with lifestyle).

Oral Leukoplakia

- A white patch or plaque on the oral mucosa that cannot be scraped off and cannot be characterized as any other diagnosable disease, considered a potentially malignant disorder. Risk factors: tobacco use (most common), alcohol, and betel nut chewing. Most common sites: buccal mucosa, alveolar ridge, and tongue.

Oral Lichen Planus

- Chronic inflammatory condition. Reticular (lace-like white lines), erosive (painful ulcers). Associated with hepatitis C. Malignant transformation rare but possible.

Oral Medicine

- Specialty diagnosing oral mucosal diseases, dental causes of facial pain, oral manifestations of systemic disease.

Oral Mucosa

- The moist stratified squamous epithelial lining of the oral cavity. It includes masticatory, lining, and specialized mucosa and provides barrier protection, sensation, lubrication, and sites for minor salivary glands.

Oral Nutritional Supplement

- A calorie-, protein-, or micronutrient-containing product consumed by mouth to supplement, not automatically replace, usual food intake. The product and dose should match the patient's needs, swallowing ability, disease restrictions, and tolerance.

Oral Pathology

- Study of diseases affecting oral tissues. Includes neoplasms, cysts, inflammatory conditions, developmental anomalies.

Oral Polio Vaccine (OPV-O)

- Sabin vaccine, a live attenuated preparation of poliovirus administered orally, historically the primary tool for global polio eradication. The bivalent formulation (bOPV) contains attenuated types 1 and 3 only; the type 2 component was withdrawn from routine use in April 2016 following the certified eradication of wild poliovirus type 2.

Oral Radiology

- Specialty of producing and interpreting images of oral and maxillofacial region.

Oral Sex (Fellatio, Cunnilingus)

- Oral stimulation of the genitals for sexual pleasure. Fellatio: oral stimulation of the penis. Cunnilingus: oral stimulation of the vulva and clitoris. Analingus (rimming): oral stimulation of the anus.

Oral Squamous Cell Carcinoma

- A malignant neoplasm arising from oral stratified squamous epithelium. Persistent induration, ulceration, red or white patches, unexplained bleeding, pain, or neck mass requires evaluation and biopsy; staging incorporates tumor, nodal, and metastatic findings.

Oral Surgery

- Surgical procedures in oral and maxillofacial region. Extractions, implant placement, biopsies, cyst removal, corrective jaw surgery.

Oral Toxicity

- See ingestion.

Oral Ulcer

- Ulceration of oral mucosa. Aphthous ulcer (canker sore) most common. Traumatic, viral, autoimmune causes.

Orbital Cellulitis

- A serious infection of the orbital tissues posterior to the orbital septum, potentially sight-threatening and life-threatening due to the risk of optic nerve compression, cavernous sinus thrombosis, and intracranial extension.

Orbital metastasis

- Secondary cancer deposits in the orbit, most commonly from breast, lung, or prostate primaries. Symptoms include proptosis, diplopia, pain, and visual impairment.

Orbital Pseudotumor

- Noninfectious, idiopathic inflammation of orbital tissues, also called idiopathic orbital inflammation. It can cause painful eye movement, proptosis, eyelid swelling, diplopia, and restricted ocular motility.

Orchiectomy

- Surgical removal of one or both testicles. Radical inguinal orchiectomy is diagnostic and therapeutic for testicular cancer; bilateral orchiectomy is a hormonal therapy for prostate cancer.

Orchitis

- Inflammation of the testis, often associated with epididymitis (epididymo-orchitis).

Orexin Receptor Antagonists

- Medications that block orexin (hypocretin) receptors to promote sleep. Suvorexant and lemborexant are approved for insomnia. They reduce wakefulness by blocking the arousal system. They have lower abuse potential than benzodiazepines and do not cause respiratory depression.

Orf

- A zoonotic viral skin infection caused by the parapoxvirus, transmitted to humans from infected sheep and goats (direct contact with lesions, fomites—virus survives in scabs for months). Orf is an occupational disease of farmers, butchers, and veterinarians. Incubation: 3–7 days.

Organ

- A structure composed of two or more different types of tissues that has a recognizable shape and performs specific functions.

Organ System

- A group of related organs that work together to perform major body functions.

Organ Transplants

- The surgical transfer of an organ from a donor to a recipient with end-stage organ failure. Solid organ transplants: kidney (most common, >90,000 per year worldwide), liver, heart, lung, pancreas, and intestine.

Organophosphate

- Acetylcholinesterase inhibitors. See organophosphate poisoning.

Organophosphate Poisoning

- Acetylcholinesterase inhibitors. SLUDGE syndrome: salivation, lacrimation, urination, defecation, GI upset, emesis. Miosis, bradycardia, bronchospasm, muscle fasciculations. Treatment: atropine, pralidoxime, supportive care.

Orogastric Tube

- Tube through mouth to stomach. For infants, unconscious patients.

Oropharyngeal Airway

- A rigid airway adjunct inserted through the mouth to displace the tongue and help maintain upper-airway patency in an unconscious patient without an intact gag reflex. It does not protect against aspiration.

Oropharyngeal Carcinoma

- Malignancy arising in the oropharynx, including the tonsil, base of tongue, soft palate, and posterior pharyngeal wall. HPV-associated tumors form an important subgroup; evaluation includes endoscopy, biopsy, imaging, and cervical-node assessment.

Oropharynx

- The posterior portion of the mouth leading to the esophagus, containing the tonsils and base of tongue. Area for swallowing and speech. Examination with tongue depressor and light source. Pathology: pharyngitis, tonsillitis, malignancy.

Orphan Drug

- A pharmaceutical agent developed specifically to treat a rare medical condition (affecting fewer than 200,000 people in the United States, as defined by the Orphan Drug Act of 1983).

Orphan Drugs

- Medicinal products developed for the diagnosis, prevention, or treatment of rare diseases (defined in the EU as affecting <5 per 10,000 people; in the US as affecting <200,000 people).

Orthodontic Bracket

- A bonded or banded attachment that transmits forces from an orthodontic archwire to a tooth. Bracket position and wire sequence guide controlled tooth movement.

Orthodontic Retainer

- A removable or fixed appliance used after active

orthodontic treatment to maintain tooth position while periodontal and alveolar tissues remodel. Retention is often needed long term because relapse can occur.

Orthodontics

- Specialty of dental management of malocclusion and dental-facial discrepancies. Appliances: fixed braces, removable appliances, clear aligners. Treatment indications: crowding, spacing, overbite, underbite, crossbite, open bite, midline discrepancy. Treatment duration: 12-24 months typically. Retention phase essential for stability.

Orthognathic Surgery

- Corrective jaw surgery for skeletal malocclusion. Le Fort I, Bilateral Sagittal Split Osteotomy. Combined with orthodontics.

Orthopedic Biologics

- Bone graft substitutes: autograft (iliac crest gold standard, osteogenic/osteoinductive/osteoconductive, donor site morbidity), allograft (osteoconductive, some osteoinductive if DBM, disease transmission risk), DBM (demineralized bone matrix, osteoinductive), synthetic (calcium phosphate, calcium

Orthopedic Implant

- Devices for bone/joint reconstruction. Types: plates, screws, rods, hinges, prostheses. Materials: titanium, cobalt-chrome, stainless steel, polyethylene, ceramic. Used for fractures, arthritis, tumors, deformities. Complications: infection, loosening, fracture, wear.

Orthopedic Infections

- Septic arthritis: see Infectious Disease. Osteomyelitis: see Infectious Disease. Prosthetic joint infection: see Periprosthetic Joint Infection. Diabetic foot infection: Wagner classification (0-5), University of Texas (grade + ischemia + infection).

Orthopedic Rehabilitation

- See orthopedics.

Orthopedic Screws

- Fixation devices for fractures. Materials: titanium, stainless steel. Types: cortical, cancellous, interfragmentary. Used with plates, rods, nails. Complications: loosening, breakage, infection.

orthopedics

- The branch of surgery concerned with the musculoskeletal system, including bones, joints, ligaments, tendons, muscles, and nerves.

Orthopnea

- Dyspnea that occurs when lying flat and is relieved by sitting upright. It is measured by the number of pillows the patient needs to sleep (two-pillow orthopnea). It is a hallmark of left heart failure and is caused by redistribution of fluid from the lower extremities to the lungs in the supine position.

Orthopnoea

- Shortness of breath that occurs when lying flat and is relieved by sitting or standing upright. Orthopnoea is a classic symptom of left-sided heart failure (elevated pulmonary venous pressure worsens pulmonary congestion in the supine position). Other causes: severe COPD, asthma exacerbation, diaphragmatic paralysis, ascites, and obesity.

Orthorexia

- Obsession with healthy eating. May lead to nutritional deficiency, social isolation.

Orthosis

- An externally applied device that supports, aligns, prevents deformity, protects, or assists movement of a body segment. It differs from a prosthesis, which replaces a missing body part.

Orthostatic Hypotension

- Decrease in SBP ≥ 20 mmHg or DBP ≥ 10 mmHg within 3 minutes of standing. Causes: dehydration, medications (antihypertensives, diuretics, alpha-blockers, antidepressants), autonomic dysfunction (diabetes, Parkinson disease), and prolonged bed rest. Symptoms: dizziness, lightheadedness, syncope, falls.

Orthostatic Intolerance

- Symptoms or signs that develop when standing because the cardiovascular and autonomic systems do not maintain adequate cerebral perfusion. Manifestations include light-headedness, palpitations, fatigue, visual dimming, presyncope, or syncope and may occur with or without orthostatic tachycardia or hypotension.

Orthotic Training

- Training for brace users: donning/doffing, skin

care, gait training with brace.

Osborn wave

- A positive deflection at the junction of the QRS complex and ST segment on electrocardiography, also known as a J wave. It is classically associated with hypothermia, where its amplitude correlates with the degree of cooling.

OSFED

- Other Specified Feeding or Eating Disorder, used when the clinical presentation causes significant distress or impairment but does not meet full criteria for any of the specific eating disorders.

Osgood-Schlatter Disease

- A traction apophysitis of the tibial tuberosity at the insertion of the patellar tendon, caused by repetitive microtrauma during periods of rapid growth.

Osler Nodes

- Painful, erythematous nodules on the pads of fingers and toes, caused by immune complex deposition in infective endocarditis. They are a minor criterion in the Duke criteria. Osler nodes are tender and may last hours to days.

Osmolality

- The concentration of osmotically active particles per kilogram of solvent. Plasma osmolality is determined mainly by sodium and its accompanying anions, glucose, and urea and governs water movement across cell membranes.

Osmosis

- The net movement of water molecules across a semipermeable membrane from a region of lower solute concentration (higher water concentration) to a region of higher solute concentration (lower water concentration). Driven by the osmotic pressure difference, which is proportional to the number of solute particles.

Osmotic Demyelination Syndrome

- Demyelinating process affecting the pons and extrapontine regions caused by overly rapid correction of chronic hyponatremia. The brain adapts to chronic hyponatremia by extruding osmoles and when sodium is corrected too quickly water shifts out of brain cells causing osmotic injury to oligodendrocytes.

Osmotic Diarrhea

- Diarrhea caused by poorly absorbed solutes retaining water in the intestinal lumen. It typically improves with fasting and may result from carbohydrate malabsorption, laxatives, or other osmotically active substances.

Osmotic Fragility Test

- A test that measures the susceptibility of red blood cells to haemolysis when placed in progressively more dilute saline solutions. Increased fragility supports hereditary spherocytosis, although the result may also be affected by other haemolytic disorders and recent transfusion.

Osmotic Pressure

- The pressure required to prevent the movement of water across a semipermeable membrane when a solute concentration gradient exists, proportional to the number of dissolved particles (van't Hoff law: $\pi = iCRT$). It is a colligative property depending on particle number, not particle identity.

Osseointegration

- Direct structural and functional connection between living bone and the surface of a load-bearing implant under appropriate conditions. It depends on implant stability, surface characteristics, bone quality, healing, and control of infection and occlusal forces.

Ossicles

- Three smallest body bones forming the ossicular chain in the middle ear: malleus (hammer, attached to tympanic membrane), incus (anvil), stapes (stirrup, footplate in oval window).

Osteoarthritis

- The most common joint disease, characterized by progressive cartilage loss, subchondral bone sclerosis, osteophyte formation, and synovial inflammation, affecting knees, hips, hands (DIP/PIP Heberden/Bouchard nodes), spine. Risk factors: age, obesity, joint injury, malalignment, genetics, female sex.

Osteoarthritis (Geriatric)

- Degenerative joint disease, most common arthritis in elderly. Affects weight-bearing joints, hands. Symptoms: pain, stiffness, swelling, decreased range of motion. Treatment: exercise, weight loss, anal-

gesics, joint replacement.

Osteoblast

- A bone-forming cell derived from mesenchymal stem cells in the bone marrow. Osteoblasts synthesise and deposit the bone matrix (osteoid), composed primarily of type I collagen and non-collagenous proteins (osteocalcin, osteonectin, alkaline phosphatase).

Osteochondritis Dissecans

- A condition characterized by separation of a fragment of articular cartilage and subchondral bone from the joint surface, most commonly in the knee (lateral aspect of the medial femoral condyle). Most common in adolescents and young adults. Etiology: repetitive microtrauma, ischemia, or genetic predisposition.

Osteochondroma

- The most common benign bone tumor, accounting for approximately 20-35 percent of all benign bone tumors. It is characterized by a bony projection with a cartilage cap arising from the surface of a bone, continuous with the underlying cortex and medullary cavity.

Osteoclast

- A large, multinucleated bone-resorbing cell derived from the monocyte-macrophage lineage (hematopoietic stem cells).

Osteocyte

- The mature, terminally differentiated bone cell embedded within the bone matrix in lacunae, derived from osteoblasts. Osteocytes are connected to each other and to bone surface cells via canaliculi, forming a mechanosensory network that senses mechanical loading and fluid flow.

Osteogenesis Imperfecta

- A connective tissue disorder caused by collagen type I mutations, ranging from mild to lethal. Features: fractures with minimal trauma, blue sclerae, dental abnormalities, and hearing loss. Severe forms (type II) cause multiple fractures in utero and perinatal death. Diagnosis: clinical features, genetic testing, and bone biopsy.

Osteoid Osteoma

- A benign bone-forming tumor that accounts for approximately 10-12 percent of all benign bone

tumors, predominantly affecting adolescents and young adults (ages 5-25 years, male-to-female ratio 2:1). It is characterized by a small (< 1.5 -2 cm) radiolucent nidus surrounded by reactive sclerotic bone.

Osteomalacia

- Impaired mineralization of newly formed osteoid in adults resulting from vitamin D deficiency, phosphate depletion, or defective mineralization.

Osteomyelitis

- An infection of bone and bone marrow, most commonly caused by bacterial pathogens, with *Staphylococcus aureus* responsible for the majority of cases.

Osteonecrosis (Avascular Necrosis)

- Osteonecrosis (AVN) is bone death from vascular compromise, affecting femoral head (most common), humeral head, femoral condyles, talus. Risk factors: glucocorticoids (high dose, cumulative), alcohol, trauma, sickle cell disease, SLE, antiphospholipid syndrome, pancreatitis, decompression sickness, radiation, organ transplantation.

Osteopathy

- A system of manual medicine and healthcare founded by Andrew Taylor Still (1828–1917) based on the principle that the body's structure (musculoskeletal system) and function are interrelated, and that the body has inherent self-regulatory and healing mechanisms.

Osteopenia

- Bone mineral density (BMD) that is lower than normal but not low enough to be classified as osteoporosis. T-score between -1.0 and -2.5 on DXA scan. A risk factor for osteoporosis and fractures.

Osteopetrosis

- A rare inherited bone disease characterized by defective osteoclast function, leading to impaired bone resorption and abnormally dense, sclerotic bones. The bone is structurally abnormal—brittle despite increased density—and prone to fracture.

Osteoporosis

- Systemic skeletal disease characterized by low bone mass and microarchitectural deterioration. Results in increased fracture risk. Diagnosis: DEXA T-score < -2.5 . Risk factors: age, female sex, Caucasian/Asian, low body weight, smoking, alcohol,

glucocorticoids, family history. Prevention: calcium, vitamin D, exercise, fall

Osteoporosis Fracture Risk Assessment

- Evaluation combining clinical risk factors, prior fragility fracture history, bone mineral density, and tools such as FRAX to estimate future fracture probability and guide treatment.

Osteosarcoma

- The most common primary malignant bone tumor in children and adolescents (excluding leukemia), with a peak incidence during the adolescent growth spurt (ages 10-20 years), and a second smaller peak in older adults (usually secondary to Paget disease or prior radiation).

Osteotomy

- A surgical procedure involving cutting and repositioning of bone to correct deformity, realign joint surfaces, alter weight-bearing forces, or change limb length.

Ostomy

- Surgical opening into bowel or urinary tract. Colostomy, ileostomy, urostomy. Appliances for collection.

Ostomy Appliance

- Collection device for feces/urine after ostomy surgery. Types: urostomy (urine), colostomy (feces), ileostomy (ileum). One-piece or two-piece systems. Open or closed end. Skin care essential. Complications: skin irritation, hernia, obstruction, prolapse.

Ostraceous psoriasis

- A severe, chronic variant of plaque psoriasis in which the lesions become thick, hard, and shell-like (ostraceous), with laminated, greyish-brown crusts resembling oyster shells. It is often refractory to standard therapies and may require systemic agents, biologics, or combination treatment.

Other Pain Conditions

- Other pain types encompass pain conditions not primarily classified as nociceptive (somatic or visceral) or classic neuropathic pain, including nociplastic pain (altered nociception without clear tissue or nerve damage, such as fibromyalgia, chronic tension-type headache, temporomandibular disorders, and chronic pelvic pain), complex

Other Specified Sexual Dysfunction

- A category for sexual dysfunction symptoms that cause distress but do not meet full criteria for any specific sexual dysfunction.

Otic capsule

- The otic capsule is the dense bony shell (otic capsule bone) that surrounds and protects the inner ear structures, including the cochlea, vestibule, and semicircular canals, forming the hardest bone in the human body.

Otitis Externa

- Otitis externa, commonly known as swimmer's ear, is an infection or inflammation of the external.

Otitis media

- A group of inflammatory conditions of the middle ear, most commonly caused by viral or bacterial infection, with *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis* being the most frequent bacterial pathogens.

Otitis Media with Effusion

- Nonpurulent fluid in the middle ear without acute signs of infection. It may cause conductive hearing loss, aural fullness, or speech and learning effects in children; many cases resolve spontaneously, but persistent or consequential effusion requires hearing assessment and ENT review.

Otocephaly

- A severe congenital craniofacial malformation characterized by absent or markedly underdeveloped mandible, ventral displacement or fusion of the ears, and associated facial and airway abnormalities. It is usually lethal because of profound impairment of breathing and feeding.

Otolaryngology (ENT)

- The surgical specialty concerned with the diagnosis and treatment of disorders of the ear, nose, throat, head, and neck. Otolaryngologists (ENT surgeons) manage conditions ranging from otitis media and sinusitis to head and neck cancer, thyroid disease, and voice disorders.

Otology

- The branch of medicine and surgery concerned with the study and treatment of ear disorders. Otology encompasses hearing loss (conductive, sensorineural, mixed), tinnitus, vertigo and balance disorders

(Meniere disease, benign paroxysmal positional vertigo, vestibular neuritis), otitis media, otosclerosis, cholesteatoma, and ear

Otorrhea

- Discharge from the ear. Classification: acute otorrhea (<6 weeks) and chronic otorrhea (>6 weeks).

Otosclerosis

- An abnormal bony growth in the middle ear that causes hearing loss.

Otoscope

- Instrument for examining ear canal and tympanic membrane. Light source, magnifying lens, speculum. Essential for otologic evaluation.

Otoscopy

- A clinical examination technique used to visualize the external auditory canal and tympanic membrane, and it is an essential component of the evaluation of ear complaints including hearing loss, ear pain, otorrhea, and tinnitus.

Outbreak

- Synonymous with epidemic. Sometimes used to refer to a local epidemic as compared to a larger, general epidemic.

Outcome Measurement

- Standardized tools: FIM, Barthel, MRC, ASIA (spinal cord), NIHSS (stroke), etc. Track progress, adjust treatment.

Outcome Research

- Study of rehabilitation outcomes. Effectiveness, cost-effectiveness, patient-reported outcomes.

Outer ear

- The external part of the ear, as well as the external ear canal and the eardrum.

Outpatient Rehabilitation

- Therapy sessions 2-3 times per week. For conditions requiring ongoing therapy: post-surgical, chronic pain, neurological conditions.

Ovarian Cancer

- See oncology.

Ovarian Cyst

- A fluid-filled sac within or on the ovary, most commonly functional (follicular or corpus luteum cysts) related to the menstrual cycle. Usually asymp-

tomatic; can present with pelvic pain, pressure, and bloating. Complications: rupture, hemorrhage, torsion (sudden severe pain with nausea/vomiting).

Ovarian Hyperstimulation Syndrome (OHSS)

- A potentially serious complication of ovarian stimulation in assisted reproduction caused by increased vascular permeability and fluid shift. It produces enlarged ovaries, abdominal distension, ascites, hemoconcentration, electrolyte abnormalities, thrombosis risk, and, in severe cases, organ dysfunction.

Ovarian Pregnancy

- A rare ectopic pregnancy (0.5-3%) implanted on the ovary. Diagnosis: Spiegelberg criteria (ipsilateral fallopian tube intact, gestational sac on the ovary, connected to the ovary by the uterovarian ligament, ovarian tissue in the sac wall). Treatment: ovarian wedge resection or oophorectomy.

Ovarian Torsion

- Twisting of the ovary (and often the fallopian tube) on its vascular pedicle, causing obstruction of blood flow—a gynecologic emergency. Risk: ovarian cysts/tumors (especially >5 cm), longer uterovarian ligament, younger women, pregnancy.

Ovariectomy

- Surgical removal of one or both ovaries.

Ovary

- The female gonad, a paired almond-shaped organ located in the pelvic cavity on each side of the uterus. Functions: production of oocytes (ova) and secretion of female sex hormones (oestrogen, progesterone, and small amounts of androgens). Each ovary contains follicles at various stages of development.

Over-the-Counter (OTC)

- Medications available without a prescription, considered safe and effective for self-care with appropriate labeling and low abuse potential. Examples: acetaminophen, ibuprofen, antacids, antihistamines, decongestants, laxatives. Regulated in the US by the FDA (OTC monograph system) and Rx-to-OTC switches (e.g., omeprazole, loratadine).

Over-the-Counter (OTC) Medications

- Nonprescription drugs approved as safe and effective for self-management of common symptoms,

available at pharmacies and retail stores without a prescription.

Over-the-Counter Drug

- A medication that can be purchased without a prescription, deemed safe and effective for self-medication when used according to label instructions. OTC drugs have established safety profiles, low toxicity, and clear dosing guidelines suitable for consumer use.

Overactive Bladder

- A clinical syndrome of urinary urgency, usually with frequency and nocturia, with or without urge incontinence, due to detrusor overactivity during filling. Causes: idiopathic (most), neurogenic (stroke, spinal injury, MS, Parkinson), bladder outlet obstruction, and aging; worsened by caffeine and diuretics.

Overall Survival

- Time from treatment to death from any cause.

Overdenture

- Removable denture supported by implants or natural teeth. Preservation of alveolar bone, improved stability.

Overdose

- Harmful effect from exposure to toxic substance. See poisoning.

Overflow Urinary Incontinence

- Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

Overgrowth Syndrome

- A group of congenital disorders characterized by excessive growth of the body, a region, or specific tissues, often with developmental or tumor-predisposition features. Causes include genomic imprinting abnormalities, mosaic variants, and alterations in growth-signaling pathways.

Overlap Myositis

- Patients with features of inflammatory myopathy (DM/PM/IMNM) plus another systemic autoimmune disease (SLE, SSc, MCTD, Sjogren). Anti-PM-Scl (Scl/PM overlap, limited SSc + myositis, good prognosis); anti-Ku (SLE/PM overlap); anti-U1-RNP (MCTD with myositis).

Ovulation

- The release of a mature egg from the ovary, at which time it is available to be fertilized by sperm.

Ovum

- The female reproductive cell (egg) after ovulation and before fertilization. The ovum is a large, non-motile cell (100 μ m diameter) surrounded by the zona pellucida (glycoprotein layer that mediates sperm binding and prevents polyspermy after fertilization) and the corona radiata (follicular cells).

Oxalate

- A metabolic end product that can form calcium oxalate crystals in the kidneys, contributing to nephrolithiasis and oxalate nephropathy, particularly in hyperoxaluric states.

Oxaloacetate

- A four-carbon citric-acid-cycle intermediate that condenses with acetyl-CoA to form citrate. It is also a precursor for gluconeogenesis and participates in amino-acid transamination to aspartate.

Oxidative Phosphorylation

- Process generating ATP using energy released by electron transfer through the electron transport chain. Protons pumped across inner mitochondrial membrane create electrochemical gradient (proton motive force). ATP synthase (Complex V) uses proton flow to phosphorylate ADP to ATP. P/O ratio: 2.5 ATP per NADH, 1.5 ATP per FADH₂.

Oxidative Stress

- Imbalance between reactive oxygen species and antioxidants. DNA damage, protein oxidation, lipid peroxidation.

Oxygen Consumption

- The volume of oxygen consumed by the body per unit time, reflecting overall metabolic rate. Resting oxygen consumption (VO₂) in a normal adult is approximately 250 mL/min (about 3.5 mL/kg/min, 1 MET). It is calculated as cardiac output $\times$ (arterial O₂ content - venous O₂ content) by the Fick principle.

Oxygen Delivery

- The rate at which oxygen is transported from the lungs to the tissues, equal to cardiac output multiplied by arterial oxygen content: $DO_2 = CO \times CaO_2$. Arterial oxygen content = $(1.34 \times \text{hemoglobin} \times$

$SaO_2) + (0.003 \times PaO_2)$. Normal oxygen delivery is approximately 1000 mL/min.

Oxygen Enhancement Ratio

- Ratio of dose under hypoxic vs. aerobic conditions. Higher for low LET radiation.

Oxygen Mask

- Delivers supplemental oxygen. Types: simple, nasal cannula, non-rebreather, Venturi.

Oxygen Saturation

- The percentage of hemoglobin saturated with oxygen in arterial blood; normal 95-100 percent. Measured noninvasively by pulse oximetry (SpO₂) and invasively by co-oximetry (SaO₂). Target levels vary by context: >94 percent general; 88-92 percent in patients with hypercapnic respiratory failure (COPD); permissive targets in ARDS.

Oxygen-Hemoglobin Dissociation Curve

- The sigmoidal relationship between partial pressure of oxygen (PO₂) and hemoglobin oxygen saturation (SO₂). The curve reflects hemoglobin's cooperative binding: oxygen binding to one subunit increases affinity for subsequent oxygen molecules. At PO₂ of 100 mmHg (alveolar), hemoglobin is approximately 98 percent saturated.

Oxytocics

- A class of medications that stimulate uterine contractions, used for induction and augmentation of labor and prevention/treatment of postpartum hemorrhage. Oxytocin (Syntocinon) — synthetic analog of endogenous oxytocin, administered IV infusion for labor induction or IM/IV for PPH prevention.

Oxytocin Infusion

- Synthetic oxytocin (Pitocin) administered intravenously for induction or augmentation of labor. Standard dosing: start at 1-2 mU/min, increase by 1-2 mU/min every 15-30 minutes to achieve 200-250 Montevideo units (MVU, measured by IUPC). Maximum dose: 40 mU/min. After achieving adequate labor, the dose may be maintained or decreased.

Ozaena

- A chronic condition characterized by atrophic rhinitis with crusting, foul-smelling nasal discharge (fetor), and a sensation of nasal obstruction despite a patent airway. Ozaena is caused by progressive atrophy of the nasal mucosa and underlying bone,

often associated with *Klebsiella ozaenae* infection.

Obstructive Nephropathy

- Kidney damage caused by blockage of urine flow anywhere from the kidney to the urethra. Causes include stones, tumors, strictures, enlarged prostate, and congenital abnormalities; untreated obstruction can lead to hydronephrosis, infection, and kidney failure.

Ophthalmoplegia

- Weakness or paralysis of one or more eye muscles, causing limited eye movement, double vision, or abnormal eye position. Causes include nerve disorders, stroke, diabetes, thyroid disease, trauma, and neuromuscular conditions.

Ocular Pain

- Pain in or around the eye caused by surface irritation, infection, inflammation, trauma, glaucoma, or orbital disease. Severe pain with vision loss, light sensitivity, nausea, or injury requires urgent ophthalmic assessment.

Oophoritis

- Inflammation of an ovary, usually associated with pelvic infection or, less commonly, autoimmune disease. It may cause lower abdominal or pelvic pain, fever, abnormal discharge, or menstrual changes; severe pain requires assessment to exclude ovarian torsion or abscess.

Chapter 16

Pocket Dictionary: P Terms

P Wave

- The first deflection on the ECG, representing atrial depolarization. Normal P waves are upright in leads I and II, inverted in aVR, and less than 0.12 seconds (3 small boxes) in duration and 2.5 mm in amplitude.

P2Y12 Inhibitor

- An antiplatelet drug that blocks the platelet P2Y12 ADP receptor, reducing activation and aggregation. Clopidogrel, prasugrel, and ticagrelor are used in selected acute coronary syndrome and percutaneous coronary intervention settings, with bleeding and drug-specific contraindications considered.

p53 Tumor Suppressor

- The p53 protein, encoded by the TP53 gene on chromosome 17p, is a transcription factor and the most commonly mutated gene in human cancer, with mutations found in over 50 percent of all malignancies.

Pacemaker

- An implanted electronic device that delivers electrical impulses to the heart to regulate heart rate and rhythm. Indications include symptomatic bradycardia, high-degree AV block, and sick sinus syndrome. Single-chamber pacemakers pace the atrium or ventricle; dual-chamber pacemakers pace both.

Pacemaker Complications

- Early: pneumothorax, hematoma, cardiac perforation/tamponade, lead dislodgement. Late: lead fracture, battery depletion, infection, tricuspid regurgitation from lead interference, venous thrombosis, and pacemaker syndrome (AV dyssynchrony in VVI pacing).

Pachygyria

- A cortical malformation in which the cerebral gyri are abnormally broad and reduced in number be-

cause of impaired neuronal migration. It may cause developmental delay, intellectual disability, seizures, and motor or neurological impairment of variable severity.

Pachymeningitis

- Inflammation of the dura mater (pachymeninx), the outermost layer of the meninges. Pachymeningitis is less common than leptomeningitis (inflammation of pia and arachnoid mater).

Pacing

- Planned regulation of activity, rest, and intensity to avoid symptom exacerbation while gradually increasing tolerance. It is used in fatigue, chronic pain, cardiopulmonary disease, and post-viral conditions.

PACS

- Picture Archiving and Communication System. Digital image storage and retrieval.

PAD Management

- Atherosclerotic occlusive disease of arteries supplying the lower limbs, causing reduced blood flow. See peripheral arterial disease.

Page kidney

- Renal compression from a perinephric hematoma or mass causing hypertension through activation of the renin-angiotensin system, typically following trauma or occurring in patients on anticoagulation.

Paget Disease of Bone

- A chronic bone disorder characterized by excessive, disorganized bone remodeling with increased osteoclastic bone resorption followed by compensatory osteoblastic bone formation, producing structurally abnormal bone that is enlarged, weak, and prone to fracture.

Paget's Disease

- A chronic bone disorder characterized by excessive, disorganized bone remodeling with increased osteoclastic bone resorption followed by compensatory osteoblastic bone formation, producing structurally abnormal bone that is enlarged, weak, and prone to fracture. See Paget Disease of Bone.

Pain

- An unpleasant sensory and emotional experience associated with actual or potential tissue damage, serving as a protective alerting system but becoming maladaptive when chronic (lasting >3 months).

Pain (Geriatric)

- Emergency pain control: NSAIDs, opioids, ketamine, lidocaine. See pain management.

Pain (Vascular)

- Claudication pain with walking. Rest pain indicates critical ischemia.

Pain Assessment Scales

- Pain assessment uses standardized scales to quantify pain intensity. Numeric rating scale (NRS) asks patients to rate pain from 0 (no pain) to 10 (worst pain imaginable). Visual analog scale (VAS) is a 10 cm line with anchors. Faces pain scale (Wong-Baker FACES) uses facial expressions for children and cognitively impaired patients.

Pain Management

- Emergency pain control: NSAIDs, opioids, ketamine, lidocaine. Assessment: location, intensity, character, timing, aggravating/alleviating factors. Pain is the fifth vital sign. Multimodal approach reduces opioid requirements. Neuropathic pain: gabapentin, pregabalin.

Pain Relief (Analgesia)

- Achieved through pharmacologic, interventional, physical, and psychologic modalities targeting nociceptive transmission and pain perception.

Pair Production

- Photon interaction producing electron-positron pair. Requires energy >1.022 MeV.

Paleness

- Unusual lightening of the skin or mucous membranes caused by reduced blood flow, vasoconstriction, anemia, shock, hypothermia, or other systemic

illness. Sudden paleness with weakness, bleeding, chest pain, or fainting requires urgent assessment.

Palindromic Rheumatism

- Characterized by recurrent, self-limited attacks of mono- or oligoarthritis (hours to days), periarticular soft tissue swelling, no radiographic erosions between attacks. 30-50% evolve to RA (anti-CCP positive predicts progression).

Palliative Care

- Specialized medical care for serious illness focusing on symptom relief and quality of life. Not limited to end-of-life. Can be provided alongside curative treatment. Components: pain management, symptom control, psychosocial support, spiritual care, advance care planning. Referral early in disease course.

Palliative Rehabilitation

- Rehabilitation for patients with terminal illness. Focus on comfort, function, quality of life.

Pallister-Hall Syndrome

- A congenital disorder usually caused by pathogenic variants in GLI3. It is characterized by hypothalamic hamartoma and mesoaxial polydactyly and may include imperforate anus, renal or airway anomalies, and pituitary hormone abnormalities.

Pallor

- Paleness of the skin and mucous membranes, caused by reduced cutaneous blood flow (vasoconstriction—cold, shock, anemia, fear) or reduced hemoglobin/pigmentation.

Palmar erythema of pregnancy

- Symmetrical redness of the palms occurring during pregnancy, thought to be related to elevated circulating oestrogen levels causing vasodilatation. It is a benign, physiological finding that typically resolves after delivery and requires no specific treatment.

Palmar Grasp Reflex

- A primitive reflex present from birth to 5-6 months, elicited by placing a finger in the infant's palm. The infant grips tightly, strong enough to support the infant's weight briefly. Bilateral symmetry assessed; asymmetry may indicate birth injury. Disappears as voluntary grasping develops around 3-4 months.

Palpation

- The second step of the physical examination, uti-

lizing the sense of touch to evaluate body structures for size, shape, consistency, mobility, tenderness, temperature, and vibration.

Palpebral Conjunctiva

- The palpebral conjunctiva is the portion of the conjunctival membrane that lines the inner surface of the upper and lower eyelids, and it is subdivided into the tarsal conjunctiva, which is firmly adherent to the tarsal plates and appears smooth and pink, and the forniceal conjunctiva, which forms loose, redundant folds at the superior an

Palpitation

- The subjective sensation of a rapid, forceful, or irregular heartbeat. Palpitations are a common symptom with a broad differential.

Palsy of the long thoracic nerve

- Weakness or paralysis of the long thoracic nerve (C5-C7), which innervates the serratus anterior muscle. Causes include trauma, compression, viral infection, or idiopathic. Results in winging of the scapula, particularly when pushing against a wall, and impaired overhead arm elevation. Most cases resolve spontaneously.

Pancoast Syndrome

- A clinical syndrome caused by an apical lung tumour invading the lower brachial plexus, sympathetic chain, or adjacent structures. It may produce severe shoulder and arm pain, hand weakness, sensory loss, and Horner syndrome.

Pancoast Tumor

- Superior sulcus tumor invading brachial plexus, subclavian vessels, and sympathetic chain. Causes Horner syndrome, shoulder/arm pain, and hand muscle atrophy. Treated with chemoradiation followed by surgery.

Pancreas

- Retroperitoneal gland (15-20 cm) behind the stomach, spanning from duodenum to spleen. Exocrine function: acinar cells secrete digestive enzymes (amylase, lipase, proteases) and bicarbonate into pancreatic duct to duodenum.

Pancreas Divisum

- A congenital anomaly in which the dorsal and ventral pancreatic ducts fail to fuse. Most people are asymptomatic, but impaired drainage through the

minor papilla can predispose to recurrent pancreatitis.

Pancreatectomy

- Removal of all (total) or part of the pancreas. Pancreaticoduodenectomy (Whipple) is used for pancreatic head and periampullary tumors; distal pancreatectomy for body and tail lesions.

Pancreatic Abscess

- A localized collection of pus within or adjacent to the pancreas, usually developing as a complication of infected pancreatic necrosis or severe pancreatitis. It may cause persistent fever, abdominal pain, sepsis, and organ failure.

Pancreatic Adenocarcinoma

- A highly lethal malignancy arising from the pancreatic ductal epithelium, with a five-year survival rate of less than 10 percent, making it one of the deadliest cancers. The most common risk factors include smoking, obesity, chronic pancreatitis, diabetes mellitus, and hereditary syndromes including BRCA2 mutations and Lynch syndrome.

Pancreatic Cancer

- Aggressive malignancy with pancreatic ductal adenocarcinoma accounting for approximately eighty-five percent of cases. It is the fourth leading cause of cancer-related death with a five-year survival of approximately ten percent. Risk factors include smoking, chronic pancreatitis, diabetes, and genetic syndromes.

Pancreatic Duct

- The main duct of the pancreas, conveying exocrine pancreatic secretions to the duodenum. It commonly joins the common bile duct before opening at the major duodenal papilla.

Pancreatic Elastase

- Measurement of fecal elastase-1 (FE-1) as a non-invasive test for pancreatic exocrine insufficiency. Values: >200 mcg/g (normal), 100-200 (mild-moderate insufficiency), <100 mcg/g (severe insufficiency). Unaffected by pancreatic enzyme replacement therapy (does not cross-react with porcine enzymes).

Pancreatic Enzyme

- Digestive enzymes secreted by pancreatic acinar cells into the duodenum, including proteases

(trypsinogen, chymotrypsinogen, proelastase, procarboxypeptidases), amylase (starch digestion), and lipase (triglyceride digestion), along with bicarbonate from ductal cells to neutralize gastric acid.

Pancreatic Exocrine Insufficiency

- Inadequate secretion or delivery of pancreatic digestive enzymes and bicarbonate, causing maldigestion and malabsorption. Chronic pancreatitis, cystic fibrosis, pancreatic resection, and advanced pancreatic disease are important causes; steatorrhea and weight loss may respond to pancreatic enzyme replacement.

Pancreatic Insufficiency

- See gastroenterology. Pancreatic enzyme replacement with meals. Fat-soluble vitamin supplementation.

Pancreatic Mass

- Abnormality in pancreas detected on imaging. Pancreatic adenocarcinoma most common malignancy. Neuroendocrine tumors also important.

Pancreatic Neuroendocrine Tumors

- Pancreatic neuroendocrine tumors (pNETs): functional (insulinoma, gastrinoma, glucagonoma, VIPoma, somatostatinoma) or non-functional (60-70%). WHO 2017 grading: NET G1 (Ki-67 <3%, mitotic <2/10 HPF), NET G2 (Ki-67 3-20%, mitotic 2-20/10 HPF), NEC G3 (Ki-67 >20%, mitotic >20/10 HPF).

Pancreatic Pseudocyst

- A pancreatic pseudocyst is a localized collection of pancreatic fluid surrounded by a wall of granulation tissue without an epithelial lining, typically occurring as a complication of acute or chronic pancreatitis. Unlike true cysts, pseudocysts lack an epithelial lining, which is why they are called pseudocysts.

Pancreatic Surgery

- Pancreaticoduodenectomy (Whipple): pancreatic head, duodenum, gallbladder, distal stomach (pylorus-preserving standard), common bile duct; reconstruction: pancreaticojejunostomy (duct-to-mucosa or invagination), hepaticojejunostomy, gastrotrojejunostomy/duodenojejunostomy.

Pancreatic Trauma

- Pancreatic injury from trauma. CT diagnosis. May

require surgery.

Pancreatitis

- Inflammation of pancreas. CT diagnosis. Edematous vs. necrotizing.

Pancreatoblastoma

- A rare malignant pancreatic epithelial tumour occurring mainly in children. It may present with an abdominal mass, pain, vomiting, weight loss, or hormonal symptoms and requires imaging, tissue diagnosis, staging, and multidisciplinary treatment.

Pancreatography

- Contrast imaging of pancreatic duct. ERCP approach. Evaluates pancreatic duct abnormalities.

Pancytopenia

- Simultaneous reduction of red cells, white cells, and platelets, indicating bone marrow failure or infiltration.

Pandemic

- Epidemic spread across multiple countries/continents. Examples: COVID-19, Spanish flu, HIV/AIDS.

Paneth Cell

- Specialized secretory epithelial cell at the base of a small-intestinal crypt that produces antimicrobial substances.

Panic Attack

- An abrupt surge of intense fear or discomfort reaching a peak within minutes.

Panic Disorder

- Recurrent unexpected panic attacks, which are abrupt surges of intense fear or discomfort peaking within minutes.

Panlobular Emphysema

- A subtype involving diffuse destruction of the entire acinus. It predominantly affects the lower lobes and is classically associated with alpha-1 antitrypsin deficiency. The destruction is more uniform compared to centriacinar emphysema.

Panniculitis

- Inflammation of the subcutaneous fat, presenting as tender, erythematous or violaceous nodules and plaques, most commonly on the lower extremities. The classification is based on whether the septa or lobules of the fat are primarily affected, and the

aetiology includes infections, autoimmune diseases, medications, and pancreatic disease.

Panoramic Radiograph

- An extraoral radiograph displaying both jaws, teeth, temporomandibular regions, and surrounding structures in one image. It is useful for surveying dentition, development, impaction, and gross osseous pathology but has less fine detail than intraoral radiographs.

Panretinal Photocoagulation

- A laser treatment for proliferative diabetic retinopathy that involves applying laser burns to the peripheral retina (typically 1,200 to 2,000 burns over three to four sessions) to ablate ischemic retinal tissue, reducing the production of vascular endothelial growth factor and other angiogenic factors that drive pathological

Pantoprazole

- A proton pump inhibitor reducing gastric acid secretion by irreversibly inhibiting the H⁺/K⁺ ATPase of parietal cells. Used for GERD, peptic ulcer disease, dyspepsia, H. pylori eradication, and acid hypersecretory states. Available oral and IV (used for stress ulcer prophylaxis in the ICU).

Panuveitis

- Inflammation involving all segments of the uveal tract, including the anterior chamber, vitreous cavity, and choroid, resulting in a severe, potentially sight-threatening condition that requires aggressive systemic immunosuppressive therapy.

Pap Smear

- Papanicolaou test (cervical cytology): a screening test collecting cervical cells (conventional or liquid-based) to detect precancerous cervical changes (CIN) and cancer, usually combined with HPV DNA testing (co-testing).

Papillary Muscle

- Muscular projections of the ventricular myocardium attached by chordae tendineae to the atrioventricular valve leaflets (anterolateral and posteromedial in the left ventricle; anterior, septal, and posterior in the right ventricle).

Papillary Thyroid Carcinoma

- Most common histological subtype of thyroid carcinoma accounting for approximately eighty percent

of cases. Characterized by nuclear features including nuclear grooves, intranuclear cytoplasmic inclusions, and Orphan Annie eye nuclei. Often multifocal and bilateral with lymph node metastases.

Papilledema

- Swelling of the optic disc caused by elevated intracranial pressure, and it is a critical clinical sign that requires urgent investigation to identify the underlying cause, which may be life-threatening.

Papilloma

- A benign epithelial neoplasm characterised by finger-like projections of fibrovascular tissue covered by hyperplastic epithelium, most commonly caused by human papillomavirus infection in cutaneous warts or mucosal sites. Cutaneous papillomas present as flesh-coloured, rough, or pedunculated growths.

Papillomatosis cutis lymphostatica

- Diffuse, warty, papillomatous thickening of the skin resulting from chronic lymphoedema, most commonly affecting the lower extremities in the setting of long-standing venous or lymphatic insufficiency. The skin develops cobblestone-like hyperkeratotic papules and plaques due to lymph stasis and chronic inflammation.

Papular

- A papule: a small, solid, elevated skin lesion <1 cm in diameter (palpable), of various colors and etiologies.

Papular Urticaria

- A hypersensitivity reaction to insect bites, especially from fleas, bedbugs, or mosquitoes, causing recurrent intensely itchy papules or small vesicles, often clustered on exposed areas in children. Diagnosis is clinical.

Papule

- A solid, raised skin lesion, conventionally less than 1 cm in diameter. Papules may arise from epidermal, dermal, inflammatory, infectious, or neoplastic processes.

Paracentesis

- Percutaneous needle aspiration of ascitic fluid from the peritoneal cavity. Used for diagnostic analysis to determine the cause of ascites and for therapeutic removal of large volumes for symptom relief.

The serum-ascites albumin gradient is calculated to differentiate portal hypertensive from non-portal hypertensive causes.

Paracetamol overdose

- Toxic ingestion of paracetamol that can cause dose-dependent hepatic necrosis; risk assessment uses the timing and serum concentration of the ingestion.

Paracetamol Toxicity

- See toxicology.

Paraesophageal hernia

- See Hiatal Hernia.

Paraganglioma

- A neuroendocrine tumor arising from extra-adrenal paraganglia, related to pheochromocytoma; may secrete catecholamines. Associations with hereditary syndromes (SDH mutations, VHL, NF1) warrant genetic counseling.

Paragonimiasis

- See Parasitic Infections.

Parainfluenza

- Infection caused by human parainfluenza viruses, commonly producing croup, laryngitis, bronchiolitis, or pneumonia in children. Adults often have a mild upper-respiratory illness.

Paranasal Sinuses

- The paranasal sinuses are four pairs of air-filled, mucosa-lined cavities within the facial bones that communicate with the nasal cavity through narrow drainage pathways called ostia, and they serve multiple functions including lightening the weight of the skull, humidifying and warming inspired air, producing mucus for nasal clearance, c

Paraneoplastic disorder

- A systemic syndrome caused by a malignancy through mechanisms other than direct invasion, metastasis, or metabolic effects of the tumor. Paraneoplastic syndromes result from hormone production, immune-mediated mechanisms, or cytokine release by the tumor, affecting distant organs.

Paraneoplastic Neurologic Syndrome

- Neurologic symptoms from immune response to cancer. Examples: Lambert-Eaton, cerebellar degeneration.

Paraneoplastic Syndrome

- Symptoms caused by cancer but not by tumor mass or metastasis. Mechanisms: hormone production, immune cross-reaction. Examples: SIADH, Cushing syndrome, Lambert-Eaton myasthenia, hypercalcemia.

Paranoia

- A thought process characterized by excessive or irrational suspiciousness and distrust of others, often manifesting as delusions of persecution, reference, or grandeur.

Paraphilia

- A condition characterized by intense and persistent sexual interest in atypical objects, situations, or individuals. Examples: exhibitionism, fetishism, frotteurism, pedophilic disorder, sexual masochism, sexual sadism, transvestic disorder, voyeurism, and paraphilia not otherwise specified.

Paraphimosis

- Urological emergency in uncircumcised males occurring when a retracted, tight foreskin becomes trapped behind the coronal sulcus of the glans penis, forming a constricting ring.

Parapneumonic Effusion

- Pleural effusion complicating pneumonia. Simple effusions resolve with antibiotics. Complicated effusions (pH < 7.2, glucose < 60 mg/dL) require chest tube drainage.

Parapneumonic Pleural Effusion

- Pleural fluid accumulation associated with bacterial pneumonia. It may be uncomplicated or progress to a complicated effusion or empyema, requiring imaging, pleural-fluid analysis, antibiotics, and sometimes drainage.

Paraquat

- Herbicide, highly toxic if ingested. Pulmonary fibrosis, renal failure. No antidote. Poor prognosis.

Paraseptal Emphysema

- A form of emphysema characterized by destruction of alveoli adjacent to the pleura. It typically affects young adults and predisposes to spontaneous pneumothorax due to rupture of subpleural blebs. It may be associated with Marfan syndrome.

Parasite

- An organism that lives on or within a host organism, deriving nutrients at the host's expense.

Parasitemia

- The presence and measurable burden of parasites in the blood. It is used particularly for malaria and some blood-borne protozoal infections and may correlate with disease severity.

Parasitic Infections

- Malaria: Plasmodium (falciparum, vivax, ovale, malariae, knowlesi). Falciparum: cerebral malaria, severe anemia, ARDS, AKI, hypoglycemia. Diagnosis: thick/thin smear, RDT (HRP2 for P. falciparum, pLDH for all species, aldolase). Amoebiasis: E. histolytica, stool antigen, PCR. Giardiasis: stool antigen, duodenal aspirate.

Parasitology

- The branch of microbiology that studies parasites, including protozoa, helminths, and ectoparasites. Parasitic infections are a major global health problem, particularly in tropical and subtropical regions. Protozoan parasites include Plasmodium (malaria), Trypanosoma (sleeping sickness, Chagas disease), and Leishmania (leishmaniasis).

Parasympathetic Nervous System

- The craniosacral division of the autonomic nervous system mediating 'rest-and-digest' functions. Preganglionic fibers from cranial nerves III, VII, IX, X, and sacral nerves S2-S4 synapse on ganglia near or within target organs (short postganglionic fibers).

Parasympatholytic

- A drug that blocks muscarinic (cholinergic) receptors, inhibiting parasympathetic nervous system activity. Agents: atropine (bradycardia, organophosphate poisoning), ipratropium/tiotropium (COPD/asthma—bronchodilation), scopolamine (motion sickness), oxybutynin/tolterodine (overactive bladder), and glycopyrrolate.

Parasympathomimetic

- A drug that mimics or enhances parasympathetic (cholinergic) activity. Direct agonists: muscarinic (pilocarpine—glaucoma, dry mouth) and nicotinic agonists.

Parathyroid

- One of the small endocrine glands behind the thy-

roid that secretes parathyroid hormone and regulates calcium and phosphate balance.

Parathyroid Crisis

- Severe hypercalcemia with calcium greater than fourteen milligrams per deciliter causing altered mental status, cardiac arrhythmias, and acute kidney injury. Most commonly occurs in patients with primary hyperparathyroidism precipitated by dehydration, immobility, or thiazide diuretics.

Parathyroid Hormone Physiology

- Parathyroid hormone is the primary regulator of calcium homeostasis secreted by the four parathyroid glands in response to low ionized calcium levels detected by the calcium-sensing receptor.

Parathyroid Hyperplasia

- Enlargement and increased cellularity of one or more parathyroid glands causing excessive parathyroid hormone secretion and primary hyperparathyroidism. It may occur sporadically or as part of multiple endocrine neoplasia.

Parathyroid Surgery

- Primary hyperparathyroidism (PHPT): single adenoma (85%), hyperplasia (15%), double adenoma (2%), carcinoma (<1%). Criteria for surgery (NIH): age <50, calcium >1 above upper limit, creatinine clearance <60, BMD T-score <-2.5, vertebral fracture, nephrolithiasis, nephrocalcinosis.

Parathyroidectomy

- Surgical removal of parathyroid glands for refractory secondary hyperparathyroidism unresponsive to medical therapy. Indicated when calcium-phosphorus product persistently elevated despite maximal medical management, symptomatic bone disease, or calciphylaxis. Subtotal or total parathyroidectomy with autotransplantation.

Parenteral Nutrition

- Intravenous nutrition bypassing GI tract. Total parenteral nutrition (TPN) provides complete nutrition. Indications: bowel failure, short bowel syndrome, prolonged NPO, malabsorption. Risks: infection, liver dysfunction, metabolic complications, line complications.

Paresthesia

- An abnormal sensation in the skin, described as tingling, prickling, numbness, burning, or a pins-

and-needles sensation, caused by dysfunction of the sensory nerves. It can be caused by any condition that affects the peripheral nerves, nerve roots, or central sensory pathways.

Parietal Bones

- Paired curved bones forming the cranial vault roof and sides. Articulate at the sagittal suture, with frontal at coronal, temporal at squamous, and occipital at lambdoid sutures. The pterion (where four bones meet) overlies the middle meningeal artery; fractures here may cause epidural hematomas.

Parietal Lobe

- The cerebral lobe posterior to the central sulcus and above the lateral sulcus, containing the primary somatosensory cortex (postcentral gyrus, Brodmann 3,1,2) processing touch, pain, temperature, and proprioception, plus the somatosensory association cortex.

Parinaud Syndrome

- A clinical condition caused by lesions of the dorsal midbrain, particularly the pretectal area and superior colliculi, resulting in a characteristic triad of vertical gaze palsy (particularly impaired upward gaze), convergence-retraction nystagmus (involuntary retraction of the eyeballs into the orbits with attempted upward gaze), and

Parkinson Disease

- A progressive neurodegenerative disorder characterized by the loss of dopaminergic neurons in the substantia nigra pars compacta of the midbrain, leading to reduced dopamine levels in the striatum and the cardinal motor features of bradykinesia, resting tremor, rigidity, and postural instability.

Parkinson's Disease

- A progressive neurodegenerative disorder caused by selective loss of dopaminergic neurons in the substantia nigra pars compacta, leading to dopamine depletion in the striatum. The pathological hallmark is the presence of Lewy bodies, intracellular inclusions composed of aggregated alpha-synuclein protein, within surviving neurons.

Parkinsonism

- A clinical syndrome characterized by bradykinesia, rigidity, tremor (resting, pill-rolling), and postural instability. Idiopathic Parkinson disease: most

common cause, responds well to levodopa.

Paronychia

- Inflammation of the nail fold. Acute paronychia is bacterial (*Staphylococcus aureus*, *Streptococcus*), presenting as painful, erythematous, swollen nail fold with possible abscess. Treatment includes warm compresses, antibiotics, and incision and drainage if abscessed.

Parotid Gland

- The largest of the three major salivary glands, located in the preauricular region, extending from the zygomatic arch superiorly to the angle of the mandible inferiorly. The parotid gland is encased in the parotid fascia (investing layer of deep cervical fascia).

Parotitis

- Inflammation of the parotid gland. Acute parotitis: most commonly caused by mumps virus (paramyxovirus—epidemic parotitis, typically in unvaccinated children, bilateral in 70%, resolves spontaneously over 7–10 days).

Paroxysmal Nocturnal Dyspnea

- Episodes of severe dyspnea that awaken the patient from sleep, typically occurring 1–2 hours after lying down. It is a classic symptom of left heart failure, caused by progressive pulmonary congestion during sleep. It is different from orthopnea in its delayed onset and nocturnal presentation.

Paroxysmal Nocturnal Hemoglobinuria

- A rare, acquired clonal disorder of hematopoietic stem cells characterized by complement-mediated intravascular hemolysis, thrombosis (particularly in unusual sites such as hepatic, portal, and cerebral veins), and bone marrow failure.

Paroxysmal Supraventricular Tachycardia

- Sudden-onset and sudden-termination episodes of rapid regular tachycardia arising above the ventricles, usually involving an atrioventricular re-entry or nodal re-entry circuit. Palpitations, dizziness, chest discomfort, or syncope may occur.

Partial Agonist

- A partial agonist is a drug that binds to a receptor and produces a biological response that is less than the maximum possible response, even when all receptors are occupied. Partial agonists have

affinity for the receptor but lower intrinsic activity compared to full agonists, producing a submaximal response regardless of dose.

Partial Denture

- Removable prosthesis replacing some teeth. Metal framework or acrylic. Requires remaining teeth for support.

Partial Response

- Significant tumor reduction but not complete.

Partial Thromboplastin Time

- aPTT/PTT: a coagulation test measuring the intrinsic and common pathways (factors XII, XI, IX, VIII, X, V, II, fibrinogen), used to monitor unfractionated heparin and detect factor deficiencies.

Partial Volume Artifact

- CT artifact from voxel containing multiple tissues. Reduced by thinner slices.

Partial-Volume Effect

- Underestimation or distortion of measured activity in small or low-contrast structures because the object size approaches the scanner's spatial resolution. It affects quantitative PET and SPECT and may require recovery or correction methods.

Particle (Vascular)

- An embolic particle used to occlude selected vessels under image guidance.

Particulate Matter

- See air pollution.

Partograph (Partogram)

- A graphic record of cervical dilatation, fetal descent, and uterine contractions over time, used to monitor labor progress and identify labor abnormalities. The WHO partograph has an alert line (1 cm/hour dilatation) and an action line (4 hours after the alert line, indicating need for intervention).

Parturition (Childbirth)

- The process of giving birth, encompassing three stages: (1) First stage: from the onset of regular uterine contractions (cervical effacement and dilation) to full dilation (10 cm). Latent phase—slow dilation (0–4 cm) over hours, variable duration. Active phase—more rapid dilation (4–10 cm), typically 1–1.5 cm/hour in nulliparous women.

Parvovirus B19

- Human parvovirus B19: a small DNA virus causing erythema infectiosum ('fifth disease') in children, transmitted by respiratory droplets.

Parvovirus B19 infection

- See Erythema Infectiosum.

Pasteurella multocida

- See Bacterial Infections.

Pasteurization

- A heat treatment reducing microbial load in liquids by heating below boiling point to inactivate vegetative pathogens and spoilage organisms while preserving quality.

Patau Syndrome

- Trisomy 13, affecting 1 in 10,000-20,000 live births. Features: holoprosencephaly (most characteristic, incomplete forebrain division), microcephaly, microphthalmia or anophthalmia, cleft lip and palate, polydactyly (postaxial), clenched fists with overlapping fingers, and congenital heart defects (VSD, PDA, dextrocardia, 80%).

Patch

- A flat, circumscribed area of altered skin color larger than approximately 1 cm. It is the larger counterpart of a macule and may be caused by pigmentary, vascular, inflammatory, or neoplastic disease.

Patch (Vascular)

- Vascular graft for patch angioplasty. See Patch Graft.

Patch Graft

- Vascular graft for patch angioplasty. Bovine pericardium, PTFE.

Patch Testing

- Patch testing identifies allergens causing allergic contact dermatitis. The standard series includes 35 allergens. Additional custom trays include occupational and personal allergens. Patches are applied to the upper back for 48 hours and removed. Reading at 48 hours (initial) and 72 to 96 hours (delayed) identifies positive reactions.

Patella

- Largest sesamoid bone, embedded in the quadriceps tendon. Articulates with the trochlear groove form-

ing the patellofemoral joint. Increases quadriceps mechanical advantage by approximately 30 percent. Triangular with inferior apex. Covered with the thickest articular cartilage (up to 7 mm).

Patella (Kneecap)

- The largest sesamoid bone in the human body, embedded within the quadriceps tendon anterior to the knee joint. The patella increases the lever arm of the quadriceps muscle (improving mechanical advantage for knee extension by 30%), protects the anterior knee joint, and distributes compressive forces across the femoral trochlea.

Patellofemoral Instability

- Patellar dislocation: lateral dislocation (95%), often reduces spontaneously. Risk factors: trochlear dysplasia (Dejour classification), patella alta (Caton-Deschamps >1.2), increased TT-TG distance (>20mm), MPFL insufficiency, femoral anteversion, genu valgum.

Patellofemoral Pain Syndrome

- Anterior knee pain related to the patellofemoral joint, common in adolescents and young adults (especially runners). Etiology: multifactorial (malalignment, quadriceps imbalance, overuse, tight lateral structures, weak VMO, and patellar instability).

Patent Ductus Arteriosus

- PDA: failure of the ductus arteriosus (fetal connection between pulmonary artery and aorta) to close after birth, causing a left-to-right shunt. Common in preterm infants; term—often small.

Patent Ductus Arteriosus (PDA)

- A congenital heart defect characterized by persistent patency of the ductus arteriosus, a vessel connecting the pulmonary artery to the proximal descending aorta, which normally closes within the first few days of life. The resulting left-to-right shunt leads to pulmonary overcirculation and left heart volume overload.

Patent Foramen Ovale

- A normal fetal communication between the right and left atria (the foramen ovale) that fails to close after birth. Present in 25-30% of the general population. Typically small and hemodynamically insignificant, with left-to-right shunting that is usually

not clinically important. Most PFOs are incidental findings on echocardiography.

Paternal uniparental disomy 14 (UDP 14 pat) thoracic dysplasia

- A rare genetic condition resulting from inheritance of both copies of chromosome 14 from the father, leading to dysregulated expression of imprinted genes on chromosome 14. It presents with severe thoracic hypoplasia, pulmonary hypoplasia, and characteristic facial features. It is a lethal condition in most cases.

Paternity Testing

- The use of genetic markers to establish or exclude a biological parent-child relationship, the reference method being short tandem repeat DNA profiling of the child, alleged father, and mother. Modern STR analysis commonly yields paternity probability exceeding 99.99 percent for true fathers.

Pathogen

- A tiny organism such as a virus, bacterium, or parasite that can invade the body and produce disease.

Pathogenesis

- Mechanism of disease development. Steps: initiation, promotion, progression.

Pathogenic Variant

- A genetic variant with sufficient validated evidence to be interpreted as causing or contributing to disease. Classification uses standardized evidence from population data, segregation, functional studies, computational analysis, and phenotype correlation.

Pathogenicity

- The capacity of a microorganism to cause disease in a susceptible host. Pathogenicity is a qualitative property, whereas virulence describes the relative degree of disease-causing capacity among strains or organisms.

Pathologic Fracture

- A fracture occurring through bone weakened by an underlying disease such as metastasis, myeloma, infection, benign tumor, or metabolic bone disorder. Evaluation includes characterization of the lesion and treatment of both the fracture and its cause.

Pathologic Hypertrophy

- The enlargement of an organ resulting from an increase in the size of its constituent cells without

an increase in cell number, typically occurring as a response to chronically increased workload or hormonal stimulation.

Pathology

- Study of disease. Includes general pathology (cell injury, inflammation, neoplasia) and systemic pathology (organ-specific diseases).

Pathology Artifact

- A distortion, alteration, or contaminant introduced during specimen collection, fixation, processing, sectioning, staining, or imaging rather than representing true disease. Artifacts must be recognized to avoid diagnostic error.

Patient

- Person receiving medical care. Also: patient zero (initial case), patient safety.

Patient Advocacy

- Acting on behalf of a patient to ensure that the patient's needs, rights, and preferences are addressed in healthcare.

Patient Communication

- Exchange of information between provider and patient. See clinical communication.

Patient Education

- Teach patient and family about condition, exercises, equipment, prevention. Improves adherence, outcomes.

Patient Positioning

- Supine, arms raised for PET. Comfort and reproducibility important.

Patient Preparation

- Fasting for FDG PET, hydration, bladder emptying. Medication review.

Patient Safety

- Prevention of harm from healthcare. Examples: medication errors, healthcare-associated infections, surgical errors. Strategies: checklists, hand hygiene, barcoding.

Patient-Centered Care

- Care respecting patient preferences, needs, values. Examples: shared decision-making, communication, coordination.

Patient-Centered Outcome

- An outcome that reflects what matters to the patient, including symptoms, function, quality of life, treatment burden, and ability to achieve personal goals, rather than relying only on laboratory or imaging measures.

Patient-Controlled Analgesia

- Patient-controlled analgesia (PCA) allows patients to self-administer small doses of analgesic medication within preset safety limits. IV PCA typically uses morphine (demand 1 mg, lockout 6 to 10 minutes, hourly limit 4 mg), hydromorphone (demand 0.2 mg, lockout 6 to 10 minutes), or fentanyl (demand 10 to 20 mcg, lockout 6 to 10 minutes).

Patient-Reported Outcome Measure (PROM)

- A validated questionnaire completed directly by a patient to assess symptoms, function, health status, or quality of life without clinician interpretation of the response.

Patlak Plot

- Linear graphical method for quantitative PET.

Pattern Recognition Receptors

- PRRs recognize PAMPs (pathogen) and DAMPs (damage). TLRs: TLR1/2 (lipoproteins), TLR2/6 (lipoteichoic acid), TLR3 (dsRNA), TLR4 (LPS, MD-2, CD14), TLR5 (flagellin), TLR7/8 (ssRNA), TLR9 (CpG DNA). Cytosolic: RIG-I/MDA5 (RNA), cGAS (DNA -> cGAMP -> STING -> IFN), NOD1/2 (peptidoglycan).

PCL Injuries

- Posterior cruciate ligament (PCL) tears: dashboard injury (posterior tibial force), hyperflexion. Clinical: posterior drawer (most sensitive), quadriceps active test, posterior sag sign. Grade I (1-5mm), II (6-10mm), III (>10mm). Isolated PCL: nonoperative (brace, rehab) for Grade I/II; surgical for Grade III or combined.

PCOS

- Common endocrine disorder affecting six to twelve percent of reproductive-aged women characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology on ultrasound. See Polycystic Ovary Syndrome.

PCR (Polymerase Chain Reaction)

- Technique amplifying specific DNA sequences exponentially. Components: template DNA, primers, DNA polymerase (Taq), nucleotides, buffer. Cycles: denaturation, annealing, extension. Applications: diagnostics, forensics, research.

PE Anticoagulation

- Initial heparin (LMWH or UFH), followed by warfarin (INR 2–3) for 3–6 months. DOACs are first-line for most patients. Duration depends on provoked vs. unprovoked status.

Peak Expiratory Flow Rate (PEFR)

- The maximum flow rate achieved during a forced expiratory manoeuvre after maximal inspiration, measured in litres per minute (L/min) using a peak flow meter. PEFR reflects the degree of airflow obstruction and is effort-dependent.

Peak Flow Meter

- See Peak Expiratory Flow Rate.

Pearly penile papules

- Small, benign, dome-shaped papules arranged circumferentially around the glans penis, representing normal anatomical variants requiring no treatment.

Pectoralis Major Muscle

- Thick fan-shaped chest muscle with clavicular and sternocostal heads, inserting on the lateral intertubercular groove lip. Innervated by medial/lateral pectoral nerves (C5–T1). Clavicular head flexes; sternocostal head extends from flexion. Adducts and medially rotates powerfully.

Pectoralis Minor Muscle

- Thin triangular muscle deep to pectoralis major, from ribs 3–5 to the coracoid process. Innervated by medial pectoral nerve (C8–T1). Draws scapula anteriorly/inferiorly, stabilizing it against the thorax. Assists forced inspiration. Tightness contributes to rounded shoulder posture and thoracic outlet syndrome.

Pectus Carinatum

- A congenital chest wall deformity characterized by anterior protrusion of the sternum (pigeon chest), caused by excessive growth of the costal cartilages pushing the sternum forward. Less common than pectus excavatum. Associated with congenital heart disease, Marfan syndrome, Noonan syndrome, and

homocystinuria.

Pectus Excavatum

- The most common congenital chest wall deformity, characterized by depression of the sternum (funnel chest), caused by excessive growth of the costal cartilages that push the sternum inward. Typically sporadic, but associated with Marfan syndrome, Ehlers-Danlos syndrome, Noonan syndrome, and Poland syndrome.

Pediatric Anesthesia

- Pediatric anesthesia requires age-appropriate dosing, equipment, and techniques. Children have higher metabolic rates, increased oxygen consumption, smaller functional residual capacity, and higher risk of desaturation. Airway anatomy differs (larger tongue, more anterior larynx, narrower subglottic area).

Pediatric Appendicitis

- Acute inflammation of the vermiform appendix, usually caused by luminal obstruction. Children may have evolving periumbilical-to-right-lower-quadrant pain, anorexia, vomiting, fever, and guarding, but presentation can be atypical in young children; timely surgical and antimicrobial assessment reduces perforation risk.

Pediatric Cancer

- Cancers in children. Common: leukemia, brain tumors, lymphoma, neuroblastoma, Wilms tumor. Often curable.

Pediatric Dentistry

- Specialty of dental care for children. Manages primary dentition, growth, behavior. Preventive focus. Special needs adaptation.

Pediatric Dermatology

- Pediatric dermatology encompasses conditions unique to children. Congenital lesions (birthmarks, hemangiomas, port-wine stains, café-au-lait macules). Infantile hemangiomas (strawberry marks) proliferate in first months of life and involute over years. Propranolol is first-line treatment for problematic hemangiomas.

Pediatric Dermatology Conditions

- Distinct from adult dermatology. Birthmarks include vascular (hemangiomas, port-wine stains) and pigmented (café-au-lait, congenital nevi, Mongolian

spots). Infantile hemangiomas proliferate in first months then involute. Congenital melanocytic nevi carry melanoma risk proportional to size.

Pediatric Diabetic Ketoacidosis

- A life-threatening complication of type 1 diabetes characterized by hyperglycemia, metabolic acidosis, and ketosis. Presents with polyuria, polydipsia, vomiting, abdominal pain, Kussmaul respirations, and fruity breath odor.

Pediatric Eczema

- Atopic dermatitis in children commonly begins in infancy with facial and extensor involvement; flexural disease predominates in older children. Pruritus, xerosis, sleep disturbance, and scratching are characteristic.

Pediatric Fever

- See pediatrics.

Pediatric Health

- Pediatric health encompasses the physical, cognitive, emotional, and social development of children from infancy through adolescence, with emphasis on preventive care, developmental surveillance, and age-appropriate management of acute and chronic conditions.

Pediatric Meningitis

- Inflammation of the meninges in children, bacterial forms being the most dangerous. Presents with fever, headache, neck stiffness, photophobia, altered mental status, and petechial rash (meningococcal). Infants may present with irritability, poor feeding, bulging fontanelle, and high-pitched cry.

Pediatric Nuclear Medicine

- Special considerations for children: lower doses, sedation often needed, distraction techniques. Dose reduction: weight-based activity, optimized protocols.

Pediatric Rehabilitation

- Rehabilitation for children with congenital or acquired disabilities. Developmental considerations, family-centered care, school collaboration.

Pediatric Shock

- Inadequate tissue oxygen delivery or utilization in a child, most often caused by hypovolemia, distributive disease, cardiogenic dysfunction, or obstruction. Children may maintain blood pressure until late;

tachycardia, abnormal perfusion, altered mental status, and oliguria are important early warning signs.

Pediatric Skin Tumors

- Pediatric skin tumors differ from adult tumors. Infantile hemangioma is the most common benign tumor of infancy. Congenital melanocytic nevus carries risk of melanoma (larger nevi have higher risk). Juvenile xanthogranuloma is a benign histiocytic lesion. Pyogenic granuloma is a common vascular lesion. Dermoid cysts are congenital.

Pediatric Toxicology

- Poisoning in children. Common: iron, medications, household cleaners, plants. Prevention: child-proof containers, education.

Pediatrician

- A medical doctor who specializes in the health and medical care of children from birth through adolescence (typically up to age 16-18).

Pediatrics

- The branch of medicine concerned with the health and medical care of infants, children, and adolescents from birth to age 18 (or up to age 16 in the UK).

Pediculosis

- Infestation with lice. Head lice spread mainly through close head-to-head contact and cause scalp pruritus with nits attached to hair shafts. Body lice live in clothing and can transmit certain infections; pubic lice infest coarse hair-bearing regions. Treatment and contact management depend on the species and local resistance patterns.

Pediculosis Capitis

- See Head lice.

Pediculus humanus capitis

- A blood-sucking parasite commonly known as the louse (plural, lice) that can cause an itchy scalp; infestations are highly contagious and especially common in school-age children.

Pedigree

- Diagram showing inheritance of traits through generations. Symbols: squares (males), circles (females), shaded (affected), half-shaded (carriers). Used to determine inheritance pattern and recurrence risk.

Pedophilic Disorder

- A paraphilic disorder characterized by intense and recurrent sexual attraction to prepubescent children (generally age 13), lasting 6 months, and causing distress or impairment or involving acting on these urges. Diagnosis requires the individual is 16 years old and 5 years older than the child.

PEG Tube

- See medical implants.

PEG Tube (Percutaneous Endoscopic Gastrostomy)

- Endoscopic placement of a feeding tube through the abdominal wall directly into the stomach. Indications: long-term enteral feeding (>4 weeks) for dysphagia (stroke, ALS, dementia, head/neck cancer) and malnutrition.

Pellagra

- A nutritional deficiency disease caused by a severe lack of niacin (Vitamin B3) or its precursor amino acid tryptophan.

Pelvic Binder

- A circumferential device applied around the pelvis to reduce motion and hemorrhage from suspected unstable pelvic-ring injury. It should be centered over the greater trochanters and used with definitive imaging and hemorrhage-control management.

Pelvic Cavity

- The lower abdominal cavity enclosed by the pelvis, containing the urinary bladder, internal reproductive organs, and rectum.

Pelvic Floor Health and Disorders

- The pelvic floor is a musculofascial sling (levator ani, coccygeus, and surrounding fascia) spanning the pelvic outlet, supporting the pelvic organs (bladder, uterus, rectum), maintaining continence, and facilitating urination, defecation, and sexual function.

Pelvic Inflammatory Disease

- Infection and inflammation of the upper female genital tract (uterus, fallopian tubes, ovaries, and peritoneum), most commonly caused by sexually transmitted infections (Chlamydia trachomatis, Neisseria gonorrhoeae). Presents with lower abdominal pain, cervical motion tenderness, adnexal tenderness, and fever.

Pelvic Inlet

- The superior opening of the true pelvis, bounded by the sacral promontory posteriorly, the arcuate lines of the ilia laterally, and the symphysis pubis anteriorly. Its anteroposterior diameter (true conjugate) normally measures at least 11 cm. The transverse diameter is typically 13 cm.

Pelvic Organ Prolapse

- POP: descent of the pelvic organs through the vagina due to pelvic floor weakness. Types: cystocele (anterior—bladder), rectocele (posterior—rectum), uterine prolapse (apical), enterocele, and vault prolapse (post-hysterectomy).

Pelvic Outlet

- The inferior opening of the true pelvis, bounded by the pubic symphysis, ischial tuberosities, sacrotuberous ligaments, and coccyx. Normal anteroposterior diameter is 11-13 cm (from the lower border of the pubic symphysis to the tip of the coccyx), and the transverse diameter (between the ischial tuberosities) is approximately 10-11 cm.

Pelvic Pain

- Pain localized to the lower abdomen, pelvis, or perineum, lasting <3 months (acute) or >3–6 months (chronic), and can arise from gynecologic, urologic, gastrointestinal, musculoskeletal, or psychologic causes.

Pelvic spleen

- An ectopic or wandering spleen that has descended into the pelvic cavity, resulting from laxity or absence of the normal ligamentous attachments that fix the spleen in the left upper quadrant. It may present as a palpable pelvic mass, with risk of torsion, infarction, or rupture. Diagnosis is by CT or ultrasound.

Pelvis

- Bony ring from 2 hip bones, sacrum, and coccyx. Each hip bone fuses from ilium, ischium, and pubis meeting at the acetabulum. Divided into false (greater, above brim) and true (lesser, below brim) pelvis. Males: narrower, deeper, heart-shaped inlet. Females: wider, shallower, oval inlet for childbirth.

Pemberton Sign

- Physical exam sign of retrosternal goiter: elevating both arms above the head for 1 minute induces

facial flushing, cyanosis, and stridor due to thoracic inlet obstruction. See Goiter.

Pemberton's sign

- A clinical sign elicited by asking the patient to raise both arms above the head for one minute, causing facial plethora, cyanosis, and distended neck veins in patients with a substernal goiter or mediastinal mass compressing the thoracic inlet.

Pemphigoid

- Bullous pemphigoid is an autoimmune subepidermal blistering disease caused by antibodies against hemidesmosomal proteins, especially BP180 and BP230. It usually causes intensely pruritic tense blisters on urticarial or erythematous skin, with less mucosal involvement than pemphigus.

Pemphigoid gestationis

- An autoimmune blistering disease of pregnancy characterised by intensely pruritic urticarial plaques and tense vesicles or bullae, typically arising in the second or third trimester around the umbilicus and spreading to the trunk and extremities.

Pemphigus

- A group of autoimmune blistering diseases with intraepidermal blisters due to antibodies against desmogleins (cell adhesion molecules): pemphigus vulgaris (PV—flaccid blisters, mucosal involvement, positive Nikolsky sign), pemphigus foliaceus (superficial, no mucosal), paraneoplastic pemphigus.

Pemphigus Vulgaris

- An autoimmune intraepidermal blistering disease caused by IgG antibodies against desmoglein 1 and 3. It presents with flaccid blisters that rupture easily, leaving painful erosions of skin and mucous membranes. Nikolsky sign is positive (lateral pressure causes epidermal sloughing).

Penetrance

- Proportion of individuals with a particular genotype who express the expected phenotype. Complete penetrance: all with genotype show phenotype. Incomplete penetrance: some with genotype do not show phenotype.

Penetrating Abdominal Trauma

- Trauma to the abdomen caused by a missile (gun-shot wound) or sharp object (stab wound). GSWs: high energy transfer, produce blast cavity, high risk

of intra-abdominal injury (90%), mandatory exploration. Stab wounds: lower energy, 30-50% do not penetrate peritoneum; local wound exploration to determine peritoneal violation.

Penicillin

- Beta-lactam antibiotic inhibiting bacterial cell wall synthesis. Targets penicillin-binding proteins (PBPs), transpeptidases that cross-link peptidoglycan. Bactericidal against growing bacteria. Spectrum: gram-positive (*Streptococcus*, *Staphylococcus*), some gram-negative (*Neisseria*, *Treponema*).

Penile Cancer

- Rare malignancy, mostly squamous cell carcinoma. Risk factors: HPV, phimosis, poor hygiene, smoking. Presentation: penile lesion, ulcer, mass. Treatment: surgery (partial/total penectomy), radiation, chemotherapy.

Penile Fracture

- Rupture of tunica albuginea during erection. "Pop" sound, immediate detumescence, swelling, ecchymosis. Surgical emergency.

Penile Implant

- Inflatable or malleative device for erectile dysfunction. See penile prosthesis.

Penile Prosthesis

- Inflatable or malleative device for erectile dysfunction. Types: two-piece, three-piece (reservoir, pump, cylinders). Surgical implantation. Complications: infection, erosion, mechanical failure, dissatisfaction.

Penis

- The male external genital organ, composed of three erectile columns: two dorsal corpora cavernosa and one ventral corpus spongiosum (containing the urethra, expanding into the glans). Root is attached to the pubic arch (crura and bulb).

Pentad of TTP

- Classic clinical pentad of Thrombotic Thrombocytopenic Purpura: (1) Thrombocytopenia, (2) Microangiopathic Hemolytic Anemia (schistocytes), (3) Neurological symptoms, (4) Renal insufficiency, and (5) Fever. See Thrombotic Thrombocytopenic Purpura (TTP).

Pentavalent Vaccine

- A WHO-prequalified five-in-one combination vac-

cine containing Diphtheria toxoid, Tetanus toxoid, whole-cell Pertussis antigen, Hepatitis B (HBsAg), and Haemophilus influenzae type b (Hib) conjugate, administered as a single intramuscular injection.

Pentose Phosphate Pathway

- A cytoplasmic glucose oxidation pathway running parallel to glycolysis, primarily in the liver, adipose tissue, adrenal cortex, and red blood cells. It has two phases: the oxidative phase (producing NADPH and ribose-5-phosphate) and the non-oxidative phase (interconverting sugar phosphates).

Peptic Ulcer

- An open sore in the stomach or duodenal lining caused most often by Helicobacter pylori infection or nonsteroidal anti-inflammatory drug use. It may cause burning upper-abdominal pain, bleeding, perforation, or obstruction. Also called peptic ulcer disease.

Peptic Ulcer Bleeding

- Haemorrhage caused by erosion of an artery or other blood vessel beneath a gastric or duodenal peptic ulcer. It is a common cause of upper gastrointestinal bleeding and may present with haematemesis, coffee-ground vomiting, melaena, or haemodynamic instability.

Peptic Ulcer Disease

- Condition characterized by ulceration of the gastrointestinal mucosa extending through the muscularis mucosae into the submucosa or deeper layers, occurring most commonly in the duodenal bulb and gastric antrum.

Peptide Bond

- A covalent amide bond formed between the carboxyl group of one amino acid and the amino group of another, releasing water. Peptide bonds link amino acids into polypeptide chains and have partial double-bond character that restricts rotation.

Percussion

- Tapping body surface to assess underlying structures. Sounds: dull (solid organ, fluid), resonant (lung), tympanic (gas-filled). Used to assess organ size, presence of fluid, air.

Percussion Note

- The sound produced by percussing the chest wall. Normal lung produces a resonant note. Dullness

suggests consolidation or fluid. Hyperresonance suggests pneumothorax or emphysema. Stony dullness suggests pleural effusion. Tympany suggests large pneumothorax.

Percutaneous Biopsy

- Biopsy performed under imaging guidance. See image-guided biopsy.

Percutaneous Coronary Intervention

- A catheter-based procedure to open blocked coronary arteries, typically involving balloon angioplasty and stent placement. Indications include acute STEMI (primary PCI), NSTEMI (early invasive strategy), and stable CAD with significant stenosis. Drug-eluting stents have reduced restenosis rates compared to bare-metal stents.

Percutaneous Drainage

- Image-guided drainage of fluid collections. See drainage procedures.

Percutaneous Nephrolithotomy

- Minimally invasive procedure for removal of large renal stones through a nephrostomy tract created in the flank. A tract is created from the skin to the collecting system under fluoroscopic guidance and dilated to allow passage of a nephroscope.

Percutaneous Nephrolithotomy (PCNL)

- Minimally invasive surgical procedure for removing large, complex, or staghorn renal calculi. Access: a needle track is established under fluoroscopic or ultrasound guidance from the flank through the renal parenchyma into the target renal calyx, dilated to allow passage of a nephroscope and working sheath.

Percutaneous Transluminal Angioplasty

- Balloon dilation of arterial stenosis. Often combined with stenting.

Percutaneous Umbilical Blood Sampling (PUBS)

- An invasive procedure for fetal blood sampling from the umbilical vein under ultrasound guidance, typically performed after 18-20 weeks.

Perforation of a viscus

- Full-thickness breach of the wall of a hollow abdominal organ, most commonly the stomach, duodenum, or colon, resulting in leakage of luminal contents into the peritoneal cavity. Causes include peptic ulcer disease, diverticulitis, appendicitis, inflamma-

tory bowel disease, ischemia, and trauma.

Performance Status

- A standardized measure of a patient's functional ability and activity level, commonly recorded with ECOG or Karnofsky scales. It helps estimate prognosis, determine treatment eligibility, and quantify change during cancer care.

Perfusion

- Blood flow through tissue. Adequate perfusion essential for organ function.

Perfusion Imaging

- Imaging tissue blood flow. CT perfusion, MR perfusion, PET perfusion. Used for stroke, tumor characterization.

Perfusion PET

- Tracers: Rb-82, N-13 ammonia, O-15 water. Measure blood flow.

Periapical Abscess

- An acute or chronic localized collection of pus at the apex of a tooth's root, arising from pulpal infection and necrosis (usually due to untreated dental caries or trauma).

Periapical Granuloma

- A chronic inflammatory lesion at the apex of a tooth, usually resulting from pulpal necrosis and persistent apical infection. It may be asymptomatic and is diagnosed through clinical, radiographic, and sometimes histopathologic assessment.

Pericardial Cavity

- The narrow, fluid-filled space between the layers of the pericardium surrounding the heart.

Pericardial Effusion

- The accumulation of excess fluid within the pericardial space. The clinical significance is determined by both the volume of fluid and the rate of accumulation; rapid accumulation of even small volumes can restrict cardiac filling, leading to cardiac tamponade. Symptoms include chest pressure, dyspnea, and orthopnea.

Pericardial Effusion Etiology

- Causes: viral pericarditis (most common in developed countries), malignancy (most common in developing countries), uremia, hypothyroidism, autoimmune diseases (SLE, RA), tuberculosis (lead-

ing cause worldwide), post-MI (Dressler syndrome), post-cardiac surgery, radiation, and idiopathic.

Pericardial Knock

- A high-pitched, early diastolic sound caused by sudden cessation of ventricular filling in constrictive pericarditis. It is heard at the left sternal border and may be confused with an S3 or opening snap. The knock is typically louder with inspiration and is a characteristic finding in constrictive pericarditis.

Pericardial metastases

- See Cardiac Metastases.

Pericardial Tamponade

- Fluid accumulation in pericardial space compromising cardiac filling. Beck triad: hypotension, jugular venous distension, muffled heart sounds. Electrical alternans on ECG. Diagnosis: echocardiography. Treatment: emergent pericardiocentesis, pericardial window.

Pericardiocentesis

- A procedure to aspirate fluid from the pericardial space, performed emergently for cardiac tamponade or diagnostically for pericardial effusion. The subxiphoid approach is most commonly used, with the needle directed toward the left shoulder under echocardiographic guidance.

Pericarditis

- Inflammation of the pericardium, the fibrous sac surrounding the heart. Acute pericarditis presents with sharp, pleuritic chest pain that improves with leaning forward, pericardial friction rub, and diffuse ST elevation on ECG.

Pericardium

- The fibroserous sac surrounding the heart and proximal great vessels. Outer fibrous pericardium (anchors to diaphragm, sternum) and inner serous pericardium (parietal layer lining the sac; visceral layer/epicardium on the heart) with the pericardial cavity (15-50 mL serous fluid) between them.

Perichondritis of the Ear

- Infection or inflammation of the auricular perichondrium, often after trauma, piercing, or surgery. *Pseudomonas aeruginosa* is an important cause; delayed treatment can produce cartilage necrosis and permanent deformity.

Pericoronitis

- Inflammation of the soft tissue around the crown of a partially erupted tooth, most often a mandibular third molar. Food retention and bacterial accumulation may cause pain, swelling, trismus, fever, or spread into deeper spaces.

Periductal mastitis

- Inflammation around the major subareolar ducts, often associated with smoking, causing a painful periareolar mass, discharge, or recurrent breast abscess.

Perilymphatic Fistula

- Abnormal communication between the inner ear fluid spaces and the middle ear or mastoid, often after trauma, barotrauma, or surgery. It may cause episodic vertigo, hearing loss, tinnitus, or pressure-induced symptoms and requires specialist evaluation.

Perimenopause

- The transition time in a woman's life that begins when ovaries produce less estrogen and menstruation becomes less frequent, and ends when the ovaries no longer produce eggs and menstruation stops.

Perinatal Psychiatry

- The subspecialty focused on psychiatric disorders during pregnancy and the postpartum period. Common conditions include perinatal depression, perinatal anxiety, perinatal OCD, and postpartum psychosis. Medication selection must balance maternal mental health with fetal or neonatal exposure risks.

Perineal Laceration

- Tearing of the perineum during vaginal delivery. First degree: skin only (perineal skin, fourchette). Second degree: perineal muscles (bulbocavernosus, transverse perineal) without anal sphincter involvement. Third degree: anal sphincter injury—3a (<50% of EAS), 3b (>50% of EAS), 3c (both EAS and IAS).

Perineal Laceration Classification

- Obstetric perineal tears are graded by depth: first degree involves skin, second degree involves perineal muscle, third degree involves the anal sphincter complex, and fourth degree extends through the rectal mucosa. Accurate examination and repair reduce long-term pelvic-floor and continence complications.

Perineal Protection

- Techniques used during the second stage to minimize perineal trauma (lacerations, episiotomy). Manual protection: the attendant's hand applies gentle pressure to the fetal head to control the speed of delivery (not too fast), supporting the perineum and flexing the fetal head.

Perineal Tear

- Laceration of the perineum during vaginal delivery, classified by degree: first degree (skin only), second degree (perineal muscles), third degree (sphincter complex involvement), and fourth degree (rectal mucosa). Most deliveries involve first or second degree tears. Third and fourth degree tears risk fecal incontinence.

Perineum

- The diamond-shaped area between the pubic symphysis, coccyx, and ischial tuberosities, containing the urogenital and anal triangles. The urogenital triangle contains the external genitalia and urethral opening; the anal triangle contains the anus and coccygeus muscles. During delivery, the perineum stretches to allow passage of the

Periodontal Abscess

- A localized accumulation of pus within periodontal tissues, commonly associated with a periodontal pocket, foreign material, or impaired drainage. It may cause pain, swelling, suppuration, tooth mobility, and systemic symptoms and requires urgent dental assessment.

Periodontal Disease

- See dentistry.

Periodontal Flap Surgery

- Surgical access for deep debridement. Flap reflected, calculus removed, bone contoured, flap repositioned.

Periodontal Ligament

- Connective tissue attaching tooth to bone. Contains collagen fibers, cells, blood vessels, nerves. Provides proprioception, shock absorption. Expanded in bruxism.

Periodontal Pocket

- A pathologically deepened gingival sulcus caused by apical migration of the junctional epithelium and loss of periodontal attachment. Pocket depth

and clinical attachment level are measured with a periodontal probe and interpreted with bleeding, inflammation, and radiographic bone loss.

Periodontal Probing

- Measurement of the depth of the gingival sulcus/pocket using a periodontal probe with millimeter markings. Normal: 1-3 mm. Periodontal pockets: 4 mm (indicates attachment loss and periodontitis). Full-mouth probing at 6 sites per tooth: mesio-buccal, mid-buccal, disto-buccal, mesio-lingual, mid-lingual, disto-lingual.

Periodontitis

- Inflammatory disease destroying supporting structures of teeth (periodontal ligament, alveolar bone). Caused by dysbiotic plaque biofilm with host immune response. Clinical signs: pocket formation, attachment loss, bone loss, tooth mobility, recession. Staging and grading guide treatment.

Perioperative Antibiotic Prophylaxis

- Administration of an appropriate antimicrobial within a defined interval before incision to reduce infection risk for selected procedures. Choice, timing, redosing, and discontinuation depend on procedure, organisms, patient factors, and local guidance.

Perioral Dermatitis

- A papulopustular eruption around the mouth, sparing the vermilion border. It is more common in young women and associated with topical corticosteroid use. Other triggers include fluoridated toothpaste, cosmetics, and inhaler steroids.

Periosteum

- The dense connective tissue membrane covering the outer surface of bones (except at articular surfaces, which are covered by cartilage). Two layers: outer fibrous (collagen, vessels, nerves) and inner cambium (osteoprogenitor cells, osteoblasts).

Peripancreatic malignancy

- Neoplastic process involving tissues surrounding the pancreas, including primary pancreatic adenocarcinoma with peripancreatic extension and metastatic disease from adjacent or distant organs. Often presents at an advanced stage with abdominal pain, weight loss, obstructive jaundice, and new-onset diabetes.

Peripartum Cardiomyopathy

- A form of dilated cardiomyopathy occurring in the last month of pregnancy or within 5 months postpartum, with no other identifiable cause. Risk factors include advanced maternal age, multiple gestations, preeclampsia, and African American race. Presents with heart failure symptoms.

Peripartum Cardiomyopathy Prognosis

- Approximately 50% of patients recover normal ventricular function within 6 months. Predictors of poor recovery include: persistent LV dilation, severe LV dysfunction at diagnosis, African American race, and delayed diagnosis.

Peripheral

- Located away from the center or toward the outer edges of the body or an organ.

Peripheral Arterial Disease

- Atherosclerotic occlusive disease of arteries supplying the lower limbs, causing reduced blood flow. Features: claudication (exertional calf pain relieved by rest), rest pain, diminished pulses, cool pale skin, ulcers/gangrene; chronic critical limb ischemia is limb-threatening.

Peripheral Arterial Disease (PAD)

- Atherosclerotic narrowing of arteries supplying limbs. Symptoms: intermittent claudication (pain with walking, relieved by rest), critical limb ischemia (rest pain, ulcers, gangrene). Diagnosis: ABI <0.9, Doppler, angiography. Treatment: risk factor modification, exercise, cilostazol, revascularization for severe disease.

Peripheral Artery Disease

- Narrowing of arteries supplying the limbs, usually caused by atherosclerosis. It may cause exertional leg pain relieved by rest, reduced pulses, nonhealing wounds, or critical limb ischemia. Also called peripheral arterial disease.

Peripheral Blood Smear

- Microscopic examination of a stained blood smear (Wright-Giemsa stain).

Peripheral Edema

- Swelling due to accumulation of fluid in the interstitial space of the limbs, most commonly the lower extremities.

Peripheral Nerve Blocks

- Peripheral nerve blocks provide targeted anesthesia and analgesia for specific surgical sites. Ultrasound-guided blocks improve success rates and reduce complications.

Peripheral Nerve Injuries

- Peripheral nerve injury classification (Sunderland): 1st degree (neuropraxia, conduction block, full recovery), 2nd degree (axonotmesis, Wallerian degeneration, endoneurium intact, recovery), 3rd degree (endoneurium disrupted, perineurium intact, partial recovery), 4th degree (perineurium disrupted, neuroma-in-continuity, no recovery with

Peripheral Nervous System

- The portion of the nervous system consisting of nerves and ganglia outside the brain and spinal cord. The PNS connects the central nervous system (CNS) to peripheral tissues and organs. Components: (1) Cranial nerves (12 pairs)—emerging from the brain/brainstem, innervating head, neck, and viscera.

Peripheral Neuropathy

- A disorder of peripheral nerves causing sensory, motor, and/or autonomic symptoms in a distal, symmetric 'stocking-glove' pattern.

Periprosthetic Joint Infection (PJI)

- Infection involving a joint prosthesis and adjacent tissues. Diagnosis integrates clinical findings, inflammatory markers, joint-fluid analysis and culture, and imaging; treatment may require debridement, implant retention, staged exchange, or resection arthroplasty.

Perirenal extramedullary hematopoiesis

- Formation of blood cells in the tissue surrounding the kidney, occurring as a compensatory response to chronic severe anemia such as myelofibrosis or thalassemia major.

Peristalsis

- Rhythmic sequential contractions of longitudinal and circular smooth muscle that propel gastrointestinal contents in an aboral (oral to anal) direction.

Peritoneal Carcinomatosis

- Diffuse metastatic seeding of the peritoneal cavity from gastrointestinal or ovarian malignancies. Presents with ascites, bowel obstruction, and

cachexia. Diagnosed by CT and paracentesis showing malignant cells. Prognosis is generally poor.

Peritoneal Dialysis

- Dialysis using peritoneum as semipermeable membrane. CAPD with four manual exchanges daily or APD using cyclor overnight. Advantages include home-based therapy and hemodynamic stability. Complications include peritonitis and catheter infection. Tenckhoff catheter is standard access.

Peritoneal Dialysis Catheter

- Flexible tube surgically placed into the peritoneal cavity for peritoneal dialysis. The Tenckhoff catheter is the most commonly used type with one or two cuffs that promote tissue ingrowth and anchor the catheter. Placement can be done surgically, laparoscopically, or percutaneously under radiological guidance.

Peritoneum

- The serous membrane lining the abdominal cavity and covering the abdominal organs. Parietal peritoneum lines the abdominal wall (innervated by somatic nerves—pain is well-localized); visceral peritoneum covers organs (innervated by visceral afferents—pain is dull, poorly localized).

Peritonitis

- Inflammation or infection of the peritoneum, commonly caused by a perforated abdominal organ, appendicitis, abdominal trauma, or infection of ascitic fluid. Severe abdominal pain, guarding, fever, vomiting, or shock requires urgent medical evaluation.

Peritonitis in PD

- Most common complication of peritoneal dialysis. Cloudy effluent, abdominal pain, and fever. Diagnosis by effluent cell count with neutrophils greater than 50/mm³. Treat with intraperitoneal antibiotics. Recurrent peritonitis may require catheter removal.

Peritonsillar Abscess

- Peritonsillar abscess, also known as quinsy, is a collection of pus in the peritonsillar space between the tonsillar capsule and the superior pharyngeal constrictor muscle, typically developing as a complication of acute tonsillitis, and it is the most common deep neck space infection.

Periungual fibroma

- Benign, flesh-colored or pink fibrous tumors arising from the nail fold or nail bed, presenting as firm, exophytic nodules that may distort the nail plate. They occur sporadically or in association with tuberous sclerosis complex. When multiple periungual fibromas are present, they raise suspicion for tuberous sclerosis.

Periventricular Leukomalacia

- Ischemic white matter injury in the premature infant brain, a major cause of cerebral palsy. Risk factors include prematurity, hypotension, and inflammation. Diagnosis by cranial ultrasound showing periventricular echodensities or cystic changes. Prevention: avoiding hypotension, maintaining normothermia, and managing infection.

Permissive Hypercapnia

- A ventilator strategy that allows PaCO₂ to rise above normal (and pH to fall) in exchange for lung-protective ventilation with lower tidal volumes and airway pressures. It is used in severe ARDS when achieving normal PaCO₂ would require harmful ventilation parameters.

Pernicious Anemia

- A type of vitamin B12 deficiency anemia caused by autoimmune destruction of gastric parietal cells, leading to lack of intrinsic factor and impaired B12 absorption in the terminal ileum.

Peroxisome

- Single-membrane organelles containing oxidative enzymes (catalase, urate oxidase, D-amino acid oxidase) that produce and degrade hydrogen peroxide (H₂O₂).

Persistent Asthma

- Asthma requiring daily controller therapy. Mild persistent: symptoms > 2 days/week but not daily. Moderate persistent: daily symptoms, nighttime awakenings > 1/week. Severe persistent: symptoms throughout the day, frequent nighttime awakenings, SABA use several times/day, extremely limited activity.

Persistent Depressive Disorder

- formerly called dysthymia, a chronic depressive mood lasting at least two years in adults (one year in children and adolescents).

Persistent Organic Pollutant

- POCPs: organochlorines, PCBs, dioxins. Bioaccumulate, long-range transport, persistent. Stockholm Convention bans/restricts.

Persistent pupillary membranes

- Remnants of the fetal tunica vasculosa lentis that fail to regress after birth, appearing as fine pigmented strands bridging across the pupil or attaching from the iris collarette to the lens capsule. They are usually benign and asymptomatic but may rarely obstruct the visual axis requiring surgical or laser treatment.

Personal Protective Equipment

- PPE: gloves, gowns, masks, eye protection. Used based on transmission-based precautions.

Personal Protective Equipment (PPE)

- Equipment for protection. Examples: gloves, gown, mask, goggles, respirator.

Personality Disorder

- A group of mental disorders characterized by enduring maladaptive patterns of thinking, feeling, and behaving that deviate from cultural expectations and cause functional impairment. Cluster A (odd/eccentric): paranoid, schizoid, schizotypal. Cluster B (dramatic/erratic): antisocial, borderline, histrionic, narcissistic.

Personalized Medicine

- Treatment tailored to individual tumor characteristics. See precision medicine.

Pertussis

- A highly contagious bacterial infection caused by *Bordetella pertussis*, producing characteristic paroxysmal cough with whooping inspiratory sound. Three stages: catarrhal (1-2 weeks of URI symptoms), paroxysmal (weeks to months of violent coughing spells), and convalescent. Most dangerous in infants under 6 months.

Pessaries

- Devices inserted into the vagina to support pelvic organs in cases of pelvic organ prolapse (cystocele, rectocele, uterine prolapse). Types: ring pessary (most common, with or without supportive membrane), shelf pessary (for severe prolapse), cube pessary (suction-based), and Gellhorn pessary (for advanced prolapse).

Pessary

- A removable device placed in the vagina to provide structural support for pelvic organ prolapse (uterine prolapse, cystocele, rectocele) or stress urinary incontinence.

PET Scan

- See nuclear medicine.

PET Scan (Positron Emission Tomography)

- Nuclear medicine imaging using radioactive tracers (most commonly FDG, a glucose analog) to detect metabolically active tissues.

PET Scanner

- Ring of detectors capturing coincidence events. BGO, LSO, or LYSO crystals. Time-of-flight PET improves signal-to-noise.

PET Tracers

- 18F-FDG: glucose metabolism (cancer, infection, inflammation). 18F-fluoride: bone turnover. 11C-metionine: protein synthesis. 68Ga-DOTATATE: somatostatin receptors. 68Ga-PSMA: prostate cancer.

PET/CT

- Combined PET and CT imaging. CT provides anatomical localization and attenuation correction. Essential for accurate interpretation. Low-dose CT for attenuation correction; diagnostic CT for detailed anatomy.

PET/MRI

- Combined PET and MRI. Superior soft tissue contrast, no ionizing radiation from CT. Useful for neurology, oncology, cardiology. Longer scan time, higher cost.

Petechiae

- Small, pinpoint, non-blanching red or purple spots on the skin or mucous membranes caused by extravasation of red blood cells from capillaries into the surrounding tissue. They are typically less than 2 millimeters in diameter and result from decreased platelet count, platelet dysfunction, or increased vascular fragility.

Petit Mal

- See Absence Seizure.

Peutz-Jeghers Syndrome

- An autosomal dominant neurocutaneous disorder caused by germline loss-of-function mutations in the STK11 (LKB1) tumor suppressor gene on chromosome 19p13.

Peutz-Jeghers's syndrome

- An autosomal dominant condition caused by STK11 gene mutations, characterized by hamartomatous polyps throughout the gastrointestinal tract and mucocutaneous hyperpigmented macules on the lips, buccal mucosa, and digits. The polyps may cause gastrointestinal bleeding, intussusception, and obstruction.

Peyronie Disease

- Fibrous plaque in penis causing curvature, pain, erectile dysfunction. Treatment: observation, medications, injection therapy, surgery.

Pfeiffer Syndrome

- An autosomal dominant craniosynostosis syndrome caused by FGFR1 or FGFR2 mutations. Features: craniosynostosis (bilateral coronal), midface hypoplasia, proptosis, hypertelorism, broad thumbs and great toes (medial deviation), and variable syndactyly.

pH

- Measure of hydrogen ion concentration: $\text{pH} = -\log[\text{H}^+]$. Normal blood pH: 7.35-7.45. Acidemia: $\text{pH} < 7.35$. Alkalemia: $\text{pH} > 7.45$. Buffers minimize pH changes.

pH Monitoring

- Measurement of esophageal pH. Evaluates gastroesophageal reflux disease.

Phacoemulsification

- The most commonly performed modern cataract surgery technique, involving the use of an ultrasonic handpiece with a titanium tip that vibrates at high frequency to fragment and emulsify the opaque lens nucleus while simultaneously irrigating and aspirating the lens material through a coaxial or bimanual system, leaving the posterior lens c

Phagocytosis

- The process by which specialized cells (phagocytes) engulf and destroy foreign particles, microorganisms, and cellular debris.

Phalanges

- 14 finger bones. Thumb: 2 (proximal, distal). Each other finger: 3 (proximal, middle, distal). MCP joints are condyloid allowing flexion, extension, abduction, adduction, circumduction. IP joints are hinge joints allowing only flexion/extension. Mallet finger and jersey finger are common tendon avulsion injuries.

Phalen Test

- Wrist flexion test for carpal tunnel syndrome. Symptoms reproduced within 60 seconds.

Phantom

- Device simulating human anatomy for quality control and calibration. Types: line source, uniform, Jaszczak, NEMA IU body phantom.

Phantom Limb Pain

- Painful sensations perceived in a limb that has been amputated, caused by persistent or reorganized activity in peripheral nerves and central sensory pathways. It may feel burning, shooting, cramping, or electric.

Phantosmia

- Perception of a smell when no corresponding odor is present. It may occur with sinonasal disease, migraine, seizures involving olfactory networks, head injury, neurodegenerative disease, infection, or other neurological conditions.

Pharmaceutical

- See pharmacology.

Pharmacodynamic Interaction

- A drug interaction in which one drug changes the effect of another at its site of action without necessarily changing its concentration. Effects may be additive, synergistic, or antagonistic, such as combined respiratory depression or bleeding risk.

Pharmacodynamics

- The study of the biochemical and physiological effects of drugs on the body and their mechanisms of action, describing what the drug does to the body.

Pharmacogenetics

- Study of genetic variation affecting drug response. Examples: TPMT (mercaptopurine toxicity), CYP2C9/VKORC1 (warfarin dosing), HLA-B*1502 (carbamazepine SJS in Asians), HLA-B*5701 (aba-

cavir hypersensitivity), UGT1A1 (irinotecan toxicity).

Pharmacogenomics

- The study of how genetic variations influence individual responses to drugs, including drug efficacy, toxicity, and metabolism. Genetic polymorphisms in drug-metabolizing enzymes, transporters, and receptors can lead to significant inter-individual variability in drug response.

Pharmacokinetic Interaction

- A drug interaction that changes absorption, distribution, metabolism, or excretion of another drug, thereby altering its concentration over time. Enzyme inhibition, enzyme induction, transporter effects, and altered gastric pH are common mechanisms.

Pharmacokinetics

- The study of drug movement through the body, describing what the body does to the drug, defined by four processes: Absorption — the process by which a drug enters the systemic circulation from its site of administration (oral, IV, IM, subcutaneous, topical, inhalation); Distribution — the movement of drug from the circulation to body tissue

Pharmacological Stress Test

- Stress testing using medications (dobutamine, adenosine, regadenoson). For patients unable to exercise.

Pharmacology

- The scientific study of drugs and their actions, properties, sources, and effects on living systems.

Pharmacovigilance

- The science and activities relating to the detection, assessment, understanding, and prevention of adverse drug reactions and other drug-related problems.

Pharyngeal Tonsil

- See Adenoids.

Pharyngitis

- Inflammation of the pharynx, the most common presentation of which is sore throat (odynophagia) with or without fever, and it is classified as bacterial or viral based on the causative organism, with viral pharyngitis accounting for approximately eighty to ninety percent of cases and bacterial pharyngitis

(primarily Group A Streptococcus)

Pharynx

- A muscular tube extending from the base of the skull (clivus) to the level of the cricoid cartilage (C6), where it continues as the esophagus.

Phase I Metabolism

- Phase I metabolism involves chemical modification of drug molecules through oxidation, reduction, and hydrolysis reactions, primarily catalyzed by the cytochrome P450 enzyme system in the liver.

Phase II Metabolism

- Phase II metabolism, also known as conjugation reactions, involves the attachment of endogenous hydrophilic molecules to drugs or their Phase I metabolites, dramatically increasing water solubility and facilitating renal excretion.

Phencyclidine (PCP)

- See hallucinogens.

Phenotype

- Observable characteristics of an individual, resulting from genotype and environmental interactions. Includes morphology, development, biochemical properties, behavior.

Phenylephrine

- A selective alpha-1 adrenergic agonist causing vasoconstriction and increased blood pressure (systemic vascular resistance) with reflex bradycardia. Used for hypotension/shock (IV bolus/infusion, common in anesthesia and sepsis), nasal congestion (topical decongestant), and mydriasis (ophthalmic).

Phenylketonuria

- See genetics.

Phenylketonuria (PKU)

- An inherited disorder, usually due to phenylalanine hydroxylase deficiency, causing accumulation of phenylalanine and reduced tyrosine availability. Untreated disease can cause intellectual disability, seizures, eczema, and a musty odor; newborn screening and dietary treatment are effective.

Pheochromocytoma

- Catecholamine-secreting tumor arising from chromaffin cells of the adrenal medulla or extra-adrenal paraganglia. Approximately ten percent are bilateral, ten percent are extra-adrenal, and ten per-

cent are malignant. Classic presentation includes episodic headache, sweating, and palpitations with sustained or paroxysmal hypertension.

Phimosis

- Inability to retract the foreskin (prepuce) proximally over the glans penis. Physiologic phimosis is normal in infants and uncircumcised young boys due to natural adhesions between glans and prepuce, resolving spontaneously in over 90% by age 3-5 years.

Phlebitis

- Inflammation of a vein, often affecting the lower extremities. Superficial thrombophlebitis: inflammation and thrombosis of a superficial vein (great saphenous vein most common), presenting with erythema, tenderness, warmth, and a palpable cord along the vein.

Phlebography

- Venography using contrast dye. Used for diagnosing venous thrombosis, malformations. Largely replaced by duplex ultrasound.

Phlegmasia cerulea dolens

- Massive deep vein thrombosis causing severe venous outflow obstruction, cyanosis, marked edema, and compromised arterial perfusion. A limb-threatening emergency with risk of venous gangrene and pulmonary embolism. Requires urgent anticoagulation and consideration of thrombolysis or surgical thrombectomy.

Phobias

- See psychiatry.

Phocomelia

- A severe congenital limb deficiency where the proximal portion of a limb (humerus/femur, radius/tibia) is absent or severely hypoplastic, and the hand or foot attaches directly to the trunk. Most famously associated with thalidomide exposure during the first trimester of pregnancy (days 20-36 after conception).

Phosphate

- An inorganic ion and component of nucleotides, phospholipids, proteins, and bone mineral. It participates in ATP energy transfer, phosphorylation-dependent signaling, intracellular buffering, and acid excretion by the kidney.

Phosphate Binders

- Calcium acetate, sevelamer, lanthanum. Bind dietary phosphate in GI tract. Target normal serum phosphorus in CKD-MBD. Avoid calcium-based binders in patients with vascular calcification.

Phosphate Homeostasis

- Serum phosphate is regulated by the interplay of PTH vitamin D and fibroblast growth factor 23. PTH promotes renal phosphate excretion. Active vitamin D stimulates intestinal phosphate absorption. FGF23 produced by osteocytes inhibits renal phosphate reabsorption and suppresses 1-alpha-hydroxylase reducing active vitamin D production.

Phosphodiesterase Inhibitor

- A class of drugs that inhibit phosphodiesterase enzymes, which break down cyclic nucleotides (cAMP and cGMP), thereby increasing intracellular levels of these second messengers and potentiating their downstream effects.

Phosphodiesterase-5 Inhibitors

- Medications that inhibit PDE-5, increasing cGMP and causing smooth muscle relaxation and vasodilation. Sildenafil and tadalafil are used for PAH and erectile dysfunction. In PAH, they reduce pulmonary vascular resistance and improve exercise capacity.

Phosphofructokinase-1

- Rate-limiting enzyme of glycolysis. Activated by AMP, fructose-2,6-bisphosphate. Inhibited by ATP, citrate.

Phospholipase C Pathway

- A GPCR signaling pathway activated primarily by Gq-protein coupled receptors (e.g., alpha-1 adrenergic, muscarinic M1/M3, vasopressin V1, angiotensin II AT1 receptors).

Phospholipid

- Lipid containing phosphate group, major component of cell membranes. Structure: glycerol backbone, two fatty acid tails (hydrophobic), phosphate group with polar head group (hydrophilic). Forms bilayers spontaneously in aqueous environment.

Phosphorus

- An essential mineral present in ATP, nucleic acids, phospholipids, and bone. Its concentration is regulated chiefly by the kidneys, parathyroid hormone,

fibroblast growth factor 23, and vitamin D; kidney failure may cause hyperphosphatemia.

Photoallergy

- See dermatology.

Photocoagulation

- Coagulation of tissue by intense light, often laser light, to treat lesions such as abnormal retinal vessels.

Photodynamic Therapy (PDT)

- A treatment that combines a photosensitizing agent with light to generate reactive oxygen species and destroy cancer cells, used for actinic keratosis, superficial skin cancers, and selected visceral tumors.

Photoelectric Effect

- Photon absorption by inner shell electron. Major interaction at low energies, high Z materials.

Photomultiplier Tube

- Converts light photons to electrical signals. Array of PMTs positioned behind crystal. Positioning circuit determines interaction location.

Photoppeak

- Full-energy peak in gamma spectrum. Used for energy window setting.

Photopheresis

- Extracorporeal treatment in which white blood cells are exposed to a photosensitizer and ultraviolet A light before being returned to the patient.

Photophobia

- Abnormal sensitivity or intolerance to light, causing discomfort or the need to avoid bright environments. It may occur with migraine, corneal disease, uveitis, meningitis, retinal disease, medication effects, or neurologic disorders.

Photosensitivity

- An abnormal skin reaction to sunlight (UVA/UVB), manifesting as rash, erythema, or burning on sun-exposed areas.

Phototherapy

- Therapeutic exposure to a controlled wavelength and dose of ultraviolet or visible light. Common modalities include narrowband UVB, broadband UVB, and PUVA; treatment requires assessment of photosensitizing drugs, cumulative exposure, eye protection, and long-term skin risk.

Phototoxic dermatitis

- An acute, dose-dependent inflammatory skin reaction occurring when certain ingested or topically applied chemicals (such as psoralens, tetracyclines, NSAIDs, or thiazides) absorb ultraviolet radiation, leading to direct cellular damage and a sunburn-like eruption in exposed areas.

Phototoxicity

- Skin reaction from UV light + chemical. Sunburn-like. Examples: tetracyclines, thiazides, NSAIDs, psoralens.

Phrenic Nerve

- Primarily from C4 (with C3 and C5: "C3, 4, 5 keeps the diaphragm alive"), traveling between anterior and middle scalene muscles, descending anterior to the anterior scalene across the first rib. Right phrenic passes anterior to the lung root; left passes anterior and lateral.

Phyllodes Tumor

- A breast tumor composed of both stromal and epithelial elements, ranging from benign through borderline to malignant. Typically a large, rapidly growing mass; treatment is surgical excision with clear margins.

Physiatrist

- Physician specialty focusing on function, quality of life for patients with impairment, disability. See rehabilitation medicine.

Physical Half-Life

- The time required for half of the radioactive nuclei in a sample to undergo nuclear decay. It is distinct from biologic half-life, which describes biological elimination, and effective half-life, which combines both processes.

Physical Therapy Assessment

- Comprehensive evaluation: history, observation, ROM, strength, sensation, coordination, balance, gait, functional tasks.

Physiologic Jaundice of Newborn

- A common, benign condition characterized by unconjugated hyperbilirubinemia appearing after 24 hours of age (peak at 3-5 days) in healthy term newborns. Caused by increased bilirubin production (high RBC turnover, short RBC lifespan) and immature hepatic conjugation (low UDP-glucuronyl

transferase activity).

Physiological Dead Space

- Volume of ventilated respiratory tract that does not participate in gas exchange, including anatomic and alveolar dead space.

Physiology

- Study of body function. Systems physiology: cardiovascular, respiratory, renal, gastrointestinal, endocrine, nervous, muscular, reproductive.

Phytomenadione

- Vitamin K is a fat-soluble vitamin essential for the post-translational gamma-carboxylation of glutamate residues to gamma-carboxyglutamate (Gla) in specific clotting factors (II, VII, IX, X, protein C, protein S) and anticoagulant proteins (protein C, protein S, protein Z). See Vitamin K: Phylloquinone.

PI-RADS

- Prostate Imaging-Reporting and Data System. Standardizes multiparametric MRI reporting for prostate cancer detection and staging.

Pia Mater

- The innermost meningeal layer, a delicate vascularized membrane firmly adherent to the surface of the brain and spinal cord, following all gyri and sulci. It carries blood vessels that penetrate the CNS and forms the tela choroidea, contributing to the choroid plexus (CSF production).

Pica

- Compulsive eating of non-nutritive substances (ice, dirt, clay). Associated with iron deficiency, pregnancy, developmental disorders.

PICC Line (Peripherally Inserted Central Catheter)

- A long, flexible catheter inserted into the basilic or cephalic vein and advanced so the tip sits in the superior vena cava. Inserted at the bedside, often by a specialized PICC nurse, with ultrasound and ECG or X-ray confirmation. Indications: long-term IV antibiotics (>2 weeks), TPN, chemotherapy, and frequent blood draws.

Pierre Robin Sequence

- A congenital anomaly triad: micrognathia (small mandible), glossoptosis (posteriorly displaced tongue causing airway obstruction), and U-shaped

cleft palate. The sequence arises from early mandibular hypoplasia, which prevents the tongue from descending, thereby impairing palatal shelf fusion.

Pierson Syndrome

- A rare autosomal-recessive disorder caused by LAMB2 variants, classically combining congenital nephrotic syndrome with microcoria and structural abnormalities of the eyes. Kidney failure, developmental impairment, and severe visual disability may occur.

Pigment Cirrhosis

- See Bronze Diabetes.

Pigmentary mosaicism

- A pattern of alternating hyperpigmented and hypopigmented skin patches arranged in a linear, segmental, or blaschkoid distribution, caused by genetic mosaicism from post-zygotic mutations affecting melanocyte development. It is a cosmetic condition without systemic involvement.

Pilar Cyst

- A benign cyst arising from the outer root sheath of hair follicles, most commonly on the scalp. It presents as a firm, mobile, subcutaneous nodule without a central punctum. Unlike epidermoid cysts, pilar cysts have smooth walls and contain dense keratin. They may be multiple in familial cases. Treatment is simple excision.

Piles (Hemorrhoids)

- Swollen, inflamed vascular cushions in the anal canal that normally aid continence. Internal hemorrhoids (above the dentate line, painless) present with painless bright red rectal bleeding and prolapse; graded I-IV by degree of prolapse.

Pilonidal Disease

- A chronic inflammatory condition of the sacrococcygeal region involving hair follicles. Pilonidal sinus: epithelialized tract containing hair and debris. Pilonidal abscess: acute infection of the sinus. Most common in young, hirsute males, exacerbated by prolonged sitting, obesity, and local trauma.

Pilonidal Sinus Disease

- A chronic sinus or cyst in the natal cleft, usually containing hair and debris. It may cause recurrent pain, swelling, drainage, or abscess formation.

Pineal Gland Tumor

- A tumor of the pineal region, including germinoma, pineocytoma, and pineoblastoma. Presents with Parinaud syndrome, hydrocephalus, and raised intracranial pressure; germinomas are highly radiosensitive.

Pinguecula

- A yellowish-white, slightly elevated benign degenerative lesion of the bulbar conjunctiva near the limbus, from elastotic degeneration of the substantia propria. Chronic ultraviolet exposure, dust, wind, and dryness are the principal risk factors.

Pink Eye

- Inflammation of the conjunctiva producing a pink or red appearance of the eye, most often indicating conjunctivitis. Causes are infectious (viral, bacterial, chlamydial) or noninfectious (allergic, irritant), with viral the most common overall.

Pink Puffer

- A clinical phenotype of emphysema in COPD. Patients present with severe dyspnea, minimal cough, thin habitus, pursed-lip breathing, accessory muscle use, and relatively normal PaO₂ with late hypercapnia. They tend to be older and have more extensive emphysema.

Pinna

- Visible external ear portion, elastic cartilage covered by skin, shaped to funnel sound into the external auditory canal. Key features: helix (outer rim), antihelix (inner ridge), tragus (canal prominence), antitragus, concha (canal-leading depression), lobule (earlobe, only non-cartilaginous part).

Pinworm

- Infection with *Enterobius vermicularis*, the most common helminth infection in developed countries, primarily in school-age children. Lifecycle: ingestion of eggs larvae hatch in small intestine adults colonize the cecum/appendix; female migrates to perianal area at night to deposit eggs.

Piriformis Syndrome

- Sciatic nerve compression by piriformis muscle. Buttock pain, radiating down leg. Diagnosis: clinical, exclude disc herniation. Treatment: stretching, injection, surgical release.

Pisiform

- Small pea-shaped sesamoid bone in the flexor carpi ulnaris tendon, in the proximal row. Articulates only with the triquetrum. Forms the medial border of Guyon's canal (ulnar nerve/artery passage). Only sesamoid carpal bone, subcutaneous and easily palpable on the ulnar palm.

Pituitary Adenoma

- Pituitary adenomas are benign neoplasms arising from the adenohypophysis (anterior pituitary gland), accounting for approximately ten to fifteen percent of all intracranial neoplasms.

Pituitary Apoplexy

- Acute hemorrhage or infarction within a pituitary adenoma causing sudden headache, visual field defects, ophthalmoplegia, and altered consciousness. Medical emergency requiring urgent glucocorticoid administration to prevent adrenal crisis from acute ACTH deficiency.

Pituitary Gland

- Pea-sized "master gland" in the sella turcica, connected to hypothalamus by infundibulum. Anterior pituitary (adenohypophysis): GH, TSH, ACTH, FSH, LH, prolactin. Posterior pituitary (neurohypophysis): ADH (vasopressin), oxytocin (from hypothalamus). Regulates other endocrine glands and bodily functions.

Pituitary tumor

- A neoplasm arising from the pituitary gland, either functioning (hormone-secreting) or non-functioning. It can cause visual field defects, headaches, and hormonal imbalances depending on size and secretory status.

Pityriasis Rosea

- An acute, self-limited eruption often beginning with a herald patch followed by oval scaly lesions distributed along skin-cleavage lines. It usually resolves within several weeks; treatment is symptomatic, and atypical, severe, recurrent, or persistent disease should be reassessed.

Placebo

- Inactive substance used as control in trials. Placebo effect: improvement from expectation.

Placebo Effect

- A psychological and physiological phenomenon

where a patient experiences a real improvement in health or symptoms after receiving an inactive treatment (like a sugar pill) due to their expectations of healing.

Placenta

- A temporary organ that develops in the uterus during pregnancy, providing oxygen and nutrients to the growing fetus while removing waste products.

Placenta increta

- Abnormal placental invasion into the myometrium without reaching the serosa, increasing the risk of massive hemorrhage at delivery and often requiring cesarean hysterectomy.

Placenta Praevia

- Placental implantation over or near the internal cervical os, affecting approximately 0.5% of pregnancies at term. Risk factors: previous caesarean section (risk increases with number of CS), placenta praevia in a prior pregnancy, multiple gestation, advanced maternal age (>35), smoking, cocaine use, and IVF.

Placental Abruption

- Premature separation of the placenta from the uterine wall before delivery, occurring in 0.5-1% of pregnancies. Risk factors: hypertension, preeclampsia, smoking, cocaine use, trauma, prior abruption, and thrombophilia.

Placental Accreta Spectrum

- Abnormal placental attachment to the myometrium, ranging from accreta (adherent to decidua basalis) to increta (invading myometrium) to percreta (invading through serosa). Major risk factor is prior caesarean section with anterior placenta previa.

Placental Separation Signs

- Clinical signs indicating the placenta has separated from the uterine wall and is ready for delivery. Classic signs: lengthening of the umbilical cord (cord lengthens as the placenta descends), a sudden gush of blood, and uterine elevation and firming (the uterus rises and becomes globular as the placenta moves into the lower segment).

Plague

- A bacterial infection caused by *Yersinia pestis*, transmitted by fleas from rodents. Three forms:

bubonic (most common, 80-90%): painful, swollen lymph nodes (buboes) after flea bite, fever, chills. Septicemic: bacteremia without bubo, endotoxic shock, DIC, purpura.

Plantar Fasciitis

- The most common cause of heel pain, caused by degeneration and microtears of the plantar fascia at its calcaneal origin. Presents with sharp, stabbing heel pain with the first steps in the morning (post-static dyskinesia), improving after a few minutes but worsening with prolonged standing or activity.

Plantar Wart (Verruca Plantaris)

- A wart (verruca) located on the sole of the foot, caused by human papillomavirus (HPV, typically types 1, 2, 4, 63). Plantar warts are endophytic (grow inward due to pressure) rather than exophytic like common warts, and are often painful with direct pressure (walking, standing).

Plantar warts

- Common warts (verrucae) occurring on the plantar surface of the foot, caused by human papillomavirus infection of the keratinocytes. They present as hard, thickened papules with a rough surface and disrupted dermatoglyphics (skin lines), often with pinpoint black dots representing thrombosed capillaries.

Plaque

- A raised, palpable lesion greater than approximately 1 cm, often formed by the coalescence of papules. Plaques are characteristic of several inflammatory, infectious, and neoplastic skin disorders.

Plaque Assay

- A method for quantifying viruses: serial dilutions of a viral suspension are added to a monolayer of host cells, forming zones of cell destruction (plaques), each plaque representing one infectious unit; results expressed as plaque-forming units per mL (PFU/mL).

Plasma Cell

- The terminally differentiated antibody-secreting cell derived from activated B lymphocytes (after antigen stimulation, class switching, and affinity maturation in germinal centers).

Plasma Cell Myeloma

- A malignant neoplasm of clonal plasma cells, usu-

ally producing a monoclonal immunoglobulin or light chain. It may cause hypercalcemia, renal impairment, anemia, and destructive bone lesions, collectively known as CRAB features, together with recurrent infection and hyperviscosity in some patients.

Plasma Concentration

- The amount of a substance present in a defined volume of plasma at a given time. It is used to interpret exposure, toxicity, therapeutic drug monitoring, and pharmacokinetic relationships.

Plasmacytoma

- A solitary mass of neoplastic plasma cells, which may occur in bone (solitary plasmacytoma of bone) or soft tissue (extramedullary plasmacytoma). It can progress to multiple myeloma.

Plasmids

- Small, circular, extrachromosomal DNA elements that replicate independently of the bacterial chromosome. Plasmids carry non-essential genes that provide selective advantages including antibiotic resistance, virulence factors, and metabolic capabilities.

Plasmodium vivax

- See Malaria.

Plastic bronchitis

- A rare condition producing large, branching mucous casts that form in the airways, causing life-threatening airway obstruction, often associated with congenital heart disease or post-fontan circulation.

Plate Method

- Simple portion control: half plate vegetables, quarter plate protein, quarter plate grains. Promotes balanced eating.

Platelet Aggregation

- The adhesion and clumping of platelets to form a hemostatic plug at sites of vascular injury.

Platelet Count

- The number of platelets per unit volume of blood (normal 150,000-400,000/ μ L), part of the CBC. Reduced (thrombocytopenia): bleeding risk (severe <20,000, spontaneous <10,000)—ITP, TTP, HIT, DIC, marrow failure, drugs, sepsis. Elevated (thrombocytosis): reactive or myeloproliferative.

Platelet Function Disorders

- Inherited: Glanzmann thrombasthenia (GPIIb/IIIa deficiency, absent platelet aggregation, mucocutaneous bleeding, recombinant FVIIa, platelet transfusions), Bernard-Soulier syndrome (GPIb/IX/V deficiency, giant platelets, thrombocytopenia, prolonged bleeding time, platelet transfusions), Storage pool deficiency (delta/dense granules, mild

Platelet Plug Formation

- The initial hemostatic response to vascular injury, involving platelet adhesion, activation, and aggregation. When the endothelium is disrupted, subendothelial von Willebrand factor and collagen are exposed to flowing blood.

Platelets (Thrombocytes)

- Small, anucleate, disc-shaped cell fragments (2–4 μ m diameter) derived from megakaryocytes in the bone marrow, circulating at $150\text{--}400 \times 10^9/\text{L}$.

Platypnea

- Dyspnea that occurs in the upright position and is relieved by lying down. It is the opposite of orthopnea. It is rare and may be associated with hepatopulmonary syndrome, intracardiac shunting, or pulmonary arteriovenous malformations.

Pleiotropy

- Single gene influencing multiple phenotypic traits. Examples: phenylketonuria (PKU) affects multiple systems; sickle cell anemia causes anemia, pain crises, organ damage.

Pleomorphic Adenoma

- Pleomorphic adenoma, also known as benign mixed tumor, is the most common benign salivary gland tumor, accounting for approximately sixty to seventy percent of all parotid gland neoplasms and a significant proportion of minor salivary gland tumors, and it is characterized by a diverse histological appearance with a mixture of epithelial,

Pleomorphism

- Variation in the size and shape of cells or their nuclei within a tissue. Marked pleomorphism is a feature of many high-grade malignancies but can also occur in some reactive or degenerative conditions.

Pleura

- The serous membrane covering the lungs (vis-

eral pleura) and lining the thoracic cavity (parietal pleura: costal, mediastinal, diaphragmatic, cervical/cupola). The pleural cavity is a potential space containing a thin film of serous fluid that couples the lung to the chest wall during ventilation.

Pleural Cavity

- The narrow, fluid-filled space between the two layers of the pleura surrounding each lung.

Pleural Effusion

- Abnormal fluid in the pleural space. Transudates result from altered pressures; exudates from inflammation, infection, or malignancy. Light's criteria differentiate the two.

Pleural Fluid Glucose

- Low pleural fluid glucose ($< 60 \text{ mg/dL}$) suggests bacterial empyema, rheumatoid pleurisy, or tuberculosis. It indicates high metabolic activity in the pleural space consuming glucose. Very low glucose is characteristic of empyema and parapneumonic effusions requiring drainage.

Pleural Fluid pH

- Pleural fluid pH < 7.20 indicates a complicated parapneumonic effusion or empyema requiring drainage. Normal pleural fluid pH is approximately 7.60. Low pH reflects high metabolic activity and anaerobic metabolism in the pleural space. It is a key decision point in managing parapneumonic effusions.

Pleural Friction Rub

- A grating, leathery sound heard on auscultation caused by inflamed pleural surfaces rubbing against each other during breathing. It is a sign of pleurisy and may be heard in pneumonia, PE, and pleural effusion. It disappears when fluid accumulates and separates the pleural surfaces.

Pleural perforation

- A breach in the pleural lining allowing air or fluid to enter the pleural space, potentially causing pneumothorax or hydropneumothorax, often from trauma, procedures, or underlying lung disease.

Pleural Plaque

- Circumscribed fibrous tissue on parietal pleura, hallmark of asbestos exposure. Bilateral, often calcified. Does not indicate asbestosis.

Pleurectomy

- Surgical removal of part or all of the pleura, used in the treatment of malignant pleural mesothelioma and recurrent malignant pleural effusion.

Pleurisy

- Pleural inflammation causing sharp, breathing-related chest pain. Causes: pneumonia, PE, autoimmune disease, malignancy. Treatment addresses the underlying cause.

Pleuritis

- Inflammation of the pleural membranes (visceral and parietal pleura). Presents with pleuritic chest pain: sharp, stabbing pain exacerbated by deep breathing, coughing, or sneezing.

Pleurodesis

- Fusion of visceral and parietal pleura to prevent recurrence of effusions or pneumothorax. Chemical pleurodesis uses talc, doxycycline, or bleomycin.

Pylorus

- The muscular outlet of the stomach that connects to the duodenum. Its sphincter regulates the passage of partially digested food into the small intestine; narrowing or blockage can cause vomiting, abdominal distension, and dehydration, especially in infants with pyloric stenosis.

Pleurodynia

- Acute, severe chest pain caused by inflammation or spasm of the intercostal muscles, often associated with viral infection.

Plexus

- A complex network of intersecting nerves or blood vessels in the body.

Plombage

- An obsolete surgical technique for treating pulmonary tuberculosis in which material was placed in the extrapleural space to compress the diseased lung and promote collapse therapy. Various materials including paraffin wax, lucite balls, and oil were used.

Plug (Vascular)

- A vascular occlusion device used to block a selected vessel under image guidance.

Plummer-Vinson Syndrome

- Triad of iron deficiency anemia, esophageal web,

and dysphagia occurring predominantly in middle-aged women. The esophageal webs are thin mucosal folds typically located in the postcricoid region of the hypopharynx and proximal esophagus.

Pneumatic Otoscopy

- A diagnostic technique that combines visualization of the tympanic membrane with assessment of its mobility in response to positive and negative air pressure changes, providing information about the presence of middle ear fluid and the condition of the Eustachian tube.

Pneumatocele

- Thin-walled, air-filled cystic spaces in the lung, often in staphylococcal pneumonia. Typically self-resolving.

Pneumatosis cystoides coli

- Condition characterized by multiple gas-filled cysts within the wall of the colon, which may be primary (idiopathic) or secondary to intestinal mucosal disruption from conditions such as chronic obstructive pulmonary disease, ischemic bowel disease, or intestinal obstruction. Often discovered incidentally on imaging or colonoscopy.

Pneumatosis Cystoides Intestinalis

- Gas-filled cysts within the bowel wall that may be primary or secondary to bowel ischemia, infection, chronic lung disease, or medications. Often an incidental finding on imaging. Primary pneumatosis is benign and asymptomatic. Secondary pneumatosis may indicate serious underlying disease. Managed by treating the underlying condition.

Pneumatosis Intestinalis

- Presence of gas within the bowel wall appearing as linear or cystic radiolucencies on imaging. Primary pneumatosis is benign and idiopathic accounting for approximately fifteen percent of cases.

Pneumocephalus

- The presence of air or gas within the cranial cavity, most commonly following trauma, surgery, or sinus disease, and potentially causing increased intracranial pressure if tension develops.

Pneumococcal Conjugate Vaccine

- PCV13 and PCV15 vaccines given at 2, 4, 6, and 12-15 months. Protects against *Streptococcus pneumoniae* serotypes causing invasive disease, pneumo-

nia, and otitis media. PCV20 now recommended for older children and adults. Has dramatically reduced rates of invasive pneumococcal disease in children since universal vaccination.

Pneumococcal sepsis

- See Bacterial Infections.

Pneumoconiosis

- ILDs from mineral dust inhalation (silicosis, asbestosis, coal worker's, berylliosis). They share interstitial fibrosis and risk of infection and malignancy.

Pneumocystis Jirovecii Pneumonia

- Opportunistic pneumonia caused by *Pneumocystis jirovecii*, especially in people with advanced HIV infection or other cellular immunodeficiency. It causes progressive exertional dyspnea, dry cough, fever, and hypoxemia.

Pneumocystis Pneumonia

- Opportunistic pneumonia from *P. jirovecii* in immunocompromised patients ($CD4 < 200$). Progressive dyspnea, dry cough, bilateral ground-glass opacities. Elevated LDH. Treatment is TMP-SMX.

Pneumography

- Contrast imaging of body cavities. Pneumoencephalography obsolete. Pneumoperitoneum for laparoscopy.

Pneumomediastinum

- Air in the mediastinum, often from alveolar rupture with air tracking along bronchovascular sheaths. Causes include asthma exacerbation, trauma, Valsalva maneuver, and esophageal rupture (Boerhaave syndrome). It presents with chest pain, subcutaneous emphysema, and Hamman's crunch (mediastinal crunch synchronous with heartbeat).

Pneumonectomy

- Removal of an entire lung, most often for centrally located lung cancer; carries higher operative morbidity and mortality than lobectomy and requires preserved contralateral lung function.

Pneumonia

- Infection of the lung parenchyma, classified as community-acquired (CAP), hospital-acquired (HAP), or ventilator-associated (VAP). Common pathogens: *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Mycoplasma pneumoniae*,

viruses. Presents with cough, fever, dyspnea, and consolidation on imaging. Severity assessment: CURB-65 or PSI.

Pneumopericardium

- Air in the pericardial space, usually from trauma (penetrating chest injury), esophageal perforation, or iatrogenic causes (central line placement, cardiac surgery). Can cause cardiac tamponade if under tension. Diagnosis by chest X-ray showing air surrounding the heart (air-fluid level if combined with effusion).

Pneumoperitoneum

- Free air within the peritoneal cavity, most commonly caused by perforation of a hollow viscus such as the stomach, duodenum, or colon. Iatrogenic causes include recent laparoscopy, endoscopy with perforation, and peritoneal dialysis. Presents with acute abdominal pain, rigidity, and signs of peritonitis.

Pneumothorax

- Air in the pleural space causing lung collapse. Primary spontaneous occurs in tall, thin young men. Secondary occurs with underlying lung disease. Tension is a medical emergency.

POEMS Syndrome

- A rare plasma-cell disorder named for polyneuropathy, organomegaly, endocrinopathy, monoclonal plasma-cell disorder, and skin changes. Other manifestations include sclerotic bone lesions, papilledema, fluid overload, thrombocytosis, and elevated vascular endothelial growth factor.

Point Mutation

- Change in single nucleotide base pair. Substitutions: transition (purine to purine, pyrimidine to pyrimidine) or transversion (purine to pyrimidine). Effects: silent (same amino acid), missense (different amino acid), nonsense (stop codon).

Point-of-Care Testing (POCT)

- Diagnostic testing performed near the patient rather than in a central laboratory. POCT can shorten turnaround time but requires validated devices, quality control, operator training, and appropriate interpretation.

Point-of-Care Ultrasound

- Bedside ultrasonography performed and interpreted

by the treating clinician to answer focused diagnostic or procedural questions. Emergency applications include cardiac activity, volume status, pulmonary pathology, abdominal free fluid, vascular thrombosis, and procedural guidance.

Poison Control

- See emergency medicine.

Poison-Center Consultation

- Specialist toxicology advice used to identify hazards, interpret exposure and laboratory data, select antidotes, guide observation or disposition, and coordinate public-health response. Consultation is appropriate for significant, uncertain, mixed, pediatric, or unusual exposures.

Poisoning

- Harmful effect from exposure to toxic substance. Routes: ingestion, inhalation, dermal, injection. Management: ABCs, decontamination, antidote if available, supportive care.

Poland Syndrome

- A congenital anomaly characterized by unilateral absence or hypoplasia of the pectoralis major muscle (specifically the sternocostal head) and ipsilateral hand anomalies (syndactyly, most commonly; also brachydactyly, ectrodactyly, or radial club hand). The right side is affected in 60-75% of cases.

Polio

- Abbreviation for poliomyelitis, an acute viral infection caused by poliovirus, an enterovirus of the Picornaviridae family transmitted via the fecal-oral route, primarily affecting children under 5 years of age.

Poliomyelitis

- See Polio.

Poliovirus

- An enterovirus that causes poliomyelitis, a paralytic illness. The virus is transmitted by the fecal-oral route. Most infections are asymptomatic, but a small percentage cause acute flaccid paralysis. The virus attacks motor neurons in the spinal cord. Diagnosis involves stool culture and PCR. There is no specific treatment.

Polish (Dental)

- Smoothing restorative surfaces to reduce plaque accumulation. Essential for longevity.

Polyarteritis Nodosa

- Necrotizing inflammation of medium-sized muscular arteries, spares lungs. Associated with hepatitis B (10-30%). Clinical: systemic symptoms, mononeuritis multiplex, livedo reticularis, renal involvement (hypertension, renal insufficiency—no glomerulonephritis), GI ischemia.

Polychondritis

- Relapsing inflammatory disorder targeting cartilage throughout the body, causing pain, swelling, and destruction of the ears, nose, larynx, trachea, and joints. May lead to airway collapse and hearing loss.

Polycystic Kidney Disease

- Inherited disorder with progressive renal cyst formation. ADPKD most common from PKD1 or PKD2 mutations. Bilateral enlarged kidneys with countless cysts. PKD1 more severe with earlier ESRD. Extrarenal manifestations include hepatic cysts and berry aneurysms. Screen family members.

Polycystic Liver Disease

- Dominant polycystic liver disease is an autosomal dominant condition characterized by multiple hepatic cysts that may cause massive hepatomegaly, abdominal pain, and compressive symptoms. Usually associated with autosomal dominant polycystic kidney disease. Liver function is preserved even with extensive cyst involvement.

Polycystic Ovary Syndrome

- Common endocrine disorder affecting six to twelve percent of reproductive-aged women characterized by hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology on ultrasound.

Polycythemia

- Elevated red blood cell mass/hematocrit. Primary: polycythemia vera (JAK2 V617F mutation, trilineage proliferation, low EPO, risk of thrombosis and myelofibrosis/leukemia). Secondary: hypoxia (COPD, high altitude, congenital heart disease, sleep apnea) with high EPO, or EPO-secreting tumors (renal cell, hepatoma).

Polycythemia Vera

- A chronic myeloproliferative neoplasm characterized by increased red blood cell mass, often with increased white blood cell and platelet counts, driven by constitutive activation of the JAK-STAT signal-

ing pathway.

Polydactylism

- Congenital presence of extra fingers or toes.

Polydactyly

- A congenital anomaly characterized by supernumerary digits on the hands or feet. Postaxial polydactyly (ulnar/fibular side, most common): more frequent in African American and Native American populations, often isolated and autosomal dominant.

Polydipsia

- Excessive, unquenchable thirst. A classic symptom of hyperosmolar states and endocrine disorders.

Polyethylene Glycol (PEG)

- An osmotic laxative used for constipation and bowel preparation for colonoscopy. PEG (e.g., Miralax, GoLYTELY) is a non-absorbable polymer that draws water into the colonic lumen, softening stool and stimulating defecation without significant electrolyte shifts. For chronic constipation: 17 g daily in water.

Polygenic Inheritance

- Trait influenced by multiple genes, each with small effect. Examples: height, skin color, common diseases (diabetes, heart disease). Produces continuous variation.

Polygenic Risk Score

- A statistical estimate of disease susceptibility calculated from the combined effects of many genetic variants, each usually contributing a small amount of risk. It modifies but does not replace clinical risk assessment or environmental factors.

Polyhydramnios

- Excess amniotic fluid volume, defined as amniotic fluid index greater than 24 cm or single deepest pocket greater than 8 cm. Occurs in 1-2% of pregnancies. Causes include gestational diabetes, fetal anomalies (GI atresia, neural tube defects), chromosomal abnormalities, and idiopathic.

Polymastia

- The presence of more than two breasts, resulting from development of accessory breast tissue along the mammary ridge, most commonly in the axillary region and present at birth or appearing at puberty.

Polymerase Chain Reaction (PCR)

- An in-vitro method that exponentially amplifies a defined DNA segment through repeated cycles of denaturation, primer annealing, and DNA synthesis by a thermostable polymerase. It is used in diagnosis, research, forensics, and pathogen detection.

Polymicrobial infections

- Infections caused by multiple microbial species. Polymicrobial infections are common in wound infections, intra-abdominal infections, and respiratory infections. The interactions between organisms can be synergistic or antagonistic. Understanding polymicrobial infections is essential for selecting appropriate antimicrobial therapy.

Polymicrogyria

- A brain malformation characterized by an excessive number of abnormally small, fused gyri, resulting in an irregular, bumpy cortical surface with shallow sulci. Caused by abnormal late neuronal migration or post-migrational cortical organization. May be focal or diffuse, unilateral or bilateral, and symmetric or asymmetric.

Polymyalgia Rheumatica

- An inflammatory syndrome of older adults (age >50, peak 70-80) causing bilateral shoulder and pelvic girdle pain and stiffness lasting >45 minutes, with elevated inflammatory markers (ESR >40, CRP elevated). Strongly associated with giant cell arteritis (15-20% develop GCA; 40-50% of GCA patients have PMR).

Polymyositis

- An idiopathic inflammatory myopathy characterized by symmetric, proximal muscle weakness (shoulders, hips) with elevated muscle enzymes (CK, aldolase, LDH). More common in women, peaks at 30-50 years.

Polyp Surveillance Colonoscopy

- Interval-based endoscopic follow-up of patients with previously identified colorectal polyps. Intervals are determined by polyp number, size, and histological features. One to two small adenomas require surveillance at seven to ten years. Three to four adenomas at three to five years.

Polyphagia

- Excessive, uncontrollable hunger or increased ap-

petite. Classic cardinal symptom of uncontrolled Diabetes Mellitus (cellular starvation due to lack of insulin or insulin resistance), Hyperthyroidism (elevated basal metabolic rate), Prader-Willi Syndrome, and atypical depression.

Polypharmacy

- Use of multiple medications by one patient, often defined numerically as five or more concurrent medicines. It increases the risk of interactions, adverse effects, prescribing cascades, nonadherence, and medication errors, although some multidrug regimens are clinically appropriate.

Polysomnography

- Sleep study measuring brain activity, breathing, oxygen saturation, muscle activity. Diagnoses sleep apnea, parasomnias.

Polysomnography (Sleep Study)

- Overnight monitoring for sleep-disordered breathing. Parameters: EEG (sleep staging: N1, N2, N3, REM), EOG (eye movements), EMG (chin and leg movements), airflow (nasal pressure, thermistor), respiratory effort (chest/abdomen belts), snore microphone, body position, oximetry (SpO₂), and ECG.

Polyunsaturated Fat

- Liquid at room temperature. Sources: fish, vegetable oils, nuts. Essential fatty acids.

Polyuria

- Excessive urine output, defined as >3 L/day in adults. Causes are divided into water diuresis and solute diuresis. Water diuresis: central diabetes insipidus (ADH deficiency), nephrogenic diabetes insipidus (ADH resistance from lithium, hypercalcemia, hypokalemia, genetic causes), and primary polydipsia (psychogenic or dipsogenic).

Pons

- Middle brainstem segment with prominent anterior bulge containing pontine nuclei (relaying cortical input to cerebellum), corticospinal fibers, and cranial nerve V, VI, VII, and VIII nuclei. Fourth ventricle floor is partly formed by the pons. Pneumotaxic center (parabrachial nucleus) regulates breathing pattern.

Popliteal Aneurysm

- Aneurysm of popliteal artery behind knee. As-

sociated with AAA. Risk: thrombosis, embolism, compression. Treatment: surgical repair or endovascular stenting.

Popliteal Artery

- Continues from femoral artery at the adductor hiatus, dividing into anterior/posterior tibial arteries at the lower popliteus border. Deepest popliteal fossa structure, related anteriorly to vein and tibial nerve. Palpable at the fossa's inferior margin.

Popliteal Fossa

- Diamond-shaped space on the posterior knee, bounded by hamstrings superiorly and gastrocnemius heads inferiorly. Contains popliteal artery/vein, tibial nerve, and common peroneal nerve. The popliteal pulse is palpated at the inferior margin.

Population Attributable Fraction

- The proportion of cases or disease burden in the total population that would theoretically be prevented if the exposure were eliminated, assuming the exposure is causal and other conditions remain unchanged.

Population Attributable Risk

- Proportion of disease in population attributable to exposure. $PAR = (\text{incidence total} - \text{incidence unexposed}) / \text{incidence total}$.

Porcelain Gallbladder

- Calcification of the gallbladder wall visible on imaging representing chronic inflammatory condition associated with cholelithiasis and chronic cholecystitis. Historically considered a strong risk factor for gallbladder carcinoma, recent studies suggest the cancer risk may be lower than previously estimated.

Porencephaly

- A cavity or cyst within the cerebral hemisphere caused by developmental malformation or destruction of previously formed brain tissue. Clinical effects depend on location and size and may include seizures, hemiparesis, developmental impairment, hydrocephalus, or cerebral palsy.

Porphyria Cutanea Tarda

- The most common porphyria, caused by a deficiency of hepatic uroporphyrinogen decarboxylase (UROD), leading to accumulation of uroporphyrinogens. Triggered by hepatitis C virus (HCV), alcohol

abuse, iron overload (hemochromatosis), and estrogen therapy.

Porphyrrias

- Metabolic disorders of heme synthesis causing cutaneous and neurological symptoms. Cutaneous porphyrias include porphyria cutanea tarda (blisters on sun-exposed skin, hypertrichosis, sclerodermoid changes), erythropoietic protoporphyria (immediate pain and burning on sun exposure without visible changes), and variegate porphyria.

Portal Hypertension

- Elevated pressure in the portal venous system defined as hepatic venous pressure gradient exceeding five millimeters of mercury with clinically significant portal hypertension occurring when the gradient exceeds ten.

Portal Hypertensive Gastropathy

- Diffuse mucosal changes in the stomach due to portal hypertension causing chronic blood loss. Characterized by mosaic pattern and cherry red spots on endoscopy representing dilated mucosal vessels. Most common in patients with advanced cirrhosis and portal hypertension.

Portal System

- Venous network draining the GI tract, spleen, and pancreas to the liver via the portal vein (formed by superior mesenteric and splenic vein union behind the pancreatic neck). Carries nutrient-rich deoxygenated blood for hepatic processing before systemic entry. Receives left gastric, right gastric, cystic, and paraumbilical veins.

Portal Vein

- The vein carrying nutrient-rich blood from the gastrointestinal tract, spleen, pancreas, and gallbladder to the liver, formed behind the pancreatic neck by the union of the superior mesenteric and splenic veins. It enters the porta hepatis with the hepatic artery and bile duct.

Portal Vein Thrombosis

- Partial or complete thrombosis of the portal vein or its branches. It may result from cirrhosis, malignancy, inflammation, thrombophilia, or abdominal surgery and can cause portal hypertension, intestinal ischemia, or variceal formation.

Portion Control

- Managing serving sizes. Larger portions available today contribute to obesity. Using smaller plates, reading labels helpful.

Portopulmonary Syndrome

- Development of pulmonary arterial hypertension in the setting of portal hypertension or liver disease. Results from increased cardiac output and splanchnic vasodilation leading to pulmonary vascular remodeling.

Positional Asphyxia

- Fatal or potentially fatal impairment of breathing caused by body position that restricts chest movement, airway patency, or venous return. Risk is increased by restraint, intoxication, obesity, and inability to reposition; findings are nonspecific and require scene and medical correlation.

Positive End-Expiratory Pressure

- Pressure maintained at end-expiration to prevent alveolar collapse. Improves oxygenation by recruiting alveoli and increasing FRC. Balances oxygenation with hemodynamic effects.

Positive Feedback

- A regulatory mechanism in which an output amplifies the initiating stimulus, producing an accelerating or self-reinforcing response that drives the system away from equilibrium and toward completion of an event.

Positive Predictive Value

- Probability that positive test indicates disease. Depends on prevalence. $PPV = \text{true positives} / \text{all positives}$.

Positive Predictive Value (PPV)

- The probability that a person with a positive test result truly has the target condition. PPV depends on test specificity and sensitivity and on disease prevalence or pretest probability in the tested population.

Positive Predictive Value in Surveillance

- The proportion of reported signals or alerts that represent true events meeting the surveillance case definition. It is influenced by the specificity of the signal and the underlying prevalence of the condition.

Positive Pressure Ventilation

- Mechanical ventilation that delivers air/oxygen under positive pressure to support or replace spontaneous breathing. Noninvasive (NIV): CPAP (continuous) and BiPAP (two levels) via mask for obstructive sleep apnea, cardiogenic pulmonary edema, and hypercapnic COPD.

Positive Symptoms

- Symptoms reflecting an excess or distortion of normal mental functions, such as hallucinations, delusions, disorganized speech, or grossly disorganized behavior. They are contrasted with negative symptoms, which reflect loss or reduction of function.

Positron

- Anti-particle of electron. Emitted by PET radionuclides. Annihilation with electron produces two 511 keV photons.

Positron Emission Mammography (PEM)

- PET imaging of breast. Higher sensitivity than mammography for dense breasts. Limited availability.

Positron Emission Tomography (PET)

- Functional imaging modality using positron-emitting radionuclides. Common isotope: fluorine-18 (110 min half-life). Detection: annihilation photons (511 keV) detected in coincidence. Tracers: FDG (metabolism), florbetapir (amyloid), flortaucipir (tau), PSMA (prostate cancer).

Post-Acute Rehabilitation

- Time-limited interdisciplinary care after an acute illness, injury, or hospitalization to restore mobility, self-care, cognition, communication, and safe community function. Intensity and setting depend on medical stability and rehabilitation potential.

Post-Cardiac Arrest Management

- Comprehensive care after ROSC: hemodynamic optimization (MAP >65 mmHg, avoid hypotension), ventilation management (target PaCO₂ 35-45 mmHg), targeted temperature management, coronary angiography for STEMI or high suspicion for ischemia, seizure monitoring and treatment, and neurological prognostication (clinical exam, SSEPs, EEG),

Post-Concussion Syndrome

- Persistent symptoms after concussion: headache,

dizziness, cognitive difficulties, sleep disturbance, mood changes. Management: symptom-directed, gradual return to activity.

Post-Dural Puncture Headache

- A positional headache occurring after accidental dural puncture during epidural placement or lumbar puncture, caused by cerebrospinal fluid leakage and intracranial hypotension. Symptoms include frontal or occipital headache worse when upright and relieved when supine, neck stiffness, tinnitus, photophobia, and nausea.

Post-exposure prophylaxis

- The administration of antimicrobial agents after exposure to a pathogen to prevent infection. PEP is used for HIV, hepatitis B, hepatitis C, rabies, and other infections. The timing of PEP is critical, with early administration being most effective. Understanding PEP is essential for managing occupational exposures and outbreaks.

Post-Exposure Prophylaxis (PEP)

- The administration of drugs after exposure to a pathogen to prevent infection.

Post-Infectious Cough

- A chronic cough persisting 3–8 weeks after a URI, caused by transient bronchial hyperresponsiveness and airway inflammation. It is self-limited and usually resolves without treatment. Inhaled corticosteroids, antitussives, and time are the mainstays of management.

Post-Mortem

- An examination of a body after death to determine the cause of death, identify disease processes, and evaluate the effects of medical or surgical treatment. Also called autopsy, necropsy.

Post-Obstructive Pneumonia

- Pneumonia occurring distal to a bronchial obstruction caused by tumor, foreign body, mucus plug, or stenosis. The obstruction prevents clearance of secretions, creating a favorable environment for infection. It does not resolve with antibiotics alone until the obstruction is relieved.

Post-Operative Atelectasis

- Collapse of alveoli after surgery, the most common post-operative pulmonary complication. Risk factors include general anesthesia, abdominal surgery,

obesity, smoking, and pre-existing lung disease. Prevention includes incentive spirometry, early ambulation, and adequate pain control.

Post-Pneumonectomy Syndrome

- A rare complication following pneumonectomy where the remaining lung herniates across the mediastinum, causing compression of airways and vessels. It presents with dyspnea, stridor, and recurrent infections. Treatment is surgical correction with mediastinal repositioning and prosthetic implant placement.

Post-Streptococcal Glomerulonephritis

- Immune-mediated glomerulonephritis following infection with nephritogenic strains of group A streptococcus typically 1 to 3 weeks after pharyngitis or 3 to 6 weeks after skin infection. Subepithelial humps on electron microscopy representing immune complex deposits.

Post-Surgical Pain

- Acute pain following operative procedures, caused by tissue incision, retraction, dissection, and inflammation, mediated by nociceptors sensitized by prostaglandins, bradykinin, and cytokines. Intensity varies by procedure (thoracotomy, joint replacement, and laparotomy produce more severe pain).

Post-Surgical Rehabilitation

- Recovery program after surgery. Phases: acute (pain control, wound care), subacute (range of motion, strengthening), functional (return to activities). Protocols vary by procedure.

Post-Term Pregnancy

- Pregnancy continuing beyond 42 weeks gestation (294 days from LMP). Risk factors include prior post-term pregnancy, nulliparity, and genetic factors. Complications include macrosomia, oligohydramnios, meconium aspiration, placental insufficiency, and stillbirth.

Post-Test Probability

- The probability of disease after incorporating a test result with the pretest probability. It can be estimated using Bayes' theorem or by converting pretest odds to post-test odds with the appropriate likelihood ratio.

Post-Thrombotic Syndrome

- Chronic pain, heaviness, edema, skin change, and

sometimes ulceration developing after deep-vein thrombosis because of persistent venous obstruction and valvular damage. Compression, exercise, skin care, and selected interventions may reduce disability.

Post-Transfusion Purpura

- A rare immune-mediated thrombocytopenic disorder occurring several days after transfusion of blood products, usually in previously sensitized individuals. It causes abrupt, severe thrombocytopenia with petechiae, mucosal bleeding, or more serious haemorrhage.

Post-Traumatic Stress Disorder

- See psychiatry.

Post-Void Residual

- Urine remaining in bladder after voiding. Normal: <50 mL. Elevated: >100-150 mL. Measured by ultrasound or catheterization. Indicates incomplete emptying.

Post-Void Residual Volume

- The amount of urine remaining in the bladder immediately after urination, measured by ultrasound or catheterization. An elevated residual suggests incomplete emptying but must be interpreted with timing, bladder volume, symptoms, and repeated measurements.

Postanesthesia Care Unit

- The postanesthesia care unit (PACU) is a monitored recovery area where patients are observed during the immediate postoperative period. Admission assessment includes vital signs, level of consciousness, pain assessment, and surgical site evaluation.

Postcholecystectomy Syndrome

- Persistence or recurrence of symptoms after cholecystectomy. Causes include retained common bile duct stones, sphincter of Oddi dysfunction, bile salt malabsorption, irritable bowel syndrome, and other gastrointestinal conditions. Evaluation includes liver function tests, MRCP, and endoscopic ultrasound.

Posterior

- Relating to the back of the body or positioned closer to the back.

Posterior Cruciate Ligament Injury

- Tear of the PCL, less common than ACL injury.

Mechanism: dashboard injury (tibia striking dashboard), hyperflexion, or fall on a flexed knee with foot plantarflexed. Presents with posterior knee pain, mild swelling, and posterior sag of the tibia. Positive posterior drawer test (most sensitive), posterior sag sign (Godfrey test).

Posterior Cruciate Ligament Tear

- PCL tear: less common, usually from direct blow to tibia (dashboard injury). Symptoms: knee pain, swelling, instability. Diagnosis: posterior drawer test, MRI. Treatment: usually conservative, surgical if combined injuries.

Posterior Descending Artery

- An artery running in the posterior interventricular groove and supplying the inferior ventricular walls and posterior interventricular septum. It usually arises from the right coronary artery in right-dominant circulation but may arise from the circumflex artery.

Posterior Drawer Test

- PCL integrity test. Knee flexed 90 degrees, posterior tibial translation.

Posterior Pituitary

- The neural lobe of the pituitary that stores and releases vasopressin and oxytocin synthesized in the hypothalamus. Vasopressin regulates renal water retention and vascular tone; oxytocin contributes to uterine contraction and milk ejection.

Posterior reversible encephalopathy syndrome

- A neurotoxic state caused by failed cerebrovascular autoregulation, typically from acute hypertension, eclampsia, immunosuppressive drugs (tacrolimus, cyclosporine), or sepsis. Presents with headache, seizures, altered consciousness, and visual disturbances.

Posterior Subcapsular Cataract

- A type of lens opacity located at the posterior aspect of the lens, just anterior to the posterior lens capsule, in the visual axis, and it is particularly visually disabling because of its central location.

Posterior Tibial Artery

- Arises from the popliteal artery, courses deep in the posterior leg, palpable behind the medial malleolus. Divides into medial/lateral plantar arteries for the

sole. Supplies posterior leg muscles. Important for peripheral vascular disease assessment. The pedal pulse behind the medial malleolus is a key clinical examination component.

Posterior Urethral Valves

- The most common cause of lower urinary tract obstruction in male infants, caused by congenital mucosal folds (valves) in the posterior urethra (prostatic urethra) that obstruct urine outflow.

Posterior Uveitis

- Inflammation primarily affecting the choroid (choroiditis), retina (retinitis), or both (chorioretinitis), and it can be caused by infectious agents including *Toxoplasma gondii*, cytomegalovirus, herpes simplex virus, tuberculosis, and syphilis, as well as non-infectious causes including sarcoidosis, Behcet disease, and Vogt-Koyanagi-Harad

Posterior Vitreous Detachment (PVD)

- Separation of the posterior vitreous cortex from the retinal surface, commonly related to age-related vitreous liquefaction. New flashes or floaters require dilated retinal examination because an associated retinal tear or detachment may be present.

Postmenopausal osteoporosis

- Bone loss caused by lower estrogen levels associated with menopause. Sometimes called type I osteoporosis.

Postmenopause

- The period in a woman's life lasting from the end of perimenopause until the end of life.

Postmortem Artifact

- Any structural, chemical, or physical change introduced into a body after death (e.g., embalming artifacts, insect activity, postmortem animal scavenging, resuscitation trauma) that mimics antemortem disease or trauma.

Postmortem Interval

- The time elapsed since death, estimated by forensic scientists.

Postmortem Interval (PMI)

- Estimated elapsed time between actual death and the time of autopsy or body discovery, calculated utilizing algor mortis, livor mortis, rigor mortis, vitreous potassium levels, and insect succession patterns.

Postmortem Toxicology

- Toxicologic analysis of specimens collected after death to identify drugs, alcohols, poisons, and metabolites. Results must account for therapeutic administration, redistribution, decomposition, specimen site, contamination, and tolerance before conclusions about causation are made.

Postnatal depression

- A depressive episode occurring after childbirth, with persistent low mood, anhedonia, sleep or appetite disturbance, guilt, impaired bonding, or suicidal thoughts.

Postoperative Care

- Structured post-surgical management encompassing pain control, fluid balance, wound care, early mobilization, and venous thromboembolism prophylaxis.

Postoperative Cognitive Dysfunction

- Decline in cognitive function (memory, concentration, executive function) after surgery and anesthesia, more common in elderly patients. POCD can be early (weeks to months) or persistent (>1 year). Incidence: 25-40% in elderly patients at hospital discharge, 10-15% at 3 months.

Postoperative Complications

- Clavien-Dindo classification: Grade I (any deviation, no treatment), Grade II (pharmacologic), Grade III (surgical/endoscopic/radiologic intervention: IIIa no GA, IIIb GA), Grade IV (ICU: IVa single organ, IVb multi-organ), Grade V (death).

Postoperative Delirium

- A common complication after major surgery, particularly in elderly patients. It is categorized as hyperactive (agitation, restlessness, hallucinations), hypoactive (lethargy, withdrawal, decreased responsiveness), or mixed. Hypoactive delirium is the most common and most missed form.

Postoperative Ileus

- Temporary impairment of gastrointestinal motility after surgery, most common after abdominal surgery. Pathophysiology: autonomic dysfunction (sympathetic overactivity), inflammatory mediators, opioids, and electrolyte abnormalities.

Postoperative Nausea and Vomiting

- Nausea, retching, or vomiting occurring within

24-48 hours after surgery. PONV is one of the most common and distressing complications. Risk factors (Apfel score): female gender, non-smoker, history of motion sickness/PONV, and postoperative opioids. 0-1 risk factors: 10-20% incidence; 2: 40%; 3: 60%; 4: 80%.

Postoperative Pain Management

- Postoperative pain management uses multimodal analgesia combining pharmacologic and regional techniques. Opioids (morphine, hydromorphone, fentanyl) provide potent analgesia but cause respiratory depression, nausea, and ileus. Acetaminophen (1000 mg IV every 6 hours) provides baseline analgesia.

Postoperative Pulmonary Complication

- Respiratory deterioration after surgery, including atelectasis, pneumonia, aspiration, respiratory failure, bronchospasm, and exacerbation of chronic lung disease. Risk reduction includes smoking cessation, lung-protective ventilation, analgesia, mobilization, and appropriate respiratory exercises.

Postpartum Blues

- A transient, self-limited condition affecting 50-80% of women, characterized by mood swings, tearfulness, irritability, and anxiety beginning 2-4 days after delivery and resolving within 2 weeks. No psychomotor disturbance or suicidal ideation. Treatment: reassurance, support, and rest. No pharmacotherapy needed.

Postpartum Cardiomyopathy (Peripartum Cardiomyopathy)

- An idiopathic dilated cardiomyopathy presenting with heart failure in the last month of pregnancy or within 5 months of delivery. Incidence: 1:1000-1:4000 live births. Risk factors: advanced maternal age, African descent, multiple gestation, hypertension, and preeclampsia. Presents: dyspnea, orthopnea, edema, fatigue, and chest pain.

Postpartum Depression

- Major depressive or manic episodes occurring during pregnancy or within four weeks of delivery (DSM-5). The term is often used more broadly to cover depressive episodes up to one year postpartum. Risk factors include prior depression, lack of social support, stressful life events, and hormonal fluctuations.

Postpartum Depression (PPD)

- A major depressive episode occurring within 4 weeks of delivery, affecting 10-15% of postpartum women. Symptoms persist >2 weeks and include depressed mood, anhedonia, sleep disturbance (insomnia despite exhaustion), appetite changes, fatigue, impaired concentration, guilt, and thoughts of harming self or the baby.

Postpartum Endometritis

- Infection of the decidua and endometrial cavity after delivery, occurring more often after cesarean birth. Fever, uterine tenderness, pelvic pain, and foul-smelling lochia are typical; treatment generally requires intravenous broad-spectrum antibiotics.

Postpartum Hemorrhage

- Blood loss >500 mL after vaginal delivery or >1000 mL after cesarean section, the leading cause of maternal mortality worldwide. Causes: uterine atony (most common, 80%), retained placenta, genital tract trauma (lacerations, episiotomy), and coagulopathy. Presents with heavy vaginal bleeding, hypotension, tachycardia, and

Postpartum Hemorrhage (Secondary)

- Excessive bleeding occurring between 24 hours and 12 weeks postpartum. Causes: retained products of conception (most common), endometritis, subinvolution of the placental site, and coagulopathy. Presents with heavy or prolonged bleeding, cramping, and passing clots. Diagnosis: ultrasound (retained tissue, thickened endometrium).

Postpartum Infection

- Infections occurring after delivery, a leading cause of maternal morbidity. Types: endometritis (most common, especially after cesarean), wound infection (cesarean incision, episiotomy, perineal laceration), urinary tract infection, mastitis, septic pelvic thrombophlebitis, and pneumonia.

Postpartum Psychosis

- A rare but severe psychiatric emergency (1-2 per 1000 deliveries) characterized by the acute onset of delusions, hallucinations, disorganized behavior, confusion, and mood swings (mania or depression) within 1-4 weeks postpartum.

Postpartum Thyroiditis

- Autoimmune thyroid inflammation presenting

within the first year postpartum (typically 2-6 months). Incidence: 5-10% of women, especially those with positive TPO antibodies (50-70% risk).

Postrenal AKI

- AKI caused by obstruction of urinary outflow at any level from renal tubules to urethra. Causes include BPH, nephrolithiasis, malignancy, retroperitoneal fibrosis, and neurogenic bladder. Bilateral ureteral obstruction or unilateral in a solitary kidney is required for significant impairment.

Poststreptococcal pustulosis

- A sterile, non-follicular pustular eruption occurring as a rare sequela of streptococcal pharyngitis, presenting with scattered pustules on an erythematous base, typically on the trunk and extremities. It is thought to be an immune-mediated hypersensitivity reaction to streptococcal antigens.

Postterm Pregnancy

- Pregnancy that extends beyond 42 weeks (294 days) of gestation, occurring in 5-10% of pregnancies. Risk factors: prior postterm pregnancy, inaccurate dating, male fetus, and obesity.

Posttraumatic Stress Disorder (PTSD)

- A psychiatric disorder that develops after exposure to a traumatic event (death, serious injury, sexual violence).

Potassium

- Major intracellular cation. Essential for cardiac function, muscle contraction, blood pressure regulation. Deficiency: hypokalemia (weakness, arrhythmias). Sources: fruits, vegetables, potatoes, beans.

Potassium Binders

- Sodium zirconium cyclosilicate and patiromer. Exchange potassium in GI tract for sodium or calcium. Treat chronic hyperkalemia allowing continued RAAS inhibitor use in CKD.

Potassium-Sparing Diuretics

- Potassium-sparing diuretics act on the distal nephron to promote sodium excretion while minimizing potassium loss. They include epithelial sodium channel blockers such as amiloride and triamterene, and mineralocorticoid receptor antagonists such as spironolactone and eplerenone.

Potency

- See pharmacology.

Potency vs Efficacy

- Potency is the amount of drug required to produce a given effect, measured by the EC₅₀ (the concentration producing 50% of maximal effect). Drugs with higher potency achieve effects at lower doses (e.g., fentanyl is more potent than morphine).

Pott Disease

- TB of the spine involving thoracic or lumbar vertebrae. Causes vertebral destruction, kyphosis (gibbus deformity), and paravertebral abscess. May cause spinal cord compression.

Potter Syndrome

- A sequence of anomalies resulting from severe oligohydramnios (low amniotic fluid volume), most commonly caused by bilateral renal agenesis, but also from autosomal recessive polycystic kidney disease, posterior urethral valves, or chronic amniotic fluid leak.

Powassan Encephalitis

- Encephalitis caused by Powassan virus, a tick-borne flavivirus.

PR Interval

- The time from the onset of the P wave to the onset of the QRS complex, representing the time for electrical conduction through the atria, AV node, and His-Purkinje system. Normal duration is 0.12-0.20 seconds (3-5 small boxes). Prolonged PR interval (>0.20 s) indicates first-degree AV block.

Prader-Willi Syndrome

- A genetic disorder caused by loss of function of genes on chromosome 15 (paternal deletion or maternal uniparental disomy). Features: neonatal hypotonia and failure to thrive, followed by hyperphagia and obesity in childhood. Intellectual disability, short stature, hypogonadism, and behavioral problems.

Pralidoxime

- An oxime that can reactivate inhibited acetylcholinesterase when administered early after organophosphate poisoning, particularly for nicotinic weakness and respiratory-muscle dysfunction. It is used with atropine and supportive care according to exposure and local guidance.

Pramipexole

- A non-ergot dopamine agonist (D₂/D₃) used for Parkinson disease (as monotherapy in early disease or adjunct to levodopa) and restless legs syndrome. Activates striatal dopamine receptors.

Prazosin

- A selective alpha-1 adrenergic antagonist causing vasodilation and reduced peripheral resistance. Used for hypertension, benign prostatic hyperplasia, and post-traumatic stress disorder-related nightmares (off-label). Marked first-dose orthostatic hypotension/syncope—start low and take at bedtime.

Prebiotics

- Non-digestible fibers feeding beneficial gut bacteria. Sources: garlic, onions, leeks, asparagus, bananas, whole grains.

Prebiotics and Probiotics

- Prebiotics are nondigestible food ingredients (fiber and oligosaccharides—inulin, fructooligosaccharides, galactooligosaccharides) that selectively stimulate the growth and activity of beneficial gut bacteria (*Bifidobacterium*, *Lactobacillus*), promoting host health.

Precision Medicine

- Treatment tailored to individual tumor characteristics. Requires molecular profiling.

Precocious Puberty

- Onset of secondary sexual characteristics before age eight in girls and age nine in boys. Central precocious puberty results from early activation of the hypothalamic-pituitary-gonadal axis. Causes include CNS tumors, hydrocephalus, and idiopathic.

Preconception Counseling

- Counseling provided to women of reproductive age before pregnancy to optimize outcomes.

Prediabetes

- Intermediate state of hyperglycemia between normal glucose homeostasis and diabetes defined by fasting glucose 100 to 125 milligrams per deciliter, two-hour glucose during oral glucose tolerance test 140 to 199 milligrams per deciliter, or A1c 5.7 to 6.4 percent.

Predictive Test

- Test predicting response to treatment.

Preeclampsia

- A pregnancy-specific hypertensive disorder defined as new-onset hypertension (blood pressure 140/90 mmHg or higher) after 20 weeks gestation with proteinuria or other end-organ damage.

Preeclampsia with Severe Features

- Preeclampsia (BP 140/90 + proteinuria/end-organ dysfunction) with one or more severe criteria: BP 160/110 mmHg, thrombocytopenia ($<100,000/\mu\text{L}$), impaired liver function (AST/ALT $>2\times$ normal, epigastric/RUQ pain), progressive renal insufficiency (creatinine >1.1 mg/dL), pulmonary edema, new-onset cerebral or visual symptoms (headache)

Pregnancy

- The physiological state in which a developing embryo or fetus grows within the female uterus, typically lasting approximately 40 weeks from the last menstrual period to childbirth.

Pregnancy Nutrition

- Nutrition care during pregnancy that supports maternal health, placental and fetal growth, and appropriate gestational weight gain. Energy, protein, folate, iron, iodine, choline, vitamin D, and other requirements vary by stage and individual clinical factors.

Pregnancy Test

- A glycoprotein hormone produced by the placenta, detected in urine and serum for pregnancy diagnosis. See hCG (Human Chorionic Gonadotropin).

Pregnancy-Induced Hypertension

- Hypertension developing after 20 weeks gestation without other features of preeclampsia. Blood pressure 140/90 mmHg or higher on two readings at least 4 hours apart. Affects 6-8% of pregnancies. Managed with antihypertensive therapy (labetalol, nifedipine, methyldopa) when blood pressure exceeds 160/110 mmHg.

Preload

- The end-diastolic volume or pressure in the ventricles, representing the degree of ventricular stretch before contraction. Preload is determined by venous return and is estimated by the pulmonary capillary wedge pressure (PCWP) or left ventricular end-diastolic pressure (LVEDP).

Premature (Early) Ejaculation

- A sexual dysfunction characterized by a persistent or recurrent pattern of ejaculation occurring within approximately 1 minute of vaginal penetration and before the individual wishes it, occurring in 75% of sexual encounters, lasting 6 months, and causing marked distress.

Premature Atrial Contraction

- A premature beat arising from an ectopic focus in the atria, producing an early P wave (often abnormal morphology) followed by a normal QRS; common and usually benign, increasing with age and caffeine/stress. ECG: early abnormal P wave, normal PR and QRS, followed by a compensatory pause.

Premature Ejaculation

- Ejaculation occurring sooner than desired. Most common sexual dysfunction in men. Treatment: behavioral techniques, topical anesthetics, SSRIs, counseling.

Premature Ovarian Insufficiency

- Loss of ovarian function before age forty characterized by amenorrhea, hypergonadotropic hypogonadism, and estrogen deficiency. Causes include genetic disorders particularly Turner syndrome and fragile X premutations, autoimmune oophoritis, iatrogenic from chemotherapy or radiation, and idiopathic.

Premature Ovarian Insufficiency (POI)

- Loss of normal ovarian activity before age 40, characterized by irregular or absent menstruation, elevated gonadotropins, and reduced estrogen production. Causes include genetic, autoimmune, iatrogenic, and idiopathic factors; hormone therapy is considered when not contraindicated.

Premature Rupture of Membranes

- PROM: spontaneous rupture of the amniotic membranes before the onset of labor; preterm PROM (PPROM) occurs before 37 weeks. Features: gush or leakage of fluid per vaginam; confirmed by sterile speculum (pooling, ferning, nitrazine pH test) or amniotic fluid biomarkers (IGFBP-1, PAMG-1).

Premature Ventricular Contractions

- Ectopic beats originating from the ventricles, appearing as wide, bizarre QRS complexes that occur

earlier than expected. PVCs are common in the general population and are usually benign. Frequent PVCs (>10% of total beats) may cause left ventricular dysfunction.

Premenstrual Dysphoric Disorder

- A severe, debilitating form of premenstrual syndrome characterized by marked mood symptoms (depression, anxiety, irritability, mood swings), physical symptoms (breast tenderness, bloating, fatigue), and functional impairment that occur during the luteal phase and remit within days of menses onset.

Premenstrual Syndrome

- A recurrent pattern of physical, emotional, and behavioral symptoms during the luteal phase of the menstrual cycle that resolve with menses. Affects up to 50% of reproductive-age women. Symptoms: irritability, mood swings, depression, anxiety, bloating, breast tenderness, fatigue, and appetite changes.

Prenatal Care

- Regular medical care during pregnancy to monitor maternal and fetal health. Includes history and physical exam, weight and blood pressure monitoring, urine testing, blood work, fundal height measurements, fetal heart rate auscultation, and ultrasound.

Prenatal Diagnosis

- Detection of genetic or structural disorders in fetus before birth. Methods: chorionic villus sampling (CVS, 10-13 weeks), amniocentesis (15-20 weeks), percutaneous umbilical blood sampling (PUBS, after 18 weeks). Non-invasive: NIPT (cell-free fetal DNA).

Prenatal Exposure

- Substance causing birth defects. See teratogen.

Prenatal Vitamins

- Multivitamin supplements formulated for pregnancy, containing folic acid (400-800 mcg), iron (27-60 mg), calcium (200-300 mg), vitamin D (400-600 IU), DHA (200 mg), and iodine (150-220 mcg). Started preconceptionally or at the first prenatal visit. Evidence supports NTD reduction from folic acid. Iron prevents maternal anemia.

Preoperative Assessment

- Preoperative assessment evaluates surgical risk and optimizes medical conditions before anesthesia.

It includes medical history, physical examination, airway assessment, cardiovascular risk stratification (Revised Cardiac Risk Index), pulmonary function assessment, and laboratory testing based on clinical indication.

Preoperative Evaluation

- Medical assessment before elective surgery to optimize outcomes and minimize perioperative risk.

Prerenal Azotemia

- Most common cause of AKI resulting from decreased effective circulating volume. Causes include dehydration, hemorrhage, heart failure, sepsis, and hepatorenal syndrome. Tubular function is intact with urine sodium less than 20 mEq/L and FENa less than 1%. Urine osmolality greater than 500 reflects avid water reabsorption.

Presbycusis

- Bilateral, usually symmetric age-related sensorineural hearing loss, initially affecting high frequencies and speech discrimination in noise. Management includes hearing rehabilitation, hearing aids, communication strategies, and cochlear implantation in selected severe cases.

Presbyopia

- Age-related loss of accommodation caused by lens hardening and loss of elasticity, beginning around age 40 and causing progressive difficulty with near vision. Corrected with reading glasses, bifocals, or progressive lenses; the most common refractive change of aging.

Prescription

- A written or electronic order by a licensed prescriber (physician, NP, PA, dentist) authorizing a pharmacy to dispense a specific drug to a specific patient. Components: patient identifiers, drug name, dose, route, frequency, quantity, refills, and prescriber signature.

Prescription Drug

- A medication that can only be dispensed with a valid prescription from a licensed healthcare provider, requiring professional oversight due to the drug's complexity, toxicity, or potential for harm. Prescription drugs undergo rigorous FDA approval through clinical trials (Phase I-IV) to establish safety and efficacy before marketing.

Prescription Medications

- Pharmaceutical drugs that require authorization from a licensed healthcare provider before they can be dispensed to a patient. These medications are regulated because they carry risks that necessitate professional oversight, including potential side effects, drug interactions, and the need for precise dosing.

Preseptal Cellulitis

- Preseptal cellulitis, also known as periorbital cellulitis, is an infection of the eyelid and periorbital tissues anterior to the orbital septum, without involvement of the orbital contents, and it is the most common orbital infection, particularly in children.

Pressure Atrophy

- The gradual wasting of tissue resulting from sustained external compression that impairs blood supply and causes cellular shrinkage and eventual death. The mechanism involves compression of blood vessels, which reduces perfusion and causes ischemia of the dependent tissue.

Pressure Injury

- Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences.

Pressure Support Ventilation

- Spontaneous breathing mode with ventilator-provided pressure support above PEEP for each breath. Reduces work of breathing. Used for weaning from mechanical ventilation.

Pressure Ulcer

- Localized damage to skin and underlying tissue caused by sustained pressure or pressure with shear, typically over a bony prominence. Staging includes non-blanchable erythema, partial-thickness loss, full-thickness loss, and full-thickness tissue loss with exposed or palpable deep structures; obscured wounds are unstageable.

Pressure Ulcer Prevention

- See geriatrics.

Pressure Urticaria

- A form of chronic inducible urticaria in which sustained pressure causes localized swelling, erythema, and pain, usually after a delay of several hours. Lesions may last 8 to 72 hours and commonly affect

the hands, feet, buttocks, or areas compressed by clothing or sitting.

Pressure-Flow Study

- Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases. See urodynamic testing.

Presyncope

- The sensation of imminent fainting (lightheadedness, weakness, visual blurring) without actual loss of consciousness, due to transient global cerebral hypoperfusion that resolves before LOC. Causes: orthostatic hypotension (volume depletion, autonomic failure, medications), vasovagal, cardiac (arrhythmias, aortic stenosis), and metabolic.

Preterm Birth

- Delivery before 37 weeks gestation, occurring in 10% of pregnancies worldwide. Subtypes: spontaneous preterm labor (50%), preterm PROM (30%), and medically indicated preterm birth (20%, for maternal or fetal indications).

Preterm Infant

- An infant born before 37 completed weeks of gestation; classified by gestational age (late preterm 34-36, moderate 32-33, very preterm <32, extremely <28 weeks) and by birth weight. Causes: preterm labor, PPRM, preeclampsia, multiple pregnancy, infection, and iatrogenic.

Preterm Labor

- Regular uterine contractions causing cervical change before 37 weeks gestation. Risk factors include prior preterm delivery, multiple gestation, infection, and cervical insufficiency. Diagnosis requires regular contractions with documented cervical change.

Preterm Premature Rupture of Membranes

- Rupture of the amniotic membranes before the onset of labor at <37 weeks gestation. Causes: infection (subclinical chorioamnionitis), cervical insufficiency, and prior PPRM. Presents with sudden gush or leakage of fluid per vagina.

Preterm Premature Rupture of

Membranes (PPROM)

- Rupture of the fetal membranes before the onset of labor and before 37 completed weeks of gestation. It increases risks of ascending infection, cord compression, placental abruption, and preterm birth.

Pretest Probability

- The estimated probability that a patient has a disease before the result of a diagnostic test is known. It is based on prevalence, symptoms, examination findings, risk factors, and prior test results and should guide test selection and interpretation.

Pretibial Myxedema

- Localized dermopathy occurring in approximately five percent of patients with Graves disease caused by glycosaminoglycan deposition in the dermis particularly over the pretibial area. Presents as non-pitting indurated plaques with a peau d'orange appearance.

Prevalence

- Total number of cases (new and existing) in population at given time. Formula: all cases / total population. Prevalence = incidence $\times$ duration.

Prevention

- Measures to prevent disease. Primary (before disease), secondary (early detection), tertiary (complication prevention).

Preventive Dentistry

- Practice of preventing dental disease. Methods: oral hygiene instruction, fluoride, sealants, diet counseling, regular check-ups.

Prevertebral air

- Gas in the prevertebral soft tissues on lateral cervical spine radiograph, indicating a breach in the aerodigestive tract. It may result from esophageal perforation, tracheal injury, retropharyngeal abscess with gas-forming organisms, or trauma.

Priapism

- Prolonged, painful erection lasting longer than 4 hours, occurring independently of sexual desire or stimulation. Medical emergency categorized into two types: ischemic (low-flow, veno-occlusive) and non-ischemic (high-flow, arterial).

Primary adrenal insufficiency (Addison's

disease)

- Destruction or dysfunction of the adrenal cortex causing deficiency of cortisol and aldosterone, presenting with fatigue, hyperpigmentation, weight loss, and salt craving. Lifelong hormone replacement is required.

Primary adrenal lymphoma

- A rare lymphoma arising primarily in the adrenal gland, most commonly diffuse large B-cell lymphoma. It often presents bilaterally and may cause adrenal insufficiency.

Primary Aldosteronism

- Overproduction of aldosterone by the adrenal glands (Conn syndrome: adrenal adenoma; or bilateral adrenal hyperplasia). Causes hypertension with hypokalemia (40% normokalemic). Presents with treatment-resistant hypertension with or without hypokalemia, metabolic alkalosis, low plasma renin, and high aldosterone-to-renin ratio

Primary Biliary Cholangitis

- Chronic autoimmune cholestatic liver disease characterized by progressive destruction of small intrahepatic bile ducts. The disease predominantly affects middle-aged women. Anti-mitochondrial antibodies are found in approximately ninety-five percent of patients.

Primary Ciliary Dyskinesia

- An autosomal recessive disorder of ciliary structure, most commonly caused by dynein arm defects. Leads to impaired mucociliary clearance, chronic sinopulmonary infections, bronchiectasis, and situs inversus (50% of cases).

Primary Hyperaldosteronism

- Autonomous aldosterone secretion most commonly caused by an aldosterone-producing adrenal adenoma termed Conn syndrome. Leads to hypertension, hypokalemia, metabolic alkalosis, and suppressed renin. Accounts for approximately five to ten percent of hypertensive patients. Screening is by elevated aldosterone-to-renin ratio.

Primary hyperoxaluria

- A group of autosomal recessive metabolic disorders characterized by overproduction of oxalate by the liver, leading to calcium oxalate deposition in the kidneys and other organs. Type 1 is caused by

alanine-glyoxylate aminotransferase deficiency. It presents with recurrent nephrolithiasis, nephrocalcinosis, and progressive renal failure.

Primary Hyperparathyroidism

- Autonomous overproduction of parathyroid hormone from one or more abnormal parathyroid glands causing hypercalcemia. Most commonly caused by a single parathyroid adenoma accounting for approximately eighty-five percent. Multigland hyperplasia accounts for approximately ten to fifteen percent and parathyroid carcinoma is rare.

Primary hypertrophic osteoarthropathy

- A rare inherited syndrome characterized by digital clubbing, periostosis of long bones, and pachydermia, presenting with joint pain, skin thickening, and visible nail-bed enlargement.

Primary Immune Response

- The adaptive immune response to first exposure to an antigen, characterized by a slow onset (lag phase: days, during which antigen processing, clonal expansion, and differentiation occur), followed by rising antibody titers (predominantly IgM, then IgG), and generation of memory cells.

Primary Immunodeficiency

- A group of genetically determined disorders of the immune system causing increased susceptibility to infection, autoimmunity, and malignancy.

Primary Lateral Sclerosis

- A rare neurodegenerative disorder confined to the upper motor neurons (corticospinal tract) of the motor cortex and brainstem. Presents with slowly progressive spasticity, hyperreflexia, Babinski sign, and pseudobulbar affect (emotional lability).

Primary Metabolism

- See hepatic metabolism.

Primary Myelofibrosis

- A chronic myeloproliferative neoplasm characterized by bone marrow fibrosis (reticulin and/or collagen fibrosis), extramedullary hematopoiesis (primarily splenic and hepatic), and a leukoerythroblastic peripheral blood picture.

Primary Open-Angle Glaucoma

- A chronic optic neuropathy with characteristic retinal ganglion-cell and optic-nerve loss, usually accompanied by progressive visual-field defects, de-

spite an anatomically open anterior-chamber angle. Intraocular pressure is an important modifiable risk factor, and treatment lowers pressure to slow progression.

Primary Ovarian Insufficiency

- Loss of normal ovarian activity before age 40, characterized by irregular or absent menses, reduced estrogen, and intermittently or persistently elevated gonadotropins. Causes include genetic, autoimmune, iatrogenic, toxic, infectious, and unknown factors.

Primary Prevention

- Preventing cancer development. Strategies: smoking cessation, vaccination (HPV, hepatitis B), sun protection, healthy diet, exercise.

Primary Sclerosing Cholangitis

- Chronic cholestatic liver disease characterized by progressive fibrosis and strictures of intrahepatic and extrahepatic bile ducts. The disease has a strong association with ulcerative colitis occurring in seventy to eighty percent of patients.

Primary Tuberculosis

- Initial M. tuberculosis infection causing Ghon focus in lower lobes with hilar lymph node involvement (Ghon complex). Most infections are contained, resulting in latent TB.

Primary Tumor

- Original cancer site. Secondary: metastatic sites.

Primordial Prevention

- Preventing the emergence of disease risk factors by addressing social, economic, behavioural, and environmental conditions early. Examples: promoting healthy diets and physical activity in children, reducing tobacco exposure, and creating safe environments for exercise.

Prion

- An infectious agent composed solely of misfolded protein (PrP^{Sc}) that propagates by converting normal cellular prion protein (PrP^C) into the abnormal conformation; contains no nucleic acid, resistant to standard sterilization (heat, radiation, formaldehyde).

Prion Disease

- Prion diseases, also known as transmissible spongiform encephalopathies, are a group of invariably

fatal neurodegenerative disorders caused by misfolding of the normal cellular prion protein PrPC into the abnormal scrapie form PrPSc.

PRK

- Photorefractive keratectomy, commonly known as PRK, is a refractive surgical procedure that corrects myopia, hyperopia, and astigmatism by reshaping the corneal stroma using an excimer laser after mechanical removal of the corneal epithelium, without creating a corneal flap.

Probable Cause

- Identification of disease. See diagnosis.

Probe

- Instrument for exploring wound or cavity.

Probiotic

- Live microorganisms administered in adequate amounts to confer a health benefit by modulating the microbiome and immune responses.

Probiotics

- Live microorganisms conferring health benefit. Strains: *Lactobacillus*, *Bifidobacterium*. Used for antibiotic-associated diarrhea, IBS, IBD. Evidence mixed.

Procalcitonin

- A biomarker for bacterial infection and sepsis, produced by thyroid C-cells and peripheral tissues in response to bacterial toxins. Levels: <0.1 ng/mL (normal), 0.1-0.25 (unlikely bacterial), 0.25-0.5 (possible), >0.5 (probable bacterial infection), >2.0 (severe sepsis/septic shock). More specific for bacterial infection than CRP.

Procalcitonin-guided antibiotic therapy

- The use of procalcitonin levels to guide initiation and discontinuation of antibiotics. Procalcitonin is a biomarker that rises in bacterial infections and falls with effective treatment. Elevated levels support the need for antibiotic therapy, while declining levels support discontinuation.

Procedural Sedation

- Administration of sedative and analgesic medicines to facilitate a painful or anxiety-provoking procedure while preserving spontaneous ventilation and protective reflexes to the intended depth. Continuous monitoring and immediate rescue capability are required.

Procedure

- See surgery.

Proctitis

- Inflammation of the rectal mucosa causing tenesmus, rectal pain, and bloody discharge. Causes include inflammatory bowel disease, sexually transmitted infections including gonorrhea, chlamydia, and herpes, radiation therapy, and ischemia. Diagnosed by proctoscopy with biopsy. Treatment is directed at the underlying etiology.

Prodrug

- A prodrug is a pharmacologically inactive compound that is converted to an active drug through metabolic processes within the body. Prodrugs are designed to overcome limitations of the parent drug including poor bioavailability, rapid metabolism, lack of tissue selectivity, or unacceptable side effects.

Product Liability

- See regulatory affairs.

Progeria

- A rare genetic premature-aging syndrome, most commonly caused by a pathogenic LMNA variant. Children develop severe growth failure, characteristic skin and facial changes, loss of subcutaneous fat, joint stiffness, and early atherosclerotic cardiovascular disease.

Progesterone

- A natural steroid hormone produced mainly by the corpus luteum after ovulation and by the placenta during pregnancy. It transforms and maintains the endometrium for implantation, reduces uterine contractility during pregnancy, supports breast glandular development, and contributes to regulation of the menstrual cycle.

Progesterone Supplementation

- Administration of progesterone to prevent preterm birth in high-risk women. Natural progesterone (17-alpha hydroxyprogesterone caproate) shown to reduce preterm birth rate in women with prior spontaneous preterm delivery. Also beneficial for short cervix detected on screening. Typically started in the second trimester.

Progestogen

- A synthetic form of progesterone, the hormone

that maintains the uterine lining in the secretory phase of the menstrual cycle and supports early pregnancy.

Prognosis

- Predicted outcome of cancer. Factors: stage, grade, molecular features, patient status. Survivability statistics.

Prognostic Test

- Test predicting disease outcome.

Progression

- Cancer growth or spread despite treatment.

Progression-Free Survival

- The length of time from a defined starting point, such as diagnosis or treatment, until cancer progression or death from any cause, whichever occurs first.

Progressive Disease

- Worsening of cancer during or after treatment, defined per RECIST as growth of target lesions, appearance of new lesions, or unequivocal progression of non-target lesions.

Progressive Supranuclear Palsy

- A neurodegenerative tauopathy characterized by the accumulation of tau protein in the basal ganglia, brainstem, and cerebral cortex, belonging to the family of Parkinson plus syndromes.

Projection

- A defense mechanism in which unacceptable thoughts, feelings, or impulses are attributed to others. For example, a patient who is angry at the therapist may accuse the therapist of being angry at them. It is common in paranoid thinking, delusional disorders, and personality disorders.

Prolactin Physiology

- Prolactin is the only pituitary hormone primarily under tonic inhibitory control by the hypothalamus through dopamine acting on D2 receptors. Prolactin stimulates lactation suppresses GnRH and is important for reproductive function. Physiological elevations occur during pregnancy lactation breast stimulation sleep and stress.

Prolactinoma

- A benign prolactin-secreting pituitary adenoma, the most common functional pituitary tumor, pre-

sented with galactorrhea, amenorrhea, infertility, and hypogonadism in women, and decreased libido, erectile dysfunction, and visual field defects in men. Prolactin levels are markedly elevated, often above 200 ng per mL.

Prolactinoma Treatment

- Dopamine agonists are first-line medical therapy for prolactinomas. Cabergoline is preferred over bromocriptine due to higher efficacy, less side effects, and twice-weekly dosing improving compliance. Normalize prolactin levels in approximately eighty to ninety percent of patients and significantly reduce tumor size in most.

Prolapse of orbital fat

- Herniation of intraorbital fat through a dehiscence in the orbital septum, most commonly presenting as a painless, soft, reducible mass in the upper eyelid or lower eyelid. It is often age-related and may be confused with a lipoma or dermoid cyst.

Prolapse of the Uterus (Uterine Prolapse)

- Descent of the uterus into the vaginal canal due to weakening of pelvic floor supports, most commonly the uterosacral and cardinal ligaments.

Proliferative Diabetic Retinopathy

- The advanced stage of diabetic retinopathy, characterized by the growth of new, fragile blood vessels (neovascularization) on the retinal surface, optic disc, or iris in response to severe retinal ischemia and the release of vascular endothelial growth factor.

Prolonged Deceleration

- A fetal heart rate deceleration lasting 2 minutes but <10 minutes. If the deceleration persists 10 minutes, it is a baseline change. Causes: maternal hypotension (epidural-induced aortocaval compression), uterine hyperstimulation, umbilical cord prolapse, placental abruption, and maternal seizure.

Prolonged Second Stage

- Second stage of labor exceeding the time limits: nulliparas — 3 hours with regional anesthesia (2 hours without); multiparas — 2 hours with regional (1 hour without). Management: reassess position, contractions, and fetal status. For inadequate contractions: oxytocin augmentation. For malposition (OP, OT): manual rotation.

Prominent iris collarette

- A visible thickening or ridge of the iris at the junction of the pupillary and ciliary zones, representing the site where the embryonic pupillary membrane was attached. It is typically a normal anatomical variant but can be a clinical sign in conditions such as Axenfeld-Rieger syndrome or iridocorneal endothelial syndrome.

Promoter

- DNA sequence upstream of gene where RNA polymerase and transcription factors bind to initiate transcription. Core promoter: TATA box, transcription start site. Regulatory elements: enhancers, silencers.

Pronation

- Rotation of the forearm so that the palm faces posteriorly in anatomical position, or downward when the elbow is flexed. The radius crosses over the ulna.

Prone Positioning

- Turning supine patients to prone for > 12 hours daily in severe ARDS. Improves oxygenation by recruiting dorsal lung segments. Reduces mortality when $\text{PaO}_2/\text{FiO}_2 < 150$.

Propafenone

- A class IC antiarrhythmic with sodium channel blockade and weak beta-blocking activity. Used for paroxysmal atrial fibrillation (in patients without structural heart disease, including 'pill-in-the-pocket'), supraventricular tachycardias, and ventricular arrhythmias.

Properdin Deficiency

- X-linked recessive (CFP gene). Properdin stabilizes alternative pathway C3 convertase. Recurrent *Neisseria* infections (meningococcemia, gonococcemia). Low AH50, normal CH50, low properdin. Treatment: meningococcal vaccination (ACWY + B), prophylactic penicillin, IVIG for severe infections.

Prophylaxis

- Preventive administration of a vaccine, antimicrobial, or immune globulin to prevent infection in exposed or at-risk individuals. Examples: post-exposure prophylaxis for rabies, HIV, and hepatitis B; perioperative antibiotic prophylaxis; and chemoprophylaxis for malaria. See also Endocarditis Pro-

phylaxis.

Propofol

- A short-acting intravenous hypnotic used for induction, maintenance, and procedural sedation. It has rapid onset and recovery but commonly causes dose-dependent hypotension and respiratory depression; it provides little intrinsic analgesia.

Propofol Infusion Syndrome

- Rare but fatal complication of prolonged high-dose propofol infusion. Features: metabolic acidosis, rhabdomyolysis, cardiac failure, hyperlipidemia. Risk: critical illness, children, high doses (>4-5 mg/kg/hr), prolonged infusion (>48 hours). Prevention: limit dose and duration.

Proprioception

- The sense of body position, movement, and joint position arising from mechanoreceptors in muscles (muscle spindles), tendons (Golgi tendon organs), and joints, as well as skin receptors. Signals ascend via the dorsal column-medial lemniscal pathway (conscious proprioception) and the spinocerebellar tract (unconscious).

Propylthiouracil (PTU)

- A thioamide antithyroid drug that inhibits thyroid peroxidase (hormone synthesis) and blocks peripheral T4-to-T3 conversion. Used for hyperthyroidism/Graves disease—preferred in the first trimester of pregnancy (lower teratogenicity than methimazole) and in thyroid storm.

Prostacyclin Analogs

- Medications that mimic prostacyclin (PGI₂), causing vasodilation and inhibiting platelet aggregation in the pulmonary vasculature. They include epoprostenol (continuous IV), treprostinil (subcutaneous, IV, inhaled, oral), and iloprost (inhaled). They are used for severe PAH.

Prostaglandin Analogs

- A class of topical medications used in the treatment of open-angle glaucoma and ocular hypertension that lower intraocular pressure by increasing the uveoscleral outflow of aqueous humor, reducing intraocular pressure by approximately twenty-five to thirty-five percent.

Prostaglandins

- Lipid compounds involved in labor initiation

and cervical ripening. Endogenous prostaglandins (PGE2, PGF2alpha) stimulate uterine contractions and promote cervical softening. Synthetic analogs used clinically: misoprostol (PGE1) and dinoprostone (PGE2 gel or insert) for cervical ripening and labor induction.

Prostate

– Walnut-sized gland (approximately 20 g) surrounding the urethra just below the bladder, producing approximately 30 percent of seminal fluid. It has peripheral, central, and transition zones.

Prostate Adenocarcinoma

– The most common non-cutaneous malignancy in men and the second leading cause of cancer death in men after lung cancer. The disease typically arises in the peripheral zone of the prostate and is hormone-dependent, requiring androgens for growth.

Prostate Biopsy

– Sampling of prostate tissue, commonly by transperineal or transrectal needle technique, for histologic diagnosis of prostate cancer. Targeted and systematic cores may be combined, with attention to infection prevention, bleeding, urinary retention, and sepsis risk.

Prostate Cancer

– Most common non-cutaneous cancer in men. Risk factors: age, family history, African American race, high red meat intake. Screening: PSA testing, DRE. Diagnosis: transrectal ultrasound-guided biopsy. Staging: TNM system, Gleason score.

Prostate-Specific Antigen (PSA)

– Glycoprotein enzyme (serine protease) produced almost exclusively by the luminal epithelial cells of the prostate gland, secreted into seminal fluid to liquefy the semen coagulum. Small amounts enter the bloodstream, where total PSA can be measured. Reference range: generally <4 ng/mL, though age-adjusted thresholds are used.

Prostatectomy

– Surgical procedure involving the partial or total removal of the prostate gland. Simple prostatectomy: removal of the inner adenoma while leaving the peripheral capsule intact, performed open, laparoscopically, or robotically for very large BPH (>80-100 grams).

Prostatitis

– Inflammation of the prostate gland, categorized by the NIH into four types: Acute bacterial prostatitis (Category I—fever, chills, dysuria, perineal pain, tender prostate; treatment: antibiotics, possible hospitalization), Chronic bacterial prostatitis (Category II—recurrent UTIs, pelvic pain; treatment: prolonged antibiotics),

Prosthesis

– Artificial replacement for missing body part. Types: upper extremity, lower extremity. Components: socket, suspension, joint, pylon, terminal device. Fitting: certified prosthetist. Training: physical therapy.

Prosthetic Limb

– Artificial replacement for missing limb. Types: above/below knee, above/below elbow. Components: socket, suspension, joint, pylon, terminal device. Materials: carbon fiber, silicone, metal. Rehabilitation: gait training, strength, ADL.

Prosthetic Training

– Training for amputees: stump care, donning/doffing, gait training, balance, stair negotiation, community ambulation.

Prosthetics

– Artificial replacement of missing body part. Elderly use: lower limb (transtibial, transfemoral), upper limb. Considerations: mobility level, cognition, vision, comorbidities, support system. Rehabilitation: physical therapy, strength training, balance.

Prosthodontics

– Dental specialty concerned with replacement of missing teeth and restoration of oral function. Types: fixed (crowns, bridges, implants), removable (partial dentures, complete dentures), maxillofacial prosthetics (ocular, nasal, auricular). Goals: aesthetics, function, phonetics, preservation of remaining structures.

Protease inhibitor

– A class of antiretroviral drugs that block the viral protease enzyme responsible for cleaving polyprotein precursors into functional proteins during HIV replication, preventing the formation of mature, infectious virions. Used in combination antiretroviral therapy for HIV infection.

Protease Inhibitors

- A class of antiviral drugs that block viral proteases, enzymes essential for processing viral polyproteins into functional components needed for viral assembly and maturation.

Protective Factor

- Attribute decreasing probability of disease. Examples: exercise (cardiovascular disease), vaccination (infectious disease).

Protein

- Polymer of amino acids linked by peptide bonds. Structure: primary (sequence), secondary (alpha helix, beta sheet), tertiary (3D folding), quaternary (multiple subunits). Functions: enzymes, structural, transport, signaling, immune defense, movement. Folding determined by amino acid sequence (Anfinsen's dogma). Chaperones assist folding.

Protein Binding

- The reversible association of drug molecules with plasma proteins, primarily albumin for acidic drugs and alpha-1 acid glycoprotein for basic drugs. Only unbound or free drug is pharmacologically active and available for distribution to tissues, metabolism by the liver, and excretion by the kidneys.

Protein C

- Natural anticoagulant. Inactivates factors Va and VIIIa. Warfarin can cause skin necrosis in deficiency.

Protein C Deficiency

- An inherited (autosomal dominant) thrombophilia causing increased VTE risk. Protein C is a vitamin K-dependent natural anticoagulant that inactivates factors Va and VIIIa and stimulates fibrinolysis. Heterozygous: 6-10 fold increased VTE risk.

Protein Folding

- The process by which newly synthesized polypeptide chains acquire their functional three-dimensional structures. The amino acid sequence (primary structure) determines the folding pathway.

Protein S

- Vitamin K-dependent plasma glycoprotein that acts as a cofactor for activated protein C in the inactivation of factors Va and VIIIa. Deficiency or dysfunction increases the risk of venous thromboembolism.

Protein S Deficiency

- An inherited (autosomal dominant) thrombophilia. Protein S is a vitamin K-dependent natural anticoagulant that acts as a cofactor for activated protein C in inactivating factors Va and VIIIa. Heterozygous: 2-10 fold increased VTE risk. Homozygous: rare, purpura fulminans.

Protein Synthesis

- Translation of mRNA code into polypeptide chain. Initiation: ribosome assembly at start codon (AUG). Elongation: aminoacyl-tRNAs bring amino acids; peptide bonds form in peptidyl transferase center; ribosome translocates. Termination: stop codon (UAA, UAG, UGA) recognized by release factors.

Protein-Energy Malnutrition (PEM)

- Severe deficiency of protein and calories. Types: marasmus (chronic, wasting), kwashiorkor (acute, edema, fatty liver), marasmic-kwashiorkor. Risk: famine, poverty, alcoholism, malignancy, malabsorption. Treatment: gradual refeeding, monitor refeeding syndrome.

Proteinuria

- Excretion of more than 150 mg of protein in urine daily. Causes include glomerular disease, tubulointerstitial disease, overflow proteinuria from multiple myeloma, and functional proteinuria from fever or exercise. Quantification by 24-hour urine or spot protein-to-creatinine ratio.

Prothrombin Time

- PT: a coagulation test measuring the extrinsic and common pathways (factors VII, X, V, II, fibrinogen), used to monitor warfarin therapy (reported as INR) and assess bleeding risk. Prolonged PT: warfarin/Vitamin K deficiency, liver disease, DIC, factor deficiencies (VII), and anticoagulant use.

Proto-Oncogene

- A normal gene that regulates cell growth, division, differentiation, or survival and can become an oncogene after activating mutation, amplification, translocation, or overexpression.

Proton Pump Inhibitor

- A class of medications that irreversibly inhibit the gastric H⁺/K⁺ ATPase (proton pump) in parietal cells, the final common pathway of gastric acid secretion, producing profound and sustained suppression

of both basal and stimulated acid production.

Proton Pump Inhibitors (PPIs)

- A class of antisecretory drugs (e.g., omeprazole, lansoprazole, pantoprazole, esomeprazole) that irreversibly inhibit the H^+/K^+ ATPase enzyme (proton pump) in gastric parietal cells. Prodrugs activated in the acidic canaliculi of parietal cells. Indications: GERD, peptic ulcer disease, H.

Proton Therapy

- Radiation using protons instead of photons. Bragg peak allows precise dose delivery. Used for pediatric tumors, skull base tumors.

Protozoa

- Single-celled eukaryotes lacking a cell wall, classified by locomotion into amoebae, flagellates, ciliates, and sporozoa.

Proximal

- Positioned closer to the trunk of the body or point of origin of a limb.

Proximal Humerus Fractures

- Proximal humerus fractures: Neer classification (4 parts: articular surface, greater tuberosity, lesser tuberosity, shaft; displaced >1 cm or angulated >45 degrees = part). 1-part (nondisplaced): sling, early motion. 2-part: surgical neck (ORIF plate, IM nail) or greater tuberosity (if displaced >5 mm or rotator cuff involvement).

Proximal Radioulnar Joint

- Pivot synovial joint between radial head circumference and ulnar radial notch, held by the annular ligament. Allows pronation/supination with approximately 150 degrees range. Parallel bones in supination; radius crosses ulna in pronation. Significant in radial head fractures, Monteggia fracture-dislocations, and nursemaid's elbow.

Proximal Tubule

- Segment of the nephron extending from Bowman capsule to the loop of Henle with a convoluted proximal straight tubule and convoluted proximal tubule. Responsible for reabsorbing approximately sixty-five percent of filtered sodium, water, bicarbonate, glucose, amino acids, and phosphate through an extensive brush border.

Proximal Tubule Reabsorption

- The proximal convoluted tubule (PCT) reabsorbs

approximately 65 percent of filtered sodium, water, glucose, amino acids, bicarbonate, and phosphate.

Proximate Cause

- See legal toxicology.

Prune Belly Syndrome

- A congenital syndrome comprising the triad: deficient abdominal wall musculature (wrinkled, prune-like appearance), bilateral cryptorchidism (undescended testes), and urinary tract anomalies (dilated ureters, dilated and trabeculated bladder, and posterior urethral valves or urethral atresia). Almost exclusively affects boys.

Pruritus

- Itching: an unpleasant sensation provoking scratching, a common symptom of skin and systemic disease.

Pruritus Ani

- Persistent itching around the anus with multiple potential causes including hemorrhoids, fissures, infections such as pinworms and candidiasis, dermatologic conditions including psoriasis and lichen sclerosus, and idiopathic causes.

Pruritus in CKD

- Severe itching in dialysis patients from uremic toxin accumulation, elevated calcium-phosphorus product, and secondary hyperparathyroidism. Difficult to treat. Include phosphate binders, dialysis optimization, gabapentin, and UV phototherapy.

PSA (Prostate-Specific Antigen)

- A serine protease produced by the prostate, used for prostate cancer screening. Normal: <4.0 ng/mL, but age-specific ranges exist. PSA >10 ng/mL: high probability of cancer. PSA 4-10: gray zone; percent free PSA ($<25\%$ suggests cancer) and PSA density help.

Pseudoaneurysm

- A contained arterial wall disruption in which blood communicates with the arterial lumen but is held by surrounding tissue rather than all layers of the arterial wall. It may follow catheterization, trauma, surgery, infection, or grafting and can rupture or embolize.

Pseudobulbar Affect

- Involuntary episodes of laughing or crying disproportionate to mood, usually due to neurologic

disease.

Pseudoclubbing

- Broadening and flattening of the nail tips without the complete loss of the angle between the nail and nail fold that characterizes true clubbing. It is seen in conditions such as hyperparathyroidism, rheumatoid arthritis, and psoriasis.

Pseudoephedrine

- A sympathomimetic (alpha-1 agonist, some beta effects) oral decongestant used for nasal congestion in colds and sinusitis. Increases nasal mucosal vasoconstriction.

Pseudogout

- Calcium pyrophosphate deposition disease (CPPD): deposition of calcium pyrophosphate dihydrate crystals in joints, causing acute arthritis resembling gout (pseudogout) or chronic osteoarthritis-like disease.

Pseudogout Flare

- Acute crystal-induced arthritis caused by calcium pyrophosphate dihydrate deposition, often affecting the knee or wrist. Identification of positively birefringent rhomboid crystals in synovial fluid supports the diagnosis.

Pseudohypoparathyroidism

- A group of inherited disorders in which target organs are resistant to parathyroid hormone, causing hypocalcaemia and hyperphosphataemia despite elevated PTH. Some forms cause Albright hereditary osteodystrophy, with short stature, round facies, brachydactyly, or subcutaneous ossifications.

Pseudomelanosis duodeni

- Benign condition characterized by dark pigment deposits in the duodenal mucosa, visible as slate-gray or black patches on endoscopy, caused by accumulation of iron-containing pigment in macrophages within the lamina propria.

Pseudomembranous Colitis

- Colonic inflammation caused by *C. difficile* infection showing yellow-white plaques on colonoscopy. The plaques consist of fibrin, inflammatory cells, and necrotic epithelial debris overlying mucosal ulceration. Presents with profuse watery diarrhea and abdominal pain.

Pseudomonas aeruginosa infection

- See Bacterial Infections.

Pseudomonas aeruginosa resistance

- *Pseudomonas aeruginosa* has intrinsic and acquired resistance mechanisms making it inherently resistant to many antibiotics. Intrinsic resistance includes low outer membrane permeability, efflux pumps, and chromosomal beta-lactamase.

Pseudomyxoma Peritonei

- Accumulation of mucinous ascites and mucin-producing tumor nodules in the peritoneal cavity, usually from a ruptured low-grade appendiceal or ovarian mucinous neoplasm; treated with cytoreductive surgery and hyperthermic intraperitoneal chemotherapy.

Pseudothrombocytosis due to red cell fragments

- Falsely elevated platelet count on automated hematology analyzers caused by fragmentation of red blood cells (schistocytes) being counted as platelets. Seen in microangiopathic hemolytic anemias such as TTP, HUS, DIC, and mechanical heart valves. Manual peripheral blood smear review confirms true platelet count.

Pseudoxanthoma elasticum

- An inherited connective tissue disorder caused by mutations in the ABCC6 gene, leading to progressive calcification and fragmentation of elastic fibres in the skin, eyes, and cardiovascular system. Cutaneous findings include yellowish, peau d'orange papules in flexural areas (neck, axillae, groin) resembling plucked chicken skin.

Psittacosis

- A zoonotic infection caused by *Chlamydia psittaci*, transmitted from infected birds (parrots, pigeons, poultry) via inhalation of dried bird droppings/secretions. Features: abrupt fever, headache, myalgia, dry cough, atypical pneumonia; complications: hepatitis, myocarditis, meningitis.

PSMA

- Prostate-specific membrane antigen. Target for PET imaging and therapy in prostate cancer.

Psoas Abscess

- Collection of pus in the iliopsoas muscle compartment. Primary psoas abscess results from hematogenous spread of *Staphylococcus aureus* typically in immunocompromised patients. Secondary psoas ab-

scess results from contiguous spread from adjacent structures including Crohn disease, diverticulitis, appendicitis, and vertebral

Psoriasis

- A chronic, immune-mediated inflammatory skin disease characterized by well-demarcated, erythematous plaques with silvery-white scale, affecting approximately 2 to 3 percent of the global population, with a bimodal age of onset (peak at 15 to 25 years and 50 to 60 years), a roughly equal male-to-female ratio, and strong genetic predisposition

Psoriatic Arthritis

- An inflammatory arthritis associated with psoriasis (precedes arthritis in 70%, simultaneous 15%, follows 15%), affecting 20-30% of psoriasis patients. CASPAR criteria: inflammatory articular disease plus psoriasis (2 points), nail dystrophy (1), negative RF (1), dactylitis (1), juxta-articular new bone formation (1); score ≥ 3 .

Psychoactive Substance

- See drugs of abuse.

Psychodermatology

- Psychodermatology addresses the mind-skin connection. Stress exacerbates psoriasis, eczema, acne, and urticaria. Psychiatric disorders (depression, anxiety, body dysmorphic disorder) are associated with skin conditions.

Psychodynamic Therapy

- A form of depth psychology that explores how unconscious processes, early life experiences, and internal conflicts influence current thoughts, feelings, and behaviors. It involves exploration of transference and countertransference, defense mechanisms, and the therapeutic relationship as a vehicle for change.

Psychogenic Non-Epileptic Seizures

- Seizure-like episodes that are not caused by abnormal electrical activity in the brain. They are a form of conversion disorder with attacks or seizures. Events may include shaking, stiffening, unresponsiveness, and other movements.

Psychological Support

- Counseling, therapy for patients with chronic illness, disability, pain. Address depression, anxiety, adjustment difficulties, coping strategies.

Psychopharmacology

- The study and clinical application of medications used to treat psychiatric disorders. It involves understanding drug mechanisms, pharmacokinetics, drug interactions, and side effect profiles.

Psychosis

- A syndrome characterized by loss of contact with shared reality, commonly involving delusions, hallucinations, disorganized thought or behavior, and negative symptoms. Causes include primary psychotic disorders, mood disorders, substances, medications, neurologic disease, and systemic illness.

Psychosocial Rehabilitation

- Support for mental illness recovery: skills training, supported employment, peer support.

Psychosocial Support

- Cancer affects mental health. Depression, anxiety common. Support groups, counseling beneficial.

Psychotherapy

- A structured treatment in which a trained therapist helps a person understand and change distressing thoughts, emotions, behaviors, or relationship patterns. Major approaches include cognitive behavioral therapy, psychodynamic therapy, interpersonal therapy, supportive therapy, exposure-based therapy, and family or group therapy.

Pterygium

- A triangular, fibrovascular proliferation of bulbar conjunctiva extending onto the cornea, usually nasal, associated with chronic ultraviolet exposure, dust, and dry climates. Most are asymptomatic or cause mild irritation, but growth toward the visual axis can induce astigmatism and obstruct vision.

PTH-Related Peptide

- A peptide that activates the parathyroid-hormone receptor and can cause humoral hypercalcemia of malignancy. It increases bone resorption and renal calcium retention while reducing phosphate reabsorption.

Pthirus pubis

- The pubic louse, a parasitic insect that infests the pubic hair and occasionally other coarse body hair, causing pediculosis pubis. Transmission is primarily through sexual contact. It causes intense pruritus and may transmit *Borrelia recurrentis* causing re-

lapsing fever. Diagnosis is by visual identification of the louse or nits.

Ptosis

- The drooping of the upper eyelid, which can be congenital or acquired, and can range from mild drooping that is primarily a cosmetic concern to severe drooping that obstructs the visual axis and causes amblyopia in children or visual impairment in adults.

PTSD

- See Post-Traumatic Stress Disorder.

Puberty

- Developmental period during which sexual maturation and reproductive capability are attained.

Pubis

- Anterior-inferior hip bone with body, superior ramus, and inferior ramus. Two pubic bones meet at the pubic symphysis (cartilaginous joint widening in childbirth). Pubic tubercle is the lateral inguinal ligament attachment. Direct hernias emerge medial to inferior epigastric vessels near it.

Public Exposure

- Radiation exposure to public from nuclear medicine patients. Guidelines for discharge after therapy. I-131 therapy: radiation safety instructions for 3-7 days.

Public Health

- The science and practice of protecting and improving the health of communities through education, policy-making, and research for disease and injury prevention.

Public Health Emergency

- Event threatening public health. Examples: pandemic, bioterrorism, natural disaster. Declaration enables special powers.

Public Health Research Bias

- Systematic error in a population-health study, including selection bias, information bias, and confounding.

Pudendal Block

- A local anesthetic block of the pudendal nerve (S2-S4) used for perineal anesthesia during the second stage of labor, outlet forceps delivery, or perineal repair. Technique: transvaginal approach — a needle

guide is placed on the ischial spine, and the needle is advanced through the sacrospinous ligament.

Puerperal Fever

- Fever higher than 38 degrees Celsius occurring during the first 10 days postpartum, not including the first 24 hours. Most common cause is endometritis, particularly after cesarean delivery. Other causes include urinary tract infection, wound infection, mastitis, and thrombophlebitis.

Puerperal psychosis

- A rare psychiatric emergency beginning soon after childbirth, characterized by rapidly changing mood, confusion, delusions, hallucinations, or disorganized behavior.

Puerperium

- The period following delivery during which the reproductive organs return to their prepregnancy state, approximately 6 weeks in duration. Characterized by uterine involution, cervical closure, lochial discharge changes, and hormonal shifts. Physical recovery includes perineal healing, breast engorgement, and metabolic readjustment.

Pugilistic Stance

- The characteristic body position adopted by fire victims—flexed arms, elbows bent, fists clenched, knees flexed—resulting from heat-induced contraction (denaturation) of muscles (mainly flexors) and tendons, not from conscious movement. It occurs with thermal injury and can be present regardless of cause (even if death preceded fire).

Pulmonary Alveolar Hemorrhage

- Bleeding into the alveolar spaces from disruption of the alveolar-capillary membrane. Causes include vasculitis (GPA, MPA, Goodpasture), connective tissue disease, medications, and coagulopathy. It presents with hemoptysis, falling hemoglobin, diffuse bilateral infiltrates, and serially bloody returns on BAL.

Pulmonary Alveolar Microlithiasis

- A rare genetic disease caused by SLC34A2 mutations, leading to accumulation of calcium phosphate microliths within the alveoli. HRCT shows diffuse bilateral calcified nodules with a “sandstorm” appearance. It is usually diagnosed in the third to fourth decade.

Pulmonary Alveolar Proteinosis

- A rare disease characterized by accumulation of surfactant-like material within the alveoli due to impaired clearance by alveolar macrophages. Most cases are autoimmune (anti-GM-CSF antibodies). Presents with progressive dyspnea and cough. HRCT shows bilateral ground-glass opacities with “crazy paving” pattern.

Pulmonary Angiography

- Contrast imaging of pulmonary arteries. Gold standard for pulmonary embolism diagnosis. Largely replaced by CTPA.

Pulmonary Arterial Hypertension (PAH)

- A precapillary pulmonary hypertension syndrome caused by increased pulmonary vascular resistance from pulmonary arteriolar disease. It may be idiopathic, heritable, drug-associated, or associated with conditions such as connective-tissue disease, congenital heart disease, portal hypertension, or HIV.

Pulmonary Arteriovenous Fistula

- An abnormal direct connection between a pulmonary artery and vein that bypasses the capillary bed. It can cause hypoxemia, cyanosis, dyspnea, paradoxical emboli, or brain abscess.

Pulmonary Arteriovenous Malformation

- An abnormal direct communication between a pulmonary artery and vein, causing a right-to-left shunt. It may be idiopathic or associated with hereditary hemorrhagic telangiectasia (Osler-Weber-Rendu syndrome). It can cause hypoxemia, paradoxical emboli (stroke, brain abscess), and hemoptysis.

Pulmonary Artery

- The blood vessel that carries deoxygenated blood from the right ventricle of the heart to the lungs.

Pulmonary Artery Catheter

- A catheter inserted through the right heart into the pulmonary artery to measure hemodynamic parameters (PA pressure, PAWP, cardiac output). Used for diagnosis and management of pulmonary hypertension, heart failure, and critical illness.

Pulmonary Atresia

- A congenital heart defect where the pulmonary valve is completely absent (atretic), obstructing right ventricular outflow. Two main types: pul-

monary atresia with intact ventricular septum (PA-IVS) and pulmonary atresia with ventricular septal defect (PA-VSD, also called tetralogy of Fallot with pulmonary atresia).

Pulmonary Capillary Wedge Pressure

- An indirect estimate of left-atrial pressure obtained by briefly occluding a small pulmonary-artery branch with a balloon-tipped catheter. It reflects left-sided filling pressure when measured correctly and interpreted in the appropriate clinical and ventilatory context.

Pulmonary Contusion

- Bruising of the lung parenchyma from blunt chest trauma, causing hemorrhage and edema without lung laceration. It impairs gas exchange and can progress to ARDS. It is often associated with rib fractures and flail chest. Treatment is supportive with lung-protective ventilation.

Pulmonary Edema

- Accumulation of fluid in the pulmonary interstitium and alveoli, most commonly due to left-sided heart failure (cardiogenic). Causes increased hydrostatic pressure leads to transudation. Presents with acute dyspnea, orthopnea, crackles, and S3 gallop. Non-cardiogenic causes include ARDS, renal failure, and high altitude.

Pulmonary Embolism

- See emergency medicine.

Pulmonary Embolism (PE)

- Occlusion of the pulmonary arterial bed by a thrombus, tumor, air, or fat, most commonly originating as a deep vein thrombosis (DVT) from the lower extremities. Presents with acute onset of dyspnea, pleuritic chest pain, cough, hemoptysis, tachycardia, and hypoxemia.

Pulmonary Embolism Treatment

- Anticoagulation: DOACs preferred (rivaroxaban, apixaban) or LMWH bridging to warfarin. Duration: 3 months for provoked (reversible risk factor), indefinite for unprovoked or recurrent. Massive PE: systemic thrombolysis (alteplase 100 mg IV over 2 hours) or surgical catheter-directed embolectomy.

Pulmonary Fibrosis

- Scarring replacing normal alveolar architecture. Causes: IPF, connective tissue disease, asbestos, ra-

diation, drugs (bleomycin, methotrexate). Features: progressive dyspnea, dry cough, digital clubbing.

Pulmonary Function Tests

- Spirometry: FVC (forced vital capacity), FEV1 (forced expiratory volume in 1 second), FEV1/FVC ratio. Obstructive pattern: FEV1/FVC <0.7 (asthma, COPD). Restrictive pattern: FVC <80% with normal or increased FEV1/FVC (pulmonary fibrosis, chest wall disorders). Lung volumes: TLC (total lung capacity), RV (residual volume).

Pulmonary Hemosiderosis

- Iron deposition in the lungs from repeated alveolar hemorrhage. It presents with hemoptysis, anemia, and diffuse bilateral infiltrates on imaging. Causes include Goodpasture syndrome, granulomatosis with polyangiitis, and idiopathic pulmonary hemosiderosis. The classic triad is hemoptysis, anemia, and diffuse pulmonary

Pulmonary Hypertension

- Elevated mean pulmonary artery pressure (>20 mmHg at rest) due to various etiologies. WHO Group 1: pulmonary arterial hypertension (idiopathic, heritable, drug-induced). Group 2: left heart disease. Group 3: lung disease. Group 4: chronic thromboembolic. Group 5: multifactorial. Diagnosis requires right heart catheterization.

Pulmonary Hypertension Classification

- WHO Classification: Group 1 (PAH: idiopathic, heritable, drug-induced, connective tissue disease-associated). Group 2 (left heart disease: systolic/diastolic dysfunction, valvular). Group 3 (lung disease/hypoxia: COPD, ILD, sleep apnea). Group 4 (chronic thromboembolic). Group 5 (multifactorial: sarcoidosis, hematologic disorders).

Pulmonary Infarction

- Lung necrosis from PE when bronchial supply is compromised. Causes pleuritic pain, hemoptysis, and wedge-shaped opacity. Most PE do not cause infarction due to dual blood supply.

Pulmonary Lobes

- The anatomical divisions of the lungs separated by interlobar fissures. The right lung has superior, middle, and inferior lobes; the left lung has superior and inferior lobes and a lingula.

Pulmonary Rehabilitation

- A supervised, multidisciplinary program combining exercise training, education, behavioral support, and self-management for chronic respiratory disease. It improves exercise capacity, dyspnea, functional status, and quality of life.

Pulmonary Sequestration

- A congenital malformation of non-functioning lung tissue that lacks normal communication with the tracheobronchial tree and receives its blood supply from a systemic artery (usually the descending aorta). Intralobar sequestration is within the normal pleural investment; extralobar is outside.

Pulmonary Stenosis

- A right ventricular outflow tract obstruction at the valvular, subvalvular, or supra-valvular level. Valvular pulmonary stenosis (most common) is often congenital, frequently associated with Noonan syndrome.

Pulmonary Surfactant

- A phospholipid-protein mixture produced by type II alveolar pneumocytes that reduces surface tension at the air-liquid interface within alveoli. The primary component is dipalmitoylphosphatidylcholine (DPPC, lecithin), which forms a monolayer that reduces surface tension to near zero during expiration.

Pulmonary tuberculosis

- See Tuberculosis (TB).

Pulmonary Valve

- Semilunar valve between right ventricle and pulmonary trunk with three cusps. Most superiorly located cardiac valve with thinner cusps than the aortic valve. Pulmonic stenosis is a congenital defect causing right ventricular hypertrophy. Less commonly affected by endocarditis than mitral/aortic valves.

Pulmonary Vascular Resistance

- The resistance to blood flow through the pulmonary circulation, calculated approximately as the transpulmonary pressure gradient divided by cardiac output. Elevated resistance contributes to pulmonary hypertension and right-ventricular afterload.

Pulmonary Vasodilator Therapy

- Medications that reduce pulmonary vascular re-

sistance and pulmonary artery pressure. Used for WHO Group 1 PAH. Includes endothelin receptor antagonists (bosentan), PDE-5 inhibitors (sildenafil), prostacyclin analogs (epoprostenol), and soluble guanylate cyclase stimulators (riociguat).

Pulmonary Vein

- The four veins (two from each lung) returning oxygenated blood from the lungs to the left atrium, a unique exception in the venous system (carrying oxygenated blood). Superior and inferior pulmonary veins enter the left atrium posteriorly.

Pulmonary Wedge Pressure

- The pressure measured by a balloon-tipped pulmonary artery catheter when the balloon is inflated, occluding a branch of the pulmonary artery. It approximates left atrial pressure. Normal is 6–12 mmHg. Elevated in left heart failure (PCWP > 18 mmHg). Low in hypovolemia.

Pulmonary-artery dissection

- Rare but life-threatening tear in the pulmonary artery wall, most commonly associated with pulmonary hypertension, Marfan syndrome, or prior pulmonary artery catheterization. Presents with acute chest pain, dyspnea, and hemodynamic collapse. Diagnosis by CT angiography or echocardiography; management requires emergent surgical repair.

Pulp

- Soft connective tissue central to tooth containing blood vessels, nerves, connective tissue. Responds to injury with inflammation, necrosis, calcification. Root canal therapy removes infected pulp.

Pulp Stone

- Calcified mass in pulp chamber or root canal. May be free, attached, or embedded. Incidental finding.

Pulpectomy

- Complete removal of pulp from primary tooth. Followed by resorbable filling material.

Pulposus

- Relating to the nucleus pulposus, the gelatinous central portion of an intervertebral disk.

Pulpotomy

- Removal of coronal pulp in primary tooth with reversible inflammation. Maintains radicular pulp vitality.

Pulse

- Rhythmic expansion of artery with each heartbeat. Normal: 60-100 bpm. Rate, rhythm, amplitude assessed.

Pulse Examination

- Assessment of arterial pulses. Absent or weak pulses indicate disease.

Pulse Oximeter

- Non-invasive device measuring oxygen saturation (SpO₂). Uses photodensitometry at two wavelengths. Normal: 95-100%. Limitations: poor perfusion, nail polish, motion artifact.

Pulse Oximetry

- A noninvasive method measuring oxygen saturation by spectrophotometry: light at two wavelengths (red and infrared) passes through a pulsatile vascular bed (finger, ear, toe); absorption difference estimates oxyhemoglobin fraction (SpO₂). Normal 95-100 percent. Used widely in monitoring (perioperative, emergency, sleep, home oxygen).

Pulseless Electrical Activity

- A cardiac arrest rhythm organized electrical activity without a detectable pulse. PEA has many reversible causes (5 H's and 5 T's): hypovolemia, hypoxia, hydrogen ion (acidosis), hypo/hyperkalemia, hypothermia; tension pneumothorax, tamponade (cardiac), toxins, thrombosis (pulmonary), thrombosis (coronary).

Pulsus Paradoxus

- A decrease in systolic blood pressure greater than 10 mmHg during inspiration, normally less than 10 mmHg. Exaggerated in cardiac tamponade (due to ventricular interdependence in the confined pericardial space), severe asthma, COPD exacerbation, and constrictive pericarditis.

Punch Biopsy

- A small, full-thickness circular biopsy obtained with a rotating trephine, used to diagnose skin and soft-tissue lesions. Suture closure may be needed.

Punctate Hemorrhages (Tardieu Spots)

- Small, pinpoint sub-ecchymotic hemorrhages (petechiae) resulting from the rupture of small capillaries under conditions of intense mechanical venous engorgement and asphyxia.

Punctate palmoplantar keratoderma

- A hereditary keratoderma characterised by multiple, small, firm, circumscribed keratotic papules and plaques scattered across the palms and soles, often with an autosomal dominant inheritance pattern. It may present in adolescence or adulthood and can be associated with malignant transformation in rare cases.

Pupil

- Central round iris aperture for light entry, size controlled by sphincter pupillae (CN III parasympathetic, constricts to 2-3 mm) and dilator pupillae (sympathetic, dilates to 8 mm). The pupillary light reflex tests CN II (afferent) and CN III (efferent): light in one eye causes bilateral constriction (direct and consensual).

Pure Tone Audiometry

- The standard behavioral test used to measure hearing sensitivity across the frequency range important for speech perception, typically from 250 to 8000 Hertz, by determining the lowest intensity level at which a pure tone can be detected approximately fifty percent of the time (the hearing threshold).

Purging

- See eating disorders.

Purging Disorder

- Recurrent purging behavior to influence weight or shape in the absence of binge eating. Self-induced vomiting, misuse of laxatives, diuretics, or other medications. It is classified under Other Specified Feeding or Eating Disorder (OSFED). It can lead to electrolyte imbalances, dental erosion, Russell's sign, and esophageal tears.

Purines

- Nitrogenous bases containing a double-ring structure (a six-membered ring fused to a five-membered imidazole ring), which serve as essential components of nucleotides in DNA and RNA. The primary purines are adenine (A) and guanine (G).

Purkinje Fibers

- The terminal branches of the cardiac conduction system, distributed throughout the ventricular myocardium. They conduct impulses rapidly (up to 4 m/s) to ensure coordinated ventricular contraction from apex to base. Purkinje fibers have an intrinsic

firing rate of 20-40 beats per minute.

Purpura

- Purple discoloration of the skin or mucous membranes caused by extravasation of red blood cells from blood vessels into the surrounding tissue. Purpura can be classified by size: petechiae are less than 2 millimeters, purpura are 2 to 10 millimeters, and ecchymoses are greater than 10 millimeters.

Purpura fulminans

- A rapidly progressive, life-threatening condition characterised by extensive, confluent, non-blanching purpuric skin necrosis with haemorrhagic bullae, resulting from acute intravascular thrombosis and disseminated intravascular coagulation.

Purulent Pericarditis

- Bacterial infection of the pericardial space, most commonly caused by *S. aureus*, *S. pneumoniae*, or gram-negative organisms. Often occurs as a complication of pneumonia, endocarditis, chest surgery, or immunosuppression. Presents with fever, pleuritic chest pain, and rapidly accumulating effusion with tamponade physiology.

Pus

- A thick, yellow or green liquid that is composed of dead cells and bacteria, most often found at the site of a bacterial infection.

Push Enteroscopy

- Endoscopic technique using a longer endoscope to visualize and intervene in the proximal small bowel beyond the reach of standard upper endoscopy. Unlike capsule endoscopy, push enteroscopy allows direct visualization with the ability to obtain biopsies and perform hemostasis.

Pustule

- A small, circumscribed elevation of skin containing pus (neutrophils) within the epidermis. Causes: bacterial (folliculitis, impetigo, furuncle), acne vulgaris (inflammatory comedones), pustular psoriasis (generalized pustular psoriasis—systemic symptoms), hidradenitis suppurativa, and drug reactions.

Putrefaction

- Postmortem decomposition of body tissues brought about by bacterial and enzymatic action, characterized by green discoloration of lower abdomen (cecum), marbling of cutaneous veins, gas disten-

sion, foul odor, and liquefaction.

Pyelonephritis

- An infection of the renal parenchyma and pelvicalyceal system, typically resulting from ascending urinary tract infection. Acute pyelonephritis most commonly affects women and is caused by gram-negative enteric organisms, particularly *Escherichia coli*, which account for approximately 80 percent of cases.

Pyelotomy

- An incision into the renal pelvis, usually made to remove an obstruction or kidney stone.

Pylephlebitis

- Septic thrombophlebitis of the portal vein or its tributaries, most commonly arising from intra-abdominal infections such as appendicitis, diverticulitis, cholangitis, or hepatic abscess. Presents with high fever, rigors, right upper quadrant pain, jaundice, and sepsis.

Pyloric Stenosis

- Hypertrophy of the pyloric sphincter muscle causing gastric outlet obstruction most common in young infants. Presents with non-bilious projectile vomiting typically beginning at two to eight weeks of age. Dehydration and metabolic alkalosis with hyponatremia and hypokalemia develop as the condition progresses.

Pyoderma Gangrenosum

- A neutrophilic dermatosis causing rapidly expanding, painful ulcers with violaceous undermined borders, pathergy phenomenon. Associated with IBD (50%), hematologic malignancy, rheumatoid arthritis, PAPA syndrome. Peristomal PG in ostomy patients.

Pyogenic Granuloma

- A benign vascular proliferation presenting as a rapidly growing, friable, red papule or nodule that bleeds easily. It commonly occurs on fingers, face, and lips, and may be triggered by trauma or pregnancy (epulis gravidarum). Diagnosis is clinical.

Pyonephrosis

- Pus in renal collecting system. Urological emergency. Requires drainage (stent or nephrostomy) and antibiotics.

Pyostomatitis vegetans

- A rare, chronic, inflammatory oral mucosal condition characterized by multiple, soft, yellowish, vegetative pustules on an erythematous base, predominantly affecting the buccal mucosa and gingiva. It is strongly associated with inflammatory bowel disease, particularly ulcerative colitis, and may precede gastrointestinal symptoms.

Pyrexia

- Body temperature $>38^{\circ}\text{C}$ (100.4°F), also known as fever. A regulated rise in body temperature mediated by pyrogens (exogenous: microbial products, endotoxins; endogenous: IL-1, IL-6, TNF, prostaglandin E_2) acting on the hypothalamic thermoregulatory center.

Pyridoxine (Vitamin B6)

- A vitamin converted to pyridoxal phosphate, a coenzyme for transamination, decarboxylation, and other amino-acid reactions. Deficiency can cause neuropathy, dermatitis, seizures, and sideroblastic anemia; excessive supplementation can also cause neuropathy.

Pyrimidines

- Nitrogenous bases containing a single six-membered heterocyclic ring, serving as structural building blocks for nucleic acids. The primary pyrimidines are cytosine (C), thymine (T, present in DNA), and uracil (U, present in RNA).

Pyruvate

- The three-carbon alpha-keto acid produced at the end of glycolysis from phosphoenolpyruvate by pyruvate kinase.

Pyruvate Dehydrogenase

- Converts pyruvate to acetyl-CoA. Regulated by phosphorylation (inactivated by PDH kinase). Requires thiamine, lipoate, CoA, FAD, NAD $^{+}$.

Pyruvate Kinase Deficiency

- An inherited red-cell glycolytic defect causing chronic hemolytic anemia of variable severity. Manifestations include jaundice, splenomegaly, gallstones, fatigue, and complications of iron overload.

Peritoneal Infection

- Infection of the lining of the abdominal cavity, often called peritonitis. Causes include a ruptured appendix or bowel, abdominal surgery, pelvic in-

fection, or contamination during dialysis. Severe abdominal pain, fever, vomiting, or a rigid abdomen requires urgent assessment.

Pelvic Floor Dysfunction

- Weakness, overactivity, poor coordination, or injury of the muscles and connective tissues supporting the bladder, bowel, and reproductive organs. Symptoms may include urinary or fecal problems, pelvic pain, constipation, pain with sex, or pelvic organ prolapse; assessment may involve pelvic-health clinicians.

Postpartum Anxiety

- Excessive, persistent, or distressing anxiety during pregnancy or after childbirth that interferes with sleep, daily function, bonding, or self-care. It may occur with panic, intrusive worries, or depression and deserves clinical assessment; thoughts of self-harm or harm to the baby require emergency help.

Postural Orthostatic Tachycardia Syndrome (POTS)

- An autonomic disorder causing an excessive increase in heart rate after standing, with symptoms such as light-headedness, palpitations, fatigue, weakness, brain fog, nausea, or fainting. Diagnosis requires clinical assessment and exclusion of other causes of orthostatic symptoms.

Pelvic Congestion Syndrome

- Chronic pelvic pain associated with enlarged or inefficient pelvic veins, often worsening with prolonged standing, before menstruation, or after intercourse. Diagnosis requires evaluation for other causes of pelvic pain and may involve pelvic imaging; treatment is individualized by gynecology or vascular specialists.

Chapter 17

Pocket Dictionary: Q Terms

Q Fever

- A zoonotic disease caused by *Coxiella burnetii*, an obligate intracellular Gram-negative bacterium. Q fever is transmitted to humans primarily by inhalation of contaminated aerosols from infected livestock (cattle, sheep, goats) and their products (placenta, birth fluids, urine, feces, milk). Incubation: 2–3 weeks.

Q Wave

- The negative deflection that follows the P wave in some electrocardiographic leads, representing initial ventricular depolarization.

QALY

- Quality-adjusted life year. Measure of health outcome incorporating quality of life.

QRS Complex

- The deflection representing ventricular depolarization, consisting of a Q wave (initial downward deflection), R wave (upward deflection), and S wave (downward deflection after R). Normal duration is 0.06–0.10 seconds (1.5–2.5 small boxes). Widened QRS (>0.12 s) indicates bundle branch block, ventricular rhythm, or hyperkalemia.

QT Interval

- The time from the onset of the QRS complex to the end of the T wave, representing total ventricular depolarization and repolarization. Normal QT is 0.35–0.44 seconds. The corrected QT (QTc) adjusts for heart rate using Bazett's formula: $QTc = QT / \sqrt{RR}$ interval.

QT Prolongation

- An electrocardiographic finding characterized by prolongation of the QT interval, representing the time from ventricular depolarization to repolarization, which predisposes to the potentially fatal ar-

rhythmia torsades de pointes. The corrected QT interval should be less than 440 milliseconds in men and 460 milliseconds in women.

Quad Screen (Maternal Serum Screening)

- Second-trimester prenatal screening (15–22 weeks) for fetal aneuploidy and neural tube defects. Measures 4 markers: AFP (alpha-fetoprotein: elevated in NTD, low in Down syndrome), hCG (elevated in Down syndrome), uE3 (unconjugated estriol: low in Down syndrome), and inhibin A (elevated in Down syndrome).

Quadrant

- One of four sections of the abdomen (RUQ, LUQ, RLQ, LLQ) used for clinical description. RUQ: liver, gallbladder, duodenum, right kidney. LUQ: stomach, spleen, left kidney, pancreas tail. RLQ: appendix, cecum, right ovary. LLQ: sigmoid colon, left ovary.

Quadrantectomy

- Surgical removal of one quadrant of an organ, most commonly a breast-conserving operation removing a breast tumour with a surrounding margin of tissue. The exact procedure and need for additional treatment depend on pathology and clinical staging.

Quadriceps Femoris

- The four-headed muscle group on the anterior thigh consisting of rectus femoris and the three vasti. It extends the knee; rectus femoris also flexes the hip, and the group contributes to standing, walking, and shock absorption.

Quadriceps Femoris Muscle

- Four-muscle anterior thigh group: rectus femoris, vastus lateralis, vastus medialis, vastus intermedius, all inserting on the tibial tuberosity via the patellar ligament. Innervated by femoral nerve (L2–L4). Pri-

mary knee extensor for walking, running, rising from seated. VMO is important for patellar tracking.

Quadriceps tendon ruptures

- A tear of the tendon connecting the quadriceps muscle to the patula, causing inability to extend the knee and requiring surgical repair, typically in middle-aged or older patients.

Quadriplegia

- Paralysis of all four limbs and trunk, resulting from a spinal cord lesion at the cervical level or a bilateral brainstem lesion. Common causes include traumatic spinal cord injury, stroke, multiple sclerosis, spinal cord tumors, and demyelinating disease.

Quadruplet Pregnancy

- A pregnancy involving four fetuses. It carries increased risks of preterm birth, fetal growth restriction, hypertensive and metabolic complications, and maternal morbidity and therefore requires specialized multidisciplinary antenatal care.

Quality Assurance

- Systematic efforts to enhance care. See quality improvement.

Quality Control

- Daily, weekly, monthly tests ensure system performance. Uniformity, energy, sensitivity, spatial resolution, count rate performance.

Quality Control (Laboratory)

- Procedures using control materials, calibration, equipment checks, and documented review to ensure that a laboratory method remains accurate and precise during routine testing.

Quality Improvement

- Systematic efforts to improve healthcare. Methods: PDSA cycles, Lean, Six Sigma. Outcomes: safety, effectiveness, patient-centeredness.

Quality of Life

- Perception of well-being including physical, psychological, social, spiritual domains. See geriatrics.

Quality-Adjusted Life Year (QALY)

- A health-economic measure combining length of life with health-related quality of life on a scale where 1 represents full health and 0 represents death. It is used to compare cost-effectiveness across interventions.

Quantification

- Measurement of radiotracer uptake or clearance. SUV (PET), clearance rates (renal), ejection fraction (cardiac). Essential for treatment monitoring.

Quantitative PET

- Measurement of absolute tracer concentration. SUV, kinetic modeling.

Quarantine

- Separation of exposed individuals to prevent disease spread. For symptomatic incubation period.

Quickening

- The first perception of fetal movement by the mother, typically felt between 16-20 weeks in multiparous women and 18-22 weeks in nulliparous women. Described initially as a subtle fluttering sensation that becomes stronger and more regular as pregnancy progresses.

Quiescent

- Inactive or dormant state of disease. Example: quiescent ulcerative colitis (no symptoms but histologic inflammation may persist).

Quinine

- An alkaloid derived from cinchona bark, with antimalarial, antipyretic, and skeletal muscle relaxant properties. Quinine is a blood schizonticide with activity against all Plasmodium species, but resistance has emerged in *P. falciparum*.

Quinsy

- Peritonsillar abscess, also known as quinsy, is a collection of pus in the peritonsillar space between the tonsillar capsule and the superior pharyngeal constrictor muscle, typically developing as a complication of acute tonsillitis, and it is the most common deep neck space infection. See Peritonsillar Abscess.

Quinupristin-dalfopristin

- A combination streptogramin antibiotic that inhibits bacterial protein synthesis by binding to the 50S ribosomal subunit. The combination is bactericidal against most gram-positive organisms including MRSA. It is indicated for vancomycin-resistant *Enterococcus faecium* infections. The drug is administered intravenously every 8-12

Quorum Sensing

- Cell-density-dependent bacterial communication

system in which bacteria produce and detect small signaling molecules (autoinducers—acyl-homoserine lactones in gram-negative, autoinducing peptides in gram-positive, AI-2 inter-species) to coordinate population-wide behaviors.

Chapter 18

Pocket Dictionary: R Terms

Rabies

- A fatal viral encephalitis caused by rabies virus, transmitted through the bite of infected animals. After inoculation, the virus travels to the CNS via peripheral nerves. Symptoms include hydrophobia, aerophobia, and progressive neurological deterioration. Diagnosis is made by direct fluorescent antibody testing of brain tissue.

Rabies vaccination

- See Prophylaxis.

Radial Artery

- The smaller terminal branch of the brachial artery, arising at the cubital fossa, running laterally along the radius to the wrist (palpable at the anatomical snuffbox) and hand. Supplies the lateral forearm and the thumb side of the hand; forms the deep palmar arch.

Radial artery thrombosis

- Formation of a thrombus in the radial artery, often at the site of arterial catheterization or in the setting of a hypercoagulable state or vasculitis. May cause hand ischemia with pain, pallor, and decreased pulse. Treatment includes anticoagulation and surgical intervention for limb-threatening ischemia.

Radial Club Hand

- A congenital longitudinal deficiency of the radial ray, ranging from thumb hypoplasia to complete absence of the radius. The hand is radially deviated due to the absent radial support, with the carpus displaced proximally and radially.

Radial keratotomy

- A refractive surgical procedure, now largely obsolete, in which radial incisions are made in the peripheral corneal stroma to flatten the central cornea and reduce myopia. It was superseded by laser refractive

surgery (LASIK, PRK) due to complications including diurnal visual fluctuation, glare, and hyperopic shift.

Radial Nerve

- From posterior cord (C5-T1), the plexus's largest branch. In the arm runs in the radial groove with the profunda brachii artery, innervating triceps and anconeus. At the elbow divides into superficial (sensory to dorsal hand) and deep (posterior interosseous) branches. Deep branch innervates forearm/hand extensors.

Radiation Biology

- Study of radiation effects on biological systems. See radiobiology.

Radiation Contamination

- Unwanted radioactive material on a person, surface, object, or body site. Contamination may be removable or fixed and requires containment, monitoring, decontamination, and documentation according to radiation-safety procedures.

Radiation Dose

- Effective dose from diagnostic procedures. PET/CT: 15-25 mSv. Bone scan: 3-5 mSv. Myocardial perfusion: 8-15 mSv. Justification: benefit should exceed risk.

Radiation Emergency

- Planning for disasters. See disaster preparedness.

Radiation Enteritis

- Inflammation and injury of the small or large intestine caused by therapeutic ionizing radiation. It may be acute with diarrhea, cramping, and nausea or delayed with strictures, malabsorption, bleeding, fistula, or obstruction.

Radiation Monitoring

- Measurement of radiation exposure. Personal dosimeters, area monitors.

Radiation Necrosis

- Irreversible tissue injury caused by ionizing radiation, often developing months or years after treatment. In the brain it may produce edema, seizures, headache, or focal deficits and can resemble recurrent tumour on imaging.

Radiation Oncologist

- Physician providing radiation therapy.

Radiation Oncology

- A medical specialty that uses high-energy radiation (such as X-rays or gamma rays) to shrink tumors and destroy cancer cells.

Radiation Physics

- Physics of radiation imaging and therapy.

Radiation Pneumonitis

- An inflammatory lung injury occurring weeks to months after thoracic radiation. It may cause dry cough, fever, dyspnea, and hypoxemia within the irradiated field and can progress to radiation fibrosis.

Radiation Poisoning

- See radiation injury.

Radiation Pregnancy Screening

- Assessment of pregnancy status before procedures involving ionizing radiation or radioactive materials when pregnancy is possible. The decision to proceed requires clinical justification, risk optimization, and consideration of gestational age and fetal dose.

Radiation Protection

- Principles: time, distance, shielding. See radiation safety.

Radiation Protection (ALARA)

- The practice of keeping radiation exposure as low as reasonably achievable while obtaining the required diagnostic or therapeutic result. It relies on justification, optimization, time reduction, distance, shielding, and appropriate contamination control.

Radiation recall

- An inflammatory reaction that occurs in previously irradiated skin or mucosa after administration of certain chemotherapeutic agents, particularly antimetabolites and anthracyclines. It manifests as

erythema, desquamation, or mucositis in the previously irradiated field, mimicking acute radiation dermatitis.

Radiation Safety

- Principles: time, distance, shielding. ALARA (As Low As Reasonably Achievable). Radioactive materials handled in designated areas. Waste disposal per regulations. Pregnancy screening for women of childbearing age.

Radiation Sickness

- Acute radiation syndrome caused by high-dose whole-body or substantial partial-body exposure to ionizing radiation. It may produce nausea, vomiting, marrow failure, infection, bleeding, gastrointestinal injury, neurovascular collapse, and death depending on dose.

Radiation Therapy

- Use of ionizing radiation to kill cancer cells. Types: external beam, brachytherapy, stereotactic. Mechanisms: DNA damage, cell death. Side effects: fatigue, skin reactions, site-specific effects.

Radiation-induced angiosarcoma

- A rare, aggressive soft tissue sarcoma arising in previously irradiated tissue, typically with a long latency period. It carries a poor prognosis due to local recurrence and metastatic potential.

Radiculitis

- Inflammation or irritation of a spinal nerve root, causing radiating pain, paresthesia, sensory loss, weakness, or reflex change in a dermatomal or myotomal distribution. Causes include infection, inflammatory disease, disc pathology, and compression.

Radioactive Iodine (I-131)

- Used for thyroid imaging and treatment. I-131 emits beta particles (therapy) and gamma rays (imaging). Half-life: 8 days. Used for hyperthyroidism (Graves, toxic nodules) and thyroid cancer ablation. I-123 for diagnostics (shorter half-life, pure gamma).

Radioactive Iodine Ablation

- Treatment of differentiated thyroid carcinoma using iodine-131 to destroy residual thyroid tissue after total thyroidectomy. Indications include intermediate and high-risk disease for remnant ablation and

treatment of residual or recurrent disease. Patients must be hypothyroid or receive recombinant TSH to maximize iodine uptake.

Radioactivity

- The spontaneous transformation of unstable atomic nuclei with emission of particles or electromagnetic radiation. Activity is the rate of nuclear transformations and is measured in becquerels.

Radiobiology

- Study of radiation effects on biological systems. DNA damage, cell death, repair mechanisms.

Radiocarpal Joint

- The wrist joint between the distal radius and articular disc proximally and the scaphoid, lunate, and triquetrum distally. It permits flexion, extension, radial deviation, ulnar deviation, and circumduction.

Radiochemical Purity

- The fraction of total radioactivity present in the intended chemical form of a radiopharmaceutical. Impurities may alter biodistribution, reduce target uptake, and increase radiation dose to nontarget tissues.

Radiochemistry

- Chemistry of radiopharmaceuticals. Labeling, purification, quality control.

Radiocontrast

- Substance enhancing image visibility. See contrast agent.

Radioembolization

- Intra-arterial delivery of radioactive microspheres (yttrium-90) to liver tumors through the hepatic artery, used for hepatocellular carcinoma and liver metastases.

Radiofrequency Ablation

- Image-guided destruction of tissue. See ablation therapy.

Radioimmunoassay (RIA)

- Sensitive immunoassay using radiolabeled antigens or antibodies. Competitive binding principle. Used for hormone measurement (thyroid, reproductive), tumor markers, drug levels. Largely replaced by non-radioactive methods (ELISA) but still used for some assays.

Radiologic Technologist

- Professional performing imaging examinations.

Radiologic-Pathologic Correlation

- Integration of imaging findings with gross and microscopic pathology to explain lesion location, extent, tissue composition, and sampling adequacy. It is important for discordant biopsy and imaging results.

Radiologist

- Physician interpreting imaging studies.

Radiology Report

- Document describing imaging findings and interpretation. Structure: indication, technique, findings, impression. Clear, concise language essential. Critical findings require immediate communication.

Radiology-Patient Identification

- Verification of the correct patient, examination, body site, indication, pregnancy status when relevant, allergies, renal function, and prior imaging before image acquisition or intervention. It is a core diagnostic and procedural safety step.

Radiolucent

- Describing a material that allows relatively greater passage of X-rays and therefore appears darker on a radiograph or CT image. Air is highly radiolucent; some stones, foreign bodies, and lesions may also be radiolucent.

Radionuclide Therapy

- Internal radiation therapy using radiopharmaceuticals. Examples: I-131 (thyroid), Lu-177 DOTATATE (neuroendocrine tumors), Lu-177 PSMA (prostate cancer), Ra-223 (bone metastases), Y-90 microspheres (liver cancer).

Radionuclidic Purity

- The fraction of radioactivity arising from the intended radionuclide rather than contaminating radionuclides. It is important because contaminants may change radiation type, energy, half-life, dosimetry, and patient risk.

Radiopaque

- Describing a material that attenuates X-rays strongly and therefore appears relatively white on a radiograph or CT image. Bone, metal, and some contrast agents are radiopaque.

Radiopharmaceutical

- Radioactive compound used for diagnostic or therapeutic purposes. Components: radionuclide + targeting molecule. Selectivity determines biodistribution and image quality.

Radiopharmaceutical Extravasation

- Inadvertent leakage of a radiopharmaceutical from the injection vessel into surrounding tissue. It can reduce diagnostic accuracy and, with therapeutic agents, cause local radiation injury; the event requires documentation and dose assessment.

Radiopharmaceutical Labeling Kit

- A prepared set of reagents used to attach a radionuclide, commonly technetium-99m, to a compound for diagnostic imaging.

Radiopharmaceutical Quality Control

- Testing performed to verify identity, radionuclidic purity, radiochemical purity, pH, sterility, endotoxin limits, and other release specifications before clinical administration. Requirements depend on the preparation and jurisdiction.

Radiopharmacy

- Preparation and dispensing of radiopharmaceuticals. Good manufacturing practice (GMP) required. Sterility, pyrogen testing, quality control.

Radiosensitization

- Increased susceptibility of cells or tissues to injury from ionizing radiation.

Radiotracer

- A compound containing a radioactive nuclide that follows a biologic pathway and permits in-vivo imaging or measurement. Its distribution reflects the target physiology, receptor, enzyme, transporter, or tissue process.

Radium-223

- Alpha emitter, half-life 11.4 days. Used for bone metastases from prostate cancer. Mimics calcium, targets bone turnover sites.

Radius

- Lateral forearm bone in the anatomical position. Proximal end: head, neck, radial tuberosity. Shaft. Distal end: styloid process, articular surfaces. The radial head articulates with the capitulum and ulnar notch, allowing pronation/supination. Bears

approximately 80 percent of wrist axial load.

Rampant Caries

- Widespread severe caries affecting multiple surfaces. Associated with sugary drinks, xerostomia, bulimia.

Ramsay Hunt Syndrome

- Reactivation of varicella-zoster virus in the geniculate ganglion causing ipsilateral facial paralysis, severe ear pain, and vesicular eruption in or around the ear or oral cavity. Hearing loss and vertigo may occur; prompt treatment is important.

Random Events

- Coincidence events from unrelated photons. Reduced by timing window and energy discrimination.

Randomization

- Random assignment to treatment groups. Reduces bias.

Randomized Controlled Trial

- RCT: experimental study randomly assigning intervention. Gold standard for evidence.

Randomized Controlled Trial (RCT)

- Gold standard for intervention studies. Participants randomly assigned to intervention or control. Minimizes bias.

Range of Motion

- ROM: joint movement capacity. Goniometer measurement. Restrictions from contracture, pain, swelling.

Ranson Criteria

- Clinical scoring system for acute pancreatitis severity incorporating eleven parameters at two time points. Admission parameters include age, white blood cell count, glucose, LDH, and AST. Forty-eight hour parameters include hematocrit drop, BUN increase, calcium, oxygen tension, base deficit, and fluid sequestration.

Ranula

- A mucocele (mucus extravasation cyst) of the sublingual salivary gland, presenting as a blue-translucent, fluctuant swelling in the floor of the mouth, typically lateral to the lingual frenulum. A ranula results from trauma or obstruction of a sublingual gland duct, causing mucus accumulation in the sublingual space.

Rape

- Sexual penetration of the mouth, vagina, or anus accomplished without consent or with a person unable to consent, often through force, threat, fraud, or impairment; many statutes subsume it under sexual assault.

Rapid Sequence Induction

- An intubation technique designed to minimize aspiration risk in patients with full stomach.

Rapid Strep Test

- Immunoassay for Group A Streptococcus (*Streptococcus pyogenes*) from throat swab. Results: 5-10 minutes. Specificity: >95%. Sensitivity: 80-90% compared to culture. Positive: treat. Negative: confirm with throat culture (especially in children).

Rapid-Sequence Intubation

- A planned emergency airway technique using a sedative-hypnotic and a neuromuscular blocker in rapid succession to facilitate tracheal intubation while minimizing aspiration risk. Preparation includes preoxygenation, backup airway plans, hemodynamic assessment, and confirmation of tube position.

Rapidly Progressive Glomerulonephritis

- Rapid renal function loss over days to weeks with crescent formation on biopsy. Type I anti-GBM with linear deposits. Type II immune complex with granular deposits. Type III pauci-immune with ANCA vasculitis. Requires aggressive immunosuppression. Without treatment progresses rapidly to ESRD.

Rasburicase

- Recombinant urate oxidase enzyme that converts uric acid to allantoin which is more soluble and easily excreted. Highly effective for prevention and treatment of hyperuricemia in tumor lysis syndrome. More effective than allopurinol for established hyperuricemia.

Rash

- A general term for any visible skin eruption, change in skin color, texture, or appearance.

Rashes

- A general term for any visible skin eruption, change in skin color, texture, or appearance. See Rash.

Rat bite fever

- A systemic bacterial infection caused by *Streptobacillus moniliformis* or *Spirillum minus*, transmitted through rat bites or contact with rat-contaminated food or water. It presents with relapsing fever, rash, polyarthritis, and lymphadenopathy. The characteristic rash is petechial or maculopapular on the extremities.

Rat-Bite Fever

- Systemic infection caused by *Streptobacillus moniliformis* or *Spirillum minus* after exposure to infected rodents. Fever, migratory arthralgia, vomiting, and a maculopapular or petechial rash may occur.

Raynaud Phenomenon

- Exaggerated vasospastic response of digital arteries to cold or stress, causing well-demarcated color changes (white-blue-red) in fingers or toes. Primary Raynaud: idiopathic, symmetric, no tissue damage, normal capillaries.

Raynaud Syndrome

- A vasospastic disorder causing episodic digital color changes (white -> blue -> red) triggered by cold or stress, due to exaggerated vasoconstriction of digital arteries. Primary Raynaud: benign, symmetric, in young women, no underlying disease.

Raynaud's phenomenon

- See Raynaud Phenomenon.

Reaction Time

- Time from stimulus to response. Declines with age, neurological disease.

Reactivation Tuberculosis

- Reactivation of latent infection, typically years later. Characteristically affects apical and posterior upper lobe segments. Features include chronic cough, hemoptysis, night sweats, weight loss, and cavitation.

Reactive Arthritis

- A post-infectious spondyloarthritis occurring 1-4 weeks after genitourinary (*Chlamydia trachomatis*) or gastrointestinal (*Salmonella*, *Shigella*, *Campylobacter*, *Yersinia*) infection. Classic triad (Reiter syndrome): arthritis, conjunctivitis/uveitis, urethritis/cervicitis (only 1/3 have full triad). HLA-B27 positive in 60-80%.

Reactive Attachment Disorder

- A pattern of inhibited, emotionally withdrawn behavior toward adult caregivers. The child rarely or minimally seeks comfort when distressed and rarely or minimally responds to comfort when distressed. The child shows a pattern of persistently irritable and sad mood that is not limited to interactions with caregivers.

Reactive Oxygen Species

- Superoxide, hydrogen peroxide, hydroxyl radical. Normal metabolism byproducts.

Reactive perforating collagenosis

- A perforating dermatosis in which dermal collagen is extruded through the epidermis, presenting as discrete, dome-shaped papules with a central keratotic plug, most commonly on the extensor extremities.

Receiver Operating Characteristic

- ROC curve: plot of sensitivity vs. 1-specificity. Area under curve measures test accuracy.

Receptor

- A protein that binds a specific signaling molecule (ligand) and initiates a cellular response. Types: (1) Cell surface: ion channel (nicotinic AChR), G-protein coupled (GPCR: adrenergic, muscarinic, opioid), enzyme-linked (RTK: insulin receptor, cytokine receptors).

Receptor Desensitization

- A rapid decrease in the response of a receptor to continuous or repeated agonist stimulation, occurring within minutes to hours.

Receptor Downregulation

- A decrease in the number or sensitivity of receptors in response to chronic receptor stimulation by agonists. When receptors are chronically activated, the cells compensate by internalizing and degrading receptors, reducing receptor density on the cell surface, and decreasing receptor responsiveness.

Receptor Tyrosine Kinases

- Single-pass transmembrane receptors with intrinsic tyrosine kinase activity in their intracellular domain. Ligand binding (e.g., insulin, EGF, PDGF, FGF) induces receptor dimerization and autophosphorylation of tyrosine residues.

Receptor Upregulation

- An increase in the number or sensitivity of receptors in response to chronic receptor blockade or decreased neurotransmitter activity. When receptors are chronically blocked by antagonists, the cells compensate by synthesizing more receptors to maintain homeostatic signaling, leading to increased sensitivity to agonists.

Recessive

- Allele whose phenotype is masked in heterozygotes. Autosomal recessive disorders: cystic fibrosis, sickle cell anemia, Tay-Sachs disease. Requires two copies for expression.

RECIST

- Response Evaluation Criteria in Solid Tumors, standardized imaging-based criteria that categorize response to anticancer therapy as complete response, partial response, stable disease, or progressive disease.

Recombinant DNA

- DNA molecule formed by combining genetic material from different sources. Basis of biotechnology. Used to produce insulin, growth hormone, vaccines, gene therapy vectors.

Recreational Therapy

- Use of recreation for therapeutic purposes. Improves quality of life, socialization, coping.

Rectal Bleeding

- Passage of bright-red or dark blood from the rectum, caused by hemorrhoids, fissures, polyps, inflammatory bowel disease, diverticular disease, or cancer. Persistent, heavy, or unexplained bleeding requires evaluation.

Rectal Prolapse

- Protrusion of rectal tissue through the anal opening. Complete prolapse involves all layers of the rectal wall and presents as a concentric cylindrical mass with visible mucosal folds. Partial prolapse involves only the mucosa. Associated with chronic straining, pelvic floor dysfunction, and neurological conditions.

Rectocele

- A weakening of the vaginal wall that allows the rectum to bulge into the vagina.

Rectovaginal Fistula

- Abnormal connection between the rectum and vagina causing passage of stool or gas through the vagina. Causes include obstetric injury particularly third and fourth degree perineal tears, Crohn disease, radiation therapy, and surgical complications. Presents with vaginal discharge of stool, flatus, or recurrent vaginal infections.

Rectum

- The final straight section of the large intestine, approximately 12-15 cm long, extending from the sigmoid colon to the anal canal. The rectum stores feces before defecation.

Rectus Abdominis Muscle

- Long flat vertical anterior abdominal wall muscle from pubic symphysis to xiphoid process/costal cartilages 5-7, divided by tendinous intersections ("six-pack"). Innervated by T7-T12. Flexes trunk, compresses abdomen, increases intra-abdominal pressure. Diastasis recti common in pregnancy/obesity.

Rectus Abdominis Sheath

- A fibrous aponeurotic envelope formed by the abdominal wall muscles and surrounding the rectus abdominis muscle. It contains the superior and inferior epigastric vessels and related nerves.

Recurrence

- Return of cancer after treatment. Can be local, regional, or distant.

Recurrence Risk

- The probability that a genetic condition will occur again in a future pregnancy or relative, estimated from the inheritance pattern, parental testing, penetrance, and observed or presumed pathogenic variant.

Recurrence-Free Survival

- Time without cancer return after treatment.

Recurrent

- Disease returning after remission. Examples: recurrent UTI, herpes.

Recurrent Caries

- Caries under or around restoration. Due to microleakage, marginal breakdown.

Red Blood Cell Labeling

- Nuclear medicine technique labeling RBCs with

Tc-99m. Used for GI bleeding detection, cardiac blood pool imaging.

Red Flag

- A symptom, sign, or test result suggesting a potentially serious condition that requires urgent assessment or escalation of care. Red flags are context-specific and should not be used as a substitute for clinical judgment.

Red Infarction

- Red infarction, also known as hemorrhagic infarction, occurs in tissues with dual blood supply, loose tissue architecture, or venous obstruction, allowing blood to accumulate in the necrotic area and produce a dark red appearance.

Redundant mitral valve

- Excess leaflet tissue of the mitral valve, most commonly the posterior leaflet, which may prolapse into the left atrium during systole. Often associated with myxomatous degeneration and mitral valve prolapse. May be asymptomatic or cause mild mitral regurgitation detected incidentally on echocardiography.

Reemerging Disease

- Previously controlled disease increasing. Examples: tuberculosis, measles, drug-resistant infections.

Reexpansion pulmonary edema

- Noncardiogenic pulmonary edema developing rapidly after rapid reexpansion of a collapsed lung, typically following chest tube drainage of a large or long-standing pneumothorax.

Refeeding Syndrome

- Life-threatening fluid and electrolyte shifts when nutrition restarted in malnourished patient. Hypophosphatemia, hypokalemia, hypomagnesemia, fluid overload, cardiac failure. Prevention: start low (25-30 kcal/kg), advance slowly, supplement electrolytes, monitor closely.

Reference Interval

- The range of laboratory results expected in a defined reference population, usually encompassing the central 95 percent of values. A result outside the interval is not by itself diagnostic and must be interpreted with clinical context, biologic variation, and the laboratory's method-specific range.

Referred Pain

- An unpleasant sensory and emotional experience

associated with actual or potential tissue damage, serving as a protective alerting system but becoming maladaptive when chronic (lasting >3 months). See pain.

Reflex anoxic seizure

- A brief seizure-like episode caused by transient cerebral hypoperfusion after a sudden painful or emotional stimulus, usually with pallor, bradycardia, and rapid recovery.

Reflex Arc

- The neural pathway that mediates a reflex action, bypassing conscious brain processing for rapid, automatic responses. A reflex arc consists of five components: (1) Receptor—sensory nerve ending detecting a stimulus (e.g., pain, stretch).

Reflux

- Gastroesophageal reflux in infants: effortless regurgitation of gastric contents, very common (physiologic ‘spitting up’) and usually benign, resolving by 12 months with maturation of the lower esophageal sphincter.

Reflux Nephropathy

- Renal scarring and chronic kidney damage caused by vesicoureteral reflux, recurrent urinary infection, or both. It may lead to hypertension, proteinuria, impaired growth, and chronic kidney disease.

Refraction

- The bending of light rays as they pass through the eye’s optical system (cornea, aqueous humour, crystalline lens, vitreous humour) to focus a sharp image on the retina. The refractive power of the eye is measured in dioptres (D). Emmetropia: normal refractive state—parallel light rays focus exactly on the retina without accommodation.

Refractive Errors and Astigmatism

- Refractive errors are vision disorders caused by the eye’s inability to properly focus light onto the retina, resulting in blurred vision.

Refractive Surgery

- Surgical procedures that reshape the cornea or alter the eye to correct refractive errors and reduce dependence on eyeglasses or contact lenses. Laser techniques include LASIK (corneal flap with excimer ablation), photorefractive keratectomy (surface ablation), and small-incision lenticule extrac-

tion (SMILE).

Refractory

- Cancer not responding to treatment. May require alternative approaches.

Refractory Period

- The period after an action potential during which a cell cannot (absolute refractory period) or can only partially (relative refractory period) generate a new action potential.

Refsum Disease

- An autosomal-recessive peroxisomal disorder of phytanic-acid metabolism. It causes retinitis pigmentosa, anosmia, peripheral neuropathy, ataxia, hearing loss, ichthyosis, and shortened metacarpals or metatarsals with variable onset.

Regadenoson Pharmacological Stress

- Selective A2A adenosine receptor agonist used as a pharmacological vasodilator for nuclear myocardial perfusion imaging. Single IV bolus injection produces coronary vasodilation with fewer side effects than adenosine (less bronchospasm, AV block). Contraindicated in severe asthma or COPD requiring oxygen.

Regional Anesthesia

- Regional anesthesia involves the administration of local anesthetic agents to specific nerves or nerve plexuses to provide surgical anesthesia and postoperative analgesia for a defined body region.

Regulatory Approval

- FDA approval process for radiopharmaceuticals. IND, NDA applications.

Regulatory T Cells

- Tregs (CD4+ CD25+ FOXP3+): essential for peripheral tolerance. Thymic (tTreg) vs peripheral (pTreg) origin. FOXP3: master transcription factor; mutations cause IPEX.

Regurgitation

- The backward flow of blood through a heart valve that fails to close properly (valvular incompetence or insufficiency).

Rehabilitation

- See rehabilitation medicine. Post-amputation or post-revascularization.

Rehabilitation Clinic

- Outpatient rehabilitation facility.

Rehabilitation Engineering

- Technology development for disability: prosthetics, orthotics, assistive devices, environmental control.

Rehabilitation Hospital

- Inpatient facility specializing in rehabilitation.

Rehabilitation Medicine

- Physician specialty focusing on function, quality of life for patients with impairment, disability. Also called physiatry.

Rehabilitation Motivation

- Patient factors that influence participation in rehabilitation, including intrinsic and extrinsic motivation and engagement with therapeutic goals.

Rehabilitation Potential

- The expected capacity to improve function, independence, participation, or quality of life through rehabilitation. It depends on diagnosis, prognosis, cognition, motivation, comorbidity, social support, and environmental resources.

Rehabilitation Unit

- Hospital ward dedicated to rehabilitation.

Reinfection

- A new episode of infection by the same or a related pathogen after the original infection has resolved. Reinfection is distinct from relapse, in which the original infection persists or returns from an incompletely eradicated focus.

Relapse

- Return of cancer after treatment. See recurrence.

Relapse-Free Survival

- The time from completion of treatment to recurrence of the cancer or death from any cause.

Relapsing Fever

- An acute febrile illness caused by *Borrelia* species, characterized by recurrent episodes of fever separated by afebrile intervals.

Relapsing Polychondritis

- A systemic autoimmune disorder causing inflammation and destruction of cartilage throughout the body. Airway involvement can cause tracheal and bronchial stenosis. Other features include auricular

chondritis, nasal chondritis, ocular inflammation, and arthritis. Treatment is corticosteroids and immunosuppressants.

Relative Afferent Pupillary Defect (RAPD)

- An asymmetric reduction in the direct pupillary light response caused by unequal afferent input from the retina or optic nerve. It is detected with the swinging-flashlight test and is also called a Marcus Gunn pupil.

Relative Biological Effectiveness (RBE)

- Ratio of dose of reference radiation to dose of test radiation for same effect. Alpha particles: RBE 10-20.

Relative Risk

- Ratio of disease risk in exposed vs. unexposed. $RR = (a/(a+b)) / (c/(c+d))$. $RR > 1$: increased risk. $RR < 1$: protective.

Reliability

- Consistency of measurement. Test-retest, inter-rater reliability.

REM Sleep Behavior Disorder

- Acting out dreams during REM sleep. Associated with Parkinson disease, Lewy body dementia. Treatment: melatonin, clonazepam.

Remifentanyl

- An ultra-short-acting synthetic opioid analgesic metabolized by non-specific plasma and tissue esterases, producing context-sensitive half-time of approximately 3 to 10 minutes regardless of infusion duration. This makes it ideal for total intravenous anesthesia (TIVA) combined with propofol, and for procedures requiring rapid emergence.

Remission

- Decrease or disappearance of cancer signs. Complete remission: no detectable disease. Partial remission: significant reduction but not complete.

Remitting Seronegative Symmetrical Synovitis with Pitting Edema

- RS3PE: an acute-onset, symmetric polyarthritides (usually in elderly) with pitting edema of the hands/feet, negative rheumatoid factor and anti-CCP, and marked response to low-dose corticosteroids.

Remodeling

- Bone adaptation to implant stress. Wolff's law: bone remodels along lines of stress.

Renal Abscess

- Collection of pus in kidney. CT diagnosis. May require drainage.

Renal Agenesis

- Congenital absence of one (unilateral, 1 in 1,000) or both (bilateral, 1 in 4,000) kidneys. Unilateral renal agenesis: usually asymptomatic, discovered incidentally — the solitary kidney undergoes compensatory hypertrophy. The contralateral kidney may have VUR, UPJ obstruction, or ectopia.

Renal Anemia

- Normocytic normochromic anemia in CKD from decreased erythropoietin production. Hemoglobin less than 10 g/dL usually requires treatment with ESAs and iron supplementation. Target hemoglobin 10 to 11.5 g/dL. HIF-PHI prolyl hydroxylase inhibitors are newer oral agents for CKD anemia.

Renal Artery

- A paired branch of the abdominal aorta supplying a kidney. It divides into segmental, interlobar, arcuate, and interlobular arteries within the renal parenchyma.

Renal Artery Aneurysm

- Dilatation of renal artery, most common visceral artery aneurysm. Usually asymptomatic. Risk of rupture if greater than 2 cm or symptomatic. Treat with surgical or endovascular repair. Higher rupture risk in pregnant women.

Renal Artery Doppler

- Ultrasound assessing renal artery stenosis. Screening for renovascular hypertension.

Renal Artery Embolism

- Acute occlusion of renal artery from cardiac emboli typically from atrial fibrillation or mural thrombus. Presents with acute flank pain and hematuria. CT angiography shows filling defect. Treat with anticoagulation and possibly thrombolysis or embolectomy if caught early.

Renal Artery Stenosis

- Narrowing of renal artery causing renovascular hypertension. Atherosclerotic most common in elderly

at ostium. Fibromuscular dysplasia in young women causing mid-distal stenosis with string-of-beads appearance. Flash pulmonary edema and declining function on ACE inhibitors indicate intervention.

Renal Biopsy

- Percutaneous needle biopsy for histological diagnosis. Indicated for unexplained AKI, nephrotic syndrome, hematuria with renal involvement, and transplant dysfunction. Ultrasound-guided for accuracy. Complications include bleeding, arteriovenous fistula, and rarely loss of kidney.

Renal Calculi (Kidney Stones)

- Crystalline mineral deposits that form within the renal pelvis and urinary tract, affecting approximately 10 percent of the population over a lifetime, with a male-to-female ratio of 2:1 and peak incidence in the 4th-6th decades. See Kidney Stones.

Renal Cell Carcinoma

- See urology.

Renal Cell Carcinoma (RCC)

- Primary malignant adenocarcinoma of the renal cortex, originating from proximal renal tubular epithelial cells. Represents 85% of all primary adult kidney cancers. Major histologic subtypes: clear cell RCC (70-80%, associated with VHL gene mutations on chromosome 3p), papillary RCC (10-15%), and chromophobe RCC (5%).

Renal Changes in Pregnancy

- Renal blood flow increases 50-80% by the second trimester, and GFR increases 50% (creatinine decreases to 0.4-0.6 mg/dL, BUN to 8-10 mg/dL). Glycosuria and aminoaciduria are normal due to decreased tubular reabsorption.

Renal Clearance

- The volume of plasma completely cleared of a substance by the kidneys per unit time, measured in mL/min. Clearance of any substance X is calculated as: $CX = (Ux \times V)/Px$, where Ux is urine concentration, V is urine flow rate, and Px is plasma concentration.

Renal Colic

- Severe, paroxysmal flank pain caused by ureteral obstruction, typically by a kidney stone, with radiation to the groin; often the worst pain experienced.

Renal Cortex

- Outer kidney portion between capsule and medulla. Granular tissue with glomeruli, convoluted tubules, and cortical collecting ducts. Extends between pyramids as renal columns. Receives approximately 90 percent of renal blood flow; primary filtration site.

Renal cryptococcosis

- See Cryptococcosis.

Renal Diet

- Restricted protein, sodium, potassium, phosphorus for kidney disease. Individualized based on stage. Dialysis increases protein need requirement.

Renal Dose Adjustment

- Modification of a medication dose, dosing interval, or both to account for reduced kidney clearance. The adjustment should use an appropriate estimate of renal function, drug-specific pharmacokinetics, treatment indication, and therapeutic monitoring when available.

Renal Drug Excretion

- The elimination of drugs by the kidney through glomerular filtration, active tubular secretion, and passive tubular reabsorption. Drugs eliminated renally (e.g., aminoglycosides, vancomycin, lithium, metformin) require dose adjustment in renal impairment.

Renal Failure

- Decline in kidney function leading to accumulation of waste products (azotemia) and electrolyte abnormalities. Acute kidney injury (AKI): rapid decline over hours to days, potentially reversible. Chronic kidney disease (CKD): progressive decline over months to years.

Renal Infarction

- Renal infarction results from occlusion of the renal artery or its branches, causing coagulative necrosis of the renal parenchyma. The kidney has end-arterial supply with limited collateral circulation, making it susceptible to ischemic injury even from small vessel occlusion.

Renal Leukemia Infiltration

- Kidney involvement by chronic lymphocytic leukemia or other hematological malignancies causing bilateral renomegaly and renal dysfunction. Leukemic cells infiltrate the interstitium and per-

itubular capillaries. May cause AKI from compression of tubules and vessels or chronic kidney disease.

Renal Mass

- Abnormality in kidney detected on imaging. Solid masses may be renal cell carcinoma. Cystic masses classified by Bosniak system.

Renal Medulla

- Inner kidney with 8-18 renal pyramids containing loops of Henle, vasa recta, and medullary collecting ducts in countercurrent arrangement for urine concentration. Pyramid bases face cortex; apices (papillae) project into minor calyces. Lower blood flow (approximately 10 percent) and higher osmolality (up to 1200 mOsm/kg at papillary)

Renal Osteodystrophy

- Bone disease in CKD including osteitis fibrosa, osteomalacia, adynamic bone disease, and osteoporosis. Results from calcium-phosphate imbalance, secondary hyperparathyroidism, and vitamin D deficiency. Manage with phosphate binders, vitamin D analogues, and parathyroidectomy for refractory hyperparathyroidism.

Renal Pelvis

- The funnel-shaped expansion of the ureter within the kidney, formed by the union of major calyces. It collects urine from the renal papillae and directs it into the ureter.

Renal Replacement Therapy

- See nephrology.

Renal Replacement Therapy (RRT)

- Treatment that substitutes for inadequate kidney excretory and regulatory function. Modalities include intermittent hemodialysis, continuous kidney replacement therapy, peritoneal dialysis, and kidney transplantation.

Renal Replacement Therapy Indications

- Urgent indications for dialysis include refractory hyperkalemia, severe metabolic acidosis, toxin exposure by a dialyzable agent, refractory fluid overload or pulmonary edema, and symptomatic uremia. The decision incorporates the patient's trajectory and overall goals of care.

Renal Scan

- Nuclear medicine imaging using Tc-99m compounds. Evaluates renal function, obstruction,

blood flow. MAG3 for function, DTPA for GFR.

Renal Scintigraphy

- Renal imaging using Tc-99m compounds. MAG3 (tubular secretion) for functional assessment, DTPA (glomerular filtration) for GFR measurement. Used for obstruction, function, transplant monitoring, renin-mediated hypertension.

Renal Transplant (Renal Transplantation)

- Best treatment for ESRD with improved survival over dialysis. See Kidney Transplantation.

Renal Trauma

- Kidney injury from trauma. CT staging guides management. Grades I-V based on injury severity.

Renal Tubular Acidosis

- Group of disorders with normal anion gap metabolic acidosis from impaired renal acid excretion or bicarbonate reabsorption. Type 1 distal with impaired H⁺ secretion and urine pH greater than 5.5. Type 2 proximal with impaired bicarbonate reabsorption. Type 4 hyperkalemic from aldosterone deficiency or resistance.

Renal Ultrasound

- Noninvasive imaging evaluating kidney size, structure, and hydronephrosis. First-line imaging for CKD evaluation. Shows cortical thickness, echogenicity, and cysts. Resistive index estimates vascular resistance. Essential for evaluating AKI to exclude obstruction.

Renal Vein

- A paired vein draining each kidney into the inferior vena cava. The left renal vein is longer and crosses anterior to the aorta, receiving the left gonadal and suprarenal veins.

Renin Inhibitor

- A class of antihypertensive drugs that directly inhibit renin, the first and rate-limiting enzyme in the renin-angiotensin-aldosterone system. Renin catalyzes the conversion of angiotensinogen to angiotensin I, which is then converted to angiotensin II by ACE. Aliskiren is the only direct renin inhibitor available.

Renin-Angiotensin-Aldosterone System

- The RAAS is a hormonal cascade regulating blood pressure, fluid volume, and electrolyte balance.

Stimulus: decreased renal perfusion pressure (detected by baroreceptors in the afferent arteriole), decreased sodium delivery to the macula densa, or increased sympathetic activity.

Renin-Angiotensin-Aldosterone System (RAAS)

- A hormonal system in which renin initiates formation of angiotensin II, promoting vasoconstriction and aldosterone secretion. Aldosterone increases distal sodium reabsorption and potassium and hydrogen-ion secretion; RAAS dysregulation contributes to hypertension, heart failure, and CKD progression.

Renin-Angiotensin-Aldosterone System Physiology

- The RAAS regulates blood pressure and fluid electrolyte balance. Renin released from juxtaglomerular cells cleaves angiotensinogen to angiotensin I. Angiotensin converting enzyme converts angiotensin I to angiotensin II which causes vasoconstriction and stimulates aldosterone secretion from the adrenal cortex.

Renography

- Nuclear medicine imaging of kidneys. Evaluates function, obstruction, blood flow.

Renovascular Hypertension

- Hypertension due to renal artery stenosis. Diagnosis: Doppler ultrasound, CTA, MRA, renography. Treatment: angioplasty, stenting, surgery.

Reperfusion Injury

- Tissue damage following restoration of blood flow after ischemia. Mechanisms: oxidative stress, calcium overload, inflammation. Examples: myocardial reperfusion injury, stroke reperfusion injury. Mitigation: antioxidants, reperfusion protocols.

Replantation of an Avulsed Tooth

- Replacement of an avulsed permanent tooth into its socket after trauma, followed by flexible stabilization and endodontic and periodontal follow-up when indicated. Prognosis depends strongly on extraoral dry time, storage conditions, root maturity, and contamination.

Repolarization

- The return of the membrane potential toward the resting value after depolarization. In neurons

and skeletal/cardiac muscle, repolarization is primarily caused by the efflux of potassium through voltage-gated potassium channels and inactivation of sodium channels.

Reporting Standards

- Standards for structured reporting. ACR guidelines, BI-RADS, LI-RADS, PI-RADS.

Reproductive Organs

- The organs involved in reproduction. Female: ovaries (produce eggs and hormones), fallopian tubes (transport eggs, site of fertilization), uterus (nurtures the developing fetus), cervix (lower part of uterus, dilates during labor), vagina (passage for sperm and baby), and vulva (external genitalia).

Reproductive Toxicity

- Effects on reproduction: infertility, teratogenesis, fetal toxicity, developmental toxicity. Testing required for pharmaceuticals.

Reproductive Years

- The period of a woman's life during which she is capable of conceiving, typically from menarche (first menstrual period, average age 12-13) to menopause (final menstrual period, average age 51).

Research

- The systematic investigation and scientific study of new materials, methods, or clinical trials to advance medical knowledge and patient care.

Research Ethics

- Moral principles guiding research involving people, animals, or health data. See medical ethics.

Reservoir

- A human, animal, or environmental habitat in which an infectious agent normally lives, multiplies, and persists. A reservoir may serve as the source from which the agent is transmitted to a susceptible host.

Residual Limb

- The portion of a limb remaining after amputation. Care includes edema control, skin inspection, shaping, strengthening, desensitization, contracture prevention, and prosthetic preparation.

Residual Volume

- The volume of air remaining in the lungs after a maximal forced expiration, approximately 1.2 L

in adults. Residual volume cannot be measured by simple spirometry because it cannot be exhaled; it is measured by body plethysmography, helium dilution, or nitrogen washout.

Resilience

- Ability to recover from stressors. Important for healthy aging.

Resolution

- Ability to distinguish small structures. Spatial, temporal, contrast resolution.

Resolution Recovery

- Correction for detector resolution degradation with distance.

Resorption Atelectasis

- Atelectasis caused by complete obstruction of an airway, leading to absorption of trapped air by the pulmonary capillaries. Common causes include mucus plugging, foreign body, tumor, and post-surgical shallow breathing. The affected area appears as increased density on CXR with volume loss.

Respiratory Acidosis

- Acid-base disturbance due to accumulation of CO₂ from hypoventilation, lowering pH (acidemia). Causes: respiratory depression (opioids, sedatives, CNS disease), airway obstruction, neuromuscular weakness (GBS, myasthenia), chest wall restriction, severe COPD/asthma, and ventilation-perfusion mismatch.

Respiratory Alkalosis

- Acid-base disturbance from excessive elimination of CO₂ by hyperventilation, raising pH (alkalemia). Causes: anxiety/hyperventilation, hypoxia, pulmonary embolism, sepsis, fever, pain, salicylate toxicity, liver failure, pregnancy, high altitude. Compensation: renal bicarbonate excretion (days).

Respiratory Changes in Pregnancy

- Functional residual capacity (FRC) decreases by 20% due to elevated diaphragm from the gravid uterus. Tidal volume increases 30-40%, minute ventilation increases 50% (due to progesterone-mediated increase in respiratory drive).

Respiratory Depression

- See emergency medicine.

Respiratory Distress

- Difficulty breathing due to various causes. Assessment: work of breathing (retractions, nasal flaring, tripod position), oxygen saturation, breath sounds. Causes: asthma, COPD, pneumonia, pulmonary embolism, pneumothorax, anaphylaxis, heart failure.

Respiratory Distress Syndrome

- A condition primarily affecting premature infants caused by pulmonary surfactant deficiency, leading to alveolar collapse and respiratory failure. Presents with tachypnea, grunting, nasal flaring, retractions, and cyanosis. Chest X-ray shows diffuse ground-glass opacity.

Respiratory Drive

- The neural output that stimulates respiratory muscles and determines ventilation. It is generated by brainstem networks and modified by chemoreceptor input, mechanoreceptors, wakefulness, metabolic state, medications, and behavioral control.

Respiratory Failure

- Inadequate gas exchange resulting in hypoxemia ($\text{PaO}_2 < 60$ mmHg), hypercapnia ($\text{PaCO}_2 > 50$ mmHg), or both. Type 1 (hypoxemic): V/Q mismatch, shunt, diffusion impairment. Type 2 (hypercapnic): hypoventilation from neuromuscular disease, COPD exacerbation, or opioid overdose.

Respiratory Health

- The proper functioning of the lungs and airways, enabling efficient gas exchange and oxygen delivery to the body. Maintaining respiratory health involves avoiding tobacco smoke, minimizing exposure to air pollutants and occupational hazards, and receiving vaccinations against respiratory infections like influenza and pneumococcal disease.

Respiratory Infections

- Illnesses caused by pathogens including viruses, bacteria, and fungi that affect the airways and lungs.

Respiratory Physiology

- The study of pulmonary function encompassing ventilation (movement of air), gas exchange (diffusion of O_2 and CO_2 across the alveolar-capillary membrane), transport of gases in blood, and regulation of breathing.

Respiratory Protection

- Health of workers. See occupational health.

Respiratory Quotient (RQ)

- The ratio of carbon dioxide production to oxygen consumption for a particular substrate or metabolic state. It differs from the respiratory exchange ratio measured at the mouth and is used in metabolic and ventilatory physiology.

Respiratory Rate

- The number of breaths per minute. Normal adult respiratory rate is 12-20 breaths/min at rest; normal newborn rate is 30-60.

Respiratory Rehabilitation

- See pulmonology.

Respiratory Syncytial Virus

- A paramyxovirus and the most common cause of bronchiolitis and pneumonia in infants and young children. Seasonal outbreaks in fall, winter, and spring. Transmission by respiratory droplets and fomites. Mild in older children and adults; can be severe or fatal in infants, premature infants, and those with chronic lung or heart disease.

Respiratory System

- The anatomical and physiological system responsible for gas exchange—delivering oxygen to the blood and removing carbon dioxide. Components: (1) Upper respiratory tract—nose, nasal cavity, paranasal sinuses, pharynx (nasopharynx, oropharynx, laryngopharynx).

Respiratory Tract Infections

- Infections affecting any part of the respiratory tract, classified by anatomical location into upper respiratory tract infections (URTIs) and lower respiratory tract infections (LRTIs).

Respiratory viruses

- Viruses that cause respiratory tract infections. Major respiratory viruses include influenza, respiratory syncytial virus (RSV), parainfluenza virus, adenovirus, and human metapneumovirus. SARS-CoV-2 causes COVID-19. Symptoms range from mild upper respiratory illness to severe pneumonia.

Response Assessment

- Evaluation of treatment response. RECIST criteria for solid tumors. PET response criteria for

lymphoma.

Response Evaluation Criteria

- RECIST criteria for tumor response. PERCIST for PET.

Resting Membrane Potential

- The steady electrical potential difference across the cell membrane of an unstimulated cell, typically -70 mV in neurons, -80 to -90 mV in skeletal and cardiac muscle, and -50 to -60 mV in smooth muscle.

Restless Legs Syndrome

- An urge to move the legs, usually accompanied by uncomfortable sensations, that begins or worsens during rest, is relieved by movement, and occurs predominantly in the evening or night. The condition causes significant distress or impairment. It is associated with iron deficiency, renal failure, pregnancy, and peripheral neuropathy.

Restorative Dentistry

- Specialty of restoring damaged teeth. Procedures: fillings, inlays, onlays, crowns, bridges, implants. Goals: function, aesthetics, durability.

Restraint

- Device or medication limiting movement. Physical restraints: vests, wrist restraints, lap traps. Chemical restraints: antipsychotics, benzodiazepines for behavior control. Should be last resort, time-limited, with monitoring. Alternatives: addressing underlying causes, environmental modification.

Restraint-Related Harm

- Physical or psychological injury associated with using a device, medication, or environmental restriction to limit a person's movement. Older adults are at risk of falls, pressure injury, delirium, aspiration, and functional decline; the least restrictive alternatives should be prioritized.

Restriction Enzyme

- Bacterial enzyme cutting DNA at specific recognition sequences. Used in molecular cloning, DNA fingerprinting, restriction fragment length polymorphism (RFLP) analysis.

Restrictive Cardiomyopathy

- A cardiomyopathy characterized by impaired ventricular relaxation and filling due to stiff, noncompliant myocardium, with preserved systolic function and normal or reduced ventricular volumes. Causes

include amyloidosis, sarcoidosis, hemochromatosis, endomyocardial fibrosis, and radiation.

Retained Placenta

- Failure of the placenta to deliver spontaneously within 30 minutes after delivery of the baby. Incidence: 1-3% of deliveries. Types: placenta adherens (failure of separation due to inadequate myometrial contractions) and trapped placenta (separated but trapped behind a closed cervix).

Retained Products of Conception (RPOC)

- Fragments of placental or fetal tissue retained in the uterus after delivery, miscarriage, or abortion. Causes: incomplete placental separation, succenturiate lobe, and placenta accreta. Presents: persistent or heavy bleeding, pain, and fever in the postpartum period.

Retention Time

- The time required for a compound to pass through a chromatographic system from injection to detection. In analytical toxicology, comparison with validated standards helps identify compounds, but retention time alone is not definitive identification.

Reticular Formation

- A network of interconnected nuclei in the brainstem (midbrain, pons, medulla) that regulates arousal, consciousness, sleep-wake transitions, and alertness (ascending reticular activating system), as well as muscle tone, posture, cardiovascular and respiratory control (vasomotor and respiratory centers), pain modulation, and attention.

Reticulocyte Count

- Percentage of young RBCs in the peripheral blood, reflecting bone marrow erythroid activity. Normal: 0.5-2.5% (absolute 25-100 K/uL). Reticulocyte production index (RPI) corrects for anemia severity: $RPI = \text{reticulocyte \%} \times (\text{Hct}/45) / \text{maturation correction factor}$.

Retina

- Innermost eye layer, a thin neural tissue layer lining the posterior two-thirds of the globe for phototransduction (light to neural signals). Ten layers, with photoreceptors (rods for low-light vision, cones for color vision) in the outer nuclear layer.

Retinal Detachment

- A separation of the neurosensory retina from the underlying retinal pigment epithelium, creating a potential space that fills with fluid and resulting in progressive vision loss in the affected area, and it is classified into three main types based on the underlying mechanism.

Retinal Hemorrhages

- Bleeding within the retina, a hallmark finding in abusive head trauma (shaken baby syndrome). Found on dilated fundoscopic examination, often bilateral and multi-layered. Also seen in severe accidental trauma, bleeding disorders, and increased intracranial pressure.

Retinal ischemia

- Reduced blood flow to the retina causing visual disturbance, most commonly from ophthalmic artery or central retinal artery occlusion. Presents as painless, sudden monocular vision loss. Emergency treatment includes ocular massage, intra-arterial thrombolysis, and reduction of intraocular pressure to restore perfusion.

Retinal Layers

- The retina consists of ten distinct layers that process visual information through a complex neural circuit.

Retinal Prosthesis

- Experimental device for blindness. Argus II system: camera to processor to electrode array stimulates retina. Indications: retinitis pigmentosa. Provides light perception, pattern recognition.

Retinal Tear

- A full-thickness break in the neurosensory retina, often caused by vitreoretinal traction. It can permit fluid to enter the subretinal space and progress to rhegmatogenous retinal detachment; symptomatic tears are generally treated urgently with laser photocoagulation or cryotherapy.

Retinal Vein Occlusion

- Obstruction of a retinal vein producing retinal hemorrhages, edema, and variable visual loss. Branch and central forms may cause macular edema, retinal ischemia, and neovascular complications.

Retinitis Pigmentosa

- A group of inherited retinal dystrophies character-

ized by progressive degeneration of rod and cone photoreceptors, leading to visual field loss and eventual blindness. Prevalence: 1 in 4,000.

Retinoblastoma

- Childhood eye cancer. RB1 gene mutation (autosomal dominant inheritance, but sporadic cases common). Hereditary form: bilateral, multifocal, earlier onset. Sporadic: unilateral, single focus. Treatment: chemotherapy, radiation, surgery, enucleation.

Retinol

- Vitamin A (retinol) is a fat-soluble vitamin essential for vision, immune function, cell differentiation, and reproduction. See Vitamin A: Retinal.

Retinopathy

- See ophthalmology.

Retinopathy of Prematurity

- Abnormal vascularization of the retina in premature infants, a leading cause of childhood blindness. Affects infants less than 32 weeks gestation and less than 1500 grams. Stages range from mild (avascular retina) to severe (retinal detachment). Screening began at 31-33 weeks corrected gestational age or 4 weeks chronological age.

Retinoschisis

- Splitting of the retinal layers, producing a separation between inner and outer retinal tissue. It may be degenerative and peripheral or inherited and juvenile, and can cause visual-field loss, reduced acuity, or complications resembling retinal detachment.

Retract (Vascular)

- See Retractor.

Retraction (Self-Retaining)

- See Self-Retaining Retractor.

Retractor

- Holds tissue aside for exposure. Types: handheld, self-retaining (Weitlaner, Gelpi).

Retractor (Abdominal)

- Instruments for surgery. See Surgical Instruments.

Retractor (Hand)

- Instruments for surgery. See Surgical Instruments.

Retractor (Surgical)

- Instruments for surgery. See surgical instruments.

Retrograde Ejaculation

- Semen enters bladder instead of exiting urethra. Causes: diabetes, surgery, medications. Dry orgasm. Treatment: medications, sperm retrieval for fertility.

Retroperitoneal Fibrosis

- Chronic fibroinflammatory tissue formation in the retroperitoneum, often encasing the ureters and great vessels. It can cause hydronephrosis, flank pain, reduced kidney function, hypertension, or lower-limb edema and may be idiopathic, immune-mediated, drug-related, or secondary to malignancy.

Retroperitoneal hemorrhage

- Bleeding into the retroperitoneal space, which may result from aortic rupture, anticoagulation therapy, retroperitoneal hematoma following femoral catheterization, trauma, or retroperitoneal neoplasms. Presents with flank pain, abdominal distension, hypotension, and anemia.

Retroperitoneum

- Space posterior to the peritoneum. Primary retroperitoneal (never intraperitoneal): kidneys, adrenals, ureters, aorta, IVC. Secondary (originally intraperitoneal): duodenum 2nd-4th parts, pancreas head/body, ascending/descending colon. Divided into anterior pararenal, perirenal, and posterior pararenal spaces.

Retropharyngeal Abscess

- A collection of pus in the retropharyngeal space, located between the posterior pharyngeal wall and the prevertebral fascia. Most common in children aged 2-4 years, arising from suppurative lymphadenitis of the retropharyngeal lymph nodes that drain the nasopharynx, adenoids, and sinuses.

Retrovirus

- An RNA virus that replicates by reverse transcribing its RNA genome into DNA, which is then integrated into the host cell genome. Family Retroviridae. Key enzyme: reverse transcriptase (RNA-dependent DNA polymerase) converts viral RNA into double-stranded DNA; integrase inserts the viral DNA (provirus) into the host chromosome.

Rett Syndrome

- An X-linked dominant neurodevelopmental disorder affecting almost exclusively females, caused by mutations in the MECP2 gene.

Return of Spontaneous Circulation

- Restoration of sustained cardiac mechanical activity after cardiac arrest, evidenced by a palpable pulse or measurable blood pressure and signs of circulation. Post-arrest care addresses oxygenation, ventilation, blood pressure, temperature, neurologic injury, and the cause of arrest.

Reverberation Artifact

- Ultrasound artifact from multiple reflections. Reduced by changing angle, using tissue harmonic imaging.

Reverse Transcription

- See molecular biology.

Reversible Cell Injury

- Cellular dysfunction caused by a harmful stimulus that can resolve when the stimulus is removed. Common changes include cellular swelling, fatty change, membrane blebbing, and reduced oxidative phosphorylation without irreversible membrane or nuclear destruction.

Revision Arthroplasty

- Surgical replacement, exchange, or repair of a failed or painful joint prosthesis. Indications include infection, aseptic loosening, instability, periprosthetic fracture, wear, and recurrent dislocation.

Reye Syndrome

- A rare acute encephalopathy with hepatic dysfunction, classically occurring in children recovering from a viral illness. It causes persistent vomiting, altered mental status, cerebral edema, and elevated liver enzymes without primary inflammatory liver failure.

Reynolds Pentad

- Extension of the Charcot triad to include hypotension and altered mental status representing suppurative cholangitis, a severe life-threatening form of ascending cholangitis. Patients present with high fever, rigors, right upper quadrant pain, jaundice, hypotension, and confusion.

Rezafungin

- A long-acting echinocandin antifungal agent used in the treatment of invasive candidiasis and candidemia. Rezafungin inhibits beta-(1,3)-D-glucan synthase with a unique pharmacokinetic profile characterized by an extremely long half-life of approxi-

mately 130 hours, allowing for once-weekly dosing after an initial loading

Rh Factor

- A system of blood group antigens (the Rh system) on the surface of red blood cells, the most important of which is the D antigen (RhD). See Rhesus (Rh) Factor.

Rh Isoimmunization

- Maternal alloantibody production against fetal Rh D-positive red blood cells in an Rh D-negative mother, occurring when fetal RBCs enter the maternal circulation (fetomaternal hemorrhage).

Rhabdoid Tumor

- A highly aggressive malignant tumor of infancy, occurring in the kidney (malignant rhabdoid tumor) or brain (atypical teratoid/rhabdoid tumor), characterized by loss of the SMARCB1 (INI1) gene.

Rhabdomyolysis and AKI

- Muscle breakdown releases myoglobin causing tubular obstruction and AKI. Elevated CK and dark urine. Aggressive IV hydration is key prevention and treatment. Alkalinize urine to increase myoglobin solubility.

Rhabdomyoma

- A benign cardiac tumour composed of striated muscle cells, occurring predominantly in infants and children and strongly associated with tuberous sclerosis. It commonly involves the ventricular myocardium and may regress spontaneously, although large or multiple tumours can cause obstruction or arrhythmias.

Rhabdomyosarcoma

- A malignant tumor showing skeletal-muscle differentiation, occurring primarily in children and commonly involving the head and neck, genitourinary tract, or extremities.

Rhegmatogenous Retinal Detachment

- The most common type of retinal detachment, caused by a full-thickness break or tear in the retina that allows liquefied vitreous humor to enter the subretinal space, separating the neurosensory retina from the underlying retinal pigment epithelium.

Rhesus (Rh) Factor

- A system of blood group antigens (the Rh system) on the surface of red blood cells, the most important

of which is the D antigen (RhD). Individuals are Rh-positive (RhD+) if the D antigen is present (85% of Caucasians, 92% of Africans, 99% of Asians) or Rh-negative (RhD-) if absent.

Rheumatic Fever

- A nonsuppurative inflammatory complication of Group A Streptococcal pharyngitis, occurring 2-4 weeks after infection. Diagnosed by Jones criteria: major manifestations (carditis, polyarthritis, chorea, erythema marginatum, subcutaneous nodules) plus evidence of prior streptococcal infection. Major criterion plus two minor criteria.

Rheumatic Heart Disease

- Valvular damage resulting from acute rheumatic fever, an autoimmune response to Group A streptococcal pharyngitis. Most commonly affects the mitral valve (stenosis/regurgitation), followed by aortic. Prevention through prompt treatment of streptococcal pharyngitis.

Rheumatoid Arthritis

- A chronic, symmetric, erosive inflammatory polyarthritis primarily affecting small joints of the hands and feet, with a female-to-male ratio of 3:1 and peak onset 40-60 years.

Rheumatoid Factor

- RF: an autoantibody (usually IgM) directed against the Fc portion of IgG, positive in 70-80 percent of rheumatoid arthritis (seropositive RA) but also in other conditions: Sjogren, SLE, chronic infection (TB, endocarditis, hepatitis C), cryoglobulinemia, sarcoidosis, and healthy elderly.

Rheumatoid Factor (RF)

- An autoantibody directed against the Fc portion of IgG, positive in 70-80% of rheumatoid arthritis patients. Also positive in other autoimmune diseases (SLE, Sjogren, mixed cryoglobulinemia), chronic infections (TB, endocarditis, hepatitis C), and 5% of healthy elderly.

Rhinencephalon

- The part of the brain concerned chiefly with the sense of smell and related limbic functions.

Rhinitis

- Inflammation or dysfunction of the nasal mucosa causing congestion, rhinorrhea, sneezing, or itching.

Rhinophyma

- A severe, end-stage subtype of phymatous rosacea occurring predominantly in elderly men. Characterized by progressive nodular thickening, hypertrophy, and lobular enlargement of the nasal skin with prominent, dilated pilosebaceous pores. Pathophysiology: sebaceous gland hyperplasia, fibrosis, and chronic inflammation.

Rhinoplasty

- A surgical procedure to alter the shape, size, or function of the nose, performed for cosmetic reasons (aesthetic rhinoplasty—improving nasal appearance) or functional reasons (functional rhinoplasty—improving nasal airflow by correcting internal/external nasal valve collapse, deviated septum, or turbinate hypertrophy).

Rhinovirus

- A genus of non-enveloped, single-stranded positive-sense RNA viruses in the Picornaviridae family, the most common cause of the common cold. Rhinoviruses have >160 identified serotypes, each with distinct surface proteins (viral capsid proteins VP1–VP4) that determine antigenicity and host immune response.

RhoGAM

- Rh immune globulin containing anti-D antibodies, used to prevent Rh isoimmunization in Rh-negative mothers. Administered intramuscularly at 28 weeks gestation and within 72 hours after delivery of an Rh-positive infant. Also given after amniocentesis, chorionic villus sampling, miscarriage, ectopic pregnancy, or abdominal trauma.

Rhonchi

- Low-pitched, continuous adventitious sounds heard primarily during expiration, caused by secretions in the large airways. They may clear with coughing. They are heard in COPD, bronchitis, and bronchiectasis. They are different from wheezes in their lower pitch and coarser quality.

Rib

- One of 12 paired, curved bones forming the thoracic cage, protecting the heart, lungs, and great vessels. True ribs (1–7) attach directly to the sternum via costal cartilage. False ribs (8–10) attach indirectly to the sternum via the costal cartilage of rib 7.

Floating ribs (11–12) have no anterior attachment.

Riboflavin (Vitamin B2)

- A water-soluble vitamin that forms the coenzymes FAD and FMN. These cofactors participate in oxidation-reduction reactions, including electron transfer, fatty-acid oxidation, and energy metabolism.

Ribosomal RNA (rRNA)

- RNA component of ribosomes, catalyzing peptide bond formation (peptidyl transferase activity). Eukaryotic ribosome: 60S (28S, 5.8S, 5S rRNAs + proteins) + 40S (18S rRNA + proteins) = 80S. Prokaryotic: 50S (23S, 5S rRNAs) + 30S (16S rRNA) = 70S. rRNA genes are highly conserved, used in phylogenetics.

Ribosome

- Ribonucleoprotein complexes (rRNA + proteins) that synthesize proteins by translating mRNA. Two subunits: 60S (large, 28S/5.8S/5S rRNA) and 40S (small, 18S rRNA) in eukaryotes (80S complete). Prokaryotes have 50S/30S (70S complete).

Ribozyme

- RNA molecule with catalytic activity. Peptidyl transferase is a ribozyme.

Rickets

- See osteomalacia. Pediatric form with growth plate changes, bowing of legs, craniotabes, rachitic rosary.

Rickettsia

- A genus of obligate intracellular Gram-negative coccobacillary bacteria transmitted by arthropod vectors (ticks, fleas, lice, mites). Rickettsia are the causative agents of the spotted fever group (SFG) and typhus group (TG) rickettsioses.

Rickettsialpox

- A mite-borne infection caused by Rickettsia akari, producing a vesicular rash, fever, headache, and an inoculation eschar. It is usually self-limited but may resemble other febrile exanthems.

Ridge Preservation

- Socket grafting after extraction to maintain bone volume. Reduces ridge resorption.

Right Atrium

- Receives deoxygenated blood from superior vena cava, inferior vena cava, and coronary sinus. Smooth

posterior part (sinus venosus) and rough anterior auricle with pectinate muscles. The fossa ovalis is the fetal foramen ovale remnant. The SA node pacemaker is near the superior vena cava opening.

Right Bundle Branch

- A continuation of the bundle of His that travels along the right side of the interventricular septum to the right ventricle. It supplies the right ventricular myocardium and Purkinje fibers. Right bundle branch block (RBBB) occurs when this branch is damaged, causing delayed right ventricular activation.

Right Coronary Artery

- Arises from the right aortic sinus and travels in the right atrioventricular groove. It supplies the right atrium, right ventricle, SA node (60% of individuals), AV node (85% of individuals), and the inferior wall of the left ventricle.

Right coronary artery occlusion

- See Right Coronary Artery.

Right Heart Failure

- Failure of the right ventricle to adequately pump blood to the lungs, causing systemic venous congestion. Causes include left heart failure (most common), pulmonary hypertension, right ventricular infarction, pulmonary embolism, and constrictive pericarditis. Signs include elevated JVD, peripheral edema, hepatomegaly, and ascites.

Right Hypochondriac Region

- One of nine abdominal regions in the upper right abdomen, bounded medially by the right midclavicular line and superiorly by the right costal margin. Contains the cecum, appendix, ascending colon, and inferior right liver lobe. Pain here may indicate appendicitis (McBurney's point), cecal pathology, or hepatic disease.

Right Iliac Region

- Lower right abdomen containing the cecum, appendix, and terminal ileum. McBurney's point (junction of lateral and middle thirds from umbilicus to ASIS) is critical for appendicitis diagnosis. Also significant for evaluating inguinal and femoral hernias, and the right ureter's course.

Right Lymphatic Duct

- Short lymphatic vessel (approximately 1.5 cm)

draining lymph from the right upper quadrant: right arm, right side of the head and neck, and right thorax. Formed by the junction of right jugular, subclavian, and bronchomediastinal trunks, emptying into the right venous angle (junction of right internal jugular and subclavian veins).

Right sixth cranial nerve palsy

- Weakness or paralysis of the right abducens nerve (CN VI), causing inability to abduct the right eye and resulting in horizontal diplopia worse at distance and on right gaze.

Right ureterovesical calculus

- A stone lodged at the junction of the right ureter and bladder, causing flank pain, hematuria, and urinary urgency or frequency. May require endoscopic or lithotripsy intervention.

Right Ventricle

- Receives deoxygenated blood from the right atrium through the tricuspid valve, pumping into the pulmonary trunk via the pulmonary valve. Thinner walls (3-5 mm) than the left ventricle due to low pulmonary resistance.

Rigor Mortis

- Post-mortem stiffening of body muscles resulting from loss of ATP synthesis following death. Without ATP, actin and myosin filaments remain permanently cross-linked in a rigid state.

Ringworm

- Dermatophytosis: superficial fungal infection of keratinized tissue (skin, hair, nails) by dermatophytes (*Trichophyton*, *Microsporum*, *Epidermophyton*). See Tinea.

Rinne Test

- The Rinne test is a screening test that compares air conduction and bone conduction hearing using a vibrating tuning fork (typically 512 Hz) to help differentiate between conductive and sensorineural hearing loss.

RIPE Therapy

- Standard 4-drug regimen for active TB: Rifampin, Isoniazid, Pyrazinamide, Ethambutol. Intensive phase 2 months, continuation phase 4 months (total 6 months). Pyridoxine prevents isoniazid-induced neuropathy.

Rippling Muscle Disease

- A rare, benign myopathy characterized by mechanically-induced muscle contractions that spread across the muscle as a visible wave (rippling), typically triggered by stretch, percussion, or movement.

RIS

- Radiology Information System. Workflow management for radiology department.

Risk Assessment

- Process: hazard identification, dose-response, exposure assessment, risk characterization. Used for regulatory decisions.

Risk Factor

- Attribute increasing probability of disease. Examples: smoking (lung cancer), hypertension (stroke), obesity (diabetes).

Risk Factor Modification

- Smoking cessation, blood pressure control, lipid management, diabetes control, exercise.

Risk-Benefit Analysis

- Weighing therapeutic benefit against potential harm.

Ristocetin-Induced Platelet Aggregation

- A platelet-function assay that evaluates platelet agglutination in the presence of ristocetin, which depends on von Willebrand factor and platelet glycoprotein Ib.

Ritalin

- A central nervous system stimulant, a piperidine derivative that blocks the reuptake of dopamine and noradrenaline (norepinephrine) by inhibiting the dopamine transporter (DAT) and noradrenaline transporter (NAT) in the prefrontal cortex, increasing their synaptic concentrations. See Methylphenidate.

Rivaroxaban

- A direct oral anticoagulant and selective factor Xa inhibitor. Used for stroke prevention in non-valvular atrial fibrillation, treatment/prophylaxis of venous thromboembolism, and thromboprophylaxis after hip/knee replacement. Fixed once-daily dosing, rapid onset, no routine monitoring.

River Blindness

- See Onchocerciasis.

RNA

- Ribonucleic acid, single-stranded nucleic acid (ribose sugar, uracil instead of thymine).

Robot-Assisted Rehabilitation

- Robotic devices for repetitive task training. Upper extremity, lower extremity, gait training. Objective measurement, high repetition.

Romberg Test

- The Romberg test is a clinical neurological test used to assess the integrity of the dorsal column-medial lemniscus pathway (proprioception) and the cerebellum in maintaining balance. The test is performed by asking the patient to stand with feet together and eyes open, and then to close the eyes.

Romberg's sign

- See Romberg Test.

Root Caries

- Caries on exposed root surfaces. Common in elderly with recession. Fluoride effective.

Root Fracture

- Vertical or horizontal fracture of root. Prognosis depends on location and direction. Vertical root fractures typically require extraction.

Rooting Reflex

- A primitive reflex present from birth to 3-4 months, triggered by stroking the infant's cheek or corner of the mouth. The infant turns toward the stimulus and opens the mouth, searching for the nipple. Facilitates breastfeeding. Absence may indicate neurologic impairment. Disappears as voluntary feeding develops.

Ropinirole

- A non-ergot dopamine agonist (D2/D3) used for Parkinson disease and restless legs syndrome. Adverse effects: nausea, dizziness, somnolence (sudden sleep attacks), orthostatic hypotension, impulse control disorders (gambling, hypersexuality), hallucinations, and leg edema. Extended-release for once-daily Parkinson dosing.

Ropivacaine

- A long-acting amide local anesthetic used for infiltration, peripheral nerve blocks, and epidural anes-

thetia. Compared with bupivacaine it generally has less cardiotoxic and motor-blocking potential at equipotent doses, but systemic toxicity remains possible.

Rosacea

- A chronic inflammatory disorder characterized by persistent or recurrent centropacial erythema, flushing, telangiectasia, inflammatory papules or pustules, phymatous change, or ocular involvement. It is classified by phenotype rather than by obligatory subtypes.

Roseola

- See Roseola Infantum.

Roseola Infantum

- A viral illness caused by Human Herpesvirus 6 (HHV-6), affecting children 6 months to 2 years. Characterized by high fever (39-40.5 degrees Celsius) for 3-5 days followed by defervescence and appearance of a maculopapular rash starting on the trunk and spreading outward. Rash appears as fever resolves. Diagnosis is clinical.

Rostral

- Positioned toward the snout or front of the head; in the brain, it means toward the forehead.

Rotator Cuff

- A group of four muscles and their tendons that stabilize the glenohumeral (shoulder) joint and allow a wide range of motion. The muscles are the supraspinatus (abduction), infraspinatus (external rotation), teres minor (external rotation), and subscapularis (internal rotation)—collectively remembered by the acronym SITS.

Rotator Cuff Arthropathy

- Glenohumeral joint degeneration associated with chronic, irreparable rotator-cuff deficiency. Superior migration of the humeral head, altered joint mechanics, pain, and loss of elevation may lead to consideration of reverse shoulder arthroplasty.

Rotator Cuff Tears

- Rotator cuff tears: partial (articular-sided PASTA, bursal-sided, intrasubstance) vs full-thickness. Ellman classification for PASTA: Grade I (<3mm), II (3-6mm), III (>6mm). Cofield sizing: small (<1cm), medium (1-3cm), large (3-5cm), massive (>5cm).

Rotavirus

- A double-stranded RNA virus that is the leading cause of severe gastroenteritis in children worldwide. The virus has a segmented genome and infects enterocytes of the small intestine. Rotavirus causes profuse watery diarrhea, vomiting, and dehydration. Diagnosis is made by antigen detection in stool.

Rotavirus Vaccine

- Oral live attenuated vaccine given at 2, 4, and 6 months (RV1) or at 2, 4, and 8 months (RV5). Protects against rotavirus gastroenteritis, a leading cause of severe diarrhea in infants. Reduced rotavirus hospitalizations by over 80% since introduction. Mild side effects include fever and diarrhea.

Roth Spots

- Retinal hemorrhages with pale centers, seen on fundoscopic examination in infective endocarditis. They are a minor criterion in the Duke criteria. Roth spots are caused by immune complex-mediated vasculitis of retinal vessels.

Rothmund-Thomson Syndrome

- A genetically heterogeneous disorder characterized by poikiloderma beginning in infancy, short stature, sparse hair, skeletal abnormalities, juvenile cataracts in some forms, and increased risk of osteosarcoma or other cancers.

Rouleau

- A stack or column of red blood cells, commonly seen in blood with increased plasma proteins.

Round Pneumonia

- Pneumonia appearing as a round opacity on CXR rather than typical consolidation. More common in children and elderly. May mimic lung mass.

Roundworm

- Ascariasis: infection by *Ascaris lumbricoides*, the most common human soil-transmitted helminth, acquired by ingesting eggs from contaminated food/soil.

Route of Administration

- The path by which a drug enters the body, influencing absorption, onset, bioavailability, and first-pass metabolism. Enteral routes: oral, sublingual, buccal, rectal. Parenteral: intravenous (100 percent bioavailability, immediate onset), intramuscular, subcutaneous, intradermal.

Rovsing Sign

- Physical exam sign of acute appendicitis: palpation of the left lower quadrant elicits pain in the right lower quadrant due to displacement of gas stretching the inflamed appendix. See Appendicitis.

RPR/VDRL (Syphilis Serology)

- Non-treponemal screening tests for syphilis. Non-treponemal: RPR (rapid plasma reagin) or VDRL (venereal disease research laboratory). Positive in primary syphilis (1-4 weeks after chancre), secondary (100% positive), and early latent.

RSV

- Respiratory syncytial virus: the most common cause of bronchiolitis and viral pneumonia in infants and young children, transmitted by droplets/fomites; seasonal (winter).

Rubella

- German measles caused by Rubivirus. Mild illness with fever, rash, and lymphadenopathy in children. Dangerous in pregnancy due to congenital rubella syndrome (cataracts, heart defects, deafness). Rash similar to measles but milder, with postauricular and suboccipital lymphadenopathy. Prevention by MMR vaccination.

Rubella virus

- A togavirus that causes rubella (German measles), a mild febrile illness with rash. The virus is transmitted by respiratory droplets. Rubella is most dangerous during pregnancy, causing congenital rubella syndrome (CRS) with deafness, cataracts, and cardiac defects. Diagnosis involves serology. Treatment is supportive.

Rubin Maneuver

- An internal rotational maneuver for shoulder dystocia: the operator inserts a hand and applies pressure to the posterior aspect of the anterior fetal shoulder, rotating it toward the fetal chest (adducting the shoulders). This reduces the bisacromial diameter and disimpacts the anterior shoulder from the pubic symphysis.

Rubinstein-Taybi Syndrome

- A genetic disorder caused by mutations in CREBBP (50-60%) or EP300 (3-8%), encoding histone acetyltransferases involved in transcriptional regulation.

Rumination Disorder

- Recurrent effortless regurgitation of recently ingested food, followed by rechewing, reswallowing, or spitting, without the retching typical of vomiting. It can cause weight loss, malnutrition, and dental complications.

Running Total

- See exposure.

Ruptured Abdominal Aortic Aneurysm

- See vascular surgery.

Russell-Silver Syndrome

- A congenital growth disorder involving prenatal and postnatal growth restriction, relative macrocephaly, triangular face, body asymmetry, and feeding difficulty. It is often associated with abnormalities of genomic imprinting on chromosomes 7 or 11.

Rx

- Prescription or treatment. Symbol derives from Latin "requirere" (to take). Also used as abbreviation for remedy. Rx symbol: Rx or %x;

Radiation Fibrosis

- Permanent scarring and stiffening of tissue caused by radiation injury, which may develop months or years after treatment. It can affect the skin, lungs, bowel, or other organs and may cause pain, reduced function, or shortness of breath.

Radiation-Recall Dermatitis

- An inflammatory skin reaction that occurs in a previously irradiated area after exposure to a triggering substance, often months or years later. The area may become red, swollen, painful, or blistered and requires clinical evaluation.

Radiculopathy

- Dysfunction of a spinal nerve root, usually from disc herniation, narrowing, or inflammation. It may cause shooting pain, tingling, numbness, weakness, or altered reflexes along the affected nerve pathway.

Rhabdomyolysis

- Rapid breakdown of skeletal muscle that releases muscle proteins into the bloodstream. Causes include crush injury, extreme exertion, seizures, heat illness, and some diseases; it can lead to acute kidney injury and requires prompt medical assessment.

Ringings in the Ears

- Perception of ringing, buzzing, hissing, or other sound without an external source, medically called tinnitus. It may be associated with hearing loss, loud-noise exposure, earwax, ear infection, medicines, or jaw disorders. Sudden hearing loss, one-sided symptoms, or pulsatile ringing requires medical evaluation.

Runny Nose

- Excess nasal drainage, medically called rhinorrhea, caused by infection, allergy, irritation, or structural nasal disease. It may occur with congestion, sneezing, cough, or sore throat. Persistent one-sided, bloody, or clear drainage after head injury requires medical assessment.

Chapter 19

Pocket Dictionary: S Terms

16S Ribosomal RNA Sequencing

- Molecular identification of bacteria and archaea using conserved and variable regions of the 16S ribosomal RNA gene. It is useful for difficult-to-culture organisms but may not distinguish closely related species or provide susceptibility data.

S3 Gallop Pathology

- A third heart sound in adults over 40 indicates ventricular dysfunction. It represents rapid ventricular filling against a dilated, noncompliant ventricle. S3 is a hallmark of systolic heart failure (HFrEF) and is associated with increased mortality.

S4 Gallop Pathology

- A fourth heart sound indicates atrial contraction against a stiff ventricle (decreased compliance). Associated with left ventricular hypertrophy from hypertension, aortic stenosis, and hypertrophic cardiomyopathy. S4 is best heard at the apex with the bell, and is accentuated by exertion, expiration, and isorhythmic AV dissociation.

Sacral Nerve Stimulation

- Neuromodulation for urinary/fecal incontinence, constipation. Sacral nerve root (S3) stimulation. Implantable pulse generator + electrode. Indications: refractory urge incontinence, fecal incontinence, non-obstructive urinary retention.

Sacral Nerve Stimulator

- Neuromodulation for urinary/fecal incontinence, constipation. See sacral nerve stimulation.

Sacral Vertebrae (S1-S5)

- Five vertebrae fused to form the sacrum. They transmit body weight from the vertebral column to the pelvic girdle and contain anterior and posterior sacral foramina for spinal nerves.

Sacroiliac Joint

- The synovial joint between the sacrum and the ilium (auricular surfaces), forming the connection between the axial skeleton and the pelvis. The sacroiliac joint is reinforced by strong ligaments (anterior, interosseous, and posterior sacroiliac ligaments, sacrotuberous and sacrospinous ligaments).

Sacroiliitis

- Inflammation of one or both sacroiliac joints, causing pain in the lower back, buttock, or posterior thigh that may worsen with prolonged rest. Causes include axial spondyloarthritis, psoriatic or enteropathic arthritis, infection, trauma, and degenerative disease.

Sacrum

- Large triangular bone from 5 fused sacral vertebrae between the hip bones. Articulates with L5 superiorly and coccyx inferiorly. The sacral promontory is an obstetric landmark. Contains 4 pairs of sacral foramina. Forms the posterior pelvic wall. Sacroiliac joints are important for force transmission between spine and lower limbs.

Safety Plan

- A collaboratively written, practical plan for responding to escalating suicidal thoughts or emotional crisis. It identifies warning signs, internal coping strategies, supportive contacts, professional resources, and methods to restrict access to lethal means.

Sagittal Plane

- A vertical plane that divides the body or an organ into right and left sides.

Salbutamol (Albuterol)

- A selective beta-2 adrenergic agonist bronchodilator used for acute relief and prevention of bron-

chospasm in asthma and COPD. Relaxes airway smooth muscle; short-acting (SABA), onset 5-15 minutes, duration 4-6 hours via inhaler or nebulizer. Adverse effects: tremor, tachycardia, palpitations, hypokalemia, and headache.

Salicylate

- See aspirin toxicity.

Salicylate Toxicity

- See toxicology.

Saline Infusion Sonohysterography

- Ultrasound evaluation of uterus using saline contrast. Evaluates intrauterine pathology.

Saliva

- Secretion of salivary glands (1-1.5 L/day). Functions: lubrication, digestion (amylase), buffering, antimicrobial, remineralization. Xerostomia (dry mouth) increases caries risk. Causes: medications, Sjogren syndrome, radiation, dehydration.

Salivary Gland

- Any exocrine gland that secretes saliva into the oral cavity. The major paired glands are the parotid, submandibular, and sublingual glands; numerous minor glands occur throughout the oral mucosa.

Salivary Gland Cancer

- A rare malignancy most often arising in the parotid gland. Histologic types include mucoepidermoid carcinoma, adenoid cystic carcinoma, and acinic cell carcinoma; facial nerve preservation is a key surgical consideration.

Salivary Gland Infection

- Infection of a salivary gland, usually bacterial and often involving the parotid or submandibular gland. Painful swelling, fever, dry mouth, and purulent drainage from the duct may occur.

Salivary Gland Scintigraphy

- Salivary gland function assessment using Tc-99m pertechnetate. Used for Sjogren syndrome, obstruction, radiation sialadenitis.

Salivary Gland Tumors

- Most salivary tumors are benign (80% in parotid, 50% in submandibular, 50% in minor glands). Benign: pleomorphic adenoma (most common overall), Warthin tumor (bilateral in 10%, smokers). Malignant: mucoepidermoid carcinoma (most common

malignant), adenoid cystic carcinoma (perineural invasion), acinic cell carcinoma.

Salivary Glands

- Exocrine glands that produce saliva.

Salivation

- The secretion of saliva by the salivary glands.

Salmonella

- A genus of Gram-negative, facultatively anaerobic, rod-shaped bacteria in the Enterobacteriaceae family, a major cause of foodborne illness worldwide. Classification: *Salmonella enterica* subsp. *enterica* serovars (>2,500 serovars). Clinically important serovars: *S. Typhi* and *S.*

Salmonellosis

- Infection with *Salmonella* bacteria (non-typhoidal), typically acquired from contaminated food (eggs, poultry, meat). Presents with acute gastroenteritis: diarrhea, fever, and abdominal cramps within 6-72 hours of ingestion. Self-limited in immunocompetent patients. Bacteremia can occur, especially in immunocompromised.

Salpingectomy

- Surgical removal of a fallopian tube, the definitive treatment for tubal ectopic pregnancy and for hydrosalpinx. Laparoscopic approach is standard (three-port technique). Indications: ruptured ectopic, hemodynamic instability, failed medical management, contralateral tube is healthy, and patient desires sterilization.

Salpingitis

- Inflammation of the fallopian tubes, most commonly caused by ascending infection from the lower genital tract. Salpingitis is a component of pelvic inflammatory disease and is often bilateral.

Salpingo-oophorectomy

- Surgical removal of one or both fallopian tubes and ovaries. Bilateral salpingo-oophorectomy treats and stages ovarian cancer and is used as risk-reducing surgery in BRCA mutation carriers.

Salpingography

- Contrast imaging of fallopian tubes. Part of hysterosalpingography.

Salpingostomy (Salpingotomy)

- A fertility-preserving surgical procedure for tubal

ectopic pregnancy: a linear incision on the antimesenteric border of the fallopian tube to remove the ectopic gestation, leaving the tube in situ. Indications: unruptured ectopic, desire for future fertility, contralateral tubal damage or absence.

Saltatory Conduction

- The rapid propagation of action potentials along myelinated nerve fibers, where the action potential appears to "jump" from one Node of Ranvier to the next. Myelin (produced by Schwann cells in the PNS and oligodendrocytes in the CNS) insulates the axon, preventing ion flow across the internodal membrane.

Salvage Therapy

- Treatment given for cancer that has not responded to or has recurred after initial (first-line) therapy.

Sandhoff Disease

- An autosomal-recessive lysosomal storage disorder caused by deficiency of beta-hexosaminidase A and B, leading to accumulation of GM2 ganglioside. It causes progressive neurodegeneration, developmental regression, seizures, hypotonia, and sometimes organ enlargement.

Sandpaper Tooth

- Rough enamel surface retains plaque. May indicate early demineralization or fluorosis.

Saphenous Vein

- Surgical procedure creating detour around blocked vessel using autologous vein (saphenous) or prosthetic graft (PTFE, Dacron). See bypass graft.

Sarcoidosis

- A multisystem granulomatous disorder of unknown etiology characterized by non-caseating granulomas, predominantly affecting lungs (90%), lymph nodes, skin, eyes, liver, heart, nervous system. Pulmonary stages (Scadding): 0 (normal), I (bilateral hilar lymphadenopathy), II (BHL + parenchymal), III (parenchymal only), IV (fibrosis).

Sarcoidosis Skin Manifestations

- Cutaneous sarcoidosis affects 25 percent of patients with systemic sarcoidosis.

Sarcoma

- Malignant tumor of mesenchymal origin (bone, cartilage, fat, muscle, blood vessels). Types: osteosarcoma, chondrosarcoma, liposarcoma, rhab-

domyosarcoma, angiosarcoma. Less common than carcinomas (1% of adult cancers).

Sarcopenia

- Age-related loss of muscle mass and function. Prevalence increases with age. Risk factors: reduced protein intake, decreased physical activity, chronic disease. Assessment: grip strength, gait speed, muscle mass measurement. Treatment: protein supplementation, resistance exercise, vitamin D, hormonal therapy if deficient.

Sarcopenic Obesity

- The coexistence of excess adiposity with reduced skeletal-muscle mass, strength, or physical performance. It increases risks of disability, falls, metabolic disease, and poor surgical outcomes and requires assessment beyond body-mass index alone.

Sarcoptes scabiei

- A mite that causes scabies, a contagious skin infestation transmitted by prolonged skin-to-skin contact. The female mite burrows into the stratum corneum, laying eggs along the burrow, which appears as a greyish, serpiginous line. The life cycle from egg to adult takes approximately 10-14 days.

SARS-CoV-2

- A coronavirus belonging to the Coronaviridae family that causes COVID-19. SARS-CoV-2 has a positive-sense single-stranded RNA genome and a spike glycoprotein that mediates attachment to ACE2 receptors. The virus replicates in the cytoplasm using its own RNA-dependent RNA polymerase.

Satiety

- Feeling of fullness after eating. Affected by meal composition, volume, fiber, protein. Satiety hormones: CCK, PYY, GLP-1.

Saturated Fat

- Solid at room temperature. Sources: meat, dairy, coconut oil, palm oil. Controversial cardiovascular effects. Current guidelines limit intake.

Saturation

- Occupancy of available binding, metabolic, transport, or elimination capacity such that further increases in concentration or dose produce a nonlinear response. Saturable metabolism can cause disproportionate rises in plasma concentration and toxic-

ity.

Scabies

- A highly contagious, intensely pruritic parasitic infestation of the skin caused by the mite *Sarcoptes scabiei* var. *hominis*. The hallmark symptom is severe, generalized pruritus that characteristically worsens at night, driven by a hypersensitivity reaction to the mite's proteins, eggs, and feces.

Scaffold

- Three-dimensional structure for tissue engineering. Porous, biodegradable. Cells populate scaffold.

Scaling and Root Planing

- Nonsurgical periodontal instrumentation that removes supra- and subgingival deposits and smooths contaminated root surfaces. It reduces inflammation and pocket depth in appropriate patients but does not regenerate lost periodontal attachment by itself.

Scalpel

- Surgical knife with interchangeable blades. Blade numbers indicate size and shape. %10: common for incisions. %11: puncture. %15: precise work.

Scalpel Handle

- Handle for holding scalpel blades. %3 handle common.

Scaphoid

- Largest proximal carpal row bone on the radial wrist side. Most commonly fractured carpal bone from falls on an outstretched hand. Unique distal blood supply makes waist fractures prone to proximal fragment avascular necrosis. Anatomical snuffbox tenderness is characteristic.

Scaphoid Fractures

- Scaphoid fractures: most common carpal fracture (young males, FOOSH). Classification: Herbert: A (incomplete), B (complete: B1 distal waist, B2 proximal waist, B3 proximal pole, B4 distal tubercle, B5 transverse). Vascularity: retrograde from distal radius -> proximal pole at risk for AVN.

Scapula

- Flat triangular bone on the posterior thorax between ribs 2-7. Key features: acromion, coracoid process, glenoid cavity, spine, supraspinous/infraspinous fossae. Stabilized by 17 muscles and serves as rotator cuff attachment.

Has 3 borders, 3 angles, and 2 surfaces. Fractures are uncommon, usually from high-energy trauma.

Scar Management

- Scar management aims to minimize hypertrophic and keloid formation. Early intervention includes silicone gel or sheeting (first-line evidence-based treatment), pressure therapy (especially for burn scars), and massage. Intralesional corticosteroids (triamcinolone) for hypertrophic scars and keloids.

Scarlatina

- A toxin-mediated illness caused by Group A *Streptococcus* producing erythrogenic toxin. See Scarlet Fever.

Scarlet Fever

- A toxin-mediated illness caused by Group A *Streptococcus* producing erythrogenic toxin. Presents with fever, pharyngitis, and a characteristic diffuse, erythematous, sandpaper-textured rash that blanches with pressure. Circumoral pallor, Pastia lines (linear petechiae in skin folds), and strawberry tongue are classic findings.

Scarring Alopecia

- Permanent hair loss caused by destruction and fibrosis of the hair follicle. Causes include lichen planopilaris, discoid lupus, central centrifugal cicatricial alopecia, folliculitis decalvans, and severe infection; early diagnosis and biopsy can prevent progression.

Scatter Correction

- Correction for scattered photons. Dual energy window method.

Scatter Events

- Photons deflected from original path. Corrected by energy windowing and scatter correction algorithms.

Scavenging System

- The anesthesia gas scavenging system collects and removes waste anesthetic gases from the operating room environment to protect staff from chronic exposure. Active scavenging uses negative pressure from vacuum source; passive scavenging vents to exhaust through gravity.

Schatzki Ring

- Thin circumferential mucosal ring located at the squamocolumnar junction of the distal esophagus at the superior margin of a hiatal hernia. The ring

is composed of mucosa and submucosa. May be asymptomatic or cause episodic dysphagia particularly to solid food and acute food bolus impaction.

Schedule I

- See controlled substances.

Schedule II

- See controlled substances.

Schedule III

- See controlled substances.

Schedule IV

- See controlled substances.

Schedule V

- See controlled substances.

Schilling Test

- A historical test of vitamin B12 absorption, formerly used to investigate pernicious anaemia and intestinal malabsorption. It has largely been replaced by serum vitamin B12, methylmalonic acid, homocysteine, and intrinsic-factor antibody testing.

Schistocytes

- Fragmented, helmet-shaped, or triangular red blood cells produced by mechanical shearing as erythrocytes pass through obstructed or fibrin-laden microvessels.

Schistosoma hematobium

- See Schistosomiasis.

Schistosomiasis

- A parasitic disease, also called bilharzia, caused by blood flukes of the genus *Schistosoma*. Infection occurs when cercariae in contaminated freshwater penetrate the skin.

Schizencephaly

- A rare brain malformation characterized by a full-thickness, gray matter-lined cleft extending from the pial surface to the lateral ventricle. Caused by early focal neuronal migration abnormalities, often associated with prenatal vascular disruption or COL4A1 mutations. Closed-lip (Type I): the cleft walls are apposed.

Schizoaffective Disorder

- A disorder characterized by an uninterrupted period of illness during which there is a major mood episode concurrent with criterion A of schizophre-

nia (delusions, hallucinations, disorganized speech, grossly disorganized or catatonic behavior, or negative symptoms).

Schizoid Personality Disorder

- A pattern of detachment from social relationships with a restricted range of emotional expression.

Schizophrenia

- A severe chronic mental illness characterized by psychotic symptoms (hallucinations, delusions, disorganized thinking) and negative symptoms (avolition, alogia, anhedonia, blunted affect), with cognitive impairment and functional decline. Diagnosis: 2 core symptoms for 1 month with continuous signs for 6 months.

Schizotypal Personality Disorder

- A pattern of social and interpersonal deficits marked by acute discomfort with and reduced capacity for close relationships, cognitive or perceptual distortions, and eccentricities of behavior.

Schnitzler Syndrome

- A rare autoinflammatory syndrome characterized by chronic urticarial rash, intermittent fever, bone pain, arthralgia or arthritis, and elevated inflammatory markers, usually with a monoclonal IgM protein and less commonly IgG.

School-Based Rehabilitation

- Rehabilitation services in schools for children.

Schwannoma

- A benign peripheral nerve sheath tumor arising from Schwann cells. It presents as a solitary, slow-growing, painless nodule along a peripheral nerve course. It can be moved laterally but not longitudinally along the nerve. Vestibular schwannoma (acoustic neuroma) arises from CN VIII. Diagnosis is by MRI.

Sciatic Nerve

- Body's largest nerve (L4-S3), formed from the lumbosacral plexus, exiting via the greater sciatic foramen below the piriformis. Descends in the posterior thigh, dividing into tibial and common peroneal nerves at the popliteal fossa. Innervates hamstrings and provides lower leg/foot motor/sensory innervation via its branches.

Sciatica

- Pain radiating along the distribution of the sciatic

nerve (L4-S3), from the lower back through the buttock and down the posterior thigh to the foot. Most commonly caused by lumbar disc herniation compressing the nerve root (L4-L5 or L5-S1). Other causes: spinal stenosis, spondylolisthesis, piriformis syndrome, and epidural abscess.

Scimitar Syndrome

- A congenital cardiopulmonary anomaly involving partial or total anomalous pulmonary venous return from the right lung to the inferior vena cava, often with right-lung hypoplasia and dextroposition of the heart. It may cause recurrent respiratory infection, pulmonary hypertension, or heart failure.

Scintillation

- Light emission from crystal when irradiated. NaI(Tl) scintillation decay time: 230 ns.

Scintillation Crystal

- Sodium iodide doped with thallium (NaI:Tl) crystal converts gamma rays to light photons. Coupled to photomultiplier tubes.

Scissors

- Cutting instruments. Types: Mayo (tissue), Metzenbaum (delicate), suture.

Scissors (Surgical)

- Instruments for surgery. See surgical instruments.

Scissors (Vascular)

- See Scissors.

Sclera

- Dense fibrous opaque outer eye coat providing structural integrity and extraocular muscle attachment. White, covering approximately five-sixths of the globe. Continuous with cornea anteriorly (limbus) and dura mater posteriorly through the optic nerve sheath. May appear yellowish in elderly or with jaundice (icterus).

Scleroderma

- Systemic sclerosis: an autoimmune connective tissue disease of skin and internal organs due to fibrosis, vasculopathy, and autoantibodies. Limited cutaneous (CREST—calcinosis, Raynaud, esophageal dysmotility, sclerodactyly, telangiectasia) and diffuse cutaneous forms.

Scleromyxedema

- A rare, chronic, progressive mucinosis characterised

by generalised papular and sclerodermoid thickening of the skin with firm, waxy papules in a "peau d'orange" pattern, most prominently on the face, neck, and hands.

Sclerosing sialadenitis

- A chronic fibrosing inflammation of the salivary glands characterized by progressive glandular enlargement and firmness, sometimes associated with IgG4-related disease.

Sclerosteosis

- A rare inherited sclerosing bone dysplasia characterized by progressive thickening of the skull and long bones, tall stature or syndactyly in some forms, and possible cranial-nerve compression. Hearing loss, facial palsy, headache, and raised intracranial pressure may occur.

Scoliosis

- A lateral spinal curvature >10 degrees on standing radiograph (Cobb angle). Idiopathic scoliosis (most common, 80%): adolescent onset (10 years), more common in girls. Congenital: due to vertebral anomalies. Neuromuscular: associated with cerebral palsy, muscular dystrophy.

Scotoma

- A focal area of diminished or absent vision within an otherwise intact visual field, classified by location and morphology.

Screening

- Detection of cancer in asymptomatic individuals. Methods: mammography (breast), colonoscopy (colon), Pap smear (cervix), PSA (prostate), low-dose CT (lung). Balance benefits vs. harms.

Screening (Geriatric)

- Detection of disease in asymptomatic elderly. Balance benefits vs. harms, life expectancy, preferences. Examples: bone density, abdominal aortic aneurysm, colorectal cancer (individualized), depression, falls.

Screening Guidelines

- Testing asymptomatic individuals for disease. See screening.

Screening Program

- An organized public-health service that systematically invites a defined population to undergo testing for an asymptomatic condition, with quality-assured

follow-up, diagnosis, treatment, and evaluation of benefits and harms.

Screening Test

- A test applied to people without symptoms or a known diagnosis to identify those at increased risk of a condition. A screening result usually requires a diagnostic or confirmatory evaluation before treatment is initiated.

Screws

- Fixation devices for fractures. See orthopedic screws.

Scrofula

- Tuberculous lymphadenitis, most commonly affecting the cervical and submandibular lymph nodes, caused by *Mycobacterium tuberculosis* or atypical mycobacteria. It presents as chronic, painless, matted lymphadenopathy that may develop into cold abscesses with draining sinuses.

Scrotal calcinosis

- The deposition of calcium within the scrotal skin, presenting as multiple, firm, whitish, painless nodules on the scrotum, which may be solitary or numerous and may ulcerate. The exact aetiology is debated and may involve dystrophic calcification of pre-existing epidermal inclusion cysts, dermal cysts, or eccrine duct abnormalities.

Scrotum

- Pouch of skin and superficial fascia containing the testes, epididymides, and lower spermatic cords. The scrotal wall has multiple layers reflecting the abdominal wall layers.

Scurvy

- A nutritional disease caused by a chronic deficiency of ascorbic acid (Vitamin C), which is essential for collagen synthesis. Vitamin C serves as a cofactor for prolyl and lysyl hydroxylases, which hydroxylate collagen fibers to allow stable cross-linking.

Seasonal Affective Disorder

- A specifier of recurrent major depressive episodes that occur at a specific time of year, most commonly winter (with remission in spring/summer). Symptoms: hypersomnia, hyperphagia with carbohydrate craving, weight gain, fatigue, and low energy.

Sebaceous Gland

- A small gland in the skin that secretes sebum (oil)

into hair follicles to lubricate and waterproof the skin and hair.

Sebaceous Glands

- Holocrine glands associated with hair follicles (pilosebaceous units) secreting sebum (triglycerides, wax esters, squalene, cholesterol). Lubricate and waterproof skin/hair, have antimicrobial properties (fatty acids), and maintain skin barrier function. Most numerous on the face and scalp.

Seborrheic Dermatitis

- A chronic inflammatory skin condition affecting sebum-rich areas: scalp, face (nasolabial folds, eyebrows), chest, and intertriginous areas. Presents with erythematous patches with greasy, yellowish scales. Dandruff is a mild form. More common and severe in Parkinson disease, HIV/AIDS, and Down syndrome.

Seborrheic keratoses

- Common, benign, epidermal skin tumors presenting as waxy, stuck-on, pigmented plaques with a rough, verrucous surface, typically appearing on the trunk and face of middle-aged and older adults. They are the most common benign skin neoplasm and are caused by keratinocyte proliferation.

Seborrheic Keratosis

- A benign epidermal tumor appearing as stuck-on, waxy, hyperpigmented plaques with a verrucous surface. They are extremely common in older adults and increase in number with age. Seborrheic keratoses are not premalignant.

Seborrhoea

- A chronic inflammatory skin condition affecting sebum-rich areas: scalp, face (nasolabial folds, eyebrows), chest, and intertriginous areas. See Seborrheic Dermatitis.

Seckel Syndrome

- A rare recessive primordial dwarfism syndrome featuring severe proportionate growth restriction, microcephaly, characteristic facial features, and developmental or intellectual impairment. Skeletal, hematological, and other congenital abnormalities may occur.

Second Malignancy

- New cancer after initial cancer treatment. Risk from radiation, chemotherapy.

Second Messenger Systems

- Intracellular signaling molecules that relay extracellular signals (first messengers: hormones, neurotransmitters, growth factors) to cellular responses.

Second Messengers

- Intracellular signaling molecules generated in response to extracellular signals (first messengers like hormones or neurotransmitters). cAMP (cyclic adenosine monophosphate): produced from ATP by adenylyl cyclase, activated by Gs-protein coupled receptors, activates protein kinase A (PKA); degraded by phosphodiesterase.

Second Opinion

- Review by another radiologist. Common for difficult cases, cancer staging.

Second Stage of Labor

- From full cervical dilatation (10 cm) to delivery of the baby. The mother feels the urge to push as the presenting part descends. Pushing: closed-glottis (Valsalva) or open-glottis. Duration: up to 3 hours in nulliparas with regional anesthesia (2 hours without), and 2 hours in multiparas (1 hour without).

Second Trimester

- The period from 13 to 27 weeks gestational age. Fetal growth accelerates, mother begins to feel quickening (fetal movement) between 16-20 weeks. Anatomy ultrasound at 18-22 weeks screens for structural anomalies. Maternal serum screening offered. Nausea typically resolves. Uterus becomes palpable above the symphysis pubis.

Second-Degree AV Block

- Intermittent failure of AV conduction. Mobitz Type I (Wenckebach): progressive PR prolongation then dropped QRS; usually benign (AV nodal, narrow QRS). Mobitz Type II: constant PR interval with sudden dropped QRS; more serious (infranodal, wide QRS), risk of progression to complete heart block and Stokes-Adams attacks.

Second-Degree Burn

- Partial-thickness burn involving the epidermis and dermis: superficial partial-thickness (blisters, moist, painful, pink) vs. deep partial-thickness (blanching, painful, may have thrombosed capillaries). Causes: thermal (scalds, flames), chemical, electrical.

Second-Hand Smoke

- See tobacco.

Secondary Attack Rate

- The proportion of susceptible contacts who develop infection after exposure to a primary case within a defined period. It helps estimate transmissibility in households, schools, healthcare settings, and other close-contact groups.

Secondary Hyperparathyroidism

- Compensatory PTH increase in CKD from decreased phosphate excretion, low calcitriol, and hypocalcemia. Leads to renal osteodystrophy and vascular calcification. Managed with dietary phosphorus restriction, phosphate binders, active vitamin D, and calcimimetics. Refractory cases may require parathyroidectomy.

Secondary Immune Response

- The rapid, enhanced immune response on re-exposure to an antigen, mediated by memory B cells and memory T cells generated during the primary response. Features: shorter lag phase, higher and faster antibody titers (predominantly high-affinity IgG), and more effective cell-mediated responses.

Secondary Immunodeficiency

- An acquired (not genetic) impairment of the immune system due to an underlying cause: HIV/AIDS (CD4+ T-cell depletion), malignancies (CLL: hypogammaglobulinemia; myeloma: hypogammaglobulinemia + impaired specific antibody; lymphoma), immunosuppressive medications (corticosteroids, chemotherapy, biologics: rituximab -> prolonged B-cell dep

Secondary Prevention

- Early detection and treatment. Examples: screening (mammography, colonoscopy, blood pressure), self-examination.

Secondary syphilis

- The systemic dissemination stage of *Treponema pallidum* infection, occurring weeks to months after the primary chancre, characterised by a generalised maculopapular rash (including the palms and soles), mucous patches, condylomata lata, lymphadenopathy, and systemic symptoms such as fever and malaise.

Secondary Tumor

- Spread of cancer cells from primary site to distant locations. See metastasis.

Secretin

- A peptide hormone produced by S-cells in the duodenal and jejunal mucosa in response to acidic chyme (pH less than 4.5) entering the duodenum from the stomach.

Secretion

- Active movement of a substance from cells into a body fluid, lumen, or external environment. In toxicokinetics, renal tubular or biliary secretion can contribute to elimination of a toxin or metabolite.

Secretory Diarrhea

- Diarrhea caused by active secretion or reduced absorption of electrolytes and water by the intestine. It persists despite fasting and may result from infections, medications, bile-acid disorders, endocrine tumors, or inflammatory disease.

Sedation Dentistry

- Pharmacological techniques for anxiety management: nitrous oxide (laughing gas), oral sedation, IV sedation, general anesthesia. Indications: dental phobia, special needs, prolonged procedures. Monitoring required for deeper sedation.

Sedation Levels

- Defined by the ASA continuum: minimal sedation (anxiolysis, normal response to verbal commands), moderate sedation (response to verbal or light tactile stimulation, airway patent, spontaneous ventilation adequate), deep sedation (purposeful response after repeated or painful stimulation, airway may require assistance, spontaneous ventilat

Sedative

- A medication that reduces central nervous system activity, producing calming effects, drowsiness, and sleep at higher doses. Sedatives are used for anxiety, procedural sedation, insomnia, and as adjuncts in anesthesia.

Sedative-Hypnotic

- CNS depressants. Benzodiazepines, barbiturates, Z-drugs. Overdose: respiratory depression, coma. Treatment: supportive, flumazenil (benzodiazepines only), airway management.

Sedatives

- A class of drugs that produce a calming or drowsy effect by depressing the central nervous system. Sedatives reduce anxiety, promote sleep, and induce relaxation. Classes: (1) Benzodiazepines—enhance GABAA receptor activity (diazepam, lorazepam, midazolam, temazepam).

Segmental Pressures

- Blood pressure measurements at different leg levels. Localizes level of obstruction.

Segmentation

- A pattern of intestinal motility involving localized, rhythmic contractions of circular muscle that segment the intestinal contents, mixing them with digestive secretions and increasing contact with the absorptive mucosal surface. Unlike peristalsis, segmentation does not propel contents distally.

Seizure

- A transient occurrence of signs and/or symptoms due to abnormal, excessive, or synchronous neuronal activity in the brain. Provoked seizures have an identifiable acute cause (fever, electrolyte disturbance, drug toxicity, withdrawal). Unprovoked seizures occur without an acute trigger.

Selective Estrogen Receptor Modulator

- A class of medications that bind to estrogen receptors (ER-alpha and ER-beta) with tissue-specific agonist or antagonist activity, depending on the target tissue and differential co-regulator recruitment.

Selective Mutism

- Consistent failure to speak in specific social situations where speaking is expected despite speaking in other situations. The disturbance interferes with educational or occupational achievement or social communication. Must last for at least one month and is not attributable to lack of knowledge of or comfort with the spoken language.

Selegiline

- A selective, irreversible MAO-B inhibitor used for Parkinson disease (mild early disease or adjunct to levodopa) and (as oral disintegrating/selegiline transdermal) for depression. Sparing MAO-A in the gut at low oral doses, reducing tyramine risk (still require dietary precautions with higher doses or patch).

Selenium

- Trace mineral, component of antioxidant enzymes. Deficiency: Keshan disease (cardiomyopathy), Kashin-Beck disease (osteoarthropathy). Sources: Brazil nuts, seafood, meat.

Selenium-75

- Calibration source. Half-life 119 days. Used for uniformity correction.

Self-Efficacy

- Patient belief in ability to manage condition. Influences adherence, outcomes.

Self-Experimentation

- See research ethics.

Self-Harm

- Intentional self-inflicted injury without suicidal intent, also termed non-suicidal self-injury; common methods include cutting, burning, and self-hitting. Prevalent in adolescents and in borderline personality disorder, depression, eating disorders, and trauma, serving affect regulation or self-punishment.

Self-Management

- Extent to which patient follows treatment recommendations. See adherence.

Semen

- The fluid ejaculated from the male reproductive tract, containing sperm (produced in the testes) suspended in seminal plasma (produced by the seminal vesicles, prostate, and bulbourethral glands). Semen provides nutrients (fructose, amino acids), protective agents (zinc, antioxidants), and a medium for sperm transport.

Semen Analysis

- Laboratory assessment of ejaculate volume, sperm concentration, total count, motility, morphology, vitality, and other characteristics. It is a key initial test in male-factor infertility and should usually be repeated because results vary biologically.

Semicircular Canals

- Three fluid-filled (endolymph) loops in each inner ear, oriented approximately orthogonally to detect angular (rotational) acceleration in three planes: horizontal (lateral), anterior (superior), and posterior. Each has an ampulla with a crista containing hair cells embedded in a gelatinous cupula.

Seminal Vesicle

- One of a paired set of male accessory reproductive glands behind the urinary bladder. Its fructose-rich secretion contributes substantially to semen and joins the vas deferens through the ejaculatory duct.

Seminal Vesicles

- Paired lobulated glands behind the bladder and anterior to the rectum, each approximately 5 cm long, producing approximately 60-70 percent of seminal volume.

Seminoma

- A malignant germ-cell tumor of the testis, typically presenting as a painless testicular mass and highly sensitive to radiation and chemotherapy.

Senna

- A stimulant laxative containing anthraquinone glycosides (sennosides) that are activated by colonic bacteria to stimulate colonic motility and fluid secretion. Onset: 6-12 hours; used for constipation and bowel preparation. Common side effects: abdominal cramps, nausea; chronic use may cause melanosis coli and electrolyte disturbances.

Sensitivity

- The proportion of people who truly have a disease who obtain a positive test result. A highly sensitive test has few false-negative results and is useful when a negative result is intended to help rule out disease in an appropriate clinical setting.

Sensorineural Hearing Loss

- Hearing loss from pathology of the cochlea or auditory (eighth cranial) nerve, characterized on pure-tone audiometry by elevated bone conduction thresholds and no air-bone gap.

Sensorium

- The state or level of consciousness and awareness of one's surroundings.

Sensory Level

- A sharply demarcated horizontal boundary below which sensation is altered on the trunk or limbs. A sensory level strongly suggests a spinal-cord lesion, although its localization requires correlation with motor, reflex, and autonomic findings.

Sensory Nerves

- Nerves that transmit sensory information from

peripheral receptors to the central nervous system. Sensory (afferent) neurons have their cell bodies in the dorsal root ganglia (spinal nerves) or cranial nerve ganglia (cranial nerves).

Sentinel Event

- Unexpected occurrence involving death or serious physical/psychological injury. Requires root cause analysis.

Sentinel Injury

- A minor but potentially significant injury in a non-mobile infant, such as unexplained bruising, oral injury, or a fracture, that may precede more severe abuse. It warrants careful history, complete examination, safeguarding assessment, and documentation.

Sentinel Lymph Node

- First lymph node draining tumor site. Biopsy assesses nodal spread.

Sentinel Lymph-Node Biopsy

- Surgical removal and pathologic examination of the first draining lymph node or nodes from a primary skin cancer. It is used for selected melanomas and other tumors to assess regional metastatic risk and guide staging.

Sentinel Surveillance

- Ongoing collection of health data from selected reporting sites, providers, or populations chosen to represent trends rather than enumerate every event. It can provide early warning and monitor changes in disease activity.

Separation Anxiety Disorder

- Developmentally inappropriate and excessive fear or anxiety concerning separation from attachment figures. The individual may have recurrent excessive distress when anticipating or experiencing separation from home or major attachment figures.

Separation of the iris

- Disinsertion or dialysis of the iris root from its attachment at the ciliary body, typically caused by blunt ocular trauma. It appears as a peripheral iridodialysis visible on slit-lamp examination and may cause monocular double vision, glare, or secondary glaucoma if extensive.

Sepsis

- Life-threatening organ dysfunction caused by dys-

regulated host response to infection. Sepsis-3 definition: infection + SOFA score increase ≥ 2 . Septic shock: sepsis with vasopressor requirement and lactate >2 mmol/L.

Sepsis Syndrome

- The spectrum of the host's dysregulated response to infection: sepsis (infection + organ dysfunction, SOFA increase ≥ 2), septic shock (sepsis + hypotension requiring vasopressors with lactate >2 mmol/L despite adequate fluid), and severe organ dysfunction/multi-organ failure.

Septal Hematoma

- Accumulation of blood between the perichondrium and the septal cartilage of the nose, typically following nasal trauma (nasal fracture). Presents with nasal obstruction, pain, and a bluish, boggy swelling of the nasal septum.

Septic Arthritis

- Joint infection, usually bacterial (Staph, Strep, Gonorrhea). Emergency: fever, severe joint pain, swelling, inability to bear weight. Diagnosis: joint aspiration (WBC $>50,000$, positive culture). Treatment: antibiotics, joint drainage (arthroscopy/open).

Septic Embolism

- Septic embolism occurs when infected material, typically thrombi containing bacteria, embolizes through the bloodstream to distant sites, causing metastatic infection. The most common source is infective endocarditis, where infected vegetations on heart valves break off and lodge in peripheral vessels.

Septic Shock

- A form of distributive shock caused by overwhelming infection and the host dysregulated immune response, leading to systemic vasodilation, increased vascular permeability, myocardial depression, and multiorgan dysfunction. It is the most common cause of death in intensive care units worldwide.

Septicaemia

- Life-threatening organ dysfunction caused by dysregulated host response to infection. See Sepsis.

Septicemia

- See Sepsis.

Septo-Optic Dysplasia

- A congenital disorder characterized by variable combinations of optic-nerve hypoplasia, midline brain abnormalities, and pituitary hormone deficiencies. It may cause visual impairment, nystagmus, developmental differences, seizures, and growth or endocrine problems.

Septoplasty

- A surgical procedure to straighten a deviated nasal septum, relieving obstruction, improving sinus drainage, and correcting related epistaxis or sleep disturbance, most often through an endonasal approach. It reshapes rather than resects the cartilaginous and bony septum while preserving support to avoid saddle nose deformity.

Sequelae

- Conditions resulting from previous disease. Examples: post-polio syndrome, stroke sequelae.

Sequential Therapy

- Treatments given in sequence. Used when combination not suitable.

Serodiagnostic

- Relating to diagnosis by detecting antibodies or other markers in blood serum.

Serology

- The laboratory evaluation of serum for antibodies, antigens, or other immune markers. Results may indicate current infection, previous exposure, vaccination, autoimmune reactivity, or passive antibody transfer and must be interpreted with timing and clinical context.

Seronegative Arthritis

- Arthritis with negative rheumatoid factor and anti-CCP. Includes spondyloarthropathies, psoriatic arthritis, reactive arthritis.

Serotonin Syndrome

- A potentially life-threatening condition caused by excessive serotonergic activity in the central and peripheral nervous systems, typically resulting from drug interactions or overdose involving serotonergic medications.

Serotonin-Norepinephrine Reuptake Inhibitors

- Antidepressants that block the reuptake of both

serotonin and norepinephrine. They include venlafaxine, duloxetine, desvenlafaxine, and levomilnacipran. SNRIs are used for major depressive disorder, generalized anxiety disorder, diabetic neuropathy, fibromyalgia, and chronic pain.

Serotonin-Norepinephrine-Dopamine Reuptake Inhibitors

- A class of antidepressants that inhibit reuptake of serotonin, norepinephrine, and dopamine. The primary agent is bupropion, which also acts as a norepinephrine-dopamine reuptake inhibitor. It is used for depression, smoking cessation, and as augmentation. It lacks sexual side effects and weight gain associated with SSRIs.

Serotype

- A group within a species of microorganisms (or viruses) distinguished by antigenic differences, usually surface antigens (capsular polysaccharides, O/H antigens, viral surface proteins).

Serpentine supravenuous hyperpigmentation

- A dermatological finding characterized by dark, wavy, serpiginous hyperpigmentation along the course of superficial veins, most commonly on the upper chest and neck. Associated with venous stasis, thrombophlebitis, and certain malignancies. The pigmentation results from hemosiderin deposition in the perivenous dermis.

Serrated Pathway

- Alternative molecular pathway to colorectal carcinogenesis accounting for fifteen to thirty percent of cases. It involves serrated polyps as precursors including sessile serrated lesions with dysplasia. Molecular features include CpG island methylator phenotype, BRAF mutations, and microsatellite instability from MLH1 silencing.

Serratus Anterior Muscle

- Fan-shaped muscle from ribs 1-8/9 to the medial scapular border. Innervated by long thoracic nerve (C5-C7). Protracts scapula, upwardly rotates glenoid for overhead elevation, stabilizes against thorax. Long thoracic nerve injury causes medial scapular winging, especially visible during wall push-ups. Commonly injured in breast surgery.

Sertoli-cell tumor

- A sex-cord stromal tumor that may produce androgens or estrogens and can cause endocrine manifestations.

Sertraline

- A selective serotonin reuptake inhibitor (SSRI) used for major depressive disorder, panic disorder, PTSD, OCD, social anxiety disorder, and premenstrual dysphoric disorder. Effective and well tolerated; mild CYP2D6 inhibition.

Serum Concentration

- The concentration of a substance measured in the liquid component of clotted blood. Interpretation depends on sampling time, exposure history, distribution, metabolism, elimination, and the relationship between concentration and toxicity.

Serum Osmolal Gap

- The difference between measured serum osmolality and calculated osmolality. An increased gap suggests unmeasured osmoles, including toxic alcohols, but interpretation must consider timing, renal function, lactate, ketones, and laboratory variation.

Serum Protein Electrophoresis (SPEP)

- Separation of serum proteins by electrophoresis into 5 fractions: albumin, alpha-1, alpha-2, beta, and gamma. Monoclonal gammopathy: M-spike in gamma region (multiple myeloma, MGUS, Waldenstrom macroglobulinemia). Immunofixation: identifies heavy chain (IgG, IgA, IgM, IgD, IgE) and light chain (kappa, lambda) isotype.

Serum Sickness

- Type III hypersensitivity reaction. Fever, rash, arthritis, proteinuria. Caused by drugs (penicillin, cephalosporins), antivenoms. Treatment: stop offending agent, antihistamines, corticosteroids.

Sever Disease

- Calcaneal apophysitis in children/adolescents. Heel pain, worse with activity. Treatment: rest, heel cups, stretching.

Severe Acute Respiratory Syndrome

- A severe respiratory illness caused by SARS coronavirus, characterized by fever, dry cough, dyspnea, and viral pneumonia that may progress to respiratory failure.

Severe Combined Immunodeficiency

- Severe combined immunodeficiency (SCID) represents a group of genetic disorders characterized by profoundly defective T cell development and function, with variable B and NK cell involvement, resulting in severe susceptibility to life-threatening infections in infancy.

Sevoflurane

- A volatile inhaled anesthetic with relatively low blood-gas solubility, permitting rapid changes in anesthetic depth and recovery. It can cause dose-dependent cardiovascular and respiratory depression and is a recognized trigger of malignant hyperthermia in susceptible patients.

Sexual Assault

- A broad legal category of any non-consensual sexual contact or penetration, encompassing rape, indecent assault, and drug-facilitated assault. Examination documents general and anogenital injuries, collects biological and trace samples, and screens for drug-facilitated assault by extended-window testing.

Sexual Dysfunction

- Persistent difficulty in sexual activity causing distress. In women: low sexual desire (hypoactive), arousal disorder, orgasmic disorder (anorgasmia), and genito-pelvic pain/penetration disorder (dyspareunia, vaginismus).

Sexual Intercourse (Coitus)

- The insertion of a penis into a partner's body for sexual pleasure, reproduction, or both. Most commonly vaginal intercourse (penile-vaginal penetration) but may also include anal intercourse. Sexual intercourse is a key component of human pair bonding and reproduction.

Sexual Masochism Disorder

- A paraphilic disorder characterized by recurrent, intense sexual arousal from the act of being humiliated, beaten, bound, or otherwise made to suffer, lasting 6 months and causing distress or impairment or involving acting on these urges.

Sexual Rehabilitation

- Address sexual function after injury, illness. Counseling, devices, medications.

Sexual Sadism Disorder

- A paraphilic disorder characterized by recurrent,

intense sexual arousal from the physical or psychological suffering of another person, lasting 6 months and causing distress or impairment or involving acting on these urges with a nonconsenting person.

Sexually Transmitted Infection

- STI: infection transmitted via sexual contact. Examples: syphilis, gonorrhea, chlamydia, HIV.

Sexually Transmitted Infections (STIs)

- Sexually transmitted infections are infections passed from one person to another through sexual contact, including vaginal, anal, and oral sex. Common STIs include chlamydia, gonorrhea, syphilis, herpes simplex virus, human papillomavirus, and HIV.

SGLT2 Inhibitor

- A drug that reduces sodium-glucose cotransport in the proximal renal tubule, increasing urinary glucose and sodium excretion.

SGLT2 Inhibitors

- Sodium-glucose co-transporter 2 inhibitors are a class of oral antidiabetic agents that block the reabsorption of glucose in the proximal convoluted tubule of the kidney, promoting glycosuria and reducing blood glucose levels.

Shadowing Artifact

- Ultrasound artifact from attenuating structures. Enhanced by harmonic imaging.

Shaken Baby Syndrome

- A form of abusive head trauma caused by violent shaking, typically in infants under 1 year. The triad: subdural hemorrhage, retinal hemorrhages, and encephalopathy. May present with irritability, lethargy, vomiting, seizures, and apnea. CT shows subdural hematoma; MRI shows diffuse axonal injury.

Shared Decision-Making

- A collaborative process in which clinicians and patients combine best available evidence with patient values, preferences, goals, and circumstances to choose among reasonable options.

Sharp Force Trauma

- Injury produced by an edged or pointed instrument that cuts or penetrates the skin, divided into incised wounds (longer than deep, clean margins) and stab wounds (deeper than wide). Wound length, depth,

angle, hesitation marks, and defense injuries infer weapon type and manner.

Sharps Container

- Puncture-resistant container for used needles and sharp instruments. Red, labeled biohazard.

Sharps Injury

- A percutaneous or mucocutaneous injury caused by a needle or other sharp instrument that may expose a person to blood or body fluids. Immediate wound care, incident reporting, risk assessment, and pathogen-specific post-exposure management are required.

Shave Biopsy

- Removal of a raised skin lesion by horizontal slicing with a scalpel at the level of the dermis; used for exophytic lesions such as seborrheic keratoses and superficial basal cell carcinomas.

Sheehan Syndrome

- Postpartum pituitary necrosis resulting from severe hemorrhage and hypotension during delivery. The pituitary gland is enlarged during pregnancy and particularly vulnerable to ischemic injury from blood loss. Presentation varies from fatigue and failure to lactate to panhypopituitarism. Growth hormone and prolactin deficiency occur first.

Sheehan Syndrome (Postpartum Pituitary Necrosis)

- Ischemic necrosis of the anterior pituitary gland due to severe postpartum hemorrhage (PPH) and hypotension. The pituitary enlarges during pregnancy and is vulnerable to ischemia from hypotensive episodes.

Shielding

- Lead, tungsten, acrylic shielding for radiation protection.

Shigellosis

- Acute infectious colitis caused by *Shigella* species (*S. sonnei* most common in developed countries). Presents with fever, abdominal cramps, tenesmus, and frequent bloody, mucoid stools. Highly contagious with low infectious dose. Diagnosis: stool culture. Treatment: azithromycin, ciprofloxacin, or ceftriaxone (resistance patterns vary).

Shingles

- Herpes zoster: reactivation of varicella-zoster virus

(VZV) from dorsal root ganglia, causing a painful, unilateral vesicular rash in a dermatomal distribution ('belt-like'), often preceded by burning pain. Risk increases with age and immunosuppression.

Shirodkar Cerclage

- A transvaginal cervical cerclage technique that involves dissection of the bladder and rectum to place the suture (Mersilene tape) as high as possible at the level of the internal os. Technically more difficult than McDonald cerclage with higher risk of hemorrhage and bladder injury.

Shivering

- Involuntary, rhythmic skeletal muscle contraction generating heat in response to cold exposure or fever, a key thermoregulatory response under hypothalamic (preoptic area) control. Shivering can increase heat production 3-5 fold.

Shock

- See hemorrhagic shock, septic shock.

Shock Wave Lithotripsy

- Non-invasive procedure using extracorporeal shock waves to fragment renal and ureteral stones. The patient is positioned over a water cushion coupling the shock wave generator to the body. Shock waves are focused on the stone under fluoroscopic or ultrasound guidance.

Shone Complex

- A group of congenital left-sided obstructive heart lesions, classically including supravulvar mitral ring, parachute mitral valve, subaortic stenosis, and coarctation of the aorta. The combination and severity vary and may lead to heart failure or systemic outflow obstruction.

Short Bowel Syndrome

- Clinical condition resulting from surgical resection or functional loss of a significant portion of the small intestine leading to inadequate absorption of nutrients.

Shoulder Dystocia

- An obstetric emergency where the fetal head delivers but the anterior shoulder is impacted behind the maternal symphysis pubis. Defined as head-to-body delivery time exceeding 60 seconds or need for ancillary maneuvers. Risk factors include macrosomia, diabetes, prior shoulder dystocia, and operative

vaginal delivery.

Shoulder Impingement

- Compression of rotator cuff tendons beneath acromion. Symptoms: shoulder pain with overhead activity, weakness. Diagnosis: clinical (Neer, Hawkins tests), ultrasound/MRI. Treatment: physical therapy, corticosteroid injection, surgical decompression.

Shoulder Instability

- Anterior instability (95%): Bankart lesion (anterior-inferior labral tear), Hill-Sachs lesion (posterolateral humeral head impression fracture), ALPSA (anterior labroligamentous periosteal sleeve avulsion), HAGL (humeral avulsion of glenohumeral ligament), GLAD (glenoid labral articular disruption).

Shunt

- Tube diverting fluid. Types: ventriculoperitoneal (CSF), ventriculoatrial, lumboperitoneal, ventriculopleural. Indications: hydrocephalus. Complications: obstruction, infection, overdrainage, migration. Valves regulate pressure.

Shunt (Vascular)

- Tube diverting fluid. See Shunt.

Shwachman Syndrome

- An inherited multisystem disorder usually involving exocrine pancreatic insufficiency, bone-marrow dysfunction, and skeletal abnormalities. Children may have steatorrhea, poor growth, recurrent infections, neutropenia, anemia, or increased risk of myeloid malignancy.

SIADH

- Syndrome of Inappropriate Antidiuretic Hormone secretion characterized by hyponatremia (dilutional) with inappropriately concentrated urine (>100 mOsm/kg) and elevated urine sodium (>40 mEq/L). Euvolemic hyponatremia.

SIADH (Syndrome of Inappropriate Antidiuretic Hormone Secretion)

- Excessive vasopressin activity causing impaired free-water excretion, euvolemic hyponatremia, and inappropriately concentrated urine after exclusion of other causes. Treatment depends on severity and cause and may include fluid restriction or carefully monitored hypertonic saline for severe symptomatic

cases.

Sialadenitis

- Inflammation of salivary gland. Acute (bacterial) or chronic (recurrent). Treatment: antibiotics, hydration, gland massage, surgery if chronic.

Sialography

- Contrast imaging of salivary ducts. Evaluates obstruction, stricture, tumors. Rarely performed now.

Sialolithiasis

- Sialolithiasis, commonly known as salivary stones, is the formation of calcified concretions within the salivary gland ducts or parenchyma, most commonly affecting the submandibular gland (approximately eighty percent of cases) due to the long, dependent course of the Wharton duct, the alkaline pH of submandibular saliva, and its higher m

Sialorrhea

- Excessive salivation. Neurological causes, medications, dental causes. Treatment: hydration, glycopyrrolate, botulinum toxin, surgery.

Sialorrhoea

- See Drooling / Excessive Salivation.

Sialuria

- A rare inherited disorder of sialic-acid metabolism caused by abnormal regulation of the GNE enzyme. Excess free sialic acid accumulates in cells and urine and may cause developmental delay, coarse facial features, hepatosplenomegaly, and variable intellectual disability.

Sicca Syndrome

- Dry eyes and dry mouth due to decreased tear and saliva production. Primary: Sjogren syndrome. Secondary: associated with RA, SLE, scleroderma, graft-versus-host disease, hepatitis C, sarcoidosis, HIV, medications (anticholinergics, antidepressants). Diagnosis: Schirmer test (<5 mm/5 min), rose bengal staining, salivary gland biopsy.

Sick Sinus Syndrome

- Sinus node dysfunction causing a spectrum of arrhythmias: sinus bradycardia, sinus arrest, sinoatrial block, and tachycardia-bradycardia syndrome. Most common in elderly patients. Presents with syncope, fatigue, palpitations, and exercise intolerance. Diagnosis: ECG, Holter monitor (pauses >3

seconds).

Sickle Cell Anemia

- The most severe form of sickle cell disease (HbSS), causing chronic hemolytic anemia and vaso-occlusive crises. Features: jaundice, splenomegaly (infancy), pain crises, acute chest syndrome, stroke, and increased infection risk from functional asplenia. Diagnosis: hemoglobin electrophoresis showing HbS.

Sickle Cell Crisis

- An acute clinical complication of sickle cell disease, most commonly a vaso-occlusive episode producing ischemic pain. Other acute crises include acute chest syndrome, splenic or hepatic sequestration, and aplastic crisis; treatment depends on the specific complication and severity.

Sickle Cell Disease

- Autosomal recessive hemoglobinopathy (HbS). Glutamic acid → valine at position 6 of beta-globin. Homozygous (HbSS) most severe. Sickling under hypoxia causes vaso-occlusion, hemolysis. Symptoms: pain crises, acute chest syndrome, stroke, splenic sequestration, aplastic crisis (parvovirus B19).

Sickle Cell Trait

- Heterozygous carrier state for the HbS mutation (HbAS), present in 8% of African Americans. Red cells contain both HbA and HbS (typically 30-40% HbS). Usually asymptomatic with normal hemoglobin and RBC indices.

Side Effect

- An unintended pharmacological effect of a drug occurring at therapeutic doses, usually predictable from the drug's mechanism of action (a Type A reaction).

Siderosis

- Benign pneumoconiosis from iron dust, seen in welders. Causes small opacities on CXR but minimal impairment. Reversible with removal of exposure.

SIDS

- See Sudden Infant Death Syndrome.

Sievert (Sv)

- The SI unit used for equivalent and effective dose, incorporating radiation type and, for effective dose, tissue sensitivity. It is used to estimate stochastic radiation risk rather than simply physical energy

deposition.

Sigmoid Colon

- S-shaped segment from descending colon to rectum in the left iliac fossa, approximately 40-50 cm, with complete mesentery (intraperitoneal, mobile). Most common diverticular disease site from narrow lumen and high pressure. Common volvulus site, particularly in elderly/institutionalized patients. Second most common colorectal cancer site.

Sigmoid volvulus

- Axial twisting of the sigmoid colon around its mesentery, the most common cause of colonic volvulus, typically occurring in elderly, institutionalized, or neuropsychiatric patients with chronic constipation and a redundant sigmoid. Presents with acute large bowel obstruction, abdominal distension, and absent bowel sounds.

Sign

- Objective evidence of disease. Examples: rash, murmur, edema. Observed by examiner.

Signal Transduction

- The process by which an extracellular signal (hormone, neurotransmitter, growth factor, drug) binding to a cell-surface or intracellular receptor is converted into a specific intracellular response.

Signal-to-Noise Ratio

- Ratio of true counts to statistical noise. Improved by longer acquisition, higher activity.

Signet Ring Cell Carcinoma

- A poorly differentiated adenocarcinoma in which intracellular mucin displaces the nucleus to the periphery, most often of the stomach or colon; associated with an aggressive course and diffuse growth pattern.

Silicone oil hyperoleon

- Emulsification of silicone oil tamponade used in vitreoretinal surgery, where the oil breaks into tiny droplets that migrate into the anterior chamber or trabecular meshwork. It can cause elevated intraocular pressure, band keratopathy, and chronic inflammation requiring oil removal and anterior chamber washout.

Silicosis

- Fibrotic lung disease caused by inhalation of crystalline silica dust, typically in mining, quarrying,

construction, or stone-cutting work. It causes progressive breathlessness and increases the risk of tuberculosis, chronic airflow limitation, and lung cancer.

Silver Diamine Fluoride

- Liquid that arrests caries and stains cavity black. Used in young children, special needs patients.

Simethicone

- See gastroenterology.

Sinding-Larsen-Johansson Syndrome

- Traction apophysitis of inferior patellar pole in adolescents. Similar to Osgood-Schlatter but at knee. Treatment: rest, activity modification.

single nucleotide polymorphism (SNP)

- Single base pair variation occurring in >1% of population. Most common genetic variation. Used in GWAS, forensics, pharmacogenetics.

Single Photon Emission Computed Tomography (SPECT)

- Nuclear medicine imaging using gamma-emitting isotopes (technetium-99m most common). Rotating gamma camera acquires images. Applications: myocardial perfusion, bone scans, thyroid imaging, renal scans, brain perfusion. Lower resolution than PET but more widely available and cost-effective.

Single Ventricle

- A group of congenital heart defects where one functional ventricle supports both the systemic and pulmonary circulations. Includes tricuspid atresia, mitral atresia, double inlet left ventricle, hypoplastic left heart syndrome, and unbalanced AV canal.

Single-Dose Toxicity

- See acute toxicity.

Sinoatrial Node

- The heart's natural pacemaker, located at the junction of the superior vena cava and right atrium. It consists of specialized cells that spontaneously depolarize at a rate of 60-100 beats per minute, setting the normal sinus rhythm.

Sinus Lift

- Procedure to increase bone height in posterior maxilla. Subperiosteal or transalveolar approach. Graft material placed.

Sinus Rhomboidalis

- A diamond-shaped depression in the floor of the fourth ventricle of the brain.

Sinuses

- Air-filled cavities within the skull bones lined by respiratory mucosa, connected to the nasal cavity via ostia.

Sinusitis

- Inflammation of the paranasal sinus mucosa. Acute sinusitis: viral (most common, resolves in <10 days) or bacterial (symptoms >10 days, double worsening, or severe: *Streptococcus pneumoniae*, *Haemophilus influenzae*, *Moraxella catarrhalis*). Chronic sinusitis: symptoms >12 weeks with radiographic evidence.

Sinusitis and Sinus Problems

- See Sinusitis. The term includes inflammatory or infectious disorders of the paranasal sinuses and should be distinguished from isolated nasal congestion or facial pain.

Sinusoidal Fetal Heart Rate Pattern

- A rare, ominous FHR pattern characterized by a smooth, sine-wave-like undulation with a fixed frequency (2-5 cycles/minute) and absent variability. Strongly associated with severe fetal anemia (Rh isoimmunization, fetomaternal hemorrhage, parvovirus B19), fetal hypoxia, and fetal acidosis.

Sirenomelia

- A rare congenital malformation in which the lower limbs are partially or completely fused. It is usually associated with severe abnormalities of the genitourinary, gastrointestinal, vascular, and spinal systems and is often fatal in the neonatal period.

Sister Mary Joseph's nodule

- A firm, hard nodule in the umbilicus resulting from metastatic malignancy, most commonly from intra-abdominal or pelvic cancers including gastric, ovarian, and colorectal carcinoma. It indicates advanced, disseminated disease with a poor prognosis. Diagnosis is by fine needle aspiration biopsy.

Sitosterolemia

- An autosomal-recessive disorder of plant-sterol transport caused by ABCG5 or ABCG8 variants. It can produce premature atherosclerosis, xanthomas, hemolytic anemia, macrothrombocytopenia, and elevated plasma plant sterols.

Situs Inversus

- A congenital condition in which the thoracic and abdominal organs are reversed or mirrored from their normal positions. Complete situs inversus (situs inversus totalis) involves all organs.

Situs inversus totalis

- A congenital condition in which the major visceral organs are mirrored from their normal positions, with the liver on the left, heart on the right, and other viscera similarly reversed.

Sjogren Syndrome

- Primary Sjogren syndrome (pSS) is a chronic autoimmune exocrinopathy characterized by lymphocytic infiltration of lacrimal and salivary glands causing dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia), with female predominance (9:1) and peak onset 40-60. Secondary SS occurs with RA, SLE, SSc.

Sjögren Syndrome

- A systemic autoimmune disease characterized by lymphocytic destruction and dysfunction of lacrimal and salivary glands, producing dry eyes and dry mouth. It may involve multiple organs and increases the risk of salivary-gland lymphoma.

Sjögren's syndrome

- A chronic autoimmune disorder characterized by lymphocytic infiltration and destruction of exocrine glands, primarily the salivary and lacrimal glands, resulting in dry eyes (keratoconjunctivitis sicca) and dry mouth (xerostomia).

Skeletal Muscle

- Voluntary, striated muscle attached to bones by tendons or directly by connective tissue. Its contraction produces movement, maintains posture, and generates heat.

Skeletal Muscle Relaxant

- A medication that reduces muscle spasm, spasticity, and associated pain.

Skeletal Survey

- A series of radiographs obtained when child abuse is suspected. Includes AP and lateral views of the skull, chest (including oblique views), abdomen, spine, and bilateral upper and lower extremities. Identifies fractures of varying ages, which is highly suggestive of non-accidental trauma.

Skeletonization

- Final stage of postmortem decomposition where soft tissues, organs, and skin have completely decayed or been scavenged, leaving only clean osseous skeletal remains.

Skilled Nursing Facility

- Inpatient facility providing skilled nursing and therapy. For recovery after hospitalization.

Skin

- The largest organ of the body (approximately 1.5–2 m²), forming a protective barrier against physical injury, pathogens, UV radiation, and water loss.

Skin Absorption

- See dermal exposure.

Skin Aging

- intrinsic (chronological) and extrinsic (photoaging) components. Photoaging causes wrinkle formation, telangiectasias, lentigines, leather-like texture, and actinic keratoses. Intrinsic aging causes thinning, wrinkling, and delayed wound healing. Prevention includes sun protection (broad-spectrum SPF 30 or greater daily).

Skin Aging and Appearance

- Skin aging is a complex biological process influenced by both intrinsic factors, such as genetics and natural aging, and extrinsic factors, particularly ultraviolet radiation exposure.

Skin Allergies and Contact Dermatitis

- Inflammatory skin reactions triggered by exposure to allergens or irritants that come into direct contact with the skin.

Skin Allergy Testing

- prick testing (immediate hypersensitivity, IgE-mediated) and intradermal testing (more sensitive, used for drug and venom allergy). Prick testing introduces allergen extract into the epidermis; wheal and flare reaction at 15 minutes indicates sensitization. Intradermal testing injects dilute allergen into the dermis.

Skin and Aging

- Skin aging involves intrinsic and extrinsic changes. Intrinsic aging causes thinning, wrinkling, and fragility. Extrinsic (photo)aging causes wrinkles, telangiectasias, lentigines, and actinic keratoses.

Collagen loss, elastin degradation, and decreased cell turnover contribute to aging skin.

Skin and Environment

- Environmental factors significantly affect skin health. Pollution causes oxidative stress, inflammation, and accelerated aging. Particulate matter penetrates skin, generating free radicals. Blue light from screens causes hyperpigmentation, particularly in darker skin tones. Indoor heating and air conditioning cause dryness.

Skin Appendages

- hair follicles, sebaceous glands, eccrine sweat glands, and apocrine sweat glands. Hair follicles cycle through anagen (growth), catagen (transition), and telogen (resting) phases. Sebaceous glands produce sebum, an oily substance lubricating skin and hair.

Skin Barrier Function

- The skin barrier prevents water loss and protects against environmental insults. The stratum corneum acts as a physical barrier, while the acid mantle (pH 4 to 6) provides chemical defense. Lipids (ceramides, cholesterol, free fatty acids) in the intercellular space form a waterproof barrier.

Skin Biopsy Methods

- Skin biopsy methods vary by lesion characteristics. Punch biopsy (2 to 6 mm circular blade) provides full-thickness specimen for most diagnoses. Shave biopsy (razor or scalpel) removes superficial lesions; deep shave (saucerization) includes subcutaneous fat for suspected melanoma.

Skin Biopsy Techniques

- Skin biopsy is selected according to the suspected diagnosis and lesion depth. Punch biopsy samples full-thickness skin; shave biopsy samples superficial lesions; excisional biopsy removes an entire small lesion; and incisional biopsy samples part of a large lesion.

Skin Cancer

- Most common cancer. Types: basal cell, squamous cell, melanoma. Prevention: sun protection. Early detection critical.

Skin Cancer Epidemiology

- Skin cancer epidemiology reveals increasing incidence worldwide. Basal cell carcinoma is the most

common cancer (1 million cases annually in the US). Squamous cell carcinoma is second (approximately 200,000 cases annually). Melanoma is less common but most deadly (about 100,000 new cases annually in the US).

Skin Cancer Prevention

- Skin cancer prevention focuses on UV radiation protection. Broad-spectrum sunscreen (SPF 30 or greater) applied daily and reapplied every 2 hours during sun exposure. Protective clothing, hats, and sunglasses. Seeking shade during peak UV hours (10 AM to 4 PM). Avoiding tanning beds.

Skin Cancer Screening

- Skin cancer screening uses the ABCDE criteria for melanoma, the ugly duckling sign, and total body skin examination. Dermoscopy (dermatoscopy) magnifies skin lesions revealing structures invisible to the naked eye, improving diagnostic accuracy. Skin self-examination teaches patients to monitor moles for changes.

Skin Cancer Staging

- Skin cancer staging guides treatment and prognosis. Melanoma: Breslow thickness (most important prognostic factor), ulceration, mitotic rate, sentinel lymph node status (AJCC 8th edition).

Skin Cancer Surgery

- Skin cancer surgical techniques maximize cure while minimizing morbidity. Wide local excision with appropriate margins (BCC: 4 mm, SCC: 4 to 6 mm, melanoma: based on Breslow thickness). Mohs micrographic surgery provides 99 percent cure rate for BCC and 97 percent for SCC with tissue preservation.

Skin Flaps

- Skin flaps transfer tissue with its blood supply. Rotation flaps rotate adjacent tissue into defect. Advancement flaps slide tissue forward. Transposition flaps move tissue across intervening skin. Random pattern flaps rely on dermal plexus; axial pattern flaps have named vascular pedicle.

Skin Graft

- Transfer of skin from a donor site to a recipient site to cover a wound. Split-thickness skin graft (STSG): epidermis + partial dermis (0.008-0.018 inches), harvested with dermatome, donor site heals

spontaneously (2-3 weeks). Full-thickness skin graft (FTSG): epidermis + full dermis, donor site requires closure, better cosmetic outcome.

Skin Grafts

- Skin grafts replace lost skin. Split-thickness grafts (epidermis and part of dermis) heal by re-epithelialization of donor site, allowing reuse. Full-thickness grafts (entire dermis) provide better cosmetic results but require primary closure of donor site.

Skin Immune System

- The skin immune system (SALT: skin-associated lymphoid tissue) provides immunological surveillance. Langerhans cells (dendritic cells in epidermis) capture antigens and present them to T cells. Dermal dendritic cells, mast cells, and resident T cells form an innate and adaptive immune network.

Skin Integrity

- Localized damage to the skin and/or underlying soft tissue caused by prolonged pressure, shear, or friction, most commonly over bony prominences. See pressure injury.

Skin Irritation

- See dermatology.

Skin Microbiome

- The skin microbiome is the community of microorganisms (bacteria, fungi, viruses) residing on the skin. *Staphylococcus aureus* colonization is increased in atopic dermatitis and psoriasis. *Cutibacterium acnes* plays a role in acne. Diversity is decreased in diseased skin.

Skin of Color

- Skin of color (Fitzpatrick IV to VI) has unique dermatologic considerations. It is more prone to hyperpigmentation disorders (post-inflammatory hyperpigmentation, melasma, acanthosis nigricans) and keloid formation. It is less prone to skin cancer but presents with more advanced disease when it occurs.

Skin Pigmentation

- Determined by melanin produced by melanocytes in the stratum basale. Eumelanin (brown-black) and pheomelanin (red-yellow) are the two types. Melanin production is stimulated by UV radiation through melanocyte-stimulating hormone (MSH).

Fitzpatrick skin type classification (I to VI) describes pigmentation and sun response.

Skin Tags

- Benign fibroepithelial polyps occurring in intertriginous areas (neck, axillae, groin, under breasts). They are associated with obesity, insulin resistance, and pregnancy. Skin tags are asymptomatic but may become irritated or bleed. Treatment includes snipping, cryotherapy, electrocautery, and shave excision.

Skull

- Bony framework of the head with 22 bones: cranium (8: frontal, 2 parietal, 2 temporal, occipital, sphenoid, ethmoid) and facial (14). Joined by sutures: coronal, sagittal, lambdoid, squamous. The skull base has foramina for cranial nerves and vessels.

Skull fracture

- A break in one or more of the cranial bones from blunt force trauma, classified as linear, depressed, basilar, or comminuted, with potential for intracranial injury and cerebrospinal fluid leak.

SLAP Lesion

- Superior labrum anterior to posterior tear. Associated with overhead activities, trauma. Treatment: rest, physical therapy, surgical repair if symptomatic.

SLAP Tears

- Superior labrum anterior-posterior (SLAP) tears: Snyder classification: I (fraying), II (detachment of biceps anchor, most common), III (bucket-handle tear of labrum, stable biceps), IV (bucket-handle extending into biceps). Clinical: O'Brien active compression test, Speed test, Yergason test, crank test. MRI arthrogram: sensitivity >90%.

Sleep

- A naturally recurring physiological state characterized by altered consciousness, reduced sensory responsiveness, and distinct brain wave patterns that is essential for physical restoration, memory consolidation, and metabolic regulation.

Sleep (Geriatric)

- See sleep disorders.

Sleep Apnea

- See pulmonology.

Sleep Apnea (Dental)

- Oral appliance for obstructive sleep apnea. Mandibular advancement device. Indications: mild to moderate OSA, CPAP intolerance.

Sleep Apnea (Obstructive Sleep Apnea - OSA)

- A sleep disorder characterized by repeated episodes of partial or complete upper airway obstruction during sleep (obstructive sleep apnea—OSA, most common) or cessation of respiratory drive (central sleep apnea—CSA), causing intermittent hypoxia, hypercapnia, and sleep fragmentation.

Sleep Bruxism

- Grinding or clenching teeth, often during sleep. See bruxism.

Sleep Hygiene

- A set of behavioral and environmental practices designed to promote consistent, uninterrupted, and restorative sleep. Key elements include maintaining a regular sleep schedule, creating a cool, dark, and quiet sleep environment, avoiding caffeine and electronic screens before bedtime, and engaging in relaxing pre-sleep routines.

Sleep Medications and Supplements

- Pharmacological agents used to treat insomnia and other sleep disorders when behavioral interventions alone are insufficient.

Sleep Studies (Polysomnography)

- Sleep tests, most commonly polysomnography, are diagnostic studies used to evaluate sleep architecture and identify sleep disorders. Polysomnography records brain waves, eye movements, heart rate, breathing patterns, blood oxygen levels, and limb movements throughout the night in a controlled laboratory setting.

Sleep-Related Hypoventilation

- Inadequate ventilation during sleep causing elevated carbon dioxide, oxygen desaturation, or both. Causes include obesity, neuromuscular disease, chest-wall disorders, chronic lung disease, medications, and central respiratory-control abnormalities.

Sleepwalking

- A parasomnia involving complex behaviors or walking during partial arousal from non-REM sleep, usu-

ally with little or no memory afterward. It is more common in children and may be triggered by sleep deprivation, stress, fever, or other sleep disorders; injury prevention and evaluation of recurrent or dangerous episodes are important.

Sleeve Gastrectomy

- Removal of major stomach, leaving sleeve. Restrictive procedure. Lower malabsorption risk than bypass. Effective weight loss.

Sliding Filament

- The mechanism of muscle contraction in which actin filaments slide past myosin filaments, shortening the sarcomere without changing filament lengths. Myosin cross-bridges bind actin, undergo a power stroke using ATP hydrolysis, detach, and reattach to new actin sites in a cyclic manner.

Slipped Capital Femoral Epiphysis

- Displacement of the femoral head relative to the femoral neck through the growth plate (physis) in adolescents. Most common hip disorder in adolescents (10-16 years). Risk factors: obesity, male sex, African American, and endocrine disorders (hypothyroidism, growth hormone deficiency).

Slit Lamp Biomicroscopy

- A fundamental ophthalmological examination technique that uses a high-intensity, adjustable slit beam of light combined with a binocular microscope to provide a magnified, stereoscopic view of the anterior segment and, with the use of auxiliary lenses, the posterior segment of the eye.

Small Airway Disease

- Disease affecting airways with diameter < 2 mm (bronchioles). It is the earliest site of damage in COPD and is often subclinical until advanced. It is characterized by narrowing, mucus plugging, and inflammation of small airways. PFTs may show air trapping (increased RV/TLC) before FEV1 decline.

Small Bowel Follow-Through

- Extended upper GI series tracking barium through small bowel.

Small Bowel Obstruction

- Mechanical blockage of the small intestine most commonly caused by adhesions from prior surgery accounting for sixty to seventy-five percent, followed by hernias and malignancy. Clinical presentation

includes colicky abdominal pain, bilious vomiting, distension, and obstipation.

Small Bowel Series

- X-ray imaging of small intestine using barium. Evaluates Crohn disease, tumors, obstructions.

Small Cell Carcinoma

- A high-grade neuroendocrine carcinoma, most often of the lung, strongly associated with smoking. Highly responsive initially to chemotherapy and radiation but prone to rapid recurrence and early brain metastases.

Small Cell Lung Cancer

- Aggressive lung cancer, strongly associated with smoking. Early metastasis, responds to chemotherapy.

Small Intestine

- Longest GI segment (approximately 6 meters) from pylorus to ileocecal valve. Primary nutrient digestion and absorption site. Three segments: duodenum (25 cm, chyme/bile/pancreatic enzymes), jejunum (2.5 m, prominent folds/villi for absorption), ileum (3.5 m, Peyer's patches for immunity).

Small-bowel volvulus

- Twisting of the small intestine around its mesentery, causing mechanical obstruction and potentially compromising mesenteric blood flow. It may be primary (idiopathic, more common in developing countries) or secondary to adhesions, malrotation, hernias, or intraluminal tumors.

Smallpox

- A highly contagious and deadly disease caused by variola virus, eradicated globally by 1980 through vaccination. Characterized by febrile prodrome followed by vesicular/pustular rash (centrifugal distribution, all lesions at same stage). Last known natural case: 1977. Bioterrorism concern due to stored virus in WHO-approved

SMART Goals for Antibiotic Stewardship

- Structured, measurable objectives for antimicrobial stewardship programs to optimize antibiotic use and reduce resistance: Specific (target a defined outcome, e.g., reducing fluoroquinolone use), Measurable (quantifiable metrics—defined daily doses, use rates), Achievable (realistic within resources), Relevant (addresses local resistance/

SMILE

- Small incision lenticule extraction, commonly known as SMILE, is a minimally invasive refractive surgical procedure that corrects myopia and myopic astigmatism by creating and removing a thin, disc-shaped piece of corneal stroma called a lenticule through a small incision, without creating a corneal flap.

Smile Design

- Planning smile based on facial features, gum display, tooth proportions. Digital smile design software available.

Smith-Lemli-Opitz Syndrome

- An autosomal recessive disorder of cholesterol metabolism caused by mutations in DHCR7, encoding 7-dehydrocholesterol reductase, the final enzyme in the cholesterol biosynthetic pathway.

Smith-Magenis Syndrome

- A genetic neurodevelopmental disorder, usually involving deletion of 17p11.2 or an RAI1 variant. It causes developmental delay, intellectual disability, distinctive facial features, sleep disturbance, behavioral dysregulation, and variable congenital abnormalities.

Smoke Evacuator

- A device that captures and filters airborne surgical plume generated by electrosurgery, laser use, or ultrasonic energy. Evacuation reduces exposure to particulate matter, toxic gases, and potentially infectious biologic material.

Smoke Inhalation Injury

- Pulmonary injury caused by inhalation of toxic products of combustion.

Smoking Cessation

- Measures to prevent disease. See prevention.

Smooth Muscle

- Involuntary, non-striated muscle found in the walls of hollow organs and blood vessels. It produces slow, sustained contractions under autonomic, hormonal, and local control.

Smothering

- Mechanical obstruction of the external nose and mouth that prevents effective ventilation. It may be caused by a covering, compression, or an obstructing

soft material; findings are often nonspecific and must be interpreted with the scene, history, and complete examination.

Snapping Hip Syndrome

- Hip pain/clicking from tendon snapping over bony prominence. Internal (iliopsoas), external (IT band), intra-articular (labral tear). Treatment: physical therapy, injection, surgery if refractory.

Sneddon Syndrome

- A non-inflammatory thrombotic vasculopathy characterized by livedo racemosa and recurrent ischemic cerebrovascular events. Headache, hypertension, cognitive impairment, and cardiac-valve abnormalities may occur; evaluation addresses vascular risk and autoimmune or thrombophilic causes.

Snoring and Sleep Apnea

- Snoring is the sound produced by vibrating soft tissues in the upper airway during sleep, often resulting from partial airway obstruction. Sleep apnea is a more serious condition characterized by repeated episodes of complete or partial airway collapse during sleep, leading to intermittent hypoxia, frequent arousal, and fragmented sleep.

SNRIs

- Serotonin-norepinephrine reuptake inhibitors are a class of antidepressant drugs that inhibit the reuptake of both serotonin and norepinephrine from the synaptic cleft, increasing both serotonergic and noradrenergic neurotransmission. Common SNRIs include venlafaxine, duloxetine, desvenlafaxine, and levomilnacipran.

Social Anxiety Disorder

- Marked fear or anxiety about one or more social situations in which the individual may be scrutinized by others. The individual fears acting in a way or showing anxiety symptoms that will be negatively evaluated.

Social Isolation

- See psychiatry.

Sodium

- Major extracellular cation. Fluid balance, nerve function, muscle contraction. Recommended intake <2300 mg/day. Excess associated with hypertension. Sources: salt, processed foods.

Sodium Bicarbonate Therapy

- Alkalinizing agent for metabolic acidosis in CKD. Target serum bicarbonate greater than 22 mEq/L. Slows CKD progression and improves muscle wasting. Dose titrated to serum levels.

Sodium Polystyrene Sulfonate

- Potassium-binding resin for hyperkalemia. Risk: colonic necrosis, especially with sorbitol.

Soft Markers for Aneuploidy

- Sonographic findings in the second trimester that are common in normal fetuses but have a statistically higher prevalence in aneuploid fetuses (especially trisomy 21).

Soft Tissue Injury

- See emergency medicine.

Soft Tissue Rheumatism

- Pain conditions affecting muscles, tendons, ligaments, bursae. Examples: frozen shoulder, lateral epicondylitis, rotator cuff tendinopathy. Treatment: rest, physiotherapy, injections, surgery if refractory.

Soft Tissue Sarcoma

- A heterogeneous group of cancers arising in muscle, fat, fibrous tissue, blood vessels, and other connective tissues, accounting for about 1% of adult cancers. Treatment: wide surgical excision with or without radiation and chemotherapy.

Soft Tissue Tumors

- Lipoma: most common benign soft tissue tumor, subcutaneous, encapsulated. Liposarcoma: most common malignant STS in adults, deep, subtypes (well-differentiated/dedifferentiated, myxoid/round cell, pleomorphic). Synovial sarcoma: t(X;18), SS18-SSX, young adults, deep extremity, "triple sign" on MRI.

Soft-tissue calcinosis

- The deposition of insoluble calcium phosphate salts in soft tissues such as skin, subcutaneous tissue, muscles, and tendons, which may be metastatic (due to hypercalcaemia or hyperphosphataemia), dystrophic (in damaged or devitalised tissue), or idiopathic.

Soil Contamination

- Environmental toxicology. Heavy metals, pesticides, industrial waste. Remediation: excavation,

bioremediation, containment.

Solar Urticaria

- A rare form of inducible urticaria triggered by ultraviolet or visible light, producing pruritic erythema and wheals within minutes on exposed skin. Symptoms usually improve after removal from the light source. Diagnosis may involve phototesting.

Soleus Muscle

- Broad flat deep posterior leg muscle from posterior tibia, fibula, and tendinous arch to the calcaneus via the Achilles tendon. Innervated by tibial nerve (S1-S2). Powerful ankle plantarflexor, the primary anti-gravity muscle for upright posture. One-joint muscle.

Solitary Pulmonary Nodule

- Single, well-circumscribed lesion < 3 cm surrounded by normal lung. Differential includes granuloma, hamartoma, and carcinoma. PET-CT and tissue sampling help determine etiology.

Solution

- A homogeneous mixture in which a drug (solute) is dissolved in a liquid vehicle (solvent, usually water, saline, alcohol, or polyethylene glycol). Solutions provide rapid absorption (oral solutions, parenteral IV solutions), ease of swallowing, and flexible dosing; they require preservatives and have limited stability.

Solvent Exposure

- See occupational medicine.

Somatic Mutation

- Acquired genetic change in cancer cells. Not inherited.

Somatic Symptom Disorder

- Characterized by one or more somatic symptoms that are distressing or result in significant disruption of daily life, accompanied by excessive thoughts, feelings, or behaviors related to the somatic symptoms.

Somatic Variant

- A DNA variant acquired after conception in a non-germline cell lineage. It is not usually transmitted to offspring and may produce mosaic disease or contribute to cancer when it affects growth-control pathways.

Somatostatin Analog

- A class of synthetic peptides that mimic the actions of native somatostatin, a hormone that inhibits the secretion of growth hormone, thyroid-stimulating hormone, insulin, glucagon, gastrin, and other GI hormones.

Somatostatinoma

- Rare pancreatic or duodenal tumor producing somatostatin causing diabetes, gallstones, steatorrhea, and hypochlorhydria from suppression of insulin, gallbladder contraction, pancreatic enzymes, and gastric acid respectively. Diagnosed by elevated somatostatin levels. Treatment is surgical resection for localized disease.

Somatotropic pituitary adenoma

- A benign neoplasm of the anterior pituitary arising from growth hormone-secreting cells, causing acromegaly in adults and gigantism in children if occurring before epiphyseal closure.

Sonographer

- A healthcare professional trained to perform ultrasound scans (sonography).

Sore Throat

- Pain, scratchiness, or irritation of the throat, most often caused by pharyngitis, typically viral but also bacterial, allergic, irritant, reflux, voice strain, traumatic, or neoplastic.

Source Control

- A procedure or intervention that eliminates, drains, debrides, or otherwise controls an anatomic focus of infection or contamination. Examples include drainage of an abscess, removal of infected tissue or device, and repair of a perforation.

Soy Protein Intolerance

- Adverse gastrointestinal reaction to soy proteins, most often seen in infants and young children. It may cause vomiting, diarrhoea, abdominal pain, blood or mucus in the stool, feeding difficulty, or poor growth.

Spasm

- A sudden, involuntary, sustained or repetitive contraction of a muscle or group of muscles.

Spasticity

- A velocity-dependent increase in muscle tone

caused by hyperexcitability of the stretch reflex after an upper motor-neuron lesion. It may impair movement, hygiene, positioning, comfort, gait, and function.

Spasticity Management

- See neurology.

Spatial Resolution

- Ability to distinguish small objects. PET: 4-6 mm. SPECT: 8-12 mm.

Special Care Baby Unit

- A ward within a hospital that provides specialist care for newborn babies who need additional support but not intensive care.

Special Stain

- A histochemical stain that highlights a particular tissue component, organism, or deposited substance not adequately demonstrated by routine hematoxylin and eosin. Examples include periodic acid-Schiff, Ziehl-Neelsen, Congo red, and Gomori methenamine silver stains.

Specific Absorbed Fraction

- Fraction of emitted energy absorbed in target organ.

Specific Activity

- Radioactivity per unit mass of compound. High specific activity essential for receptor targeting (low receptor density).

Specific Gravity

- Ratio of urine density to water density reflecting solute concentration. Normal 1.005 to 1.030. Less precise than osmolality but faster to measure. Elevated in dehydration, glycosuria, proteinuria, and contrast. Decreased in diabetes insipidus and fluid overload.

Specific Learning Disorder

- Persistent difficulties in reading, written expression, or mathematics, despite adequate intelligence and educational opportunity. Diagnosis requires academic skills significantly below expected for age, difficulties for at least six months, and no better explained by intellectual disability, sensory impairment, or neurological

Specific Phobia

- Marked fear or anxiety about a specific object or

situation that is out of proportion to the actual danger posed. Exposure to the phobic stimulus almost invariably provokes an immediate anxiety response. The object or situation is actively avoided or endured with intense anxiety.

Specificity

- The proportion of people who do not have a disease who obtain a negative test result. A highly specific test has few false-positive results and a positive result can provide strong evidence for disease when applied to the intended population.

Specimen Adequacy

- The degree to which a collected clinical specimen is sufficient in quantity, correctly identified, properly transported, and representative of the site being tested. Inadequacy can cause invalid, misleading, or falsely negative results.

Specimen Container

- Container for collecting specimens. Types: urine cup, stool container, sputum cup, biohazard bag.

Specimen Cup

- Container for urine or stool samples.

Specimen Transport Medium

- A formulated medium that preserves the viability and diagnostic characteristics of microorganisms during transport from the collection site to the laboratory. The required medium, temperature, container, and transport time depend on the specimen and suspected pathogen.

SPECT Scan (Single Photon Emission CT)

- Nuclear medicine imaging using gamma-emitting radioisotopes (Tc-99m, Tl-201, I-123) with a rotating gamma camera to create 3D images.

Spectral Analysis

- Energy discrimination to reject scatter. Energy window typically 20% centered on photopeak.

Speculum

- Device for opening body passage for examination. Types: vaginal, nasal, ear, anal.

Speech

- Phonation and articulation. Teeth, tongue, lips, palate involved. Dental prostheses may affect speech initially.

Speech Delay

- A delay in language development compared to peers: by 12 months—no single words; 18 months—<10-20 words/no word combinations by 24 months; or regression.

Speech Discrimination

- Speech discrimination testing, also known as word recognition scoring, measures the ability to understand speech at a comfortable listening level, and it is an important component of the audiometric evaluation that provides information about cochlear and neural function beyond what pure tone audiometry alone can reveal.

Speech Therapy

- See speech-language pathology.

Speech-Language Pathology Assessment

- Evaluate communication, swallowing, cognition. Standardized tests, clinical observation.

Sperm

- The male gamete (reproductive cell), produced in the seminiferous tubules of the testis through spermatogenesis. Sperm consists of: head (contains the haploid nucleus and acrosome containing enzymes for egg penetration), midpiece (contains mitochondria for energy), and tail (flagellum for motility).

Spermatocele

- Benign, painless, fluid-filled cystic retention cavity arising from the epididymis (usually the head/caput), containing avascular fluid and non-viable spermatozoa. Etiology: obstruction of an efferent ductule of the epididymis.

Spermatogonium

- A germ cell in the testis that divides and develops into primary spermatocytes.

Spermatorrhea

- Involuntary discharge of semen, especially without orgasm.

Sphenoid Bone

- Butterfly-shaped bone at the skull base, with body (sphenoid sinuses), greater/lesser wings, and pterygoid processes. The sella turcica houses the pituitary gland.

Sphenoidal Sinus

- The sphenoid sinuses are paired, air-filled cavities

located within the body of the sphenoid bone, the most posterior of the paranasal sinuses, situated deep within the skull at the center of the cranial base, and they are in close proximity to critical neurovascular structures including the optic nerves, the internal carotid arteries, th

Spherocytes

- Small, dark, sphere-shaped red blood cells lacking central pallor. Formed when RBC membrane is lost without loss of volume, reducing the surface area-to-volume ratio.

Sphygmomanometer

- Blood pressure measuring device. Components: cuff, gauge (mercury, aneroid, digital), inflation bulb. Mercury: gold standard. Digital: convenient but less accurate. Korotkoff sounds: phase I (systolic), phase V (diastolic).

Spider angioma

- A benign vascular lesion consisting of a central arteriole with radiating small vessels, resembling a spider, most commonly found on the face, neck, and upper trunk. They are associated with hyperestrogenic states including pregnancy, oral contraceptive use, and chronic liver disease, particularly cirrhosis.

Spider bite

- A cutaneous envenomation resulting from the bite of a spider, ranging from minor local reactions to severe systemic envenomation. The brown recluse spider may cause loxoscelism with necrotic skin lesions, while the black widow spider may cause latrodectism with muscle cramps, hypertension, and autonomic dysfunction.

Spina Bifida

- A neural tube defect: incomplete closure of the neural tube (days 22-28), causing spinal cord/meningeal malformation.

Spina Bifida Meningocele

- A form of spina bifida where the meninges (but not neural tissue) herniate through a vertebral defect, forming a CSF-filled sac. The spinal cord remains in its normal position and is not involved. Presents as a visible, fluid-filled, translucent sac on the back, typically without neurologic deficits.

Spina Bifida Myelomeningocele

- The most severe form of spina bifida, where both

the meninges and neural tissue (spinal cord and nerve roots) herniate through a vertebral defect. Presents as a visible, flat or sac-like lesion on the back with neurologic deficits below the level of the lesion (paralysis, sensory loss, bowel/bladder dysfunction).

Spina Bifida Occulta

- The mildest form of spina bifida, characterized by incomplete closure of the vertebral arch without herniation of neural elements. Often asymptomatic; may be detected by skin findings (dimple, hair tuft, lipoma) over the lower spine. Incidence approximately 1-2% of the population.

Spinal Analgesia

- Intrathecal administration of a local anesthetic and/or opioid primarily to provide analgesia, often with limited motor block. Intrathecal opioids can extend postoperative pain relief but may cause pruritus, urinary retention, respiratory depression, and other opioid effects.

Spinal Anesthesia

- Injection of local anesthetic into the lumbar subarachnoid space to produce temporary sensory, motor, and sympathetic blockade below the level of injection. Complications include hypotension, bradycardia, urinary retention, post-dural puncture headache, high or total spinal block, infection, and bleeding.

Spinal Anesthesia for Cesarean Delivery

- A neuraxial anesthetic technique used for elective and many emergency cesarean deliveries. A small dose of local anesthetic (hyperbaric bupivacaine 12-15 mg) with opioid (fentanyl 15-25 mcg, morphine 0.15-0.25 mg) is injected into the subarachnoid space. Produces rapid surgical anesthesia (T4 sensory level for cesarean).

Spinal Angiography

- Contrast imaging of spinal vessels. Evaluates vascular malformations.

Spinal Cord

- The cylindrical bundle of nerve fibers extending from the medulla oblongata (at the foramen magnum) to the conus medullaris (L1-L2 in adults), surrounded by the vertebral column and meninges (dura mater, arachnoid mater, pia mater). The

spinal cord is divided into 31 segments: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal.

Spinal Cord Compression

- Oncologic emergency. Back pain, weakness, sensory loss, bowel/bladder dysfunction. MRI diagnostic. Treatment: steroids, radiation, surgery.

Spinal Cord Injury

- Spinal cord injury: ASIA impairment scale (A-E). Complete (ASIA A) vs incomplete.

Spinal dysraphism

- A group of congenital malformations resulting from incomplete closure of the neural tube during embryonic development, affecting the spinal cord and overlying vertebral structures. Includes open defects (myelomeningocele) and closed defects (spina bifida occulta, lipomyelomeningocele, dermal sinus).

Spinal Epidural Abscess

- A collection of pus in the spinal epidural space, usually caused by haematogenous or contiguous infection. It may present with severe local back pain, fever, spinal tenderness, radicular pain, neurological deficit, or sphincter disturbance, although the classic triad is often incomplete.

Spinal Fusion

- Surgical union of two or more vertebrae using bone graft or bone graft substitutes with instrumentation (rods, screws, cages). Approaches: posterior (posterolateral fusion, transforaminal lumbar interbody fusion TLIF), anterior (anterior lumbar interbody fusion ALIF), and lateral (extreme lateral interbody fusion XLIF).

Spinal Nerve

- A mixed nerve formed by the union of dorsal and ventral roots as it exits the vertebral canal. There are 31 pairs, carrying sensory, motor, and autonomic fibers to and from the spinal cord.

Spinal Precautions

- Immobilization of spine in trauma patients to prevent neurological injury. Indications: mechanism (fall >3 feet, MVC, gunshot), altered mental status, focal neurological deficit, midline spinal tenderness. Technique: cervical collar, backboard, log roll. Clearance: NEXUS criteria or Canadian C-spine rule.

Spinal Shock

- A state of transient physiological areflexia and flaccid paralysis occurring immediately after acute spinal cord injury, below the level of the lesion.

Spinal Stenosis

- Narrowing of the spinal canal causing compression of the spinal cord or nerve roots. Lumbar stenosis (most common): neurogenic claudication (leg pain/weakness with walking, relieved by sitting/leaning forward), often bilateral.

Spindle-cell sarcoma

- A group of sarcomas characterized by elongated, spindle-shaped malignant cells. It includes entities like leiomyosarcoma, fibrosarcoma, and malignant peripheral nerve sheath tumor, with variable behavior.

Spine

- See Spinal Cord and Vertebral Column.

Spine Disorders

- The spine, or vertebral column, is a complex bony and soft-tissue structure that provides axial support, protects the spinal cord, and enables flexible movement of the trunk. It consists of 33 vertebrae divided into cervical, thoracic, lumbar, sacral, and coccygeal regions, separated by intervertebral discs that act as shock absorbers.

Spinocerebellar Ataxias

- A group of autosomal dominant neurodegenerative disorders characterized by progressive cerebellar ataxia, often with additional neurological features. Over forty different SCA types have been identified, with SCA1, SCA2, SCA3 (Machado-Joseph disease, the most common worldwide), SCA6, and SCA7 being the most prevalent.

Spirochaete

- A slender, spiral-shaped bacterium of the group Spirochaetes.

Spirochete

- A thin, coiled gram-negative bacterium with internal periplasmic flagella conferring corkscrew motility, requiring darkfield or special stains. Pathogenic genera include *Treponema* (syphilis), *Borrelia* (Lyme disease, relapsing fever), and *Leptospira* (leptospirosis), diagnosed mainly by serology and dark-field microscopy.

Spirometry

- Most common PFT measuring air volume and flow during forced breathing. Key measurements: FVC, FEV1, FEV1/FVC ratio. Used to diagnose and classify obstructive and restrictive diseases.

Spiroglactone

- A nonselective aldosterone (mineralocorticoid) receptor antagonist and potassium-sparing diuretic. Used for heart failure with reduced ejection fraction (NYHA II-IV—reduces mortality, often with ACE/ARB + beta-blocker), resistant hypertension, primary hyperaldosteronism, ascites in cirrhosis, and acne/hirsutism in women (off-label).

Spleen

- Largest lymphoid organ in the left upper quadrant under ribs 9-11, approximately 12 cm long and 150-200 g. Red pulp filters blood (removing old/damaged RBCs and platelets); white pulp mounts immune responses (lymphoid follicles and periarteriolar lymphoid sheaths). Enclosed by a fibroelastic capsule with trabeculae.

Spleen Hilum

- Concave medial surface where the tortuous splenic artery (celiac trunk branch) enters and splenic vein exits, with lymphatics and nerves. Artery divides into 5 segmental branches at the hilum (basis for partial splenectomy). Splenic vein joins the superior mesenteric vein forming the portal vein.

Splenectomy

- Indications: trauma (grade IV/V, hemodynamic instability), ITP (refractory), hereditary spherocytosis/elliptocytosis, thalassemia major, sickle cell (sequestration crisis), TTP (refractory), lymphoma staging (historical), hypersplenism, splenic cysts/tumors, Wandering spleen. Laparoscopic preferred.

Splenic cyst

- A fluid-filled lesion within the spleen, classified as true cysts with an epithelial lining, including congenital, epidermoid, and dermoid cysts, or pseudocysts without a lining, typically resulting from trauma, infection, or pancreatitis. Most are asymptomatic and discovered incidentally on imaging.

Splenic Infarction

- Splenic infarction occurs when the splenic artery or

its branches are occluded, typically by emboli from the heart. The spleen has end-arterial supply with limited collateral circulation, making it vulnerable to infarction even with small vessel occlusion.

Splenic Rupture

- Traumatic or spontaneous splenic injury. CT diagnosis. May require splenectomy.

Splenomegaly

- Enlargement of the spleen beyond its normal size (12 cm, weight 150–200 g). Splenomegaly can be detected by percussion (Castell sign—dullness in the lowest intercostal space on inspiration) and palpation (normal spleen is not palpable; a palpable spleen indicates at least 2–3 times enlargement).

Splice-Site Variant

- A variant affecting the sequences required for accurate removal of introns and joining of exons during RNA processing. It may cause abnormal transcripts, exon skipping, intron retention, or reduced functional protein.

Splint

- Immobilizes injured body part. Types: aluminum, plastic, canvas, pneumatic.

Splint (Dental)

- Fixation of mobile teeth or fractured tooth. Wire and composite, orthodontic bracket splint.

Splinter Hemorrhages

- Linear hemorrhages under the nail bed, oriented longitudinally. In infective endocarditis, they are caused by septic microemboli to the nail bed capillaries. They are a minor Duke criterion. Splinter hemorrhages can also result from trauma, psoriasis, and systemic vasculitis.

Splitting

- A primitive defense mechanism in which objects or self-representations are viewed as entirely good or entirely bad, with no integration of positive and negative qualities. It is a core feature of borderline personality disorder and contributes to idealization-devaluation patterns in relationships. It reflects impaired object

Spondyloarthritis

- Group of inflammatory arthritis associated with HLA-B27. Includes ankylosing spondylitis, psoriatic arthritis, reactive arthritis, enteropathic arthritis.

Features: axial involvement, enthesitis, dactylitis.

Spondylolisthesis

- Forward slippage of one vertebral body relative to the vertebra below it. Isthmic (Type II, most common in children/adolescents): defect in pars interarticularis (spondylolysis), often L5-S1. Degenerative (Type III, most common in adults): facet joint and disc degeneration, often L4-L5.

Sponge (Vascular)

- See Sponge.

Spontaneous Bacterial Peritonitis

- Infection of ascitic fluid without an identifiable intra-abdominal source representing a serious complication of cirrhosis with ascites. The pathogenesis involves impaired immune function, bacterial translocation from the intestinal lumen, and seeding of ascitic fluid which provides a favorable medium for bacterial growth.

Spontaneous Breathing Trial

- A period of unsupported breathing to assess a patient's readiness for extubation. The patient breathes through a T-piece or with minimal pressure support for 30–120 minutes. Successful SBT predicts successful extubation.

Spontaneous Pneumothorax

- Pneumothorax without trauma. Primary in healthy individuals; secondary with underlying lung disease. Treatment depends on size.

Spore

- A highly resistant, dormant form of certain bacteria (*Bacillus*, *Clostridium*) and fungi, enabling survival under adverse conditions. Bacterial endospores resist heat, desiccation, chemicals, and radiation; germination occurs in favorable conditions. Clinically important: *Clostridium difficile*, *C. botulinum*, *C. tetani*, *C.*

Sporotrichosis

- A subcutaneous fungal infection caused by *Sporothrix schenckii*, a dimorphic fungus found in soil, moss, and plant matter (rose thorns, sphagnum moss).

Sports Rehabilitation

- See sports medicine.

Sprain

- An injury to a ligament (connective tissue connecting bone to bone) caused by excessive stretching or tearing. Grading: Grade I (mild stretching, microscopic tears), Grade II (partial tear with laxity), Grade III (complete tear with instability).

Sprengel Deformity

- A congenital elevation of the scapula due to failure of the scapula to descend from its embryonic cervical position to its normal thoracic position during weeks 9-12 of gestation. The scapula is higher, smaller, and rotated inferiorly compared to the normal side. Range of motion is limited, especially in shoulder abduction.

Spur-cell hemolytic anemia

- A severe hemolytic anemia seen in advanced liver disease where acanthocytosis develops due to altered membrane cholesterol, causing premature red cell destruction in the spleen.

Squamous Cell Carcinoma

- Cutaneous squamous cell carcinoma is a malignant keratinocyte tumor commonly arising on chronically sun-exposed skin or in areas of chronic inflammation or immunosuppression. It may present as a scaly, indurated, hyperkeratotic papule, plaque, or ulcer.

Squint

- A misalignment of the visual axes of the two eyes, where one eye is deviated inward (esotropia), outward (exotropia), upward (hypertropia), or downward (hypotropia) relative to the other, disrupting binocular vision and potentially leading to amblyopia in children. See Strabismus.

SSRIs

- Selective serotonin reuptake inhibitors are a class of antidepressant drugs that selectively inhibit the reuptake of serotonin from the synaptic cleft into the presynaptic neuron by blocking the serotonin transporter, increasing serotonergic neurotransmission.

ST Segment

- The isoelectric period between ventricular depolarization (QRS) and repolarization (T wave). It corresponds to the plateau phase of ventricular action potentials. ST segment elevation (>1 mm in limb leads or >2 mm in precordial leads) suggests

acute myocardial infarction, pericarditis, or early repolarization.

ST-Segment Elevation

- Elevation of the ECG ST segment above the isoelectric baseline, reflecting abnormal ventricular repolarization. It may indicate acute coronary occlusion but can also occur in pericarditis, early repolarization, ventricular aneurysm, or other conditions; interpretation requires clinical context and serial testing.

St. John's Wort

- Herbal supplement. CYP3A4 inducer, reduces efficacy of many drugs (oral contraceptives, warfarin, antiretrovirals).

Stab Wound

- A penetrating sharp-force injury in which wound track depth exceeds skin defect length, produced by a pointed or sharp-edged implement. The skin defect approximates blade width only when perpendicular to Langer's lines, and a single-edged blade produces one sharp and one blunt end.

Stable Disease

- Cancer neither growing nor shrinking significantly.

Stage I

- Early cancer, localized. Good prognosis.

Stage II

- Localized but larger or with local spread.

Stage III

- Regional spread to lymph nodes or nearby structures.

Stage IV

- Distant metastasis. Poor prognosis.

Stages of Labor

- Four stages describing the progression from uterine contractions to delivery of the placenta. First stage: onset of regular contractions to full cervical dilatation (10 cm). Second stage: full dilatation to delivery of the baby. Third stage: delivery of the baby to delivery of the placenta.

Staghorn Calculus

- A large renal stone that occupies the renal pelvis and extends into at least two calyces, creating a cast of the collecting system. Composition: struvite (magnesium ammonium phosphate), associated

with urease-producing bacteria (*Proteus mirabilis*, *Klebsiella*, *Pseudomonas*) causing urinary tract infection.

Staining

- A laboratory technique applying dyes to microbial smears or tissue sections to visualise microorganisms and reveal structural characteristics. Major methods include Gram, acid-fast, Giemsa, and fluorescent stains, providing rapid presumptive identification from clinical specimens and guiding initial empiric therapy.

Stainless Steel

- Alloy for preformed crowns, orthodontic bands, instruments. Corrosion-resistant.

Standard of Care

- The conduct, skill, diligence, and judgment a reasonably prudent practitioner of the same specialty would exercise under similar circumstances; the benchmark for medical negligence. It is established through guidelines, peer practice, and expert testimony rather than fixed statute.

Standard Precautions

- Precautions for all patient care. Examples: hand hygiene, PPE, safe injection, respiratory hygiene.

Standardized Uptake Value (SUV)

- Quantitative PET measure: tissue activity divided by injected dose per body weight. Normalizes for body size, injection dose. SUVmax: maximum pixel value in lesion. SUVmean: average lesion uptake. Background: blood pool, liver.

Stapedectomy

- A microsurgical procedure performed to treat otosclerosis, a condition characterized by abnormal bone remodeling in the otic capsule that fixes the stapes footplate in the oval window, preventing its normal vibratory movement and causing progressive conductive hearing loss.

Stapes

- The stapes, commonly known as the stirrup, is the smallest bone in the human body and the most medially positioned ossicle in the middle ear, articulating with the incus at the incudostapedial joint and transmitting sound vibrations to the inner ear through its footplate, which is attached to the oval window by the annular ligament.

Staph Infection

- Infections caused by *Staphylococcus aureus* (and coagulase-negative staphylococci), a gram-positive coccus in clusters.

Staphylococcal Infection

- Infections caused by *Staphylococcus aureus* and other staphylococci. Manifestations range from superficial skin infections (folliculitis, furuncles, carbuncles, impetigo) to deep-seated infections (osteomyelitis, septic arthritis, endocarditis, pneumonia, bacteremia, sepsis). Methicillin-resistant *S.*

Staphylococcal Pneumonia

- Severe pneumonia from *S. aureus*, often post-influenza. May cause necrotizing pneumonia, lung abscess, and empyema.

Staphylococcal scalded skin

- See staphylococcal scalded skin syndrome. A toxin-mediated disease caused by exfoliative toxins of *Staphylococcus aureus*, resulting in widespread erythema and flaccid bullae that rupture easily, leaving painful, moist, erythematous denuded areas resembling a scald burn. It predominantly affects infants and young children.

Staphylococcus aureus

- A gram-positive, catalase-positive, coagulase-positive coccus that causes a wide range of infections including skin infections, pneumonia, endocarditis, and osteomyelitis. *S. aureus* produces numerous virulence factors including toxins, enzymes, and surface proteins. MRSA strains carry the *mecA* gene encoding PBP2a.

Staple (Vascular)

- See Stapler.

Stargardt Disease

- The most common inherited macular dystrophy, usually autosomal recessive and caused by ABCA4 gene mutations. Presents in childhood or adolescence with progressive central vision loss, retinal flecks, and bull's-eye maculopathy; no curative treatment exists.

Starling Forces

- Starling forces describe the balance of hydrostatic and oncotic pressures that govern fluid movement across capillary walls, determining the distribution of fluid between the intravascular and extravascular

compartments.

Statin

- An inhibitor of HMG-CoA reductase that lowers hepatic cholesterol synthesis and increases hepatic LDL-receptor expression. Statins reduce atherosclerotic cardiovascular events; adverse effects include muscle symptoms, rare severe myopathy, and liver-enzyme abnormalities.

Statins in Diabetes

- HMG-CoA reductase inhibitors are first-line therapy for cardiovascular risk reduction in diabetic patients. Most adults with diabetes aged forty to seventy-five benefit from moderate-intensity statin therapy regardless of baseline LDL cholesterol.

Station

- The relationship of the fetal presenting part to the level of the ischial spines, measured in centimeters.

Status Asthmaticus

- A severe, life-threatening asthma exacerbation unresponsive to standard bronchodilator therapy. Characterized by severe airflow obstruction, hypoxemia, and hypercapnia. Treatment includes continuous nebulized albuterol, IV magnesium sulfate, systemic corticosteroids, and possibly intubation.

Status Epilepticus

- A seizure lasting at least 5 minutes or recurrent seizures without recovery to baseline consciousness between them, requiring urgent treatment. Ongoing convulsive or nonconvulsive status can cause neuronal injury, systemic complications, and death.

Steady State

- The condition where the rate of drug administration equals the rate of elimination, resulting in constant average plasma concentrations with repeated dosing. During regular dosing at fixed intervals, steady state is achieved after approximately 4 to 5 half-lives, regardless of the dosing frequency or dose size.

Steal Syndrome

- Ischemia of distal extremity due to blood flow diversion to bypass graft or AV fistula. Most common: upper extremity after AV fistula creation. Symptoms: hand pain, pallor, numbness, weakness. Treatment: banding, ligation, revision.

Steatorrhea

- The passage of excess fat in the stool, causing greasy, pale, foul-smelling, floating stools—a hallmark of fat malabsorption.

Stem Cell Transplantation

- Hematopoietic stem cell transplantation (HSCT) replaces diseased or ablated bone marrow with healthy donor or autologous stem cells.

STEMI

- ST-elevation myocardial infarction: transmural myocardial necrosis from acute total occlusion of a coronary artery, an emergency requiring immediate reperfusion. ECG: ST-segment elevation in contiguous leads (or new LBBB); troponin elevated.

Stenosis

- Narrowing of vessel lumen. Causes: atherosclerosis, thrombosis, dissection, external compression. Symptoms depend on location and severity. Treatment: medical, angioplasty, stenting, bypass.

Stenosis of the left anterior descending artery

- See Left Anterior Descending Artery.

Stenotrophomonas maltophilia resistance

- *Stenotrophomonas maltophilia* is a gram-negative bacterium with intrinsic resistance to many antibiotics including carbapenems. The organism produces a chromosomal beta-lactamase (L1) that confers resistance to beta-lactams. Treatment options include trimethoprim-sulfamethoxazole, fluoroquinolones, and minocycline. S.

Stent

- Tubular scaffold maintaining vessel/duct patency. Materials: metal (stainless steel, cobalt-chrome, nitinol), polymer. Types: bare-metal, drug-eluting. Applications: coronary, peripheral, biliary, ureteral, tracheal. Complications: restenosis, thrombosis, migration, fracture.

Stent (Vascular)

- Tubular scaffold maintaining vessel/duct patency. See Stent.

Stent Balloon (Vascular)

- See Stent Balloon.

Stent Graft

- Stent-graft for aortic aneurysm repair

(EVAR/TEVAR). See endovascular graft.

Stent Placement

- Image-guided deployment of stent. Vascular, biliary, ureteral, esophageal. Maintains patency.

Stent-Graft (Vascular)

- See Stent-Graft.

Stenting

- Placement of mesh tube to keep vessel open. Drug-eluting stents reduce restenosis.

Stepping Reflex

- A primitive reflex present from birth to 2 months, elicited by holding the infant upright with feet touching a flat surface. The infant makes stepping-like movements as if walking. Disappears by 2 months as voluntary motor control develops. Reappears around 9-10 months as voluntary walking emerges.

Stereotactic Biopsy

- Biopsy using stereotactic guidance. Breast, brain, lung.

Stereotactic Body Radiation Therapy (SBRT)

- A highly precise form of external radiation delivering high-dose ablative treatment in few fractions, used for early-stage lung cancer and selected liver, pancreatic, spinal, and kidney tumors.

Stereotactic Radiosurgery

- Single high-dose radiation to small target. Used for brain metastases, AVMs, acoustic neuroma.

Sterile Field

- A designated area maintained free of viable microorganisms during an invasive procedure. Only sterile equipment and materials may contact the field or the sterile portions of instruments.

Sterile Site

- A body site or specimen type expected to contain no viable microorganisms in health, such as blood, cerebrospinal fluid, synovial fluid, or tissue from a closed space. Recovery of a pathogen from a sterile site is generally clinically significant, subject to contamination assessment.

Sterilisation

- A permanent method of contraception. Female sterilization (tubal occlusion): the fallopian tubes are blocked or cut to prevent the egg from reach-

ing the uterus, performed by laparoscopy (clip or diathermy) or at the time of cesarean delivery.

Sterilization

- A process eliminating all microbial life including spores, achieving a sterility assurance level of at least 10^{-6} .

Sternoclavicular Joint

- A saddle-type synovial joint between the clavicle and manubrium of the sternum. It is the only bony connection between the upper limb and axial skeleton and permits movement of the shoulder girdle.

Sternocleidomastoid Muscle

- Strap-like anterior neck muscle from manubrium and medial clavicle to the mastoid process. Innervated by CN XI and C2-C3. Divides neck into anterior/posterior triangles. Unilaterally: laterally flexes to opposite side, rotates face to same side. Bilaterally: flexes neck, assists forced inspiration.

Sternum

- Flat bone in the anterior midline thorax, with manubrium, body, and xiphoid process. The sternal angle (of Louis) at the manubriosternal junction is a key landmark at the 2nd costal cartilage, T4-T5, and tracheal bifurcation. Common site for bone marrow biopsy. Median sternotomy is the standard cardiac surgical approach.

Stethoscope

- Acoustic device for listening to internal body sounds. Components: chest piece (bell and diaphragm), tubing, earpieces. Bell: low-frequency sounds. Diaphragm: high-frequency sounds. Essential for cardiac, pulmonary, abdominal auscultation.

Stevens-Johnson Syndrome

- A severe, life-threatening mucocutaneous reaction characterized by widespread skin erythema, necrosis, and bullae formation with mucosal involvement (oral, ocular, genital). It is most commonly caused by drug reactions (sulfonamides, anticonvulsants, allopurinol, NSAIDs) and infections (*Mycoplasma pneumoniae*).

Stickler Syndrome

- An autosomal dominant connective tissue disorder caused by mutations in collagen genes (COL2A1 most common, also COL11A1, COL11A2, COL9A1).

Stillbirth

- Intrauterine fetal death (IUFD) at or after 20-24 weeks (definition varies by country); fetal death before viability (<20 weeks) is miscarriage.

Stimulant Use Disorder

- A problematic pattern of stimulant use (cocaine, amphetamines, methamphetamine, MDMA) leading to significant impairment. Stimulants increase dopamine and norepinephrine activity. Withdrawal is characterized by fatigue, depression, increased appetite, hypersomnia, psychomotor retardation, and vivid unpleasant dreams.

Stippling (Tattooing)

- Patterned dermal abrasions and punctate hemorrhages surrounding a firearm entrance wound, caused by unburned and partially burned gunpowder grains striking the skin at close range (15–75 cm).

Stochastic Effects

- Random radiation effects with no threshold. Cancer, genetic effects. Probability increases with dose.

Stoma Complications

- Complications related to intestinal or urinary stomas.

Stomach

- J-shaped muscular organ in the left upper quadrant, connecting esophagus to duodenum. Regions: cardia, fundus, body, antrum, pylorus. Functions: food storage, mechanical churning, chemical digestion (pepsin, HCl), intrinsic factor secretion (B12 absorption). Mucosa has rugae (folds) allowing expansion.

Stomach Cancer

- Gastric adenocarcinoma, a malignancy of the stomach; a leading cause of cancer death worldwide (more common in East Asia). Risk factors: *Helicobacter pylori* infection (major), diet (smoked/preserved foods, nitrates), smoking, pernicious anemia, familial syndromes (HDGC, Lynch). Types: intestinal (distal, H.

Stomach Ulcer

- Peptic ulcer in the gastric mucosa associated with H. See Gastric Ulcer.

Stomatitis

- Inflammation of the oral mucosa, presenting as erythema, ulceration, pain, and difficulty eating.

STOPP/START

- Screening Tool of Older Person's Prescriptions (STOPP) and Potential Older Person's Prescriptions (START). Identifies inappropriate prescribing.

Storage Diseases

- A group of inherited metabolic disorders characterized by accumulation of substrates within cells due to specific enzyme deficiencies. Lysosomal storage diseases result from defects in lysosomal hydrolytic enzymes, leading to accumulation of undigested macromolecules within lysosomes.

Strabismus

- A misalignment of the visual axes of the two eyes, where one eye is deviated inward (esotropia), outward (exotropia), upward (hypertropia), or downward (hypotropia) relative to the other, disrupting binocular vision and potentially leading to amblyopia in children.

Strain

- An injury to a muscle or tendon (connective tissue connecting muscle to bone) caused by excessive stretching, contraction, or overload. Grading: Grade I (mild stretching with few muscle fibers torn), Grade II (moderate partial tear), Grade III (complete muscle/tendon rupture).

Strain Imaging

- Echocardiographic or MRI technique measuring myocardial deformation (shortening and thickening). Global longitudinal strain (GLS) by speckle tracking is more sensitive than ejection fraction for detecting subclinical myocardial dysfunction. GLS < -18% is normal.

Strangulation

- Compression of the neck causing occlusion of the airway (asphyxia) and/or cervical vessels (carotid/jugular), leading to unconsciousness and death. Types: ligature strangulation (cord, rope, cloth—external mark above the larynx, often horizontal), manual (manual throttling—fingertip bruises, thumb imprint, often vertical), and

Strength Testing

- Manual muscle testing, dynamometry, isokinetic

testing. Quantifies weakness, tracks progress.

Strep Throat

- Acute streptococcal pharyngitis caused by group A Streptococcus (*S. pyogenes*), most common in children 5-15 years. Features: sudden sore throat, fever, tonsillar exudate, palatal petechiae, anterior cervical lymphadenopathy, and headache/abdominal pain; cough, coryza, and oral ulcers suggest viral.

Streptococcal cellulitis with lymphangitis

- See Cellulitis.

Streptococcal Infection

- Infections caused by Streptococcus pyogenes (Group A Strep, GAS). Common presentations: pharyngitis (strep throat), scarlet fever (pharyngitis with erythematous rash), impetigo, cellulitis, and necrotizing fasciitis.

Streptococcal pharyngitis

- See Group A Strep.

Streptococcus agalactiae

- A gram-positive, beta-hemolytic streptococcus (Group B Streptococcus) that is a leading cause of neonatal sepsis, meningitis, and pneumonia. The organism colonises the gastrointestinal and genital tracts. Transmission occurs during birth. Risk factors include prematurity, prolonged rupture of membranes, and maternal colonisation.

Streptococcus pneumoniae

- A gram-positive, encapsulated diplococcus that causes pneumonia, otitis media, sinusitis, meningitis, and sepsis.

Streptococcus pyogenes

- A gram-positive, beta-hemolytic streptococcus (Group A Streptococcus) that causes pharyngitis, skin infections, and post-infectious complications including rheumatic fever and glomerulonephritis. The organism produces streptolysins, erythrogenic toxin, and streptokinase.

Streptokinase

- A thrombolytic agent that forms a plasminogen activator complex, converting plasminogen to plasmin and lysing fibrin clots. Used for STEMI, massive pulmonary embolism, and deep vein thrombosis (largely replaced by tissue plasminogen activator, which is more fibrin-specific and less antigenic).

Stress

- A physiological and psychological response to perceived threats or demands that activates the sympathetic nervous system and the hypothalamic-pituitary-adrenal axis.

Stress Cardiomyopathy

- See Takotsubo Cardiomyopathy.

Stress Echocardiography

- Echocardiography during exercise or pharmacological stress. Detects coronary artery disease by identifying wall motion abnormalities.

Stress Fracture

- A bone injury caused by repetitive submaximal loading that exceeds the rate of remodeling and repair. It often presents with gradually increasing, localized activity-related pain and may progress to a complete fracture if loading continues.

Stress Shielding

- Reduction in physiologic loading of adjacent bone caused by an implant that carries a disproportionate share of mechanical force. Chronic stress shielding may lead to bone resorption and compromise fixation.

Stress Ulcer Prophylaxis

- Prevention of stress-related mucosal disease in critically ill patients. Risk factors: mechanical ventilation >48 hours, coagulopathy, trauma, burns, sepsis. Agents: H2 blockers, proton pump inhibitors. Duration: while risk factors present.

Stress Urinary Incontinence

- Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

Stretch Reflex

- A monosynaptic spinal reflex in which stretching a muscle (e.g., a tap on the patellar tendon) excites muscle spindles, generating an afferent signal (Ia fibers) that directly excites the alpha motor neurons of the same muscle, causing contraction and resistance to stretch.

Striae Gravidarum

- Stretch marks developing on the abdomen, breasts, thighs, and buttocks during pregnancy due to mechanical stretching of the dermis combined with

hormonal effects on collagen. Initially appear as red or purple striae (striae rubrae) and eventually fade to silver-white (striae albae). Occur in 50-90% of pregnancies.

Stridor

- A harsh, often high-pitched respiratory sound caused by turbulent airflow through a narrowed upper airway. Inspiratory stridor usually indicates supraglottic or glottic obstruction; biphasic or expiratory patterns help localize disease, but any severe or progressive stridor is an airway emergency.

Stripping of Membranes (Membrane Sweep)

- A procedure performed during a cervical exam where the examiner inserts a finger through the internal os and sweeps the membranes circumferentially, separating them from the lower uterine segment. Releases endogenous prostaglandins, promoting cervical ripening and potentially initiating labor.

Stroke

- A medical emergency caused by the interruption of blood flow to the brain, resulting in neuronal death and neurological deficits.

Stroke Volume

- The volume of blood ejected by the left ventricle with each heartbeat, normally 60-100 mL. It is determined by three factors: preload, afterload, and myocardial contractility. Stroke volume equals end-diastolic volume minus end-systolic volume. Reduced stroke volume occurs in heart failure, hypovolemia, and cardiogenic shock.

Strongyloides stercoralis

- See Strongyloidiasis.

Strongyloidiasis

- Infection with *Strongyloides stercoralis*, a unique helminth capable of autoinfection. Most infections asymptomatic or mild (abdominal pain, diarrhea, urticaria). Larvae migration causes larva currens (rapidly migrating serpiginous rash, perianal).

Structured Reporting

- Reporting using standardized templates. Improves consistency, completeness.

Struvite Stones

- Infection stones composed of magnesium ammo-

nium phosphate. Form in alkaline urine with urease-producing organisms including *Proteus* and *Klebsiella*. Staghorn calculi fill renal pelvis almost exclusively from *Proteus*. Treatment requires complete stone removal with antibiotics and urinary acidification. High recurrence if infection persists.

Stupor

- A state of reduced consciousness in which the patient is unresponsive to most environmental stimuli but can be temporarily aroused by vigorous, repeated stimulation (painful stimuli, loud shouting).

Sturge-Weber Syndrome

- A neurocutaneous syndrome (phakomatosis) characterized by a facial capillary malformation (port-wine stain) in the V1 distribution of the trigeminal nerve (forehead and upper eyelid) and ipsilateral leptomeningeal angiomas (venous malformations of the pia mater). Caused by a somatic activating mutation in *GNAQ*.

Stye

- A common infection of the eyelid. External hordeolum: acute bacterial infection of the Zeis or Moll glands at the eyelash follicle, typically caused by *Staphylococcus aureus*. Presents as a tender, red, well-localized swelling at the lid margin. Internal hordeolum: infection of the meibomian gland, larger and more painful than external.

Subacute

- Between acute and chronic. Examples: subacute thyroiditis.

Subacute Rehabilitation

- Less intensive than acute rehab. For patients needing continued therapy but not as intensive. Skilled nursing facility setting.

Subacute Thyroiditis

- Inflammatory condition of the thyroid most likely viral in etiology presenting with painful tender thyroid enlargement, fever, and malaise. Associated with elevated erythrocyte sedimentation rate and C-reactive protein.

Subarachnoid

- Situated beneath the arachnoid membrane surrounding the brain and spinal cord.

Subarachnoid Hemorrhage

- Bleeding into the subarachnoid space, the area

between the arachnoid membrane and the pia mater surrounding the brain, most commonly caused by rupture of a cerebral (berry) aneurysm.

Subclavian Artery

- The artery supplying the upper limb, arising on the right from the brachiocephalic trunk and on the left directly from the aortic arch, passing over the first rib and becoming the axillary artery. Branches: vertebral, internal thoracic, thyrocervical trunk, costocervical trunk.

Subclavian Vein

- The continuation of the axillary vein beyond the lateral border of the first rib. It joins the internal jugular vein behind the sternoclavicular joint to form the brachiocephalic vein.

Subclavian vein thrombosis

- Formation of a blood clot in the subclavian vein, often associated with central venous catheterization, thoracic outlet syndrome, or Paget-Schroetter syndrome (effort-related thrombosis in young, active individuals). Presents with arm swelling, pain, and cyanosis.

Subcorneal pustular dermatosis

- See Sneddon-Wilkinson disease. A chronic, relapsing, sterile pustular eruption characterised by flaccid, pus-filled blisters that accumulate beneath the stratum corneum, typically on the trunk and intertriginous areas. It is often associated with IgA gammopathy or monoclonal gammopathy and may be mistaken for pustular psoriasis.

Subculture

- The transfer of microorganisms from one culture to fresh culture medium, used to maintain pure cultures, isolate colonies, increase viability, prepare standardized inocula, and perform identification/susceptibility testing. On solid media, subculture aims to obtain isolated single colonies from a mixed or stock culture.

Subcutaneous Emphysema

- Air in the subcutaneous tissues, often from pneumothorax, pneumomediastinum, or chest trauma. It causes a characteristic crepitus on palpation. It is usually benign and resolves when the underlying air leak is treated. Severe cases may cause skin necrosis and compartment syndrome.

Subcutaneous Tissue

- Subcutaneous tissue (hypodermis, superficial fascia) lies beneath the dermis and connects skin to underlying structures. It contains adipose tissue (fat), blood vessels, nerves, and the superficial muscular aponeurotic system (SMAS). Subcutaneous fat provides insulation, cushioning, and energy storage.

Subdural Hematoma

- A collection of blood between the dura mater and the arachnoid mater, caused by tearing of bridging veins that traverse the subdural space.

Subgaleal Hemorrhage

- A life-threatening accumulation of blood in the subgaleal space (between the periosteum and the epicranial aponeurosis). Can contain up to 50% of the infant's blood volume. Risk factors include vacuum-assisted delivery.

Sublingual epidermoid cyst

- A benign cyst lined by stratified squamous epithelium beneath the tongue mucosa, presenting as a slow-growing midline swelling that may require surgical excision if symptomatic.

Sublingual Gland

- The sublingual glands are the smallest of the major salivary glands, located in the floor of the mouth beneath the tongue, and they produce primarily mucous saliva that is delivered into the oral cavity through multiple small ducts (ducts of Rivinus) that open along the sublingual fold, and through the major sublingual duct (Bartholin's duct).

Submandibular Gland

- One of the three paired major salivary glands, located in the submandibular triangle beneath the mandible. Mixed gland (predominantly serous) secreting saliva via Wharton's duct opening at the sublingual caruncle (floor of mouth). Saliva contains amylase, mucins, lysozyme, IgA.

Subscapular Artery

- An artery arising from the axillary artery that supplies the scapular region and adjacent muscles.

Substance Abuse

- A chronic, relapsing brain disorder characterized by compulsive substance use or behavior despite harmful consequences. See addiction.

Substance Abuse Psychiatry

- The subspecialty focused on the evaluation and treatment of substance use disorders and co-occurring psychiatric conditions. Dual diagnosis patients require integrated treatment addressing both disorders simultaneously.

Substance Dependence

- A chronic, relapsing brain disorder characterized by compulsive substance use or behavior despite harmful consequences. See addiction.

Substance Intoxication

- A reversible substance-specific syndrome caused by recent use that produces clinically significant behavioral or physiologic changes. Diagnosis requires assessing timing, dose or exposure, expected pharmacology, co-ingestants, and medical complications.

Substance Use

- The consumption of alcohol, drugs, or other chemical substances, which can lead to health, social, and psychological issues if used excessively.

Substance Use Disorder

- A problematic pattern of substance use leading to clinically significant impairment or distress, manifested by at least two of eleven criteria within a twelve-month period.

Substance Use in Pregnancy

- Use of alcohol, tobacco, marijuana, opioids, cocaine, and other drugs during pregnancy. Tobacco: increased risk of IUGR, preterm birth, placental abruption, SIDS, and low birth weight. Opioids: neonatal abstinence syndrome (NAS, withdrawal in 60-80% of exposed newborns), treated with morphine or methadone. Alcohol: FASD.

Substance Withdrawal

- A substance-specific syndrome occurring after reduction or cessation of heavy or prolonged use. Withdrawal can range from discomfort to life-threatening complications such as seizures, delirium tremens, severe autonomic instability, or respiratory depression depending on the substance.

Substance/Medication-Induced Sexual Dysfunction

- Sexual dysfunction directly caused by substance use or medication effects.

Substernal goiter

- Extension of thyroid tissue into the mediastinum, typically arising from the thyroid gland in the neck and growing downward through the thoracic inlet. It may cause compressive symptoms including dyspnea, dysphagia, and superior vena cava syndrome. Most are multinodular goiters. Diagnosis is by CT or MRI.

Succenturiated Lobe

- An accessory lobe of the placenta, connected to the main placenta by blood vessels that traverse the membranes. If undetected after delivery, the succenturiated lobe may be retained in the uterus, causing postpartum hemorrhage or infection. Diagnosis: careful inspection of the placenta and membranes after delivery.

Succinate Dehydrogenase

- Complex II of electron transport chain. Also TCA cycle enzyme. Produces FADH₂.

Succinylcholine

- A depolarizing neuromuscular-blocking drug that produces rapid, brief skeletal-muscle paralysis by activating nicotinic acetylcholine receptors at the neuromuscular junction. It can cause hyperkalemia, bradycardia, myalgia, masseter spasm, prolonged apnea in pseudocholinesterase deficiency, and malignant hyperthermia.

Sucking Reflex

- A primitive reflex present from birth, elicited by placing an object in the infant's mouth. The infant rhythmically sucks on the object. Coordinated sucking and swallowing typically develop by 34-36 weeks gestational age. Premature infants may lack coordinated suck-swallow, requiring gavage feeding.

Sucrose Lysis Test

- A historical screening test for paroxysmal nocturnal haemoglobinuria in which patient red blood cells are exposed to serum in a low-ionic-strength sucrose solution, allowing complement-mediated lysis to be detected.

Suction (Tip)

- See Suction Tip.

Suction (Vascular)

- See Suction Tip.

Suction Device

- Creates negative pressure for fluid removal. Components: canister, tubing, suction catheter. Used for airway clearance, surgical field, wound drainage.

Suction Tip

- Attachment for suction device. Types: Yankauer (oropharyngeal), Frink (surgical).

Sudden Cardiac Death

- Unexpected death within 1 hour of symptom onset. Most common cause: ventricular fibrillation. Risk factors: coronary artery disease, cardiomyopathy, channelopathies. Prevention: ICD implantation, antiarrhythmics.

Sudden Infant Death Syndrome

- SIDS: the sudden, unexplained death of an infant <1 year (peak 2-4 months) that remains unexplained after a thorough investigation (autopsy, scene examination, and review of clinical history). Risk factors: prone sleep, bed-sharing, soft bedding, overheating, maternal smoking, prematurity/low birth weight, and co-sleeping.

Sudden Infant Death Syndrome (SIDS)

- Sudden, unexpected death of an infant under 1 year of age that remains unexplained after a thorough medico-legal investigation, including a complete autopsy, crime scene investigation, and review of clinical history. Peak incidence: 2-4 months of age.

Sudden Sensorineural Hearing Loss

- Rapid sensorineural hearing loss, conventionally at least 30 dB over three contiguous frequencies within 72 hours. It requires urgent audiologic and otologic assessment because early treatment may improve recovery in selected patients.

Sudden Unexpected Infant Death (SUID)

- The sudden and unexpected death of an infant younger than 1 year, whether or not the cause is known before investigation. Sudden infant death syndrome is the unexplained subtype after complete scene, history, and autopsy evaluation.

Suffocation

- Death or injury from failure of oxygen to reach the lungs, a category of asphyxia distinct from strangulation. Subtypes include smothering (airway obstruction), choking or foreign-body obstruction, me-

chanical asphyxia (positional, chest compression), and entrapment in oxygen-deficient environments.

Sugammadex

- A selective relaxant binding agent that reverses aminosteroid neuromuscular blockers (rocuronium and vecuronium) by encapsulating them in a cyclodextrin molecule, forming a 1:1 complex that is renally excreted. Onset of action is within 2 to 3 minutes.

Suicidal Ideation

- Thoughts about killing oneself, ranging from passive (wish to be dead, wishes to fall asleep and not wake) to active (thoughts of killing oneself with varying degrees of intent, plan, and means). The C-SSRS grades severity from 1 (wish to be dead) to 5 (active suicidal ideation with specific plan and intent).

Suicide

- The intentional act of taking one's own life. Suicide is a major public health problem worldwide, causing 700,000 deaths annually.

Suicide Risk Assessment

- A clinical evaluation of suicidal thoughts, intent, plan, access to means, past attempts, self-harm, psychiatric and medical illness, substance use, protective factors, and immediate safety. It informs but does not replace ongoing clinical judgment and safety planning.

Sulfa Drugs

- A class of bacteriostatic antibiotics that competitively inhibit dihydropteroate synthase, an enzyme essential for folic acid synthesis in bacteria. See sulfonamides.

Sulfasalazine

- A DMARD combining sulfapyridine and 5-aminosalicylic acid, used in rheumatoid arthritis, psoriatic arthritis, and inflammatory bowel disease (ulcerative colitis). Slows disease progression in RA; onset over 1-3 months.

Sulfatide

- A sulfated glycolipid found particularly in myelin and nervous tissue.

Sulfonamide

- A class of antibiotics (sulfamethoxazole, sulfadiazine, sulfasalazine) that inhibit bacterial di-

hydropteroate synthase, blocking folate synthesis—bacteriostatic. Often combined with trimethoprim (co-trimoxazole: TMP-SMX). Used for UTIs, PCP (pneumocystis pneumonia), toxoplasmosis, *Neisseria*, and MRSA.

Sulfonamides

- A class of bacteriostatic antibiotics that competitively inhibit dihydropteroate synthase, an enzyme essential for folic acid synthesis in bacteria. They are structural analogs of para-aminobenzoic acid and compete with PABA for the active site of the enzyme.

Sulfonylureas

- Oral antidiabetic agents that stimulate insulin secretion from pancreatic beta cells by blocking ATP-sensitive potassium channels on the beta cell membrane. This causes membrane depolarization, opening of voltage-gated calcium channels, calcium influx, and exocytosis of insulin-containing granules.

Sulfur Colloid Scan

- Nuclear medicine imaging using Tc-99m sulfur colloid. Evaluates liver, spleen, bone marrow, lymph nodes.

Sulphonamides

- A class of antimicrobial agents containing the sulfonamide group (SO_2NH_2). Sulphonamides were the first effective systemic antibacterial agents (Prontosil, 1935—Gerhard Domagk). Mechanism: competitive inhibition of dihydropteroate synthase, blocking folate synthesis (para-aminobenzoic acid analogue)—bacteriostatic.

Sun Protection (Photoprotection)

- Sun protection involves measures taken to reduce exposure to ultraviolet radiation from the sun, which is the primary environmental risk factor for skin cancer and photoaging.

Sunburn

- Acute erythema (and later edema, pain, blistering, peeling) of skin from excessive ultraviolet (UV) radiation (UVB predominantly), causing DNA damage and inflammatory response. Risk: repeated sunburn increases skin cancer risk (melanoma, basal/squamous cell carcinoma) and photoaging.

Superantigen

- A protein that bypasses antigen processing and

non-specifically activates large T-cell populations by binding MHC class II and the T-cell receptor V region, triggering massive cytokine release. Staphylococcal TSST-1 and enterotoxins, and streptococcal pyrogenic exotoxins, cause toxic shock and scarlet fever.

Superficial

- Located near or on the surface of the body.

Superficial Fascia

- A layer of loose connective tissue beneath the dermis, also called subcutaneous tissue or the hypodermis.

Superficial thrombophlebitis

- See Phlebitis.

Superimposed Preeclampsia

- Preeclampsia developing in a woman with pre-existing chronic hypertension. Diagnosis: sudden worsening of hypertension, new-onset proteinuria or a sudden increase in proteinuria, or new-onset thrombocytopenia, elevated liver enzymes, or symptoms (headache, visual changes). Occurs in 20-50% of women with chronic hypertension.

Superinfection

- A new infection that develops during or after treatment of a previous infection, often involving an antimicrobial-resistant organism or an organism not susceptible to the original therapy, such as *Clostridioides difficile* after antibiotic exposure.

Superior

- Positioned higher or closer to the head.

Superior Mesenteric Artery

- Second major abdominal aortic branch at L1, the midgut artery. Supplies distal duodenum through proximal two-thirds of transverse colon. Branches: inferior pancreaticoduodenal, jejunal, ileal, ileocolic, right colic, middle colic. Arises anterior to the pancreatic uncinate process, crosses the third duodenal part.

Superior mesenteric artery syndrome

- Compression of the duodenum between the aorta and the superior mesenteric artery, causing proximal duodenal obstruction. Presents with postprandial nausea, vomiting, and epigastric pain, often after significant weight loss that narrows the SMA-aortic angle. Treatment includes nutritional rehabilitation

and surgical duodenal mobilization.

Superior Semicircular Canal Dehiscence

- A defect or marked thinning of the bony covering of the superior semicircular canal that can produce sound- or pressure-induced vertigo, autophony, and conductive-appearing hearing findings. Diagnosis combines characteristic symptoms, physiologic testing, and high-resolution temporal-bone CT.

Superior Vena Cava

- Formed by brachiocephalic vein junction at the first right costal cartilage, returning upper body deoxygenated blood to the right atrium. Approximately 7 cm, valveless. Related to right lung/pleura, right phrenic nerve, and azygos vein.

Superior Vena Cava Syndrome

- Obstruction of SVC by tumor. Facial swelling, dyspnea, dilated neck veins. Emergency: radiation, stenting.

Superior vena caval obstruction

- Compression or thrombosis of the superior vena cava impairing venous return from the head, neck, and upper extremities. Most commonly caused by malignancy (lung cancer, lymphoma) or indwelling central venous devices. Presents with facial swelling, plethora, distended neck veins, and upper extremity edema.

Supernumerary rib

- An extra rib beyond the typical twelve pairs, most commonly a cervical rib above the first thoracic rib, which may compress the brachial plexus or subclavian vessel causing thoracic outlet syndrome.

Supernumerary Teeth

- Extra teeth beyond normal complement. Most common: mesiodens between maxillary centrals. May cause impaction or malocclusion.

Supervised Exercise Therapy

- A structured walking program for peripheral artery disease that improves claudication distance and functional capacity, commonly prescribed for 30–45 minutes three times weekly.

Supination

- Rotation of the forearm so that the palm faces anteriorly in anatomical position, or upward when the elbow is flexed. The radius and ulna become parallel.

Supine Hypotensive Syndrome

- Compression of the inferior vena cava by the gravid uterus when the pregnant woman lies flat on her back after 20 weeks gestation. Results in decreased venous return, reduced cardiac output, and maternal hypotension, dizziness, pallor, and syncope.

Supplement Safety

- Supplements not strictly regulated. Contamination, mislabeling risks. Consult healthcare provider before use.

Supplemental Oxygen Therapy

- Administration of oxygen above atmospheric concentration. Indicated for hypoxemia ($\text{PaO}_2 < 60$ mmHg or $\text{SpO}_2 < 90\%$). Delivered via nasal cannula (1–6 L/min), simple mask, non-rebreather mask, or mechanical ventilation. Target SpO_2 88–92% in COPD patients to avoid CO_2 retention.

Supplementation

- Use of a vitamin, mineral, protein, fiber, or other nutrient preparation to correct a deficiency or meet a clinically identified increased requirement. The indication, dose, interactions, product quality, and risk of toxicity should be assessed before use.

Supportive Care

- Mainstay of poison management. Airway, breathing, circulation, IV access, monitoring, treat complications.

Suppository

- A solid, single-dose dosage form designed to be inserted into a body orifice (rectum, vagina, or urethra) where it melts, softens, or dissolves, releasing the active pharmaceutical ingredient for local or systemic effects.

Supraglottic Airway Device

- An airway device positioned above the glottis to maintain oxygenation and ventilation without passing through the vocal cords. It is used for selected anesthetized or unconscious patients and is not a definitive substitute for a cuffed tracheal tube when aspiration protection is required.

Supraglottic Airway Devices

- Supraglottic airway devices (SADs) sit above the vocal cords, providing a seal around the laryngeal inlet without tracheal intubation. First-generation SADs (classic LMA, i-gel) provide airway seal but

do not protect against aspiration.

Suprascapular Nerve

- A branch of the brachial plexus that supplies the supraspinatus and infraspinatus muscles.

Supraventricular Tachycardia

- A rapid rhythm (150–250 bpm) originating above the ventricles, characterized by narrow QRS complexes. Most common forms are AV nodal reentrant tachycardia (AVNRT) and AV reentrant tachycardia (AVRT). Episodes are typically paroxysmal with sudden onset and offset. Vagal maneuvers and adenosine are diagnostic and therapeutic.

Surfactant

- A phospholipid-protein complex produced by type II pneumocytes that reduces alveolar surface tension, preventing atelectasis. Deficiency causes respiratory distress syndrome in premature infants. Exogenous surfactant (beractant, poractant alfa) administered via endotracheal tube.

Surgery

- Surgical removal of tumor. Curative when complete resection possible. Palliative for symptom relief.

Surgical Drain

- A device placed to remove blood, pus, bile, urine, lymph, or other fluid from a surgical site or body cavity. Indications, output characteristics, duration, and removal criteria should be documented because drains can also cause infection, bleeding, or tissue injury.

Surgical Hemostasis

- Control of bleeding during an operation using pressure, ligation, sutures, clips, electrosurgery, topical agents, vessel sealing, or other techniques. Effective hemostasis requires identification of the bleeding source and correction of coagulopathy when present.

Surgical Instruments

- Instruments for surgery. Categories: cutting (scalpel, scissors), grasping (forceps, clamps), retraction, hemostasis (hemostat, electrocautery).

Surgical Margin

- The edge of tissue removed during an excision or resection, assessed microscopically for residual disease. A negative margin has no tumor at the examined edge; a positive margin indicates tumor extends to the inked or designated margin.

Surgical Repair

- See repair.

Surgical Risk Stratification

- Estimation of the likelihood of perioperative complications using patient factors, urgency, procedure risk, functional capacity, and disease-specific assessment. It informs optimization, monitoring level, shared decisions, and postoperative planning but cannot eliminate uncertainty.

Surgical Safety Checklist

- A structured checklist used at defined points before anesthesia, before incision, and before leaving the operating room to verify patient identity, procedure, site, consent, equipment, specimens, and team communication.

Surgical Site Infection

- SSI classification (CDC): superficial incisional (skin/subcutaneous, <30 days), deep incisional (fascia/muscle, <30/90 days with implant), organ/space (any part opened/manipulated, <30/90 days with implant).

Surgical Site Infection (SSI)

- Infection involving the incision, organ, or operative space within the surveillance period after surgery. It may be superficial incisional, deep incisional, or organ-space and requires appropriate culture, antimicrobial therapy, drainage, debridement, or reoperation when indicated.

Surgical Time-Out

- A deliberate team pause immediately before incision to verify patient identity, procedure, site, consent, allergies, antibiotics, imaging, implants, equipment, and anticipated critical events. It is a core surgical-safety intervention.

Surgical Wound Classification

- Categorization of an operative wound as clean, clean-contaminated, contaminated, or dirty/infected according to the degree of microbial exposure and existing infection. The class helps estimate surgical-site infection risk and guide prophylaxis.

Surrogate Decision-Maker

- Legal document expressing healthcare wishes. See advance directive.

Surveillance

- Systematic collection, analysis, interpretation of health data. Types: passive (reporting), active (proactive case finding), sentinel (selected sites).

Surveillance Imaging

- Repeat imaging to detect recurrence. Protocol varies by cancer type, stage, treatment.

Survival

- Length of life after diagnosis or event. In elderly, consider comorbidities, functional status, preferences. Life expectancy estimates based on age, sex, health status. Shared decision-making essential.

Survival Rate

- Percentage of patients surviving for specified time after diagnosis. 5-year survival commonly used.

Survivorship

- Care for cancer survivors. Addresses late effects, surveillance, quality of life.

Survivorship Care Plan

- Document summarizing treatment, follow-up, late effects. Provided to cancer survivors.

Susac Syndrome

- An autoimmune microangiopathy affecting the brain, retina, and inner ear. The characteristic triad is encephalopathy, branch retinal-artery occlusions, and sensorineural hearing loss, though all features may not be present initially.

Susceptibility Testing

- Laboratory tests determining the sensitivity of a bacterial isolate to antimicrobials to guide therapy. Methods: disk diffusion (Kirby-Bauer—zone sizes), broth dilution (MIC—minimum inhibitory concentration, MBC—minimum bactericidal concentration), E-test (gradient strips), and automated systems (VITEK).

Suspected Cause

- Identification of disease. See diagnosis.

Suspension

- A pharmaceutical formulation in which fine solid drug particles are dispersed (not dissolved) in a liquid vehicle, often with suspending agents and stabilizers. Suspensions are used when a drug is poorly soluble, for oral (antibiotic suspensions, antacids), topical, ophthalmic, and parenteral (depot) admin-

istration.

Suture

- A thread or strand of material used to approximate tissues (bring wound edges together) and hold them in place until healing occurs.

Suture (Perineal)

- A thread or strand of material used to approximate tissues (bring wound edges together) and hold them in place until healing occurs. See Suture.

Suture Anchor

- A small implant that secures a tendon, ligament, or other soft tissue to bone. It may be made from metal, polymer, or bioabsorbable material and can fail through pullout, breakage, migration, or inflammatory reaction.

Sutures

- Surgical threads or stitches used to close wounds or surgical incisions. In obstetrics, sutures are used for: episiotomy repair, perineal tear repair (1st-4th degree), cesarean section closure (uterine, fascial, and skin layers), and cervical cerclage.

SUV

- See standardized uptake value.

SVC Syndrome

- Obstruction of the superior vena cava causing facial swelling, plethora, and distended neck veins. Commonly from lung cancer (SCLC, squamous) and lymphoma. Oncologic emergency.

Swallow Study

- X-ray imaging of swallowing using barium. Evaluates dysphagia, aspiration, structural abnormalities.

Swallowing

- Difficulty swallowing, classified as oropharyngeal (difficulty initiating a swallow, choking, coughing, nasal regurgitation) or esophageal (sensation of food sticking retrosternally). See dysphagia.

Swallowing Assessment

- Clinical and, when indicated, instrumental evaluation of oral, pharyngeal, and esophageal swallowing to identify dysphagia, aspiration, and appropriate food or liquid consistency. Videofluoroscopic or endoscopic studies may be required when bedside assessment is insufficient.

Swallowing Therapy

- See speech-language pathology.

Sweat Chloride Test

- The gold standard diagnostic test for cystic fibrosis. Measures chloride concentration in sweat after pilocarpine iontophoresis. Normal: <30 mmol/L. Intermediate: 30-59. Positive: 60 mmol/L. False positives: malnutrition, adrenal insufficiency, hypothyroidism, and ectodermal dysplasia.

Sweat Gland

- A gland in the skin that produces sweat, playing a critical role in thermoregulation and waste excretion.

Sweat Glands

- Exocrine glands that secrete sweat onto the skin surface or into hair follicles. Eccrine glands are widely distributed and produce a watery secretion for thermoregulation; their ducts open directly onto the surface and are activated mainly by sympathetic cholinergic fibers.

Sweating

- The secretion of sweat by eccrine glands, the primary heat-loss mechanism in humans, controlled by sympathetic cholinergic innervation (in contrast to the catecholaminergic sympathetic response elsewhere).

Sweet Syndrome

- Sweet syndrome (acute febrile neutrophilic dermatosis) presents with fever, neutrophilia, and tender erythematous plaques/nodules with dense dermal neutrophilic infiltrate. Classical (idiopathic), malignancy-associated (AML, solid tumors), drug-induced (G-CSF, granulocyte colony-stimulating factor). Associated with VEXAS

Sweet's syndrome

- Acute febrile neutrophilic dermatosis, characterized by fever, neutrophilia, and painful, erythematous plaques or nodules on the skin, histologically showing dense neutrophilic infiltration of the dermis. It may be idiopathic, drug-induced, or associated with malignancy, particularly hematological cancers.

Swine Flu

- A contagious respiratory illness caused by influenza A or B viruses, characterized by sudden onset of fever, myalgia, headache, malaise, nonproductive

cough, sore throat, and nasal congestion. See Influenza.

Symblepharon

- Adhesion of the palpebral conjunctiva (inner eyelid) to the bulbar conjunctiva or cornea, most commonly resulting from chemical burns, Stevens-Johnson syndrome, ocular cicatricial pemphigoid, or repeated surgical intervention. It restricts eye movement and may require surgical lysis with amniotic membrane or mucous membrane grafting.

Sympathetic Nervous System

- The division of the autonomic nervous system responsible for the 'fight or flight' response, preparing the body for intense physical activity or stress. Sympathetic preganglionic neurons originate in the intermediolateral column of the spinal cord gray matter (T1–L2), making it the thoracolumbar system.

Sympatholytic

- An agent that blocks the sympathetic nervous system, used to treat hypertension, sympathetic overactivity, and catecholamine excess. Classes: beta-blockers (propranolol, metoprolol), alpha-blockers (prazosin, doxazosin, phenoxybenzamine), centrally acting agents (clonidine, methyldopa), and neuron blockers (reserpine, guanethidine).

Symptom

- Subjective experience of disease. Examples: pain, fatigue, nausea. Reported by patient.

Synapse

- The specialized junction between a neuron and another neuron (or effector cell—muscle, gland) where information is transmitted from one cell to another.

Synaptic Plasticity

- Activity-dependent change in the strength or structure of synaptic transmission. Long-term potentiation and long-term depression are important cellular mechanisms underlying learning, memory, adaptation, and some forms of chronic pain.

Synaptic Transmission

- The fundamental process by which neurons communicate with each other through specialized junctions called synapses.

Syncope

- Transient loss of consciousness due to cerebral hypoperfusion. Causes: vasovagal, orthostatic, cardiac

(arrhythmia, structural), neurologic. Assessment: history, examination, ECG, orthostatic BP. Risk stratification: San Francisco Syncope Rule, OESIL score.

Syncope (Fainting)

- A transient loss of consciousness (TLOC) due to global cerebral hypoperfusion, characterized by rapid onset, short duration, and spontaneous complete recovery. Syncope is common (lifetime prevalence 40%).

Syndactyly

- A congenital anomaly where two or more digits are fused together by skin (simple syndactyly) or by both skin and bone (complex syndactyly). Most common congenital hand anomaly, often bilateral. May be isolated (autosomal dominant, incomplete penetrance) or syndromic (Poland, Apert, Crouzon, Holt-Oram syndromes).

Syndrome

- Group of symptoms and signs occurring together. Examples: metabolic syndrome, irritable bowel syndrome.

Syndrome of inappropriate ADH secretion

- Excessive antidiuretic hormone activity causing euvolemic hyponatremia with inappropriately concentrated urine and elevated urinary sodium.

Syndrome Surveillance

- The public health monitoring of pre-diagnostic clinical indicators (symptoms, syndromes—fever, rash, GI illness, influenza-like illness) in real time to detect outbreaks, bioterrorism, or emerging infections earlier than laboratory confirmation.

Syndromic Surveillance

- Monitoring of symptoms, clinical encounters, laboratory requests, or other near-real-time signals before definitive diagnoses are available. It supports early detection of outbreaks and public-health threats but may have lower specificity.

Synovial chondromatosis

- A benign, metaplastic condition characterized by the formation of multiple cartilaginous loose bodies within joint synovium, bursae, or tendon sheaths. It presents with joint pain, swelling, locking, and restricted range of motion, most commonly affecting

the knee, hip, and elbow.

Synovial Fluid Analysis

- Laboratory examination of aspirated joint fluid for appearance, viscosity, leukocyte count and differential, crystals under polarized light, Gram stain, and culture. It helps distinguish noninflammatory, inflammatory, crystal-associated, and septic arthritis.

Synovial Sarcoma

- A soft tissue sarcoma typically arising near joints of the extremities in young adults, characterized by the t(X;18) chromosomal translocation.

Synovitis

- Inflammation of the synovial membrane lining a joint, causing swelling, pain, warmth, and joint effusion. Causes: inflammatory arthritis (RA, psoriatic, gout, reactive, septic), autoimmune (SLE), and trauma/overuse.

Synovium

- The inner layer of the joint capsule in synovial joints, composed of synovial lining cells (type A macrophage-like, type B fibroblast-like) on a vascularized connective tissue base. Produces synovial fluid (hyaluronic acid, lubricin, proteinases) for lubrication and chondrocyte nutrition.

Synovography

- Contrast imaging of joint cavity. Evaluates synovial pathology.

Synthesis Module

- Automated system for radiopharmaceutical preparation. PET tracer synthesis: FDG, Florbetapir. Quality control: radiochemical purity, pH, sterility.

Syphilis

- A bacterial STI caused by *Treponema pallidum*. Primary: chancre at inoculation site. Secondary: rash, mucous patches, condylomata lata. Latent: asymptomatic. Tertiary: gummas, cardiovascular, neurosyphilis. Diagnosis: non-treponemal tests (RPR/VDRL) confirmed by treponemal tests. Treatment: penicillin G benzathine IM.

Syphilitic alopecia

- Non-scarring hair loss occurring as a manifestation of secondary syphilis, presenting with diffuse thinning or patchy, moth-eaten alopecia of the scalp, eyebrows, or beard. It is thought to result from direct spirochaetal invasion of hair follicles during

the spirochaetaemic phase.

Syphilitic gumma

- A destructive, granulomatous lesion occurring in tertiary syphilis, characterized by a center of necrotic tissue surrounded by plasma cells, lymphocytes, and fibroblasts. Gummas may affect any organ, most commonly the skin, bones, liver, and central nervous system. They present as painless, firm nodules that may ulcerate.

Syringe

- Cylindrical device for injecting or withdrawing fluids. Components: barrel, plunger, luer lock. Various sizes: 1 mL to 60 mL.

Syringomyelia

- A condition characterized by the formation of a fluid-filled cyst (syrinx) within the spinal cord, which can expand and elongate over time, compressing and destroying the surrounding spinal cord tissue.

Syrinx

- A syrinx is a fluid-filled cavity within the spinal cord, typically containing cerebrospinal fluid or a similar fluid, that develops as a result of abnormal CSF flow dynamics within the spinal cord. Syringomyelia: a syrinx in the central spinal cord. Hydromyelia: dilation of the central canal.

Systematic Review

- Comprehensive summary of research on specific question. Uses explicit methods to minimize bias. Cochrane reviews example.

Systemic Antifungals

- Systemic antifungals treat extensive or refractory fungal infections. Terbinafine is first-line for onychomycosis (6 weeks for fingernails, 12 weeks for toenails) and tinea capitis. Itraconazole pulse therapy (200 mg twice daily for 1 week per month) treats onychomycosis. Fluconazole treats tinea corporis, cruris, and versicolor.

Systemic Effects

- Whole-body effects of toxins. Cardiac, neurological, renal, hepatic, hematological.

Systemic Inflammatory Response Syndrome

- SIRS: a systemic inflammatory response defined by two or more of: temperature $>38^{\circ}\text{C}$ or $<36^{\circ}\text{C}$,

heart rate >90, respiratory rate >20 or PaCO₂ <32 mmHg, WBC >12,000 or <4,000 or >10 percent bands. May result from infection (sepsis), trauma, burns, pancreatitis, surgery, or other insults.

Systemic Juvenile Idiopathic Arthritis

- sJIA: a subtype of juvenile idiopathic arthritis with prominent systemic inflammation—daily spiking fevers, evanescent salmon-pink rash, arthritis, lymphadenopathy, hepatosplenomegaly, and elevated inflammatory markers (ferritin).

Systemic Lupus Erythematosus

- Multisystem autoimmune disease, more common in women. Features: malar rash, photosensitivity, oral ulcers, arthritis, serositis, renal disorder, neurologic disorder, hematologic disorder, immunologic disorder. Diagnosis: ANA positive plus clinical/immunologic criteria.

Systemic Sclerosis

- Systemic sclerosis, also known as scleroderma, is a chronic autoimmune disease characterized by progressive fibrosis of the skin and internal organs, vasculopathy, and immune dysfunction.

Systemic Therapy

- Treatment that reaches the whole body, including chemotherapy, endocrine therapy, targeted therapy, and immunotherapy, as opposed to local therapies such as surgery and radiation.

Systemic Vascular Resistance

- The resistance against which the left ventricle must pump, calculated as $(MAP - CVP) / CO \times 80$. Normal: 800-1200 dynes/sec/cm⁵. Elevated SVR occurs in cardiogenic shock (compensatory vasoconstriction), aortic stenosis, and systemic hypertension. Decreased SVR occurs in septic shock, anaphylaxis, and cirrhosis.

Systemic Vascular Resistance (SVR)

- The resistance to blood flow offered by the systemic circulation, determined mainly by arteriolar tone. It is calculated from the pressure gradient across the systemic circulation and cardiac output and is a major determinant of arterial pressure.

Systolic Murmurs

- Murmurs occurring between S1 and S2. Ejection systolic murmurs (crescendo-decrescendo) are caused by flow across semilunar valves (aortic steno-

sis, pulmonary stenosis) or increased flow across normal valves.

Sézary Syndrome

- The leukemic variant of cutaneous T-cell lymphoma, characterized by erythroderma, lymphadenopathy, pruritus, and circulating atypical lymphocytes (Sézary cells).

Scleritis

- Painful, deep inflammation of the white outer coat of the eye. It may cause redness, light sensitivity, blurred vision, and tenderness and can be associated with autoimmune disease; vision changes or severe eye pain require urgent ophthalmic assessment.

Shoulder Pain

- Pain in or around the shoulder caused by injury, tendon or bursa disorders, arthritis, nerve disease, or referred pain. Sudden weakness, deformity, chest symptoms, fever, or pain after major trauma requires prompt evaluation.

Scrotal Infection

- Infection of the scrotal skin or underlying structures, which may cause redness, swelling, warmth, pain, or drainage. Rapidly worsening pain, fever, discoloration, or severe swelling is an emergency because serious soft-tissue infection must be excluded.

Scrotal Pain

- Pain in the scrotum caused by infection, trauma, hernia, fluid collections, varicocele, or testicular torsion. Sudden severe pain, nausea, a high-riding testis, or abnormal testicular position requires immediate assessment for torsion.

Sinus Pain

- Pressure or pain around the forehead, cheeks, nose, or eyes caused by sinus inflammation, infection, migraine, or other conditions. Severe headache, eye swelling, vision change, high fever, or neurologic symptoms requires urgent assessment.

Sputum

- Mucus or phlegm produced in the lower airways and expelled by coughing. Its amount, color, odor, or blood content may provide clues to infection, inflammation, or bleeding, but appearance alone does not establish a diagnosis.

Sphenoid Sinusitis

- Inflammation or infection of the sphenoid sinus,

which lies deep behind the nasal cavity. It may cause deep headache, facial pain, or visual symptoms and can rarely spread to nearby nerves or the brain; persistent or severe symptoms require medical assessment.

Sickle Cell Dactylitis

- Painful swelling of the hands or feet caused by blockage of small blood vessels during a sickle-cell episode, especially in infants and young children. It may be an early sign of sickle-cell disease and requires appropriate clinical evaluation.

Shoulder Dislocation

- Complete displacement of the upper arm bone from the shoulder socket, usually caused by a fall, collision, or forceful twisting. It causes severe pain, deformity, and inability to move the arm; urgent reduction and assessment for associated fractures or nerve and blood-vessel injury are required.

Skin Infection

- Infection of the skin or underlying soft tissue, commonly caused by bacteria, fungi, or viruses. Redness, warmth, swelling, pain, drainage, or fever may occur; rapidly spreading redness, severe pain, blisters, or systemic illness requires prompt medical assessment.

Sleep Paralysis

- Temporary inability to move or speak when falling asleep or waking, while awareness is partly or fully preserved. Episodes usually last seconds to minutes and may include frightening hallucinations. Recurrent episodes can be associated with sleep deprivation, irregular sleep, narcolepsy, or other sleep disorders.

Spinal Muscular Atrophy

- An inherited disorder causing progressive loss of motor neurons and muscle weakness, often beginning in infancy or childhood but sometimes appearing in adulthood. Severity varies; breathing, swallowing, and mobility may be affected. Early diagnosis allows genetic counseling and disease-modifying care.

Small Fiber Neuropathy

- Damage to small sensory and autonomic nerve fibers, often causing burning pain, tingling, altered temperature or pain sensation, abnormal sweating, or gastrointestinal and cardiovascular symptoms.

Diabetes, autoimmune disease, vitamin deficiency, toxins, and inherited disorders may contribute; evaluation seeks an underlying cause.

Small Intestinal Bacterial Overgrowth

- An abnormal increase or change in bacteria within the small intestine, often causing bloating, abdominal pain, nausea, diarrhea, fullness, weight loss, or nutrient deficiency. Causes include impaired intestinal movement, structural problems, surgery, and some systemic diseases; persistent symptoms require medical evaluation.

Chapter 20

Pocket Dictionary: T Terms

T Wave

- The deflection representing ventricular repolarization. Normal T waves are upright in leads I, II, and V3-V6, and inverted in aVR. They should be at least 10% of the QRS amplitude in limb leads and 25% in precordial leads. Abnormal T waves may indicate ischemia (inversion), hyperkalemia (peaked), or pericarditis (diffuse elevation).

T-Cell Lymphomas

- Peripheral T-cell lymphomas (PTCL) are a heterogeneous group of aggressive lymphomas including PTCL-NOS (not otherwise specified, most common), angioimmunoblastic T-cell lymphoma (AITL, T-follicular helper origin, RHOA mutations, dysimmune features), anaplastic large cell lymphoma (ALCL, ALK+ vs ALK-), enteropathy-associated T-cell

T-Cell Receptor (TCR)

- A clonally distributed receptor on T lymphocytes that recognizes a peptide antigen presented by a major histocompatibility complex molecule. TCR signaling requires accessory and co-stimulatory signals for effective T-cell activation.

T-Cells (T-Lymphocytes)

- A type of lymphocyte that matures in the thymus and is the central effector of cell-mediated immunity. T-cells express the T-cell receptor (TCR) on their surface, which recognises peptide antigens presented by major histocompatibility complex (MHC) molecules on antigen-presenting cells.

T3

- Triiodothyronine, the most biologically active thyroid hormone, produced by deiodination of T4 in peripheral tissues (liver, kidney) and secreted by the thyroid. Regulates metabolism (BMR), pro-

tein/carbohydrate/lipid metabolism, growth, and development.

T4

- Thyroxine, the main hormone secreted by the thyroid gland (thyroxine, 80-90 percent of secretion), converted to the active T3 in tissues. Bound largely to thyroxine-binding globulin (TBG); free T4 is the active unbound fraction. Regulates metabolism, growth, and development.

Tabes dorsalis

- See Locomotor Ataxia (Tabes Dorsalis).

Tablet

- A solid dosage form of compressed drug plus excipients (binders, fillers, disintegrants, lubricants, coatings). Types: immediate-release, enteric-coated (resistant to gastric acid), sustained/controlled-release, sublingual, buccal, effervescent, chewable, and orally disintegrating.

Tachycardia

- A heart rate exceeding 100 beats per minute in adults at rest. Tachycardia is a sign, not a diagnosis, and requires investigation to determine the underlying rhythm and cause.

Tachyphylaxis

- A rapid, acute decrease in response to a drug after repeated administration over a short period, distinct from tolerance which develops more gradually. Tachyphylaxis results from rapid depletion of neurotransmitter stores, receptor desensitization, or receptor downregulation.

Tachypnea

- Abnormally rapid respiratory rate (adult >20 breaths/min). Physiologic (exercise, anxiety) or pathologic: hypoxia, fever, sepsis, metabolic acido-

sis (Kussmaul), pulmonary embolism, pneumonia, heart failure, pain, and compensation for metabolic derangement.

Tachypnoea

- Abnormally rapid breathing, defined as a respiratory rate >20 breaths per minute in adults (age-dependent: >60 in neonates, >40 in 1–12 months, >30 in 1–5 years, >20 in >5 years). Tachypnoea is a non-specific sign of respiratory or metabolic distress.

Tachysystole (Uterine Hyperstimulation)

- More than 5 contractions in 10 minutes averaged over 30 minutes, or contractions lasting 2 minutes. Tachysystole with fetal heart rate changes (late decelerations, prolonged deceleration, loss of variability) is more concerning. Causes: oxytocin (most common), prostaglandins (misoprostol > dinoprostone), and cocaine.

TACO

- Transfusion-Associated Circulatory Overload, a complication of blood transfusion due to volume overload causing pulmonary edema. Risk factors: pre-existing cardiac or renal disease, extremes of age, large transfusion volume, rapid infusion rate.

Tacrolimus

- A calcineurin inhibitor immunosuppressant (macrolide) that blocks T-cell activation (NFAT/IL-2 inhibition). Used for solid organ transplant rejection prophylaxis/treatment and graft-versus-host disease, and topically for atopic dermatitis.

Tactile Fremitus

- Vibrations transmitted through the lung to the chest wall, felt during palpation. It is increased in consolidation (pneumonia) due to better sound transmission through solid tissue. It is decreased in pleural effusion, pneumothorax, and atelectasis due to impaired transmission.

Taeniasis (Tapeworm Infection)

- Intestinal infection caused by adult tapeworms of the genus *Taenia* (cestodes). Two important species: *Taenia solium* (pork tapeworm) and *Taenia saginata* (beef tapeworm). Transmission: ingestion of undercooked meat containing cysticerci (larval cysts). Adult tapeworms attach to the small intestinal wall via scolex (armed with hooks in *T.*

Takayasu Arteritis

- A chronic granulomatous vasculitis primarily affecting the aorta and its major branches, most commonly diagnosed in women under 50 years of age, particularly in Asian populations.

Takayasu's arteritis

- A chronic granulomatous vasculitis of the aorta and its major branches, predominantly affecting young women of Asian descent. Causes arterial stenosis, occlusion, or aneurysm formation. Presents with limb claudication, absent pulses, bruit, hypertension, and constitutional symptoms. Treatment: corticosteroids and immunosuppressives.

Takotsubo Cardiomyopathy

- Transient left ventricular apical ballooning resembling a Japanese octopus trap (takotsubo), typically triggered by emotional or physical stress. ECG may show ST elevation mimicking STEMI. Coronary angiography shows no obstructive disease. Most patients recover normal ventricular function within weeks.

Talcosis

- Pneumoconiosis from talc dust inhalation or IV injection of crushed oral medications. Causes upper lobe nodular fibrosis. HRCT may show “galaxy sign” of calcified nodules.

Talus

- Second largest tarsal bone articulating with tibia/fibula proximally at the ankle. Approximately 60 percent of its surface is articular cartilage with no muscular attachments. Transmits entire body weight to the calcaneus. Talar neck fractures may disrupt blood supply causing avascular necrosis in up to 50 percent.

Tangier Disease

- An autosomal-recessive disorder of HDL metabolism caused by ABCA1 dysfunction. It produces very low HDL cholesterol, enlarged yellow-orange tonsils, hepatosplenomegaly, peripheral neuropathy, thrombocytopenia, and premature atherosclerosis in some patients.

Tanner Staging

- A scale for assessing sexual maturity during puberty. Stage 1: prepubertal. Stage 2: breast buds (girls), testicular enlargement (boys). Stage 3: breast en-

largement, pubic hair. Stage 4: areolar development, more pubic hair. Stage 5: adult breasts and genitalia, adult hair pattern.

Tapeworm

- Cestode infection: segmented flatworms (tapeworms) transmitted by ingestion of larvae (pork—*Taenia solium*; beef—*Taenia saginata*; fish—*Diphyllobothrium latum*, causing B12 deficiency) or eggs (*Echinococcus*—hydatid cysts; *Hymenolepis*). Intestinal tapeworm: abdominal discomfort, passage of proglottids in stool; T.

Tapeworm Infection

- Intestinal infection caused by adult tapeworms of the genus *Taenia* (cestodes). See Taeniasis (Tapeworm Infection).

Tardive Dyskinesia

- Involuntary, repetitive, purposeless movements caused by prolonged use of antipsychotic medications, typically developing after months to years of exposure. Movements include choreoathetoid movements of the tongue, face, and extremities (lip smacking, tongue protrusion, grimacing, choreiform movements of limbs).

Target-Controlled Infusion

- A computerized infusion pump system that maintains a predetermined drug concentration in plasma or effect site using pharmacokinetic modeling. TCI systems use three-compartment pharmacokinetic models to calculate infusion rates that achieve and maintain target concentration.

Targeted Temperature Management

- Controlled therapeutic hypothermia for comatose survivors of cardiac arrest. Target: 32-36 degrees C for 24-72 hours followed by gradual rewarming. Mechanism: reduces metabolic demand, limits reperfusion injury, and reduces free radical generation. TTM2 trial showed 33 degrees C was not superior to 36 degrees C.

Targeted Therapy

- Drugs targeting specific molecular abnormalities in cancer. Examples: imatinib (BCR-ABL), trastuzumab (HER2), rituximab (CD20), tyrosine kinase inhibitors. Requires molecular testing.

Tarlov Cyst

- A cerebrospinal-fluid-filled perineural cyst, usually

located near the dorsal root ganglia in the sacral spine. Many are incidental, but larger or symptomatic cysts can cause sacral pain, radiculopathy, sensory change, bowel or bladder dysfunction, or sexual dysfunction.

Tarsal Tunnel Syndrome

- Tibial nerve compression at ankle. Symptoms: burning pain, tingling in foot sole. Tinel sign at ankle. Treatment: orthotics, corticosteroid injection, surgical decompression.

Tarsus

- The collection of seven bones forming the posterior part of the foot, between the tibia/fibula and the metatarsals.

Taste

- The sensation produced by the interaction of chemical substances with taste receptor cells (TRCs) on the tongue, palate, pharynx, epiglottis, and upper esophagus. Taste is one of the five primary senses (gustation).

Tau PET

- Tracers: Flortaucipir, MK-6240. Detect tau tangles in neurodegenerative diseases.

Taurodontism

- Enlarged pulp chamber, short roots. Molars most affected. Associated with genetic conditions.

Tay-Sachs Disease

- Autosomal recessive lysosomal storage disorder (HEXA gene, hexosaminidase A). Accumulation of GM2 ganglioside in neurons. Ashkenazi Jewish prevalence higher. Symptoms: developmental regression, cherry-red spot on macula, seizures, paralysis, death by age 4. No cure. Carrier screening recommended.

TCA Cycle

- The tricarboxylic acid (Krebs) cycle occurs in the mitochondrial matrix, completing the oxidation of acetyl-CoA (from carbohydrates, fats, and proteins) to CO₂ while generating reducing equivalents (NADH, FADH₂) for the electron transport chain.

TCAs

- Tricyclic antidepressants are a class of drugs that inhibit the reuptake of serotonin and norepinephrine, similar to SNRIs, but also block histamine H1 recep-

tors, muscarinic acetylcholine receptors, and alpha-1 adrenergic receptors, producing numerous side effects.

TCAs (Tricyclic Antidepressants)

- Antidepressants that block reuptake of serotonin and norepinephrine, with additional anticholinergic, antihistaminic, and alpha-adrenergic blocking effects. Examples include amitriptyline, nortriptyline, imipramine, and desipramine. They are effective for depression, neuropathic pain, and migraine prophylaxis.

Tdap Vaccination in Pregnancy

- Administration of tetanus, diphtheria, and acellular pertussis (Tdap) vaccine during each pregnancy, optimally at 27-36 weeks. The goal: boost maternal antibodies to protect the newborn from pertussis (whooping cough) before the infant can be vaccinated at 2 months.

Tdap Vaccine

- Tetanus, Diphtheria, and acellular Pertussis booster given at 11-12 years and during each pregnancy (27-36 weeks gestation). Replaces one dose of Td. Important for pertussis protection, particularly for protecting newborns through maternal antibody transfer. Also given to adolescents and adults to maintain tetanus and diphtheria immunity.

Technetium-99m

- Most widely used radionuclide in nuclear medicine. Half-life: 6 hours. Emits 140 keV gamma rays. Produced from molybdenum-99/technetium-99m generator. Used with various ligands: MDP (bone), sestamibi (myocardium), DTPA (renal), nanocolloid (lymphatic), HMPAO (brain perfusion).

Technique

- Description of imaging protocol in radiology report.

Tedizolid pharmacokinetics

- Tedizolid has a longer half-life than linezolid, allowing for once-daily dosing. The drug has approximately 90% oral bioavailability. Tedizolid is metabolized to an active metabolite (tzd-M1) that contributes to efficacy. The drug achieves high concentrations in skin and soft tissues.

Teenage Pregnancy

- Pregnancy in adolescents (usually 10-19 years), associated with higher maternal and fetal risks:

preterm birth, low birth weight, preeclampsia, anemia, and increased neonatal mortality; also socioeconomic and educational consequences. Causes: unprotected intercourse, lack of contraception/education, poverty, and peer/family factors.

Teeth

- Hard, mineralized structures in the mouth used for cutting, tearing, and grinding food.

Teething

- The eruption of primary (deciduous) teeth, usually starting around 6 months (range 3-12), with lower central incisors first; 20 teeth by 2-3 years.

Teicoplanin

- A glycopeptide antibiotic related to vancomycin with a longer half-life allowing for less frequent dosing. Teicoplanin inhibits cell wall synthesis by binding to D-Ala-D-Ala precursors. It has activity against gram-positive organisms including MRSA.

Tele radiology

- Remote interpretation of imaging studies.

Tele-Rehabilitation

- Remote rehabilitation via technology. Video consultations, remote monitoring, home exercise programs. Increases access, especially rural.

Telemedicine

- Remote delivery of healthcare using technology. Examples: video visits, remote monitoring, store-and-forward. Increases access.

Telogen Effluvium

- Diffuse hair shedding caused by premature entry of hair follicles into the telogen (resting) phase. Triggers include severe illness, surgery, childbirth, rapid weight loss, emotional stress, medications (beta-blockers, retinoids, anticoagulants), and nutritional deficiencies (iron, zinc, vitamin D).

Telomere

- Protective DNA sequence at chromosome ends. Shortens with cell division. Associated with aging.

Temperature

- Measure of body heat. Normal: 36.5-37.5°C. Sites: oral, rectal, axillary, temporal.

Temperature (Body Temperature)

- The balance between heat production (metabolism, muscle activity) and heat loss (radiation, conduc-

tion, convection, evaporation). Normal core body temperature: 36.5–37.5°C (97.7–99.5°F), regulated by the hypothalamus (preoptic anterior hypothalamus—thermoregulatory center).

Temple Syndrome

- A genomic-imprinting disorder involving chromosome 14q32 that causes prenatal and postnatal growth restriction, hypotonia, developmental delay, feeding difficulty, short stature, and abnormal pubertal development.

Temporal Arteritis

- A vasculitic disorder affecting medium and large arteries, particularly the temporal artery, predominantly occurring in adults over 50 years of age.

Temporal Bones

- Paired bones forming the cranial floor and lateral wall, with squamous, petrous, mastoid, tympanic, and styloid parts. Contain the external auditory canal, middle and inner ear, and foramina: carotid canal (internal carotid), jugular foramen (CN IX–XI, IJV), and foramen lacerum.

Temporal Lobe

- One of the four major lobes of the cerebral hemisphere, located beneath the lateral fissure (Sylvian fissure) on the lateral surface of the brain, extending medially to the parahippocampal gyrus. The temporal lobe is involved in auditory processing, language comprehension, memory, and emotion.

Temporal Lobe Epilepsy

- The most common form of focal (partial) epilepsy, arising from the medial temporal lobe structures (hippocampus, amygdala, parahippocampal gyrus).

Temporomandibular Joint

- Synovial ginglymoarthrodial joint between mandibular condyle and temporal bone's mandibular fossa, separated by an articular disc. Only bilateral body joint, allowing hinge and sliding movements. Disc divides into superior/inferior compartments. Temporomandibular disorders cause pain, clicking, limited movement, common with bruxism.

Temporomandibular Joint Disorder (TMD)

- Pain and dysfunction of TMJ and masticatory muscles. Symptoms: jaw pain, clicking, limited opening, headache, ear pain. Etiology: multifactorial (brux-

ism, stress, trauma, occlusion). Treatment: conservative (reassurance, self-care, splint, NSAIDs), physical therapy, rare surgery.

Temporomandibular Joint Replacement

- Total joint replacement for severe TMJ pathology. Indications: ankylosis, severe degenerative disease, failed previous surgery.

Tendinitis

- Inflammation of a tendon, typically at its insertion (enthesis) or at the musculotendinous junction.

Tendon

- Dense regular connective tissue connecting muscle to bone, composed primarily of type I collagen fibers arranged in parallel bundles. Transmits muscle force to bone to produce movement. Poor vascularity (slower healing). Enclosed in tendon sheath (synovial) at friction points.

Tendon Transfer

- Surgical relocation of a functioning tendon to restore movement or balance when the original muscle-tendon unit is absent, paralyzed, or irreparable. Successful transfer requires an adequate donor, appropriate excursion, and a stable recipient joint.

Tendonitis

- Tendinopathy: inflammation or degeneration of a tendon from overuse, repetitive strain, or trauma, causing localized pain, swelling, and tenderness with activity. Common sites: rotator cuff (supraspinatus), lateral elbow (tennis elbow), medial elbow (golfer's elbow), Achilles, patellar (jumper's knee), biceps, and De Quervain.

Tenecteplase

- A modified tissue plasminogen activator (TNK-tPA) thrombolytic with longer half-life, higher fibrin specificity, and resistance to PAI-1. Used for acute ischemic stroke and STEMI. Given as a single IV bolus (weight-based), simplifying administration. Adverse effects: hemorrhage (intracranial—major concern), allergic reactions.

Tenesmus

- A sensation of incomplete defecation, characterized by a constant feeling of needing to pass stool accompanied by straining, pain, and cramping, despite little or no stool being passed. Tenesmus is a symptom, not a diagnosis.

Tennis Elbow

- Lateral epicondylitis, a tendinopathy of the common extensor tendon at the lateral epicondyle (extensor carpi radialis brevis). Caused by repetitive wrist extension and overuse, not limited to tennis. Presents with lateral elbow pain worsened by wrist extension and grip. Tenderness over the lateral epicondyle.

Tenosynovitis

- Inflammation of a tendon and its synovial sheath, causing pain, swelling, and crepitus along the tendon.

Tension Headache

- The most common headache type, characterized by bilateral, pressing/tightening head pain of mild-moderate intensity (like a band around the head), lasting 30 min to days, without nausea or vomiting; photophobia or phonophobia may be present but not both.

Tension Pneumothorax

- Life-threatening pneumothorax with mediastinal shift. Emergency needle decompression (2nd intercostal space, midclavicular line) followed by chest tube. Signs: respiratory distress, hypotension, tracheal deviation, distended neck veins, unilateral absent breath sounds.

Teratogen

- Substance causing birth defects. Examples: thalidomide, isotretinoin, alcohol, warfarin, valproate, ACE inhibitors. Prenatal exposure critical.

Teratogenicity

- The ability of a drug, chemical, or infectious agent to cause birth defects or developmental abnormalities when exposure occurs during pregnancy. The risk of teratogenic effects depends on the timing of exposure, with the embryonic period from 3 to 8 weeks after fertilization being the most critical for organogenesis.

Teratoma

- A type of germ cell tumor composed of tissues derived from two or three germ layers (ectoderm, endoderm, mesoderm), reflecting pluripotent differentiation.

Terazosin

- A selective alpha-1 adrenergic blocker causing vas-

cular and prostatic smooth muscle relaxation. Used for benign prostatic hyperplasia (improves urinary flow) and hypertension. Adverse effects: first-dose orthostatic hypotension/syncope (start low, at bedtime), dizziness, nasal congestion, and fatigue.

Terbinafine

- An allylamine antifungal agent that inhibits squalene epoxidase, an early enzyme in the ergosterol biosynthesis pathway, causing accumulation of squalene and depletion of ergosterol in the fungal cell membrane. The accumulation of squalene is fungicidal, while ergosterol depletion disrupts membrane function.

Teriparatide for Osteoporosis

- Recombinant human parathyroid hormone fragments one through thirty-four that stimulate osteoblast-mediated bone formation. Indicated for severe osteoporosis with fractures or very low T-scores and for patients who have failed or are intolerant to bisphosphonates. Daily subcutaneous injection for up to two years.

Terminal Delirium

- Delirium occurring near the end of life, often caused by advanced organ failure, medication effects, metabolic disturbance, infection, or reduced cerebral perfusion. Management prioritizes comfort, reversible contributors when consistent with goals, and support for the patient and family.

Terminal Illness

- Predicted disease outcome. See prognosis.

Tertiary Prevention

- Reducing complications from established disease. Examples: rehabilitation, chronic disease management, palliative care.

Test Strips

- Reagents for diagnostic testing. Types: urine dipstick, blood glucose, ketone.

Testicular Cancer

- Most common solid malignancy in young men 15-35 years. Types: germ cell tumors (seminoma, non-seminoma), sex cord-stromal tumors. Presentation: painless testicular mass, heaviness, ache. Diagnosis: scrotal ultrasound, tumor markers (AFP, beta-hCG, LDH).

Testicular Disorders

- Conditions affecting the testes, including testicular torsion (spermatic cord twisting causing ischemic pain and swelling, a surgical emergency requiring detorsion within 6 hours for testicular salvage), epididymitis (inflammation of the epididymis, often from bacterial infection—STI in young men, urinary pathogens in older men—treated with antibiotics).

Testicular Microlithiasis

- Multiple tiny calcifications in testes. Associated with testicular cancer risk. Surveillance recommended.

Testicular Torsion

- Surgical emergency caused by twisting of the spermatic cord around its longitudinal axis, cutting off venous return and leading to arterial occlusion, ischemia, and infarction of the testicle.

Testicular Torsion Emergency

- Rotation of the spermatic cord that compromises testicular blood flow, causing sudden unilateral scrotal pain, nausea, a high-riding testis, and an abnormal cremasteric reflex. Immediate surgical exploration and detorsion are required; imaging must not delay treatment when suspicion is high.

Testis

- The male reproductive gland that produces spermatozoa in seminiferous tubules and secretes testosterone from interstitial Leydig cells. It is normally located in the scrotum.

Testis (Testicle)

- The male gonad, a paired ovoid organ (4–5 cm × 2–3 cm, weight 15–20 g) located in the scrotum, responsible for sperm production (spermatogenesis) and testosterone secretion. Each testis is covered by the tunica vaginalis (peritoneal remnant) and tunica albuginea (dense fibrous capsule).

Testosterone

- The primary male sex hormone, an androgen produced mainly in the testicles that is essential for the development of male reproductive tissues, secondary sexual characteristics, and overall health.

Tetanus

- An acute neurological disease caused by tetanospasmin toxin from *Clostridium tetani*, introduced through wounds. Characterized by generalized rigidity

and painful muscle spasms (trismus/lockjaw, risus sardonicus, opisthotonos). Autonomic instability can occur in severe cases. Diagnosis: clinical.

Tetany

- A clinical syndrome of neuromuscular hyperexcitability caused by hypocalcaemia (most common—total calcium <2.1 mmol/L, ionised calcium <1.1 mmol/L), but also caused by hypomagnesaemia, hypokalaemia, and alkalosis (respiratory or metabolic).

Tethered Cord Syndrome

- A congenital or acquired condition where the spinal cord is abnormally attached (tethered) to surrounding tissues, restricting its normal upward movement within the spinal canal. Caused by a thickened filum terminale, lipomeningomyelocele, diastematomyelia, or previous myelomeningocele repair.

Tetracycline Stain

- Intrinsic discoloration from tetracycline antibiotics during tooth development. Gray or brown bands. Treatment: bleaching, veneers, crowns.

Tetracyclines

- A class of bacteriostatic antibiotics that bind reversibly to the 30S ribosomal subunit and block the attachment of aminoacyl-tRNA to the acceptor site, inhibiting protein synthesis. They include tetracycline, doxycycline, and minocycline.

Tetralogy of Fallot

- The most common cyanotic congenital heart defect, accounting for 7–10% of congenital cardiac anomalies.

Th1/Th2/Th17/Tfh/Treg Balance

- CD4⁺ T helper subsets defined by master transcription factors and signature cytokines: Th1 (Tbet, IFN- γ , IL-12, STAT4): intracellular pathogens, autoimmunity (RA, MS, type 1 diabetes). Th2 (GATA3, IL-4/5/13, IL-4/STAT6): helminths, allergy, asthma.

Thalamus

- Bilateral ovoid gray matter structure forming the third ventricle's lateral walls, the brain's relay station for nearly all sensory (except olfaction), motor, and limbic information to the cortex. Numerous nuclei: VPL (spinothalamic, dorsal column), VPM (trigeminal), LGN (visual), MGN (auditory),

VL/VA (motor).

thalassemia

- A group of inherited hemoglobin disorders caused by reduced or absent synthesis of one or more globin chains, leading to microcytic hypochromic anemia.

Thalassemia Major

- Beta-thalassemia, an autosomal recessive hemoglobin disorder with absent or minimal beta-globin chain production. Presents in infancy with severe anemia, failure to thrive, hepatosplenomegaly, and skeletal deformities. Diagnosis by hemoglobin electrophoresis showing absent HbA and elevated HbF and HbA2.

Thallium Scan

- Nuclear medicine imaging using Tl-201. Myocardial perfusion, tumor detection. Largely replaced by Tc-99m agents.

Thallium-201

- SPECT radionuclide, half-life 73 hours. Used for myocardial perfusion imaging. Mechanism: K⁺ analog, taken up by Na⁺/K⁺ ATPase.

Thanatology

- Scientific study of death, dying, post-mortem phenomena, and the psychological, social, and ethical aspects of bereavement. In forensic medicine, thanatology focuses on establishing clinical and legal definitions of death (brain death vs.

Thanatophoric Dysplasia

- A severe skeletal dysplasia, usually caused by an FGFR3 variant, with markedly shortened limbs, narrow thorax, macrocephaly, and characteristic skull and vertebral abnormalities. The severe thoracic restriction commonly causes neonatal respiratory failure.

Theranostic Pair

- Combined therapy and diagnostics using matched radiopharmaceutical pairs. See theranostics.

Theranostics

- Combined therapy and diagnostics using matched radiopharmaceutical pairs. Diagnostic isotope (e.g., Ga-68) identifies target, therapeutic isotope (e.g., Lu-177) delivers treatment. Examples: Ga-68/Lu-177 DOTATATE (neuroendocrine tumors), PSMA theranostics (prostate cancer).

Therapeutic Alliance

- The collaborative, trusting relationship between therapist and patient that is foundational to effective psychotherapy. It includes agreement on goals, tasks, and the bond between therapist and patient. It is the strongest predictor of psychotherapy outcome across all therapeutic modalities.

Therapeutic Drug Monitoring

- The clinical practice of measuring specific drug concentrations in blood or plasma to individualize drug dosing and ensure therapeutic effectiveness while minimizing toxicity. TDM is particularly important for drugs with narrow therapeutic indices, where small changes in concentration can lead to therapeutic failure or toxicity.

Therapeutic Exercise

- Exercise prescribed to improve function, reduce pain, prevent disability. Types: stretching, strengthening, aerobic, balance, functional. Progression: low to high intensity.

Therapeutic Hypothermia

- Controlled cooling of the body temperature used in psychiatric practice for severe catatonia and neuroleptic malignant syndrome. In NMS, cooling is a key treatment alongside discontinuation of the offending agent and dantrolene administration.

Therapeutic Index

- The therapeutic index is a quantitative measure of the relative safety of a drug, defined as the ratio of the dose that produces toxicity in 50 percent of the population to the dose that produces the desired therapeutic effect in 50 percent of the population.

Therapeutic Window

- The range of plasma drug concentrations between the minimum effective concentration (producing a therapeutic effect) and the minimum toxic concentration (producing toxicity), in which efficacy is achieved with acceptable toxicity.

Therapeutics

- The branch of medicine concerned with the treatment of disease, encompassing all interventions used to prevent, manage, or cure illness.

Therapy

- Medical intervention for disease. See treatment.

Therapy Monitoring

- PET for early response assessment. Chang criteria in lymphoma.

Thermal Injury

- Tissue damage caused by heat, flame, hot liquid, steam, or cold. Burn severity depends on depth, total body surface area, location, inhalation injury, associated trauma, and patient factors; assessment should include airway and circulatory complications.

Thermal Injury (Burns)

- Tissue damage from heat. Management: stop burning, remove clothing, cool with water (not ice), cover with sterile dressing. Assess depth and extent. Fluid resuscitation for >15% BSA burns. Tetanus prophylaxis, pain control, wound care.

Thermal Therapy

- Heat and cold therapy. Heat: relax muscles, increase circulation. Cold: reduce inflammation, pain.

Thermogenesis

- Heat production from food digestion, muscle activity, brown fat. Diet-induced thermogenesis 10% of TDEE.

Thermometer

- Device measuring body temperature. Types: mercury (phased out), digital, tympanic, temporal artery, infrared. Normal oral: 36.5-37.5°C.

Thermoneutral Zone

- The range of ambient temperatures over which body temperature is maintained by basal metabolic heat production alone, without additional thermoregulatory responses such as shivering or sweating. In humans the zone is approximately 27-32 degrees C for a resting nude adult. Within the zone, vasomotor tone adjusts heat loss.

Thermoplasty

- A bronchoscopic procedure that uses radiofrequency energy to ablate smooth muscle in the airways, reducing bronchospasm. It is approved for severe, persistent asthma in adults not controlled with ICS/LABA. It reduces exacerbations and emergency department visits but does not improve lung function.

Thermoregulation

- The physiological maintenance of core body tem-

perature near the set point (37 degrees C) via a balance of heat production and heat loss, controlled by the preoptic area of the hypothalamus.

Thiamine

- See biochemistry.

Thiamine (Vitamin B1)

- A water-soluble vitamin whose active coenzyme form, thiamine pyrophosphate, is required by pyruvate dehydrogenase, alpha-ketoglutarate dehydrogenase, transketolase, and branched-chain ketoacid dehydrogenase. Deficiency causes beriberi and Wernicke encephalopathy.

Thiazide Diuretics

- Thiazide diuretics act on the distal convoluted tubule by inhibiting the sodium-chloride cotransporter, reducing sodium and chloride reabsorption and producing natriuresis and diuresis. They are less potent than loop diuretics but have a longer duration of action, typically 6 to 12 hours.

Thiazolidinediones

- Insulin sensitizers activating peroxisome proliferator-activated receptors primarily in adipose tissue and skeletal muscle. Agents include pioglitazone and rosiglitazone. Mechanisms include enhanced insulin-mediated glucose uptake, altered adipokine secretion, and reduced hepatic glucose output.

Thiopental Sodium

- An ultra-short-acting barbiturate, used for induction of general anesthesia (IV, onset 30–60 seconds, duration 5–10 minutes) and as a third-line agent for raised intracranial pressure (cerebroprotective effect: reduces cerebral metabolism, cerebral blood flow, and intracranial pressure).

Third Stage of Labor

- From delivery of the baby to delivery of the placenta. The placenta separates from the uterine wall and is expelled spontaneously or with gentle cord traction. Normal duration: 5-30 minutes.

Third Trimester

- The period from 28 weeks to delivery. Rapid fetal weight gain occurs, lungs mature with surfactant production. Fetal movements become regular and countable. Braxton Hicks contractions begin. Group B Streptococcus screening at 36 weeks. Fetal

positioning assessed for delivery. Viability generally considered at 23-24 weeks.

Third-Degree AV Block

- Complete heart block with no conduction between atria and ventricles. Atria and ventricles beat independently (AV dissociation). Ventricular escape rhythm: junctional (40-60, narrow QRS) or ventricular (20-40, wide QRS).

Third-Degree Burn

- Full-thickness burn destroying the epidermis and dermis (into subcutaneous tissue): wound is white, leathery, charred, painless (nerve destruction), and non-blanching; no spontaneous healing—requires skin grafting. Assessment: percent BSA (rule of nines), inhalation injury (facial burns, soot, hoarseness—intubation), and depth.

Thoracentesis

- Removal of pleural fluid via needle insertion through the chest wall. Performed for diagnosis and symptom relief. Ultrasound guidance improves safety.

Thoracic Aortic Aneurysm

- Aneurysm of thoracic aorta. Associated with hypertension, Marfan syndrome, bicuspid aortic valve.

Thoracic Cage

- The bony and cartilaginous framework formed by the thoracic vertebrae, ribs, costal cartilages, and sternum. It protects thoracic organs and assists respiratory movements.

Thoracic Cavity

- The chest cavity, enclosed by the ribs and chest muscles, containing the lungs, heart, esophagus, and trachea.

Thoracic Duct

- Body's largest lymphatic vessel, approximately 40 cm long, draining lymph from the entire body below the diaphragm and left side above it (approximately 75 percent of total lymph).

Thoracic endometriosis

- Endometrial tissue within the thoracic cavity causing catamenial pneumothorax, hemothorax, or lung nodules, recurring cyclically with menstruation and often requiring hormonal suppression.

Thoracic Outlet Syndrome

- Compression of neurovascular bundle at thoracic outlet. Types: neurogenic (most common), venous, arterial. Symptoms: neck/shoulder/arm pain, paresthesia, weakness, vascular symptoms. Diagnosis: clinical, imaging, nerve conduction. Treatment: physical therapy, surgical decompression.

Thoracic Vertebrae (T1-T12)

- Have costal facets on bodies and transverse processes for rib articulation. Heart-shaped bodies, long downward spinous processes, coronally-oriented facets allowing rotation and lateral flexion but limiting flexion/extension. Small circular vertebral foramen.

Thoracolumbar Spine Trauma

- Thoracolumbar trauma: T10-L2 (thoracolumbar junction), L3-L5. AO classification: A (compression), B (distraction), C (translation/rotation). TLICS (Thoracolumbar Injury Classification and Severity): morphology (1-3), PLC (0-2), neurologic (0-3). <4 nonop, >4 op. Neurologic: complete (ASIA A) vs incomplete.

Thoracotomy

- A surgical incision into the chest wall to access the thoracic cavity (pleural space, lungs, heart, great vessels, esophagus, diaphragm, mediastinum).

Thought Blocking

- An abrupt interruption in the train of thought before it is completed, as if the thought was removed or pushed out of the patient's mind. The patient may pause mid-sentence, appear confused, and report that the thought was "taken out" or "blocked." It is a formal thought disorder seen in schizophrenia.

Threadworm (Enterobiasis)

- A common intestinal parasitic infection caused by *Enterobius vermicularis* (pinworm). Prevalence: most common helminth infection in developed countries, primarily affecting children aged 5–10 years. Transmission: fecal-oral (ingestion of eggs from contaminated hands, food, bedding, or fomites).

Three-Dimensional Ultrasound

- Ultrasound creating 3D images. Used in obstetrics, gynecology.

Thrombectomy

- A surgical or interventional radiology procedure

to remove a blood clot (thrombus) from a blood vessel, restoring blood flow. Types: (1) Endovascular thrombectomy (mechanical thrombectomy)—for acute ischemic stroke due to large vessel occlusion (LVO—internal carotid artery, middle cerebral artery M1 segment).

Thrombectomy (Vascular)

- See Thrombectomy Device.

Thromboangiitis Obliterans (Buerger Disease)

- Segmental inflammatory disease of small/medium arteries and veins, strongly associated with smoking. Affects upper and lower extremities. Symptoms: ischemic rest pain, Raynaud, superficial thrombophlebitis, ulceration, gangrene. Treatment: complete smoking cessation (only effective intervention), sympathectomy, amputation for gangrene.

Thrombocytes

- Small, anucleate, disc-shaped cell fragments (2–4 μ m diameter) derived from megakaryocytes in the bone marrow, circulating at 150–400 $\times 10^9$ /L. See Platelets (Thrombocytes).

Thrombocytopathy

- A disorder of platelet function that impairs primary hemostasis despite a normal or near-normal platelet count. It may be inherited or acquired and can cause mucocutaneous bleeding, easy bruising, prolonged bleeding after procedures, or menorrhagia.

Thrombocytopenia

- A reduced platelet count ($<150,000/\mu\text{L}$), causing bleeding risk when severe ($<10\text{--}20,000$).

Thrombocytosis

- An elevated platelet count ($>450,000/\mu\text{L}$). Reactive (secondary): infection, inflammation, iron deficiency, malignancy, post-splenectomy, hemorrhage, exercise—usually mild, no thrombotic risk.

Thromboelastography (TEG)

- A viscoelastic whole-blood assay that measures the initiation, formation, strength, and breakdown of a clot. It provides a functional overview of coagulation and fibrinolysis and may guide targeted blood-component therapy in major bleeding.

Thromboembolism

- See Venous Thrombosis.

Thrombolysis

- Pharmacological dissolution of thrombus. tPA, streptokinase, alteplase. Used for acute arterial occlusion, PE, stroke.

Thrombolysis for PE

- Systemic thrombolytics (alteplase) for massive PE with hemodynamic instability. Rapid clot dissolution but significant bleeding risk.

Thrombolysis for Stroke

- Thrombolysis with intravenous alteplase (tPA) is the primary acute treatment for ischemic stroke when administered within the time window of symptom onset. Alteplase is a recombinant tissue plasminogen activator that converts plasminogen to plasmin, which degrades fibrin in the thrombus, restoring blood flow to the ischemic brain.

Thrombolytic

- A drug that dissolves blood clots by activating plasminogen to plasmin, which degrades fibrin. Agents: alteplase (tPA), tenecteplase (TNK-tPA), reteplase, streptokinase, urokinase. Indications: acute ischemic stroke (within 4.5 hours), STEMI, massive pulmonary embolism, and acute limb ischemia.

Thrombophilia

- A tendency to develop venous thromboembolism (VTE) due to inherited or acquired hypercoagulable states. Inherited: factor V Leiden, prothrombin G20210A mutation, antithrombin deficiency, protein C deficiency, protein S deficiency.

Thrombophilia Testing

- Thrombophilia testing evaluates for inherited and acquired hypercoagulable states in patients with venous thromboembolism, recurrent pregnancy loss, or strong family history of VTE.

Thrombopoiesis

- The production of platelets from megakaryocytes in the bone marrow. It is regulated principally by thrombopoietin and involves megakaryocyte maturation, proplatelet formation, and release of circulating platelets.

Thrombosed intracranial aneurysm

- An intracranial arterial aneurysm that has undergone partial or complete thrombosis of its lumen, which may reduce rupture risk but can cause mass effect on adjacent brain structures. Presents with

headache, cranial nerve palsies, or seizures depending on location. Diagnosis: CT, MRI, or conventional angiography.

Thrombosis

- The formation of a blood clot within the cardiovascular system during life, potentially obstructing blood flow and causing tissue ischemia or infarction. Arterial thrombi typically form on ruptured atherosclerotic plaques, are composed primarily of platelets and fibrin, and appear pale and grossly attached to the vessel wall.

Thrombosis of the inferior vena cava

- Formation of a blood clot within the inferior vena cava, which may occur in the setting of IVC filters, malignancy, hypercoagulable states, or extension from iliofemoral deep vein thrombosis. May cause bilateral lower extremity edema, abdominal pain, and hepatic or renal dysfunction. Requires anticoagulation and possible thrombectomy.

Thrombotic Microangiopathy

- Endothelial damage with platelet-rich microthrombi causing MAHA with schistocytes, thrombocytopenia, and organ ischemia. TTP from ADAMTS13 deficiency treated with plasmapheresis. HUS from Shiga toxin managed supportively. Atypical HUS from complement dysregulation responds to eculizumab.

Thrombotic Thrombocytopenic Purpura

- A life-threatening thrombotic microangiopathy caused by severe deficiency of ADAMTS13 (a disintegrin and metalloproteinase with thrombospondin type 1 motif member 13), a von Willebrand factor-cleaving protease.

Thrombotic Thrombocytopenic Purpura (TTP)

- A life-threatening microangiopathic hemolytic anemia caused by severe deficiency (<10% activity) of ADAMTS13, a plasma metalloprotease that cleaves ultra-large von Willebrand factor (vWF) multimers. Acquired (autoantibody against ADAMTS13) or congenital (Upshaw-Schulman syndrome).

Thrombotic vasculopathy

- A spectrum of disorders characterized by thrombus formation within blood vessels of various sizes, often associated with antiphospholipid syndrome,

disseminated intravascular coagulation, thrombotic thrombocytopenic purpura, or malignancy.

Thrombus Organization

- The process by which a thrombus is gradually replaced by granulation tissue and eventually fibrous scar tissue. The process begins when inflammatory cells, particularly macrophages, infiltrate the thrombus and begin digesting the fibrin and dead cells through fibrinolysis.

Thrush

- A common vaginal infection caused by the yeast *Candida albicans* (also called candidiasis, vaginal thrush, or candida vulvovaginitis). Symptoms: itching, soreness, white discharge (like cottage cheese), dyspareunia, and external dysuria. Risk factors: pregnancy, antibiotic use, diabetes, immunosuppression, and high estrogen states.

Thumb CMC Arthritis

- Thumb carpometacarpal (CMC) arthritis: Eaton-Littler classification: I (normal), II (slight narrowing, osteophytes <2mm), III (narrowing, osteophytes >2mm, subluxation), IV (pantrapezial arthritis). Clinical: base of thumb pain, pinch weakness, thenar wasting, squaring (subluxation).

Thymoma

- A tumor of thymic epithelial cells, associated with myasthenia gravis, pure red cell aplasia, and hypogammaglobulinemia. Malignant behavior is determined by capsular invasion and stage.

Thymus

- Primary lymphoid organ in the anterior mediastinum, largest in childhood, involutes with age. Site of T-cell maturation (thymopoiesis): thymocytes undergo positive and negative selection to produce self-tolerant, MHC-restricted T cells. Epithelial cells produce thymic hormones (thymosin, thymopoietin).

Thyroglobulin Surveillance

- Thyroglobulin produced only by thyroid follicular cells serves as a tumor marker after total thyroidectomy for differentiated thyroid carcinoma. In patients who have undergone complete thyroidectomy and radioactive iodine ablation thyroglobulin should be undetectable.

Thyroglossal Duct Cyst

- Congenital midline neck mass resulting from persistence of the thyroglossal duct the embryological tract of thyroid descent. Most common congenital neck mass presenting in childhood. Located at or near the hyoid bone and moves with swallowing and tongue protrusion. Diagnosis is by ultrasound demonstrating a cystic midline neck mass.

Thyroid Cancer

- Usually indolent. Types: papillary, follicular, medullary, anaplastic.

Thyroid Cancer Staging

- American Joint Committee on Cancer TNM staging for differentiated thyroid carcinoma incorporates tumor size, extent of invasion, lymph node metastases, and distant metastases. Age is a critical prognostic factor with patients under fifty having more favorable outcomes. Stage I is any tumor without metastases in patients under fifty.

Thyroid Cartilage

- The thyroid cartilage is the largest of the laryngeal cartilages, consisting of two laminae that fuse anteriorly at an angle of approximately ninety degrees in males and one hundred twenty degrees in females, creating the laryngeal prominence (commonly known as the Adam's apple), which is more prominent in males due to the acute angle and

Thyroid Disorders

- Conditions affecting the thyroid gland, a butterfly-shaped endocrine organ in the neck that produces hormones regulating metabolism, growth, and development.

Thyroid Eye Disease

- Autoimmune inflammatory disorder of the orbital tissues associated with Graves disease characterized by expansion of extraocular muscles and orbital fat. Pathogenesis involves orbital fibroblasts expressing the TSH receptor targeted by the same thyroid-stimulating immunoglobulins affecting the thyroid.

Thyroid Fine-Needle Aspiration

- Image-guided sampling of thyroid-nodule cells for cytologic classification. Results are commonly reported using the Bethesda system and guide surveillance, molecular testing, surgery, or other management.

Thyroid Follicular Carcinoma

- approximately 10 to 15 percent of thyroid cancers and is the second most common differentiated thyroid malignancy. Unlike papillary carcinoma, follicular carcinoma typically spreads hematogenously to distant sites including bone, lung, and liver, rather than to regional lymph nodes.

Thyroid Function Tests

- Laboratory assessment of the hypothalamic-pituitary-thyroid axis. TSH is the most sensitive screening test with elevated levels indicating primary hypothyroidism and suppressed levels indicating hyperthyroidism. Free T4 measures the biologically active unbound thyroxine.

Thyroid Gland

- An endocrine gland located in the anterior neck, consisting of two lobes (right and left) connected by an isthmus, lying anterior to the trachea between the cricoid cartilage and the suprasternal notch.

Thyroid Hormone

- T3 and T4. Increase basal metabolic rate. Required for normal development.

Thyroid Nodule

- A thyroid nodule is a discrete lesion within the thyroid gland that is palpably or radiographically distinct from the surrounding thyroid parenchyma, and it is a common clinical finding, present in approximately five percent of palpable neck examinations and up to fifty percent of individuals on ultrasound imaging, with the vast majority

Thyroid Nodule Evaluation

- Systematic approach to evaluation of thyroid nodules detected by physical examination or imaging. Thyroid ultrasound characterizes nodule features including size, composition, echogenicity, margins, calcifications, and shape. TI-RADS classification stratifies malignancy risk and guides biopsy decisions.

Thyroid Papillary Carcinoma

- The most common type of thyroid cancer, accounting for approximately 80 percent of all thyroid malignancies, and has an excellent prognosis with over 95 percent 10-year survival.

Thyroid Peroxidase Antibody

- An autoantibody against thyroid peroxidase, an

enzyme required for thyroid-hormone synthesis. It supports autoimmune thyroid disease, especially Hashimoto thyroiditis, but may be present in some euthyroid individuals.

Thyroid Scan

- Nuclear medicine imaging using I-123 or Tc-99m. Evaluates thyroid function and nodules. Hot nodules rarely malignant. Cold nodules require further evaluation.

Thyroid Scintigraphy

- Thyroid imaging using I-123, I-131, or Tc-99m pertechnetate. Assesses function, nodules, ectopic tissue. Hot nodules (functioning) rarely malignant; cold nodules require ultrasound/FNA.

Thyroid Stimulating Hormone

- TSH (thyrotropin): the anterior pituitary hormone that stimulates the thyroid to produce and release T4 and T3, controlled by TRH (hypothalamus) and negative feedback from thyroid hormones.

Thyroid Storm

- Life-threatening exacerbation of thyrotoxicosis presenting with hyperpyrexia, tachycardia, heart failure, altered mental status, and multi-organ dysfunction. Precipitated by infection, surgery, trauma, or iodine load in patients with untreated hyperthyroidism. Mortality rates range from twenty to thirty percent.

Thyroid Surgery

- lobectomy, subtotal thyroidectomy, and total thyroidectomy. Indications: Bethesda V/VI nodules on cytology, thyroid cancer (papillary >1.5 cm, follicular, medullary, anaplastic), Graves disease (compressive symptoms, failed medical therapy, severe ophthalmopathy), and multinodular goiter with compressive symptoms or substernal

Thyroid Ultrasound

- High-frequency sonographic imaging used to evaluate thyroid size, nodules, cysts, cervical lymph nodes, and structural features associated with malignancy. Suspicious nodules may require ultrasound-guided fine-needle aspiration.

Thyroiditis

- Inflammation of the thyroid gland with various etiologies. Hashimoto thyroiditis: autoimmune, most common cause of hypothyroidism in iodine-sufficient

areas. Subacute (De Quervain) thyroiditis: post-viral, presents with painful thyroid and transient hyperthyroidism followed by hypothyroidism.

Thyroiditis Classification

- Inflammatory conditions of the thyroid classified by clinical presentation and etiology. Hashimoto thyroiditis is chronic autoimmune with hypothyroidism. Graves disease is autoimmune with hyperthyroidism. Subacute thyroiditis is painful and likely viral.

Thyrotoxic Periodic Paralysis

- Recurrent episodes of transient flaccid weakness associated with thyrotoxicosis, usually accompanied by hypokalemia caused by intracellular potassium shift. Attacks often affect the proximal limbs and are triggered by rest after exercise or carbohydrate-rich meals.

TI-RADS

- The Thyroid Imaging Reporting and Data System is a standardized ultrasound-based risk stratification system developed by the American College of Radiology to evaluate thyroid nodules and guide recommendations for fine-needle aspiration biopsy, reducing unnecessary biopsies of benignappearing nodules while ensuring adequate sampling of sus

Tibia

- Larger leg bone, the main weight-bearer transmitting body weight from knee to ankle. Proximal: tibial plateau (medial/lateral condyles) and tibial tuberosity (patellar ligament insertion). Shaft: sharp subcutaneous anterior border. Distal: medial malleolus articulating with the talus at the ankle.

Tibial Nerve

- Sciatic division (L4-S3) continuing from posterior thigh through the popliteal fossa into the deep posterior leg, innervating gastrocnemius, soleus, tibialis posterior, flexor digitorum longus, and flexor hallucis longus. Passes behind the medial malleolus as the tarsal tunnel, dividing into medial/lateral plantar nerves for the foot.

Tibial Shaft Fractures

- Tibial shaft fractures: most common long bone fracture. AO/OTA: 42-A (simple), 42-B (wedge), 42-C (complex). Soft tissue critical: Gustilo for open.

Tibialis Anterior Muscle

- Superficial anterior leg muscle from lateral tibial condyle and shaft to medial cuneiform and first metatarsal base. Innervated by deep peroneal nerve (L4-L5). Primary dorsiflexor for foot clearance during gait swing phase; inverts foot. Weakness causes foot drop with steppage gait.

Tibialis Posterior Muscle

- Deepest posterior leg muscle from posterior tibia, fibula, and interosseous membrane to navicular tuberosity, medial cuneiform, and sustentaculum tali. Innervated by tibial nerve (L4-L5). Plantarflexes and inverts foot; primary dynamic medial longitudinal arch stabilizer.

Tic Douloureux

- A chronic pain condition characterized by recurrent, sudden, severe, electric shock-like pain in the distribution of the trigeminal nerve (typically V2/V3). See Trigeminal Neuralgia.

Tick Paralysis

- Acute, progressive flaccid paralysis caused by neurotoxin released by an attached tick. Weakness usually begins in the legs and ascends; respiratory failure can occur and symptoms improve after prompt tick removal.

Tickborne relapsing fever

- A vector-borne illness caused by *Borrelia hermsii* and related species, transmitted by *Ornithodoros* ticks. It presents with recurrent episodes of fever, headache, myalgia, and arthralgia, separated by afebrile intervals. The relapsing pattern results from antigenic variation of the spirochete's surface lipoproteins.

Tics

- Sudden, rapid, recurrent, nonrhythmic stereotyped movements (motor tics) or vocalizations (vocal tics) that are usually brief and suppressible. Transient tics (most common in childhood) vs. chronic tic disorder; Tourette syndrome: multiple motor + at least one vocal tic for >1 year beginning before age 18.

Tidal Volume

- The volume of air inspired or expired during each normal breath, approximately 500 mL (6-8 mL/kg) in a resting adult. Tidal volume is measured by

spirometry. Minute ventilation = tidal volume x respiratory rate (6 L/min). Only a portion reaches the alveoli (alveolar ventilation) after subtracting dead space (150 mL).

Tigecycline

- A glycylicycline antibiotic derived from minocycline with expanded spectrum of activity. Tigecycline inhibits bacterial protein synthesis by binding to the 30S ribosomal subunit. It has activity against gram-positive organisms including MRSA and VRE, gram-negative organisms including ESBL-producing Enterobacteriaceae, and anaerobes.

Tilt Table Test

- Evaluation of unexplained syncope by reproducing orthostatic stress. Patient is strapped to a motorized table tilted from supine to 60-80 degrees for 20-45 minutes. Positive: reproduction of symptoms with hypotension and/or bradycardia (vasovagal syncope, delayed orthostatic hypotension).

Time in Range

- Percentage of time that continuous glucose monitoring readings remain within the target glucose range of seventy to one hundred eighty milligrams per deciliter for most adults with diabetes. General target is greater than seventy percent corresponding to approximately seventeen hours per day.

Time of Death

- The moment or estimated time at which biological death occurred, a key medicolegal question. Estimated from postmortem changes: body cooling (algor mortis), rigor mortis, lividity (livor mortis), decomposition/putrefaction, insect colonization, and chemical changes (vitreous potassium, vitreous glucose, body temperature).

Time-of-Flight PET

- A PET acquisition and reconstruction technique that uses the measured arrival-time difference of coincident annihilation photons to improve localization of the annihilation event. It can improve signal-to-noise performance, especially in larger patients.

Timed Up-and-Go Test

- A functional mobility test measuring the time required to stand from a chair, walk a short distance, turn, return, and sit. Prolonged performance sug-

gests impaired mobility and increased fall risk but must be interpreted with gait, vision, cognition, and environment.

Timothy Syndrome

- A rare multisystem disorder caused by pathogenic variants in *CACNA1C*, characterized by marked QT prolongation and congenital heart disease with syndactyly. Developmental, neurological, immune, and facial features may also occur.

Tin

- Component of dental amalgam. Reduces strength, increases creep.

Tincture

- Alcohol extract of plant or chemical substance. Common concentration: 10-40% alcohol. Examples: tincture of iodine (antiseptic), tincture of benzoin (skin protector). Also used for liquid medications.

Tinea

- Dermatophytosis: superficial fungal infection of keratinized tissue (skin, hair, nails) by dermatophytes (*Trichophyton*, *Microsporum*, *Epidermophyton*).

Tinea (Dermatophytosis)

- Superficial fungal infection by dermatophytes. *Tinea corporis* (ringworm): annular, scaly, erythematous plaque with central clearing. *Tinea pedis* (athlete's foot): interdigital scaling, maceration, and fissuring. *Tinea cruris* (jock itch): erythematous, pruritic rash in groin with raised border.

Tinea barbae

- Dermatophyte infection of the beard and mustache area, presenting with erythematous, scaly, follicular papulopustules or deep inflammatory nodules, often acquired from contact with infected animals or through grooming. It may be inflammatory (kerion-like) or superficial, and differential diagnosis includes bacterial folliculitis and acne.

Tinea Capitis

- A dermatophyte infection of the scalp and hair shafts, most common in children. It may cause scaling, alopecia, broken hairs, black dots, or an inflammatory kerion.

Tinea circinata

- An older term for *tinea corporis* (ringworm) of the body, presenting as annular, erythematous, scaly plaques with active, raised borders and central clear-

ing. It is caused by dermatophyte fungi and may be acquired through direct contact with infected humans, animals, or fomites.

Tinea Corporis

- A dermatophyte infection of glabrous skin, presenting as annular, erythematous, scaly plaques with raised, active borders and central clearing. It is contagious through direct contact with infected persons, animals, or contaminated objects. Diagnosis is by KOH preparation showing branching septate hyphae.

Tinea faciei

- Dermatophyte infection of the facial skin, often presenting with erythematous, scaly, annular or ill-defined plaques that may mimic eczema or rosacea, leading to frequent misdiagnosis. It is commonly caused by *Trichophyton* or *Microsporum* species and may be acquired from pets, other humans, or self-inoculation from other body sites.

Tinea imbricata

- A chronic dermatophyte infection caused by *Trichophyton concentricum*, endemic in the South Pacific and Southeast Asia, characterised by widespread, concentric, overlapping, polycyclic scaly rings producing a "wood-grain" pattern on the trunk and limbs.

Tinea Incognito

- A dermatophyte infection whose clinical appearance has been altered by inappropriate topical corticosteroid or immunosuppressive treatment. It may become less scaly and more extensive or atypical; fungal testing is often needed before directed therapy.

Tinea nigra

- A superficial fungal infection of the stratum corneum caused by *Hortaea werneckii*, presenting as a painless, brown to black, non-scaly macule or patch, most commonly on the palms or soles. It is acquired in tropical and subtropical regions and is often confused with acral melanoma.

Tinea Pedis

- A dermatophyte infection of the feet presenting in interdigital, moccasin, vesiculobullous, or ulcerative patterns. It may coexist with *tinea unguium*. Diagnosis is clinical or mycologic when uncertain;

treatment uses topical antifungals for limited disease and oral therapy for extensive, refractory, or nail-associated infection.

Tinea Versicolor

- A superficial fungal infection caused by *Malassezia* species, presenting as hypo- or hyperpigmented macules with fine scale on the trunk and proximal extremities. It is more noticeable after sun exposure due to unaffected tanning of surrounding skin.

Tinel Sign

- Tapping over nerve produces tingling in distribution. Indicates nerve irritation/regeneration. Used for carpal tunnel, cubital tunnel, tarsal tunnel.

Tinnitus

- The perception of sound in the absence of an external acoustic stimulus, commonly described as ringing, buzzing, hissing, or roaring in the ears, and it affects approximately fifteen to twenty percent of the global population, with a significant impact on quality of life in severe cases.

Tiotropium Bromide

- A long-acting muscarinic antagonist (LAMA) bronchodilator used for maintenance therapy in COPD. Once-daily dosing provides sustained bronchodilation. It reduces exacerbations and improves lung function. It is often combined with a LABA for additive bronchodilation.

TIPS (Transjugular Intrahepatic Portosystemic Shunt)

- A radiologically placed shunt connecting the portal vein to a hepatic vein within the liver, creating a portosystemic shunt to reduce portal pressure. Indications: variceal bleeding (refractory or recurrent), refractory ascites, and hepatic hydrothorax.

Tissue

- A group of similar cells and their surrounding materials that work together to perform a specific function.

Tissue Distribution

- Reversible movement of a chemical from blood into tissues and organs. Distribution is influenced by perfusion, lipid solubility, protein binding, tissue affinity, barriers, body composition, and time after exposure.

Tissue Expander

- Balloon implanted under skin/muscle, gradually filled with saline to expand tissue. Used for breast reconstruction, scalp reconstruction, burn coverage. Removed after expansion, replaced with permanent implant or flap.

Tissue Fixation

- Preservation of tissue morphology and molecular components by chemical or physical methods before processing. Neutral buffered formalin is commonly used for routine histopathology, while fixation choice affects antigen detection, nucleic-acid testing, and tissue architecture.

Tissue plasminogen activator-associated angioedema

- Angioedema occurring as a complication of tissue plasminogen activator therapy for acute ischemic stroke or myocardial infarction, affecting approximately 1 to 5 percent of patients. It typically involves the face, tongue, and upper airway and may be life-threatening.

Titanium

- Biocompatible metal for implants and frameworks. Osseointegration capability.

Titer

- A laboratory measure of the concentration of an antibody (or other substance) in serum, expressed as the highest dilution that still produces a detectable reaction (e.g., 1:128).

TLC

- Thin-layer chromatography for radiochemical purity.

TNF Alpha

- Tumor necrosis factor alpha is a potent pro-inflammatory cytokine produced primarily by activated macrophages, as well as by T cells, NK cells, mast cells, and endothelial cells. TNF-alpha exists as a 26-kilodalton transmembrane protein that is cleaved by the metalloprotease TACE to release a soluble 17-kilodalton active form.

TNM Staging

- Classification of cancer extent. See cancer staging.

Tobacco

- See nicotine.

Tocolysis

- The use of medications to suppress uterine contractions and delay preterm labor, providing time for corticosteroid administration (fetal lung maturity) and in utero transfer to a unit with appropriate neonatal facilities. First-line: nifedipine (calcium channel blocker) or atosiban (oxytocin receptor antagonist).

Tocolytic Therapy

- Medications used to inhibit uterine contractions temporarily to prolong pregnancy, allowing time for corticosteroid administration and transport to a tertiary care center. Indications: preterm labor with intact membranes, 24-34 weeks.

Tocolytics

- Medications used to suppress preterm labor (contractions before 37 weeks gestation), to delay delivery for corticosteroid administration (fetal lung maturity) and in utero transport to a tertiary care center.

Toe Pressure

- Pressure in toe digits. Useful when ankle pressures non-compressible (diabetes, renal disease).

Toe-Brachial Index (TBI)

- The ratio of systolic toe pressure to brachial systolic pressure. It can support diagnosis of peripheral arterial disease when ankle arteries are noncompressible because of medial arterial calcification.

Tolbutamide

- A first-generation sulfonylurea oral hypoglycaemic agent used in the management of type 2 diabetes. Mechanism: stimulates insulin secretion from pancreatic β -cells by binding to the sulfonylurea receptor (SUR1) on ATP-sensitive potassium (KATP) channels, causing membrane depolarisation, calcium influx, and exocytosis of insulin granules.

Tolerance

- A decreased response to a drug after repeated administration, requiring dose escalation to achieve the same effect. Unlike tachyphylaxis, tolerance develops over weeks to months of chronic use.

Toll-Like Receptor (TLR)

- A class of single-pass membrane-spanning pattern recognition receptors (PRRs) expressed on innate immune cells (macrophages, dendritic cells). Rec-

ognize evolutionary conserved microbial structures (PAMPs).

Tongue

- A muscular organ in the mouth that aids in tasting, chewing, swallowing, and speech.

Tongue Cleaning

- Removal of tongue coating using scraper or brush. Reduces bad breath.

Tonic Neck Reflex

- A primitive reflex present from birth to 6 months, also called fencer's pose. When the head is turned to one side, the arm on that side extends while the opposite arm flexes. Present in both directions. Asymmetry may indicate neurologic injury. Disappears as voluntary head control develops.

Tonic-clonic seizure

- A generalized seizure with a tonic stiffening phase followed by rhythmic clonic jerking, impaired consciousness, and a postictal state.

Tonsillar cancer

- A malignancy arising from the palatine tonsils, most commonly squamous cell carcinoma associated with HPV infection. It often presents with throat pain, otalgia, and cervical lymphadenopathy.

Tonsillectomy

- The surgical removal of the palatine tonsils, and it is one of the most commonly performed surgical procedures in children, traditionally indicated for recurrent acute tonsillitis (seven episodes in one year, five per year for two years, or three per year for three years), chronic tonsillitis, and obstructive sleep apnea caused by tonsill

Tonsillitis

- Inflammation of the tonsils, most commonly the palatine tonsils, which are lymphoid tissue structures located in the oropharynx between the palatoglossal and palatopharyngeal.

Tonsils

- Aggregated lymphoid tissue forming Waldeyer's ring at the pharyngeal entrance. Pharyngeal tonsils (adenoids) in the nasopharynx, palatine tonsils at the oropharyngeal pillars, lingual tonsils at the tongue base, and tubal tonsils near the Eustachian tube orifices. They are the first line of defense against inhaled or ingested pathogens.

Tooth

- A mineralized organ composed primarily of enamel, dentin, cementum, and dental pulp. Teeth cut and grind food, contribute to speech and facial support, and are anchored in alveolar bone by the periodontal ligament.

Tooth Eruption

- Process by which tooth moves from intraosseous position to functional occlusion. Primary (deciduous) dentition: 20 teeth, eruption 6-30 months. Permanent dentition: 32 teeth (including third molars), eruption 6-21 years. Delayed eruption may indicate systemic conditions, local obstruction, or normal variant.

Tooth Luxation

- Traumatic displacement of a tooth without complete avulsion. Concussion and subluxation involve periodontal injury with little or no displacement, whereas extrusive, lateral, and intrusive luxation involve increasingly marked displacement and require prompt dental evaluation.

Tooth Replantation

- See Replantation of an Avulsed Tooth.

Tooth Sensibility Testing

- Clinical testing of a tooth's response to thermal, electric, or dentin stimulation. It assesses neural response rather than directly measuring pulp blood flow, so results must be interpreted with history, examination, and imaging.

Tooth Sensitivity

- Pain in response to thermal, sweet, acidic, or tactile stimulus. Caused by dentin exposure (recession, erosion, abrasion, abfraction). Management: desensitizing toothpaste, fluoride varnish, restorative coverage, gum graft for recession.

Tooth Whiteness

- Aesthetic concern driving bleaching. Natural shade varies. Over-bleaching causes sensitivity and translucency.

Toothache

- Common dental symptom indicating underlying pathology. Causes: dental caries, pulpitis (reversible or irreversible), periapical abscess, cracked tooth, periodontal disease, sinusitis (referred pain). Evaluation: history, clinical exam, radiographs. Manage-

ment: treat underlying cause, analgesics, antibiotics if infection present.

Tophaceous gout

- Chronic tophaceous gout represents advanced gout with deposition of large, visible tophi consisting of monosodium urate crystals surrounded by granulomatous inflammation. Tophi typically develop after years of inadequately treated gout and occur on the ears, fingers, toes, and olecranon bursa.

Tophus

- Deposits of monosodium urate crystals in soft tissues, characteristic of chronic gout. Appear as subcutaneous nodules over joints, ears, olecranon, Achilles tendon. Can erode through skin, forming draining sinus tracts. Radiographically: punched-out erosions with overhanging edges.

Topical Antibiotics

- Topical antibiotics treat superficial bacterial skin infections. Mupirocin 2 percent ointment is first-line for impetigo and folliculitis (inhibits bacterial protein synthesis). Retapamulin 1 percent ointment is an alternative. Fusidic acid is used outside the US.

Topical Antifungals

- Topical antifungals treat superficial fungal infections. Allylamines (terbinafine, naftifine) inhibit squalene epoxidase, fungicidal. Azoles (clotrimazole, miconazole, ketoconazole, econazole) inhibit ergosterol synthesis, fungistatic. Ciclopirox has broad-spectrum activity. Tolnaftate is effective against dermatophytes.

Topical Calcineurin Inhibitors

- Non-steroidal anti-inflammatory agents used for atopic dermatitis and facial eczema. They inhibit T-cell activation by blocking calcineurin, preventing cytokine production. Advantages over corticosteroids: no skin atrophy, safe for face and intertriginous areas. Side effects include transient burning and itching at application site.

Topical Corticosteroid Potency

- Topical corticosteroids are classified into seven potency groups. Group 1 (ultra-high): clobetasol propionate 0.05 percent. Group 2 (high): betamethasone dipropionate 0.05 percent. Group 3 to 4 (medium): triamcinolone acetonide 0.1 per-

cent. Group 5 to 6 (low): hydrocortisone 1 percent. Group 7 (lowest): hydrocortisone 0.5 percent.

Topical Retinoids

- Vitamin A derivatives used for acne, photoaging, and hyperpigmentation. Tretinoin (all-trans retinoic acid) is the most studied. Adapalene is more stable and less irritating. Tazarotene is the most potent. Mechanism: normalizes follicular keratinization, increases cell turnover, and stimulates collagen production.

TORCH Panel

- Serologic screening for congenital infections: Toxoplasma, Rubella, CMV, Herpes simplex (and others: syphilis, parvovirus B19, VZV, Zika). IgM: indicates recent/active infection (maternal or fetal). IgG: past infection or passive immunity. Paired maternal and neonatal testing.

Torsades de Pointes

- A specific form of polymorphic ventricular tachycardia with QRS complexes that twist around the baseline, occurring in the setting of QT prolongation. It may degenerate to ventricular fibrillation.

Torsemide

- A potent loop diuretic inhibiting the Na-K-2Cl cotransporter in the loop of Henle. Used for edema in heart failure, cirrhosis, and renal disease, and hypertension. More consistently absorbed than furosemide; long duration; oral-to-IV conversion is nearly 1:1. May have less mortality effect than furosemide in heart failure trials.

Torticollis

- A condition in which the head is held tilted to one side because of neck-muscle spasm or contracture.

Torticollis (Wryneck)

- A condition in which the head is tilted to one side (ear toward the shoulder) and rotated to the opposite side (chin toward opposite shoulder), due to contraction or shortening of the sternocleidomastoid muscle.

Torus mandibularis

- A bony outgrowth arising from the lingual surface of the mandible, typically bilateral and located in the premolar region above the mylohyoid ridge. It is a common, asymptomatic developmental anomaly, more prevalent in women and certain ethnic popu-

lations.

Torus palatinus

- A bony exostosis arising from the midline of the hard palate, typically presenting as a slow-growing, asymptomatic, hard, bony protuberance discovered incidentally. It is more common in women and certain ethnic groups.

Total Anomalous Pulmonary Venous Return

- A cyanotic congenital heart defect where all four pulmonary veins drain into the systemic venous circulation (right atrium or its tributaries) rather than the left atrium. Supracardiac (most common, 45%): pulmonary veins drain via a vertical vein into the left innominate vein or SVC.

Total Body Surface Area

- The percentage of body surface involved by a burn or other extensive skin process. It is estimated using methods such as the rule of nines, Lund-Browder chart, or the patient's palm and is used to guide burn severity assessment and resuscitation.

Total Breech Extraction

- A maneuver where the operator grasps the fetal feet and delivers the entire baby in breech presentation. The most common indication is delivery of the second twin (malpresentation). Contraindicated for term singleton breech due to high risk of head entrapment, nuchal arms, and birth trauma.

Total Daily Energy Expenditure (TDEE)

- Total calories burned in 24 hours: BMR + thermic effect of food + physical activity. Estimation guides dietary recommendations. Activity factor multiplies BMR.

Total Haemolytic Complement Test

- A functional assay of the classical complement pathway, commonly reported as CH50.

Total Hip Arthroplasty

- THA indications: end-stage OA, inflammatory arthritis, AVN, fracture, dysplasia. Approaches: posterior (most common, dislocation risk), direct anterior (DAA, muscle-sparing, fluoroscopy), anterolateral (Watson-Jones), lateral (Hardinge).

Total Hip Arthroplasty (THA)

- Replacement of the hip joint with prosthetic components. Acetabular component: metal shell +

polyethylene liner. Femoral component: metal stem + femoral head (cobalt-chrome or ceramic). Fixation: cemented (PMMA, for elderly, osteoporotic bone), cementless (porous-coated, for younger patients, biologic fixation), and hybrid.

Total Intravenous Anesthesia

- Total intravenous anesthesia (TIVA) uses intravenous agents exclusively for induction and maintenance of anesthesia, avoiding volatile inhalation agents. The most common TIVA regimen combines propofol infusion (50 to 200 mcg/kg/min) with remifentanyl infusion (0.1 to 0.5 mcg/kg/min).

Total Iron Binding Capacity

- TIBC: a measure of the maximum iron-binding capacity of transferrin in serum, reflecting total iron-transport capacity. Calculated from transferrin or measured directly.

Total Knee Arthroplasty

- TKA indications: end-stage OA, inflammatory arthritis, post-traumatic. Components: femoral (cobalt-chrome), tibial (titanium tray + polyethylene insert), patellar (polyethylene, resurfacing vs not). Alignment: mechanical (neutral), kinematic (patient-specific), restricted kinematic. Ligament balancing: measured resection, gap

Total Knee Arthroplasty (TKA)

- Replacement of the knee joint with prosthetic components. Femoral component (cobalt-chrome), tibial component (titanium tray + polyethylene insert), and patellar component (polyethylene, optional). Fixation: cement (PMMA) most common in older adults; cementless in younger.

Total Lesion Glycolysis

- Product of metabolic tumor volume and mean SUV. Prognostic marker.

Total Lung Capacity

- Total lung air volume after maximal inspiration. Measured by body plethysmography. Increased in obstructive disease, decreased in restrictive disease. TLC < 80% confirms restriction.

Total Parenteral Nutrition

- Intravenous administration of nutrients sufficient to meet essentially all nutritional requirements when the gastrointestinal tract cannot be used adequately.

Tourette Disorder

- A neurodevelopmental disorder characterized by multiple motor tics and one or more vocal tics lasting >1 year, with onset before age 18. Motor tics: eye blinking, head jerking, shoulder shrugging, facial grimacing. Vocal tics: throat clearing, sniffing, grunting, barking, coprolalia (obscene words, occurs in <10%).

Tourette Syndrome

- A neurodevelopmental disorder characterized by multiple motor tics and at least one vocal tic occurring over at least one year. Tics are sudden, rapid, recurrent, stereotyped movements or vocalizations. They are often preceded by premonitory urges. Tics typically worsen in adolescence and may improve in adulthood.

Tourniquet

- A device applied proximally to a limb to compress arterial inflow and control life-threatening extremity hemorrhage when direct pressure is inadequate or impractical. Application time, bleeding control, distal perfusion, and the need for definitive hemorrhage management must be documented.

Townes Syndrome

- A congenital disorder caused by pathogenic variants in SALL1, classically involving anal atresia and characteristic thumb abnormalities. Renal, cardiac, hearing, foot, and genital anomalies are common and variable.

Toxemia

- An older term for pre-eclampsia (also called toxemia of pregnancy), reflecting the now-outdated theory that the condition was caused by toxins in the blood. Pre-eclampsia is a multisystem disorder of pregnancy characterized by hypertension and proteinuria developing after 20 weeks gestation, affecting 2-8% of pregnancies.

Toxic Dose

- Dose causing toxicity. Varies by substance, individual, route, duration.

Toxic Encephalopathy

- See toxicology.

Toxic Epidermal Necrolysis

- A severe, life-threatening drug reaction characterized by widespread full-thickness epidermal necrosis

and detachment, affecting >30% of body surface area. It represents a more severe form of Stevens-Johnson syndrome. Common causative drugs include anticonvulsants (carbamazepine, phenytoin), sulfonamides, allopurinol, and NSAIDs.

Toxic Megacolon

- Acute life-threatening dilation of the colon associated with systemic toxicity, representing one of the most serious complications of inflammatory bowel disease, particularly ulcerative colitis. It can also occur in infectious colitis caused by *Clostridioides difficile* and cytomegalovirus.

Toxic Shock Syndrome

- An acute, life-threatening illness caused by bacterial exotoxins. Staphylococcal TSS: associated with tampon use, surgical wounds, and nasal packing; presents with fever, hypotension, diffuse rash, and multi-organ involvement. Streptococcal TSS: associated with invasive GAS infections. Diagnosis: CDC case definition.

Toxicokinetics

- The study of absorption, distribution, metabolism, and excretion of a toxic substance and its relationship to concentration over time. Toxicokinetics helps estimate exposure, timing, accumulation, and expected duration of toxicity.

Toxicology

- Study of poisons and their effects. Branches: clinical, forensic, environmental, occupational, regulatory.

Toxicology Specimen Preservation

- Collection, labeling, storage, and transport of biological specimens to preserve analytes and prevent contamination or degradation. Appropriate specimens may include peripheral blood, urine, vitreous humor, bile, liver, hair, or gastric contents depending on the suspected exposure and timing.

Toxidrome

- A recognizable pattern of signs and symptoms produced by a class of toxins. Important patterns include opioid, sympathomimetic, anticholinergic, cholinergic, sedative-hypnotic, and serotonin toxidromes; treatment combines supportive care, targeted antidotes when available, and decontamination decisions.

Toxin

- A biological poison produced by a microorganism, plant, or animal that damages host tissues at low concentrations.

Toxocariasis

- Infection by dog roundworm (*Toxocara canis*) or cat (*T. cati*), acquired by ingesting infective eggs from soil contaminated by animal feces (pica in children a risk).

Toxoid vaccines

- Vaccines that protect against harmful bacterial toxins. These vaccines contain toxins that have been detoxified and rendered inactive.

Toxoplasma gondii

- An obligate intracellular protozoan parasite that causes toxoplasmosis. The definitive host is the cat (family Felidae), which sheds environmentally resistant oocysts in feces.

Toxoplasmosis

- Infection caused by the intracellular protozoan *Toxoplasma gondii*. Definitive host: cats (shed oocysts in feces). Humans are intermediate hosts, infected via ingestion of undercooked meat (containing tissue cysts), cat feces, or transplacentally. Immunocompetent: usually asymptomatic or mild flu-like illness with lymphadenopathy.

TPMT Polymorphism

- Thiopurine S-methyltransferase is an enzyme that metabolizes thiopurine drugs including azathioprine, 6-mercaptopurine, and thioguanine, which are used in the treatment of leukemia, inflammatory bowel disease, and autoimmune conditions.

Trabeculectomy

- A conventional glaucoma filtration surgery that creates a new drainage pathway for aqueous humor by removing a small piece of the trabecular meshwork and creating a partial-thickness scleral flap under which aqueous humor filters from the anterior chamber into the subconjunctival space, forming a filtering bleb.

Trace Elements

- Minerals needed in small amounts. Zinc, copper, manganese, selenium, chromium, molybdenum, iodine. Deficiency or toxicity possible.

Tracer Principle

- Administered in amounts too small to produce pharmacological effect. Dose optimized for imaging, not therapeutic.

Trachea

- The windpipe, a 10-12 cm fibrocartilaginous tube from the larynx (C6) to the carina (T4/T5) where it bifurcates into main bronchi. Sixteen to twenty C-shaped hyaline cartilage rings (open posteriorly, completed by trachealis muscle) maintain patency.

Tracheal agenesis with bronchoesophageal fistula

- Congenital malformation characterized by absent or severely hypoplastic trachea with an abnormal communication between the bronchus and esophagus, a rare and typically fatal condition presenting in neonates with respiratory distress, cyanosis, and inability to intubate.

Tracheal Foreign Body

- A foreign object lodged in the trachea, causing severe respiratory distress, stridor, and complete airway obstruction. It is a medical emergency requiring immediate bronchoscopic removal. In complete obstruction, emergency cricothyrotomy may be needed.

Tracheal Stenosis

- Narrowing of the trachea, often a complication of prolonged endotracheal intubation or tracheostomy. It presents with progressive dyspnea, stridor, and wheezing that may be mistaken for asthma. Diagnosis is by bronchoscopy or CT. Treatment is balloon dilation, stenting, or surgical resection.

Tracheitis

- Inflammation of the tracheal mucosa. Acute bacterial tracheitis: a rare but life-threatening cause of upper airway obstruction in children, typically presenting as a superinfection following viral croup (parainfluenza).

Tracheobronchomalacia

- Abnormal softening of the trachea and/or bronchi causing excessive airway collapse during breathing. It may be congenital or acquired. Symptoms include chronic cough, recurrent infections, and expiratory airflow obstruction. Diagnosis is by dynamic CT or bronchoscopy. Treatment includes CPAP, stenting,

or surgical reconstruction.

Tracheobronchomegaly

- Abnormal enlargement of the trachea and main bronchi, usually associated with connective-tissue weakness. It may cause recurrent lower-respiratory infection, bronchiectasis, ineffective cough, dyspnea, and difficulty clearing airway secretions.

Tracheoesophageal Fistula

- An abnormal connection between the trachea and esophagus, often associated with esophageal atresia. Presents with recurrent aspiration, coughing with feedings, and cyanosis during meals. The H-type fistula may present later in infancy with chronic aspiration and failure to thrive. Diagnosis by esophagram or bronchoscopy.

Tracheomalacia

- Softening of the tracheal cartilage causing excessive collapse of the trachea during expiration. It may be congenital or acquired (prolonged intubation, relapsing polychondritis). Symptoms include expiratory stridor and difficulty clearing secretions. Diagnosis is by dynamic CT or bronchoscopy.

Tracheostomy

- Creation of a surgical opening from the anterior neck into the trachea with placement of a tube. Indications include prolonged airway support, upper-airway obstruction, and selected secretion-management needs; complications include bleeding, infection, obstruction, displacement, and tracheal injury.

Tracheostomy Tube

- See medical instruments.

Tracheotomy

- Surgical creation of an airway opening in the anterior neck into the trachea. See Tracheostomy.

Trachoma

- A chronic keratoconjunctivitis caused by *Chlamydia trachomatis* (serovars A-C), the leading infectious cause of blindness worldwide. Repeated infection causes conjunctival scarring, trichiasis (in-turned eyelashes), corneal abrasion, opacification, and blindness. Transmission: direct contact, flies, fomites.

Traction

- Pulling force on limb. Types: skin, skeletal. Used

for fractures, deformities.

Tractional retinal detachment

- Separation of the neurosensory retina caused by contraction of fibrovascular membranes, commonly associated with proliferative diabetic retinopathy.

Tractotomy

- Surgical interruption of a nerve tract in the brain or spinal cord.

Train-of-Four Monitoring

- Train-of-four monitoring assesses neuromuscular blockade by delivering four supramaximal electrical stimuli at 2 Hz to a peripheral nerve, commonly the ulnar nerve at the wrist. The ratio of the fourth twitch height to the first twitch height estimates recovery from non-depolarizing blockade.

Trained Immunity

- Innate immune memory: epigenetic/metabolic reprogramming of monocytes/macrophages/NK cells after initial stimulus (BCG, beta-glucan, Candida, oxidized LDL) -> enhanced response to secondary challenge (heterologous protection).

Trajectory

- The path a projectile follows in flight and, in wound ballistics, the course it traverses through tissue from entrance to exit or rest. Internal trajectory is reconstructed from entrance and exit defects, the organ track, recovered fragments, and clothing damage; angles disclose shooter position and victim posture.

TRALI

- Transfusion-Related Acute Lung Injury, a life-threatening complication of blood transfusion characterized by acute hypoxemia and bilateral pulmonary edema within 6 hours of transfusion, without evidence of volume overload.

Tram-Track Appearance

- Histological finding on silver stain of MPGN representing GBM duplication. Results from mesangial cell interposition into capillary wall with new basement membrane formation. Creates double-contour pattern on Jones methenamine silver stain.

Tramadol

- An atypical opioid analgesic (weak mu-agonist plus serotonin and norepinephrine reuptake inhibition) used for moderate pain. Prodrug (M1, via CYP2D6) contributes analgesia; efficacy varies with CYP2D6

genotype.

Tranexamic Acid

- An antifibrinolytic agent that inhibits lysine-binding sites on plasminogen and reduces clot breakdown. It is used in selected trauma, obstetric, cardiac, orthopedic, and bleeding indications according to current protocols. Dose and route depend on the indication and renal function.

Tranquillisers

- A class of drugs that have a calming or sedative effect, reducing anxiety, tension, and agitation. Historically classified as major tranquilisers (antipsychotics/neuroleptics—for psychosis) and minor tranquilisers (anxiolytics—for anxiety). The term is now less common in clinical practice.

Trans Fat

- Artificially hydrogenated fat. Increases cardiovascular risk. Banned in many countries. Read labels.

Transabdominal Scan

- An ultrasound scan performed with the transducer placed on the maternal abdomen. A full bladder improves visualization of pelvic structures. Used in early pregnancy to confirm viability and gestational age, and throughout pregnancy to assess fetal growth, anatomy, amniotic fluid volume, and placental location.

Transarterial Chemoembolization (TACE)

- A procedure in which chemotherapy and embolizing particles are infused selectively into tumor-feeding hepatic arteries, combining regional chemotherapy with ischemic tumor necrosis; used for unresectable hepatocellular carcinoma.

Transbronchial Biopsy

- Bronchoscopic biopsy of lung parenchyma obtained through the bronchial wall. It is used to diagnose interstitial lung diseases (sarcoidosis, HP), reject lung transplant rejection, and sample peripheral lung lesions. It carries a higher risk of pneumothorax than endobronchial biopsy.

Transcervical Foley Catheter for Cervical Ripening

- A mechanical method for cervical ripening: a Foley catheter (16-30 Fr) is inserted through the cervix, the balloon is inflated with 30-60 mL of saline, and

traction is applied (or the catheter is taped to the thigh with gentle tension). The balloon applies direct pressure to the internal os, stimulating endogenous prostaglandin release.

Transcranial Doppler

- Ultrasound measuring blood flow velocity in cerebral arteries. Used for vasospasm detection, stroke risk assessment.

Transcription

- Synthesis of RNA from DNA template by RNA polymerase. Eukaryotic: RNA Pol II transcribes mRNA, rRNA, snRNA; RNA Pol I transcribes rRNA; RNA Pol III transcribes tRNA, 5S rRNA. Steps: initiation (promoter recognition, transcription factors), elongation (5' to 3'), termination. Promoter elements: TATA box, CAAT box, GC box.

Transcription Factor

- Protein binding to specific DNA sequences, regulating transcription. Activators enhance transcription; repressors inhibit. Examples: p53, NF-kB, steroid hormone receptors.

Transdermal

- A route of drug administration delivering drug through intact skin into the systemic circulation via a patch or gel, bypassing first-pass metabolism and providing steady plasma concentrations over hours to days. Examples: nitroglycerin, fentanyl, nicotine, estrogen, clonidine, rotigotine.

Transfer Training

- Teaching safe transfer techniques: bed to chair, chair to toilet, floor to standing. Use of transfer boards, lift equipment.

Transferase

- An enzyme that transfers a functional group from one molecule to another.

Transference

- A psychoanalytic concept in which feelings, attitudes, and expectations originally directed toward a significant figure from childhood are unconsciously redirected onto the therapist. In psychodynamic therapy, transference is a therapeutic tool for understanding the patient's relational patterns and resolving conflicts.

Transferrin

- An iron transport protein synthesized by the liver,

carrying iron in plasma (2 iron-binding sites). In iron deficiency: transferrin is elevated with low ferritin, giving high TIBC and low transferrin saturation; in iron overload/inflammation/chronic disease: transferrin is low (acute phase).

Transfusion Medicine

- Blood components: PRBCs (leukoreduced, irradiated for at-risk), FFP (coagulation factors), platelets (apheresis pooled, leukoreduced), cryoprecipitate (fibrinogen, vWF, factor VIII, XIII). ABO/Rh compatibility.

Transfusion of Neonate

- Administration of packed red blood cells to neonates for symptomatic anemia or hematocrit below transfusion threshold. Indications include acute blood loss, severe anemia (hematocrit less than 20%), and respiratory distress. Volume: 10-20 mL/kg over 2-4 hours. Blood products must be CMV-negative, irradiated, and crossmatched.

Transfusion Reaction

- An adverse reaction to transfusion of blood products. Acute hemolytic transfusion reaction: ABO incompatibility, immediate fever/chills, hypotension, back pain, DIC, renal failure; management: STOP transfusion, maintain BP/urine output, supportive care.

Transient Ischemic Attack

- A transient ischemic attack (TIA) is a temporary episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia without acute infarction.

Transition to Adulthood

- Support for youth with disabilities transitioning to adult services.

Transitional Care

- See geriatrics.

Translation

- Decoding of mRNA sequence into polypeptide chain by ribosomes. Occurs in cytoplasm. tRNAs carry specific amino acids and recognize codons via anticodons. Steps: initiation, elongation, termination. Energy: GTP hydrolysis. Antibiotics targeting translation: tetracyclines (30S), chloramphenicol (50S), erythromycin (50S), streptomycin (30S).

Translocation

- Movement of chromosomal segment to non-homologous chromosome. Reciprocal: exchange between chromosomes. Robertsonian: fusion of two acrocentric chromosomes. Balanced: no genetic material lost; unbalanced: gain or loss.

Transplacental

- Passing through or occurring across the placenta.

Transplant Immunology

- Hyperacute rejection: preformed anti-donor antibodies (ABO, HLA) -> immediate graft failure (minutes-hours); prevented by crossmatch.

Transplant Rejection

- Immune-mediated destruction of a transplanted organ or tissue, classified by timing and mechanism. Hyperacute: preformed donor-specific antibodies, minutes to hours post-transplant, immediate graft thrombosis, irreversible; prevention by cross-matching.

Transplant Surgery

- Transplant surgery involves replacing a failing organ with a healthy donor organ. Kidney transplant: living donors (laparoscopic nephrectomy, left kidney preferred) or deceased donors (standard criteria, expanded criteria, DCD).

Transport

- Regulations for shipping radioactive materials. Package types: excepted, A, B.

Transposition of the Great Arteries

- A cyanotic congenital heart defect where the aorta arises from the right ventricle and the pulmonary artery arises from the left ventricle (ventriculoarterial discordance). The result: two parallel circulations (systemic and pulmonary) with no mixing.

Transposons

- Mobile genetic elements that can move between different locations within the bacterial genome or between the chromosome and plasmids. Transposons carry genes including antibiotic resistance genes and can facilitate the spread of resistance.

Transudate

- A transudate is a type of edema fluid with low protein content and few cells, resulting from imbalances in hydrostatic and oncotic pressures across

capillary walls rather than from inflammation or increased vascular permeability.

Transudative Pleural Effusion

- From systemic factors altering hydrostatic/oncotic pressures. Most commonly from heart failure. Low protein, low LDH. Treatment addresses the underlying cause.

Transurethral Resection of the Prostate (TURP)

- Gold standard surgical procedure for benign prostatic hyperplasia (BPH) refractory to medical management or complicated by urinary retention, renal insufficiency, or recurrent hematuria.

Transverse Colon

- Longest, most mobile colonic segment from hepatic to splenic flexure, approximately 45 cm, with complete mesentery (transverse mesocolon). Crosses anterior to duodenum, pancreas, and major vessels. Splenic flexure is the highest colon point and common gas trapping site. Most common diverticular disease site along with the sigmoid.

Transverse Lie

- A fetal lie where the long axis of the fetus is perpendicular to the long axis of the mother. The presenting part is a shoulder or arm. Occurs in approximately 1% of term pregnancies. Risk of cord prolapse and obstructed labor. Cesarean delivery required as vaginal delivery is not possible.

Transverse Myelitis

- An inflammatory disorder of the spinal cord causing acute or subacute motor, sensory, and autonomic dysfunction.

Transverse Plane

- A horizontal plane that divides the body or an organ into upper (superior) and lower (inferior) portions.

Transversus Abdominis Muscle

- Deepest flat abdominal muscle with horizontal fibers, from inguinal ligament, iliac crest, thoracolumbar fascia, and inner lower six costal cartilages. Innervated by T7-T12 and L1. Compresses abdomen without trunk movement, providing core stability and increased intra-abdominal pressure during coughing, lifting, Valsalva.

Transvestic Disorder

- A paraphilic disorder characterized by recurrent, intense sexual arousal from cross-dressing, lasting 6 months and causing marked distress or impairment. Almost exclusively in heterosexual males. Transvestic disorder is distinguished from gender dysphoria by the primary motivation: sexual arousal (transvestic) vs.

TRAP Sequence

- Twin Reversed Arterial Perfusion sequence, a complication of monochorionic twin gestations where one twin (the pump twin) perfuses the other (the acardiac twin) via arterioarterial anastomoses.

Trapezium

- Saddle-shaped distal row carpal bone on the radial side, articulating with the first metacarpal at the CMC joint of the thumb. This saddle joint allows thumb opposition essential for grip and fine motor function. Basal joint arthritis is common in postmenopausal women, causing pinch grip pain and weakness.

Trapezius Muscle

- Large triangular muscle from occipital bone and C7-T12 spinous processes to clavicle, acromion, and scapular spine. Innervated by CN XI and C3-C4. Upper fibers elevate scapula, middle fibers retract, lower fibers depress. Essential for shoulder stability and scapulothoracic motion.

Trapezoid

- Smallest distal row carpal bone between trapezium and capitate, articulating with the second metacarpal. Wedge shape provides stability for the firmly fixed second metacarpal. Fractures rare (less than 1 percent of carpal fractures). Contributes to the radial column wrist stability.

Trauma

- Physical injury from external force. Primary survey: ABCDE (Airway, Breathing, Circulation, Disability, Exposure). Secondary survey: head-to-toe examination, history (AMPLE), diagnostic tests. Injuries: fractures, dislocations, internal bleeding, organ damage. ATLS protocol guides management.

Trauma (Vascular)

- See vascular trauma.

Trauma Center

- Hospital designated to provide comprehensive care for injured patients. Levels: Level I (complete care, research, prevention), Level II (similar but limited resources), Level III (stabilization, transfer), Level IV (advanced trauma life support, transfer). Transfer criteria: need for higher level of care, specific injuries.

Trauma Surgery

- Trauma surgery encompasses the acute care of injured patients, guided by ATLS (ABCDE) principles: Airway maintenance with cervical spine protection, Breathing and ventilation, Circulation with hemorrhage control, Disability/neurologic assessment, and Exposure/environmental control.

Traumatic bleb

- A filtering bleb formed after glaucoma filtration surgery (trabeculectomy) that develops complications from trauma, or a thin-walled, elevated bleb that leaks or is at risk of rupture. Traumatic bleb failure or overfiltering can lead to hypotony, choroidal detachment, and endophthalmitis risk.

Traumatic Brain Injury

- Disruption of brain function caused by an external mechanical force. Severity: mild (GCS 13-15, concussion), moderate (GCS 9-12), severe (GCS 3-8). Primary injury: direct damage at moment of impact (contusion, diffuse axonal injury, intracranial hemorrhage).

Traumatic spondylolisthesis of the axis (hangman's fracture)

- A bilateral fracture of the pars interarticularis of the C2 vertebra (axis), typically caused by hyperextension of the head. Classically associated with judicial hanging but also occurs in motor vehicle accidents and falls. May present with neck pain, torticollis, or neurological deficits if the spinal cord is compressed.

Travel medicine

- The branch of medicine dealing with health problems associated with international travel. Pre-travel consultation includes vaccinations, malaria prophylaxis, and travel advice. Common travel-related illnesses include traveller's diarrhea, malaria, and hepatitis A.

Treacher Collins Syndrome

- An autosomal dominant craniofacial disorder caused by mutations in TCOF1 (most common), POLR1C, or POLR1D genes, encoding ribosomal RNA transcription factors.

Treat-to-Target Strategy

- A management approach in which treatment is adjusted at regular intervals toward a predefined goal such as remission or low disease activity. It requires validated disease-activity measures, shared decisions, monitoring, and timely escalation or modification.

Treatment

- Medical intervention for disease. Includes pharmacological, surgical, lifestyle, supportive.

Treatment-Resistant Depression

- See psychiatry.

Tremor

- Involuntary, rhythmic oscillatory movement of a body part, classified by context: resting tremor (Parkinson disease—'pill-rolling', reduced with action), action/kinetic tremor (cerebellar, essential tremor—bilateral postural/action tremor, worsened by holding position), postural tremor (essential tremor, physiologic tremor, hyperthyroidis

Trephining (Trepanation)

- A surgical procedure in which a hole is drilled or scraped into the skull (cranium), exposing the dura mater. Historically, trephining dates back to Neolithic times (10,000 BCE) and was performed for ritual, therapeutic (releasing 'evil spirits'), and post-traumatic indications.

Treponema pallidum

- A spirochete that causes syphilis, a sexually transmitted infection. The organism is transmitted through sexual contact or transplacentally (congenital syphilis). Primary syphilis presents with a painless genital ulcer (chancre). Secondary syphilis presents with rash, mucous patches, and condylomata lata.

Treponema pallidum infection

- See Sexually Transmitted Infections (STIs).

Trepopnea

- Dyspnea that occurs in one lateral decubitus posi-

tion but not the other. It is typically worse when lying on the side of the affected lung. It may be caused by unilateral lung disease, pleural effusion, or mediastinal mass.

Triage

- The rapid sorting of patients according to urgency, clinical risk, and expected resource needs so that those requiring immediate intervention are prioritized. Triage is dynamic and must be repeated when a patient's condition changes.

Trial of Labor After Cesarean (TOLAC)

- The attempt at vaginal birth in a woman with a prior cesarean, which may result in VBAC or a failed TOLAC (requiring repeat cesarean). Candidates: one prior low-transverse cesarean, clinically adequate pelvis, no contraindications. Counseling: success rate 60-80%, uterine rupture risk 0.5-0.9%, perinatal mortality from rupture <0.1%.

Triceps Brachii Muscle

- Three-headed posterior arm muscle. Long head from infraglenoid tubercle, lateral head from posterior humerus above radial groove, medial head below. All insert on the olecranon. Innervated by radial nerve (C7-C8). Primary elbow extensor; assists shoulder extension/adduction. Triceps reflex (C7) tests radial nerve/spinal cord integrity.

Trichinosis

- A parasitic infection caused by *Trichinella spiralis*, acquired by eating undercooked pork or wild game (bear, boar) containing encysted larvae. Larvae are released in the intestine (enteric phase: diarrhea, abdominal pain), then migrate to skeletal muscle (parenteral phase: myositis, periorbital edema, fever, eosinophilia).

Trichobezoar

- See Gastric Bezoar.

Trichomonas

- A protozoan parasite, *Trichomonas vaginalis*, causing trichomoniasis—the most common non-viral sexually transmitted infection worldwide (156 million cases annually). *T. vaginalis* is a flagellated, motile protozoan that infects the urogenital tract. Transmission: sexual intercourse. Incubation: 4–28 days.

Trichomoniasis

- A sexually transmitted infection caused by the pro-

tozoan *Trichomonas vaginalis*. Often asymptomatic in men; women present with vaginal discharge (frothy, yellow-green, malodorous), itching, dysuria, and inflammation. Associated with preterm birth and increased HIV acquisition risk. Diagnosis: wet mount microscopy, culture, NAAT.

Trichothiodystrophy Syndrome

- A group of inherited disorders characterized by brittle sulfur-deficient hair with a distinctive banded appearance, often accompanied by photosensitivity, ichthyosis, developmental delay, short stature, or recurrent infection.

Trichotillomania

- Recurrent pulling out of one's hair, resulting in hair loss. The individual has made repeated attempts to decrease or stop hair pulling. The hair pulling causes clinically significant distress or impairment. It is not attributable to another medical condition.

Tricuspid Atresia

- Absence of the tricuspid valve, resulting in no direct communication between the right atrium and right ventricle. Blood flow depends on an atrial septal defect and ventricular septal defect for mixing. Presents with cyanosis.

Tricuspid Regurgitation

- Incomplete closure of the tricuspid valve allowing blood to flow backward from the right ventricle to the right atrium during systole. Causes include right ventricular dilation (from pulmonary hypertension or left heart failure), endocarditis, rheumatic disease, and carcinoid syndrome.

Tricuspid Stenosis

- Narrowing of the tricuspid valve opening, most commonly caused by rheumatic heart disease. It is often associated with mitral stenosis. Symptoms include fatigue, edema, and hepatic congestion. Murmur: low-pitched mid-diastolic rumble at the left lower sternal border that increases with inspiration.

Tricuspid Valve

- Right atrioventricular valve with anterior, posterior, and septal cusps supported by chordae tendineae and three papillary muscles. Largest cardiac valve. Most commonly affected in IV drug use-associated endocarditis. Tricuspid regurgitation from right ventricular dilation, endocarditis, or carcinoid. Stenosis

is rare, mostly rheumatic.

Tricyclic Antidepressant Overdose

- Anticholinergic effects, cardiac toxicity (QRS prolongation, arrhythmias), seizures, hypotension. Treatment: sodium bicarbonate (QRS >100 ms), seizure control, supportive care.

Tricyclic Antidepressants

- Older antidepressants that non-selectively inhibit the reuptake of serotonin and norepinephrine. They include amitriptyline, nortriptyline, imipramine, clomipramine, and desipramine.

Trigeminal Nerve

- Cranial nerve V (CN V), the largest cranial nerve, providing sensory innervation to the face and anterior scalp (ophthalmic V%1, maxillary V%2, mandibular V%3 divisions) and motor innervation to the muscles of mastication (masseter, temporalis, medial and lateral pterygoids, tensor veli palatini, tensor tympani, mylohyoid, anterior belly

Trigeminal Neuralgia

- A chronic pain condition characterized by recurrent, sudden, severe, electric shock-like pain in the distribution of the trigeminal nerve (typically V2/V3). Often triggered by light touch, chewing, or brushing teeth. Most cases are caused by vascular compression of the trigeminal nerve root.

Trigger Finger

- Trigger finger (stenosing tenosynovitis): A1 pulley thickening, flexor tendon nodule -> catching/locking. Clinical: triggering, palpable nodule at distal palmar crease, morning stiffness. Grades: I (pain only), II (catching, active extension), III (catching, passive extension), IV (locked flexed).

Triglyceride

- See biochemistry.

Trimethoprim-sulfamethoxazole

- A fixed-dose combination of trimethoprim and sulfamethoxazole that synergistically inhibits folate synthesis. TMP-SMX is indicated for urinary tract infections, *Pneumocystis jirovecii* pneumonia, and community-acquired pneumonia. The combination is bactericidal against many gram-positive and gram-negative organisms.

Trimethylaminuria

- A metabolic disorder in which impaired oxidation

of trimethylamine causes a characteristic fish-like body odor in sweat, urine, and breath. It may be inherited or acquired and can cause substantial psychosocial distress despite limited physical illness.

trinucleotide repeat

- Expansion of three-nucleotide sequences causes diseases: Huntington (CAG), Fragile X (CGG), myotonic dystrophy (CTG), Friedreich ataxia (GAA). Anticipation common.

Tripe palms

- Diffuse, velvety, thickened hyperkeratosis of the palmar skin with a cerebriform, corrugated appearance resembling the lining of a cow's stomach (tripe), which is strongly associated with underlying malignancy, most commonly gastric adenocarcinoma. It may occur alone or in conjunction with acanthosis nigricans.

Triple-Negative Breast Cancer

- Breast cancer lacking ER, PR, HER2. Aggressive, limited treatment options.

Triplets

- A multiple pregnancy with three fetuses (trichorionic, dichorionic, or monochorionic combinations); commonly results from assisted reproductive technology. High-risk: preterm delivery (average 32-33 weeks), low birth weight, preeclampsia, gestational diabetes, growth restriction, and complications of monochorionicity (TTTS).

Triploidy Syndrome

- A chromosomal condition in which cells contain three complete haploid chromosome sets. It causes severe fetal growth restriction and multiple congenital anomalies and is usually incompatible with prolonged survival; partial or mosaic forms vary.

Triptan

- A class of medications used for acute migraine treatment, acting as 5-HT_{1B/1D} receptor agonists. 5-HT_{1B} receptors are located on cranial blood vessels; activation causes vasoconstriction of dilated meningeal arteries.

Triquetrum

- Pyramidal proximal row carpal bone on the ulnar side, articulating with lunate, hamate, pisiform, and TFCC. Second most commonly fractured carpal bone. Contributes to proximal carpal row and ulnar

wrist stability. Pisiform articulates with its volar surface.

Trisomy

- Presence of three copies of a chromosome instead of two. Trisomy 21 (Down syndrome): intellectual disability, characteristic facies, heart defects. Trisomy 18 (Edwards): severe intellectual disability, multiple anomalies. Trisomy 13 (Patau): severe intellectual disability, holoprosencephaly, cleft lip/palate.

Trisomy 21

- Down syndrome: the most common chromosomal disorder, caused by an extra chromosome 21 (trisomy 21; 95 percent meiotic nondisjunction, mosaicism, translocation).

Trophoderm

- The outer cellular layer of a blastocyst that contributes to formation of the placenta.

Tropical Sprue

- Chronic diarrhea and malabsorption syndrome of uncertain etiology occurring predominantly in tropical and subtropical regions. The condition is characterized by villous atrophy of the small intestinal mucosa leading to malabsorption of nutrients including folate, iron, and vitamin B12.

Troponin

- Cardiac-specific troponin I and T, the preferred biomarkers for myocardial injury (MI). Highly sensitive and specific. Rise 2-4 hours after symptom onset, peak at 12-24 hours, remain elevated for 7-10 days. High-sensitivity troponin (hs-cTn) allows earlier detection (1-2 hours).

Troponins

- Cardiac-specific proteins (troponin I and T) released into the bloodstream during myocardial cell injury. They are the most sensitive and specific biomarkers for myocardial infarction. High-sensitivity troponin assays can detect very low levels, allowing early diagnosis.

Trousseau Sign of Latent Tetany

- A sensitive and specific clinical sign of hypocalcemia. Elicited by inflating a blood pressure cuff on the upper arm to a pressure greater than systemic systolic blood pressure for 3 minutes.

Trousseau Syndrome

- Recurrent or migratory venous thrombosis associ-

ated with active malignancy, particularly mucin-producing adenocarcinoma. It reflects tumor-associated activation of coagulation and is distinct from Trousseau sign of latent tetany.

True Ribs

- Ribs 1-7 articulating directly with the sternum via their own costal cartilages. Each has a head, neck, tubercle, and shaft. The first two are short and strongly curved for neck/shoulder muscle attachment. Costal cartilages provide thoracic cage elasticity for respiratory expansion.

Truncus Arteriosus

- A congenital heart defect where a single large vessel arises from both ventricles, receiving a mixed blood supply. Associated with a large ventricular septal defect and abnormal semilunar valve. Presents with cyanosis and heart failure early in life.

Trypanosoma brucei

- A protozoan parasite that causes African trypanosomiasis (sleeping sickness), transmitted by the bite of the tsetse fly (*Glossina* species). Two subspecies cause human disease: *T. b. gambiense* (West African form, chronic, accounting for over 95% of reported cases) and *T. b. rhodesiense* (East African form, acute).

Trypanosomiasis

- Two forms: African trypanosomiasis (sleeping sickness, transmitted by tsetse fly, caused by *Trypanosoma brucei gambiense* – chronic West African form, or *T. b. rhodesiense* – acute East African form) and American trypanosomiasis (Chagas disease, transmitted by triatomine bugs, caused by *Trypanosoma cruzi*).

TSH-Receptor Antibody

- An antibody that binds the thyrotropin receptor. Stimulating antibodies cause Graves disease, while blocking antibodies can cause hypothyroidism; measurement supports diagnosis and can help assess fetal or neonatal risk in pregnancy.

Tubal Ligation

- Surgical sterilization by occluding the fallopian tubes, preventing the ovum from reaching the uterus. Postpartum tubal ligation: after vaginal or cesarean delivery (Pomeroy technique, Parkland). Interval: laparoscopic (bands, clips, bipolar cautery) or mini-

laparotomy (Hulka or Filshie clip).

Tubal Pregnancy

- An ectopic pregnancy implanted in the fallopian tube, accounting for >95% of ectopic pregnancies. Most common site: ampulla (70%). Risk factors: prior ectopic, PID (chlamydia), tubal surgery, smoking, IUD, and ART. Presents with lower abdominal pain, amenorrhea, and vaginal bleeding.

Tuberculin Skin Test

- Intradermal injection of purified protein derivative (PPD) to test for *Mycobacterium tuberculosis* infection. Read at 48-72 hours measuring induration (not erythema). Positive: greater than 5 mm in HIV or close contacts; greater than 10 mm in high-risk groups; greater than 15 mm in low-risk populations.

Tuberculin Skin Test (PPD)

- Intradermal injection of 0.1 mL of purified protein derivative (PPD) on the forearm (Mantoux method). Read at 48-72 hours measuring induration (not erythema).

Tuberculoid leprosy

- The paucibacillary pole of the leprosy spectrum characterised by strong cell-mediated immunity to *Mycobacterium leprae*, resulting in well-formed granulomas and few or no organisms in tissue.

Tuberculosis

- See Tuberculosis (TB).

Tuberculosis (TB)

- A chronic infection caused by *Mycobacterium tuberculosis*, transmitted by airborne droplets; pulmonary TB is most common, extrapulmonary forms occur (lymph node, pleural, meningeal, miliary, skeletal, renal). Features: chronic cough (>3 weeks), hemoptysis, fever, night sweats, weight loss; Ghon complex, apical cavitation on imaging.

Tuberculosis in Children

- Infection with *Mycobacterium tuberculosis*, often from household contacts. Latent TB infection (LTBI) positive tuberculin skin test or interferon-gamma release assay with no symptoms or radiographic findings. Active TB presents with cough, fever, weight loss, and lymphadenopathy. Diagnosis by PPD, chest X-ray, and sputum culture.

Tuberous sclerosis

- An autosomal dominant neurocutaneous disorder.

der caused by mutations in TSC1 (hamartin) or TSC2 (tuberin) genes, leading to benign tumor growth in multiple organs. Neurological manifestations include cortical tubers, subependymal nodules, subependymal giant cell astrocytomas (SEGAs), and epilepsy (often infantile spasms).

Tuberous Sclerosis Complex

- An autosomal dominant genetic disorder caused by mutations in TSC1 (hamartin) or TSC2 (tuberin) genes, leading to mTOR pathway activation. Features: cortical tubers, subependymal nodules, cardiac rhabdomyomas, angiomyolipomas, facial angiofibromas, and seizures (often infantile spasms).

Tubography

- Contrast imaging of fallopian tubes. Evaluates patency.

Tubular Reabsorption

- The process by which the renal tubules return filtered water and solutes from the tubular lumen to the blood, reclaiming 99 percent of the glomerular filtrate. Proximal tubule: reabsorbs 65 percent of sodium, most glucose, amino acids, water, bicarbonate, phosphate, and urate.

Tubular Secretion

- The transport of substances from the peritubular capillary blood into the tubular lumen, a second mechanism (besides filtration) by which the kidney eliminates waste and regulates ions.

Tubuloglomerular Feedback

- A feedback mechanism at the juxtaglomerular apparatus (JGA) that adjusts GFR based on tubular fluid NaCl concentration. The macula densa cells (specialized tubular epithelial cells in the thick ascending limb) sense NaCl concentration via the NKCC2 cotransporter.

Tularemia

- A zoonotic bacterial infection caused by *Francisella tularensis*, transmitted through tick bites, handling infected animals (rabbits, rodents), contaminated water, or aerosolization. Ulceroglandular (most common): ulcer at inoculation site + tender lymphadenopathy.

Tumor

- Abnormal mass of tissue, may be benign or malignant. Tumors classified by tissue of origin. Ter-

minology: -oma usually benign, but exceptions (melanoma, lymphoma, seminoma are malignant).

Tumor Board

- A multidisciplinary conference of oncologists, surgeons, radiologists, and pathologists who review individual cancer cases to coordinate diagnosis and treatment planning.

Tumor Burden

- The total amount of cancer in the body, estimated from the number, size, and extent of tumor deposits; higher burden is generally associated with worse prognosis.

Tumor Differentiation

- The degree to which neoplastic cells morphologically and functionally resemble their normal counterparts of origin.

Tumor Embolism

- Tumor embolism occurs when malignant cells, tumor fragments, or tumor thrombi enter the venous circulation and travel to distant sites. The pulmonary vasculature is the most common site of embolization, as venous blood from most organs drains to the lungs through the pulmonary arteries.

Tumor Grade

- A histologic measure of how much tumor cells resemble normal tissue; low-grade (well-differentiated) tumors behave less aggressively than high-grade (poorly differentiated) tumors. Grading is a prognostic, not staging, feature.

Tumor Grading

- A description of a tumor based on how abnormal the cancer cells look under a microscope compared to normal cells; a lower grade indicates slower-growing cancer.

Tumor Heterogeneity

- The presence of genetically, epigenetically, or phenotypically distinct subpopulations within the same tumor or between primary and metastatic sites. Heterogeneity can influence prognosis, sampling, treatment response, and resistance.

Tumor Lysis Syndrome

- Metabolic emergency from rapid cell death. Hyperuricemia, hyperkalemia, hyperphosphatemia, hypocalcemia. Prevention: hydration, allopurinol, rasburicase.

Tumor Marker

- Substance produced by cancer or body in response to cancer. Used for diagnosis, monitoring, prognosis. Examples: PSA (prostate), CA-125 (ovarian), CEA (colon), AFP (liver), CA 19-9 (pancreas).

Tumor Markers

- Substance produced by cancer or body in response to cancer. See tumor marker.

Tumor Microenvironment

- Surrounding tissue influencing tumor growth. Includes immune cells, blood vessels, fibroblasts, signaling molecules.

Tumor Mutational Burden (TMB)

- The number of somatic mutations per unit of coding DNA in a tumor specimen. TMB may correlate with production of neoantigens and response to some immune checkpoint inhibitors, but interpretation requires a validated assay and disease-specific context.

Tumor Necrosis Factor

- Pro-inflammatory cytokine. Insulin resistance in obesity.

Tumor Necrosis Factor (TNF)

- A proinflammatory cytokine (TNF- α) produced mainly by macrophages, acting on TNFR1/TNFR2 receptors. Functions: activation of endothelium and leukocytes, fever, cachexia, apoptosis, and coordination of inflammation and the acute-phase response.

Tumor Necrosis Factor Inhibitor

- TNF inhibitor (biologic): monoclonal antibodies/fusion proteins blocking TNF- α —infliximab, adalimumab (monoclonals), etanercept (soluble receptor), certolizumab, golimumab. Used in rheumatoid arthritis (after methotrexate failure), ankylosing spondylitis, psoriatic arthritis, psoriasis, IBD (Crohn, UC).

Tumor Staging

- A system of describing the size of a cancer and how far it has spread throughout the body from its original location (e.g., using the TNM system).

Tumor Suppressor

- Gene that prevents uncontrolled cell growth. See tumor suppressor gene.

Tumor Suppressor Gene

- Gene that prevents uncontrolled cell growth. Examples: TP53, RB1, BRCA1, BRCA2, APC. Loss of function contributes to cancer.

Tumor Suppressor Genes

- Tumor suppressor genes encode proteins that normally inhibit cell proliferation, promote DNA repair, or induce programmed cell death in response to cellular damage. They function as brakes on the cell cycle, and loss of their function removes critical checkpoints that normally prevent uncontrolled growth.

Tumor-Suppressor Gene

- A gene whose normal product restrains cell proliferation, promotes DNA repair, induces apoptosis, or maintains genomic stability. Loss or inactivation of both functional copies is common in the development of many cancers, although mechanisms vary by gene.

Tungiasis

- See Parasitic Infections.

Tunica Adventitia

- The outer connective-tissue layer of a blood-vessel wall. It contains collagen and elastic fibers that provide structural support, along with nerves and, in larger vessels, small vessels called vasa vasorum that supply the outer wall. It is relatively prominent in veins and helps anchor vessels to surrounding tissues.

Tunica Media

- The middle layer of a blood-vessel wall, located between the tunica intima and tunica adventitia. It consists primarily of smooth muscle and elastic fibers; its contraction or relaxation changes vascular diameter, regulating blood pressure and tissue perfusion. It is thickest in arteries and thinner in veins.

Turbinates

- The nasal turbinates, also known as conchae, are three pairs of scroll-like bony projections on the lateral wall of the nasal cavity that are covered by a highly vascularized respiratory mucosa, and they function to increase the surface area available for air conditioning, create turbulent airflow for optimal filtration and

Turbinates Hypertrophy

- Chronic enlargement of the nasal turbinates, most commonly the inferior turbinates, resulting in nasal airway obstruction, which can be caused by chronic inflammation from allergic rhinitis, non-allergic rhinitis, chronic sinusitis, hormonal changes, or vasomotor dysfunction.

Turner Hypoplasia

- Enamel defect on permanent tooth from infection of overlying primary tooth. Localized hypoplasia.

Turner Syndrome

- Chromosomal abnormality in females with complete or partial absence of one X chromosome most commonly with a karyotype of 45,X occurring in one in two thousand five hundred live female births.

Twin-to-Twin Transfusion Syndrome

- See complete definition in the Multiple Gestation section above.

Twin-Twin Transfusion Syndrome

- A complication of monochorionic diamniotic twin pregnancies where placental vascular anastomoses (arteriovenous connections) cause unbalanced blood flow from one twin (the donor) to the other (the recipient). The donor twin becomes hypovolemic, anemic, and growth-restricted with oligohydramnios ("stuck twin").

Tylenol

- See acetaminophen.

Tympanic Cavity

- See Middle Ear.

Tympanic hyperectasis

- Abnormal outward bulging or retraction of the tympanic membrane from chronic middle ear pressure changes, potentially causing conductive hearing loss and requiring tympanostomy tube placement.

Tympanic membrane

- The tympanic membrane (eardrum) is a thin, cone-shaped, translucent membrane at the medial end of the external auditory canal that separates the external ear from the middle ear and serves as the interface for sound transmission by converting acoustic energy into mechanical vibrations of the ossicular chain.

Tympanometry

- An objective, quantitative test that measures the compliance (mobility) of the tympanic membrane and the pressure within the middle ear as a function of air pressure in the external auditory canal, providing information about middle ear function that supplements otoscopic examination.

Tympanostomy Tubes

- Small tubes inserted through the tympanic membrane to ventilate the middle ear and prevent fluid accumulation. Indicated for recurrent acute otitis media (3 episodes in 6 months or 4 in 12 months) or persistent otitis media with effusion greater than 3 months with hearing loss. Performed as outpatient procedure under general anesthesia.

Type 1 Diabetes Mellitus

- Type 1 diabetes is an autoimmune disease in which the immune system attacks and destroys the insulin-producing beta cells of the pancreas, resulting in absolute insulin deficiency.

Type 2 Diabetes Mellitus

- Type 2 diabetes is a chronic metabolic disorder characterized by insulin resistance and progressive pancreatic beta-cell dysfunction, leading to hyperglycemia.

Type I Hypersensitivity

- An immediate (anaphylactic) hypersensitivity reaction mediated by allergen-specific IgE bound to mast cells and basophils; re-exposure cross-links IgE, triggering degranulation and release of histamine and mediators within minutes. Manifestations: allergic rhinitis, asthma, urticaria, food allergy, and anaphylaxis.

Type II Hypersensitivity

- An antibody-mediated (cytotoxic) hypersensitivity reaction in which IgG or IgM binds to cell-surface or matrix antigens, causing cell destruction by complement activation (MAC), opsonization/phagocytosis, or antibody-dependent cellular cytotoxicity.

Type III Hypersensitivity

- An immune complex-mediated hypersensitivity reaction in which circulating antigen-antibody complexes deposit in tissues and vessels, activating complement and attracting neutrophils, causing inflammation and tissue damage. Local forms: Arthus

reaction, serum sickness.

Type III hypersensitivity reaction

- See Hypersensitivity Type III.

Type IV Hypersensitivity

- A delayed-type hypersensitivity (cell-mediated) reaction mediated by sensitized T cells (not antibody), occurring 24-72 hours after antigen exposure.

Typhlitis

- Inflammation of the cecum in immunocompromised patients particularly those with neutropenia from chemotherapy. Also known as neutropenic enterocolitis. Presents with right lower quadrant pain, fever, and diarrhea. Diagnosis is by CT showing cecal wall thickening.

Typhoid

- See infectious disease.

Typhoid Fever

- A systemic infection caused by *Salmonella typhi*, acquired through contaminated food or water. Incubation 7-14 days. Presents with prolonged fever, headache, abdominal pain, rose spots, and relative bradycardia. Diagnosis: blood culture (first week), stool culture (second week), Widal test.

Typhus

- A group of acute febrile illnesses caused by *Rickettsia* species, transmitted by arthropod vectors. Three main forms: (1) Epidemic (louse-borne) typhus—*Rickettsia prowazekii*, transmitted by the human body louse (*Pediculus humanus corporis*). Associated with overcrowding, poverty, cold climate, and refugees ('war typhus').

Typical Antipsychotics

- Typical antipsychotics, also known as first-generation antipsychotics, primarily block dopamine D2 receptors in the mesolimbic pathway, reducing positive symptoms of schizophrenia including hallucinations, delusions, and disorganized thinking.

Tyramine

- See pharmacology.

Tyrosine Kinase Inhibitor

- A class of targeted cancer therapy drugs that inhibit tyrosine kinases, enzymes that phosphorylate tyrosine residues on proteins and activate signaling pathways involved in cell growth, proliferation,

differentiation, and survival. Tyrosine kinase mutations or overexpression drive many cancers.

Tyrosinemia

- An autosomal recessive inborn error of tyrosine metabolism.

Tachyarrhythmia

- An abnormally fast heart rhythm, which may arise from the atria, ventricles, or the heart's conduction system. Symptoms can include palpitations, dizziness, chest discomfort, shortness of breath, or fainting; fainting or severe symptoms require urgent assessment.

Tarsal Coalition

- An abnormal connection between two bones in the rear or middle foot, usually present from birth. It can cause a stiff, painful flatfoot, recurrent ankle sprains, and limited foot motion, often becoming more noticeable during adolescence.

Throat Cancer

- Cancer arising in the pharynx or nearby structures of the throat. Persistent hoarseness, difficulty or pain with swallowing, a neck lump, unexplained weight loss, or persistent sore throat requires medical evaluation, especially in people who use tobacco or alcohol.

Thrombophlebitis

- Inflammation of a vein associated with a blood clot, commonly affecting a superficial vein in the leg. It may cause a tender, red, firm cord under the skin. Deep-vein involvement can lead to pulmonary embolism and requires prompt medical assessment.

Tooth Infection

- Infection of the tooth, its pulp, or surrounding tissues, commonly caused by untreated decay, cracks, or gum disease. It may cause toothache, sensitivity, swelling, fever, or drainage. Facial swelling, difficulty swallowing or breathing, or spreading infection requires emergency assessment.

Chapter 21

Pocket Dictionary: U Terms

U Wave

- A small deflection following the T wave, best seen in precordial leads V2-V3. U waves are thought to represent repolarization of the Purkinje fibers or papillary muscles. Prominent U waves are associated with hypokalemia, bradycardia, and certain drugs (digitalis, quinidine).

Ubiquinone

- Lipid-soluble electron carrier in mitochondrial membrane. Also called coenzyme Q.

Uhl Anomaly

- A rare congenital absence or severe deficiency of right-ventricular myocardium, leaving a thin-walled right ventricle. It may cause right-sided heart failure, arrhythmia, cyanosis, or sudden deterioration in infancy or childhood.

Ulcer

- Full-thickness loss of epidermis extending into the dermis or deeper tissue, often healing with a scar. Causes include vascular disease, pressure, infection, neuropathy, inflammatory disease, trauma, and malignancy.

Ulcer (Vascular)

- See arterial ulcer, venous ulcer, diabetic foot ulcer.

Ulcerative Colitis

- Chronic inflammatory disease limited to the colon and rectum, characterized by continuous mucosal and submucosal inflammation that invariably begins in the rectum and extends proximally in an uninterrupted pattern.

Ulna

- Medial forearm bone, longer than the radius. Proximal end: trochlear notch, olecranon, coronoid process. Shaft. Distal end: head, styloid process. The

trochlear notch articulates with the trochlea forming the primary elbow hinge. Bears approximately 20 percent of wrist axial load.

Ulnar

- Relating to the ulna bone of the forearm or the ulnar side (medial side) of the hand. Ulnar nerve compression causes paresthesia in medial one and a half fingers.

Ulnar Artery

- The larger terminal branch of the brachial artery at the cubital fossa, running medially down the forearm to the wrist (palpable medial to the pisiform) and hand. Supplies the medial forearm and ulnar side of the hand; forms the superficial palmar arch. Provides the common interosseous artery.

Ulnar Hypoplasia

- Underdevelopment or partial absence of the ulna, usually presenting as a congenital forearm and hand anomaly. It may cause forearm shortening, radial deviation of the hand, limited elbow or wrist movement, and associated digital or systemic abnormalities.

Ulnar Nerve

- From medial cord (C8-T1), passing behind the medial epicondyle (cubital tunnel), into the forearm innervating flexor carpi ulnaris and medial FDP half. In the hand innervates hypothenar muscles, interossei, medial two lumbricals, and adductor pollicis. Sensation to medial 1.5 digits.

Ulnar Nerve Dysfunction

- Impairment of the ulnar nerve, commonly caused by compression at the elbow (cubital tunnel) or wrist (Guyon canal). Symptoms include numbness or tingling in the little and ring fingers, weakened grip, and hand-muscle wasting; severe lesions may

produce an ulnar claw.

Ultrasonography

- See Ultrasound (Ultrasonography).

Ultrasound

- Imaging using high-frequency sound waves (2-18 MHz). No ionizing radiation. Right upper quadrant (RUQ): gallbladder (gallstones, cholecystitis), liver (steatosis, cysts), kidneys, and pancreas. Pelvic: uterus, ovaries, adnexa. Renal: hydronephrosis, stones, cysts, tumors. Scrotal: testicular torsion, epididymitis, varicocele, hydrocele.

Ultrasound Contrast

- Substance enhancing image visibility. See contrast agent.

Ultrasound Therapy

- Therapeutic ultrasound for tissue healing, pain, scar tissue.

Ultrasound Transducer

- A device that emits and receives high-frequency sound waves to generate diagnostic images or guide procedures. Linear, curvilinear, phased-array, and endocavitary transducers are selected according to depth and anatomic application.

Umbilical Artery Doppler

- Ultrasound assessment of blood flow in the umbilical arteries to evaluate placental resistance. Normal flow shows a pulsatility index decreasing throughout pregnancy with forward diastolic flow. Absent or reversed end-diastolic flow indicates severe placental dysfunction and is associated with increased perinatal morbidity and mortality.

Umbilical Cord

- The connection between the fetus and the placenta, containing two umbilical arteries (carrying deoxygenated blood from the fetus to the placenta) and one umbilical vein (carrying oxygenated blood from the placenta to the fetus). The vessels are embedded in Wharton jelly, a mucopolysaccharide matrix that prevents compression.

Umbilical Cord Blood Gas

- Analysis of arterial and venous cord blood collected immediately after delivery. Umbilical artery (UA): reflects fetal (baby's) acid-base status. Normal UA pH >7.20, base excess < -10. Umbilical vein (UV): reflects maternal/placental status.

Umbilical Cord Prolapse

- Descent of the umbilical cord through the cervix alongside or past the presenting fetal part, occurring when the membranes rupture. Risk factors: malpresentation (transverse lie, breech), preterm/low birth weight, polyhydramnios, multiparity, and artificial ROM with high presenting part.

Umbilical Hernia

- A hernia through the umbilical ring, common in infants (most umbilical hernias close spontaneously by age 4-5) and adults (risk factors: obesity, pregnancy, ascites, cirrhosis). Presents as a reducible bulge at the umbilicus, worse with increased intra-abdominal pressure. In adults: risk of incarceration and strangulation is higher.

Umbilical Region

- Central region around the umbilicus containing small intestine, transverse colon, and great vessels. The umbilicus is a landmark for assessing ascites, caput medusae in portal hypertension, and umbilical hernias. Percussion and auscultation here are essential components of the abdominal examination.

Umbilication

- Formation of a central depression or navel-like indentation in a lesion or structure.

Umbilicus (Navel)

- The scar on the anterior abdominal wall marking the site of attachment of the umbilical cord in fetal life, located at the level of the L3-L4 intervertebral disc, approximately midway between the xiphoid process and the pubic symphysis.

Uncompetitive Inhibition

- A type of enzyme inhibition where the inhibitor binds only to the enzyme-substrate (ES) complex at a site distinct from the active site, forming an inactive ESI complex. The inhibitor cannot bind to the free enzyme.

Unconsciousness

- See Coma.

Underweight

- BMI < 18.5 kg/m². Associated with osteoporosis, infertility, anemia, weakened immunity, surgical complications. Causes: hypermetabolism, malabsorption, eating disorders, malignancy, chronic infection.

Undescended Testis

- Cryptorchidism: failure of one or both testes to descend into the scrotum, the most common congenital male genital anomaly (preterm 30 percent, term 3-4 percent); often descends spontaneously in the first months. Complications: impaired fertility, increased risk of testicular cancer, torsion, and hernia.

Undescended Testis (Cryptorchidism)

- Congenital failure of one or both testes to descend along the normal path from the retroperitoneum into the scrotum during fetal development. Most common congenital anomaly of the male genitalia, affecting 3% of full-term and up to 30% of premature male infants. Spontaneous descent usually occurs within the first 3-4 months of life.

Undulant Fever

- A zoonotic infectious disease caused by gram-negative coccobacilli of the genus *Brucella*, transmitted to humans through contact with infected animals (cattle, goats, pigs, sheep) or consumption of unpasteurized dairy products. See Brucellosis.

Unguentum

- A semi-solid, greasy topical preparation for application to the skin, mucous membranes, or eyes. See Ointment.

Uniparental Disomy

- Inheritance of both copies of a chromosome or chromosome segment from one parent and none from the other. It can cause disease through abnormal genomic imprinting, homozygosity for a recessive variant, or chromosomal dosage effects.

Unresponsive Coma

- See coma.

Unstable Angina

- Myocardial ischemia at rest, new-onset severe angina, or a crescendo pattern (more frequent, longer, lower-threshold) without evidence of myocardial necrosis (normal troponin). Pathophysiology: acute plaque rupture with thrombosis causing transient severe coronary obstruction.

Upper Airway Obstruction

- Narrowing of the airway above the carina, causing inspiratory stridor, dyspnea, and characteristic changes on flow-volume loops (flattening of the in-

spiratory limb). Causes include tumors, stenosis, goiter, and vocal cord dysfunction. It is a medical emergency when severe.

Upper Endoscopy

- Upper endoscopy: flexible endoscopic examination of the esophagus, stomach, and duodenum. See Esophagogastroduodenoscopy (EGD).

Upper Extremity Venous Doppler

- Ultrasound assessing upper extremity DVT. Common in catheter-related thrombosis.

Upper Gastrointestinal Bleeding

- Bleeding from esophagus, stomach, or duodenum. Causes: peptic ulcer, varices, Mallory-Weiss tear, gastritis. Presentation: hematemesis, melena, hematochezia. Management: ABCs, IV access, type and crossmatch, PPI infusion, endoscopy within 24 hours. Variceal bleeding: octreotide, antibiotics, band ligation.

Upper GI Series

- Fluoroscopic examination of the esophagus, stomach, and duodenum after oral barium contrast. Single-contrast: looks for gross abnormalities (obstruction, masses). Double-contrast (barium + gas): better mucosal detail for ulcers, erosions, and early malignancies.

Upper Motor Neuron (UMN) Syndrome

- A pattern caused by injury to corticospinal pathways above the anterior horn cell or cranial motor nucleus. Findings may include weakness with spasticity, hyperreflexia, clonus, and an extensor plantar response, although acute lesions can initially produce flaccidity.

Upright Labor Positions

- Maternal positions during labor that use gravity to promote fetal descent: walking, standing, squatting, kneeling, and sitting upright on a birth stool. Benefits of upright vs. recumbent positions: shorter first stage (1 hour), less epidural use, fewer operative deliveries, less perineal trauma, and less severe pain.

Uptake

- Cellular accumulation of radiotracer. Depends on blood flow, membrane transport, metabolism, binding.

Urachal Cyst

- A cystic remnant of the embryonic urachus between

the bladder and umbilicus. It may be asymptomatic or become infected, causing lower-abdominal pain, fever, umbilical discharge, or urinary symptoms.

Urachal mucinous cystic tumor

- A rare tumor arising from the urachal remnant between the bladder dome and umbilicus. It may be benign or malignant, with mucinous histology, and requires surgical excision.

Uraemia

- The clinical syndrome resulting from accumulation of urea and other nitrogenous waste products in the blood in advanced chronic kidney disease (CKD stage 4-5) and acute kidney injury.

Urate-Lowering Therapy

- Long-term treatment that reduces serum urate to dissolve monosodium urate deposits and prevent gout flares and structural damage. Xanthine oxidase inhibitors and uricosuric medicines are selected according to renal function, comorbidity, and treatment goals.

Urea Cycle

- A metabolic pathway occurring in both the mitochondria and cytoplasm of hepatocytes that converts toxic ammonia (NH_3), derived from amino acid catabolism, into nontoxic, water-soluble urea for excretion by the kidneys.

Urea Cycle Disorder

- A group of inborn errors of ammonia metabolism caused by deficiency of any of the six enzymes or two transporters in the urea cycle. X-linked: ornithine transcarbamylase deficiency (OTC, most common, 50%).

Uremia

- The clinical syndrome of advanced renal failure resulting from the accumulation of nitrogenous waste products and toxins (urea, creatinine, others), usually with $\text{GFR} < 15\text{--}20 \text{ mL/min}$ (CKD stage 5).

Uremic Encephalopathy

- Neurological syndrome in severe renal failure. Confusion, asterixis, seizures, coma. Results from uremic toxin accumulation. Diagnosis is clinical in context of advanced renal failure. Emergent dialysis indicated with rapid neurological improvement typically.

Uremic Frost

- Urea crystallization on skin in severe untreated uremia. Rare finding indicating extremely advanced uremia. White crystalline deposits on skin. Indicates immediate need for dialysis. Extremely rare in developed countries due to early detection.

Uremic Pericarditis

- Pericardial inflammation in advanced uremia. Pleuritic chest pain, friction rub, diffuse ST elevation. Indicates need for emergent dialysis. Does not respond to anti-inflammatory therapy. Complications include tamponade. Emergent hemodialysis produces resolution within 2 to 3 weeks.

Uremic Platelet Dysfunction

- Impaired platelet adhesion and aggregation in uremia causing bleeding tendency. Easy bruising, epistaxis, GI bleeding. Improves with dialysis and desmopressin. Cryoprecipitate provides temporary benefit before procedures.

Uremic stomatitis

- Oral mucosal inflammation occurring in advanced kidney failure, presenting as white plaques, ulceration, or bleeding of the gums and tongue due to accumulation of uremic toxins.

Ureter

- The paired muscular tubes (25-30 cm) conveying urine from the renal pelvis to the urinary bladder, crossing the pelvic brim at the pelvic inlet. Three anatomical narrowings (pelviureteric junction, pelvic brim, vesicoureteric junction) are common sites of ureteric stone impaction.

Ureter Duplex

- A congenital duplication of the ureter, complete or partial, draining one kidney. It may be asymptomatic or associated with vesicoureteral reflux, ureterocele, obstruction, recurrent urinary infection, or impaired renal drainage.

Ureteral Obstruction

- Blockage of urine flow causing proximal dilation and increased pressure. Acute causes renal colic with flank pain. Chronic leads to hydronephrosis and CKD. Emergent decompression with stent or nephrostomy for infected obstructed kidney. Diagnosis by CT or ultrasound showing hydronephrosis.

Ureteral Stent

- Double-J stent maintaining ureteral patency. Indications: obstruction (stone, stricture, malignancy), post-ureteroscopy. Materials: silicone, polyurethane. Sizes: 4.7-7 Fr, lengths 24-30 cm. Complications: irritation, hematuria, encrustation, migration.

Ureteral Stent (Double-J Stent)

- A tube placed in the ureter from the renal pelvis to the bladder to maintain patency, with pigtail coils at both ends (hence double-J). Indications: ureteral obstruction (stones, stricture, malignancy), after ureteroscopy (prevent post-op obstruction), and as a pre-treatment for complex ureteral stones.

Ureteral Stricture

- Narrowing of the ureter caused by fibrosis, surgery, instrumentation, radiation, infection, malignancy, or congenital disease. It can produce upstream hydronephrosis, pain, infection, and declining renal function.

Ureteritis

- Inflammation of the ureter, usually related to infection, obstruction, stone passage, instrumentation, or adjacent inflammatory disease. It may cause flank or abdominal pain, urinary symptoms, hematuria, and fever.

Ureterocele

- A cystic dilatation of the distal ureter within the bladder wall or bladder lumen. It can obstruct urine flow and cause urinary infection, hydronephrosis, reflux, stone formation, or impaired renal function.

Ureterolithiasis

- The presence of kidney stones (calculi) within the ureter, causing ureteral obstruction and renal colic. Stones form from calcium oxalate/phosphate (most), uric acid, struvite (infection), and cystine.

Ureteropelvic Junction Obstruction (UPJO)

- Impaired drainage from the renal pelvis into the proximal ureter, caused by intrinsic narrowing, aberrant vessels, scarring, or functional peristaltic abnormality. It may cause hydronephrosis, flank pain, infection, stones, or loss of renal function.

Ureteroscopy

- Endoscopic procedure utilizing a thin, flexible or

semirigid ureteroscope passed through the urethra, bladder, and into the ureter or renal collecting system. Used primarily for the diagnostic evaluation and therapeutic management of ureteral and renal calculi, ureteral strictures, and upper tract urothelial carcinoma (UTUC).

Urethra

- The tube conveying urine from the bladder to the exterior. Male urethra (18-20 cm): prostatic, membranous, and spongy (penile) parts; carries urine and semen; longer and more tortuous than female. Female urethra (3-4 cm): short, straight, predisposing to ascending urinary tract infections.

Urethracele

- A herniation or prolapse of the urethra into the anterior vaginal wall, often occurring together with a cystocele (bladder prolapse). Urethracele is caused by weakening of the pubocervical fascia, often from childbirth, aging, or increased intra-abdominal pressure.

Urethral Diverticulum

- A periurethral outpouching that communicates with the urethral lumen, usually arising from infected or obstructed periurethral glands. It may cause dysuria, post-void dribbling, recurrent urinary infection, dyspareunia, or a palpable anterior-vaginal-wall mass.

Urethral Meatus

- External opening of urethra. Position abnormal in hypospadias, epispadias.

Urethral Pressure Profile

- Measurement of urethral closure pressure along urethra. Used in incontinence evaluation.

Urethral Pressures

- Measurement of urethral closure pressure along urethra. See urethral pressure profile.

Urethral Sling

- Synthetic or fascial tape supporting urethra for stress incontinence. Mid-urethral sling (TVT, TOT) standard. Minimally invasive. Complications: erosion, urgency, retention.

Urethral Stricture

- Narrowing of the urethra due to fibrosis of the corpus spongiosum, most commonly caused by trauma (pelvic fracture, straddle injury, iatrogenic from

catheterisation or urethral instrumentation), infection (gonococcal urethritis leading to stricture years later, now less common in the antibiotic era), and lichen sclerosus (Balanitis Xerotica)

Urethritis

- Inflammation of the urethra, most commonly caused by sexually transmitted infections (*Chlamydia trachomatis*, *Neisseria gonorrhoeae*) or non-infectious causes. Presents with dysuria, urethral discharge, and meatal itching. Diagnosis: urine nucleic acid amplification testing (NAAT).

Urethrography

- Contrast imaging of urethra. Evaluates urethral strictures, trauma, anomalies.

Urge Urinary Incontinence

- Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being. See urinary incontinence.

Urgency

- A sudden, compelling desire to urinate that is difficult to defer—a lower urinary tract symptom. Causes: overactive bladder (detrusor overactivity, often idiopathic), urinary tract infection (urgency + dysuria), bladder outlet obstruction (BPH), neurogenic bladder, interstitial cystitis, and pelvic floor disorders.

Uric Acid Nephropathy

- Acute kidney injury from uric acid crystal precipitation in tubules. Occurs in tumor lysis syndrome and severe hyperuricemia. Prevent with hydration and rasburicase. Alkalinization of urine may be harmful as uric acid precipitates at alkaline pH unlike other stone types.

Uric Acid Stones

- 5 to 10% of kidney stones. Form in acidic urine with pH less than 5.5 and hyperuricosuria. Risk factors include gout, high purine diet, and chronic diarrhea. Radiolucent but visible on CT. Treat with urine alkalinization with potassium citrate and allopurinol. May dissolve with alkalinization which is unique among stone types.

Urinal

- Male bedside container.

Urinalysis

- Analysis of urine. Dipstick: specific gravity, pH,

leukocyte esterase (WBCs), nitrite (bacteria), protein, glucose, ketones, bilirubin, urobilinogen, blood, and hemoglobin.

Urinary Bladder

- A hollow, distensible muscular organ located in the pelvic cavity posterior to the pubic symphysis, serving as a reservoir for urine.

Urinary Incontinence

- Involuntary leakage of urine, significantly affecting psychological, social, and physical well-being.

Urinary Retention

- Inability to completely empty the bladder, which can be acute (sudden, painful, inability to void) or chronic (gradual, often painless, with overflow incontinence). Causes: benign prostatic hyperplasia, urethral stricture, neurogenic bladder, medications (anticholinergics), and pelvic organ prolapse.

Urinary Retraining

- See urology.

Urinary Tract Infection

- Infection of the urinary tract, classified as lower (cystitis, urethritis) or upper (pyelonephritis). Most common pathogen: *Escherichia coli* (80%). Cystitis: dysuria, urinary frequency, urgency, suprapubic pain. Pyelonephritis: fever, flank pain, nausea/vomiting, costovertebral angle tenderness.

Urinary Tract Infection (UTI)

- Inflammatory response of the urothelium to microbial invasion, ranging from lower tract infection (cystitis, urethritis) to upper tract infection (pyelonephritis, renal abscess).

Urinary Tract Infection in Pregnancy

- Asymptomatic bacteriuria (2-10% of pregnancies) and symptomatic UTI (cystitis, pyelonephritis). Pregnancy increases the risk of pyelonephritis due to hormonally mediated ureteral dilatation and stasis.

Urinary Tract Infections

- See Urinary Tract Infection.

Urinary Tract Infections (UTIs)

- See Urinary Tract Infection.

Urination

- The biological process of releasing urine from the bladder through the urethra out of the body; also known as micturition.

Urine Albuminuria

- Abnormal urinary loss of albumin resulting from increased glomerular permeability or impaired tubular reclamation. Persistent albuminuria is independently associated with CKD progression and cardiovascular risk, even when eGFR is preserved.

Urine Analysis

- See nephrology.

Urine Drug Screen

- Immunoassay-based screening for common drugs of abuse in urine. Detects: amphetamines, benzodiazepines, cocaine (benzoylecgonine), opiates (morphine, codeine, heroin), oxycodone, methadone, fentanyl, barbiturates, PCP, THC (cannabinoids), and alcohol. Pros: rapid (minutes at POC), inexpensive.

Urine Formation

- The process by which the kidney produces urine through three steps: glomerular filtration (formation of 180 L/day ultrafiltrate), tubular reabsorption (return of 99 percent of water and most solutes), and tubular secretion (active addition of waste and ions).

Urine hCG (Pregnancy Test)

- Rapid point-of-care test detecting human chorionic gonadotropin in urine. Sensitivity: detects as low as 25 mIU/mL. Positive as early as 12-14 days after conception (before missed period). Results in 3-5 minutes. False negative: dilute urine, very early pregnancy, ectopic pregnancy.

Urine Osmolality

- Concentration of urine relative to plasma normally 50 to 1200 mOsm/kg. High greater than 500 in prerenal states reflecting intact concentrating ability. Low less than 350 indicates impaired concentrating ability in diabetes insipidus, chronic kidney disease, and excessive fluid intake. More precise than specific gravity.

Urine Sediment

- Microscopic examination of centrifuged urine providing diagnostic information about renal pathology. Normal findings include few RBCs, WBCs, and hyaline casts.

Urine Sodium

- Concentration of sodium in urine measured in mEq/L. Low less than 20 in prerenal states reflect-

ing avid tubular reabsorption. High greater than 40 in intrinsic renal disease, diuretic use, and salt-wasting nephropathies. Should be interpreted in context of volume status, medications, and concurrent therapies.

Urine Toxicology

- Urine, blood, hair testing for substances of abuse. See drug screening.

Urodynamic Testing

- Suite of diagnostic procedures designed to evaluate the physiological function of the lower urinary tract (bladder and urethral sphincter) during filling, storage, and emptying phases.

Urodynamics

- A set of tests that assess the function of the lower urinary tract (bladder and urethra).

Uroflow

- Measurement of urine flow rate. See uroflowmetry.

Uroflowmetry

- Measurement of urine flow rate. Parameters: max flow rate (Qmax), mean flow rate, flow time, pattern. Normal Qmax: >15 mL/s. Reduced in obstruction, weak detrusor.

Urolithiasis

- Stone formation in the urinary tract (kidneys, ureters, bladder, urethra). Types: calcium oxalate (most common, 60-70%), calcium phosphate (15-20%), struvite (infection-related, 10-15%), uric acid (5-10%), and cystine (1-2%).

Urologic Diagnostic Yield

- The proportion of urologic diagnostic procedures or tests that provide a definitive and clinically useful result.

Urology

- Surgical and medical specialty concerned with the study, diagnosis, and management of diseases affecting the male and female urinary tract (kidneys, ureters, urinary bladder, urethra) and the male reproductive system (prostate, testes, epididymis, seminal vesicles, penis).

Urothelial Carcinoma

- Urothelial carcinoma, previously called transitional cell carcinoma, is the most common malignancy of the urinary tract, arising from the urothelium that

lines the renal pelvis, ureters, bladder, and proximal urethra.

Urothelium

- The specialized transitional epithelium lining the renal pelvis, ureters, bladder, and part of the urethra.

Urticaria

- See Hives (Urticaria). Urticaria may occur with angioedema, which involves deeper dermal and subcutaneous swelling, often affecting the lips, eyelids, tongue, or extremities.

Urticaria multiforme

- A hypersensitivity reaction in children, often triggered by infections or medications, characterised by widespread, annular or polycyclic erythematous plaques with dusky, ecchymotic, or targetoid centres and pseudovesication, differing from true urticaria by the presence of dusky discolouration and prolonged lesion duration.

Urticaria Pigmentosa

- A cutaneous form of mastocytosis in which excessive mast cells accumulate in the skin, producing tan or brown macules or papules that may swell and itch when rubbed. Flushing, abdominal symptoms, or systemic involvement can occur.

Usher Syndrome

- An inherited disorder combining sensorineural hearing loss with retinitis pigmentosa, the most common cause of combined deaf-blindness. Three types are distinguished by severity and age of onset; management requires coordinated audiological, ophthalmological, and genetic care.

Usual Interstitial Pneumonia

- The histopathological pattern in IPF with temporal heterogeneity, fibroblast foci, and honeycombing. HRCT shows basal-predominant, subpleural reticular opacities with traction bronchiectasis.

Uterine Atony

- Failure of the uterus to contract adequately after delivery, resulting in postpartum hemorrhage. The most common cause of PPH. Risk factors include prolonged labor, multiple gestation, macrosomia, oxytocin augmentation, and uterine fibroids. Diagnosis is clinical: boggy, soft uterus with excessive bleeding.

Uterine Balloon Tamponade

- A minimally invasive technique for managing postpartum hemorrhage due to uterine atony when uterotonics fail. A balloon (Bakri Postpartum Balloon, 500-1000 mL capacity) is inserted transvaginally into the uterine cavity and inflated with saline until bleeding stops. The balloon provides mechanical compression against the uterine wall.

Uterine Fibroid (Leiomyoma)

- Benign smooth muscle tumors of the uterus, very common (70-80% of women by age 50). Submucosal: most symptomatic, cause heavy bleeding. Intramural: most common. Subserosal: cause pressure symptoms. Symptoms: menorrhagia, pelvic pain/pressure, urinary frequency, and reproductive dysfunction (infertility, miscarriage).

Uterine Fibroids

- Benign smooth muscle tumors of the uterus, very common (70-80% of women by age 50). See Uterine Fibroid (Leiomyoma).

Uterine Inversion

- A rare but life-threatening obstetric emergency where the uterus turns inside out, protruding through the cervix or vaginal introitus. Occurs in 1:2000-1:20,000 deliveries. Causes: fundal pressure, excessive cord traction (especially with fundal placenta), uterine atony, and fundal fibroids.

Uterine perforation

- A full-thickness breach of the uterine wall, most commonly occurring during instrumentation such as dilation and curettage or hysteroscopy, potentially causing hemorrhage and requiring surgical repair.

Uterine Prolapse

- Descent of the uterus into the vaginal canal due to weakening of the uterosacral/cardinal ligaments and pelvic floor, most commonly after childbirth and menopause. Grades 1-3 based on descent. Risk: vaginal delivery, obesity, aging, chronic straining, heavy lifting.

Uterine Rupture

- A full-thickness tear through the uterine wall, potentially life-threatening to both mother and fetus. Risk factors include prior caesarean section (especially classical incision), oxytocin augmentation, and operative vaginal delivery. Presents with sudden ab-

dominal pain, loss of fetal heart rate, and maternal hemorrhage.

Uterine Tachysystole

- More than five uterine contractions in 10 minutes averaged over 30 minutes, whether or not accompanied by fetal heart-rate abnormalities. It can reduce uteroplacental perfusion, particularly when caused by oxytocin or prostaglandins.

Uterotonic

- A medication that stimulates uterine contraction, used for induction/augmentation of labor, prevention and treatment of postpartum hemorrhage, and medical termination of pregnancy.

Uterus

- Hollow muscular organ in the pelvis between the bladder and rectum, approximately 7.5 cm long in adult females, responsible for menstruation, implantation, and fetal development during pregnancy.

UTI

- Infection of the urinary tract, classified as lower (cystitis, urethritis) or upper (pyelonephritis). See Urinary Tract Infection.

UV Light

- Skin reaction from UV light + chemical. See phototoxicity.

Uvea

- The middle vascular coat of the eye, consisting of three parts from anterior to posterior: the iris (the coloured part of the eye, controlling pupil size), the ciliary body (producing aqueous humour and controlling lens shape via the zonular fibers), and the choroid (the vascular layer supplying the outer retina, between the sclera and th

Uveal melanoma

- The most common primary intraocular malignancy in adults, arising from melanocytes within the uveal tract (choroid, ciliary body, or iris), presenting with visual symptoms, a pigmented intraocular mass, or incidental finding on examination.

Uveitis

- An inflammatory condition of the uveal tract, which includes the iris, ciliary body, and choroid, and it is classified anatomically based on the location of the inflammation: anterior uveitis (iritis and iridocyclitis), intermediate uveitis (pars planitis and pe-

riphlebitis), posterior uveitis (choroiditis and chorioretinitis), and panuvei

Uveoparotid fever

- A manifestation of sarcoidosis characterized by the triad of bilateral parotid gland enlargement, anterior uveitis, and fever, sometimes associated with facial nerve palsy. It reflects granulomatous inflammation and is treated with systemic corticosteroids and potentially disease-modifying agents.

Uvula

- A small, fleshy flap of tissue that hangs from the back of the throat over the root of the tongue.

Uvulitis

- Inflammation and swelling of the uvula, often caused by infection, trauma, allergy, dehydration, or irritation. It may cause throat pain, difficulty swallowing, gagging, or a foreign-body sensation; marked swelling with breathing difficulty requires urgent assessment.

Ureteral Perforation

- A tear or hole in the ureter, sometimes caused by a stone, trauma, surgery, or an endoscopic procedure. Urine may leak into surrounding tissues, causing pain, infection, or inflammation; prompt imaging and urologic assessment are required.

Urethral Infection

- Infection involving the urethra, commonly caused by bacteria, including sexually transmitted organisms. Symptoms may include burning urination, discharge, frequency, or pelvic discomfort; testing and appropriate treatment help prevent complications and transmission.

Urinary Urgency

- A sudden, difficult-to-delay need to urinate. It may occur with urinary infection, bladder irritation, overactive bladder, stones, neurologic disease, or other conditions. Urgency with fever, flank pain, blood in the urine, or inability to urinate requires medical assessment.

Uterine Cancer

- Cancer arising in the uterus, most commonly from the endometrium. Abnormal vaginal bleeding, bleeding after menopause, pelvic pain, or unusual discharge requires evaluation; early diagnosis often permits effective treatment.

Uterine Hemorrhage

- Heavy or abnormal bleeding from the uterus, occurring during pregnancy, after childbirth, or at other times. Causes include pregnancy complications, fibroids, hormonal disorders, and malignancy; severe bleeding with dizziness or weakness is an emergency.

Chapter 22

Pocket Dictionary: V Terms

Vaccination

- Administration of a vaccine to stimulate immune response and provide protection against infectious disease. Types: live attenuated, inactivated, subunit, mRNA, viral vector. Schedule varies by country. Herd immunity achieved when sufficient proportion immune.

Vaccination Schedule

- The recommended immunization schedule for infants, children, and adolescents (e.g., WHO/CDC): birth—HepB; 2 months—DTaP, IPV, Hib, PCV13, RV; 4 months—same; 6 months—HepB, IPV, PCV, RV, influenza (seasonal); 12 months—MMR, varicella, Hib, PCV; 18 months—DTaP, HepA; 4-6 years—DTaP, IPV, MMR, varicella; 9-16 years—HPV; 11-12 years—Tdap, men

Vaccine

- A biological preparation that induces protective immunity against a specific pathogen, administered before exposure (prophylactic) or therapeutically (e.g., rabies, some cancer vaccines).

Vaccine Immunology

- Live attenuated (MMR, varicella, rotavirus, yellow fever, oral typhoid, BCG, intranasal flu): replicate in host -> strong humoral + cellular immunity; contraindicated in immunocompromised. Inactivated/killed (polio IPV, hepatitis A, rabies, injectable flu): safer, weaker immunity, need boosters.

Vaccines and Immunizations

- Biological preparations that stimulate the immune system to develop protection against specific infectious diseases. By exposing the body to harmless components of a pathogen, vaccines enable the immune system to generate memory B and T cells that can mount a rapid response upon future exposure.

VACTERL Association

- A non-random association of congenital anomalies, diagnosed when at least three of the following are present: Vertebral anomalies (hemivertebrae, fused vertebrae, scoliosis, 70%), Anorectal malformations (imperforate anus, 80%), Cardiac defects (VSD, tetralogy of Fallot, 50-80%), Tracheo-Esophageal fistula (with or without esophageal atre

Vacuum Delivery

- Assisted vaginal delivery using a vacuum extractor (cup applied to the fetal scalp with negative pressure) to expedite delivery for prolonged second stage, fetal distress, or maternal indications.

Vacuum-Assisted Delivery

- Operative vaginal delivery using a vacuum extractor applied to the fetal scalp. Types: soft cup (Kiwi OmniCup, Mityvac, for outlet and low deliveries) and rigid cup (metal Kiwi, for posterior positions and midpelvic deliveries).

Vagina

- Muscular fibromembranous canal extending from the cervix to the vulva, approximately 7-9 cm long in adult females. Functions include receiving the penis during intercourse, serving as the birth canal, and channeling menstrual flow.

Vaginal Birth After Cesarean (VBAC)

- Vaginal delivery in a woman who has had a prior cesarean delivery. Success rate: 60-80% in appropriately selected candidates. Contraindications: prior classical or T-incision, prior uterine rupture, vertical uterine incision, more than 2 prior cesareans, and other contraindications to vaginal delivery.

Vaginal Cancer

- A rare gynecologic malignancy, most commonly squamous cell carcinoma linked to human papillo-

mavirus. Treatment depends on stage and includes surgery, radiation, and chemotherapy.

Vaginal Delivery (Spontaneous)

- The spontaneous passage of the fetus through the birth canal without the use of operative instruments (vacuum or forceps). Typically occurs in the second stage of labor with maternal pushing.

Vaginal Dryness

- Reduced vaginal lubrication, commonly associated with menopause, breastfeeding, hormonal changes, medications, or irritation. It may cause discomfort, itching, painful intercourse, or minor bleeding.

Vaginal Dryness and Atrophy (Genitourinary Syndrome of Menopause)

- Vaginal dryness and atrophy, collectively termed genitourinary syndrome of menopause, result from declining estrogen levels that cause thinning, drying, and loss of elasticity of the vaginal tissues. Symptoms include vaginal dryness, irritation, burning, dyspareunia, and urinary symptoms such as frequency and recurrent infections.

Vaginal Intercourse

- The insertion of the penis into the vagina for sexual pleasure or reproduction. The most common form of sexual intercourse. Vaginal intercourse can lead to pregnancy if sperm are deposited in the vagina.

Vaginismus

- Involuntary spasms of the vaginal muscles.

Vaginitis

- Inflammation of the vaginal mucosa, one of the most common gynecologic complaints.

Vaginospasm

- Involuntary contraction of the pelvic-floor muscles surrounding the vagina, causing pain or difficulty with vaginal penetration. It may be associated with fear of pain, prior trauma, pelvic-floor dysfunction, infection, hormonal factors, or other genitopelvic pain conditions.

Vagus Nerve (Cranial Nerve X)

- The tenth cranial nerve (CN X), the longest and most widely distributed cranial nerve in the body. It contains sensory, motor, and parasympathetic fibers, exiting the skull via the jugular foramen.

Vagus Nerve Stimulation

- A neuromodulation treatment involving electrical stimulation of the left vagus nerve via an implanted device. It is FDA-approved for treatment-resistant depression and epilepsy. It requires surgical implantation and delivers intermittent stimulation. Improvement may take months to develop.

Valgus/Varus Stress Test

- Collateral ligament integrity. Valgus (MCL), varus (LCL).

Valinemia

- A rare inherited disorder of branched-chain amino-acid metabolism causing elevated plasma valine. It may present with developmental delay, feeding difficulty, hypotonia, or neurological symptoms, although some individuals are minimally affected.

Valley Fever

- The common name for coccidioidomycosis, a fungal infection caused by *Coccidioides* species and acquired by inhaling airborne spores from soil. It usually causes a mild respiratory illness but can disseminate to skin, bone, joints, or meninges, especially in people at higher risk.

Valproate (Sodium Valproate)

- Valproate (valproic acid, sodium valproate, divalproex sodium) is a broad-spectrum antiepileptic drug effective against all seizure types (generalised tonic-clonic, absence, myoclonic, and partial seizures).

Valsalva's Manoeuvre

- Forced expiratory effort against a closed airway (glottis closed or mouth and nose pinched shut), typically performed by bearing down as if straining during a bowel movement. The manoeuvre has four phases: Phase I (onset of straining): increased intrathoracic pressure compresses the aorta, causing a transient rise in blood pressure.

Valvular Heart Disease

- Valvular heart disease involves dysfunction of one or more heart valves, leading to stenosis, regurgitation, or a combination of both. The most common causes are degenerative calcification of aortic and mitral valves, rheumatic heart disease, infective endocarditis, and ischemic papillary muscle dysfunction.

Vancomycin-resistant Enterococcus

- Enterococci resistant to vancomycin through acquisition of *vanA* or *vanB* gene clusters encoding altered cell wall precursors. VRE is a significant cause of healthcare-associated infections, particularly in immunocompromised patients. Treatment options include linezolid, daptomycin, and quinupristin-dalfopristin (for *E. faecium*)

Variable Decelerations

- A common fetal heart rate pattern characterized by an abrupt decrease in FHR (onset to nadir <30 seconds) with a variable shape, typically caused by umbilical cord compression. Variable decelerations are classified as mild (<30 sec duration, >80 bpm), moderate (30-60 sec, 70-80 bpm), or severe (>60 sec, <70 bpm).

Variceal Hemorrhage

- Life-threatening complication of portal hypertension involving rupture of dilated esophageal or gastric varices. The risk is determined by variceal size, red color signs, liver disease severity, and portal hypertension degree.

Varicella

- Varicella-zoster virus (VZV) infection: primary infection causes chickenpox, characterized by a generalized vesicular rash (macules papules vesicles crusts, 'dew drops on a rose petal'), fever, and malaise; highly contagious.

Varicella Vaccine

- Live attenuated vaccine given at 12-15 months and 4-6 years. Protects against chickenpox (varicella-zoster virus). Two doses provide greater than 90% protection. Contraindicated in immunocompromised and pregnant individuals. Also reduces risk of herpes zoster later in life.

Varicella-zoster virus

- A herpesvirus that causes chickenpox (varicella) and shingles (herpes zoster). VZV establishes latent infections in dorsal root ganglia and can reactivate decades later. Chickenpox is highly contagious and typically occurs in childhood. Reactivation as shingles causes painful dermatomal vesicular rash.

Varicocele

- Abnormal dilation and tortuosity of the pampiniform venous plexus within the spermatic cord, re-

sulting from incompetent or absent venous valves.

Varicose Ulcer

- Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. See chronic venous insufficiency.

Varicose Veins

- Dilated, tortuous superficial veins, most commonly in the lower extremities. Caused by venous valve incompetence leading to venous reflux. Risk factors: family history, prolonged standing, obesity, pregnancy, and older age. Symptoms: leg heaviness, swelling, aching. Complications: skin changes, ulceration, thrombophlebitis.

Variegate Porphyria

- An inherited hepatic porphyria caused by reduced protoporphyrinogen oxidase activity. It can cause acute neurovisceral attacks with abdominal pain and neuropathy and/or photosensitive blistering skin disease.

Variola

- A highly contagious and deadly disease caused by variola virus, eradicated globally by 1980 through vaccination. See Smallpox.

Vas Deferens

- Muscular duct (approximately 45 cm long) transporting sperm from the epididymal tail to the ejaculatory duct. It ascends through the spermatic cord, passes through the inguinal canal, enters the pelvis, crosses over the ureter, and joins the seminal vesicle duct to form the ejaculatory duct (in the prostate).

Vasa Previa

- Fetal blood vessels traverse the membranes over or near the internal cervical os without protection from placental tissue or the umbilical cord. Rupture of membranes can cause rapid fetal exsanguination, so antenatal diagnosis and planned cesarean delivery are important.

Vascular Access

- Vascular access for hemodialysis includes arteriovenous fistulas (AVF), arteriovenous grafts (AVG), and central venous catheters (CVC). The fistula first initiative prioritizes AVF as the gold standard due to best long-term patency and lowest complication rates.

Vascular Access for Hemodialysis

- A surgically created or catheter-based route that permits high-flow blood removal and return during hemodialysis. Arteriovenous fistulae are generally preferred when feasible, followed by grafts; central venous catheters have higher infection and thrombosis risks.

Vascular Dementia

- Second most common dementia. Caused by cerebrovascular disease. Stepwise decline. Treatment: vascular risk factor modification.

Vascular Diagnostic Yield

- The proportion of vascular diagnostic procedures or tests that provide a clinically useful result.

Vascular Graft

- Synthetic or biologic conduit for vascular reconstruction. PTFE, Dacron, saphenous vein.

Vascular Graft Patency

- Continued openness and functional blood flow through a vascular graft or bypass. Patency is monitored clinically and with duplex or other imaging because thrombosis, stenosis, infection, intimal hyperplasia, and anastomotic problems can cause failure.

Vascular Imaging

- Techniques for visualizing blood vessels: duplex ultrasound, CT angiography, MR angiography, conventional angiography.

Vascular Malformation

- Abnormal blood vessel development. Arteriovenous malformation, venous malformation, lymphatic malformation.

Vascular Ring

- A congenital abnormality of the aortic arch or its branches that encircles and may compress the trachea, esophagus, or both. Symptoms include stridor, wheezing, recurrent respiratory infection, dysphagia, feeding difficulty, or respiratory distress.

Vascular Trauma

- Injury to an artery or vein from blunt, penetrating, iatrogenic, or crush mechanisms. Assessment prioritizes hemorrhage control, distal perfusion, ischemia, expanding hematoma, bruit, thrill, and associated nerve or bone injury.

Vascularity

- Blood supply to tissue. Implants require vascularization for integration. Angiogenesis essential.

Vasculitis

- Inflammation of blood vessels causing tissue ischemia and necrosis. Classification by vessel size. Large vessel: giant cell arteritis, Takayasu arteritis. Medium vessel: polyarteritis nodosa (PAN), Kawasaki disease.

Vasculitis (Systemic)

- See vasculitis. Large vessel (GCA, Takayasu), medium vessel (PAN, Kawasaki), small vessel (ANCA-associated, IgA vasculitis).

Vasectomy

- Minor surgical procedure for permanent male contraception involving surgical interruption, occlusion, or excision of a segment of both vasa deferentia, preventing spermatozoa from entering the ejaculate. Performed under local anesthesia using traditional incisional or no-scalpel vasectomy (NSV) techniques.

Vasoconstriction

- Narrowing of a blood vessel caused by contraction of its muscular wall.

Vasodilation

- The widening of blood vessels during inflammation, resulting in increased blood flow to the affected area. This process is mediated primarily by histamine released from mast cell granules acting on H1 receptors, nitric oxide produced by endothelial nitric oxide synthase, and prostaglandins particularly PGE2.

Vasomotor Rhinitis

- See Non-Allergic Rhinitis.

Vasopressin (Antidiuretic Hormone, ADH)

- See Antidiuretic Hormone (ADH).

Vasopressors in Anesthesia

- Vasopressors maintain blood pressure during anesthesia-induced vasodilation. Phenylephrine is a pure alpha-1 agonist causing vasoconstriction without cardiac effects, used for pure vasodilatory hypotension. Dose: bolus 50 to 100 mcg IV or infusion 0.05 to 0.5 mcg/kg/min.

Vasovagal Syncope

- Reflex syncope mediated by increased vagal tone and withdrawal of sympathetic activity, causing bradycardia and/or vasodilation resulting in transient cerebral hypoperfusion. Triggers: prolonged standing, heat, pain, emotional stress. Preceded by prodrome (lightheadedness, nausea, diaphoresis, visual tunneling).

Vaughan Williams Classification

- The Vaughan Williams classification categorizes antiarrhythmic drugs into four classes based on their primary mechanism of action.

VEAL CHOP Mnemonic

- A memory aid for interpreting fetal heart rate patterns. VEAL: Variable decelerations Cord compression, Early decelerations Head compression, Accelerations OK (fetal well-being), Late decelerations Placental insufficiency.

Vector

- A living carrier, commonly an arthropod, that transmits an infectious agent between hosts. Biological vectors support replication or development of the agent; mechanical vectors transfer it without such biologic change.

Vector Control

- Reduction of vector populations. Examples: insecticides, larvicides, environmental modification, personal protection.

Vector-Borne Disease

- Disease transmitted by vector (insect, tick). Examples: malaria, dengue, Lyme disease.

Vector-borne diseases

- Diseases transmitted by arthropod vectors including mosquitoes, ticks, fleas, and sandflies. Major vector-borne diseases include malaria, dengue, Zika, chikungunya, Lyme disease, and leishmaniasis. Prevention through vector control, personal protection, and chemoprophylaxis is essential.

Vegan Diet

- Excludes all animal products. Risk of deficiencies: vitamin B12, iron, calcium, vitamin D, omega-3, zinc, iodine. Requires planning or supplementation. Associated with lower BMI, reduced cardiovascular risk.

Vegetarian Diet

- Excludes meat, fish, poultry. Lacto-ovo includes dairy and eggs. Vegan excludes all animal products. Nutrient adequacy requires attention.

Vegetation

- A vegetation is a mass composed of platelets, fibrin, and microorganisms that forms on a heart valve or endocardial surface in the setting of infective endocarditis. The vegetation typically attaches to the valve leaflet on the low-pressure side and may progressively destroy the underlying valve tissue, causing valvular regurgitation.

Vein

- A blood vessel that carries deoxygenated blood toward the heart (except pulmonary veins, which carry oxygenated blood). Veins have thinner walls and larger lumens than arteries, with less smooth muscle and elastic tissue.

Vein-graft aneurysms

- Focal dilation of a venous bypass graft, most commonly a saphenous vein graft used in coronary artery bypass surgery. Rare but may cause graft thrombosis, distal embolization, compression of adjacent structures, or rupture. Diagnosis: coronary angiography or CT angiography. Treatment: re-intervention or medical management.

Vena Cava

- The two largest systemic veins returning deoxygenated blood to the right atrium. Superior Vena Cava (SVC): formed by brachiocephalic veins, drains head, neck, upper limbs, thorax (no valves). Inferior Vena Cava (IVC): formed by common iliac veins at L5, drains abdomen, pelvis, lower limbs (receives hepatic, renal, gonadal, lumbar veins).

Venereal Diseases

- Sexually transmitted infections are infections passed from one person to another through sexual contact, including vaginal, anal, and oral sex. See Sexually Transmitted Infections (STIs).

Venesection (Phlebotomy)

- The controlled removal of blood from a vein for therapeutic or diagnostic purposes.

Venipuncture

- Puncture of a vein to obtain blood specimens or administer IV therapy. Common sites: median

cubital vein (most common), cephalic vein, basilic vein, and dorsal hand veins.

Venography

- Contrast imaging of veins. Evaluates deep vein thrombosis, venous obstruction.

Venom

- A biological poison produced by a microorganism, plant, or animal that damages host tissues at low concentrations. See toxin.

Venomous Bite

- See emergency medicine.

Venous Insufficiency

- Impaired venous return from the lower extremities due to venous valve incompetence or obstruction. See chronic venous insufficiency.

Venous Stasis Ulcer

- A chronic wound caused by chronic venous insufficiency, most common type of leg ulcer (70-80%). Located on the medial gaiter area (medial malleolus). Presents with shallow, irregular ulcer with granulation tissue, surrounded by hemosiderin deposition (brownish discoloration), lipodermatosclerosis, edema, and varicose veins.

Venous Thromboembolism in Pregnancy

- Pregnancy increases the risk of VTE (DVT and PE) by 5-10 fold due to a hypercoagulable state (increased fibrinogen, factors VII-XII, vWF, decreased protein S, and venous stasis from the gravid uterus).

Venous Thromboembolism Prophylaxis

- Measures used to reduce perioperative deep-vein thrombosis and pulmonary embolism, including early mobilization, mechanical compression, and pharmacologic anticoagulation when bleeding risk permits.

Venous Thrombosis

- The formation of a blood clot (thrombus) within a vein, most commonly in the deep veins of the lower extremities (deep vein thrombosis, DVT).

Venous Ulcer

- A chronic wound caused by chronic venous insufficiency, most common type of leg ulcer (70-80%). See Venous Stasis Ulcer.

Ventilation-Perfusion (V/Q) Scan

- Lung imaging assessing ventilation and perfusion.

Mismatch indicates pulmonary embolism. Alternative to CTPA in renal impairment or contrast allergy.

Ventilation-Perfusion Mismatch

- The most common cause of hypoxemia. Areas of lung are ventilated but not perfused (dead space) or perfused but not ventilated (shunt). It is the mechanism of hypoxemia in COPD, pneumonia, and PE.

Ventilation-Perfusion Ratio (V/Q)

- The relationship between alveolar ventilation and pulmonary capillary perfusion. Regional mismatch is a major cause of impaired oxygenation; low V/Q produces venous admixture, while high V/Q reflects wasted ventilation.

Ventilation/Perfusion (V/Q) Scan

- Pulmonary embolism diagnosis. Ventilation: inhaled Tc-99m DTPA aerosol. Perfusion: injected Tc-99m MAA. Mismatch (perfusion defect with preserved ventilation) suggests PE. High single-photon emission computed tomography (SPECT) sensitivity.

Ventilator

- Mechanical device providing ventilatory support. Modes: assist-control, SIMV, CPAP, PRVC. Parameters: tidal volume, respiratory rate, FiO₂, PEEP.

Ventilator-Associated Pneumonia

- Pneumonia > 48 hours after intubation. Most common ICU-acquired infection. Diagnosis uses quantitative BAL or protected specimen brush cultures.

Ventilator-Induced Lung Injury (VILI)

- Lung injury caused or amplified by mechanical ventilation, including volutrauma, barotrauma, atelectrauma, and biotrauma. Lung-protective ventilation limits tidal volume and excessive airway pressures while maintaining appropriate oxygenation and carbon-dioxide targets.

Ventral

- Relating to the belly or front side of the body; synonym for anterior in human anatomy.

Ventricle

- One of the two lower, thicker-walled chambers of the heart that pump blood out of the heart. The right ventricle pumps deoxygenated blood into the

pulmonary artery (low-pressure system); the left ventricle pumps oxygenated blood into the aorta (high-pressure system, generating systemic blood pressure).

Ventricle (Brain)

- The interconnected system of fluid-filled cavities within the brain producing and containing cerebrospinal fluid: two lateral ventricles (cerebrum), third ventricle (diencephalon), and fourth ventricle (pons/medulla, continuous with the central canal).

Ventricular Assist Device

- A mechanical pump that supports heart function in patients with advanced heart failure. LVADs (left ventricular assist devices) are most common, supporting the failing left ventricle. The HeartMate 3 is the current standard with fully magnetically levitated rotors.

Ventricular Assist Device (VAD)

- Mechanical pump supporting heart function. Types: left (LVAD), bi-ventricular (BiVAD), right (RVAD). Indications: end-stage heart failure, bridge to transplant/cardiomyectomy. Complications: bleeding, infection, stroke, pump thrombosis. Continuous-flow devices now standard.

Ventricular Fibrillation

- Rapid, chaotic ventricular electrical activity resulting in no effective cardiac output, leading to cardiac arrest. ECG shows irregular, disorganized waveforms with no identifiable QRS complexes. VFib is a medical emergency requiring immediate defibrillation.

Ventricular Fibrillation (VF)

- Chaotic electrical activity in ventricles, causing no cardiac output. Immediate defibrillation is treatment. Causes: coronary artery disease, cardiomyopathy, electrolyte disturbances, drug toxicity. Post-resuscitation care: targeted temperature management, coronary angiography, ICU admission.

Ventricular Septal Defect

- The most common congenital heart defect (30% of all CHD), characterized by an abnormal opening in the interventricular septum allowing left-to-right shunting.

Ventricular Tachycardia

- A wide-complex tachycardia (QRS >0.12 s) origi-

nating from the ventricles, with a rate of 100-250 bpm. Monomorphic VT has uniform QRS complexes; polymorphic VT has varying QRS morphology. VT is life-threatening and may degenerate to ventricular fibrillation.

Ventriculoatrial

- Relating to or connecting a ventricle of the brain or heart with an atrium; often describing a cerebrospinal-fluid shunt to the right atrium.

Ventriculography

- Imaging ventricular function. Radionuclide ventriculography (MUGA scan) accurately measures ejection fraction. Cardiac MRI also accurate.

Ventriculoperitoneal Shunt

- Tube diverting fluid. See shunt.

Venturi Mask

- Precise FiO₂ delivery. Color-coded adapters.

Venule

- A small vessel that drains blood from capillary beds into veins. Postcapillary venules are important sites for leukocyte migration and inflammatory fluid exchange.

Vernal conjunctivitis

- A bilateral, recurrent, allergic conjunctivitis affecting children and young adults, typically occurring in spring and summer. It presents with intense ocular itching, tearing, photophobia, and mucous discharge. Palpable cobblestone papillae on the upper tarsal conjunctiva are characteristic. It is associated with atopy and elevated IgE.

Vernal keratoconjunctivitis

- See Vernal conjunctivitis.

Verruca (Wart)

- Benign epidermal hyperplasia caused by human papillomavirus (HPV). Common warts (verruca vulgaris): rough papules on hands/fingers. Plantar warts: painful lesions on soles, grow inward due to pressure. Flat warts (verruca plana): smooth, flat-topped papules on face, hands, legs. Filiform warts: finger-like projections, often on face.

Verruciform Xanthoma

- A rare benign lesion characterized by a verrucous epithelial surface with lipid-laden macrophages in the underlying connective tissue. It most often

occurs in oral or genital mucosa and may resemble a wart or malignant lesion clinically.

Verrucous carcinoma

- A low-grade, well-differentiated variant of squamous cell carcinoma with a warty, exophytic appearance. It is locally invasive but rarely metastasizes, commonly affecting the oral cavity, skin, or genitalia.

Vertebra

- The individual bone of the vertebral column (33 vertebrae: 7 cervical, 12 thoracic, 5 lumbar, 5 sacral fused, 4 coccygeal). Parts: vertebral body, vertebral arch (pedicles, laminae), spinous and transverse processes, articular processes/facets, and vertebral foramen (spinal canal).

Vertebral Canal

- The bony canal formed by the vertebrae that encloses and protects the spinal cord.

Vertebral Column

- 33 vertebrae: 7 cervical, 12 thoracic, 5 lumbar, 5 fused sacral, 4 fused coccygeal. Protects the spinal cord, supports the head/trunk, provides muscle/rib attachment. Each typical vertebra has a body, vertebral arch, and 7 processes. Intervertebral discs (nucleus pulposus/annulus fibrosus) provide cushioning.

Vertebrobasilar Insufficiency

- A condition of reduced blood flow through the vertebral arteries or basilar artery to the posterior circulation of the brain, including the brainstem, cerebellum, occipital lobes, and thalami.

Vertebroplasty

- Percutaneous injection of bone cement into fractured vertebra. Provides pain relief, stabilization.

Vertical Talus

- A congenital rigid flatfoot deformity in which the talus is positioned vertically and the midfoot is dorsiflexed. It may be isolated or associated with neuromuscular or genetic syndromes and usually requires early orthopedic assessment and treatment.

Vertigo

- A false sensation of rotational or linear movement of the self or the environment, and it is a symptom rather than a disease, requiring differentiation from dizziness, disequilibrium, and presyncope.

Vesical

- Relating to bladder. Vesical carcinoma, vesical diverticulum, vesical neck.

Vesicle

- A small (<1 cm) fluid-filled blister within the epidermis, containing clear fluid (serum) or, if infected, pus. Causes: viral (herpes simplex, herpes zoster, varicella, hand-foot-mouth disease), dermatitis (contact, dyshidrotic eczema), autoimmune (pemphigus, bullous pemphigoid early), and friction burns.

Vesicoureteral Reflux

- Retrograde flow of urine from the bladder into the ureters and potentially the kidneys, most commonly due to incompetent ureterovesical valve. Predisposes to pyelonephritis and renal scarring. More common in children, presenting with recurrent febrile UTIs. Diagnosis: voiding cystourethrogram (VCUG). Grading I-V.

Vestibular Migraine

- Episodic vestibular symptoms associated with migraine biology, with or without simultaneous headache. Episodes may include vertigo, motion sensitivity, visual aura, or sound sensitivity; diagnosis is clinical after exclusion of other vestibular and neurologic causes.

Vestibular Neuritis

- Acute vestibular dysfunction due to inflammation of the vestibular nerve, typically viral in origin. Presents with acute onset of vertigo, nausea, vomiting, and nystagmus lasting days. Hearing is normal (unlike labyrinthitis). Diagnosis: clinical, HINTS exam (Head Impulse, Nystagmus, Test of Skew) to differentiate from central causes.

Vestibular papillomatosis

- A normal anatomical variant of the female vulva characterised by multiple, soft, finger-like or filiform papules on the labia minora and vestibule, often mistaken for genital warts (condylomata acuminata).

Vestibular Rehabilitation

- See neurology.

Vestibular Schwannoma

- Vestibular schwannoma, also known as acoustic neuroma, is a benign, slow-growing tumor arising from the Schwann cells of the vestibular portion of the eighth cranial nerve (vestibulocochlear nerve),

typically from the superior vestibular nerve.

Vestibule

- Central bony labyrinth between cochlea and semi-circular canals, containing utricle (horizontal linear acceleration and head tilt detection) and saccule (vertical linear acceleration detection). Each has a macula with hair cells covered by gelatinous membrane embedded with otoconia (calcium carbonate crystals).

VEXAS Syndrome

- VEXAS (Vacuoles, E1 ubiquitin-activating enzyme, X-linked, Autoinflammatory, Somatic) syndrome is an adult-onset autoinflammatory disorder caused by somatic UBA1 mutations in hematopoietic stem cells, affecting men >50.

Viagra (Sildenafil)

- Sildenafil is a phosphodiesterase type 5 (PDE5) inhibitor used primarily for erectile dysfunction (ED) and pulmonary arterial hypertension (PAH, as Revatio).

Vibrio cholerae

- A gram-negative, curved rod that causes cholera, a severe diarrhoeal illness. The organism produces cholera toxin, which causes profuse watery diarrhea (rice-water stools). Transmission occurs through contaminated water and food. The infectious dose varies depending on stomach acidity.

Vibrio vulnificus infections

- Infections caused by *Vibrio vulnificus*, a gram-negative bacterium found in warm seawater and raw seafood. *V. vulnificus* can cause wound infections, septicemia, and gastroenteritis. The most severe form is necrotizing fasciitis and primary septicemia, which can be fatal.

Video Laryngoscope

- An airway device with a camera near the blade tip that displays the laryngeal view on a monitor. It may improve visualization during tracheal intubation, particularly when direct laryngoscopy is difficult, but does not eliminate aspiration or airway-trauma risk.

Video Laryngoscopy

- Video laryngoscopy uses a camera-equipped blade to display an indirect view of the glottis on a screen.

Videofluoroscopic Swallow Study

- Real-time X-ray imaging of swallowing. Detailed assessment of oropharyngeal phase.

Videostroboscopy

- A diagnostic technique used to evaluate the structure and function of the vocal cords by producing a slow-motion, composite image of vocal cord vibration using stroboscopic light flashes synchronized to the frequency of vocal cord vibration, allowing the assessment of vocal cord mucosal wave, amplitude, symmetry, phase closure, and.

Villous atrophy

- Flattening and loss of the normal small-intestinal villi, reducing absorptive surface area and occurring in conditions such as coeliac disease.

VIPoma

- A rare pancreatic neuroendocrine tumor secreting vasoactive intestinal peptide (VIP), causing WDHA syndrome: watery diarrhea (large volume, 3-5 L/day), hypokalemia, and achlorhydria/acidosis. Most are malignant and located in the pancreatic tail. Diagnosis: elevated VIP levels, pancreatic tumor on imaging.

Viral assembly

- The process of packaging newly synthesised viral genomes and proteins into complete virions. Assembly typically occurs in the nucleus or cytoplasm, depending on the virus type. The process involves self-assembly of capsid proteins around the viral genome.

Viral attachment

- The initial step in viral infection where viral surface proteins bind to specific receptors on host cells. Attachment determines viral tropism and host range. The viral attachment protein is called the ligand, and the host cell receptor is typically a protein or glycan.

Viral Conjunctivitis

- The most common form of infectious conjunctivitis, caused predominantly by adenoviruses (serotypes 3, 4, 7, 8, 19, and 37), and it is highly contagious, spreading through direct contact with infected ocular secretions or contaminated surfaces.

Viral culture

- The growth of viruses in cell culture for identi-

cation and characterisation. Viral culture remains the gold standard for many viral infections. The technique involves inoculating cell lines with clinical specimens and observing for cytopathic effects.

Viral cytopathic effects

- Observable changes in host cells caused by viral infection. CPE include cell rounding, detachment, syncytia formation (fusion of infected cells), inclusion bodies, and cell lysis. CPE are used in viral diagnosis and can be observed by light microscopy.

Viral entry

- The process by which viruses cross the host cell membrane to deliver their genome into the cytoplasm. Entry occurs through membrane fusion (enveloped viruses) or endocytosis (enveloped and non-enveloped viruses). For membrane fusion, viral fusion proteins mediate merger of the viral envelope with the host cell membrane.

Viral gastroenteritis

- Gastrointestinal inflammation caused by viral infections. The major causes include rotavirus, norovirus, adenovirus, and astrovirus. Symptoms include watery diarrhea, vomiting, and abdominal cramps. Diagnosis is made by antigen detection or PCR in stool. Treatment is supportive with rehydration.

Viral hepatitis

- Inflammation of the liver caused by viral infections. The major hepatitis viruses include A, B, C, D, and E. Hepatitis A and E are transmitted through the fecal-oral route and cause acute, self-limited infections.

Viral immune evasion

- Mechanisms by which viruses evade host immune responses. Strategies include antigenic variation, inhibition of interferon signalling, downregulation of MHC molecules, and encoding viral homologues of cytokines. Understanding viral immune evasion is essential for developing effective vaccines and antiviral therapies.

Viral Infections

- Illnesses caused by viruses, which are intracellular pathogens that hijack host cellular machinery to replicate. They range from self-limited conditions such as the common cold and gastroenteritis to serious diseases including influenza, hepatitis, HIV,

and COVID-19.

Viral latency

- A state in which viral genomes persist in host cells without producing infectious particles. Latent viruses can reactivate under certain conditions including immunosuppression. Herpesviruses establish latent infections in specific cell types (HSV in neurons, EBV in B cells, CMV in myeloid cells).

Viral latency mechanisms

- The strategies by which viruses establish and maintain latent infections. Herpesviruses maintain their genomes as episomes in the nucleus. HIV integrates into the host genome as a provirus. Latent viruses express minimal proteins to avoid immune detection. Reactivation can be triggered by immunosuppression, stress, or other stimuli.

Viral Load (HIV RNA)

- Quantitative measurement of HIV-1 RNA in plasma by RT-PCR (Roche COBAS, Abbott Real-Time). Normal: undetectable (<20-40 copies/mL). Treatment goal: undetectable viral load (U=U: Undetectable = Untransmittable). Initiation: start ART regardless of viral load or CD4. Monitoring: at baseline, 4-6 weeks after ART start, then q3-6 months.

Viral Meningitis

- Inflammation of the meninges caused by viruses, usually Enteroviruses. Presents similarly to bacterial meningitis but generally less severe. CSF shows lymphocytic pleocytosis, normal glucose, and mildly elevated protein. Self-limited in most cases; supportive treatment.

Viral pathogenesis

- The mechanisms by which viruses cause disease. Pathogenesis involves attachment, entry, replication, and spread within the host. Viral damage can be direct (cytopathic effects) or indirect (immune-mediated). Direct effects include cell death, altered cell function, and cell transformation.

Viral pneumonia

- See Viral Infections.

Viral release

- The process by which newly assembled virions exit the host cell to infect other cells. Release occurs through budding (enveloped viruses) or cell lysis

(non-enveloped viruses). Budding allows the virus to acquire an envelope from host cell membranes without killing the cell.

Viral replication

- The process of synthesising new viral genomes and proteins within host cells. The mechanism varies by virus type. DNA viruses typically replicate in the nucleus using host cell DNA polymerases (except poxviruses). RNA viruses typically replicate in the cytoplasm using viral RNA-dependent RNA polymerase.

Viral replication cycle

- The series of steps by which viruses reproduce within host cells.

Viral structure

- The basic components of viral particles (virions). Viruses consist of a nucleic acid genome (DNA or RNA) surrounded by a protein coat called a capsid. Some viruses have an outer lipid envelope derived from host cell membranes. The capsid is composed of structural units called capsomeres.

Viral uncoating

- The process by which the viral capsid is disassembled to release the viral genome into the host cell cytoplasm. Uncoating occurs after entry and before replication. The mechanism varies by virus type and can involve pH-dependent conformational changes, proteolytic cleavage, or disassembly of capsid proteins.

Virchow's Triad

- Virchow's triad describes three broad categories of factors that contribute to thrombosis: endothelial injury or dysfunction, stasis or turbulent blood flow, and hypercoagulability. Endothelial injury exposes subendothelial collagen and tissue factor, initiating platelet adhesion and the coagulation cascade.

Viremia

- The presence of virus in the bloodstream. Viremia may be transient or persistent and can enable dissemination to target organs; it may be detected by culture, antigen testing, or nucleic-acid amplification.

Virilisation

- The development of male secondary sexual characteristics in females, resulting from excessive andro-

gen production or exposure.

Virilism Foetal

- Prenatal or neonatal masculinization of a genetically female fetus or infant due to excess androgen exposure. Causes include congenital adrenal hyperplasia, maternal androgen-producing disease, or placental and fetal steroidogenic disorders.

Virology

- The branch of microbiology concerned with viruses: structure, classification (DNA/RNA, enveloped/nonenveloped, Baltimore scheme), replication, pathogenesis, epidemiology, diagnosis, treatment, and prevention. Diagnostic methods: cell culture, antigen detection, nucleic acid amplification (PCR), and serology (IgM/IgG).

Virtual Reality

- Immersive technology for rehabilitation. Engaging, motivates exercise. Used for balance, coordination, cognitive training.

Virulence

- The relative capacity of a microorganism to cause disease or tissue injury in a host. It depends on microbial factors such as toxins, adhesins, invasiveness, and immune evasion, together with host susceptibility.

Virulence Factor

- A microbial product or trait that enhances pathogenicity and ability to cause disease. Examples: adhesins (pili, fimbriae—*E. coli*, *Neisseria*), toxins (exotoxins, endotoxin), capsules (antiphagocytic—*S. pneumoniae*, *N. meningitidis*, *H.*

Viscera

- The internal organs of the body, particularly those within the thoracic and abdominal cavities. The thoracic viscera include the heart, lungs, esophagus, trachea, and thymus.

Vision

- The sensory ability to perceive light and interpret visual information processed by the eyes and brain, enabling recognition of shapes, colors, depth, and motion. Normal vision depends on the proper function of the cornea, lens, retina, and optic nerve, as well as intact neural pathways to the visual cortex.

Vision and Hearing

- The two primary sensory modalities through which

humans perceive and interact with their environment. Age-related declines in both senses are common, with presbyopia and cataracts affecting vision and presbycusis affecting hearing, often beginning as early as the fourth decade of life.

Vision Impairment

- Reduced visual acuity or field. Common causes: cataract, AMD, glaucoma, diabetic retinopathy, stroke. Screening: annual eye exam after 65. Interventions: cataract surgery, glaucoma treatment, low vision aids. Impact: functional limitation, fall risk, isolation.

Vision Loss

- A decrease in the ability to see that cannot be fully corrected with eyeglasses, contact lenses, or medical treatment. It ranges from partial impairment to complete blindness and can result from conditions such as macular degeneration, glaucoma, diabetic retinopathy, cataracts, and traumatic injury.

Vision Problems

- Changes in visual acuity, field, color, or clarity caused by refractive error, eye disease, neurologic disease, or systemic illness. Sudden vision loss is an emergency.

Vision Screening

- Screening for visual problems in children: newborn—red reflex examination (cataract, retinoblastoma); 3-5 years—visual acuity, strabismus screening (cover test, photostrengthening), and amblyopia risk factors.

Visual Acuity

- The resolving power of the eye, defined as the ability to distinguish two points as distinct, and the most common clinical measure of visual function.

Visual Agnosia

- Inability to recognize or identify visually presented objects, faces, or symbols despite adequate vision and attention. It results from dysfunction of higher-order visual association cortex and may follow stroke, trauma, neurodegenerative disease, or other brain injury.

Visual Field Defect

- Loss or alteration of a portion of the normally perceived visual field caused by dysfunction of the retina, optic nerve, chiasm, optic tract, radiations,

or visual cortex. The pattern can help localize the lesion.

Visual Field Defects

- Areas of reduced or absent vision within the visual field that correspond to damage at specific points along the visual pathway, and their pattern provides crucial localizing information for diagnosis.

Visual Field Perimetry

- A systematic method of measuring the sensitivity of the retina across the entire visual field using standardized light stimuli, providing a quantitative map of visual function that can detect, characterize, and monitor visual field loss caused by diseases of the retina, optic nerve, and visual pathways.

Vital Signs

- Key physiological measurements: temperature, pulse, respiration rate, blood pressure, oxygen saturation, pain. Normal ranges: T 36.5-37.5°C, HR 60-100, RR 12-20, BP <120/80, SpO2 >95%.

Vital Tooth Bleaching

- Whitening of natural teeth with peroxide gel. In-office or take-home trays. Effect temporary.

Vitamin A

- Fat-soluble vitamin. Functions: vision (rhodopsin), immunity, reproduction, cellular differentiation. Sources: liver, dairy, orange vegetables (beta-carotene). Deficiency: night blindness, xerophthalmia. Toxicity: teratogenic, liver damage.

Vitamin A deficiency

- A nutritional disorder resulting from inadequate intake or absorption of retinol, causing xerophthalmia, night blindness, follicular hyperkeratosis, and impaired immunity. Severe deficiency leads to corneal ulceration and blindness, and is a leading cause of preventable childhood blindness worldwide.

Vitamin A: Retinal

- Vitamin A (retinol) is a fat-soluble vitamin essential for vision, immune function, cell differentiation, and reproduction. In the eye, retinol is oxidized to retinal (retinaldehyde), which binds to opsin in the photoreceptor outer segments forming rhodopsin (in rods) and photopsins (in cones).

Vitamin B Complex

- The vitamin B complex comprises eight water-soluble B vitamins that function primarily as coen-

zymes or cofactors in cellular metabolism: Vitamin B1 (thiamine, a cofactor in carbohydrate metabolism and nerve function), Vitamin B2 (riboflavin, precursor of FAD/FMN in oxidation-reduction reactions), Vitamin B3 (niacin, precursor of NAD⁺/NADP

Vitamin B1 (Thiamine)

- Water-soluble vitamin, coenzyme in carbohydrate metabolism. Deficiency: beriberi (wet, dry), Wernicke-Korsakoff. Risk: alcoholism, bariatric surgery, dialysis. Treatment: supplementation.

Vitamin B12

- Cobalamin, essential for DNA synthesis, red blood cell production, and neurologic function. Normal: 200-900 pg/mL. Deficiency: <200 pg/mL (but borderline 200-300 may still be symptomatic).

Vitamin B12 (Cobalamin)

- Required for nerve function, red blood cell formation, DNA synthesis. Deficiency: megaloblastic anemia, neurological damage. Risk: vegans, elderly, pernicious anemia, metformin use, bariatric surgery. Requires intrinsic factor for absorption.

Vitamin B12 Deficiency

- Megaloblastic anemia, neurological symptoms. Pernicious anemia, vegan diet. Treated with B12 supplementation.

Vitamin B12 Deficiency in Older Adults

- Deficiency caused by inadequate intake, malabsorption, autoimmune intrinsic-factor deficiency, gastric surgery, or medications. It may cause megaloblastic anemia, neuropathy, cognitive changes, gait impairment, and glossitis; neurologic injury can occur without anemia.

Vitamin B12: Cobalamin

- Vitamin B12 (cobalamin) is a water-soluble vitamin containing cobalt, essential for two enzymatic reactions: (1) methionine synthase (cytoplasmic, requiring methylcobalamin), converting homocysteine to methionine (linking to folate metabolism), and (2) methylmalonyl-CoA mutase (mitochondrial, requiring adenosylcobalamin), converting methy

Vitamin B1: Thiamine

- Thiamine (thiamine pyrophosphate, TPP) is a water-soluble B vitamin essential as a cofactor for several key metabolic enzymes: pyruvate dehydro-

genase (linking glycolysis to the TCA cycle), alpha-ketoglutarate dehydrogenase (TCA cycle), transketolase (pentose phosphate pathway), and branched-chain alpha-ketoacid dehydrogenase (branched-chain

Vitamin B2 (Riboflavin)

- Precursor for FAD and FMN. Deficiency: cheilosis, glossitis, seborrheic dermatitis. Common with B-complex deficiency.

Vitamin B3 (Niacin)

- Precursor for NAD⁺ and NADP⁺. Deficiency: pellagra (diarrhea, dermatitis, dementia). Sources: meat, fish, poultry, grains.

Vitamin B3: Niacin

- Niacin (nicotinic acid and nicotinamide) is a water-soluble vitamin that serves as the precursor for the coenzymes NAD⁺ (nicotinamide adenine dinucleotide) and NADP⁺ (nicotinamide adenine dinucleotide phosphate), which are essential for over 400 enzymatic reactions including glycolysis, the TCA cycle, oxidative phosphorylation, and

Vitamin B5 (Pantothenic Acid)

- Component of coenzyme A. Deficiency rare. Involved in fatty acid metabolism.

Vitamin B6 (Pyridoxine)

- Coenzyme in amino acid metabolism. Deficiency: seizures, anemia, dermatitis. Risk: isoniazid therapy, alcoholism.

Vitamin B6: Pyridoxine

- Pyridoxine (pyridoxal phosphate, PLP, the active form) is a water-soluble B vitamin essential as a cofactor for over 100 enzymatic reactions, particularly in amino acid metabolism.

Vitamin B7 (Biotin)

- Coenzyme for carboxylation reactions. Deficiency rare, may occur with raw egg consumption (avidin binds biotin).

Vitamin B9 (Folate)

- Essential for DNA synthesis, red blood cell formation. Deficiency: megaloblastic anemia, neural tube defects in pregnancy. Sources: leafy greens, legumes, fortified grains. Supplementation recommended before and during pregnancy.

Vitamin B9: Folate

- Folate (folic acid, tetrahydrofolate/THF) is a water-

soluble B vitamin essential for one-carbon transfer reactions in nucleotide synthesis (thymidylate synthesis for DNA, purine synthesis), amino acid metabolism (methionine synthesis from homocysteine), and methylation reactions.

Vitamin C (Ascorbic Acid)

- A water-soluble vitamin and a powerful antioxidant, essential for collagen synthesis (required for proline and lysine hydroxylation in collagen triple helix formation), carnitine synthesis, neurotransmitter synthesis (dopamine to norepinephrine), and iron absorption (reduces ferric to ferrous iron in the gut).

Vitamin D

- 25-hydroxyvitamin D, the storage form of vitamin D. Normal: >30 ng/mL (75 nmol/L). Insufficiency: 20-30 ng/mL. Deficiency: <20 ng/mL (<50 nmol/L). Extremely deficient: <10 ng/mL (high risk of osteomalacia/rickets). Sources: sun exposure (80%), diet, and supplements.

Vitamin D Deficiency

- Inadequate vitamin-D status causing impaired calcium and phosphate homeostasis and, in severe cases, defective bone mineralization. It may result from low exposure or intake, malabsorption, liver or kidney disease, or altered metabolism.

Vitamin D: Calcitriol

- Vitamin D is a fat-soluble vitamin (actually a prohormone) essential for calcium and phosphorus homeostasis and bone health. Cholecalciferol (D3) is synthesized in the skin from 7-dehydrocholesterol by UVB radiation, or obtained from diet (ergocalciferol/D2 from plants, D3 from animal sources).

Vitamin E

- A group of lipid-soluble antioxidants, principally tocopherols, that protect cell membranes from lipid peroxidation. Deficiency is uncommon but may cause hemolysis, neuropathy, ataxia, and muscle weakness.

Vitamin E: Tocopherol

- Vitamin E (alpha-tocopherol) is a fat-soluble vitamin and the body's primary lipid-soluble antioxidant, protecting polyunsaturated fatty acids in cell membranes from lipid peroxidation by free radicals (reactive oxygen species). It donates a hydrogen atom to lipid peroxyl radicals, breaking the chain

reaction of lipid peroxidation.

Vitamin K

- Fat-soluble vitamin for clotting factors II, VII, IX, X. Gut bacteria produce K2. Deficiency: bleeding, especially in newborns (prophylaxis). Sources: leafy greens (K1), fermented foods (K2). Antibiotics can cause deficiency.

Vitamin K Deficiency

- Coagulopathy, bleeding. Newborns receive prophylaxis. Antibiotics can cause deficiency.

Vitamin K Prophylaxis

- Intramuscular administration of vitamin K (1 mg, 0.5 mg for preterm) to all newborns at birth to prevent vitamin K deficiency bleeding (VKDB, formerly hemorrhagic disease of the newborn). Newborns have low stores of vitamin K (limited placental transfer, sterile gut, low in breast milk).

Vitamin K: Phylloquinone

- Vitamin K is a fat-soluble vitamin essential for the post-translational gamma-carboxylation of glutamate residues to gamma-carboxyglutamate (Gla) in specific clotting factors (II, VII, IX, X, protein C, protein S) and anticoagulant proteins (protein C, protein S, protein Z).

Vitamins

- Organic micronutrients required in small amounts for metabolic function, cannot be synthesized in sufficient quantities. Fat-soluble (A, D, E, K) absorbed with lipids, stored in liver/fat. Water-soluble (B complex, C) not stored, excess excreted.

Vitiligo

- An acquired disorder of depigmentation characterized by well-demarcated pale or depigmented macules and patches caused by loss or dysfunction of epidermal melanocytes.

Vitiligo and Pigmentation Disorders

- Vitiligo is a chronic skin condition characterized by the loss of melanocytes, resulting in well-defined white patches on the skin and sometimes mucous membranes.

Vitiligo Treatment

- Vitiligo treatment options include topical corticosteroids (first-line for limited disease), topical calcineurin inhibitors (tacrolimus, pimecrolimus for face and intertriginous areas), narrowband UVB

phototherapy (first-line for widespread disease), excimer laser for localized lesions, and systemic immunosuppressants for rapidly

Vitrectomy

- Surgical removal of the vitreous gel, typically via the pars plana approach, using a cutting probe under microscopic visualization. Indications include vitreous hemorrhage, complicated retinal detachment, macular holes, epiretinal membranes, intraocular foreign bodies, and endophthalmitis.

Vitreous Body

- The vitreous humor, or vitreous body, is a transparent, gel-like substance that fills the space between the lens and the retina, occupying approximately eighty percent of the volume of the eye, and it consists primarily of water (approximately ninety-nine percent) with a network of collagen fibrils and hyaluronic acid that give it its gel

Vitreous Chemistry

- Analysis of the fluid from the eye after death for substances such as electrolytes, glucose, urea, creatinine, and ethanol. It may support assessment of postmortem interval or metabolic disturbance but is not independently diagnostic of cause or time of death.

Vitreous Hemorrhage

- Bleeding into the vitreous cavity that causes sudden painless floaters, cobwebs, haze, or loss of vision. Common causes include proliferative diabetic retinopathy, retinal tear or detachment, trauma, and posterior vitreous separation.

Vitreous Humor

- The vitreous humor, or vitreous body, is a transparent, gel-like substance that fills the space between the lens and the retina, occupying approximately eighty percent of the volume of the eye, and it consists primarily of water (approximately ninety-nine percent) with a network of collagen fibrils and hyaluronic acid that give it its gel

Vitriolage

- The deliberate throwing or application of a corrosive chemical (vitriol—sulfuric acid, historically; now any strong acid/base) onto a person to disfigure, disable, or kill; a form of assault. Injuries: severe facial and body burns, scarring, blindness,

respiratory injury from fumes, and psychological trauma.

VLDL

- Very low-density lipoprotein. Transports endogenous triglycerides.

Vocal Cord Dysfunction

- Abnormal adduction of the vocal cords during inspiration, causing inspiratory stridor and dyspnea. It may mimic asthma but does not respond to bronchodilators. It is often misdiagnosed as refractory asthma. Diagnosis is by laryngoscopy showing paradoxical vocal cord motion. Treatment is speech therapy and avoiding triggers.

Vocal Cord Nodules

- Vocal cord nodules, also known as singer's nodules or contact nodules, are bilateral, symmetric, benign epithelial thickenings that develop on the anterior third of the vocal cords at the junction of the membranous and cartilaginous portions, caused by chronic vocal abuse, misuse, or overuse that results in repeated mechanical trauma to t

Vocal Cord Paralysis

- Immobility of one or both vocal cords due to disruption of the recurrent laryngeal nerve (RLN). Unilateral vocal cord paralysis: hoarseness, breathy voice, aspiration, and ineffective cough.

Vocal Cord Paresis

- Partial paralysis or weakness of one or both vocal cords due to impaired function of the recurrent laryngeal nerve (RLN) or the vagus nerve. Unilateral paresis is more common and presents with hoarseness, breathy dysphonia, vocal fatigue, and sometimes aspiration.

Vocal Cord Polyp

- Vocal cord polyps are benign, unilateral, sessile or pedunculated lesions that arise from the vocal cord mucosa, typically as a result of a single traumatic vocal event such as shouting, screaming, or coughing, causing hemorrhage and edema in the superficial lamina propria (Reinke space).

Vocal Cords

- The vocal cords, also known as vocal folds, are two pairs of horizontal folds of tissue within the.

Vocal Folds

- Paired mucosal folds within the larynx that vibrate

as air passes between them to produce sound. They also close to protect the airway and assist in raising intrathoracic pressure.

Vocal Resonance

- The transmission of vocal sounds to the chest wall, assessed by having the patient speak while auscultating. It is increased in consolidation and decreased in pleural effusion or pneumothorax. It correlates with bronchophony and whispered pectoriloquy.

Vocal-Cord Dysfunction

- Episodic inappropriate adduction of the vocal folds, especially during inspiration, causing throat tightness, noisy breathing, and apparent asthma-like attacks. Diagnosis is supported by laryngoscopy during symptoms; speech therapy and trigger management are central treatments.

Voice and Speech

- The products of complex coordinated activity of the respiratory, laryngeal, and articulatory systems.

Voiding Cystourethrogram (VCUG)

- Fluoroscopic imaging of the bladder and urethra during voiding, after filling the bladder with contrast via a catheter. Indications: evaluation of vesicoureteral reflux (VUR) in children with recurrent UTIs, posterior urethral valves in males, and post-surgical assessment of the bladder/urethra.

Volatile Anesthetics

- Halogenated inhaled anesthetic agents, including sevoflurane, desflurane, and isoflurane, that reversibly depress central nervous system function. They provide dose-dependent hypnosis and immobility but may cause hypotension, respiratory depression, postoperative nausea, and malignant hyperthermia in susceptible individuals.

Volkmann Contracture

- Permanent flexion deformity of the wrist, hand, and fingers caused by ischemic fibrosis of forearm muscles, usually after compartment syndrome or severe vascular injury. It can produce pain, weakness, sensory deficits, and major loss of hand function.

Volume of Distribution

- A theoretical pharmacokinetic parameter that relates the total amount of drug in the body to the plasma concentration, calculated as V_d equals amount of drug in the body divided by plasma con-

centration. It represents the apparent volume into which a drug distributes and is not a true anatomical volume.

Volutrauma

- Lung injury from excessive tidal volume causing alveolar overdistension. Prevented by low tidal volume ventilation (6 mL/kg IBW). Key component of ARDS management.

Volvulus

- Twisting of a loop of intestine around its mesenteric attachment point causing luminal obstruction and compromise of the mesenteric blood supply which can rapidly progress to bowel ischemia.

Vomiting

- Reflex expulsion of gastric contents. Causes: GI, neurological, metabolic, toxic, psychological. Treatment: antiemetics, identify cause, prevent aspiration.

Von Hippel-Lindau Disease

- Autosomal dominant syndrome caused by mutations in the VHL tumor suppressor gene on chromosome three predisposing to hemangioblastomas of the central nervous system and retina, clear cell renal cell carcinoma, pheochromocytoma, pancreatic cysts and neuroendocrine tumors, and endolymphatic sac tumors.

von Willebrand Disease

- The most common inherited bleeding disorder, caused by quantitative or qualitative deficiency of von Willebrand factor (VWF), a multimeric glycoprotein that mediates platelet adhesion to subendothelium and carries factor VIII. Type 1 (partial quantitative deficiency, 70-80%): mild mucocutaneous bleeding, easy bruising,

Voriconazole

- A triazole antifungal agent used in the treatment of invasive aspergillosis, candidemia, and other serious fungal infections. Voriconazole inhibits the fungal cytochrome P450 enzyme lanosterol 14-alpha demethylase, preventing ergosterol synthesis and disrupting fungal cell membrane integrity.

Voyeuristic Disorder

- A paraphilic disorder involving recurrent, intense sexual arousal from observing an unsuspecting person who is naked, disrobing, or engaging in sexual

activity. The individual has acted on these urges or they cause marked distress or interpersonal difficulty. Duration 6 months. Most common in males with onset in adolescence.

VRE

- Vancomycin-resistant *Enterococcus*, resistant to vancomycin through acquisition of *vanA* or *vanB* gene clusters. VRE is a significant cause of healthcare-associated infections, particularly in immunocompromised patients. Treatment options include linezolid, daptomycin, and quinupristin-dalfopristin (for *E. faecium* only).

Vulva

- External female genitalia, including the mons pubis (fatty mound over the pubic symphysis), labia majora (outer fatty folds covered with hair), labia minora (inner hairless folds), clitoris (erectile tissue homologous to the penis, with the glans being the most sensitive erogenous zone), vestibular glands (Bartholin's glands lubricating t

Vulvar Cancer

- Malignancy of the vulva, most commonly squamous cell carcinoma (90 percent), related to HPV (younger women, with VIN) or lichen sclerosus (older women); also melanoma, basal cell, Paget disease, Bartholin gland adenocarcinoma. Risk: age, HPV infection, smoking, immunosuppression, chronic vulvar dermatoses.

Vulvar vestibulitis

- Pain and inflammation of the vulvar vestibule.

Vulvitis

- Inflammation of the vulva (external female genitalia), often presenting with pruritus, burning, irritation, erythema, edema, and discharge. Vulvitis is rarely isolated and is most commonly part of vulvovaginitis (inflammation of both vulva and vagina).

Vulvodynia

- Pain in the vulva that may or may not be brought on by touch or pressure.

Vulvovaginal Candidiasis

- Symptomatic infection of the vulva and vagina, usually caused by *Candida albicans*. Pruritus, soreness, erythema, and thick discharge are common; diagnosis is clinical with microscopy or culture when recurrent, severe, atypical, or treatment-resistant.

Uveal Disease

- A disorder affecting the uvea, the pigmented middle layer of the eye containing the iris, ciliary body, and choroid. Inflammation, infection, tumors, or vascular disease may cause pain, floaters, light sensitivity, or vision loss and require ophthalmic assessment.

Ventricular Arrhythmia

- An abnormal heart rhythm arising from the ventricles. It may cause palpitations, dizziness, fainting, chest discomfort, or sudden cardiac arrest; fainting or severe symptoms require emergency care.

Visual Impairment

- Reduced vision that cannot be fully corrected by ordinary glasses or contact lenses. Causes include refractive error, cataract, glaucoma, retinal disease, optic-nerve injury, and neurologic disease; new or sudden loss of vision requires urgent assessment.

Vaginal Itching

- Itching or irritation of the vulva or vagina caused by candidiasis, bacterial vaginosis, sexually transmitted infections, dermatitis, hormonal changes, or other conditions. It may occur with discharge, odor, soreness, or burning; recurrent or unexplained symptoms require examination and appropriate testing rather than self-diagnosis.

Chapter 23

Pocket Dictionary: W Terms

Waardenburg Syndrome

- A group of inherited disorders characterized by sensorineural hearing loss and pigmentary abnormalities of the eyes, hair, or skin. Dystopia canthorum, limb abnormalities, or intestinal aganglionosis may occur in particular genetic forms.

WAGR Syndrome

- A contiguous-gene deletion syndrome involving Wilms tumor predisposition, aniridia, genitourinary anomalies, and variable developmental delay. It usually involves chromosome 11p13 and requires tumor surveillance and multidisciplinary care.

Waist Circumference

- Measure of central obesity. Thresholds: men >102 cm, women >88 cm (increased risk). Better predictor of metabolic risk than BMI. Waist-to-hip ratio also used.

Waldenstrom Macroglobulinemia

- A rare B-cell lymphoplasmacytic lymphoma characterized by bone marrow infiltration by lymphoplasmacytic cells and production of monoclonal IgM paraprotein. It accounts for less than 2 percent of hematologic malignancies and typically affects older adults (median age 63-65 years).

Waldenstrom's macroglobulinemia

- See Waldenström's macroglobulinemia.

Waldenström's macroglobulinemia

- An indolent B-cell lymphoplasmacytic lymphoma characterized by the production of a monoclonal IgM paraprotein, causing hyperviscosity syndrome with visual disturbances, bleeding, and neurological symptoms. It typically affects elderly patients and involves the bone marrow, lymph nodes, spleen, and liver.

Walker

- Walking aid for stability. Types: standard, rollator (wheels), knee walkers.

Walking Speed

- Gait speed: important functional marker. <0.8 m/s indicates disability risk.

Wandering Spleen

- Abnormal mobility of the spleen caused by laxity or absence of its supporting ligaments. It may be asymptomatic or cause abdominal pain, splenic enlargement, torsion, infarction, or acute abdomen.

Warfarin Management

- Warfarin (vitamin K antagonist) inhibits gamma-carboxylation of factors II, VII, IX, X, proteins C and S. Target INR: 2-3 (most indications), 2.5-3.5 (mechanical mitral valve, recurrent VTE on therapeutic INR). Initiation: 5 mg daily x2-3 days, then adjust per INR; genotype-guided (CYP2C9, VKORC1) algorithms.

Warthin Tumor

- A benign salivary-gland neoplasm composed of oncocytic epithelial and lymphoid tissue, usually arising in the parotid gland. It is strongly associated with tobacco exposure, may be multifocal or bilateral, and is generally managed according to symptoms, diagnosis, and growth.

Warts

- Benign epidermal hyperplasia caused by human papillomavirus (HPV). See Verruca (Wart).

Warty Dyskeratoma

- A benign follicular keratinizing tumor presenting as a solitary umbilicated papule or nodule, often on the head or neck. Histology shows a cup-shaped invagination filled with keratin and acantholytic

dyskeratotic cells.

Watchful Waiting

- Conservative observation of a cancer without immediate treatment, reserving therapy for symptom onset or progression; used in low-risk prostate cancer and indolent lymphomas.

Water Birth (Underwater Delivery)

- Delivery of a baby in a tub of warm water. Proponents claim: reduced pain, fewer perineal lacerations, and a gentler transition for the baby. Water birth is used in many birth centers and some hospitals. Criteria: low-risk pregnancy, term, clear fluid, and no signs of fetal distress.

Water Flosser

- Device using water stream to clean between teeth. Adjunct to brushing and flossing.

Water Intoxication

- A potentially life-threatening condition of hyponatremia (low serum sodium) caused by excessive water intake exceeding the kidneys' maximal free water excretion capacity (0.7-1 L/hour in healthy adults).

Water Soluble Toxins

- Excreted by kidneys. Dialysis effective for small molecules with low protein binding (ethanol, methanol, ethylene glycol, lithium, salicylates).

Water-Borne Disease

- Disease transmitted via contaminated water. Examples: cholera, typhoid, hepatitis A.

Water-Deprivation Test

- A supervised dynamic test assessing the kidney's ability to concentrate urine when free-water intake is withheld, followed when appropriate by desmopressin. It helps distinguish central diabetes insipidus, nephrogenic diabetes insipidus, and primary polydipsia but requires careful monitoring for hyponatremia and dehydration.

Watery Eyes

- Excessive tearing caused by irritation, allergy, infection, dry-eye reflex tearing, or blockage of the tear-drainage system. Persistent unilateral tearing or pain requires ocular assessment.

Waxy Casts

- Broad refractile cylindrical structures with smooth appearance indicating severe chronic kidney dis-

ease. Form from degeneration of granular casts in dilated tubules. Broad diameter reflects formation in dilated collecting ducts. Relatively specific for advanced CKD with significant tubular atrophy.

Weakness

- Reduced muscle strength or ability to generate force, which may be generalized or limited to one body region. Causes include nerve or muscle disease, stroke, spinal disorders, metabolic abnormalities, infection, and deconditioning; sudden one-sided weakness is an emergency.

Wear Debris

- Particles generated by mechanical abrasion or degradation of implant materials. Macrophage-mediated inflammation around debris can produce osteolysis, implant loosening, and failure of an arthroplasty.

Weaver Syndrome

- A congenital overgrowth and neurodevelopmental disorder associated with pathogenic variants in EZH2. It may cause prenatal overgrowth, advanced bone age, characteristic facial features, hypotonia, developmental delay, and variable intellectual disability.

Weber Test

- The Weber test is a screening test for lateralization of hearing that uses a vibrating tuning fork (typically 512 Hz) placed on the midline of the skull (vertex or forehead) to determine whether the sound is heard equally in both ears or louder in one ear, providing information about the type and side of hearing loss.

Wedge resection

- Surgical removal of a wedge-shaped piece of tissue, typically lung or breast. Used for diagnosis or treatment of localized lesions. Less extensive than lobectomy.

Wegener's Granulomatosis

- Wegener's granulomatosis, now called granulomatosis with polyangiitis, is a systemic vasculitis characterized by necrotizing granulomatous inflammation of the upper and lower respiratory tracts and pauci-immune necrotizing glomerulonephritis.

Weight Gain in Pregnancy

- Recommended gestational weight gain according

to the Institute of Medicine (IOM) guidelines, based on prepregnancy BMI. Underweight (BMI <18.5): 28-40 lbs (12.5-18 kg). Normal weight (BMI 18.5-24.9): 25-35 lbs (11.5-16 kg). Overweight (BMI 25-29.9): 15-25 lbs (7-11.5 kg). Obese (BMI 30): 11-20 lbs (5-9 kg).

Weight Loss

- The intentional reduction of body weight, typically through a combination of dietary modification, increased physical activity, and behavioral changes. Achieving sustainable weight loss requires creating a consistent calorie deficit while preserving lean muscle mass through adequate protein intake and strength training.

Weight Loss and Management

- Weight loss and management encompasses the ongoing process of achieving and maintaining a healthy body weight through sustainable lifestyle modifications.

Weight Loss Medications

- Pharmacologic agents approved for the treatment of obesity when used in conjunction with lifestyle modifications.

Weil's Disease

- A zoonotic bacterial infection caused by spirochaetes of the genus *Leptospira*, transmitted through contact with water or soil contaminated with the urine of infected animals (rodents, cattle, dogs). See Leptospirosis.

Wen (Epidermoid Cyst)

- A common benign cyst lined by stratified squamous epithelium with keratin debris. See Epidermal Inclusion Cyst.

Werner Syndrome

- An autosomal-recessive progeroid disorder caused most often by pathogenic variants in the WRN gene. It typically presents after puberty with premature ageing, short stature, bilateral cataracts, scleroderma-like skin changes, thin limbs, diabetes, osteoporosis, vascular disease, and increased risk of malignancy.

Wernicke's Encephalopathy

- See toxicology.

Wernicke-Korsakoff Syndrome

- A severe neurological disorder caused by thiamine

(Vitamin B1) deficiency, most commonly associated with chronic alcoholism, severe malnutrition, or hyperemesis gravidarum.

West Nile Virus

- A mosquito-borne flavivirus, endemic in the US, Africa, Europe, and Asia. Most infections (80%) are asymptomatic. West Nile fever (20%): fever, headache, myalgia, arthralgia, rash. Neuroinvasive disease (<1%): meningitis, encephalitis, acute flaccid paralysis (anterior horn cell involvement).

West Syndrome

- An infantile epileptic encephalopathy characterized by epileptic spasms, hypsarrhythmia on electroencephalography, and developmental regression or arrest. Causes include structural brain disease, genetic disorders, metabolic disease, or acquired brain injury.

Westermark Sign

- Focal area of oligemia (decreased vascularity) distal to an occluded pulmonary artery segment on chest radiography in acute Pulmonary Embolism. See Pulmonary Embolism (PE).

Wharton Jelly

- A gelatinous substance (mucopolysaccharide matrix) surrounding the umbilical cord vessels, composed primarily of hyaluronic acid and chondroitin sulfate. It protects the umbilical vessels from compression and kinking during fetal movement and uterine contractions.

Wheal

- Raised, itchy swelling of skin (urticaria/hives). Caused by mast cell degranulation and histamine release. Examples: insect bite, allergic reaction, dermatographism.

Wheelchair

- Mobility device for patients with walking difficulties. Types: manual, power. Components: seat, backrest, wheels, footrests, armrests. Assessment: seating evaluation, pressure relief, posture. Training: mobility, transfers, safety.

Wheelchair Seating and Positioning

- Assessment and configuration of a wheelchair, cushion, supports, and posture to optimize stability, propulsion, comfort, function, pressure distribution, and prevention of musculoskeletal deformity.

Wheelchair Skills

- Training for wheelchair users: propulsion, maneuvering, obstacles, slopes, stairs, transfers.

Wheezes

- High-pitched, continuous, musical sounds heard primarily during expiration, caused by airway narrowing. They are a hallmark of asthma and COPD. Expiratory wheezing indicates intrathoracic airway obstruction; inspiratory wheezing indicates extrathoracic obstruction.

Wheezing

- A continuous, high-pitched, musical adventitious lung sound produced by turbulent airflow through narrowed or compressed lower airways during expiration (and occasionally inspiration). Pathophysiology: bronchospasm, mucosal edema, airway inflammation, or intraluminal secretions.

Whiplash Injury

- An acceleration-deceleration injury to the cervical spine, most commonly resulting from rear-end motor vehicle collisions. Mechanism: sudden hyperextension followed by hyperflexion of the neck causes stretching or tearing of cervical muscles, ligaments, facet joint capsules, and annulus fibrosus of intervertebral discs.

Whipple Disease

- A rare systemic bacterial infection caused by *Tropheryma whipplei*, affecting the small intestine and multiple organ systems. Presents with arthralgias (prodromal, years before GI symptoms), weight loss, chronic diarrhea, steatorrhea, and abdominal pain.

Whipple Procedure

- Pancreaticoduodenectomy for cancers of the pancreatic head, distal bile duct, or ampulla. Involves removal of the pancreatic head, duodenum, distal bile duct, and gallbladder with three anastomoses: hepaticojejunostomy, pancreaticojejunostomy, and gastrojejunostomy. The pylorus-preserving variant maintains the entire stomach.

Whipple Triad

- Diagnostic criteria for Insulinoma: (1) Symptoms of hypoglycemia (diaphoresis, confusion), (2) Documented low plasma glucose (<55 mg/dL), and (3) Prompt relief of symptoms following glucose administration. See Insulinoma.

Whipworm (Trichuriasis)

- Intestinal infection caused by the soil-transmitted helminth *Trichuris trichiura* (whipworm), endemic in tropical and subtropical regions with poor sanitation. Humans acquire infection by ingesting embryonated eggs from contaminated soil, food, or water.

Whispered Pectoriloquy

- The ability to hear whispered words clearly through the stethoscope over consolidated lung tissue. It is a sign of pulmonary consolidation. Normally, whispered words are barely audible over the lung fields. It is one of the classic signs of pneumonia on physical examination.

White Blood Cell Count

- WBC count: the total number of leukocytes per unit volume (normal $4.0\text{--}11.0 \times 10^9/\text{L}$), part of the CBC with differential (neutrophils, lymphocytes, monocytes, eosinophils, basophils). Leukocytosis: infection (neutrophilia), inflammation, stress, steroids, leukemia (marked), corticosteroids.

White Blood Cell Labeling

- Nuclear medicine technique labeling WBCs with Tc-99m. Used for infection localization.

White Cell

- The cells of the immune system that defend the body against infection and foreign substances. Produced in the bone marrow from hematopoietic stem cells.

White Cell Count

- A laboratory measurement of the number of white blood cells (leukocytes) in a given volume of blood, part of the complete blood count (CBC). Normal range: $4,000\text{--}11,000$ cells/L ($4\text{--}11 \times 10^9/\text{L}$). Differential white cell count measures the proportion of each leukocyte type: neutrophils, lymphocytes, monocytes, eosinophils, basophils.

White Infarction

- White infarction, also known as anemic infarction, occurs in solid organs with end-arterial circulation where collateral blood supply is limited, resulting in pale, well-demarcated areas of coagulative necrosis.

White Matter

- Regions of the brain and spinal cord composed primarily of myelinated nerve fibers, responsible for

transmitting signals between brain regions.

Whites (Leucorrhoea)

- A non-bloody, physiological or pathological vaginal discharge. Physiological leucorrhoea: normal, clear to whitish, odourless discharge that varies with the menstrual cycle (increased at ovulation, pre-menstrually, during pregnancy, sexual arousal, and with oral contraceptive use).

Whole Exome Sequencing

- Sequencing only protein-coding regions (1-2% of genome). More cost-effective than whole genome. Detects 85% of disease-causing variants.

Whole Genome Sequencing

- Determining complete DNA sequence of an individual's genome. Cost decreasing, clinical applications expanding. Identifies coding and non-coding variants.

Whole-Body Scan

- Acquisition of images covering entire body. Used for oncology staging, infection localization. PET/CT typically neck-to-mid-thigh or full body.

Whole-Brain Radiation Therapy (WBRT)

- Radiation directed at the entire brain, used for multiple brain metastases, leptomeningeal disease, and prophylactic cranial irradiation in small cell lung cancer and acute lymphoblastic leukemia.

Whole-Genome Sequencing

- Sequencing of nearly the complete nuclear genome, including coding and noncoding regions, together with mitochondrial DNA when covered by the assay. It can detect a broad range of variants but generates substantial interpretation and incidental-finding challenges.

Whooping Cough

- A highly contagious bacterial infection caused by *Bordetella pertussis*, producing characteristic paroxysmal cough with whooping inspiratory sound. See Pertussis.

Williams Syndrome

- A microdeletion syndrome involving the elastin gene on chromosome 7q11.23. Features: elfin facies, supravalvular aortic stenosis, hypercalcemia, intellectual disability with strong verbal skills, and overly friendly personality. Often associated with connective tissue abnormalities.

Wilms Tumor

- Most common renal malignancy in children. Large abdominal mass. Associated with WAGR, Beckwith-Wiedemann, and hemihypertrophy. Treated with surgery and chemotherapy. Excellent prognosis exceeding 90%.

Wilms' tumor (Nephroblastoma)

- The most common renal malignancy in children, accounting for 95% of pediatric renal tumors (peak incidence 2-5 years). It is an embryonal tumor arising from metanephric blastemal cells.

Wilson Disease

- Autosomal recessive copper metabolism disorder (ATP7B gene). Copper accumulation in liver, brain, cornea. Kayser-Fleischer rings, hepatopathy, movement disorder, psychiatric symptoms. Treatment: chelation (penicillamine, trientine), zinc, liver transplant.

Wilson's disease

- An autosomal recessive disorder of copper metabolism caused by mutations in the ATP7B gene, leading to impaired biliary copper excretion and toxic copper accumulation in the liver, brain, cornea, and other organs.

Windowing

- Adjusting CT image brightness and contrast to optimize visualization of specific tissues. Bone window: wide window width, high level. Soft tissue window: narrow width, mid level. Lung window: wide width, low level.

Windpipe

- The windpipe, a 10-12 cm fibrocartilaginous tube from the larynx (C6) to the carina (T4/T5) where it bifurcates into main bronchi. See Trachea.

Wireless Motility Capsule

- Ingestible capsule that measures pH, pressure, and temperature throughout the gastrointestinal tract. Provides regional transit times through the stomach, small bowel, and colon in a single test. The pH sensor identifies transition points between segments based on characteristic pH changes.

Wiskott-Aldrich Syndrome

- X-linked recessive disorder (WAS gene mutation, encodes WASP protein, regulates actin polymerization in hematopoietic cells). Triad: microthrom-

bocytopenia (small platelets, <70 fl), eczema, recurrent infections (encapsulated bacteria, viruses). Combined immunodeficiency with progressive T-cell lymphopenia.

Withdrawal Reflex

- A protective polysynaptic spinal reflex in which a painful stimulus causes flexion (withdrawal) of the stimulated limb, removing it from the source of injury, with reciprocal extension of the contralateral limb (crossed extensor reflex) to maintain posture.

Withdrawal Symptoms

- See Withdrawal Syndrome. used primarily for erectile dysfunction (ED) and pulmonary arterial hypertension (PAH, as Revatio).

Withdrawal Syndrome

- A predictable set of signs and symptoms that occur when a drug is discontinued or the dose is reduced after chronic use, reflecting the physiological rebound from drug adaptation.

Wolff-Parkinson-White Syndrome

- A pre-excitation syndrome caused by an accessory pathway (bundle of Kent) connecting atria and ventricles, bypassing the AV node. ECG shows short PR interval (<0.12 s), delta wave (slurred upstroke of QRS), and widened QRS.

Wolfram Syndrome

- Autosomal recessive disorder caused by mutations in the WFS1 gene encoding wolframin. Classic triad of diabetes mellitus, optic atrophy, and diabetes insipidus. Additional features include sensorineural deafness, neurodegeneration, and psychiatric illness. Onset of diabetes typically in the first decade.

Wolman's Disease

- A severe lysosomal acid-lipase deficiency presenting in infancy with hepatosplenomegaly, adrenal calcification, malabsorption, vomiting, diarrhea, and failure to thrive. It is a severe form of lysosomal acid-lipase deficiency and can be fatal early without treatment.

Wood Lamp Examination

- Inspection of skin under long-wave ultraviolet A light in a darkened room. It can accentuate pigmentary differences and produce characteristic fluorescence in selected infections, but findings are not independently diagnostic.

Woods Corkscrew Maneuver

- An internal rotation maneuver for shoulder dystocia where the operator inserts a hand and rotates the fetal shoulders 180 degrees by applying pressure to the posterior aspect of the anterior shoulder. This disimpacts the shoulder from behind the symphysis pubis. Can be repeated if the shoulder re-impacts in the new position.

Woods Screw Maneuver

- See Woods Corkscrew Maneuver in the main section above.

Word Recognition Scoring

- See Speech Discrimination.

Word Salad

- Severe disorganized speech consisting of a jumble of words and phrases that lack meaning or logical connection. It is a hallmark of formal thought disorder in schizophrenia and represents the most severe form of disorganized speech. It reflects profound disruption of language processing.

Wound

- An injury to living tissue caused by mechanical force; in forensic medicine, wounds are classified by mechanism: abrasion (scrape—loss of epidermis, no bleeding), laceration (tear from blunt force—ragged edges, tissue bridges), incised/incision (clean cut by sharp edge—linear, edges gape), stab/puncture (penetrating injury by pointed weap

Wound Care

- Management of traumatic wounds: irrigation, debridement, closure (sutures, staples, glue), dressing. Tetanus prophylaxis based on vaccination history and wound type. Antibiotics for contaminated wounds, animal/human bites, or signs of infection. Healing: primary, secondary, tertiary intention.

Wound Dehiscence

- Separation of the layers of a surgical wound after closure, most commonly occurring within 3-14 days postoperatively.

Wound Healing

- Complex process of tissue repair. Phases: hemostasis, inflammation, proliferation, remodeling. Implant integration depends on healing.

Wound Healing Phases

- Wound healing occurs in overlapping phases: hemostasis (immediate, platelet aggregation and clot formation), inflammation (days 1 to 4, neutrophils and macrophages clear debris and bacteria), proliferation (days 4 to 21, granulation tissue, angiogenesis, epithelialization, collagen deposition), and remodeling (day 21 to 2 years, collagen

hydrocele, and elephantiasis.

Wound Infection

- See infection.

Wound VAC

- Vacuum-assisted closure for wound healing. Foam dressing connected to vacuum pump. Removes exudate, reduces edema, promotes granulation. Used for complex wounds, surgical sites, pressure ulcers.

Wrisberg Ganglion

- A sensory ganglion associated with the intermediate nerve of the facial nerve.

Wrist Drop

- See Radial Nerve Palsy (Wrist Drop).

Wrist Joint

- Condylodisc synovial joint between distal radius and articular disc proximally and scaphoid/lunate/triquetrum distally. Allows flexion, extension, radial/ulnar deviation, circumduction. The triangular fibrocartilage complex absorbs ulnar side load.

Wrist Pain

- Pain in or around the wrist, commonly caused by sprain, fracture, tendinitis, arthritis, or nerve compression such as carpal tunnel syndrome. Persistent pain, deformity, numbness, or weakness requires assessment.

Wound Complication

- A problem that develops during healing, such as infection, separation of wound edges, bleeding, fluid collection, tissue death, or excessive scarring. Increasing pain, redness, swelling, drainage, fever, or wound opening should be assessed promptly.

Wuchereria

- A genus of filarial nematodes that includes *Wuchereria bancrofti*, a major cause of lymphatic filariasis. Infection is transmitted by mosquitoes and may cause lymphangitis, chronic lymphedema,

Chapter 24

Pocket Dictionary: X Terms

X Chromosome

- One of the two human sex chromosomes, carrying genes involved in sex development and many non-sex-specific biological functions. Individuals typically have one or two X chromosomes, and dosage is partly regulated by X-chromosome inactivation.

X-Inactivation

- Process inactivating one X chromosome in female mammals, forming Barr body. Occurs early in development, random in most tissues. Ensures dosage compensation.

X-Linked Agammaglobulinemia

- Caused by mutations in the BTK gene (Bruton tyrosine kinase), essential for B-cell development and maturation. Results in absent or very low mature B cells (<1% CD19+ cells), profoundly low all immunoglobulin isotypes, and absent antibody responses.

X-Linked Inheritance

- Transmission of a pathogenic variant on the X chromosome. Phenotypes differ by sex, X-inactivation, and variant mechanism; fathers do not transmit an X-linked variant to their sons.

X-linked retinoschisis

- An inherited vitreoretinal dystrophy caused by mutations in the RS1 gene, characterized by splitting of the retinal nerve fibre layer producing a spoke-wheel pattern of cystic spaces at the macula and peripheral retinal schisis.

X-ray

- See radiography.

X-Ray (Plain Radiography)

- The most common form of medical imaging, using X-rays to create a 2D projection image. Uses: chest

(pneumonia, pneumothorax, effusion, cardiomegaly, mass), abdomen (obstruction, perforation, stones), musculoskeletal (fractures, dislocation, arthritis, osteomyelitis), and spine (fracture, alignment, degeneration).

X-ray (Radiography)

- Projectional imaging using ionizing radiation. Dense structures (bone, contrast) appear white; air appears black. Fast, inexpensive, widely available. Two-dimensional projection of three-dimensional structures. Mammography: specialized breast X-ray for screening.

X-ray Attenuation

- See CT.

X-ray Shield

- Lead apron, thyroid shield for radiation protection during imaging.

X-Rays (Röntgen Rays)

- The most common form of medical imaging, using X-rays to create a 2D projection image. See X-Ray (Plain Radiography).

Xanax

- A class of drugs that enhance the effect of the inhibitory neurotransmitter gamma-aminobutyric acid at the GABA-A receptor, increasing the frequency of chloride channel opening and producing anxiolytic, sedative, anticonvulsant, and muscle relaxant effects. See benzodiazepines.

Xanthelasma

- Xanthelasma palpebrarum are soft, yellow, lipid-filled papules or plaques on the medial canthus of the eyelids (upper more commonly than lower), the most common form of cutaneous xanthoma. They are composed of foam cells (lipid-laden macrophages)

in the superficial dermis.

Xanthine Oxidase Inhibitor

- A class of medications that inhibit xanthine oxidase, the enzyme that catalyzes the final steps of purine metabolism: hypoxanthine to xanthine and xanthine to uric acid. By inhibiting this enzyme, XOIs reduce uric acid production and lower serum uric acid levels, used for management of chronic gout and tumor lysis syndrome.

Xanthogranulomatous

- Chronic destructive renal infection with replacement of parenchyma by lipid-laden macrophages. See xanthogranulomatous pyelonephritis.

Xanthogranulomatous Pyelonephritis

- Chronic destructive renal infection with replacement of parenchyma by lipid-laden macrophages. Associated with obstructing stones and Proteus infection. Enlarged kidney with scattered low-attenuation areas on CT resembling a bear paw. Treated with nephrectomy as the kidney is non-functional and harbors chronic infection.

Xanthoma

- Cutaneous deposits of lipids (cholesterol, triglycerides) in the dermis or tendons, presenting as yellow-orange papules, nodules, or plaques.

Xenobiotic

- Foreign chemical substance in body. Drugs, pollutants, pesticides, industrial chemicals. Metabolized by liver enzymes.

Xenograft

- Transplanted tissue or synthetic material. See graft.

Xenograft (Heterograft)

- A surgical graft taken from a donor of a different species (xenos = foreign, Greek). Xenografts are used as temporary biological dressings or as tissue replacements.

Xeroderma

- Abnormally dry skin, common in elderly patients and winter months. See Xerosis.

Xeroderma Pigmentosum

- A rare autosomal recessive disorder of nucleotide excision repair (NER) characterized by extreme sun sensitivity and a 1000-fold increased risk of skin

cancers (basal cell carcinoma, squamous cell carcinoma, and melanoma, often in childhood). Mutations occur in one of eight genes (XPA-XPG and XP-variant/polymerase eta).

Xerophthalmia

- See clinical nutrition.

Xerosis

- Abnormally dry skin, common in elderly patients and winter months. It results from decreased sebaceous and sweat gland activity, impaired skin barrier function, and environmental factors. Treatment includes regular emollient application (cream or ointment rather than lotion), gentle cleansing, and humidification.

Xerostomia

- Dry mouth due to decreased saliva. Common in elderly from medications (anticholinergics, diuretics, antidepressants), Sjogren syndrome, radiation. Symptoms: difficulty swallowing, speaking, tasting, dental caries. Treatment: hydration, sugarless gum/candy, saliva substitutes, pilocarpine.

Xerostomia (Dry Mouth)

- A subjective sensation of oral dryness resulting from reduced or absent saliva production (salivary gland hypofunction). Xerostomia affects approximately 20% of the general population, increasing with age and with the number of medications taken.

Xylitol Toxicity

- See toxicology.

Xanthomatosis

- A disorder in which fatty deposits called xanthomas develop in the skin, tendons, or organs. It may be associated with inherited or acquired lipid disorders and can signal increased cardiovascular or organ-specific risk requiring medical evaluation.

Chapter 25

Pocket Dictionary: Y Terms

Y Chromosome

- The sex chromosome that typically carries the SRY gene and other genes involved in testicular development and male sex differentiation. It is present in most individuals with a typical 46,XY karyotype and is transmitted through the paternal line.

YAG Capsulotomy

- YAG laser posterior capsulotomy is a minimally invasive laser procedure used to treat posterior capsule opacification, the most common late complication of cataract surgery, in which the posterior lens capsule becomes opacified by proliferation of residual lens epithelial cells, causing decreased vision.

Yaws

- A chronic infectious disease caused by *Treponema pallidum pertenue* and transmitted mainly through skin-to-skin contact in children. It begins with a skin lesion and may progress to destructive lesions of skin, bone, and cartilage if untreated.

Yaws (Frambesia)

- A chronic, non-venereal treponematoses caused by *Treponema pallidum* subspecies *pertenue*, closely related to the syphilis spirochete (*T. pallidum pallidum*).

Yeast

- Unicellular fungi that reproduce by budding (blastoconidia formation) or fission. The clinically most important yeasts are *Candida* species (*Candida albicans*, *C. glabrata*, *C. parapsilosis*, *C. tropicalis*, *C. krusei*, *C. auris*) and *Cryptococcus* species (*C. neoformans*, *C. gattii*).

Yeast Infection

- Vulvovaginal candidiasis: infection by *Candida albicans* (most common), causing vulvar and vagi-

nal itching, burning, thick white 'cottage cheese' discharge, dyspareunia, and erythema; more common in pregnancy, diabetes, antibiotic use, and immunosuppression. Diagnosis: clinical; wet mount (budding yeast/hyphae), KOH, or culture.

Yellow Fever

- A mosquito-borne viral hemorrhagic fever caused by yellow fever virus (Flavivirus), endemic in tropical Africa and South America. Incubation 3-6 days. Presents with fever, myalgia, headache, and vomiting. A proportion of patients progress to toxic phase with jaundice, hemorrhagic fever, and multi-organ failure (hepatorenal).

Yellow-nail syndrome

- A rare triad of yellow, thickened, slow-growing nails, lymphedema, and chronic respiratory disease including bronchiectasis and pleural effusions. The nail changes result from impaired lymphatic drainage affecting nail growth. It typically affects middle-aged adults.

Yersinia Enterocolitica

- A gram-negative coccobacillus of the family Enterobacteriaceae, a common cause of enterocolitis and mesenteric lymphadenitis, especially in children and young adults in temperate climates.

Yersinia pestis

- A gram-negative rod that causes plague, a severe and potentially fatal infection. The organism is transmitted by flea bites or direct contact with infected animals. Bubonic plague presents with painful lymphadenopathy (buboes). Pneumonic plague involves the lungs and is transmitted by respiratory droplets.

Yolk Sac

- The first extraembryonic structure visible on ul-

trasound at 5-6 weeks gestation. It provides early nutrition to the embryo before the placental circulation is established, produces blood cells until the fetal liver takes over, and forms the primitive gut. A normal yolk sac is 3-6 mm in diameter.

Yolk-sac tumor

- A malignant germ-cell tumor that commonly occurs in children and produces alpha-fetoprotein.

Yttrium-90

- Beta emitter, half-life 64 hours. Used in radioembolization (Y-90 microspheres) for liver cancer. Bremsstrahlung imaging possible.

Yersinia Infection

- Infection caused by Yersinia bacteria, commonly producing diarrhea, abdominal pain, fever, or vomiting. Some cases mimic appendicitis; severe illness, dehydration, bloody diarrhea, or persistent symptoms requires medical assessment.

Chapter 26

Pocket Dictionary: Z Terms

Z-Plasty

- Surgical technique for contracture release using Z-shaped incision. Lengthens tissue, changes scar orientation.

Zalcitabine

- A nucleoside reverse transcriptase inhibitor (NRTI) that was previously used in combination antiretroviral therapy for HIV-1 infection but is now rarely used because more effective and better-tolerated agents are available.

Zavanelli Maneuver

- The final, rarely performed maneuver for an irreducible shoulder dystocia: the fetal head is replaced into the vagina (cephalic replacement) and the patient undergoes emergency cesarean delivery.

Zellweger Syndrome

- The severe end of the Zellweger spectrum of peroxisomal biogenesis disorders, usually caused by PEX-gene variants. It presents in neonates with hypotonia, seizures, feeding difficulty, distinctive facial features, liver dysfunction, neuronal migration defects, and profound developmental impairment.

Zenker Diverticulum

- Pulsion diverticulum at the junction of the thyropharyngeus and cricopharyngeus muscles in the proximal esophagus. It presents with dysphagia, regurgitation of undigested food, halitosis, and aspiration. Diagnosis is by barium swallow showing the characteristic outpouching.

Zenker's diverticulum

- See Esophageal Diverticulum.

Zidovudine (AZT)

- A thymidine-analogue nucleoside reverse-transcriptase inhibitor that is phosphorylated

intracellularly and incorporated into HIV DNA, causing chain termination. It is now used selectively because combination antiretroviral regimens have better tolerability and resistance profiles.

Ziehl-Neelsen Stain

- An acid-fast stain retaining carbol fuchsin in waxy mycolic-acid cell walls despite acid-alcohol decolorization, with methylene blue counterstain rendering acid-fast organisms red. Clinically important for *Mycobacterium tuberculosis*, *M. leprae*, *Nocardia*, and *Cryptosporidium* oocysts, it is a rapid initial sputum test for tuberculosis.

Zieve's Syndrome

- A triad of haemolytic anemia, hyperlipidaemia, and jaundice occurring in the setting of chronic alcohol use disorder and alcoholic liver disease (acute alcoholic steatohepatitis or alcoholic cirrhosis).

Zika

- Zika virus: a mosquito-borne (*Aedes*) flavivirus, also sexually transmitted. Usually asymptomatic or mild (fever, rash, conjunctivitis, arthralgia). Major concern: congenital Zika syndrome if acquired in pregnancy—microcephaly, brain abnormalities, eye defects; also Guillain-Barré syndrome.

Zika virus

- A positive-sense RNA virus belonging to the Flaviviridae family that causes mild febrile illness. Zika virus is transmitted by *Aedes* mosquitoes and can also be transmitted sexually. Infection during pregnancy can cause congenital Zika syndrome including microcephaly. Most infections are asymptomatic or mild.

Zinc

- An essential trace element that acts as a structural or catalytic component of numerous proteins and en-

zymes, including DNA-binding zinc-finger proteins. Deficiency may impair growth, wound healing, immunity, taste, and reproductive function.

Zinc Deficiency

- A nutritional or malabsorptive disorder causing impaired wound healing, dermatitis, alopecia, diarrhoea, taste disturbance, and increased infection risk. Causes include inadequate intake, malabsorption, chronic diarrhoea, excessive losses, and inherited zinc-transport disorders.

Zinc Ointment (Zinc Oxide)

- Zinc oxide is a topical agent with mild antiseptic, astringent, and barrier-protective properties, used in a wide range of dermatological preparations. Mechanism: zinc is an essential trace element and cofactor for numerous metalloenzymes involved in wound healing (matrix metalloproteinases, DNA polymerases, superoxide dismutase).

Zirconia

- Strong ceramic material for dental restorations. High flexural strength. Can be monolithic or layered with porcelain.

Zollinger-Ellison Syndrome

- Gastrin-secreting tumor called gastrinoma causing gastric acid hypersecretion, severe peptic ulcers, and diarrhea. Most tumors are located in the pancreas or duodenum with approximately sixty percent being malignant. Presents with multiple or refractory peptic ulcers, diarrhea, and esophageal reflux.

Zooglea

- A gelatinous mass of bacteria embedded in a polysaccharide matrix.

Zoon Balanitis

- A chronic benign inflammatory disorder of the glans penis, usually presenting as a sharply demarcated shiny orange-red or rust-colored patch in uncircumcised men.

Zoonoses

- Infectious diseases naturally transmitted between vertebrate animals and humans. Transmission may occur through direct contact, animal products, food, water, or arthropod vectors. Examples include rabies, brucellosis, leptospirosis, Lyme disease, psittacosis, and toxoplasmosis.

Zoonosis

- Disease transmitted from animals to humans. Examples: rabies, Lyme disease, avian influenza, COVID-19.

Zoonotic infections

- Infections that are naturally transmitted between animals and humans. Major zoonotic infections include rabies, brucellosis, leptospirosis, and psittacosis. The animal reservoir can be domestic or wild animals. Transmission occurs through direct contact, ingestion, or vectors.

Zoster

- A painful, vesicular rash caused by reactivation of latent varicella-zoster virus (VZV, human herpesvirus 3) in the dorsal root ganglia. See herpes zoster (shingles).

Zygoma (Zygomatic Bone)

- The cheekbone, a paired, quadrangular bone of the facial skeleton (viscerocranium) that forms the prominence of the cheek and the lateral wall and floor of the orbit.

Zygomatic Bone

- A paired facial bone forming the prominence of the cheek and part of the lateral wall and floor of the orbit. It articulates with the frontal, sphenoid, temporal, and maxillary bones and transmits forces from the face to the cranial skeleton.

Zygomycosis

- An infection caused by fungi historically classified in the class Zygomycetes, including organisms now placed among mucormycetes. It can produce rapidly invasive disease of the nose, sinuses, orbit, lungs, skin, or gastrointestinal tract, especially in immunocompromised patients.

Zygote

- The single diploid cell formed by the fusion of male and female gametes (spermatozoon and secondary oocyte) at fertilization. The zygote contains 46 chromosomes (23 from each parent) and represents the beginning of a new, genetically unique individual.

Zymogen

- An inactive enzyme precursor that is converted into an active enzyme, often by cleavage.

Zymogen (Proenzyme)

- An inactive precursor of an enzyme that requires proteolytic cleavage to become catalytically active.